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2020-21 EDITION

Pathfinder

CDS

COMBINED DEFENCE SERVICES

Entrance Examination

More than 8000 MCQs

Complete Coverage of Syllabus

Chapterwise Division of Previous Years' Questions

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Compiled & Edited by
Arihant 'Expert Team'

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ARIHANT PUBLICATIONS (INDIA) LIMITED

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'Ramchhaya' 4577/15, Agarwal Road, Darya Ganj, New Delhi -110002
Tele: 011- 47630600, 43518550

Head Office
Kalindi, TP Nagar, Meerut (UP) - 250002
Tel: 0121-7156203, 7156204

Sales & Support Offices
Agra, Ahmedabad, Bengaluru, Bareilly, Chennai, Delhi, Guwahati, Hyderabad, Jaipur, Jhansi, Kolkata, Lucknow, Nagpur & Pune.

ISBN 978-93-24196-24-8

PO No : TXT-XX-XXXXXXX-X-XX

Published by Arihant Publications (India) Ltd.

For further information about the books published by Arihant, log on to www.arihantbooks.com or e-mail at info@arihantbooks.com

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COMBINED DEFENCE SERVICES

ABOUT THE EXAMINATION

A career in Defence services is the much sought after occupation today for those young and courageous youths of the country who are willing to dedicate their lives to defend the country and its people, the Combined Defence Services Examination is the first test before they can join one of the best defence forces in the world i.e. the Indian Armed Forces.

UPSC conducts the Combined Defence Services Exam twice every year generally in February and August for recruiting officers for the Indian Military Academy, Naval Academy, Air Force Academy and Officers Training Academy. Male candidates can join IMA, Naval and Air Force but for female candidates OTA in army is available.

The examination comprises of two stages-the first stage consists of written test and those who qualify the written test are called for interview by Service Selection Board (SSB) for Intelligence and Personality Test.

NATIONALITY

A candidate must be either

(i) Indian citizen, or

(ii) A subject of Bhutan, or

(iii) A subject of Nepal, or

(iv) A Tibetan refugee who came over to India before 1st January, 1962 with the intention of permanently setting in India, or

(v) A person of Indian origin who has migrated from Pakistan, Burma, Sri Lanka and East African countries of Kenya, Uganda, the United Republic of Tanzania, Zambia, Malawi, Zaire and Ethiopia and Vietnam with the intention of permanently setting in India.

Provided that a candidate belonging to categories (ii), (iii), (iv) and (v) above shall be a person in whose favour a certificate of eligibility has been issued by the Government of India. Certificate of eligibility, will not , however, be necessary in the case of candidate who are Gorkha subjects of Nepal.

AGE LIMIT, SEX AND MARITAL STATUS

(i) For IMA and Indian Naval Academy-unmarried male candidates having age not less than 18 years on 1st July and not more than 23 years on 2nd July in accordance with the year of examination are eligible.

(ii) For Air Force Academy-male candidates having age not less than 19 years as on 1st July and not more than 23 years as on 2nd July in accordance with the year of examination. (upper age relation of upto 26 years for candidates holding Commercial Pilot License issued by DGCA).

(iii) For Officer's Training Academy-male candidates (married or unmarried) and female candidates (unmarried and issueless widows or divorces who have not remarried) having age not less than 18 years as on 1st July and not more than 24 years as on 2nd July in accordance with the year of examination are eligible.

EDUCATIONAL QUALIFICATION

(i) For IMA/OTA a degree from a recognised university or equivalent.

(ii) For Naval Academy degree in engineering from a recognised university.

(iii) For Air Force Academy degree from a recognised university (with Physics and Mathematics at 10+2 level) or Bachelor of Engineering.

EXAMINATION PATTERN

(i) The written examination will be as follows:

Subject	Duration	No. of Questions	Maximum Marks
English	2 hours	120	100
General Knowledge	2 hours	120	100
Elementary Mathematics	2 hours	100	100

Note- For admission in OTA, candidates are required to give only English and General knowledge papers of 200 marks.

One third of the marks will be deducted for each wrong answer.

(ii) The SSB procedure consists of two stage selection process. Only those candidates who clear the stage I are permitted to appear for stage II.

(a) Stage I comprises of Officer Intelligence Rating (OIR) tests and Picture Perception, Description Test (PP and DT)

(b) Stage II comprises of Interview, Group Testing Officer tasks, Psychology Tests and the Conference. These tests are conducted over 4 days.

SYLLABUS

PAPER I ENGLISH

The question paper will be designed to test the candidates' understanding of English and workman-like use of words. Questions in English are from Synonyms, Antonyms, Reading Comprehension, Para jumbles, Error spotting, Jumbled sentences, Sentence connection and till in the blanks.

PAPER II GENERAL KNOWLEDGE

General Knowledge, including knowledge of current events and such matters of everyday observation and experience in their scientific aspects as may be expected of an educated person who has not made a special study of any scientific subject. The paper will also include questions on history of India and geography of a nature which candidates should be able to answer without special study.

PAPER III ELEMENTARY MATHEMATICS

Arithmetic

Number system-Natural numbers, Integers, Rational and real numbers.
Fundamental Operations-addition, subtraction, multiplication, division, square roots, decimal fractions.

Unitary Method

Time and distance, time and work, percentages, applications to simple and compound interest, profit and loss, ratio and proportion, variation.

Elementary Number

Theory Division algorithm, prime and composite numbers, Tests of divisibility by 2, 3, 4, 5, 9 and 11 Multiples and factors, Factorisation Theorem, HCF and LCM. Evaluation algorithm, logarithms to base 10, laws of logarithm, use of logarithms tables.

Algebra

Basic operations, simple factors, Remainder Theorem, HCF, LCM, Theory of polynomials, solutions of quadratic equations, relation between its roots and coefficients (only real roots to be considered). Simultaneous linear equations in two unknowns-analytical and graphical solutions. Simultaneous linear in-equations in two variables and their solutions. Practical problems leading to two simultaneous linear equations
or in-equations in two variables or quadratic equations in one variable and their solutions. Set language and set notation, Rational expressions and conditional identities, laws of indices.

Trigonometry

Sine x, cosine x, tangent x when $0° \leq x \leq 90°$. Values of sin x, cos x and tan x, for x = 0°, 30°, 45°, 60° and 90°. Simple trigonometric identities. Use of trigonometric tables. Simple cases of height and distance.

Geometry

Lines and angles, Plane and plane figures theorems on

- Properties of angles at a point
- Parallel lines
- Sides and angles of a triangle
- Congruency of triangles
- Similar triangles
- Concurrence of medians and altitudes
- Properties of angles, sides and diagonals of a parallelogram, rectangle and square
- Circle and its properties, including tangents and normals
- Locus point

Mensuration

Areas of squares, rectangles, parallelograms, triangle and circle. Areas of figures,
which can be split up into the figures (Field Book). Surface area and volume of cuboids,
lateral surface and volume of right circular cones and cylinders. Surface area and volume of spheres.

Statistics

Collection and tabulation of statistical data. Graphical representation of frequency polygons, histograms, bar charts, pie charts, etc. Measures of Central Tendency.

COMBINED DEFENCE SERVICES

SOLVED PAPER

2020 (II & I)

CDS

Combined Defence Service

SOLVED PAPER 2020 (II)

PAPER I Elementary Mathematics

1. $x^3 + x^2 + 16$ is exactly divisible by x, where x is a positive integer. The number of all such possible values of x is

(a) 3 (b) 4
(c) 5 (d) 6

(c) We have, $x^3 + x^2 + 16$ is exactly divisible by x.

$\therefore \frac{x^3 + x^2 + 16}{x} = x^2 + x + \frac{16}{x}$

$x^2 + x + \frac{16}{x}$ is positive integer

$\therefore$ x is factor of 16.

i.e., 1, 2, 4, 8, 16

$\therefore$ Number of all possible value of x is 5.

2. The number of (a, b, c), where a, b, c are positive integers such that $abc = 30$, is

(a) 30 (b) 27
(c) 9 (d) 8

(b) We have,

$abc = 30$, where a, b, c are positive integer.

$\therefore 30 = 1 \times 1 \times 30 = 3$ ways

$30 = 1 \times 2 \times 15 = 6$ ways

$30 = 1 \times 3 \times 10 = 6$ ways

$30 = 1 \times 5 \times 6 = 6$ ways

$30 = 2 \times 3 \times 5 = 6$ ways

$\therefore$ Total number of (a, b, c)

$= 3 + 6 + 6 + 6 + 6$

$= 27$ ways

3. If the roots of the quadratic equation $x^2 - 4x - \log_{10} N = 0$ are real, then what is the minimum value of N?

(a) 1 (b) $\frac{1}{10}$
(c) $\frac{1}{100}$ (d) $\frac{1}{10000}$

(d) Given equation $x^2 - 4x - \log_{10} N = 0$ has real roots

$\therefore (4)^2 + 4(\log_{10} N) \geq 0$

$\log_{10} N \geq -4$

$N \geq (10)^{-4}$

$N \geq \frac{1}{10^4}$

$\therefore$ Minimum value of $N = \frac{1}{10000}$

4. The number of different solutions of the equation $x + y + z = 12$, where each of x, y and z is a positive integer, is

(a) 53 (b) 54
(c) 55 (d) 56

(c) We have,

$x + y + z = 12$, where x, y, z is a positive integer.

$\therefore$ Total number of different solution

$= {}^{12-1}C_{3-1} = {}^{11}C_2 = \frac{11 \times 10}{2} = 55$

5. If $I = a^2 + b^2 + c^2$, where a and b are consecutive integers and $c = ab$, then I is

(a) an even number and it is not a square of an integer
(b) an odd number and it is not a square of an integer.
(c) square of an even integer.
(d) square of an odd integer.

(d) We have, $I = a^2 + b^2 + c^2$

a, b are consecutive integer and $c = ab$

$\therefore \quad b = a + 1$

and $c = a(a+1) = a^2 + a$

$\therefore I = a^2 + (a+1)^2 + (a^2 + a)^2$

$I = a^2 + a^2 + 2a + 1 + a^4 + a^2 + 2a^3$

$I = a^4 + 2a^3 + 3a^2 + 2a + 1$

$I = a^4 + a^2 + 1 + 2a^3 + 2a^2 + 2a$

$I = (a^2 + a + 1)^2$

$a^2 + a + 1$ is an odd integer.

$\therefore$ I is a square of an odd integer.

6. If the number $23P62971335$ is divisible by the smallest odd composite number, then what is the value of P?

(a) 4 (b) 5
(c) 6 (d) 7

(a) Given,

$23P62971335$ is divisible by smallest odd composite number.

$\because$ Smallest odd composite number is 9.

$\therefore$ Given number is divisible by 9.

For divisibility by 9, sum of its digits is multiple of 9.

$\therefore 2 + 3 + P + 6 + 2 + 9 + 7 + 1 + 3 + 3 + 5 = 41 + P$

$\therefore$ The possible value of P is 4.

7. What is the remainder when the sum $1^5 + 2^5 + 3^5 + 4^5 + 5^5$ is divided by 4?

(a) 0 (b) 1 (c) 2 (d) 3

(b) Given number,

$1^5 + 2^5 + 3^5 + 4^5 + 5^5$

$1 + 2^5 + (4-1)^5 + 4^5 + (4+1)^5$

$1 + 2^5 + 4^5 + 4m - 1 + 4n + 1$

where m and n are integer.

$1 + 2^5 + 4^5 + 4m + 4n = 1 + 4k$

$2^5 + 4^5 + 4m + 4n$ is divisible by 4.

$\therefore$ Remainder = 1

8. What is the digit in the unit place of 3^{99}?

(a) 1 (b) 3
(c) 7 (d) 9

(c) Given number

$3^{99} = 3^3 \cdot 3^{96} = 27(3^4)^{24}$

$= 27(81)^{24}$

$= 27(80+1)^{24}$

$= 27(80k+1)$

$= 27 \times 80k + 27$

$\because 27 \times 80k$ has unit digit is 0

and unit digit of 27 is 7.

$\therefore$ Unit digit of $3^{99} = 7$

9. LCM of two numbers is 28 times their HCF. The sum of the HCF and the LCM is 1740. If one of these numbers is 240, then what is the other number?

(a) 420 (b) 640
(c) 820 (d) 1040

(a) Let LCM of two number be x and HCF of two number be y.

Given,

$x = 28y$ … (i)

and $x + y = 1740$ … (ii)

From Eqs. (i) and (ii), we get

$x = 1680$ and $y = 60$

We know that,

LCM × HCF

= Product of two numbers

$\therefore$ Other number $= \frac{1680 \times 60}{240} = 420$

10. $(x^n - a^n)$ is divisible by $(x-a)$, where $x \neq a$, for every

(a) natural number n
(b) even natural number n only
(c) odd natural number n only
(d) prime number n only

(a) $x^n = (x + a - a)^n$

$x^n = (x-a)^n + {}^nC_1(x-a)^{n-1}(a) + {}^nC_2(x-a)^{n-2}a^2 + \ldots + a^n$

$\Rightarrow x^n - a^n = (x-a)[(x-a)^{n-1} + {}^nC_1(x-a)^{n-2}a + \ldots]$

Clearly, $x^n - a^n$ is divisible by $x - a$ for every natural number n.

11. If 17^{2020} is divided by 18, then what is the remainder?

(a) 1 (b) 2 (c) 16 (d) 17

(a) We have,

$(17)^{2020} = (18-1)^{2020}$

$= 18k + (-1)^{2020} = 18k + 1$

$\therefore$ If 17^{2020} is divided by 18,
then remainder = 1

12. What is the value of

$$\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \ldots + \frac{1}{\sqrt{99}+\sqrt{100}}?$$

(a) 1 (b) 5
(c) 9 (d) 10

(c) Given,

$$\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \ldots + \frac{1}{\sqrt{99}+\sqrt{100}}$$

$$= \frac{\sqrt{2}-1}{2-1} + \frac{\sqrt{3}-\sqrt{2}}{3-2} + \frac{\sqrt{4}-\sqrt{3}}{4-3} + \ldots + \frac{\sqrt{100}-\sqrt{99}}{100-99}$$

$$= \sqrt{2} - 1 + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} + \ldots + \sqrt{100} - \sqrt{99}$$

$= \sqrt{100} - 1 = 10 - 1 = 9$

13. If $x^m = \sqrt[14]{x\sqrt{x\sqrt{x}}}$, then what is the value of m?

(a) $\frac{1}{8}$ (b) $\frac{1}{4}$ (c) $\frac{3}{4}$ (d) $\frac{7}{4}$

(a) We have,

$x^m = \sqrt[14]{x\sqrt{x\sqrt{x}}}$

$\Rightarrow x^m = \sqrt[14]{x\sqrt{x \cdot x^{1/2}}}$

$\Rightarrow x^m = \sqrt[14]{x(x^{3/2})^{1/2}}$

$\Rightarrow x^m = \sqrt[14]{x \cdot x^{3/4}}$

$\Rightarrow x^m = (x^{7/4})^{\frac{1}{14}}$

$\Rightarrow x^m = x^{\frac{1}{8}}$

$\Rightarrow m = \frac{1}{8}$

14. The sum of all possible products taken two at a time out of the numbers $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$ is

(a) 0 (b) – 30
(c) – 55 (d) 55

(c) Given number, $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$

We know that,

$$(x_1 + x_2 + x_3 \ldots x_n)^2 = \sum_{i=1}^{n} x_i^2 + 2\sum_{1 \le i < j \le n}^{n} x_i x_j$$

$\therefore (-1 + 1 - 2 + 2 + 3 - 3 + 4 - 4 + 5 - 5)^2$

$= 2(1^2 + 2^2 + 3^2 + 4^2 + 5^2) + 2(\Sigma x_i x_j)$,

$0 = 55 + \Sigma x_i x_j$

$\Sigma x_i x_j = -55$

15. A train of length 110 m is moving at a uniform speed of 132 km/h. The time required to cross a bridge of length 165 m is

(a) 6.5 s (b) 7 s
(c) 7.5 s (d) 8.5 s

(c) Length of train = 110 m

Length of bridge = 165 m

Total length = 110 + 165

= 275 m

Speed of train = 132 km / h

$= 132 \times \frac{5}{18}$ m / s

Time required to cross the bridge

$= \frac{\text{Total distance}}{\text{Speed}}$

$= \frac{275 \times 18}{132 \times 5}$

= 7.5 s

16. The simple interest on a certain sum is one-fourth of the sum. If the number of years and the rate of annual interest are numerically equal, then the number of years is

(a) 2.5 (b) 3
(c) 3.5 (d) 5

(d) Given, SI $= \frac{1}{4}P$

and $R = T$

$\therefore$ SI $= \frac{PRT}{100}$

$\Rightarrow \frac{1}{4}P = \frac{PT^2}{100}$

$\Rightarrow T^2 = 25$

$\therefore T = 5$

$\because$ Number of years = 5

17. A 60-page book has n lines per page. If the number of lines were reduced by 3 in each page, the number of pages would have to be increased by 10 to give the same writing space. What is the value of n?

(a) 18 (b) 21
(c) 24 (d) 30

(b) Total number of line in 60 page $= 60n$ ... (i)

Total number of line when number of line were reduced 3 in each page and number of pages increased by $10 = (n-3)(70)$ ... (ii)

From Eqs. (i) and (ii),

$60n = (n-3)70$

$60n = 70n - 210$

$n = 21$

18. If x men working x hours per day can do x units of work in x days, then y men working y hours per day in y days would be able to do k units of work. What is the value of k?

(a) x^2y^{-3} (b) x^3y^{-2}
(c) y^2x^{-3} (d) y^3x^{-2}

(d) Given,

x men working x hours per day can do x units of work in x days

$\therefore \quad x^3 = x$

$\Rightarrow \quad x^2 = 1$ ... (i)

y men working y hours per day in y day would be able to do k units of work.

$\therefore \quad y^3 = k$

$\Rightarrow \quad \dfrac{y^3}{k} = 1$ ...(ii)

From Eqs. (i) and (ii),

$$x^2 = \frac{y^3}{k} \Rightarrow k = x^{-2}y^3$$

19. Let $d(n)$ denote the number of positive divisors of a positive integer n. Which of the following are correct?

1. $d(5) = d(11)$
2. $d(5)\cdot d(11) = d(55)$
3. $d(5) + d(11) = d(16)$

Select the correct answer using the code given below:

(a) 1 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

(b) Here, $d(n)$, denotes the number of positive divisors of a positive integer n.

$\therefore d(5) = 2, d(11) = 2,$

$d(55) = d(5 \times 11) = 2 \times 2 = 4$

$d(16) = d(2^4) = (4+1) = 5$

$\therefore d(5) = d(11)$ 1st is true.

$d(5)\,d(11) = 2 \times 2 = 4$

$= d(55)$ 2nd is true.

$d(5) + d(11) = 2 + 2 = 4 \neq d(16)$ ... 3 rd is false.

$\therefore$ 1 and 2 are correct.

20. If $A_n = P_n + 1$, where P_n is the product of the first n prime numbers, then consider the following statements :

1. A_n is always a composite number.
2. $A_n + 2$ is always an odd number.
3. $A_n + 1$ is always an even number.

Which of the above statements is/are correct?

(a) only 1 (b) only 2
(c) only 3 (d) 2 and 3

(d) Given $A_n = P_n + 1$, where P_n is the product of first n prime number

P_n is always an even number. [$\because P_1 = 2$]

$\therefore A_n$ is an odd number.

$\therefore A_n + 1$ is always an even number.

$\therefore A_n + 2$ is always an add number.

21. A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000 gm for 800 gm. What is the actual profit percentage?

(a) 20% (b) 25% (c) 30% (d) 40%

(b) Here, 200 gm is gained in 800 gm.

$$\therefore \text{Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100$$

$$= \frac{200}{800} \times 100 = 25\%$$

22. A river 3 m deep and 40 m wide is flowing at the rate of 2 km/h and falls into the sea. What is the amount of water in litres that will fall into the sea from this river in a minute?

(a) 4000000 L (b) 400000 L
(c) 40000 L (d) 4000 L

(a) Speed of flow of water = 2 km/h

Width of river = 40 m

Depth of river = 3 m

Volume of water in one minutes

$$= 40 \times 3 \times 2000 \times \frac{1}{60}$$

$= 4000\ m^3$

$= 4000000$ L

23. If a television set is sold at ₹ x, a loss of 28% would be incurred. If it is sold at ₹ y, a profit of 12% would be incurred. What is the ratio of y to x ?

(a) 41 : 9 (b) 31 : 9
(c) 23 : 9 (d) 14 : 9

(d) Let the cost of television be ₹ 100.

If television set sold at 28% loss,

then selling price of television $(x) = 100 - 28 = 72$

If television sold at 12% profit

then selling price of television

i.e., $(y) = 100 + 12 = 112$

$$\therefore \quad \frac{y}{x} = \frac{112}{72}$$

$$= \frac{14}{9}$$

$y : x = 14 : 9$

24. By increasing the speed of his car by 15 km/h, a person covers a distance of 300 km by taking an hour less than before. What was the original speed of the car?

(a) 45 km/h
(b) 50 km/h
(c) 60 km/h
(d) 75 km/h

(c) Let the speed of car = x km/h

Speed of car after increasing 15 km/h $= (x + 15)$ km/h

Time taken by car on original speed

$$= \frac{300}{x}$$

Time taken by car after increasing speed

$$= \frac{300}{x + 15}$$

Difference between time = 1 h

$$\therefore \quad \frac{300}{x} - \frac{300}{x+15} = 1$$

$\Rightarrow 300(x + 15 - x) = x(x + 15)$

$\Rightarrow x^2 + 15x - 4500 = 0$

$\Rightarrow (x + 75)(x - 60) = 0$

$x = 60,\ x \neq -75$

$\therefore$ Original speed of car = 60 km / h

25. Three persons start a business with capitals in the ratio $\frac{1}{3}:\frac{1}{4}:\frac{1}{5}$. The first person withdraws half his capital after 4 months. What is his share of profit if the business fetches an annual profit of ₹ 96800?

(a) ₹ 32000 (b) ₹ 34500
(c) ₹ 36000 (d) ₹ 36800

(a) Three person start a business with capital in the ratio

$\frac{1}{3}:\frac{1}{4}:\frac{1}{5} = 20:15:12$

Total share of capital in a year by 1st person is

$= (20 \times 4)x + (10 \times 8)x = 160x$

2nd person $= (15 \times 12)x = 180x$

3rd person $= (12 \times 12)x = 144x$

$\therefore$ Ratio of their profits $= 160:180:144$

Profit share of 1st person

$= \left(\frac{160}{160+180+144}\right) \times 96800$

$= \frac{160}{484} \times 96800 =$ ₹ 32000

26. If x varies as y, then which of the following is/are correct?

1. $x^2 + y^2$ varies as $x^2 - y^2$
2. $\frac{x}{y^2}$ varies inversely as y
3. $\sqrt[n]{x^2 y}$ varies as $\sqrt[2n]{x^4 y^2}$

Select the correct answer using the code given below :

(a) 1 and 2 (b) 2 and 3
(c) only 3 (d) 1, 2 and 3

(b) Given x varies as y

$\therefore \quad x = ky$

1. $x^2 + y^2 = k^2y^2 + y^2 = y^2(k^2+1)$

and $x^2 - y^2 = k^2y^2 - y^2 = y^2(k^2-1)$

$\therefore x^2 + y^2$ not varies with $x^2 - y^2$.

2. $\frac{x}{y^2} = \frac{ky}{y^2} = \frac{k}{y}$

$\therefore$ Clearly, $\frac{x}{y^2}$ varies inversely as y.

3. $\sqrt[n]{x^2y} = (x^2y)^{\frac{1}{n}} = (k^2y^3)^{1/n}$

and $\sqrt[2n]{x^4y^2} = (x^4y^2)^{\frac{1}{2n}}$

$= (x^2\ y)^{\frac{1}{n}} = (k^2y^3)^{1/n}$

$\because \sqrt[n]{x^2y}$ is varies as $\sqrt[2n]{x^4y^2}$.

27. Ena was born 4 yr after her parents marriage. Her mother is 3 yr younger than her father and 24 yr older than Ena, who is 13 yr old. At what age did Ena's father get married?

(a) 25 yr (b) 24 yr
(c) 23 yr (d) 22 yr

(c) Present age of Ena = 13 yr

Present age of her mother

$= (13 + 24) = 37$ yr

Age of Ena mother's when Ena born

$= 37 - 13 = 24$ yr

Age of Ena mother get married

$= 24 - 4 = 20$ yr

$\therefore$ Age of Ena father's

$= (20 + 3) = 23$ yr.

28. Mahesh is 60 yr old. Ram is 5 yr younger to Mahesh and 4 yr elder to Raju. Babu is a younger brother of Raju and he is 6 yr younger. What is the age difference between Mahesh and Babu?

(a) 18 yr (b) 15 yr (c) 13 yr (d) 11 yr

(b) Here, age of Mahesh = 60 yr

Ram is 5 yr younger to Mahesh.

$\therefore$ Age of Ram $= 60 - 5 = 55$ yr

Ram is 4 yr elder to Raju

$\therefore$ Raju's age $= 55 - 4 = 51$ yr

Babu's is 6 yr younger to Raju

$\therefore$ Babu's age $= 51 - 6 = 45$

$\therefore$ Difference of age between Mahesh and Babu $= 60 - 45 = 15$ yr

29. The number of items in a booklet is N. In the first year there is an increase of $x\%$ in this number and in the subsequent year there is a decrease of $x\%$. At the end of the two year, what will be the number of items in the booklet?

(a) Less than N
(b) Equal to N
(c) More than N
(d) It depends on the value of N

(a) Number of items in a booklet $= N$

Number of items in booklet after first year

Increase at rate of $x\% = N + x\%$ of N

$= N\left(1 + \frac{x}{100}\right)$

Number of items in booklet in 2nd year

Decrease at rate of $x\%$

$= N\left(1+\frac{x}{100}\right) - N\left(1+\frac{x}{100}\right)\frac{x}{100}$

$= N\left(1+\frac{x}{100}\right)\left(1-\frac{x}{100}\right) = N\left(1-\frac{x^2}{10000}\right)$

Which is less than N.

30. If $ab + xy - xb = 0$ and $bc + yz - cy = 0$, then what is $\frac{x}{a} + \frac{c}{z}$ equal to?

(a) $\frac{y}{b}$ (b) $\frac{b}{y}$ (c) 1 (d) 0

(c) Given, $ab + xy - xb = 0$ and $bc + yz - cy = 0$

$\Rightarrow x(y - b) = -ab$ and $c(b - y) = -yz$

$\Rightarrow \frac{x}{a} = \frac{-b}{y-b}$ and $\frac{c}{z} = \frac{-y}{b-y} = \frac{y}{y-b}$

$\therefore \frac{x}{a} + \frac{c}{z} = \frac{-b}{y-b} + \frac{y}{y-b} = \frac{y-b}{y-b} = 1$

31. What is the HCF of the polynomials $x^6 - 3x^4 + 3x^2 - 1$ and $x^3 + 3x^2 + 3x + 1$?

(a) $(x+1)$ (b) $(x+1)^2$
(c) x^2+1 (d) $(x+1)^3$

(d) Given polynomial, $x^6 - 3x^4 + 3x^2 - 1$

$= (x^2-1)^3 = (x+1)^3(x-1)^3$

and $x^3 + 3x^2 + 3x + 1 = (x+1)^3$

HCF $= (x+1)^3$

32. The HCF and the LCM of two polynomials are $3x + 1$ and $30x^3 + 7x^2 - 10x - 3$ respectively. If one polynomial is $6x^2 + 5x + 1$, then what is the other polynomial?

(a) $15x^2 + 4x + 3$ (b) $15x^2 + 4x - 3$
(c) $15x^2 - 4x + 3$ (d) $15x^2 - 4x - 3$

(d) HCF and LCM of two polynomials are $3x + 1$ and $30x^3 + 7x^2 - 10x - 3$ and one polynomial is $6x^2 + 5x + 1$.

We know,

HCF × LCM = Product of two polynomials

$\therefore$ Other polynomial

$= \frac{(3x+1)(30x^3 + 7x^2 - 10x - 3)}{6x^2 + 5x + 1}$

$= \frac{(3x+1)(30x^3 + 7x^2 - 10x - 3)}{(3x+1)(2x+1)}$

$= \frac{30x^3 + 7x^2 - 10x - 3}{2x+1}$

$= \frac{(15x^2 - 4x - 3)(2x+1)}{2x+1}$

$= 15x^2 - 4x - 3$

33. If $(p+2)(2q-1) = 2pq - 10$ and $(p-2)(2q-1) = 2pq - 10$, then what is pq equal to?

(a) – 10 (b) – 5
(c) 5 (d) 10

(c) Given,

$$(p+2)(2q-1) = 2pq - 10$$
and $(p-2)(2q-1) = 2pq - 10$
$$\therefore 2pq + 4q - p - 2 = 2pq - 10$$
$$\Rightarrow \quad 4q - p = -8 \quad \ldots \text{(i)}$$
and $2pq - 4q - p + 2 = 2pq - 10$
$$-4q - p = -12$$
$$4q + p = 12 \quad \ldots \text{(ii)}$$
From Eqs. (i) and (ii), $p = 10, q = \dfrac{1}{2}$
$$\therefore pq = 10 \times \frac{1}{2} = 5$$

34. What is the value of
$$\frac{a^2 + ac}{a^2c - c^3} - \frac{a^2 - c^2}{a^2c + 2ac^2 + c^3} - \frac{2c}{a^2 - c^2} + \frac{3}{a + c}?$$

(a) 0 (b) 1
(c) $\dfrac{ac}{a^2 + c^2}$ (d) $\dfrac{6}{a + c}$

(d) We have,
$$\frac{a^2 + ac}{a^2c - c^3} - \frac{a^2 - c^2}{a^2c + 2ac^2 + c^3} - \frac{2c}{a^2 - c^2} + \frac{3}{a + c}$$
$$= \frac{a(a+c)}{c(a+c)(a-c)} - \frac{(a+c)(a-c)}{c(a+c)(a+c)} - \frac{2c}{(a+c)(a-c)} + \frac{3}{a+c}$$
$$= \frac{a}{c(a-c)} - \frac{a-c}{c(a+c)} - \frac{2c}{(a+c)(a-c)} + \frac{3}{a+c}$$
$$= \frac{a(a+c) - (a-c)(a-c) - 2c^2 + 3c(a-c)}{c(a+c)(a-c)}$$
$$= \frac{a^2 + ac - a^2 - c^2 + 2ac - 2c^2 + 3ac - 3c^2}{c(a+c)(a-c)}$$
$$= \frac{6ac - 6c^2}{c(a+c)(a-c)}$$
$$= \frac{6c(a-c)}{c(a+c)(a-c)}$$
$$= \frac{6}{a+c}$$

35. What is the square root of $4x^4 + 8x^3 - 4x + 1$?

(a) $2x^2 - 2x - 1$
(b) $2x^2 - x - 1$
(c) $2x^2 + 2x + 1$
(d) $2x^2 + 2x - 1$

(d) We have,
$$4x^4 + 8x^3 - 4x + 1$$
$$= 4x^4 + 4x^2 + 8x^3 - 4x + 1 - 4x^2$$
$$= 4x^4 + 4x^2 + 1 + 2(2x^2)(2x) - 2(2x) - 2(2x^2)$$
$$= (2x^2)^2 + (2x)^2 + (1)^2 + 2(2x^2)(2x) - 2(2x)(1) - 2(2x^2)(1)$$
$$= (2x^2 + 2x - 1)^2$$
$$\therefore \sqrt{4x^4 + 8x^3 - 4x + 1}$$
$$= \sqrt{(2x^2 + 2x - 1)^2}$$
$$= 2x^2 + 2x - 1$$

36. The sum of the digits of a two digit number is 13 and the difference between the number and that formed by reversing the digits is 27. What is the product of the digits of the number?

(a) 35 (b) 40
(c) 45 (d) 54

(b) Two digits number be $10x + y$.

Given, $x + y = 13 \quad \ldots$ (i)
and $(10x + y) - (10y + x) = 27$
$$x - y = 3 \quad \ldots \text{(ii)}$$
From Eqs. (i) and (ii),
$$x = 8, y = 5$$
$$\therefore \quad xy = 8 \times 5 = 40$$

37. If $\dfrac{x}{b+c} = \dfrac{y}{c+a} = \dfrac{z}{b-a}$, then which one of the following is correct?

(a) $x + y + z = 0$
(b) $x - y - z = 0$
(c) $x + y - z = 0$
(d) $x + 2y + 3z = 0$

(b) Let $\dfrac{x}{b+c} = \dfrac{y}{c+a} = \dfrac{z}{b-c} = k$
$$\therefore \quad x = k(b+c)$$
$$y = k(c+a)$$
$$z = k(b-a)$$
$$x - y - z = k$$
$$(b + c - c - a - b + a) = 0$$
$$\therefore \quad x - y - z = 0$$

38. X, Y and Z travel from the same place with uniform speeds 4 km/h, 5 km/h and 6 km/h respectively. Y starts 2 h after X. How long after Y must Z start in order that they overtake X at the same instant?

(a) $\dfrac{3}{2}$ h (b) $\dfrac{4}{3}$ h
(c) $\dfrac{9}{8}$ h (d) $\dfrac{11}{8}$ h

(b) Distance travelled by X in t hours $= 4t$

Distance travelled by Y starts 2 h after X
$$= 5(t - 2)$$
Distance travelled by X and Y are same.
$$\therefore \quad 4t = 5t - 10$$
$$\Rightarrow \quad t = 10 \text{ h}$$
Let Z starts n hours after Y.

$\therefore$ Distance travelled by $Z = 6(8 - n) = 48 - 6n$

Distance travelled by X and Z are same.
$$\therefore \quad 40 = 48 - 6n$$
$$\Rightarrow 6n = 8 \Rightarrow n = \frac{4}{3} \text{ h}$$

39. $1 - x - x^n + x^{n+1}$, where n is a natural number, is divisible by

(a) $(1 + x)^2$ (b) $(1 - x)^2$
(c) $1 - 2x - x^2$ (d) $1 + 2x - x^2$

(b) Given,
$$1 - x - x^n + x^{n+1}$$
$$= 1 - x^n - x + x^{n+1}$$
$$= 1(1 - x^n) - x(1 - x^n) = (1 - x)(1 - x^n)$$
$$= (1 - x)(1 - x)(1 + x + x^2 \ldots + x^{n-1})$$
$$= (1 - x)^2(1 + x + x^2 + \ldots + x^{n-1})$$
$\therefore (1 - x)^2$ is a factor of $1 - x - x^n + x^{n-1}$.

40. A person sold an article for ₹ 75 which cost him ₹ x. He finds that he realised x% profit on his outlay. What is x equal to?

(a) 20% (b) 25%
(c) 50% (d) 100%

(c) Given cost price of articles = ₹ x

Profit % = x%

Selling price = ₹ 75
$$\text{Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100\%$$
$$x\% = \frac{75 - x}{x} \times 100\%$$
$$\Rightarrow x^2 + 100x - 7500 = 0$$
$$\Rightarrow (x + 150)(x - 50) = 0$$
$$x = 50$$

41. A car did a journey in t hours. Had the average speed been x km/h greater, the journey would have taken y hours less. How long was the journey?

(a) $x(t-y)t\,y$ (b) $x(t-y)t\,y^{-1}$
(c) $x(t-y)t\,y^{-2}$ (d) $x(t+y)t\,y$

(b) Let the speed of car $= S$ km / h

Distance travelled by car $= D$

Time taken $= t$ hour

$$\therefore \quad S = \frac{D}{t} \qquad \ldots \text{(i)}$$

When speed of car increased x km/h.

$\therefore$ Speed of car $= (S + x)$

Distance be same $= D$

Time $= (t - y)$

$$\therefore S + x = \frac{D}{t-y} \qquad \ldots \text{(ii)}$$

From Eqs. (i) and (ii),

$$x = \frac{D}{t-y} - \frac{D}{t}$$

$$x = \frac{D(t - t + y)}{t(t-y)}$$

$$D = x(t-y)ty^{-1}$$

42. When a ball is allowed to fall, the time it takes to fall any distance varies as the square root of the distance and it takes 4 s to fall 78.40 m. How long would it take to fall 122.50 m?

(a) 5 s (b) 5.5 s (c) 6 s (d) 6.5 s

(a) Given, $t = k\sqrt{d}$

When, $t = 4$ and $d = 78.40$

$$\therefore \quad k = \frac{4}{\sqrt{78.40}}$$

When, $d = 122.50$

$$\therefore \quad t = \frac{4}{\sqrt{78.40}} \times \sqrt{122.50}$$

$$= 4\sqrt{\frac{122.50}{78.40}}$$

$$t = 4 \times \sqrt{\frac{1225}{784}}$$

$$= 4 \times \frac{35}{28} = 5 \text{ sec}$$

43. If $6^{3-4x}4^{x+5} = 8$ (Given $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$), then which one of the following is correct?

(a) $0 < x < 1$ (b) $1 < x < 2$
(c) $2 < x < 3$ (d) $3 < x < 4$

(c) Given, $6^{3-4x}4^{x+5} = 8$

$$\frac{6^3 \cdot 4^x \cdot 4^5}{6^{4x}} = 2^3$$

$$\Rightarrow \quad \frac{2^3 \cdot 3^3 \cdot 2^{2x} \cdot 2^{10}}{2^{4x} \cdot 3^{4x}} = 8$$

$$\Rightarrow \quad 3^{4x} \cdot 2^{2x} = 2^{10} \cdot 3^3$$

Taking log both sides,

$$4x\log 3 + 2x\log 2 = 10\log 2 + 3\log 3$$

$$\frac{(2x-10)}{3-4x} = \frac{\log 3}{\log 2} = \frac{0.477}{0.301}$$

$$\Rightarrow \frac{2x-10}{3-4x} = 1.58$$

$$\Rightarrow 2x - 10 = (3-4x)(1.58)$$

$$\Rightarrow 2x - 10 = 4.74 - 6.32x$$

$$\Rightarrow \quad 8.32x = 14.74$$

$$\Rightarrow \quad x = \frac{14.74}{8.32} = 1.77$$

$$\therefore \quad 1 < x < 2$$

44. The Euclidean algorithm is used to calculate the

(a) square root of an integer
(b) cube root of an integer
(c) square of an integer
(d) HCF of two integers

(d) Euclidean algorithm is used to calculate the HCF of two integers.

45. If radius of a sphere is rational, then which of the following is/are correct?

1. Its surface area is rational.
2. Its volume is rational.

Select the correct answer using the code given below

(a) only 1
(b) only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

(d) Radius of sphere is rational

Surface area of sphere $= 4\pi r^2$, which is irrational

Volume of sphere $= \frac{4}{3}\pi r^3$, which is also irrational.

46. If $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, then what is $m^{\frac{4}{3}}n^{\frac{2}{3}} + m^{\frac{2}{3}}n^{\frac{4}{3}}$ equal to?

(a) 0 (b) 1
(c) mn (d) m^2n^2

(b) We have, $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$

$$\frac{1-\sin^2\theta}{\sin\theta} = m$$

$$\text{and } \frac{1-\cos^2\theta}{\cos\theta} = n$$

$$\frac{\cos^2\theta}{\sin\theta} = m \text{ and } \frac{\sin^2\theta}{\cos\theta} = n$$

$$m^{4/3}n^{2/3} + m^{2/3}n^{4/3}$$

$$= m^{2/3}n^{2/3}[m^{2/3} + n^{2/3}]$$

$$\left(\frac{\cos^2\theta}{\sin\theta}\right)^{2/3}\left(\frac{\sin^2\theta}{\cos\theta}\right)^{2/3}\left[\frac{\cos^{4/3}\theta}{\sin^{2/3}\theta} + \frac{\sin^{4/3}\theta}{\cos^{2/3}\theta}\right]$$

$$= (\sin\theta\cos\theta)^{2/3}\left[\frac{\cos^2\theta + \sin^2\theta}{(\sin\theta\cos\theta)^{2/3}}\right]$$

$$= 1$$

47. If $\cos\theta + \sec\theta = k$, then what is the value of $\sin^2\theta - \tan^2\theta$?

(a) $4 - k$ (b) $4 - k^2$
(c) $k^2 - 4$ (d) $k^2 + 2$

(b) Given, $\cos\theta + \sec\theta = k$

$$(\cos\theta + \sec\theta)^2 = k^2$$

$$\cos^2\theta + \sec^2\theta + 2 = k^2$$

$$1 - \sin^2\theta + 1 + \tan^2\theta + 2 = k^2$$

$$4 - k^2 = \sin^2\theta - \tan^2\theta$$

$$\therefore \quad \sin^2\theta - \tan^2\theta = 4 - k^2$$

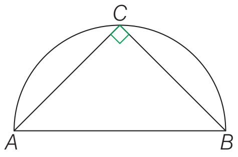
48. ABC is a triangle inscribed in a semicircle of diameter AB. What is $\cos(A+B) + \sin(A+B)$ equal to?

(a) 0 (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) 1

(d) We have,

ABC is a triangle inscribed in a semi-circle of diameter AB.

$\therefore \angle ABC$ is a right angle at C.



$\therefore A + B = 90^\circ$

$$\cos(A+B) + \sin(A+B)$$

$$= \cos 90^\circ + \sin 90^\circ$$

$$= 0 + 1$$

$$= 1$$

49. Consider the following statements :

1. $\sin\theta = x + \frac{1}{x}$ is possible for some real value of x.
2. $\cos\theta = x + \frac{1}{x}$ is possible for some real value of x.

Which of the above statements is/are correct?

(a) only 1 (b) only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(d) We have,

1. $\sin\theta = x + \frac{1}{x}$

$\sin\theta \in [-1, 1]$

$x + \frac{1}{x} \in (-\infty, -2] \cup [2, \infty]$

$\therefore$ 1st is incorrect.

2. $\cos\theta = x + \frac{1}{x}$

$\cos\theta \in [-1, 1]$

$x + \frac{1}{x} \in [-\infty, -2] \cup [2, \infty)$

$\therefore$ 2nd is also incorrect.

50. What is the magnitude (in radian) of the interior angle of a regular pentagon?

(a) $\frac{\pi}{5}$ (b) $\frac{2\pi}{5}$ (c) $\frac{3\pi}{5}$ (d) $\frac{4\pi}{5}$

(c) Interior angle of a regular pentagon

$= \frac{(5-2)}{5} \times \pi = \frac{3\pi}{5}$

51. The difference between two angles is 15° and the sum of the angles in radian is $\frac{5\pi}{12}$. The bigger angle is k times the smaller angle. What is k equal to?

(a) $\frac{4}{3}$ (b) $\frac{3}{2}$ (c) $\frac{6}{5}$ (d) $\frac{7}{6}$

(b) Let the angle be x and kx.

$\therefore \quad kx - x = 15^\circ \quad \ldots(i)$

and $kx + x = \frac{5\pi}{12}$

$\Rightarrow \quad x(k+1) = 75^\circ \quad \ldots(ii)$

From Eqs. (i) and (ii),

$\frac{k-1}{k+1} = \frac{15}{75}$

$\Rightarrow \quad 5k - 5 = k + 1$

$\Rightarrow \quad 4k = 6 \Rightarrow k = 3/2$

52. Consider the following statements :

1. The equation $2\sin^2\theta - \cos\theta + 4 = 0$ is possible for all θ.
2. $\tan\theta + \cot\theta$ cannot be less than 2, where $0 < \theta < \frac{\pi}{2}$.

Which of the above statements is/are correct?

(a) only 1
(b) only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

(b) We have,

1. $2\sin^2\theta - \cos\theta + 4 = 0$

Put $\theta = 90^\circ$

$2 \times 1 - 0 + 4 = 6 \neq 0$

$\because$ 1st is incorrect.

2. $\tan\theta + \cot\theta$

$\tan\theta + \frac{1}{\tan\theta} \geq 2$

$\therefore$ 2nd is correct.

53. A road curve is to be laid out on a circle. What radius should be used if the track is to change direction by 42° in distance of 44 m?

(Assume $\pi = \frac{22}{7}$)

(a) 60 m (b) 66 m
(c) 75 m (d) 80 m

(a) Given, $\theta = 42^\circ$

Length of an arc = 44

We know, $\quad l = \frac{\theta}{360} \times 2\pi r$

$\therefore r = \frac{360^\circ \times l}{\theta \times 2\pi}$

$= \frac{360 \times 44 \times 7}{42 \times 2 \times 22} \quad \left[\pi = \frac{22}{7}\right]$

$r = 60$ m

54. What is the maximum value of $3\sin\theta - 4$?

(a) -4 (b) -1 (c) 0 (d) 1

(b) We know that,

$-1 \leq \sin\theta \leq 1$

$-3 \leq 3\sin\theta \leq 3$

$-3 - 4 \leq 3\sin\theta - 4 \leq 3 - 4$

$-7 \leq 3\sin\theta - 4 \leq -1$

$\therefore$ Maximum value of $3\sin\theta - 4 = -1$

55. If $\sin\theta + \cos\theta = \sqrt{2}$, then what is $\sin^6\theta + \cos^6\theta + 6\sin^2\theta\cos^2\theta$ equal to?

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) 1 (d) $\frac{7}{4}$

(d) Given,

$\sin\theta + \cos\theta = \sqrt{2}$

It is possible only $\theta = 45^\circ$

$\therefore \sin^6\theta + \cos^6\theta + 6\sin^2\theta\cos^2\theta$

$= \sin^6 45^\circ + \cos^6 45^\circ + 6\sin^2 45^\circ \cos^2 45^\circ$

$= \left(\frac{1}{\sqrt{2}}\right)^6 + \left(\frac{1}{\sqrt{2}}\right)^6 + 6\left(\frac{1}{\sqrt{2}}\right)^2\left(\frac{1}{\sqrt{2}}\right)^2$

$= \frac{1}{8} + \frac{1}{8} + 6 \times \frac{1}{2} \times \frac{1}{2}$

$= \frac{1+1+12}{8} = \frac{14}{8} = 7/4$

56. What is the least value of $9\sin^2\theta + 16\cos^2\theta$?

(a) 0 (b) 9
(c) 16 (d) 25

(b) We have,

$9\sin^2\theta + 16\cos^2\theta$

$= 9\sin^2\theta + 9\cos^2\theta + 7\cos^2\theta$

$= 9(\sin^2\theta + \cos^2\theta) + 7\cos^2\theta$

$= 9 + 7\cos^2\theta$

$\therefore 0 \leq \cos^2\theta \leq 1$

$\therefore$ Minimum value of $9 + 7\cos^2\theta = 9$

57. If $\cos 47^\circ + \sin 47^\circ = k$, then what is the value of $\cos^2 47^\circ - \sin^2 47^\circ$?

(a) $k\sqrt{2-k^2}$ (b) $-k\sqrt{2-k^2}$
(c) $k\sqrt{1-k^2}$ (d) $-k\sqrt{1-k^2}$

(a) Given,

$\cos 47^\circ + \sin 47^\circ = k$

$\Rightarrow \cos^2 47^\circ + \sin^2 47^\circ + 2\sin 47^\circ \cos 47^\circ = k^2$

$\Rightarrow \sin 2(47^\circ) = k^2 - 1$

$\Rightarrow \sin^2(2\theta) = (k^2-1)^2 \quad$ [let $47^\circ = \theta$]

$\Rightarrow 1 - \cos^2 2\theta = (k^2-1)^2$

$\Rightarrow \quad \cos^2 2\theta = 1 - (k^2-1)^2$

$\Rightarrow \quad \cos 2\theta = \sqrt{1 - (k^4 - 2k^2 + 1)}$

$\Rightarrow \quad \cos 2\theta = \sqrt{2k^2 - k^4}$

$= k\sqrt{2-k^2}$

$\cos^2\theta - \sin^2\theta = k\sqrt{2-k^2}$

$\therefore \cos^2 47^\circ - \sin^2 47^\circ = k\sqrt{2-k^2}$

58. If $\operatorname{cosec}\theta - \sin\theta = p^3$ and $\sec\theta - \cos\theta = q^3$, then what is the value of $\tan\theta$?

(a) $\frac{p}{q}$ (b) $\frac{q}{p}$ (c) pq (d) p^2q^2

(b) We have,

$\operatorname{cosec}\theta - \sin\theta = p^3$ and

$\sec\theta - \cos\theta = q^3$

$\frac{1-\sin^2\theta}{\sin\theta} = p^3$ and $\frac{1-\cos^2\theta}{\cos\theta} = q^3$

$\frac{\cos^2\theta}{\sin\theta} = p^3$ and $\frac{\sin^2\theta}{\cos\theta} = q^3$

$\therefore \frac{\cos^2\theta}{\sin\theta} \times \frac{\cos\theta}{\sin^2\theta} = \frac{p^3}{q^3}$

$\Rightarrow (\cot\theta)^3 = \frac{p^3}{q^3} \Rightarrow \cot\theta = \frac{p}{q}$

$\therefore \quad \tan\theta = \frac{q}{p}$

59. If $0 \le \alpha, \beta \le 90^\circ$ such that $\cos(\alpha - \beta) = 1$, then what is $\sin\alpha - \sin\beta + \cos\alpha - \cos\beta$ equal to?

(a) – 1 (b) 0 (c) 1 (d) 2

(b) Given,

$\cos(\alpha - \beta) = 1$, where $0 \le \alpha, \beta \le 90^\circ$

$\therefore \alpha - \beta = 0, \alpha = \beta$

$\therefore \sin\alpha - \sin\beta + \cos\alpha - \cos\beta$

$= \sin\alpha - \sin\alpha + \cos\alpha - \cos\alpha = 0$

60. Consider the following statements.

1. The value of $\cos 61^\circ + \sin 29^\circ$ cannot exceed 1.
2. The value of $\tan 23^\circ - \cot 67^\circ$ is less than 0.

Which of the above statements is/are correct?

(a) only 1 (b) only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(a) We have,

1. $\cos 61^\circ + \sin 29^\circ$

$= \cos(90^\circ - 29^\circ) + \sin 29^\circ$

$= \sin 29^\circ + \sin 29^\circ = 2\sin 29^\circ$

$\because \sin 29^\circ < \sin 30^\circ$

$\Rightarrow 2\sin 29^\circ < 2\sin 30^\circ$

$\Rightarrow 2\sin 29^\circ < 2 \times \frac{1}{2} < 1$

1st is correct.

2. $\tan 23^\circ - \cot 67^\circ$

$= \tan 23^\circ - \cot(90^\circ - 23^\circ)$

$= \tan 23^\circ - \tan 23^\circ = 0$

$\therefore$ 2nd is incorrect.

61. In a quadrilateral $ABCD$, $\angle B = 90^\circ$ and $AB^2 + BC^2 + CD^2 - AD^2 = 0$, then what is $\angle ACD$ equal to?

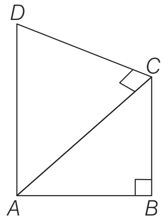
(a) 30° (b) 60°
(c) 90° (d) 120°

(c) Given, in quadrilateral $ABCD$

$\angle B = 90^\circ$

and $AB^2 + BC^2 + CD^2 - AD^2 = 0$

In ΔABC, $\angle B = 90^\circ$



$\therefore AB^2 + BC^2 = AC^2$

$\therefore AB^2 + BC^2 + CD^2 - AD^2 = 0$

$\Rightarrow AC^2 + CD^2 = AD^2$

$\therefore \angle C = 90^\circ$

[By converse of Pythagoras theorem]

$\therefore \angle ACD = 90^\circ$

62. In a ΔABC, $AC = 12$ cm, $AB = 16$ cm and AD is the bisector of $\angle A$. If $BD = 4$ cm, then what is DC equal to?

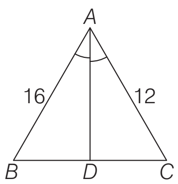
(a) 2 cm (b) 3 cm
(c) 4 cm (d) 5 cm

(b) Given, in ΔABC

$AC = 12$ cm, $AB = 16$ cm

$BD = 4$, AD is the bisector of $\angle A$.

By angle bisector,



$\frac{AB}{AC} = \frac{BD}{CD}$

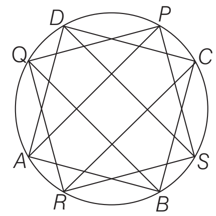
$\Rightarrow \quad \frac{16}{12} = \frac{4}{CD}$

$\therefore \quad CD = 3$

63. $ABCD$ is a cyclic quadrilateral. The bisectors of the angles A, B, C and D cut the circle at P, Q, R and S respectively. What is $\angle PQR + \angle RSP$ equal to?

(a) 90° (b) 135°
(c) 180° (d) 270°

(c) Here, $ABCD$ is cyclic quadrilateral, the bisector of the angles A, B, C and D cut the circle at P, Q, R and S respectively.



$\therefore PQRS$ is also cyclic quadrilateral

$\therefore \angle PQR + \angle RSP = 180^\circ$

64. ABC is an equilateral triangle. The side BC is trisected at D such that $BC = 3\,BD$. What is the ratio of AD^2 to AB^2?

(a) 7 : 9 (b) 1 : 3
(c) 5 : 7 (d) 1 : 2

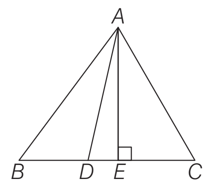
(a) Here, ABC is an equilateral triangle.

$\therefore AB = BC = AC$

and $BC = 3BD$

Draw an altitude AE on BC

$\therefore BE = EC$



In ΔABE,

$AE^2 + BE^2 = AB^2$

$AE^2 = AB^2 - BE^2 \quad \ldots(i)$

In ΔADE,

$AE^2 = AD^2 - DE^2 \quad \ldots(ii)$

From Eqs. (i) and (ii),

$AB^2 - BE^2 = AD^2 - DE^2$

$AB^2 - AD^2 = BE^2 - DE^2$

$= (BE + DE)(BE - DE)$

$\Rightarrow AB^2 - AD^2 = CD \cdot BD$

$[\because BE = CE]$

$\therefore \quad BE + DE = CD$

$\Rightarrow \quad AB^2 - AD^2 = \frac{2BC}{3} \times \frac{BC}{3}$

$\Rightarrow \quad AB^2 - AD^2 = \frac{2BC^2}{9}$

$\Rightarrow 9AB^2 - 9AD^2 = 2AB^2$

$\Rightarrow \quad 7AB^2 = 9AD^2$

$\Rightarrow \quad \frac{AD^2}{AB^2} = \frac{7}{9}$

$\therefore \quad AD^2 : AB^2 = 7 : 9$

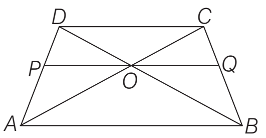
65. Consider the following statements :

1. The diagonals of a trapezium divide each other proportionally.
2. Any line drawn parallel to the parallel sides of a trapezium divides the non-parallel sides proportionally.

Which of the above statements is/are correct?

(a) only 1 (b) only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(c) 1. Diagonals of a trapezium divide each other proportionally are true



In trapezium $ABCD$

AB is parallel to CD

$\therefore$ In ΔAOB and ΔCOD

$\angle COD = \angle AOB$

[Vertically opposite angles]

$\angle OCD = \angle OAB$

[Alternate interior opposite angles]

$\therefore \Delta AOB \sim \Delta COD$

$$\frac{AO}{CO} = \frac{OB}{OD}$$

$\therefore$ Diagonals of trapezium divide each other proportionally.

2. In ΔDAB

PO is parallel to AB.

$$\therefore \frac{DP}{PA} = \frac{DO}{OB} \quad \ldots \text{(i) [By B.P.T]}$$

In ΔBCD,

OQ is parallel to DC.

$$\therefore \frac{DO}{OB} = \frac{CQ}{QB} \quad \ldots \text{(ii)}$$

[By B.P.T]

From Eqs. (i) and (ii),

$$\frac{DP}{PA} = \frac{CQ}{QB}$$

$\therefore$ Any line drawn parallel to the parallel sides of a trapezium divides non-parallel sides proportionally.

66. If H, C and V are respectively the height, curved surface area and volume of a cone, then what is $3\pi VH^3 + 9V^2$ equal to?

(a) C^2H^2 (b) $2\,C^2H^2$
(c) $5\,C^2H^2$ (d) $7C^2H^2$

(a) Here, H, C and V are respectively the height, curved surface area and volume of a cone.

$\therefore 3\pi VH^3 + 9V^2$

$$= 3\pi\left(\frac{1}{3}\pi r^2 H\right)H^3 + 9\left(\frac{1}{3}\pi r^2 H\right)^2 \quad \left[\because V = \frac{1}{3}\pi r^2 H\right]$$

$= \pi^2 r^2 H^4 + \pi^2 r^4 H^2$

$= \pi^2 r^2 H^2 [r^2 + H^2] = H^2 \pi^2 r^2 l^2$

$= H^2(\pi r l)^2 = H^2 C^2 \quad [\because C = \pi r l]$

$= C^2H^2$

67. How many solid lead balls each of diameter 2 mm can be made from a solid lead ball of radius 8 cm?

(a) 512 (b) 1024
(c) 256000 (d) 512000

(d) Here, $R = 8$ cm, $r = 0.1$ cm

Volume of sphere whose radius is 8 cm, $V_1 = \frac{4}{3}\pi(8)^3$

Volume of sphere whose radius is (0.1) cm, $V_2 = \frac{4}{3}\pi(0.1)^3$

$\because V_1 = nV_2$

$$\Rightarrow \frac{4}{3}\pi(8)^3 = n \times \frac{4}{3}\pi(0.1)^3$$

$\Rightarrow \quad n = 512 \times 1000$

$\therefore \quad n = 512000$

68. The two sides of a triangle are 40 cm and 41 cm. If the perimeter of the triangle is 90 cm, what is its area?

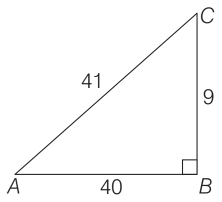
(a) 90 cm^2 (b) 135 cm^2
(c) 150 cm^2 (d) 180 cm^2

(d) Here, sides of triangle are 40 cm, 41 cm and perimetre of triangle is 90 cm.

$\therefore$ Third sides of triangle

$= 90 - (40 + 41) = 9$

$\therefore 40^2 + 9^2 = 1600 + 81 = 1681 = 41^2$



$\therefore \Delta ABC$ is a right angle.

Area of $\Delta ABC = \frac{1}{2} \times AB \times BC$

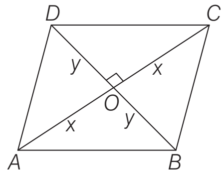
$= \frac{1}{2} \times 40 \times 9 = 180$ cm^2

69. The diagonals of a rhombus differ by 2 units and its perimeter exceeds the sum of the diagonals by 6 units. What is the area of the rhombus?

(a) 48 sq units (b) 36 sq units
(c) 24 sq units (d) 12 sq units

(c) Let the diagonal of rhombus.

$AC = 2x$ and $BD = 2y$



$\therefore OA = OC = x, \quad OB = OD = y$

$\therefore AC - BD = 2$

$\Rightarrow \quad x - y = 1 \quad \ldots$(i)

$DC^2 = OC^2 + OD^2 = x^2 + y^2$

Given perimetre of rhombus exceeds the sum of diagonal by 6 units.

$\therefore 4\,DC = AC + BD + 6$

$\Rightarrow 4\sqrt{x^2 + y^2} = 2x + 2y + 6$

$\Rightarrow 2\sqrt{x^2 + y^2} = x + y + 3 \Rightarrow 4(x^2 + y^2)$

$= x^2 + y^2 + 9 + 6x + 6y + 2xy$

$\Rightarrow 3(x^2 + y^2) = 6(x + y) + 9 + 2xy$

$\Rightarrow 3[(x - y)^2 + 2xy] - 2xy = 6(x + y) + 9$

$\Rightarrow 3 + 4xy = 6(x + y) + 9 \quad [\because x - y = 1]$

$\Rightarrow \quad 2xy = 3x + 3y + 3 \quad \ldots$ (ii)

From Eqs. (i) and (ii), we get $x = 4$, $y = 3$

Area of rhombus $= \frac{1}{2}(2x)(2y) = 2xy$

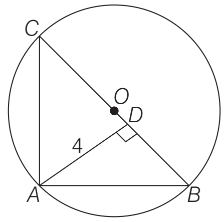
$= 2 \times 4 \times 3 = 24$ sq units

70. What is the area of a right-angled triangle, if the radius of the circumcircle is 5 cm and altitude drawn to the hypotenuse is 4 cm?

(a) 20 cm^2 (b) 18 cm^2 (c) 16 cm^2 (d) 10 cm^2

(a) ABC is a right angle triangle.

BC is hypotenuse.



Given, $OB = OC = r = 5$ cm

$\therefore \quad BC = 2r = 10$ cm

Altitude $\quad AD = 4$ cm

$\therefore$ Area of $\Delta ABC = \frac{1}{2} \times AD \times BC$

$= \frac{1}{2} \times 4 \times 10 = 20$ cm^2

71. In a triangle, values of all the angles are integers (in degree measure). Which one of the following cannot be the proportion of their measures?

(a) 1 : 2 : 3 (b) 3 : 4 : 5
(c) 5 : 6 : 7 (d) 6 : 7 : 8

(d) In triangle, angle of triangle are integers.

$\therefore$ Sum of angle of triangle is 180°.

(a) $x + 2x + 3x = 180^\circ$

$6x = 180^\circ$

$\Rightarrow x = 30^\circ$ is integer

(b) $3x + 4x + 5x = 180^\circ$

$12x = 180^\circ$

$\Rightarrow x = 15^\circ$ is integer

(c) $5x + 6x + 7x = 180^\circ$

$18x = 180^\circ$

$\Rightarrow x = 10^\circ$ is integer

(d) $6x + 7x + 8x = 180^\circ$

$21x = 180^\circ$

$x = \frac{180^\circ}{21}$ is not integer

$\therefore$ Only option (d) is not integer.

72. The length of a rectangle is increased by 10% and breadth is decreased by 10%. Then, the area of the new rectangle is

(a) neither increased nor decreased
(b) increased by 1%
(c) decreased by 1%
(d) decreased by 10%

(c) Let the length of rectangle $= x$ m

and breadth of rectangle $= y$ m

$\therefore$ Area of rectangle $= xy$

When length is increased by 10%

$\therefore$ New length $= \frac{110x}{100}$

When breadth is decreased by 10%

$\therefore$ New breadth $= \frac{90y}{100}$

Area of new rectangle

$= \frac{9900}{10000}xy = \frac{99}{100}xy$

Difference $= xy - \frac{99xy}{100} = \frac{1}{100}xy$

Percentage decrease in area of new rectangle $= \frac{xy/100}{xy} \times 100\%$

$= 1\%$

$\therefore$ Area of new rectangle is decreased by 1%.

73. The surface areas of two spheres are in the ratio 1 : 4. What is the ratio of their volumes?

(a) 1 : 16 (b) 1 : 12
(c) 1 : 10 (d) 1 : 8

(d) Let the surface areas of two spheres is s_1 and s_2 and their radii is r_1 and r_2 respectively.

$\therefore s_1 = 4\pi r_1^2,\ s_2 = 4\pi r_2^2$

$\frac{s_1}{s_2} = \frac{r_1^2}{r_2^2}$

$\Rightarrow \frac{1}{4} = \left(\frac{r_1}{r_2}\right)^2$ $[\because s_1 : s_2 = 1 : 4]$

$\Rightarrow \frac{r_1}{r_2} = \frac{1}{2} \therefore \frac{V_1}{V_2} = \frac{4/3\pi r_1^3}{4/3\pi r_2^3}$

$= \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1}{2}\right)^3$

$= \frac{1}{8}$

$\therefore V_1 : V_2 = 1 : 8$

74. The length, breadth and height of a brick are 20 cm, 15 cm and 10 cm respectively. The number of bricks required to construct a wall with dimensions 45 m length, 0.15 m breadth and 3 m height is

(a) 12450 (b) 11250
(c) 6750 (d) None of these

(c) Given, $l = 20$ cm, $b = 15$ cm

$h = 10$ cm

Volume of one brick
$= 20 \times 15 \times 10 = 3000\ \text{cm}^3$

Dimension of walls is 45m, 0.15 m and 3 m,

$\therefore$ Volume of wall
$= (45 \times 0.15 \times 3)\ \text{m}^3$

$= (20.25)\ \text{m}^3 = 20.25 \times 10^6 \text{cm}^3$

Number of bricks

$= \frac{\text{Volume of wall}}{\text{Volume of one brick}}$

$= \frac{20.25 \times 10^6}{3000}$

$= \frac{2025 \times 10^4}{3 \times 10^3} = 6750$

75. If the sum of all interior angles of a regular polygon is twice the sum of all its exterior angles, then the polygon is

(a) Hexagon (b) Octagon
(c) Nonagon (d) Decagon

(a) Interior angle of regular polygon whose side n is $\left(\frac{n-2}{n}\right)\pi$.

Exterior angle of regular polygon whose side n is $\pi\left(1 - \frac{n-2}{n}\right) = \pi\left(\frac{2}{n}\right)$

$\therefore \left(\frac{n-2}{n}\right)\pi = 2\left(\frac{2\pi}{n}\right)$

$\Rightarrow n - 2 = 4$

$\Rightarrow n = 6$

Hence, the polygon is hexagon.

76. A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel? (Assume $\pi = \frac{22}{7}$)

(a) 17.5 cm (b) 35 cm
(c) 70 cm (d) 140 cm

(b) We have,

A bicycle wheel makes 5000 revolution in moving 11 km.

$\therefore$ In one revolutions distance covered by wheel of bicycle is $\frac{11}{5000}$ km

$= \frac{11000}{5000} = \frac{11}{5}$ m

Radius of wheel of bicycle $= r$ cm

$\therefore 2\pi r = \frac{11}{5} \times 100$ cm

$r = \frac{11 \times 20 \times 7}{2 \times 22}$

$= 35$ cm

77. The volumes of two cones are in the ratio 1 : 4 and their diameters are in the ratio 4 : 5. What is the ratio of their heights?

(a) 25 : 64 (b) 16 : 25
(c) 9 : 16 (d) 5 : 9

(a) Given, $\frac{V_1}{V_2} = \frac{1}{4}$ and $\frac{r_1}{r_2} = \frac{4}{5}$

$\therefore \frac{\frac{1}{3}\pi r_1^2 h_1}{\frac{1}{3}\pi r_2^2 h_2} = \frac{1}{4}$

$\Rightarrow \left(\frac{r_1}{r_2}\right)^2 \times \frac{h_1}{h_2} = \frac{1}{4}$

$\Rightarrow \left(\frac{4}{5}\right)^2 \times \frac{h_1}{h_2} = \frac{1}{4}$

$\Rightarrow \frac{h_1}{h_2} = \left(\frac{5}{4}\right)^2 \times \frac{1}{4} = \frac{25}{64}$

$\therefore h_1 : h_2 = 25 : 64$

78. In a triangle ABC, if $2\angle A = 3\angle B = 6\angle C$, then what is $\angle A + \angle C$ equal to?

(a) 90° (b) 120°
(c) 135° (d) 150°

(b) Given, in triangle ABC

$$2\angle A = 3\angle B = 6\angle C$$

$\because \quad \angle A + \angle B + \angle C = 180^\circ$

$\Rightarrow \frac{3}{2}\angle B + \angle B + \frac{1}{2}\angle B = 180^\circ$

$\Rightarrow \quad 3\angle B = 180^\circ \Rightarrow \angle B = 60^\circ$

$\therefore \angle A + \angle C = 180^\circ - \angle B$

$= 180^\circ - 60^\circ = 120^\circ$.

79. If the perimeter of a circle and a square are equal, then what is the ratio of the area of the circle to that of the square?

(a) $1:\pi$ (b) $2:\pi$ (c) $3:\pi$ (d) $4:\pi$

(d) Given,

Perimetre of a circle and square are equal.

$\therefore 2\pi r = 4x \Rightarrow \pi r = 2x$

Area of circle $= \pi r^2$

Area of square $= x^2 = \left(\frac{\pi r}{2}\right)^2 = \frac{\pi^2 r^2}{4}$

Ratio of area of circle to square

$$= \frac{\pi r^2}{\frac{\pi^2 r^2}{4}} = \frac{4}{\pi} = 4:\pi$$

80. The lengths of the sides of a right- angled triangle are consecutive even integers (in cm). What is the product of these integers?

(a) 60 (b) 120 (c) 360 (d) 480

(d) Let the sides of a right angle triangle is $x-2, x, x+2$.

$\therefore \quad (x+2)^2 = x^2 + (x-2)^2$

$x^2 + 4x + 4 = x^2 + x^2 - 4x + 4$

$x = 8$

$\therefore$ Sides of triangle are 6, 8, 10.

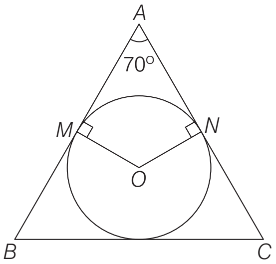
Product of sides of triangle
$= 6 \times 8 \times 10 = 480$.

81. A circle is inscribed in a triangle ABC. It touches the sides AB and AC at M and N respectively. If O is the centre of the circle and $\angle A = 70^\circ$, then what is $\angle MON$ equal to?

(a) 90° (b) 100° (c) 110° (d) 120°

(c) Given,

A circle inscribed in a triangle ABC.



Circle touches sides AB and AC at M and N respectively.

$\therefore MA$ and NA is tangent of circle

$\therefore \quad \angle OMA = \angle ONA = 90^\circ$

$\therefore \angle A + \angle MON = 180^\circ$

$70^\circ + \angle MON = 180^\circ$

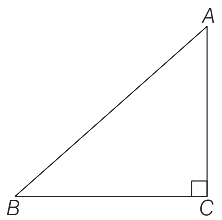
$\angle MON = 110^\circ$

82. The sum of the squares of sides of a right-angled triangle is 8450 square units. What is the length of its hypotenuse?

(a) 50 units (b) 55 units
(c) 60 units (d) 65 units

(d) In right angle triangle ABC, $\angle C = 90^\circ$

Given, $AB^2 + BC^2 + AC^2 = 8450$



$AB^2 + AB^2 = 8450 \Rightarrow 2AB^2 = 8450$

$\therefore \quad AB = \sqrt{4225} = 65$ units

83. A triangle and a parallelogram have equal areas and equal bases. If the altitude of the triangle is k times the altitude of the parallelogram, then what is the value of k?

(a) 4 (b) 2 (c) 1 (d) $\frac{1}{2}$

(b) Let base of triangle and parallelogram be b and altitude of triangle and parallelogram be h_1 and h_2 respectively.

Given, area of triangle = Area of parallelogram and $h_1 = kh_2$

$\Rightarrow \quad \frac{h_1}{h_2} = k$

$\therefore \frac{1}{2}bh_1 = bh_2$

$\frac{h_1}{h_2} = 2$

84. Areas of two squares are in the ratio $m^2 : n^4$. What is the ratio of their perimeters?

(a) $m:n$ (b) $n:m$
(c) $m:n^2$ (d) $m^2:n$

(c) Given, ratio of area of two squares is $m^2 : n^4$.

$$\frac{A_1}{A_2} = \frac{m^2}{n^4} \Rightarrow \frac{x_1^2}{x_2^2} = \frac{m^2}{n^4} \Rightarrow \frac{x_1}{x_2} = \frac{m}{n^2}$$

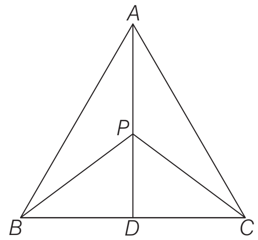
Ratio of perimeter of square
$= 4x_1 : 4x_2 = x_1 : x_2 = m : n^2$

85. AD is the median of the triangle ABC. If P is any point on AD, then which one of the following is correct?

(a) Area of triangle PAB is greater than the area of triangle PAC.
(b) Area of triangle PAB is equal to area of triangle PAC.
(c) Area of triangle PAB is one-fourth of the area of triangle PAC.
(d) Area of triangle PAB is half of the area of triangle PAC.

(b) We have,
AD is median of the ΔABC.

P is any point on AD.



Area of ΔABD = Area of ΔADC

In ΔPBC, PD is also median of ΔPBC

$\therefore$ Area of ΔPBD = Area of ΔPCD

Area of ΔPAB = Area of ΔABD – Area of ΔPBD

Area of ΔPAC = Area of ΔABD – Area of ΔPDC

$\therefore$ Area of ΔPAB is equal to Area of ΔPAC.

86. What is the area of a segment of a circle of radius r subtending an angle θ at the centre?

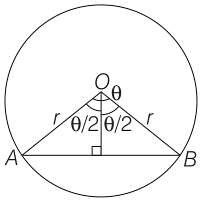
(a) $\frac{1}{2}r^2\theta$

(b) $\frac{1}{2}r^2\left(\theta - 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)$

(c) $\frac{1}{2}r^2\left(\theta - \sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)$

(d) $\frac{1}{2}r^2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$

(b) Area of segment of a circle of radius r subtending an angle θ at the centre



$= $ Area of sector – Area of ΔOAB

$= \frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$

$= \frac{1}{2}r^2(\theta - \sin\theta)$

$= \frac{1}{2}r^2\left(\theta - 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right)$

$\left[\because \sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\right]$

87. ABC is a triangle right-angled at C. Let P be any point on AC and Q be any point on BC. Which of the following statements is/are correct?

1. $AQ^2 + BP^2 = AB^2 + PQ^2$
2. $AB = 2PQ$

Select the correct answer using the code given below :

(a) only 1 (b) only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(a) Given, ABC is a right angled at C and P and Q any point on AC and BC respectively.

In ΔAQC

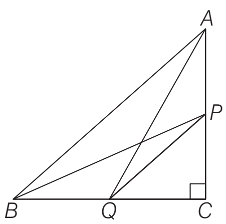
$AQ^2 = QC^2 + AC^2$ ...(i)

In ΔBPC,

$BP^2 = BC^2 + PC^2$ ...(ii)

Adding Eqs. (i) and (ii),

$AQ^2 + BP^2 = QC^2 + AC^2 + BC^2 + PC^2$



$AQ^2 + BP^2 = AC^2 + BC^2 + QC^2 + PC^2 = AB^2 + PQ^2$

$\therefore$ Ist statement is correct.

If P and Q are mid-point of AC and BC respectively, then $AB = 2PQ$

$\therefore$ 2nd statement is incorrect.

88. Four circular coins of equal radius are placed with their centres coinciding with four vertices of a square. Each coin touches two other coins. If the uncovered area of the square is 42 cm^2, then what is the radius of each coin? (Assume $\pi = \frac{22}{7}$)

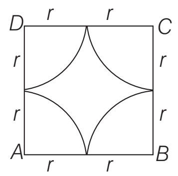
(a) 5 cm (b) 7 cm
(c) 10 cm (d) 14 cm

(b) Let radius of coins $= r$

$\therefore$ Sides of square $= 2r$

Area of shaded region

= Area of square – Area of four quadrants



$= (2r)^2 - \pi r^2$

$= 4r^2 - \pi r^2$

$= r^2\left(4 - \frac{22}{7}\right) = \frac{6r^2}{7}$

$\Rightarrow \frac{6r^2}{7} = 42$

$\Rightarrow r^2 = 49$

$\Rightarrow r = 7$ cm

89. The radii of the flat circular faces of a bucket are x and $2x$. If the height of the bucket is $3x$, what is the capacity of the bucket? (Assume $\pi = \frac{22}{7}$)

(a) $11x^3$ (b) $22x^3$ (c) $44x^3$ (d) $55x^3$

(b) Given,

Radius of flat circular faces of a bucket are x and $2x$ and height of bucket is $3x$.

Volume of bucket

$= \frac{1}{3}\pi h(R^2 + r^2 + Rr)$

$\therefore$ Capacity of bucket

$= \frac{1}{3}\pi(3x)\{(2x)^2 + (x)^2 + (2x)(x)\}$

$= \pi x(4x^2 + x^2 + 2x^2)$

$= 7\pi x^3$

$= 7 \times \frac{22}{7}x^3 = 22x^3$

90. If p, q, r, s and t represent length, breadth, height, surface area and volume of a cuboid respectively, then what is $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$ equal to?

(a) $\frac{s}{t}$ (b) $\frac{2t}{s}$ (c) $\frac{s}{2t}$ (d) $\frac{2s}{t}$

(c) We have, p, q, r, s and t represent the length, breadth, height, surface area and volume of cuboid

$\therefore s = 2(pq + qr + rp),\ t = pqr$

$\frac{s}{t} = 2\left(\frac{pq + qr + rp}{pqr}\right)$

$= 2\left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r}\right)$

$\Rightarrow \frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{s}{2t}$

91. Fifteen candidates appeared in an examination. The marks of the candidates who passed in the examination are 9, 6, 7, 8, 8, 9, 6, 5, 4 and 7. What is the median of marks of all the fifteen candidates?

(a) 6 (b) 6.5
(c) 7 (d) 7.5

(a) The marks of the candidates who passed in examination are 9, 6, 7, 8, 8, 9, 6, 5, 4, 7

Rearranging marks in ascending order of all the fifteen candidates

–, –, –, –, –, 4, 5, 6, 6, 7, 7, 8, 8, 9, 9

Median of marks of all the fifteen candidates

$= \frac{(15 + 1)\text{th observation}}{2}$

= 8 th observation

= 6

92. If the yield (in gm) of barley from 7 plots of size one square yard each, were found to be 180, 191, 175, 111, 154, 141 and 176, then what is the median of yield?

(a) 111 gm (b) 154 gm
(c) 175 gm (d) 176 gm

(c) Given data

180, 191, 175, 111, 154, 141, 176

Rearranging in ascending order

111, 141, 154, 175, 176, 180, 191

Median $= \left(\frac{7 + 1}{2}\right)$th observations

= 4 th observations

= 175

93. Which one of the following measures of central tendency will be used to determine the average size of the shoe sold in the shop?

(a) Arithmetic mean
(b) Geometric mean
(c) Median
(d) Mode

(a) Average size of the shoe sold in shop by Arithmetic mean.

94. When the class intervals have equal width, the height of a rectangle in a histogram represents

(a) Width of the class
(b) Lower class limit
(c) Upper class limit
(d) Frequency of the class

(d) The height of a rectangle in a histogram represents frequency of the class.

95. The ages of 7 family members are 2, 5, 12, 18, 38, 40 and 60 yr respectively. After 5 yr a new member aged x year is added. If the mean age of the family now goes up by 1.5 yr, then what is the value of x?

(a) 1 (b) 2
(c) 3 (d) 4

(b) We have,

Ages of 7 family members are

2, 5, 12, 18, 38, 40, 60

Mean $= \dfrac{2+5+12+18+38+40+60}{7}$

$= \dfrac{175}{7}$

$= 25$

Ages of 7 family members after 5 yr

7, 10, 17, 23, 43, 45, 65

After 5 yr new member ages x years is add

$\therefore$ Mean $= \dfrac{7+10+17+23+43+45+65+x}{8}$

$= \dfrac{210+x}{8}$

New mean $= 25 + 1.5 = 26.5$

$\therefore 26.5 = \dfrac{210+x}{8}$

$\Rightarrow 212 = 210 + x \Rightarrow x = 2$

96. The mean weight of 100 students in a class is 46 kg. The mean weight of boys is 50 kg and that of girls is 40 kg. The number of boys exceeds the number of girls by

(a) 10 (b) 15 (c) 20 (d) 25

(c) We have, 100 student in a class.

Let number of boys $= x$

and number of girls $= y$

$\therefore \quad x + y = 100 \quad \ldots$(i)

Mean of weight of boys = 50 kg

$\therefore$ Total weight of boys $= 50\,x$ kg

Mean of weight of girls = 40 kg

$\therefore$ Total weight of girls $= 40y$ kg

Mean of weight of 100 students

= 46 kg

$\therefore$ Total weight of 100 students = 4600 kg

$\therefore \quad 50x + 40y = 4600 \quad \ldots$(ii)

From Eqs. (i) and (ii),

$x = 60,\ y = 40$

$\therefore \quad x - y = 60 - 40$

$= 20$

97. What is the algebraic sum of the deviations from the mean of a set of values 25, 65, 73, 75, 83, 76, 17, 15, 7, 14?

(a) – 1 (b) 0
(c) 1 (d) 2

(b) We know that, algebraic sum of the deviation from the mean of any data is zero.

98. The mean of five observations x, $x+2$, $x+4$, $x+6$, $x+8$ is m. What is the mean of the first three observations?

(a) m (b) $m-1$ (c) $m-2$ (d) $m-3$

(c) Given data x, $x+2$, $x+4$, $x+6$, $x+8$

Mean $= m$

$\therefore m = \dfrac{x+x+2+x+4+x+6+x+8}{5}$

$5m = 5x + 20$

$x = m - 4$

$\therefore$ Mean of x, $x+2$, $x+4$

$= \dfrac{x+x+2+x+4}{3}$

$= \dfrac{3x+6}{3} = x + 2 = m - 4 + 2 = m - 2$

99. What is the median of 2, 4, 6, ... , 100?

(a) 48 (b) 49 (c) 50 (d) 51

(d) Given, data

2, 4, 6, ... , 100

Here, $n = 50$

$\therefore$ Median $= \dfrac{(25\text{th} + 26\text{th}) \text{ Observation}}{2}$

$\therefore$ 25 th observation $= 25 \times 2 = 50$

26 th observation $= 26 \times 2 = 52$

$\therefore$ Median $= \dfrac{50+52}{2} = \dfrac{102}{2} = 51$

100. The harmonic mean and the geometric mean of two numbers are 10 and 12 respectively. What is their arithmetic mean?

(a) $\dfrac{25}{3}$ (b) $\sqrt{120}$
(c) 11 (d) 14.4

(d) Given, HM = 10,

GM = 12

We know that,

$\text{AM.HM} = (\text{GM})^2 \Rightarrow \text{AM}\,(10) = (12)^2$

$\Rightarrow \quad \text{AM} = \dfrac{144}{10} = 14.4$

PAPER II English

Directions (Q. Nos. 1-10) *In this section you have two short passages. After each passage, you will find some items based on the passage. Read the passages and answer the items based on them. You are required to select your answers based on the content of the passage and opinion of the author only.*

PASSAGE - I

Post colonial cultural analysis has been concerned with the elaboration of theoretical structures that contest the previous dominant western ways of seeing things. A simple analogy would be with feminism, which has involved a comparable kind of project: there was a time when any book you might read, any speech you might hear, any film that you saw, was always told from the point of view of male.

The woman was there, but she was, always an object, never a subject. From what you would read, or the films you would see, the woman was always the one who was looked at. She was never the observing eye. For centuries it was assumed that women were less intelligent than men and that they did not merit the same degree of education. They were not allowed a vote in the political system.

By the same token, any kind of knowledge developed by women was regarded as non-serious, trivial, gossip or alternatively as knowledge that had been discredited by science, such as superstition or traditional practices of childbirth or healing. All these attitudes were part of a larger system in which women were dominated, exploited, and physically abused by men. Slowly, but increasingly, from the end of 18th century, feminists began to contest this situation. The more they contested it, the more it became increasingly obvious that these attitudes extended into the whole of the culture; social relations, politics, law, medicine, the arts, popular and academic knowledge.

1. Post colonialism is

(a) a contestation of the then existing dominant western practices
(b) a contestation of western practices in colonial states
(c) a contestation of the superstitious practices
(d) an approval of indigenous practices

(a) Post colonialism as defined in the passage is 'that contest the previous dominant western ways of seeing things'. Accordingly, option (a) is the correct answer.

2. What does '........... she was always an object, never a subject' mean ?

(a) Women were given respect and worshipped
(b) Women were not given any right equal to men
(c) Women were treated at par with men
(d) Women liked to be treated inferior to men

(b) The given phrase refers to the inequality that was meted out against women wherein she was never given any right as equal to men.
Hence, option (b) correctly elucidates this fact and thus is the correct answer.

3. Why was 'she never the observing eye'?

(a) She was beautiful, so she was observed by men
(b) She liked to be observed by men
(c) Women were assumed to be less intelligent than men
(d) Women were assumed to be more intelligent than men

(b) The word 'observing eye' in the phrase refers to the qualities of wit and intelligence. Then the statement that women were never the observing eye indicates that women were assumed to be less intelligent than men.

4. The contestation to dominance of the male resulted in

(a) participation of women in social relations, politics, law, medicine, the arts, popular and academic knowledge
(b) participation of men in social relations, politics, law, medicine, the arts, popular and academic knowledge
(c) participation of women in social movements
(d) contestations with males in life leading to divorce

(a) The given passage clearly states that to contradict or fight against the male dominance, women began coming out of their homes and started participating in social relations, politics, law, medicine, the arts, popular and academic knowledge.

5. Which word in the passage is opposite of 'contrast'?

(a) Contestations (b) Trivial
(c) Discredited (d) Analogy

(d) The word contrast means different. Its opposite is analogy which means similarity.

PASSAGE - II

How wonderful is the living world! The wide range of the living types is amazing. The extraordinary habitats in which we find living organisms, be it cold mountains, deciduous forests, oceans, fresh water lakes; deserts or hot springs, leave us speechless. The beauty of a galloping horse, or a migrating bird, the valley of flowers or the attacking shark evokes awe and a deep sense of wonder. The ecological conflict and cooperation among members of a population and among populations of a community or even a molecular traffic inside a cell make us deeply reflect on - what indeed is life? This question has two implicit questions within it. The first is a technical one and seeks answer to what living is as opposed to the non-living, and the second is the philosophical one, and seeks answer to what the purpose of life is. What is living? When we try to define 'living', we conventionally look for distinctive characteristics exhibited by living organisms. Growth, reproduction, ability to sense environment and mount a suitable response come to our mind immediately as unique features of living organisms. One can add a few more features like metabolism, ability to self-replicate, self-organise, interact and emergence to this list.

6. Why are the living types amazing?

(a) The extraordinary diversity of habitats makes it amazing
(b) The living organisms are acting as per their interests

(c) The human thinking makes the living types amazing
(d) The evolution of life makes it amazing

(a) The given passage states "The wide range of the living types is amazing". This indicates that the diversity of organism and habitats makes the living types amazing.

7. Why does the author say, 'ecological conflict and cooperation' ?
(a) Because living organisms are structured this way
(b) Because ecological mechanism works with conflict and cooperation
(c) Because humans want to fight and live together
(d) Because living organisms sometimes fight and sometimes live together

(d) The author states this to highlight the tendency of all living organisms to fight in certain occasions and to live peacefully in others.

8. Which of the following statements is true of the passage?
(a) Meaning of life could be reflected as to what living is as opposed to the non-living and what the purpose of life is
(b) Meaning of life could be reflected as to how living organisms live and non-living organisms exist
(c) Meaning of life could be reflected as to where the life begins and where it ends
(d) Meaning of life could be reflected on how various living organisms differ

(a) The given passage states that the meaning of life could be reflected as to what living is as opposed to the non-living and what the purpose of life is.

9. Distinctive characteristics exhibited by organisms indicate that
(a) they are living organisms
(b) they are non-living organisms
(c) they can be either living organisms or non-living organisms
(d) they know the purpose of life

(a) According to the given passage, the distinctive characteristics indicate that an organism is a living being.

10. Which word in the passage means 'unique'?
(a) common (b) characteristics
(c) distinctive (d) general

(c) The synonym of unique is distinctive.

Directions (Q.Nos. 11-20) *Each item in this section has a sentence with three underlined parts labelled as (a), (b) and (c). Read each sentence to find out whether there is any error in any underlined part and indicate your response on the Answer Sheet against the corresponding letter, i.e., (a) or (b) or (c). If you find no error, your response should be indicated as (d).*

11. He has been (a) one the most revered member (b) of the committee of enquiry (c). No error. (d)

(b) Preposition 'of' will be added after "one…the" to make the sentence grammatically correct.

12. Rahul asked me (a) whether I was interested (b) to joining the group for the trip (c). No error. (d)

(c) Preposition 'in' will replace 'to' to make the sentence grammatically correct.

13. 'Where there is a will (a) then there is a way' (b) is an old epithet (c). No error. (d)

(b) Conjunction 'then' will be removed to make the sentence grammatically correct.

14. Indian feminism grew out of the women's movements (a) of the late nineteenth century, (b) reached full maturity in the early twentieth century (c). No error. (d)

(c) Article 'the' will be removed to make the sentence grammatically correct.

15. The greatest merit of democracy (a) is that everyone feels free (b) and can pursues his/her interest (c). No error. (d)

(c) 'Pursues' will be replaced by 'pursue' to make the sentence grammatically correct.

16. All stake holders of education (a) have the right to ask for accountability (b) in every aspects of its implementation (c). No error. (d)

(c) 'Aspects' will be replaced by 'aspect' to make the sentence grammatically correct.

17. Learning many languages (a) promotes linguistic, cultural and social harmonies (b) among people speaking different languages (c). No error. (d)

(b) 'Harmonies' will be replaced by 'harmony' to make the sentence grammatically correct.

18. One should not act (a) according to one's (b) whims and fancies on public places (c). No error. (d)

(c) Preposition 'on' will be replaced by 'in' to make the sentence grammatically correct.

19. Economists believe that (a) India had taken a new turn in 1990 (b) with the liberalisation to her economy (c). No error. (d)

(c) Preposition 'to' will be replaced by 'of' to make the sentence grammatically correct.

20. Irrigation works have a special importance (a) in an agricultural countries like India, (b) where rainfall is unequally distributed throughout the seasons. (c) No error. (d)

(b) Article 'an' will be removed to make the sentence grammatically correct.

Directions (Q. Nos. 21-30) *Each of the following items, in this section consists of a sentence, parts of which have been jumbled. These parts have been labelled as* P, Q, R *and* S. *Given below each sentence are four sequences namely (a), (b), (c) and (d). You are required to re-arrange the jumbled parts of the sentence and mark your response accordingly.*

21. history of life (P) evolutionary Biology is (Q) forms on earth (R) the study of (S)

The correct sequence should be

(a) S P Q R (b) Q S P R (c) R P Q S (d) P S Q R

(b) The correct and meaningful sentence is given by QSPR.

22. life is considered (P) the origin of (Q) the history of universe (R) a unique event in (S)

The correct sequence should be

(a) Q P S R (b) P S Q R (c) S Q P R (d) R S P Q

(a) The correct and meaningful sentence is given by QPSR.

23. productive resources is (P) how we manage (Q) and competitiveness (R) critical to strategic growth (S)

The correct sequence should be

(a) P Q R S (b) R S P Q (c) S R P Q (d) Q P S R

(d) The correct and meaningful sentence is given by QPSR.

24. in service firms (P) operations strategy (Q) from the corporate strategy (R) is generally inseparable (S)

The correct sequence should be

(a) S R Q P (b) Q P S R (c) R S P Q (d) P S Q R

(b) The correct and meaningful sentence is given by QPSR.

25. are travelling, (P) a recent survey has revealed (Q) that they are worried about their safety (R) even as more and more Indians (S)

The correct sequence should be

(a) S P Q R (b) Q S R P (c) P R S Q (d) R P S Q

(a) The correct and meaningful sentence is given by SPQR.

26. the imagination of children (P) stories can exercise. (Q) more than the stories (R) because they tell (S)

The correct sequence should be

(a) Q R S P (b) S P Q R
(c) Q P S R (d) R S Q P

(c) The correct and meaningful sentence is given by QPSR.

27. as a record of (P) and suffering of humans (Q) the achievements, experiments (R) history is considered (S)

The correct sequence should be

(a) S P R Q (b) R Q S P
(c) P Q R S (d) Q R S P

(a) The correct and meaningful sentence is given by SPRQ.

28. can be invented (P) it appears (Q) has been invented (R) that all that (S)

The correct sequence should be

(a) Q S P R (b) Q R S P
(c) R S Q P (d) S P Q R

(a) The correct and meaningful sentence is given by QSPR.

29. during the last century (P) Indian social, political and cultural life (Q) as a testimony of (R) Indian cinema stands (S)

The correct sequence should be

(a) S P Q R (b) Q R S P
(c) P Q R S (d) S R Q P

(d) The correct and meaningful sentence is given by SRQP.

30. of all searches for knowledge (P) should be the beginning (Q) an exploration into truth (R) and experiments of life (S)

The correct sequence should be

(a) R Q P S (b) S P Q R (c) R S P Q (d) Q R S P

(a) The correct and meaningful sentence is given by RQPS.

Directions (Q. Nos. 31-39) *Given below are some idioms/phrases followed by four alternative meanings to each. Choose the response (a), (b), (c) or (d) which is the most appropriate expression and mark your response in the Answer Sheet accordingly.*

31. Get the jitters

(a) Feeling anxious (b) Feeling happy
(c) Stammering (d) Feeling exposed

(a) The idiom 'get the jitters' means 'to feel anxious'.

32. French leave
(a) Absent from work without asking for permission in French
(b) Asking for permission before leaving work
(c) Work for permission to get leave
(d) Absent from work without asking for permission

(d) The idiom 'French leave' means 'to be absent from work without informing or asking for permission'.

33. Take a stand
(a) To publicly express an opinion about something
(b) To make a stand for one to sit
(c) To be firm on your work
(d) To be part of the work

(a) The idiom 'take a stand' means 'to publicly express opinions'.

34. Cut and run
(a) To avoid a difficult situation by leaving suddenly
(b) To avoid an event suddenly
(c) To meet some danger suddenly
(d) To ask for sudden meeting with someone

(a) The idiom 'cut and run' means 'to avoid a situation by leaving suddenly'.

35. Cut the cord
(a) To stop needing your parents for money
(b) To stop needing someone else to look after you and start acting independently
(c) To be safe on your own
(d) To be a married person

(b) The idiom 'cut the cord' means ' to stop needing or relying on someone to look after you and start acting independently'.

36. Cupboard love
(a) Loving someone to get something from the person
(b) Loving the cupboards
(c) Innocent love
(d) Loving to be free of all conditions

(a) The idiom 'Cupboard love' means 'to love someone for ones own benefit or to love someone to get something from the person'.

37. Around the corner
(a) A thing which is at the end of the corner
(b) An event or thing which is going to happen soon
(c) An event that corners someone for his wrong
(d) An event that happens in the corner of powerful place.

(b) The phrase 'around the corner' means 'an event that is going to take place soon'.

38. With Heavy Heart
(a) With heavy weight
(b) With joy and humour
(c) With sense of shame
(d) With pain and regret

(d) The idiom 'With heavy heart' means 'with pain and regret'.

39. Cost a bomb
(a) To be very arrogant
(b) To be with rich people
(c) To be very expensive
(d) To be stingy

(c) The idiom 'cost a bomb' means 'to be very expensive'.

40. Roll your sleeves up
(a) To prepare for wrestling
(b) To prepare for hard work
(c) To make someone work for you
(d) To work with others

(b) The idiom 'roll your sleeves up' means 'to prepare for hard work'.

Directions (Q.Nos. 41-50) *In this section each item consists of six sentences of a passage. The first and sixth sentences are given in the beginning as* S1 *and* S6. *The middle four sentences in each have been jumbled up and labelled as* P, Q, R *and* S. *You are required to find the proper sequence of the four sentences and mark your response accordingly on the Answer Sheet.*

41. S1: The country's economy is growing and would continue to grow at a rapid pace in the coming years.
S6 : The market share of electrical vehicles increases with increasing availability of infrastructure.
P : It also provides us an opportunity to grow as manufacturer of electric vehicles.
Q : According to NITI Aayog (2019), if India reaches an electric vehicles sales penetration, emission and oil savings can be achieved.
R : Given the commitments that India has made on the climate front as a nation and on environmental aspects, it is likely that larger and larger share of automobile sector would be in the form of electric vehicles.
S : This presents a great opportunity for the automobile industry as the demand for automobiles would only increase.

The correct sequence should be
(a) S R Q P (b) R Q S P
(c) Q P S R (d) Q S R P

(a) The correct and meaningful order of sentences is SRQP.

42. S1: Central government receipts can broadly be divided into non-debt and debt receipts.
S6 : This is also evident from the composition of non-debt receipts.
P : Debt receipts mostly consist of market borrowing and other liabilities which the government is obliged to repay in the future.
Q : The non-debt receipts comprise of tax revenue, non-tax revenue, recovery of loans and disinvestment receipts.
R : The outcomes as reflected in the Provisional Actual figures is lower than the budget estimate owing to reduction in the net tax revenue.
S : The Budget 2018-19 targeted significantly high growth in non debt receipts of the Central Government, which was driven by robust growth.

The correct sequence should be
(a) S R P Q (b) R S Q P
(c) P Q R S (d) Q P R S

(d) The correct and meaningful order of sentences is QPRS.

43. S1: Palaeontology is the study of the remains of dead organisms over enormous spans of time.
S6 : Faunal analysis gives information about the animal people hunted and domesticated, the age of animal at death, and the diseases that afflicted them.

P : Bones provide a great information.
Q : The distribution of faunal remains (animal bones) at a site can indicate which areas were used for butchering, cooking, eating, bone tool making and refuse dumping.
R: Within this discipline, molecular biology and DNA studies have been used to understand hominid evolution.
S : Hominid evolution answers the questions about what ancient people looked like, and to plot patterns of migration.

The correct sequence should be
(a) Q P R S (b) S P Q R
(c) R S P Q (d) P Q R S

(c) The correct and meaningful order of sentences is RSPQ.

44. S1 : Hormones have several functions in the body.
S6 : The two hormones together regulate the glucose level in the blood.
P : They help to maintain the balance of biological activities in the body.
Q : Insulin is released in response to the rapid rise in blood glucose level.
R : On the other hand hormone glucagon tends to increase the glucose level in the blood.
S : The role of insulin in keeping the blood glucose level within the narrow limit is an example of this function.

The correct sequence should be
(a) P S R Q (b) R S P Q
(c) S R Q P (d) Q R S P

(a) The correct and meaningful order of sentences is PSRQ.

45. S1 : All living things affect the living and non-living things around them.
S6 : This interdependability needs to be understood when we, humans, consume much morethan required and abuse nature.
P : This can also affect the population of fox, if foxes depend on rabbits for food.
Q : For example, earthworms, make burrows and worm casts.
R : This act of earthworms affects the soil, and therefore the plants growing in it.
S : Rabbit's fleas carry the virus which causes myxomatosis, so they can affect the size of the rabbit population.

The correct sequence should be
(a) R S Q P (b) P S R Q
(c) Q R S P (d) S Q R P

(c) The correct and meaningful order of sentences is QRSP.

46. S1 : The ecosystem of water is complex and many environmental factors are intricately linked.
S6 : The trees slowly transfer rainwater into the sub-soil and this is critical for sustaining water for months after the rains.
P : Thick forests make for excellent catchments.
Q : The problems we see are because we have undermined, these links over decades.
R : First, rain and snowfall are the only sources of water-about 99%.
S : In the four months of monsoon, there are about 30-35 downpours and the challenge is to hold this water in systems that can last us over 365 days.

The correct sequence should be
(a) Q R S P (b) P S R Q
(c) S R Q P (d) R Q S P

(a) The correct and meaningful order of sentences is QRSP.

47. S1: Politics is exciting because people disagree.
S6 : It is not solitary people who make politics and a good society; it is the people together which make good politics and society.
P : For Aristotle politics is an attempt to create a good society because politics is, above all, a social activity.
Q : They also disagree about how such matters should be resolved, how collective decision should be made and who should have a say.
R : They disagree about how they should live.
S : Who should get what? How should power and other resource be distributed? Should society be based on cooperation or conflict? And so on.

The correct sequence should be
(a) R S Q P (b) P Q S R
(c) Q S R P (d) R S P Q

(a) The correct and meaningful order of sentences is RSQP.

48. S1: Regular exercise makes many of the organ systems become more efficient.
S6 : Different activities require different levels of fitness.
P : It can improve your strength; make your body more flexible and less likely to suffer from sprain.
Q : It can also improve your endurance.
R : It also uses up energy and helps to prevent large amounts of fat building up in the body.
S : Exercise can increase your fitness in three ways.

The correct sequence should be
(a) Q R S P (b) R S P Q
(c) P S Q R (d) S Q R P

(b) The correct and meaningful order of sentences is RSPQ.

49. S1: On increasing the temperature of solids, the kinetic energy of the particles increases.
S6 : The temperature at which a solid melts to become a liquid at the atmospheric pressure is called its melting point.
P : A stage is reached when the solid melts and is converted to a liquid.
Q : Due to the increase in kinetic energy, the particles start vibrating with greater speed.
R : The particles leave their fixed positions and start moving more freely.
S : The energy supplied by heat overcomes the forces of attraction between the particles.

The correct sequence should be
(a) Q S R P (b) Q R S P
(c) P R S Q (d) S P R Q

(a) The correct and meaningful order of sentences is QSRP.

50. S 1 : Things are often not what they seem.

S6 : This happened without you even knowing it. So imagine the changes that occur to this earth and humanity.

P : But you are really not, because the Milky Way galaxy, of which you are a part, is moving through space at 2.1 million kilometre an hour.

Q : So in roughly twenty second that it would have taken you to read this paragraph, you have already moved thousands of kilometre.

R : And that is without taking into account the effects of Earth's rotation on its own axis, its orbiting around the Sun and Sun's journey around the Milky Way.

S : As you read this sentence, perhaps sitting in a comfortable chair in your study, you would probably consider yourself at rest.

The correct sequence should be

(a) Q R P S (b) R Q P S
(c) P Q R S (d) S P R Q

(d) The correct and meaningful order of sentences is SPRQ.

Directions (Q. Nos. 51-60) *Each of the following sentences in this section has a blank space and four words or group of words are given after the sentence. Select the most appropriate word or group of words for the blank space and indicate your response on the Answer Sheet accordingly.*

51. If I .... a good match I would have got married.

(a) had found (b) have found
(c) found (d) have

(a) As the given sentence is in past perfect form, 'had found' will be the correct filler for the sentence.

52. The lady has been declared as one of top ten .... of the community.

(a) more powerful members
(b) most powerful members
(c) most powerful member
(d) more powerful member

(b) The correct adjectival phrase to be filled in the blank is 'most powerful members'.

53. When I visited the villages nearby the city I .... many water bodies intact.

(a) came across (b) come across
(c) came (d) came in

(a) As the given sentence is in past tense, 'came across' will be the correct filler for the sentence.

54. He has lost all his investments and he is ....

(a) broke (b) broken
(c) discredited (d) defunct

(a) Adjective broke will be the correct filler for the sentence.

55. He .... whether he could get any certificate for the course.

(a) said (b) told
(c) thought of (d) asked

(d) The tense form 'asked' will be correct filler for the sentence.

56. I .... farewell to all my course mates last year.

(a) bid (b) bade
(c) said (d) bad

(b) The word Bade is correctly used with the word 'farewell'.

57. Very few of the texts from very early Vedic period are .... now.

(a) extant (b) exit
(c) exempt (d) redundant

(a) The word 'Extant' meaning surviving will be the correct filler for the sentence.

58. A speech is a .... address, delivered to an audience that seeks to convince, persuade, inspire or inform.

(a) formal (b) informal
(c) humorous (d) political

(a) The word 'formal' is the correct filler for the sentence.

59. All that .... is not gold.

(a) glitter (b) glitters
(c) glittering (d) gliding

(b) The word 'glitters' is the correct filler for the sentence.

60. Having been in politics for about 40 years, the party now treats him like ....

(a) a have-been
(b) a had-been
(c) a has-been
(d) would have been

(c) The phrase 'a has-been' meaning a person who is outdated is the correct filler.

Directions (Q. Nos. 61-70) *Each item in this section consists of a sentence with an underlined word(s) followed by four words/group of words. Select the option that is nearest in meaning to the underlined word and mark your response on the Answer Sheet accordingly.*

61. Emboldened by its success, the leader now plans to go ahead with the plan and implementation.

(a) Encouraged (b) Disgruntled
(c) Succeeded (d) Failed

(a) The given word 'Emboldened' means encouraged.

62. It is encouraging to see India's indigenous cinema is going places.

(a) homogenous
(b) classical
(c) home-grown
(d) Non-native language

(c) The given word 'Indigenous' means originating or occurring naturally in a particular place; native. Therefore, its synonym will be home-grown.

63. The ability to imagine and conceive a common good is inconsistent with what is known as 'pleonexia' is a major struggle for a good democracy to realise.

(a) Greed to grab everything for oneself
(b) Greed to accumulate more and more wealth
(c) Dislike for others
(d) Over ambitious

(b) The given word 'pleonexia' means extreme greed for wealth or material possessions.

64. He tried to avoid saying something that would implicate him further.

(a) reward (b) incriminate
(c) encourage (d) incite

(b) The given word 'Implicate' means to introduce some to crime or to incriminate.

65. The statutory corporate tax which forms the major income of the government has not changed this year.

(a) legislature (b) unlawful
(c) government (d) legal

(d) The given word 'statutory' means legal.

66. He has been part of the all dissident activities.

(a) rebellious (b) supportive
(c) conformist (d) legal

(a) The given word 'dissident' means rebellious.

67. Advocacy is one major component of any new programme.

(a) promotion (b) opposition
(c) critique (d) liking

(a) The given word 'advocacy' means to promote someone.

68. People avoided him for his high mindedness.

(a) toughness (b) strong principles
(c) anger (d) whims

(b) The given word 'high-mindedness' refers to someone with very high and strong principles.

69. There is a tendency to treat social changes as mere development in terms of accumulation of wealth.

(a) position (b) predisposition
(c) thinking (d) idea

(b) The given word 'tendency' means an inclination towards a particular characteristic or type of behaviour. The word 'predisposition' means the same.

70. During the ancient period poets were patronised through various institutions.

(a) supported (b) respected
(c) opposed (d) scolded

(a) The given word 'patronised' means to be supported.

Directions (Q.Nos. 71-80) *Each item in this section consists of sentences with an underlined word followed by four words or group of words. Select the option that is opposite In meaning to the underlined word and mark your response on the Answer Sheet accordingly.*

71. The archaic thinking leads to unfounded beliefs.

(a) antiquated
(b) outmoded
(c) beyond the times
(d) modern

(d) The word 'archaic' means old fashioned. Its antonym is modern.

72. Police had to resort to tear gas to diffuse tension among the crowd.

(a) concentrate (b) scatter
(c) disperse (d) strew

(a) The word 'diffuse' means spread-out. Its antonym is concentrate.

73. Unrest in some pockets made the city dwellers confine themselves at home.

(a) Turbulence (b) Unease
(c) Apprehension (d) Calm

(d) The word 'unrest' means 'a feeling of disturbance'. Its antonym is calm.

74. Peace and tranquility are instruments which would boost the development of society.

(a) uproar (b) calm
(c) serenity (d) sound

(a) The word 'tranquility' means peace. Its antonym is uproar which means disturbance.

75. Barring a decision of such disputes, other matters relating to the election of President or Vice-President may be regulated by law made by Parliament.

(a) excepting (b) without
(c) including (d) excluding

(c) The word 'barring' means exceptions. Its antonym is including.

76. His speech was full of emotions and it was an extempore.

(a) prepared (b) ready made
(c) unrehearsed (d) ad lib

(a) The word 'extempore' means spontaneous or unprepared. Its antonym is prepared.

77. The teacher asked her students to understand the ensuing problems and address them suitably .

(a) subsequent (b) consequent
(c) retrospective (d) en suite

(c) The word 'ensuing' means 'as a result of'. Its antonym is retrospective which deals with past.

78. All the allegations against the actor were expunged by the committee of inquiry.

(a) got rid of (b) part of
(c) accepted (d) rejected

(c) The word 'expunged' means to remove. Its antonym is accepted.

79. His relatives dissuaded him from giving up the job.

(a) persuaded (b) discouraged
(c) advised against (d) deter

(a) The word 'dissuaded' means to discourage. Its antonym is persuaded which means to convince.

80. He is one of the confidants of the leader and can influence the decision of the government.

(a) opponents (b) intimate
(c) close friend (d) colleague

(a) The word 'confidants' refers to a friend or a companion. Its antonym is opponents.

Directions (Q. Nos. 81-90) *Each of the following sentences has a word or phrase underlined. Read the sentences carefully and find which part of speech the underlined word is. Indicate your response on the Answer Sheet accordingly.*

81. All the pilgrims rested for a while under the banyan tree.

(a) Adverb (b) Place value
(c) Preposition (d) Verb

(c) The word 'under' is a preposition.

82. The wonderful statue, of the leader welcomes all people to city.

(a) Object (b) Adjective
(c) Noun phrase (d) Noun

(d) The word 'statue' is a noun.

83. This is his pen.

(a) Possessive pronoun
(b) Possessive adjective
(c) Adverb
(d) Verb

(b) The word 'his' is a possessive adjective.

84. When people found that the jewel was in records of Rahim, they' gave it to him.

(a) Pronoun (b) Nominative
(c) Noun (d) Adverb

(a) The word 'it' is a pronoun.

85. It is eleven O'clock now and all of us should retire to bed.

(a) Personal pronoun
(b) Relative pronoun
(c) Impersonal pronoun
(d) Verb

(c) The word 'it' is an impersonal pronoun.

86. The flower is very beautiful.

(a) Adjective
(b) Adverb
(c) Preposition
(d) Conjunction

(b) The word 'very' is an adverb.

87. This boy is stronger than Ramesh.
(a) Pronoun (b) Adjective
(c) Article (d) Adverb

(b) The word 'this' is an adjective.

88. I hurt myself.
(a) Noun
(b) Pronoun
(c) Demonstrative preposition
(d) Adjective

(b) The word 'myself' is a pronoun.

89. The ants fought the wasps.
(a) Intransitive verb
(b) Transitive verb
(c) Demonstrative verb
(d) Adjective

(c) The word 'fought' is a transitive.

90. I can hardly believe it.
(a) Adjective
(b) Preposition
(c) Adverb
(d) Verb

(c) The word 'hardly' is an adverb.

Directions (Q.Nos. 91-110) *Each of the following sentences in this section has a blank space with four words or group of words given. Select whichever word or group of words you consider the most appropriate for the blank space and indicate your response on the Answer Sheet accordingly.*

The difficult thing about .... **91.** (a) studying (b) study (c) studies (d) are studying the science of habits is that most people, when they hear

(a) The correct filler is studying.

about this field of research .... **92.** (a) wanting (b) wanted (c) wants (d) want to know the secret formula for quickly changing any habit.

(d) The correct filler is want.

If scientists have discovered how .... **93.** (a) those (b) this (c) these (d) that patterns work, then it stands to reason that they

(c) The correct filler is these.

.... **94.** (a) must (b) will (c) could (d) might have also found a recipe for rapid change, right? If only it ....

(a) The correct filler is must.

.... **95.** (a) are (b) were (c) was (d) will be that easy. It's not .... **96.** (a) these (b) this (c) that (d) which formulas don't

(b) The correct filler is were.

(c) The correct filler is that.

exist. The problem is that there isn't one formula for .... **97.** (a) changing (b) changed (c) having changed (d) changes for

(a) The correct filler is changing.

habits. There are thousands. Individuals and habits are .... **98.** (a) full (b) all (c) complete (d) most

(b) The correct filler is all.

different, and so the specifics of diagnosing and changing the patterns in our lives differ from person to ....

99. (a) people (b) persons (c) personnel (d) person and behaviour to behaviour. Giving up

(d) The correct filler is person.

cigarettes is different .... **100.** (a) from (b) since (c) to (d) into curbing overeating, which is different

(a) The correct filler is from.

from changing how you communicate with your spouse, ... **101.** (a) it
(b) this
(c) what
(d) which

(d) The correct filler is which.

is different from how you prioritise tasks at work. What's more, each person's habits are .... **102.** (a) broken by different cravings. As a result, this book does not
(b) given
(c) driven
(d) prescribed

(c) The correct filler is driven.

.... **103.** (a) contain one prescription. Rather, I hoped to deliver something
(b) contains
(c) contained
(d) containing

(a) The correct filler is contain.

else : a framework for understanding .... **104.** (a) how habits work and a
(b) what
(c) where
(d) whose

(a) The correct filler is how.

guide to experimenting with how they .... **105.** (a) might change. Some habits yield easily
(b) would
(c) will
(d) must

(a) The correct filler is might.

to analysis and influence. Others are .... **106.** (a) quiet complex and obstinate, and require prolonged study. And for
(b) most
(c) better
(d) more

(d) The correct filler is more.

others, change is a .... **107.** (a) process that never fully concludes. But that does not
(b) processing
(c) processed
(d) processes

(a) The correct filler is process.

.... **108.** (a) means it can't occur. Each chapter in this book explains
(b) meant
(c) meaning
(d) mean

(d) The correct filler is mean.

a different aspect of why habits exist and how they function. The framework

.... **109.** (a) describing in this section is an attempt to distil, in
(b) described
(c) will describe
(d) description

(b) The correct filler is described.

.... **110.** (a) a very basic way, the tactics that researchers have found
(b) any
(c) the
(d) rather

(a) The correct filler is a .

for diagnosing and shaping habits within our own lives.

Directions (Q. Nos. 111-120) *In this section a word is spelt in four different ways. Identify the one which is correct. Choose the correct response (a), (b), (c) or (d) and indicate on the Answer Sheet accordingly.*

111. Which one of the following alternatives has the correct spelling?
(a) Mountaneous (b) Mountenous
(c) Mountaineous (d) Mountainous

(d) Mountainous.

112. Which one of the following alternatives has the correct spelling?
(a) Etiquette (b) Etiquete
(c) Etieqютte (d) Etequtte

(a) The correct spelling is Etiquette.

113. Which one of the following alternatives has the correct spelling?
(a) Curriculam (b) Curiculum
(c) Curiculeum (d) Curriculum

(d) The correct spelling is Curriculum.

114. Which one of the following alternatives has the correct spelling?
(a) Magnificent (b) Magnificant
(c) Magneficent (d) Magenficient

(a) The correct spelling is Magnificent.

115. Which one of the following alternatives has the correct spelling?
(a) Felecitation (b) Felicitation
(c) Falicitation (d) Felicitasion

(b) The correct spelling is Felicitation.

116. Which one of the following alternatives has the correct spelling?
(a) Twelth (b) Twelfth
(c) Tweluth (d) Twelthe

(b) The correct spelling is Twelfth.

117. Which one of the following alternatives has the correct spelling?
(a) Snobbery (b) Snoberry
(c) Snabbery (d) Snobbory

(a) The correct spelling is Snobbery.

118. Which one of the following alternatives has the correct spelling?
(a) Neurasis (b) Nuroesis
(c) Neurosis (d) Neuresis

(c) The correct spelling is Neurosis.

119. Which one of the following alternatives has the correct spelling?
(a) Dipthteria (b) Diptheria
(c) Diphtheria (d) Diphthria

(c) The correct spelling is Diphtheria.

120. Which one of the following alternatives has the correct spelling?
(a) Meagre
(b) Megare
(c) Meagr
(d) Megear

(a) The correct spelling is Meagre.

PAPER III General Studies

1. As per the Budget Estimates of 2019-20, the following are some of the important sources of tax receipts for the Union Government:

1. Corporation Tax
2. Taxes on Income other than Corporation Tax
3. Goods and Services Tax
4. Union Excise Duties

Which one of the following is the correct descending order of the foresaid tax receipts as a percentage of GDP?
(a) 1, 2, 3, 4 (b) 1, 3, 2, 4
(c) 3, 2, 1, 4 (d) 2, 4, 3, 1

(b) As per the Budget Estimates of 2019-20, the important sources of tax receipts for the Union Government are as follows-

Sources	Budgeted 2019-20 (₹ crore)
Corporation Tax	766000
Goods and Services Tax	663343
Taxes on Income	569000
Union Excise Duties	300000
Customs	155904

Hence, option (b) is correct.

2. As per the World Bank's Ease of Doing Business Ranking, India's rank has improved from 142 in 2014 to 63 in 2019. During this period, in which of the following parameters has India's rank deteriorated?
(a) Ease of starting a business
(b) Getting electricity
(c) Registering property
(d) Paying taxes

(c) As per the World Bank's Ease of Doing Business Report, India's rank has improved from 142 in 2014 to 63 in 2019. During this period India's rank has deteriorated in Registering Property.

S.No.	Parameters	2014	2019
1.	Starting a business	156	136
2.	Dealing with construction permits	183	27
3.	Getting electricity	134	22
4.	Registering property	115	154
5.	Getting credit	30	25
6.	Protecting investors	21	13
7.	Paying taxes	154	115
8.	Trading across borders	122	68
9.	Enforcing contracts	186	163
10.	Resolving insolvency	135	52

3. Which one of the following statements with regard to the National Food Security Act is not correct?
(a) The Act was enacted in the year 2013.
(b) The Act was rolled out in the year 2014.
(c) The Act legally entitles 67% of the population to receive highly subsidised food grains.
(d) The Act is not being implemented in all the States/Union Territories.

(d) The National Food Security (NFS) Act, 2013 aims to provide subsidised food grains to 67% of India's population. It was rolled out in the year 2014. It is implemented in all the State/Union Territories.

Salient Features of this Act

- Beneficiaries of the PDS are entitled to 5 kg per person per month of cereals at the following prices: Rice at ₹ 3 per kg, Wheat at ₹ 2 per kg and Coarse grains (millet) at ₹ 1 per kg.
- Pregnant women, lactating mothers and certain categories of children are eligible for daily free cereals.
- It includes the Midday Meal Scheme, Integrated Child Development Services (ICDS) scheme and the Public Distribution System (PDS).

4. Which one of the following statements about Indian economy during 2019-20 is not correct?

(a) There has been deceleration in growth rate.
(b) There has been sluggish growth in tax revenue relative to the Budget Estimates.
(c) Fiscal deficit as percentage of GDP has been as per the Budget Estimates.
(d) The non-tax revenue registered a considerably higher growth.

(c) There has been deceleration in growth rate in Indian Economy during 2019-20. There has been sluggish growth in tax revenue relative to the Budget Estimates. Fiscal deficit as percentage of GDP has been more than the Budget Estimates. The non-tax revenue registered a considerable growth.

5. As per the Budget Estimates of expenditure on major subsidies during 2019-20, the maximum expenditure was likely to be on

(a) urea subsidy
(b) petroleum subsidy
(c) food subsidy
(d) fertilizer subsidy

(c) In 2019-20, the total expenditure on subsidies is estimated to increase to ₹ 3,38,949 crore (13.3%) over the revised estimate of 2018-19. This is owing to an increase in expenditure on petroleum, fertilizer, food and other interest subsidies.

Subsidies in 2019-20 (₹ crore)

Subsidy	Budget 2019-20
Food subsidy	184220
Fertilizer subsidy	79996
Petroleum subsidy	37478
Other subsidies	37255
Total	338949

Hence, option (c) is correct.

6. Which one of the following Indian places receives minimum rainfall in a year?

(a) Jodhpur
(b) Leh
(c) New Delhi
(d) Bengaluru

(b) Leh receives minimum rainfall in a year among the given option. The average annual rainfall is only 102 mm (4.02 inches).

The annual average rainfall of the other given places are

Jodhpur – 363 mm

New Delhi – 617 mm

Bengaluru – 970 mm

7. Timber vegetation is generally not found in which of the following regions?

(a) Subtropical region
(b) Temperate region
(c) Alpine region
(d) Tundra region

(d) Timber vegetation is generally not found in Tundra region. The Tundra is a treeless polar desert found in the high latitudes in the polar regions. Tundra lands are covered with snow for much of the year.

The Tundra regions are-Alaska, Canada, Russia, Greenland, Iceland.

The characteristics of the Tundra Regions are:

- Extremely cold climate
- Low biotic diversity
- Limitation of drainage
- Simple vegetation structure

The Arctic Tundra's temperature is – 34° C to – 6° (c)

8. Decadal growth rate of population in percentage was highest in India in the year

(a) 1991 (b) 1981
(c) 1971 (d) 1961

(c) The decadal growth rate of population in India in percentage was highest in India in the year 1971 which was 24.80. The decadal growth of other given year is as follows.

- 1961-21.64%
- 1981-24.66%
- 1991-27.87%

The Decadal growth of 2011 is 17.64.

9. The Isotherm Line, which divides India North-South in almost two equal parts in the month of January, is

(a) 10 °C (b) 25 °C
(c) 15 °C (d) 20 °C

(d) The Isotherm Line, which divides India North-South in almost two equal parts in the month of January is 20°C

Isotherm is a line that connect points of equal temperature on map, so at every point along a given isotherm the temperature values are same. It is used to observe the distribution of air temperature over a vast area.

10. Which one of the following indicates the Tropical Savannah climate?

(a) Aw (b) Dfc (c) Cwg (d) Am

(a) Aw or the tropical wet and dry climate, also known as the Savannah climate, where there is an extended dry season during the winter.

During the wet season, rainfall is less than 1000 mm, occurring mainly in the summer time.

Other Climatic Types

Aw - Monsoonal Type Climate

Dfc - Cold Humid winter with Short summer.

Cwg - Monsoon Type with Dry winters.

11. The largest geographical area of India is covered by which one of the following types of soils?

(a) Inceptisols (b) Entisols
(c) Alfisols (d) Vertisols

(a) The largest geographical area of India is covered by Inceptisols, which is 39.74%. Inceptisols are usually the weakly developed young soil though they are more developed than entisols.

Other soils cover an area of

Entisols – 28.08%

Albisols – 13.55%

Vertisols – 8.52%

12. Which one of the following cities is closest to the Equator?

(a) Mogadishu (b) Singapore
(c) Colombo (d) Manila

(b) Singapore is closest to the equator.

Other cities which are near to the equator are

Quito (Equador)
Kampala (Uganda)
Pekanbaru (Indonesia)
Padang (Indonesia)
Maccapa (Brazil)
Kisumu (Kenya)
Libreville (Gabon)

13. Who among the following gave evidence before the Joint Select Committee on the Government of India Bill, 1919 in favour of female franchise?

1. Annie Besant 2. Sarojini Naidu
3. Hirabai Tata

Select the correct answer using the codes given below.

(a) Only 1 (b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

(d) Annie Besant, Sarojini Naidu and Hirabai Tata gave evidence before the Joint Select Committee on the Government of India Bill, 1919 in favour of Female Franchise.

The Government of India Act, 1919, was an Act of the Parliament of the United Kingdom. It was passed to expand participation of Indians in the Government of India. The Act embodied the reforms recommended in the report of the Secretary of State of India, Edwin Montagu, and the Viceroy Lord Chelmsford.

14. In which one of the following places was the Ahmadiyya Movement started by Mirza Ghulam Ahmad?

(a) Patna (b) Aligarh
(c) Bhopal (d) Gurdaspur

(d) The Ahmadiyya Movement was launched by Mirza Ghulam Ahmad of Qadiyan in 1889 from Gurudaspur.

He began his work as a defender of Islam against the Polemics (Religions Propoganda) of the Arya Samaj and the Christian missionaries.

15. With whom did Subhas Chandra Bose form an alliance to destroy the Holwell Monument in Calcutta during 1939-40?

(a) The Communist Party of India
(b) The Muslim League
(c) The Hindu Mahasabha
(d) The Unionist Party

(b) Subhas Chandra Bose with an alliance with the Muslim League destroyed the Holwell Monument in Calcutta during 1939-40.

The Holwell Monument was constructed to commemorate the victims of the 'Black Hole Tragedy 1956. It was named after John Zephaniah Holwell, one of the survivor of this incident.

16. Who among the following created the first All India Trade Union Congress in 1920?

(a) BP Wadia (b) SA Dange
(c) NM Joshi (d) BT Ranadive

(c) NM Joshi created the first All India Trade Union Congress (AITUC) in 1920.

The AITUC was founded on October 31, 1920 in Mumbai .

- It is associated with the Communist Party of India. According to provisional statistics from the Ministry of Labour, AITUC had a membership of 14.2 million in 2013.

17. Which one among the following was India's first trade union in the proper sense of the term?

(a) Bombay Labour Union
(b) Ahmedabad Labour Union
(c) Madras Labour Union
(d) Allahabad Labour Union

(c) Madras Union Labour was India first trade Union in the proper sense of term. Madras Labour Union was formed on April 27, 1918 by the workers of the then Buckingham and Carnatic Mills at Perambur, Chennai. The two prominent traders involved in forming the union were Selvapathi Chettiyar and Ramanujulu Naidu.
BP Wadia was the founding member of the MLU. Selvapathi and Ramanujulu became the General Secretaries of the union while **Thiru Vi Ka** its Vice President.

18. Who among the following formed the Seva Samiti Boy Scouts Association in 1914?

(a) Hriday Nath Kunzru
(b) SG Vaze
(c) Annie Besant
(d) Shri Ram Bajpai

(a) Hriday Nath Kunzru and Madan Mohan Malviya formed the Seva Samiti Boy Scouts Association. On the line of the Worldwide Baden-Powell Organisation, which at the time refused to allow Indians to join it.

They were assisted by Shri Ram Bajpai.

19. Which one among the following is not correct about the Secretary General of the Lok Sabha?

(a) The Secretary General is the advisor of the Speaker.
(b) The Secretary General acts under the authority in the name of the Speaker.
(c) The Secretary General works under the Speaker with delegated authority.
(d) The Secretary General passes orders in the name of the Speaker.

(d) The Secretary General of the Lok Sabha passes orders in the name of the speaker is not correct statement.

The Secretary General of the Lok Sabha is the Administrative Head of the Lok Sabha Secretariat. He/she is appointed by the Speaker of the Lok Sabha. The post of Secretary General is of the rank of the Cabinet Secretary in the Government of India, who is the senior most civil servant to the Indian Government.
In the discharge of his constitutional and statutory responsibilities, the Speaker of the Lok Sabha is assisted by the Secretary-General, Lok Sabha, functionaries of the level of the Additional Secretary, Joint Secretary and other officers and staff of the Secretariat at various levels.
The Secretary General remains in office till his/her retirement at the age of 60. He/she is answerable only to the Speaker, his/her action cannot be discussed or criticised in or outside the Lok Sabha. On behalf of the President of India, he/she summons members to attend session of Parliament and authenticates Bills in the absence of the Speaker.

20. Who among the following moved the motion of Secret Sitting Session of the Assembly (1942)?

(a) MS Aney (b) GV Mavalankar
(c) CM Stephen (d) A. Ayyangar

(a) Madhav Shrihari Aney (MS Aney) moved the motion of Secret Sitting Session of the Assembly in 1942.
The objective of this Session was to discuss the 2nd World War situation.
He was member of the Viceroy's Executive Council from 1941-1944.
He is popularly referred as Loknayak Bapujj Aney. He was also Governor of Bihar from 12th January, 1948 to 14th June, 1952.

21. Which one among the following statements pertaining to the President's term of Office is not correct?

(a) The President holds Office for a term of five years.
(b) The President may be removed from the Office by way of impeachment.
(c) The President may resign before the expiration of his/her term by writing to the Speaker of the Lok Sabha.
(d) The President shall, not withstanding the expiration of his/her term, continue to hold Office until his/her successor enters upon his/her Office.

(c) "The President may resign before the expiration of his/her term by writing to the speaker of the Lok Sabha", is not correct. The President submits his resignation to the Vice-President.
The Indian President is the Head of the State and he is also called the First Citizen of India. He is a part of Union Executive, provisions of which are dealt with Articles 52-78 including Articles related to President (Articles 52-62).

Supreme Court shall inquire and decide regarding all doubts and disputes arising out of or in connection with the election of a President as per Article 71(1) of the Constitution. Supreme Court can remove the President for the electoral malpractices or upon being not eligible to be Lok Sabha member under the Representation of the People Act, 1951.

The President may also be removed before the expiry of the term through impeachment for violating the Constitution of India by the Parliament of India. The process may start in either of the two Houses of the Parliament.

22. Which of the following Articles in the Constitution of India are exceptions to the Fundamental Rights enumerated in Article 14 and Article-19?

(a) Article 31A and Article 31C
(b) Article 31B and Article 31D
(c) Article 12 and Article 13
(d) Article 16 and Article 17

(a) Article 31A and Article 31C in the Constitution of the India are exceptions to the Fundamental Rights enurmated in Article 14 and Article 19. By the 1st Constitutional Amendment of 1951, the Parliament added Article 31A to the Indian Constitution.

Article 31A of Indian Constitution was immune to Article-14 and 19 of Indian Constitution that provide for Right to Equality and the Right to Freedom, respectively.
Article 31 (C) of Indian Constitution was included through the 25th Amendment Act of 1971 through which the government gave primacy to some Directive Principles of State Policy over the Fundamental Rights.

23. Which one of the following is not the necessary condition for the issue of a Writ of Quo Warranto?

(a) The Office must be a Public Office.
(b) The Office must be created by the Statute or by the Constitution itself.
(c) The Office must not be a substantive one.
(d) There has been a contravention of the Constitution or a Statute in appropriating such person to that Office.

(c) "The Office must not be a substantive one" is not the necessary condition for the issue of a Writ of Quo Warranto. Orders, warrants, directions, etc. Issued under authority are examples of Writ.

There are five major types of Writ viz. Habeas Corpus, Mandamus, Prohibition, Quo Warranto and Certiorari.

Quo warranto means 'by what warrant?' This writ is issued to enquire into legality of the claim of a person or Public Office. It restrains the person or authority to act in an Office which he/she is not entitled to; and thus stops usurpation of Public Office by anyone. This Writ is applicable to the public offices only and not to Private Offices.

24. Which one of the following Commissions is related to Article 338A?

(a) The National Commission for Scheduled Castes
(b) The National Commission for Scheduled Tribes
(c) The National Commission for Backward Classes
(d) The National Commission for Women

(b) Article 338A of the Indian Constitution deals with the National Commission for Scheduled Tribes.

This article was added in the Constitution through the 89th Constitutional Amendment Act, 2003.

The Chairperson, Vice-Chairperson and other members of the Commission shall be appointed by the President by Warrant under his hand and seal. The Commission shall have the power to regulate its own procedure. Article 338 of the Indian Constitution deals with National Commission for Scheduled Castes.

25. August 12 is celebrated as

(a) the World Environment Day
(b) the World No-Tobacco Day
(c) the International Day against Drug Abuse and Illicit Trafficking
(d) the International Youth Day

(d) The August 12 is celebrated as the International Youth Day, each year to recognise efforts of the World's youth in enhancing Global Society.

It also the day aims to raise awareness about the problems faced by youth. The first International Youth Day was observed on August 12, 2000.

The theme of the International Youth Day, 2020 was 'Youth Engagement for Global Action'.

26. Which one of the following countries had chosen the name 'Nisarga' for the cyclone which devastated the coastline of Maharashtra and Gujarat in June 2020?

(a) Maldives (b) Bangladesh
(c) Thailand (d) Japan

(b) Bangladesh had chosen the name 'Nisarga' for the cyclone which devastated the coastline of Maharashtra and Gujarat in June 2020. The Nisarga cyclone is formed because of the depression in the Arabian Sea. It is a tropical cyclone formed because of exceptional warm surface ocean temperatures.

27. Who among the following played the role of Shakuntala Devi in the biopic movie based on the life of the famous mathematician?

(a) Madhuri Dixit (b) Rani Mukherjee
(c) Tabu (d) Vidya Balan

(d) Indian bollywood actress Vidya Balan played the role of Shakuntala Devi in the biopic movie based on the life of the famous Mathematician.

Shakuntala Devi (1929 -2013) was an Indian writer and mental calculator, popularly known as the 'Human Computer'.

She strove to simplify numerical calculations for students. Her talent earned her a place in the 1982 edition of The Guinness Book of World Records.

28. GC Murmu, who was appointed as the Comptroller and Auditor General of India in August 2020, was the Lieutenant Governor/ Administrator of which one of the following Union Territories prior to this appointment?

(a) Ladakh
(b) Jammu and Kashmir
(c) Chandigarh (d) Puducherry

(b) Girish Chandra Murmu is appointed as the 14th Comptroller and Auditor General of India on August 8, 2020. He was the first Lieutenant Governor of the Union Territory of Jammu and Kashmir.

He is a 1985 batch Indian Administrative Service Officer of Gujarat Cadre and was Principal Secretary of Narendra Modi during his tenure as Chief Minister of Gujarat.

29. Which one of the following States is planned to host the Khelo India Youth Games (4th Edition)?

(a) Kerala (b) Haryana
(c) Gujarat (d) Manipur

(b) On July 25, 2020, the Union Minister of Youth Affairs and Sports, Shri Kiren Rijiju announced that Haryana will host the Fourth Edition of Khelo India Youth Games. The games has been scheduled to be conducted after Tokyo Olympics.

The Khelo India Scheme was launched to improve sports culture in India. The Scheme is implemented by the Ministry of Youth Affairs and Sports. The Scheme aims to make India a good sporting nation.

30. Which one among the following is not a Coral Reef Island?

(a) Great Barrier Reef (Australia) (b) Rainbow Reef (Fiji)
(c) Swaraj Island (India) (d) Kyushu Island (Japan)

(d) **Kyushu Isalnd** (Japan) is not a Coral Reef Island. A coral island is a type of island formed from coral detritus and associated organic material. It occur in tropical and sub-tropical areas, typically as part of coral reefs which have grown to cover a far larger area under the sea.

The **Great Barrier Reef** is the world's largest Coral Reef System composed of over 2900 individual reefs and 900 islands stretching for over 2,300 km over an area of approximately 344,400 km^2. The reef is located in the Coral Sea, off the Coast of Queensland, Australia.

The **Rainbow Reef** is a reef in the Somosomo Strait between the Fijian islands of Taveuni and Vanua Levu. It is one of the most famous dive sites in the South Pacific.

Swaraj Island is part of Ritchie's Archipelago, in India's Andaman Islands. It's known for its dive sites and beaches, like Elephant Beach, with its coral reefs.

31. Since 2014-15, India has consistently run trade surplus with which one among the following countries?

(a) China (b) Saudi Arabia (c) USA (d) Germany

(d) Since 2014-15, India has consistently run trade surplus with USA.

Trade Status	Country	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20 (April-Nov.)
Trade Surplus Country	USA	20.63	18.55	19.90	21.27	16.86	10.91
Trade Deficit Countires	UAE	6.89	9.67	10.87	6.41	0.34	0.25
	China PRP	–48.48	–52.70	–51.11	–63.05	–53.57	–35.32
	Saudi Arabia	–16.95	–13.94	–14.86	–16.66	–22.92	–14.32
	Iraq	–13.42	–9.83	–10.60	–16.15	–20.58	–13.98
	Germany	–5.25	–5.00	–4.40	–4.61	–6.26	–3.09
	Korea RP	–8.93	–9.52	–8.37	–11.90	–12.05	–7.80
	Indonesia	–10.96	–10.31	–9.94	–12.48	–10.57	–6.99
	Switzerland	–21.06	–18.32	–16.27	–17.84	–16.90	–11.97
	Hong Kong	8.03	6.04	5.84	4.01	–4.99	–3.88
	Singapore	2.68	0.41	2.48	2.74	–4.71	–3.15

Bilateral Trade Surplus/Deficit **(Values in US $ Billion)**

32. Arrange the following countries in descending order as per the Global Human Development Index, 2019 :

1. Germany 2. USA 3. South Africa 4. India

Select the correct answer using the codes given below.

(a) 1, 2, 3, 4 (b) 1, 3, 2, 4
(c) 3, 2, 1, 4 (d) 4, 3, 2, 1

(a) Countries in descending order as per the Global Human Development Index, 2019 are- Germany (4) USA (15), South Africa (113) and India (129). In HDI 2019, Norway, Switzerland, Ireland and Germany topped the HDI ranking of 189 countries and territories, while Niger, the Central African Republic, South Sudan, and Burundi have the lowest scores in the Index. The United Nations Development Programme (UNDP) ranks countries into four tiers of human development by combining measurements of life expectancy, education and per-capita income into the Human Development Index (HDI) in its annual Human Development Report.

33. As per the use-based classification of the Index of Industrial Production (IIP), the maximum weight has been assigned to

(a) primary goods
(b) intermediate goods
(c) consumer durables
(d) consumer non-durables

(a) The Index of Industrial Production (IIP) is an index that indicates the performance of various industrial sectors of the Indian economy. It is calculated and published by the Central Statistical Organisation (CSO), every month. The maximum weight has been assigned to primary goods.Weights of the different sectors under the used based classification (2011-12 series)

Sectors	Number of Groups	Weights
Primary Goods	15	34.05
Capital Goods	67	8.22
Intermediate Goods	110	17.22
Infrastructure/Construction goods	29	12.34
Consumer durables	86	12.84
Consumer Non-durables	100	15.33
Total	**407**	**100**

34. Normally, there will not be a shift in the demand curve when

(a) price of a commodity falls.
(b) consumers want to buy more at any given price.
(c) average income rises.
(d) population grows.

(a) The demand curve is a graphical representation of the relationship between the price of a good or service and the quantity demanded for a given period of time. If the price of a commodity falls then there will be no shift in the demand curve.

35. A market, in which there are a large number of firms, homogeneous product, infinite elasticity of demand for an individual firm and no control over price by firms, is termed as

(a) oligopoly
(b) imperfect competition
(c) monopolistic competition
(d) perfect competition

(d) **Perfect competition** is a market form in which there are a large number of firms, homogenous product, infinite elasticity of demand for an individual firm and no control over price rise. Perfect competition provides both allocative efficiency and productive efficiency.

In **oligopoly**, small group of large sellers (oligopolists) dominate the market. **Monopolistic** competition occurs when an industry has many firms offering products that are similar but not identical. Imperfect competition refers to a situation where the characteristics of an economic market do not fulfil all the necessary conditions of a perfectly competitive market.

36. Which one of the following statements with regard to ozone is not correct?

(a) Ozone is found mostly at 15-55 km in the atmosphere.
(b) Ozone is produced by gaseous chemical reactions.
(c) 16th November is celebrated as the International Day for the Preservation of the Ozone Layer.
(d) Ozone is a form of oxygen in which three oxygen atoms are bounded together.

(c) September 16 was designated by the United Nations General Assembly as the International Day for the Preservation of the Ozone Layer. This designation had been made on December 19, 2000, in commemoration of the date, in 1987, on which nations signed the Montreal Protocol on Substances that Deplete the Ozone layer. The theme for 2020 World Ozone Day was 'Ozone for Life: 35 years of Ozone Layer Protection'.

37. Sea of Azov is connected to

(a) Black Sea (b) Baltic Sea
(c) Mediterranean Sea
(d) North Sea

(a) The Sea of Azov is a sea in Eastern Europe connected to the Black Sea by the narrow (about 4 km) Strait of Kerch, and is sometimes regarded as a Northern Extension of the Black Sea. The sea is bounded in the North-West by Ukraine, in the south-east by Russia. The Don River and Kuban River are the major rivers that flow into it. There is a constant outflow of water from the Sea of Azov to the Black Sea. The Sea of Azov is the shallowest sea in the world, with the depth varying between 0.9 and 14 m (2 ft 11 in and 45 ft 11 in).

38. Climax Mine, the largest producer of Molybdenum, is located in

(a) Canada (b) USA
(c) Australia (d) South Africa

(b) The Climax mine, located in Climax, Colorado, United States, is a major Molybdenum mine. Shipments from the mine began in 1915. At its highest output, the Climax mine was the largest Molybdenum mine in the world, and for many years It supplied three-fourths of the world's supply of Molybdenum. The mine is owned by Climax Molybdenum Company, a subsidiary of Freeport-McMoRan.

Molybdenum is a chemical element with the symbol Mo and atomic number 42.

39. Which one among the following Union Territories of India is the smallest in geographical area?

(a) Chandigarh
(b) Puducherry
(c) Dadra and Nagar Haveli and Daman and Diu
(d) Lakshadweep

(d) The Union Territory of Lakshadweep is smallest in area. It has total area of 32 km^2. It is an archipelago, situated in Arabian Sea (Indian Ocean). Its capital is Kavaratti. It is divided into two group of islands i.e. Amindivi (Northern group of islands) and Cannanore (Southern group of Islands). The Islands are made up of coral reef.

The geographical areas of other given territories are

Chandigarh	- 14 km^2
Puducherry	- 483 km^2
Dadra and Nagar Haveli and Daman and Diu	- 603 km^2

40. Buenos Aires and Montevideo are situated across the banks of

(a) River Plate (b) Orinoco River
(c) Purus River (d) Madeira River

(a) Buenos Aires, the capital and most populous city of Argentina, located on the western bank of the River Plate. Buenos Aires is one of the safest cities to visit In South America. Montevideo is situated on the Southern coast of the country, on the North-Eastern bank of the River Plate. Montevideo is the capital and largest city of Uruguay.
The city was established in 1724 by a Spanish soldier, Bruno Mauricio de Zabala, as a strategic move amidst the Spanish-Portuguese dispute over the platine region.

41. The largest Barrier Reef System in the world is found at

(a) East Australian Coast
(b) West Australian Coast
(c) North Australian Coast
(d) South Australian Coast

(a) The Great Barrier Reef is the world's largest coral reef. It is found off the East Australian coast. It is made up of nearly 2900 coral reefs and over 600 islands. It is 3,27,800 km^2 big and 2600 km long. It has been listed as an important World Heritage Site by UNESCO.

It is home to over 1600 species of fish, 411 species of hard coral and 150 species of soft coral, more than 30 species of whales and dolphins and six of the world's seven species of marine turtles.

The Great Barrier Reef is the largest structure made by living things. It can be seen from outer space. The biggest threat to the Great Barrier Reef today is coral bleaching caused by high sea water temperatures as a result of global warming.

42. During the 19th century, who among the following wrote *Satapatra* Series?

(a) MG Ranade
(b) BG Tilak
(c) Bankim Chandra Chatterjee
(d) GH Deshmukh

(d) 'Lokahitavadi' Gopal Hari Deshmukh wrote Satapatra Series (1848-50) in which he called for social reforms, advocated indigenous enterprise, but also welcomed British rule. Gopal Hari Deshmukh (1823 - 1892) was an activist, thinker, social reformer and writer from Maharashtra.

43. Which one of the following was not a demand made by the Congress moderates?

(a) Universal Adult Franchise
(b) Repeal of the Arms Act
(c) Extension of Permanent Settlement
(d) Higher Jobs for Indians in the army

(a) Universal Adult Franchise was not a demand made by the Congress moderates.

The early phase of the Congress was dominated by the 'moderates'. They believed in British rule and were loyal to them.

Prominent moderate leaders-Dadabhai Naoroji, Womesh Chandra Bonnerjee, G. Subramania Aiyer, Sir Surendranath Banerjee and Gopal Krishna Gokhale.

Major Popular Demands of the Moderates

- Decreased land revenue tax and ending peasant oppression.
- Education of the masses and organising public opinion, make people aware of their rights.
- Indian representation in the Executive Council and in the Indian Council in London.
- Separation of the Executive from the judiciary.
- Decreased land revenue tax and ending peasant oppression.
- After 1892, raised the slogan, ‘‘No taxation without representation’’.
- Abolishing salt tax and duty on sugar.
- Freedom of speech and expression.
- Freedom to form associations.
- Development of modern capitalist industries in India.
- End of economic drain of India by the British.
- Repealing the Arms Act of 1878.

44. Who among the following founded the Mohammedan Anglo-Oriental Defence Association (1893)?

(a) Auckland Colvin
(b) Badruddin Tyabji
(c) Theodore Beck
(d) Sir Syed Ahmad Khan

(d) Mohammedan Anglo-Oriental Defence Association was established in 1893 by Sir Syed Ahmad Khan as a part of the Aligarh Movement. He established it as he considered competence in English and Western Sciences necessary skills for maintaining Muslims' political influence, especially in Northern India. Also, the growing influence and popularity of the Congress became a cause of concern for the British. In order to counter the growing influence of the Congress, the British encouraged the formation of the Mohammedan Anglo-Oriental (M.A.O.) Defence Association.

Sir Syed Ahmed Taqvi bin Syed Muhammad Muttaqi (1817-1898), commonly known as Sir Syed Ahmed Khan, was an Islamic pragmatist, reformer, and philosopher of nineteenth century. He is considered as the pioneer of Muslim nationalism in India.

45. After the First World War, the Triveni Sangh was formed by

(a) the Jats and Gujjars
(b) the Rajputs and Yadavs
(c) the Jats and Yadavs
(d) the Ahirs and Kurmis

(d) The Triveni Sangh was formed in 1934 by the members of three backward castes of Bihar, namely Yadavs (Ahirs), Koer and Kurmi. It was formed to fight against the political solidarity of ‘middle peasant castes’ as well as to carve a space in democratic politics for the lower castes. Its nomenclature was derived from the confluence of three mighty rivers *viz.* the Ganga, Yamuna and the mythical Saraswati at Allahabad. The Sangh claimed of having atleast one million dues-paying members. Its formation was countered by the formation of Indian National Congress's Backward class Federation, which was established at the same time.

46. Who among the following was the first to accept a ministerial position in the Central Provinces in October 1925?

(a) BS Moonje (b) MR Jayakar
(c) SB Tambe (d) BN Sasmal

(c) Shripad Balwant Tambe accepted a ministerial position in the central provinces in October, 1925. He was a pledger from Amravati in Berar division of Central Provinces. He was a member of the Swaraj Party and President of the Central Provinces Legislative Council. His appointment created political interest throughout India and the Swaraj Party.

Balakrishna Shivram Moonje was a leader of the Hindu Mahasabha in India.

Makund Ramrao Jayakar was the first Vice-Chancellor of the University of Poona.

Birendra Nath Sasmal was a lawyer and political leader. He was known as Deshpran because of his work for the country and for his efforts in the Swadeshi Movement.

47. Who among the following formed the National Liberation Federation (Liberal Party)?

(a) Motilal Nehru and CR Das
(b) Muhammad Ali and CR Das
(c) TB Sapru and MR Jayakar
(d) MR Jayakar and CR Das

(c) The National Liberation Federation (Liberal Party) was formed by Surendra Nath Banerjee and some of its prominent leaders were Tej Bahadur Sapru, VS Srinivasa Sastri and MR Jayakar. The Liberal Party was formed in 1910, and British intellectuals and British officials were often participating members of its committees.

Tej Bahadur Sapru emerged as the most important leader among the Liberals.

48. The National Disaster Management Authority functions under the Ministry of

(a) Environment, Forest and Climate Change
(b) Home Affairs
(c) Commerce and Industry
(d) Finance

(b) The National Disaster Management Authority (NDMA) functions under the Ministry of Home Affairs. It was established through the Disaster Management Act enacted by the Parliament on December 23rd 2005.

NDMA is responsible for framing policies, laying down guidelines and best practices for coordinating with the State Disaster Management Authorities (SDMAs) to ensure a holistic and distributed approach to disaster management. Prime Minister Narendra Modi is the Chairman of the NDMA. The members enjoy the rank of Secretary of the Union government.

49. The socialist idea of *Sapta Kranti* (Seven Revolutions) was proposed by

(a) Ram Manohar Lohia
(b) Jawaharlal Nehru
(c) MG Ranade
(d) Jayaprakash Narayan

(a) The socialist idea of Sapta Kranti (Seven Revolutions) was proposed by great socialist leader Ram Manohar Lohia.

His Seven Revolutions include

1. For equality between man and woman.
2. Against political, economic and race-based inequalities.
3. For the destruction of castes.
4. Against foreign domination.
5. For economic equality, planned production and against private property.
6. Against interference in private life.
7. Against arms and weapons and for Satyagraha.

He provided the ideological and organisational base for OBC empowerment. Much of his career was devoted to combating injustice through the development of a distinctly Indian version of socialism.

50. Which one among the following is not a character of a Secular State?

(a) It refuses theocracy.
(b) It separates religion from the State.
(c) A State in order to be secular must be democratic.
(d) It must prevent religious conflict and promote religious harmony.

(c) For being a Secular State, it is not necessary that State must be democratic. There are many autocratic states which are secular. *The characteristics of Secular State are as follows*

- The term 'Secular' means being 'separate' from religion, or having no religious basis.
- It refuses theocracy.
- It must prevent religious conflict and promote religious harmony.
- Religion is kept separate from the social, political, economic and cultural spheres of life.

51. A special address by the Governor refers to the address delivered by the Governor

(a) when President's Rule is called for
(b) when a national emergency necessitates dissolution of Legislative Assembly
(c) at the commencement of the first session after general election and at the first session of each year
(d) whenever he/she has concluded that such is necessary

(c) Article-176(1) of the Constitution of India provides that the Governor shall Address State's Assembly at the commencement of the first Session after each general election to the Assembly and at the commencement of the first session of each year and inform the Legislature of the causes of its Summons.
The aforementioned address of the Governor is referred as Special Address. The executive power of the State is vested in the Governor. He is appointed by the President by warrant under his hand and seal.

52. Which one of the following statements in relation to Panchayats is not correct?

(a) Legislature of a State may, by law, make provisions with respect to the composition of Panchayats.
(b) Panchayat area means the territorial area of a Panchayat.
(c) Gram Sabha includes all persons in the electoral rolls of village within a Panchayat.
(d) Reservation of seats for SCs and STs has nothing to do with proportion of their population.

(d) Seats reserved for Scheduled Castes (SCs) and Scheduled Tribes (STs) and the chairpersons of the Panchayats at all levels also shall be reserved for SCs and STs in proportion to their population. Hence, statement (d) is incorrect. The 73rd Constitutional Amendment Act of 1992 provided Constitutional status to the Panchayats. *The salient features of Panchayati Raj System are follows*

- Three-tier system of panchayats at village, intermediate block/taluk/mandal and district.
- levels except States with population below 20 lakhs (Article-243B).
- Panchayat area means the territorial area of a Panchayat.
- Gram Sabha includes all people in the electoral rolls of village within a Panchayat.
- Legislature of a state may, by law, make provision with respect to the composition of Panchayat.

53. Which one of the following statements with regard to 'protective democracy' is not correct?

(a) It propounds that citizen participation is essential in democracies.
(b) Citizens must be able to protect themselves from governmental encroachments.
(c) It is compatible with laissez-faire capitalism.
(d) Political equality is understood in formal terms as equal voting rights.

(c) John Locke (1631-1704) is regarded as the great apostle of protective democracy.
The basic features of protective democracy are as follows

- It propound that the citizen participation is essential in democracies.
- Both the popular sovereignty and representative form of government are legitimate.
- Citizens must be able to protect themselves from governmental encroachments.
- Political equality is understood in formal terms as equal voting rights.
- The authority is accountable to the People and in order to establish it elections are held on regular basis.
- Constitution is the source of power for all and is the guarantor of rights and liberties.

54. In August 2020, a blast has taken place at Beirut killing about one hundred people and thousand wounded. The blast was caused by

(a) dynamite
(b) ammonium nitrate
(c) RDX
(d) mercury nitride

(b) In August 2020, a blast had taken place at Lebanon's capital Beirut. At least 100 people were killed and nearly 4000 injured in this blast. The blast was occurred due to 2700 tonnes of ammonium nitrate stored for six years in a warehouse in the port. Ammonium nitrate (NH_4NO_3) is a white, crystalline chemical which is soluble in water. It is the main ingredient in the manufacture of commercial explosives used in mining and constructions.

55. What is 'Little Boy'?

(a) The fission bomb dropped at Hiroshima
(b) The fusion bomb dropped at Nagasaki
(c) The first nuclear bomb tested by America
(d) The first nuclear bomb tested by North Korea

(a) Little Boy is a codename of the fission bomb dropped at Hiroshima (a city in Japan) on August 6, 1945 by US. Uranium was used in it as a fissile material. On August 9, 1945, US dropped another bomb codenamed Fat Man, on Nagasaki. These bombing marked the end of World War II, with Japan surrendering to the Allies on August 14, 1945. Thousands more died in the following years due to the exposure to radiation from the blast and also from the black rain that fell in the aftermath of the explosions.

56. Which one of the following Indian institutes was approved by the Drugs Controller General of India for conducting human trials of the Oxford-Astra Zeneca COVID-19 vaccine candidate?

(a) Bharat Biotech
(b) AIIMS
(c) Serum Institute of India
(d) National Institute of Epidemiology

(c) Serum Institute of India (SII) was approved by the Drug Controller General of India for conducting trials of the Oxford-Astra Zeneca Coronavirus Vaccine. ChAdOx1 nCoV-19 or AZD1222 has been developed by the Oxford University and drug maker Astra Zeneca (Anglo-Swedish pharma major).

SII is the world's largest vaccine manufacturer by the number of doses produced and sold globally. It has a manufacturing partnership with Astra Zeneca to produce the Oxford- Astra Zeneca vaccine.

57. The recent explosion near OIL well in Baghjan is due to

(a) removing the spool during the blowout control operations
(b) transfer of oil from its depot to a pipeline
(c) the leakage of methyl isocyanide
(d) the leakage of radiations from radioactive substance

(a) On May 27th, 2020, an explosion was occurred near OIL well in Baghjan (Assam) at the time of removing spool during the blowout control operation. The blowout occurred at Well No. 5 in the Oil Field, resulting in a leak of natural gas.

The leaking well subsequently caught fire and has caused environmental damage to the nearby Dibru-Saikhowa National Park.

58. Who among the following is the architect of the Ram Temple being constructed at Ayodhya?

(a) PO Sompura
(b) Chandrakant Sompura
(c) Brinda Somaya
(d) BV Doshi

(b) Chandrakant Sompura is the architect of the Ram Temple, being constructed at Ayodhya (UP). He is a Ahmedabad-based architect whose design was approved by Shri Ram Janmabhoomi Teerth Kshetra.The stone laying ceremony of the Ram Temple was done by the Prime Minister Narendra Modi in August 2020.

59. Which one of the following countries is not located on the Tropic of Capricorn?

(a) Chile
(b) Brazil
(c) Paraguay
(d) Uruguay

(d) The Tropic of Capricorn is one of the five major circles of latitude marked on maps of Earth. Its latitude is 23°26'11.7'' South of the Equator. There are 10 countries, 3 continents and 3 water bodies on it.

Continent	Country
South America	Argentina, Brazil, Chile Paraguay
Africa	Namibia, Botswana, South Africa, Mozambique, Madagascar
Australia	Australia

Uruguay is not located on Tropic of Capricorn.

60. Match List-I with List-II and select the correct answer using the codes given below the Lists.

List-I (Active Volcano)	List-II (Location)
A. Mount Merapi	1. Hawaii
B. Sakurajima	2. Italy
C. Mount Vesuvios	3. Japan
D. Mauna Loa	4. Indonesia

Codes

	A	B	C	D
(a)	1	2	3	4
(b)	1	3	2	4
(c)	4	2	3	1
(d)	4	3	2	4

(d) Correct match are as follows

Active Volcano	Location
Mount Merapi	Indonesia
Sakurajima	Japan
Mount Vesuvius	Italy
Mouna Loa	Hawaii

61. Which one of the following is not a major tectonic plate?

(a) Saudi Arabian plate
(b) Antarctica and the surrounding oceanic plate
(c) India-Australia-New Zealand plate
(d) Pacific plate

(a) Lithosphere is divided in different tectonic plates. These plates are of two types- Major and Minor. There 7 major plates and 20 minor plates.

Major Plates are as follows

1. North American plate
2. South American plate
3. Pacific plate
4. Antarctica and the surrounding oceanic plate
5. Eurasia and the adjacent oceanic plate
6. Africa with the eastern Atlantic floor plate
7. India-Australia-New Zealand plate

Saudi Arabian plate is a minor plate. Hence, option (a) is correct.

62. Which one of the following is considered as the deepest point of the oceans?

(a) Tonga Trench
(b) Mariana Trench
(c) Philippine Trench
(d) Kermadec Trench

(b) The Mariana Trench is considered as deepest point of the oceans. It is located in the western Pacific Ocean, East of the Mariana Islands. The maximum known depth is 10,984 m (36,037 ft) at the southern end of a small slot-shaped valley its floor known as the Challenger Deep.

Other Important Trenches of the world

Trench	Ocean
Tonga Trench	Pacific Ocean
Philippine Trench	Pacific Ocean
Kuril-Kamchatka Trench	Pacific Ocean
Kermadec Trench	Pacific Ocean
Izu-Bonin Trench	Pacific Ocean
Japan Trench	Pacific Ocean
Puerto Rico Trench	Atlantic Ocean
South Sandwich Trench	Atlantic Ocean
Peru-Chile Trench or Atacama Trench	Pacific Ocean
Java Trench	Indian Ocean

63. The four planets closest to the Sun are called

(a) Terrestrial planets (b) Giant planets
(c) Dwarf planets (d) Gas planets

(a) The four planet closest Sun are called Terrestrial planets. It includes Mercury, Venus, Earth, and Mars. Terrestrial planets are Earth-like planets made up of rocks or metals with a hard surface. Terrestrial planets also have a molten heavy-metal core, and topological features such as valleys, volcanoes and craters.

64. Which one of the following countries does not have Tundra vegetation?

(a) Belarus (b) USA
(c) Russia (d) Canada

(a) Belarus do not have Tundra vegetation. The tundra is the coldest biome. It also receives low amounts of precipitation, making the tundra similar to a desert. Scattered trees grow in some tundra regions. The tundra soil is rich in nitrogen and phosphorus. Tundra is found in the regions just below the ice caps of the Arctic, extending across North America, to Europe, and Siberia in Asia.

65. Which one of the following is not a fluvial landform?

(a) Cirque (b) Gorge
(c) Braids (d) Canyon

(a) Cirque is a glacial landform. The landform created by the glacier are known as glacial landform. Aretes, Nunatak, Esker etc. are important glacial landforms.

66. In the Gandhara School of Art, initially blue schist and green phyllite were used. When did stucco completely replace stone as main material used by Gandhara School sculptors?

(a) 1st Century CE
(b) 2nd Century CE
(c) 3rd Century CE
(d) 5th Century CE

(c) The Gandhara school of Art flourished in the areas of Afghanistan and present North-Western India. It flourished from 1st Century BCE to 4th Century BCE. The material used for in the sculpture making were green phyllite and gray blue schist and Stucco, which was used increasingly after the 3rd Century BCE. This school of Art is associated with the Greco-Roman style of Art. The main theme of this art were derived from the Mahayana Buddhism.

67. Which one of the following statements about Gupta coins is not correct?

(a) Gupta kings issued large number of gold coins known as Dinar.
(b) Chandragupta II, Kumaragupta I, Skandagupta and Budhagupta issued silver coins.
(c) The obverses of coins are carved with the images of the kings and on the reverse are carved deities.
(d) The largest number of coins issued by the Guptas were of copper.

(d) Gupta king issued large number of gold coins known as 'Dinars'. Gupta coins were usually minted in gold and silver. The obverses of coins were carved with the images of the kings in the reverse with deities. Gupta kings like Chandragupta II, Kumaragupta I, Skandagupta and Budhagupta issued silver coins. Gupta kings also depicted socio-political event such as marriage of the kings and queens, Ashvmedha Yagya etc.

Gupta dynasty reigned from 319 to 467 CE. It covered much of the Indian subcontinent.

68. Who among the following wrote The *Philosophy of the Bomb?*

(a) Sukhdev
(b) Chandrashekhar Azad
(c) Bhagwati Charan Vohra
(d) Bhagat Singh

(c) The Philosophy of the Bomb was written by the revolutionary freedom fighter Bhagwati Charan Vohra. He was associated with Hindustan Socialist Republican Association. He died in Lahore on May 28, 1930 while testing a bomb.

69. At which one of the following Sessions of the Indian National Congress was the resolution on Fundamental Rights and Economic Policy passed?

(a) Tripuri Session
(b) Lahore Session
(c) Lucknow Session
(d) Karachi Session

(d) At the Karachi Session (1931), the Indian National Congress passed to resolution on Fundamental Rights (FR) and Economic policy.

70. Which one of the following towns was not a centre of the Revolt of 1857?

(a) Ayodhya (b) Agra
(c) Delhi (d) Kanpur

(b) Agra was not a centre of the Revolt of 1857. In 1857, people of country started a revolt against the exploitative British government. The revolt began on 10th May 1857, at Meerut in the form of sepoy mutiny. The political economic and socio-religious activities of British government were responsible for the revolt.

71. Consider the following statements :

The Azamgarh Proclamation refers to

1. the declaration by the rebels of 1857.
2. the statement by the leader of the underground movement in the Revolt of 1942.

Which of the statements given above is/are not correct?

(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

(b) The Azamgarg proclamation was declared by the Feroz Shah, (Grandson of the Mughal Emperor) in 1857. He fought against the British forces in the Awadh region of Uttar Pradesh. It was like a manifesto in which aims were set out for what rebels were fighting for. This proclamation appealed both Hindus and Muslims to cooperate with each other in fight against the Britishers.

72. Ibn Batuta went to China as the envoy of which one of the following Delhi Sultans?

(a) Alauddin Khilji
(b) Muhammad bin Tughluq
(c) Iltutmish
(d) Firoz Shah Tughluq

(b) Muhammad bin Tughlaq (AD 1324-51) had sent Ibn Batuta (Moroccan Traveller) to China as the envoy. Ibn Batuta arrived India in AD 1334. He wrote about the history of the Tughlaq dynasty.

73. Al-Biruni's Kitab-ul-Hind was written in which language?

(a) Arabic (b) Persian
(c) Urdu (d) Turkish

(a) Al-Biruni's Kitab-ul-Hind was written in Arabic language. He came to India with Mahmud Ghazni in 11th Century AD and lived here for many years. He also learnt Sanskrit language. In this book, he described the socio-economic, political, religious and economic condition of the then India.

74. Which of the following statements with regard to the privileges of the Members of the Parliament are correct?

1. Privileges would not be fettered by the Article-19(1)(a) of the Constitution of India.
2. Privileges must be read subject to the Articles-20-22 and Article-32 of the Constitution of India.
3. Immunity is available in relation to both civil and criminal prosecution.
4. Immunity is available in relation to freedom of speech even in his/her private or personal capacity.

Select the correct answer using the codes given below.

(a) 1, 2 and 4 (b) 1 and 2
(c) 2 and 3 (d) 1 and 4

(b) Every Member of the Parliament enjoys certain privileges so that they can carry out their duty without interference. Privileges are of two type-Collective and Individual.

Individual Privileges of Member of Parliament include.

- Immunity from assest during the session of Parliament. This privilege is available in civil case only.
- A MP may refuse to appear in court or present any evidence, during a Parliamentary session.
- No member is liable to any proceeding in any given court for anything said or any vote by him/her in the Parliament or its committees.

Privileges would not be fettered by the Article-191(a) of the Constitution. And it must be read subject to the Article 20-22 and Article-32 of the Constitution. Privileges does not cover freedom of speech in his/her private or personal capacity.

75. Which one of the following statements with regard to the appointment of the Members of the Parliamentary Committees is correct?

(a) The Members are only appointed
(b) The Members are only elected
(c) The Members are only nominated.
(d) The Members are appointed or elected on a motion made and adopted or nominated by the Speaker of the Lok Sabha or the Chairman of the Rajya Sabha

(c) Parliamentary Committees are constituted to perform various functions. Members are appointed or elected on a motion made and adopted or nominated by the Speaker of Lok Sabha or the Chairman of the Rajya Sabha. It has a secretariat provided by the Lok Sabha/Rajya Sabha secretariat. Indian Constitution mentions two kinds of Parliamentary Committees - Standing Committees and Ad Hoc Committees.

Standing Committees are permanent committees and are constituted for a fixed tenure. Ad hoc Committees are appointed for a specific purpose and they cease to exist when they finish the task assigned to them after submitting the report.

76. Which one of the following is not a classified category of political parties as outlined by the Election Commission of India?

(a) National Parties
(b) State Recognised Parties
(c) Regional Parties
(d) Registered Unrecognised Parties

(b) State Recognised Parties is not a classified category of political parties in India as outlined the Election Commission. The Commission classifies parties into three main heads: National Parties, State Parties, and Registered (unrecognised) Parties.

Political parties in India are classified for the allocation of symbols. Currently, there are 8 National Parties. It includes the Indian National Congress (INC), the Nationalist Congress Party (NCP), the Bharatiya Janata Party (BJP), the Communist Party of India (CPI), the Communist Party of India, Marxists (CPI-M), the Bahujan Samaj Party (BSP), and the Rashtriya Janata Dal (RJD).

77. Which of the following terms were added to the Preamble of the Constitution of India by the Constitutional Amendment, 1976?

1. Socialist 2. Secular
3. Integrity 4. Fraternity

Select the correct answer using the codes given below.

(a) 1 and 2 (b) 1, 2 and 3
(c) 2 and 4 (d) 1, 3 and 4

(b) The 42nd Amendment Act, 1976 added words ‘Socialist’, ‘Secular’ and ‘Integrity’ in the Preamble of the Constitution of India. This Amendment is termed as Mini Constitution. *Other Provision of this Amendment-*

- Added Fundamental Duties by the citizens (new Part IV A).
- Made the President bound by the advice of the cabinet.
- Provided for administrative tribunals and tribunals for other matters (Added Part-XIV A).
- Frozen the seats in the Lok Sabha and State Legislative Assemblies on the basis of 1971 census till 2001- Population Controlling Measure
- Made the constitutional amendments beyond judicial scrutiny.
- Curtailed the power of judicial review and writ jurisdiction of the Supreme Court and High Courts.
- Raised the tenure of Lok Sabha and State Legislative Assemblies from 5 to 6 years.

78. Who among the following was the advisor to the Constituent Assembly?

(a) BN Rau
(b) BR Ambedkar
(c) Pattabhi Sitaramayya
(d) Alladi Krishnaswamy

(a) BN Rau was appointed as the Constitutional Adviser to the Constituent Assembly in 1946. He created general draft of the Constitution of India. He was also India's representative to the United Nations Security Council from 1950 to 1952.
BR Ambedkar was the President of the Drafting Committee of the Constituent Assembly.
Pattabhi Sitaramayya played instrumental role in the creation of Andhra Pradesh in 1953.
Alladi Krishnaswamy Iyer was the member of the Drafting Committee of the Constituent Assembly.

79. In the Indian judicial system, writs are issued by

(a) the Supreme Court only
(b) the High Courts only
(c) the Supreme Court and High Courts only
(d) the Supreme Court, High Courts and Lower Courts

(c) The Constitution of India under Article-32 and 226 empowers the Supreme Court of India High Court to issue writs respectively. Writs are a written order that commands constitutional remedies for Indian Citizens against the violation of their Fundamental Rights. Writs are of five types, namely, Habeas Corpus, Prohibition, Mandamus, Certiorari and Quo-Warranto. The High Courts can issue writs not only for the preservation of Fundamental Rights, but also for any other purpose. However, the Supreme Court can issue writs only for the preservation of Fundamental Rights.

80. The Citizenship (Amendment) Act falls under which one of the following parts of the Constitution of India?

(a) Part I (b) Part II
(c) Part IV (d) Part VI

(b) The Citizenship (Amendment) Act falls under Part II of the Constitution of India. This part contains Articles-5-11.

Article	Provision
5	Citizenship at the commencement of the Constitution
6	Citizenship of certain persons who have migrated from Pakistan
7	Citizenship of certain migrants to Pakistan
8	Citizenship of certain persons of Indian origin residing outside India
9	People voluntarily acquiring citizenship of a foreign country will not be citizens of India.

Article	Provision
10	Any person who is considered a citizen of India under any of the provisions of this Part shall continue to be citizens and will also be subject to any law made by the Parliament.
11	The Parliament has the right to make any provision concerning the acquisition and termination of citizenship and any other matter relating to citizenship.

81. Recently a rare kind of yellow turtle was discovered in India. The State in which it was seen is

(a) Uttarakhand
(b) Odisha
(c) Tamil Nadu
(d) Arunachal Pradesh

(b) A rare Yellow Turtle was discovered in India from Balasore district of Odisha. The colour of turtle was due to albinism. This turtle is known as the Indian flap shell turtle. This turtle is commonly found in Pakistan, Sri Lanka, India, Nepal, Bangladesh, and Myanmar. Albinism is a type of genetic disorder where it is little or no production of pigment in the skin, eyes, and hair or in other species in the fur, feathers or scales.

82. ASEEM is

(a) Aatmanirbhar Skilled Employee Employer Measurement
(b) Aatmanirbhar Skilled Employee Employer Mapping
(c) Aatmanirbhar Skilled Employee Enterprises Medium
(d) Automatic Skilled Employee Employer Mission

(b) ASEEM is an acronym of 'Aatamanirbhar Skilled Employee Employer Mapping' portal. It was launched by the Ministry of Skill Development and Entrepreneurship (MSDE) in July 2020 to help skilled people in fighting sustainable livelihood opportunities.

It is Managed by the National Skill Development Corporation (NSDC) in collaboration with Bengaluru-based company 'Betterplace'.

The ASEEM digital platform consists of three IT-based interfaces

Employer Portal Employer onboarding, Demand Aggregation, candidate selection

Dashboard Reports, Trends, analytics, and highlight gaps

Candidate Application Create and Track candidate profile, share job suggestion

83. The Government of India programme regarding 'Stay in India and Study in India' is initiated by

(a) the Ministry of Youth Affairs and Sports
(b) the Ministry of Culture
(c) the Ministry of Education
(d) the Ministry of Tourism

(c) A programme called 'Study in India - Stay in India' was initiated by the Ministry of Education in August, 2020. The objective of this plan is to prevent students from leaving the country seeking higher education abroad and also to bring back Indian students studying abroad.

For this initiative a committee has been constituted under the leadership of DK Singh to prepare guidelines and measures to strengthen both Study in India - Stay in India.

84. Which one of the following lakes in India has a large quantity of a substance found in the Moon?

(a) Lonar Lake (Maharashtra)
(b) Pangong Lake (Ladakh)
(c) Chilika Lake (Odisha)
(d) Loktak Lake (Manipur)

(a) Lonar Lake is in Indian State of Maharashtra and it has a large quantity of a substance found in the Moon. In March 2019, IIT-Bombay had found in research that mineral contents of Lonar lake is similar to Moon rocks. This lake was formed after a meteorite crash around 50,000 years ago.

The Lonar Crater is also a wildlife sanctuary and a notified Geo-Heritage monument. It is home to a variety of flora and fauna.

85. The Pragyan rover installed in Chandrayaan-2 mission had how many wheels?

(a) 2
(b) 3
(c) 4
(d) 6

(d) Indian Space Research Organisation (ISRO) launched Mission Chandrayaan-2 in July 2019 (Mission was unsuccessful). This Mission was comprised of Vikram, the lander and Pragyan,the rover. Pragyan was a six-wheeled solar-powered vehicle. It was sent to study the composition of the surface near the lunar landing site and determine the abundance of various elements.

86. If the linear momentum of a moving object gets doubled due to application of a force, then its kinetic energy will

(a) remain same
(b) increase by four times
(c) increase by two times
(d) increase by eight times

(b) Let m be the mass and v be the velocity of the moving object.

Kinetic energy of object,

$$K = \frac{1}{2}mv^2 = \frac{1}{2m}(m^2v^2)$$

$$K = \frac{1}{2m}(mv)^2 \quad \ldots(i)$$

Now, linear momentum,

$$p = mv \quad \ldots(ii)$$

Substituting Eq (ii) in Eq (i), we get

$$K = \frac{p^2}{2m} \Rightarrow K \propto p^2$$

If linear momentum gets doubled, then new kinetic energy will become

$$K' = \frac{(2p)^2}{2m} = \frac{4p^2}{2m}$$

$$K' = 4K$$

Hence, if linear momentum of a moving object gets doubled, its kinetic energy will increase by four times.

87. If the distance between two objectsi s increased by two times, the gravitational force between them will

(a) remain same
(b) increase by two times
(c) decrease by two times
(d) decrease by four times

(d) The gravitational force between two objects of masses m_1 and m_2, separated by a distance r is given by

$$F = \frac{Gm_1m_2}{r^2} \quad \ldots(i)$$

where, G is the universal gravitational constant.

If the distance between the objects is increased by two times, the gravitational force will become,

$$F' = \frac{Gm_1m_2}{(2r)^2} = \frac{Gm_1m_2}{4r^2}$$

$$\Rightarrow F' = \frac{F}{4} \quad \text{[Using Eq (i)]}$$

Hence, the gravitational force will decrease by four times.

88. Which one of the following statements about the properties of neutrons is not correct?

(a) Neutron mass is almost equal to proton mass.
(b) Neutrons possess zero charge.
(c) Neutrons are located inside the atomic nuclei.
(d) Neutrons revolve around the atomic nuclei.

(d) Statement (d) is not correct about the properties of neutrons, whereas all other statements are correct.

Correct Statement Electrons revolve around the atomic nuclei, not neutrons.

89. Which one of the following statements with regard to the ultraviolet light is not correct?

(a) It is an electromagnetic wave
(b) It can travel through vacuum
(c) It is a longitudinal wave
(d) Its wavelength is shorter/smaller than that of visible light

(c) Statement in option (c) is not correct about the ultraviolet light.

The correct statement is :

Ultraviolet light is a transverse wave.

90. If the speed of light in air is represented by c and the speed in a medium is v, then the refractive index of the medium can be calculated using the formula

(a) v/c (b) c/v
(c) $c/(2.v)$ (d) $\frac{(c-v)}{c}$

(b) Refractive index of medium

$$= \frac{\text{Speed of light in air}}{\text{Speed of light in medium}} = \frac{c}{v}$$

91. Under the kingdom– Plantae, which of the following individuals are predominantly aquatic?

(a) Bryophytes (b) Algae
(c) Pteridophyta (d) Gymnosperms

(b) Under the kingdom– plantae, Phyllum-Algae have individuals that are predominantly aquatic (both fresh water and marine).

These are chlorophyll bearing simple, thalloid, autotrophic organism. They occurs in a variety of habitats like standing water, moist stones, soils and woods.

92. All the individuals of a particular organism, such as rose plants, belong to a taxonomic category called

(a) Species (b) Genus
(c) Family (d) Order

(a) All the individuals of a particular organism, such as rose plants belongs to a taxonomic category called species because species is a group of individual organisms with fundamental similarities.

93. Pearls are harvested from

(a) prawn (b) pila
(c) tuna (d) oyster

(d) Pearls are harvested from oyster. A pearl is an ulcer formed in the oysters when an unknown outside particle get into the shell. It is due to the stress that the oysters produce nacre that will eventually become the pearl in a couple of years. So, basically the pearl is the result of a disease in the oyster's body.

94. Wings of birds and bats are considered analogous structures because they have

(a) common origin and common function
(b) different origin and common function
(c) common origin and different function
(d) different origin and different function

(b) Wings of birds and bats looks alike and perform same function of flying. But anatomically they are different in origin. Birds have feathers projecting back from lightweight, fused arms and hand bones. Bats have flexible, relatively short wings with membranes stretched between elongated fingers. Thus, these are analogous.

95. Apart from hyper acid secretion, peptic ulcers are also developed due to bacterial infection. The causative agent is

(a) *Helicobacter pylori*
(b) *E. coli*
(c) *Streptococcus pneumoniae*
(d) *Salmonella typhimurium*

(a) *Helicobacter pylori* is the causative agent of peptic ulcer. It is a sore that develops on the living of the oesophagus, stomach or small intestine. Ulcer occurs when stomach acid damages the living of the digestive tract. Common causes of ulcer include the bacteria *H. pylori* and anti- inflammatory pain relievers including aspirin.

96. A mixture of sodium chloride (salt) and ammonium chloride can be separated by

(a) sublimation (b) filtration
(c) chromatography (d) distillation

(a) A mixture of sodium chloride (salt) and ammonium chloride can be separated by sublimation. This technique involves the conversion of a solid direct into vapour. On heating, ammonium chloride changes from solid state to ammonia vapours. Here, ammonium chloride can be sublimed whereas sodium chloride cannot be sublimed.

97. Symbol of element was introduced by

(a) John Dalton
(b) Antoine Lavoisier
(c) Jons Jacob Berzelius
(d) Robert Boyle

(a) Symbol of element was introduced by John Dalton. Symbols are the representation of an element. It is simple to use the symbol of an element rather than writing a whole word.

Some symbols along with elements are as follows :

⊙ Hydrogen ● Carbon
Ⓟ Phosphorus ⊕ Sulphur
© Copper Ⓛ Lead

98. Identify the correct pair of elements among the following which are liquid at room temperature and standard pressure.

(a) Bromine and Fluorine
(b) Mercury and Rubidium
(c) Bromine and Thallium
(d) Bromine and Mercury

(d) Bromine and Mercury is the pair of element which are liquid at room temperature and standard pressure. Bromine is a liquid non-metal whereas mercury is a liquid metal.

99. Which one of the following oxides shows both acidic and basic behaviour?

(a) Zinc oxide (b) Copper oxide
(c) Magnesium oxide
(d) Calcium oxide

(a) Zinc oxide (ZnO) shows both acidic and basic behaviour. It reacts with both acids and bases to form salt and water.

$$\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Hydrochloric acid}}{2HCl(aq)} \longrightarrow \underset{\text{Zinc chloride}}{ZnCl_2(aq)} + \underset{\text{Water}}{H_2O(l)}$$

$$\underset{\text{Zinc oxide}}{ZnO(s)} + \underset{\text{Sodium hydroxide}}{2NaOH(aq)} \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2(aq)} + \underset{\text{Water}}{H_2O}$$

100. Silver articles turn black when kept in the open for longer time due to the formation of

(a) H_2S (b) AgS (c) $AgSO_4$ (d) Ag_2S

(d) Silver articles turn black when kept in the open for longer time due to the formation of silver sulphide (Ag_2S). Silver metal reacts with sulphur present in the atmosphere and forms Ag_2S. Complete reaction is as follows

$$2Ag + S \longrightarrow Ag_2S$$

101. Which of the following lenses will bend the light rays through largest angle?

(a) Lens with power + 2.0 D
(b) Lens with power + 2.5 D
(c) Lens with power – 1.5 D
(d) Lens with power – 2.0 D

(b) The power of a lens is defined as its ability to bend the light rays falling on it. So, the lens with more power will bend the light rays through largest angle.

∴ The lens with + 2.5 D will bend the light rays through largest angle.

102. A luminous object is placed at a distance of 40 cm from a converging lens of focal length 25 cm. The image obtained in the screen is

(a) erect and magnified
(b) erect and smaller
(c) inverted and magnified
(d) inverted and smaller

(c) Given,

Object distance, $u = -40$ cm

Focal length, $f = 25$ cm

Let v be the image distance.

From lens formula,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\Rightarrow \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{25} + \frac{1}{(-40)}$$

$$\Rightarrow \quad \frac{1}{v} = \frac{1}{25} - \frac{1}{40} = \frac{8-5}{200} = \frac{3}{200}$$

$$\Rightarrow \quad v = \frac{200}{3} = 66.7 \text{ cm}$$

Now, magnification produced by lens,

$$m = \frac{v}{u} = \frac{66.7}{(-40)} = -1.66$$

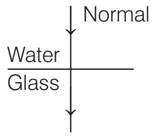
Since, the magnification is negative, so the image formed will be real and inverted.

Also, the magnitude of magnification is greater than unity, so the image will be magnified.

103. When a light ray enters into glass medium from water at an angle of incidence 0°, what would be the angle of refraction?

(a) 90° (b) 45°
(c) 0°
(d) The ray will not enter at all

(c) When a light ray enters into glass medium from water at angle of incidence 0°, it will fall normally on the glass medium.



If the incident ray falls normally to the surface of glass medium, then there will be no bending of the light ray. It goes straight without any deviation.

∴ Angle of refraction = 0°

104. Which one of the following phenomena verifies the fact that light travels much faster than sound?

(a) Twinkling of stars in night sky
(b) Lighting of a matchstick
(c)Thunderstorm
(d) Mirage

(c) In case of a thunderstorm, the lightning is seen first and then we hear the thunder after some time.

This is because light travels with speed 3×10^8 ms^{-1} in air while sound travels at a speed of 340 ms^{-1} in air. This phenomenon proves the fact that light travels much faster than sound.

105. Which one of the following combinations of source and screen would produce sharpest shadow of an opaque object?

(a) A point source and an opaque screen
(b) An extended source and an opaque screen
(c) A point source and a transparent screen
(d) An extended source and a transparent screen

(c) To form the sharpest shadow of an opaque object, a point source and a transparent screen are required.

106. Which one among the following is a non-conventional source of energy?

(a) Petroleum
(b) Coal
(c) Radioactive elements
(d) Solar energy

(d) Non-conventional sources of energy are those sources of energy which are being produced continuously in nature and are inexhaustible. Amongst the given options, solar energy is a non-conventional source of energy.

107. Which one of the following statements about phloem is correct?

(a) Phloem transports water and minerals.
(b) Phloem transports photosynthetic products.
(c) Phloem is a simple tissue.
(d) Phloem gives support to the plant.

(d) Option (b) is correct as the phloem is the means by which the products of photosynthesis are transported to non-photosynthetic parts of the plant, such as the roots and to developing leaves, nectaries, fruits and seeds.
Other statements can be corrected as

- Xylem transports water and minerals.
- Phloem is a complex tissue.
- Primary function of phloem is transport of sugars not support.

108. Mature sclerenchyma cells have

(a) cellulose wall and are living
(b) lignified wall and are living
(c) suberized wall and are dead
(d) lignified wall and are dead

(d) Mature Sclerenchyma consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits.
They are usually dead and without protoplasts.

109. In human beings, the chromosomes that determine birth of a normal female child are

(a) one X chromosome from mother and one X chromosome from father
(b) one X chromosome from mother and one Y chromosome from father
(c) two X chromosomes from mother and one X chromosome from father
(d) one X chromosome and one Y chromosome from father and one X chromosome from mother

(a) The chromosome number of a normal female is 44 + XX. It has one X-chromosomes which comes from mother and one X-chromosome which comes from father.
Thus option 'A' is correct.

- Fusion of two X from mother and one X from father will lead to birth of superfemale (XXX).
- Fusion of one X-chromosome and one Y-chromosome from father and one X-chromosome from mother will lead to birth of a child with Klienfelter syndrome (XXY).

110. Antibiotic such as penicillin blocks

(a) cell wall formation in bacteria
(b) RNA synthesis in bacteria
(c) DNA synthesis in bacteria
(d) division in bacteria

(a) Antibiotics such as penicillin kills bacteria by inhibiting or blocking the proteins which cross-link peptidoglycans in the cell wall. When these bacterium divides in the presence of penicillin, it cannot fill in the 'holes' left in its cell wall.

111. The radioactive isotope of hydrogen is

(a) protium (b) deuterium
(c) tritium (d) hydronium

(c) Tritium is the radioactive isotope of hydrogen. The nucleus of tritium contains one proton and two neutrons. It is represented as ^3H or T. It is used as a radioactive tracer, in radioluminescent light sources for watches and instruments.

112. Which one of the following is used for storing biological tissues?

(a) Liquid nitrogen (b) Liquid helium
(c) Liquid argon (d) Liquid bromine

(a) Liquid nitrogen is used for storing biological tissues. This process is known as Cryo preservation.

In this process, organelles, cells tissues or other biological lonsbuets susceptible to damage caused by unregulate Chemical kinetics are preserved by cooling to very low temperatures.

113. Which one of the following does not form oxide on reaction with oxygen?

(a) Magnesium (b) Lead
(c) Tin (d)Silver

(c) Tin (Sn) does not form oxide on reaction with oxygen. It readily form peroxide. Complete reaction is

$$\underset{}{Sn} + O_2 \longrightarrow \underset{\text{Tin peroxide}}{SnO_2}$$

All other options such as magnesium (Mg), lead (Pb) and Silver (Ag) forms oxide. Reactions are as follows

$$2Mg + O_2 \longrightarrow 2MgO$$
$$2Pb + O_2 \longrightarrow 2PbO$$
$$4Ag + O_2 \longrightarrow 2Ag_2O$$

114. The valency of phosphorus is

(a) 2, 3 (b) 3, 4 (c) 4, 5 (d) 3, 5

(d) Valency of phosphorus is 3, 5-orbits are stable when they are fully filled or half filled. So, to attain a stable configuration it can take 3 electrons or remove 5 electrons. So, valency of 'P' is 3 and 5.

115. Lead nitrate on heating gives

(a) PbO_2 and NO_2
(b) PbO and NO_2
(c) PbO and NO
(d) PbO_2 and NO

(b) Lead nitrate on heating gives PbO (lead oxide) and NO_2 (nitrogen dioxide). Complete reaction is as follows

$$\underset{\substack{\text{Lead Nitrate}\\\text{Ceohite salt}}}{2Pb(NO_3)_2(s)} \xrightarrow{\text{heat}} \underset{\substack{\text{Lead Oxide}\\\text{Residue}}}{2Pbo(s)}$$
$$+ \underset{\substack{\text{Nitrogen dioxide}\\\text{(brown)}}}{4NO_2(g)} + O_2(g)$$

116. Which one of the following is a decommissioned aircraft carrier?

(a) INS Rajput (b) INS Chakra
(c) INS Khanderi (d) INS Viraat

(d) Amongst the given options, INS Viraat is a decommissioned aircraft carrier. It was a Centaur-class aircraft carrier of the Indian Navy. India acquired this aircraft carrier from England and commissioned into the Indian Navy on May 12th, 1987. Viraat was formally decommissioned on March 6th, 2017.

INS Khanderi is a Kalvari-class submarine, built in India. **INS Rajput** is a guided-missile destroyer. **INS Chakra** is a nuclear-powered attack submarine.

117. Who among the following has won the Singles Title in Wimbledon Tennis Championship (Women) in the year 2019?

(a) Karolina Pliskova (b) Simona Halep
(c) Serena Williams (d) Naomi Osaka

(b) Simona Halep (Romania) won the Women's Singles title in Wimbledon Tennis Championship in the year 2019.

Other Winners

Category	Winner
Men's Singles	N. Djokovic (Serbia)
Men's Doubles	Juan Sebastian Cabal (Colombia) and Robert Farah (Colombia)
Women's Doubles	SW. Hsieh (Taiwan) and B. Strycova (Czech Republic)
Mixed Doubles	I. Dodig (Croatia) and L. Chan (Taiwan)

118. Recently islands of Andaman and Nicobar were connected with mainland by Submarine Optical Fibre Cable.

Which one of the following islands was not connected initially?

(a) Shaheed Island
(b) Swaraj Island
(c) Little Andaman
(d) Port Blair

(a) In August 2020, submarine Optical Fibre Cable (OFC) connecting Andaman and Nicobar Islands to the mainland was inaugurated. Services began from Chennai to Port Blair, Port Blair to Little Andaman and Port Blair to Swaraj Island. Hence, Shaheed Island was not connected initially. The project of laying 2300 km optical fibre started in December, 2018.

A submarine optical fibre communications cable is a cable laid on the seabed between land-based stations to transmit telecommunication signals across stretches of ocean and sea.

119. In August 2020, who among the following was administered the oath as the Prime Minister of Sri Lanka for the fourth time?

(a) Gotabaya Rajapaksa
(b) Basil Rajapaksa
(c) Mahinda Rajapaksa
(d) Namal Rajapaksa

(c) Mahinda Rajapaksa was administered oath as the Prime Minister of Sri Lanka for the fourth time in August 2020. He is leader of Sri Lanka People's Party (SLPP) leader. His party registered victory in 145 constituencies, bagging a total of 150 seats with its allies, a two-thirds majority in the 225-member Parliament. He has replaced Ranil Wickremesinghe.Mahinda Rajapaksa was sworn in as the Prime Minister for the first time on 6th April, 2004.

120. Which one of the following Indian Ocean island nations has recently declared a State of environmental emergency due to oil spill from a grounded ship?

(a) Maldives (b) Mauritius
(c) Madagascar (d) Sri Lanka

(a) Mauritius is an island nation in the Indian Ocean, which declared a state of emergency due to oil spill in August 2020. A japanese bulk-carrier ship MV Wakashio which was carrying fuel oil had split into two parts near Blue Bay Marine Park in South-East Mauritius. It caused leakage of over 1000 tonnes of oil. It endangers the already endangered coral reefs, seagrasses, mangroves, the fishes and other aquatic fauna.

CDS

Combined Defence Service

SOLVED PAPER 2020 (I)

PAPER I Elementary Mathematics

1. The number $2 \times 3 \times 5 \times 7 \times 11 + 1$ is
(a) a prime number
(b) not a prime, but power of a prime
(c) not a power of a prime, but a composite even number
(d) not a power of a prime, but a composite odd number

(a) Given, number

$2 \times 3 \times 5 \times 7 \times 11 + 1$
$= 2310 + 1 = 2311$

Which is a prime number.

2. Two unequal pairs of numbers satisfy the following conditions
(i) The product of the two numbers in each pair is 2160.
(ii) The HCF of the two numbers in each pair is 12.

If x is the mean of the numbers in the first pair and y is the mean of the numbers in the second pair, then what is the mean of x and y?

(a) 60 (b) 72 (c) 75 (d) 78

(b) Let two unequal pairs of numbers are $(12a, 12b)$ and $(12c, 12d)$.

According to the question,

$12a \times 12b = 12c \times 12d = 2160$

$\Rightarrow \quad a \times b = c \times d = 15$

Here factors of 15 are 1, 3, 5, 15

$\therefore a = 1, b = 3, c = 5, d = 15$

Now, $x = \dfrac{12a + 12b}{2}, y = \dfrac{12c + 12d}{2}$

$\therefore$ Mean of x and $y = \dfrac{x + y}{2}$

$= \dfrac{12(a + b + c + d)}{4}$

$= 3(1 + 3 + 5 + 15)$

$= 3 \times 24 = 72$

3. How many digits are there in $(54)^{10}$? (Given that $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$)

(a) 16 (b) 18
(c) 19 (d) 27

(b) Let $x = (54)^{10}$

$\Rightarrow \log x = 10 \log 54$
$= 10 \log(2 \times 3^3)$
$= 10\{\log 2 + 3 \log 3\}$
$= 10\{0.301 + 3 \times 0.477\}$
$= 10\{0.301 + 1.431\}$
$= 10 \times 1.732 = 17.32$

Since, the characteristic of $\log x$ is 17.

$\therefore$ Number of digits is $(54)^{10}$ is

$17 + 1 = 18$

4. Which one of the following is a set of solutions of the equation $x^{\sqrt{x}} = \sqrt[n]{x^x}$, if n is a positive integer?

(a) $\{1, n^2\}$ (b) $\{1, \sqrt{n}\}$
(c) $\{1, n\}$ (d) $\{n, n^2\}$

(a) Given, $x^{\sqrt{x}} = \sqrt[n]{x^x}$ ...(i)

$\Rightarrow \quad x^{x^{1/2}} = (x^x)^{1/n}$

$\Rightarrow \quad x^{x^{1/2}} = x^{x/n}$

$\Rightarrow \quad x^{1/2} = x/n$

$\Rightarrow \quad x^{1/2} = n$

$\Rightarrow \quad x = n^2$

On putting $x = 1$ in Eq. (i), we get

$(1)^{\sqrt{1}} = (1^1)^{1/n}$

$\Rightarrow \quad 1 = 1$

$\Rightarrow x = 1$ also satisfies the given equation.

So, solution set is $\{1, n^2\}$

5. In a competitive examination, 250 students have registered. Out of these, 50 students have registered for Physics, 75 students for Mathematics and 35 students for both Mathematics and Physics. What is the number of students who have registered neither for Physics nor for Mathematics?

(a) 90 (b) 100 (c) 150 (d) 160

(d) Total number of students have registered = 250

Students have registered for Physics = 50

Students have registered for Mathematics = 75

And students have registered for both Mathematics and Physics = 35

$\therefore$ Number of students who registered for atleast one subject

$= (50 - 35) + (75 - 35) + 35$
$= 15 + 40 + 35 = 90$

$\therefore$ The number of students who have registered neither for Physics nor for Mathematics $= 250 - 90 = 160$

6. If the sum of the digits of a number $10^n - 1$, where n is a natural number, is equal to 3798, then what is the value of n?

(a) 421 (b) 422 (c) 423 (d) 424

(b) Given, number $= 10^n - 1$

For $n = 1$, $10^1 - 1 = 9$
For $n = 2$, $10^2 - 1 = 99$
For $n = 3$, $10^3 - 1 = 999$

$\therefore$ Sum of digits of $10^n - 1 = 9n$, where n is a natural number.

$\therefore \quad 9n = 3798 \Rightarrow n = 422$

7. Which one of the following is the largest number among 2222^2, 222^{22}, 22^{222}, 2^{2222}?

(a) 2^{2222} (b) 22^{222} (c) 222^{22} (d) 2222^2

(a) Given, $2222^2, 222^{22}, 22^{222}, 2^{2222}$

Taking log

$= \log(2222)^2, \log(222)^{22}, \log(22)^{222}, \log(2)^{2222}$

$= 2\log(2222), 22\log(222), 222\log(22), 2222\log(2)$

$= 2\log(2\times1111), 22\log(2\times111), 222\log(2\times11), 2222\log(2)$

$= 2(\log2+\log1111), 22(\log2+\log111), 222(\log2+\log11), 2222\log2$

$= 2(0.301+3), 22(0.301+2), 222(0.301+1), 2222(0.301)$

Since, $\log2 = 0.301$, $\log1111 = 3$ (approx.)

$\log111 = 2$ (approx.), $\log11 = 1$ (approx.)

$= 2(3.301), 22(2.301), 222(1.301), 2222(0.301)$

$= 6.602, 55.622, 288.822, 668.822$

Thus, 2^{2222} is the largest number.

8. If m is the number of prime numbers between 0 and 50; and n is the number of prime numbers between 50 and 100, then what is $(m-n)$ equal to?

(a) 4 (b) 5 (c) 6 (d) 7

(b) Prime numbers between 0 and 50 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 19, 31, 37, 41, 43, 47.

$\therefore$ Total number of prime numbers between 0 to 50.

$\therefore m = 15$

And Prime numbers between 50 to 100 are

53, 59, 61, 67, 71, 73, 79, 83, 89, 97

$\therefore$ Total number of prime numbers between 50 to 100

$\therefore n = 10$

$\therefore m - n = 15 - 10 = 5$

9. If $\frac{a}{b}=\frac{1}{3}, \frac{b}{c}=2, \frac{c}{d}=\frac{1}{2}, \frac{d}{e}=3$ and $\frac{e}{f}=\frac{1}{4}$, then what is the value of $\frac{abc}{def}$?

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{3}{8}$ (d) $\frac{27}{4}$

(c) Given, $\frac{a}{b}=\frac{1}{3}$ ... (i)

$\frac{b}{c}=2$ ... (ii)

$\frac{c}{d}=\frac{1}{2}$ ... (iii)

$\frac{d}{e}=3$ ... (iv)

and $\frac{e}{f}=\frac{1}{4}$ ... (v)

On multiplying Eqs. (i), (ii) and (iii), we get

$\frac{a}{b}\times\frac{b}{c}\times\frac{c}{d}=\frac{1}{3}\times2\times\frac{1}{2}=\frac{1}{3}$

$\Rightarrow \frac{a}{d}=\frac{1}{3}$ ... (vi)

On multiplying Eqs. (ii), (iii) and (iv), we get

$\frac{b}{c}\times\frac{c}{d}\times\frac{d}{e}=2\times\frac{1}{2}\times3=3$

$\Rightarrow \frac{b}{e}=3$ ... (vii)

On multiplying Eqs. (iii), (iv) and (v), we get

$\frac{c}{d}\times\frac{d}{e}\times\frac{e}{f}=\frac{1}{2}\times3\times\frac{1}{4}$

$\Rightarrow \frac{c}{f}=\frac{3}{8}$ ... (viii)

Again multiplying Eqs. (vi), (vii) and (viii), we get

$\frac{abc}{def}=\frac{1}{3}\times3\times\frac{3}{8}=\frac{3}{8}$

10. Which one of the following is the largest divisor of $3^x+3^{x+1}+3^{x+2}$, if x is any natural number ?

(a) 3 (b) 13 (c) 39 (d) 117

(c) Given, $3^x+3^{x+1}+3^{x+2}$

$= 3^x(1+3+3^2)$

$= 3^x(13)$ Where x is any natural number.

Put $x=1$, then, $3^1(13)=39$

Put $x=2$, then $3^2(13)=117$

$\therefore$ Largest divisor of $3^x+3^{x+1}+3^{x+2}$ is 39.

11. A two-digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two-digit number is 8, then what is the number?

(a) 81 (b) 42 (c) 24 (d) 18

(a) Let two digit number $= 10x + y$

According to the question,

$10x + y = 4(10y + x) + 9$

$\Rightarrow 10x + y = 40y + 4x + 9$

$\Rightarrow 6x - 39y = 9$

$\Rightarrow 2x - 13y = 3$ ... (i)

Also, $xy = 8 \Rightarrow y = 8/x$ ... (ii)

From Eqs. (i) and (ii), we get

$2x - 13(8/x) = 3$

$\Rightarrow 2x^2 - 3x - 104 = 0$

$\Rightarrow x = \frac{3\pm\sqrt{9+4\times2\times104}}{2\times2}$

$\Rightarrow x = \frac{3\pm\sqrt{9+832}}{4}=\frac{3\pm\sqrt{841}}{4}$

$\Rightarrow x = \frac{3\pm29}{4}$

$\Rightarrow x = \frac{3+29}{4}, \frac{3-29}{4}$

$\Rightarrow x = 8, -\frac{13}{2}$

From Eq. (ii), we get

$y = \frac{8}{8} = 1$

$\therefore$ Required number $= 10x + y$
$= 10\times8+1$
$= 81$

12. If α and β are the roots of the quadratic equation $x^2+kx-15=0$ such that $\alpha-\beta=8$, then what is the positive value of k?

(a) 2 (b) 3
(c) 4 (d) 5

(a) Given quadratic equation,

$x^2+kx-15=0$

$\therefore \alpha+\beta=-\frac{k}{1}=-k$ ... (i)

and $\alpha\beta=-15$ ... (ii)

and also, given $\alpha-\beta=8$ ... (iii)

Since, $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$

$\Rightarrow (8)^2=(-k)^2-4(-15)$

$\Rightarrow 64=k^2+60$

$\Rightarrow k^2=4$

$\Rightarrow k=\pm2$

$\therefore k=2>0$

13. What are the values of p and q respectively, if $(x-1)$ and $(x+2)$ divide the polynomial x^3+4x^2+px+q?

(a) 1, – 6 (b) 2, – 6
(c) 1, 6 (d) 2, 6

(a) Given polynomial is

x^3+4x^2+px+q.

Since, $(x-1)$ divide the given polynomial.

$\therefore (1)^3+4(1)^2+p(1)+q=0$

$\Rightarrow 1+4+p+q=0$

$\Rightarrow p+q=-5$ ... (i)

and $(x+2)$ divide the polynomial

$\therefore (-2)^3+4(-2)^2+p(-2)+q=0$

$\Rightarrow -8+16-2p+q=0$

$\Rightarrow -2p+q=-8$ ... (ii)

On solving Eqs. (i) and (ii), we get

$p=1, q=-6$

14. If $(x+k)$ is the HCF of x^2+5x+6 and $x^2+8x+15$, then what is the value of k?

(a) 5 (b) 3
(c) 2 (d) 1

(b) Since, $(x+k)$ is the HCF of x^2+5x+6 and $x^2+8x+15$.

$\therefore (-k)^2+5(-k)+6=0$

$\Rightarrow \quad k^2-5k+6=0 \quad \dots$ (i)

and $(-k)^2+8(-k)+15=0$

$\Rightarrow \quad k^2-8k+15=0 \quad \dots$ (ii)

From Eqs. (i) and (ii), we get

$3k-9=0 \Rightarrow k=3$

15. If $5^{x+1}-5^{x-1}=600$, then what is the value of 10^{2x}?

(a) 1 (b) 1000
(c) 100000 (d) 1000000

(d) Given, $5^{x+1}-5^{x-1}=600 \quad \dots$ (i)

$\Rightarrow 5^x\left(5-\frac{1}{5}\right)=600$

$\Rightarrow \quad 5^x\left(\frac{24}{5}\right)=600$

$\Rightarrow \quad 5^x=\frac{600\times 5}{24}=125=5^3$

$\Rightarrow \quad x=3$

$\therefore \quad 10^{2x}=10^6=1000000$

16. A number divides 12288, 28200 and 44333 so as to leave the same remainder in each case. What is that number?

(a) 272 (b) 232
(c) 221 (d) 120

(c) Let divisor is x and remainder is r. Then, according to the question,

$12288=x\times q_1+r \quad \dots$ (i)

$28200=x\times q_2+r \quad \dots$ (ii)

and $44333=x\times q_3+r \quad \dots$ (iii)

From Eqs. (i) and (ii), we get

$(q_2-q_1)x=28200-12288=15912$

From Eqs. (ii) and (iii), we get

$(q_3-q_2)x=44333-28200=16133 \quad \dots$ (v)

From Eqs. (i) and (iii), we get

$(q_3-q_1)x=44333-12288=32045 \quad \dots$ (vi)

Since Eqs. (iv), (v) and (vi) are divisible by x.

$\therefore$ HCF of (15912,16133, 32045) = 221

$\therefore \quad x=221$

17. If $x^2+9y^2=6xy$, then what is $y:x$ equal to?

(a) 1 : 3 (b) 1 : 2
(c) 2 : 1 (d) 3 : 1

(a) Given, $x^2+9y^2=6xy$

$\Rightarrow x^2+9y^2-6xy=0$

$\Rightarrow \quad (x-3y)^2=0$

$\Rightarrow \quad x-3y=0$

$\Rightarrow \quad x=3y$

$\Rightarrow \quad y/x=1/3$

$\Rightarrow \quad y:x=1:3$

18. If m and n are positive integers such that $m^n=1331$, then what is the value of $(m-1)^{n-1}$?

(a) 1 (b) 100 (c) 121 (d) 125

(b) Given, $m^n=1331$

$\Rightarrow \quad m^n=11^3$

$\Rightarrow \quad m=11, n=3$

$\therefore \quad (m-1)^{n-1}=(11-1)^{3-1}$

$=10^2=100$

19. What is $\frac{1}{a^{m-n}-1}+\frac{1}{a^{n-m}-1}$ equal to?

(a) 1 (b) -1
(c) 0 (d) $2a^{m-n}$

(b) $\frac{1}{a^{m-n}-1}+\frac{1}{a^{n-m}-1}$

$=\frac{1}{\frac{a^m}{a^n}-1}+\frac{1}{\frac{a^n}{a^m}-1}$

$=\frac{a^n}{a^m-a^n}+\frac{a^m}{a^n-a^m}$

$=\frac{a^n}{a^m-a^n}-\frac{a^m}{a^{m-a^n}}$

$=\frac{a^n-a^m}{a^m-a^n}=-\frac{a^n-a^m}{a^n-a^m}=-1$

20. If $x=\sqrt{2}$, $y=\sqrt[3]{3}$ and $z=\sqrt[6]{6}$, then which one of the following is correct?

(a) $y<x<z$ (b) $z<x<y$
(c) $z<y<x$ (d) $x<y<z$

(b) Given, $x=\sqrt{2}$, $y=3\sqrt{3}$ and $z=6\sqrt{6}$

$\Rightarrow x=2^{1/2}$, $y=3^{1/3}$ and $z=6^{1/6}$

$\Rightarrow x=2^{6/12}$, $y=3^{4/12}$ and $z=6^{2/12}$

$\Rightarrow x=(64)^{1/12}$, $y=(81)^{1/12}$

and $z=(36)^{1/12}$

Clearly, $z<x<y$.

21. If $\log x=1.2500$ and $y=x^{\log x}$, then what is $\log y$ equal to?

(a) 4.2500 (b) 2.5625
(c) 1.5625 (d) 1.2500

(c) Given, $\log x=1.2500$

and $\quad y=x^{\log x} \Rightarrow$

$\log y=(\log x)(\log x)$

$\Rightarrow \log y=(\log x)^2$

$\Rightarrow \log y=(1.25)^2=1.5625$

22. If $f(x)$ is divided by $(x-\alpha)(x-\beta)$, where $\alpha\neq\beta$, then what is the remainder?

(a) $\frac{(x-\alpha)f(\alpha)-(x-\beta)f(\beta)}{\alpha-\beta}$

(b) $\frac{(x-\alpha)f(\beta)-(x-\beta)f(\alpha)}{\alpha-\beta}$

(c) $\frac{(x-\beta)f(\alpha)-(x-\alpha)f(\beta)}{\alpha-\beta}$

(d) $\frac{(x-\beta)f(\beta)-(x-\alpha)f(\alpha)}{\alpha-\beta}$

(c) Given, $f(x)$ is divided by $(x-\alpha)(x-\beta)$.

$\therefore$ Let quotient is $q(x)$ and remainder is $(ax+b)$.

$\therefore f(x)=(x-\alpha)(x-\beta)q(x)+(ax+b) \quad \dots$ (i)

On putting $x=\alpha,\beta$ in Eq. (i), we get

$f(\alpha)=a\alpha+b \quad \dots$ (ii)

and $f(\beta)=a\beta+b \quad \dots$ (iii)

From Eqs. (i) and (ii), we get

$a(\alpha-\beta)=f(\alpha)-f(\beta)$

$\Rightarrow \quad a=\frac{f(\alpha)-f(\beta)}{\alpha-\beta} \quad \dots$ (iv)

From Eqs. (ii) and (iv), we get

$f(\alpha)=\left(\frac{f(\alpha)-f(\beta)}{\alpha-\beta}\right)\alpha+b$

$\Rightarrow b=\frac{\alpha f(\beta)-\beta f(\alpha)}{\alpha-\beta}$

$\therefore$ Remainder $ax+b$

$=\frac{f(\alpha)x-f(\beta)x}{\alpha-\beta}+\frac{\alpha f(\beta)-\beta f(\alpha)}{\alpha-\beta}$

$=\frac{(x-\beta)f(\alpha)-(x-\alpha)f(\beta)}{\alpha-\beta}$

23. If the area of a square is $2401x^4+196x^2+4$, then what is its side length?

(a) $49x^2+3x+2$ (b) $49x^2-3x+2$
(c) $49x^2+2$ (d) $59x^2+2$

(c) Given area of square

$=2401x^4+196x^2+4$

$=(49x^2)^2+2\times 49x^2\times 2+(2)^2$

$=(49x^2+2)^2$

$\therefore$ Length of side $=49x^2+2$

24. If x varies as yz, then y varies inversely as

(a) xz (b) $\frac{x}{z}$
(c) $\frac{z}{x}$ (d) $\frac{1}{(xz)}$

(c) Given, x varies as yz.

$\therefore x\propto yz$

$\Rightarrow \frac{x}{z}\propto y \Rightarrow y\propto\frac{1}{(z/x)}$

25. If the points P and Q represent real numbers $0\cdot7\overline{3}$ and $0\cdot5\overline{6}$ on the number line, then what is the distance between P and Q?

(a) $\frac{1}{6}$ (b) $\frac{1}{5}$ (c) $\frac{16}{45}$ (d) $\frac{11}{90}$

(a) Let $x = 0.7\overline{3} = 0.73333....$

$\Rightarrow 10x = 7.3333$ .... (i)

$\Rightarrow 100x = 73.333....$ .... (ii)

Subtracting Eq. (i) from Eq. (ii),

$90x = 66$

$\Rightarrow x = \frac{66}{90} = \frac{11}{15}$

Similarly, $y = 0.5\overline{6}$

$\Rightarrow y = \frac{56-5}{90}$

$\Rightarrow y = \frac{51}{90} = \frac{17}{30}$

$\therefore$ Distance between P and Q

$= \frac{11}{15} - \frac{17}{30} = \frac{22-17}{30} = \frac{5}{30} = \frac{1}{6}$

26. What is the point on the xy-plane satisfying $5x + 2y = 7xy$ and $10x + 3y = 8xy$?

(a) $\left(-1, \frac{1}{6}\right)$ (b) $\left(\frac{1}{6}, -1\right)$

(c) $\left(1, \frac{1}{6}\right)$ (d) $\left(-\frac{1}{6}, -1\right)$

(b) Given, $5x + 2y = 7xy$

$\Rightarrow \frac{5}{y} + \frac{2}{x} = 7$ ... (i)

and $10x + 3y = 8xy$

$\Rightarrow \frac{10}{y} + \frac{3}{x} = 8$ ... (ii)

From 2 × Eq. (i) – Eq.(ii), we get

$\frac{4}{x} - \frac{3}{x} = 14 - 8$

$\Rightarrow \frac{1}{x} = 6 \Rightarrow x = \frac{1}{6}$

From Eqs. (i), we get

$\frac{5}{y} + 2 \times 6 = 7 \Rightarrow \frac{5}{y} = 7 - 12 = -5$

$\Rightarrow y = -1$

27. The price of an article X increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was ₹ 5000 and price of article Y was ₹ 2000. In which year the difference in their prices exceeded ₹ 5000 for the first time?

(a) 2012 (b) 2013 (c) 2014 (d) 2015

(b) In year 2010,

Price of article X is ₹ 5000.

and price of article Y is ₹ 2000.

Difference of price of X and Y

$= 5000 - 2000 =$ ₹ 3000

In 2013,

price of article

$X = 5000\left(1 + \frac{20}{100}\right)^3$

$= 5000 \times \left(\frac{120}{100}\right)^3$

$= 5000 \times \frac{6}{5} \times \frac{6}{5} \times \frac{6}{5}$

= ₹ 8640

Price of article $Y = 2000\left(1 + \frac{10}{100}\right)^3$

$= 2000\left(\frac{110}{100}\right)^3$

$= 2000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10} =$ ₹ 2662

$\therefore$ Difference of price of X and Y

$= 8640 - 2662 =$ ₹ 5978

28. The ratio of speeds of X and Y is 5 : 6. If Y allows X a start of 70 m in a 1.2 km race, then who will win the race and by what distance?

(a) X wins the race by 30 m
(b) Y wins the race by 90 m
(c) Y wins the race by 130 m
(d) The race finishes in a dead heat

(c) Speed of $X = 5k$

and speed of $Y = 6k$

Distance travelled by

$X = 1.2 \times 1000 - 70$

$= 1200 - 70$

$= 1130$ m

Distance travelled by $Y = 1200$ m

$\therefore$ Time taken by $X = \frac{1130}{5k} = \frac{226}{k}$

Time taken by $Y = \frac{1200}{6k} = \frac{200}{k}$

Since, $\frac{226}{k} > \frac{200}{k}$

$\therefore$ Y wins the race.

$\therefore$ Difference of distance to win the race $= \left(\frac{226}{k} - \frac{200}{k}\right) \times 5k$

$= 26 \times 5 = 130$m

29. A train takes two hours less for a journey of 300 km if its speed is increased by 5 km/h from its usual speed. What is its usual speed?

(a) 50 km/h (b) 40 km/h
(c) 35 km/h (d) 25 km/h

(d) Let usual speed of train $= x$ km/h

According to the question,

$\frac{300}{x} - \frac{300}{x+5} = 2$

$\Rightarrow \frac{300(x+5-x)}{x(x+5)} = 2$

$\Rightarrow 300 \times 5 = 2x^2 + 10x$

$\Rightarrow 2x^2 + 10x - 1500 = 0$

$\Rightarrow x^2 + 5x - 750 = 0$

$\Rightarrow x^2 + 30x - 25x - 750 = 0$

$\Rightarrow x(x+30) - 25(x+30) = 0$

$\Rightarrow (x+30)(x-25) = 0$

$\Rightarrow x + 30 = 0$ or $x - 25 = 0$

$\Rightarrow x = -30$ or $x = 25$

$\Rightarrow x = 25$ km/h $[\because x \neq -25]$

30. If 6 men and 8 women can do a piece of work in 10 days, and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman?

(a) 2 : 1 (b) 1 : 2 (c) 4 : 3 (d) 3 : 4

(a) Let a man's one day work $= x$ unit

and a woman's one day work $= y$ unit

$\therefore$ 6 men and 8 women one day work

$\Rightarrow 8x + 8y = \frac{1}{10}$ ... (i)

and 13 men and 24 women one day work

$\Rightarrow 13x + 24y = \frac{1}{4}$ ... (ii)

On solving Eqs. (i) and (ii), we get

$x = \frac{1}{100}$ and $y = \frac{1}{200}$

$\therefore$ Ratio of x and $y = \frac{1}{100} : \frac{1}{200} = 2 : 1$

31. Students of a class are made to sit in rows of equal number of chairs. If number of students is increased by 2 in each row, then the number of rows decreases by 3. If number of students is increased by 4 in each row, then the number of rows decreases by 5. What is the number of students in the class?

(a) 100 (b) 105 (c) 110 (d) 120

(d) Let the numbers of row $= x$

Number of students in each row $= y$

$\therefore$ Number of students $= xy$

According to the question,

$(x-3)(y+2) = xy$

$\Rightarrow xy + 2x - 3y - 6 = xy$

$\Rightarrow 2x - 3y = 6$ ... (i)

and $(x-5)(y+4) = xy$

$\Rightarrow xy + 4x - 5y - 20 = xy$

$\Rightarrow 4x - 5y = 20$ ... (ii)

On solving Eqs. (i) and (ii), we get

$x = 15$ and $y = 8$

$\therefore$ Number of students in the class

$= xy = 15 \times 8 = 120$

32. A sum was put at simple interest at certain rate for 2 yr. Had it been put at 1% higher rate of interest, it would have fetched ₹ 24 more. What is the sum?

(a) ₹ 500 (b) ₹ 600
(c) ₹ 800 (d) ₹ 1200

(d) Let ₹ x was put at simple interest at rate of r% per annum for 2 yr.

According to the question,

$$SI = \frac{x \times r \times 2}{100} \quad \dots (i)$$

and $$(SI + 24) = \frac{x \times (r+1) \times 2}{100} \quad \dots (ii)$$

From Eqs. (i) and (ii), we get

$$24 = \frac{2x}{100}$$

$$\Rightarrow x = \frac{24 \times 100}{2}$$

$$\Rightarrow x = 1200$$

$\therefore$ Required sum = ₹ 1200

33. The population of two villages is 1525 and 2600 respectively. If the ratio of male to female population in the first village is 27 : 34 and the ratio of male to female population in the second village is 6 : 7, then what is the ratio of male to female population of these two villages taken together?

(a) 33 : 41 (b) 85 : 82
(c) 71 : 90 (d) 5 : 6

(d) Given population of first village = 1525

Let male population in first village is $27x$.

And female population in first village $= 34x$

Then, $27x + 34x = 1525$

$\Rightarrow 61x = 1525 \Rightarrow x = 25$

$\therefore$ Male population in first village

$= 27x$

$= 27 \times 25 = 675$

and Female population in first village

$= 34x$

$= 34 \times 25 = 850$

Population of second village = 2600

Let male population in second village $= 6y$

and female population in second village $= 7y$

Then, $6y + 7y = 2600$

$\Rightarrow 13y = 2600 \Rightarrow y = 200$

$\therefore$ Male population in second village

$= 6y = 1200$

and Female population in second village $= 7y = 1400$

Now, total number of male in both villages $= 675 + 1200 = 1875$

Total number of female in both villages $= 850 + 1400 = 2250$

$\therefore$ Required ratio $= 1875 : 2250 = 5 : 6$

34. In a class room the ratio of number of girls to that of boys is 3 : 4. The average height of students in the class is 4.6 feet. If the average height of the boys in the class is 4.8 feet, then what is the average height of the girls in the class?

(a) Less than 4.2 feet
(b) More than 4.2 feet but less than 4.3 feet
(c) More than 4.3 feet but less than 4.4 feet
(d) More than 4.4 feet but less than 4.5 feet

(c) Let Number of girls $= 3x$

and number of boys $= 4x$

Total number of students

$= 3x + 4x = 7x$

Sum of heights of all students

$= 4.6 \times 7x$

$= 32.2x$

sum of heights of all the boys

$= 4.8 \times 4x$

$= 19.2x$

$\therefore$ Sum of heights of all the girls

$= 32.2x - 19.2x = 13x$

$\therefore$ Average height of the girls in the class $= \frac{13x}{3x} = \frac{13}{3} = 4.33$

Hence, Average height of the girls in the class more than 4.3 feet but less than 4.4 feet.

35. What is the median of the data 3, 5, 9, 4, 6, 11, 18?

(a) 6 (b) 6.5 (c) 7 (d) 7.5

(a) Given data 3, 5, 9, 4, 6, 11, 18.

Now, arranging in ascending order

3, 4, 5, 6, 9, 11, 18

$\therefore$ Median = 6

36. In a pie-diagram there are three sectors. If the ratio of the angles of the sectors is 1 : 2 : 3, then what is the angle of the largest sector?

(a) 200° (b) 180° (c) 150° (d) 120°

(b) Let angles of three sectors are x, $2x$ and $3x$.

In a pie–diagram,

$x + 2x + 3x = 360$

$\Rightarrow 6x = 360° \Rightarrow x = 60°$

$\therefore$ Largest angle

$= 3x = 3 \times 60° = 180°$

37. The maximum marks in a test are converted from 250 to 50 for the purpose of an Internal Assessment. The highest marks scored were 170 and lowest marks were 70. What is the difference between the maximum and minimum marks scored in the Internal Assessment?

(a) 15 (b) 17 (c) 20 (d) 24

(c) Since maximum marks in a test are converted from 250 to 50 for the Internal Assessment.

$$\therefore \quad \frac{50}{250} = \frac{1}{5}$$

$\Rightarrow$ Marks to reduced to $\frac{1}{5}$ th for the purpose of an Internal Assessment.

$\therefore$ Highest marks $= \frac{150}{5} = 34$

and Lowest marks $= \frac{70}{5} = 14$

$\therefore$ Required difference $= 34 - 14 = 20$

Directions (Q. Nos. 38-40) *Read the following information and answer the given questions follow.*

The following data presents count of released convicts who have served prison terms (X), those who have received some educational or technical training during their term (Y) and those who were offered Company placement (Z) respectively, from six different jails A, B, C, D, E and F, in the year 2010.

	X	*Y*	*Z*
A	86	45	25
B	1305	903	461
C	2019	940	474
D	1166	869	416
E	954	544	254
F	1198	464	174

38. Jails with highest and smallest percentage of trained convicts are respectively

(a) F and D (b) D and F
(c) C and A (d) D and A

(b) Total released convicts from point A

$= 86 + 45 + 25 = 156$

$\therefore$ Trained convicts from jail A

$= \frac{45}{156} \times 100 = 28.84\%$

Total released convicts from jail B

$= 1305 + 903 + 461 = 2669$

Trained convicts from jail B

$= \frac{903}{2669} \times 100 = 33.83\%$

Total released convicts from jail C

$= 2019 + 940 + 474 = 3433$

Trained convicts from jail C

$= \frac{940}{3433} \times 100 = 27.38\%$

Total released convicts from jail D

$= 1166 + 869 + 416 = 2451$

Trained convicts from jail D

$= \frac{869}{2451} \times 100 = 35.45\%$

Total released convicts from jail E

$= 954 + 544 + 254 = 1752$

Trained convicts from jail E

$= \frac{544}{1752} \times 100 = 51.05\%$

Total released convicts from jail F

$= 1198 + 464 + 174 = 1836$

Trained convicts from jail F

$= \frac{464}{1836} \times 100 = 25.27\%$

$\therefore$ Jails with highest and smallest percentage of trained convicts are respectively 35.45% and 25.27%.

39. Jail with highest placement rate of trained convicts is

(a) F (b) D (c) B (d) A

(d) Placement rate of trained convicts of jail A $= \frac{25}{45} \times 100 = 55.55$

Placement rate of trained convicts of jail B $= \frac{461}{903} \times 100 = 51.05$

Placement rate of trained convicts of jail C $= \frac{474}{940} \times 100 = 50.43$

Placement rate of trained convicts of jail D $= \frac{416}{869} \times 100 = 47.87$

Placement rate of trained convicts of jail E $= \frac{254}{544} \times 100 = 46.69$

Placement rate of trained convicts of jail F $= \frac{174}{464} \times 100 = 37.5$

$\therefore$ Jail with highest placement rate of trained convicts = 55.55

40. Jails from which more than half of the trained convicts are offered jobs, are

(a) A, B and C (b) A, B and D
(c) A, D and E (d) A, E and F

(a) Trained convicts from jail A = 45

$\therefore$ Half of the trained convicts

$= \frac{45}{2} = 22.5$

Convicts who offered company placement from jail A = 25

Trained convicts from jail B = 903

$\therefore$ Half of the trained convicts

$= \frac{903}{2} = 451.5$

Convicts who offered company placement from jail B = 461

Trained convicts from jail C = 940

$\therefore$ Half of the trained convicts

$= \frac{940}{2} = 470$

Convicts who offered company placement from jail C = 474

Trained convicts from jail D = 869

$\therefore$ Half of the trained convicts = 434.5

Convicts who offered company placement from jail D = 416

Trained convicts of jail E = 544

$\therefore$ Half of the trained convicts = 272

Convicts who offered company placement from jail E = 254

Trained convicts from jail F = 464

$\therefore$ Half of the trained convicts = 232

Convicts who offered company placement from jail F = 174

Hence, jails from which more than half of the trained convicts are offered jobs are A, B and C.

41. The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order of the digits, is

(a) Six (b) Five (c) Four (d) Three

(c) Let the three digits number $= xyz$

and the reverse of it will be $= zyx$

Since, both the numbers are divisible by 7.

$\therefore \quad 100x + 10y + z = 7m \quad \dots \text{(i)}$

and $\quad 100z + 10y + x = 7n \quad \dots \text{(ii)}$

From Eqs. (i) and (ii), we get

$\Rightarrow \quad 99(x - z) = 7(m - n)$

Here, 99 is not divisible by 7 then it is must that $(x - z)$ is a factor of 7.

$\therefore$ Possible that $x = 2,\ z = 9$;

$x = 1,\ z = 8$

$\therefore$ Possible numbers are 168, 861, 259 and 952.

42. How many integral values of x and y satisfy the equation $5x + 9y = 7$, where $-500 < x < 500$ and $-500 < y < 500$?

(a) 110
(b) 111
(c) 112
(d) None of the above

(b) Given equation $5x + 9y = 7$,

where $-500 < x,\ y < 500$.

Now, $y = \frac{7 - 5x}{9}$

x	y
−1	12/9
−4	3
−13	8
−22	13
−31	18

Here, when

$x = -4, -13, -22, -31, \dots$

i.e. common difference with 9, then value of y = integer value

$\therefore \quad 495 = -495 + (n - 1)(9)$

$\Rightarrow n - 1 = \frac{495 + 495}{9}$

$= \frac{990}{9} = 110 \Rightarrow n = 111$

43. Let XYZ be a 3-digit number. Let $S = XYZ + YZX + ZXY$. Which of the following statements is/are correct?

1. S is always divisible by 3 and $(X + Y + Z)$.
2. S is always divisible by 9.
3. S is always divisible by 37.

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) 1 and 2 (d) 1 and 3

(d) Given, $S = XYZ + YZX + ZXY$

$\Rightarrow S = (100X + 10Y + Z) + (100Y + 10Z + X) + (100Z + 10X + Y)$

$\Rightarrow S = 111(X + Y + Z)$

$= 3 \times 37\ (X + Y + Z)$

$\Rightarrow S$ is always divisible by 3, 37 and $(X + Y + Z)$.

44. In covering certain distance, the average speeds of X and Y are in the ratio 4 : 5. If X takes 45 min more than Y to reach the destination, then what is the time taken by Y to reach the destination?

(a) 135 min (b) 150 min
(c) 180 min (d) 225 min

(c) When the speed is same, then speed is inversely proportional to time.

$\therefore t_Y : t_X = 4 : 5$

$\therefore$ 1 unit difference = 45 min

$\therefore$ 4 unit $= 4 \times 45 = 180$ min

45. For two observations, the sum is S and product is P. What is the harmonic mean of these two observations?

(a) $\frac{2S}{P}$ (b) $\frac{S}{(2P)}$
(c) $\frac{2P}{S}$ (d) $\frac{P}{(2S)}$

(c) Given that, $S = x + y$

$P = xy$

$\therefore$ Harmonic Mean $= \frac{2xy}{x+y} = \frac{2P}{S}$

46. If the annual income of X is 20% more than that of Y, then the income of Y is less than that of X by p%. What is the value of p?

(a) 10 (b) $16\frac{2}{3}$
(c) $17\frac{1}{3}$ (d) 20

(b) Let annual income of Y = ₹ 100

Then annual income of X = ₹ 120

$\therefore \quad p\% = \frac{120-100}{120} \times 100$

$= \frac{20}{120} \times 100$

$= \frac{20 \times 5}{6} = 16\frac{2}{3}$

47. What is the least perfect square which is divisible by 3, 4, 5, 6 and 7?

(a) 1764 (b) 17640
(c) 44100 (d) 176400

(c) A number which is divisible by 3, 4, 5, 6 and 7 should be a multiple of 3, 4, 5, 6 and 7.

i.e. $2^2 \times 3 \times 5 \times 7$

For least perfect square, then it should be a perfect square i.e.

$2^2 \times 3^2 \times 5^2 \times 7^2 = 44100$

48. In a water tank there are two outlets. It takes 20 min to empty the tank if both the outlets are opened. If the first outlet is opened, the tank is emptied in 30 min. What is the time taken to empty the tank by second outlet?

(a) 30 min (b) 40 min
(c) 50 min (d) 60 min

(d) Let x is the time taken when both outlets are open and y is the time taken by 1st outlet then time taken by IInd outlet

$= \frac{xy}{y-x} = \frac{20 \times 30}{30-20} = \frac{600}{10}$

$= 60$ min

49. If $(x^2 - 1)$ is a factor of $ax^4 + bx^3 + cx^2 + dx + e$, then which one of the following is correct?

(a) $a + b + c = d + e$
(b) $a + b + e = c + d$
(c) $b + c + d = a + e$
(d) $a + c + e = b + d$

(d) Given, $(x^2 - 1)$ is a factor of $ax^4 + bx^3 + cx^2 + dx + e$

$\Rightarrow (x-1)(x+1)$ is a factor of $ax^4 + bx^3 + cx^2 + dx + e$

$\Rightarrow x = -1, 1$ roots of $ax^4 + bx^3 + cx^2 + dx + e$

$\therefore \quad a(-1)^4 + b(-1)^3 + c(-1)^2 + d(-1) + e = 0$

$\Rightarrow \quad a - b + c - d + e = 0$

$\Rightarrow \quad a + c + e = b + d$

50. If $\left(x^8 + \frac{1}{x^8}\right) = 47$, what is the value of $\left(x^6 + \frac{1}{x^6}\right)$?

(a) 36 (b) 27 (c) 18 (d) 9

(c) Given, $x^8 + \frac{1}{x^8} = 47$

$\Rightarrow x^8 + \frac{1}{x^8} + 2 = 47 + 2$

$\Rightarrow \left(x^4 + \frac{1}{x^4}\right)^2 = 49 \Rightarrow x^4 + \frac{1}{x^4} = 7$

$\Rightarrow \quad x^4 + \frac{1}{x^4} + 2 = 7 + 2 = 9$

$\Rightarrow \quad \left(x^2 + \frac{1}{x^2}\right) = 3$

$\therefore \quad x^6 + \frac{1}{x^6} = (x^2)^3 + \left(\frac{1}{x^2}\right)^3$

$= \left(x^2 + \frac{1}{x^2}\right)^3 - 3(x^2)\left(\frac{1}{x^2}\right)\left(x^2 + \frac{1}{x^2}\right)$

$= (3)^3 - 3(3) = 27 - 9 = 18$

51. A wheel makes 360 revolutions in one minute. What is the number of radians it turns in one second?

(a) 4π (b) 6π (c) 12π (d) 16π

(c) 60 sec = 360 revolutions

$\therefore$ 1 sec = 6 revolutions

$= 6 \times 2\pi$ radians $= 12\pi$ radians

52. What is the least value of $(25\operatorname{cosec}^2 x + \sec^2 x)$?

(a) 40 (b) 36 (c) 26 (d) 24

(b) $25\operatorname{cosec}^2 x + \sec^2 x$

$\therefore$ Minimum value of $25\operatorname{cosec}^2 x + \sec^2 x$

$= (\sqrt{25} + \sqrt{1})^2 = (5+1)^2 = 36$

53. Let $0 < \theta < 90^\circ$ and $100\theta = 90^\circ$. If $\alpha = \sum_{n=1}^{99} \cot n\theta$, then which one of the following is correct?

(a) $\alpha = 1$ (b) $\alpha = 0$
(c) $\alpha > 1$ (d) $0 < \alpha < 1$

(a) Given that,

$100\theta = 90^\circ$ and $\alpha = \sum_{n-1}^{99} \cot n\theta$

$= \cot\theta . \cos 2\theta . \cot 3\theta \ldots \cot 99\theta$

$= \cot\theta . \cos 2\theta . \cot 3\theta \ldots . \tan(90^\circ - 99\theta)$

$= \cot\theta . \cot 2\theta . \cot 3\theta \ldots . \tan(100\theta - 99\theta)$

$= \cot\theta . \cot 2\theta . \cot 3\theta \ldots . \tan\theta$

$= \cot\theta . \tan\theta . \cot 2\theta . \tan 2\theta \ldots .$

$\Rightarrow \alpha = 1$

54. If $\tan 6\theta = \cot 2\theta$, where $0 < 6\theta < \frac{\pi}{2}$, then what is the value of $\sec 4\theta$?

(a) $\sqrt{2}$ (b) 2 (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{4}{3}$

(a) Given, $\tan 6\theta = \cot 2\theta$

$\Rightarrow \quad \tan 6\theta = \tan(90^\circ - 2\theta)$

$\Rightarrow \quad 6\theta = 90^\circ - 2\theta$

$\Rightarrow 6\theta + 2\theta = 90^\circ \Rightarrow 8\theta = 90^\circ$

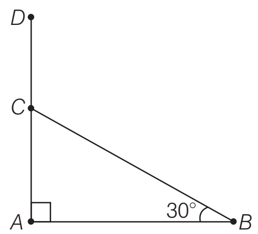
$\Rightarrow \quad 4\theta = 45^\circ$

$\therefore \quad \sec 4\theta = \sec 45^\circ = \sqrt{2}$

55. A tree of height 15 m is broken by wind in such a way that its top touches the ground and makes an angle 30° with the ground. What is the height from the ground to the point where tree is broken?

(a) 10 m (b) 7 m
(c) 5 m (d) 3 m

(c)



Let AD be the tree such that $AD = 15$ m

And let $AC = x$

Then, $CD = BC = 15 - x$

$\therefore \quad \sin 30^\circ = \frac{AC}{BC}$

$= \frac{x}{15 - x}$

$\Rightarrow \quad \frac{1}{2} = \frac{x}{15 - x}$

$\Rightarrow \quad 2x = 15 - x$

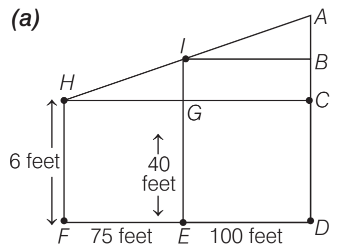
$\Rightarrow \quad 3x = 15$

$\therefore \quad x = 5$ m

56. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75 feet, then what is the height of the taller tower (approximately)?

(a) 85 feet (b) 110 feet
(c) 125 feet (d) 140 feet

(a)



Here,

$\Delta ABI \sim \Delta IGH$

$\therefore \dfrac{IG}{AB} = \dfrac{HG}{IB} \Rightarrow \dfrac{40-6}{AB} = \dfrac{75}{100}$

$\Rightarrow AB = \dfrac{34 \times 100}{75} = 45.33$

$\therefore AD = AB + BD = AB + IE$

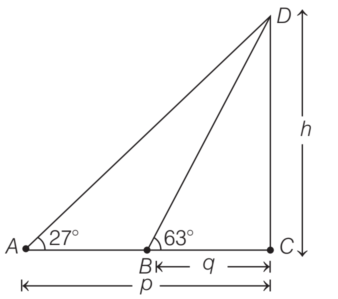
$= 45.33 + 40 = 85.33$ feet

$= 85$ feet (approx.)

57. The angles of elevation of the top of a tower from two points at distances p and q from the base and on the same straight line are 27° and 63° respectively. What is the height of the tower?

(a) pq (b) $\sqrt{pq}$
(c) $\dfrac{pq}{2}$ (d) $\dfrac{\sqrt{pq}}{2}$

(b) Let height of the tower CD be h.



$\therefore$ In ΔACD,

$\tan 27° = \dfrac{CD}{AC}$

$\Rightarrow \tan 27° = \dfrac{h}{p}$ ... (i)

and In ΔBCD, $\tan 63° = \dfrac{CD}{BC}$

$\Rightarrow \tan(90° - 27°) = \dfrac{h}{q}$

$\Rightarrow \cot 27° = \dfrac{h}{q}$ ... (ii)

From Eqs. (i) and (ii), we get

$\tan 27° \cot 27° = \dfrac{h}{p} \cdot \dfrac{h}{q}$

$\Rightarrow h^2 = pq$

$\Rightarrow h = \sqrt{pq}$

58. What is the value of $\sin^2 6° + \sin^2 12° + \sin^2 18° + \ldots + \sin^2 84° + \sin^2 90°$?

(a) 1 (b) 2 (c) 4 (d) 8

(d) $\sin^2 6 + \sin^2 12° + \sin^2 18° + \ldots$

$+ \sin^2 84° + \sin^2 90°$

$= (\sin^2 6° + \sin^2 84°)$

$+ (\sin^2 12° + \sin^2 78°) + (\sin^2 18° + \sin^2 72°)$

$+ \ldots + (\sin^2 42 + \sin^2 48°) + 1$

$= (\sin^2 6° + \cos^2 6°)$

$+ (\sin^2 12° + \cos^2 12°)$

$+ (\sin^2 18° + \cos^2 18°) + \ldots + (\sin^2 42° + \cos^2 42°) + 1$

$= (1 + 1 + 1 + \ldots 7 \text{ times}) + 1$

$= 7 + 1 = 8$

59. What is $\dfrac{\cos\theta}{1+\sin\theta} + \dfrac{1}{\cot\theta}$ equal to?

(a) $\operatorname{cosec}\theta$ (b) $\sec\theta$
(c) $\sec\theta + \operatorname{cosec}\theta$ (d) $\operatorname{cosec}\theta - \cot\theta$

(b) $\dfrac{\cos\theta}{1+\sin\theta} + \dfrac{1}{\cot\theta}$

$= \dfrac{\cos\theta}{1+\sin\theta} + \dfrac{1}{\cos\theta/\sin\theta}$

$= \dfrac{\cos\theta}{1+\sin\theta} + \dfrac{\sin\theta}{\cos\theta}$

$= \dfrac{\cos^2\theta + \sin\theta + \sin^2\theta}{\cos\theta(1+\sin\theta)}$

$= \dfrac{(\cos^2\theta + \sin^2\theta) + \sin\theta}{\cos\theta(1+\sin\theta)}$

$= \dfrac{1+\sin\theta}{\cos\theta(1+\sin\theta)} = \sec\theta$

60. What is $\dfrac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos\theta - 1} - \dfrac{\sin\theta + 1}{\cos\theta}$ equal to?

(a) 0 (b) 1
(c) $2\sin\theta$ (d) $2\cos\theta$

(a) $\dfrac{\sin\theta - \cos\theta + 1}{\sin\theta + \cos - 1} - \dfrac{\sin\theta + 1}{\cos\theta}$

$= \dfrac{\sin\theta/\cos\theta - \cos\theta/\cos\theta + 1/\cos\theta}{\sin\theta/\cos\theta + \cos\theta/\cos\theta - 1/\cos\theta} - \dfrac{\sin\theta/\cos\theta + 1/\cos\theta}{\cos\theta/\cos\theta}$

$= \dfrac{\tan\theta - 1 + \sec\theta}{\tan\theta + 1 - \sec\theta} - \dfrac{\tan\theta + \sec\theta}{1}$

$= \dfrac{\tan\theta + \sec\theta - 1}{(\tan\theta - \sec\theta) + 1} - (\tan\theta + \sec\theta)$

$= \dfrac{(\tan\theta + \sec\theta - 1)(\tan\theta + \sec\theta)}{(\tan\theta - \sec\theta + 1)(\tan\theta + \sec\theta)} - \tan\theta - \sec\theta$

$= \dfrac{(\tan\theta + \sec\theta - 1)(\tan\theta + \sec\theta)}{(\tan^2\theta - \sec^2\theta) + (\tan\theta + \sec\theta)} - \tan\theta - \sec\theta$

$= \dfrac{(\tan\theta + \sec\theta - 1)(\tan\theta + \sec\theta)}{-1 + (\tan\theta + \sec\theta)} - \tan\theta - \sec\theta$

$= \tan\theta + \sec\theta - \tan\theta - \sec\theta$

$= 0$

61. What is $(\tan x + \tan y)(1 - \cot x \cot y) + (\cot x + \cot y)(1 - \tan x \tan y)$ equal to?

(a) 0 (b) 1 (c) 2 (d) 4

(a) $(\tan x + \tan y)(1 - \cot x \cot y)$

$+ (\cot x + \cot y)(1 - \tan x \tan y)$

$= \tan x + \tan y - \cot y - \cot x$

$+ \cot x + \cot y - \tan y - \tan x = 0$

62. What is $\sqrt{\dfrac{\sec x - \tan x}{\sec x + \tan x}}$ equal to?

(a) $\dfrac{1}{\sin x + \cos x}$

(b) $\dfrac{1}{\tan x + \cot x}$

(c) $\dfrac{1}{\sec x + \tan x}$

(d) $\dfrac{1}{\operatorname{cosec} x + \cot x}$

(c) $\sqrt{\dfrac{\sec x - \tan x}{\sec x + \tan x}}$

$= \sqrt{\dfrac{(\sec x - \tan x)(\sec x + \tan x)}{(\sec x + \tan x)(\sec x + \tan x)}}$

$= \sqrt{\dfrac{\sec^2 x - \tan^2 x}{(\sec x + \tan x)^2}}$

$= \dfrac{\sqrt{\sec^2 x - \tan^2 x}}{\sqrt{(\sec x + \tan x)^2}}$

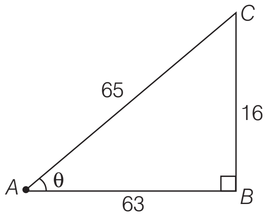
$= \dfrac{1}{\sec x + \tan x}$

63. If θ lies in the first quadrant and $\cot\theta = \frac{63}{16}$, then what is the value of $(\sin\theta + \cos\theta)$?

(a) 1 (b) $\frac{69}{65}$ (c) $\frac{79}{65}$ (d) 2

(c) Given, $\cot\theta = \frac{63}{16}$

$\Rightarrow \quad \tan\theta = \frac{16}{63}$



In right angled triangle ABC,

$AC^2 = 63^2 + 16^2 = 3969 + 256$

$\Rightarrow \quad AC^2 = 4225$

$\Rightarrow \quad AC = 65$

$\therefore \sin\theta + \cos\theta = \frac{16}{65} + \frac{63}{65} = \frac{79}{65}$

64. What is the value of $\frac{1 - 2\sin^2\theta\cos^2\theta}{\sin^4\theta + \cos^4\theta} + 4$ equal to?

(a) 0 (b) 1 (c) 2 (d) 5

(d) $\frac{1 - 2\sin^2\theta\cos^2\theta}{\sin^4\theta + \cos^4\theta} + 4$

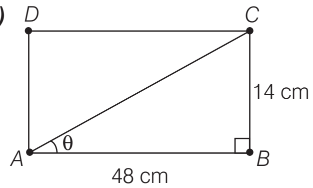
$= \frac{1 - 2\sin^2\theta\cos^2\theta}{(\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta} + 4$

$= \frac{1 - 2\sin^2\theta\cos^2\theta}{1 - 2\sin^2\theta\cos^2\theta} + 4 = 1 + 4 = 5$

65. A rectangle is 48 cm long and 14 cm wide. If the diagonal makes an angle θ with the longer side, then what is $(\sec\theta + \text{cosec}\,\theta)$ equal to?

(a) $\frac{775}{168}$ (b) $\frac{725}{168}$ (c) $\frac{375}{84}$ (d) $\frac{325}{84}$

(a)



Diagonal, $AC = \sqrt{48^2 + 14^2}$

$= \sqrt{2304 + 196}$

$= \sqrt{2500} = 50$

$\therefore \quad \sec\theta = \frac{AC}{AB} = \frac{50}{48} = \frac{25}{24}$

and $\text{cosec}\,\theta = \frac{AC}{BC} = \frac{50}{14} = \frac{25}{7}$

$\therefore \quad \sec\theta + \text{cosec}\,\theta = \frac{25}{24} + \frac{25}{7}$

$= \frac{175 + 600}{168}$

$= \frac{775}{168}$

66. What is the area of the triangle having side lengths $\frac{y}{z} + \frac{z}{x}, \frac{z}{x} + \frac{x}{y}, \frac{x}{y} + \frac{y}{z}$?

(a) $\frac{(x + y + z)^2}{xyz}$ (b) $\frac{\sqrt{xyz}}{x + y + z}$

(c) $\sqrt{\frac{x}{y} + \frac{y}{z} + \frac{z}{x}}$ (d) $\sqrt{\frac{xy + yz + zx}{xyz}}$

(c) Let $a = \frac{y}{z} + \frac{z}{x}$

$b = \frac{z}{x} + \frac{x}{y}$ and $c = \frac{x}{y} + \frac{y}{z}$

$\therefore \quad s = \frac{a + b + c}{2} = \frac{2\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x}\right)}{2}$

$= \frac{x}{y} + \frac{y}{z} + \frac{z}{x}$

$\therefore$ Area of triangle

$= \sqrt{s(s - a)(s - b)(s - c)}$

$= \sqrt{\left[\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x}\right)\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x} - \frac{y}{z} - \frac{z}{x}\right)\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x} - \frac{z}{x} + \frac{x}{y}\right)\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x} - \frac{x}{y} - \frac{y}{z}\right)\right]}$

$= \sqrt{\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x}\right)\left(\frac{x}{y}\right)\left(\frac{y}{z}\right)\left(\frac{z}{x}\right)}$

$= \sqrt{\frac{x}{y} + \frac{y}{z} + \frac{z}{x}}$

67. If the angles of a triangle are 30° and 45° and the included side is $(\sqrt{3} + 1)$ cm, then what is the area of the triangle?

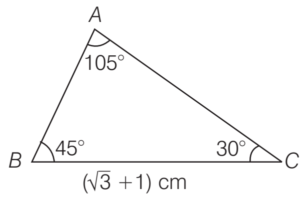
(a) $(\sqrt{3} + 1)\,\text{cm}^2$ (b) $(\sqrt{3} + 3)\,\text{cm}^2$

(c) $\frac{1}{2}(\sqrt{3} + 1)\,\text{cm}^2$ (d) $2(\sqrt{3} + 1)\,\text{cm}^2$

(c) Third angle of the triangle

$= 180° - (30° + 45°)$

$= 180° - 75° = 105°$



Now, $\frac{\sqrt{3} + 1}{\sin 105°} = \frac{AB}{\sin 30°}$

$\Rightarrow \quad \frac{\sqrt{3} + 1}{\left(\frac{\sqrt{3} + 1}{2\sqrt{2}}\right)} = \frac{x}{1/2}$

$\Rightarrow \quad 2\sqrt{2} = 2x$

$\Rightarrow \quad x = \sqrt{2}$

$\therefore$ Required area of triangle

$= \frac{1}{2}(AB \times BC) \times \sin 45°$

$= \frac{1}{2} \times \sqrt{2} \times (\sqrt{3} + 1) \times \frac{1}{\sqrt{2}}$

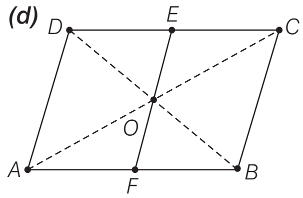
$= \frac{1}{2}(\sqrt{3} + 1)\,\text{cm}^2$

68. $ABCD$ is a plate in the shape of a parallelogram. EF is the line parallel to DA and passing through the point of intersection O of the diagonals AC and BD. Further, E lies on DC and F lies on AB. The triangular portion DOE is cut out from the plate $ABCD$. What is the ratio of area of remaining portion of the plate to the whole?

(a) $\frac{5}{8}$ (b) $\frac{5}{7}$

(c) $\frac{3}{4}$ (d) $\frac{7}{8}$

(d)



We know that, the diagonal of a parallelogram cut the parallelogram in the four equal parts.

$\therefore \text{ar}(AOB) = \text{ar}(BOC) = \text{ar}(DOC)$

$= \text{ar}(AOD)$

And $\text{ar}(DOE) = \text{ar}(EOC)$

$= \frac{1}{2}\text{ar}(COD)$

$= \frac{1}{2} \times \frac{1}{4}\text{ar}(ABCD)$

$= \frac{1}{8}\text{ar}(ABCD)$

$\therefore$ ar (Remaining portion of the plate after cutting portion DOE)

$= \frac{7}{8}\text{ar }(ABCD)$

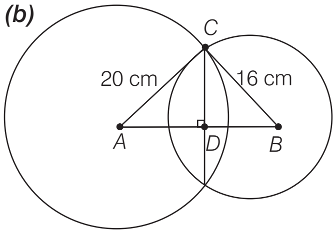
$\therefore$ The ratio of area of remaining portion of the plate to the whole

$= \frac{7/8 \text{ ar }(ABCD)}{\text{ar }(ABCD)} = 7/8$

69. Two circles of radii 20 cm and 16 cm intersect and the length of common chord is 24 cm. If d is the distance between their centres, then which one of the following is correct?

(a) $d < 26$ cm
(b) 26 cm $< d <$ 27 cm
(c) 27 cm $< d <$ 28 cm
(d) $d > 28$ cm

(b)



Here, $AB = d$, $CD = 12$ cm

In ΔADC, $\angle ADC = 90°$

$\therefore \quad AC^2 = AD^2 + CD^2$

$\Rightarrow \quad 20^2 = AD^2 + 12^2$

$\Rightarrow \quad AD^2 = 400 - 144 = 256$

$\Rightarrow \quad AD = 16$

And in ΔBCD, $\angle BDC = 90°$

$BC^2 = BD^2 + CD^2$

$\Rightarrow \quad 16^2 = BD^2 + 12^2$

$\Rightarrow \quad BD^2 = 256 - 144 = 112$

$\Rightarrow \quad BD = \sqrt{112} = 10.58$

$\therefore \quad AB = AD + DB$

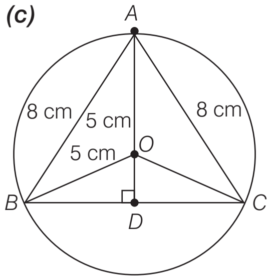
$\Rightarrow \quad d = 16 + 10.58 = 26.58$ cm

$\therefore 26 < d < 27$ cm

70. In a circle of radius 5 cm, AB and AC are two chords such that $AB = AC = 8$ cm. What is the length of chord BC?

(a) 9 cm (b) 9.2 cm
(c) 9.6 cm (d) 9.8 cm

(c)



Let $OD = x$ cm

and $BD = y$ cm

In right angle triangle BOD,

$BO^2 = OD^2 + BD^2$

$\Rightarrow \quad 5^2 = x^2 + y^2 \quad \ldots$ (i)

And in right angled triangle ABD,

$AB^2 = AD^2 + BD^2$

$\Rightarrow 8^2 = (5 + x)^2 + y^2$

$\Rightarrow 64 = (5 + x)^2 + (5^2 - x^2)$ [from Eq. (i)]

$\Rightarrow \quad 64 = 25 + x^2 + 10x + 25 - x^2$

$\Rightarrow \quad 64 = 50 + 10x$

$\Rightarrow 10x = 14 \Rightarrow x = 1.4$

Now, from Eq. (i), $25 = (1.4)^2 + y^2$

$\Rightarrow \quad 25 = 1.96 + y^2$

$\Rightarrow \quad y = \sqrt{25 - 1.96}$

$= 4.8$

$\therefore$ Length of chord $BC = 2y$

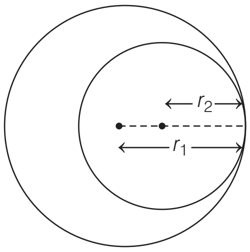
$= 2 \times 4.8$

$= 9.6$ cm

71. Two circles touch internally. The sum of their areas is 136π cm^2 and distance between their centres is 4 cm. What are the radii of the circles?

(a) 11 cm, 7 cm
(b) 10 cm, 6 cm
(c) 9 cm, 5 cm
(d) 8 cm, 4 cm

(b)



Let radius of larger circle be r_1 and radius of smaller circle be r_2.

$\therefore \quad r_1 - r_2 = 4$

$\Rightarrow \quad r_1 = 4 + r_2 \quad \ldots$ (i)

and $\quad \pi r_1^2 + \pi r_2^2 = 136\pi$

$\Rightarrow \quad \pi(4 + r_2)^2 + \pi r_2^2 = 136\pi$

$\Rightarrow \quad 16 + r_2^2 + 8r_2 + r_2^2 = 136$

$\Rightarrow \quad 2r_2^2 + 8r_2 = 136 - 16 = 120$

$\Rightarrow \quad r_2^2 + 4r_2 - 60 = 0$

$\Rightarrow \quad r_2^2 + 10r_2 - 6r_2 - 60 = 0$

$\Rightarrow r_2(r_2 + 10) - 6(r_2 + 10) = 0$

$\Rightarrow \quad (r_2 + 10)(r_2 - 6) = 0$

$\Rightarrow \quad r_2 = 6$

$[\because r_2 \neq -10]$

$\therefore r_1 = 4 + 6 = 10$ cm

72. If area of a circle and a square are same, then what is the ratio of their perimeters?

(a) $2\sqrt{\pi} : 1$ (b) $\sqrt{\pi} : 1$
(c) $\sqrt{\pi} : 2$ (d) $\sqrt{\pi} : 4$

(c) Let radius of circle be r and side of square be x.

Then, according to the question,

$\pi r^2 = x^2$

$\Rightarrow \quad \sqrt{\pi} r = x \Rightarrow \dfrac{r}{x} = \dfrac{1}{\sqrt{\pi}}$

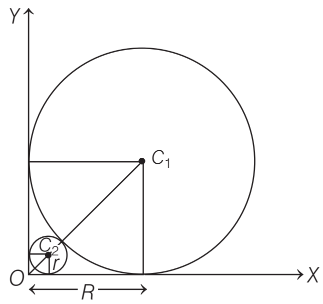
$\therefore$ Ratio of their perimeters $= \dfrac{2\pi r}{4x}$

$= \dfrac{\pi}{2} \cdot \dfrac{1}{\sqrt{\pi}} = \dfrac{\sqrt{\pi}}{2} = \sqrt{\pi} : 2$

73. A circle of diameter 8 cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle?

(a) $4(3 - \sqrt{2})$ cm (b) $4(3 - 2\sqrt{2})$ cm
(c) $8(3 - \sqrt{2})$ cm (d) $8(3 - 2\sqrt{2})$ cm

(d)



Let the distance between the small circle and the origin O is x and radius of small circle and larger circle are r and R respectively.

$\therefore OC_2 = \sqrt{r^2 + r^2} = r + x$

$\Rightarrow \quad r + x = \sqrt{2}\, r$

$\Rightarrow \quad x = r(\sqrt{2} - 1) \quad \ldots$ (i)

And also,

$OC_2 = \sqrt{R^2 + R^2}$

$= (x + 2r + R)$

$\Rightarrow \quad x + 2r + R = \sqrt{2}R$

$\Rightarrow \quad r(\sqrt{2} - 1) + 2r + 4 = \sqrt{2}(4)$

$[\because 2R = 8$ cm$]$

$\Rightarrow r(\sqrt{2} - 1 + 2) = 4(\sqrt{2} - 1)$

$\Rightarrow \quad r(\sqrt{2} + 1) = 4(\sqrt{2} - 1)$

$\Rightarrow \quad r = \dfrac{4(\sqrt{2} - 1)}{\sqrt{2} + 1} \times \dfrac{\sqrt{2} - 1}{\sqrt{2} - 1}$

$\Rightarrow \quad r = \dfrac{4(\sqrt{2} - 1)^2}{2 - 1}$

$\Rightarrow \quad r = 4(2 + 1 - 2\sqrt{2})$

$\Rightarrow \quad r = 4(3 - 2\sqrt{2})$

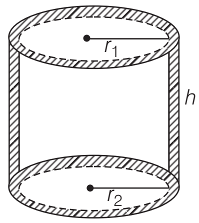
$\therefore$ Diameter of the smaller circle $= 2r$

$= 8(3 - 2\sqrt{2})$ cm

74. The thickness of a cylinder is 1 foot, the inner radius of the cylinder is 3 feet and height is 7 feet. To paint the inner surface it requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surfaces of the cylinder?

(a) $\frac{7}{3}$ L (b) $\frac{3}{2}$ L (c) $\frac{8}{3}$ L (d) $\frac{10}{3}$ L

(c) Given, thickness of a cylinder is 1 feet and inner radius of the cylinder $(r_2) = 3$ feet



$\therefore$ Outer radius of the cylinder

$(r_1) = (3 + 1) = 4$ feet

Now, Inner surface area of cylinder

$= 2\pi r_2 h$
$= 2\pi(3)(7)$ feet^2
$= 42\pi$ feet^2

$\therefore \quad 42\pi$ feet$^2 \equiv 1$ L

$\Rightarrow$ 1 feet$^2 \equiv \frac{1}{42\pi}$ L

Total surface area of the cylinder

$= 2\pi r_1 h + 2\pi r_2 h + 2(\pi r_1^2 - \pi r_2^2)$
$= 2\pi h(r_1 + r_2) + 2\pi(r_1 - r_2)(r_1 + r_2)$
$= 2\pi \times 7(4 + 3) + 2\pi(4 - 3)(4 + 3)$
$= 2\pi(49) + 14\pi$
$= 112\pi$ feet^2

$\therefore$ Required quantity of same colour to paint all the surface of cylinder

$= 112\pi \times \frac{1}{42\pi}$ L $= \frac{8}{3}$ L

75. A square and a rectangle have equal areas. If one side of the rectangle is of length numerically equal to the square of the length of the side of the square, then the other side of the rectangle is

(a) square root of the side of the square
(b) half the side of the square
(c) of unit length
(d) double the side of the square

(c) Let side of square be x, then one side of rectangle be x^2 and let other side of rectangle be y.

Then, according to the question,

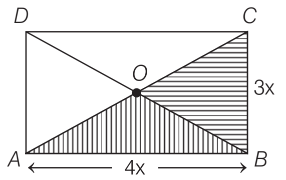
$x^2 = x^2 \times y$ (area of square = area of rectangle)

$\Rightarrow y = 1$

76. The length and breadth of a rectangle are in the ratio 4 : 3. Then, what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side?

(a) 3 : 4 (b) 4 : 3 (c) 16 : 9 (d) 1 : 1

(d) Let length of rectangle be $4x$ and breadth of rectangle be $3x$.



$\therefore$ Diagonal of rectangle,

$AC = \sqrt{(4x)^2 + (3x)^2} = \sqrt{25x^2} = 5x$

Here, side of triangle OAB formed by the parts of the diagonal with long side are $2.5x$, $2.5x$ and $4x$.

And side of triangle OBC formed by the parts of the diagonal with shord side are $2.5x$, $2.5x$ and $3x$.

$$\therefore \frac{\text{ar}(\Delta OAB)}{\text{ar}(\Delta OBC)} = \frac{\frac{1}{2} \times 4x \times \frac{3x}{2}}{\frac{1}{2} \times 3x \times \frac{4x}{2}} = \frac{4 \times 3}{3 \times 4}$$

$\Rightarrow \text{ar}(\Delta AOB) : \text{ar}(\Delta BOC) = 1 : 1$

77. Suppose a region is formed by removing a sector of 20° from a circular region of radius 30 feet. What is the area of this new region?

(a) 150π sq feet (b) 550π sq feet
(c) 650π sq feet (d) 850π sq feet

(d) Area of new region

$= \pi r^2 - \left(\frac{\theta}{360°}\right)\pi r^2 = \pi r^2\left(1 - \frac{\theta}{360°}\right)$

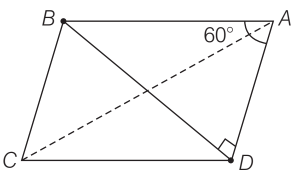
$= \pi(30)^2\left(1 - \frac{20°}{360°}\right) = \pi(900)\left(\frac{17}{18}\right)$

$= 850\pi$ sq feet

78. $ABCD$ is a parallelogram where AC and BD are the diagonals. If $\angle BAD = 60°$, $\angle ADB = 90°$, then what is BD^2 equal to?

(a) $\frac{3}{5}AB^2$ (b) $\frac{3}{4}AB^2$ (c) $\frac{1}{2}AB^2$ (d) $\frac{2}{3}AB^2$

(b) Given, $ABCD$ is a parallelogram where, AC and BD are the diagonals and $\angle BAD = 60°$, $\angle ADB = 90°$



In right angle triangle ADB

$\sin 60° = \frac{BD}{AB}$

$\Rightarrow \quad \frac{\sqrt{3}}{2} = \frac{BD}{AB}$

$\Rightarrow \quad BD^2 = \frac{3}{4}AB^2$

79. A line through the vertex A of a parallelogram $ABCD$ meets DC in P and BC produced in Q. If P is the mid-point of DC, then which of the following is/are correct?

1. Area of ΔPDA is equal to that of ΔPCQ.
2. Area of ΔQAB is equal to twice that of ΔPCQ.

Select the correct answer using the code given below:

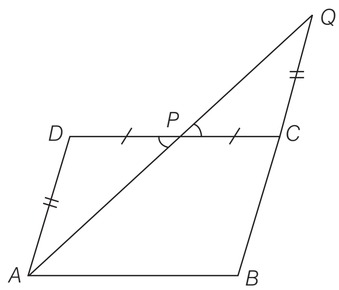
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(a) Since, $BC || AD$

$\therefore \angle PAD = \angle CQP$

$CP = PD$ [P is the middle point of CD]

and $\angle CPQ = \angle APD$



$\therefore \quad \Delta PDA \cong \Delta PCQ$ [by AAS]

$\therefore$ ar(ΔPDA) = ar(ΔPCQ)

Now, $\angle BQA = \angle CQP$

Since, the ΔPCQ is a part of the ΔBQA.

So, there area cannot be equal.

80. How many cubic metre of earth is to be dug out to dig a well of radius 1.4 m and depth 5 m?

(a) 30.2 m^3 (b) 30.4 m^3
(c) 30.6 m^3 (d) 30.8 m^3

(d) Volume of well with radius 1.4 m and depth 5 m.

$= \pi \times (1.4)^2 \times 5 = \frac{22}{7} \times 1.4 \times 1.4 \times 5$

$= 30.8$ cubic metre

81. If the diagonals of a rhombus are x and y, then what is its area?

(a) $\frac{xy}{2}$ (b) $\frac{xy}{4}$
(c) xy (d) $x^2 - y^2$

(a) Area of rhombus

$= \frac{1}{2} \times$ diagonal I $\times$ diagonal II

$= \frac{1}{2} xy$

82. The lengths of sides of a triangle are $3x$, $4\sqrt{y}$, $5\sqrt[3]{z}$, where $3x < 4\sqrt{y} < 5\sqrt[3]{z}$. If one of the angles is 90°, then what are the minimum integral values of x, y and z respectively?

(a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 1, 1 (d) 3, 4, 5

(c) Lengths of sides of a triangle are $3x$, $4\sqrt{y}$, $5\sqrt[3]{z}$, where $3x < 4\sqrt{y} < 5\sqrt[3]{z}$ and one angle is 90°.

$\therefore (5\sqrt[3]{z})^2 = (3x)^2 + (4\sqrt{y})^2$

$\Rightarrow 25z^{2/3} = 9x^2 + 16y$

$\Rightarrow$ Value of $x, y, z = 1$ which satisfied the equation.

83. What is the maximum number of circumcircles that a triangle can have?

(a) 1 (b) 2 (c) 3 (d) Infinite

(a) A triangle has maximum one circumcircles since it touches all the vertex of the triangle.

84. If an arc of a circle of radius 6 cm subtends a central angle measuring 30°, then which one of the following is an approximate length of the arc?

(a) 3.14 cm (b) 2.15 cm
(c) 2.14 cm (d) 2 cm

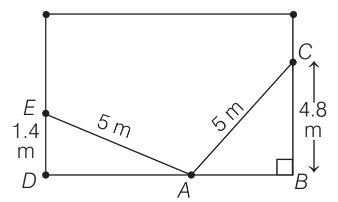
(a) Length of arc $= \frac{\theta}{360°} \times 2\pi r$

$= \frac{30}{360°} \times 2 \times \pi \times 6 = \pi = 3.14$ cm

85. A ladder 5 m long is placed in a room so as to reach a point 4.8 m high on a wall and on turning the ladder over to the opposite side of the wall without moving the base, it reaches a point 1.4 m high. What is the breadth of the room?

(a) 5.8 m (b) 6 m (c) 6.2 m (d) 7.5 m

(c) Here, AC and AE are portion of ladder.



In right angled ΔABC,

$AB = \sqrt{5^2 - (4.8)^2} \Rightarrow AB = 1.4$ m

and in right angled triangle ADE

$AD = \sqrt{5^2 - (1.4)^2} = 4.8$

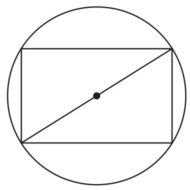
$\therefore$ The breadth of the room

$BD = AB + AD = 1.4 + 4.8 = 6.2$ m

86. What is the area of the largest square plate cut from a circular disk of radius one unit?

(a) 4 sq units (b) $2\sqrt{2}$ sq units
(c) π sq units (d) 2 sq units

(d) For largest square plate cut from a circular disk



Diagonal of square

= Diameter of circle

$\Rightarrow \sqrt{2} \times$ side of square

$= 2 \times 1 = 2$ unit

$\Rightarrow$ side of square $= \frac{2}{\sqrt{2}} = \sqrt{2}$ unit.

$\therefore$ Area of the largest square plate

$= (\text{side})^2 = (\sqrt{2})^2 = 2$ sq. units

87. Out of 4 identical balls of radius r, 3 balls are placed on a plane such that each ball touches the other two balls. The 4th ball is placed on them such that this ball touches all the three balls. What is the distance of centre of 4th ball from the plane?

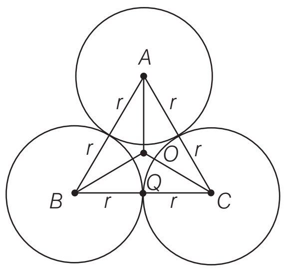
(a) $2\sqrt{\frac{2}{3}}\, r$ unit

(b) $\frac{\sqrt{3} + 2\sqrt{2}}{\sqrt{2}}\, r$ unit

(c) $\frac{r}{3 - 2\sqrt{2}}$ unit

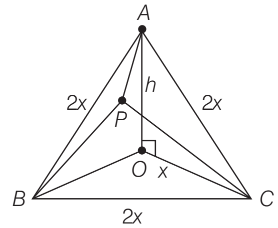
(d) $\frac{\sqrt{3} + 2\sqrt{2}}{\sqrt{3}}\, r$ unit

(a) Let Q be the mid-point of BC and since ΔABC is an equilateral triangle.



$\therefore \cos 30° = \frac{CQ}{OC} = \frac{\sqrt{3}}{2} = \frac{r}{x\,(\text{let})}$

$\Rightarrow x = \frac{2r}{\sqrt{3}}$ ... (i)



Let P be the position of 4th ball and OP is the distance of centre of 4th ball from the ground.

In ΔPOC, $\angle POC = 90°$

$CP^2 = OP^2 + OC^2$

$\Rightarrow OP^2 = CP^2 - OC^2 = (2r)^2 - x^2$

$= 4r^2 - \left(\frac{2r}{\sqrt{3}}\right)^2$ [By using Eq. (i)]

$= 4r^2 - \frac{4r^2}{3} = \frac{8r^2}{3}$

$\Rightarrow OP = \frac{2\sqrt{2}r}{\sqrt{3}} = 2\sqrt{\frac{2}{3}}\, r$ unit

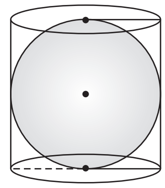
88. A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct?

(a) $p = q$
(b) $p = 2q$
(c) $2p = q$
(d) $2p = 3q$

(a) Let radius of sphere be r.

$\therefore$ Surface area of sphere,

$p = 4\pi r^2$ ... (i)



Now, height of cylinder, $h = 2r$

$\therefore$ Surface area of the cylinder

$q = 2\pi rh = 2\pi r(2r)$

$= 4\pi r^2$ ... (ii)

From Eqs. (i) and (ii), we get $p = q$

89. $ABCD$ is a quadrilateral such that $AD = DC = CA = 20$ units, $BC = 12$ units and $\angle ABC = 90°$. What is the approximate area of the quadrilateral $ABCD$?

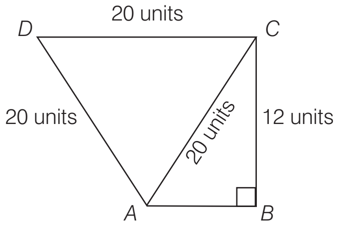
(a) 269 sq units (b) 300 sq units
(c) 325 sq units (d) 349 sq units

(a) In right angled triangle $ABCD$,

$AB^2 = AC^2 - BC^2$

$= (20)^2 - (12)^2 = 400 - 144$

$\Rightarrow AB = \sqrt{256} = 16$ units



$\therefore$ Area of quadrilateral $ABCD$

$= \text{ar}(\Delta ACD) + \text{ar}(\Delta ABC)$

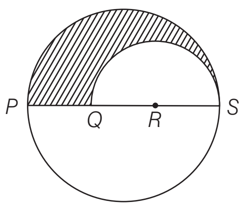
$= \frac{\sqrt{3}}{4}(\text{side})^2 + \frac{1}{2} \times \text{Base} \times \text{height}$

$= \frac{\sqrt{3}}{4} \times (20)^2 + \frac{1}{2} \times 16 \times 12$

$= 100\sqrt{3} + 96$

$= 269$ sq units (approx.)

90. Let $PQRS$ be the diameter of a circle of radius 9 cm. The length PQ, QR and RS are equal. Semi-circle is drawn with QS as diameter (as shown in the given figure). What is the ratio of the shaded region to that of the unshaded region?



(a) 25 : 121
(b) 5 : 13
(c) 5 : 18
(d) 1 : 2

(b) We have, $PS = 2 \times 9 = 18$ cm

Since, $PQ = QR = RS$

$\therefore PQ = QR = RS = \frac{PS}{3} = \frac{18}{3} = 6$ cm

$\therefore$ Area of the shaded region

$= \frac{1}{2}(\pi \times 9^2 - \pi \times 6^2)$

$= \frac{\pi}{2}(81 - 36) = \frac{45}{2}\pi \text{cm}^2$

And area of unshaded region

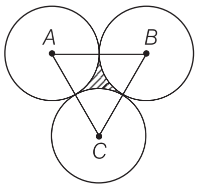
$= \frac{1}{2}(\pi \times 9^2 + \pi \times 6^2)$

$= \frac{\pi}{2}(81 + 36) = \frac{\pi}{2} \times 117$

$\therefore$ Ratio of the shaded region to that of the unshaded region

$= \frac{\frac{45}{2}\pi}{\frac{117}{2}\pi} = \frac{45}{117} = 5 : 13$

91. What is the area of the shaded region in the given figure, if the radius of each of the circles is 2 cm?



(a) $4\sqrt{3} - 2\pi \text{ cm}^2$
(b) $\sqrt{3} - \pi \text{ cm}^2$
(c) $\sqrt{3} - \frac{\pi}{2} \text{ cm}^2$
(d) $2\pi - 2\sqrt{3} \text{ cm}^2$

(a) Side of the equilateral triangle $ABC = 2 \times 2 = 4$ cm

$\therefore$ Area of shaded region

= Area of equilateral triangle ABC

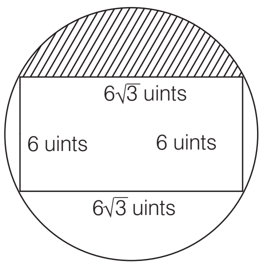
$- 3 \times$ Area of sector with angle $60°$ and radius 2 cm

$= \frac{\sqrt{3}}{4} \times (4)^2 - 3 \times \left(\frac{60°}{360°}\right)\pi(2)^2$

$= \sqrt{3} \times 4 - 3 \times \frac{1}{6} \times \pi(4)$

$= 4\sqrt{3} - 2\pi \text{ cm}^2$

92. In the given figure, what is the area of the shaded region?



(a) $9(\pi - \sqrt{3})$ sq units
(b) $3(4\pi - 3\sqrt{3})$ sq units
(c) $3(3\pi - 4\sqrt{3})$ sq units
(d) $9(\sqrt{3} - \pi)$ sq units

(b) Diameter of circle

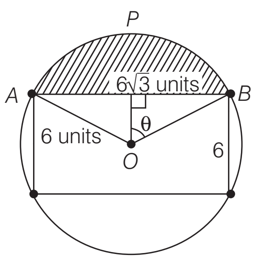
= diagonal of rectangular part

$= \sqrt{(6\sqrt{3})^2 + (6)^2}$

$= \sqrt{108 + 36} = \sqrt{144} = 12$

$\therefore$ Radius of circle $= \frac{12}{2} = 6$ cm

In ΔOAB,



$\sin\theta = \frac{BC}{OB} = \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2}$

$\Rightarrow \sin\theta = \sin 60° \Rightarrow \theta = 60°$

$\therefore \angle AOB = 2\theta = 120°$

$\therefore$ Area of shaded region $APBA$

= Area of sector $APBOA$ – area of ΔABO

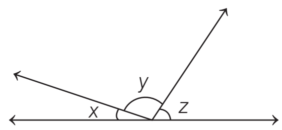
$= \left(\frac{120°}{360°}\right)\pi(6)^2 - \frac{1}{2} \times 3 \times 6\sqrt{3}$

$[\because OC = 3 \text{ units}]$

$= \frac{1}{3} \times \pi \times 36 - 9\sqrt{3} = 12\pi - 9\sqrt{3}$

$= 3(4\pi - 3\sqrt{3})$ sq units

93. In the given figure, if $\frac{y}{x} = 6$ and $\frac{z}{x} = 5$, then what is the value of x ?



(a) 45°
(b) 30°
(c) 15°
(d) 10°

(c) Given, $\frac{y}{x} = 6$ and $\frac{z}{x} = 5$

$\Rightarrow y = 6x$ and $z = 5x$

Now, $x + y + z = 180°$ [straight angle]

$\Rightarrow x + 6x + 5x = 180°$

$\Rightarrow 12x = 180 \Rightarrow x = 15°$

94. $ABCD$ is a trapezium, where AB is parallel to DC. If $AB = 4$ cm, $BC = 3$ cm, $CD = 7$ cm and $DA = 2$ cm, then what is the area of the trapezium?

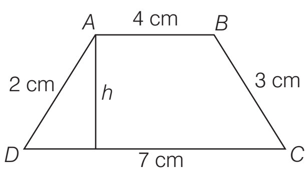
(a) $22\sqrt{\frac{2}{3}} \text{ cm}^2$
(b) $22\sqrt{\frac{3}{2}} \text{ cm}^2$
(c) $22\sqrt{3} \text{ cm}^2$
(d) $\frac{22\sqrt{2}}{3} \text{ cm}^2$

(d) Area of trapezium, when the lengths of parallel and non-parallel sides are given

$= \frac{a + b}{k}\sqrt{s(s-k)(s-c)(s-d)}$

Where $k = b - a = (7 - 4) = 3$ cm

and $s = \frac{k + c + d}{2} = \frac{3 + 2 + 3}{2} = 4$ cm



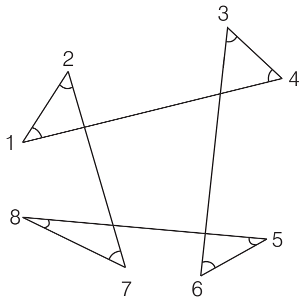
$\therefore$ Required area of trapezium

$= \frac{(4 + 7)}{3}\sqrt{4(4-3)(4-2)(4-3)}$

$= \frac{11}{3}\sqrt{4(1)(2)(1)} = \frac{11 \times 2\sqrt{2}}{3}$

$= \frac{22\sqrt{2}}{3} \text{ cm}^2$

95. Angles are shown in the given figure. What is the value of $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8$?



(a) 240° (b) 360°
(c) 540° (d) 720°

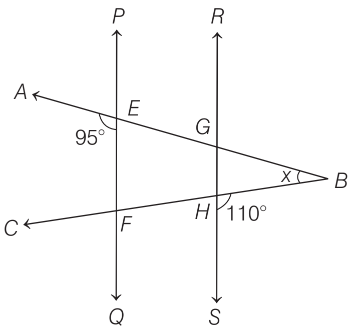
(b) Sum of angles of a quadrilateral $= 360^\circ$

$\Rightarrow (180^\circ - (\angle 1 + \angle 2)) + (180^\circ - (\angle 3 + \angle 4)) + (180^\circ - (\angle 5 + \angle 6)) + (180^\circ - \angle 7 + \angle 8)) = 360^\circ$

$\Rightarrow 720^\circ - (\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8) = 360^\circ$

$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 = 360^\circ$

96. In the given figure PQ is parallel to RS, $\angle AEF = 95^\circ$, $\angle BHS = 110^\circ$, and $\angle ABC = x^\circ$. Then, what is the value of x?



(a) 15 (b) 25
(c) 30 (d) 35

(b) $\angle BEF = 180^\circ - 95^\circ = 85^\circ$

and $\angle FHG = 110^\circ$ [vertically opposite angle]

$\therefore$ Since $PQ || RS$

$\therefore \quad \angle EFH + \angle GHF = 180^\circ$

$\Rightarrow \quad \angle EFH + 110^\circ = 180^\circ$

$\Rightarrow \quad \angle EFH = 180 - 110^\circ = 70^\circ$

In ΔBEF, $\angle B + \angle F + \angle E = 180^\circ$

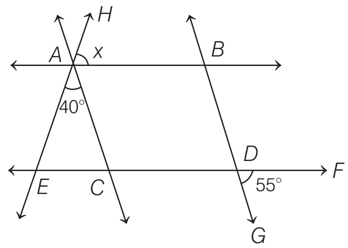
$\Rightarrow \quad x + 70^\circ + 85^\circ = 180^\circ$

$\Rightarrow \quad x + 155^\circ = 180^\circ$

$\Rightarrow \quad x = 180^\circ - 155^\circ$

$\Rightarrow \quad x = 25^\circ$

97. In the given figure AB is parallel to CD and AC is parallel to BD. If $\angle EAC = 40^\circ$, $\angle FDG = 55^\circ$, $\angle HAB = x^\circ$, then what is the value of x?



(a) 85 (b) 80
(c) 75 (d) 65

(a) Given, $AB || CD$ and $AC || BD$.

$\therefore \angle BAC = \angle BDC = 55^\circ$

Now,

$\angle HAB + \angle BAC + \angle CAE = 180^\circ$

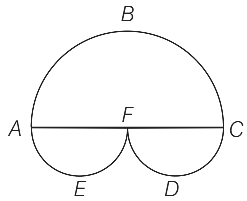
$\Rightarrow x + 55^\circ + 40^\circ = 180^\circ$

$\Rightarrow \quad x + 95^\circ = 180^\circ$

$\Rightarrow \quad x = 180^\circ - 95^\circ$

$\Rightarrow \quad x = 85^\circ$

98. In the given figure, there are three semi-circles ABC, AEF and CDF. The distance between A and C is 28 units and F is the mid-point of AC. What is the total area of the three semi-circles?



(a) 924 sq units (b) 824 sq units
(c) 624 sq units (d) 462 sq units

(d) Given, $AC = 28$ units

$\therefore AF = \dfrac{28}{2} = 14$ units

$\therefore$ Area of three semi-circles ABC, AEF and CDF

$= \dfrac{\pi(14)^2}{2} + \dfrac{\pi(7)^2}{2} + \dfrac{\pi(7)^2}{2}$

$= \dfrac{\pi}{2}(14^2 + 7^2 + 7^2)$

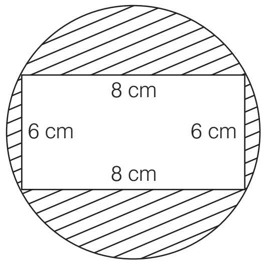
$= \dfrac{\pi}{2}(196 + 49 + 49)$

$= \dfrac{\pi}{2}(294)$

$= \dfrac{22}{7} \times \dfrac{1}{2} \times 294$

$= 462$ sq units

99. What is the approximate area of the shaded region in the figure given?



(a) 15.3 cm^2 (b) 25.5 cm^2
(c) 28.4 cm^2 (d) 30.5 cm^2

(d) Diameter of circle = diagonal of rectangular region $= \sqrt{8^2 + 6^2}$

$= \sqrt{64 + 36} = \sqrt{100} = 10$ cm

$\therefore$ Radius of circle $= \dfrac{10}{2} = 5$ cm

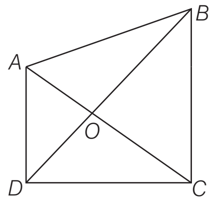
$\therefore$ Area of shaded region

= Area of circle – area of rectangular region

$= \pi(5)^2 - (8 \times 6) = 3.14 \times 25 - 48$

$= 78.5 - 48 = 30.5 \text{ cm}^2$

100. Consider the following statements with reference to the given figure.



1. The sum of the areas of ΔAOD and ΔBOC is equal to the sum of the areas of ΔAOB and ΔDOC.
2. $\angle AOD = \angle BOC$
3. $AB + BC + CD + DA > AC + BD$

Which of the above statements are correct?

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

(b) $\angle AOD = \angle BOC$ [vertically opposite angles]

In ΔADC, $AD + DC > AC$ … (i)

In ΔABC, $AB + BC > AC$ … (ii)

In ΔBCD, $BC + CD > BD$ … (iii)

and ΔBCD, $AB + AD > BD$ … (iv)

On adding Eqs. (i), (ii), (iii) and (iv), we get

$2(AB + BC + CD + AD) > 2(AC + BD)$

$\Rightarrow AB + BC + CD + AD > AC + BD$

It is not necessary that

$\text{ar}(\Delta AOD) + \text{ar}(\Delta BOC) = \text{ar}(\Delta AOB) + \text{ar}(\Delta DOC)$

PAPER II English

Directions (Q. Nos. 1-10) *In this section you have a few short passages. After each passage, you will find some items based on the passage. First, read a passage and answer the items based on it. You are required to select your answers based on the contents of the passage and opinion of the author only.*

PASSAGE-I

Not all agricultural societies become civilisations, but no civilisation can become one without passing through the stage of agriculture. This is because at some stage in the development of agriculture, as productivity improves, not all people would need to be engaged in producing or procuring food.

A significant number of people could be freed up to pursue other activities such as building walls or monuments for new cities; making new tools, weapons and jewellery; organising long-distance trade; creating new artistic masterpieces; coming up with new inventions; keeping accounts; and perhaps constructing new public infrastructure such as irrigation canals that further improve the productivity of agriculture, thus realising even more people to do new things.

This can happen, of course, only if a society that has transitioned to high-productivity agriculture has also, at some stage in its evolution, found a way to channel the bonanza of free time into other work fruitfully.

In the ancient world, this often involved creating new ideologies and new hierarchies or power structures to coerce or otherwise convince large groups of people to devote their time to the new tasks for very little reward.

1. Which one of the following statements is true according to the author?

(a) Agriculture has always been part of all civilisations.
(b) Not all civilisations have undergone the processes of agriculture.
(c) Agriculture gave birth to new civilisations.
(d) Communities discontinued agriculture to become civilisations.

(a) According to the given passage, all civilisations have developed after going through a phase of agriculture. Hence, agriculture had been as part of all civilisations.

2. A significant number of people were sent to carry out other work from agriculture because

(a) there were insufficient agricultural products.
(b) people were needed to build monuments, weapons, jewellery, etc.
(c) there were sufficient agricultural products.
(d) this enabled the development of civilisations.

(c) The given passage states that as the agriculture production was bulk, people had to time to invest their time in other activities.

3. What kind of agriculture based societies would emerge as civilisations?

(a) Societies which achieved high productivity in agriculture had the opportunity to find time for other work.
(b) Societies which depended on agriculture completely moved to other fruitful work so as to move to many places.
(c) Societies which transitioned from one stage of agriculture to another.
(d) Societies which could not do agriculture for lack of resources moved to other work.

(a) According to the given passage, those agricultural societies where agricultural production was high and in bulk, those societies emerged as civilisations.

4. People as groups were convinced to do new work through

(a) reward, force and community persuasions.
(b) ideologies, hierarchies and power structures.
(c) excessive agricultural products.
(d) very high rewards.

(b) The last line of the passage states that it was 'ideologies, hierarchies and power structures' that convinced or forced people to devote their time to activities other than agriculture.

5. Which word in the passage means 'changeover'?

(a) Transitioned
(b) Channel
(c) Coerce
(d) Hierarchies

(a) The synonym of changeover is transitioned. Both word represent a shift to a new way.

PASSAGE-II

When we pick up a newspaper, a book or an article, we come to our task with certain preconceptions and predispositions. We expect to find a specific piece of information or be presented with an argument or an analysis of something, say, the likelihood of recession in the next six months or the reasons why children can't read. We probably know a little about the book or article we are reading even before we start. There was, after all, some reason why we chose to read one piece of writing rather than another.

Our expectations and predispositions may, however, blind us to what the article and its author is actually saying. If, for example, we are used to disagreeing with the author, we may see only what we expect to see and not what is actually there. Day after day in our routine pattern of life we expose ourselves to the same newspaper, the same magazine, even books by authors with the same perspectives. In order to reflect on our reading habits and improve our skills we need to break out of this routine, step back and look at what we are doing when we read.

6. According to the author, which one of the following statements is not true?

(a) Reader's preconceptions influence their reading.
(b) Readers have expectations when they read an article or a book.
(c) Readers look for specific information in any of their readings.
(d) Readers assume that everything they read will have new information.

(d) As per the given passage, readers have certain presumptions and ideas that govern what and how they would read any literary work. In fact, it also points that readers look for some specific pieces of information in any literature. Hence, option (d) is not given in the passage.

7. Our expectations and predispositions may, however, blind us because

(a) we may not get the actual ideas of the author.
(b) we will get the actual ideas of the author.
(c) we may disagree with the author.
(d) we will agree with all the ideas of the author.

(a) The passage points out that whenever we have preconceived ideas or viewpoints, it blinds us or does not let us understand what the author truly wants to say. We often assume the meaning to be something else than what the author actually wanted to say.

8. One of the ways to improve our reading habits is to

(a) break the routine by changing the time of reading.
(b) change the types of topics we read.
(c) break the routine of reading the same newspaper.
(d) stop reading for some time and then restart reading.

(c) According to the given passage, the best way to improve our reading habit is to break the routine of reading the same literary work again and again.

9. Which quality does the author here advocate, to be a good reader?

(a) Being objective to the ideas of the author.
(b) Having preconceptions and predispositions.
(c) Having continuous routines.
(d) Disagreeing with the author.

(a) The passage clearly states that we must be open and objective to the ideas of the author to become an effective reader.

10. Which word in the passage means 'viewpoints'?

(a) Preconceptions
(b) Predispositions
(c) Pattern
(d) Perspectives

(d) The synonym of 'viewpoints' is 'perspectives', as both words means a persons opinion or point of view.

Directions (Q. Nos. 11-20) *Each item in this section has a sentence with three underlined parts labelled as (a), (b) and (c). Read each sentence to find out whether there is any error in any underlined part and indicate your response on the Answer Sheet against the corresponding letter i.e., (a) or (b) or (c). If you find no error, your response should be indicated as (d).*

11. After mysteriously expanding for decades (a) / Antarctica's sea ice cover (b) / starting melting. (c) / No error (d)

(c) Replace 'starting' with 'started' to make the given sentence grammatically and contextually meaningful.

12. The auction, conducted by the bank (a) / will be price based (b) / using multiple priced method. (c) / No error (d)

(c) Replace 'method' with 'methods' to make the given sentence grammatically and contextually meaningful.

13. If the scheme would have been implemented effectively (a) / all affected (b) / would have benefitted. (c) / No error (d)

(a) Replace 'would have been' with 'had been' to make the given sentence grammatically and contextually meaningful.

14. Government Stock offers (a) / safety, liquidity and attractive returned (b) / for long duration. (c) / No error (d)

(b) Replace 'returned' with 'returns' to make the given sentence grammatically and contextually meaningful.

15. Scrolling thorough my social media timeline, (a) / I hovered over a video (b) / of a minor road traffic accident. (c) / No error (d)

(a) Replace 'thorough' with 'through' to make the given sentence grammatically and contextually meaningful. Thorough means in detail.

16. The fascination with gold at least (a) / seems to be a case were traditional belief and (b) / modern finance would point the same way. (c) / No error (d)

(c) Replace 'were' with 'where' to make the given sentence grammatically and contextually meaningful.

17. Evolutionary biology leave us (a) / distinctly pessimistic about the possibility (b) / that altruism can arise naturally among humans. (c) / No error (d)

(a) Replace 'leave' with 'leaves' to make the given sentence grammatically and contextually meaningful.

18. When everything starts working for you (a) / you will find (b) / things are achieve and delivered. (c) / No error (d)

(c) Replace 'achieve' with 'achieved' to make the given sentence grammatically and contextually meaningful.

19. If I were you (a) / I would not go for (b) / change of job. (c) / No error (d)

(c) Add 'a' before change to make the given sentence grammatically and contextually meaningful.

20. At the beginning of the nineteenth century, (a) / female literacy was extremely lowed (b) / in comparison to male literacy. (c) / No error (d)

(b) Replace 'lowed' with 'low' to make the given sentence grammatically and contextually meaningful.

Directions (Q. Nos. 21-30) *Each of the following items in this section consists of a sentence, parts of which have been jumbled. These parts have been labelled as P, Q, R and S. Given below each sentence are four sequences, namely (a), (b), (c) and (d). You are required to re-arrange the jumbled parts of the sentence and mark your response accordingly.*

21. for long and (P) / the backbone of India (Q) / will continue to be the same (R) / agriculture has been (S)

(a) SPQR
(b) SQPR
(c) QRSP
(d) QSRP

(b) The correct and meaningful order of the parts of sentence is SQPR.

22. the cry of general public (P) / agenda in any country (Q) / public policy making (R) / is generally driven by (S)

(a) RQSP (b) RPSQ
(c) PSRQ (d) QRSP

(a) The correct and meaningful order of the parts of sentence is RQSP.

23. before it starts (P) / of the government is (Q) The essential power (R) / the power to manage conflict (S)

(a) RSPQ (b) SQRP
(c) RQSP (d) QRSP

(c) The correct and meaningful order of the parts of sentence is RQSP.

24. a majority of the vote (P) / the party that received (Q) / of the government (R) / must take control (S)

(a) QPSR (b) PSRQ
(c) RSPQ (d) SQPR

(a) The correct and meaningful order of the parts of sentence is QPSR.

25. can express a view on (P) / in which the electorate (Q) / a particular issue of public policy (R) / a referendum is a vote (S)

(a) SQPR (b) RPQS
(c) QRSP (d) PQRS

(a) The correct and meaningful order of the parts of sentence is SQPR.

26. in modern societies (P) / or merely suppressed (Q) / has class conflict (R) / been resolved (S)

(a) RPSQ (b) RSPQ
(c) PRSQ (d) QRSP

(a) The correct and meaningful order of the parts of sentence is RPSQ.

27. several of our food (P) / are being extensively cultivated (Q) / and vegetable crops (R) / hybrid varieties of (S).

(a) QRSP (b) SPQR
(c) QPRS (d) SPRQ

(d) The correct and meaningful order of the parts of sentence is SPRQ.

28. against the officer (P) / reason for the accusation (Q) / there should have been / (R) who was in-charge at that time (S)

(a) RPSQ (b) RQPS
(c) PQRS (d) SPRQ

(b) The correct and meaningful order of the parts of sentence is RQPS.

29. poetry is (P) / and ideas (Q) / powerful feelings (R) / the spontaneous overflow of (S)

(a) SRQP
(b) PQRS
(c) RSQP
(d) PSRQ

(d) The correct and meaningful order of the parts of sentence is PSRQ.

30. historical identity and a common descent (P) / a group of people (Q) / is called an ethnic group (R) / who share a common culture (S)

(a) QSPR
(b) QRPS
(c) PSQR
(d) RQPS

(a) The correct and meaningful order of the parts of sentence is QSPR.

Directions (Q. Nos. 31-40) *Given below are some idioms/phrases followed by four alternative meaning to each. Choose the response (a), (b), (c) or (d) which is the most appropriate expression and mark your response in the Answer Sheet accordingly.*

31. A paper tiger

(a) Person or organisation that appears powerful, but actually is not.
(b) Person or organisation that acts like a tiger.
(c) People who campaign for the protection of tigers.
(d) A daredevil.

(a) The idiom 'a paper tiger' means 'a person or organisation that appears to be powerful but actually is not'.

32. Lily-livered

(a) Brave and courageous
(b) Not brave
(c) Comical
(d) Outrageous

(b) The idiom 'lily livered' means 'coward or not brave.

33. Eat like a bird

(a) Eat fast
(b) Eat very little
(c) Eat a lot
(d) Pretending to be eating

(b) The idiom 'Eat like a bird' means 'eat very little'.

34. The dog days

(a) Days celebrating dogs
(b) The bitter days
(c) The hottest days
(d) The coldest days

(b) The idiom 'the dog days' means 'The bitter days'.

35. A banana republic

(a) A small or poor country with a weak government.
(b) A small or poor country which produces banana.
(c) A country which has been occupied a big country.
(d) A country without any government.

(a) The idiom 'a banana republic' means 'a small or poor country with a weak government'.

36. The pros and cons

(a) The good and bad parts of a situation.
(b) Like and dislike of a situation.
(c) A bad experience in an event.
(d) A good moment of an event.

(a) The idiom 'The pros and cons' means 'the good and bad parts of a situation'.

37. Prime the pump

(a) To do something in order to make something succeed.
(b) To do good things to succeed in life.
(c) To do something in order to get bad things done.
(d) Asking people to do things to make something succeed.

(a) The idiom 'prime the pump' means 'to do something in order to make something succeed'.

38. The green-eyed monster

(a) Feeling of being joyous
(b) Feeling of being jealous
(c) Feeling bad about happenings
(d) Feeling lucky about something

(b) The idiom 'the green-eyed monster' means 'feeling of jealousy'.

39. Rise to the occasion

(a) To celebrate a success in a difficult situation.
(b) To regret a situation which ended in failure.
(c) To succeed in dealing with a difficult situation.
(d) To motivate people to succeed in a difficult situation.

(c) The idiom 'rise to the occasion' means 'to succeed in dealing with a difficult situation'.

40. Call it a day
(a) End of the day
(b) Completion of work
(c) Stop doing something
(d) A beautiful day

(b) The idiom 'call it a day' means 'the completion of work'.

Directions (Q. Nos. 41-50) *In this section each item consists of six sentences of a passage. The first and the sixth sentences are given in the beginning are S1 and S6. The middle four sentences in each have been jumbled up and labelled as P, Q, R and S. You are required to find out the proper sequence of the four sentences and mark your response accordingly on Answer Sheet.*

41. S1 : Chinua Achebe was born in 1930 and educated at the Government College in Umuahia, Nigeria.
S6 : Chinua Achebe has written over twenty books, including novels, stories, essays and collections of poetry, and won the Nobel Prize for literature.
P : During the Civil War in Nigeria, he worked for the Biafran government service.
Q : After the War, he was appointed Senior Research Fellow at the University of Nigeria, Nsukka.
R : He joined the Nigerian Broadcasting Company in Lagos in 1954, later becoming its Director of External Broadcasting.
S : He received a BA from London University in 1953 and in 1956 he studied broadcasting in London at the BBC.

The correct sequence should be
(a) SRPQ (b) RPQS
(c) PQRS (d) QRSP

(a) The correct and meaningful paragraph is given by SRPQ.

42. S1 : "Every person carries in his head a mental model of the world-a subjective representation of external reality," writes Alvin Toffler in Future Shock.
S6 : When we begin to think we can do so only because our mind is already filled with all sorts of ideas with which to think.
P : It organises our knowledge and gives us a place from which to argue.
Q : This mental model is, he says, like a giant filing cabinet.
R : It contains a slot for every item of information coming to us.
S : As E.F. Schumacher says, "When we think, we do not just think; we think ideas."

The correct sequence should be
(a) PSRQ (b) SPRQ
(c) QRPS (d) RQPS

(c) The correct and meaningful paragraph is given by QRPS.

43. S1 : Biology is the study of life in its entirety.
S6 : Classical descriptive and clueless biology found a theoretical framework in the evolutionary theory of Darwin.
P : In later years, the focus was physiology and internal morphology or anatomy.
Q : Darwinian ideas of evolution by natural selection changed the perception completely.
R : The growth of biology as a natural science during the last 1000 years is interesting from many points of view.
S : One feature of this growth is changing emphasis from mere description of life forms to identification and classification of all recorded living forms.

The correct sequence should be
(a) RSPQ (b) SPRQ
(c) QRPS (d) PQRS

(a) The correct and meaningful paragraph is given by RSPQ.

44. S1 : Biology is the youngest of the formalised disciplines of natural science.
S6 : Life expectancy of human beings has dramatically changed over the years.
P : However, the twentieth century and certainly the twenty-first century has demonstrated the utility of biological knowledge in furthering human welfare, be it in health sector or agriculture.
Q : The discovery of antibiotics, and synthetic plant-derived drugs, anaesthetics have changed medical practice on one hand and human health on the other hand.
R : Applications of physics and chemistry in our daily life also have a higher visibility than those of biology.
S : Progress in physics and chemistry proceeded much faster than in biology.

The correct sequence should be
(a) QPRS (b) PRQS
(c) RPQS (d) SRPQ

(d) The correct and meaningful paragraph is given by SRPQ.

45. S1 : People in society need many goods and services in their everyday life including food, clothing, shelter, transport, etc.
S6 : The teacher in the local school has the skills required to impart education to the students.
P : A weaver may have some yarn, some cotton and other instruments required for weaving cloth.
Q : A family farm may own a plot of land, some grains, farming implements, may be a pair of bullocks and also the labour services of the family members.
R : Every individual has some amount of the goods and services that one would like to use.
S : In fact, the list of goods and services that any individual needs is so large that no individual in society, to begin with, has all the things one needs.

The correct sequence should be
(a) PQRS (b) RSPQ
(c) QPSR (d) SRQP

(d) The correct and meaningful paragraph is given by SRQP.

46. S1 : Farming is the main production activity in the village.
S6 : The new ways of farming need less land, but much more capital.
P : These have allowed the farmers to produce more crops from the same amount of land.
Q : Over the years there have been many important changes in the way farming is practised.
R : But in raising production, a great deal of pressure has been put on land and other natural resources.
S : This is an important achievement, since land is fixed and scarce.

The correct sequence should be
(a) QPSR (b) RSPQ
(c) SRPQ (d) PRSQ

(a) The correct and meaningful paragraph is given by QPSR.

47. S1 : Britain was the first country to experience modern industrialisation.

S6 : This gave people a wider choice for ways to spend their earnings and expanded the market for the sale of goods.

P : This meant that the kingdom had common laws, a single currency and a market that was not fragmented by local authorities and uneven taxation.

Q : It had been politically stable since the seventeenth century, with England, Wales and Scotland unified under a monarchy.

R : By then a large section of the people received their income in the form of wages and salaries than in goods.

S : By the end of the seventeenth century, money was widely used as the medium of exchange.

The correct sequence should be

(a) QPSR (b) PSQR
(c) RSQP (d) SRQP

(b) The correct and meaningful paragraph is given by PSQR.

48. S1 : For several million years, humans lived by hunting wild animals and gathering wild plants.

S6 : As a result, conditions were favourable for the growth of grasses such as wild barley and wheat.

P : This led to the development of farming and pastoralism as a way of life.

Q : This change took place because the last ice age came to an end about 13,000 years ago and with that warmer, wetter conditions prevailed.

R : Then, between 10,000 and 4,500 years ago, people in different parts of the world learnt to domesticate certain plants and animals.

S : The shift from foraging to farming was a major turning point in the human history.

The correct sequence should be

(a) QSPR (b) SPQR
(c) PSQR (d) RPSQ

(d) The correct and meaningful paragraph is given by RPSQ.

49. S1 : All governments claim eternal consistency and success.

S6 : Diplomacy offers choices, and those choices must be negotiated with other sovereign actors.

P : Choices involved uncertainty, risk and immediacy; those who must take the choices operate in the contemporary political milieu.

Q : And yet the essence of governance is choice.

R : Nowhere is this more true than in foreign policy decision-making.

S : Some even claim omniscience.

The correct sequence should be

(a) SQPR (b) QSRP
(c) SRPQ (d) RSPQ

(a) The correct and meaningful paragraph is given by SQPR.

50. S1 : Buddhism continued to spread into many lands of Asia during the period of 5th and 6th century.

S6 : He translated several scriptural commentaries into Pali and wrote a work called the Visuddimagga, which soon attained the status of a classic work on Theravada doctrine and meditation.

P : While this can be understood as a part of larger processes of cultural interaction, especially trade, a key role was played by monks.

Q : We know a little bit about some of them, but there must have been countless men whose commitment to the Buddhist path gave them the courage and determination to persevere in the face of the long, hard journey to India and back.

R : Buddhism had made its way to Sri Lanka many centuries earlier, during the time of Ashoka, and a thriving Buddhist community soon took root.

S : In the 5th century, the monk Buddhaghosha travelled to Sri Lanka.

The correct sequence should be

(a) RQPS (b) QRPS
(c) PQRS (d) PSRQ

(c) The correct and meaningful paragraph is given by PQRS.

Directions (Q. Nos. 51-60) *Each of the following sentences in this section has a blank space and four words or group of words are given after the sentence. Select the most appropriate word or group of words for the blank space and indicate your response on the Answer Sheet accordingly.*

51. On his way to the capital, the minister .... the eminent social worker at his residence.

(a) called on (b) called
(c) calling for (d) call off

(a) The word 'called on' makes the sentence grammatically correct and contextually meaningful.

52. The fire brigade fought for four hours to .... the fire in the building.

(a) put in (b) put out
(c) put on (d) put off

(b) The word 'put out' makes the sentence grammatically correct and contextually meaningful.

53. Ravi has proved that he can .... on his promise by winning the match.

(a) carry through (b) carry out
(c) carry (d) carry off

(b) The word 'carry out' makes the sentence grammatically correct and contextually meaningful.

54. It is best to .... politics when in the classroom.

(a) keep out (b) keep on
(c) keep off (d) keeping

(a) The word 'keep out' makes the sentence grammatically correct and contextually meaningful.

55. It shows that she has .... many years of service.

(a) put in (b) put out
(c) put (d) put on

(a) The word 'put in' makes the sentence grammatically correct and contextually meaningful.

56. The chairperson said that the group was .... of time.

(a) running out (b) running
(c) running with (d) run out

(a) The word 'running out' makes the sentence grammatically correct and contextually meaningful.

57. If I .... an angel, I would solve the problems of people.

(a) am (b) were (c) was (d) have

(b) The word 'were' makes the sentence grammatically correct and contextually meaningful.

58. Where there is a ...., there is a way.

(a) way (b) road (c) wing (d) will

(d) The word 'will' makes the sentence grammatically correct and contextually meaningful.

59. The police could not establish how the accident ....

(a) came off (b) came about
(c) came on (d) came out

(b) The word 'came about' makes the sentence grammatically correct and contextually meaningful.

60. I .... my old friend after twenty years.

(a) ran into (b) ran in
(c) run in (d) run on

(a) The word 'ran into' makes the sentence grammatically correct and contextually meaningful.

Directions (Q. Nos. 61-70) *Each item in this section consists of a sentence with an underlined word followed by four words/group of words. Select the option that is nearest in meaning to the underlined word and mark your response on the Answer Sheet accordingly.*

61. All the developments that took place in the 20th century have had implications for the next century.

(a) consequences (b) interferences
(c) feedback (d) planning

(a) The word implications means 'conclusion'. From the given options, 'Consequences' means the same and hence is the correct answer.

62. He is such a leader that his actions are contagious.

(a) complicated (b) transmittable
(c) effective (d) unthinkable

(b) The word contagious means 'capable of being transmitted'. From the given options, 'transmitted' is the correct answer.

63. The budget incorporated a number of tax reforms which included higher taxes for the very rich.

(a) excluded (b) integrated
(c) laid down (d) removed

(b) The word incorporated means 'included'. From the given options, 'integrated' means the same and hence is the correct answer.

64. His thesis makes all generic statements which have already been proved.

(a) specific (b) crude
(c) broad (d) non-standard

(c) The word generic means 'general or common'. From the given options, 'broad' means the same and hence is the correct answer.

65. The captain produced yet another stellar show to make her team enter the semi-finals.

(a) extraordinary (b) eclipse
(c) poor (d) not a great

(a) The word stellar means 'extraordinary'. Hence, option (a) is the correct answer.

66. A new show is trying to change the cliched depictions of women in animation.

(a) original (b) hackneyed
(c) crony (d) artificial

(b) The word cliched means 'a lack of originality'. From the given options, 'hackneyed' means the same and hence is the correct answer.

67. Not everyone finds a vocation which suits one's aptitude.

(a) attitude (b) approach
(c) liking (d) occupation

(d) The word vocation means 'profession or occupation'. Hence, option (d) is the correct answer.

68. Uninterrupted rain had fatigued the commuters from the outskirts to the city and work suffered.

(a) excited (b) refreshed
(c) slowed (d) exhausted

(d) The word fatigued means 'exhausted'.

69. The leader said, "I am aghast with the developments so far. I will take time to understand this".

(a) satisfied (b) sad
(c) amused (d) horrified

(d) The word aghast means 'horrified'.

70. The cause of the accident is yet to be ascertained, but police officials suspect the driver of the vehicle allegedly fell asleep.

(a) determined (b) curtained
(c) thought of (d) being known

(a) The word ascertained means 'to find out or discover'. From the given options, 'determined' means the same and hence is the correct answer.

Directions (Q. Nos. 71-80) *Each item in this section consists of sentences with an underlined word followed by four words or group of words. Select the option that is opposite in meaning to the underlined word and mark your response on the Answer Sheet accordingly.*

71. Early medieval period was not a combination of urban and rural civilisation. It was not a period of urban decay as claimed by some.

(a) survival (b) waste away
(c) decomposition (d) spoil

(a) The word decay means 'to decompose or die'. From the given options, 'survival' meaning to continue to live is its antonym.

72. He speaks eloquently and can pull crowds.

(a) confusingly
(b) expressively
(c) powerfully
(d) fluently

(a) The word eloquently means 'fluently or persuasively'. From the given options, 'confusingly' is its antonym.

73. Everyone has to fight the inertia in the system.

(a) sluggishness (b) indolence
(c) activity (d) torpor

(c) The word inertia means 'lazy or inactive'. From the given options, 'inactivity' is its antonym.

74. There is a need to promote philanthropy in education.

(a) charity
(b) benevolence
(c) nastiness
(d) likeliness

(c) The word philanthropy means 'the desire to promote the welfare of others'. From the given options, 'nastiness' is its antonym.

75. What we lack in the current times is compassion.

(a) empathy (b) carefulness
(c) indifference (d) hardship

(c) The word compassion means 'kindness'. From the given options, 'indifference' is its antonym.

76. Tempestuous behaviour would not yield much in any place.
(a) relaxed (b) passionate
(c) intense (d) windy

(c) The word tempestuous means 'characterised by strong and turbulent or conflicting emotion'. From the given options, 'relaxed' is its antonym.

77. Wooing everyone over an issue for support will not serve much purpose.
(a) discouraging (b) encouraging
(c) pursuing (d) persuading

(a) The word wooing means 'seek the favour' or 'support'. From the given options, 'discouraging' its antonym.

78. The highest award was bestowed upon her for her yeoman service.
(a) conferred (b) withdrawn
(c) imparted (d) imbibed

(b) The word bestowed means 'to give'. From the given options, 'withdrawn' is its antonym.

79. One feels elated when someone praises one's work.
(a) feels good (b) excited
(c) depressed (d) sober

(c) The word elated means 'happy and joyful'. From the given options, 'depressed' meaning 'sorrowful and sad' is its antonym.

80. All business activities need not result in profit-making. There is a need to be charitable.
(a) lenient (b) malevolent
(c) unforeseen (d) gracious

(b) The word charitable means 'to assist or help someone in need'. From the given options, 'malevolent' meaning 'a wish to do something bad or evil' is its antonym.

Directions (Q. Nos. 81-90) *Each of the following sentences has a word or phrase underlined. Read the sentences carefully and find which part of speech the underlined word belongs to. Indicate your response on the Answer Sheet accordingly.*

81. He has been working in the Department of Foreign Affairs since 2002.
(a) Adverb (b) Adjective
(c) Intensifier (d) Noun

(a) The given highlighted word is an Adverb.

82. The man in dark blue is the one who made us win the match.
(a) Relative clause
(b) Interrogative pronoun
(c) Relative pronoun
(d) Affirmative

(c) The given highlighted word is a Relative Pronoun.

83. The most beautiful actor of the industry was awarded today.
(a) Adjective (b) Numeral
(c) Adverb (d) Noun

(a) The given highlighted word is an Adjective.

84. "What is the latest news?" asked the Captain.
(a) Relative pronoun
(b) Adjective
(c) Adverb
(d) Adjectival clause

(b) The given highlighted word is an Adjective.

85. Noticing the change in the behaviour of the officer, the cadets returned to their position.
(a) Participle
(b) Present continuous
(c) Noun phrase
(d) Noun

(a) The given highlighted word is a Participle.

86. When he reached the department, the officials had left for the meeting.
(a) Past perfect verb
(b) Past tense
(c) Dependent clause
(d) Independent clause

(a) The given highlighted word is a Past Perfect verb.

87. He has offered her another chance.
(a) Intransitive verb (b) Past tense
(c) Perfect tense (d) Transitive verb

(c) The given highlighted word is a verb in perfect tense.

88. The building is very ancient.
(a) Transitive verb
(b) Intransitive verb
(c) Phrasal verb
(d) Auxiliary verb

(b) The given highlighted word is an Intransitive verb.

89. Hurrah! What a scintillating beauty the landscape is !
(a) Conjunction (b) Adjective
(c) Adverb (d) Interjection

(d) The given highlighted word is an Interjection.

90. Ravi was declared as the winner in the tie because he had hit the most number of fours and sixes.
(a) Conjunction (b) Interjection
(c) Adverb (d) Cause

(a) The given highlighted word is a Conjunction.

Directions (Q. Nos. 91-100) *In this section a word is spelt in four different ways. Identify the one which is correct. Choose the correct response (a), (b), (c) or (d) and indicate on the Answer Sheet accordingly.*

91. (a) Continuum (b) Continuem
(c) Contuneim (d) Continueiam

(a) The correctly spelt word is 'Continuum' which means 'a continuing sequence'.

92. (a) Stretegy (b) Stretagy
(c) Stratagy (d) Strategy

(d) The correctly spelt word is 'Strategy' which means 'a plan'.

93. (a) Commisionor (b) Commisioner
(c) Commissioner (d) Comissioner

(c) The correctly spelt word is 'Commissioner' which refers to 'a person appointed by a commission'.

94. (a) Vacum (b) Vacuum
(c) Vacuem (d) Vacam

(b) The correctly spelt word is 'Vacuum'.

95. (a) Psephology (b) Psefoloagy
(c) Sephology (d) Psyphology

(a) The correctly spelt word is 'Psephology' which means 'the statistical study of elections and trends in voting'.

96. (a) Neuphrology (b) Nephrology
(c) Neprology (d) Neaprology

(b) The correctly spelt word is 'Nephrology' which means 'the study of a specialty of medicine and pediatric medicine that concerns with study of the kidneys'.

97. (a) Psudonym (b) Pseudonym
(c) Pseudanym (d) Seeudonym

(b) The correctly spelt word is 'Pseudonym' which means 'another name'.

98. (a) Pnumonia (b) Neumonia
(c) Pneumonia (d) Numania

(c) The correctly spelt word is 'Pneumonia' which refers to 'a disease of lungs'.

99. (a) Resilient (b) Resilint
(c) Risilient (d) Realisent

(a) The correctly spelt word is 'Resilient' which means 'strong and tough'.

100. (a) Suplementary
(b) Supplementary
(c) Supplementery
(d) Supplemantory

(b) The correctly spelt word is 'Supplementary' which means 'completing or enhancing something'.

Directions (Q. Nos. 101-110) *In this section two sentences are given and you are required to find the correct sentence which combines both the sentences. Choose the correct response (a), (b), (c) or (d) and indicate on the Answer Sheet accordingly.*

101. Which is the correct combination of the given two sentences? The officer will return from China on Monday. You can meet him.

(a) You can meet the officer when he returned from China on Monday.
(b) You can meet the officer when he will return from China on Monday.
(c) You can meet the officer when he returns from China on Monday.
(d) The officer will meet you when you return from China on Monday.

(c) The grammatically correct combination of the sentences is given in option (c).

102. Which is the correct combination of the given two sentences? He is hard-working. He is honest too.

(a) He is not only hard-working, but also honest.
(b) He is only hard-working and honest.
(c) He is hard-working but honest too.
(d) He is not hard-working but also honest.

(a) The grammatically correct combination of the sentences is given in option (a).

103. Which is the correct combination of the given two sentences? Parents have been waiting since morning. They want to meet the counsellor.

(a) The counsellor has been waiting to meet the parents since morning.
(b) Parents had been waiting to meet the counsellor since the morning.
(c) Parents are waiting to meet the counsellor in the morning.
(d) Parents have been waiting since morning to meet the counsellor.

(b) The grammatically correct combination of the sentences is given in option (b).

104. Which is the correct combination of the given two simple sentences using 'If' clause? Minchi should have worked hard. She would have cleared the test.

(a) If Minchi had worked hard, she would have cleared the test.
(b) Had not Minchi worked hard, she could not have cleared the test.
(c) If Minchi has worked hard, she would have cleared the test.
(d) If Minchi had worked hard, she will have cleared the test.

(a) The grammatically correct combination of the sentences is given in option (a).

105. Which one of the following is the correct statement combining the two statements using 'though'? He has been trying his level best to win. He could not succeed.

(a) Though he is trying his level best to win, he could not succeed.
(b) He is trying his level best to win, though he could not succeed.
(c) Though he has been trying his level best to win, he could not succeed.
(d) Though he had been trying his level best to win, he could not succeed.

(d) The grammatically correct combination of the sentences is given in option (d).

106. Which is the correct combination of the given two sentences using 'relative clause'?

Gandhiji preached peace. He is an apostle of peace.

(a) Gandhiji who preached peace is an apostle of peace.
(b) Gandhiji preached peace because he is an apostle of peace.
(c) Gandhiji who preached peace is called an apostle of peace.
(d) Gandhiji is an apostle of peace because he preached peace.

(c) The grammatically correct combination of the sentences is given in option (c).

107. Which is the correct combination of the given two sentences?

Priya reached the station. The bus left before her.

(a) When Priya reached the station, the bus had already left.
(b) When Priya had reached the station, the bus already left.
(c) Priya reached the station, when the bus already left.
(d) When Priya had reached the station, the bus had already left.

(a) The grammatically correct combination of the sentences is given in option (a).

108. Which is the correct combination of the given two sentences?

He is too tired. He could not stand.

(a) He is so tired that he could scarcely stand.
(b) He is too tired and cannot stand.
(c) He will not stand and he is very tired.
(d) He is so tired that he could not be standing.

(a) The grammatically correct combination of the sentences is given in option (a).

109. Which is the correct combination of the given two sentences?

The teacher entered the classroom. All students stopped talking.

(a) No sooner did the teacher enter the classroom than the students stopped talking.
(b) As soon as the teacher entered the classroom all students were asked to stop talking.
(c) All students stopped talking as the teacher enters the classroom.
(d) No sooner did the students stop talking than the teacher entered the classroom.

(a) The grammatically correct combination of the sentences is given in option (a).

110. Which one of the following is the correct statement of the combination of the two sentences given below using 'whereas'?

Kavya is interested in reading books. Her sister shows interest in outdoor games.

(a) Kavya is interested in reading books whereas her sister's interest is outdoor games.
(b) Kavya is interested in reading books whereas her sister is not interested in it.

(c) Kavya is interested in reading whereas her sister's interest is outdoor games.
(d) Kavya is interested in reading books whereas her sister's interest is to play outside.

(a) The grammatically correct combination of the sentences is given in option (a).

Directions (Q. Nos. 110-120) *In this section direct speech sentences are given and you are required to find the correct indirect speech sentence of the same. Choose the correct response (a), (b), (c) or (d) and indicate on the Answer Sheet accordingly.*

111. Rahul said to his teacher, "Madam, what is the way to solve the question"?
(a) Rahul asked his teacher what the way to solve the question was.
(b) Rahul told his teacher what was the way to solve the question.
(c) Rahul asked to his teacher what the way was to solve the question.
(d) Rahul told his teacher what the way was to solve the question.

(c)

112. He said to his friend, "Could you please close the door?"
(a) He requested his friend to close the door.
(b) He requested his friend to please close the door.
(c) He ordered his friend to close the door.
(d) He wanted his friend to close the door for him.

(a)

113. Raj said to Sheela, "The Sun rises in the East."
(a) Raj told Sheela that the Sun rose in the East.
(b) Raj told Sheela that the Sun rises in the East.
(c) Raj asked Sheela that the Sun rises in the East.
(d) Raj said to Sheela that the Sun has arisen in the East.

(b)

114. Navanitha said to her friends, "What a scintillating beauty it is !"
(a) Navanitha told to her friends that it was a scintillating beauty.
(b) Navanitha exclaimed to her friends what a scintillating beauty it was.
(c) Navanitha asked her friends whether it was a scintillating beauty.
(d) Navanitha exclaimed to her friends that it was a scintillating beauty.

(b)

115. The Captain said to the soldiers, "March forward and aim at the peak of the hill today."
(a) The Captain requested the soldiers to march forward and aim at the peak of the hill that day.
(b) The Captain ordered the soldiers to march forward and aim at the peak of the hill today.
(c) The Captain ordered the soldiers to march forward and aim at the peak of the hill that day.
(d) The Captain told the soldiers that they should march forward and aim at the peak of the hill that day.

(c)

116. "Where were you last evening?" said the lady to her maid.
(a) The lady asked her maid where she had been the previous evening.
(b) The lady asked her maid where she had been in the last evening.
(c) The lady asked her maid where had she been the evening before.
(d) The lady told her maid where she had been to the last evening.

(a)

117. "Those who sowed the seeds last season will reap the harvest this season," said the leader to her followers.
(a) The leader said to her followers that those who sowed the seeds the previous season would reap the harvest that season.
(b) The leader addressed her followers that those who have sown the seeds the previous season would reap the harvest this season.
(c) The leader addressed her followers that those who had sown the seeds the previous season would reap the harvest that season.
(d) The leader advised her followers that those who sow the seeds the previous season would reap the harvest this season.

(c)

118. He said to his manager, "Could you please pass the bill this week?"
(a) He told his manager that bill to be passed.
(b) He requested his manager to pass the bill that week.
(c) He ordered his manager to pass the bill that week.
(d) He requested his manager to pass the bill this week.

(b)

119. The village chief said to the villagers, "All of us need to adopt new regulations. We will protect our Earth forever."
(a) The village chief ordered the villagers that all of them needed to adopt new regulations and they would protect their Earth forever.
(b) The village chief told the villagers that all of them need to adopt new regulations and they will protect their Earth forever.
(c) The village chief wanted the villagers needed to adopt new regulations and they would protect their Earth forever.
(d) The village chief told the villagers that all of them needed to adopt new regulations and they would protect their Earth forever.

(d)

120. The grandfather said to the baby, "May you live long with all good things of life".
(a) The grandfather blessed to the baby with long life and all good things of life.
(b) The grandfather asked the baby that she would live long with all good things of life.
(c) The grandfather wanted the baby to live long with all good things of life.
(d) The grandfather blessed the baby that she would live long with all good things of life.

(a)

PAPER III General Studies

1. Scattering of α-particles by a thin gold foil suggests the presence of

(a) electron in an atom
(b) proton in an atom
(c) positively charged nucleus at the centre of an atom
(d) isotopes of gold

(c) Rutherford proposed the model which was based upon the α-particle scattering experiment. In this experiment, Rutherford concluded that the atom consists of a heavy and positively charged part at its centre, called the nucleus. The entire mass of an atom resides in its nucleus and is equal to the sum of masses of protons and neutrons, since the mass of electron is negligible.

2. The elements of which of the following pairs are isobars?

(a) ${}^{1}_{1}H$ and ${}^{3}_{1}H$ (b) ${}^{1}_{1}H$ and ${}^{2}_{1}H$
(c) ${}^{12}_{6}C$ and ${}^{14}_{6}C$ (d) ${}^{40}_{18}Ar$ and ${}^{40}_{20}Ca$

(d) Isobars are the atoms of different elements that have same mass number but different atomic number. Therefore, the elements of ${}^{40}_{18}Ar$ and ${}^{40}_{20}Ca$ are isobars.

3. Which one of the following chemical reaction is not feasible?

(a) $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
(b) $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
(c) $Cu + PbCl_2 \rightarrow CuCl_2 + Pb$
(d) $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$

(c)
- $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$
  In this reaction, iron being more reactive than Cu, displaces Cu from $CuSO_4$ solution and forms new product, iron sulphate and Cu metal. So, this reaction is feasible.
- $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$
  In this reaction, zinc being more reactive than Cu, displaces Cu from. $CuSO_4$ solution and forms zinc sulphate and Cu metal. So, this reaction is also feasible.
- $Cu + PbCl_2 \longrightarrow Cu\ Cl_2 + Pb$
- This reaction is not feasible because lead chloride ($PbCl_2$) is less soluble. Thus, in this reaction, Pb not displaced by $PbCl_2$.
- $Mg + CuSO_4 \longrightarrow MgSO_4 + Cu$
  In this reaction, magnesium is more reactive than Cu.
  It displaces Cu from $CuSO_4$ solution and form magnesium sulphate and Cu metal. Hence, this reaction is feasible.

4. A solution having pH equal to zero is known as

(a) Highly alkaline solution
(b) Highly acidic solution
(c) Weakly acidic solution
(d) Neutral solution

(b) pH is a number which indicates the acidic or basic nature of a solution. The solutions having pH between 0 to 2 are strongly acidic, those with pH between 2 to 4 are moderately acidic, while others having pH between 4 to 7 are weakly acidic. When the solutions having pH equal to 7 are neutral whereas others which have pH between 12 to 14 are strongly alkaline.

5. Which one of the following acids is produced in human stomach?

(a) Formic acid (b) Sulphuric acid
(c) Nitric acid (d) Hydrochloric acid

(d) Stomach acid or gastric juice, is a digestive acid i.e., formed in the stomach and is composed of hydrochloric acid (HCl), potassium chloride (KCl) and sodium chloride (NaCl). Hydrochloric acid (HCl) plays a key role in digestion of proteins and enzymes.

6. Match List-I with List-II and select the correct answer using the codes given below the Lists

List I (Compound)	List II (Use)
(A) Boric acid	1. Antiseptic
(B) Citric acid	2. Food preservative
(C) Magnesium hydroxide	3. Antacid
(D) Acetic acid	4. Pickle

Codes

	A	B	C	D
(a)	1	2	3	4
(b)	1	3	2	4
(c)	4	3	2	1
(d)	4	2	3	1

(a) Boric acid is used as an antiseptic and insecticide.
- Citric acid is used as a preservative and flavouring agent in food and beverages such as soft drinks.
- Magnesium hydroxide reduces stomach acid, hence it is used as an antacid.
- Acetic acid is used in pickle as Vinegar.

Thus, the correct code is given in option (a).

7. The property of the sound waves that determines the pitch of the sound is its

(a) frequency (b) amplitude
(c) wavelength (d) intensity

(a) The property of the sound waves that determines the pitch of the sound is its frequency. The pitch of a sound depends on the frequency of vibration. Greater the frequency of a sound, the higher will be its pitch.

8. Which one of the following is not a property of the X-rays?

(a) They are deflected by electric fields.
(b) They are not deflected by magnetic fields.
(c) They have high penetration length in matter.
(d) Their wavelength is much smaller than that of visible light.

(a) X-rays are not deflected by electric fields because X-rays do not carry any charge. These waves are electromagnetic radiations. X-rays were discovered by Roentgen in 1895. Their wavelength is of the order of 10^{-12}m to 10^{-8}m. They are mainly used in detecting the fracture of bones, hidden bullet, needle, costly material etc, inside the body and also used in the study of crystal structure.

Properties of X-rays are :
- They are not deflected by electric and magnetic fields.
- They have high penetration length in matter.
- Their wavelength is much smaller than that of visible light.

9. Which one of the following is not true about the image formed by a plane mirror?

(a) It is of the same size as the subject.
(b) It is laterally inverted.
(c) It is real image.
(d) It is formed as far behind the mirror as the object is in front.

(c) Image formed by a plane mirror has following properties :
- It is always virtual and erect.
- The size of image is equal to the size of the object.
- The image formed is as far behind the mirror as the object is in front of it.
- The image is laterally inverted (i.e., left seems to be right and *vice-versa*)

10. In a periscope, the two plane mirrors are kept

(a) parallel to each other
(b) perpendicular to each other
(c) at an angle of 60° with each other
(d) at an angle of 45° with each other

(a) In a periscope, the two plane mirrors are kept parallel to each other at an 45° angle with its surface, which helps to reflect the light rays from the upper end of the periscope to lower end of the periscope.

Periscope is especially used in a submarine to see above the surface of the sea.

11. If the speed of light in air is 3×10^8 m/s, then the speed of light in a medium of refractive index $\frac{3}{2}$ is

(a) 2×10^8 m/s (b) $\frac{9}{4} \times 10^8$ m/s
(c) $\frac{3}{2} \times 10^8$ m/s (d) 3×10^8 m/s

(a) Given, speed of light in air

$= 3 \times 10^8$ m/s

Refractive index, $n = \frac{3}{2}$

Refractive index of a medium,

$$n = \frac{c}{v}$$

where,

c = speed of light in air and

v = speed of light in medium.

Using the formula,

$$n = \frac{c}{v}$$

$$\Rightarrow \quad v = \frac{c}{n} = \frac{3 \times 10^8}{3/2}$$

$$\therefore \quad v = 2 \times 10^8 \text{ m/s}$$

Therefore, speed of light in a medium of refractive index $\frac{3}{2}$ is 2×10^8 m/s.

12. Which one of the following types of radiation has the shortest wavelength ?

(a) Radio waves
(b) Visible light
(c) Infrared (IR)
(d) Ultraviolet (UV)

(d) Decreasing order of wavelength of Electromagnetic waves are **:**

Radio waves > Infrared (IR)> Visible light> Ultraviolet (UV)

Electromagnetic Spectrum

Name	**Frequency range** (Hz)	**Wavelength range** (m)
Gamma (γ) rays	5×10^{22} to 5×10^{18}	0.6×10^{-14} to 10^{-10}
X-rays	3×10^{21} to 1×10^{16}	10^{-13} to 3×10^{-8}
Ultraviolet rays (UV)	8×10^{14} to 8×10^{16}	4×10^{-9} to 4×10^{-7}
Visible light	4×10^{14} to 8×10^{14}	4×10^{-7} to 8×10^{-7}
Thermal or Infrared rays (IR)	3×10^{11} to 4×10^{14}	8×10^{-9} to 3×10^{-3}
Microwaves	3×10^{8} to 3×10^{11}	10^{-3} to 1
Radiowaves	3×10^{3} to 3×10^{11}	10^{-3} to 10^{5}

13. The unit of the force constant k of a spring is

(a) N-m (b) N/m (c) N-m^2 (d) N/m^2

(b) The unit of force constant k of a spring is N/m. According to Hooke's law, force applied by spring,

$$F = -kx \quad \text{or, } k = -\frac{F}{x}$$

where, k is a constant that is a characteristic of the spring known as spring constant or force constant and x is the final elongation of the spring.

14. Which one of the following is not an epidemic disease?

(a) Cholera (b) Malaria
(c) Smallpox (d) Elephantiasis

(c) Smallpox is not an epidemic disease. An epidemic is the rapid spread of an infectious disease to a large number of people in a given population within a short period of time, usually two weeks or less. Malaria and elephantiasis both are vector (Mosquito) borne diseases, thus during favourable conditions like flood or heavy rain, the sudden increase in numbers of mosquito will also cause outbreak of these diseases.

Similarly, cholera is also an epidemic disease as it is water borne disease, thus presence of bacteria *Vibrio cholerae* in local water source like pond or well will cause outbreak of cholera in every member of locality in a short period of time.

A pandemic is an epidemic of world-wide proportions i.e. outbreaks of these diseases occur across international borders, e.g. smallpox, which throughout history, has killed between 300-500 million people in all over the world.

15. Which one of the following animals has a three-chambered heart?

(a) *Scoliodon* (b) Salamander
(c) Pigeon (d) Human being

(b) Amphibians and reptiles, such as salamanders, frogs, toads, lizards and snakes, have three-chambered heart, one ventricle and two atria. In three-chambered heart, blood from the ventricle travels to the lungs and skin where it is oxygenated and also to the body. In the ventricle, deoxygenated and oxygenated blood are mixed before being pumped out of the heart.

Scoliodon has two-chambered heart, while pigeon and human beings have four-chambered heart.

16. Which one of the following is the correct sequence of events during sexual reproduction in plants?

(a) Seedling, formation of embryo, pollination, fertilisation, division of zygote
(b) Formation of embryo, seedling, pollination, fertilisation, division of zygote
(c) Pollination, fertilisation, division of zygote, formation of embryo, seedling
(d) Seedling, formation of embryo, division of zygote, pollination, fertilisation

(c) In flowering plants during sexual reproduction following events occur

- **Formation of Flower** which is the sexual organ of plant.
- **Gametogenesis** in which pollen grains are formed in anther, while ovules are formed in ovary.
- **Pollination** in which pollen grains transferred to style of gynoecium/Pistil through various means like air, water, insects, etc.
- **Fertilisation** Where a haploid male gamete fuses with another haploid egg cell of embryo sac of ovule.
- **Embryogenesis** After fertilisation a diploid zygote is formed, which develops into a multicellular embryo through repetitive cell divisions.
- **Formation of Seed and Seedling** After maturation the embryo develops into seed. The dormant seed after germination is known as seedling, which give rise to a new young plant.

17. When air is blown from mouth into a test-tube containing limewater, the limewater turns milky. This is due to the presence of

(a) water vapour
(b) oxygen
(c) carbon dioxide
(d) carbon monoxide

(c) The lime water [$Ca(OH)_2$] is turned into milky due to formation of calcium carbonate when the lime water reacts with carbon dioxide. The products formed in this reaction is calcium carbonate ($CaCO_3$) and water. The reaction is given below

$$\underset{\text{Calcium hydroxide (Lime water)}}{Ca(OH)_2} + \underset{\text{Carbon dioxide}}{CO_2} \longrightarrow \underset{\text{Calcium Carbonate}}{CaCO_3} + \underset{\text{Water}}{H_2O}$$

18. Which one of the following is the 'energy currency' for cellular processes?

(a) Glucose (b) ATP
(c) ADP (d) Pyruvic acid

(b) During the process of oxidation of food within a cell, all the energy contained in the respiratory substrates is not released free into the cell, or in a single step. Instead, it is released in a series of stepwise reactions controlled by enzymes and is trapped as chemical energy in the form of Adenosine Triphosphate (ATP).

Thus, it is said that ATP acts as the energy currency of the cell. The energy trapped in ATP is used in many energy requiring biological processes of the organisms such as locomotion, digestion, thinking, etc. During aerobic respiration, most of the ATPs are formed in mitochondria.

19. Which one of the following is the first enzyme to mix with food in the digestive tract?

(a) Trypsin (b) Cellulose
(c) Pepsin (d) Amylase

(d) Amylase is the first enzyme which mix with food in the digestive tract. The mastication of food in mouth is the first event of digestion in alimentary canal or digestive tract. During this process, digestion starts in the oral cavity by hydrolytic action of enzyme salivary amylase.

The saliva secreted into oral cavity contains two enzymes, i.e. amylase and lysozyme. About 30% of the starch gets hydrolysed in the oral cavity by the action of salivary amylase (at optimum pH 6.8) into a disaccharide, i.e. maltose.

Lysozyme acts as an antibacterial agent. It kills bacteria thus prevents infections.

20. Mahatma Gandhi's Dandi March, a great event in Indian freedom struggle, was associated with

(a) iron
(b) sodium chloride
(c) sulphur
(d) aluminium

(b) Mahatma Gandhi's Dandi March, a great event in Indian freedom struggle was associated with sodium chloride.

21. Match List I with List II and select the correct answer using the code given below the Lists.

	List I (Name)		**List II** (Formula)
(A)	Bleaching powder	1.	$NaHCO_3$
(B)	Baking soda	2.	$Na_2CO_3 \cdot 10H_2O$
(C)	Washing soda	3.	$Ca(OH)_2$
(D)	Slaked lime	4.	$CaOCl_2$

Code

	A	B	C	D
(a)	4	1	2	3
(b)	4	2	1	3
(c)	3	2	1	4
(d)	3	1	2	4

(a) The chemical formula of bleaching powder is $Ca(OCl)Cl$. It is used for water treatment and as a bleaching agent.

- Sodium bicarbonate ($NaHCO_3$) is commonly known as baking soda. It is widely used in baking.
- Sodium Carbonate ($Na_2CO_3 \cdot 10H_2O$) is the inorganic compound which is used as washing soda.
- Calcium hydroxide ($Ca(OH)_2$) is a colorless crystal i.e., known as slaked lime. Slaked lime is used as a pH–regulating agent and acid neutraliser in soil and water.

Hence, the correct codes are shown in option (a).

22. The number of water molecules associated with copper sulphate molecule to form crystals is

(a) 2
(b) 4
(c) 5
(d) 6

(c) Copper sulphate ($CuSO_4 \cdot 5H_2O$) is the inorganic compound in which four water molecules are directly attached to Cu atom while one is present as water of crystallisation.

23. Silver articles become black after sometime when exposed to air because

(a) silver gets oxidised to silver oxide
(b) silver reacts with moist carbon dioxide in the air to form silver carbonate
(c) silver reacts with sulphur in the air to form a coating of silver sulphide
(d) silver reacts with nitrogen oxides in the air to form silver nitrate

(c) When the silver articles react with the hydrogen sulphide gas that is present in the air, a black coating of silver sulphide (Ag_2S) is formed. Silver sulphide are insoluble in all the solvents.

$$\underset{\text{Silver}}{2Ag(s)} + \underset{\text{Hydrogen sulphide}}{H_2S(g)} \longrightarrow \underset{\text{Silver sulphide}}{Ag_2S\ (s)} + \underset{\text{Hydrogen gas}}{H_2(g)}$$

24. If x the temperature of a system in kelvin and y is the temperature of the system in °C, then the correct relation between them is

(a) $x = 273 - y$ (b) $x = 273 + y$
(c) $x = 173 + y$ (d) $x = 173 - y$

(b) Relation among the temperatures measured by Celsius scale and Kelvin scale.

$$K = 273 + C$$

where, K and C are the temperature of a system in kelvin and celsius (°C) respectively.

According to question, $K = x$ and $C = y$

Hence, $x = 273 + y$

25. The resistivity ρ of a material may be expressed in units of

(a) ohm (b) ohm/cm
(c) ohm-cm (d) ohm-cm^2

(c) The resistivity ρ of a material may be expressed in units of ohm-cm.

Resistivity of a conductor is defined as the resistance of a conductor of unit length and unit area of cross-section.

Resistivity, $\rho = \frac{RA}{l}$

where, l is the length of the conductor, A is the area of cross-section of the conductor and R is the resistance of the conductor.

26. The electromagnetic waves, which are used for satellite communication, are

(a) infrared radiations
(b) ultraviolet radiations
(c) radio waves
(d) visible lights

(c) The electromagnetic waves, which are used for satellite communication, are radio waves.

In the space wave propagation, the radiowaves emitted from the transmitter antenna reaches the receiving antenna through space. These radiowaves are called space waves. The space waves are the radiowaves of frequency range from 54 MHz to 4.2 GHz.

The space wave propagation is used for television broadcast microwave link and satellite communication.

27. Which one of the following is the correct sequence of organs that occur in the path of urine flow in human body?

(a) Kidney, ureter, urinary bladder, urethra
(b) Kidney, urinary bladder, ureter, urethra
(c) Kidney, ureter, urethra, urinary bladder
(d) Urinary bladder, kidney, urethra, ureter

(a) Urine is formed in the kidneys through the filtration of blood. The urine is then passed through the two separate tubes connected to each kidney, known as ureters which open to the urinary bladder, where urine is stored. During urination, the urine is passed from the urinary bladder through the urethra to the outside of the body.

28. Which of the following endocrine glands is not found in pair in humans?

(a) Adrenal (b) Pituitary
(c) Testis (d) Ovary

(b) Pituitary is a single gland, which is not found in pair like adrenal gland, testie and overies. It is the smallest endocrine gland but serves very important role in the human endocrine system. It directly or indirectly controls almost all other endocrime glands of the body. It is also known as master gland.

It is reddish grey in colour and is roughly oval in shape and about a size of a pea seed. It is located in a small bony cavity of the brain called sella tursica.

29. Which one of the following cell organelles contains DNA?

(a) Golgi apparatus
(b) Mitochondrion
(c) Lysosome
(d) Endoplasmic reticulum

(b) Mitochondrion is a semi-autonomous cell organelle. Mitochondria were first observed by Kollikar in 1850. They contain DNA as well as ribosomes and are able to synthesise proteins. According to scientific observation, the new mitochondira are originated by the growth and division of pre-existing mitochondria, which can't occur without the synthesis of proteins.

30. Who among the following scientists introduced the concept of immunisation to the medical world?

(a) Edward Jenner
(b) Robert Koch
(c) Robert Hooke
(d) Carl Linnaeus

(a) Edward Jenner (17th May, 1749-26th January, 1823) was an English physician who invented the smallpox vaccine. Jenner is often called a pioneer of immunisation or father of vaccination as he invented first vaccine and also proposed the concept of immunisation.

Robert Koch investigated the anthrax disease cycle in 1876 and studied the bacteria that cause tuberculosis in 1882 and cholera in 1883. He also formulated Koch's postulates.

Robert Hooke looked at a sliver of cork through a microscope lens and first time discovered cells in 1635.

Carl Linnaeus is known as the 'Father of Taxonomy'. He proposed binominal nomenclature for uniform and unique identification of organisms.

31. On 31st December, 1929 in which one of the following Congress Sessions was proclamation of Purna Swaraj made?

(a) Ahmedabad (b) Calcutta
(c) Lahore (d) Lucknow

(c) On 31st, December, 1929, in the Lahore Session of the Congress, proclamation of Purna Swaraj made. During this session, Pt. Jawaharlal Nehru was the President of the Congress. The resolution of Purna Swaraj proclaimed 26th January, 1930 as the Independence Day. This session was historic because for the first time in the history of freedom struggle, the national leaders demanded complete independence. Later, 26th, January was selected as the Republic Day of Independent India.

32. Which one of the following acts reserved seats for women in Legislatures in accordance with the allocation of seats for different communities?

(a) The Government of India Act, 1858
(b) The Indian Councils Act, 1909
(c) The Government of India Act, 1919
(d) The Government of India Act, 1935

(d) The Government of India Act, 1935 reserved seats for women in Legislatures in accordance with the allocation of seats for different communities.

According to this Act, the council of states (upper house) had 156 members for British India and upto 104 members for the princely states. All members from British India were to be directly elected, except for six members who were to be nominated by the Governor-General, to secure due representation for the scheduled classes, women and minority communities.

33. Which one among the following was demanded by the All India Depressed Classes Leaders' Conference at Bombay in 1931?

(a) Universal adult suffrage
(b) Separate electorates for untouchables
(c) Reserved seats for the minorities
(d) A unitary state in India

(b) 'All India Depressed Classes Leaders' conference at Bombay in 1931 demanded for separate electorates for untouchables. This demand arose because their leaders were dissatisfied with the provisions of Joint electorates provided in the Simon Commission report. This report empowered Governor to authorise a non–depressed class member to contest on behalf of depressed classes.

Finally, this demand led to the issuance of MacDonald Award or Communal Award (1932). This award provided for separate electorate for Scheduled Castes.

34. Who among the following was one of the founders of the Indian Society of Oriental Art?

(a) Rabindranath Tagore
(b) Abanindranath Tagore
(c) Dwarkanath Tagore
(d) Bankim Chandra Chattopadhyaya

(b) Abanindranath Tagore alongwith his brother Gaganendranath Tagore founded of Indian Society of Oriental Art (ISOA) in 1907. The primary objective of ISOA was to promote the rich artistic heritage of India. This initiative went a long way in demolishing the myth of the cultural supremacy of the western world.

35. Who among the following Sultans succeeded in finally breaking and destroying the power of Turkan-i-Chihalgani?

(a) Iltutmish
(b) Balban
(c) Alauddin Khalji
(d) Muhammad bin Tughluq

(b) After Balban ascended the throne, he realised that the real threat to the monarchy was from Chahalgani. Thus, he broke its power.

Turkan-i-Chihalgani was a class of ruling elite of fourty powerful military leaders. This class was created by Slave dynasty ruler Iltutmish.

36. Who among the following Mongol leaders/commanders did not cross Indus to attack India?

(a) Chenghiz Khan
(b) Tair Bahadur
(c) Abdullah
(d) Qutlugh Khwaja

(a) Among the following leaders/commanders, Chenghiz Khan was the Mongol ruler who did not cross the Indus to attack India.

Tair Bahadur invaded Punjab in 1241. Abdullah invaded Punjab with his force in 1292.

Qutlugh Khwaja attacked Delhi in 1299. This led to Battle of Kili.

37. In the region of Eastern shore of Adriatic sea, a cold and dry wind blowing down from the mountain is known as

(a) Mistral (b) Bora
(c) Bise (d) Blizzard

(b) Bora is a cold and dry wind blowing down the mountain in the region of Eastern shore of Adriatic sea.

Mistral is a cold and dry strong wind in Southern France that blows down from the North along the lower Rhone river valley towards Mediterranean sea.

Bise is a cold, vigourous and persistent North or North-Easterly wind blowing from the alpine mountains, that affects Switzerland and Eastern France.

Blizzard is a cold, violent, powdery polar wind. They are prevalent in the North and South polar regions like Canada, the USA, Siberia, etc.

38. Nyishi tribe is found mainly in

(a) Andaman and Nicobar
(b) Arunachal Pradesh
(c) Nilgiri-Kerala
(d) Kashmir Valley

(b) Nyishi tribe is found mainly in Arunachal Pradesh. Nyishi tribe is the largest ethnic community in Arunachal Pradesh. Nyishi tribe people are primarily agriculturalists who practice Jhum (Shifting) cultivation.

The attire of Nyishis uses the creast of a hornbill beak. As a result, they have adversely affected the population of hornbill bird. Boori Boot Yollo is an important festival.

39. In the field of tourism, which one of the following Indian States is described as 'One State Many Worlds'?

(a) Assam (b) West Bengal
(c) Karnataka (d) Rajasthan

(c) In the field of tourism, Karnataka is described as 'One State Many Worlds'. This is because of its diverse culture and topography.

Assam is known as 'The Awesome Assam'. West Bengal is known as 'Beautiful Bengal'. Rajasthan is known as 'the Incredible State of India.'

40. The number of people per unit area of arable land is termed as

(a) agricultural density
(b) arithmetic density
(c) physiological density
(d) economic density

(c) The number of people per unit area of arable land is termed as **physiological density**. A higher physiological density suggests that the available agricultural land is being used by more people.

Agricultural density is defined as the number of farmers per unit area of farmland. **Arithmetic density** is same as population density (Number of people per unit area of land).

41. Which one of the following rivers joins Ganga directly?

(a) Chambal (b) Son
(c) Betwa (d) Ken

(b) Among the following rivers, Son river directly joins river Ganga.

Son river originates near Amarkantak in Anuppur district of Madhya Pradesh and joins Ganga river near Patna (Bihar).

Chambal, Betwa and Ken rivers join Yamuna river in Uttar Pradesh. Yamuna then joins Ganga at Prayagraj (Uttar Pradesh).

42. Which one of the following is not a type of commercial agriculture?

(a) Dairy farming (b) Grain farming
(c) Livestock ranching
(d) Intensive subsistence agriculture

(b) Among the following, intensive subsistence agriculture is not a type of commercial agriculture.

Commercial agriculture is a cropping method in which crops and livestock are raised in order to sell the products in the market for remuneration. It is capital intensive agriculture in which crops are grown on large farms, using modern technologies like farm machinery, fertilizers, insecticides, high yeilding variety of seeds, etc. Dairy farming, grain farming, livestock ranching, plantation agriculture, etc. are few examples of this type of agriculture.

Intensive subsistence agriculture is practiced in areas where farmers have small fields. The crops are grown to be consumed by farmers themselves.

43. Which one of the following is not a land use category?

(a) Forest land
(b) Pasture land
(c) Marginal land
(d) Barren and wasteland

(d) Among the following Barren and wasteland is not a land use category. Barren and wastelands include mountains and hill slopes, deserts and rocky areas. These areas cannot be brought under plough except at a high input cost with possible low returns.

44. Which one of the following is not an objective of the MGNREGA?

(a) Providing up to 100 days of skilled labour in a financial year
(b) Creation of productive assets
(c) Enhancing livelihood security
(d) Ensuring empowerment to women

(b) Creation of productive assets is not an objective of the MGNREGA. It is an accronym of Mahatma Gandhi National Rural Employment Guarantee Act. It was enacted in the year 2005.

The objectives of MGNREGA include

- Ensuring social protection for the most vulnerable in rural India.
- Ensuring livelihood security for poor through creation of durable assets.
- Strengthening drought proofing and flood management in rural India.
- Aiding in empowerment of the marginalised communities, especially women, SCs and STs, through the rights based legislation.
- Strengthening decentralisation etc.

45. Which one of the following statements with regard to the functioning of the Panchayats is not correct?

(a) Panchayats may levy, collect and appropriate taxes, duties, tolls, etc.
(b) A person who has attained the age of 25 years will be eligible to be a member of a Panchayat.
(c) Every Panchayat shall ordinarily continue for five years from the date of its first meeting.
(d) A Panchayat reconstituted after premature dissolution shall continue only for the remainder of the full period.

(b) Among the given options, only option (b) is not correct. According to Article-243 of the Indian Constitution, a person who has attained the age of 21 years (and not 25 years) will be eligible to be a member of a Panchayat. Rest of the statements are correct.

46. Which one of the following is the earliest launched scheme of the Government of India?

(a) Deendayal Antyodaya Yojana
(b) Pradhan Mantri Gram Sadak Yojana
(c) Saansad Adarsh Gram Yojana
(d) Deendayal Upadhyaya Grameen Kaushalya Yojana

(b) Among the given schemes of the Government of India, Pradhan Mantri Gram Sadak Yojana was launched earliest. *The correct order of the launch dates of the schemes is given below*

- Pradhan Mantri Gram Sadak Yojana (December, 2000)
- Deendayal Antyodaya Yojana (June, 2011)
- Deendayal Upadhyaya Grameen Kaushalya Yojana (September, 2014).
- Saansad Adarsh Gram Yojana (October, 2014)
  Pradhan Mantri Gram Sadak Yojana was launched with an objective to provide single all–weather road connectivity to eligible unconnected habitation of designated population of 500+ in plain areas and 250+ in North–East, hill, tribal and desert areas.

47. Which of the following statements is not correct regarding the Members of Parliament Local Area Development Scheme (MPLADS)?

(a) Members of the Parliament (MPs) sanction, execute and complete works under the scheme
(b) Nominated Members of the Parliament can recommend works for implementation anywhere in the country
(c) The scheme is fully funded by the Government of India
(d) The annual entitlement per MP is ₹ 5 crore

(a) Under MPLADS, the MPLAD division is responsible for the implementation of the scheme. Under the scheme, each MP has the choice to suggest to the District Collector for works to the tune of ₹ 5 crore per annum to be taken up in his/her constituency.

48. Saubhagya, a Government of India scheme, relates to which of the following areas?

(a) Achieving universal household electrification
(b) Providing clean cooking fuel to poor households
(c) Rationalising subsidies on LPG
(d) Stopping female foeticide

(a) Pradhan Mantri Sahaj Bijli Har Ghar Yojana (Also known as Saubhagya) was launched in September, 2017. Under this scheme, free electricity connections to all households (both APL and BPL) in rural areas and poor families in urban areas was provided. Rural Electrification Corporation (REC) has been designated as nodal agency for this scheme.

49. The 'Basel Convention' is aimed at protecting human health and environment against adverse effects of which of the following?

(a) Hazardous wastes
(b) Persistent Organic Pollutants
(c) Mercury
(d) Chemicals and pesticides

(a) The 'Basel Convention' is aimed at protecting human health and environment against adverse effects 'hazardous waste'. The convention is a multilateral agreement on the control of transboundry movement of hazardous waste and their disposal. It was adopted in 1989 in Basel (Switzerland).

The **Stockholm Convention (2001)** is related to Persistent Organic Pollutants (POPs). 'The **Minamata convention** (2013) on mercury is related to protection of human health and environment from the adverse effects of mercury.

The **Rotterdam Convention** (2004) is related to protection of human health and environment from adverse effect of chemicals and pesticides.

50. Which one of the following is the biggest cause of incidence of migration of female persons in India?

(a) Employment (b) Education
(c) Marriage (d) Business

(c) The biggest cause of incidence of migration of female persons in India is marriage. According to Census Data of 2011, out of total women migrants, 64.9% migrated due to marriage, 3.2% due to employment, 1.3% due to education and 0.3% due to business.

51. Which one of the following is not correct about Repo rate?

(a) It is the interest rate charged by the Central Bank on overnight loan.
(b) It is the interest rate paid by the commercial banks on overnight borrowing.
(c) It is the interest rate agreed upon in the loan contract between a commercial bank and the Central Bank.
(d) It is the cost of collateral security.

(d) Among the given options, only option (d) is incorrect. Repo Rate (or repurchase rate) is the key policy rate of interest at which Central bank (or RBI) lends short-term money to banks. It helps in managing the liquidity of the economy.

52. The Cash Reserve Ratio refers to

(a) the share of Net Demand and Time Liabilities that banks have to hold as liquid assets
(b) the share of Net Demand and Time Liabilities that banks have to hold as balances with the RBI
(c) the share of Net Demand and Time Liabilities that banks have to hold as part of their cash reserves
(d) the ratio of cash holding to reserves of banks

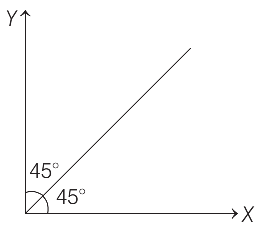
(b) The Cash Reserve Ratio (CRR) refers to the share of Net Demand and Time Liabilities that banks have to hold as balances with the RBI. CRR is set according to guidelines of the Central Bank of a country. Currently, CRR is 4% of total deposits of banks in India.

CRR is also an instrument used by RBI to control the liquidity in the system.

53. In economics, if a diagram has a line passing through the origin and has 45° angle with either axis and it is asserted that along the line $X = Y$ what is tacitly assumed?

(a) Both variables are pure numbers.
(b) Both variables are in the same unit.
(c) Both variables are in different units.
(d) At least one variable is a pure number.

(b)



It is assumed that along the line $X = Y$. Then, tacitly assumed that variables of X and Y are in the same units. Because an angle of 45° degree means that value of X equal the value of Y.

54. Who among the following won the Best Men's Player Award of FIFA Football Awards, 2019?

(a) Cristiano Ronaldo
(b) Virgil van Dijk
(c) Lionel Messi
(d) Xavi

(c) Lionel Messi won the best Men's player Award of FIFA Football Awards, 2019.

The Best FIFA Football Award is an association football award presented annually by the sport's governing body, FIFA. The first awarding ceremony was held on 9th January, 2017 in Zurich (Switzerland).

55. In September, 2019, which one of the following travel giants declared itself bankrupt?

(a) Expedia (b) Cox and Kings
(c) SOTC (d) Thomas Cook

(d) In September, 2019, travel giant Thomas Cook declared itself bankrupt.

Thomas Cook is a British tour operator, one of the world's oldest travel brands. One of the main reasons for it going bankrupt was not moving its market on digital platform.

56. Greta Thunberg, a teenage environment activist who was in news recently hails from

(a) Sweden (b) Germany
(c) The USA (d) Canada

(a) Greta Thunberg, a teenage environment activist hails from Sweden. She addressed the 2018 UN Climate Change Conference and took part in 2019 UN Climate Action Summit.

Due to her activism, she received awards like Ambassador of Conscience Award, Right Livelihood Award, International Children's Peace Prize, etc in 2019.

57. Recently the Reserve Bank of India has imposed limitations, initially for a period of six months, on the withdrawal of amount by account holders of which one of the following banks?

(a) IndusInd Bank
(b) Dhanlaxmi Bank
(c) Punjab and Maharashtra Cooperative Bank
(d) South Indian Bank

(c) In September, 2019 the RBI has imposed limitations on the withdrawal amount by the account holders of Punjab and Maharashtra Cooperative Bank (PMC), initially for a period of six months. This was done in the light of recently exposed scam in the bank.

PMC bank was founded in 1984 and has 137 branches across seven States. Its customers include small business, housing societies and institutions.

58. Amitabh Bachchan was recently conferred with the prestigious Dada Saheb Phalke Award. Who among the following was the first recipient of the award?

(a) Prithviraj Kapoor (b) Devika Rani
(c) Sohrab Modi (d) Naushad

(b) The first Dada Saheb Phalke Award was conferred to the actress Devika Rani in 1969. She is widely acknowledged as the 'First Lady of Indian Cinema'.

The Dada Saheb Phalke Award is India's highest award in cinema presented annually at the National Film Awards Ceremony by the Directorate of Film Festivals (an organisation under the Ministry of Information and Broadcasting).

59. The famous Mughal painting, depicting Jahangir embracing the Safavid king Shah Abbas, was painted by which one of the following Mughal painters?

(a) Abd al-Samad (b) Abul Hasan
(c) Dasavant (d) Bishandas

(b) The famous Mughal Painting, depicting Jahangir embracing the Safavid king Shah Abbas was painted by Abul Hasan in 1615 CE.

Abul Hasan (1589–1630 CE) was a Mughal painter of miniatures under the reign of Jahangir. Jahangir bestowed on him the title 'Nadir–uz–Saman' (Wonder of the Age).

Abd al–Samad is one of the founding masters of the Mughal miniature tradition. He headed the Mughal imperial workshop during reign of Akbar.

Dasavant was also a painter in Mughal Court during Akbar.

Bishandas was a Mughal painter during emperor Jahangir.

60. Which of the following statements about 'Mughal Mansab' system are correct?

1. 'Zat' rank was an indicator of a Mansabdar's position in the imperial hierarchy and the salary of the Mansabdar.
2. 'Sawar' rank indicated the number of horsemen the Mansabdar was required to maintain.
3. In the seventeenth century, Mansabdars holding 1000 or above 'Sawar' rank were designated as nobles (Umara).

Codes

(a) 1 and 2 (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3

(d) According to the Mughal Mansab or Mans ab dari system 'Zat' rank was an indicator of a Mansabdar's position in the imperial hierarchy and the Salary.

'Sawar' rank indicated the number of horsemer the Mansabdar was required to maintain.

In the 17th century, Mansabdars holding 1000 or above 'Samar' rank were designated as nobels (Umara).

Hence, all the statements are correct.

The Mansabdari system was introduced by Mughal emperor Akbar as new administrative machinery and revenue system.

61. Which one of the following pairs is not correctly matched?

(a) Kuddapah-kar	Rocky wastelands
(b) Nancai	Wet fields
(c) Puncai	Dry fields
(d) Tottakal	Garden lands

(a) Among the following options, only option (a) is not correctly matched.

In ancient India, Kuddapah–kar was one of the two main seasons of rice cultivation (the other being Samba-peshanam). Rest all the options are correctly matched.

62. Which of the following rulers were identified through matronymics (names derived from that of the mother)?

(a) Mallas of Pava
(b) Videhas of Mithila
(c) Yaudheyas
(d) Satavahanas

(d) Satavahanas rulers were identified through matronymics. For example, the name of greatest ruler of Satavahanas was Gauthamiputra Satakarni, which meant Gauthami's son Satakarni.

The Satavahanas (or Andhras) were ancient Indian dynasty based in Deccan region. They reigned from late 2nd century BCE to early 3rd century CE and ruled over present day Telangana, Maharashtra, Andhra Pradesh and parts of Gujarat, Madhya Pradesh and Karnataka.

63. Which one of the following statements about the famous text of *Panchatantra* is correct?

(a) It is a philosophical text reflecting the debates of the time and refuting rival positions
(b) It is a text ushering in linguistics as a formal science
(c) It is a text discussing developments in various spheres of natural sciences
(d) It is a text showing through illustration what should and should not be done

(d) Panchatantra is a text showing through illustration what should and should not be done. It is a 'Niti-Shastra' (or textbook of Niti) which roughly means 'the book on wise conduct of life'.

Vishnu Sharma is considered to be the author of the text. Original text was written in Sanskrit language.

64. Geomorphic factors influencing plant and animal distributions are

(a) slope angle and relief only
(b) slope aspect and relative relief
(c) slope angle, slope aspect and relief
(d) slope angle, slope aspect and relative relief

(d) Geomorphic factors influencing plant and animal distribution are slope angle, slope aspect and the distribution is also influenced by climatic factors (like temperature, water, light, etc.) and Edopic factors (like salinity, soil pH, mineral nutrients, etc.)

65. Which one of the following groups of cities does not have Sclerophyll as its natural vegetation cover?

(a) Valparaiso and Cape Town
(b) Lisbon and Perth
(c) Los Angeles and Adelaide
(d) Las Vegas and Queensland

(d) Among the following groups of cities, Las Vegas and Queensland does not have Sclerophyll as its natural vegetation cover.

Sclerophyll is a type of vegetation that has hard leaves, short internodes and leaf orientation parallel or oblique to direct sunlight. This vegetation is mainly found in Mediterranean type of climate.

Thus, Sclerophyll vegetation is found in areas like Perth, Sydney and Adelaide region of Australia, the Mediterranean basin, Californian region, Cape province of South Africa, etc.

66. Which of the following are warm ocean currents?

(a) Kuroshio and California Current
(b) North Atlantic Drift and Brazil Current
(c) Canaries and Benguela Current
(d) West Wind Drift and Falkland Current

(b) North Atlantic Drift (North Atlantic Ocean) and Brazil current (South Atlantic Ocean) are warm ocean currents.

67. In India, how many States/Union Territories have more than two international boundaries?

(a) 1 (b) 2 (c) 3 (d) 4

(d) In India, 4 States/Union Territories have more than two international boundaries. *These are*

- **Ladakh** with Pakistan, Afghanistan and China.
- **Sikkim** with Nepal, Bhutan and China.
- **West Bengal** with Nepal, Bhutan and Bangladesh.
- **Arunachal Pradesh** with Bhutan, China and Myanmar.

68. In the Hadley cell thermal circulation, air rises up and finally descends at

(a) intertropical convergence zone
(b) doldrums
(c) sub-tropical high-pressure cells
(d) equatorial troughs

(c) In the Hadley cell thermal circulation, air rises up (at equator) and finally descends at sub-tropical high- pressure cells.

According to tri-cellular model of atmospheric circulation, circulation takes place in three cells namely:

Hadley Cell Air rises at equator (0°) and decends near sub-tropical high-pressure zone (30° N and S).

Ferrel Cell Air decends at sub-tropical high pressure zone (30° N and S) and ascends at sub-polar low pressure zone (60° N and S).

Polar Cell Air rises in sub–polar low pressure zone (60° N and S) and descends at poles in both hemisphere.

69. Which one of the following soils is characterised by very high content of organic matter?

(a) Vertisol (b) Histosol
(c) Gelisol (d) Spodosol

(b) Histosol soils are characterised by very high content of organic matter. They contain at least 20-30% of organic matter by weight. Histosols are often very difficult to cultivate because of poor drainage and due to low chemical fertility.

Vertisols are rich in clay content.

Gelisols are permafrost soils in which organic content is limited to uppermost layer.

Spodosols are acidic soils rich in iron and aluminium.

70. Overseas Indians can exercise franchise in an election to the Lok Sabha under which of the following conditions?

1. They must be citizens of India.
2. Their names must figure in the electoral roll.
3. They must be present in India to vote.

Select the correct answer using the codes given below.

(a) 1, 2 and 3 (b) 2 and 3
(c) 1 and 2 (d) Only 1

(a) All the statements regarding franchise of overseas Indians are correct.

According to Section 20A of the Representation of People Act, 1950, overseas Indians can exercise franchise in an election to the Lok Sabha only under the conditions

- They must be citizens of India.
- Their names must figure in electoral roles.
- They must be present in India to vote.

71. Which one of the following is not a feature of the Ayushman Bharat Scheme?

(a) There is no cap on family size and age
(b) The scheme includes pre-and post hospitalisation expenses
(c) A defined transport allowance per hospitalisation will also be paid to the beneficiary
(d) The scheme provides a benefit cover of ₹ 10 lakh per family

(d) Among the following options, only option (d) is incorrect.

Ayushman Bharat Scheme provides a benefit cover of ₹ 5 lakh (and not ₹10 lakh) per family per year.

Ayushman Bharat Scheme is world's largest health insurance scheme fully financed by the Government of India.

Eligibility criteria for the scheme include

- No restrictions on family size, age or gender.
- All pre-existing conditions are covered from day one.
- Covers upto 3 days of pre-hospitalisation and 15 days of post-hospitalisation expenses.
- Benefits of the scheme are portable across country.
- Public hospitals are reimbursed for healthcare services at par with private hospitals.

72. Which of the following are considered to be the four pillars of human development?

(a) Equity, inclusion, productivity and empowerment
(b) Equity, productivity, empowerment and sustainability
(c) Productivity, gender, inclusion and equity
(d) Labour, productivity, inclusion and equity

(b) Human development is defined as the process of enlarging people's freedoms and opportunities, and improving their well-being. This concept was propounded by economist Mahbub-ul-haq in 1970s.

The four pillars of human development are

Equity It refers to creating equal access to opportunities and ensure that it is available to all.

Sustainability It means durableness in the availability of opportunities.

Productivity It means productivity in terms of human work.

Empowerment It means to have power to make choices.

73. Which one of the following was added as a Fundamental duty through the Constitution (86th Amendment) Act, 2002?

(a) To strive towards excellence in individual and collective activity
(b) To provide opportunities for education to one's child between the age of 6 and 14 years
(c) To work for the welfare of women and children
(d) To promote peace and harmony

(b) The Fundamental Duty "To provide opportunities for education to one's child between the age of 6 and 14 years", was added through the Constitution (86th Amendment) Act of 2002. This was done by amending the Article-51 A of the Constitution.

This amendment further added Article-21A to the Constitution.

Article-21A states that, "the State shall provide free and compulsory education to all children of the age of 6 to 14 years in such manner as the state may, by law, determine."

Moreover, 86th Constitutional Amendment Act substituted new article for Article-45. New article states that, "The state shall endeavour to provide early childhood care and education for all children until they complete the age of 6 years."

74. Which one of the following Articles of the Constitution of India protects a person against double jeopardy?

(a) Article-20 (b) Article-21
(c) Article-22 (d) Article-23

(a) Article-20 of the Constitution of India protects a person against double jeopardy. This means that the Constitution prohibits the prosecution and punishment for the same offence more than once.

Other provisions of the Article-20 are

- Protection against Ex-post Facto legislation, i.e. an individual cannot be convicted for actions that were committed before the enactment of the law.
- Immunity from self-incrimination, i.e. no person accused of any offence shall be compelled to be a witness against himself.

75. Consider the following statements about Stone Age in India :

1. Different periods are identified on the basis of the type and technology of stone tools.
2. There are no regional variations in the type and technology of tools in different periods.
3. Stone Age cultures of different periods evolved uniformly in a neat unilinear fashion all over the subcontinent.

Which of the statements given above is/are correct?

(a) Only 1 (b) 1 and 2
(c) Only 3 (d) 1, 2 and 3

(a) Among the given statements, only statement 1 is correct about Stone Age in India.

Stone Age in India was characterised by different periods that were identified on the basis of the type and technology of store tools. For instance, during palaeolithic age, tools were made of quartzite and sharpened by chipping technique; Mesolithic age was characterised by use of 'microlith' store tools; while during neolithic age tools were more specialised (like knife, dagger, etc.) and were completely or partially polished.

There was large regional variations in the type and technology of tools in different periods. For example, in the foothills of Himalayas, tools were majorly made up of pebbles, while in central India basaltic rocks were main type of stone used in making tools.

Stone Age in India did not evolve uniformly in a unilinear fashion over the sub-continent. For instance, stone tools in Shivalik hills date back to about 2 to 1.2 million years, while in Bori (Maharashtra) they data back to 1 million years back.

76. From which one of the following factory sites were limestone and chert blades mass produced and sent to various Harappan settlements in Sindh?

(a) Sukkur and Rohri Hills
(b) Khetri in Rajasthan
(c) Chagai Hills
(d) Hills of Baluchistan

(a) Sukkur and Rohri hills were the factory sites from where limestone and chert blades were mass produced and sent to various Harappan settlements in Sindh.

Khetri in Rajasthan was known for exploitation of copper. Chagai hills were known for availability of Lapis Lazuli stone during the period.

77. The work *Siyar-ul- Mutakherin*, which describes the Battle of Plassey, 1757, was written by

(a) Salabat Jung (b) Qasim Khan
(c) Ghulam Husain (d) Ram Mohan Roy

(c) The work Siyar-ul-Mutakherin, which describes the Battle of Plassey, 1757 was written by Ghulam Husain. This work was completed in 1781 CE and has three volumes.

78. Who believed that the Russian designs were 'an imminent peril to the security and tranquility' of the Indian Empire in 1836?

(a) Lord Auckland
(b) Lord Palmerston
(c) Lord Canning
(d) Alexander Burnes

(b) Lord Palmerston believed that the Russian designs were 'an imminent peril to the security and tranquility' of the Indian empire in 1836.

Lord Palmerston was foreign secretary during the Whig Cabinet of Lord Melborne. It was this belief that guided the policy of British India towards Afghanistan. This finally led to Afghan wars.

79. The 'Tattvabodhini Sabha' was established by

(a) Devendranath Tagore in 1839
(b) Keshab Chandra Sen in 1857
(c) Akshay Kumar Datta in 1850
(d) Dwarakanath Tagore in 1840

(a) The 'Tattvabodhini Sabha' was established by Devendranath Tagore in 1839. It was established as Tattvaranjini Sabha and later renamed as Tattvabodhini Sabha. This Sabha was a particular reform movement organisation, aimed to popularise Brahmodharma (or Brahmo faith). Its primary objective was to propagate the spirit of Hindu Scriptures, including the Vedas.

80. What was the code name given to the first ever tri-service military exercise between India and USA?

(a) Lion Triumph
(b) Elephant Triumph
(c) Tiger Triumph
(d) Bison Triumph

(c) Tiger Triumph was the code name given to the first ever tri-service military exercise between India and USA. This exercise was a Humanitarian Assistance and Disaster Relief (HADR) exercise. It was held in November, 2019 off the Visakhapatnam and Kakinada coasts in Andhra Pradesh.

Other joint exercises *between India and USA are*

- Military exercise : Yudh Abhyas and Vajra Prahar
- Air Forces Exercise : Cope India
- Naval Exercise (includes Japan) : Malabar.

81. 'Naropa' is an annual festival of

(a) Sikkim
(b) Ladakh
(c) Arunachal Pradesh
(d) Nagaland

(b) Naropa is an annual festival of Ladakh. It is one of the biggest festivals in the Himalayas that celebrates the life and legacy of the Buddhist Scholar Naropa.

In 2019, the festival was organised near Hemis Monastry of Ladakh.

The shordol dance (folk dance of Ladakh) is performed during this festival.

82. Which one of the following is India's official entry for the Best International Feature Film category in the 92nd Academy Awards?

(a) Bulbul Can Sing
(b) Super Deluxe
(c) Gully Boy
(d) And The Oscar Goes To

(c) *Gully Boy* was selected as India's official entry for the Best International Feature Film category in the 92nd Academy Awards.

The Academy Award for Best International Feature Film Award is one of the Academy Awards given annually by the USA based Academy of Motion Picture Arts and Sciences (AMPAS).

83. The Global Goalkeeper Award is given by

(a) the Bill and Melinda Gates Foundation
(b) the United Nations Environment Programme
(c) the Kellogg School of Management
(d) the World Meteorological Organisation

(a) The Global Goalkeeper Award is given by the Bill and Melinda Gates Foundation. In 2019, it was given to the Prime Minister of India for India's massive progress under the Swachh Bharat Abhiyan.

The Goalkeepers, Global Goals Awards aim to highlight the extraordinary stories of remarkable individuals who are taking action to give life to the global goals and help achieve them by 2030. In 2019, fourth edition of these awards was presented.

These awards were presented in 5 categories in 2019–Progress Award (age 16-30), Changemaker Award (age 16-30), Campaign Award (age 16-30), Goalkeepers Voice Award (any age) and the Global Goalkeeper Award (any age).

The Global Goal Keeper Award aims to recognise a political leader for his/her commitment to global goals through impactful work in their country.

84. 'Gandhi Solar Park' is located at

(a) New York (b) Vladivostok
(c) Thimphu (d) Houston

(a) 'Gandhi Solar Park' is located at New York (UN headquarters). This park was inaugurated by PM Narendra Modi alongwith other world leaders, commemorating the 150th birth anniversary of Mahatma Gandhi.

Gandhi Solar Park is a first of its kind symbolic effort by India at the UN. It has 50 KW capacity, and a gesture by India that it is willing to go beyond the talk on climate change.

85. Who among the following was the first to arrive in Africa as traders that eventually led to European colonisation of Africa?

(a) French (b) Spanish
(c) Portuguese (d) Dutch

(c) Portuguese were the first to arrive in Africa as traders that eventually led to European colonisation of Africa. They first reached Africa as explorers under Henry the Navigator. Initially, the Europeans colonised the areas which were un-inhabited like cape varde region.

86. The college of Military Engineering affiliated to Jawaharlal Nehru University is situated at

(a) New Delhi (b) Dehradun
(c) Nainital (d) Pune

(d) The college of Military Engineering affiliated to Jawaharlal Nehru University is situated at Pune. It was established in 1943 as School of Military Engineering at Roorkee and later moved to Pune in 1948. This was done in view of increased responsibilities and in keeping with the higher status of the degree of Engineering courses being conducted by the school. The institute is the premier technical training institution of the Corps of Engineers of the Indian Army.

87. Which of the following statements with regard to Coal India Limited (CIL) is/are true?

1. CIL has its headquarters at Kolkata.
2. CIL operates through 82 mining area spread over twenty provincial States of India.
3. CIL is the single largest coal producing company in the world.

Select the correct answer using the codes given below.

(a) Only 1 (b) 1 and 3
(c) 2 and 3 (d) 1, 2 and 3

(b) Only statements 1 and 3 regarding Coal India Limited (CIL) are correct.

CIL is a state owned coal mining corporate, which came into being in November, 1975. CIL's headquarters is at Kolkata. CIL is the single largest coal-producing company in the world. CIL operates through 82 mining areas spread over eight (and not 20) provincial states of India.

88. Which one of the following climatic types is found in Central Spain?

(a) Subarctic
(b) Mediterranean dry hot summer
(c) Subtropical Steppe
(d) Humid continental warm summer

(b) Climate of Spain is mainly Mediterranean with the extremes of temperatures being found in Central Spain (due to continentality).

Mediterranean type of climate is characterised by seasonal shifting of pressure belts. In this climate, summers are hot and dry while winters are cold and wet. It is best developed in the European region bordering Mediterranean sea.

89. Which one of the following is not among the principal languages of Jammu and Kashmir?

(a) Urdu (b) Gujari
(c) Koshur (d) Monpa

(d) Monpa is not among the principal languages of Jammu and Kashmir. Monpa language is spoken by the Monpa ethnic group of Arunachal Pradesh.

According to the department of tourism of Jammu and Kashmir, major languages of Jammu and Kashmir are Urdu, Hindi and Koshur. Further, the Gujari language spoken by Gurjar tribes which are spread across North India, including Jammu and Kashmir.

90. The major part of Central Asia is dominated by which one of the following language families?

(a) Indo-European (b) Sino-Tibetan
(c) Austric (d) Altaic

(d) The major part of Central Asia is dominated by Altaic family of languages. Altaic group of language, mainly includes Turkie, Mangolian and Tungusic family of languages.

Apart from Altaic, Indo-European is the second major family of language in this region. Central Asia is the region which stretches from Caspian sea in the West, China in the East, Iran and Afghanistan in the South and Russia in the North. It consists of Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan and Uzbekistan.

91. Which one of the following Articles was defended by Dr. BR Ambedkar on the plea that it would be used as 'a matter of last resort'?

(a) Article-352 (b) Article-359
(c) Article-356 (d) Article 368

(c) Article-356 was defended by Dr. BR Ambedkar on the plea that it would be used as a matter of last resort.

Article-356 imposes President's rule in the states, if the government of the state cannot carry on the governance according to the provisions of the Constitution.

Since, this article empowers Union Government over State Government it weakens federalism. Hence, it was opposed by various members of the Constituent Assembly. As a result, Dr. Ambedkar defended this provision by stating the above statement.

92. What is the ground on which the Supreme Court can refuse relief under Article-32?

(a) The aggrieved person can get remedy from another court.
(b) That disputed facts have to be investigated.
(c) That no Fundamental Right has been infringed.
(d) That the petitioner has not asked for the proper writ applicable to his/her case.

(c) The Supreme Court can refuse relief under 32 on the ground that no Fundamental Right has been infringed.

Article-32 provides the right to constitutional remedies which means that a person has a right to move to Supreme Court for getting his/her Fundamental Rights protected. Supreme Court has the power to issue writs, homely habeas corpus, mandamus, prohibition, quo warranto and certiorari.

93. The Ministry of Heavy Industries and Public Enterprises consists of

(a) the Department of Heavy Industry and the Department for Promotion of Industry and Internal Trade
(b) the Department of Public Enterprises and the Department for Promotion of Industry and Internal Trade
(c) the Department of Scientific and Industrial Research and the Department of Heavy Industry
(d) the Department of Heavy Industry and the Department of Public Enterprises

(d) The Ministry of Heavy Industries and Public Enterprises consists of the department of Heavy Industry and the department of Public Enterprises.

Department for promotion of Industry and Internal Trade is under the Ministry of Commerce and Industry. Department of Scientific and industrial research is under the Ministry of Science and Technology.

94. The First Delimitation Commission in India was constituted in

(a) 1949 (b) 1950 (c) 1951 (d) 1952

(d) The first Delimitation Commission in India was constituted in 1952 under the Delimitation Commission Act, 1952.

Delimitation Commission is a statutory body concerned with fixing the limits of territorial constituencies in the country. Till now, Delimitation Commission has been constituted four times in 1952, 1963, 1973 and 2002, respectively.

95. Who among the following stated in the Constituent Assembly that on 26th January, 1950, India was going to enter a life of contradictions?

(a) Dr. BR Ambedkar
(b) Jawaharlal Nehru
(c) Mahatma Gandhi
(d) SP Mukherjee

(a) Dr. BR Ambedkar was the one who stated in the Constituent Assembly that on 26th January, 1950, India was going to enter a life of contradictions. This because, with the enactment of Constitution, India would have political equality (one person, one vote), but in social and economic life, India will have inequality.

96. The power of the Supreme Court to decide in the case of a dispute between two or more States is called

(a) original jurisdiction
(b) inherent jurisdiction
(c) plenary jurisdiction
(d) advisory jurisdiction

(a) The power of the Supreme Court to decide in the case of a dispute between two or more states is called original jurisdiction.

Original jurisdiction of Supreme Court is under Article-131. It refers to matters for which Supreme Court can be approached directly. *Few matters covered under original jurisdiction are*

- Any dispute between Indian Government and one or more states.
- Any dispute between Indian Government and one or more states on one side and one or more states on the other side.
- Any dispute between two or more states, etc.

97. Who among the following is the Chairman of the Economic Advisory Council to the Prime Minister (EAC-PM)?

(a) Ratan P. Watal (b) Bibek Debroy
(c) Ashima Goyal (d) Sajjid Chinoy

(b) Bibek Debroy is the Chairman of the Economic Advisory Council to the Prime Minister (EAC-PM).

EAC-PM is an independent body constituted to give advice on economic and related issues to the Government of India, specifically to the Prime Minister.

Ratan P. Watal is the Member Secretary of EAC-PM, while Ashima Goyal and Sajjid Chinoy are part-time members.

98. Hilsa is the national fish of

(a) Pakistan (b) India
(c) Bangladesh (d) Nepal

(c) Hilsa is the national fish of Bangladesh. Around 70% of the total Hilsa fish production in the world is from Bangladesh.

Hilsa is also the State fish of West Bengal. Masheer is the national fish of Pakistan. Dolphin is the national fish of India.

99. The Vijaynagar Advanced Landing Ground of the Indian Air Force, which was reopened recently is located in

(a) Jammu and Kashmir
(b) Arunachal Pradesh
(c) Karnataka
(d) Himachal Pradesh

(b) The Vijaynagar Advanced Landing Ground of the Indian Air Force is located in Arunachal Pradesh. Earlier, it was declared unfit for fixed wing aircraft such as transport planes as the runway got damaged. It was reopened in September, 2019.

100. Rustom-2, which crashed in Karnataka recently, was a/an

(a) fighter aircraft
(b) helicopter
(c) transport aircraft
(d) unmanned aerial vehicle

(d) Rustom-2, which crashed in Karnataka September, 2019, was an unmanned aerial vehicle. Aeronautical Development Establishment (ADE), a laboratory of DRDO based in Bangalore was responsible for its design and development. It is a Medium-Altitude Long Endurance Unmanned Aerial Vehicle (MALE-UAV) that flies at an altitude of 3000-9000 m for extended durations of time.

101. The maiden trilateral naval exercise involving India, Singapore and Thailand was held at

(a) Port Blair (b) Chennai
(c) Panaji (d) Kochi

(a) The maiden trilateral exercise involving the Republic of Singapore's Navy (RSN), Royal Thailand Navy (RTN) and Indian Navy was held at Port Blair, in Andaman sea in September, 2019. The name of exercise was SITMEX-19 (Singapore-India Thailand Maritime Exercise). The objective of exercise as to bolster the maritime inter-relationship among them and contribute significantly to enhance the overall security in the region.

102. The creation of a Federal Court in India was advocated by which of the following Acts/Commissions?

(a) The Government of India Act, 1919
(b) The Lee Commission, 1923
(c) The Government of India Act, 1935
(d) The Indian Councils Act, 1909

(c) The Government of India Act, 1935 advocated for the creation of a Federal Court in India. This act was watershed moment in the evolution of Indian Constitutional. This act was enacted during the period of Governor-General, Lord Wellingto.
Other important provisions of this act are as follows

- Establishment of federation of India with British India territories and Princely states.
- Elaborate safeguards and protective instruments for minorities.

103. Who founded the 'Seva Samiti' at Allahabad in 1914?

(a) Hridayanath Kunzru
(b) GK Gokhale
(c) Shri Ram Bajpai
(d) TB Sapru

(a) Seva Samiti was founded by the Hridayanath Kunzru at Allahabad (Prayagraj) in 1914. He had set-up this samiti to organise social service during natural disaster like flood, to promote education, sanitation, to uplift depressed class and reform criminals. He was one of the most prominent member of the servants of India society, founded by GK Gokhale in 1905.

104. The State of Hyderabad in the Deccan officially acceded to the Indian Union in the year

(a) 1948 (b) 1950
(c) 1949 (d) 1947

(a) After getting Independence from Britisher's in 1947, the Princely State of Hyderabad in the Deccan, declared itself as an independent province and denied to amalgamate with India. The Nizam Osman Alikhan of Hyderabad formed an irregular army known as Razakars. When talks were not resulted into any solution, then military action was taken under the code name 'Operation Polo' in 1948.

Razakar's were defeated very badly and State of Hyderabad was merged with India.

105. The Hunter Commission (1882) appointed to survey the State of education in India

(a) deprecated University education
(b) overruled the Despatch on 1854
(c) endorsed the Despatch of 1854 with greater emphasis on primary education
(d) criticised the grants-in-aid system of schooling

(c) The Hunter Commission was appointed by the then Governor-General of India in 1882 to survey the state of education in India. The report of Hunter Commission endorsed the Despatch of 1854 with greater emphasis on primary education. This commission gave suggestion regarding secondary education, grant-in-aid for indigenous schools, encouragement of primary education and also emphasised on the female education in the country.

Charles Wood had sent a despatch to Lord Dalhousie, the then Governor- General of India for disseminating education in India. The Woods despatch is considered as 'Magna-Carta' of English Education in India.

106. The power to legislate on all matters relating to elections to Panchayats lies with

(a) the Parliament of India
(b) the State Legislatures
(c) the State Election Commission
(d) the Election Commission of India

(b) According to the 73rd Constitutional Amendment Act, 1992 the power to legislate on all matters relating to election to Panchayat lies with the State Legislature. Under the Article-329, an election to a Panchayat can be called in question only by an election petition which should be presented to such authority and in such manner as may be prescribed by or under any law made by the State Legislature.

107. The 11th Schedule of the Constitution of India distributes powers between

(a) the Union and the State Legislatures
(b) the State Legislatures and the Panchayat
(c) the Municipal Corporation and the Panchayat
(d) the Gram Sabha and the Panchayat

(b) The 11th Schedule of the Constitution of India distribute powers between the State Legislature and Panchayat. There are overall 12 Schedules in the Constitution. *The Schedules are as follows*

Schedule	Subject
1st	List of State and Union Territories
2nd	Salary and Emolument
3rd	Form of oaths and affirmation
4th	Allocation of seats to State and UTs in the Rajya Sabha
5th	Administration and Control of Scheduled Areas
6th	Administration of Assam, Meghalaya, Tripura and Mizoram
7th	Distribution of Power between the Union and the State List
8th	List of Recognised Language
9th	Validation of Certain Act and Regulation
10th	Provision also disqualification on ground of defection
11th	It has 29 matters. This schedule was added by the 73rd Amendment Act, 1992
12th	Contains the Powers, Authority and Responsibilities of Panchayat

108. The provisions of the Constitution of India pertaining to the institution of Panchayat do not apply to which one of the following States?

(a) Meghalaya (b) Tripura
(c) Assam (d) Goa

(a) Panchayati Raj is the basic unit of administration in a system of governance. The Constitutional (73rd Amendment) Act, 1992 came into force in India on 24th April, 1993 to provide constitutional status to the Panchayati Raj institutions. This act was extended to the Panchayats in the tribal areas of eight states, namely Andhra Pradesh, Gujarat, Himachal Pradesh, Maharashtra, Madhya Pradesh, Odisha and Rajasthan from 24th December, 1996. Currently, the Panchayati Raj system exists in all States of India except Nagaland, Meghalaya, Mizoram and in all Union Territories except Delhi and Jammu and Kashmir.

109. Which one of the following rivers does not drain into Black sea?

(a) Volga (b) Dnieper
(c) Don (d) Danube

(a) Volga river does not drain into Black sea. Black sea is a body of water and marginal sea of Atlantic Ocean between Eastern Europe, the Car casus and Western Asia. Many rivers drain into it such as the Danube, Dnieper, Southern Bug, Dniester, Don and the Rioni.

The Volga river is the longest river in Europe. It is also Europe's largest river in terms of discharge and drainage basin. The Volga river flows through Central Russia and drains into Caspian sea.

110. The National Water Academy (NWA) is located at

(a) Dehradun
(b) Hyderabad
(c) Bhopal
(d) Khadakwasla

(d) National Water Academy (NWA) is located at Khadakwasla, Pune. NWA was formerly known as Central Training Unit. It was formed in 1988 to impart training to the in-service engineers of various Central/State organisations involved in the development and management of water resources.

111. Which one of the following is the correct sequence of formation of the Commissions starting from the earliest?

(a) Finance Commission, Planning Commission, Investment Commission, Election Commission
(b) Election Commission, Planning Commission, Finance Commission, Investment Commission
(c) Planning Commission, Election Commission, Finance Commission, Investment Commission
(d) Investment Commission, Finance Commission, Planning Commission, Election Commission

(b) In option (b) the commissions are arranged in the correct sequence. The Election Commission of India was formed on 25th January, 1950. The Planning Commission was formed on 15th March, 1950. The Finance Commission was first time formed on 22nd November, 1951. The Investment Commission of India (ICI) was formed in December 2004.

112. The formulation of policy in respect to Intellectual Property Rights (IPRs) is the responsibility of

(a) the Ministry of Law and Justice
(b) the Department of Science and Technology
(c) the Department for Promotion of Industry and Internal Trade
(d) the Ministry of Human Resource Development

(c) The Comptroller General of Patents, Designs and Trade Marks (CGPDTM) under the Department for Promotion of Industry and Internal Trade (DPIIT) under the Ministry of Commerce and Industry is entrusted with the responsibility of administering the laws relating to Patents, Designs, Trade Marks and Geographic Indications with the Territory of India. The DPIIT is also entrusted with matters concerning the UN agency (World Intellectual Property Organisation, WIPO) on Intellectual Property Rights (IPRs).

113. Which one of the following is the latest addition to the AYUSH group of healthcare system?

(a) Unani (b) Siddha
(c) Sowa-Rigpa (d) Reiki

(c) Sowa-Rigpa is the latest addition to the AYUSH group of healthcare systems. The Union Cabinet in November, 2019 also approved for setting up of National Institute of Sowa-Ragpa (NISR) at Leh (Ladakh). Sowa-Rigpa is a traditional system of medicine in the Himalayan Belt of India. It originated in Tibet and popularly practised in countries namely, India, Nepal, Bhutan, Mongolia and Russia.

The other AYUSH systems are, Ayurveda, Yoga, Naturopathy, Unani, Siddha and Homeopathy. Reiki has not been included in AYUSH group of healthcare systems.

114. Which one of the following is the nodal agency in India for the United Nations Environment Programme?

(a) The Ministry of Environment, Forest and Climate Change
(b) The Ministry of Science and Technology
(c) The Ministry of Earth Sciences
(d) The Ministry of Home Affairs

(a) The Ministry of Environment, Forest and Climate Change (MoEFCC) is the nodal agency in the administrative structure of the Central Government for the planning, promotion, coordination and overseeing the implementation of India's environmental and forestry policies and programmes.

The Ministry also serves as the nodal agency in the country for the United Nations Environment Programme (UNEP).
The United Nations Environment Programme (UNEP) is the leading global environmental authority that sets the global environmental agenda, promotes the coherent implementation of the environmental dimension of sustainable development within the United Nations system and serves as an authoritative advocate for the global environment.

115. According to the Census 2011, in India, what is the percentage of people (approximately) considered to be migrants (internal), i.e., now settled in a place different from their previous residence?

(a) 25% (b) 35%
(c) 45% (d) 55%

(b) According to census data 2011, 45.36 crore Indians (35%) in India are internal migrants, now settled in a place different from their previous residence. In 2001, the figure stood at 31.45 crore. Most of the migrants, around 70%, are females who migrate for marriage.
Migrants by place of birth are those who are enumerated at a village/town at the time of census other than their place of birth. A person is considered as migrant by place of last residence, if the place in which he is enumerated during the census is other than his place of immediate last residence.

116. Suppose an agricultural labourer earns ₹ 400 per day in her village. She gets a job to work as babysitter in a nearby town @ ₹ 700 per day. She chose to work as agricultural labourer. Which one of the following is the opportunity cost of the agricultural labourer?

(a) ₹ 1100 (b) ₹ 700
(c) ₹ 400 (d) ₹ 300

(d) The opportunity cost is the loss of potential gain from other alternatives when one alternative is chosen. The opportunity cost is the 'cost' incurred by not enjoying the benefit associated with the best alternative choice. Opportunity cost is a key concept in economics and has been described as expressing "the basic relationship between scarcity and choice".

Opportunity cost requires sacrifices. If there is no sacrifice involved in a decision, there will be no opportunity cost. In this regard the opportunity costs not involving cash flows are not recorded in the books of accounts, but they are important considerations in business decisions. So, when an agricultural labourer refuses to work as babysitter, she has incurred an opportunity cost of ₹ 300.

117. Match List-I with List-II and select the correct answer using the codes given below the Lists :

	List-I (Market structure)		List-II (Characteristic)
A	Perfect competition	1.	Only one producer selling one commodity
B	Monopoly	2.	Few producers selling similar or almost similar products
C	Monopolistic Competition	3.	Many producers selling differentiated products
D	Oligopoly	4.	Many producers selling similar products

Codes

	A	B	C	D
(a)	4	3	1	2
(b)	4	1	3	2
(c)	2	1	3	4
(d)	2	3	1	4

(b) Perfect competition is the situation prevailing in a market in which buyers and sellers are so numerous and well informed that all elements of monopoly are absent and the market price of a commodity is beyond the control of individual buyers and sellers.

A monopoly is a specific type of economic market structure. A monopoly exists when a specific person or enterprise is the only supplier of a particular good. As a result, monopolies are characterised by a lack of competition within the market producing a good or service.

Monopolistic competition is a type of imperfect competition such that many producers sell products that are differentiated from one another and hence are not perfect substitutes. An oligopoly is a market form wherein a market or industry is dominated by a small number of large sellers.

118. Which one of the following was the host country for World Tourism Day, 2019?

(a) The USA
(b) India
(c) Russia
(d) Canada

(b) World Tourism Day is observed every year on 27th September with the objective to raise awareness about the importance of tourism among the global community and to encourage its social, cultural, political and economic values. The World Tourism Day celebrations have been led by the United Nations World Tourism Organisation (UNWTO) since 1970.

The World Tourism Day is hosted by a different country every year. World Tourism Day, 2019 was hosted by India for the very first time at New Delhi. This year the World Tourism Day theme was "Tourism and Jobs: A better future for all".

Djibouti and Addis Ababa hosted the World Tourism Day, 2020 on the theme "Tourism: Building Peace! Fostering Knowledge !"

119. BRICS Summit, 2020 will be hosted by

(a) India (b) China
(c) Russia (d) Brazil

(c) BRICS is the acronym coined for an association of five major emerging national economies: Brazil, Russia, India, China and South Africa. Since 2009, the BRICS nations have met annually at formal summits. Russia hosted the 12th BRICS Summit in July, 2020 at St Petersburg. China hosted the 9th BRICS Summit at Xiamen in September, 2017, while The 11th BRICS Summit was convened in November 2020 at Brasilia, Brazil.

The 2019 BRICS Summit was focused on the theme, 'BRICS: Economic Growth for an Innovative Future'.

120. The Government of India has recently constituted a civilian award in the name of Sardar Vallabhbhai Patel in the field of contribution to

(a) unity and integrity of India
(b) art and culture
(c) social work
(d) entrepreneurship

(a) On 20th September, 2019 Government of India has instituted Sardar Vallabhbhai Patel Award. It is highest civilian award in the field of contribution to the unity and integrity of India. The award seeks to recognise notable and inspiring contributions to promote the cause of national unity and integrity and to reinforce the value of a strong and United India. The award will be announced on the occasion of the National Unity Day. i.e., the birth anniversary of Sardar Patel on 31st October.

CDS

Combined Defence Service

SOLVED PAPER 2019 (II)

PAPER I Elementary Mathematics

Directions (Q. Nos. 1 and 2) *Read the following frequency distribution for two series of observations and answer the given question in below.*

Class interval	Frequency	
	Series-I	Series-II
10 - 20	20	4
20 - 30	15	8
30 - 40	10	4
40 - 50	x	2x
50 - 60	y	y
Total	**100**	**100**

1. What is the mean of frequency distribution of Series-I?

(a) 33.6 (b) 35.6 (c) 37.6 (d) 39.6

(c) First of all we have to find x and y from the given table

$20 + 15 + 10 + x + y = 100$

$x + y = 55$ …(i)

$4 + 8 + 4 + 2x + y = 100$

$2x + y = 84$ …(ii)

On solving Eqs. (i) and (ii), we get

$x = 29, y = 26$

Class interval	Class mean x_i
10-20	15
20-30	25
30-40	35
40-50	45
50-60	55

Mean of frequency distribution of series-I

$\therefore \quad \bar{x} = \dfrac{\Sigma f_1 x_i}{\Sigma f_1} = \dfrac{3760}{100} = 37.6$

2. What is the mode of the frequency distribution of Series-II?

(a) 26 (b) 36 (c) 46 (d) 56

(c)

Since, the highest frequency is 58. Therefore, modal class is 40-50.

$\therefore \quad l = 40, h = 10, f_1 = 58,$

$f_0 = 4, f_2 = 26$

$\therefore \quad \text{Mode} = l + \left(\dfrac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$

$= 40 + \left(\dfrac{58 - 4}{2 \times 58 - 4 - 26}\right) \times 10$

$= 40 + \dfrac{54}{86} \times 10$

$= 40 + 6.2$

$= 46.2 \approx 46$

Directions (Q. No. 3-6) *Read the following information and answer the given question in below.*

Let the distribution of number of scooters of companies X and Y sold by 5 showrooms (A, B, C, D and E) in a certain year be denoted by S1 and the distribution of number of scooters of only company X sold by the five show-rooms in the same year be denoted by S2.

Showroom	A	B	C	D	E	Total number of scooters sold
S1 (in %)	19	21	15	33	12	6400
S2 (in%)	24	18	20	30	8	3000

3. Number of scooters of company Y sold by showroom E is what per cent of the number of scooters of both companies sold by showroom C ?

(a) 52 (b) 54
(c) 55 (d) 56

(c) Scooters of both companies sold by showroom $E = \dfrac{12}{100} \times 6400 = 768$

Scooters of company X sold by showroom $E = \dfrac{8}{100} \times 3000 = 240$

Scooters of company Y sold by

$E = 768 - 240 = 528$

Scooters of both company is i.e., sold by showroom

$C = \dfrac{15}{100} \times 6400 = 960$

$\therefore$ Required % $= \dfrac{528}{960} \times 100 = 55\%$

4. Number of scooters of both the companies sold by showroom B is what percent more than the number of scooters of company X sold by showroom A?

(a) $78\dfrac{2}{3}$ (b) $83\dfrac{1}{3}$
(c) $86\dfrac{2}{3}$ (d) $88\dfrac{1}{3}$

(c) Number of scooters of both the companies X and Y sold by showroom $B = 21\%$ of 6400

$= 6400 \times \dfrac{21}{100} = 1344$

Number of scooters of company X sold by showroom $A = 24\%$ of 3000

$= 3000 \times \frac{24}{100} = 720$

$\therefore$ Required percent

$= \frac{1344 - 720}{720} \times 100$

$= \frac{624}{720} \times 100$

$= \frac{260}{3} = 86\frac{2}{3}\%$

5. What is the average number of scooters of company Y sold by the showrooms A, C and E?

(a) $461\frac{1}{3}$ (b) $431\frac{1}{3}$
(c) $426\frac{1}{3}$ (d) $416\frac{1}{3}$

(**a**) Number of scooters of both the companies X and Y sold by showrooms A, C and E
$= (19 + 15 + 12)\%$ of 6400

$= 6400 \times \frac{46}{100} = 2944$

Number of scooters of company X sold by showrooms A, C and E
$= (24 + 20 + 8)\%$ of 3000

$= 3000 \times \frac{52}{100} = 1560$

$\therefore$ Number of scooters of company Y sold by showrooms A, C and E

$= 2944 - 1560$

$= 1384$

$\therefore$ Required average

$= \frac{1384}{3} = 461\frac{1}{3}$

6. What is the difference between the number of scooters of both companies sold by showroom A and total number of scooters of company X sold by showrooms B and E together?

(a) 416 (b) 426
(c) 432 (d) 436

(**d**) Number of scooters of both the companies X and Y sold by showroom $A = 19\%$ of 6400

$= 6400 \times \frac{19}{100} = 1216$

Total number of scooters of company X sold by showroom B and E together
$= (18 + 8)\%$ of 3000

$= 3000 \times \frac{26}{100} = 780$

$\therefore$ Required difference

$= 1216 - 780 = 436$

Directions (Q. Nos. 7-10) *Read the following information and answer the given question in below*

The data shows that Indian roads are turning deadlier over the years.

Year	2014	2015	2016	2017
Number of bikers killed	40957	46070	52750	48746
Number of pedestrians killed	12330	15746	15746	20457
Number of cyclists killed	4037	2585	2585	3559

7. What was the average number of pedestrians killed per day in the year 2017?

(a) 51 (b) 53 (c) 54 (d) 56

(***d***) Since, 2017 is not a leap year, total number of days in 2017 = 365 days.

$\therefore$ Required average $= \frac{20457}{365} = 56.05$

≈ 56 pedestrians.

8. What is the approximate percentage change in the pedestrains fatalities during the period 2014-17?

(a) 66% (b) 68% (c) 71% (d) 76%

(**a**) Required percentage

$= \frac{20457 - 12330}{12330} \times 100$

$= \frac{81270}{1233} = 65.91\% \approx 66\%$

9. What is the average number of bikers killed daily in road accidents in the year 2017?

(a) 163 (b) 152
(c) 147 (d) 134

(***d***) Since, 2017 is not a leap year, total number of days in 2017 = 365 days

$\therefore$ Required average $= \frac{48746}{365}$

$= 133.55$

≈ 134 bikers.

10. What is the average number of cyclists killed daily in road accidents in 2017?

(a) 10 (b) 12 (c) 19 (d) 21

(***a***) Since, 2017 is not a leap year, total number of days in 2017 = 365 days

$\therefore$ Required average $= \frac{3559}{365} = 9.75$

≈ 10 cyclists.

11. Let a and b be two positive real numbers such that $a\sqrt{a} + b\sqrt{b} = 32$ and $a\sqrt{b} + b\sqrt{a} = 31$. What is the value of $\frac{5(a+b)}{7}$?

(a) 5 (b) 7
(c) 9
(d) Cannot be determined

(***a***) Given, $a\sqrt{a} + b\sqrt{b} = 32$ ...(i)

$a\sqrt{b} + b\sqrt{a} = 31$ ...(ii)

On squaring both sides Eqs. (i) and (ii), we get

$(a\sqrt{a} + b\sqrt{b})^2 = 32^2$

$[\because (a + b)^2 = a^2 + b^2 + 2ab]$

$\therefore a^3 + b^3 + 2ab\sqrt{ab} = 1024$ ...(iii)

$(a\sqrt{b} + b\sqrt{a})^2 = 31^2$

$a^2b + b^2a + 2ab\sqrt{ab} = 961$ ...(iv)

On subtracting Eq. (iv) from Eq. (iii), we get

$(a^3 + b^3 + 2ab\sqrt{ab})$

$- (a^2b + b^2a + 2ab\sqrt{ab})$

$= 1024 - 961$

$a^3 + b^3 - a^2b - b^2a = 63$

$a^3 - a^2b - b^2a + b^3 = 63$

$a^2(a - b) - b^2(a - b) = 63$

$(a^2 - b^2)(a - b) = 63$

$(a + b)(a - b)(a - b) = 63$

$\{\because a^2 - b^2 = (a + b)(a - b)\}$

$(a + b)(a - b)^2 = 63$

$(a + b)$ and $(a - b)^2$ must be a co-prime number, by factorising 63 we get 7 and 9 that can satisfy the above equation.

$\because \quad a + b = 7$

$(a - b)^2 = 9 \Rightarrow a - b = \pm 3$

If we take, $a - b = -3$, we will get b negative that can not be possible,

$\because \quad a + b = 7$

$a - b = 3$

On solving this, we get

$a = 5$

$\Rightarrow \quad b = 2$

Value of $\frac{5(a + b)}{7} = \frac{5(5 + 2)}{7} = 5$

12. If $x = \frac{1 + \sqrt{3}}{2}$ and $y = x^3$, then y satisfies which one of the following equations?

(a) $8y^2 - 20y - 1 = 0$
(b) $8y^2 + 20y - 1 = 0$
(c) $8y^2 + 20y + 1 = 0$
(d) $8y^2 - 20y + 1 = 0$

(*a*) Given, $y = x^3$ ...(i)

and $x = \frac{1+\sqrt{3}}{2}$ ...(ii)

On putting $x = \frac{1+\sqrt{3}}{2}$ in Eq. (i), we get

$$y = \left[\frac{1+\sqrt{3}}{2}\right]^3$$

$$y = \frac{(1+\sqrt{3})^3}{2^3}$$

$$= \frac{1+3\sqrt{3}+3\sqrt{3}(1+\sqrt{3})}{8}$$

$[\because (a+b)^3 = a^3+b^3+3ab(a+b)]$

$$y = \frac{1+3\sqrt{3}+3\sqrt{3}+9}{8}$$

$$y = \frac{5+3\sqrt{3}}{4}$$

$4y = 5+3\sqrt{3}$ ...(iii)

On squaring both sides in Eq. (iii), we get

$(4y)^2 = (5+3\sqrt{3})^2$

$16y^2 = 25+27+30\sqrt{3}$

$[\because (a+b)^2 = a^2+b^2+2ab]$

$16y^2 = 52+30\sqrt{3}$

$8y^2 = 26+15\sqrt{3}$ ...(iv)

On multiplying Eq. (iii) by 5, we get

$20y = 25+15\sqrt{3}$ ...(v)

Eq. (v) substract from Eq. (iv),

$\Rightarrow 8y^2 - 20y = 1$

$\Rightarrow 8y^2 - 20y - 1 = 0$

Option (a) will satisfy '*y*'.

13. HCF of two numbers is 12. Which one of the following can never be their LCM?

(a) 80 (b) 60 (c) 36 (d) 24

(*a*) HCF of two numbers is 12.

LCM is common multiple for the two numbers.

So, the required answer is 80 because all other are multiple of 12.

14. Consider the following statements

1. Unit digit in 17^{174} is 7.
2. Difference of the squares of any two odd numbers is always divisible by 8.
3. Adding 1 to the product of two consecutive odd numbers makes it a perfect square.

Which of the above statements are correct?

(a) 1, 2 and 3 (b) Only 1 and 2
(c) Only 2 and 3 (d) Only 1 and 3

(*c*)

1. The last digit of a number depends on the unit digit, so 17 as 7 will show same characteristic.

We know, the cyclicity of 7 is 5.

While solving this problem we will take the power and divide by 5.

174 can be written as $5n+4$, where $n = 34$

Hence, $17^{174} = 17^{(5\times 34+4)}$

Hence, 7^4 and 17^{174} will have same digit in the unit place, so answer is 1.

2. Let any two odd number be 5 and 3.

According to the question,

$\Rightarrow 5^2 - 3^2 = 25 - 9 = 16$

16 which is divisible by 8.

Let check once more with non consecutive odd number be 7 and 3.

According to the question,

$\Rightarrow 7^2 - 3^2 = 49 - 9 = 40$

40 which is divisible by 8.

So, difference of the square of any two odd number is divisible by 8.

3. Let two consecutive number be x and $x+2$.

According to the question, $x(x+2)+1$

$= x^2+2x+1 = (x+1)^2$

Thus, adding 1 to the product of two consecutive odd numbers makes it perfect square.

Hence, statements 2 and 3 are correct and statement 1 is incorrect.

15. The rate of interest on two different schemes is the same and it is 20%. But in one of the schemes, the interest is compounded half yearly and in the other the interest is compounded annually. Equal amounts are invested in the schemes. If the difference of the returns after 2 yr is ₹ 482, then what is the principal amount in each scheme?

(a) ₹ 10000 (b) ₹ 16000
(c) ₹ 20000 (d) ₹ 24000

(*c*) Let the principal be $100x$.

In 1st scheme $P = 100x$,

Rate $(R) = \frac{20}{2}\% = 10\%$,

Time $(T) = 4$ half yearly.

$$CI_1 = \left[P\left(1+\frac{r}{100}\right)^T - P\right]$$

$$= P\left[\left(1+\frac{r}{100}\right)^T - 1\right]$$

$$= 100x\left[\left(1+\frac{10}{100}\right)^4 - 1\right]$$

$$= 100x\left[\left(\frac{11}{10}\right)^4 - 1\right]$$

$$= 100x\left[\frac{14641-10000}{10000}\right]$$

$$= 100x \times \frac{4641}{10000}$$

$\Rightarrow CI_1 = 46.41x$

In 2nd scheme

$P = 100x$, $R = 20\%$, $T = 2$ yr

$$CI_2 = P\left[\left(1+\frac{r}{100}\right)^T - 1\right]$$

$$= 100x\left[\left(1+\frac{20}{100}\right)^2 - 1\right]$$

$$= 100x\left[\left(\frac{6}{5}\right)^2 - 1\right]$$

$$= 100x\left[\frac{36-25}{25}\right]$$

$CI_2 = 44x$

According to the question,

$CI_1 - CI_2 = 482$

$46.41x - 44x = 482$

$2.41x = 482$

$x = \frac{482}{2.41}$

$\Rightarrow x = 200$

$\therefore$ Principal $= 100 \times 200$
$= ₹20000$

16. For what value of k can the expression x^3+kx^2-7x+6 be resolved into three linear factors?

(a) 0 (b) 1 (c) 2 (d) 3

(*a*) For a particular value of k we must get 3 different value of x.

We can satisfy the option in this type of question, and by solving we will get that only option (a) will satisfy the equation.

$\Rightarrow x^3+kx^2-7x+6 = 0$

On putting $k = 0$,

$\Rightarrow x^3+x^2-x^2-7x+6 = 0$

$\Rightarrow x^3-x^2+x^2-x-6x+6 = 0$

$\Rightarrow x^2(x-1)+x(x-1)-6(x-1) = 0$

$\Rightarrow (x-1)(x^2+x-6) = 0$

$\Rightarrow (x-1)(x^2+3x-2x-6) = 0$

$\Rightarrow (x-1)(x(x+3)-2(x+3)) = 0$

$\Rightarrow (x-1)(x-2)(x+3) = 0$

We will get, $x = 1, 2$, and -3.

17. X, Y and Z start at same point and same time in the same direction to run around a circular stadium. X completes a round in 252 s, Y in 308 s and Z in 198 s. After what time will they meet again at the starting point?

(a) 26 min 18 s (b) 42 min 36 s
(c) 45 min (d) 46 min 12 s

(d) X completes round in 252 s.
Y completes round in 308 s.
Z completes round in 198 s.
They all meet again at starting together after, LCM of 252, 308 and 198

$252 = 2 \times 2 \times 3 \times 3 \times 7$
$308 = 2 \times 2 \times 7 \times 11$
$198 = 2 \times 3 \times 3 \times 11$

$\therefore$ Required LCM
$= 2 \times 2 \times 3 \times 3 \times 7 \times 11$
$= 2772$ s $= 46$ min 12 s.

18. What is the LCM of $\frac{1}{3}, \frac{5}{6}, \frac{2}{9}, \frac{4}{27}$?

(a) $\frac{5}{18}$ (b) $\frac{1}{27}$
(c) $\frac{10}{27}$ (d) $\frac{20}{3}$

(d) LCM of $\frac{1}{3}, \frac{5}{6}, \frac{2}{9}$ and $\frac{4}{7}$

$\Rightarrow \frac{\text{LCM of numerator}}{\text{HCF of denominator}}$

$\therefore \frac{\text{LCM of 1, 5, 2 and 4}}{\text{HCF of 3, 6, 9, 27}} = \frac{20}{3}$

19. If the equations $x^2 + 5x + 6 = 0$ and $x^2 + kx + 1 = 0$ have a common root, then what is the value of k ?

(a) $-\frac{5}{2}$ or $-\frac{10}{3}$ (b) $\frac{5}{2}$ or $\frac{10}{3}$
(c) $\frac{5}{2}$ or $-\frac{10}{3}$ (d) $-\frac{5}{2}$ or $\frac{10}{3}$

(b) Given, equations

$x^2 + 5x + 6 = 0$ ...(i)
$x^2 + kx + 1 = 0$ ...(ii)

If both equations have common roots then,

$\Rightarrow x^2 + 5x + 6 = 0$
$\Rightarrow x^2 + 3x + 2x + 6 = 0$
$\Rightarrow x(x + 3) + 2(x + 3) = 0$
$\Rightarrow (x + 2)(x + 3) = 0$
$\Rightarrow x = -2, -3$

The value of x must satisfy the Eq. (ii)

On putting $x = -2$
$\Rightarrow x^2 + kx + 1 = 0$
$\Rightarrow (-2)^2 + k(-2) + 1 = 0$
$\Rightarrow 4 - 2k + 1 = 0$
$\Rightarrow 2k = 5 \Rightarrow k = \frac{5}{2}$

On putting $x = -3$
$\Rightarrow x^2 + kx + 1 = 0$
$\Rightarrow (-3)^2 + k(-3) + 1 = 0$
$9 - 3k + 1 = 0 \Rightarrow 3k = 10$
$\therefore k = \frac{10}{3}$

20. A lent ₹ 25000 to B and at the same time lent some amount to C at same 7% simple interest. After 4 yr A received ₹ 11200 as interest from B and C. How much did A lend to C?

(a) ₹ 20000 (b) ₹ 25000
(c) ₹ 15000 (d) ₹ 10000

(c) Let A lent x amount to C.
According to the question,

$\Rightarrow \frac{25000 \times 7 \times 4}{100} + \frac{x \times 7 \times 4}{100} = 11200$ $\left[\because \text{SI} = \frac{P \times R \times T}{100}\right]$

$7000 + \frac{28x}{100} = 11200$
$28x = 420000$
$\therefore x = 15000$

21. A trader sells two computers at the same price, making a profit of 30% on one and a loss of 30% on the other. What is the net loss or profit percentage on the transaction?

(a) 6% loss (b) 6% gain
(c) 9% loss (d) 9% gain

(c) Let the selling price of each computers be $100x$.

Cost price of computer 1
$= \text{selling price} \times \left(\frac{100}{100 + \text{profit } \%}\right)$
$= 100x \times \frac{100}{130} = \frac{1000x}{13}$

Cost price of computer 2
$= \text{selling price} \times \left(\frac{100}{100 - \text{loss } \%}\right)$
$= 100x \times \frac{100}{70} = \frac{1000x}{7}$

Total cost price of computers 1 and 2
$= \frac{1000x}{13} + \frac{1000x}{7} = \frac{20000x}{91}$

Total selling price of both computers
$= 100x + 100x = 200x$

Net loss = Total CP – Total SP
$= \frac{20000x}{91} - 200x$
$= \frac{20000x - 18200x}{91} = \frac{1800}{91}x$

Net loss% $= \frac{\frac{1800}{91}x}{\frac{20000x}{91}} \times 100$
$= 9\%$

Alternate Method

When two article sell at same selling price,one at a profit of a% another at a loss of a%, then there is always a loss of $\left(\frac{a}{10}\right)^2$%.

Here, $a = 30\%$

Required loss percentage
$= \left(\frac{a}{10}\right)^2 = \left(\frac{30}{10}\right)^2 = 9\%$

22. The monthly incomes of A and B are in the ratio 4 : 3. Each save ₹ 600. If their expenditures are in the ratio 3 : 2, then what is the monthly income of A?

(a) ₹ 1800 (b) ₹ 2000
(c) ₹ 2400 (d) ₹ 3600

(c) Let monthly income of A and B be $4x$ and $3x$.

According to the question,

$\frac{A\text{'s expenditure}}{B\text{'s expenditure}} = \frac{3}{2}$
[$\because$ Expenditure = Income – Saving]

$\frac{4x - 600}{3x - 600} = \frac{3}{2}$
$8x - 1200 = 9x - 1800$
$x = 600$

A's monthly income $= 4 \times 600$
= ₹ 2400

23. The train fare and bus fare between two stations is in the ratio 3 : 4. If the train fare increases by 20% and bus fare increases by 30%, then what is the ratio between revised train fare and revised bus fare?

(a) $\frac{9}{13}$ (b) $\frac{17}{12}$
(c) $\frac{32}{43}$ (d) $\frac{19}{21}$

(a) Let the fare of train and bus be $3x$ and $4x$.

According to the question,

New fare of train $= 3x \times \frac{120}{100} = 3.6x$

New fare of bus $= 4x \times \frac{130}{100} = 5.2x$

$\therefore$ Required ratio $= \frac{3.6x}{5.2x} = \frac{36}{52} = \frac{9}{13}$

24. When N is divided by 17, the quotient is equal to 182. The difference between the quotient and the remainder is 175. What is the value of N?

(a) 2975 (b) 3094 (c) 3101 (d) 3269

(c) Let the remainder be x.

According to the question,

$182 - x = 175 \Rightarrow x = 7$

$\therefore$ Number

$= \text{Divisor} \times \text{Quotient} + \text{Remainder}$

$= 17 \times 182 + 7$

$= 3094 + 7 = 3101$

25. A stock of food grains is enough for 240 men for 48 days. How long will the same stock last for 160 men?

(a) 72 days (b) 64 days
(c) 60 days (d) 54 days

(a) Let total stock of food grains be 1 unit.

240 men eat 1 units food grains in

$= 48$ days

1 man eat 1 unit food grains in

$= 48 \times 240$ days

(By unitary method)

$\therefore$ 160 men eat 1 unit food grains in

$= \dfrac{48 \times 240}{160}$

$= 72$ days

26. A hollow right circular cylindrical vessel of volume V whose diameter is equal to its height, is completely filled with water. A heavy sphere of maximum possible volume is then completely immersed in the vessel. What volume of water remains in the vessel?

(a) $\dfrac{V}{2}$ (b) $\dfrac{V}{3}$
(c) $\dfrac{2V}{3}$ (d) $\dfrac{V}{4}$

(b) Let the radius of cylinder $= r$

Height of cylinder $(h) = 2r$

Maximum possible sphere can be immersed of radius r.

Volume of cylinder $(V) = \pi r^2 h$

$= \pi r^2 (2r) = 2\pi r^3$

Volume of sphere $= \dfrac{4}{3}\pi r^3$

After immersing sphere in the cylinder the remain water

$= 2\pi r^3 - \dfrac{4}{3}\pi r^3$

$= \dfrac{6\pi r^3 - 4\pi r^3}{3} = \dfrac{2\pi r^3}{3}$

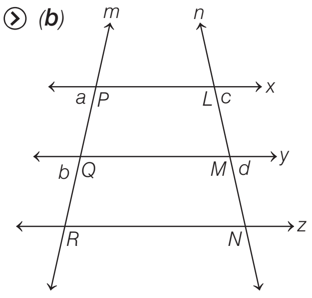
Volume of cylinder $(V) = 2\pi r^3$

$\therefore$ Remaining volume $= \dfrac{V}{3}$

27. Three parallel lines x, y and z are cut by two transversals m and n. Transversal m cuts the lines x, y, z at P, Q, R, respectively and transversal n cuts the lines x, y, z at L, M, N, respectively. If $PQ = 3$ cm, $QR = 9$ cm and $MN = 10.5$ cm, then what is the length of LM?

(a) 3 cm (b) 3.5 cm
(c) 4 cm (d) 4.5 cm

(b)



$x \parallel y \parallel z$

Then, by Basic proportionality theorem,

$\dfrac{a}{b} = \dfrac{c}{d}$ or $\dfrac{a}{c} = \dfrac{b}{d}$

Here, $a = PQ = 3$ cm

$b = QR = 9$ cm

$c = LM = ?$

$d = MN = 10.5$

$\Rightarrow \dfrac{a}{b} = \dfrac{c}{d}$

$\Rightarrow \dfrac{3}{9} = \dfrac{c}{10.5}$

$c = 3.5 = LM$

$LM = 3.5$ cm

28. The area of a sector of a circle of radius 4 cm is 25.6 cm^2. What is the radian measure of the arc of the sector?

(a) 2.3 (b) 3.2 (c) 3.3 (d) 3.4

(b) Area of a sector $= 25.6$ cm^2

Length of arc $(l) = \dfrac{A \times 2}{r}$

$l = \dfrac{25.6 \times 2}{4} \Rightarrow l = 12.8,\ \theta = \dfrac{l}{r}$

θ = angle in radian

$\therefore \theta = \dfrac{12.8}{4} = \dfrac{128}{40} = \dfrac{16}{5} = 3.2$

29. Which one of the following is correct in respect of a right angled triangle?

(a) Its orthocentre lies inside the triangle
(b) Its orthocentre lies outside the triangle
(c) Its orthocentre lies on the triangle
(d) It has no orthocentre

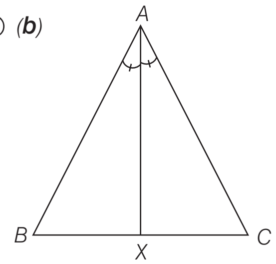
(c) Orthocentre of a right angle triangle lies on the triangle.

It is a property of a right angle triangle.

30. Let the bisector of the angle BAC of a triangle ABC meet BC in X. Which one of the following is correct?

(a) $AB < BX$ (b) $AB > BX$
(c) $AX = CX$ (d) None of these

(b)



$\angle BXA$ is exterior angle of ΔAXC

$\angle BXA = \angle XAC + \angle ACX$

$\therefore \angle BXA > \angle XAC \quad (\angle XAC = \angle BAX)$

$\angle BXA > \angle BAC$

$\therefore AB > BX$ (side opposite to greater angle is greater)

31. What is the value of
$\log_{10}(\cos\theta) + \log_{10}(\sin\theta)$
$+ \log_{10}(\tan\theta) + \log_{10}(\cot\theta)$
$+ \log_{10}(\sec\theta) + \log_{10}(\operatorname{cosec}\theta)$?

(a) –1 (b) 0 (c) 0.5 (d) 1

(b) $\log_{10}(\cos\theta) + \log_{10}(\sin\theta)$
$+ \log_{10}(\tan\theta)$
$+ \log_{10}(\cot\theta) + \log_{10}(\sec\theta)$
$+ \log_{10}(\operatorname{cosec}\theta)$

$\Rightarrow \log_{10}(\cos\theta \cdot \sin\theta)$
$+ \log_{10}(\tan\theta \cdot \cot\theta)$
$+ \log_{10}(\sec\theta \cdot \operatorname{cosec}\theta)$

$[\because \log_a b + \log_a c = \log_a b \cdot c]$

$\Rightarrow \log_{10}(\cos\theta \cdot \sin\theta)$
$+ \log_{10}(\sec\theta \cdot \operatorname{cosec}\theta) + \log_{10}(1)$

$\Rightarrow \log_{10}(\cos\theta \cdot \sin\theta \cdot \sec\theta \cdot \operatorname{cosec}\theta) + 0$

$[\because \log_{10} 1 = 0]$

$\Rightarrow \log_{10}(1) \Rightarrow 0$

32. If $\cos^2 x + \cos x = 1$, then what is the value of $\sin^{12} x + 3\sin^{10} x + 3\sin^8 x + \sin^6 x$?

(a) 1 (b) 2 (c) 4 (d) 8

(a) Given, $\cos^2 x + \cos x = 1$

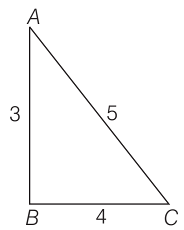
$$\cos x = 1 - \cos^2 x$$
$$\cos x = \sin^2 x$$
$$\Rightarrow \sin^{12} x + 3\sin^{10} x + 3\sin^8 x + \sin^6 x$$
$$\Rightarrow (\sin^4 x)^3 + 3(\sin^4)^2 \sin^2 x + 3(\sin^2 x)^2 \cdot \sin^4 x + (\sin^2 x)^3$$
$$[\because (a+b)^3 = a^3 + b^3 + 3a^2b + 3b^2a)]$$
$$\Rightarrow (\sin^4 x + \sin^2 x)^3$$
$$[\because \sin^2 x = \cos x, \sin^4 x = \cos^2 x]$$
$$\Rightarrow (\cos^2 x + \cos x)^3$$
$$[\text{given}, \cos^2 x + \cos x = 1]$$
$$\Rightarrow (1)^3 = 1$$

Option (a) is correct.

33. If $0 < \theta < 90°$, $\sin\theta = 3/5$ and $x = \cot\theta$, then what is the value of $1 + 3x + 9x^2 + 27x^3 + 81x^4 + 243x^5$?

(a) 941 (b) 1000 (c) 1220 (d) 1365

(d) Given, $\sin\theta = \dfrac{3}{5}$



In a right angle triangle,

$$\sin\theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}}, \frac{P}{H} = \frac{3}{5}$$

Using Pythagoras theorem,

$$AC^2 = AB^2 + BC^2,\ 5^2 = 3^2 + BC^2$$
$$\Rightarrow BC^2 = 5^2 - 3^2 \Rightarrow BC = 4$$
$$\cot\theta = \frac{\text{Base}}{\text{Perpendicular}} = \frac{BC}{AB}$$
$$\cot\theta = \frac{4}{3} \Rightarrow x = \frac{4}{3}\ [\text{given}, \cot\theta = x]$$
$$\Rightarrow 1 + 3x + 9x^2 + 27x^3 + 81x^4 + 243x^5$$
$$\Rightarrow 1 + 3x + 9x^2 + 27x^3 (1 + 3x + 9x^2)$$
$$\Rightarrow (1 + 3x + 9x^2)(1 + 27x^3)$$

Now, on putting $x = \dfrac{4}{3}$

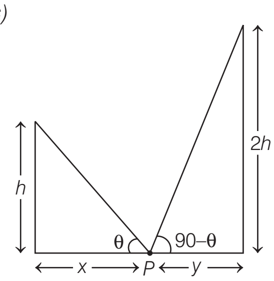
$$\left[1 + 3 \times \frac{4}{3} + 9 \times \left(\frac{4}{3}\right)^2\right]\left[1 + 27 \times \left(\frac{4}{3}\right)^3\right]$$
$$\Rightarrow \left(1 + 3 \times \frac{4}{3} + 9 \times \frac{16}{9}\right)\left(1 + 27 \times \frac{64}{27}\right)$$
$$\Rightarrow (1 + 4 + 16)(1 + 64)$$
$$= 21 \times 65 = 1365$$

Option (d) is correct.

34. The angles of elevation of the tops of two pillars of heights h and $2h$ from a point P on the line joining the feet of the two pillars are complementary. If the distances of the foot of the pillars from the point P are x and y respectively, then which one of the following is correct?

(a) $2h^2 = x^2 y$ (b) $2h^2 = x y^2$
(c) $2h^2 = xy$ (d) $2h^2 = x^2 y^2$

(c)



According to the question,

$$\tan\theta = \frac{h}{x} \quad \ldots(i)$$
$$\tan(90 - \theta) = \frac{2h}{y}$$
$$\cot\theta = \frac{2h}{y} \quad \ldots(ii)$$
$$[\because \tan(90° - \theta) = \cot\theta]$$

On multiplying Eqs. (i) and (ii), we get

$$\tan\theta \cdot \cot\theta = \frac{h}{x} \times \frac{2h}{y}$$
$$1 = \frac{2h^2}{xy} \quad [\tan\theta.\cot\theta = 1]$$
$$\therefore \quad 2h^2 = xy$$

35. What is the value of $\dfrac{\sin 19°}{\cos 71°} + \dfrac{\cos 73°}{\sin 17°}$?

(a) 0 (b) 1 (c) 2 (d) 4

(c) Given, $\dfrac{\sin 19°}{\cos 71°} + \dfrac{\cos 73°}{\sin 17°}$

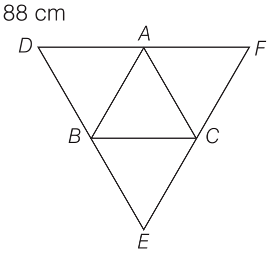
$$= \frac{\sin 19°}{\cos(90 - 19°)} + \frac{\cos 73°}{\sin(90 - 73°)}$$
$$= \frac{\sin 19°}{\sin 19°} + \frac{\cos 73°}{\cos 73°} = 1 + 1 = 2$$
$$\left[\begin{array}{l}\because \sin(90° - \theta) = \cos\theta, \\ \cos(90° - \theta) = \sin\theta\end{array}\right]$$

Option (c) is correct.

36. The perimeter of a triangle is 22 cm. Through each vertex of the triangle, a straight line parallel to the opposite side is drawn. What is the perimeter of triangle formed by these lines?

(a) 33 cm (b) 44 cm (c) 66 cm
(d) 88 cm

(b)



As, $AC \parallel BE$ and $AB \parallel CE$, $ABEC$ forms a parallelogram.

where, $AC = BE$ and $AB = CE$

Similarly,

As, $AC \parallel DB$ and $BC \parallel AD$, $ADBC$ forms a second parallelogram.

where, $AC = BD$ and $AD = BC$

With this we can conclude that B is mid-point of DE and AC is half of DE.

Similarly, it will be true for FD and EF also.

The ratio of the perimeter of new triangle to that original triangle is 2 : 1.

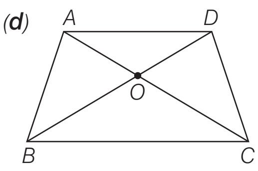
$\therefore$ Required perimeter

$$= 22 \times \frac{2}{1} = 44 \text{ cm}$$

37. The sides AD, BC of a trapezium $ABCD$ are parallel and the diagonals AC and BD meet at O. If the area of triangle AOB is 3 cm^2 and the area of triangle BDC is 8 cm^2, then what is the area of triangle AOD?

(a) 8 cm^2 (b) 5 cm^2
(c) 3.6 cm^2 (d) 8 cm^2

(d)



ΔABC and ΔBDC

Lie on the same base BC and between same parallel AD and BC.

$\therefore ar(\Delta ABC) = ar(BDC)$

Subtracting $ar(\Delta BOC)$ both sides, we get $ar(\Delta ABC) - ar(\Delta BOC)$

$ar(BDC) - ar(BOC)$,

$ar(\Delta AOB) = ar(DOC)$

Then, $ar(\Delta DOC) = 3 \text{ cm}^2$

So, $ar(\Delta BOC) = ar(BDC) - ar(\Delta DOC)$

$= 8\text{ cm}^2 - 3\text{ cm}^2 = 5\text{ cm}^2$

In ΔABO and ΔBOC,

$$\frac{ar(\Delta ABO)}{ar(\Delta BOC)} = \frac{\frac{1}{2} \times AO \times \text{height}}{\frac{1}{2} \times CO \times \text{height}}$$

$$\frac{3}{5} = \frac{AO}{CO}$$

$\Delta AOD \sim \Delta BOC$ {AAA}

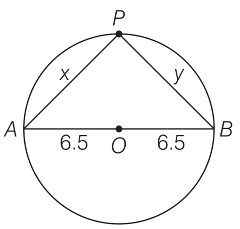
$$\frac{ar(\Delta AOD)}{ar(BOC)} = \left(\frac{AO}{CO}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$$

$$ar(\Delta AOD) = \frac{9}{25} \times 5 = 1.8\text{ cm}^2$$

38. A line segment AB is the diameter of a circle with centre at O having radius 6.5 cm. Point P is in the plane of the circle such that $AP = x$ and $BP = y$. In which one of the following cases the point P does not lie on the circle?

(a) $x = 6.5$ cm and $y = 6.5$ cm
(b) $x = 12$ cm and $y = 5$ cm
(c) $x = 5$ cm and $y = 12$ cm
(d) $x = 0$ cm and $y = 13$ cm

(a)



Circle with centre O having AB as a diameter.

We know that, diameter create 90° angle on the circumference of the circle.

$\therefore \quad x^2 + y^2 = 13^2$

$x^2 + y^2 = 169$

Now, by putting option we can get the required answer,

On putting option (a)

$x = 6.5, y = 6.5$

$\Rightarrow (6.5)^2 + (6.5)^2 = 169$

$\Rightarrow 42.25 + 42.25 = 169$

$\Rightarrow 84.50 \neq 169$

So, option (a) cannot be possible.

39. The perimeters of two similar triangles ABC and PQR are 75 cm and 50 cm, respectively. If the length of one side of the triangle PQR is 20 cm, then what is the length of corresponding side of the triangle ABC?

(a) 25 cm (b) 30 cm
(c) 40 cm (d) 45 cm

(b) ΔABC and ΔPQR are two similar triangles.

Then, $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = k$

$\therefore \; AB = kPQ, BC = kQR, AC = kPR$

$AB + BC + AC = K(PQ + QR + PR)$

$75 = 50 \times k$

$k = \frac{75}{50}$

$\Rightarrow \quad k = \frac{3}{2}$

$\frac{AB}{PQ} = \frac{3}{2}$

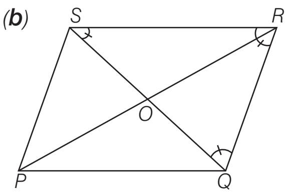
$AB = \frac{3}{2} \times PQ = \frac{3}{2} \times 20$

$\therefore \quad AB = 30$ cm

40. Let $PQRS$ be a parallelogram whose diagonals PR and QS intersect at O. If triangle QRS is an equilateral triangle having a side oflength 10 cm, then what is the length of the diagonal PR?

(a) $5\sqrt{3}$ cm (b) $10\sqrt{3}$ cm
(c) $15\sqrt{3}$ cm (d) $20\sqrt{3}$ cm

(b)



In triangle QRS, each angle is 60°. AO is altitude of triangle QRS. Altitude in equilateral triangle $= \frac{\sqrt{3}}{2}a$

Altitude $= \frac{\sqrt{3}}{2} \times 10 \quad [\because a = 10 \text{ cm}]$

$= 5\sqrt{3}$ cm

ΔQSR is similar to ΔPQS and both are equilateral triangle with side 10 cm.

Then, diagonal $PR = 2 \times$ Altitude

$= 2 \times 5\sqrt{3}$

$= 10\sqrt{3}$ cm

41. The areas of three adjacent faces of a cuboid are x, y and z. If V is the volume of the cuboid, then which one of the following is correct?

(a) $V = xyz$ (b) $V^2 = xyz$
(c) $V^3 = xyz$ (d) $V = (xyz)^2$

(b) Area of three faces of cuboid are x, y and z. Let l, b and h be the length, breadth and height of cuboid.

According to the question,

$x = l \times b$

$y = b \times h$

$z = h \times l$

$xyz = l \times b \times b \times h \times h \times l$

$= l^2 b^2 h^2 = (lbh)^2$

$\therefore xyz = V^2 \quad (\because V = l \times b \times h)$

42. If l is the length of the median of an equilateral triangle, then what is its area?

(a) $\frac{\sqrt{3}l^2}{3}$ (b) $\frac{\sqrt{3}l^2}{2}$ (c) $\sqrt{3}\,l^2$ (d) $2l^2$

(a) Median of a equilateral triangle

$= \frac{\sqrt{3}}{2}a$

$l = \frac{\sqrt{3}}{2}a \Rightarrow a = \frac{2l}{\sqrt{3}}$

Area of equilateral triangle $= \frac{\sqrt{3}}{4}a^2$

$= \frac{\sqrt{3}}{4} \times \left(\frac{2l}{\sqrt{3}}\right)^2$

$= \frac{\sqrt{3}}{4} \times \frac{4l^2}{3} = \frac{\sqrt{3}l^2}{3}$

43. A piece of wire is in the form of a sector of a circle of radius 20 cm, subtending an angle 150° at the centre. If it is bent in the form of a circle, then what will be its radius?

(a) $\frac{19}{3}$ cm (b) 7 cm
(c) 8 cm (d) None of these

(d) Radius = 20 cm

Length of arc $(l) = \frac{\theta}{360°} \times 2\pi r$

$= \frac{150°}{360°} \times 2 \times \pi \times 20$

$= \frac{50}{3}\pi$

Perimeter of sector

= Length of are + $2r$

$= \frac{50}{3}\pi + 40$

According to the question,

$2\pi r = \frac{50}{3}\pi + 40$

$\Rightarrow \quad r = \frac{25}{3} + \frac{20}{\pi}$

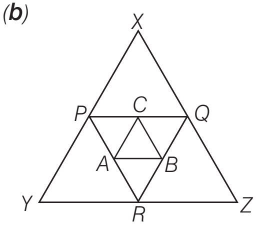
$\Rightarrow \quad r = \frac{25}{3} + \frac{70}{11}$

$\therefore \quad r = \frac{485}{33} = 14.7$ [approx.]

44. Suppose P, Q and R are the mid-points of sides of a triangle of area 128 cm^2. If a triangle ABC is drawn by joining the mid-points of sides of triangle PQR, then what is the area of triangle ABC?

(a) 4 cm^2 (b) 8 cm^2
(c) 16 cm^2 (d) 32 cm^2

(b)



We know, by mid-point theorem, in ΔXYZ and ΔPQR

$$\frac{1}{4}\text{ar}(\Delta XYZ) = \text{ar}(\Delta PQR)$$

Similarly, $\frac{1}{4}\text{ar}(\Delta PQR) = \text{ar}(\Delta ABC)$

$$\frac{1}{16}\text{ar}(\Delta XYZ) = \text{ar}(\Delta ABC)$$

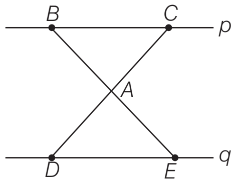
$$\text{ar}(\Delta ABC) = \frac{128}{16}\text{ cm}^2$$

$$\therefore \quad \text{ar}(\Delta ABC) = 8\text{ cm}^2$$

45. Let two lines p and a be parallel. Consider two points B and C on the line p and two points D and E on the line q. The line through B and E intersects the line through C and D at A in between the two lines p and q. If $AC : AD = 4 : 9$, then what is the ratio of area of triangle ABC to that of traingle ADE?

(a) 2 :.3 (b) 4 : 9
(c) 16 : 81 (d) 1: 2

(c)



In ΔABC and ΔADE,

$\angle BAC = \angle DAE$ [opposite angle]

$\angle ACB = \angle ADE$ [alternate interior angle]

$\angle ABC = \angle AED$ [alternate interior angle]

$\because \quad \Delta ABC \approx \Delta AED$

$$\frac{Ar(\Delta ABC)}{Ar(\Delta AED)} = \frac{AC^2}{AD^2} = \frac{4^2}{9^2}$$

$$\therefore \frac{Ar(\Delta ABC)}{Ar(\Delta AED)} = \frac{16}{81}$$

46. An equilateral triangle and a square are constructed using metallic wires of equal length. What is the ratio of area of triangle to that of square?

(a) 3 : 4 (b) 2 : 3
(c) $4\sqrt{3}$: 9 (d) $2\sqrt{3}$: 9

(c) Let side of triangle be 'x' unit.

Let the side of square be 'a' unit.

Perimeter of square = $4 \times a = 4a$ unit

According to the question,

Perimeter of triangle = Perimeter of square

$$3x = 4a$$

$$x = \frac{4a}{3}$$

$$\text{Required ratio} = \frac{\text{Area of triangle}}{\text{Area of square}} = \frac{\frac{\sqrt{3}}{4}x^2}{a^2} = \frac{\frac{\sqrt{3}}{4} \times \frac{4a}{3} \times \frac{4a}{3}}{a^2} = \frac{4\sqrt{3}}{9}$$

$\therefore$ $4\sqrt{3}$: 9

47. All the four sides of a parallelogram are of equal length. The diagonals are in the ratio 1 : 2. If the sum of the lengths of the diagonals is 12 cm, then what is the area of the parallelogram?

(a) 9 cm^2 (b) 12 cm^2
(c) 16 cm^2 (d) 25 cm^2

(c) If all the sides of a parallelogram is equal that mean its a rhombus.

Let diagonal of rhombus are x and $2x$.

According to the question,

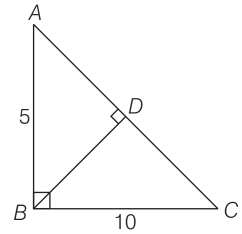
$$x + 2x = 12$$
$$3x = 12$$
$$x = 4$$

$$\therefore \text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 4 \times 8 = 16\text{ cm}^2$$

48. ABC is a triangle right angled at B. If $AB = 5$ cm and $BC = 10$ cm, then what is the length of the perpendicular drawn from the vertex B to the hypotenuse?

(a) 4 cm (b) $2\sqrt{5}$ cm
(c) $\frac{4}{\sqrt{5}}$ cm (d) 8 cm

(b)



In ΔABC, $AB = 5$ cm, $BC = 10$ cm

Using Pythagoras theorem, we get

$$AB^2 + BC^2 = AC^2,\ 5^2 + 10^2 = AC^2$$

$$AC = 5\sqrt{5}$$

$$BD = \frac{AB \times BC}{AC} = \frac{5 \times 10}{5\sqrt{5}}$$

$$BD = \frac{10}{\sqrt{5}} = \frac{10}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \therefore BD = 2\sqrt{5}\text{ cm}$$

49. Two cylinders of equal volume have their heights in the ratio 2 : 3. What is the ratio of their radii?

(a) $\sqrt{3}$: 1 (b) $\sqrt{3}$: $\sqrt{2}$
(c) 2 : $\sqrt{3}$ (d) $\sqrt{3}$: 2

(b) Let radius of the cylinders be r_1 and r_2 and their heights be $2x$ and $3x$, respectively.

According to the question,

$$\pi r_1^2 \times 2x = \pi r_2^2 \times 3x$$

[$\because$ Volume = $\pi r^2 h$]

$$\Rightarrow \quad \frac{r_1^2}{r_2^2} = \frac{3}{2}$$

$$\Rightarrow \quad \frac{r_1}{r_2} = \frac{\sqrt{3}}{\sqrt{2}}$$

$$\therefore \quad r_1 : r_2 = \sqrt{3} : \sqrt{2}$$

50. The length and breadth of a rectangle are increased by 20% and 10%, respectively. What is the percentage increase in the area of the rectangle?

(a) 32% (b) 30% (c) 25% (d) 15%

(a) Let length and breadth of rectangle be x and y.

According to the question,

$$\text{New length} = \frac{120}{100} \times x = 1.2x$$

$$\text{New breadth} = \frac{110}{100} \times y = 1.1y$$

Area of rectangle = $x \times y$

Area of new rectangle = $1.2x \times 1.1y = 1.32xy$

$\therefore$ Required percentage

$$= \frac{1.32xy - xy}{xy} \times 100 = \frac{0.32xy}{xy} \times 100 = 32\%$$

Alternate method

Here, $a = 20\%$, $b = 10\%$

Required percentage increase in the area of rectangle

$$= \left(a + b + \frac{a \times b}{100}\right)\%$$

$$= \left(20 + 10 + \frac{20 \times 10}{100}\right)\%$$

$$= (30 + 2)\% = 32\%$$

51. If the length of the hypotenuse of a right angled triangle is 10 cm, then what is the maximum area of such a right angled triangle?

(a) 100 cm^2 (b) 50 cm^2

(c) 25 cm^2 (d) 10 cm^2

(c) $(10)^2 = x^2 = y^2$

For maximum value, $x = y$,

Then, $(10)^2 = x^2 + x^2$

$\Rightarrow \quad 2x^2 = 100$

$x = 5\sqrt{2}$ cm

$\therefore$ Maximum area possible $= \frac{1}{2} \times x \times y$

$$= \frac{5\sqrt{2} \times 5\sqrt{2}}{2}$$

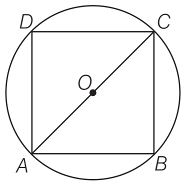
$$= 25 \text{ cm}^2$$

52. A square is drawn such that its vertices are lying on a circle of radius 201 mm. What is the ratio of area of circle to that of square?

(a) 11 : 7 (b) 7 : 11

(c) 20 : 19 (d) 19 : 20

(a)



Let side fo square be x cm.

Diagonal of square

$= 201 \times 2 = 402$ cm

Using Pythagoras theorem,

$x^2 + x^2 = (402)^2$

$\Rightarrow \quad 2x^2 = 402 \times 402$

$x^2 = 201 \times 402$

$\Rightarrow \quad x = 201\sqrt{2}$

$\therefore$ Required ratio $= \frac{\pi r^2}{x^2}$

$$= \frac{\frac{22}{7} \times 201 \times 201}{201\sqrt{2} \times 201\sqrt{2}}$$

$$= \frac{11}{7} = 11 : 7$$

53. A right circular cylinder has a diameter of 20 cm and its curved surface area is 1000 cm^2. What is the volume of the cylinder?

(a) 4000 cm^3 (b) 4500 cm^3

(c) 5000 cm^3 (d) 5200 cm^3

(c) Radius of circle $= \frac{\text{Diameter}}{2}$

$= \frac{20}{2} = 10$cm

Curved surface area = 1000 cm^2

($\because h$ = height of cylinder)

$\Rightarrow \quad \pi rh = 500 \text{ cm}^2$

Volume of cylinder $= \pi r^2 h$

$= \pi rh \times r = 500 \times 10$

$= 5000 \text{ cm}^3$

54. A piece of wire of length 33 cm is bent into an arc of a circle of radius 14 cm. What is the angle subtended by the arc at the centre of the circle?

(a) 75° (b) 90° (c) 135° (d) 150°

(c) Length of arc (l) = 33 cm

Radius (r) = 14 cm

$$l = \frac{\theta}{360°} \times 2\pi r$$

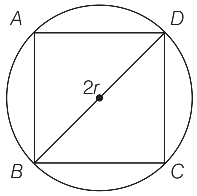
$$33 = \frac{\theta}{360°} \times 2 \times \frac{22}{7} \times 14$$

$\therefore \quad \theta = 135°$

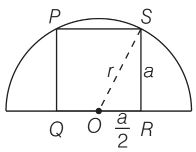
55. What is the ratio of the area of a square inscribed in a semicircle of radius r to the area of square inscribed in a circle of radius r ?

(a) 1 : 2 (b) 2 : 5 (c) 2 : 3 (d) 3 : 5

(b)



(i)



(ii)

Let r be the radius of circle and side of square 'a'. In fig. (i),

Diagonal of square $= 2r$

Using Pythagoras theorem in ΔBCD,

$BC^2 + CD^2 = BD^2$

$a^2 + a^2 = (2r)^2 \Rightarrow 2a^2 = 4r^2$

$a = r\sqrt{2}$

In fig. (ii),

Using Pythagoras theorem in ΔORS,

$(OS)^2 = (OR)^2 + (RS)^2$

$$r^2 = a^2 + \frac{a^2}{4} \Rightarrow r^2 = \frac{5a^2}{4}$$

$$a^2 = \frac{4r^2}{5} \Rightarrow a = \frac{2r}{\sqrt{5}}$$

$$\therefore \text{ Required Ratio} = \frac{\left(\frac{2r}{\sqrt{5}}\right)^2}{(r\sqrt{2})^2}$$

[$\because$ area of square = $(\text{side})^2$]

$$= \frac{\frac{4r^2}{5}}{r^2 \times 2}$$

$$= \frac{2}{5} \Rightarrow 2 : 5$$

56. The quotient when $x^4 - x^2 + 7x + 5$ is divided by $(x + 2)$ is $ax^3 + bx^2 + cx + d$. What are the values of a, b, c and d, respectively?

(a) 1, – 2, 3, 1 (b) –1, 2, 3, 1

(c) 1, – 2, – 3, – 1 (d) –1, 2, – 3, – 1

(a) Lets divide $x^4 - x^2 + 7x + 5$ with long division method,

$$x + 2\ )\ x^4 - x^2 + 7x + 5\ (\ x^3 - 2x^2 + 3x + 1$$

$$\underline{\overset{-}{x^4} \overset{-}{+ 2x^3}}$$

$$- 2x^3 - x^2$$

$$\underline{\overset{+}{-2x^3} \overset{+}{-} 4x^2}$$

$$3x^2 + 7x$$

$$\underline{\overset{-}{3x^2} \overset{-}{+ 6x}}$$

$$x + 5$$

$$\underline{\overset{-}{x} \overset{-}{+ 2}}$$

$$3$$

According to the question,

$ax^3 + bx^2 + cx + d$

$= x^3 - 2x^2 + 3x + 1$

By comparing it, we will get

$a = 1, b = -2, c = 3$ and $d = 1$

57. The sides of a triangle are 30 cm, 28 cm and 16 cm, respectively. In order to determine its area, the logarithm of which of the quantities are required?

(a) 37, 11, 28, 16 (b) 21, 30, 28, 7

(c) 37, 21, 11, 9 (d) 37, 21, 9, 7

(d) Sides of triangle are 30, 28 and 16 cm

Semi perimeter $= \frac{30 + 28 + 16}{2} = 37$

Area

$(A) = \sqrt{37(37 - 30)(37 - 28)(37 - 16)}$

$A = \sqrt{37 \times 7 \times 9 \times 21}\, cm^2$

On taking log both sides, we get

$\log A = \log(37 \times 7 \times 9 \times 21)^{1/2}$

$= \frac{1}{2}\log(37 \times 7 \times 9 \times 21)$ $[\because \log a^b = b\log a]$

$= \frac{1}{2}[\log 37 + \log 7 + \log 9 + \log 21]$ $[\because \log(a \times b) = \log a + \log b]$

Logarithmic of 37, 7, 9 and 21 is required.

58. If $\log_{10} 1995 = 3.3000$, then what is the value of $(0.001995)^{\frac{1}{8}}$?

(a) $\frac{1}{10^{0.3475}}$ (b) $\frac{1}{10^{0.3375}}$

(c) $\frac{1}{10^{0.3275}}$ (d) $\frac{1}{10^{0.3735}}$

(b) Let $x = (0.001995)^{1/8}$

On taking $\log_{10}$ both sides, we get

$\log_{10} x = \log_{10}(0.001995)^{1/8}$

$= \frac{1}{8}\log_{10}(0.001995)$ $[\because \log a^b = b\log a]$

$= \frac{1}{8}\log_{10}\left[\frac{1995}{1000000}\right]$

$= \frac{1}{8}[\log_{10}1995 - \log_{10}10^6]$

$= \frac{1}{8}[\log_{10}1995 - 6\log_{10}10]$

$= \frac{1}{8}[3 \cdot 3 - 6] = \frac{1}{8}[-2.7] = -\frac{1}{8}[2.7]$

$= -0.3375$

On taking antilog both sides, we get

$x = \frac{1}{10^{0.3375}}$

59. What is $(x-a)(x-b)(x-c)$ equal to ?

(a) $x^3 - (a+b+c)x^2 + (bc + ca + ab)x - abc$

(b) $x^3 + (a+b+c)x^2 + (bc + ca + ab)x + abc$

(c) $x^3 - (bc + ca + ab)x^2 + (a + b + c)x - abc$

(d) $x^3 + (bc + ca + ab)x^2 - (a + b + c)x - abc$

(a) Let $y = (x-a)(x-b)(x-c)$

$= (x^2 - ax - bx + ab)(x - c)$

$= x^3 - ax^2 - bx^2 + abx - cx^2 + acx + bcx - abc$

$= x^3 - x^2(a+b+c) + x(ab + bc + ac) - abc$

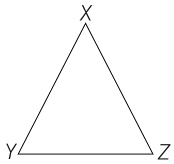
After solving this we get option (a) is correct.

60. Let XYZ be an equilateral triangle in which $XY = 7$ cm. If A denotes the area of the triangle, then what is the value of $\log_{10} A^4$? (given that $\log_{10} 1050 = 3.0212$ and $\log_{10} 35 = 1.5441$)

(a) 5.3070 (b) 5.3700

(c) 5.5635 (d) 5.6535

(a) Given, XYZ is an equilateral triangle and $XY = 7$ cm



Area (A) of equilateral triangle $= \frac{\sqrt{3}}{4}(XY)^2$

$A = \frac{\sqrt{3}}{4}(7)^2 = \frac{\sqrt{3}}{4} \times 49\, cm^2$ ...(i)

$= \log_{10} A^4 = \log_{10}(A^2)^2 = 2\log_{10} A^2$

$= 2\log_{10}\left(\frac{49\sqrt{3}}{4}\right)^2$ by Eq. (i)

$= 2\log_{10}\left[\frac{49 \times 49 \times 3}{4 \times 4}\right]$

$= 2[\log_{10}(49 \times 49 \times 3) - \log_{10}16]$ $\left[\because \log\frac{a}{b} = \log a - \log b\right]$

$= 2[\log_{10}(7^4 \times 3) - \log_{10}2^4]$

$= 2[\log_{10}7^4 + \log_{10}3 - 4\log_{10}2]$ $[\because \log(a \times b) = \log a + \log b]$

$= 2\left[4\log_{10}7 + \log_{10}3 - 4\log_{10}\frac{2 \times 5}{5}\right]$ $[\because \log a^b = b\log a]$

[5 divided and multiply in last number]

$= 2[4\log_{10}7 + \log_{10}3 - 4(\log_{10}10 - \log_{10}5)]$

$= 2[4\log_{10}7 + 4\log_{10}5 + \log_{10}3 - 4\log_{10}10]$

$= 2[4\log_{10}35 + \log_{10}3 - 4\log_{10}10]$

$= 2[4 \times \log_{10}35 + \log_{10}3 - 4]$ ...(ii) $[\because \log_{10}10 = 1]$

$\Rightarrow \log_{10}1050 = 3.0212$

$\log_{10}(105 \times 10) = 3.0212$

$\log_{10}10 + \log_{10}(35 \times 3) = 3.0212$

$\log_{10}10 + \log_{10}35 + \log_{10}3 = 3.0212$

$\log_{10}3 = 3.0212 - \log_{10}35 - \log_{10}10$

$= 3.0212 - 1.5441 - 1 = 0.4771$ ...(iii)

On putting the value of $\log_{10}3$ in Eq. (ii), we get

$= 2[4 \times 1.5441 + 0.4771 - 4]$ $[\because \log_{10}35 = 1.5441]$

$= 2 \times [(6.1764 + 0.4771) - 4]$

$= 2 \times (6.6535 - 4)$

$= 2 \times (2.6535)$

$= 5.3070$

Option (a) is correct.

61. A hollow sphere of external and internal diameters 6 cm and 4 cm, respectively is melted into a cone of base diameter 8 cm. What is the height of the cone?

(a) 4.75 cm (b) 5.50 cm

(c) 6.25 cm (d) 6.75 cm

(a) The external radius of hollow sphere

$(r_1) = \frac{\text{Diameter}}{2} = \frac{6}{2} = 3$ cm

The internal radius of hollow sphere

$r_2 = \frac{4}{2} = 2$ cm

Volume of hollow sphere $= \frac{4}{3}\pi(r_1^3 - r_2^3)$

$= \frac{4}{3}\pi(3^3 - 2^3) = \frac{19}{3} \times 4\pi$

According to the question, volume of hollow sphere = Volume of cone

$\frac{19}{3} \times 4\pi = \frac{1}{3}\pi(4)^2 \times h$

(h = height of cone, radius of cone = 4 cm)

$\therefore h = \frac{19}{4} = 4.75$ cm

62. A solid metallic cylinder of height 10 cm and radius 6 cm is melted to make two cones in the ratio of volume 1 : 2 and of same height as 10 cm. What is the percentage increase in the flat surface area?

(a) 25% (b) 50% (c) 75% (d) 100%

(b) Radius of cylinder $(r) = 6$ cm

Height of cylinder $(h) = 10$ cm

Let volume of cone be v_1 and v_2.

Ratio of volume of two cones

$v_1 : v_2$ is 1 : 2

If v_1 is x, v_2 will be $2x$.

Hence, volume of cylinder

$= v_1 + v_2 = 3x$

Volume of cylinder $= \pi r^2 h$

$= \frac{22}{7} \times 6 \times 6 \times 10$

$= \frac{7920}{7}\, cm^3$

Surface area of flat surface of cylinder

$= 2\pi r^2 = 2 \times 6 \times 6 \times \pi = 72\pi$

Now, this cylinder is recast into 2 cones in the ratio 1 : 2.

So, volume of cone (v_1) = volume of cylinder $\times \dfrac{x}{x + 2x} = \dfrac{x}{3x} \times \dfrac{7920}{7}$

$$= \frac{2640}{7} \text{ cm}^3$$

Volume of cone (V_2) $= \dfrac{2x}{3x} \times \dfrac{7920}{7}$

$$= \frac{5280}{7} \text{ cm}^3$$

Let the radii of the 2 cones be R_1 and R_2, respectively.

The height of both cones is 10 cm.

$$\pi R_1^2 h = \frac{2640}{7}$$

and $\quad \pi R_2^2 h = \dfrac{5280}{7}$

$$\pi R_1^2 = \frac{264}{7}$$

and $\quad \pi R_2^2 = \dfrac{528}{7} \quad [\because h = 10]$

$$R_1^2 = \frac{84}{7} \text{ and } R_2^2 = \frac{168}{7}$$

$\pi R_1^2 = 12\pi$ and $\pi R_2^2 = 24\pi$

Combined surface area of flat surface of both cones

$= 12\pi + 24\pi = 36\pi.$

Change in flat surface area

$$= \frac{72\pi - 36\pi}{72\pi} \times 100$$

$$= \frac{36\pi}{72\pi} \times 100 = 50\%$$

63. If one side of a right-angled triangle (with all sides integers) is 15 cm, then what is the maximum perimeter of the triangle?

(a) 240 cm (b) 225 cm
(c) 113 cm (d) 112 cm

*(**a**)* Let 2 other sides be x and y.

So, $\quad x^2 = y^2 + 15^2$

[for maximum perimeter]

$$x^2 - y^2 = 15^2$$

$$(x - y)(x + y) = 225$$

To make perimeter maximum we need to minimize the difference between sides.

So, on comparing

$x + y = 225 \quad$ ...(i)

$x - y = 1 \quad$ ...(ii)

By solving Eqs. (i) and (ii), we get

$x = 113$, $y = 112$

$\therefore$ Maximum perimeter $= x + y + 15$

$= 113 + 112 + 15$

$= 240$ cm

64. A thin rod of length 24 feet is cut into rods of equal size and joined, so as to form a skeleton cube. What is the area of one of the faces of the largest cube thus constructed?

(a) 25 sq feet (b) 24 sq feet
(c) 9 sq feet (d) 4 sq feet

*(**d**)* Length of rod = 24 feet

Let the side of cube = x feet

Cube consist 12 edges,

so, $\quad 12x = 24$ feet

$x = 2$ feet

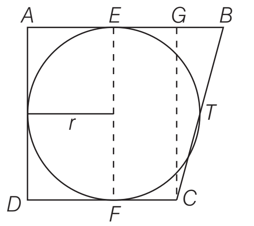
Edge of the largest cube that can be constructed of 24 feet rod is 2 feet.

Now, the area of one of the face $= 2 \times 2 = 4$ sq feet.

65. Consider a trapezium $ABCD$, in which AB is parallel to CD and AD is perpendicular to AB. If the trapezium has an incircle which touches AB at E and CD at F, where $EB = 25$ cm and $FC = 16$ cm, then what is the diameter of the circle?

(a) 16 cm (b) 25 cm
(c) 36 cm (d) 40 cm

*(**d**)*



$ABCD$, is trapezium in which AB is parallel to CD. Let BC touches circle at T.

$FC = CT = 16$ cm

$EB = BT = 25$ cm [tangent of circle]

$BC = CT + BT = 16 + 25 = 41$

Let radius of the circle is r,

ΔCGB, is right angled triangle,

From Pythagoras theorem,

$$CB^2 = CG^2 + BG^2$$

Here, $CB = 2r$, $BC = 41$

$BG = BE - FC = 25 - 16 = 9$

$41^2 = (2r)^2 + 9^2$

$(2r)^2 = 41^2 - 9^2$

$4r^2 = 50 \times 32$

$\therefore \quad r = 20$ cm

Diameter of the circle $= 2 \times 20 = 40$ cm

66. Three copper spheres of radii 3 cm, 4 cm and 5 cm are melted to form a large sphere. What is its radius?

(a) 12 cm (b) 10 cm (c) 8 cm (d) 6 cm

*(**d**)* Sphere of radii 3 cm, 4 cm and 5 cm are melted to form a large sphere of radius r.

According to the question,

$$\frac{4}{3}\pi 3^3 + \frac{4}{3}\pi 4^3 + \frac{4}{3}\pi 5^3 = \frac{4}{3}\pi r^3$$

$\Rightarrow \quad 3^3 + 4^3 + 5^3 = r^3$

$\Rightarrow \quad 27 + 64 + 125 = r^3$

$\Rightarrow \quad r^3 = 216$

$\Rightarrow \quad r = 6$ cm

67. The volume of a hemisphere is 155232 cm^3. What is the radius of the hemisphere?

(a) 40 cm (b) 42 cm
(c) 38 cm (d) 36 cm

*(**b**)* Let the radius of hemisphere be 'r' cm.

According to the question,

$$\frac{2}{3}\pi r^3 = 155232$$

$\Rightarrow \quad \dfrac{2}{3} \times \dfrac{22}{7} \times r^3 = 155232$

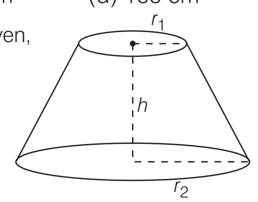
$\Rightarrow \quad r^3 = 74088$

$\Rightarrow \quad r = 42$ cm

68. A bucket is in the form of a truncated cone. The diameters of the base and top of the bucket are 6 cm and 12 cm, respectively. If the height of the bucket is 7 cm, what is the capacity of the bucket?

(a) 535 cm^3 (b) 462 cm^3
(c) 234 cm^3 (d) 166 cm^3

*(**b**)* Given,



Here, $r_1 = 3$ cm, $r_2 = 6$ cm, $h = 7$ cm

Volume of bucket

$$= \frac{1}{3}\pi h(r_1^2 + r_2^2 + r_1 r_2)$$

$$= \frac{1}{3} \times \frac{22}{7} \times 7(3^2 + 6^2 + 3 \times 6)$$

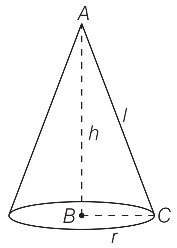
$$= \frac{1}{3} \times \frac{22}{7} \times 7 \times 63$$

$$= 22 \times 21 = 462 \text{ cm}^3$$

69. A right circular cone has height 8 cm. If the radius of its base is 6 cm, then what is its total surface area?

(a) 96π cm^2 (b) 69π cm^2
(c) 54π cm^2 (d) 48π cm^2

(a) Given,



Here, $r = 6$ cm, $h = 8$ cm

ΔABC is right angled triangle,

Using Pythagoras theorem,

$r^2 + h^2 = l^2$

$6^2 + 8^2 = l^2$

$36 + 64 = l^2$

$l^2 = 100 = 10$

According to the question,

Total surface area $= \pi r^2 + \pi rl$

$= \pi r(r + l)$

$= \pi \times 6(6 + 10)$

$= \pi \times 6 \times 16$

$= 96\pi$ cm^2

70. Six cubes, each with 12 cm edge are joined end to end. What is the surface area of resulting cuboid?

(a) 3000 cm^2 (b) 3600 cm^2
(c) 3744 cm^2 (d) 3777 cm^2

(c) When six cubes are joined end to end, then resulting solid is a cuboid.

Length $(l) = 6 \times 12 = 72$ cm

Breadth $(b) = 12$ cm

Height $(h) = 12$ cm

Surface area of cuboid

$= 2(lb + bh + lh)$

$= 2(72 \times 12 + 12 \times 12 + 12 \times 72)$

$= 2 \times 1872$ cm^2 $= 3744$ cm^2

71. If the sum of a real number and its reciprocal is $\frac{26}{5}$, then how many such numbers are possible?

(a) None (b) One
(c) Two (d) Four

(b) Let the number be x.

Reciprocal of number $= \frac{1}{x}$

According to the question,

$$x + \frac{1}{x} = \frac{26}{5} \Rightarrow \frac{x^2 + 1}{x} = \frac{26}{5}$$

$\Rightarrow \quad 5(x^2 + 1) = 26x$

$\Rightarrow \quad 5x^2 + 5 - 26x = 0$

$\Rightarrow \quad 5x^2 - 25x - x + 5 = 0$

$\Rightarrow \quad 5x(x - 5) - 1(x - 5) = 0$

$\Rightarrow \quad (5x - 1)(x - 5) = 0$

$\therefore \quad x = 5, x = \frac{1}{5}$

72. Consider the following statements :

1. If p is relatively prime to each of q and r, then p is relatively prime to the product qr.
2. If p divides the product qr and if p divides q, then p must divide r.

Which of the above statements is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(a)

1. Let $p = 4, q = 3, r = 5$

Here, p is relatively prime to q and r.

The product of $qr = 3 \times 5 = 15$

In this case also p is relatively prime to product of qr.

2. Let $p = 4, q = 8$ and $r = 3$

Here, product of $qr = 8 \times 3 = 24$

p divides the product of qr but cannot divide r.

Let $p = 4, q = 8$ and $r = 12$

Product of $qr = 96$

p divides the product of qr and also divide r.

We cannot say that p will divide r.

Hence, statement 1 is correct and statement 2 is incorrect.

73. Radha and Rani are sisters. Five years back, the age of Radha was three times that of Rani, but one year back the age of Radha was two times that of Rani. What is the age difference between them?

(a) 8 (b) 9 (c) 10 (d) 11

(a) Let the present age of Radha be x Rani be y.

Five years ago,

Age of Radha $= x - 5$

Age of Rani $= y - 5$

According to the question,

$x - 5 = 3(y - 5) \Rightarrow x - 5 = 3y - 15$

$3y - x = 10 \quad \ldots$(i)

One year ago,

Age of Radha $= x - 1$

Age of Rani $= y - 1$

According to the question,

$x - 1 = 2(y - 1) \Rightarrow x - 1 = 2y - 2$

$\Rightarrow \quad 2y - x = 1 \quad \ldots$(ii)

On solving Eqs. (i) and (ii), we get

$y = 9, x = 17$

$\therefore$ Required difference $= 17 - 9 = 8$ yr

74. A person carries ₹ 500 and wants to buy apples and oranges out of it. If the cost of one apple is ₹ 5 and the cost of one orange is ₹ 7, then what is the number of ways in which a person can buy both apples and oranges using total amount?

(a) 10 (b) 14 (c) 15 (d) 17

(b) Let there are x number of apples and y number of oranges

So, $\quad 5x + 7y = 500$

Now, we have to check for $x = 1, 2, \ldots$

When, $\quad x = 2$

$10 + 7 \times y = 500, y = 70$

When, $x = 9$, $45 + 7y = 500$, $y = 65$

Similarly, we can go ahead and find all the ways or we can check at which value of y oranges will be minimum.

For $y = 5$, x (s) will be minimum

$5x + 35 = 900, x = 93$

This forms an arithmetic series, where $a = 2, d = 9 - 2 = 7$ and $a_n = 93$

$a_n = a + (n - 1)d, \quad 93 = 2 + (n - 1)7$

$n - 1 = 13, n = 14$

There are 14 number of ways by which a person can by apples and oranges.

75. Given y is inversely proportional to $\sqrt{x}$, and $x = 36$ when $y = 36$. What is the value of x when $y = 54$?

(a) 54 (b) 27
(c) 16 (d) 8

(c) According to the question,

$$y \propto \frac{1}{\sqrt{x}}$$

$$y = \frac{k}{\sqrt{x}}$$

$x = 36, y = 36$

$\Rightarrow \quad 36 = \frac{k}{\sqrt{36}} \Rightarrow k = 216$

Now, $y = 54, k = 216$

$$54 = \frac{216}{\sqrt{x}}$$

$\therefore \quad \sqrt{x} = 4$

$\Rightarrow \quad x = 16$

76. What is the square root of $16 + 6\sqrt{7}$?

(a) $4 + \sqrt{7}$ (b) $4 - \sqrt{7}$
(c) $3 + \sqrt{7}$ (d) $3 - \sqrt{7}$

(c) Let $x = 16 + 6\sqrt{7}$

On square rooting both sides, we get

$$\sqrt{x} = \sqrt{16 + 6\sqrt{7}}$$
$$= \sqrt{9 + 7 + 2 \times 3 \times \sqrt{7}}$$
$$= \sqrt{(3)^2 + (\sqrt{7})^2 + 2 \times 3 \times \sqrt{7}}$$
$$= \sqrt{(3 + \sqrt{7})^2}$$
$[\because (a + b)^2 = a^2 + b^2 + 2ab]$
$$\sqrt{x} = 3 + \sqrt{7}$$

77. What is the number of digits in $7^{25}, 8^{23}$ and 9^{20}, respectively? [given, $\log_{10} 2 = 0.301$, $\log_{10} 3 = 0.477$, $\log_{10} 7 = 0.845$]

(a) 21, 20, 19 (b) 20, 19, 18
(c) 22, 21, 20 (d) 22, 20, 21

(c) Let $x = 7^{25}$

On taking log both sides, we get

$\log_{10} x = \log_{10} 7^{25}$
$= 25 \log_{10} 7$ $[\because \log a^b = b \log a]$
$= 25 \times 0.845 \Rightarrow = 21.125$

$\therefore$ Number of digits $= 21 + 1 = 22$

Let $y = 8^{23}$

On taking log both sides, we get

$\log_{10} y = \log_{10} 8^{23}$
$= \log_{10} (2^3)^{23} = \log_{10} 2^{69}$
$= 69 \log_{10} 2$ $[\because \log a^b = b \log a]$
$= 69 \times 0.301$
$= 20.769$

$\therefore$ Number of digits $= 20 + 1 = 21$

Let $z = 9^{20}$

On taking log both sides, we get

$\log_{10} z = \log_{10} 9^{20}$
$= \log_{10} (3^2)^{20}$
$= \log_{10} 3^{40} = 40 \log_{10} 3$
$[\because \log a^b = b \log a]$
$= 40 \times 0.477$
$= 19.08$

Number of digits $= 19 + 1 = 20$

78. Let x be the smallest positive integer such that when 14 divides x, the remainder is 7; and when 15 divides x, the remainder is 5. Which one of the following is correct?

(a) $20 < x < 30$ (b) $30 < x < 40$
(c) $40 < x < 50$ (d) $x > 50$

(b) Dividend = (Integer quotient) × divisor + remainder

$x = 14p + 7$
$\Rightarrow x = 15q + 5$

Where, p and q are quotients.

How, we can put in the values of p and q into the equation, starting with 0.

When $x = 14p + 7$
$p = 0$ $x = 14 \times 0 + 7 = 7$
$p = 1$ $x = 14 \times 1 + 7 = 21$
$p = 2$ $x = 14 \times 2 + 7 = 35$ etc.
When $x = 15q + 5$
$q = 0$ $x = 15 \times 0 + 5 = 5$
$q = 1$ $x = 15 \times 1 + 5 = 20$
$q = 2$ $x = 15 \times 2 + 5 = 35$

Here, we found the common value for x. Hece, the least possible value for x for which both statements are true, is 35.

79. Two taps X and Y are fixed to a water tank. If only X is opened, it drains out the full tank of water in 20 min. If both X and Y are opened, then they drain out the full tank of water in 15 min. If only Y is opened, how long does it take to drain out the full tank of water?

(a) 30 min (b) 45 min
(c) 60 min (d) 90 min

(c) X drains out full tank in 20 min.

X 1 min work $= \frac{1}{20}$

X and Y drain out full tank in 15 min.

X and Y 1 min work $= \frac{1}{15}$

Y 1 min work $= \frac{1}{15} - \frac{1}{20}$
$= \frac{1}{60}$

Y will take 60 min to drain out full tank.

80. Consider the following statements

1. $\sqrt{75}$ is a rational number.
2. There exists at least a positive integer x such that $-\frac{4x}{5} < -\frac{7}{8}$.
3. $\frac{x - 2}{x} < 1$ for all real value of x.
4. 4.232323... can be expressed in the form p/q where p and q are integers.

Which of the above statements are correct?

(a) 1 and 2
(b) 2 and 3
(c) 3 and 4
(d) 2 and 4

(d)
1. $\sqrt{75}$ is not a rational number.
Only root of perfect square is rational number, here 75 is not a perfect square.
2. $-\frac{4x}{5} < -\frac{7}{8}$
$\Rightarrow -32x < -35$
$\Rightarrow 32x > 35$
It is possible, if $x \geq 2$.
3. $\frac{x - 2}{x} < 1$ does not follow for all real values, It only follow for natural numbers,
For example, If $x = -1$
$\frac{-1 - 2}{-1} < 1$
$\Rightarrow \frac{-3}{-1} < 1$
$3 < 1$ (does not follow)
if $x = 3$
$\frac{3 - 2}{3} < 1, \frac{1}{3} < 1$ (follow)
4. $x = 4.232323$ ...(i)
On multiply by 100 both sides, we get
$100x = 423.2323$ ...(ii)
On subtracting Eq. (i) from Eq. (ii), we get
$100x - x = 423.2323 \ldots - 4.2323 \ldots\ldots$
$99x = 419$
$\Rightarrow x = \frac{419}{99}$
$\therefore \frac{419}{99}$ is in $\frac{p}{q}$ form and both p and q are integers.

Hence, statements 2 and 4 are correct and statements 1 and 3 are incorrect.

81. A library has an average number of 510 visitors on Sunday and 240 on other days. What is the average number of visitors per day in a month of 30 days beginning with Saturday?

(a) 276
(b) 282
(c) 285
(d) 375

(c) Since, the months begin with Saturday, there will be five Sunday,

Required average

$$= \frac{\text{Total number of visitors}}{\text{Number of days}}$$
$$= \frac{510 \times 5 + 240 \times 25}{30}$$
$$= \frac{8550}{30} = 285$$

82. If $\dfrac{36}{11} = 3 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$, where x, y and z are natural numbers, then what is $(x + y + z)$ equal to

(a) 6 (b) 7
(c) 8 (d) 9

(*a*) Given, $\dfrac{36}{11} = 3 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$

$\Rightarrow \quad 3 + \dfrac{3}{11} = 3 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$

By comparing, we get

$\dfrac{3}{11} = \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$

$x + \cfrac{1}{y + \cfrac{1}{z}} = \dfrac{11}{3}$

$\Rightarrow x + \cfrac{1}{y + \cfrac{1}{z}} = 3 + \dfrac{2}{3}$

By comparing, we get

$x = 3$,

$y + \dfrac{1}{z} = \dfrac{3}{2}$

$y + \dfrac{1}{z} = 1 + \dfrac{1}{2}$

By comparing, we get $y = 1$, $z = 2$

Then, $(x + y + z) = 3 + 2 + 1 = 6$

83. A person sells two items each at Rs. 990, one at a profit of 10% and another at a loss of 10%. What is the combined percentage of profit or loss for the two items?

(a) 1% loss
(b) 1% profit
(c) No profit no loss
(d) 0.5% profit

(*a*) Let CP of 1st and 2nd items are $100x$ and $100y$ respectively.

According to the question,

SP_1 of 1st item

$= CP \times \left(\dfrac{100 + \text{profit}\,\%}{100}\right)$

$= 100x \times \dfrac{110}{100} = 110x$

SP_2 of 2nd item

$= CP \times \left(\dfrac{100 - \text{Loss}\%}{100}\right)$

$= 100y \times \dfrac{90}{100} = 90y$

Selling price of both items is 990

$\therefore \quad 110x = 990$

$\Rightarrow \quad x = 9$

$90y = 990$

$\Rightarrow \quad y = 11$

CP_1 of 1st item $= 100 \times 9 = 900$

CP_2 of 2nd item $= 100 \times 11 = 1100$

Total cost price $= CP_1 + CP_2$

$= 900 + 1100$

$= 2000$

$\therefore$ Required loss

$\% = \dfrac{2000 - 1980}{2000} \times 100$

$= 1\%$ loss

Alternate Method

Here, $a = 10\%$

$\therefore$ Required loss percentage

$= \left(\dfrac{a}{10}\right)^2 = \left(\dfrac{10}{10}\right)^2 = 1\%$

84. It takes 11 h for a 600 km journey if 120 km is done by train and the rest by car. It takes 40 min more if 200 km are covered by train and the rest by car. What is the ratio of speed of the car to that of the train?

(a) 3 : 2 (b) 2 : 3
(c) 3 : 4 (d) 4 : 3

(*a*) Let the speed of train be x km/h and car y km/h.

According to the question,

$\dfrac{120}{x} + \dfrac{480}{y} = 11 \quad \ldots\text{(i)}$

$\dfrac{200}{x} + \dfrac{400}{y} = 11\dfrac{2}{3} \quad \ldots\text{(ii)}$

To multiply Eq. (i) by 5 and Eq. (ii) by 3, and subtracting, we get

$\dfrac{600}{x} + \dfrac{2400}{y} - \dfrac{600}{x} - \dfrac{1200}{y} = 55 - 35$

$\dfrac{1200}{y} = 20$

$\Rightarrow \quad y = 60$ km/h

On putting $y = 60$ in Eq. (i), we get

$\dfrac{120}{x} + \dfrac{480}{60} = 11$

$\Rightarrow \quad \dfrac{120}{x} = 11 - 8$

$x = 40$ km/hr

$\therefore \quad$ Required ratio $= \dfrac{y}{x} = \dfrac{60}{40}$

$= \dfrac{3}{2}$

$= 3 : 2$

85. A real number x is such that $(x - x^2)$ is maximum. What is x equal to ?

(a) – 1.5 (b) – 0.5 (c) 0.5 (d) 1.5

(*c*) x is a real number such that $(x - x^2)$ is maximum.

Let $\quad y = x - x^2$

On differential both sides w.r.t. x, we get

$\dfrac{dy}{dx} = \dfrac{d}{dx}(x) - \dfrac{d}{dx}(x^2)$

$= x^{1-1} - 2x^{2-1}$

$\dfrac{dy}{dx} = 1 - 2x$

To be the function maximum $\dfrac{dy}{dx} = 0$

$1 - 2x = 0 \Rightarrow 2x = 1$

$\therefore \quad x = \dfrac{1}{2} \Rightarrow x = 0.5$

Alternate method : On solving the given question with respect of option :

From option (a),

$(x - x^2) =$
$-1.5 - (-1.5)^2 = -1.5 - 2.25 = -3.75$

From option (b),

$(x - x^2) = -0.5 - (-0.5)^2$
$= 0.5 - 0.25 = -0.75$

From option (c),

$(x - x^2) = 0.5 - (0.5)^2$
$= 0.5 - 0.25 = 0.25$

From option (d),

$(x - x^2) = 1.5 - (-1.5)^2$
$= 1.5 - 2.25 = -0.75$

Hence, $x = 0.5$

86. If 10^n divides $6^{23} \times 75^9 \times 105^2$, then what is the largest value of n?

(a) 20 (b) 22 (c) 23 (d) 28

(*a*) Let $x = 6^{23} \times 75^9 \times 105^2$

We can also write this expression as

$x = 2^{23} \times 3^{23} \times 5^{18} \times 3^9$
$\times 5^2 \times 3^2 \times 7^2 = 2^{23} \times 3^{34} \times 7^2 \times 5^{20}$

If 10^n divides this expression, then n must be equal to the number of zero in the expression, from above expression we can conclude that there will be 20 number of zero $(5^{20} \times 2^{20})$ because only 5 and 2 make zero.

So, $\quad n = 20$

87. What is the digit in the unit's place of the number represented by $3^{98} - 3^{89}$?

(a) 3 (b) 6
(c) 7 (d) 9

(b) $x = 3^{98} - 3^{89} = 3^{89}(3^9 - 1)$

We know that, the cyclicity of 3 is 4.

So, we can write above expression as

$x = 3^{4\times 22} \times 3^1(3^{2\times 4} \times 3 - 1)$

It will be equal to $= 3^1(3^1 - 1) = 3 \times 2 = 6$

Unit digit of the expression is 6.

88. The sum of the squares of four consecutive natural numbers is 294. What is the sum of the numbers?

(a) 38 (b) 34 (c) 30 (d) 26

(b) Let the four consecutive natural numbers are $x-1, x, x+1, x+2$

According to the question,

$(x-1)^2 + x^2 + (x+1)^2 + (x+2)^2 = 294$

$x^2 + 1 - 2x + x^2 + x^2 + 1 + 2x + x^2 + 4x + 4 = 294$

$4x^2 + 4x - 288 = 0$

$\Rightarrow x^2 + x - 72 = 0$

$x^2 + 9x - 8x - 72 = 0$

$\Rightarrow x(x+9) - 8(x+9) = 0$

$(x-8)(x+9) = 0$

$x = 8$ and -9 ($x = -9$)[ignore]

The numbers are, 7, 8, 9, 10

Sum of numbers

$= 7 + 8 + 9 + 10 = 34$

89. The equation $x^2 + px + q = 0$ has roots equal to p and q where $q \neq 0$. What are the values of p and q, respectively?

(a) 1, – 2 (b) 1, 2
(c) – 1, 2 (d) –1, – 2

(a) $x^2 + px + q = 0$

So, $p + q = -p$

$q = -2p$ ...(i)

$pq = q \Rightarrow p = 1$

Now, on putting $p = 1$ in Eq. (i), we get

$q = -2 \times 1$

$q = -2$

$p = 1$ and $q = -2$

90. How many pairs of natural numbers are there such that the difference of their squares is 35?

(a) 1 (b) 2 (c) 3 (d) 4

(b) Let the number be x and y.

$x^2 - y^2 = 35$

$(x-y)(x+y) = 35$

Thus, the factor of 35 possibles are 7, 5, 35, 1

If $x - y = 5$

$x + y = 7$

$x = 6$ and $y = 1$ [1st pair]

If $x + y = 35$

$x - y = 1$

$x = 18$ and $y = 17$ [2nd pair]

Only 2 pair are possible.

91. If $(b-6)$ is one root of the quadratic equation $x^2 - 6x + b = 0$, where b is an integer, then what is the maximum value of b^2?

(a) 36 (b) 49 (c) 64 (d) 81

(d) $(b-6)$ is one root of expression $x^2 - 6x + b = 0$

So, $x = b - 6$ should satisfy the equation.

$(b-6)^2 - 6(b-6) + b = 0$

$b^2 + 36 - 12b - 6b + 36 + b = 0$

$b^2 - 17b + 72 = 0$

$b^2 - 8b - 9b + 72 = 0$

$b(b-8) - 9(b-8) = 0$

$(b-9)(b-8) = 0$

$b = 8$ and 9

To get maximum value of b^2

$b = 9 \Rightarrow b^2 = 81$

92. If $a = \sqrt{7 + 4\sqrt{3}}$, then what is the value of $a + \frac{1}{a}$?

(a) 2 (b) 3
(c) 4 (d) 7

(c) $a = \sqrt{7 + 4\sqrt{3}}$

$= \sqrt{4 + 3 + 2 \times 2\sqrt{3}}$

$= \sqrt{2^2 + (\sqrt{3})^2 + 2 \times 2 \times \sqrt{3}}$

$= \sqrt{(2 + \sqrt{3})^2}$

$[\because (a+b)^2 = a^2 + b^2 + 2ab]$

$a = 2 + \sqrt{3}$

So, $\frac{1}{a} = \frac{1}{2 + \sqrt{3}}$

$\frac{1}{a} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$

$\frac{1}{a} = \frac{2-\sqrt{3}}{4-3} = 2 - \sqrt{3}$

Now, $a + \frac{1}{a} = 2 + \sqrt{3} + 2 - \sqrt{3} = 4$

93. What is the maximum value of the expression $\frac{1}{x^2 + 5x + 10}$?

(a) $\frac{15}{4}$ (b) $\frac{15}{2}$ (c) 1 (d) $\frac{4}{15}$

(d) $\frac{1}{x^2 + 5x + 10}$

Take denominator $x^2 + 5x + 10$.

$y = x^2 + 5x + 10$

$= x^2 + 2 \times \frac{5}{2}x + \frac{25}{4} - \frac{25}{4} + 10$

$= \left(x + \frac{5}{2}\right)^2 + \frac{15}{4}$

To make the expression $\frac{1}{x^2 + 5x + 10}$ maximum, then we must minimize the value of denominator.

To minimize denominator of expression $\frac{1}{\left(x + \frac{5}{2}\right)^2 + \frac{15}{4}}$, we can take $x = -\frac{5}{2}$

The maximum value of expression will be $\frac{1}{\left(-\frac{5}{2} + \frac{5}{2}\right)^2 + \frac{15}{4}} = \frac{4}{15}$

94. If the ratio of the work done by $(x+2)$ workers in $(x-3)$ days to the work done by $(x+4)$ workers in $(x-2)$ days is 3 : 4, then what is the value of x?

(a) 8 (b) 10
(c) 12 (d) 15

(b) Number of workers $(x+2)$ worked for $(x-3)$ days and $(x+4)$ workers worked for $(x-2)$ days.

According to the question,

$\frac{(x+2)(x-3)}{(x+4)(x-2)} = \frac{3}{4}$

$\frac{x^2 - x - 6}{x^2 + 2x - 8} = \frac{3}{4}$

$4(x^2 - x - 6) = 3(x^2 + 2x - 8)$

$4x^2 - 4x - 24 = 3x^2 + 6x - 24$

$x^2 - 10x = 0$

$x^2 = 10x \therefore x = 10$

95. Which one of the following is not correct?

(a) 1 is neither prime nor composite
(b) 0 is neither positive nor negative
(c) If $p \times q$ is even, then p and q are always even
(d) $\sqrt{2}$ is an irrational number

(c) If $p \times q$ is even, then p and q can or cannot be even, Let illustrate with example,

Let $p = 2$ and $q = 4$, then

$p \times q = 2 \times 4 = 8$ (even)

Let $p = 2$ and $q = 3$, then

$p \times q = 2 \times 3 = 6$ (even)

Let $p = 3$ and $q = 4$, then

$p \times q = 3 \times 4 = 12$ (even)

With above example we can conclude that if any one of p and q is even we will get $p \times q$ even.

96. What is the sum of all integer values of n for which $n^2 + 19n + 92$ is a perfect square?

(a) 21 (b) 19 (c) 0 (d) –19

(*d*) Let $k^2 = n^2 + 19n + 92$

On multiplying both sides by 4, we get

$$4n^2 + 76n + 368 = 4k^2$$

$\Rightarrow (2n)^2 + 2 \times 19 \times 2n + 361 + 7 = 4k^2$

$\Rightarrow (2n + 19)^2 = (4k^2 - 7)$ ...(i)

Let $(2n + 19) = m$

Then, $4k^2 - 7 = m^2$

$4k^2 - m^2 = 7$

$(2k - m)(2k + m) = 7$

$[\because a^2 - b^2 = (a + b)(a - b)]$

Factor of 7 are 7 and 1

So, $2k + m = 7$

$2k - m = 1$

By solving this, we get $k = 2$

Now, putting $k = 2$ in Eq. (i), we get

$(2n + 19)^2 = (4(2)^2 - 7) = 16 - 7$

$\Rightarrow (2n + 19)^2 = 9$

$\Rightarrow 2n + 19 = \pm 3$

$\Rightarrow 2n + 19 = +3 \Rightarrow n = -8$

$\Rightarrow 2n + 19 = -3 \Rightarrow n = -11$

Sum of integer value of $n = -8 + (-11) = -19$

97. What is the LCM of the polynomials $x^3 + 3x^2 + 3x + 1$, $x^3 + 5x^2 + 5x + 4$ and $x^2 + 5x + 4$?

(a) $(x + 1)^3(x + 4)(x^2 + x + 1)$

(b) $(x + 4)(x^2 + x + 1)$

(c) $(x + 1)(x^2 + x + 1)$

(d) $(x + 1)^2 (x + 4)(x^2 + x + 1)$

(*a*) Let $p = x^3 + 3x^2 + 3x + 1$

$= (x + 1)^3$

$[\because (a + b)^3 = a^3 + b^3 + 3a^2b + 3b^2a]$

Let $q = x^3 + 5x^2 + 5x + 4$

$= x^3 + 4x^2 + x^2 + 4x + x + 4$

$= x^2(x + 4) + x(x + 4) + 1(x + 4)$

$= (x + 4)(x^2 + x + 1)$

Let $r = x^2 + 5x + 4$

$= x^2 + 4x + x + 4$

$= x(x + 4) + 1(x + 4)$

$= (x + 1)(x + 4)$

LCM of p, q and r

$= (x + 1)^3(x + 4)(x^2 + x + 1)$

98. What is the value of

$$\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{9(x-y)(y-z)(z-x)}$$

(a) 0 (b) $\frac{1}{3}$

(c) $\frac{1}{9}$ (d) 1

(*b*) Let

$$p = \frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{9(x-y)(y-z)(z-x)}$$

Let $a = x - y$, $b = y - z$, $c = z - x$

$a + b + c = x - y + y - z + z - x = 0$If $a + b + c = 0$,

then $a^3 + b^3 + c^3 = 3abc$

$$\therefore p = \frac{3(x-y)(y-z)(z-x)}{9(x-y)(y-z)(z-x)}$$

$$= \frac{1}{3}$$

99. If $X = \{a, \{b\}, c\}$, $Y = \{\{a\}, b, c\}$ and $Z = \{a, b, \{c\}\}$, then $(X \cap Y) \cap Z$ equals to

(a) $\{a, b, c\}$ (b) $\{\{a\}, \{b\}, \{c\}\}$

(c) $\{\phi\}$ (d) ϕ

(*d*) Given, $X = \{a, \{b\}, c\}$

$Y = \{\{a\}, b, c\}$, $Z = \{a, b, \{c\}\}$

Then, $X \cap Y = \{c\}$

$(X \cap Y) \cap Z = \{c\} \cap \{a, b, \{c\}\} = \phi$

100. Two numbers p and q are such that the quadratic equation $px^2 + 3x + 2q = 0$ has -6 as the sum and the product of the roots. What is the value of $(p - q)$?

(a) –1 (b) 1 (c) 2 (d) 3

(*c*) $px^2 + 3x + 2q = 0$

Sum of roots $(\alpha + \beta) = -\frac{3}{p}$

Product of root $(\alpha \cdot \beta) = \frac{2q}{p}$

According to the question, $\frac{-3}{p} = -6$

$$p = \frac{1}{2}$$

$\Rightarrow \frac{2q}{p} = -6$

$\Rightarrow q = \frac{-6 \times p}{2}$

$\Rightarrow q = \frac{-3}{2}$

Then, $p - q = \frac{1}{2} - \left(-\frac{3}{2}\right) = \frac{1}{2} + \frac{3}{2}$

$= \frac{4}{2} = 2$

PAPER II English

Directions (Q. Nos. 1-7) *In this section, a word is spelled in four different ways. You are to identify the one which is correct. Choose the alternative bearing the correct spelling from (a), (b), ©) and (d).*

1. (a) Accommodate
(b) Acommodate
(c) Accomdate
(d) Acomodait

(*a*) 'Accommodate' is the correct spelling and it means to make an adjustment to suit a particular purpose.

2. (a) Recommand
(b) Reccommed
(c) Recommend
(d) Reccomand

(*c*) 'Recommend' is the correct spelling. The word means to advocate or to suggest that a particular person will be suitable for a job.

3. (a) Argyument (b) Argument
(c) Arguement (d) Argyooment

(*b*) 'Argument' is the correct spelling. The word means a set of reasons given in support of an idea.

4. (a) Decisive (b) Desisive
(c) Descisive (d) Desicive

(*a*) 'Decisive' is the correct spelling. The word means able to make decisions quickly.

5. (a) Aggressive (b) Agresive
(c) Agressive (d) Aggresive

(*a*) 'Aggressive' is the correct spelling. The world means behaving or done in a determined and forceful way.

6. (a) Assassination (b) Asassination
(c) Asasination (d) Assasination

(*a*) 'Assassination' is the correct spelling. The word means the act of killing a prominent person for either political or other reason.

7. (a) Embarassment
(b) Embbarasment
(c) Embrasement
(d) Embarrassment

(*d*) 'Embarrassment' is the correct spelling. The word means a feeling of self-consciousness, shame or awkwardness.

Directions (Q. Nos. 8-17) *Given below are some idioms/phrases followed by four alternative meanings to each. Choose the response (a), (b), (c) or (d) which is the most appropriate meaning.*

8. Dirt cheap
(a) Extremely cheap
(b) Extremely costly
(c) Very cheap person
(d) Very cheap item

(*a*) Idiom 'Dirt cheap' means 'very inexpensive'. So, option (a) 'extremely cheap' express its correct meaning.

9. A shrinking violet
(a) A lean person
(b) A shy person
(c) A happy person
(d) A sad person

(*b*) Idiom 'A shrinking violet' means 'an extremely shy person'. So, option (b) 'a shy person' express its correct meaning.

10. Gordian knot
(a) Undoable job
(b) A difficult problem
(c) A different problem
(d) Doable job

(*b*) Idiom 'Gordian knot' means 'an extremely difficult problem'. So, option (b) express the correct meaning of given idiom.

11. Fall in a heap
(a) To be at the mercy of someone else
(b) To be thinking about someone
(c) To lose control of one's own feelings
(d) To be in control of one's own feelings

(*c*) The idiom 'Fall in a heap' means 'to lose control of one's own feelings'. So, option (c) express the correct meaning of given idiom.

12. Have a conniption fit
(a) To be very angry
(b) To be very happy
(c) To be very sad
(d) To be a jubilant person

(*a*) The idiom 'Have a conniption fit' means 'to become unreasonably angry or upset'. So, option (a) express the correct meaning of given idiom.

13. Be in seventh heaven
(a) To be extremely happy
(b) To be extremely upset
(c) To be extremely adventurous
(d) To be extremely silent

(*a*) The idiom 'Be in seventh heaven' means 'to be extremely happy'. So, option (a) is the correct answer of given idioms.

14. Hand in glove
(a) Working separately
(b) Working together
(c) Working for someone
(d) Not willing to work

(*b*) The idiom 'Hand in glove' means 'working together'. So, option (b) express the correct meaning of given idiom.

15. Nip in the bud
(a) Prevent a small problem before it becomes severe
(b) Prevent the big problems
(c) Make it severe
(d) Beating the problem

(*a*) The idiom 'Nip in the bud' means 'prevent a small problem before it becomes severe'. So, option (a) express the correct meaning of given idiom.

16. Like a shag on a rock
(a) Completely alone
(b) Completely idle
(c) Complete silence
(d) Complete happy

(*a*) Idiom 'Like a shag on a rock' means 'completely alone or isolated'. So, option (a) express the correct meaning of given idiom.

17. A pearl of wisdom
(a) An important piece of news
(b) An important person
(c) An important thing for life
(d) An important piece of advice

(*d*) Idiom 'A pearl of wisdom' means 'an important piece of advice'. So, option (d) is the correct choice.

Directions (Q. Nos. 18-36) *Each of the following passage in this section has some blank spaces with four words or groups of words given. Select whichever word or group of words you consider most appropriate for the* *blank space and indicate your response on the Answer Sheet accordingly.*

CLOZE COMPREHENSION 1

The founders of the Indian Republic .... **18.** (a) had
(b) has
(c) has had
(d) were

the farsightedness and the courage to commit .... **19.** (a) them
(b) themselves
(c) the people
(d) The course

to two innovations of historical significance in nation-buiding and social engineering : first, to .... **20.** (a) build
(b) building
(c) constructing
(d) built

a democratic and civil ...... **21.**
(a) libertarian
(b) liberation
(c) liberating
(d) liberty

society among illiterate people and, second, to undertake economic development .... **22.** (a) with a
(b) within a
(c) for the
(d) without a

democratic political structure, Hitherto, in all societies in which an economic takeoff or an early industrial and agricultural ....
23. (a) breakthrough
(b) breakout
(c) breaking
(d) investment

had occurred, effective democracy, especially from the working people, had been extremely limited. On the other hand, ... **24.**
(a) with
(b) from
(c) within
(d) for

the beginning, India was committed to .... **25.** (a) few
(b) some
(c) a
(d) an

democratic and civil libertarian political order and a representative system of government .... **26.**
(a) basing
(b) basis of
(c) based
(d) function

on free and fair elections to be conducted on the basis of universal adult franchise.

Solutions (Q. Nos. 18-26)

18. (a) Here, in the given blank, 'had' is an appropriate choice as the given sentence is in Past Tense.

19. (b) Here, reflexive pronoun 'themselves' will be used as it is used for plural noun 'founders'.

20. (a) With 'to', base form of verb is used. So, option (a) 'build' (V_1) is the correct choice for the given blank.

21. (a) Here, 'Libertarian' is the correct word according to the given context as 'Libertarian' means 'a person or society who believes in free will'.

22. (b) In the given blank option (b) 'within a' is the correct choice as the word 'within' is used to indicate enclosure or containment.

23. (a) 'Break through' means 'an important development or achievement' and it is best suited for the given blank.

24. (b) In the given blank, preposition 'from' will be used. 'From' is used to show the starting point of an action.

25. (c) Article 'a' should be used before the word which starts with a consonant sound (democratic).

26. (c) The suitable word for the given blank in 'based'. 'Based on something' means 'to use particular ideas or facts to make a decision'.

CLOZE COMPREHENSION 2

Ecology, in a very simple term, is a science that .... **27.** (a) studies
(b) study
(c) studying
(d) exploring

the interdependent, mutually reactive interconnected relationships .... **28.** (a) among
(b) between
(c) to
(d) for

the organisms and .... **29.**
(a) their
(b) its
(c) theirs
(d) all

physical environment on the one hand and among the organisms on the other hand. .... **30.**
(a) Through
(b) In spite of
(c) Though
(d) Because

the term 'ecology' was first coined and used by the German biologist Ernst Haeckel in 1869, a few conceptual terms .... **31.** (a) are
(b) were
(c) have been
(d) have

already proposed to reveal relationships .... **32.** (a) among
(b) those
(c) of
(d) between

organisms and their environment. For example, French zoologist IG Hilaire used the term 'ethology' .... **33.** (a) for
(b) to
(c) with
(d) in

the study of the relations of ....
34. (a) the
(b) a
(c) live
(d) dead

organisms within the family and society in the aggregate and in the community. British naturalist St. George Jackson Mivart proposed the term 'hexicology' with regard to the study of the relations ....
35. (a) for
(b) of
(c) within
(d) in

living creatures to other organisms and their environment as regards the nature of the locality they frequency, the temp erature and the .... **36.** (a) amount
(b) focus
(b) share
(d) quality

of light which suit them, and their relations to other organisms as enemies, rivals, or accidental and involuntary benefactors.

Solutions (Q. Nos. 27-36)

27. (a) Here, the general fact is given so Present Indefinite Tense will be used. Hence, option (a) 'studies' is the correct choice.

28. (b) Here, Preposition 'between' will be used to fill the given blank. 'Between' is used with two persons or things or group of things.

29. (a) Here, the suitable word is 'their' to indicate the physical environment of the organisms.

30. (c) Conjunction 'though' is the appropriate choice to fill the given blank as the rest of the sentence appear to contradict.

31. (b) The given sentence is in Past Tense, so Past form 'were' will be used to fill the given blank.

32. (d) To fill the given blank, preposition 'between' is the suitable choice. 'Between' is used with two persons or things or group of things.

33. (a) Here, preposition 'for' is the correct choice to fill the given blank. 'For' is used to show the purpose of something.

34. (a) Here, article 'the' should be used to make definite or specific reference that has been already referred to.

35. (b) In the given blank preposition 'of' is the suitable choice. Preposition 'of' is used to show the possession.

36. (a) With the word 'light', option (a) 'amount' is the suitable choice in the given context.

Directions (Q. Nos. 37-46) *Each item in the following questions consists of a sentence with an underlined word followed by four words. Select the option that is nearest in meaning to the underlined word and mark your response on your answer sheet accordingly.*

37. The properties of the family have been impounded by the order of the court.

(a) Confiscated
(b) Permitted
(c) Sold
(d) Put on hold

(a) The word 'Confiscated' is the correct synonym of the given word 'Impounded'. Both words mean to seize and take legal custody of something.

38. The officer in charge of the operations has been impugned for the excesses.

(a) Expelled
(b) Rewarded
(c) Challenged
(d) Given allowance

(c) 'Challenged' is correct synonym of 'Impugned'. Both means to challenge or oppose as false or lacking integrity impugned the defendant's character.

39. Cognitivist and linguists believe that every child is born with innate qualities.

(a) Biological (b) Intrinsic
(c) Extrinsic (d) Unnatural

(b) 'Intrinsic' is the correct synonym of the given word 'Innate' both words mean inborn or natural.

40. It was obligatory for the board to implement the rule.

(a) Compulsory (b) Unnecessary
(c) By chance (d) Problematic

(a) 'Compulsory' is the appropriate synonym of the given underlined word 'Obligatory'. Both means mandatory or required by a legal, moral or other rule.

41. They describe the act as a blatant betrayal of faith.

(a) Loyal (b) Faithfulness
(c) Treachery (d) Honesty

(c) 'Treachery' is the appropriate synonym of the given underlined word 'Betrayal'. Both words mean disloyalty or to expose to an enemy by treachery.

42. However, if it must decide, then it should do so on the narrowest ground possible.

(a) Widest (b) Slightly
(c) Smallest (d) Thick

(b) 'Slightly' is the appropriate synonym of the word 'Narrowest'. Both words mean limited in extent, amount or scope.

43. This is akin to a contractual relationship that places obligations on the entities entrusted with data.

(a) Removed (b) Narrow
(c) Similar (d) Unparallel

(c) 'Similar' is the appropriate synonym of the given underlined word 'Akin'. Both mean corresponding or alike.

44. Many communication problems can be attributed directly to misunderstanding and inaccuracies.

(a) Disapproved (b) Unofficial
(c) Ascribed (d) Tribute

(c) 'Ascribed' is the correct synonym of the underlined word 'Attributed'. Both words mean regard something as being caused by.

45. The exemptions granted to state institutions for acquiring informed consent from processing personal data in many cases appear to be too blanket.

(a) Obtain (b) Lose
(c) Giving (d) Thinking

(a) 'Obtain' is the correct synonym of the underlined word 'Acquiring'. Both words mean to obtain or to begin to have.

46. The manner in which this exercise has been undertaken leaves much to be desired.

(a) Disliked (b) Unlikely
(c) Wish for (d) Asked for

(c) 'Wish for' is the appropriate synonym of the underlined word 'Desire'. Both means strongly wished for or intended.

Directions (Q. Nos. 47-55) *In this section, each item consists of six sentences of a passage. The first and sixth sentences are given in the beginning as S_1 and S_6. The middle four sentences in each item have been jumbled up and labelled as P, Q, R and S. You are required to find the proper sequence of the four sentences and mark you response accordingly.*

47. S_1 : The master always says, "Refuse to be miserable".

S_6 : His is the art of right contact in life.

P : Before you fall into self-pity and blame games, remember that responsibility comes to only those who feel responsible.

Q : Challenges are faced by the strong and courageous, and if life brings you such opportunities, then turn failures into success.

R : Life can be painful, but it need not be sorrowful.

S : If you wan to be happy, find occasions to be cheerful.

The correct sequence should be

(a) RSPQ
(b) SQPR
(c) QRSP
(d) RQSP

(a) RSPQ is the correct sequence.

48. S_1 : Gandhiji reached Newcastle and took charge of the agitation.

S_6 : The treatment that was meted out to these brave men and women in jail included starvation and whipping and being forced to work in the mines by mounted military police.

P : During the course of the march, Gandhiji was arrested twice, released, arrested a third time and sent to jail.

Q : The employers retaliated by cutting off water and electricity to the workers' quarters, thus forcing them to leave their homes.

R : Gandhiji decided to march this army of over two thousand men, women and children over the border and thus see them lodged in Transvaal jails.

S : The morale of the workers, however, was very high and they continued to march till they were prosecuted and sent to jail.

The correct sequence should be

(a) QRPS (b) SRQP
(c) QPSR (d) RQSP

(d) RQSP is the correct sequence.

49. S_1 : One of the most important forces in the modern world, socialism was a direct result of the Industrial Revolution.

S_6 : This is how socialism as a theory and practice came into being.

P : Socialism was a direct challenge to capitalism and sought to put and end to such an exploitative economic structure.

Q : The gulf between the 'haves' and the 'have nots' continued to increase and out of this gap between the rich and poor sprang disputes.

R : It generated new wealth but as this new wealth only went to a minority, it could not solve the question of distribution.

S : The Industrial Revolution solved the question of production.

The correct sequence should be

(a) PQRS (b) SRQP
(c) SRPQ (d) RQSP

(b) SRQP is the correct sequence.

50. S_1 : Institutions define and play a regulatory role with regard to human behaviour.

S_6 : It shows how important it is for a nation to build institutions for nurturing democracy.

P : Once established, institutions set a dynamic relationship with the members constituting them and they mutually affect each other.

Q : They shape preferences, power and privilege.

R : At the same time, institutions themselves can be transformed by the politics they produce and such transformation can affect social norms and behaviours.

S : They also provide a sense of order and predictability.

The correct sequence should be

(a) RPQS (b) QRSP
(c) PSRQ (d) QSRP

(b) QRSP is the correct sequence.

51. S_1 : Idioms are a colourful and fascinating aspect of language.

S_6 : Idioms may also suggest a particular attitude of the person using them, for example, disapproval, humour, exasperation or admiration, so you must use them carefully.

P : Your language skills will increase rapidly if you can understand idioms and use them confidently and correctly.

Q : They are commonly used in all types of language, informal and formal, spoken and written.

R : In additions, idioms often have a stronger meaning than non-idiomatic phrases.

S : One of the main problems students have with idioms is that it is often impossible to guess the meaning of an idiom from the words it contains.

The correct sequence should be

(a) RQPS (b) RSPQ
(c) SRQP (d) QPSR

(d) QPSR is the correct sequence.

52. S_1 : Each organism is adapted to its environment.

S_6 : What can be taken in and broken down depends on the body design and functioning.

P : There is a range of strategies by which the food is taken in and used by the organism.

Q : For example, whether the food source is stationary (such as grass) or mobile (such as deer), would allow for differences in how the food is accessed and what is nutritive apparatus used by a cow or a lion.

R : The form of nutrition differs depending on the type and availability of food material as well as how it is obtained by an organism.

S : Some organisms break down the food material outside the body and them absorb it and others take in the whole material and break it down inside their bodies.

The correct sequence should be

(a) RQPS (b) QPSR
(c) SQPR (d) QPRS

(a) RQPS is the correct sequence.

53. S_1 : "When I was alive and had a human heart", answered the statue, "I did not know what tears were, for I lived in the Palace of Sans-Souci where sorrow is not allowed to enter.

S_6 : And now that I am dead they have set me up here so high that I can see all the ugliness and all the misery of may city, and though my heart is made of lead yet I cannot choose but weep.''

P : So I lived, and so I died.

Q : Round the garden ran a very lofty wall, but I never cared to ask what lay beyond it, everything about me was so beautiful.

R : My courtiers called me the Happy Prince, and happy indeed I was, if pleasure be happiness.

S : In the daytime I played with my companions in the garden and in the evening I led the dance in the Great Hall.

The correct sequence should be

(a) QSRP (b) PQRS
(c) PRQS (d) RPQS

(*) SQRP is the correct sequence.

54. S_1 : One day her mother, having made some cakes, said to her, ''Go, my dear, and see how your grandmother is doing, for I hear she has been very ill. Take her a cake, and this little pot of butter.''

S_6 : ''Does she live far off?'' said the wolf.

P : He asked her where she was going.

Q : The poor child who did not know that it was dangerous to stay and talk to a wolf, said to him, ''I am going to see my grandmother and carry her a cake and a little pot of butter from my mother''.

R : As she was going through the wood, she met with a wolf, who had a very great mind to eat her up, but he dared not, because of some woodcutters working nearby in the forest.

S : She set out immediately to go to her grandmother, who lived in another village.

The correct sequence should be

(a) PRQS (b) SRPQ
(c) PRSQ (d) RPQS

(b) SRPQ is the correct sequence.

55. S_1 : I had spent many nights in the jungle looking for game, but this was the first time I has ever spent a night lookign for a man-eater.

S_6 : It was in this position my men an hour later found me fast asleep; of the tiger I had neither heard nor seen anything.

P : I bitterly regretted the impulse that had induced me to place myself at the man-eather's mercy.

Q : The length of road immediately in front of me was brilliantly lit by the, moon, but to right and left the overhanging trees cast dark shadows and when the night wind agitated the branches and the shadows moved, I saw a dozen tigers advancing on me.

R : As the grey dawn was lighting up the snowy range which I was facing, I rested my head to my drawn-up knees.

S : I lacked the courage to return to the village and admit I was too frightened to carry out my self-imposed task, and with teeth chattering, as much from fear as from cold, I sat out the long night.

The correct sequence should be

(a) QPSR (b) SRPQ
(c) PRSQ (d) RPQS

(a) QPSR is the correct sequence.

Directions (Q. Nos. 56-65) *Given below are a few sentences. Identify the part of speech of the underlined word. Choose the response (a), (b), (c), or (d) which is the most appropriate answer.*

56. Rita eats her dinner quicky.

(a) Verb (b) Preposition
(c) Adjective (d) Adverb

(a) 'Eats' is a verb as it show action.

57. He thought the movie ended abruptly.

(a) Noun (b) Adverb
(c) Verb (d) Adjective

(b) 'Abruptly' is an adverb, which is modifying the verb 'ended' in the sentence.

58. I will meet you in the third week of August.

(a) Pronoun (b) Verb
(c) Preposition (d) Noun

(c) 'In' is a simple preposition and used to describe a location, time or place is used as preposition.

59. Jasmines and roses are my favourite flowers.

(a) Verb (b) Preposition
(c) Conjunction (d) Interjection

(c) 'And' is a conjunction as it joins two subjects.

60. She truthfully answered the detective's questions.

(a) Verb (b) Adjective
(c) Noun (d) Adverb

(d) Here, the word 'truthfully' is an adverb which is modifying the verb 'answered'.

61. Hurrah! We won the game!

(a) Interjection (b) Conjunction
(c) Noun (d) Pronoun

(a) The underlined word 'hurrah' is an Interjection or exclamation. In this part of speech, different words show different emotions.

62. The son writes meaningless letters to his father.

(a) Adverb (b) Verb
(c) Pronoun (d) Adjective

(d) The word meaningless is an 'adjective' that telling more about the noun (letters).

63. The secretary himself visited the affected families.

(a) Verb (b) Noun
(c) Adverb (d) Pronoun

(d) The word 'himself' is possessive pronoun, used for noun 'secretary'.

64. The children were walking, through the forest.

(a) Verb (b) Adverb
(c) Adjective (d) Preposition

(d) Here, 'through' is preposition. 'Through' is a simple preposition and can be used with respect to 'by' means of or from one side to another.

65. The Presiding Officer walked slowly to the dais.

(a) Adverb (b) Adjective
(c) Verb (d) Noun

(a) Slowly is an 'adverb' that modifying the verb 'walked'.

Directions (Q. Nos. 66-75) *Each item in this section consists of a sentence with an underlined word followed by four words/group of words. Select the option that is opposite in meaning to the underlined word and mark your response on your answer sheet accordingly.*

66. Beauty lies is the eyes of the beholder.

(a) Allure (b) Charm
(c) Inelegance (d) Ideal

(c) From the given options 'inelegance' is the correct antonym of underlined word 'beauty'. It means ungraceful or ugly.

67. Reading details about suicide cases can push vulnerable people taking the extreme step.

(a) Imperious (b) Impervious
(c) Helpless (d) Defenseless

(b) The underlined word 'vulnerable' means exposed to the possibility of being attacked or harmed. Its correct antonym is 'impervious' which means unable to be affected by anything.

68. Standing before a judge in a courtroom can be daunting for anyone.

(a) Uncomfortable (b) Encouraging
(c) Demoralising (d) Off-putting

(b) The underline word 'daunting' means to make (some one) feel intimidated or apprehensive. Its correct antonym is 'encouraging' which means to persuade someone to do something.

69. He has been facing a kind of intimidation by his friends for last two years.

(a) Wiles (b) Conviction
(c) Persuasion (d) Support

(c) The underlined, word 'intimidation' means to frighten or threaten someone. Its correct antonym is 'persuasion' which means the act of reasoning or pleading with someone.

70. There are many factors that constrain the philosophy of job enrichment in practice.

(a) Oblige (b) Pressure
(c) Restrict (d) Support

(d) The underlined word 'constrain' means compel or force (someone) to follow a particular course of action.

So, among the given options its correct antonym is 'support'.

71. People look for plausible remedies to the problems which they do not know.

(a) Acceptable (b) Unthinkable
(c) Solvable (d) Believable

(b) The underlined word 'plausible' means believable or probable. So, its correct antonym would be 'unthinkable' or incredulous.

72. The departing speech of the Chairperson ended with a plaintive note.

(a) Melancholic
(b) Gleeful
(c) Doleful
(d) Adventurous

(b) The underlined word 'plaintive' means sad or mournful.Its correct antonym would be 'gleeful' which means full of happiness.

73. The members have taken a unanimous decision to discord some of the rulings of the Managing Committee on problems relating to maintenance.

(a) Accord (b) Dissension
(c) Dispute (d) Friction

(a) The underlined word 'discord' means disagreement between people. Its correct antonym would be 'accord' means an official agreement or treaty.

74. The insolent nature of the speaker had provoked the members of the house and this led to pandemonium.

(a) Respectful
(b) Autocratic
(c) Impudent
(d) Thought provoking

(a) The underlined word 'insolent' means showing a rude and arrogant behaviour. Its correct antonym is 'respectful' which means showing regards or respect.

75. Incessant rains have resulted in failure of crops during this season.

(a) Sporadic
(b) Persistent
(c) Continual
(d) Ceaseless

(a) The underlined word 'incessant' means continuous or non-stop.

Its correct antonym would be 'sporadic' which means happening only sometimes, not regular or continuous.

Directions (Q. Nos. 76-88) *Each of the following items in this section consists of a sentence, the parts of which have been jumbled. These parts have been labelled as P, Q, R and S. Given below each sentence are four sequence namely (a), (b), (c) and (d). You are required to rearrange the jumbled parts of the sentence and mark your response accordingly.*

76. the company are often asked
(P)
the formal or informal interviews
(Q)
employees who are leaving
(R)
for their opinions during
(S)

The correct sequence should be

(a) RPSQ (b) RQPS
(c) PSQR (d) PQSR

(a) RPSQ is the correct sequence.

77. a hailstorm activity in the evenings
(P)
there is a possibility of
(Q)
while there could be
(R)
heavy rain towards the weekend
(S)

The correct sequence should be

(a) SQPR
(b) QSRP
(c) QRPS
(d) SPRQ

(b) QSRP is the correct sequence.

78. has been below normal since last week
(P)
the minimum temperature
(Q)
in some part of the city
(R)
when rain and hailstorm activity recorded
(S)

The correct sequence should be

(a) RSPQ (b) SPRQ
(c) QPSR (d) PSQR

(c) QPSR is the correct sequence.

79. for guest teachers (P)
in the Department of Biotechnology (Q) was also held (R)
a Selection Committee meeting (S)

The correct sequence should be

(a) SPRQ (b) QRSP
(c) PRQS (d) RSPQ

(a) SPRQ is the correct sequence.

80. for contractual assignment at Cultural Centres abroad (P)
as Teacher of Indian Culture for two years (Q)
applications are invited in a prescribed format (R)
from Indian Nationals for deployment (S)

The correct sequence should be

(a) QPRS (b) SRPQ
(c) PQRS (d) RSQP

(d) RSQP is the correct sequence.

81. while they are small (P) and do the (Q)
great things while they are easy (R)
do the difficult things (S)

The correct sequence should be

(a) SRQP (b) PSQR
(c) SRPQ (d) QPSR

(a) SRQP is the correct sequence.

82. then you sure (P) if you can't (Q)
don't deserve me at my best (R) handle me at my worst (S)

The correct sequence should be

(a) PRQS (b) QSPR
(c) RQSP (d) PSRQ

(b) QSPR is the correct sequence.

83. you will be more disappointed (P)
than by the ones you did do (Q)
by the things you didn't do (R) twenty years from now (S)

The correct sequence should be

(a) PRSQ (b) PRQS
(c) PQSR (d) SPRQ

(d) SPRQ is the correct sequence.

84. man is one who can lay (P)
a firm foundation with the bricks (Q)
a successful (R)
others have thrown at him (S)

The correct sequence should be

(a) PQSR (b) RQSP
(c) RPQS (d) QSPR

(c) RPQS is the correct sequence.

85. what we may be (P)
but not we not know (Q)
we know (R) what we are (S)

The correct sequence should be

(a) RSQP (b) QPRS
(c) QRPS (d) RQPS

(a) RSQP is the correct sequence.

86. for the ordinary (P) not willing to risk (Q)
the unusual if you are (R)
you will have to settle (S)

The correct sequence should be

(a) PRQS (b) SPQR
(c) RQSP (d) QSRP

(c) RQSP is the correct sequence.

87. as mere stepping stones (P)
his major achievements (Q)
for the next advance (R) he regarded (S)

The correct sequence should be

(a) SPQR (b) SQPR
(c) SPRQ (d) RPQS

(b) SQPR is the correct sequence.

88. have a great influence (P)
and they often shape our personality (Q)
on our adult lives (R)
events in our childhood (S)

The correct sequence should be

(a) SPRQ (b) SQRP
(c) SRQP (d) PQRS

(a) SPRQ is the correct sequence.

Directions (Q. Nos. 89-103) *Each item in this section has a sentence which has multiple parts. Find out the error part from the given option and indicate your response from the options (a), (b), (c) and (d) on the answer sheet.*

89. Experience has shown that the change-over from a closed economy to a mercantile economy has presented in human society innumerable problems.

(a) Experience has shown that
(b) The change over from a closed economy
(c) to a mercantile economy has presented
(d) in human society innumerable problems

(d) In the given sentence, part (d) has an error. 'In' should be replaced by article 'the' to make the sentence grammatically correct.

90. A closed economy is identified as a human community which produces all it consumes and consumed all it produces.

(a) A closed economy is identified
(b) as a human community
(c) which produces all it consumes
(d) and consumed all it produces

(d) Here, part (d) of the given sentence has an error. The word 'consumed' should be replaced with 'consumes' as the whole sentence is in simple present tense.

91. Iron is the most useful against all metals.

(a) Iron is (b) the most useful
(c) against all metals
(d) No error

(c) Here, part (c) of the given sentence has an error. The word 'against' should be replaced with 'among' because the sentence talks about all metals. 'Among' is for more than two things.

92. Mumbai is largest cotton centre in the country.
(a) Mumbai is
(b) largest cotton centre
(c) in the country (d) No error

(b) Here, part (b) has an error. Before superlative word 'largest' we should use definite article 'the.'

93. While every care have been taken in preparing the results, the company reserves the right to correct any inadvertent errors at a later stage.
(a) While every care have been taken
(b) in preparing the results
(c) the company reserves the right to correct
(d) any inadvertent errors at a later stage

(a) Here, part (a) has an error. The subject (every care) of the given sentence is singular. So, verb should also be used in singular form. Therefore, we must use 'has been' in place of 'have been' to make the given sentence grammatically correct.

94. My sister and me are planning a trip from Jaipur to Delhi.
(a) My sister and me are
(b) planning a trip
(c) from Jaipur to Delhi
(d) No error

(a) Here, part(a) has an error of pronoun with noun (my sister), use 'I' in place of 'me' to make the given sentence grammatically correct.

95. Despite the thrill of winning the lottery last week, my neighbour still seems happy.
(a) Despite the thrill of winning
(b) the lottery last week,
(c) my neighbour
(d) still seems happy

(b) In the given sentence, part (b) is an erroneous part. Article 'a' should be used in place of 'the' before lottery. Because lottery is a non-specific or non-particular noun and with non-specific or indefinite nouns article 'a' should be used.

96. Children are not allowed to use the swimming pool unless they are with an adult.
(a) Children are not allowed
(b) to use the swimming pool
(c) unless they are with an adult
(d) No error

(d) There is no error. Sentence is grammatically correct.

97. Here knowledge of Indian languages are far beyond the common.
(a) Her knowledge
(b) of Indian languages
(c) are far beyond the common
(d) No error

(c) Here, part (c) has an error. Use 'is' in place of 'verb' are as the subject knowledge is singular, to make the given sentence grammatically correct.

98. The care, as well as the love of a father, were missing in her life.
(a) The care, as well as the love
(b) of a father
(c) were missing in her life
(d) No error

(c) Here, part (c) is an erroneous part. As we know that with singular subject, singular verb should be used. Therefore, 'was' should be used in place of 'were' as the subject of the given sentence (care) is singular.

99. You look as if you have ran all the way home.
(a) you look as if
(b) you have ran
(c) all the way home
(d) No error

(b) Here, part (b) has an error. With present perfect verb have, we use third form of verb, so run should be used in place of 'ran' to make the given sentence grammatically correct.

100. The real voyage of discovery consist not in seeking new landscapes, but in having new eyes.
(a) The real voyage of discovery
(b) consist not in seeking new landscapes,
(c) but in having new eyes
(d) No error

(b) Here, part (b) has an error. Write 'consists' in place of 'consist' to make the given sentence correct.

101. No struggle can ever succeeded without women participating side by side with men.
(a) No struggle can ever succeeded
(b) without women participating
(c) side by side with men
(d) No error

(a) Here, part (a) has an error. Remove succeeded and write 'succeed' as sentence is in simple present tense.

102. Education is the passport to the future, for tomorrow belong to those who prepare for it today.
(a) Education is the passport to the future,
(b) for tomorrow belong to those
(c) who prepare for it today
(d) No error

(b) Here, part (b) has an error, 'belongs' should be used in place of 'belong' to make the given sentence grammatically correct.

103. There come a time when you have to choose between turning the page and closing the book.
(a) There come a time
(b) when you have to choose
(c) between turning the page
(d) and closing the book

(a) Here, part (a) has an error. Replace 'there come' with 'there comes' to make the given sentence grammatically correct.

Directions (Q.Nos. 104-120) *In this section, you have few short passage. After each passage, you will find some items based on the passage. First, read a passage and answer the items based on it. You are required to select your answers based on the contents of the passage and opinion of the author only.*

PASSAGE 1

Mankind's experience of various evolutionary changes from primitive times to the present day has been extensive and varied. However, man's problems were never before as complicated as they seem to be today. Man's economic activity centres primarily around production. Labour is said to be the primary factor of production; its role, therefore, has been given a lot of importance. It should be useful to have an overall view of the economic history of man from the nomadic times to the modern factory system and study its relevance to the various labour problems of today.

Initially, man passed through 'the hunting and fishing stage'. During this period, his basic needs were adequately met by Nature. Wild animals, birds and fruits satisfied his hunger, and his thirst was quenched by the waters of springs and rivers.

Caves gave him shelter and barks of trees were used as clothing. During this stage of man's progress, labour problems did not exist because of the absence of any economic, political and social systems.

Then came 'the pastoral stage', which was marked by a certain amount of economic activity. The nomadic and migratory nature of man persisted and together with his goats and cattle, he moved on to fresh pastures and meadows. Some conflicts would sometimes take place among herd-owners, for, during this period, the institution of nominal private property ownership was not known.

This stage paves the way for 'the agricultural stage', during which the class system began to develop. There was a small artisan class mostly self-employed; and there were also landed proprietors or Zamindars as well as slaves.

During the fourth stage of these developments, the handicrafts stage', a number of social and economic changes took place which marked the beginning of the labour problem in the world. The self-sufficient economy of the village underwent a drastic change. The community of traders and merchants emerged.

104. Humanity's evolution from primitive stage to the present has been

(a) static and smooth
(b) huge and diversified
(c) always violent
(d) always peaceful

(b) According to the passage, humanity's evolution has been very huge as well as diversified. It has evolved extensively and in various field.

105. .... "man's problems were never before as complicated as they seem to be today" means

(a) the present times are the best times of humanity
(b) the present times are the crucial period for humanity
(c) the present times pose much more challenges to humans than the previous times
(d) the present times provide much more facilities than the previous times

(c) The given sentence means that the present times pose much more challenges to humans that it was in primitive times. Man faces more complications and challenges in present time.

106. Why does the author say that labour problems did not exist during 'the hunting and fishing stage'?

(a) There was no nation existing at that time
(b) There were no economic, political and social systems
(c) There was no capitalism and market
(d) There was no labour law

(b) The author says that labour problem did not exist during the hunting and fishing state as society was not divided and did not have any economic, political and social system. Everybody lived in the same simple way.

107. "The pastoral stage was marked by a certain amount of economic activity." How?

(a) Humans started migrating and held goat-herds
(b) Humans started owning land
(c) Conflicts started as humans owned goats
(d) Humans started doing agriculture

(a) The pastoral stage was marked by a certain amount of economic activity as humans started migrating from one place to another alongwith his goats and cattle.

108. Which word in the passage means 'surfaced'?

(a) Quenched (b) Emerged
(c) Nomadic (d) Adequately

(b) The word 'surfaced' means 'having risen or emerged'. So, option (b) 'emerged' is the correct choice.

PASSAGE 2

Ever since independence, land reforms have been a major instrument of state policy to promote both equity and agricultural investment. Unfortunately, progress on land reforms has been slow, reflecting the resilience of structures of power that gave rise to the problem in the first place. The main instrument for realising more equitable distribution of land is the land ceiling laws. These laws were enacted by several states during the late 1950s and 1960s, and the early 1970s saw more stringent amendments in the laws to plug loopholes in the earlier laws.

But the record of implementation has not been satisfactory. Around 3 million hectares of land has been declared surplus so far, which is hardly 2 percent of net sown area in India. About 30 percent of this land has not yet been distributed as it is caught up in the litigations.

Besides, a number of Benami and clandestine transactions have resulted in illegal possession of significant amounts of land above ceiling limits. There are widespread reports of allotment of inferior, unproductive, barren and wasteland to landless household, many of whom have been forced to sell it off, in the absence of resources to make it productive.

In many instances, lands allotted to the rural poor under the ceiling laws are not in their possession. In some cases, Pattas were issued to the beneficiaries, but possession of land shown in the Pattas was not given, or corresponding changes were not made in the records of right.

The balance of power in rural India is so heavily weighed against the landless and the poor that implementing land ceiling laws is difficult. It is clear that without massive mobilisation of the rural poor and depending on democratic governance in rural India very little can be achieved in this direction.

Although half of India's population continues to depend on agriculture as its primary source of livelihood, 83 percent of farmers operate holdings of less than 2 hectares in size and the average holding size is only 1.23 hectares. This is often in fragments and unirrigated. There are also those who are entirely landless, although agriculture is their main source of livelihood.

They have inadequate financial resources to purchase and often depend on leasing in small plots, on insecure terms, for short periods, sometimes only for one season. Hence, many face insecurity of tenure and the growing threat of land alienation and pressure from urbanisation, industrialisation and powerful interest.

109. Why does the land reform prove to be slow?

(a) Because of the disparity in power structure
(b) Because of the power of the government
(c) Because states have different laws
(d) Because of the scarcity of land in the country

(*a*) As per the passage land reforms are very slow due to disparity in power structure and also 'land ceiling law' has not been implemented successfully.

110. Which of the following statements is/are correct?

1. Land ceiling laws have proved to be unsatisfactory.
2. The democratic structure of the government cannot provide solution to the problem of land reforms.
3. The owners of land have abundant natural resources.
4. Identified land for distribution has not been distributed due to court cases against it.

Select the correct answer using the codes given below

(a) 1 and 4 (b) Only 1
(c) 3 and 4 (d) 2 and 4

(*a*) Statement 1 and 4 are correct. Land ceiling laws have proved to be unsatisfactory and land which has been identified for distribution has court cases or litigations so it could not be distributed.

111. One of the reasons of selling off the lands by the allottees is that the lands

(a) unproductive and barren
(b) salty, not getting water
(c) fertile,but uncultivable
(d with the powerful people

(*a*) Land alloted to landless household was unproductive and barren, so they sell off the unproductive land in the absence of resources.

112. Which word/group of words in the passage means 'lawsuit'?

(a) Amendments
(b) Litigations
(c) Illegal possession
(d) Fragments

(*b*) Litigations also means 'Law suit'. Both words means 'a claim or dispute brought to a law court for adjudication'.

113. According to the author, what is the primary source of livelihood of majority of India's population?

(a) Industry (b) Forest
(c) Agriculture (d) None

(*c*) The primary source of livelihood of majority of India's population is 'agriculture'.

114. "There are also those who are entirely landless, although agriculture is their main source of livelihood" means

(a) they do not have money to buy lands
(b) they have sold off their lands to others
(c) most of them are agriculture labourers
(d) they are migrant labourers from other places

(*c*) The given statement means that although some farmers are totally dependent on agriculture for livelihood but they do not have enough money to buy lands.

Passage 3

Despite downsizings, workers' overall job satisfaction actually improved between 1988 and 1994. Some reasons given were improved work flow, better cooperation between departments, and increased fairness in supervision. Many firms today rely on attitude surveys to monitor how employees feel about working in their firms.

The use of employee attitude surveys had grown since 1944 when the National Industrial Conference Board "had difficulty finding fifty companies that had conducted opinion surveys". Today, most companies are aware of the need for employees' anonymity the impact of both the design of the questions and their sequence, the importance of effective communication, including knowing the purpose of the survey before it is taken and getting feedback to the employees after it is completed. Computerisation of surveys can provide anonymity, if there is no audit trail to the user, especially for short answers that are entered rather than written or typed on an identifiable machine.

Survey software packages are available that generate questions for a number of standard topics and can be customised by modifying existing questions or by adding questions. If the survey is compu- terised, reports can be generated with ease to provide snapshots of a given period of time, trend analysis and breakdowns according to various demographics. You may be interested in responses by age, sex, job categories, departments, division, functions or geography. The survey can be conducted by placing microcomputers in several locations convenient for employees' use. Employees are advised where the computers will be, for how long, and when the data will be collected (for instance, daily at 5:00 p.m. for three weeks). The screens should not be viewable to supervisors or passers-by. While there may be some risk that employees will take the survey more than once, there are comparable risks with other methods too. Managers may be interested in knowing how they are perceived by their peers and subordinates. Packages are available that can be customised, which allow the manager to complete a self-assessment tool used to compare self-perceptions to the anonymous opinions of others. This comparison may assist in the development of a more effective manager.

115. Which one of the following is not the reason for improved job satisfaction of employees?

(a) Improved work flow
(b) Better cooperation between departments
(c) Supervisors fairness
(d) Increased remuneration

(*d*) No where in the passage it has been mentioned that increased remuneration is reason for improved job satisfaction.

116. Companies feel that it is necessary to

(a) maintain anonymity of the employees and to have effective design and sequence of questions and effective communication
(b) maintain the fairness of the managers to be part of the survey
(c) conduct surveys from their employees
(d) maintain anonymity of the employees and not to have effective design and sequence of questions and effective communication

(*c*) Many companies felt that it is necessary to conduct surveys from their employees.

117. One major benefit of using survey software packages is

(a) reports can be generated easily
(b) privacy of a person is exposed to the supervisors
(c) employees would like to take up the test on computer
(d) employer can get to know the information immediately

(*a*) If companies use survey software packages, they can generate reports easily, can provide snapshots of a given period of time, trend analysis and breakdowns according to various demographics.

118. Which word in the passage means 'tendency'?

(a) Trend (b) Breakdowns
(c) Convenient (d) Perceptions

(*a*) 'Trend' also means 'tendency'. Both words means 'an inclination or likelihood'.

119. "The screens should not be viewable to supervisors or passers by." Why?

(a) To maintain the secrecy of a person
(b) The main problem is to enable everyone to participate
(c) The manager has to be fair enough
(d) To maintain the problems faced by women in job market

(*a*) The screens should not be viewable to supervisors or passers by in order to keep the surveys secret.

120. What does the word 'customised' means here?

(a) Adapted
(b) Take as it is
(c) Fixed
(d) Mass produced

(*a*) **The word 'customise' also means 'adapted'. Both word means 'to modify according to a customer's individual requirements'.**

PAPER III General Studies

1. Which one of the following is the motto of NCC?

(a) Unity and Discipline
(b) Unity and Integrity
(c) Unity and Command
(d) Unity and Service

(*a*) National Cadet Corps (NCC) is youth wing of armed forces with its headquarters at New Delhi (India). It is open to school and college students on voluntary basis.

The discussion for motto of NCC was started in 11th Central Advisory Meeting (CAD) held on 11th August, 1978, at that time there were many motto in mind like 'duty and discipline', 'duty, unity and discipline', 'duty and unity'. Later, at the 12th CAD meeting 'unity and discipline' was selected as the motto of NCC.

2. Which one of the following departments is not under the Ministry of Home Affairs?

(a) Department of Official Languages
(b) Department of Border Management
(c) Department of Jammu and Kashmir Affairs
(d) Department of Legal Affairs

(*d*) The Ministry of Home Affairs is a Ministry of the Government of India, it is mainly responsible for the maintenance of internal security and domestic policy. The Ministry has 6 departments, namely– Department of Border Management, Department of Internal Security, Department of Jammu and Kashmir Affairs, Department of Home, Department of Official Languages and Department of States.

Ministry of Home Affairs do not have Department of Legal Affairs under it. It is under Ministry of Law and Justice.

3. Which of the following statements is/are correct?

1. India is a signatory to the United Nations Convention to Combat Desertification (UNCCD).
2. Ministry of Home Affairs is the nodal ministry in the Government of India for the UNCCD.

Select the correct answer using the codes given below

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(*a*) United Nations Convention to Combat Desertification (UNCCD) is a convention to mitigate the effects of drought through national action programs that incorporate long-term strategies supported by international cooperation and partnership arrangements.

The convention was adopted in France (Paris) on 17th June, 1994 and entered into force in December, 1996. India became a signatory to UNCCD on 14th October, 1994 and ratified it on 17th December, 1996. Ministry of Environment, Forest and Climate Change is the nodal ministry for the convention, not Ministry of Home Affairs.

Hence, option (a) is correct.

4. Under which one of the following Articles of the Constitution of India, a statement of estimated receipts and expenditure of the government of India has to be laid before the Parliament in respect of every financial year?

(a) Article-110 (b) Article-111
(c) Article-112 (d) Article-113

(*c*) *Article-112 in the Constitution of India 1949 says*

The President shall in respect of every financial year cause to be laid before both the Houses of Parliament a statement of the estimated receipts and expenditure of the government of India for that year, in this Part referred to as the annual financial statement.

- Article 110 (3) of the Constitution of India categorically states that if any question arises whether a Bill is a Money Bill or not, the decision of the speakers of the house of the people thereon shall be final.
- Article 111 of the Constitution of India 'Assent to Bills' when a Bill has been passed by the Houses of Parliament, it shall be presented to the President and the President shall declare either that he assent to the Bill, or that he withholds assent there from.
- According to Article 113 of the Indian Constitution estimates expenditure from the Consolidated Fund of India in the Annual Financial Statement are to be voted the Lok Sabha.

5. The South Asian Association for Regional Cooperation (SAARC) was founded in

(a) Colombo (b) Islamabad
(c) Kathmandu (d) Dhaka

(*d*) The South Asian Association for Regional Cooperation (SAARC) is the regional inter-governmental organisation in South Asia.

SAARC was founded in Dhaka on 8th December, 1985. Its secretariat is based in Kathmandu (Nepal). The organisation promotes development of economic and regional integration.

Its member states include Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan and Sri Lanka.

6. Which one of the following countries is not a founding member of the New Development Bank?

(a) Brazil (b) Canada
(c) Russia (d) India

(*b*) New Development Bank, formerly, referred to as the BRICS Development Bank, is a multilateral development bank established by the BRICS States (Brazil, Russia, India, China, South Africa).

The bank is headquartered in Shanghai (China). The first regional office of the NDB is in Johannesburg (South Africa).

As Canada is not a member of BRICS, it is not a member of the New Development Bank.

7. The Public Financial Management System (PFMS) is a web-based online software application designed, developed, owned and implemented by the

(a) Department of Financial Services
(b) Institute of Government Accounts and Finance
(c) Controller General of Accounts
(d) National Institute of Financial Management

(*c*) The Public Financial Management System (PFMS), earlier known as Central Plan Schemes Monitoring System (CPSMS), is a web-based online software application developed and implemented by the office of Controller General of Accounts (CGA).

PFMS was initially started during 2009 as a Central Sector Scheme of Planning Commission with the objective of tracking funds released under all plan schemes of Government of India and real time reporting of expenditure at all levels of programme implementation. From 2014, it has been envisaged that digitisation of accounts shall be achieved through PFMS. Controller General of Accounts (CGA) is the Principal Accounting Adviser to Government of India in the department of expenditure, Ministry of Finance.

8. Match List I with List II and select the correct answer using the codes given below the lists.

List I (Institute)	**List II** (Location)
A. National Institute of Ayurveda	1. Chennai
B. national Institute of Homoeopathy	2. Bengaluru
C. National Institute of Unani Medicine	3. Kolkata
D. national Institute of Siddha	4. Jaipur

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	1	3	2	4
(c)	4	3	2	1	(d)	4	2	3	1

(*c*) **National Institute of Ayurveda**, established in 1976 at Jaipur by the Ministry of Health and Family Welfare, Government of India. It is the lone institute of Ayurveda under the AYUSH.

National Institute of Homoeopathy is an autonomous organisation under Ministry of AYUSH, Government of India. It was established on 10th December, 1975 in Kolkata.

National Institute of Unani Medicine is an autonomous organisation for research and training in Unani medicine in India. It was established in 1984 at Bengaluru. Under Ministry of AYUSH.

National Institute of Siddha is a institute for study and research of Siddha medicine. It was established in 2005 at Chennai.

Hence, option (c) is the correct match.

9. Which of the following statements about 'Invest India' is/are correct ?

1. It is a joint venture (not for profit) company.
2. It is the National Investment Promotion and Facilitation Agency of India.

Select the correct answer using the codes given below

(a) Only 1 (b) Only 2
(c) Both 1and 2 (d) Neither 1 nor 2

(*c*) Invest India was operationalised in early 2010, as a joint venture company between the Department of Industrial Policy and Promotion (DIPP) (35% equity), Federation of Indian Chambers of Commerce and Industry (FICCI) (51% equity) and State Govenment of India (0.5% each).

The core mandate of invest India is investment promotion and facilitation. It provides sector-specific and state-specific information to a foreign investor, assists in expediting regulatory approvals and offers hand-holding services.

Hence, both statements are correct.

10. The National Dope Testing Laboratory functions under

(a) Ministry of Health and Family Welfare
(b) Ministry of Science and Technology
(c) Ministry of Youth Affairs and Sports
(d) Ministry of Home Affairs

(*c*) National Dope Testing Laboratory (NDTL) is a premier analytical testing and research organisation established as an autonomous body under the Ministry of Youth Affairs and Sports, Government of India. It is the only laboratory in the country responsible for human sports dope testing. It was established in 2008 with an aim to get permanently accredited by International Olympic Committee and world anti-doping agency to do the testing for the banned drugs in human sports.

Hence, option (c) is correct.

11. In how many phases was the general election 2019 conducted in India?

(a) 6 phases (b) 7 phases
(c) 8 phases (d) 9 phases

(*b*) The 2019 Indian general election was held in 7 phases from 11th April to 19th May, 2019 to constitute the 17th Lok Sabha. The votes were counted and result was declared on 23rd May, 2019. The voting turnout was over 67% the highest ever as well as the highest participation by women voters.

The BJP-led National Democratic Alliance (NDA) won 353 seats whereas Congress-led United Progressive Alliance (UPA) won 91 seats. Since, no party was able to secure at least 10% of seats of total strength of Lok Sabha i.e. 55, there is no opposition leader in 17th Lok Sabha.

12. Which one of the following statements about the Organisation of Islamic Cooperation is not correct?

(a) Its permanent secretariat is located at Jeddah.

(b) It endeavours to safeguard and protect interests of the, Muslim world in the spirit of promoting international peace and harmony among various people of the world.
(c) It is the largest inter-governmental organisation of the world.
(d) It has consultative and cooperative relations with the UN.

(*c*) The Organisation of Islamic Cooperation (OIC) is an international organisation founded in 1969, consisting of 57 member states, with 53 being Muslim majority countries. The permanent secretariat is the executive organ of the organisation, located in Jeddah (Saudi Arabia).

The organisation has consultative and cooperative relations with the UN and other inter-governmental organisations to protect the vital interest of the Muslims. It endeavours to safeguard and protect interest of the Muslim world in the spirit of promoting international peace and harmony among various people of the world.

The OIC is the second largest inter-government organisation in the world after the UN.

Hence, option (c) is not correct.

13. Who among the following won the Italian Open Women's Tennis Singles Title 2019?

(a) Karolina Pliskova
(b) Johanna Konta
(c) Naomi Osaka (d) Serena Williams

(*a*) Karolina Pliskova won the Italian Open Women's Tennis Singles Title 2019, defeating Johanna Konta in the final. Naomi Osaka lost the tennis single title.

14. Which among the following IN ship(s) participated in the SIMBEX-19?

1. INS Kolkata 2. INS Shakti
3. INS Vikrant

Select the correct answer using the codes given below

(a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) Only 1

(*b*) SIMBEX (Singapore India Maritime Bilateral Exercise) is an annual bilateral naval exercise between India and Singapore. On the Indian side, two vessels 'INS Kolkata' and 'INS Shakti' participated in the exercise. Meanwhile, Singapore was represented by 'RSN Steadfast' and 'RSN Valiant'. INS Vikrant is an Indigenous Aircraft Carrier 1 which is expected to enter the service in 2023.

15. 'Triples' is a new format of

(a) Boxing (b) Judo
(c) Chess (d) Badminton

(*d*) The world body governing badminton, BFA has launched two new formats of the game-'Air Badminton' and 'Triples' with new dimension of the court and an innovative shuttlecock called Airshuttle. In triples format, the match will be played between a team of three players each with presence of at least one female. Traditional competitive badminton has been an indoor game. But in the new format launched Air Badminton will be outdoor sports.

16. Who among the following was the Chairman of the Committee on Deepening Digital Payments appointed by the RBI?

(a) HR Khan (b) Nandan Nilekani
(c) NR Narayana Murthy
(d) Sanjay Jain

(*b*) The Reserve Bank of India had constituted a high-level committee on Deepening of Digital Payments under the Chairmanship of 'Nandan Nilekani', former Chairman, UIDAI, in January, 2019. SEBI constituted a committee under the Chairmanship of 'HR Khan' (Retd Deputy Governor of RBI) to review Foreign Portfolio Investors (FPI) regulations 2014. 'Sanjay Jain' and 'HR Khan' are Member of Committee on Deepening of Digital Payments, not chairman. Committee on Deepening Digital Payments submitted its report in May 2019.

17. 'The Sasakawa Award' of United Nations is given in recognition of the work done in the field of

(a) disaster reduction
(b) peace keeping
(c) health services
(d) poverty alleviation

(*a*) United Nations Office for Disaster Risk Reduction (UNDRR) conferred 'Sasakawa Award 2019' for Disaster Reduction to Dr. Pramod Kumar Mishra, Additional Principal Secretary to Prime Minister of India.

The award was announced at the 6th session of global platform for Disaster Risk Reduction, 2019 at Geneva. The award was given in recognition of his long-term dedication to improve the resilience of communities most exposed to disasters. The UN Sasakawa Award is the most prestigious international award for disaster risk management.

18. Why was India's GS Lakshmi in news?

(a) She was the first Indian to play cricket for an English County Club.
(b) She became the first female ICC match referee.
(c) She was awarded the Ramon Magsaysay Award for the year 2019.
(d) She was the recipient of the Booker Prize in the year 2019.

(*b*) The ICC appointed India's GS Lakshmi as first ever female match referee. Lakshmi was a match referee in domestic women's cricket in 2008-09 and has also overseen three women's ODI matches and three women's T-20 matches in capacity of a match referee.

19. Who among the following was elected as the President of Indonesia for the second term?

(a) Joko Widodo
(b) Prabowo Subianto
(c) Sandiaga Uno (d) Jusuf Kalla

(*a*) Joko Widodo (better known as Jokowi) won a second term in office as Indonesian President. He got 55% of votes against his opponent Prabowo Subianto. In 2014, also Jokowi won by a slightly narrower margin. Voters enduring support for Jokowi can be partly explained his sensible policies.

20. In India, 21st May is observed as

(a) NRI Day
(b) National Youth Day
(c) National Technology Day
(d) National Anti-Terrorism Day

(*d*) Every year 21st May is observed as Anti-terrorism Day in India to wean away the youth from terrorism and showing as to how it is prejudicial to the national interest. The date is chosen to commemorate the death anniversary of one of the most eminent PM of India, 'Rajiv Gandhi' (20th August, 1941-21st May, 1991).

21. Arrange the following in the chronological order of their implementation :

1. The Indian Factory Act (First)
2. The Vernacular Press Act
3. The Morley-Minto Reforms
4. The Cornwallis Code

Select the correct answer using the codes given below

(a) 4, 2, 1, 3 (b) 2, 4, 1, 3
(c) 3, 4, 1, 2 (d) 2, 1, 3, 4

(***a***) Option (a) is the correct sequence. **The Cornwallis code** is a body of legislation enacted in 1793 by the East India Company to improve the governance of its territories India.

The Vernacular Press Act (1878) was enacted by the Britishers to curtail the freedom of Indian press and prevent the expression of criticism towards British policies.

During Lord's Ripon time, the **'First Factory Act'** was adopted in 1881 to improve the service conditions of the factory workers in India. The act banned the appointment of children below the age of seven in factories. It also reduced the working hours of children.

The Morley-Minto Reforms (1909) was an act of the Parliament of the UK that brought about a limited increase in the Involvement of Indians in the governance of British India.

22. Article-371A of the Constitution of India provides special privileges to which states?

(a) Nagaland (b) Mizoram
(c) Sikkim (d) Manipur

(***a***) Article 371A of the Constitution of India provides special privileges to the state of Nagaland with an aim to preserve their tribal culture.

Article 371 also confers special privileges to the state of Assam, Meghalaya, Sikkim, Manipur and Arunachal Pradesh. Article 371 gives the power to the President of India to establish separate development boards for Vidarbha, Marathwada regions of Maharashtra and the rest of the State and Saurashtra, Kutch and rest of Gujarat. Special provisions with respect to Andhra Pradesh, Karnataka, Goa are dealt in Articles 371D and 371E, 371J, 371I respectively.

23. How many Zonal Councils were set-up vide Part-III of the States Re-organisation Act,1956?

(a) Eight (b) Seven
(c) Six (d) Five

(***d***) Zonal Councils are Advisory Councils and are made up of the States of India that have been grouped into five zones to foster cooperation among them. These were set-up vide Part III of the States Reorganisation Act, 1956.

The present composition of each these Zonal Council is under Northern Zonal Council, Central Zonal Council, Eastern Zonal Council, Western Zonal Council and Southern Zonal Council. The North Eastern States are under another statutory body, the North Eastern Council.

24. Which provision of the Constitution of India provides that the President shall not be answerable to any Court in India for the exercise of powers of his office?

(a) Article-53 (b) Article-74
(c) Article-361 (d) Article-363

(***c***) **Article-361** of the Constitution of India provides that the President or the Governor of a State, shall not be answerable to any court for the exercise and performance of the powers and duties of his office or for any act done or purporting to be done by him in the exercise and performance of these powers and duties.

Article 53 Executive Power of the Union The executive power of the union shall be vested in the President and shall be exercised by him either directly or through officers subordinate to him in accordance with this Constitution.

Article 74 of the Constitution of the Republic of India provides for a Council of Ministers which shall aid the President in the exercise of his functions.

Article 363 of the Constitution bars the jurisdiction of all courts in any dispute arising out of any agreement which was entered into or executed before the commencement of the Constitution by any ruler of an Indian state to which the Government of India was party.

25. Which law prescribes that all proceedings in the Supreme Court shall be in English language?

(a) Article-145 of the Constitution of India
(b) Article-348 of the Constitution of India
(c) The Supreme Court Rules, 1966
(d) An Act Passed by the Parliament

(***b***) Article-348 of the Constitution of India prescribes language to be used in the Supreme Court and in the High Court and for acts, bills, etc.

All proceeding in the Supreme Court and in every High Court, the authoritative texts of all bills to be introduced or amendment there to be moved in either House of Parliament or either House of State Legislature, shall be in English.

26. The total number of members in the Union Council of Ministers in India shall, not exceed

(a) 10% of the total number of members of the Parliament
(b) 15% of the total number of members of the Parliament
(c) 10% of the total number of members of the Lok Sabha
(d) 15% of the total number of members of the Lok Sabha

(***d***) The Union Council of Ministers Exercises Executive Authority in the Republic of India. It consists of senior ministers, called Cabinet Ministers, Minister of State or Deputy Ministers. According to the Constitution of India, the total numbers of ministers in the Council of Ministers must not exceed 15% of the total numbers of member of the Lok Sabha.

27. Which one of the following is the most noticeable characteristic of the mediterranean climate?

(a) Limited geographical extent
(b) Dry summer
(c) Dry winter
(d) Moderate temperature

(***b***) Mediterranean climate is characterised by dry summer, mild and wet winters. The climate receive its name from the mediterranean basin, where this climate type is most common. Climate zones are located along the Western sides of continents, between 30° and 45° North and South of the equator.

28. Which one of the following rivers takes a 'U' turn at Namcha Barwa and enters India?

(a) Ganga (b) Tista
(c) Barak (d) Brahmaputra

(***d***) Brahmaputra originates on the Angsi Glacier located on the Northern side of the Himalayas as Yarlung Tsangpo river and flows in Southern Tibet to break through the Himalayas in great Gorges.

Tsangpo (Brahmaputra) enters India after taking a 'U' turn at Namcha Barwa and flows in Arunachal Pradesh where it is known as 'Dihang' or 'Siang' rivers.

29. What was the Dutt-Bradley Thesis?

(a) The Working Committee of the Indian National Congress decided that Congress should play a crucial role in realising the independence of India

(b) The Socialist Party decided to play foremost part in anti-imperialist struggle
(c) Revolutionary socialist Batukeshwar Dutt put forth a ten-point plan to work for the success of anti-imperialist front
(d) It was a Communist Party document, according to which the National Congress could play a great part and a foremost part in realising the anti-imperialist people's front

*(**d**)* Dutt-Bradley Thesis was a Communist Party document, according to which the National Congress could play a great part and a foremost part in realising the anti-imperialist people's front.

The Anti-imperialistic people's front in India written by Rajni Palme Dutt and Ben Bradley, popularly known as the Dutt-Bradley Thesis.

Anti-imperialist people are opposed to colonialism, colonial empires, hegemony, imperialism and the territorial expansion of a counter beyond its established borders.

Hence, option (d) is the correct answer.

30. The Khuntkatti tenure was prevalent in which one of the following regions of India during the British Colonial Rule?

(a) Bundelkhand (b) Karnataka
(c) Chota Nagpur
(d) Madras Presidency

*(**c**)* Tribal people in Chota Nagpur area, which is present day in Jharkhand, practised Khuntkatti Tenure (system) (joint holding tribal lineages) till mid-19th century.

31. Who was the author of the book '*Plagues and Peoples*'?

(a) William H McNeil
(b) WI Thomas
(c) Rachel Carson
(d) David Cannadine

(***a***) Plagues and Peoples is a book on epidemiological history by William Hardy McNeil published in New York City in 1976. It was a critical and popular success, offering a radically new interpretation of the extra ordinary impact of infectious disease on culture as a means of enemy attack.

The book ranges from examining the effects of small pox in Mexico, the bubonic plague in China, to the typhoid epidemic in Europe.

32. Who among the following started the Indian Agriculture Service?

(a) Lord Curzon (b) William Bentinck
(c) Lord Minto (d) Lord Rippon

*(**c**)* The Indian Agriculture Service was started in 1906 under the All India Board of Agriculture. It was during the viceroyalty of Lord Minto. However, the training of staff could not be taken out due to lack of funds and primary aim remained as merely education.

33. Chandimangal was composed in which one of the following languages during the 16th century CE?

(a) Sanskrit (b) Tamil
(c) Bengali (d) Oriya

*(**c**)* The Chandimangal is an important subgenre of Mangal Kavya, the most significant genre of medieval Bengali literature.

The text belonging to this subgenre praise Chandi or Abhaya, primarily a folk goddess, but subsequently identified with puranic goddess Chandi.

This identification was probably completed a few centuries before the earliest composition of the Chandimangal Kavya.

34. In December, 1962, which Soviet leader declared that China was responsible for the Sino-Indian War of 1962?

(a) Khrushchev (b) Bulganin
(c) Suslov (d) Malenkov

*(**a**)* The Sino-Indian War, also known as the Indo-China war and Sino-Indian border conflict, was a war between China and India that occurred in 1962. On 2nd October, Soviet leader Nikita Khrushchev declared that China was responsible for the Sino-Indian War of 1962.

35. Which of the following statements with regard to the 'Make in India' initiative is/are correct?

1. It was launched in the year 2018.
2. Its objective is to foster innovation.

Select the correct answer using the codes given below

(a) Only 1 (b) Only 2
(c) Both 1and 2 (d) Neither 1 nor 2

(***d***) 'Make in India' initiative was lunched by Prime Minister Modi in September, 2014 as part of a wider set of nation building initiatives. It aim to transform India into a global design and manufacturing hub. It covers twenty five sectors of the economy.

Hence, option (d) is correct.

36. Which one of the following states does not have a Legislative Council?

(a) Karnataka
(b) Telangana
(c) Jammu and Kashmir
(d) Arunachal Pradesh

(***d***) Legislative Council is the Upper House in those States of India that have a bicameral legislature its establishment is defined in Article-169 of the Constitution of India.

As of 2019, 7 out of 29 States have Legislative Council namely—Andhra Pradesh, Bihar, Karnataka, Maharashtra, Jammu and Kashmir, Telangana and Uttar Pradesh.

37. What is SWAYAM?

(a) Study Webs of Active-Learning for Young Aspiring Minds
(b) Study Webs of Active-Learning for Youth Aspiring Minds
(c) Study Webs of Active-Learning for Young Aspiration Minds
(d) Study Webs of Active-Learning for Youth of Aspiration Minds

(***a***) 'SWAYAM' stands for Study Webs of Active-Learning for Young Aspiring Minds. It is a programme of Ministry of Human Resource Development, Government of India. It provides opportunities for life long learning. Here, learner can choose from hundred of courses, virtually every course that is taught at the university, college, school level and these shall be offered by best of the teachers in India and elsewhere.

38. Which one of the following is not enumerated in the Constitution of India as a Fundamental Duty of citizens of India?

(a) To safeguard public property
(b) To protect and improve the natural environment
(c) To develop the scientific temper and spirit of inquiry
(d) To promote international peace and security

(***d***) The Fundamental Duties of citizens were added to the Constitution by the 42nd Amendment in 1976, upon the recommendation of the Swaran Singh Committee that was constituted by the government.

Under Article 51, it mentions to promote international peace and security. It is not enumerated in the Constitution of India as Fundamental Duty of Citizens of India.

39. Who among the following in his book 'The Managerial Revolution' argued that a managerial class dominated all industrial societies, both capitalist and communist, by virtue of its technical and scientific knowledge and its administrative skills?

(a) James Burnham
(b) Robert Michels
(c) Gaetano Mosca
(d) Vilfredo Pareto

(***a***) James Burnham in his book 'The Managerial Revolution' argued that a managerial class dominated all industrial societies, both capitalist and communist, by virtue of its technical and scientific knowledge and its administrative skills.

40. Which one of the following conditions laid down in the Constitution of India for the issue of a writ of Quo-Warranto is not correct?

(a) The office must be public and it must be created by a statute
(b) The office must be a substantive one
(c) There has been a contravention of the Constitution or a state in appointing such person to that office
(d) The appointment is in tune with a statutory provision

(***d***) Writ of 'Quo-Warranto' in literal sense means by what authority or warrant it is issued by the court to enquire into legality of claim of a person to public office. Hence, it prevents illegal **usurpation** of public office by a person. Hence, option (d) is not correct.

41. Which one of the following equals Personal Disposable Income?

(a) Personal Income – Direct taxes paid by households and miscellaneous fees, fines, etc.
(b) Private Income – Saving of Private Corporate Sectors – Corporation Tax
(c) Private Income – Taxes
(d) Total expenditure of households – Income Tax gifts received

(***a***) Personal Income measures the income that is received by individuals, but not necessarily earned.

But personal disposable income represents what people actually have that they can spend. It is also a result of consumer spending as well as private saving. Personal Disposable Income = Personal Income – Direct taxes paid by households and miscellaneous fees, fines, etc.

Hence, option (a) is correct.

42. The working of the price mechanism in a free-market economy refers to which one of the following?

(a) The interplay of the forces of demand and supply.
(b) Determination of the inflation rate in the economy.
(c) Determination of the economy's propensity to consume.
(d) Determination of the economy's full employment output.

(***a***) Free market is an economic system based on supply and demand with little or no government control.

Its depends on interplay of the forces of demand and supply.

It is a summary description of all voluntary exchanges that take place in a given economic environment.

43. Indexation is a method whose use can be associated with which one of the following?

(a) Controlling inflation
(b) Nominal GDP estimation
(c) Measurement of savings rate
(d) Fixing of wage compensation

(***a***) Indexation is a method or technique used by organisations or government to control Inflation by connecting prices and asset value to inflation.

This is done by linking adjustments made to the value of a good, service or another metric, to a predetermined index.

44. A car undergoes a uniform circular motion. The acceleration of the car is

(a) zero
(b) a non-zero constant
(c) non-zero but not a constant
(d) None of the above

(***c***) Circular motion is a movement of an object along the circumference of a circle or rotation along a circular path. It can be uniform with constant angular rate of rotation and constant speed or non-uniform with a changing rate of rotation.

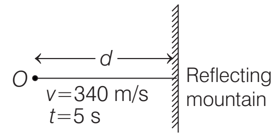
Since, the object's velocity vector is constantly changing direction, the moving object is undergoing acceleration by a centripetal force in the direction of the centre of rotation without this acceleration, the object would move in a straight line, according to Newton's laws of motion.

Hence, the acceleration of the car is a non-zero but not a constant.

45. An echo is heard after 5 s of the production of sound which moves with a speed of 340 m/s. What is the distance of the mountain from the source of sound which produced the echo?

(a) 0.085 km (b) 0.85 km
(c) 0.17 km (d) 1.7 km

(***b***) Given, speed of sound = 340 m/s



Let the distance of the mountain from the source of sound is d.

To heard echo after 5 s, sound will travel $2d$ distance. Since Echo is repection of sound which is reflected from any surface.

$\therefore$ Speed $= \dfrac{\text{Distance}}{\text{Time}}$

$\Rightarrow \quad 340 = \dfrac{2d}{5}$

$\Rightarrow \quad d = 170 \times 5 = 850\text{ m} = 0.85\text{ km}$

46. A 100 W electric bulb is used for 10 h/day. How many units of energy are consumed in 30 days?

(a) 1 unit (b) 10 units
(c) 30 units (d) 300 units

(***c***) Given, power of electric bulb = 100W

Duration used each day = 10h

Number of days = 30

Total energy consumed by the electric bulb = Power × Duration

$= 100 \times 10\text{ h/day} \times 30\text{ days}$

$= 30000 = 30\text{ kWh}$

So, energy consume is 30 units.

47. Which of the following statements relating to the Fifth Schedule of the Constitution of India is not correct?

(a) It relates to the special provision for administration of certain areas in the States other than Assam, Meghalaya, Tripura and Mizoram.
(b) Tribal Advisory Councils are to be constituted to give advice under the Fifth Schedule.
(c) The Governor is not authorised, to make regulations to prohibit or

restrict the transfer of land by or among members of the Scheduled Tribes.

(d) The Governors of the States in which there are scheduled areas have to submit reports to the President regarding the administration of such areas.

(*c*) The Fifth Schedule of the Constitution deals with the administration and control of scheduled areas and Scheduled Tribe in any State except the four States of Assam, Meghalaya, Tripura and Mizoram. It says, Tribal Advisory Councils are to be constituted to give advice under the Fifth Schedule.

The Governor is empowered to make regulations to prohibit or restrict the transfer of land by or among members of the Scheduled Tribes.

Hence, option (c) is not correct.

48. Consider the following statements with regard to the formation of new states and alteration of boundaries of existing states.

1. Parliament may increase the area of any state.
2. Parliament may diminish the area of any state.
3. Parliament cannot alter the boundary of any state.
4. Parliament cannot alter the name of any state.

Which of the statements given above is/are not correct?

(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 4 only

(*a*) According to Article 3 of Indian constitutions a state has no say over the formation of new states beyond communicating its views to Parliament

Article 3 assigns to parliament the power to enact legislation for the formation of new states.

Parliament may create new states in a number of ways-(a) separating territory from any state (b) uniting two or more states (c) uniting part of states and (d) uniting any territory to a part of any state.

49. Consider the following statements:

1. The Advocate General of a State in India is appointed by the President of India upon the recommendations of the Governor of the concerned state.
2. As provided in the Code of Civil Procedure, High Courts have original appellate advisory jurisdiction at the state level.

Which of the statement(s) given above is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(*d*) The Advocate General of a State in India is appointed by the Governor of a State.

A High Court is primarily a Court of Appeal. It hears appeals against the Judgements of Subordinate Courts functioning in its territorial jurisdiction. It has appellate jurisdiction in both civil and criminal matters.

Under Act 226 original appellate jurisdiction under Article 226 of Constitution, not civil procedure code. Hence, both the statements are incorrect.

50. Which one of the following forms of Constitution contains the features of both the Unitary and Federal Constitution?

(a) Unitary (b) Federal
(c) Quasi-Federal (d) Quasi-Unitary

(*c*) Quasi-Federal forms of Constitution contains the features of both the unitary and Federal Constitution.

The Constitution of India has both the unitary as well as federal features. Federal features are dual polity, written Constitution, division of powers, supremacy of the Constitution, Rigid Constitution, independent judiciary, Bicameralism. Unitary features are strong centre, single Constitution, flexibility of the Constitution etc.

51. Which one of the following Indian States has no international boundary?

(a) Bihar
(b) Chhattisgarh
(c) Uttarakhand
(d) Meghalaya

(*b*) India is the 7th largest country in the world and it is the only country in the Indian subcontinent to share its land frontiers with every member country of the subcontinent .

Afghanistan, Bhutan, Bangladesh, China, Myanmar, Nepal and Pakistan are bordering countries of India. Chhattisgarh do not have any international boundary.

52. Which one of the following Indian cities is not located on a river bank?

(a) Agra
(b) Bhagalpur
(c) Bhopal
(d) Kanpur

(*c*) Agra is a city in the State of Uttar Pradesh in India situated on the banks of river Yamuna.

Bhagalpur is one of the important city of the State of Bihar. It is situated on the banks of the river Ganga.

Kanpur is situated in State of Uttar Pradesh, it is situated on Ganga river. Hence, Bhopal is not situated on a river bank.

53. Where are Jhumri Telaiya and Mandar Hills situated?

(a) Jharkhand
(b) Bihar
(c) Assam
(d) West Bengal

(*) Jhumri Telaiya is a city in Koderma district of Jharkhand (India). Geographically, it is located in the Damodar River valley. The city is known for Mica Production and Telaiya dam. The dam was constructed on River Barakar by the Damodar Valley Corporation (DVC) and opened in the year 1953. It is a part of Damodar Valley Project.

Mandar Hill is a small mountain situated in Banka district under Bhagalpur division of state of Bihar. This hill has many references in Hindu mythology as Mandarachal Parvat.

As per references found from Puranas and Mahabharata this hill was used for 'Samudra Manthan'.

54. Which one of the following is not correct regarding South India?

(a) Diurnal range of temperature is less
(b) Annual range of temperature is less
(c) Temperature is high throughout the year
(d) Extreme climatic conditions are found

(*d*) South Indian regions have a tropical climate with the monsoons playing a major part. Diurnal range of temperature is less in South India.

Annual range of temperature is also due to proximity to equator. Temperature is high throughout the year. Climatic conditions are not extreme here as they are in Northern part of India.

Hence, option (d) is not correct.

55. Which one of the following statements regarding sex composition is not correct?

(a) In some countries, sex ratio is expressed as number of males per thousand females.
(b) In India, sex ratio is expressed as number of females per thousands males.
(c) At world level, sex ratio is about 102 males per 100 females.
(d) In Asia, there is high sex ratio.

(*d*) In some countries, sex ratio is expressed as number of males per thousand females. Whereas, in India, it is expressed as number of females per thousand males.

At world level, sex ratio is about 102 males per 100 females. Europe has highest sex ratio in the world whereas Asia has lowest sex ratio.

Hence, option(d) is incorrect.

56. Who among the following has given the concept of Human Development?

(a) Amartya Sen
(b) Mahbub-ul-Haq
(c) Sukhamoy Chakravarty
(d) GS Chaddha

(*b*) The concept of Human Development was developed by economist Mahbub-ul-Haq.

Human Development is defined as the process of enlarging people's freedom and opportunities and improving their well-being.

57. Which one of the following regions is an important supplier of citrus fruits?

(a) Equatorial region
(b) Mediterranean region
(c) Desert region
(d) Sub-humid region

(*b*) Citrus fruits are the highest value fruit crop in terms of international trade. They are produced all over the world. Mediterranean region is an important supplier of citrus fruits.

Oranges account for the majority of citrus production, but the industry also sees significant quantities of grape fruits, pomelos, lemons.

58. Who were the Nayanars?

(a) Those who were immersed in devotion to Vishnu
(b) Those who were devotees of Buddha
(c) Leaders who were devotees of Shiva
(d) Leaders who were devotees of Basveshwara

(*c*) Nayanars were a group of 63 saints in the 6th to 8th century who were devoted to the Hindu God Shiva in Tamil Nadu.

They along with the Alwars, influenced the Bhakti movement in Tamil. The names of the Nayanars were first compiled by Sundarar.

59. Match List I with List II and select the correct answer using the codes given below the lists.

List I (Ethnic Territorial Segment)	List II (Related Occupational Pattern)
A. Maruta Makkal	1. Pastoralists
B. Kuravan Makkal	2. Fishing people
C. Mullai Makkal	3. Ploughmen
D. Neytal Makkal	4. Hill people

Codes

	A	B	C	D
(a)	3	1	4	2
(b)	2	1	4	3
(c)	3	4	1	2
(d)	2	4	1	3

(*c*) According to Tamil literature, the basic unit of ethnic identification in Dravidian culture was 'Nadu' and these were divided into **five** types on the basis of natural sub-region.

The ethnic territorial segments are

Maruta Makkal	Ploughmen inhabiting fertile land.
Kuravan Makkal	Hill people who leave the forest to work in Panai.
Mullai Makkal	Pastoralist.
Neytal Makkal	Fishing people living in coastal villages.
Palai Makkal	People of dry plains

60. Who among the following Mughal emperors was a follower of the Naqshbandiyya leader, Khwaja Ubaydullah Ahrar?

(a) Babur
(b) Humayun
(c) Akbar
(d) Jahangir

(*a*) Mughal emperor Babur was the follower of Khwaja Ubaydullah Ahrar, a member of Naqshbandi Sufi order. This order was introduced in India by Khwaja Baqi Billah from the beginning the mystics of this order stressed on the observance of the shariat and denounced all innovations or biddat.

61. Which one of the following statements about the Government of India Act, 1919 is not correct?

(a) It extended the practice of communal representation
(b) It made the Central Executive responsible to the Legislature
(c) It is also known as the Montague-Chelmsford Reforms
(d) It paved the way for federalism by clearly separating the responsibilities of the Centre and the Provinces

(*d*) Government of India Act 1919, also known as Montague-Chelmsford Reforms. It extended the principle of communal representation to Sikhs, Indian Christians, Anglo-Indians and Europeans. It divided the provincial subjects into 2 parts transferred and reserved. Transferred subjects were to be administered by the Governor with the aid of ministers responsible to the Legislative Council.

Reserved subjects were to be administered by the Governor and his Executive Council without being responsible to the Legislative Council.

Hence, option (d) is incorrect.

62. The concept of 'Four Pillar State', free from district magistracy for India was suggested by

(a) Lala Lajpat Rai
(b) Ram Manohar Lohia
(c) Raja Ram Mohan Roy
(d) Subhash Chandra Bose

(*b*) Ram Manohar Lohia believes in decentralisation of economic and political powers. So he gave the concept of Four Pillar State, which is based on the principle of division of powers.

According to him four pillar constitute four limbs of the State. They are the village, the district, the province and the centre with sovereign powers.

63. Which one among the following is not a part of the Fundamental Rights (Part III) of the Constitution of India?

(a) Prohibition of traffic in human beings and forced labour
(b) Prohibition of employment of children in factories
(c) Participation of workers in management of industries
(d) Practice any profession or to carry on any occupation, trade or business

(*c*) Participation of workers in management of industries is not a part of the Fundamental Rights (Part III) of the Constitution of India.

Participation of workers in management of industries is directive principle of State policy.

Fundamental Rights are enshrined in Part-III of the Constitution from Article-12-35. Prohibition of traffic in human beings and forced labour is Article-23.

Prohibition of employment of children in factories is Article-24. Practice any profession, or to carry on any occupation, trade business is Article 19(1)g.

64. Which one of the following is not geographical requirement for cultivation of cotton?

(a) Temperature reaching 25°C or more in summer.
(b) Moderate to light rainfall.
(c) Medium laom soil with good drainage.
(d) A growing period of at least 100 frost free days.

(*d*) Cultivation of cotton requires a growing period of atleast 200 frost free days, and not just 100 frost free days.

Other requirements for cotton cultivation include, temperature reaching 25°C or more in summer, moderate to light rainfall (55-100 cm), medium loam soil with good drainage and high water retention capacity.

Rainfall during harvesting season is harmful for its production. India is the largest producer of cotton in the world.

65. Which one of the following statements regarding temperate coniferous forest biome is not correct?

(a) They are characterised by very little undergrowth.
(b) They have a growing period of 50 to 100 days in a year.
(c) There is low variation in annual temperature.
(d) There is high range in spatial distribution of annual precipitation.

(*c*) Temperate coniferous forest are found in areas with cool winters, warm summers and abundant rainfall. Trees like spruce, pine and cedar grows here, but they have little undergrowth.

They have a growing period of 50 to 100 days in a year. Rainfall is abundant but there is high range in spatial distribution of annual precipitation. Thus, option (a), (b) and (d) are correct. These areas have moderate temperature but experiences seasonal changes and considerable valuation.

Temperate coniferous forest biome has high (and not low) variation in annual temperature. It ranges from an average of –40°C in winters to 10°C in summers.

Hence, option (c) is not correct.

66. Match List I with List II and select the correct answer using the codes given below the lists.

List I (Peak)	List II (Name of Hill)
A. Anamudi	1. Nilgiri
B. Doddabetta	2. Satpura
C. Dhupgarh	3. Aravalli
D. Guru Shikhar	4. Annamalai

Codes

	A	B	C	D
(a)	3	2	1	4
(b)	3	1	2	4
(c)	4	1	2	3
(d)	4	2	1	3

(*c*) **Anamudi** is the highest peak in the Western Ghats in India. It is located at junction of Cardamom hills, Annamalai hills and Palani hills.

Doddabetta is the highest peak in the Nilgiri mountains, situated in Nilgiris distric of TamilNadu.

Dhupgarh is highest point on the Satpura mountains, situated in Panchmaehi, Madhya Pradesh.

Guru Shikhar, a peak in Rajasthan is the highest point of the Aravalli range. Hence, option (c) is correct.

67. Coral reefs are not found in which one of the following regions?

(a) Lakshadweep Islands
(b) Gulf of Kutch
(c) Gulf of Mannar
(d) Gulf of Cambay

(*d*) In India, coral reefs are found in Andaman and Nicobar Islands, Gulf of Kutch, Gulf of Mannar, Lakshadweep Islands, Gulf of Khambat etc. It is not found in Gulf of Cambay.

Hence, option (d) is not correct.

68. In which one of the following states is jute not significantly cultivated?

(a) Assam
(b) West Bengal
(c) Odisha
(d) Andhra Pradesh

(*d*) The cultivation of jute is mainly confined to the Eastern region of the country. The jute crop is grown in nearly 83 districts of West Bengal, Assam, Odisha, Bihar, Uttar Pradesh, Tripura and Meghalaya.

Jute is not cultivated in Andhra Pradesh significantly.

Hence, option (d) is not correct.

69. Consider the following statements:

1. According to Mahavamsa, Ashoka turned to the Buddha's dhamma when his nephew Nigrodha preached the doctrine to him.
2. Divyavadana ascribes Ashoka being drawn to the Buddha's teaching to the influence of Ashokavandana Samudra, a merchant-turned monk.
3. Dipavamsa speaks of Samudra, the 12 year old son of a merchant, as the key figure in Ashoka's coming under the influence of the Buddhist dhamma.

Which of the statements given above is/are correct?

(a) Only 1 (b) Only 2
(c) 1 and 2 (d) 1 and 3

(*a*) According to Mahavamsa, Ashoka turned to the Budha's dhamma when his nephew Nigrodha preached the doctrine to him.

Divyavadana ascribes Ashoka being drawn to the Budha's teaching to the influence of Upagupta (a Buddhist monk).

Dipavamsa refers to three visits to the Sri Lanka by the Buddha, the places being Kelaniya, Deegavapi Raja Maha Viharaya. The Dipavamsa lauds the Theravada as a great banyan tree.

Hence, statement 2 and 3 are incorrect and 1 is correct.

70. Name the site that gives us valuable information about India's maritime links on the Coromandel coast.

(a) Bharukachchha
(b) Karur
(c) Arikamedu
(d) Anuradhapura

(*c*) Arikamedu gives us valuable information about India's maritime links on Coromandel coast. It is situated in South India's Puducherry.

Arikamedu was an important port of Chola Kingdom. It also helps in trade with Roman people.

71. Where are the largest quantity of cichlids found in India?

(a) Backwaters of Kerala
(b) Sunderbans
(c) Narmada
(d) Godavari

(***d***) Cichlids are fish from the family cichlidae in the order *cichlid forums*. They are the largest verteberate families in the world.

They are found mostly in Africa and South America. They are foundless in brackish and saltwater habitats.

They are largest in number in the Godavari river of Indian subcontinent.

72. Which Greek philosopher coined the term "Geography" in the 3rd century B.C.E.?

(a) Euclid (b) Plato
(c) Eratosthenes (d) Clio

(***c***) Geography derives from the Greek Word 'geographia', which would be 'to describe or write about the earth'.

Eratosthenes (3rd century BCE) was the first person to use the word geography. Geography is a field of science devoted to the study of lands, features, in habitants and governance.

73. Who is the author of the 16th century Sanskrit text, the *Vraja Bhakti Vilasa* which focuses on the Braj region in North India?

(a) Todar Mal
(b) Narayana Bhatta
(c) Chaitanya
(d) Rupa Goswami

(***b***) Narayana Bhatta Goswami was Brahmin born in South India. Later, he came to Vraja and took initiation from Krishna Das Brahmachari. He wrote the 16th Century Sanskrit text Vraja Bhakti Vilasa which focuses on the Braj region in North India.

74. Bose-Einstein condensate is a

(a) solid state of matter
(b) fifth state of matter
(c) plasma
(d) state of condensed matter

(***b***) Bose-Einstein Condensate (BEC) is a fifth state of matter. The BEC is formed by working a gas of extremely low density, about one- hundred tons and the density of normal air to super low temperatures. Other states of matter : Solid, liquid, gas and plasma.

75. The rate of evaporation of liquid does not depend upon

(a) temperature
(b) its surface area exposed to the atmosphere
(c) its mass
(d) humidity

(***c***) The rate of evaporation of liquid does not depend upon its mass, because it is a surface phenomenon. Factors affecting evaporation are:
An increase in surface area
An increase in temperature
Humidity

76. Rutherford's alpha particle scattering experiment on thin gold foil was responsible for the discovery of

(a) electron (b) proton
(c) atomic nucleus (d) neutron

(***c***) Rutherford's alpha particle scattering experiment leads to the discovery of atomic nucleus. He proposed that, there is a positively charged spherical centre in an atom, called nucleus.

77. Food chain is

(a) relationship between autotrophic organisms
(b) exchange of genetic material between two organisms
(c) passage of food (and thus energy) from one organism to another
(d) modern entrepreneur establishment providing food outlets

(***c***) Food chain is the linear sequence of transfer of matter and energy in the form of food from organism to organism. It begins from the producer organisms, i.e. plants, and ends with decomposer species like bacteria, fungi, etc.

78. Which one of the following is active transport?

(a) It is the movement of a substance against a diffusion gradient with the use of energy from respiration
(b) It is the movement of a substance against a diffusion gradient without the use of energy
(c) It is the movement of a substance against a diffusion gradient with the use of energy from photosynthesis
(d) It is the movement of a substance along a diffusion gradient with the use of energy from respiration

(***a***) Active transport is the movement of molecules across a membrane, from a region of their lower concentration to a region of their higher concentration, i.e. aganist the diffusion or concentration gradient. Active transport always requires cellular energy to achieve this movement.

79. Chlorophyll in photosynthetic prokaryotic bacteria is associated with

(a) plastids
(b) membranous vesicles
(c) nucleoids (d) chromosomes

(***b***) Chlorophyll in photosynthetic prokaryotic bacteria is associated with membranous vesicles. These organisms lack membrane bound organelles (such as chloroplast). They have infoldings of the plasma membrane present in the scattered form in cytoplasm for attachment of chlorophyll. These carryout photosynthesis.

80. What do you mean by 'Demographic Dividend'?

(a) A rise in the rate of economic growth due to a higher share of working age people in a population.
(b) A rise in the rate of literacy due to development of educational institutions in different parts of the country.
(c) A rise in the standard of living of the people due to the growth of alternative livelihood practices.
(d) A rise in the gross employment ratio of a country due to government policies.

(***a***) 'Demographic Dividend' means a rise in the rate of economic growth due to a higher share of working age people in a population.

The change in working age people is typically brought by a decline in fertility and mortality rates.

81. Which of the following organisms is responsible for sleeping - sickness?

(a) *Leishmania* (b) *Trypanosoma*
(c) *Ascaris* (d) *Helicobacter*

(***b***) *Trypanosoma* is responsible for sleeping-sickness. It is a vector borne disease, transmitted to humans by the tse-tse fly. Its symptoms include, fever, headache, joint pains and poor mental and physical coordination.

82. Which one of the following body parts/organs of the human body does not have smooth muscles?

(a) Ureters (b) Iris of eye
(c) Bronchi of lungs
(d) Biceps

(*d*) Amongst the options that are given, 'biceps' does not have smooth muscles. Smooth muscles are involuntary muscles as their functioning cannot be directly controlled by our own will.

The muscle present in the biceps as well as triceps are called skeletal or striated or voluntary muscles. We can control the movements of these muscles.

83. What is Inter-cropping?

(a) It is the time period between two cropping seasons.
(b) It is growing two or more crops in random mixture.
(c) It is growing two or more crops in definite row patterns.
(d) It is growing of different crops on a piece of land in a pre-planned succession.

(*c*) Inter-cropping is growing two or more crops simultaneously on the same field in a definite pattern. A few rows of one crop alternate with a few rows of a second crop, for example Soyabean + Maize.

The crops are selected such their nutrient requirements are different. This ensures maximum utilisation of nutrients supplied and both the cultivations give better returns.

84. Magnification is

(a) actual size of specimen/observed size
(b) observed size of specimen/actual size
(c) actual size of specimen-observed size
(d) observed size of specimen actual size

(*b*) The ratio of the observed size of specimen by a spherical mirror to the size of the actual, is called the **linear magnification** by the spherical mirror, it is denoted by m.

i.e. $$m = \frac{I}{O}$$

where, I = observed size of specimen and O = actual size of the object.

85. Which one of the following cell organelles is known as 'suicidal bags' of a cell?

(a) Lysosomes (b) Plastids
(c) Endoplasmic reticulum
(d) Mitochondria

(*a*) Lysosomes are the cell organelles which are known as 'suicidal bags' of a cell. They contain powerful hydrolytic enzymes which are capable of breaking down all organic material present inside the cell.

During the disturbance in cellular metabolism or when the cell gets damaged, lysosomes may burst and the enzymes digest their own cell. Therefore, they are also known as suicidal bags of a cell.

86. Which one of the following statements with regard to economic models is not correct?

(a) They involve simplification of complex processes.
(b) They represent the whole or a part of a theory.
(c) They can be expressed only through equations.
(d) They help in gaining an insight into cause and effect.

(*c*) Economic model is a theoretical construct representing economic processes by a set of variables and a set of logical relationship between them. This can be expressed through equations, models etc.

They involve simplification of complex processes. They help in gaining an insight into cause and effect.

87. The value of the slope of a normal demand curve is

(a) positive (b) negative
(c) zero (d) infinity

(*b*) In a market equilibrium, whenever price of a commodity rises, the demand of commodity declines, in the same way when prices fall, demand increases. So, demand and prices have negative relationship with each other.

Therefore, the value of slope of a normal demand curve is negative.

88. Which one of the following is an example of a price floor?

(a) Minimum Support Price (MSP) for Jowar in India.
(b) Subsidy given to farmers to buy fertilizers.
(c) Price paid by people to buy goods from ration shops.
(d) Maximum Retail Price (MRP) printed on the covers/packets of goods sold in India.

(*a*) A price floor is a minimum price. It is a regulatory tool. Price floor is defined as an intervertion to raise market prices if the government feels that price is too low MSP for Jowar or minimum wages are example of price floor.

89. Which one of the following factors is not considered in determining the Mini mum Support Price (MSP) in India?

(a) Cost of production
(b) Price trends in international and domestic markets
(c) Cost of living index
(d) Inter-crop price parity

(*c*) In formulating Minimum Supports Price (MSP), the Commission for Agricultural Cost and Prices (CACP) considers
Cost of production
Changes in input prices
Input-output price parity
Trends in market prices
Demand and supply
Inter-crop price parity
Effect in industrial cost structure
Effect on cost of living
Effect on general price level
International price situation
Parity between prices paid and prices received by the farmers
Effect on issue prices and implication for subsidy
Hence, option (c) is not consider in determing the MSP.

90. Which one of the following is not a dimension of the human development index?

(a) A long and healthy life
(b) Knowledge
(c) Access to banking and other financial provisions
(d) A decent standard of living

(*c*) Human Development Index is an index of life expectancy, education and per capita income indicators.

There are three dimensions of index—Long and healthy life, knowledge and a decent standard of living.

Hence, option (c) is not a dimension of the Human Development Index.

91. Gini Coefficient or Gini Ratio can be associated with which one of the following measurements in an economy?

(a) Rate of inflation (b) Poverty index
(c) Income inequality
(d) Personal income

(*c*) In economics, Gini Coefficient represents income or wealth distribution of a nation's residents and most commonly used to measure income inequality.

It was developed by the Italian statistician and sociologist Corrado Gini and published in his 1912 paper 'Variability and Mutability'.

92. Consider the following statements :
1. Particles of matter intermix on their own.
2. Particles of matter have force acting between them.

Which of the statements given above is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(*c*) Particles of matter intermix on their own. Particles of matter attract each other. The force of attraction, responsible for keeping them close is inter-molecular force of attraction.

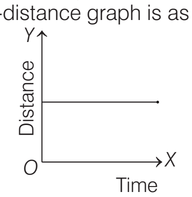
Hence, statement 1 and 2 both are correct.

93. Rate of evaporation increases with
(a) an increase of surface area
(b) an increase in humidity
(c) a decrease in wind speed
(d) a decrease of temperature

(*a*) The evaporation is a surface phenomenon. The rate of evaporation increases on increasing the surface area of the liquid. A substance that has a larger surface area will evaporate faster, as there are more surface molecules that are able to escape.

94. If an object is at rest, then the time (X-axis) *versus* distance (Y-axis) graph
(a) is vertical (b) is horizontal
(c) has 45° positive slope
(d) has 45° negative slope

(*b*) If an object is at rest, then the time-distance graph is as below



Hence, graph is horizontal.

95. Consider the following statements about mixture :
1. A substance can be separated into other kinds of matter by any physical process.
2. Dissolved sodium chloride can be separated from water by the physical process of evaporation.

Which of the statements given above is/are correct?
(a) Only 1
(b) Only 2
(c) Both 1 and 2
(d) Neither 1 nor 2

(*b*) In order to separate the components of mixture, single or a combination of methods are used to separation of mixture can be done by both physical and chemical process. So, statement 1 is incorrect.

Dissolved sodium chloride can be separated from water by evaporation. This method is used to separate a volatile component from a non-volatile component of a mixture.

So, statement 2 is correct.

96. Which one of the following option is incorrect?
(a) Elements are defined by the number of protons they possess.
(b) Isobars are atoms having the same atomic number, but different mass number.
(c) The mass number of an atom is equal to the number of nucleons in its nucleus.
(d) Valency is the combining capacity of an atom.

(*b*) Atoms of different elements with different atomic numbers, but same mass number are known as isobars.

Hence, option (b) is in correct.

97. If the speed of a moving magnet inside a coil increases, the electric current in the coil
(a) increases (b) decreases
(c) reverses
(d) remains the same

(*a*) If the speed of a moving magnet inside a coil increases, the electric current in the coil increases. Whenever, a magnetic field and an electric conductor move relative to one another, so the conductor crosses lines of force in the magnetic field.

Hence, the current produced by electromagnetic induction is greater.

98. The frequency (in Hz) of a note that is one octave higher than 500 Hz is
(a) 375 (b) 750
(c) 1000 (d) 2000

(*c*) An octave or perfect octave is the interval between one musical pitch and another with double its frequency.

Hence, option (c) is correct.

99. Which of the following statements as per the Constitution of India are not correct?
1. The President tenders his resignation to the Chief Justice of India.
2. The Vice-President tenders his resignation to the President of India.
3. The Comptroller and Auditor General of India is removed from his office in the like manner as the President of India.
4. A Judge of the Supreme Court can resign his office by writing under his hand addressed to the Chief Justice of India.

Select the incorrect answer using the codes given below
(a) 1 and 2 (b) 2 and 3
(c) 1, 2 and 3 (d) 1, 3 and 4

(*d*) According to Constitution of India, President can resign from his office at any time by addressing the resignation letter to the Vice-President.

The Vice-President of India can resign anytime by writting to President of India.

The Comptroller and Auditor General of India (CAG) can be removed by the President on the same grounds and in same manner as a judge of Supreme Court (not President).

A Judge of Supreme Court can resign from his office by writting to President of India.

Hence, statements 1, 3 and 4 are incorrect and statement 2 is correct.

100. Rajya Sabha has exclusive jurisdiction in
(a) creation of new states
(b) declaring a war
(c) financial emergency
(d) authorising Parliament to legislate on a subject in the state list

(*d*) Due to its federal character, the Rajya Sabha has been given two exclusive or special powers that are not enjoyed by the Lok Sabha.
(i) It can authorise the Parliament to make a law on a subject enumerated in the state list.
(ii) It can authorise the Parliament to create new all India services common to both the Centre and States.

101. The term 'soil impoverishment' relates to which one of the following?
(a) Soil erosion
(b) Soil deposition
(c) Soil getting very deficient in plant nutrients
(d) Soil getting enriched with plant nutrients

(*c*) The term 'soil impoverishment' relates to soil getting very deficient in plant nutrients.

Soil gets impoverished due to reasons like : over-grazing mono-cropping, leaching, erosion, land use charge etc.

An impoverished soil leads to increase in input cost of crop cultivation in the form of fertilizers cost, pesticide cost, etc. Soil erosion refers to removal of top fertile layers of the soil. Erosion may be due to natural reasons (like : river erosion, glacial erosion etc.) or anthropogenic factor (like : mining activities, urbanisation, etc). Usually, soil erosion leads to soil impoverishment.

102. Which one of the following is the correct sequential phase in the successional development of vegetation community in a habitat?

(a) Migration, reaction, stabilisation and nudation
(b) Migration, stabilisation, reaction and nudation
(c) Nudation, migration, reaction and stabilisation
(d) Reaction, migration, stabilisation and nudation

(*c*) The successional development of vegetation community in a habit at depends on many factors. In the begining an area is bare, with time there begins migration of people to that area. This is followed by reaction of people to changing environment condition and lastly people try to stabilise in a given area with certain set of climatic condition. Therefore, the correct sequence is nudation, migration, reaction and stabilisation.

103. Match List I with List II and select the correct answer using the codes given below the lists.

List I (Soil Type)	Lilst II (Major Characteristic)
A. Oxisols	1. Very rich in organic matter
B. Vertisols	2. Soil lacking horizons
C. Histosols	3. Very old and highly weathered
D. Entisols	4. Rich in clay content and highly basic

Codes

	A	B	C	D
(a)	3	1	4	2
(b)	3	4	1	2
(c)	2	1	4	3
(d)	2	4	1	3

(*a*) **Oxisols** are very old soil and they are highly weathered.

Vertisols are very rich in organic matter.

Histosols are rich in clay content and highly basic.

Entisols are soils which lack horizons.

104. Which one of the following mountains separates Black Sea and Caspian Sea?

(a) Urals (b) Caucasus
(c) Carpathians
(d) Balkan mountains

(*b*) The Caucasus mountains are a mountain system at the intersection of Europe and Asia stretching between Black sea and Caspian sea. It is home to mount Elbrus, highest peak in Europe.

It includes greater Caucasus in the North and lesser Caucasus in the South.

105. Rains caused by thunderstorms during the hot weather season (mid-March to mid-June) in Karnataka are called

(a) Kalbaisakhi
(b) Mango showers
(c) Loo
(d) Cherry blossoms

(*b*) These pre-monsoon rains are as called 'Mango Showers'. Mango showers are common in state of Kerala, Karnataka, Maharashtra and some part of Tamil Nadu.

These showers arrive generally in late April and May and are usually very difficult to predict.

106. Which one of the following is the largest fresh water lake in India?

(a) Chilika (b) Loktak
(c) Dal (d) Wular

(*d*) Wular lake is the largest fresh water lake in India. It is sited in Bandipura district in Jammu and Kashmir. The lake basin was formed as a result of tectonic activity and is fed by the Jhelum River.

The area of this lake is seasonally from 30 to 260 sq. km.

Directions (Q. Nos. 107-110) *The following items consists of two statements, Statement I and Statement II. Examine these two statements carefully and select the correct answer using the codes given below.*

Codes

(a) Both the statements are individually true and statement II is the correct explanation of statement I
(b) Both the statements are individually true, but statement II is not the correct explanation of statement I
(c) Statement I is true, but statement II is false
(d) Statement I is false, but statement II is true

107. Statement I The Greek travellers were most impressed by the fertility of India's soil and the energy and ability of her cultivators.

Statement II Ancient India knew the use of manure.

(*a*) Ancient India knew the use of manure. Because of this Greek travellers were most impressed by the fertility of India's soil and the energy and ability of India's cultivators.

From Maurya period to Gupta period, the arrival of Greeks in India is common.

Greek travellers visited India and impressed by India's natural beauty in terms of plants, animals, minerals.

Hence, both the statements are correct and statement II is correct explanation of statements I.

108. Statement I Non-cooperation began in Punjab with the student movement inspired by Lala Lajpat Rai in January 1921.

Statement II The Sikh dominaed central Punjab countryside was stirred by the powerful Akali upsurge.

(*d*) Non-cooperation movement was started to demand Swaraj from Britishers. The main goal is not cooperate with British government and to create barriers in their daily work. Gandhiji started non-cooperation movement on 1st August, 1920 and ended till Febraury, 1922. The Sikh dominated central Punjab countryside was stirred by the powerful Akali upsurge. Hence, statement I is false, but statemen II is true.

109. Statement I The Oudh Kisan Sabha established in 1920 failed to bring under its wing any Kisan Sabhas.

Statement II The Oudh Kisan Sabha asked the Kisans to refuse to till Bedakhli land, not to offer hari and begar.

(*d*) In 1917, Jawaharlal Nehru, Madan Mohan Malviya and Gauri Shankar Mishra etc. established Uttar Pradesh Farmers Organisation, but due to Khilafat Movement, there were disputes among farmers leaders. This resulted in creation of Awadh farmers organisation in 1920 under Ramchandra. Awadh farmers were asked to refuse to till Bedakhli land by Kisan organisations. The Oudh Kisan Sabha established in 1920 was not failed to bring any Kisan Sabha under its wing. Hence, statement I is false, but statement II is true.

110. **Statement I** The united provinces during non-cooperation became one of the strongest bases of the Congress.

Statement II The literary outcrop of non-cooperation in Bengal was quite meagre compared to the days of the Swadeshi agitation.

(*b*) Non-cooperation movement started under the guidance of Mahatma Gandhi. As Swadeshi agitation began due to Partition of Bengal, many people from Bengal participated in the movement, but the literary outcrop of non-cooperation in Bengal was quite meagre. Hence, both the statements are correct.

111. Who were Alvars?

(a) Those who immersed in devotion to Vishnu
(b) Devotees of Shiva
(c) Those who worshipped abstract form of God
(d) Devotees of Shakti

(*a*) The Alvars, were Tamil Poets–Saints of South India who favour support Net Bhakti to the Hindu God Vishnu or his Avatar Krishna in their songs of longing, ecstasy and service.

112. Which one of the following is mono atomic?

(a) Hydrogen (b) Sulphur
(c) Phosphorus (d) Helium

(*d*)

Elements	Atomicity
Hydrogen	Diatomic
Sulphur	Polyatomic
Phosphorus	Tetra atomic
Helium	Mono atomic

Hence, helium is mono atomic among given options.

113. In graphite, each carbon atom is bonded to three other carbon atoms

(a) forming a three-dimensional structure
(b) in the same plane giving a hexagonal array
(c) in the same plane giving a square array
(d) in the same plane giving a pentagonal array

(*b*) In graphite, each carbon atom is bonded to three other carbon atoms in the same plane given a hexagonal array. The fourth electron of each carbon atom is free.

114. Soap solution used for cleaning purpose appears cloudy. This is due to the fact that soap micelles can

(a) refract light (b) scatter light
(c) diffract light (d) polarise light

(*b*) Soap solution used for cleaning purpose. When soap is mixed in water, a colloidal solution is formed. The soap solution has soap micelles which are an aggregate of soap molecules. These micelles are large and they scatter light. That is why the soap solution appears cloudy.

115. People prefer to wear cotton clothes in summer season. This is due to the fact that cotton clothes are

(a) good absorbers of water
(b) good conveyors of heat
(c) good radiators of heat
(d) good absorbers of heat

(*a*) People prefer to wear cotton clothes in summer season to keep them cool and comfortable. In summer, we sweat more, cotton being a good absorber of water helps in absorbing the sweat and exposes it to the atmosphere for evaporation.

So, we can say that cotton clothes are good absorbers of water.

116. Employing chromatography, one cannot separate

(a) radioisotopes
(b) colours from a dye
(c) pigments from a natural colour
(d) drugs from blood

(*a*) Chromatography is the technique for the separation of those solvents, that dissolve in the same solvent. Radioisotopes cannot be dissolve in the same solvent to separate it.

117. Consider the following statements :

''Atomic number of an element is a more fundamental property than its atomic mass.'' Who among the following scientists has made the above statement?

(a) Dmitri Mendeleev
(b) Henry Moseley
(c) JJ Thomson
(d) Ernest Rutherford

(*b*) Given statement was stated by Henry Moseley. Mendeleev's periodic table was, therefore accordingly modified.

118. Which one of the following acids is also known as vitamin-C?

(a) Methanoic acid
(b) Ascorbic acid
(c) Lactic acid
(d) Tartaric acid

(*b*) Vitamin-C is known as ascorbic acid. It is found in all citrus fruits. It's deficiency in diet comes scurvy.

119. Which one of the following is not found in animal cells?

(a) Free ribosomes
(b) Mitochondria
(c) Nucleolus
(d) Cell wall

(*d*) Animal cell does not possess cell wall. It is present in the plant cells, outside the plasma membrane as a rigid outer covering. It maintains the shape of the cell and protects it. Cell wall is also present in fungi, algae and bacteria.

120. *Marsilea,* fern and horse-tail are examples of which one of the following plant groups?

(a) Pteridophyta
(b) Bryophyta
(c) Gymnosperms
(d) Angiosperms

(*a*) *Marsilea,* fern and horse-tail are examples of division–Pteridophyta of plant kingdom. In this group, the plant body is differentiated into roots, stem and leaves and has specialised tissue for the conduction of water and other substances from one part of the plant body to another.

CDS

Combined Defence Service

SOLVED PAPER 2019 (I)

PAPER I Elementary Mathematics

1. What is the remainder when $(17^{29}+19^{29})$ is divided by 18?

(a) 6 (b) 2
(c) 1 (d) 0

(d) $= \dfrac{17^{29}+19^{29}}{18}$

$= \dfrac{(18-1)^{29}}{18} + \dfrac{(18+1)^{29}}{18}$

$\therefore$ Remainder $= (-1)^{29} + (1)^{29}$

$= -1 + 1$

$= 0$

2. What is the largest value of n such that 10^n divides the product $2^5 \times 3^3 \times 4^8 \times 5^3 \times 6^7 \times 7^6 \times 8^{12} \times 9^9 \times 10^6 \times 15^{12} \times 20^{14} \times 22^{11} \times 25^{15}$?

(a) 65 (b) 55
(c) 50 (d) 45

(a)
$2^5 \times 3^3 \times 4^8 \times 5^3 \times 6^7 \times 7^6 \times 8^{12} \times 9^9$
$\times 10^6 \times 15^{12} \times 20^{14} \times 22^{11} \times 25^{15}$

10 makes by = (2×5)

Find the number of five (5) in the product

$5^3 \times 10^6 \times 15^{12} \times 20^{14} \times 25^{15}$

Number of 5

$3+6+12+14+30=65$

[$\because (10)^6 = (2\times 5)^6$ 6 five in this product]

10^{65} will divide the product $n = 65$

3. How many pairs (A, B) are possible in the number $479865AB$ if the number is divisible by 9 and it is given that the last digit of the number is odd?

(a) 5 (b) 6
(c) 9 (d) 11

(a) Divisible rule of 9

Sum of digit of the numbers is divisible by 9.

The number $= 479865AB$

Sum of the digit

$= 4+7+9+8+6+5+A+B$

$= 39 + A + B$

Pairs (A, B) also the number is odd.

45 and 54 are two numbers divisible by 9.

Pairs [(1, 5), (5, 1), (3, 3), (6, 9), (8, 7)]

Option (a) is correct.

4. Consider the multiplication $999 \times abc = def\ 132$ in decimal notation, where a, b, c, d, e and f are digits. What are the values of a, b, c, d, e and f, respectively?

(a) 6, 6, 8, 6, 8, 7 (b) 8, 6, 8, 6, 7, 8
(c) 6, 8, 8, 7, 8, 6 (d) 8, 6, 8, 8, 6, 7

(d) $999 \times abc = def\ 132$

$$\begin{array}{r} 999 \\ \times\ a\ b\ c \\ \hline def\ 132 \end{array}$$

In first step $9 \times c =$ unit digit 2

$c = 8$

$$\begin{array}{r} 9\ 9\ 9 \\ \times\ a\ b\ 8 \\ \hline 7\ 9\ 9\ 2 \\ \times \\ \hline def\ 132 \end{array}$$

In second step $9 \times b = 9 +$ (unit digit 3)

$b = 6$

$$\begin{array}{r} 9\ 9\ 9 \\ \times\ a\ 6\ 8 \\ \hline 7\ 9\ 9\ 2 \\ 5\ 9\ 9\ 4\ \times \\ \times\ \times \\ \hline def\ 132 \end{array}$$

Similarly, $a = 8$

$$\begin{array}{r} 9\ 9\ 9 \\ \times\ 8\ 6\ 8 \\ \hline 7\ 9\ 9\ 2 \\ 5\ 9\ 9\ 4\ \times \\ 7\ 9\ 9\ 2\ \times\ \times \\ \hline 8\ 6\ 7\ 1\ 3\ 2 \end{array}$$

The values of a, b, c, d, e and $f = 8, 6, 8, 8, 6, 7$

Option (d) is correct.

5. Three cars A, B and C started from a point at 5 pm, 6 pm and 7 pm, respectively and travelled at uniform speeds of 60 km/h, 80 km/h and x km/h, respectively in the same direction. If all the three meet at another point at the same instant during their journey, then what is the value of x?

(a) 120 (b) 110 (c) 105 (d) 100

(a) Speed of car A = 60 km/h

Speed of car B = 80 km/h

Speed of car C = x km/h

Cars A, B and C started from a point (M)

Car A started at 5 pm from point M

distance coverd by car A in 1h = speed × time = $60 \times 1 = 60$ km

Then, distance between point M and N $= 60$ km

Car B started at 6 pm from point (M)

In same direction

Relative speed of car A and car B

$= 80 - 60 = 20$ km/h

Time taken by car B meet to car A

$= \frac{60}{20} = 3$h $\left[\because \text{Time} = \frac{\text{Distance}}{\text{Speed}}\right]$

After three hour car B meet car A at point Z at 9 : 00 pm.

The distance of M to Z

$= 3 \times 80 = 240$ km

Car C started at 7 : 00 pm [from point (M)]

Car C has to travel 240 km in 2 h to meet car A and car B.

Speed of car $C = \frac{240}{2} = 120$ km/h

Option (a) is correct. $\left[\because \text{Speed} = \frac{\text{Distance}}{\text{Time}}\right]$

6. Priya's age was cube of an integral number (different from 1) four years ago and square of an integral number after four years. How long should she wait so that her age becomes square of a number in the previous year and cube of a number in the next year?

(a) 7 yr (b) 12 yr
(c) 14 yr (d) 21 yr

(c) Let say Priya age 4 yr ago $= x$ yr

Current age $= x + 4$ yr

Age after 4 yr $= x + 4 + 4 = x + 8$

x is cube and $x + 8$ is an square

8 and 16 are possible solution.

Cube $(x)^3$: 8, 27, 64, 125

$(x^3 + 8)$: 16, 35, 72, 133

Square : 4, 9, 16, 25, 36, 49, 64, 81, 100 is square

So, $x = 8$ and $x + 8 = 16$

So, present age = 12 yr

Her age becomes square of the number in the previous year and cube or the number in next year.

$\Rightarrow$ Cube = square +2

Cube = 27

Square = 25

Age = 26 yr

$26 - 12 = 14$ yr

Then, she has to wait for 14 yr

Option (c) is correct.

7. Which of the following statements is not true?

(a) The difference of two prime numbers, both greater than 2, is divisible by 2

(b) For two different integers m, n and a prime number p, if p divides the product $m \times n$, then p divides either m or n

(c) If a number is of the form $6n - 1$ (n being a natural number), then it is a prime number

(d) There is only one set of three prime numbers such that there is a gap of 2 between two adjacent prime numbers

(c) Every prime number is of the form $(6n - 1)$ or $(6n + 1)$ but every number which is of form $(6n - 1)$ or $(6n + 1)$ not necessarily prime.

Examples showing $(6n - 1)$

Let $n = 6 \; (6 \times 6 - 1) = 36 - 1 = 35$ (It is not a prime number)

Let $n = 4 \; (6 \times 4 - 1) = 24 - 1 = 23$ (It is a prime number)

So, option (c) is correct answer.

8. For $x > 0$, what is the minimum value of $x + \frac{x+2}{2x}$?

(a) 1
(b) 2
(c) $2\frac{1}{2}$
(d) Cannot be determined

(c) $x + \frac{x+2}{2x}, x > 0$ ...(i)

$= x + \frac{x}{2x} + \frac{2}{2x} = x + \frac{1}{2} + \frac{1}{x}$

Differentiation of the Eq. (i), we get

$$0 = 1 + 0 - \frac{1}{x^2}$$

$$\frac{1}{x^2} = 1 \Rightarrow x^2 = 1$$

$$x = \pm 1$$

$$x = 1 \quad (x > 0)$$

On putting $x = 1$ in Eq. (i), we get

$$= (1) + \frac{(1) + 2}{2(1)}$$

$$= 1 + \frac{3}{2} = \frac{5}{2} = 2\frac{1}{2}$$

Option (c) is correct.

9. If $\frac{1 + px}{1 - px}\sqrt{\frac{1 - qx}{1 + qx}} = 1$, then what are the non-zero solutions of x?

(a) $\pm \frac{1}{p}\sqrt{\frac{2p - q}{q}}, 2p \neq q$

(b) $\pm \frac{1}{pq}\sqrt{p - q}, p \neq q$

(c) $\pm \frac{p}{q}\sqrt{p - q}, p \neq q$

(d) $\pm \frac{q}{p}\sqrt{2p - q}, 2p \neq q$

(a) $\frac{1 + px}{1 - px}\sqrt{\frac{1 - qx}{1 + qx}} = 1$

$$\frac{1 + px}{1 - px} = \sqrt{\frac{1 + qx}{1 - qx}}$$

On squaring both sides, we get

$$\frac{(1 + px)^2}{(1 - px)^2} = \frac{1 + qx}{1 - qx}$$

Use Componendo and Dividendo rule,

$$\frac{(1 + px)^2 + (1 - px)^2}{(1 + px)^2 - (1 - px)^2} = \frac{1 + qx + 1 - qx}{1 + qx - 1 + qx}$$

$$\Rightarrow \frac{(1 + p^2x^2 + 2px) + (1 + p^2x^2 - 2px)}{(1 + p^2x^2 + 2px) - (1 + p^2x^2 - 2px)} = \frac{1 + qx + 1 - qx}{1 + qx - 1 + qx}$$

$[\because (a + b)^2 = a^2 + b^2 + 2ab, (a - b)^2 = a^2 + b^2 - 2ab]$

$$\Rightarrow \frac{2 + 2p^2x^2}{4px} = \frac{2}{2qx}$$

$$\Rightarrow \frac{1 + p^2x^2}{2px} = \frac{1}{qx}$$

$$\Rightarrow 1 + p^2x^2 = \frac{2p}{q} \Rightarrow p^2x^2 = \frac{2p}{q} - 1$$

$$\therefore \quad px = \pm\sqrt{\frac{2p - q}{q}}$$

$$x = \pm\frac{1}{p}\sqrt{\frac{2p - q}{q}}$$

Option (a) is correct.

10. In a hostel the rent per room is increased by 20%. If number of rooms in the hostel is also increased by 20% and the hostel is always full, then what is the percentage change in the total collection at the cash counter?

(a) 30% (b) 40% (c) 44% (d) 48%

(c) In a hostel the rent per room is increased by 20%.

The number of rooms in the hostel is also increased by 20%.

Then, the percentage change in the total collection at the cash counter

$$= a + b + \frac{a \times b}{100}$$

$\therefore \quad a = b = 20\%$

$$= 20 + 20 + \frac{20 \times 20}{100} = 44\%$$

Option (c) is correct.

11. Radha and Hema are neighbours and study in the same school. Both of them use bicycles to go to the school. Radha's speed is 8 km/h whereas Hema's speed is 10 km/h. Hema takes 9 min less than Radha to reach the school. How far is the school from the locality of Radha and Hema?

(a) 5 km (b) 5.5 km
(c) 6 km (d) 6.5 km

(c) Radha's speed = 8 km/h

Hema's speed = 10 km/h

Let the distance of the school from the locality of Radha and Hema = x km

According to the question,

$$\frac{x}{8} - \frac{x}{10} = \frac{9}{60} \quad \left[\because \text{Time} = \frac{\text{Distance}}{\text{Speed}}\right]$$

$$\frac{10x - 8x}{8 \times 10} = \frac{3}{20} \left[\because 1 \text{ minute} = \frac{1}{60} \text{hour}\right]$$

$$2x = 12, x = 6 \text{ km}$$

Option (c) is correct.

12. Which of the following pair of numbers is the solution of the equation $3^{x+2} + 3^{-x} = 10$?

(a) 0, 2 (b) 0, –2
(c) 1, –1 (d) 1, 2

(b) $3^{x+2} + 3^{-x} = 10$

$\Rightarrow \quad 3^x \cdot 3^2 + \frac{1}{3^x} = 10$

Let $3^x = y$, then

$\Rightarrow \quad 9y + \frac{1}{y} = 10$

$\Rightarrow \quad 9y^2 + 1 = 10y$

$\Rightarrow \quad 9y^2 - 10y + 1 = 0$

$\Rightarrow 9y^2 - 9y - y + 1 = 0$

$\Rightarrow 9y(y-1) - 1(y-1) = 0$

$\Rightarrow \quad (y-1)(9y-1) = 0$

$\therefore \quad y = 1, \frac{1}{9}$

when $3^x = 1 \Rightarrow 3^x = 3^0 \Rightarrow x = 0$

when $3^x = \frac{1}{9} \Rightarrow 3^x = 3^{-2} \Rightarrow x = -2$

So, $\quad x = 0, -2$

13. It is given that $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$. How many digits are there in $(108)^{10}$?

(a) 19 (b) 20
(c) 21 (d) 22

(c) Given, $\log_{10} 2 = 0.301$

$\log_{10} 3 = 0.477$

$x = (108)^{10}$

On taking log both sides, we get

$\log x = \log 108^{10}$

$\log x = 10 \log 108$

$\log x = 10[\log(2^2 \times 3^3)]$

$[\because 108 = 2^2 \times 3^3]$

$\log x = 10[\log 2^2 + \log 3^3]$

$[\because \log(a \times b) = \log a + \log b]$

$\log x = 10[2\log 2 + 3\log 3]$

$[\because \log a^b = b \log a]$

$\log x = 10[2 \times 0.301 + 3 \times 0.477]$

$\log x = 10 \times 2.033 = 20.33$

On taking antilog both sides, we get

$x = e^{20.33}$

Number of digit are there is $= 20.33 \approx 21$

Option (c) is correct.

14. The sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the number is

(a) 17 (b) 29
(c) 43 (d) None of these

(d) The sum of the three prime number $(P_1 + P_2 + P_3) = 100$ ...(i)

Solve by option

$P_1 - P_2 = 36$ ...(ii)

Option (a)

$P_2 = 17$

$P_1 - 17 = 36$

$P_1 = 36 + 17$

$P_1 = 53$

From Eq. (i), we get

$P_1 + P_2 + P_3 = 100$

$53 + 17 + P_3 = 100$

$P_3 = 30$ (It is not a prime)

Option (b)

$P_2 = 29$

$P_1 - 29 = 36$

$P_1 = 36 + 29$

$P_1 = 65$ (It is not a prime)

Option (c)

$P_2 = 43$

$P_1 - 43 = 36$

$P_1 = 36 + 43 = 79$

On putting in Eq. (i), we get

$P_1 + P_2 + P_3 = 100$

$79 + 43 + P_3 = 100$

$122 + P_3 = 100$

$P_3 = -22$

(It is not a prime)

Then, option (d) is correct answer.

15. If a, b and c are positive integers such that $\dfrac{1}{a + \dfrac{1}{b + \dfrac{1}{c + \dfrac{1}{2}}}} = \dfrac{16}{23}$, then what is the mean of a, b and c?

(a) 1 (b) 2
(c) 1.33 (d) 2.33

(b) $\dfrac{1}{a + \dfrac{1}{b + \dfrac{1}{c + \dfrac{1}{2}}}} = \dfrac{16}{23} \Rightarrow \dfrac{16}{23}$

$$= \frac{1}{23/16} = \frac{1}{1 + \dfrac{7}{16}}$$

$$= \frac{1}{1 + \dfrac{1}{16/7}} = \frac{1}{1 + \dfrac{1}{2 + \dfrac{2}{7}}}$$

$$= \frac{1}{1 + \dfrac{1}{2 + \dfrac{1}{7/2}}} = \frac{1}{1 + \dfrac{1}{2 + \dfrac{1}{3 + \dfrac{1}{2}}}}$$

Then, $a = 1, b = 2, c = 3$

Mean of a, b and $c = \dfrac{a + b + c}{3}$

$$= \frac{1 + 2 + 3}{3} = 2$$

Option (b) is correct.

Directions (Q. Nos. 16-18) *Read the given information carefully and answer the given questions below.*

In a certain town of population size 100000 three types of newspapers (I, II and III) are available. The percentages of the people in the town who read these papers are as follows.

Newspaper	Proportion of readers
I	10%
II	30%
III	5%

Both I and II	8%
Both II and III	4%
Both I and III	2%
All the three (I, II and III)	1%

16. What is the number of people who read only one newspaper?

(a) 20000 (b) 25000
(c) 30000 (d) 35000

(*a*) Percentage of people who read only one newspaper.

= I + II + III − 2 [I and II + II and III + I and III] + 3 [I, II and III]

= 10 + 30 + 5 − 2 [8 + 4 + 2] + 3(1)

= 45 − 28 + 3 = 20%

Population of town = 100000

Number of people who read only one newspaper $= 100000 \times \frac{20}{100} = 20000$

Option (a) is correct.

17. What is the number of people who read atleast two newspapers?

(a) 12000 (b) 13000
(c) 14000 (d) 15000

(*a*) Percentage of people who read atleast two newspapers.

= I and II + II and III + I and III − 2 [I, II and III]

= 8 + 4 + 2 − 2(1) = 14 − 2 = 12%

Number of people who read atleast two newspapers $= 100000 \times \frac{12}{100} = 12000$

Option (a) is correct.

18. What is the number of people who do not read any of these three newspapers?

(a) 62000 (b) 64000
(c) 66000 (d) 68000

(*d*) The percentage of people who read newspaper

I + II + III − [I and II + II and III + I and III] + [I, II and III]

= 10 + 30 + 5 − [8 + 4 + 2] + 1

= 32%

Percentage of people who do not read any of these three newspapers.

= 100% − 32% = 68%

The number of people who do not read any of these three newspapers

$= 100000 \times \frac{68}{100} = 68000$

Option (d) is correct.

19. What is the unit place digit in the expansion of 7^{73}?

(a) 1 (b) 3
(c) 7 (d) 9

(*c*) The unit place digit = 7

$$\begin{bmatrix} 7^1 = 7 \\ 7^2 = 9 \\ 7^3 = 3 \\ 7^4 = 1 \\ 7^5 = 7 \end{bmatrix}$$

Then, cyclicity of 7 is 4

$7^{73} = 7^{\frac{73}{4}} \Rightarrow$ remainder $= \frac{73}{4} = 1 = 7^1$

unit digit.

Option (c) is correct.

20. Suppose n is a positive integer such that $(n^2 + 48)$ is a perfect square. What is the number of such n?

(a) One (b) Two (c) Three (d) Four

(*c*) $x = \sqrt{n^2 + 48}$

On squaring both side, we get

$$x^2 = n^2 + 48$$

$$x^2 - n^2 = 48$$

$$(x - n)(x + n) = 48$$

n is an integer

x − n	x + n
2	24
4	12
6	8

Option (c) is correct.

21. For $x = \frac{4\sqrt{6}}{\sqrt{2} + \sqrt{3}}$, what is the value of $\frac{x + 2\sqrt{2}}{x - 2\sqrt{2}} + \frac{x + 2\sqrt{3}}{x - 2\sqrt{3}} = ?$

(a) 1 (b) $\sqrt{2}$ (c) $\sqrt{3}$ (d) 2

(*d*) $x = \frac{4\sqrt{6}}{\sqrt{2} + \sqrt{3}}$

$$x = 4\sqrt{6}(\sqrt{3} - \sqrt{2})$$

$$\frac{x + 2\sqrt{2}}{x - 2\sqrt{2}} + \frac{x + 2\sqrt{3}}{x - 2\sqrt{3}}$$

[divided by $\sqrt{2}$ and divided by $\sqrt{3}$]

$$= \frac{\frac{x}{\sqrt{2}} + 2}{\frac{x}{\sqrt{2}} - 2} + \frac{\frac{x}{\sqrt{3}} + 2}{\frac{x}{\sqrt{3}} - 2}$$

$$= \frac{\frac{4\sqrt{6}}{\sqrt{2}}(\sqrt{3} - \sqrt{2}) + 2}{\frac{4\sqrt{6}}{\sqrt{2}}(\sqrt{3} - \sqrt{2}) - 2} + \frac{\frac{4\sqrt{6}}{\sqrt{3}}(\sqrt{3} - \sqrt{2}) + 2}{\frac{4\sqrt{6}}{\sqrt{3}}(\sqrt{3} - \sqrt{2}) - 2}$$

$$= \frac{4\sqrt{3}(\sqrt{3} - \sqrt{2}) + 2}{4\sqrt{3}(\sqrt{3} - \sqrt{2}) - 2} + \frac{4\sqrt{2}(\sqrt{3} - \sqrt{2}) + 2}{4\sqrt{2}(\sqrt{3} - \sqrt{2}) - 2}$$

$$= \frac{4 \times 3 - 4\sqrt{6} + 2}{4 \times 3 - 4\sqrt{6} - 2} + \frac{4\sqrt{6} - 8 + 2}{4\sqrt{6} - 8 - 2}$$

$$= \frac{14 - 4\sqrt{6}}{10 - 4\sqrt{6}} + \frac{4\sqrt{6} - 6}{4\sqrt{6} - 10}$$

$$= \frac{7 - 2\sqrt{6}}{5 - 2\sqrt{6}} + \frac{3 - 2\sqrt{6}}{5 - 2\sqrt{6}}$$

[negative sign common in both]

$$= \frac{10 - 4\sqrt{6}}{5 - 2\sqrt{6}} = \frac{2(5 - 2\sqrt{6})}{5 - 2\sqrt{6}} = 2$$

Option (d) is correct.

22. x, y and z are three numbers such that x is 30% of z and y is 40% of z. If x is p% of y, then what is the value of p?

(a) 45 (b) 55
(c) 65 (d) 75

(*d*) x, y and z are three numbers,

$$x = \frac{30}{100} \times z \Rightarrow x = \frac{3z}{10} \quad \text{... (i)}$$

$$y = \frac{40}{100} \times z \Rightarrow y = \frac{4z}{10} \quad \text{...(ii)}$$

$$x = \frac{p}{100} \times y$$

$$P = \frac{x \times 100}{y} = \frac{\frac{3z}{10}}{\frac{4z}{10}} \times 100$$

[from Eqs. (i) and (ii)]

$$= \frac{3 \times 100}{4} = 75\%$$

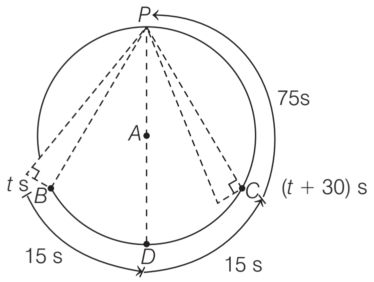
Option (d) is correct.

23. A plane is going in circles around an airport. The plane takes 3 min to complete one round. The angle of elevation of the plane from a point P on the ground at time t is equal to that at time $(t + 30)$ s.

At time $(t + x)$ s, the plane flies vertically above the point P. What is x equal to?

(a) 75 s (b) 90 s (c) 105 s (d) 135 s

(c) "A" is airport and a plane is going in circles around it.



The plane takes 3 min to complete one round.

In s = 3 × 60 = 180 s

Plane reaches at B in t s ...(i)

Plane reaches at C in $(t + 30)$ S ...(ii)

The angle of elevation of the plane from a point P on the ground at time t s is equal to that at time $(t + 30)$ s.

Plane is going to D from point B distance increases respect to P.

[because PD is longest chord of circle or diameter of circle]

After point D the distance of the plane decreases with respect to point P and at point C the distance is equal to arc PC and PB.

Then, it take equal time from B to D and from D to C.

i.e. plane takes total 30 s

15 s from B to D, 15 s from D to C.

The plane takes total time to complete a round = 180 sec

$$PB + BD + DC + CP = 180 \text{ sec}$$
$$CP + 15 + 15 + CP = 180 \text{ sec}$$
$$2CP = 180 - 30 = 150$$
$$CP = 75 \text{ sec}$$

Then, total time is $(t + x)$ s, when plane is at point P.

Major arc from B to P.

Total time = $(t + 75 + 15 + 15)$

Total time = $(t + 105)$ s

24. Consider the following statements in respect of two integers p and q (both > 1) which are relatively prime.

1. Both p and q may be prime numbers.
2. Both p and q may be composite numbers.
3. One of p and q may be prime and the other composite.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

(d) p and q are two integers.

p and q are also relative prime,
HCF = 1

Condition 1

Both p and q may be prime numbers

$(p, q) = (3, 5)$ HCF = 1

It is possible.

Condition 2

Both p and q may be composite numbers.

Let $(p, q) = (9, 25)$, HCF = 1

It is also possible.

Condition 3

One of p and q may be prime and the other composite.

Let $(p, q) = (5, 8)$ HCF = 1

It is also possible.

Then, condition (1), (2) and (3) are possible.

Option (d) is correct.

25. In a class of 100 students, the average weight is 30 kg. If the average weight of the girls is 24 kg and that of the boys is 32 kg, then what is the number of girls in the class?

(a) 25 (b) 26 (c) 27 (d) 28

(a) Let the number of girls = x

The number of boys = $100 - x$

Total weight of girls = average weight × number of girls = $24 \times x = 24x$

Total weight of boys = $32 \times (100 - x)$

Total weight of students
= $30 \times 100 = 3000$

$$\therefore\ 24x + 32(100 - x) = 3000$$
$$24x + 3200 - 32x = 3000$$
$$8x = 200$$
$$x = 25$$

Hence, the number of girls in the class is 25.

Alternate method

A class of 100 students.

The average weight of the class = 30 kg.

The average weight of the girls = 24 kg.

The average weight of the boys = 32 kg.

By Alligation

Girls Boys
24 32
30
2 6
2 : 6 = 1 : 3

Ratio of girls and boys = 1 : 3

The number of girls = $100 \times \dfrac{1}{(1+3)}$

$= \dfrac{100}{4} = 25$

Option (a) is correct.

26. For any two real numbers a and b, $\sqrt{(a-b)^2} + \sqrt{(b-a)^2}$ is

(a) always zero (b) never zero
(c) positive only if $a \neq b$
(d) positive if and only if $a > b$

(c) a and b are any two real numbers.

$a = 3,\ b = 2$

$$\sqrt{(a-b)^2} + \sqrt{(b-a)^2}$$
$$= \sqrt{(3-2)^2} + \sqrt{(2-3)^2}$$
$$= \sqrt{(1)^2} + \sqrt{(-1)^2} = 1 + 1 = 2$$

positive only if $a \neq b$

Option (c) is correct.

27. If $a : b = c : d = 1 : 6$, then what is the value of $\dfrac{a^2 + c^2}{b^2 + d^2} = ?$

(a) $\dfrac{1}{600}$ (b) $\dfrac{1}{60}$ (c) $\dfrac{1}{36}$ (d) $\dfrac{1}{6}$

(c) $a : b = c : d = 1 : 6$

$$\frac{a}{b} = \frac{c}{d} = \frac{1}{6}$$

Let $a = x$, $c = x$, $b = 6x$, $d = 6x$

Now, $\dfrac{a^2 + c^2}{b^2 + d^2} = \dfrac{(x)^2 + (x)^2}{(6x)^2 + (6x)^2}$

$$= \frac{x^2 + x^2}{36x^2 + 36x^2}$$
$$= \frac{2x^2}{72x^2} = \frac{1}{36}$$

Option (c) is correct.

28. What is $0.\overline{53} + 0.5\overline{3}$ equal to?

(a) $1.0\overline{68}$ (b) $1.0\overline{68}$
(c) $1.\overline{068}$ (d) 1.068

(a) $0.\overline{53} + 0.5\overline{3}$

In first number bar on two-digit and in second number bar on one-digit

LCM of bar digits (2, 1) = 2

Last two-digits have bar

$$\begin{array}{r} 0.535 \\ +\ 0.533 \\ \hline 1.068 \\ \hline \end{array}$$

$1.0\overline{68}$

Alternative Method

$$0.\overline{53} + 0.\overline{53} = \frac{53}{99} + \frac{53-5}{90}$$
$$\Rightarrow \frac{53}{99} + \frac{48}{90} = \frac{53 \times 90 + 48 \times 99}{99 \times 90}$$
$$= \frac{4770 + 4752}{8910} = \frac{9522}{8910} = 1.0\overline{68}$$

Option (a) is correct.

29. The inequality $3^N > N^3$ holds when
(a) N is any natural number
(b) N is a natural number greater than 2
(c) N is a natural number greater than 3
(d) N is a natural number except 3

(*d*) $3^N > N^3$

Let $N = 2$, $3^2 > 2^3$
$9 > 8$ (It is correct)
Let $N = 3$, $3^3 > 3^3$ (It is not correct)
Let $N = 4$, $3^4 > 4^3$
$81 > 64$ (It is correct)
Then, N is a natural number except 3.
Hence, option (d) is correct.

30. Which one of the following is an irrational number?
(a) $\sqrt{59049}$ (b) $\frac{231}{593}$
(c) 0.45454545....
(d) 0·12112211122211112222....

(*d*) The number that cannot be expressed in the form of p/q are called irrational number.

For example $\sqrt{2}, \sqrt{3}, \sqrt{7}, \sqrt{11}$ etc.

Option (a)
$\sqrt{59049} = 243$
Option (b) $\frac{231}{593}$ (it is p/q form)
Option (c) 0.454545........
$\frac{45}{99}$ (it is p/q form)
Option (d) 0.121122111222 .....
[it can not expressed in form of p/q]
Then, option (d) is correct answer.

31. A race has three parts. The speed and time required to complete the individual parts for a runner is displayed on the following chart

	Part I	Part II	Part III
Speed (km/h)	9	8	7.5
Time (min)	50	80	100

What is the average speed of this runner?
(a) 8.17 km/h (b) 8 km/h
(c) 7.80 km/h (d) 7.77 km/h

(*b*)

	Part I	Part II	Part III
Speed (km/h)	9	8	7.5
Time (min)	50	80	100

Distance of Part I $= 9 \times \frac{50}{60} = \frac{15}{2}$ km

[$\because$ Distance = Speed × Time]

Distance of part II $= 8 \times \frac{80}{60} = \frac{32}{3}$ km

Distance of part III $= 7.5 \times \frac{100}{60} = \frac{75}{6}$

Total Distance $= \frac{15}{2} + \frac{32}{3} + \frac{75}{6}$
$= \frac{45 + 64 + 75}{6}$
$= \frac{184}{6}$ km

Total time = 50 + 80 + 100 = 230 min
$= \frac{230}{60} = \frac{23}{6}$ h

The average speed of this runner

$= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{\frac{184}{6}}{\frac{23}{6}} = \frac{184 \times 6}{6 \times 23}$

= 8 km/h
Option (b) is correct.

32. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$, then which one of the following statements is correct?
(a) Each fraction is equal to 1 or – 1
(b) Each fraction is equal to $\frac{1}{2}$ or 1
(c) Each fraction is equal to $\frac{1}{2}$ or – 1
(d) Each fraction is equal to $\frac{1}{2}$ only

(*d*) $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = k$ (let)

$\frac{a}{b+c} = k$
$a = k(b + c)$ ...(i)
$\frac{b}{c+a} = k$
$b = k(c + a)$ ...(ii)
$\frac{c}{a+b} = k$
$c = k(a + b)$ ...(iii)

On adding Eqs. (i), (ii) and (iii), we get
$a + b + c = 2k(a + b + c)$
$k = 1/2$
Option (d) is correct.

33. The number 3^{521} is divided by 8. What is the remainder?
(a) 1 (b) 3 (c) 7 (d) 9

(*b*) $\frac{3^{521}}{8}$

Remainder $= \frac{3 \cdot 3^{520}}{8} = \frac{3 \cdot (3^2)^{260}}{8}$
$= \frac{3 \cdot (8+1)^{260}}{8} = \frac{3(1)^{260}}{8} = \frac{3}{8}$

$\therefore$ 3 is remainder
Option (b) is correct.

34. A prime number contains the digit X at unit's place. How many such digits of X are possible?
(a) 3 (b) 4 (c) 5 (d) 6

(*b*) The digit x at unit's place in prime number.

Then, such digit of x are possible
$x = [1, 3, 7, 9]$
Option (b) is correct.

35. If an article is sold at a gain of 6% instead of a loss of 6%, the seller gets ₹ 6 more. What is the cost price of the article?
(a) ₹ 18 (b) ₹ 36
(c) ₹ 42 (d) ₹ 50

(*d*) Let cost price of an article = ₹ x

If an article sold at a gain of 6%, then selling price of an article
$= \text{Cost price} \times \left(\frac{100 + \text{gain\%}}{100}\right)$
$= \frac{106}{100} \times x =$ ₹ $\frac{106x}{100}$

If an article sold at a loss of 6%, then selling price = Cost price
$\times \left(\frac{100 - \text{Loss\%}}{100}\right)$
$= \frac{94}{100} \times x =$ ₹ $\frac{94x}{100}$

Then, $\frac{106x}{100} - \frac{94x}{100} = 6$
$\Rightarrow$ $12x = 600$
$\therefore$ $x =$ ₹ 50

36. If $3^x = 4^y = 12^z$, then z is equal to
(a) xy (b) $x + y$
(c) $\frac{xy}{x+y}$ (d) $4x + 3y$

(*c*) $3^x = 4^y = 12^z = k$ (let)

$3 = k^{\frac{1}{x}}$ ...(i)
$4 = k^{1/y}$ ...(ii)
$12 = k^{1/z}$ ...(iii)

$\because$ $3 \times 4 = 12$
$k^{1/x} \times k^{1/y} = k^{1/z}$
$k^{\frac{1}{x} + \frac{1}{y}} = k^{\frac{1}{z}}$
[by Eqs. (i), (ii) and (iii)]

$$\Rightarrow \quad \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

$$\therefore \quad z = \frac{xy}{x+y}$$

Option (c) is correct.

37. A field can be reaped by 12 men or 18 women in 14 days. In how many days can 8 men and 16 women reap it?

(a) 26 days (b) 24 days
(c) 9 days (d) 8 days

(c) A field can be reaped by 12 men or 18 women in 14 days.

$$12M = 18W$$
$$2M = 3W$$
$$M = 3/2W \quad \ldots(i)$$

Total work $= 18W \times 14$ days

Let 8 men and 16 women can reap the field $= x$ days

$(8M + 16W) \times x$ days
$= 18W \times 14$ days

$$\left(8 \times \frac{3}{2}W + 16W\right) \times x = 18W \times 14$$
[by Eq. (i)]

$$D = \frac{18W \times 14}{28W} = 9 \text{ days}$$

Then, Option (c) is correct.

38. If $(4a+7b)(4c-7d) = (4a-7b)(4c+7d)$, then which one of the following is correct?

(a) $\frac{a}{b} = \frac{c}{d}$ (b) $\frac{a}{d} = \frac{c}{b}$
(c) $\frac{a}{b} = \frac{d}{c}$ (d) $\frac{4a}{7b} = \frac{c}{d}$

(a) $(4a+7b)(4c-7d)$
$= (4a-7b)(4c+7d)$

$$\Rightarrow \quad \frac{4a+7b}{4a-7b} = \frac{4c+7d}{4c-7d}$$

Use componendo and dividendo rule,

$$\frac{(4a+7b)+(4a-7b)}{(4a+7b)-(4a-7b)} = \frac{(4c+7d)+(4c-7d)}{(4c+7d)-(4c-7d)}$$

$$\frac{8a}{14b} = \frac{8c}{14d}$$

$$\frac{a}{b} = \frac{c}{d}$$

Option (a) is correct.

39. Given that the polynomial $(x^2 + ax + b)$ leaves the same remainder when divided by $(x-1)$ or $(x+1)$. What are the values of a and b, respectively?

(a) 4 and 0
(b) 0 and 3
(c) 3 and 0
(d) 0 and any integer

(d) The polynomial $(x^2 + ax + b)$ leaves the same remainder when divided by $(x-1)$ or $(x+1)$.

$$(x-1) = 0$$
$$x = 1 \quad \ldots(i)$$
$$(x+1) = 0$$
$$x = -1 \quad \ldots(ii)$$

On putting these value in polynomial

$$(1)^2 + a(1) + b = R \quad \text{[remainder]}$$
$$a + b + 1 = R \quad \ldots(iii)$$
$$(-1)^2 + a(-1) + b = R$$
$$b - a + 1 = R \quad \ldots(iv)$$

Eq. (iii) is equal to Eq. (iv), we get

$$a + b + 1 = b - a + 1$$
$$2a = 0$$
$$a = 0$$

b any integer.

Hence, option (d) is correct.

40. Tushar takes 6 h to complete a piece of work, while Amar completes the same work in 10 h. If both of them work together, then what is the time required to complete the work?

(a) 3 h (b) 3 h 15 min
(c) 3 h 30 min (d) 3 h 45 min

(d) Tushar takes 6 h to complete a piece of work. While Amar complete the same work in 10 h.

Then, both complete the same work

$$= \frac{1}{6} + \frac{1}{10} = \frac{16}{60}$$

$\therefore$ Required time $= \frac{60}{16} = 3$ h 45 min

Option (d) is correct.

41. What is the value of $2 + \sqrt{2 + \sqrt{2 + \sqrt{\ldots\ldots\ldots\ldots}}}$?

(a) 1 (b) 2
(c) 3 (d) 4

(d) $x = 2 + \sqrt{2 + \sqrt{2 + \sqrt{\ldots\ldots\ldots}}}$

$$y = \sqrt{2 + \sqrt{2 + \sqrt{2 \ldots}}}$$

On squaring both sides, we get

$$y^2 = 2 + \sqrt{2 + \sqrt{2 \ldots}}$$
$$y^2 = 2 + y$$
$$y^2 - y - 2 = 0$$
$$y^2 - 2y + y - 2 = 0$$
$$y(y-2) + 1(y-2) = 0$$
$$(y+1)(y-2) = 0$$
$$y = 2, -1$$

$(y = -1)$ not possible because all terms in equation are positive.

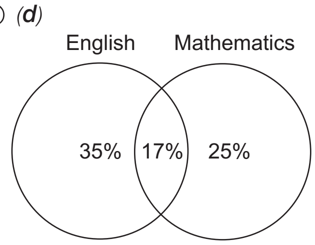
$$y = 2, x = 2 + y$$
$$x = 2 + 2 = 4$$

Option (d) is correct.

42. In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% failed in both the subjects, then what per cent passed in both the subjects?

(a) 77 (b) 58 (c) 48 (d) 23

(d)



Total number of students fail
$= 35 + 17 + 25 = 77\%$

Total number of students passed in both the subjects $= 100 - 77 = 23\%$

Option (d) is correct.

43. A man who recently died left a sum of ₹ 390000 to be divided among his wife, five sons and four daughters. He directed that each son should receive 3 times as much as each daughter receives and that each daughter should receive twice as much as their mother receives. What was the wife's share?

(a) ₹ 14000 (b) ₹ 12000
(c) ₹ 10000 (d) ₹ 9000

(c) Let wife's of man receives = ₹ x

Each daughter receives = ₹ $2x$

and each son receives = ₹ $6x$

According to the question,

$$x + 4(2x) + 5(6x) = 390000$$
$$39x = 390000$$
$$\therefore \quad x = 10000$$

Option (c) is correct.

44. What is the least number of complete years in which a sum of money put out at 40% annual compound interest will be more than tripled?

(a) 3 (b) 4
(c) 5 (d) 6

(*b*) Let $P = 100, R = 40\%$

$T = ?$ [the least number of complete years in which amount will be more than tripled]

$P = 100$

$A > 300$

$$300 < P\left(1 + \frac{R}{100}\right)^n$$

$$300 < 100\left(1 + \frac{40}{100}\right)^n$$

$$3 < (7/5)^n$$

Let $n = 3$, $3 < (1.4)^3$

$3 < 2.744$ [it is not true]

Let $n = 4$, $3 < (1.4)^4$

$3 < 3.8416$ [it is true]

Hence, the least number of complete years in which amount will be more than tripled is 4 yr.

45. A person divided a sum of ₹ 17200 into three parts and invested at 5%, 6% and 9% per annum simple interest. At the end of two years, he got the same interest on each part of money. What is the money invested at 9%?

(a) ₹ 3200 (b) ₹ 4000
(c) ₹ 4800 (d) ₹ 5000

(*b*) A person divided a sum of ₹ 17200 into three parts.

$R_1 = 5\%$, $R_2 = 6\%$, $R_3 = 9\%$, Time $= 2$yr

Let the money invested at 5%, 6% and 9% be x_1, x_2 and x_3 respectively.

We got the same interest

$$\frac{x_1 \times 5 \times 2}{100} = \frac{x_2 \times 6 \times 2}{100} = \frac{x_3 \times 9 \times 2}{100} \quad \left[\because \text{SI} = \frac{P \times R \times T}{100}\right]$$

$$x_1 \times \frac{10}{100} = x_2 \times \frac{12}{100} = x_3 \times \frac{18}{100}$$

Ratio of sum

$$x_1 \times \frac{10}{100} = x_2 \times \frac{12}{100}$$

$$\frac{x_1}{x_2} = \frac{6}{5}$$

$$\Rightarrow \quad x_1 : x_2 = 6 : 5$$

$$x_2 \times \frac{12}{100} = x_3 \times \frac{18}{100}$$

$$\frac{x_2}{x_3} = \frac{3}{2}$$

$$\Rightarrow \quad x_2 : x_3 = 3:2$$

$x_1 : x_2 : x_3$

$= 6 \times 3 : 5 \times 3 : 2 \times 5 = 18 : 15 : 10$

$\therefore$ The money invested at

$$9\% = x_3 = \frac{17200 \times 10}{(18 + 15 + 10)}$$

$$= \frac{172000}{43} = ₹\, 4000$$

46. What is $\dfrac{(x-y)^3 + (y-z)^3 + (z-x)^3}{3(x-y)(y-z)(z-x)}$ equal to?

(a) 1 (b) 0 (c) $\frac{1}{3}$ (d) 3

(*a*) $\dfrac{(x-y)^3 + (y-z)^3 + (z-x)^3}{3(x-y)(y-z)(z-x)}$

If $a + b + c = 0$

$a^3 + b^3 + c^3 = 3abc$

$x - y + y - z + z - x = 0$

Then, $\dfrac{3(x-y)(y-z)(z-x)}{3(x-y)(y-z)(z-x)} = 1$

Option (a) is correct answer.

47. If $a^x = b^y = c^z$ and $b^2 = ac$, then what is $\frac{1}{x} + \frac{1}{z}$ equal to?

(a) $\frac{1}{y}$ (b) $-\frac{1}{y}$ (c) $\frac{2}{y}$ (d) $-\frac{2}{y}$

(*c*) Given, $a^x = b^y = c^z$ and $b^2 = ac$

$a^x = b^y = c^z = k$ (let)

$a = k^{1/x}$

$b = k^{1/y}$

$c = k^{1/z}$

$\because \quad b^2 = ac$

$(k^{1/y})^2 = k^{1/x} \times k^{1/z}$

$k^{\frac{2}{y}} = k^{\left(\frac{1}{x} + \frac{1}{z}\right)}$

$\frac{2}{y} = \frac{1}{x} + \frac{1}{z}$

Option (c) is correct.

48. If p and q are the roots of the equation $x^2 - 15x + r = 0$ and $p - q = 1$, then what is the value of r?

(a) 55 (b) 56 (c) 60 (d) 64

(*b*) p and q are the roots of the equation.

$x^2 - 15x + r = 0 \quad p - q = 1 \quad$ ...(i)

$$\left[\begin{array}{c} ax^2 + bx + c = 0 \\ x^2 - \left(-\frac{b}{a}\right)x + \frac{c}{a} = 0 \\ \alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a} \end{array}\right]$$

[α and β are the root of Eq. (i)]

Then, $x^2 - 15x + r = 0$

$p + q = 15$ ...(ii)

$r = pq$ ...(iii)

$p - q = 1$ ...(iv)

On solving Eqs. (ii) and (iv)

$p = 8$

$q = 7$

$r = 8 \times 7 = 56$

Hence, option (b) is correct.

49. For the inequation $x^2 - 7x + 12 > 0$, which one of the following is correct?

(a) $3 < x < 4$
(b) $-\infty < x < 3$ only
(c) $4 < x < \infty$ only
(d) $-\infty < x < 3$ or $4 < x < \infty$

(*d*) $x^2 - 4x - 3x + 12 > 0$

$x(x-4) - 3(x-4) > 0$

$(x-4)(x-3) > 0$

$\therefore \quad x > 4$ or $x < 3$

$-\infty < x < 3$

or $\quad 4 < x < \infty$

Hence, option (d) is correct.

50. The expression $5^{2n} - 2^{3n}$ has a factor

(a) 3 (b) 7
(c) 17 (d) None of these

(*c*) $5^{2n} - 2^{3n}$ has a factor,

Let $n = 1$

$\Rightarrow \quad 5^2 - 2^3 = 25 - 8 = 17$

17 is factor of $5^{2n} - 2^{3n}$

Option (c) is correct.

51. If $\tan x = 1$, $0 < x < 90°$, then what is the value of $2 \sin x \cos x$?

(a) $\frac{1}{2}$ (b) 1 (c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$

(*b*) Given, $\tan x = 1, 0 < x < 90°$

$x = \tan^{-1}(1)$

$x = \tan^{-1}(\tan 45°)$

$\therefore \quad x = 45° = 2 \sin x \cos x$

$= 2 \sin 45° \cos 45°$

$= 2 \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = 1$

Option (b) is correct.

52. What is the value of $\sin 46° \cos 44° + \cos 46° \sin 44°$?

(a) $\sin 2°$ (b) 0
(c) 1 (d) 2

(*c*) Given,

$\sin 46° \cos 44° + \cos 46° \sin 44°$

$= \sin 46^\circ \cos(90^\circ - 46^\circ)$
$+ \cos 46^\circ \sin(90^\circ - 46^\circ)$

$[\because \sin(90^\circ - \theta) = \cos\theta,$
$\cos(90^\circ - \theta) = \sin\theta]$

$= \sin 46^\circ \sin 46^\circ + \cos 46^\circ \cos 46^\circ$

$= \sin^2 46^\circ + \cos^2 46^\circ = 1$

$[\because \sin^2\theta + \cos^2\theta = 1]$

Option (c) is correct.

Alternate Method

$\sin 46^\circ \cos 44^\circ + \cos 46^\circ \sin 44^\circ$

$= \sin(46^\circ + 44^\circ)$

$[\because \sin A \cos B + \cos A \sin B = \sin(A + B)]$

$= \sin 90^\circ$

$= 1$

53. Suppose $0 < \theta < 90^\circ$, then for every θ, $4\sin^2\theta + 1$ is greater than or equal to

(a) 2 (b) $4\sin\theta$
(c) $4\cos\theta$ (d) $4\tan\theta$

(a)

$4\sin^2\theta + 1$	$0 < \theta < 90^\circ$
Let $\theta = 30^\circ$	$\theta = 45^\circ$
$= 4\sin^2 30^\circ + 1$	$= 4\sin^2 45^\circ + 1$
$= 4 \times \frac{1}{4} + 1 = 2$	$= 4 \times \frac{1}{2} + 1 = 3$

Then, this equation greater than or equal to 2.

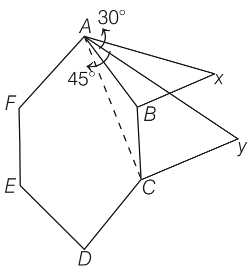
Option (a) is correct.

54. Consider a regular hexagon $ABCDEF$. Two towers are situated at B and C. The angle of elevation from A to the top of the tower at B is 30°, and the angle of elevation to the top of the tower at C is 45°. What is the ratio of the height of towers at B and C?

(a) $1 : \sqrt{3}$ (b) 1 : 3
(c) 1 : 2 (d) $1 : 2\sqrt{3}$

(b) A regular hexagon $ABCDEF$.

Two towers are situated at B and C.



Let side of hexagon = a

BX tower at B and CY tower at C

ΔABX

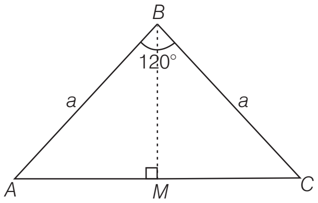
$AB = a$

$\tan 30^\circ = \frac{BX}{BA}$

$\frac{1}{\sqrt{3}} = \frac{BX}{a}$

$BX = \frac{a}{\sqrt{3}}$ ...(i)

ΔABC



$[\angle B = 120^\circ$ each internal angle of hexagon $120^\circ]$

$BM \perp AC$

ΔABM

$\angle ABM = 60^\circ$

$\angle AMB = 90^\circ$

$\sin 60^\circ = \frac{AM}{AB}$

$\frac{\sqrt{3}}{2} = \frac{AM}{a}$

$AM = \frac{\sqrt{3}a}{2}$

Similarly,

$MC = \frac{\sqrt{3}}{2}a$

$AC = AM + MC$

$= \frac{\sqrt{3}}{2}a + \frac{\sqrt{3}}{2}a$

$= \sqrt{3}a$

ΔAYC, $\tan 45^\circ = \frac{CY}{AC}$

$1 = \frac{CY}{\sqrt{3}a}$

$\sqrt{3}a = CY$

Ratio of tower B and C

$BX : CY$

$\frac{a}{\sqrt{3}} : \sqrt{3}a$

1 : 3

Option (b) is correct.

55. What is the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \ldots \tan 89^\circ$?

(a) 0 (b) 1
(c) 2 (d) ∞

(b) Given, $\tan 1^\circ \tan 2^\circ \tan 3^\circ \ldots \tan 89^\circ$

$= (\tan 1^\circ \tan 89^\circ)(\tan 2^\circ \tan 88^\circ)\ldots$

$\ldots(\tan 44^\circ \tan 45^\circ)\tan 45^\circ$

$= 1 \times 1 \ldots 1 \times 1$

$[\because \tan\theta \tan(90^\circ - \theta) = 1]$

$= 1$ $[\because \tan 45^\circ = 1]$

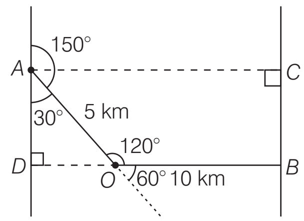
Option (b) is correct.

56. There are two parallel streets each directed North to South.

A person in the first street travelling from South to North wishes to take the second street which is on his right side. At some place, he makes a 150° turn to the right and he travels for 15 min at the speed of 20 km/h. After that he takes a left turn of 60° and travels for 20 min at the speed of 30 km/h in order to meet the second street. What is the distance between the two streets?

(a) 7.5 km (b) 10.5 km
(c) 12.5 km (d) 15 km

(c) There are two parallel streets.



AO = travells for 15 min at the speed of 20 km/h

Distance = speed × time

$= \frac{15}{60} \times 20 = 5$ km

OB = travells for 20 min at the speed of 30 km/h

$= \frac{20}{60} \times 30 = 10$ km

ΔADO, $\angle DAO = 180^\circ - 150^\circ = 30^\circ$

$\sin 30^\circ = \frac{OD}{OA} = \frac{OD}{5}$

$\frac{1}{2} = \frac{OD}{5}$

$OD = 2.5$ km

The distance between the two streets.

$DB = DO + OB$

$= 2.5 + 10 = 12.5$ km

Option (c) is correct.

57. If $3\tan\theta = \cot\theta$ where $0 \le \theta < \frac{\pi}{2}$, then what is the value of θ?

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

(a) Given, $3\tan\theta = \cot\theta$ $0 \le \theta < \frac{\pi}{2}$

$3\tan\theta = \frac{1}{\tan\theta}$ $\left[\because \cot\theta = \frac{1}{\tan\theta}\right]$

$\tan^2\theta = \frac{1}{3}$

$\tan\theta = \frac{1}{\sqrt{3}}$

[$\tan\theta$, value positive in first quadrant]

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \tan^{-1}(\tan 30^\circ)$$

$$\theta = 30^\circ \Rightarrow \theta = \frac{\pi}{6}$$

Option (a) is correct.

58. What is the value of $\sin^2 25^\circ + \sin^2 65^\circ$?

(a) 0 (b) 1 (c) 2 (d) 4

(*b*) Given, $\sin^2 25^\circ + \sin^2 65^\circ$

$= \sin^2 25^\circ + \sin^2(90^\circ - 25^\circ)$

$= \sin^2 25^\circ + \cos^2 25^\circ$ $[\because \sin(90^\circ - \theta) = \cos\theta]$

$= 1$ $[\because \sin^2\theta + \cos^2\theta = 1]$

Option (b) is correct.

59. What is the value of $\sin^6\theta + \cos^6\theta + 3\sin^2\theta\cos^2\theta - 1$?

(a) 0 (b) 1 (c) 2 (d) 4

(*a*) Given,

$\sin^6\theta + \cos^6\theta + 3\sin^2\theta\cos^2\theta - 1$

$= (\sin^2\theta)^3 + (\cos^2\theta)^3 + 3\sin^2\theta\cos^2\theta(\sin^2\theta + \cos^2\theta) - 1$

$[\because (a+b)^3 = a^3 + b^3 + 3ab(a+b)]$

$= (\sin^2\theta + \cos^2\theta)^3 - 1$ $[\because \sin^2\theta + \cos^2\theta = 1]$

$= (1)^3 - 1 = 0$

Option (a) is correct.

60. Consider the following for real numbers α, β, γ and δ

1. $\sec\alpha = 1/4$ 2. $\tan\beta = 20$
3. $\mathrm{cosec}\,\gamma = 1/2$ 4. $\cos\delta = 2$

How many of the above statements are not possible?

(a) One (b) Two
(c) Three (d) Four

(*c*) α, β, γ and δ are real numbers

1. $\sec\alpha = 1/4$ [$\because$ Value of $\sec\alpha$ lies between $(-\infty, -1] \cup [1, \infty)$]
   It is not possible.
2. $\tan\beta = 20$ [$\because$ value of $\tan\beta$ lies between $[-\infty, \infty]$]
   It is possible.
3. $\mathrm{cosec}\,\gamma = 1/2$ [$\because$ value of cosec γ lies between $(-\infty, -1] \cup [1, \infty)$]
   It is not possible.
4. $\cos\delta = 2$ [$\because$ value of $\cos\delta$ lies between $[-1, 1]$]
   It is not possible.

Three statements are not correct.

Then,

Option (c) is correct.

61. Consider the following grouped frequency distribution

x	f
0-10	8
10-20	12
20-30	10
30-40	p
40-50	9

If the mean of the above data is 25.2, then what is the value of p?

(a) 9 (b) 10
(c) 11 (d) 12

(*c*)

x	x_i	f_i	$x_i f_i$
0-10	5	8	40
10-20	15	12	180
20-30	25	10	250
30-40	35	p	35p
40-50	45	9	405
		$\Sigma f_i = 39 + p$	$\Sigma x_i f_i = 875 + 35p$

$$\bar{x}(\text{Mean}) = \frac{\Sigma x_i f_i}{\Sigma f_i}$$

$$25.2 = \frac{875 + 35p}{39 + p}$$

$25.2(39 + p) = 875 + 35p$

$982.8 + 25.2p = 875 + 35p$

$982.8 - 875 = 35p - 25.2p$

$107.8 = 9.8p$

$$p = \frac{107.8}{9.8} = 11$$

Option (c) is correct.

62. Consider the following frequency distribution

x	f
8	6
5	4
6	5
10	8
9	9
4	6
7	4

What is the median for the distribution?

(a) 6 (b) 7 (c) 8 (d) 9

(*c*) Arrange in ascending order

(x)	(f)
4	6
5	4
6	5
7	4
8	6
9	9
10	8
	n = 42

Here, $\frac{N}{2} = \frac{42}{2} = 21$

21th observation = 8

22th observation = 8

Median

$$= \frac{\left(\frac{n}{2}\right)\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}}{2}$$

$$= \frac{21\text{th term} + 22\text{th term}}{2}$$

$$= \frac{8+8}{2} = \frac{16}{2} = 8$$

63. The average of 50 consecutive natural numbers is x. What will be the new average when the next four natural numbers are also included?

(a) $x + 1$ (b) $x + 2$
(c) $x + 4$ (d) $x + (x/54)$

(*b*) The average of 50 consecutive natural numbers is x.

New average

$$= \text{Old average} + \frac{\text{Number of next terms}}{2}$$

$$= x + \frac{4}{2} = x + 2$$

64. Consider two-digit numbers which remain the same when the digits interchange their positions. What is the average of such two-digit numbers?

(a) 33 (b) 44 (c) 55 (d) 66

(*c*) Two-digit numbers which remain the same when the digits interchange their positions.

11, 22, 33, 44, 55, 66, 77, 88, 99

$$\text{Average} = \frac{\text{Sum of numbers}}{\text{Number of terms}}$$

$$= \frac{[11 + 22 + 33 + 44 + 55 + 66 + 77 + 88 + 99]}{9}$$

$$= \frac{495}{9} = 55$$

Alternative method

$$\text{Average} = \frac{\text{First term} + \text{Last term}}{2}$$

$$= \frac{11 + 99}{2} = 55$$

Option (c) is correct.

65. Diagrammatic representation of data includes which of the following?
1. Bar-diagram
2. Pie-diagram
3. Pictogram

Select the correct answer using the code given below
(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

(*d*) Diagrammatic representation of data includes
1. Bar-diagram
2. Pie-diagram
3. Pictogram

Option (d) is correct.

66. The data collected from which one of the following methods is not a primary data?
(a) by direct personal interviews
(b) by indirect personal interviews
(c) by schedules sent through enumerators
(d) from published thesis

(*d*) The data collected from published thesis is not a primary data.
Option (d) is correct.

67. The monthly expenditure of a person is ₹ 6000. The distribution of expenditure on various items is as follows

	Item of expenditure	Amount (in ₹)
1.	Food	2000
2.	Clothing	660
3.	Fuel and rent	1200
4.	Education	480
5.	Miscellaneous	1660

If the above data is represented by a percentage bar-diagram of height 15 cm, then what are the lengths of the two segments of the bar-diagram corresponding to education and miscellaneous, respectively?
(a) 1.25 cm and 5 cm
(b) 1.2 cm and 4.15 cm
(c) 1.2 cm and 3.5 cm
(d) 4.15 cm and 6 cm

(*b*) The monthly expenditure of a person is 6000.

Percentage of educations expenditure

$$= \frac{480}{6000} \times 100 = 8\%$$

15 cm height of bar-diagram.

Length of the education in bar-diagram

$$= 15 \times \frac{8}{100} = 1.2 \text{ cm}$$

Percentage of miscellaneous's expenditure

$$= \frac{1660}{6000} \times 100 = \frac{166}{600}\%$$

Length of the miscellaneous's in bar-diagram

$$= 15 \times \frac{166}{600} = 4.15 \text{ cm}$$

Option (b) is correct.

68. If the mean of m observations out of n observations is n and the mean of remaining observations is m, then what is the mean of all n observations?
(a) $2m - \frac{m^2}{n}$
(b) $2m + \frac{m^2}{n}$
(c) $m - \frac{m^2}{n}$
(d) $m + \frac{m^2}{n}$

(*a*) The mean of m observations $= n$

The sum of observations $= mn$

The mean of $(n - m)$ observations $= m$

The sum of observations $= (n - m)\, m$

The mean of all observations

$$= \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

$$= \frac{mn + (n-m)m}{n}$$

$$= \frac{m(n + n - m)}{n}$$

$$= \frac{2nm - m^2}{n} = 2m - \frac{m^2}{n}$$

Option (a) is correct.

69. Which one of the following pairs is correctly matched?

(a)	Median	= Graphical location
(b)	Mean	= Graphical location
(c)	Geometric Mean	= Ogive
(d)	Mode	= Ogive

(*a*) Median is graphical location.
Then, Option (a) is correct.

70. The following pairs relate to frequency distribution of a discrete variable and its frequency polygon. Which one of the following pairs is not correctly matched?

(a)	Base line of the polygon	X-axis
(b)	Ordinates of the vertices of the polygon	Class frequencies
(c)	Abscissa of the vertices of the polygon	Class marks of the frequency distribution
(d)	Area of the polygon	Total frequency of the distribution

(*d*) Area of the polygon is not related to the total frequency of the distribution.
Then, option (d) is correct.

71. In a rectangle, length is three times its breadth. If the length and the breadth of the rectangle are increased by 30% and 10% respectively, then its perimeter increases by
(a) $\frac{40}{3}\%$ (b) 20%
(c) 25% (d) 27%

(*c*) Breadth of the rectangle $= x$

Length of the rectangle $= 3x$

Perimeter of rectangle $= 2(x + 3x)$

$$= 8x \quad \ldots(i)$$

$[\because P = 2\,(l + b)]$

Increases length 30%, then

$$\text{Length} = \frac{130}{100} \times 3x = \frac{39x}{10}$$

Increases breadth 10%, then

$$\text{Breadth} = \frac{110}{100} \times x = \frac{11x}{10}$$

New perimeter of rectangle

$$= 2\left(\frac{39x}{10} + \frac{11x}{10}\right) = 10x \quad \ldots(ii)$$

Increases in perimeter
Eqs. (i) and (ii), we get

$$10x - 8x = 2x$$

Percentage of change in perimeter

$$= \frac{2x}{8x} \times 100 = 25\%$$

Option (c) is correct.

72. What is the percentage decrease in the area of a triangle if its each side is halved?

(a) 75% (b) 50%
(c) 25% (d) No change

(**a**) Let we have an equilateral triangle and side of triangle $= x$

Area of triangle $= \frac{\sqrt{3}}{4}x^2$

Now, Triangle each side is halved $= x/2$

New area of triangle

$$= \frac{\sqrt{3}}{4}\left(\frac{x}{2}\right)^2 = \frac{\sqrt{3}}{4}\frac{x^2}{4}$$

Change in area

$$= \frac{\sqrt{3}}{4}x^2 - \frac{\sqrt{3}}{4}\frac{x^2}{4} = \frac{\sqrt{3}}{4}\frac{3x^2}{4}$$

Change in percentage of area

$$= \frac{\frac{\sqrt{3}}{4} \times \frac{3x^2}{4}}{\frac{\sqrt{3}}{4}x^2} \times 100 = 75\%$$

Option (a) is correct.

Alternate method

Here, $a = 50\%$

Required percentage decrease in area of equilateral triangle

$$= \left(2a - \frac{a^2}{100}\right)\% = 2 \times 50 - \frac{(50)^2}{100}$$

$$= 100 - \frac{2500}{100} = 100 - 25 = 75\%$$

73. The volume of a spherical balloon is increased by 700%. What is the percentage increase in its surface area?

(a) 300% (b) 400%
(c) 450% (d) 500%

(**a**) Let the radius of spherical balloon $= r$

Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of balloon $= 4\pi r^2$

According to question,

Volume of balloon is increased by 700%

$\therefore$ New Volume $= \frac{4}{3}\pi r^3 \times \frac{800}{100} = \frac{4}{3}\pi(2r)^3$

$\therefore$ New radius $= 2r$

$\therefore$ New surface area $= 4\pi(2r)^2$

$$= 4\pi \times 4r^2 = 16\pi r^2$$

$\therefore$ Required percentage increases

$$= \frac{16\pi r^2 - 4\pi r^2}{4\pi r^2} \times 100$$

$$= \frac{12}{4} \times 100 = 300\%$$

74. If the lengths of two parallel chords in a circle of radius 10 cm are 12 cm and 16 cm, then what is the distance between these two chords?

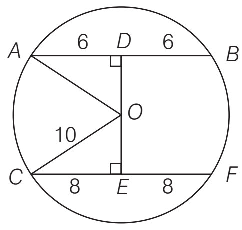
(a) 1 cm or 7 cm (b) 2 cm or 14 cm
(c) 3 cm or 21 cm (d) 4 cm or 28 cm

(**b**) Radius of circle = 10 cm

Length of chords = 12 cm and 16 cm

If chords opposite side from centre.

O is centre of circle.



AB and CD are two chords 12 cm and 16 cm, respectively.

In ΔAOD,

$AO = 10$ cm [radius of circle]

$AD = 6$ cm

$$(AO)^2 = (AD)^2 + (OD)^2$$
$$(10)^2 = (6)^2 + (OD)^2$$
$$(10)^2 - (6)^2 = OD^2$$
$$\sqrt{100 - 36} = OD$$
$$OD = 8 \text{ cm}$$

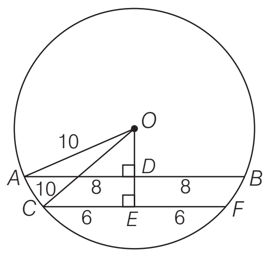
In ΔCEO, $(CO)^2 = (CE)^2 + (OE)^2$

$$(10)^2 = (8)^2 + (OE)^2$$
$$\sqrt{100 - 64} = OE, OE = 6 \text{ cm}$$

Distance between two chords

$(DE) = DO + OE = 8 + 6 = 14$ cm

If both chords are in same side from centre.



O is centre of circle.

$AB = 16$ cm

$CD = 12$ cm

In ΔAOD,

$$(AO)^2 = (AD)^2 + (OD)^2$$
$$(10)^2 = (8)^2 + OD^2$$
$$OD^2 = 100 - 64$$
$$OD = \sqrt{36}$$
$$OD = 6 \text{ cm}$$

In ΔCOE,

$$(CO)^2 = (OE)^2 + (CE)^2$$
$$(10)^2 = (OE)^2 + (6)^2$$
$$OE = \sqrt{100 - 36}$$
$$OE = 8 \text{ cm}$$

The distance between two chords

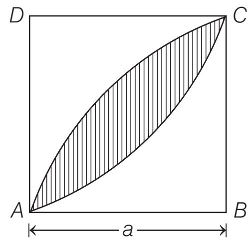
$DE = OE - OD = 8 - 6 = 2$ cm

75. Considering two opposite vertices of a square of side 'a' as centres, two circular arcs are drawn within the square joining the other two vertices, thus forming two sectors. What is the common area in these two sectors?

(a) $a^2\left(\pi + \frac{1}{2}\right)$ (b) $a^2\left(\pi - \frac{1}{2}\right)$
(c) $a^2\left(\frac{\pi}{2} - 1\right)$ (d) $a^2\left(\frac{\pi}{2} + 1\right)$

(**c**) $ABCD$ is a square

and Side of a square $= a$



Shaded region is the common area of the sectors.

Area of shaded region

= 2 × Area of sector – Area of square

$$= 2 \times \pi a^2 \times \frac{90°}{360°} - a^2$$

[$\because$ area of sector $\pi r^2 \frac{\theta}{360°}$]

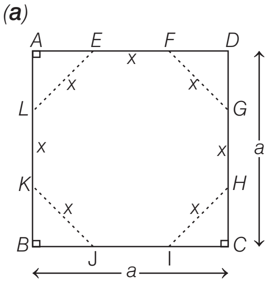
$$= 2 \times \frac{\pi a^2}{4} - a^2 = a^2\left(\frac{\pi}{2} - 1\right)$$

Option (c) is correct.

76. The corners of a square of side 'a' are cut away so as to form a regular octagon. What is the side of the octagon?

(a) $a(\sqrt{2} - 1)$ (b) $a(\sqrt{3} - 1)$
(c) $\frac{a}{\sqrt{2} + 2}$ (d) $\frac{a}{3}$

(**a**)



$ABCD$ is square.

The corners of square are cut away by doted line.

EFGHIJKL is octagon.

Let side of octagon = x

$AE + FD = a - x$ [$\because AE = FD$]

$2AE = a - x$

$AE = \left(\frac{a-x}{2}\right)$ ...(i)

Similarly, $AL = \left(\frac{a-x}{2}\right)$

ΔAEL $(AL)^2 + (AE)^2 = (LE)^2$

$\left(\frac{a-x}{2}\right)^2 + \left(\frac{a-x}{2}\right)^2 = x^2$

$\sqrt{2\left(\frac{a-x}{2}\right)^2} = x$

$x = \sqrt{2}\left(\frac{a-x}{2}\right)$

$\sqrt{2}x = a - x$

$\Rightarrow \sqrt{2}\,x + x = a$

$x(\sqrt{2} + 1) = a$

$x = \frac{a}{\sqrt{2}+1}$

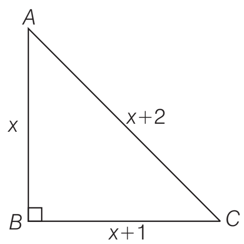
$x = a(\sqrt{2} - 1)\left[\because \frac{1}{\sqrt{2}+1} = \sqrt{2} - 1\right]$

Then, option (a) is correct.

77. Three consecutive integers form the lengths of a right angled triangle. How many sets of such three consecutive integers is/are possible?

(a) Only one (b) Only two
(c) Only three (d) Infinitely many

(*a*) Three consecutive integers form the lengths of a right-angled triangle.



$(x)^2 + (x+1)^2 = (x+2)^2$

$x^2 + x^2 + 2x + 1 = x^2 + 4x + 4$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0 \left[\begin{array}{r} x = 3 \\ x + 1 = 4 \\ x + 2 = 5 \end{array}\right]$

$x = 3$ or $x = -1$

Only one set have consecutive integers.

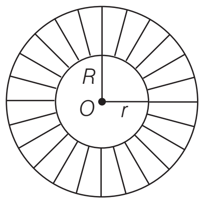
[not possible length never be negative]

Option (a) is correct.

78. Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles?

(a) 154 cm^2 (b) 308 cm^2
(c) 462 cm^2 (d) 616 cm^2

(*c*) Two circle are drawn with the same centre (*O*).



Circumference of smaller circle = 44 cm

and Radius of smaller circle = r

Circumference of circle = $2\pi r$

$\Rightarrow 2\pi r = 44$

$r = 7$ ...(i)

Circumference of big circle

= 2 × smaller circle

= 2 × 44 = 88

$2\pi R = 88$

$R = 14$

Area between two circles

$A = \pi R^2 - \pi r^2$

$A = \pi(R^2 - r^2)$

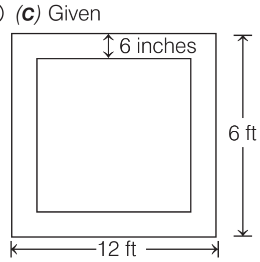
$= \frac{22}{7} \times [14^2 - 7^2] = \frac{22}{7} \times 21 \times 7$

$= 462\ cm^2$

79. A rectangular red carpet of size 6 ft × 12 ft has a dark red border 6 inches wide. What is the area of the dark red border?

(a) 9 sq feet (b) 15 sq feet
(c) 17 sq feet (d) 18 sq feet

(*c*) Given



Width of red border = 6 inches

$= \frac{6}{12} = 0.5$ feet {$\because$ 1 ft = 12 inches}

Length of outer rectangle = 12 feet

Breadth of outer rectangle = 6 feet

Area of outer rectangle

$= l \times b = 12 \times 6 = 72$ sq feet

Length of inner rectangle

$= 12 - 0.5 - 0.5 = 11$ feet

Breadth of inner rectangle

$= 6 - 0.5 - 0.5 = 5$ feet

Area of inner rectangle

$= 11 \times 5 = 55$ sq feet

Area of darkred border

= (area of outer rectangle)

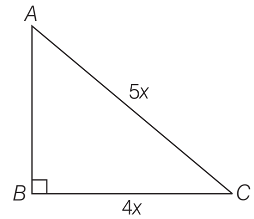
− (area of inner rectangle)

$= 72 - 55 = 17$ sq feet

80. The perimeter of a right-angled triangle is k times the shortest side. If the ratio of the other side to hypotenuse is 4 : 5, then what is the value of k?

(a) 2 (b) 3 (c) 4 (d) 5

(*c*) ΔABC is right angle triangle at B



$AC = 5x$, $BC = 4x$

AB is shortest side of triangle.

$(AB)^2 + (BC)^2 = (AC)^2$

$(AB)^2 + (4x)^2 = (5x)^2$

$AB = \sqrt{25x^2 - 16x^2} = \sqrt{9x^2}$

$AB = 3x$

Perimeter of triangle is k times the shortest side

$AB + BC + CA = k \times (AB)$

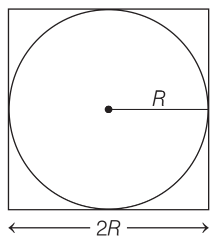
$3x + 4x + 5x = k \times (3x)$

$k = \frac{12x}{3x} = 4 \Rightarrow k = 4$

81. A 12 m long wire is cut into two pieces, one of which is bent into a circle and the other into a square enclosing the circle. What is the radius of the circle?

(a) $\frac{12}{\pi + 4}$ (b) $\frac{6}{\pi + 4}$
(c) $\frac{3}{\pi + 4}$ (d) $\frac{6}{\pi + 2\sqrt{2}}$

(*b*) A square enclosing the circle



Radius of circle = R

and Side of square = $2R$

Circumference of circle = $2\pi R$

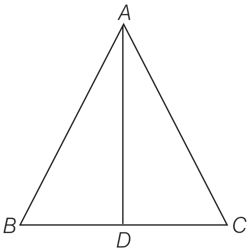
Perimeter of square = $4(2R) = 8R$

Total length of wire

$$2\pi R + 8R = 12$$
$$2(\pi R + 4R) = 12$$
$$R(\pi + 4) = 6$$
$$R = \frac{6}{\pi + 4}$$

Radius of circle $= \dfrac{6}{\pi + 4}$

82. The angles of a triangle are in the ratio 1 : 1 : 4. If the perimeter of the triangle is k times its largest side, then what is the value of k?

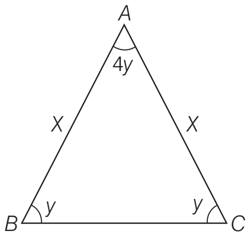


(a) $1 + \dfrac{2}{\sqrt{3}}$ (b) $1 - \dfrac{2}{\sqrt{3}}$

(c) $2 + \dfrac{2}{\sqrt{3}}$ (d) 2

(a) The angle of triangle are in the ratio 1 : 1 : 4.

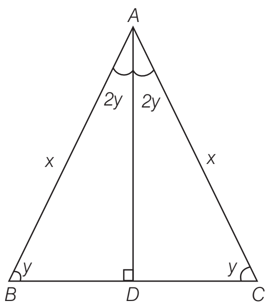
Then, two angle are equal.



Then, ABC is isosceles triangle

Let $AB = x$

A perpendicular drawn from vertex A to BC is D.



$BD = DC$ [isosceles triangle]

AD is also angle bisector

In ΔADB,

$2y + y + 90 = 180^\circ$

[sum of triangle's interior angles]

$$y = 30^\circ$$
$$\cos 30^\circ = \frac{BD}{AB}$$
$$\frac{\sqrt{3}}{2} = \frac{BD}{x}$$
$$\Rightarrow \quad BD = \frac{\sqrt{3}x}{2}$$

Similarly, $DC = \dfrac{\sqrt{3}x}{2}$

$$BC = BD + DC$$
$$BC = \frac{\sqrt{3}x}{2} + \frac{\sqrt{3}x}{2}$$
$$= \sqrt{3}x$$

Perimeter of triangle equal to k times larger side of triangle.

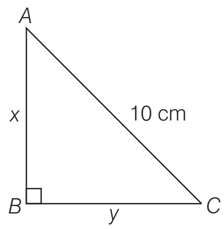
$$x + x + \sqrt{3}x = k\sqrt{3}x$$
$$\sqrt{3}xk = 2x + \sqrt{3}x$$
$$k = \frac{\sqrt{3}x + 2x}{\sqrt{3}x}$$
$$k = 1 + \frac{2}{\sqrt{3}}$$

83. The hypotenuse of a right angled triangle is 10 cm and its area is $24\ \text{cm}^2$. If the shorter side is halved and the longer side is doubled, the new hypotenuse becomes

(a) $\sqrt{245}$ cm (b) $\sqrt{255}$ cm

(c) $\sqrt{265}$ cm (d) $\sqrt{275}$ cm

(c) ABC is right angle triangle at B.



$$AC = 10\ \text{cm}$$
$$AB = x$$
$$BC = y$$
$$(AB)^2 + (BC)^2 = (AC)^2$$
$$x^2 + y^2 = 100 \quad \ldots(i)$$

Area of triangle $= \dfrac{1}{2} \times AB \times BC$

$$24 = \frac{1}{2} \times x \times y$$
$$xy = 48 \quad \ldots(ii)$$

We know, $(x + y)^2 = x^2 + y^2 + 2xy$

$$(x + y)^2 = 100 + 2 \times 48 = 196$$
$$x + y = 14 \quad \ldots(iii)$$

and $(x - y)^2 = x^2 + y^2 - 2xy$

$$(x - y)^2 = 100 - 2 \times 48 = 4$$
$$x - y = 2 \quad \ldots(iv)$$

From Eqs. (iii) and (iv), we get

$$x = 8 \Rightarrow y = 6$$

The shorter side is halved

$$y_1 = \frac{6}{2} = 3\ \text{cm}$$

The longer side is doubled

$$x_2 = 8 \times 2 = 16\ \text{cm}$$

Now, new hypotenuse

$$= \sqrt{x_1^2 + y_1^2} = \sqrt{16^2 + 3^2}$$
$$= \sqrt{256 + 9} = \sqrt{265}\ \text{cm}$$

84. In a circle of radius 8 cm, AB and AC are two chords such that $AB = AC = 12$ cm. What is the length of chord BC?

(a) $2\sqrt{6}$ cm (b) $3\sqrt{6}$ cm

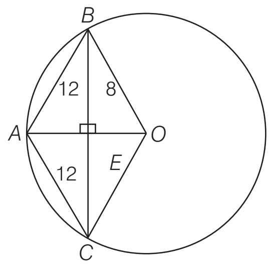
(c) $3\sqrt{7}$ cm (d) $6\sqrt{7}$ cm

(d) In a circle,

Radius = 8 cm

$AB = AC = 12$ cm

AB and AC are chords of circle.



$AO = 8$ cm is also radius

$$s = \frac{12 + 8 + 8}{2} = \frac{28}{2} = 14\ \text{cm}$$

Area of ΔABO

$$= \sqrt{s(s - a)(s - b)(s - c)}$$
$$= \sqrt{14 \times (14 - 12)(14 - 8)(14 - 8)}$$
$$= \sqrt{14 \times 2 \times 6 \times 6} = 12\sqrt{7}\ \text{cm}^2$$

BE is height of ΔBAO

Area of $\Delta ABO = \dfrac{1}{2} \times BE \times AO$

$$12\sqrt{7} = \frac{1}{2} \times BE \times 8$$
$$BE = 3\sqrt{7}\ \text{cm}$$

Length of $BC = 2 \times BE$

$$= 2 \times 3\sqrt{7} = 6\sqrt{7}\ \text{cm}$$

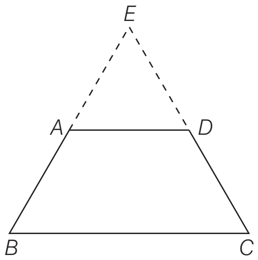
85. Consider the following statements:

1. An isosceles trapezium is always cyclic.
2. Any cyclic parallelogram is a rectangle.

Which of the above statements is/are correct?`

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

(*c*)



We produce *BA* and *CD* and meet them at *E*.

$AB = CD$ (given)…(1)

$AD \parallel BC$ (given)

them,

$AE = DE$…(2)

On adding Eq. (1) and (2), we get

$AB + AE = CD - DE$

$BE = EC$

$\therefore \quad \angle EBC = \angle ECB = \theta$

then,

$\angle BAD = 180 - \angle ABC$

$\angle BAD = 180 - \theta$ …(3)

Similarly

$\angle CDA = 180^\circ - \theta$ …(4)

then

$\angle ABC + \angle CDA = \theta + 180 - \theta$

$= 180^\circ$

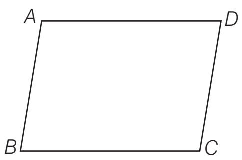
and $\angle DAB + \angle DCB = 180^\circ - \theta + \theta$

$= 180^\circ$

Then, an isosceles trapezium is always cyclic.

$\because$ Statement I is true.

(2) Any cyclic parallelogram is rectangle.



In parallelogram $\angle B = \angle D = x$ … (i)

In cyclic parallelogram,

$\angle B + \angle D = 180^\circ$

$x + x = 180^\circ$ {from Eq. (i)}

$2x = 180^\circ \Rightarrow x = 90^\circ$

Then,

It is a rectangle.

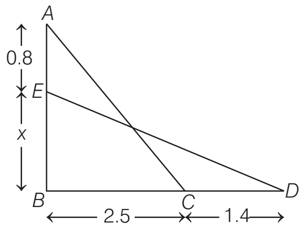
Statement 2nd also correct.

86. A ladder is resting against a vertical wall and its bottom is 2.5 m away from the wall. If it slips 0.8 m down the wall, then its bottom will move away from the wall by 1.4 m.

What is the length of the ladder?

(a) 6.2 m (b) 6.5 m
(c) 6.8 m (d) 7.5 m

(*b*) A ladder is resting against a vertical wall.



AC is length of ladder.

After slip,

DE is length of ladder

$AC = DE$ …(i)

ΔABC using pythogoras theorem

$(0.8 + x)^2 + (2.5)^2 = (AC)^2$

$x^2 + 0.64 + 1.6x + 6.25 = AC^2$…(ii)

In ΔBDE using pythogoras theorem

$x^2 + (2.5 + 1.4)^2 = (DE)^2$

$x^2 + 15.21 = DE^2$ …(iii)

From Eqs. (ii) and (iii) equal

by ($AC = DE$) Eq. (i)

Then,

$x^2 + 15.21 = x^2 + 0.64 + 1.6x + 6.25$

$x = \frac{15.21 - 6.89}{1.6}$

$x = 5.2$ m …(iv)

On putting $x = 5.2$ in Eq. (iii), we get

$(5.2)^2 + 15.21 = (DE)^2$

$DE = \sqrt{27.04 + 15.21}$

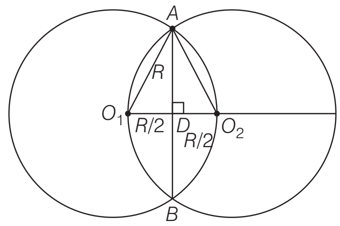
$DE = \sqrt{42.5} = 6.5$ m

Option (b) is correct.

87. Two equal circles intersect such that each passes through the centre of the other. If the length of the common chord of the circles is $10\sqrt{3}$ cm, then what is the diameter of the circle?

(a) 10 cm (b) 15 cm
(c) 20 cm (d) 30 cm

(*c*) Two equal circles intersect such that each passes through the centre of the other.



Radius of circles = *R*

AB is common chord = $10\sqrt{3}$ cm

$AD = \frac{AB}{2} = \frac{10\sqrt{3}}{2} = 5\sqrt{3}$ cm

In ΔAO_1D,

$(O_1A)^2 = (O_1D)^2 + (AD)^2$

$R^2 = \left(\frac{R}{2}\right)^2 + (5\sqrt{3})^2$

$R^2 - \frac{R^2}{4} = 25 \times 3$

$\frac{3R^2}{4} = 25 \times 3, R = \sqrt{25 \times 4}$

$R = 10$ cm

Then, diameter of circle

$= 2R = 2 \times 10 = 20$ cm

Option (c) is correct.

Alternate Method

We know that, Length of common chord

$= \sqrt{3}r$

$10\sqrt{3} = \sqrt{3}r$

Diameter of circle = $2 \times 10 = 20$ cm

88. Consider the following statements.

I. The number of circles that can be drawn through three non-collinear points is infinity.

II. Angle formed in minor segment of a circle is acute.

Which of the above statements is/are correct?

(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

(*d*) I. Only one circle can be drawn from three non-collinear points.

II. (i) Angle formed in minor segment of a circle is obtuse.

(ii) Angle formed in major segment of a circle is acute.

Then, Option (d) is correct.

89. Consider the following inequalities in respect of any triangle *ABC*:

1. $AC - AB < BC$
2. $BC - AC < AB$
3. $AB - BC < AC$

Which of the above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

(*d*) Property of triangle

Difference of two side of triangle is less than third side of the triangle.

Then, all the statement is correct.

Option (d) is correct.

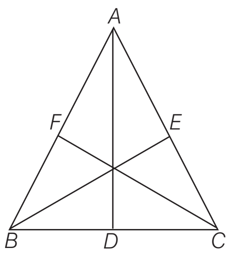
90. Consider the following statements :

1. The perimeter of a triangle is greater than the sum of its three medians.
2. In any triangle ABC, if D is any point on BC, then $AB + BC + CA > 2AD$.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

(*c*) (1) The perimeter of a triangle is greater than the sum of its three medians.



It is property of triangle.

AD, BE and CF are three medians of triangle.

(2) ABC is triangle and D is a point on BC.

By triangle properties,

ΔABD, $AB + BD > AD$ ...(i)

ΔADC, $AC + DC > AD$ ...(ii)

On adding Eqs. (i) and (ii), we get

$AB + AC + BD + DC > 2AD$

$AB + AC + BC > 2AD$

$[\because BD + DC = AD]$

Option (c) is correct.

Directions (Q. Nos. 91-92) *Read the given information carefully and answer the given questions below.*

A cube is inscribed in a sphere. A right circular cylinder is with in the cube touching all the vertical faces. A right circular cone is inside the cylinder. Their heights are same and the diameter of the cone is equal to that of the cylinder.

91. What is the ratio of the volume of the sphere to that of the cone?

(a) $6\sqrt{3} : 1$ (b) $7 : 2$
(c) $3\sqrt{3} : 1$ (d) $5\sqrt{3} : 1$

(*a*)

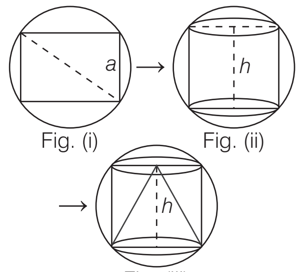


Fig. (i) Fig. (ii) Fig. (iii)

Radius of sphere = R

Then, 2 × (Radius of sphere)

= Diagonal of cube

$2 \times R = \sqrt{3}a$ [a = side of cube]

$$a = \frac{2R}{\sqrt{3}}$$

Height of cylinder (h) = Side of cube [fig. (ii)]

$$h = \frac{2R}{\sqrt{3}}$$

Radius of cylinder = $\frac{1}{2} \times$ Side of cube

$$r = \frac{1}{2} \times \frac{2R}{\sqrt{3}} = \frac{R}{\sqrt{3}}$$

Radius of cone = Radius of cylinder

$$r = \frac{R}{\sqrt{3}}$$

Height of cone (h) = Height of cylinder

$$h = \frac{2R}{\sqrt{3}}$$

Volume of sphere : Volume of cone

$$= \frac{4}{3}\pi R^3 : \frac{1}{3}\pi \frac{R^2}{3} \times \frac{2R}{\sqrt{3}} = 6\sqrt{3} : 1$$

$\because$ Volume of sphere $V = \frac{4}{3}\pi r^3$,

Volume of curve = $\frac{1}{3}\pi r^2 h$

Option (a) is correct.

92. What is the ratio of the volume of the cube to that of the cylinder?

(a) 4 : 3 (b) 21 : 16
(c) 14 : 11 (d) 45 : 32

(*c*) Volume of cube : Volume of cylinder

$$= \left(\frac{2R}{\sqrt{3}}\right)^3 : \pi \left(\frac{R}{\sqrt{3}}\right)^2 \frac{2R}{\sqrt{3}}$$

$$= a^3 : \pi r^2 h$$

$$= \frac{8R^3}{3\sqrt{3}} : \frac{22}{7} \times \frac{2R^3}{3\sqrt{3}} = 14 : 11$$

Option (c) is correct.

93. Consider the following statements :

1. The surface area of the sphere is $\sqrt{5}$ times the curved surface area of the cone.
2. The surface area of the cube is equal to the curved surface area of the cylinder.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

(*d*) (1) Surface area of the sphere is $\sqrt{5}$ times of the curved surface area of cone.

Surface area of the sphere : Curved surface area of cone

$$= 4\pi R^2 : \pi r l \quad [\because l = \sqrt{r^2 + h^2}]$$

$$l = \sqrt{\left(\frac{R}{\sqrt{3}}\right)^2 + \left(\frac{2R}{\sqrt{3}}\right)^2}$$

$$\left[l = \sqrt{\frac{5R^2}{3}} = \frac{\sqrt{5}}{\sqrt{3}}R\right]$$

$$= 4\pi R^2 : \pi \frac{R}{\sqrt{3}} \times \frac{\sqrt{5}}{\sqrt{3}} R = 12 : \sqrt{5}$$

Surface area of sphere is not $\sqrt{5}$ times of the curved surface area.

(2) Surface area of the cube

$$= 6 \times a^2 = 6 \times \frac{2R}{\sqrt{3}} \times \frac{2R}{\sqrt{3}} = 8R^2$$

Curved surface area of the cylinder

$$= 2\pi r h = 2 \times \frac{22}{7} \times \frac{R}{\sqrt{3}} \times \frac{2R}{\sqrt{3}} = \frac{44R^2}{21}$$

The surface area of the cube is not equal to curved surface area of the cylinder.

Then, option (d) is correct.

Directions (Q. Nos. 94-96) *Read the given information carefully and answer the given questions below.*

$ABCD$ is a quadrilateral with $AB = 9$ cm, $BC = 40$ cm, $CD = 28$ cm, $DA = 15$ cm and angle ABC is a right-angle.

94. What is the area of triangle ADC?

(a) 126 cm^2 (b) 124 cm^2
(c) 122 cm^2 (d) 120 cm^2

95. What is the area of quadrilateral $ABCD$?

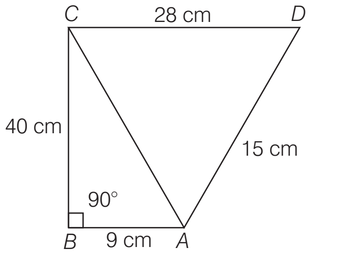
(a) 300 cm^2 (b) 306 cm^2
(c) 312 cm^2 (d) 316 cm^2

96. What is the difference between perimeter of triangle ABC and perimeter of triangle ADC?

(a) 4 cm (b) 5 cm

(c) 6 cm (d) 7 cm

Solution (Q.Nos. 94-96)



94. (*a*) In ΔABC, $\angle B = 90°$

$$(AC)^2 = AB^2 + BC^2$$

$$AC = \sqrt{(9)^2 + (40)^2}$$

$$AC = \sqrt{81 + 1600}$$

$$AC = 41 \text{ cm}$$

Area of ΔADC

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$S = \frac{a+b+c}{2}$$

[a, b and c side of triangle]

$$S = \frac{41 + 15 + 28}{2} = \frac{84}{2} = 42 \text{ cm}$$

$$A = \sqrt{42(42-28)(42-41)(42-15)}$$

$$= \sqrt{42 \times 14 \times 1 \times 27}$$

$$A = 7 \times 3 \times 2 \times 3 = 126 \text{ cm}^2$$

95. (*b*) Area of quadrilateral $ABCD$

= Area of ΔABC + Area of ΔADC

Area of $\Delta ABC = \frac{1}{2} \times 40 \times 9 = 180 \text{ cm}^2$

Area of quadrilateral = 180 + 126

$= 306 \text{ cm}^2$

96. (*c*) Difference between perimeter of ΔABC and ΔADC.

Perimeter of $\Delta ABC = 40 + 41 + 9$

$= 90$ cm

Perimeter of $\Delta ADC = 28 + 41 + 15$

$= 84$ cm

Difference of perimeter of ΔABC

and $\Delta ADC = 90 - 84 = 6$ cm

Directions (Q. Nos. 97 and 98) Read the given information carefully and answer the given question below. Anequilateral triangle ABC is inscribed in a circle of radius $20\sqrt{3}$ cm.

97. What is the length of the side of the triangle?

(a) 30 cm (b) 40 cm

(c) 50 cm (d) 60 cm

(d) Let length of the side of the triangle $= a$

$$\text{Circumradius of triangle} = \frac{a}{\sqrt{3}}$$

$$20\sqrt{3} = \frac{a}{\sqrt{3}}$$

$$a = 60 \text{ cm}$$

Option (d) is correct.

98. The centroid of the triangle ABC is at a distance d from the vertex A. What is d equal to?

(a) 15 cm

(b) 20 cm

(c) $20\sqrt{3}$ cm

(d) $30\sqrt{3}$ cm

(c) Centroid of the triangle ABC is at a distance "d" from the vertex A.

All the centres of triangle at one point in equilateral triangle.

Then, circumradius of circumcircle is equal to "d" distance $= 20\sqrt{3}$ cm

Option (c) is correct.

Directions (Q. Nos. 99 and 100) *Read the given information carefully and answer the given question below.*

The sum of length, breadth and height of a cuboid is 22 cm and the length of its diagonal is 14 cm.

99. What is the surface area of the cuboid?

(a) 288 cm^2

(b) 216 cm^2

(c) 144 cm^2

(d) Cannot be determined due to insufficient data

(*a*) Length of cuboid $= l$

Breadth of cuboid $= b$

Height of cuboid $= h$

$$l + b + h = 22 \text{ cm} \quad \ldots\text{(i)}$$

Length of diagonal is $= 14$ cm

$$\sqrt{l^2 + b^2 + h^2} = 14$$

$$l^2 + b^2 + h^2 = (14)^2$$

$$= 196 \text{ cm} \quad \ldots\text{(ii)}$$

Surface area of cuboid

$= 2(lb + bh + hl)$

Taking square of Eq. (i),

$$(l + b + h)^2 = l^2 + b^2 + h^2 + 2(lb + bh + hl)$$

$$(22)^2 = 196 + 2(lb + bh + hl)$$

$$484 - 196 = 2(lb + bh + hl)$$

$$2(lb + bh + hl) = 288 \text{ cm}^2$$

Option (a) is correct.

100. If S is the sum of the cubes of the dimensions of the cuboid and V is its volume, then what is $(S - 3V)$ equal to?

(a) 572 cm^3

(b) 728 cm^3

(c) 1144 cm^3

(d) None of the above

(*c*) S is the sum of the cube of the dimensions of the cuboid.

$$S = l^3 + b^3 + h^3$$

$$V = lbh$$

$$\therefore (S - 3V) = l^3 + b^3 + h^3 - 3lbh$$

$$[\because (a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2) - (ab + bc + ca)]$$

$$= (l + b + h)[l^2 + b^2 + h^2 - (lb + bh + hl)]$$

$$= 22[196 - 144] = 22 \times 52 \quad \left[\begin{array}{l}\because 2(lb + bh + hl) = 288 \\ lb + bh + hl = 144\end{array}\right]$$

$$= 1144 \text{ cm}^3$$

Option (c) is correct.

PAPER II English

Directions (Q.Nos. 1-10) *Each of the following sentences in this section has a blank space and four words or group of words are given after the sentence. Select the word or group of words you consider the most appropriate for the blank space and indicate your response in the answer sheet accordingly.*

1. How we.........to ageing is a choice we must make wisely?

(a) respond (b) absolve
(c) discharge (d) overlook

(a) 'respond' is correct word to be used for the given blank as respond means to react to something.

2. Complementary medicine .... fewer risks; since it is used alongwith standard remedies, often to lessen side -effects and enhance feelings of well-being.

(a) reacts (b) releases
(c) ejects (d) carries

(d) 'carries' is appropriate word for the given blank. The word means an act of carrying something.

3. Stress many ....fertility, in man and women.

(a) engage (b) reduce
(c) inject (d) deduce

(b) 'reduce' is the aporopriate word for the blank. The word means to become diminished.

4. The football match had to be ....because of the weather.

(a) called on (b) called off
(c) called out (d) called over

(b) 'called off' is the appropriate phrase for the given blank. The phrase means to cancel or abandon something.

5. Nobody believed Ram at first but he ....to be right.

(a) came out (b) carried out
(c) worked out (d) turned out

(d) 'turned out' is the appropriate phrase for the blank as it means to happen is a particular way.

6. How are you......in your new job? Are you enjoying it?

(a) keeping on (b) going on
(c) getting on (d) carrying on

(c) 'getting on' is an appropriate phrase. for the give blank. The word means to have a good relationship.

7. We live.........a tower block. Out apartment is on the fifteenth floor.

(a) at (b) in
(c) over (d) above

(b) Here, preposition 'in' is an appropriate word for the blank, as in is used for an enclosed space.

8. You were going to apply for the job and then you decided not to so what.......?

(a) put you off (b) put you out
(c) turned you off (d) turned you away

(a) 'put you off' is the most appropriate phrase for the given blank. The phrase means to cause someone to lose interest or enthusiasm.

9. ....it was raining, he went out without a raincoat.

(a) Even (b) Since
(c) Unless (d) Although

(d) According to the given sentence, 'Although' is the most appropriate word to be used for the given blank. .

10. I parked my car in a no-parking zone, but I.........it.

(a) came up with (b) got away with
(c) made off with (d) got on with

(b) 'got away with' is the most appropriate phrase to be used for the blank. The phrase means to escape blame punishment or undesirable' consequence for an act that is wrong or mistaken.

Directions (Q.Nos. 11-20) *Each item in this sectioin consists of a sentence with an underlined word followed by four words/group of words. Select the option that is the nearest in meaning to the underlined word and mark your response on your answer sheet accordingly.*

11. A <u>provocative</u> message had been doing rounds on social media to instigate the mob against migrants.

(a) Dexterous (b) Inflammatory
(c) Valiant (d) Prudent

(b) 'Inflammatory' and 'Provocative' are synonyms. Both words mean arousing or intended to arouse angry or violent feelings (especially of speech or writing).

12. The differences include increase in mean temperature and heavy <u>precipitation</u> in several regions.

(a) Drought (b) Oasis
(c) Rainfall (d) Snowing

(c) 'Rainfall' is the correct synonym of the word 'Precipitation'. Precipitation means rain, snow or hail that falls to or condenses on the ground.

13. The portal will help victims and complainants to <u>anonymously</u> report cyber crime.

(a) Incognito (b) Directly
(c) Unfailingly (d) Is Situ

(a) 'Incognito' is the correct synonym of the word 'Anonymously'. Both words mean unknown or having one's true identity concealed.

14. He is suffering from a <u>terminal</u> disease.

(a) Sublunary (b) Terrific
(c) Chronic (d) Incurable

(d) 'Terminal and Incurable' are synonyms. Both words mean predicted to lead to death (of a disease) especially slowly.

15. Doctors are <u>reluctant</u> to take rural postings despite big salary offers.

(a) Disinclined
(b) Eager
(c) Fervent
(d) Unrepentant

(a) 'Disinclined' is synonym of the word 'Reluctant'. Both words mean unenthusiastic on unwilling.

16. The authorities have <u>reprimanded</u> to subordinate officer for violating the protocol.

(a) Extolled
(b) Purported
(c) Admonished
(d) Required an apology

(c) 'Admonished' is the correct synonym of the word 'Reprimanded'. Both words mean meaning is blamed or scolded.

17. For Gandhiji, India's religious and linguistic diversity was an <u>asset</u>, not a liability.

(a) Obligation (b) Advantage
(c) Attribute (d) Reinforcement

(*b*) 'Advantage' is the closest synonym to the word 'Asset'. Both words mean benefit or advantage.

18. How hysterical he is!
(a) Berserk (b) Inconsistent
(c) Duplicitous (d) Insincere.

(*a*) 'Berserk' is the correct synonym of the word 'Hysterical'. Both words mean crazy or out of control with anger excitement.

19. Mahesh is mostly prejudiced in his political opinion.
(a) Objectionable (b) Predatory
(c) Jaundiced (d) Intimate

(*c*) The word 'Jaundiced' means the same as 'Prejudiced'. Both words mean biased or influenced.

20. Do not indulge in tautology.
(a) Truth telling (b) Prolixity
(c) Foretelling
(d) Telepathic conversation

(*b*) The word 'Prolixity' means the same as 'Tautology'. Both words mean a phrase or expression in which the same thing is said twice in different words.

Directions (Q.Nos. 21-30) *each item in this sectioin consists of a sentence with an underlined word followed by four words. Select the option that is opposite in meaning to the undeerlined word and mark your response on your answer sheet accordingly.*

21. His religious views are rather fanatical.
(a) Bigoted (b) Rabid
(c) Moderate (d) Militant

(*c*) The underlined word 'fanatical' means extremely interested in something, to a degree that some people find unreasonable. So, among the given options, 'moderate' would be its correct antonym. Moderate means make or become less extreme interested in something.

22. Religious fundamentalists often consider the followers of other religions to be heretics.
(a) Dissenter (b) Believer
(c) Renegade (d) Apostate

(*b*) The underlined word 'heretics' means a person who belongs to a particular religion but whose beliefs or actions seriously disagree with the principles of that religion. So, among the given options, 'believer' is the correct antonym of the word 'heretics. Believer is an adherent of a particular religion.

23. According to GB Shaw, mean have become inert. Therefore, life force has chosen women to perform its functions.
(a) Lively (b) Quiescent
(c) Dormant (d) Apathetic

(*a*) The underlined word 'inert' means motion less or still. Its correct antonym from the given options would be 'lively' and it means full of energy, moving or active.

24. Some of the men are highly misanthropic.
(a) Anti-social (b) Philosophic
(c) Atrophic (d) Philanthropic

(*d*) The underlined word 'misanthropic' means unsociable or unfriendly'. Its correct antonym would be 'philanthropic' which means humanitarian, liberal, unselfish etc.

25. The teacher was a very profound man.
(a) Sincere (b) Erudite
(c) Scholarly (d) Superficial

(*d*) The underlined word 'profound' means felt or experienced very strongly or in an extreme way. Its correct antonym from the given options is 'superficial' which means concerned only what is obvious or apparent, not deep or serious.

26. His hand-writing is readable.
(a) Well-written
(b) Decipherable
(c) Illegible
(d) Comprehensible

(*c*) The word 'readable' means which can be read clearly. Its correct antonym is 'illegible' which means the writing which is not clear or hard to read.

27. Mohan is his steadfast friends.
(a) Committed (b) Unwavering
(c) Unfaltering (d) Unreliable

(*d*) The word 'steadfast' here means loyal or dedicated. Its correct antonym is 'unreliable' which means untrustworthy or undependable.

28. Radha often goes tempestuous while debating.
(a) Calm (b) Violent
(c) Fierce (d) Vehement

(*a*) The word 'tempestuous' means angry or violent. Its correct antonym is 'calm' which means peaceful.

29. The thief had very vital information to pass on to the police.
(a) Crucial (b) Inessential
(c) Indispensable (d) Fundamental

(*b*) The underlined word 'vital' means absolutely necessary or essential. Its correct antonym is 'inessential' which means not necessary.

30. His lectures are often wordy and pointless.
(a) Diffuse (b) Concise
(c) Garrulous (d) Voluble

(*b*) The underlined word 'wordy' means unnecessary repetition of language. Its antonym is 'concise' which means very brief or to the point.

Directions (Q. Nos. 31-40) *Each of the following items in this section consists of a sentence, the parts of which have been jumbled. these parts have been labelled as P, Q, R and S. given below each sentence are four sequences namely (a), (b), (c) and 9d). You are required to rearrange the jumbled parts ofthe sentence and mark your response accordingly.*

31. the prize money (P) for refusing her (Q)
pepsico was ordered (R)
to compensate the woman (S)
The correct sequence should be
(a) RSQP (b) SPQR
(c) RPSQ (d) QRSP

(*a*) RSQP is the correct sequence.

32. trade operating from a colony (P)
held a meeting (Q)
demanding a probe into the illegal drug (R)
the residents of the city (S)
The correct sequence should be
(a) QRSP (b) SPQR
(c) SQRP (d) RSQP

(*c*) SQRP is the correct sequence.

33. the university authorities cancelled the ongoing students' union election and (P)
following students' unrest on campus (Q)

closed till further orders
(R)
declared the institution
(S)
The correct sequence should be
(a) QRSP (b) QPSR
(c) SQRP (d) RSQP

(*b*) QPSR is the correct sequence.

34. brushed past the latter's pet dog
(P)
stabbed to death by a man
(Q)
after his vehicle accidentally
(R)
a cargo van driver was allegedly
(S)
The correct sequence should be
(a) QRSP (b) QPSR
(c) SQRP (d) SQPR

(*c*) SQRP is the correct sequence.

35. an earthquake and tsunami
(P)
the disaster mitigation agency
(Q)
said that the death toll from
(R)
in Indonesia has crossed 1500
(S)
The correct sequence should be
(a) PQSR (b) RPSQ (c) SQRP (d) QRPS

(*d*) QRPS is the correct sequence.

36. scientists say they have developed a new
(P)
illnesses such as heart disease and cancer
(Q)
DNA tool that uses machine learning to accurately
(R)
predict people's height and assess their risk for serious
(S)
The correct sequence should be
(a) PRSQ (b) RPSQ
(c) PSRQ (d) QRPS

(*a*) PRSQ is the correct sequence.

37. a rare evergreen tree in the Southern-Western Ghats
(P)
researchers have found that
(Q)
common white-footed ants are the best pollinators of
(R)
bees might be the best known pollinators but
(S)
The correct sequence should be
(a) PRSQ (b) SQRP
(c) QSRP (d) PQRS

(*b*) SQRP is the correct sequence.

38. say from their forties onwards
(P)
it is thus a good idea
(Q)
and continue to exercise early enough
(R)
for senior citizens to start
(S)
The correct sequence should be
(a) PRSQ (b) QRSP
(c) QSRP (d) PQRS

(*c*) QSRP is the correct sequence.

39. scientists have determined
(P)
injury in animals and humans
(Q)
that is linked to the severity of spinal cord
(R)
a gene signature
(S)
The correct sequence should be
(a) PSRQ (b) QRPS
(c) QSPR (d) PQRS

(*a*) PSRQ is the correct sequence.

40. like a muscle and repeating the process
(P)
and stable reading circuit
(Q)
helps the child build a strong
(R)
the brain works
(S)
The correct sequence should be
(a) QSRP (b) SPRQ
(c) QSPR (d) RQPS

(*b*) SPRQ is the correct sequence.

Directions (Q. Nos. 41-50) *In this section each item consists of six sentences of a passage. The first and sixth sentences are given in the begining as S1 and S6. The middle four sentences in each have been jumbled up and labelled as P, Q, R and S you are required to find the proper sequence of the four sentences and mark your response accordingly on the answer sheet.*

41. S_1: He no longer dreamed of storms, nor of women, nor of great occurrences, nor of great fish, nor fights, nor contests of strength, nor of his wife.
S_6 : He urinated outside the shack and then went up the road to wake the boy.
P : He never dreamed about the boy.
Q : He only dreamed of places and of the lions on the beach now.
R : He simply woke, looked out through the open door at the moon and unrolled his trousers and put them on.
S : They played like young cats in the dusk and he loved them as he loved the boy.
The correct sequence should be
(a) RQPS (b) SRQP
(c) QSPR (d) RRSQ

(*c*) QSPR is the correct sequence.

42. S_1: We do not know, after 60 years of education, how to protect ourselves against epidemics like cholera and plague.
S_6 : This is the disastrous result of the system under which we are educated.
P : If our doctors could have started learning medicine at an earlier age, they would not make such as poor show as they do.
Q : I have seen hundreds of homes. I cannot say that I have found any evidence in them of knowledge of hygiene.
R : I consider it a very serious blot on the state of our education that our doctors have not found it possible to eradicate these diseases.
S : I have the greatest doubt whether our graduates know what one should do in case one is bitten by a snake.

The correct sequence should be
(a) RQSP (b) PRQS
(c) QRPS (d) PQSR

(c) QRPS is the correct sequence.

43. S_1: The weak have no place here, in this life or in any other life. Weakness leads to slavery.

S_6 : This is the great fact : strength is life weakness is death. Strength is felicity, life eternal; immortal; weakness is constant strain and misery; weakness is death.

P : They dare not approach us, they have no power to get a hold on us, until the mind is weakened.

Q : Weakness leads to all kinds of misery, physical and mental. Weakness is death.

R : But they cannot harm us unless we become weak, until the body is ready and predisposed to receive them.

S : There are hundreds of thousands of microbes surrounding us.

The correct sequence should be
(a) PQRS (b) PRQS
(c) QRSP (d) QSRP

(d) QSRP is the correct sequence.

44. S_1: The Nobel Prize for Economics in 2018 was awarded to Paul Romer and William Nordhaus for their work in two separate areas : economic growth and environmental economics respectively.

S_6 : Among recent winners of Nobel Prize in Economics, it's hard to think of one issue which is more topical and relevant to India.

P : But there is a common thread in the work.

Q : In economic jargon it's termed as externality.

R : Productive activity often has spillovers, meaning that is can impact and unrelated party.

S : Romer and Nordhaus both studied the impact of externalities and came up with profound insights and economic models.

The correct sequence should be
(a) PQRS (b) PRQS
(c) QSPR (d) QSRP

(b) PRQS is the correct sequence.

45. S_1: India's museums tend to be dreary experiences.

S_6 : Because it's better to attract crowds than dust.

P : Even the Louvre that attracted an eye-popping 8.1 million visitors last year compared to India's 10.18 million foreign tourists, has hooked up with Beyonce and Jay-Z for promotion, where they take a selfie with Mona Lisa.

Q : Our museums need to get cool too.

R : A change of approach is clearly called for.

S : Troops of restless schoolchildren are often the most frequent visitors, endlessly being told to lower their voices and not touch ther art.

The correct sequence should be
(a) PQRS (b) PRSQ
(c) SRPQ (d) QSRP

(c) SRPQ is the correct sequence.

46. S_1: A decade ago UN recognised that rape can constitute a war crime and a constitutive act of genocide.

S_6 : The fact that these two peace laureates come from two different nations underlines that this problem has been widespread, from Rwanda to Myanmar.

P : This year's Nobel Peace Prize has been awarded to two exceptional individuals for their fight to end the use of sexual violence as the weapon of war.

Q : Denis Mukwege is a doctor who has spent decades treating rape survivors in the Democratic Republic of Congo, where along civil war has repeatedly witnessed the horror of mass rapes.

R : Nadia Murad is herself a survivor of sexual war crimes, perpetuated by IS against the Yazidis.

S : Today she campaigns tirelessly to put those IS leaders in the dock in international courts.

The correct sequence should be
(a) PQRS (b) PRQS
(c) SRQP (d) QRSP

(a) PQRS is the correct sequence.

47. S_1: Few scientists manage to break down the walls of the so-called ivory tower of academia and touch and inspire people who may not otherwise be interested in science.

S_6 : Not many would have 6 survived this, let alone excelled in the manner he did.

P : Stephen Hawking was one of these few.

Q : Around this time he was diagnosed with Amyotrophic Lateral Sclerosis, an incurable motor neuron disease and given two years to live.

R : Judging by the odds he faced as a young graduated student of physics at Cambridge University, nothing could have been a more remote possibility.

S : When he was about 20 years old, he got the shattering news that he could not work with the great Fred Hoyle for his PhD, as he had aspired to.

The correct sequence should be
(a) PQSR (b) PRQS
(c) SRPQ (d) PRSQ

(d) PRSQ is the correct sequence.

48. S_1: The climate question presents a leapfrog era for India's development paradigm.

S_6 : This presents a good template for India, building on its existing plans to introduce electric mobility through buses first and cars by 2030.

P : It is aimed at achieving a shift to sustainable fuels, getting cities to commit to eco-friendly mobility and delivering more walkable communities, all of which will improve the quality of urban life.

Q : At the Bonn conference, a new Transport Decerbonisation Alliance has been declared.

R : This has to be resolutely pursued, breaking down the barriers to wider adoption of rooftop solar energy at every level and implementing net metering systems for all categories of consumers.

S : Already, the country has chalked out an ambitious policy on renewable energy, hoping to generate 175 gigawatts of power from green sources by 2022.

The correct sequence should be

(a) SRQP (b) SPRQ
(c) PRSQ (d) QRSP

(a) SRQP is the correct sequence.

49. S_1: The dawn of the information age opened up great opportunities for the beneficial use of data.

S_6 : To some, in this era of big data analytics and automated, algorithm based processing of zettabytes of information, the fear that their personal data may be unprotected may conjure up visions of a dystopian world in which individual liberties are compromised.

P : But it is the conflict between the massive scope for progress provided by the digital era and the fear of loss of individual autonomy that is foregrounded in any debates about data protection laws.

Q : It also enhanced that perils of unregulated and arbitrary use of personal data.

R : It is against this backdrop that the white paper made public to elicit views from the public on the shape and substance of a comprehensive data protection laws assumes significance.

S : Unauthorised leaks, backing and other cyber crimes have rendered data bases vulnerable.

The correct sequence should be

(a) SQRP (b) QPRS
(c) SRPQ (d) QSPR

(d) QSPR is the correct sequence.

50. S_1: In a globalised world, no country can hope to impose tariffs without affecting its own economic interests.

S_6 : The ongoing trade war also threatens the rules-based global trade order which has managed to amicably handle trade disputes between countries for decades.

P : So both the US and China, which have blamed each other for the ongoing trade war, are doing no good to their own economic fortunes by engaging in this tit-for-tat tariff battle.

Q : Apart from disadvantaging its consumers, who will have to pay higher prices for certain goods, tariffs will also disrupt the supply chain of producers who rely on foreign imports.

R : China, which is fighting an economic slowdown, will be equally affected.

S : The minutes of the US Federal Reserve June policy meeting show that economic uncertainty due to the trade war is already affecting private investment in the US, with many investors deciding to scale back or delay their investment plans.

The correct sequence should be

(a) SQPR (b) QPSR
(c) QRPS (d) PSRQ

(b) QPSR is the correct sequence.

Directions (Q. Nos. 51-70) *In this section you have few short passages. After each passage, you will find some items based on the passage. First, read a passage and answer the items based on it you are required to select your answer based on the contents of the passage and opinion of the author only.*

PASSAGE 1

From 1600 to 1757 the East India Company's role in India was that of a trading corporation which brought goods or precious metals into India and exchanged them for Indian goods like textiles and spices, which it sold abroad. Its profits came primarily from the sale of Indian goods abroad. Naturally, it tried constantly to open new markets for Indian goods in Britain and other countries.

Thereby, it increased the export of Indian manufacturers and thus encouraged their production. This is the reason why Indian rulers tolerated and even encouraged the establishment of the Company's factories in India. But, from the very beginning, the British manufacturers were jealous of the popularity that India textiles enjoyed in Britain. All of a sudden, dress fashions changed and light cotton textiles began to replace the coarse woolens of the English. Before, the author of the famous novel, Robinson Crusoe, complained that Indian cloth had "crept into houses, our closets and bed chambers curtains, cushions, chairs and at last beds themselves were nothing but calicos or India stuffs".

The British manufacturers put pressure on their government, to restrict and prohibit the sale of Indian goods in England. By 1720, laws had been passed forbidding the wear or use of printed or dyed cotton cloth. In 1760 a lady had to pay a fine of 200 for possessing an imported handkerchief ! Moreover, heavy duties were imposed on the import of plain cloth. Other European countries, except Holland, also either prohibited the import of Indian cloth or imposed heavy import duties. In spite of these laws, however, Indian silk and cotton textiles still held their own in foreign markets, until the middle of the eighteenth century when the English textile industry began to develop on the basis of new and advanced technology.

51. The East India Company was encouraging the export of Indian manufacturers because

(a) it was a philanthropic trading corporation
(b) it wanted Indian manufacturers to prosper in trade and commerce
(c) it profited from the sale of Indian goods in foreign markets
(d) it feared Indian Kings who would not permit them trade in India

(c) East India Company was encouraging the export of Indian goods as it got huge profits by the sale of Indian goods in foreign markets.

52. The people of England used Indian cloths because

(a) they loved foreign and imported clothes
(b) the Indian textile was light cotton
(c) the Indian cloths were cheaper
(d) the Indian cloths could be easily transported

(*b*) The British people used Indian clothes because the clothes were light cotton and of breathable fabric.

53. What did the British manufacturer do to compete with the Indian manufacturers?

(a) They pressurised the government to levy heavy duties on export of Indian clothes
(b) They pressurised the government to levy heavy duties on import of Indian clothes
(c) They requested people to change their fashion preferences
(d) They lowered the prices of the Britain made textile

(*b*) The British manufacturers were jealous of Indian manufacturers and they pressurised the British Government to levy heavy duties on import of Indian clothes.

54. Which source is cited by the author to argue that Indian textile was in huge demand in 18th century England?

(a) The archival source
(b) The scientific source
(c) The journalistic source
(d) The literary source

(*d*) The literary source, novel 'Robinson Crusoe' is cited by the author to argue that Indian textile was in huge demand in 18th century England.

55. "New and advanced technology" in the paragraph refers to

(a) the French Revolution
(b) the Glorious Revolution of England
(c) the Industrial Revolution
(d) the Beginning of Colonialism

(*c*) New and Advanced technology in the paragraph refers to the Industrial Revolution.

PASSAGE 2

Zimbabwe's prolonged political crisis reached the boiling point earlier this month when President Robert Mugabe dismissed the Vice-President, Emmerson Mnangagwa. A battle to succeed the 93-years-old liberation hero-turned. President had already been brewing within the ruling Zimbabwe African National Union-Patriotic Front (Zanu-PF), with the old guard backing Mr Mnangagwa, himself a freedom fighter and 'Generation 40', a grouping of younger leaders supporting Mr Mugabe's 52-years-old wife, Grace. Ms Mugabe, known for her extravagant lifestyle and interfering ways, has been vocal in recent months about her political ambitions. 'Mr Mugabe was seen to have endorsed her when on November 6 he dismissed Mr Mnangagwa. But Mr Mugabe, who has ruled Zimbabwe since its independence in 1980, erred on two counts: he underestimated the deep connections Mr Mnangagwa has within the establishment and overestimated his own power in a system he has helped shape. In the good old days, Mr Mugabe was able to rule with an iron grip. But those days are gone. Age and health problems have weakened his hold on power, while there is a groundswell of anger among the public over economic mismanagement. So, when he turned against a man long seen by the establishment as his successor, Mr Mugabe left little doubt that he was acting from a position of political, weakness. This gave the security forces the confidence to turn against him and make it clear they didn't want a Mugabe dynasty. The military doesn't want to call its action a coup detat, for obvious reasons.

A coup would attract international condemnation, even sanctions. But it is certain that the army chief, Gen Constantino Chiwenga, is in charge, His plan, as it emerges, is to force Mr Mugabe to resign and install a transitional government, perhaps under Mr Mnangagwa, until elections are held.

56. In the paragraph, who has been called liberation hero?

(a) Constantino Chiwenga
(b) Emmerson Mnangagwa
(c) Robert Mugabe
(d) Army Chief

(*c*) As stated in the passage 93-years-old Robert Mugabe. the President of Zimbabwe has been called a liberation hero.

57. Mrs Mugabe is supported by

(a) Mr Mnangagwa
(b) Mr Mugabe
(c) Generation 40
(d) Zanu-PF

(*c*) As stated in the passage Mrs Mugabe was supported by Generation 40, a group of younger leaders.

58. Mr Mugabe's political weakness became apparent when

(a) he endorsed his wife
(b) he turned against the army
(c) he suffered from health issues
(d) he dismissed Mr Mnangagwa

(*d*) Mr Mugabe's political weakness became apparent when he dismissed Mr Mnangagwa, who was the Vice-President of Zimbabwe.

59. The secuirty forces of Zimbabwe staged a coup against the President because

(a) they wanted Mrs Mugabe as the President
(b) they were aware of Mugabe's failing wealth
(c) they disliked Mugabe's extravagant lifestyle
(d) they did not want a Mugabe dynasty

(*d*) The security forces of Zimbabwe staged a coup against the President because they did not want a Mugabe's dynasty.

60. Why does the military not want to call it a coup d'etat?

(a) Because coup is immoral
(b) Because coup is illegal
(c) Because coup would lead to international censure and sanctions
(d) Because it would make the public revolt

(*c*) The military does not want to call it a coup d'etat as coup would lead to international censure and sanctions.

PASSAGE 3

Over-eating is one of the most wonderful practices among those who think that they can afford it. In fact, authorities say that nearly all who can get as much as they desire, over-eat to their disadvantage. This class of people could save a great more food then they can save by missing one meal per week and at the same time they could improve their health. A heavy meal at night, the so-called 'dinner', is the fashion with many and often is taken shortly before retiring. It is unnecessary and could be forgone, not only once a week but daily without loss of strength. From three to five hours are needed to digest food. While sleeping, this food not being required to give energy for work, is in many cases converted into excess fat, giving rise to over-weight. The evening meal should be light, taken three or four hours before retiring. This prevents over-eating, conserves energy and reduces the cost of food.

61. Why should those who over-eat refrain from doing so?
(a) Because over-eating leads to loss of wealth
(b) Because over-eating is bad for health
(c) Because over-eating conserves food
(d) Because over-eating is immoral and unhealthy

(b) Over-eating should be refrained as it is bad for health and a waste of food.

62. Over-eating is more prevalent among
(a) the rich
(b) the poor
(c) everybody
(d) the bourgeoisie

(a) Actually over-eating is more common among those who can afford the food.

63. The writer is asking the readers
(a) to skip the heavy dinner and take light evening meal instead
(b) to stop eating anything at night
(c) to take food only during the day
(d) to eat food before the sunset

(a) The writer is asking the readers to skip the heavy dinner just before retiring.They should, instead take light evening meals.

64. What is the most appropriate time for having evening meal?
(a) An hour after the sunset
(b) Three or four hours before sleeping
(c) Before the sunset
(d) Just before sleeping

(b) Evening meal, ideally, should be taken three or four hours before sleeping.

65. According to the passage, how many times a day should we have food?
(a) Three times
(b) Two times
(c) Once
(d) Has not been specified

(d) It has not been specified in the passage as how many times a day should we have food?

66. According to the passage, people over-eat
(a) because they can afford to
(b) because they are hungry
(c) because they have to work more
(d) because they have to conserve energy

(a) According to the passage, people over eat because they can afford and can get as much as they desire generally over-eat the food.

PASSAGE 4

Much has been said of the common ground of religious unity. I am not going just now to venture my own theory. But if anyone here hopes that this unity will come by the triumph of anyone of the religions and the destruction of the others, to him I say, "Brother, yours' is an impossible hope." Do I wish that the Christian would become Hindu? God forbid. Do I wish that the Hindu or Buddhist would become Christian? God forbid.

The seed is put in the ground and earth and air and water are placed around it. Does the seed become the earth, or the air, or the water? No. It becomes a plant.

It develops after the law of its own growth, assimilates the air, the earth and the water, converts them into plant substance and grows into a plant.

Similar is the case with religion. The Christian is not to become a Hindu or a Buddhist, nor a Hindu or a Buddhist to become a Christian. But each must assimilate the sprit of the others and yet preserve his individuality and grow according to his own law of growth.

If the Parliament of Religions has shown anything to the world, it is this: it has proved to the world that holiness, purity and charity are not the exclusive possessions of any church in the world and that every system has produced men and women of the most exalted character.

In the face of this evidence, if anybody dreams of the exclusive survival of his own religion and the destruction of the others, I pity him from the bottom of my heart and point out to him that upon the banner of every religion will soon be written in spite of resistance: 'Help and not fight,' 'Assimilation and not Destruction,' 'Harmony and Peace and not Dissension.'

67. According to the author of the passage, people should
(a) change their religions
(b) follow their religions and persuade others to follow it
(c) follow their own religions and respect other religions
(d) disrespect other religions

(c) According to the passage, people should follow their own religions and respect other religions also.

68. The Parliament of Religions is
(a) a Christian organisation
(b) a Buddhist organisation
(c) a Hindu organisation
(d) a platform for discussion about every religion of the world

(d) It is a platform for discussion about every religion of the world.

69. What does the author think about those who dream about the exclusive survival of their own religions and the destruction of the others?
(a) He hates them
(b) He desires to imprison them
(c) He pities them
(d) He praises them

(c) There is nothing like 'My Religions' only. The author takes a pity on those people who dream about the exclusive survival of their own religions and the destruction of the others. All religions are same and all basics of any religion are also the same.

70. According to the passage, what is 'impossible hope'?
(a) One day, all the people of the world will follow only one religion
(b) One day, there will be no religion
(c) Purity and charity are the exclusive possessions
(d) Banner of every religion will soon be written

(a) According to the passage, the impo- ssible hope is that one day, all the people of the world will follow only one religion.

Directions (Q. Nos. 71-90) *Each item in this section has a sentence with three parts labelled as (a), (b) and (c). Read each sentence to find out whether there is any error in any part and indicate you response on the answer sheet against the corresponding letter i.e., (a) or (b) or (c). If you find no error, your response should be indicated as (d)*

71. Except for few days(a)/in a year during the monsoon(b)/ the river cannot flow on its own.(c)/ No error.(d)

(b) There is no error, sentence is grammatically correct.

72. Being apprised with our approach (a)/ the whole neighbourhood (b)/ came out to meet the minister.(c)/ No error(d)

(a) Here, in part (a) 'of' should be used in place of 'with' because the word 'apprised' is followed by preposition 'of'.

73. The celebrated grammarian Patanjali(a)/ was(b)/ a contemporary to Pushyamitra Sunga.(c)/No error(d)

(c) Here, remove 'to' and use 'of' before pushyamitra to make the given sentence grammatically correct.

74. His appeal for funds(a)/ met(b)/ a poor responses. (c)/ No error (d)/

(b) Here, in part(b) 'met with' should be used in place of 'met' to make the given sentence grammatically correct.

75. Buddhism teaches that(a)/ freedom from desires(b)/ will lead to escape suffering. (c)/ No error (d)

(c) Write 'from' between escape and suffering to make the given sentence grammatically correct.

76. This hardly won liberty(a)/ was not to be (b)/ lightly abandoned. (c)/ No error (d)

(a) Here use of adverb hardly is incorrect. It should be 'This hard' won liberty to make the sentence grammatically correct.

77. My friend said(a)/ he never remembered(b)/ having read a more enjoyable book.(c)/ No error(d)/

(d) No error, the sentence is grammatically correct.

78. With a population of over one billion, (a)/ India is second most popular country(b)/ in the world ofter China.(c)/ No error(d)

(b) Here, article 'the' should be used before second as 'most' is the superlative degree word. Before superlative degree, use of 'the' is must.

79. There are hundred of superstitions(a)/ which survive(b)/ in the various parts of the country. (c)/ No error. (d)

(d) There is no error in the sentence as it is grammatically correct sentence.

80. It is(a)/ in the temperate countries of Northern Europe(b)/ that the beneficial effects of cold is most clearly manifest.(c)/ No error(d)

(c) With plural noun 'effects' verb 'are' should be used in place of 'is' to make the given sentence grammatically correct.

81. The effects of female employment(a)/ on gender equality(b)/ now appear to be trickling at the next generation.(c)/No error(d)

(c) Here in part (c) preposition 'to' is used in place of 'at' after the word 'trickling' to make the given sentence correct.

82. Since the 15 minutes that she drives,(a)/ she confesses that she feels like(b)/ a woman with wings.(c)/ No error(d)

(a) Remove 'since' and use 'in' here. Use of since is incorrect according to the given syntax.

83. India won(a)/ by an innings(b)/ and three runs.(c)/No error(d)

(d) The sentence has no error .

84. Each one(a)/ of these chairs(b)/ are broken.(c)/ No error(d)

(c) Here, in part (c) singular verb 'is' should be used in place of 'are' because the subject of the sentence (Each one) is singular, so verb also be used in singular form.

85. Few creature(a)/ outwit(b)/ the fox in Aesop's Fables.(c)/ No error(d)

(a) Here, in part(a) 'Few creatures' should be used in place of 'Few creature' to make the given syntax correct.

86. Anywhere in the world(a)/ when there is conflict(b)/ women and children suffer the mos.(c)/No error(d)

(b) For place, 'where' should be used instead of 'when'. So, here in part (b) replace 'when' with 'where' to make the given sentence grammatically correct.

87. The man is(a)/ the foundational director (b)/ of this company.(c)/ No error(d)

(b) Here, in part (b) of the given sentence, article 'a' will be used in place of 'the' before 'foundational' to make the sentence more appropriate.

88. Parents of LGBT community members(a)/ are coming in(b)/ with a little help from NGOs.(c)/ No error(d)

(b) Here, 'coming in' should be replaced with 'coming up' to make the given syntax correct.

89. To love one art form is great(a)/ but to be able to appreciate another(b)/ and find lateral connections are priceless.(c)/No error(d)

(d) There is no error and the sentence is grammatically correct.

90. Female literacy rate has gone up by11%(a) in the past decade as opposed to(b)/ a 3% increase in male literacy.(c)/No error(d)

(c) Write 'literacy rate' in place of 'literacy' to make the sentence meaningfully and grammatically correct.

Directions (Q. Nos. 91-100) *Each of the following sentences in this section has a blank space with four words or group of words given. Select whichever word or group of words you consider most appropriate for the blank space and indicate your response on the answer sheet accordingly.*

CLOZE COMPREHENSION 1

The question whether war is ever justified, and if so under what circumstances, is one which has been forcing itself ....

91. (a) upon (b) on
(c) at (d) over

the attention of all thoughtful men. On this question I find myself in the somewhat ....

92. (a) delightful position of
(b) painful
(c) pleasant
(d) lovely

holding that no single one of the combatants is justified in the present war, while not taking the extreme Tolstoyan view that war is under all circumstances a ....

93. (a) duty (b) obligation
(c) responsibility (d) crime

Opinions on such a subject as war are the outcome of ....

94. (a) feeling (b) sentiment
(c) reason (d) patriotism

rather than of thought : given a man's emotional temperament, his convictions, ...

95. (a) however (b) as well as
(c) both (d) despite

on war in general and on any particular war which may occur during his lifetime, can be ....

96. (a) thought (b) intimated
(c) suggested (d) held

with tolerable certainty. The arguments used will be mere reinforcements to convictions otherwise reached. The fundamental facts in this as in all ethical ....

97. (a) questions (b) answer
(c) statements (d) experiences

are feelings; all that thought can do is to clarify and systematise the expression of those feelings and it is such clarifying and systematising of my own feelings that I wish to ....

98. (a) engage (b) praise
(c) attempt (d) commend

in the present article. In fact, the question of rights and wrongs of a particular war is generally ....

99. (a) considered
(b) observed
(c) transferred
(d) opined

from a juridical or quasi-juridical ....

100. (a) possibility.
(b) formula.
(c) force.
(d) standpoint

Solutions (Q.Nos. 91-100)

91. (a) With the word 'itself', preposition 'upon' is the suitable choice.

92. (b) In the given context, 'painful' is the appropriate word to fill the given blank.

93. (d) With respect to the word 'war', option (d) 'crime' is the appropriate choice.

94. (a) 'Feeling' is the appropriate word to fill the given blank according to given context. 'Feeling' is an emotional state or reaction.

95. (c) Here, conjunction 'both' will be used to fill the given blank as 'both …… and' is a conjunction pair used to join one statement or fact to another.

96. (c) Here, 'suggested' is the appropriate word to fill the given blank as it means 'put-forward for consideration'.

97. (a) With the word 'ethical', option (a) 'questions' is the suitable choice. 'Ethical questions' are those questions that involve consideration of conflicting moral choices and dilemmas, with several alternative solutions.

98. (c) In the given context, option (a) 'considered' is the appropriate choice.

99. (a) In the given context, 'considered' is the appropriate word to fill the given blank. 'Considered' means 'having been thought about carefully'.

100. (d) With the word 'juridical or quasi-juridical' option (d) 'stand point' is the appropriate choice. The word 'stand point' means 'a set of beliefs and ideas from which opinions and decisions are formed'.

CLOZE COMPREHENSION 2

The Nobel Prize for Chemistry this year is a tribute to the power of ....

101. (a) evolution
(b) devolution.
(c) revolution.
(d) involution.

The laureates harnessed evolution and used it in the ....

102. (a) microscope
(b) field (c) market
(d) laboratory

with amazing results. Frances H Arnold, an American who was given one-half of the prize, used 'directed evolution' to ....

103. (a) inhibit
(b) synthesize
(c) hamper
(d) hold back

variants of naturally occurring enzymes that could be used to ....

104. (a) constitute
(b) sink
(c) manufacture
(d) resolve

biofuels and pharmaceuticals. The other half went to George P Smith, also of the US and Sir Gregory P Winter, from the UK, who evolved antibodies to ....

105. (a) combat
(b) support
(c) observe
(d) invite

autoimmune diseases and even metastatic cancer through a process called phage display.

Solutions (Q.Nos. 101-105)

101. (a) The whole passage is about evolution and its scientific usage. So, option (a) 'evolution' is the correct word to fill the given blank. 'Evolution' is the gradual development of something over a period of time.

102. (d) Experiments are done is the laboratory to get amazing results. So, option (d) 'laboratory' is the suitable choice in the given context.

103. (b) The word 'synthesize' means 'to produce a substance by combining other substances chemically'. Hence, it is best suited for the given blank in the given context.

104. (c) 'Biofuels and pharmaceuticals' are manufactured with the help of naturally occurring enzymes. So, option (c) 'manufacture' is the suitable choice in the given context.

105. (a) In context of autoimmune diseases, 'combat' is the correct word to fill the given blank as it means 'to take action to reduce or prevent something bad or undesirable'.

Directions *(Q. Nos. 106-120) given below are some idioms/phrases followed by four alternatives meaning to each. choose the response (a), (b), (c) or (d) which is the most appropriate expression.*

106. A match made in heaven
(a) a marriage that is solemnised formally
(b) a marriage that is unsuccessful
(c) a marriage that is likely to be happy and successful
(d) a marriage of convenience

(c) The idiom 'A match made in heaven' means 'a very successful marriage as

both partners are compatible with each other'. So, option (c) express the correct meaning of given idiom.

107. A culture vulture

(a) someone who is very keen to experience art and literature
(b) someone who wants to defend ancient culture
(c) someone who is ashamed of one's own culture
(d) someone who looks at her/his culture critically

(a) Idiom 'A culture vulture' means 'a person who is very keen to experience art and literature'. So, option (a) express the correct meaning of given idiom.

108. A death blow

(a) to be nearly dead
(b) to be deeply afraid of death
(c) to beat someone to death
(d) an action or event which causes something to end or fail

(d) The idiom 'A death blow' means 'an action or event which causes something to end or fail'. So, option (d) express the correct meaning of given idiom.

109. The jewel in the crown

(a) someone who has many skills
(b) something that one wants
(c) the most valuable thing in a group of things
(d) the jewel in the crown of the king

(c) Idiom 'The jewel in the crown' means 'the most valuable thing in a group of similar things'. So, option (c) express the correct meaning of given idiom.

110. To live in a fool's paradise

(a) to live a life that is dishonest
(b) to be happy because you will not accept how bad a situation really is
(c) to believe that things you want will happen
(d) to enjoy yourself by spending a lot of money

(b) Idiom 'To live in a fool's paradise' means 'to be happy because one do not know or will not accept how bad a situation really is'. So, option (b) express the correct meaning of given idiom.

111. A rotten apple

(a) to remove something which is rotten
(b) one bad person in a group of good people
(c) a loving and kind person
(d) a disorganised person with bad habits

(b) The idiom 'A rotten apple' means 'one bad person in a group of good people'. So, option (b) express the correct meaning of given idiom.

112. To vote with your feet

(a) to show that you do not support something
(b) to replace something important
(c) to change something you must do
(d) to express a particular opinion

(a) Idiom 'To vote with your feet' means 'to show that you do not support something'. So, option (a) express the correct meaning of given idiom.

113. Verbal diarrhea

(a) to be sick
(b) to talk to much
(c) to be in a difficult situation
(d) to be a good orator

(b) The idiom 'Verbal diarrhea' means 'to talk to much'. So, option (b) express the correct meaning of given idiom.

114. To sail close to the wind

(a) to pretend to be something that you are not
(b) to be in some unpleasant situation
(c) to be destroyed by a belief
(d) to do something that is dangerous

(d) The idiom 'To sail close to the wind' means 'to do something that is dangerous or that may be illegal or improper'. So, option (d) express the correct meaning of given idiom.

115. A double entendre

(a) to look at someone or something twice
(b) a situation in which you cannot succeed
(c) a word which has two meanings
(d) something that causes both advantages and problems

(c) The idiom 'A double entendre' means 'a word or phrase that has two meanings'. So, option (c) express the correct meaning of given idiom.

116. To cut your own throat

(a) to stop doing something
(b) to do something because you are angry
(c) to behave in a relaxed manner
(d) to allow someone to do something

(b) The idiom 'To cut you own throat' means 'to do something which may cause problem for oneself'. So, option (b) express the correct meaning of given idiom.

117. Cook the books

(a) to record false information in the accounts of an organisation
(b) to do something that spoils someone's plan
(c) to tell a false story
(d) to be very angry

(a) The idiom 'Cook the books' means 'to record false information in the accounts of an organisation'. So, option (a) express the correct meaning of given idiom.

118. Change your tune

(a) to listen to good music
(b) to do things that you are not willing to
(c) to change your opinion completely because it will bring you an advantage
(d) to pretend to be very friendly

(c) The idiom 'Change your tune' means 'to change one's opinion completely as it will bring you an advantage'. So, option (c) express the correct meaning of given idiom.

119. Blue blood

(a) to swallow poison
(b) to be overly interested in someone
(c) to suddenly become jealous
(d) to belong to a family of the highest social class

(d) The idiom 'Blue blood' means 'a member of a wealthy upper class family or ancestry'. So, option (d) express the correct meaning of given idiom.

120. Cut the crap

(a) an impolite way of telling someone to stop saying things that are not true
(b) to stop needing someone else to look after you
(c) to talk about something important
(d) to upset someone by criticising them

(a) The idiom 'Cut the crap' means 'a rude way of telling someone to stop saying things that are not true or not important'. So, option (a) express the correct meaning of given idiom.

PAPER III General Studies

1. Henry T Colebrooke was a Professor of Sanskrit in which one of the following institutions?

(a) Fort William College
(b) Serampore Mission
(c) Kashi Vidyapith
(d) Asiatic Society

(a) Henry Thomas Colebrooke was a Sanskrit scholar and orientalist. He was appointed to a writership in India during that time he learnt Sanskrit.

He was a Professor of Sanskrit at college of Fort William College in 1805. He was elected as President of the Asiatic Society of Calcutta. He also translated two treatises, the Mitacshara of Vijnaneshwara and the Dayabhaga of Jimutavahana in English.

2. The Deccan Agriculturalists' Relief Act of 1879 was enacted with which one of the following objectives?

(a) Restore lands to the dispossessed peasants
(b) Ensure financial assistance to peasants during social and religious occasions
(c) Restrict the sale of land for indebtedness to outsiders
(d) Give legal aid to insolvent peasants

(d) Deccan Agriculturalists Relief Act in 1879 was enacted with the objective to give legal aid to insolvent peasants.

In Deccan region of Maharashtra, money lenders from Gujarat exploited peasants, usurped their land.

This led to large scale riots in 1875.

3. The Damin-i-Koh was created by the British government to settle which one of the following communities?

(a) Santhal (b) Mundas
(c) Oraons (d) Saoras

(a) Damin-i-Koh was the name given to the forested hilly areas of Rajmahal hills (Present day Jharkhand), inhabited by Santhal tribe. The Britishers intruded in their areas and interfered in their socio-economic life, thus, restricting their movement in forest.

All atrocities on Santhal's resulted into large scale rebellion in 1855. Later, Santhal Pargana was constituted by the Britishers.

4. The Limitation Law, which was passed by the British in 1859, addressed which one of the following issues?

(a) Loan bonds would not have any legal validity
(b) Loan bonds signed between moneylender and ryots would have validity only for three years
(c) Land bonds could not be executed by moneylenders
(d) Loan bonds would have validity for ten years

(b) The Limitation Law was passed by the British in 1859 with an objective that loan bond signed between money lenders and ryots would have validity only for three years.

It took final shape when government of India enacted Limitation Act in 1963.

5. Who among the following was known during the days of the Revolt of 1857, as 'Danka Shah'?

(a) Shah Mal
(b) Maulvi Ahmadullah Shah
(c) Nana Sahib
(d) Tantia Tope

(b) Maulvi Ahmadullah Shah was known as 'Danka Shah', was one of the leading figures of the great Revolt of 1857.

He gave tough fight to the Britishers in the Awadh region of Uttar Pradesh, even liberated Faizabad from British rule.

He was symbol of Hindu-Muslim unity and fought until his death with the hand of British agent in 1858.

6. The Summary Settlement of 1856 was based on which one of the following assumptions?

(a) The Talukdars were the rightful owners of the land
(b) The Talukdars were interlopers with no permanent stakes in the land
(c) The Talukdars could evict the peasants from the lands
(d) The Talukdars would take a portion of the revenue which flowed to the State

(b) The Settlement Act of 1856 was introduced under the guidance of Lord Dalhousie in Awadh region, this settlement was not in favour of Talukdar as it gave priority to village Zamindar.

As a result of this act, Talukdars were left as a interlopers with no permanent stake in the land.

7. The Inter-State Council was set-up in 1990 on the recommendation of

(a) Punchhi Commission
(b) Sarkaria Commission
(c) Rajamannar Commission
(d) Mungerilal Commission

(b) Article-263 in Part XI of the Constitution provides for the establishment of an Inter-State Council for better coordination among the states and between Centre and States.

In 1990, the Inter-State Council was established on the recommendation of Sarkaria Commission (1983-87).

It consists of the following members.

(i) Prime Minister as the Chairman
(ii) Chief Minister of all the States and Union Territories
(iii) Administrators of Union Territories not having Assemblies
(iv) Governor's of States under President rule
(v) Six Central Cabinet Ministers

8. Which among the following writs is issued to quash the order of a Court or Tribunal?

(a) Mandamus (b) Prohibition
(c) Quo Warranto (d) Certiorari

(d) Certiorari is issued to quash the order of a Court or Tribunal. There are five major types of writs Habeas Corpus, Mandamus, Prohibition, Quo Warranto and Certiorari. In India, both Supreme Court and High Court are empowered with writ jurisdiction.

Certiorari It is issued by Supreme Court and High Court to transfer a particular matter or quash the order of lower court.

Mandamus It is a command issued by a court to a public authority, lower court or tribunal to perform their duties.

Prohibition Supreme Court and High Court may prohibit lower courts who exceeds their jurisdiction are act in country to the rule of natural justice.

Quo Warranto This unit is issued to enquire into legality of the claim of a person or public office.

Habeas Corpus is used by the courts to find out if a person has been illegally detained.

9. Which among the following statements about the power to change the basic structure of the Constitution of India is/are correct?

1. It falls outside the scope of the amending powers of the Parliament.
2. It can be exercised by the people through representatives in a Constituent Assembly.
3. It falls within the constituent powers of the Parliament.

Select the correct answer using the codes given below

(a) 1 and 3 (b) 1 and 2
(c) Only 1 (d) 2 and 3

(*c*)

1. The concept of basic structure was laid down by the Supreme Court in the Keshavanand Bharti Case (1973). The basic structure doctrine states that the Constitution of India has certain basic features that cannot be altered or destroyed through amendments by the Parliament.

   Article 368 does not give absolute powers to the Parliament to amend any part of the Constitution.

   Thus, statement (1) is correct.
2. The amendment in basic structure can also not be brought via Constituent Assembly since it was dissolved in 24th January, 1950.

   Hence, statement (2)is incorrect.
3. Constituent power is the power to formulate a Constitution or to propose amendments to or revisions of the Constitution and to ratify such proposal. The Constitution of India vests constituent power upon the Parliament subject to the special procedure laid down therein. However, scope to amend the basic structure is limited.

   Hence, statement (3) is incorrect.

10. When a proclamation of emergency is in operation, the right to move a court for the enforcement of all Fundamental Rights remains suspended, except

(a) Article-20 and Article-21
(b) Article-21 and Article-22
(c) Article-19 and Article-20
(d) Article-15 and Article-16

(*a*) The President can proclam emergency under Article-352 when security of nation or a part of it is threatened by war as external aggression or armed rebellion.

Article-20 and Article-21 which ensure protection of life and personal liberty cannot be suspended at any cost.

- Article-22 Protection against arrest and detention in certain cases.
- Article-19 Protection of certain sights regarding freedom of speech etc.
- Article-15 Prohibition of discrimination on grounds of religion, race, caste, sex or place of birth.
- Article 16 Equality of opportunity in matters of public employment.

11. Which one of the following Articles of the Constitution of India lays down that no citizen can be denied the use of wells, tanks and bathing ghats maintained out of state funds?

(a) Article-14
(b) Article-15
(c) Article-16
(d) Article-17

(*b*) Article-15 of the Constitution abolishes untouchability and forbids its practice in any form. So, no person would be denied to use the wells, tanks and bathing ghats maintained out of state funds.

In 1955, government enacted protection of Civil Right Act to abolish untouchability from society.

12. Who amongst the following organised the All India Scheduled Castes Federation?

(a) Jyotiba Phule
(b) Periyar
(c) BR Ambedkar
(d) MK Karunanidhi

(*c*) All India Scheduled Castes Federation was founded by BR Ambedkar in 1942 to organise schedule caste people against Brahmanical ideology.

He also established Bahishkrit Hitkarni Sabha, Independent Labour Party and started weekly paper 'Mooknayak'. He was appointed as the first Law Minister of India.

13. Paul Allen, who died in October, 2018, was the co-founder of

(a) Oracle (b) IBM
(c) Microsoft (d) SAP

(*c*) Paul Allen was co-founder of Microsoft along with the Bill Gates. Microsoft is an American Multinational Company with its headquarter is in Washington. The CEO of Microsoft as on 2019 is Satya Nadella.

14. The mobile app 'eVIGIL' is helpful in

(a) conducting free and fair e-tendering process in government offices.
(b) fighting against corruption in public services.
(c) removing garbage from the municipal areas.
(d) reporting violation of model code of conduct in election-bound states.

(*d*) The Election Commission of India had launched eVIGIL app for citizens to report any violation of the Model Code of Conduct (MCC) during elections. The android based app aimed at empowering people across the country to share evidence of malpractice by political parties, their candidates and activists directly to ECI.

15. 'Prahaar' is

(a) a battle tank
(b) a surface-to-surface missile
(c) an aircraft carrier
(d) a submarine

(*b*) Prahaar is a surface-to-surface short range tactical ballistic missile. **Prahaar** has been indigenously developed by Defence Research and Development Organisation (DRDO).

It is a quick-reaction, all-weather, all-terrain, highly accurate battle field support tactical missile with advance manoeuvring capability. It is capable of carrying multiple types of warheads weighing around 200 kg and neutralising different types of targets.

16. Who among the following is/are the recipient/recipients of Rajiv Gandhi Khel Ratna Award, 2018?

(a) Virat Kohli
(b) S Mirabai Chanu and Virat Kohli
(c) Neeraj Chopra
(d) Hima Das and Neeraj Chopra

(*b*) The Rajiv Gandhi Khel Ratna Award is the highest sporting honour in our country. It was started in 1991-92. In 2018 Virat Kohli and Mirabai Chanu were bestowed with this award in the field of Cricket and Weightlifting.

In 2019, Deepa Malik and Bajrang Punia were awarded Rajiv Gandhi Khel Ratna in the field of Paralympic and Freestyle Wrestling respectively.

17. Pakyong Airport is located in

(a) Sikkim
(b) jammu and Kashmir
(c) Arunachal Pradesh
(d) Mizoram

(*a*) Recently Prime Minister inaugurated Pakyong airport in the State of Sikkim. It is Sikkim's first airport constructed by the Airport Authority of India.

18. The United Nations has been observing International Day of Rural Women on

(a) 15th July (b) 15th August
(c) 15th September (d) 15th October

(*d*) The United Nations observes International Day of Rural Women on 15th October.

International Day of Rural Women is observed to celebrate the crucial role that women and girls play in ensuring the sustainability of rural households and communities, thus, improving rural livelihoods and overall well-being. The First International Day of Rural Women was observed on 15th October, 2008.

19. Who among the following is the first Indian to win Pulitzer Prize?

(a) Arundhati Roy
(b) Gobind Bihari Lal
(c) Vijay Seshadri
(d) Jhumpa Lahiri

(*b*) The Pulitzer Prize is awarded for the achievement in the field of journalism, literature and musical composition, presented by Columbia University.

Govind Bihari Lal was first Indian to win Pulitzer prize in 1937. Vijay Seshadri and Jhumpa Lahiri won the Pulitzer Prize in the year of 2014 and 2006 respectively.

20. Saurabh Chaudhary excels in which one of the following sports?

(a) Archery (b) Shooting
(c) Boxing (d) Judo

(*b*) Saurabh Chaudhary is an Indian shooter. He won the gold medal at 2018 Asian Games in 10m Air pistol. He created history by becoming youngest Indian to win gold medal at the Asian Games.

21. Which one of the following is not an assumption in the law of demand?

(a) There are no changes in the taste and preferences of consumers
(b) Income of consumers remains constant
(c) Consumers are affected by demonstration effect
(d) There are no changes in the price of substitute goods

(*c*) Consumers are affected by demonstration effect, is not an assumption in the law of demand.

The law of demand is a fundamental concept of economics which states that at a higher price consumers will demand a lower quantity of goods and vice-versa.

22. Which one of the following statements is not correct?

(a) When total utility is maximum, marginal utility is zero.
(b) When total utility is decreasing, marginal utility is negative.
(c) When total utility is increasing marginal utility is positive.
(d) When total utility is maximum, marginal and average utility are equal to each other.

(*d*) Total utility means total benefit obtained by a person from consumption of goods and services, whereas, marginal utility means the amount of utility, a person gains from the consumption of each successive unit of a commodity.

Relation between total utility and marginal utility

(i) When total utility is maximum, marginal utility is zero
(ii) When total utility increase, marginal utility is positive
(iii) When total utility decrease, marginal utility is negative

Hence, option (d) is not correct.

23. Consider the following statements about indifference curves.

1. Indifference curves are convex to the origin.
2. Higher indifference curve repre sents higher level of satisfaction.
3. Two indifference curves cut each other.

Which of the statements given above is/are correct?

(a) Only 1 (b) 1 and 2
(c) 2 and 3 (d) Only 3

(*b*) An indifference curve are convex of combinations of goods which derive the same level of satisfaction so that the consumer is indifferent to any of the combination he consumes.

Hence statement (1) is correct.

Higher indifference curve means higher level of satisfaction as higher indifference curve consists of more of two goods or the same quantity of one good and more quantity of the other good. Hence statement (2) is correct. Two indifference curves cannot cut each other.

Thus, statement 3 is incorrect.

24. Consider the following statements about a joint-stock company.

1. It has a legal existence.
2. There is limited liability of shareholders.
3. It has a democratic management.
4. It has a collective ownership.

Which of the statements given above are correct?

(a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 1, 2, 3 and 4

(*d*) A joint-stock company is a business entity in which shares of the company's stock can be bought and sold by share holders.

Each shareholder owns company stock in proportion, evidenced by their shares.

All the above are features of a joint-stock company

(i) It has a legal existence .
(ii) There is limited liability of shareholders for satisfaction of the debt of the company the personal property of the shareholders cannot be used.
(iii) It has a democratic management.
(iv) It has a collective ownership.

Hence, all the statements are correct.

25. When some goods or productive factors are completely fixed in amount, regardless of price, the supply curve is

(a) horizontal
(b) downward sloping to the right
(c) vertical
(d) upward sloping to the right

(*d*) When some goods or productive factors are completely fixed in amount regardless of price, the supply curve is upward sloping to the right.

Supply curve is a graphic representation of the correlation between the cost of a good or service and the quantity supplied for a given period.

The upward sloping means that the firms will be willing to increase the production in response to a higher market price because the higher price may make additional production profitable.

26. Who designed the Bombay Secretariat in the 1870s?

(a) Henry St Clair Wilkins
(b) Sir Cowasjee Jehangir Readymoney
(c) Purushottamdas Thakurdas
(d) Nusserwanji Tata

(***a***) The Bombay Secretariat was completed in 1874 and designed by Captain Henry St Clair Wilkins in the Venetian Gothic style.

With its arcaded verandahs and huge gable over the West facade, it was a monument to the civic pride of Bombay's British rulers.

27. Who was the founder of Mahakali Pathshala in Calcutta?

(a) Her Holiness Mataji Maharani Tapaswini
(b) Sister Nivedita
(c) Madame Blavatsky
(d) Sarojini Naidu

(***a***) Mataji Maharani Tapaswini founded Mahakali Pathshala in Calcutta in 1893 to foster women education.

28. Which European ruler had observed, ''Bear in mind that the commerce of India is the commerce of the world ... he who can exclusively command it is the dictator of Europe''?

(a) Queen Victoria
(b) Peter the Great of Russia
(c) Napoleon Bonaparte
(d) Gustav II Adolf

(***b***) Peter the great was a Russian Czar (emperor) in the late 17th century. He was best known for his extensive reforms in an attempt to establish Russia as a great nation.

As India was one of the main centres of the world trade and industry, Peter observed, "Bear in mind that the commerce of India is the commerce of the world.''

29. Which European traveller had observed, ''A Hindu woman can go anywhere alone, even in the most crowded places and she need never fear the impertinent looks and jokes of idle loungers''?

(a) Francois Bernier
(b) Jean-Baptiste Tavernier
(c) Thomas Roe
(d) Abbe JA Dubois

(***d***) Abbe Jean Antoine Dubois was a French Catholic missionary in India. He was an Indologist and authored the book Hindu manners, custom and ceremonies. In his book, he wrote about position of women in the society.

He had observed that a Hindu woman can go anywhere alone, even in the most crowded places and she need never fear the impertinent looks and jokes of idle loungers. He was called as Dadda Swami by the local people.

30. Who was the author of the book '*Plagues and Peoples*'?

(a) William H McNeil
(b) WI Thomas
(c) Rachel Carson
(d) David Cannadine

(***a***) 'Plagues and Peoples' is a book on epidemiological history by William Hardy McNeil published in New York City in 1976. It was a critical and popular success, offering a radically new interpretation of the extra ordinary impact of infectious disease on culture as a means of enemy attack.

The book ranges from examining the effects of small pox in Mexico, the bubonic plague in China, to the typhoid epidemic in Europe.

31. Which Indian social theorist had argued that the idea of a homogenised Hinduism was constructed through the 'cultural arrogance of post-enlightenment Europe?'

(a) Ashis Nandy
(b) Partha Chatterjee
(c) TK Oommen
(d) Rajni Kothari

(***a***) Ashis Nandy is an Indian political psychologist, social theorist and critic. He has given theoretical criticism of European colonialism, development, modernity, and secularism among many others.

According to him, the idea homogenised Hindusim emnates from the 'cultural arrogance of post-enlightenment Europe'.

32. 'Sub-prime crisis' is a term associated with which one of the following events?

(a) Economic recession
(b) Political instability
(c) Structural adjustment programmes
(d) Growing social inequality

(***a***) Sub-prime crisis is Economic recession that took place in the USA from 2007 to 2010.

Economic recession is a slow down or massive contraction in economic activities. The primary cause was the collapse of the housing bubble. As a result of this crisis, job growth, saving and investment reduced to abysmal.

33. Which one of the following is not a change brought about by the Indian Independence Act of 1947?

(a) The Government of India Act, 1935 was amended to provide an interim Constitution.
(b) India ceased to be a dependency.
(c) The Crown was the source of authority till new Constitution was framed.
(d) The Governor-General was the Constitutional Head of Indian Dominion.

(***b***) India ceased to be a dependency is not a charge brought about by the Indian Independence Act of 1947.

On 20th February, 1947 British PM declared that British rule will end by 30th June 1948, but on 3rd June, 1947 Mountabatten Plan was accepted by Congress which led to enaction of Indian Independence Act.

As per Act India was declared as an independent and Sovereign State from 15th August, 1947.

Hence, option (b) is correct.

34. Which one of the following is not a correct statement regarding the provision of Legislative Council in the State Legislature?

(a) The States of Bihar and Telangana have Legislative Councils
(b) The total number of members in the Legislative Council of a State shall not exceed one-third of the total number of members in the Legislative Assembly
(c) One-twelfth of all members shall be elected by electorates consisting of local bodies and authorities

(d) One-twelfth of all members shall be elected by graduates residing in the state

(*c*) The Constitution as provided autonomy to state to have unicameral and bicameral legislature, 7 out of 28 states have bicameral legislature, Legislative Assembly and Council.

The Legislative Council is the Upper House where members are indirectly elected.

The maximum strength of the council is fixed at one-third of the total strength of assembly and minimum is fixed at 40.

Manner of Election

One-third	By Local Bodies and Authorities
One-third	By Legislative Assemblies
One-twelfth	By Graduates
One twelfth	By Teachers
One-sixth	Nominated by Governor

Hence, option (c) is incorrect.

35. Which one of the following is not correct about the Panchayats as laid down in Part IX of the Constitution of India?

(a) The Chairperson of a Panchayat needs to be directly elected by people in order to exercise the right to vote in the Panchayat meetings
(b) The State Legislature has the right to decide whether or not offices of the Chairpersons in the Panchayats are reserved for SCs, STs or women
(c) Unless dissolved earlier, every Panchayat continues of a period of five years
(d) The State Legislature may by law make provisions for audit of accounts of the Panchayats

(*a*) The Part IX was inserted in Constitution through the 73rd Amendment Act.

Important Features

The Chairman of a Panchayat shall be elected in such a manner as the State Legislature determines. So statement (a) is not correct.

Rest all statements are true. It also provided one-third reservation of seats for women in State Finance Commission, State Election Commission etc.

36. Which one of the following is not correct about Administrative Tribunals?

(a) The Parliament may be law Constitute Administrative Tribunals both at the Union and State levels
(b) Tribunals may look into disputes and complaints with respect to recruitment and conditions of service of persons appointed to public services
(c) Tribunals established by a law of the Parliament can exclude the jurisdiction of all courts to allow for special leave to appeal
(d) The law establishing the Tribunals may provide for procedures including rules of evidence to be followed

(*c*) The 42nd Constitutional Amendment Act of 1976 added a new Part XIV-A entitled as 'Tribunals'. It consists of two Article-323A and 323B.

Article-323A empowers the Parliament to establish Administrative Tribunal, in 1985 the Administrative Tribunal Act was passed to establish Central Administrative Tribunal.

All the statements are true except (c), as they are not empowered to exclude the jurisdiction of all court to allow for special leave to appeal.

Article-323B gives power to both Parliament and State Legislature to establish tribunal for different disputes.

37. A market situation when many firms sell similar, but not identical products is termed as

(a) perfect competition
(b) imperfect competition
(c) monopolistic competition
(d) oligopoly

(c) **Monopolistic competition** is situation where many firms are competing against each other but selling products that are similar, this type of situation leads to imperfect competition in the market.

This market situation is against the interest of consumers.

Perfect competition is the situation prevailing in a market in which buyers and sellers are so numerous and well informed that all elements of monopoly are absent and the market price of a commodity is beyond the control of individual buyers and sellers.

Imperfect competition is the situation prevailing in a market in which elements of monopoly allow individual producers or consumers to exercise some control over market prices.

An oligopoly is a market form wherein a market or industry is dominated by a small number of large sellers.

38. Consider the following statements.

1. Inflation in India continued to be moderate during 2017-18.
2. There was significant reduction in food inflation, particularly pulses and vegetables during the period.

Which of the statements given above is/are correct?

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(c) As per economic survey. Inflation in india continued to be moderate 3.3% during 2017-18.

It was mainly due to the significant reduction in food inflation, particularly pulses and vegetables during the period.

Hence, both the statements are true.

39. Which one of the following hypothesis postulates that individual's consumption in any time period depends upon resources available to the individual, rate of return on his capital and age of the individual?

(a) Absolute Income Hypothesis
(b) Relative Income Hypothesis
(c) Life-Cycle Hypothesis
(d) Permanent Income Hypothesis

(c) The life-cycle hypothesis postulates that individual consumption in anytime period depends on.

(i) The resources available to the individual.
(ii) The rate of return on his capital.
(iii) The age of the individual.

It was developed by Ando and Modigliani.

40. According to John Maynard Keynes, employment depends upon

(a) aggregate demand
(b) aggregate supply
(c) effective demand
(d) rate of interest

(c) John Maynard Keynes was the founder of modern macro economics. He was a British economist.

As per his book the general theory of employment, interest and money, employment depends on effective demand. With the increase in demand, manufacturing and other services require more person to manufacture goods and provide services thus leading to job growth.

Hence, option(c) is true.

41. Which one of the following canons of taxation was not advocated by Adam Smith?

(a) Canon of equality
(b) Canon of certainty
(c) Canon of convenience
(d) Canon of fiscal adequacy

(***d***) Adam Smith was an economist, best known for his book *'Wealth of Nation'*.

He presented four canon of taxation, later many other canon were developed. Smith's canon of taxation include

(i) Canon of equality
(ii) Canon of certainty
(iii) Canon of convenience
(iv) Canon of economy

So, option (d) is correct answer.

42. Which Arab scientist could be given the credit of christening the mathematical discipline of algorithm?

(a) Al-Khwarizmi (b) Ibn al-Haytham
(c) Ibn Rushd (d) Ibn Sina

(***a***) The credit of christening the mathematical discipline of algorithm goes to Arab scientist Muhammad Ibn Musa Al-Khwarizmi. He was a Persian scholar who produced works in mathematic, astronomy and geography. In 820 AD, he was appointed as the astronomer and head of the library of House of Wisdom in Baghdad.

43. Which one of the following developments took place because of the Kansas-Nebraska Act of 1854?

(a) The Missouri Compromise was repealed and people of Kansas and Nebraska were allowed to determine whether they should own slaves or not
(b) The Act did not permit the territories the right to vote over the question of slavery
(c) The voice of the majority in regard to the issue of slavery was muzzled
(d) The Federal Government had the sole authority to decide on slavery

(***a***) The Kansas-Nebraska Act was passed by the US Congress in 1854. It allowed people in the territories of Kansas and Nebraska to decide for themselves whether or not allow slavery with in their borders.

This act repealed the Missouri compromise of 1820 which prohibited slavery.

44. Which one of the following issues was included in the Indo-US Nuclear Agreement of 2007?

(a) India has 'advance right to reprocess' US-origin safeguarded spent fuel.
(b) India did not have the right to build a strategic fuel reserve with the help of the other supplier countries.
(c) India should not test a nuclear device.
(d) The US will impede the growth of India's nuclear weapons programme.

(***c***) The Indo-US nuclear agreement also known as 123 agreement was watershed moment in Indo-US relationship. The deal would have indirectly bring India under purview of NSG and US laws that would not allow India to conduct nuclear test in the future. The issue of nuclear test was later clarified as the moratorium on nuclear test was unilateral and voluntary and there was no pressure on India from outside. Thus, (c) is the correct answer.

45. Which of the following statements about Alladi Krishnaswami Ayyar, as a drafting member of the Constitution of India, are correct?

1. He favoured the role of the Supreme Court in taking important decisions related to the interpretation of the Constitution of India.
2. He felt that the Supreme Court had to draw the line between liberty and social control.
3. He believed in the dominance of the executive over the judiciary.
4. He favoured a dictatorial form of governance.

Select the correct answer using the codes given below

(a) 1 and 2 (b) 1, 2 and 3
(c) 3 and 4 (d) 1, 2 and 4

(***a***) Only statement 1 and 2 are correct. Alladi Krishnaswami was eminent lawyer from Chennai, in Constituents Assembly he favoured the role of Supreme Court to interpret the Constitution when required. He also wanted that the Supreme Court had to draw the line between liberty and social control.

46. Which of the following are the core functions of the United Nations multidimensional peacekeeping operations?

1. Stabilisation
2. Peace consolidation
3. To extend support to a losing state in a war

Select the correct answer using the codes given below

(a) 1, 2 and 3 (b) 2 and 3
(c) 1 and 3 (d) 1 and 2

(***d***) The United Nation peacekeeping was established in 1948, to help countries to cross the difficult path from conflict to peace. Their main work peace building, peace making and peace enforcement.

They do not take part in war with the objective winning or helping loosing state.
Hence, option (d) is correct.

47. The South China Sea dispute involves which of the following countries?

1. China 2. Vietnam
3. Malaysin 4. Indonesia

Select the correct answer using the codes given below

(a) 1 and 4 (b) 1 and 2
(c) 1, 2 and 3 (d) 2, 3 and 4

(***c***) The South China disputes involve both island and maritime claims among several sovereign states within the region, namely, Brunei, the People's Republic of China (PRC), Republic of China (Taiwan), Malaysia, the Phillippines, and Vietnam.

Hence, option (c) is correct.

48. The 'Kyoto Protocol' is an international treaty that commits State parties to reduction in

(a) poverty
(b) greenhouse gases emission
(c) nuclear armaments
(d) agricultural subsidy

(**b**) The 'Kyoto Protocol' is an international treaty which extends the 1992 United Nations Framework Convention on Climate Change (UNFCCC) that commits State parties to reduce greenhouse gases emission which are responsible for Global Warming and Climate Change. The Kyoto Protocol was adopted in Kyoto, Japan on December 11, 1997 and entered into force on February 16, 2005. There are currently 192 parties singed the protocol.

49. The 'Beijing Declaration' is concerned with which one of the following issues?

(a) Rights of Children
(b) Rights of Women
(c) Right to Development
(d) Reduction of Tariffs

(**b**) The 'Beijing Declaration' is concerned with the 'Right of Women', this declaration was outcome of fourth world conference on women in 1995.

It advances the goals of equality, development and peace for all women.

50. The 'Gujral Doctrine' relates to which one of the following issues?

(a) Build trust between India and its neighbours
(b) Initiate dialogue with all insurgent groups in India
(c) Undertake development activities in Naxal-dominated areas
(d) Ensure food security

(**a**) The main objective of 'Gujaral doctrine' to build trust between India and its neighbours, was initiated in 1996 by IK Gujaral, the then Finance Minister.

51. Match List-I with List-II and select the correct answer using the code given below the Lists

	List-I (Compound/Molecule)		List-II (Shape of Molecule)
A.	CH_3F	1.	Trigonal planar
B.	HCHO	2.	Tetrahedral
C.	HCN	3.	Trigonal pyramidal
D.	NH_3	4.	Linear

Code

	A	B	C	D
(a)	2	4	1	3
(b)	2	1	4	3
(c)	3	4	1	2
(d)	3	1	4	2

(**b**) A. CH_3F — Its molecular shape is tetrahedral
B. HCHO — Its molecular shape is trigonal planar
C. HCN — Its molecular shape is linear
D. NH_3 — Its molecular shape is trigonal pyramidal

52. Very small insoluble particles in a liquid may be separated from it by using

(a) crystallisation
(b) fractional distillation
(c) centrifugation
(d) decantation

(***d***) There are some mixtures which contain insoluble solid particles suspended in a liquid. The solid particles which are insoluble in a liquid can be separated by using **decantation** method.

53. Which one of the following elements cannot be detected by 'Lassaigne's test'?

(a) I (b) Cl (c) S (d) F

(***d***) Fluorine cannot be detected by Lassaigne's test because silver fluoride is soluble in water and does not precipitate thus this method cannot be used for detection of fluorine.

54. In which of the following, functional group isomerism is not possible?

(a) Alcohols (b) Aldehydes
(c) Alkyl halides (d) Cyanides

(***c***) In alkyl halides, functional group isomerism is not possible. Rest of all options show functional group isomerism.

55. Which one of the following statements is not correct?

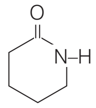
(a) Fischer projection represents the molecule in an eclipsed conformation.
(b) Newman projection can be represented in eclipsed, staggered and skew conformations.
(c) Fischer projection of the molecule is its most stable conformation.
(d) In Sawhorse projections the lines are inclined at an angle of 120° to each other.

(***c***) Statement is given in option (c) is correct. Correct statements is Fischer projection always depicts, the molecule in eclipsed form, hence it is most stable conformation.

56. The monomer/monomers used for the synthesis of nylon 6 is/are

(a) hexamethylenediamine and adipic acid
(b) caprolactam
(c) urea and formaldehyde
(d) phenol and formaldehyde

(***b***) Nylon 6 is only made from one kind of monomer, a monomer is called caprolactam.



Caprolactam

57. Which one among the following stars is nearest to the Earth?

(a) Sirius
(b) Arcturus
(c) Spica
(d) Proxima Centauri

(***d***) The nearest star to the Earth is Sun followed by the Proxima Centauri.

Proxima Centauri is part of the star system known as 'Alpha Centauri'.

58. Which of the following planets of our solar system has least mass?

(a) Neptune (b) Jupiter
(c) Mars (d) Mercury

(***d***) Mercury is the smallest and closest planet to Sun. It has the least mass in the solar system. The planet with the highest mass is Jupiter.

59. Two identical solid pieces, one of gold and other of silver, when immersed completely in water exhibit equal weights. When weighed in air (given that, density of gold is greater than that of silver),

(a) the gold piece will weigh more
(b) the silver piece will weigh more
(c) Both silver and gold pieces weigh equal
(d) weighing will depend on their masses

(***c***) Since, gold is more denser than silver. But it is given that both pieces have identical and equal weight in water, it means both have equal volume and mass. (May be gold piece hollow inside.)

(As we know, force of gravity is determined by mass and buoyant force is determined by volume)

So, both silver and gold pieces weight equal in air.

60. If the wavelengths corresponding to ultraviolet, visible and infrared radiations are given as $\lambda_{\text{ultraviolet}}$, λ_{visible} and $\lambda_{\text{infrared}}$ respectively, then which one of the following gives the correct relationship among these wavelengths?

(a) $\lambda_{\text{ultraviolet}} < \lambda_{\text{infrared}} < \lambda_{\text{visible}}$
(b) $\lambda_{\text{ultraviolet}} > \lambda_{\text{visible}} > \lambda_{\text{infrared}}$
(c) $\lambda_{\text{ultraviolet}} > \lambda_{\text{infrared}} > \lambda_{\text{visible}}$
(d) $\lambda_{\text{ultraviolet}} < \lambda_{\text{visible}} < \lambda_{\text{infrared}}$

*(**d**)* As we know that, infrared light has a lower frequency than visible light and visible light has a lower frequency than ultraviolet light.

i.e. $\nu_{\text{ultraviolet}} > \nu_{\text{visible}} > \nu_{\text{infrared}}$

Since, wavelength of a light is inversely proportional to its frequency, therefore

$$\lambda_{\text{ultraviolet}} < \lambda_{\text{visible}} < \lambda_{\text{infrared}}$$

61. An electron and a proton starting from rest get accelerated through potential difference of 100 kV. The final speeds of the electron and the proton are v_e and v_p, respectively. Which one of the following relation is correct?

(a) $v_e > v_p$
(b) $v_e < v_p$
(c) $v_e = v_p$
(d) Cannot be determined

*(**a**)* As we know that,

$$KE = \frac{1}{2}mv^2 = qV$$

For proton and electron, the relation of charge and mass are $q_p = -q_e$

and $m_p = m_e \cdot 1840$

$$v \propto \sqrt{\frac{2qV}{m}} \Rightarrow v \propto \sqrt{\frac{1}{m}}$$

Here, we can see that mass of proton is very high than mass of electron, so $v_e > v_p$.

62. If two vectors **A** and **B** are at an angle $\theta \neq 0°$, then

(a) $|\mathbf{A}| + |\mathbf{B}| = |\mathbf{A} + \mathbf{B}|$
(b) $|\mathbf{A}| + |\mathbf{B}| > |\mathbf{A} + \mathbf{B}|$
(c) $|\mathbf{A}| + |\mathbf{B}| < |\mathbf{A} + \mathbf{B}|$
(d) $|\mathbf{A}| + |\mathbf{B}| = |\mathbf{A} - \mathbf{B}|$

*(**b**)* Since, any one side of a triangle is less than the sum of the other two sides individually, so for any two vectors **A** and **B**,

$$|\mathbf{A} + \mathbf{B}| \leq |\mathbf{A}| + |\mathbf{B}|$$

If $\theta \neq 0$, then

$$|\mathbf{A}| + |\mathbf{B}| > |\mathbf{A} + \mathbf{B}|.$$

63. Which one of the following functions is not carried out by smooth endoplasmic reticulum?

(a) Transport of materials
(b) Synthesis of lipid
(c) Synthesis of protein
(d) Synthesis of steroid hormone

*(**c**)* Endoplasmic reticulum has two types–Smooth Endoplasmic Reticulum (SER) and Rough Endoplasmic Reticulum (RER). SER have smooth surface due to lack of ribosomes. Thus, they do not help in protein synthesis unlike RER which contain ribosomes (the site of protein synthesis).

SER help in synthesis of steroid hormones and lipids. They also form vesicles for the transportation of materials.

64. Which one of the following cell organelles mainly functions as storehouse of digestive enzymes?

(a) Desmosome (b) Ribosome
(c) Lysosome (d) Vacuoles

*(**c**)* Lysosomes are called storehouse of digestive enzymes. These are single membrane bound cell organelles which contain various digestive or hydrolytic enzymes.

Due to the presence of hydrolytic enzymes these are also called as suicidal bags of cell. These enzymes take part in hydrolysis of extracellular materials.

65. Which one of the following tissues is responsible for increase of girth in the stem of a plant?

(a) Tracheid
(b) Pericycle
(c) Intercalary meristem
(d) Lateral meristem

*(**d**)* The lateral meristem tissues are responsible for increase of girth in the stem of a plant. This tissue is present along the side of organs, e.g. vascular cambium and cork cambium, which are responsible for the secondary growth of plants. Due to this secondary growth the girth of stems and roots increases.

66. Which one of the following organisms is dependent on saprophytic mode of nutrition?

(a) *Agaricus* (b) *Ulothrix*
(c) *Riccia* (d) *Cladophora*

*(**a**)* *Agaricus* is an organism which is dependant on saprophytic mode of nutrition. They obtain their nutrition from dead and decaying organic matter.

Rest organisms, i.e. *Ulothrix, Riccia* and *Cladophora* have autotrophic mode of nutrition.

67. Which one of the following has a bilateral symmetry in its body organisation?

(a) *Asterias* (b) *Sea anemone*
(c) *Nereis* (d) *Echinus*

*(**c**)* *Nereis* is a member of phylum–Annelida, which has bilateral symmetry in its body organisation.

In bilateral symmetry, a lengthwise vertical plane divided the animals into two equal and opposite halves. Sea anemone of phylum–Cnidaria and *Asterias* and *Echinus* of phylum–Echinodermata have radial symmetry.

68. Which one of the following pairs of animals is warm- blooded?

(a) Crocodile and ostrich
(b) Hagfish and dogfish
(c) Tortoise and ostrich
(d) Peacock and camel

*(**d**)* Peacock and camel are warm-blooded animals. These animals regulate their body temperature in any kind of environment. On the other hand, crocodile, hagfish, dogfish, tortoise are cold-blooded animals.

69. Which one of the following States of India is not covered by flood forecasting stations set-up by the Central Water Commission?

(a) Rajasthan
(b) Jammu and Kashmir
(c) Tripura
(d) Himachal Pradesh

(**d**) Central Water Commission is the nodal organisation for flood forecasting in the country. CWC comes under Ministry of Water Resource, River Development and Ganga Rejuvenation.

It is the modal agency for flood forecasting in India.

Except Himachal Pradesh all other mentioned states are covered by flood forecasting stations setup by the CWC.

Hence, option (d) is not correct.

70. The city of Cartagena, which is famous for Protocol on Biosafety, is located in

(a) Colombia

(b) Venezuela
(c) Brazil
(d) Guyana

(**a**) Cartagena Protocol on Biosafety is an international treaty that seeks to protect biodiversity from the potential risks posed by Genetically Modified Organisms (GMO).

It was singed on 15th May, 2000 in Cartagena which is in Colombia.

71. Which one among the following is the most populated state in India as per Census 2011?

(a) Goa (b) Mizoram
(c) Meghalaya (d) Sikkim

(**c**) The Census of 2011 was 15th Census held in two phases.

Among the above state Meghalaya is most populated and Sikkim is least populated.

The state of Uttar Pradesh (First rank), Maharashtra (Second rank) and Bihar (Third rank) are among top three populated states according to the Census 2011.

72. Which among the following countries of South America does the Tropic of Capricorn not pass through?

(a) Chile (b) Bolivia
(c) Paraguay (d) Brazil

(**b**) The Tropic of Capricorn is an imaginary line of latitude at 23.5° South of the equator. This latitude runs through following countries: Namibia, Botswana, South Africa, Mozambique Madagascar , Australia, Chile, Argentina, Paraguay and Brazil. It does not pass through Bolivia.

Hence, option (b) is not correct.

73. Which one of the following is not correct about Sargasso Sea?

(a) It is characterised with anti-cyclonic circulation of ocean currents.
(b) It records the highest salinity in Atlantic ocean.
(c) It is located West of Gulf stream and East of Canary current.
(d) It confined in gyre of calm and motionless water.

(**a**) Sargasso sea is the calm centre of the anti-cylonic gyre in the North Atlantic, comprising a large eddy of surface water, the boundaries of which are demarcated by major current systems such as the Gulf Stream, Canaries current, and North Atlantic Drift.

The Sargasso sea is a large, warm (18° C), saline region which is characterised by an abundance of floating brown seaweed (*Sargassum*).

Hence, option (a) is not correct.

74. Match List I with List II and select the correct answer using the codes given below the lists.

	List I (City)		**List II** (Product)
A.	Detroit (USA)	1.	Motarcar
B.	Antwerp (Belgium)	2.	Diamond Cutting
C.	Tokyo (Japan)	3.	Steel
D.	Harbin (China)	4.	Ship Building

Codes

	A	B	C	D		A	B	C	D
(a)	3	4	2	1	(b)	3	2	4	1
(c)	1	4	2	3	(d)	1	2	4	3

(**d**)

City	**Product**
Detroit (USA)	Motarcar
Antwerp (Belgium)	Diamond Cutting
Tokyo (Japan)	Ship Building
Harbin (China)	Steel

75. Which one of the following is not situated on Varanasi- Kanyakumari National Highway?

(a) Satna (b) Rewa
(c) Katni (d) Jabalpur

(**a**) The Varanasi-Kanyakumari National highway is Part of NH-44. Major cities such as Rewa, Katni, Jabalpur, Hyderabad, Bengaluru etc. are located on this highway.

Satna does not lie on this Highway.

76. Which one of the following methods is not suitable for urban rainwater harvesting?

(a) Rooftop recharge pit
(b) Recharge wells
(c) Gully plug
(d) Recharge trench

(**c**) Except Gully plug, all the others such as Rooftop recharge pit, Recharge wells and Recharge trench are viable in urban rainwater harvesting.

Gully plugs or Checkdams are mainly built to prevent erosion and to settle sediments and pollutant.

77. If one plots the tank irrigation in India and superimposes it with map of well irrigation, one may find that the two are negatively related. Which of the following statements explain the phenomenon?

1. Tank irrigation predates well irrigation.
2. Tank irrigation is in the areas with impervious surface layers.
3. Well irrigation requires sufficient groundwater reserves.
4. Other forms of irrigation are not available.

Select the correct answer using the codes given below

(a) 1, 2 and 3 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

(**b**) The tank irrigation is more in the rocky plateau area of the country, where the rainfall is not even and highly seasonal and the rocks are impervious.

The Eastern Madhya Pradesh, Chhattisgarh, Odisha, Tamil Nadu and some parts of Andhra Pradesh have areas under tank irrigation.

On the other hand, well irrigation is most common in alluvial plains of the country except the desert of Rajasthan.

Plains of UP, Bihar, Gujarat, Karnataka and Tamil Nadu are the states which are mostly under well irrigation. These areas have large ground water reserves.

Thus, only statements 2 and 3 correctly explains this phenomenon.

78. When hot water is placed into an empty water bottle, the bottle keeps its shape and does not soften. What type of plastic is the water bottle made from ?

(a) Thermoplastic (b) PVC
(c) Polyurethane (d) Thermosetting

(**d**) Water bottle made from thermosetting plastics. Thermosetting plastic is a polymer that irreversibly becomes rigid when heated. Also known as thermoset, thermosetting polymer.

79. Which of the following is/are state function/functions?

1. $q + W$ 2. q
3. W 4. $H - TS$

Select the correct answer using the codes given below.

(a) 1 and 4 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) Only 1

(**a**) Internal energy $(\Delta E) = q + W$.

It is a state function because it is independent of the path.

Gibbs energy $(G) = H - TS$

It is also a state function because it is independent of the path.

Heat (q) and work (W) are not state functions being path dependent.

80. For a certain reaction,
$\Delta G^{-} = -45$ kJ/mol and $\Delta H^{-} = -90$ kJ/mol at 0°C.
What is the minimum temperature at which the reaction will become spontaneous, assuming that ΔH^{-} and ΔS^{-} are independent of temperature?

(a) 273 K (b) 298 K
(c) 546 K (d) 596 K

(a) $\because \Delta G = \Delta H - T\Delta S$ and ΔH and ΔS are independent of temperature,

Also, given, $\Delta G^{-\theta} = (-)$ ve at 273 K,

Thus, reaction is spontaneous at 273 K.

Hence, (a) is the correct option.

81. The PCl_5 molecule has trigonal bipyramidal structure. Therefore, the hybridisation of p-orbitals should be

(a) sp^2 (b) sp^3 (c) dsp^2 (d) dsp^3

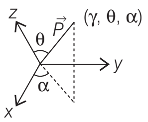
(d) Phosphorus has five electrons in valence orbitals, which can hybridised and give five hybrid orbitals, *viz.* Three p-orbitals one each of s-and d-orbitals.

Thus, hybridisation of P in PCl_5 is sp^3d.

82. In spherical polar coordinates (γ, θ, α), θ denotes the polar angle around z-axis and α denotes the azimuthal angle raised from x-axis. Then, the y-component of **P** is given by

(a) $P \sin\theta \sin\alpha$ (b) $P \sin\theta \cos\alpha$
(c) $P \cos\theta \sin\alpha$ (d) $P \cos\theta \cos\alpha$

(a) Here, the y-component of $\vec{P}$ is given by, $y = (\vec{P}\sin\theta)\sin\alpha$



83. For an ideal gas, which one of the following statements does not hold true?

(a) The speed of all gas molecules is same.
(b) The kinetic energies of all gas molecules are not same.
(c) The potential energy of the gas molecules is zero.
(d) There is no interactive force between the molecules.

(a) Statement (a) is incorrect, because an ideal gas consist of a very large number of molecules which are in a state of continuous, rapid and random motion. They move in all direction with different speeds, ranging from zero to infinity, this is a postulate of an ideal gas.

84. What is a constellation?

(a) A particular pattern of equidistant stars from the Earth in the sky.
(b) A particular pattern of stars that may not be equidistant from the Earth in the sky.
(c) A particular pattern of planets of our solar system in the sky.
(d) A particular pattern of stars, planets and satellites in the sky due to their position in the space.

(***b***) A constellation is a group of stars usually in a recognisable shape or pattern.

In 1922, the International Astronomical Union (IAU) formally accepted the modern list of 88 constellations.

85. The Hooke's law is valid for

(a) only proportional region of the stress-strain curve
(b) entire stress-strain curve
(c) entire elastic region of the stress-strain curve
(d) elastic as well as plastic region of the stress-strain curve

(a) Hooke's law is valid for only proportional region of the stress- strain curve.

$\because$ Hooke's law $\rightarrow$ stress $\propto$ strain

86. Which one of the following statements regarding histone proteins is correct?

(a) Histones are proteins that are present in mitochondrial membrane
(b) Histones are proteins that are present in nucleus in association with DNA
(c) Histones are proteins associated with lipids in the cytosol
(d) Histones are proteins associated with carbohydrates in the cytosol

(b) Statement in option (b) regarding histone protein is correct. These are protein that are present in nucleus in association with DNA.

87. Which one of the following statements regarding haemoglobin is correct?

(a) Haemoglobin present in RBCs can carry only oxygen, but not carbon dioxide
(b) Haemoglobin of RBC can carry both oxygen and carbon dioxide
(c) Haemoglobin of RBCs can carry only carbon dioxide
(d) Haemoglobin is only used for blood clotting and not for carrying gases

(b) Haemoglobin of RBCs can carry both oxygen and carbon dioxide and it not used for clotting of blood. Hence, option (b) is the correct answer.

88. Which one of the following is the correct sequence of passage of light in a compound microscope?

(a) Condenser–Objective lens–Eye piece–Body tube
(b) Objective lens–Condenser–Body tube–Eye piece
(c) Condenser–Objective lens–Body tube–Eye piece
(d) Eye piece–Objective lens–Body tube–Mirror

(c) The correct sequence of passage of light in a compound microscope is condenser–Objective lens–Body tube-Eye piece.

Compound microscope is a common light microscope usually used in labs of schools and colleges. In this type of microscope, light reflected by mirror goes to condenser which condense light on object. From here, light goes to objective lens, body tube and eye piece, respectively.

89. Which one of the following statements is correct?

(a) Urea is produced in liver
(b) Urea is produced in blood
(c) Urea is produced from digestion of starch
(d) Urea is produced in lung and kidney

(a) The statement given in option (a) is correct, i.e. urea is produced in liver. As a result of metabolism of nitrogenous compounds (amino acids, proteins, nucleic acids), ammonia is produced.

It combines with CO_2 through ornithine cycle and forms urea.

90. Which one of the following river valleys of India is under the influence of intensive gully erosion?

(a) Kosi (b) Chambal
(c) Damodar (d) Brahmaputra

(*b*) Chambal river valley is under the influence of intensive gully erosion. Chambal forms ravines or badlands due to intensive gully erosion.

It occurs when water is channeled across unprotected land and washes away the soil along drainage lines. Chambal rises from Mhow district in Vindhya Ranges and it is a chief tributary of Yamuna river.

91. Which one of the following may be the true characteristic of cyclones?

(a) Temperate cyclones move from West to East with Westerlies whereas tropical cyclones follow trade winds.
(b) The front side of cyclone is known as the 'eye of cyclone'.
(c) Cyclones possess a centre of high pressure surrounded by closed isobars.
(d) Hurricanes are well-known tropical cyclones which develop over mid-latitudes.

(*a*) Temperate cyclones are cyclones of mid-latitudes and hence are primairly under influence of permanent winds of mid-latitudes, i.e Westerlies.

Their movement is therefore Eastwards of their origin with average velocity of
32 km per hour in summers and 48 km per hour in winters.

Normally, tropical cyclones move from East to West under the influence of trade winds because trade winds are permanent winds of tropical latitudes.

The general direction is therefore Westwards from their origin. They advance with varying velocities. Weak cyclones move at the speed of about 32 km per hour while hurricanes attain the velocity of 180 km per hour or more.

Hence, option (a) is correct.

92. The headquarters of the International Tropical Timber Organisation is located at

(a) New Delhi (b) Yokohama
(c) Madrid (d) Jakarta

(*b*) The International Tropical Timber Organisation was established in 1986 with a objective to conserve and manage tropical forests resources.

Its headquarter is located at Yokohama (Japan).

93. Atmospheric conditions are well-governed by humidity. Which one among the following may best define humidity?

(a) Form of suspended water droplets caused by condensation.
(b) Deposition of atmospheric moisture.
(c) Almost microscopically small drops of water condensed from and suspended in air.
(d) The moisture content of the atmosphere at a particular time and place.

(*d*) Humidity refers to the moisture content of the atmosphere at a particular time and place. It is measured in either relative terms (relative humidity) or absolute terms (absolute humidity).

It is relative humidity which indicate the likelihood of precipitation.

Fog is a form of suspended water droplets caused by condensation.

Fog can be considered a type of cloud at very low altitude. It reduces visibility and is a hazard.

Almost microscopically small drops of water condensed from and suspended in air is initial stage of cloud formation. When these droplets combine together, clouds are formed.

94. The Shompens are the vulnerable tribal group of

(a) Jharkhand
(b) Odisha
(c) West Bengal
(d) Andaman and Nicobar Islands

(*d*) The Shompens are inhabitant of Andaman and Nicobar Islands. Many more tribes live on this island such as the Great Andamanes, Onge, Jarawa and Sentinelese.

95. Which one of the following cities was not included in the list of smart cities in India?

(a) Silvassa (b) Jorhat
(c) Itanagar (d) Kavaratti

(*b*) The smart city mission was launched by Government of India to develop smart cities in country.

Silvassa and Kavaratti is included in the list of smart cities. Itanagar (Arunachal Pradesh) was also included as a special case in Smart City Mission in August, 2017.

Shillong became 100th city, added to the smart city mission. Jorhat (Assam) is not included among the smart cities.

96. Find the correct arrangement of the following urban agglomerations in descending order as per their population size according to Census 2011.

(a) Delhi–Mumbai–Kolkata–Chennai
(b) Mumbai–Delhi–Kolkata–Chennai
(c) Mumbai–Kolkata–Delhi–Chennai
(d) Kolkata–Chennai–Mumbai–Delhi

(*b*) According to the Census of 2011 Mumbai is the most populated urban agglomerations followed by Delhi, Kolkata and Chennai.

There are total 53 million plus cities in India as per Census of 2011. Least populated among them is Kota.

97. Match List I with List II and select the correct answer using the codes given below the lists.

List I (Type of Lake)		List II (Example)	
A.	Tectonic	1.	Lonar Lake
B.	Crater	2.	Gangabal Lake
C.	Glacial	3.	Purbasthali Lake
D.	Fluvial	4.	Bhimtal Lake

Codes

	A	B	C	D		A	B	C	D
(a)	4	1	2	3	(b)	4	2	1	3
(c)	3	1	2	4	(d)	3	2	1	4

(*a*)

Type of Lake	Example
Tectonic	**Bhimtal Lake (Uttarakhand)**
Crater	Lonar Lake (Maharashtra)
Glacial	Gangabal Lake (Jammu and Kashmir)
Fluvial	Purbasthali Lake (W. Bengal)

98. The Andaman group of islands and the Nicobar group of islands are separated by which one of the following latitudes?

(a) 8° N latitude (b) 10° N latitude
(c) 12° N latitude (d) 13° N latitude

(*b*) The Andaman and Nicobar group are situated in Bay of Bengal. The Islands are an extension of Arakan mountain ranges. Both the group are separated by 10°N latitude.

The Saddle peak is highest peak and India's only active volcano on these Islands. Barren island is also located in Andaman sea.

99. Daman Ganga Reservoir Project with about 115 km of minor canals and distributaries is located in

(a) NCT
(b) Dadra and Nagar Haveli
(c) Puducherry
(d) Goa

(*b*) The Daman Ganga Reservoir Project is located in Dadra and Nagar Haveli. The Daman Ganga (also known as Daman river) flows through Maharashtra, Gujarat, Daman and Diu and Dadra and Nagar Haveli.

100. Consider the following statements relating to Coal India Limited:
1. It is designated as a 'Maha Ratna' company under the Ministry of Coal.
2. It is the single largest coal producing company in the world.
3. The Headquarters of Coal India Limited is located at Ranchi (Jharkhand).

Which of the statement(s) given above is/are correct?
(a) Only 1(b) 1 and 2
(c) 2 and 3 (d) 1, 2 and 3

(*b*) Coal India Limited is a public sector company under the Ministry of Coal.

Its salient features are
(i) It is designated as a 'Maha Ratna' company under the Ministry of coal.
(ii) It is the single largest coal producing company in the world. Thus statements 1 and 2 are correct.
(iii) Coal India is headquartered at Kolkata, West Bengal.

Thus, statement 3 is incorrect.

101. Afro-Asian solidarity as a central element of India's foreign policy was initiated by which of the following Prime Ministers?
(a) Narendra Modi
(b) IK Gujaral
(c) JL Nehru
(d) Manmohan Singh

(c) Afro-Asian solidarity as a central element of India's foreign policy was initiated by PM Jawaharlal Nehru.

JL Nehru believed that Asia had a certain responsibility toward the people of Africa. In this regard, he followed Afro-Asian solidarity as a central element of India's foreign policy.

102. The Prime Minister's National Relief Fund is operated by which one of the following bodies?
(a) The Prime Minister's Office (PMO)
(b) The National Disaster Management Authority
(c) The Ministry of Finance
(d) The National Development Council (NDC)

(*a*) The Prime Minister's National Relief Fund (PMNRF) was established in 1948 to assist displaced person from Pakistan, now resources of PMNRD are utilised to render relief to families of those killed in natural calamities and man-made disaster. This fund is operated by the Prime Minister's Office (PMO).

103. Which one of the following statements with regard to India's surgical strike mission inside Pakistan occupied Kashmir is correct?
(a) It was conducted in the year 2018
(b) It was led by the Indian Air Force
(c) It was not given any name
(d) It was sanctioned by the United Nations

(*c*) On 29th September, 2016 India announced that it conducted "surgical strikes" against the launch pads of militants across the line of control in Pakistan occupied Kashmir (PoK). The operation was not given any name. In 2019, Balakot strike were conducted by Indian Air Force in response of Pulwama attack.

Hence, option (c) is correct.

104. Which one of the following statements about the National Green Tribunal is not correct?
(a) It was set-up in the year 2010.
(b) It is involved in effective and expeditious disposal of cases relating to environmental protection and conservation of forests.
(c) It may consider giving relief and compensation for damages to persons and property.
(d) It is bound by the procedures laid down under the Code of Civil Procedure, 1908.

(*d*) The Legislate Act of Parliament defines the National Green Tribunal Act, 2010 as follows, ''An act to provide for the establishment of a National Green Tribunal for the effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources including enforcement of any legal right relating to environment and giving relief and compensation for damages to persons and property and for matters connected therewith or incidental thereto."

The tribunal shall not be bound by the procedure laid down under the Code of Civil Procedure, 1908, but shall be guided by principles of natural justice.

Hence, option (d) is not correct.

105. Which one of the following statements about the provisions of the Constitution of India with regard to the State of Jammu and Kashmir is not correct. Prior to arogational of Article 370.
(a) The Directive Principles of State Policy do not apply.
(b) Article-35A gives some special rights to the permanent residents of the State with regard to employment, settlement and property.
(c) Article-19(1)(f) has been omitted.
(d) Article-368 is not applicable for the Amendment of Constitution of the State.

(*c*) The Government of India abrogated the Article 370 in August 2019, thus making all the provisions of the Indian Constitution applicable to Jammu and Kashmir.

Prior to its abrogation, Article 370 provided the State of Jammu and Kashmir, power to have a separate Constitution, a state flag and autonomy over the internal administration of the state.

According to question, option (c) is not correct.

106. In 1921, during which one of the following tours, Gandhiji shaved his head and began wearing loincloth in order to identify with the poor?
(a) Ahmedabad (b) Champaran
(c) Chauri Chaura (d) South India

(d) Gandhiji was on Journey down to South India (Madurai) in 1921.

He asked the people to wear Khadi, but they replied that they are too poor, cannot afford Khadi clothes, after this incident Gandhiji decided to shave his head and began wearing loincloth in order to identify with the poor.

107. Simla was founded as a hill station to use as strategic place for billeting troops, guarding frontier and launching campaign during the course of
(a) Anglo-Maratha War
(b) Anglo-Burmese War
(c) Anglo-Gurkha War
(d) Anglo-Afghan War

(*c*) Simla was founded as a hill station by the Britisher during Anglo-Gurkha or Anglo-Nepalese war fought between 1814-16.

The war ended with the signing of the treaty of Sugauli in 1816. The strategic location of Simla decided its for times.

108. Which politician in British India had opposed to a Pakistan that would mean ''Muslim Raj here and Hindu Raj elsewhere''?
(a) Khan Abdul Ghaffar Khan
(b) Sikandar Hayat Khan
(c) Maulana Abul Kalam Azad
(d) Rafi Ahmed Kidwai

(*b*) Premier of Punjab Sir Sikandar Hyat Khan predicted to told the Punjab Legislative Assembly.

On 11th March 1941 "We do not ask for freedom that there may be Muslim Raj here and Hindu Raj elsewhere. If that is what Pakistan means I will have nothing to do with it.

109. Match List I with List II and elect the correct answer using the codes given below the lists.

	List I (Author)		List II (Book)
A.	Sekhar Bandyopadhyay	1.	*Jawaharlal Nehru : A Biography, Vol-I, 1889-1947*
B.	Sarvepalli Gopal	2.	*From Plassey to Partition : A History of Modern India*
C.	David Hardiman	3.	*The Ascendancy of the Congress in Uttar Pradesh, 1926-1934*
d.	Gyanendra Pandey	4.	*Gandhi in His Time and Ours*

Codes

	A	B	C	D		A	B	C	D
(a)	2	4	1	3	(b)	2	1	4	3
(c)	3	1	4	2	(d)	3	4	1	2

(*b*)

Sekhar Bandyopadhyay	*From Plassey to Partition: A History of Modern India*
Sarvepalli Gopal	*Jawaharlal Nehru : A Biography, Vol-I,* 1889-1947
David Hardiman	*Gandhi in His Time and Ours*
Gyanendra Pandey	*The Ascendancy of the Congress in Uttar Pradesh,* 1926 - 1934

110. Eight states have achieved more than 99% household electrification prior to the launch of 'Saubhagya Scheme'.
Which one of the following is not among them?
(a) Kerala
(b) Punjab
(c) Himachal Pradesh
(d) Madhya Pradesh

(*d*) Eight states which achieved more than 99% household electrification prior to the launch of Saubhagya Scheme are Andhra Pradesh, Gujarat, Goa, Haryana, Himachal Pradesh, Kerala, Punjab and Tamil Nadu.

So, Madhya Pradesh is not among them. Pradhan Mantri Sahaj Bijli Har Ghar Yojna (Saubhagya) to ensure electrification of all willing household in the country.

111. In October, 2018, India was elected as a member to the United Nations Human Rights Council for a period of
(a) five years (b) four years
(c) three years (d) two years

(*c*) India was elected for three years (till October 2021) as a member of UNHRC. India received 188 votes, the highest polled by any of the 18 countries elected in the voting.

This is the 5th time India is elected to the Geneva-based Council, the main body of the UN charged with promoting and monitoring human rights. India will join China and Nepal, besides Pakistan, which were elected to the 47 Member Council in previous years to serve three year terms.

112. Consider the following statements about the Bureau of Pharma PSUs of India (BPPI)
1. It is the implementing agency of Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP).
2. It has been registered as an independent society under the Societies Registration Act, 1860.

Which of the statements given above is/are correct?
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

(*c*) Both the statement are true. Pradhan Mantri Bhartiya Janaushadhi Pariyojana was launched in 2015 with an objective to make quality education available of affordable prices for all particularly, poor and disadvantage.

It is implemented by Bureau of Pharma PSUs of India (BPPI) which is registered as an independents society under the Societies Registration Act, 1860.

Hence, both the statements are correct.

113. Consider the following statements about a scheme launched by the Government of India.
It was launched to provide social security during old age and to protect elderly persons aged 60 years and above against a future fall in their interest income due to uncertain market conditions. The scheme enables old age income security for senior citizens through provision of assured pension/return linked to the subscription amount based on government guarantee to Life Insurance Corporation of India (LICI). Identify the scheme.
(a) Pradhan Mantri Swasthya Suraksha Yojana
(b) Pradhan Mantri Vaya Vandana Yojana
(c) Liveability Index Programme
(d) Rashtriya Vayoshri Yojana

(*b*) Pradhan Mantri Vaya Vandana Yojana (PMVVY) provides an assured pension based on a guaranteed rate of return of 8% per annum for 10 year. Its salient feature
(i) Investment limit extended upto 15 lakh.
(ii) Time period for subscription up to 21st March, 2020.
(iii) On premature exit 98% purchase price will be refunded.
(iv) Senior citizen will get a pension upto 10000 per month.

114. Who among the following won India's first ever gold medal in the International Youth Olympic Games (2018) held in Argentina?
(a) Neeraj Chopra
(b) Praveen Chitravel
(c) Jeremy Lalrinnunga
(d) Suraj Panwar

(*c*) Jeremy Lalrinnunga becomes India's first ever gold medalist after winning gold in the mens 62 kg weightlifting competition.

He hails from State of Mizoram. India ranked 17th with 13 medals which include 3 Gold, 9 Silver and 1 Bronze.

115. EK Janaki Ammal National Award on Taxonomy is administered by the

(a) Ministry of Agriculture and Farmers Welfare
(b) Ministry of New and Renewable Energy
(c) Ministry of Health and Family Welfare
(d) Ministry of Environment, Forest and Climate Change

(***d***) EK Janaki Ammal National Award on Taxonomy is administered by the Ministry of Environment, Forest and Climate change.

This award is categorised in three fields Plants, Animals and Microbial Taxonomy.

The award is named after EK Janaki Ammal legendary Botanist from Kerala.

116. Which one of the following pairs of military training institute of India and location is not correctly matched?

(a) Army War College : Mhow
(b) High Altitude Warfare School : Gulmarg
(c) Army Air Defence College : Pune
(d) Rashtriya Indian Military College : Dehradun

(***c***)

Institute	Location
Army War College	Mhow (Madhya Pradesh)
High Altitude Warfare School	Gulmarg (Jammu and Kashmir)
Army Air Defence College	Gopalpur (Odisha)
Rashtriya Indian Military College	Dehradun (Uttarakhand)

Army War College is located in Mhow (Madhya Pradesh). It is a Tactical training and research institution established in 1971.

High Altitude Warfare School is located in Gulmarg (Jammu and Kashmir). It is a training and research establishment of Indian Army set up in 1948.

Army Air Defence College is located in Gopalpur (Odisha). It is a training academy of Indian Army established in 1940.

Rashtriya Indian Military College is located in Dehradun (Uttarakhand). It is a military school for boys established in 1922.

117. Which one of the following viruses is responsible for the recent death of lions in Gir National Park?

(a) Canine Distemper Virus
(b) Nipah Virus
(c) Hendra Virus
(d) Foot–and–Mouth Disease Virus

(***a***) Canine Distemper Virus is the main cause of recent death of lions in Gir National Park. It is a member of Paramyxoviridae family that causes canine distemper disease which affects a wide variety of animal families, i.e., dogs, cats, etc.

This disease is highly contagious *via* inhalation. It affects gestrointestinal and respiratory tracts and sometimes nervous system. High fever, eye inflammation, coughing, diarrhea, loss of appetite, hardening of nose, etc, are its common symptoms.

118. Till 2018, which of the following countries have legalised the possession and use of recreational cannabis?

1. America
2. Canada
3. Nigeria
4. Uruguay

Select the correct answer using the codes given below

(a) 1, 2 and 3 (b) 2 and 4
(c) 1 and 4 (d) 1, 2 and 4

(***b***) Cannabis (Marijuana), is a psycho active drug used for medical and recreational purpose.

Canada became the second country after Uruguay to Legalise possession and use of recreational cannabis.

In US, it is legalised at federal level, but few State allow use of cannabis for recreational purpose.

119. Which of the following are the benefits of the Pradhan Mantri Jan Arogya Yojana (PMJAY)?

1. Free treatment available at all public and empanelled private hospitals in times of need.
2. Cashless and paperless access to quality health-care services.
3. Government provides health insurance cover of up to ₹ 5,00,000 per family per year.
4. Pre-existing diseases are not covered.

Select the correct answer using the codes given below

(a) 1 and 3 (b) 1, 2 and 3
(c) 2 and 4 (d) 2, 3 and 4

(***b***) Benefits of PMJAY Scheme have been enumerated below

(i) **Ayushman Bharat** Pradhan Mantri Jan Arogya Yojana (PMJAY) will provide a cover of up to ₹ 5 lakhs per family per year, for secondary and tertiary care hospitalisation.
(ii) When fully implemented, PMJAY will become the world's largest fully government-financed health protection scheme.
It is a visionary step towards advancing the agenda of Universal Health Coverage (UHC).
(iii) PMJAY will provide cashless and paperless access to service for the beneficiary at the point of service.
(iv) PMJAY will help reduce catastrophic expenditure for hospitalisations, which impoverishes people and will help mitigate the financial risk arising out of catastrophic health episodes.
(v) Over 10.74 crore vulnerable entitled families (approximately 50 crore beneficiaries) will be eligible for these benefits.
(vi) Entitled families will be able to use the quality health services they need without facing financial hardships.

Hence, option (b) is the correct.

120. The 11th BRICS Summit in 2019 hosted by

(a) China (b) Russia
(c) Brazil (d) India

(***c***) The 11th BRICS Summit in 2019 was held in Brazil. BRICS is an association of five major emerging national economies. It include Brazil, Russia, India, China and South Africa. The 12th edition of BRICS Summit will be hosted by Russia at Saint Petersburg in July 2020.

CDS

Combined Defence Service

SOLVED PAPER 2018 (II)

PAPER I Elementary Mathematics

1. The highest four-digit number which is divisible by each of the numbers 16, 36, 45, 48 is

(a) 9180 (b) 9360

(c) 9630 (d) 9840

(b) Number which is divisible by 16, 36, 45 and 48 = LCM of (16, 36, 45, 18)

LCM of 16, 36, 45, 48 = 720

Now, highest four digit number

= 9999

i.e. 9999 = 720 × 13 + 639

∴ Highest four digit number exactly divisible by 16, 36, 45 and 48

= 9999 − 639 = 9360

2. If $x = y^a$, $y = z^b$ and $z = x^c$, then the value of abc is

(a) 1 (b) 2

(c) −1 (d) 0

(a) Given, $x = y^a$

Or, $x = (z^b)^a$ (as $y = z^b$)

Or, $x = \left((x^c)^b\right)^a$ (as $z = x^c$)

Or, $x = x^{abc}$

Or, $abc = 1$

3. If $x = 2 + 2^{2/3} + 2^{1/3}$, then the value of the expression $x^3 - 6x^2 + 6x$ will be

(a) 2 (b) 1

(c) 0 (d) −2

(a) Given, $x = 2 + 2^{2/3} + 2^{1/3}$

$\Rightarrow \quad x - 2 = 2^{2/3} + 2^{1/3}$

Cubing both sides, we get

$(x-2)^3 = (2^{2/3} + 2^{1/3})^3$

$\Rightarrow \quad x^3 - 2^3 - 3(x)(2)(x-2)$

$= (2^{2/3})^3 + (2^{1/3})^3 + 3(2)^{2/3}$

$(2)^{1/3}(2^{2/3} + 2^{1/3})$

$(\because (a-b)^3 = a^3 - b^3 - 3ab\,(a-b)$

$(a+b)^3 = a^3 + b^3 + 3ab\,(a+b)$

$\Rightarrow x^3 - 8 - 6x^2 + 12x$

$= 2^2 + 2 + 6(x-2)$

$\Rightarrow x^3 - 6x^2 + 12x = 6 + 8 + 6x - 12$

$\Rightarrow x^3 - 6x^2 - 6x = 2$

4. How many five-digit numbers of the form $XXYXX$ is/are divisible by 33?

(a) 1 (b) 3

(c) 5 (d) Infinite

(b) For a number to be divisible by 33 it must be divisible by 3 and 11.

Now, a number is divisible by 11, if the difference of the sum of digits at odd places and the sum of digits at even places is either zero or divisible by 11.

∴ For $XXYXX$ to be divisible by 11

$2X + Y - 2X$ i.e. Y must either be 0 or divisible by 11.

Now, Y cannot be divisible by 11 because it is a single digit number.

∴ $Y = 0$

Now, for a number to be divisible by 3 the sum of digits must be divisible by 3.

i.e. $X + X + 0 + X + X$ or 4X must be divisible by 3.

This would happen only when X is either 3, 6 or 9.

Hence, 3 five digit numbers of the form $XXYXX$ are divisible by 33 viz. 33033, 66066, 99089.

5. A five-digit number $XY235$ is divisible by 3 where X and Y are digits satisfying $X + Y \le 5$. What is the number of possible pairs of values of (X, Y)?

(a) 5 (b) 6 (c) 7 (d) 9

(c) Given, $XYZ35$ is divisible by 3 and $X + Y \le 5$

Now, for a number to be divisible by 3, the sum of digits must be divisible by 3.

i.e. $X + Y + 2 + 3 + 5$ or $10 + X + Y$ is divisible by 3.

Now, two cases are possible, i.e. 12 and 15.

∴ Possible values of

$(X, Y) = (1, 1), (2, 0), (1, 4), (2, 3),$

$(3, 2), (4, 1)$ *and* $(5, 0)$

i.e. 7 possible values are there.

6. If $x^2 - 6x - 27 > 0$, then which one of the following is correct?

(a) $-3 < x < 9$ (b) $x < 9$ or $x > -3$

(c) $x > 9$ or $x < -3$ (d) $x < -3$ only

(c) Given, $x^2 - 6x - 27 > 0$

Or $\quad x^2 - 9x + 3x - 27 > 0$

$x(x-9) + 3(x-9) > 0$

$(x+3)(x-9) > 0$

Now, for $(x+3)\ (x-9)$ to be greater than 0.

Either $(x+3)$ and $(x-9)$ must be greater than 0.

or both $(x+3)$ and $(x-9)$ must be less than 0.

$\therefore x > 9$ and $x < -3$

7. The number of divisors of the number 38808 exclusive of the divisors 1 and itself, is

(a) 74 (b) 72

(c) 70 (d) 68

(c) To find the number of divisors of a number we must write it in the form of $a^m b^n c^o d^p \ldots$ where $a, b, c, d \ldots$ are prime numbers.

Now,
$38808 = 2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 7 \times 11$

$= 2^3 \times 3^2 \times 7^2 \times 11^1$

Now, number of divisors of 38808 except 1 and 38808

$= (3+1)(2+1)(2+1)(1+1) - 2$
$= 4 \times 3 \times 3 \times 2 - 2$
$= 72 - 2 = 70$

8. HCF and LCM of two polynomials are $(x+3)$ and $(x^3 - 9x^2 - x + 105)$ respectively. If one of the two polynomials is $x^2 - 4x - 21$, then the other is

(a) $x^2 + 2x - 21$ (b) $x^2 + 2x + 15$
(c) $x^2 - 2x - 15$ (d) $x^2 - x - 15$

(c) Given, HCF of polynomials
$= x + 3$

LCM of polynomials
$= x^3 - 9x^2 - x + 105$

One polynomial $= x^2 - 4x - 21$

We know that, (HCF × LCM) of two numbers = Product of two number

∴ Other polynomial

$$= \frac{(x+3)(x^3 - 9x^2 - x + 105)}{x^2 - 4x - 21}$$
$$= \frac{(x+3)(x^3 - 9x^2 - x + 105)}{(x-7)(x+3)}$$
$$= \frac{x^3 - 9x^2 - x + 105}{x - 7}$$

$\Rightarrow$ Other polynomial $= x^2 - 2x - 15$

9. If α and β are two real numbers such that $\alpha + \beta = -\frac{q}{p}$ and $\alpha\beta = \frac{r}{p}$, where $1 < p < q < r$, then which one of the following is the greatest?

(a) $\frac{1}{\alpha+\beta}$ (b) $\frac{1}{\alpha} + \frac{1}{\beta}$

(c) $-\frac{1}{\alpha\beta}$ (d) $\frac{\alpha\beta}{\alpha+\beta}$

(c) Given, $\alpha + \beta = \frac{-q}{p}, \alpha\beta = \frac{r}{p}$

and $1 < p < q < r$

From option(a) $\frac{1}{\alpha+\beta} = \frac{-p}{q}$

Option (b)
$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-q}{p} \times \frac{p}{r} = \frac{-q}{r}$$

Option (c) $\frac{-1}{\alpha\beta} = \frac{-p}{r}$

Option (d) $\frac{\alpha\beta}{\alpha+\beta} = \frac{r}{p} \times \left(\frac{-p}{q}\right) = \frac{-r}{q}$

Now, except (d) in all other options the numerator is smaller than denominator.

∴ Option (b) is smallest.

Now, from option (a) and (c), i.e. $\frac{-p}{q}$ and $\frac{-p}{r}$, $\frac{-p}{r}$ is greater as $r > q$

Now, from option (c) and (b), i.e. $\frac{-p}{r}$ and $\frac{-q}{r}$

$\frac{-p}{r}$ is greater as $q > p$

∴ $-\frac{1}{\alpha\beta}$ is greatest.

10. Two workers 'A' and 'B' working together completed a job in 5 days. Had 'A' worked twice as efficiently as he actually did and 'B' worked one-third as efficiently as he actually did, the work would have completed in 3 days. In how many days could 'A' alone complete the job?

(a) $3\frac{1}{2}$ days (b) $4\frac{1}{6}$ days

(c) $5\frac{1}{2}$ days (d) $6\frac{1}{4}$ days

(d) Let the efficiency of A and B be 'a' and 'b' respectively.

According to the question,

$$(a+b)5 = \left(2a + \frac{b}{3}\right)(3)$$

$\Rightarrow 5a + 5b = 6a + b$

$\Rightarrow \quad a = 4b$

or $\quad \frac{a}{b} = \frac{4}{1}$

Now, efficiency $\propto \frac{1}{\text{Time}}$

∴ Ratio of time taken A and B = 1 : 4

Let time taken by A to finish the job be x days

∴ Time taken by B to finish the job = $4x$ days

Now, according to the question,

$$\frac{1}{x} + \frac{1}{4x} = \frac{1}{5}$$
$$\Rightarrow \quad \frac{4x + x}{4x^2} = \frac{1}{5}$$
$$\Rightarrow \quad \frac{5x}{4x^2} = \frac{1}{5}$$
$$\Rightarrow \quad x = \frac{25}{4} = 6\frac{1}{4} \text{ days}$$

∴ Number of days taken by A to complete job $6\frac{1}{4}$ days

11. If $x^6 + \frac{1}{x^6} = k\left(x^2 + \frac{1}{x^2}\right)$, then k is equal to

(a) $\left(x^2 - 1 + \frac{1}{x^2}\right)$ (b) $\left(x^4 - 1 + \frac{1}{x^4}\right)$

(c) $\left(x^4 + 1 + \frac{1}{x^4}\right)$ (d) $\left(x^4 - 1 - \frac{1}{x^4}\right)$

(b) We know that,
$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

$$\therefore \quad x^6 + \frac{1}{x^6} = (x^2)^3 + \left(\frac{1}{x^2}\right)^3$$
$$= \left(x^2 + \frac{1}{x^2}\right)\left(x^4 - 1 + \frac{1}{x^4}\right)$$
$$\therefore \quad k = x^4 - 1 + \frac{1}{x^4}$$

12. If the sum of the squares of three consecutive natural numbers is 110, then the sum of their cubes is

(a) 625 (b) 654 (c) 684 (d) 725

(c) Let the three consecutive natural numbers be

$(x-1), x, (x+1)$

According to the question,

$\Rightarrow (x-1)^2 + x^2 + (x+1)^2 = 110$

$\because \quad (a-b)^2 = a^2 + b^2 - 2ab$
$\quad (a+b)^2 = a^2 + b^2 + 2ab$

$\Rightarrow x^2 + 1 - 2x + x^2 + x^2 + 1 + 2x = 110$

$\Rightarrow \quad 3x^2 = 110 - 2$

$\Rightarrow \quad x^2 = \frac{108}{3} = 36$

$\therefore \quad x = 6$

∴ The numbers are (6 – 1), 6, (6 + 1)
i.e. 5, 6 and 7

Now, sum of their cubes
$= 5^3 + 6^3 + 7^3$
$= 125 + 216 + 343 = 684$

13. The product of two integers p and q, where $p > 60$ and $q > 60$, is 7168 and their HCF is 16. The sum of these two integers is

(a) 256 (b) 184 (c) 176 (d) 164

(c) Given, p and q are integers where $p > 60$ and $q > 60$ also $p.q = 7168$

And HCF of p and $q = 16$

Now, the HCF is 16 that means both 'p' and 'q' are divisible by 16

So, p and q might be 64, 80, 96, 112, ...

Now, $64 \times 112 = 7168$

$\therefore$ Required integers = 64, 112

And sum of these two integers $= 64 + 112 = 176$

14. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then the value of $\log_{10}(0.72)$ is equal to

(a) 0.9286 (b) 1.9286

(c) 1.8572 (d) 1.9572

(b) We have, $\log_{10} 2 = 0.3010$

and $\log_{10} 3 = 0.4771$

$$\log_{100}(0.72) = \log_{10^2}(0.72)$$
$$= \frac{1}{2}\log_{10}(0.72)$$
$$= \frac{1}{2}\log_{10}\frac{72}{100}$$
$$= \frac{1}{2}[\log_{10}72 - \log_{10}100]$$
$$= \frac{1}{2}[\log_{10}9 \times 8 - \log_{10}10^2]$$
$$= \frac{1}{2}[\log_{10}9 + \log_{10}8 - 2\log_{10}10]$$
$$= \frac{1}{2}[\log_{10}3^2 + \log_{10}2^3 - 2]$$
$$= \frac{1}{2}[2\log_{10}3 + 3\log_{10}2 - 2]$$
$$= \frac{1}{2}[2 \times 0.4771 + 3 \times 0.3010 - 2]$$
$$= \frac{1}{2}[0.9542 + 0.9030 - 2]$$
$$= \frac{1}{2}[1.8572 - 2] = 0.9286 - 1 = \bar{1}.9286$$

15. If $a^x = b^y = c^z$ and $abc = 1$, then the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ will be equal to

(a) –1 (b) 0 (c) 1 (d) 3

(b) Given, $a^x = b^y = c^z = k$ (let)

and $abc = 1$

Now, $a^x = k$

or $x = k^{1/x}$

Similarly, $b = k^{1/y}$ and $c = k^{1/z}$

Now, $abc = 1$ or $k^{1/x}.k^{1/y}.k^{1/z} = 1$

or $k^{\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)} = 1$

or $k^{\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)} = k^0$

or $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$

16. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b}$ is

(a) $\frac{a}{bc}$ (b) $\frac{b}{ac}$ (c) $\frac{c}{ab}$ (d) $\frac{1}{abc}$

(b) We have, α, β are the roots of the equation

$$ax^2 + bx + c = 0$$

$$\therefore \alpha + \beta = \frac{-b}{a}, \alpha\beta = \frac{c}{a}$$

Now, $\frac{1}{a\alpha + b} + \frac{1}{a\beta + b}$

$$= \frac{a\beta + b + a\alpha + b}{(a\alpha + b)(a\beta + b)}$$
$$= \frac{a(\alpha + \beta) + 2b}{a^2(\alpha\beta) + ab(\alpha + \beta) + b^2}$$
$$= \frac{a\left(\frac{-b}{a}\right) + 2b}{a^2\left(\frac{c}{a}\right) + ab\left(\frac{-b}{a}\right) + b^2}$$
$$= \frac{b}{ac}$$

17. Consider the following statements in respect of three 3-digit numbers XYZ, YZX and ZXY:

1. The sum of the numbers is not divisible by $(X + Y + Z)$.
2. The sum of the numbers is divisible by 111.

Which of the above statements is / are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

(b) We have, 3-digits numbers are XYZ, YZX, ZXY

$XYZ = 100X + 10Y + Z$

$YZX = 100Y + 10Z + X$

$ZXY = 100Z + 10X + Y$

$XYZ + YZX + ZXY$
$= 100(X + Y + Z) + 10(X + Y + Z) + X + Y + Z$
$= (X + Y + Z)(100 + 10 + 1)$
$= (X + Y + Z)(111)$

$\therefore$ The sum of number is divisible by $X + Y + Z$ and 111.

18. The number of all pairs (m, n), where m and n are positive integers, such that $\frac{1}{m} + \frac{1}{n} - \frac{1}{mn} = \frac{2}{5}$ is

(a) 6 (b) 5 (c) 4 (d) 2

(c) We have, $\frac{1}{m} + \frac{1}{n} - \frac{1}{mn} = \frac{2}{5}$

$$\Rightarrow \frac{n + m - 1}{mn} = \frac{2}{5}$$
$$5n + 5m - 5 = 2mn$$
$$5m - 5 = 2mn - 5m$$

Here, m and n such positive integer

$\therefore$ Possible value are

(3, 10), (4, 5), (5, 4), (10, 3)

19. If $a = xy^{p-1}$, $b = yz^{q-1}$, $c = zx^{r-1}$, then $a^{q-r}b^{r-p}c^{p-q}$ is equal to

(a) abc (b) xyz

(c) 0 (d) None of these

(d) We have,

$$a = xy^{p-1}, b = yz^{q-1}, c = zx^{r-1}$$

Now, $a^{q-r}b^{r-p}c^{p-q}$

$$= (xy^{p-1})^{q-r}(yz^{q-1})^{r-p}(zx^{r-1})^{p-q}$$
$$= x^{q-r}y^{(p-1)(q-r)}y^{r-p}z^{(q-1)(r-p)}z^{p-q}x^{(r-1)(p-q)}$$
$$= x^{(q-r)+(r-1)(p-q)}y^{(p-1)(q-r)+r-p}z^{(q-1)(r-p)+p-q}$$
$$= x^{q-r+pr-qr-p+q}y^{pq-pr-q+r+r-p}z^{qr-pq-r+p+p-q}$$
$$= x^{2q-p+r(p+q-1)}y^{2r-q+p(q-r-1)}z^{2p-r+q(r-p-1)}$$

20. The number of sides of two regular polygons are in the ratio 5 : 4. The difference between their interior angles is 9°. Consider the following statements

1. One of them is a pentagon and the other is a rectangle.
2. One of them is a decagon and the other is an octagon.
3. The sum of their exterior angles is 720°.

Which of the above statements is/are correct?

(a) 1 only

(b) 2 only

(c) 1 and 3

(d) 2 and 3

(d) Let the numbers of sides of two regular polygons is $5x$ and $4x$

$\therefore$ Interior angles are $\frac{5x-2}{5x}\times 180^\circ$

and $\frac{4x-2}{4x}\times 180^\circ$

respectively

According to the question,

$$\frac{5x-2}{5x}\times 180^\circ - \frac{4x-2}{4x}\times 180^\circ = 9^\circ$$

$$\Rightarrow \frac{180^\circ}{x}\left[\frac{5x-2}{5}-\frac{4x-2}{4}\right] = 9^\circ$$

$$\Rightarrow \frac{180^\circ}{x}\left[\frac{20x-8-20x+10}{20}\right] = 9^\circ$$

$$\Rightarrow x = 2$$

$\therefore$ Number of sides of polygons are 10 and 8 respectively

$\therefore$ One of them decagon and other is octagon

We know that, sum of exterior angle of regular polygon is 360°

$\therefore$ Sum of exteriors angles of polygons are $360^\circ + 360^\circ = 720^\circ$

21. The minimum value of the expression $2x^2+5x+5$ is

(a) 5 (b) 15/8

(c) $-15/8$ (d) 0

(b) Let $y = 2x^2+5x+5$

$$y = 2\left(x^2+\frac{5}{2}x+\frac{5}{2}\right)$$

$$y = 2\left(x^2+\frac{5}{2}x+\frac{25}{16}\right)+5-\frac{25}{8}$$

$$y = 2\left(x+\frac{5}{4}\right)^2+\frac{15}{8}$$

$\therefore$ Minimum value of y is $\frac{15}{8}$

22. If H is the harmonic mean P and Q, then the value of $\frac{H}{P}+\frac{H}{Q}$ is

(a) 1 (b) 2

(c) $\frac{P+Q}{PQ}$ (d) $\frac{PQ}{P+Q}$

(b) We know that, harmonic mean of a and b is

$$H.M = \frac{2ab}{a+b}$$

$$\therefore \quad H = \frac{2PQ}{P+Q}$$

$$\Rightarrow \quad \frac{2}{H} = \frac{P+Q}{PQ}$$

$$\Rightarrow \quad 2 = H\left(\frac{1}{P}+\frac{1}{Q}\right)$$

$$\Rightarrow H\left(\frac{1}{P}+\frac{1}{Q}\right) = 2$$

$$\Rightarrow \quad \frac{H}{P}+\frac{H}{Q} = 2$$

23. The sum of all possible products taken two at a time out of the numbers $\pm 1, \pm 2, \pm 3, \pm 4$ is

(a) 0 (b) -30 (c) 30 (d) 55

(b) Let ± 1, ± 2, ± 3, ± 4 are the roots of the polynomials

$$\therefore (x+1)(x-1)(x+2)(x-2)(x+3)(x-3)(x+4)(x-4) = 0$$

$$\Rightarrow (x^2-1)(x^2-4)(x^2-9)(x^2-16) = 0$$

$$\Rightarrow (x^4-5x^2+4)(x^4-25x^2+144) = 0$$

$$\Rightarrow x^8-30x^6+148x^4-820x^2+576 = 0$$

The sum of the roots taken two at a time is coefficient of x^6 is -30

24. The remainder when $3x^3-2x^2y-13xy^2+10y^3$ is divided by $(x-2y)$ is equal to

(a) 0 (b) y

(c) $y-5$ (d) $y+3$

(a) Let $f(x)$

$$= 3x^3-2x^2y-13xy^2+10y^3$$

And $\quad g(x) = x-2y$

When $f(x)$ is divided by $g(x)$

$$\therefore \quad f(2y) = 3(2y)^3-2(2y)^2y-13(2y)y^2+10y^3$$

$$f(2y) = 24y^3-8y^3-26y^3+10y^3$$

$$f(2y) = 0$$

Hence, remainder = 0

25. If $ab+bc+ca=0$, then the value of $\frac{(b^2-ca)(c^2-ab)+(a^2-bc)(c^2-ab)+(a^2-bc)(b^2-ca)}{(a^2-bc)(b^2-ca)(c^2-ab)}$ is

(a) -1 (b) 0 (c) 1 (d) 2

(b) We have, $ab+bc+ca=0$

Now,

$$\frac{(b^2-ca)(c^2-ab)+(a^2-bc)(c^2-ab)+(a^2-bc)(b^2-ca)}{(a^2-bc)(b^2-ca)(c^2-ab)}$$

$$= \frac{1}{a^2-bc}+\frac{1}{b^2-ca}+\frac{1}{c^2-ab}$$

$$= \frac{1}{a^2+ab+ac}+\frac{1}{b^2+ab+bc}+\frac{1}{c^2+bc+ca}$$

$$= \frac{1}{a(a+b+c)}+\frac{1}{b(a+b+c)}+\frac{1}{c(a+b+c)}$$

$$= \frac{1}{(a+b+c)}\left[\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right]$$

$$= \frac{1}{(a+b+c)}\left(\frac{ab+bc+ca}{abc}\right) = 0$$

26. What is the principal amount which earns ₹ 210 as compound intrest for the second year at 5% per annum?

(a) ₹ 2000 (b) ₹ 3200

(c) ₹ 4000 (d) ₹ 4800

(c) Let principal be ₹ P

$\because$ Amount = Principal $\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$

$\therefore$ Amount after one year

or, $= P\left(1+\frac{5}{100}\right)^1 = \frac{21P}{20}$

or CI after one year $= \frac{21P}{20} - P = \frac{P}{20}$

Similarly, amount after two years

$$= P\left(1+\frac{5}{100}\right)^2 = \frac{441}{400}P$$

and CI after two years $= \frac{441P}{400} - P = \frac{41P}{400}$

According to the question,

$$\frac{41P}{400}-\frac{P}{20} = 210$$

$$\Rightarrow \quad \frac{41P-20P}{400} = 210$$

$$\Rightarrow \quad 21P = 210\times 400$$

or $\quad$ Principal = ₹ 4000

27. In an examination, 50% of the candidates failed in English, 40% failed in Hindi and 15% failed in both the subjects. The percentage of candidates who passed in both English and Hindi is

(a) 20% (b) 25%

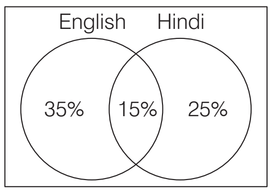
(c) 60% (d) 75%

(b) Given,

50% candidates failed in English

40% candidates failed in Hindi

and 15% candidates failed in both



$\therefore$ Total percentage of candidates who failed either in one or two subjects $= (35+15+25)\% = 75\%$

$\therefore$ Percentage of candidates who passed in both subjects

$= (100 - 75)\% = 25\%$

28. A train 100 m long passes a platform 100 m long in 10 s. The speed of the train is

(a) 36 km/h (b) 45 km/h

(c) 54 km/h (d) 72 km/h

(d) Given, length of train = 100 m

Length of platform = 100 m

Time taken to pass platform = 10 s

Now, distance covered by train 100 m + 100 m = 200 m

$\because$ Speed $= \dfrac{\text{Distance}}{\text{Time}}$

$\therefore$ Speed $= \dfrac{200}{10} = 20$ m/s

$\Rightarrow$ speed of train $= 20 \times \dfrac{3600}{1000}$ km/h

($\because$ 1 km = 1000 m, 1 h = 3600 s)

= 72 km/h

29. A cyclist covers his first 20 km at an average speed of 40 kmp/h, another 10 km at an average speed of 10 kmp/h and the last 30 km at an average speed of 40 kmp/h. Then, the average speed of the entire journey is

(a) 20 km/h

(b) 26.67 km/h

(c) 28.24 km/h

(d) 30 km/h

(b) We know that, Time $= \dfrac{\text{Distance}}{\text{Speed}}$

Time taken by cyclist to cover first 20km $= \dfrac{20}{40} = \dfrac{1}{2}$ h

Time taken by cyclist to cover 10 km

$= \dfrac{10}{10} = 1$ h

And time taken by cyclist to cover remaining 30 km $= \dfrac{30}{40} = \dfrac{3}{4}$ h

Now, total distance travelled

= (20 + 10 + 30) km = 60 km

Total time taken $= \left(\dfrac{1}{2} + 1 + \dfrac{1}{3}\right)$ h

$= \left(\dfrac{2+4+3}{4}\right) = \dfrac{9}{4}$ h

$\therefore$ Required average speed

$= \dfrac{60}{9} \times 4$

$= \dfrac{80}{3}$ km/h

= 26.67 km/hr

30. In a race of 1000 m, A beats B by 150 m, while in another race of 3000 m, C beats D by 400 m. Speed of B is equal to that of D. (Assume that A, B, C and D run with uniform speed in all the events). If A and C participate in a race of 6000 m, then which one of the following is correct?

(a) A beats C by 250 m

(b) C beats A by 250 m

(c) A beats C by 115.38 m

(d) C beats A by 115.38 m

(c) In a race of 1000 m, A beats B by 150 m

So, if A covers 1000 m in 't' s,

then B covers 850m , 't' s

Similarly, if C covers 3000 m in 'T' s, then D covers 2600 m in 'T' s

Now, speed of B is equal to speed of D.

Now, time taken by B cover 2600 m

$$T = \frac{t}{850} \times 2600$$

or $\dfrac{T}{t} = \dfrac{52}{17}$

Let $T = 52x$ and $t = 17x$

Now, A will cover 6000 m in $6t$ s or $17 \times 6x = 102x$

and C will cover 6000 m in $2t$ s

or $2 \times 52x = 104x$

A takes less time it means A beats C

Now, C takes $104x$ to cover 6000 m

So, C will cover distance in

$$2x = \frac{6000}{104} \times 2$$

= 115.38 m

So, A beats C by 115.38 m

31. The sum of ages of a father, a mother, a son Sonu and daughters Savita and Sonia is 96 yr. Sonu is the youngest member of the family. The year Sonu was born, the sum of the ages of all the members of the family was 66 yr. If the father's age now is 6 times that of Sonu's present age, then 12 yr. Hence, the father's age will be

(a) 44 yr (b) 45 yr

(c) 46 yr (d) 48 yr

(d) Sum of ages of father, mother, Sonu, Savita and Sonia is 96 yr and when Sonu was born the sum was 66 yr.

$\because$ There are 5 members in the family.

$\therefore$ Present age of Sonu $= \dfrac{96-66}{5} = 6$ yr

$\therefore$ Present age of father $= 6 \times 6 = 36$ yr

and father's age =12 yr

Hence, = 36 + 12 = 48 yr

32. 'A' is thrice as good a workman as 'B' and takes 10 days less to do a piece of work than 'B' takes. The number of days taken by 'B' alone to finish the work is

(a) 12 (b) 15

(c) 20 (d) 30

(b) A is thrice as good a work man as B

Let efficiency of B be b

$\therefore$ Efficiency of $A = 3b$

Now, A takes 10 days less to do a piece of work.

$\therefore \quad \dfrac{1}{b} - \dfrac{1}{3b} = 10$

$\Rightarrow \quad \dfrac{3b-b}{3b^2} = 10$

$\Rightarrow \quad \dfrac{2b}{3b^2} = 10$

$\Rightarrow \quad b = \dfrac{1}{15}$

$\therefore$ Time taken by B to finish work = 15 days

33. Out of 85 children playing badminton or table tennis or both, the total number of girls in the group is 70% of the total number of boys in the group. The number of boys playing only badminton is 50% of the number of boys and the total number of boys playing badminton is 60% of the total number of boys. The number of children playing only table tennis is 40% of the the total number of children and a total of 12 children play badminton and table tennis both. The number of girls playing only badminton is

(a) 14

(b) 16

(c) 17

(d) 35

(a) Total children = 85

Let number of boys be B

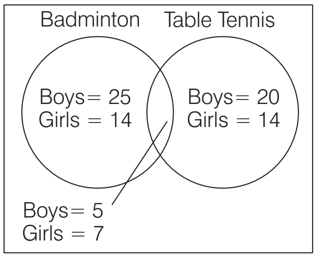
$\therefore$ Number of girls $= \dfrac{70}{100}B$

Now, $B + \dfrac{70}{100}B = 85$

$\therefore \quad B = 50$

And number of girls = 85 – 50 = 35

Now, according to the question,



$\therefore$ Number of girls playing only badminton = 14

34. A person bought two articles X and Y from a departmental store. The sum of prices before sales tax was ₹ 130. There was no sales tax on the article X and 9% sales tax on the article Y. The total amount the person paid, including the sales tax was ₹ 136.75. What was the price of the article Y before sales tax ?

(a) ₹ 75 (b) ₹ 85
(c) ₹ 122 (d) ₹ 125

(a) Let the sum of article X and Y before sales tax be ₹ x and ₹ y respectively

According to the question,

$$x + y = 130 \quad \ldots\text{(i)}$$

Now, after 9% sales tax on item Y

$$x + \frac{109}{100}y = 136.75$$

$$100x + 109y = 13675 \quad \ldots\text{(ii)}$$

Solving Eqs. (i) and (ii), we get

$$x = 55$$
$$y = 75$$

$\therefore$ Cost of item Y before sales tax ₹ 75

35. According to Mr. Sharma's will, half of his property goes to his wife and the rest in equally divided between his two sons, Ravi and Raj. Some years later, Ravi dies and leaves half of his property to his brother Raj. When Raj dies has leaves half of his property to his widow and remaining to his mother, who is still alive. The mother now owns ₹ 88,000 worth of the property. The total worth of the property of Mr. Sharma was

(a) ₹ 1,00,000 (b) ₹ 1,24,000
(c) ₹ 1,28,000 (d) ₹ 1,32,000

(c) Let total property of Mr. Sharma be ₹$16x$

After Mr. Sharma's death

Property of Mrs. Sharma = $8x$

Property of Ravi and Raj = $4x$ and $4x$

After Ravi's death

Property of Raj = $4x + 2x = 6x$

After Raj's death

Property of Mrs. Sharma

$$= 8x + 3x = 11x$$

Now, according to the question,

$$11x = 88000 \text{ or } x = 8000$$

$\therefore$ Property of Mr. Sharma

$$= 16 \times 8000 = ₹\ 1{,}28{,}000$$

36. X bought 4 bottles of lemon juice and Y bought one bottle of orange juice. Orange juice per bottle costs twice the cost of lemon juice per bottle. Z bought nothing but contributed ₹ 50 for his share of the drink which they mixed together and shared the cost equally. It Z's ₹ 50 is covered from his share, then what is the cost of one bottle of orange juice?

(a) ₹ 75 (b) ₹ 50 (c) ₹ 46 (d) ₹ 30

(b) Let the cost of 1 bottle of lemon juice be ₹ x

$\therefore$ Cost of 1 bottle of orange juice

$$= ₹ 2x$$

Total share = ₹ 50 × 3 = ₹ 150

$\therefore \quad 4x + 2x = 150$

or $x = \frac{150}{6} = 25$

$\therefore$ Cost of 1 bottle of orange juice

$$= 25 \times 2 = ₹\ 50$$

37. Ten (10) years before, the ages of a mother and her daughter were in the ratio 3 : 1. In another 10 yr from now, the ratio of their ages will be 13 : 7. What are their present ages?

(a) 39 yr, 21 yr (b) 55 yr, 25 yr
(c) 75 yr, 25 yr (d) 49 yr, 31 yr

(b) Let the age of mother 10 yr ago be $3x$ yr

$\therefore$ Age of daughter 10 yr ago = x yr

According to the question,

$$\frac{3x + 20}{x + 20} = \frac{13}{7}$$

$$\Rightarrow \quad 21x + 140 = 13x + 260$$

$$\Rightarrow \quad 8x = 120$$

Or $\quad x = \frac{120}{8} = 15$

$\therefore$ Present age of mother = $3x + 10$

$= 3 \times 15 + 10 = 55$ yr and present age of daughter = $x + 10 = 25$ yr

38. In a class of 60 boys, there are 45 boys who play chess and 30 boys who play carrom. If every boy of the class plays atleast one of the two games, then how many boys play carrom only?

(a) 30 (b) 20 (c) 15 (d) 10

(c) Total boys = 60

Boys who play chess = 45

Boys who play carrom = 30

$\therefore$ Boys who play both chess and carrom

$$= (45 + 30) - 60 = 15$$

Boys who play only carrom = 30 − 15 = 15

39. Two equal amounts were borrowed at 5% and 4% simple interest. The total interest after 4 yr amounted to ₹ 405. What was the total amount borrowed?

(a) ₹ 1075 (b) ₹ 1100
(c) ₹ 1125 (d) ₹ 1150

(c) Let the amount borrowed be ₹ P each for both the rates.

$\because$ Simple interest

$$= \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

$\therefore$ According to the question,

$$\frac{P \times 5 \times 4}{100} + \frac{P \times 4 \times 4}{100} = 405$$

$$\Rightarrow \quad 20P + 16P = 40500$$

$$\therefore \quad P = \frac{40500}{36} = 1125$$

$\therefore$ Amount borrowed = ₹ 1125

40. Twelve (12) men work 8 h per day and require 10 days to build a wall. If 8 men are available, how many hours per day must they work to finish the work in 8 days?

(a) 10 h (b) 12 h (c) 15 h (d) 18 h

(c) 12 men can do a work in 10 days if they work 8 h per day.

$\therefore$ 12 men can do the work in 80 h

1 man will do the same work in (12×80) h

$$= 960 \text{ h}$$

And 8 men will do the same work = $\frac{960}{8}$ h

Now, to complete the work in 8 days, each men would have to work for $\left(\frac{120}{8}\right)$ h each day.

$\therefore$ Required time = $\frac{120}{8}$ h = 15 h

41. A milk vendor bought 28 L of milk at the rate of ₹ 8.50 /L. After adding some water he sold the mixture at the same price. If his gain is 12.5%, how much water did he add?

(a) 4.5 L (b) 4 L

(c) 3.5 L (d) 3 L

(c) Cost price of 1 L of milk = ₹ 850

Selling price of l L of mixture = ₹ 8.50

Profit percentage = 12.5%

∴ Cost price of 1 L of mixture

$$= \frac{100}{112.5} \times 8.50$$

$$= ₹\ \frac{68}{9}$$

By rule of alligation,

Water ₹ 0 Milk ₹ $\frac{17}{2}$

Mean Price ₹ $\frac{68}{9}$

₹ $\frac{17}{18}$ ₹ $\frac{68}{9}$

Ratio of milk and water in the mixture

$$= \frac{17}{18} : \frac{68}{9} = 1 : 8$$

Now, quantity of milk in mixture

= 28 L

$$\therefore \text{Quantity of water in mixture} = \frac{28}{8} \times 1$$

$$= 3.5 \text{ L}$$

42. The minute hand of a clock overtakes the hour hand after every 72 min of correct time. How much time does the clock lose or gain in a day of normal time?

(a) Lose $130\frac{10}{11}$ min

(b) Lose $157\frac{1}{11}$ min

(c) Gain $121\frac{9}{11}$ min

(d) Gain $157\frac{1}{11}$ min

(a) As we know that in a correct clock, the min hand gains 55 min spaces over the hour hand in 60 min

To be together again, the minute hand must gain 60 min over the hour hand.

∵ 55 min are gained in

$$\left(\frac{60}{55} \times 60\right) = 65\frac{5}{11} \text{ min}$$

But they are together after 72 min

$$\therefore \text{Lose in 72 min} = \left(72 - 65\frac{5}{11}\right) \text{min}$$

$$= 6\frac{6}{11} \text{ min} = \frac{72}{11} \text{ min}$$

$$\therefore \text{Lose in 24 h} = \left(\frac{72}{11} \times \frac{60 \times 24}{72}\right) \text{min}$$

$$= \frac{1440}{11} \text{ min}$$

$$= 130\frac{10}{11} \text{ min}$$

43. A thief steals a car parked in a house and goes away with a speed of 40 km/h. The theft was discovered after half an hour and immediately the owner sets off in another car with a speed of 60km/h. When will the owner meet the thief?

(a) 55 km from the owner's house and one hour after the theft

(b) 60 km from the owner's house and 1.5 h after the theft

(c) 60 km from the owner's house and 1.5 h after the discovery of the theft

(d) 55 km from the owner's house and 1.5 h after the theft

(b) Speed of thief = 40 km/h

Speed of owner = 60 km/h

∵ Distance = Speed × Time

∴ Distance travelled by thief in half hour $= 40 \times \frac{1}{2}$

= 20 km

Now, relative speed of owner and thief = (60 – 40) km/h = 20 km/h

∴ Time taken by owner to catch thief

$$= \frac{20}{20} = 1 \text{ h}$$

Distance covered by thief after the ft

$$= 40 \times \left(1 + \frac{1}{2}\right)$$

$$= 40 \times \frac{3}{2} = 60 \text{ km}$$

∴ The owner will meet the thief at 60 km from the owner's hour and 1.5 h after the theft.

44. *X* and *Y* together can finish a job in 6 days. *X* can alone do the same job in 12 days. How long will *Y* alone take to do the same job?

(a) 16 days (b) 12 days

(c) 10 days (d) 8 days

(b) *X* can do the job in 12 days

$$\therefore \text{Work done by } X \text{ in 1day} = \frac{1}{12}$$

Similarly, work done by *X* and *Y* in 1 day

$$= \frac{1}{6}$$

$$\therefore \text{1 day work of } Y = \frac{1}{6} - \frac{1}{12}$$

$$= \frac{2-1}{12} = \frac{1}{12}$$

∴ *Y* will complete the work in 12 days.

45. Twelve (12) persons can paint 10 identical rooms in 16 days. In how many days can 8 persons paint 20 such rooms?

(a) 12 (b) 24 (c) 36 (d) 48

(d) 12 person can paint 10 identical rooms in 16 days

∴ Time taken by 12 persons to paint 20 such identical rooms = 16 × 2 = 32 days

And time taken by 1 person to paint 20 identical rooms = (32 × 12) days

∴ Time taken by 8 persons to paint 20 such rooms $= \left(\frac{32 \times 12}{8}\right)$ days = 48 days

46. There are n zeros appearing immediately after the decimal point in the value of $(0.2)^{25}$. It is given that the value of $\log_{10} 2 = 0.30103$. The value of n is

(a) 25 (b) 19

(c) 18 (d) 17

(c) Let $x = (0.2)^{25}$

Taking log both sides, we get

$$\log x = \log(0.2)^{25}$$

$$\Rightarrow \log x = 25\log\left(\frac{2}{10}\right)$$

$$\Rightarrow \log x = 25(\log_{10} 2 - \log_{10} 10)$$

$$\Rightarrow \log x = 25(0.3010 - 1)$$

$$\Rightarrow \log x = 7.525 - 25$$

$$\Rightarrow \log x = 0.525 - 18$$

$$\Rightarrow \quad x = \text{Antilog}(0.525) \times 10^{-18}$$

∴ 18 zero appearing immediately after the decimal point.

47. The ratio of the sum and difference of the ages of the father and the son is 11 : 3. Consider the following statements

1. The ratio of their ages is 8 : 5.
2. The ratio of their ages after the son attains twice the present age will be 11 : 8.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

(b) From statement 1

Ratio of ages of father and son

$= 8:5$

Let father's age $= 8x$

and son's age $= 5x$

$\therefore$ According to the question, the ratio of the sum and difference of the ages of father and son is 11 : 3.

$\therefore \quad \frac{8x+5x}{8x-5x} = \frac{11}{3}$

But it is not equal to $\frac{11}{3}$ as $\frac{13}{3} \neq \frac{11}{3}$

$\therefore$ Statement 1 is not true.

From statement 2

Ratio of the age of father and son after the son attains twice the present age will be 11 : 8

Let father's age be F and son's age be S

$\therefore \quad \frac{F+S}{S+S} = \frac{11}{8}$

$\Rightarrow \quad 8F + 8S = 22S$

$\Rightarrow \quad 8F = 14S$ or $\frac{F}{S} = \frac{7}{4}$

Let father's age be $7x$ and son's age be $4x$

$\therefore$ According to the question,

$\frac{7x+4x}{7x-4x} = \frac{11x}{3x} = \frac{11}{3}$

$\therefore$ Statement 2 is correct

48. The solution of linear inequalities $x+y \geq 5$ and $x-y \leq 3$ lies

(a) Only in the first quadrant

(b) In the first and second quadrants

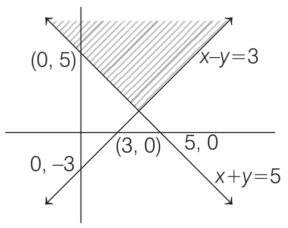
(c) In the second and third quarants

(d) In the third and fourth quadrants

(b) We have,

$x + y \geq 5$ and $x - y \leq 3$

The graph of linear equalities are below



Clearly, from graph solution lies in first and second quadrant

49. It is given that equations $x^2 - y^2 = 0$ and $(x-a)^2 + y^2 = 1$ have single positive solution. For this, the value of 'a' is

(a) $\sqrt{2}$ (b) 2 (c) $-\sqrt{2}$ (d) 1

(a) We have, $x^2 - y^2 = 0$

and $\quad (x-a)^2 + y^2 = 1$

$\Rightarrow x^2 - y^2 = 0 \Rightarrow x^2 = y^2$

Put the value y^2 in, we get

$(x-a)^2 + y^2 = 1$

$\Rightarrow \quad (x-a)^2 + x^2 = 1$

$\Rightarrow 2x^2 - 2ax + a^2 - 1 = 0$

Equation have single positive solution

$\therefore \quad D = 0$

$\Rightarrow 4a^2 - 4(a^2-1)(2) = 0$

$\Rightarrow \quad 4a^2 - 8a^2 + 8 = 0$

$\Rightarrow \quad a^2 = 2$

$\Rightarrow \quad a = \sqrt{2}$

50. If α, β and γ are the zeros of the polynomial $f(x) = ax^2 + bx^2 + cx + d$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to

(a) $\frac{b^2 - ac}{a^2}$ (b) $\frac{b^2 - 2ac}{a}$

(c) $\frac{b^2 + 2ac}{b^2}$ (d) $\frac{b^2 - 2ac}{a^2}$

(d) We have,

$f(x) = ax^3 + bx^2 + cx + d$

α, β, γ are the zero of the given polynomials

$\therefore \quad \alpha + \beta + \gamma = -\frac{b}{a}$

$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

$\alpha\beta\gamma = -\frac{d}{a}$

$(\alpha+\beta+\gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$

$\Rightarrow \quad \left(-\frac{b}{a}\right)^2 = \alpha^2 + \beta^2 + \gamma^2 + \frac{2c}{a}$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2$

$= \frac{b^2}{a^2} - \frac{2c}{a}$

$= \frac{b^2 - 2ac}{a^2}$

Directions (Q. Nos. 51-54) Consider the following for the next 04 (four) items that follow :

In an examination of class XII, 55% students passed in Biology, 62% passed in Physics, 60% passed in Chemistry, 25% passed in Physics and Biology, 30% passed in Physics and Chemistry, 28% passed in Biology and Chemistry. Only 2% failed in all the subjects.

51. What percentage of students passed in all the three subjects?

(a) 6 (b) 5 (c) 4 (d) 3

(c) Given =

$n(B) = 55\%$

$n(P) = 62\%$

$n(C) = 60\%$

$n(P \cap B) = 25\%$

$n(P \cap C) = 30\%$

$n(B \cap C) = 28\%$

$n(P \cup B \cup C)' = 2\%$

$\therefore n(P \cup B \cup C) = 100\% - n(P \cup B \cup C)'$

$= 100\% - 2\% = 98\%$

$n(P \cup B \cup C) = n(B) + n(P) + n(C) - n(P \cap B) - n(P \cap C) - n(B \cap C) - n(P \cap B \cap C)$

$\Rightarrow \quad 98 = 55 + 62 + 60 - 25 - 28 - 30 + n(P \cap B \cap C)$

$\Rightarrow n(P \cap B \cap C) = 4\%$

$\because$ Percentage of student passed in all three subjects is 4%

52. What percentage of students passed in exactly one subject?

(a) 21 (b) 23 (c) 25 (d) 27

(b) percentage of students passed in exactly one subject

$= n(B) + n(P) + n(C) - 2\{n(P \cap B) + n(B \cap C) + n(P \cap C)\} + 3n(P \cap B \cap C)$

$= 55 + 62 + 60 - 2(25 + 28 + 30) + 3(4)$

$= 177 - 166 + 12 = 23\%$

53. If the number of students in 360, then how many passed in atleast two subjects?

(a) 270 (b) 263 (c) 265 (d) 260

(a) Percentage of student passed in exactly two subject

$= n(P \cup B) + n(P \cap C) + n(B \cap C) - 3n(P \cap B \cap C)$

$= 25 + 28 + 30 - 12 = 71\%$

% age of passed atleast two subjects

$= 71\% + 4\% = 75\%$

$\therefore$ Number of students passed

$= \frac{75}{100} \times 360 = 270$

54. What is the ratio of number of students who passed in both Physics and Chemistry to number of students who passed in both Biology and Physics but not Chemistry?

(a) 7 : 10 (b) 10 : 7

(c) 9 : 7 (d) 7 : 9

(b) Percentage of student passed in both Physics and Chemistry = 30%

Percentage of student passed in both Biology and Physics but not Chemistry

$= (25-4)\% = 21\%$

$\therefore$ Ratio $= \dfrac{30\%}{21\%} = 10:7$

55. Data on ratings of hotels in a city is measured on

(a) Nominal scale (b) Ordinal scale
(c) Interval scale (d) Ratio scale

(b) Data on rating of hotels in a city is measured on ordinal scale.

56. The average marks of section A are 65 and that of section B are 70. If the average marks of both the sections combined are 67, then the ratio of number of students of section A to that of section B is

(a) 3 : 2 (b) 1 : 3 (c) 3 : 1 (d) 2 : 3

(a) Let the number of students in section A = x

and the number of students in Section B = y

$\therefore$ Total number of student $= x + y$

$\therefore 65x + 70y = 67(x + y)$

$\Rightarrow \quad 2x = 3y$

$\Rightarrow \quad x : y :: 3 : 2$

57. The median of 19 observations is 30. Two more observations are made and the values of these are 8 and 32. What is the median of the 21 observations?

(a) 32 (b) 30 (c) 20
(d) Cannot be determined due to insufficient data

(b) The median of the 21 observation is no charge because 8 is less than 30 and 32 is greater than 30

58. As the number of observations and classes increases, the shape of a frequency polygon

(a) Tends to become jagged
(b) Tends to become increasingly smooth
(c) Stays the same
(d) Varies only if data become more reliable

(b) We know that, when the number of observations and classes increases then the shape of a frequency polygon tends to become increasingly smooth.

59. Let $\bar{x}_1$ and $\bar{x}_2$ (where $\bar{x}_2 > \bar{x}_1$) be the means of two sets comprising n_1 and n_2 (where $n_2 < n_1$) observations respectively. If $\bar{x}$ is the mean when they are pooled, then which one of the following is correct?

(a) $\bar{x}_1 < \bar{x} < \bar{x}_2$
(b) $\bar{x} > \bar{x}_2$
(c) $\bar{x} < \bar{x}_1$
(d) $(\bar{x}_1 - \bar{x}) + (\bar{x}_2 - \bar{x}) = 0$

(a) $\bar{x} = \dfrac{\bar{x}_1 n_1 + \bar{x}_2 n_2}{n_1 + n_2}$

$\Rightarrow n_1\bar{x} + n_2\bar{x} = n_1\bar{x}_1 + n_2\bar{x}_2$

$\Rightarrow n_1(\bar{x} - \bar{x}_1) = n_2(\bar{x}_2 - \bar{x})$

$\Rightarrow \quad \dfrac{\bar{x}_2 - \bar{x}}{\bar{x} - \bar{x}_1} = \dfrac{n_1}{n_2}$

$\Rightarrow \quad \dfrac{\bar{x}_2 - \bar{x}}{\bar{x} - \bar{x}_1} > 1 \; (n_1 > n_2)$

$\because \quad \bar{x}_2 - \bar{x} > \bar{x} - \bar{x}_1$

$\therefore \quad \bar{x}_2 > \bar{x} > \bar{x}_1$

60. Consider the following statements:

Statements I Median can be computed even when the end intervals of a frequency distribution are open

Statements II Median is a positional average.

Which one of the following is correct in respect of the above statements?

(a) Both Statement I and Statement II are true and Statement II is the correct explanation of Statement I
(b) Both Statement I and Statement II are true and Statement II is not the correct explanation of Statement I
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

(a) We know that, median always lies in the center of the distribution therefore it is called a positional average and so we can compute it even when the end intervals of a frequency distribution are open.

61. If $\cos\theta = \dfrac{1}{\sqrt{5}}$, where $0 < \theta < \dfrac{\pi}{2}$, then $\dfrac{2\tan\theta}{1 - \tan^2\theta}$ is equal to

(a) 4 / 3 (b) –4 / 3 (c) 1 / 3 (d) –2 / 3

(b) We have,

$\cos\theta = \dfrac{1}{\sqrt{5}}$

$\Rightarrow \quad \tan\theta = \sqrt{\sec^2\theta - 1} = \sqrt{5-1} = 2$

$\therefore \dfrac{2\tan\theta}{1-\tan^2\theta} = \dfrac{2(2)}{1-(2)^2} = \dfrac{4}{1-4} = \dfrac{-4}{3}$

62. If $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$ and $\cos\theta < \cos\phi$, then which one of the following is correct ?

(a) $\theta < \phi$
(b) $\theta > \phi$
(c) $\theta + \phi = 90^\circ$
(d) No conclusion can be drawn

(b) $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$

and $\cos\theta < \cos\phi$

$\theta > \phi$ [$\cos x$ is decreasing function in $\left[0, \dfrac{\pi}{2}\right]$]

63. On the top of a hemispherical dome of radius r, there stands a flag of height h. From a point on the ground, the elevation of the top of the flag is 30°. After moving a distance d towards the dome, when the flag is just visible, the elevation is 45°. The ratio of h to r is equal to

(a) $\sqrt{2} - 1$ (b) $\dfrac{\sqrt{3}+1}{2\sqrt{2}}$

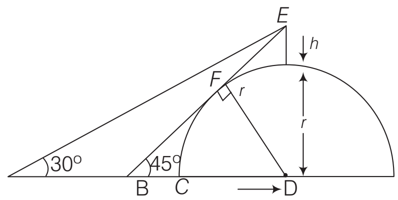
(c) $\dfrac{\sqrt{3}+1}{2\sqrt{2}}d$ (d) $\dfrac{(\sqrt{3}+1)(\sqrt{2}-1)}{2\sqrt{2}}d$

(a) According to given information, we have the following figure.

In ΔBDE, we have $\tan 45^\circ = \dfrac{DE}{BD}$

$\Rightarrow \quad BD = DE = h + r$

$\Rightarrow \quad BC + r = h + r$



$\Rightarrow \quad BC = h$

Now, as $\angle EBD = 45°$, therefore $\angle BED = 45°$

Also, In ΔBFD

$$\tan 45° = \frac{FD}{FB}$$

$\Rightarrow \quad FB = FD = r$

Similarly, $\quad EF = r$

Now, consider ΔBDE, then we have

$(BE)^2 = (BD)^2 + (DE)^2$

$\Rightarrow \quad (2r)^2 = (h+r)^2 + (h+r)^2$

$\Rightarrow \quad 4r^2 = 2(h+r)^2$

$\Rightarrow \quad 2r^2 = (h+r)^2$

$\Rightarrow \quad h + r = \sqrt{2}r$

$\Rightarrow \quad h = (\sqrt{2}-1)r$

$\Rightarrow \quad \frac{h}{r} = \sqrt{2} - 1$

64. Let $\sin(A+B) = \frac{\sqrt{3}}{2}$ and $\cos B = \frac{\sqrt{3}}{2}$, where A, B are acute angles. What is $\tan(2A - B)$ equal to ?

(a) 1/ 2 (b) $\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) 1

(c) We have,

$\sin(A+B) = \frac{\sqrt{3}}{2}$

$\Rightarrow A + B = 60°$ ... (i)

and $\cos B = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad B = 30°$ ... (ii)

Solving Eqs. (i) and (ii), we get

$A = 30°$

$\therefore \tan(2A - B) = \tan(60 - 30°)$

$= \tan 30° = \frac{1}{\sqrt{3}}$

65. Consider the following statements

1. If $\frac{\cos\theta}{1-\sin\theta} + \frac{\cos\theta}{1+\sin\theta} = 4$, where $0 < \theta < 90°$, then $\theta = 60°$.
2. If $3\tan\theta + \cot\theta = 5\,\text{cosec}\,\theta$, where $0 < \theta < 90°$, then $\theta = 60°$.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

(c) We have,

1. $\frac{\cos\theta}{1-\sin\theta} + \frac{\cos\theta}{1+\sin\theta} = 4$

$\Rightarrow \frac{\cos\theta\,(1+\sin\theta+1-\sin\theta)}{1-\sin^2\theta} = 4$

$\Rightarrow \quad \frac{2\cos\theta}{\cos^2\theta} = 4$

$\Rightarrow \quad \cos\theta = \frac{1}{2} = \cos 60°$

$\Rightarrow \quad \theta = 60°$

$\therefore$ Statement 1 is correct

2. $3\tan\theta + \cot\theta = 5\,\text{cosec}\,\theta$

put $\quad \theta = 60°$

$\because 3\tan 60° + \cot 60° = 5\,\text{cosec}\,60°$

$3\sqrt{3} + \frac{1}{\sqrt{3}} = 5 \times \frac{2}{\sqrt{3}}$

$\Rightarrow \quad \frac{9+1}{\sqrt{3}} = \frac{10}{\sqrt{3}}$

$\Rightarrow \quad \frac{10}{\sqrt{3}} = \frac{10}{\sqrt{3}}$

$\therefore$ Statement 2 is also correct

66. Consider the following statements

1. $\cos^2\theta = 1 - \frac{p^2+q^2}{2pq}$, where p, q are non-zero real numbers, is possible only when $p = q$.
2. $\tan^2\theta = \frac{4pq}{(p+q)^2} - 1$, where p, q are non-zero real numbers, is possible only when $p = q$.

Which of the statements given above is/are correct ?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

(c) $\cos^2\theta = 1 - \frac{p^2+q^2}{2pq}$

$0 \le \cos^2\theta \le 1$

$\theta \le 1 - \frac{p^2 \times q^2}{2pq} \le 1$

$0 \le 2pq - (p^2+q^2) \le 2pq$

$p^2 + q^2 - 2pq \le 0$

$\Rightarrow \quad (p-q)^2 \le 0$

It is possible only when $p = q$

$\therefore$ Statement 1 is correct

$\tan^2\theta = \frac{4pq}{(p+q)^2} - 1$

$\Rightarrow \quad \tan^2\theta \ge 0$

$\Rightarrow \quad \frac{4pq}{(p+q)^2} - 1 \ge 0$

$\Rightarrow \quad 4pq - (p+q)^2 \ge 0$

$\Rightarrow \quad (p+q)^2 - 4pq \le 0$

$\Rightarrow p^2 + q^2 + 2pq - 4pq \le 0$

$\Rightarrow \quad (p-q)^2 \le 0$

It is possible only $p = q$

$\therefore$ Statement 2 is correct

67. Consider the following statements

1. $\cos\theta + \sec\theta$ can never be equal to 1.5
2. $\sec^2\theta + \text{cosec}^2\theta$ can never be less than 4.

Which of the statements given above is / are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

(c) $\cos\theta + \sec\theta$

$\Rightarrow \quad \cos\theta + \frac{1}{\cos\theta}$

$\Rightarrow \quad \cos\theta + \frac{1}{\cos\theta} \ge 2$

$\therefore$ Statement 1 is correct

$\sec^2\theta + \text{cosec}^2\theta$

$= \frac{1}{\cos^2\theta} + \frac{1}{\sin^2\theta}$

$= \frac{1}{\sin^2\theta \cdot \cos^2\theta}$

$= \frac{4}{(2\sin\theta\cos\theta)^2} = \frac{4}{\sin^2 2\theta}$

$\therefore \quad \frac{4}{\sin^2 2\theta} \ge 4$

Statement 2 is also correct.

68. If $\sin^2 x + \sin x = 1$, then what is the value of $\cos^{12} x + 3\cos^{10} x + 3\cos^8 x + \cos^6 x$?

(a) –1 (b) 0 (c) 1 (d) 8

(c) We have,

$\sin^2 x + \sin x = 1$

$\Rightarrow \sin x = 1 - \sin^2 x = \cos^2 x$

$\Rightarrow \quad \sin^2 x = \cos^4 x$

$\Rightarrow \quad 1 - \cos^2 x = \cos^4 x$

$\Rightarrow \cos^4 x + \cos^2 x = 1$

cubing both sides,

$(\cos^4 x + \cos^2 x)^3 = (1)^3$

$\Rightarrow \cos^{12} x + 3\cos^{10} x + 3\cos^8 x + \cos^6 x = 1$

69. If $3\sin\theta + 5\cos\theta = 4$, then what is the value of $(3\cos\theta - 5\sin\theta)^2$?

(a) 9 (b) 12

(c) 16 (d) 18

(d) We have,

$3\sin\theta + 5\cos\theta = 4$

squaring both sides,

$9\sin^2\theta + 25\cos^2\theta + 30\sin\theta\cos\theta = 16$

$\Rightarrow 9 - 9\cos^2\theta + 25 - 25\sin^2\theta + 30\sin\theta\cos\theta = 16$

$\Rightarrow \dfrac{9\cos^2\theta + 25\sin^2\theta - 30}{\sin\theta\cos\theta} = 9 + 25 - 16$

$\Rightarrow (3\cos\theta - 5\sin\theta)^2 = 18$

70. If $\cot\theta(1+\sin\theta) = 4m$ and $\cot\theta(1-\sin\theta) = 4n$, then which one of the following is correct?

(a) $(m^2+n^2)^2 = mn$
(b) $(m^2-n^2)^2 = mn$
(c) $(m^2-n^2)^2 = m^2n^2$
(d) $(m^2+n^2)^2 = m^2n^2$

(b) We have,

$\cot\theta(1+\sin\theta) = 4m$ ... (i)

$\cot\theta(1-\sin\theta) = 4n$ ... (ii)

$16mn = \cot^2\theta(1-\sin^2\theta)$

$16mn = \cot^2\theta\cos^2\theta$ ...(iii)

Now, add eqs. (i) and (ii), we get

$\cot\theta = 2(m+n)$

Subtracting eqs. (ii) from (i), we get

$\cot\theta\sin\theta = 2(m-n)$

$\Rightarrow \cos\theta = 2(m-n)$

Putting the value of $\cot\theta$ and $\cos\theta$ in eq. (iii), we get

$16mn = (2^2(m-n)^2)(2^2(m+n)^2)$

$= 16(m^2-n^2)^2$

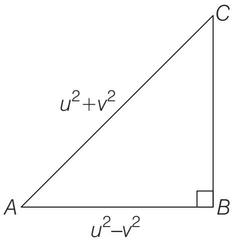
$\Rightarrow mn^2(m^2-n^2)^2$

71. If base and hypotenuse of a right triangle are (u^2-v^2) and (u^2+v^2) respectively and the area of the triangle is 2016 square units, then the perimeter of the triangle may be

(a) 224 units (b) 288 units
(c) 448 units (d) 576 units

(b) According to the question,

In right angle ΔABC



Here,

$AC^2 = AB^2 + BC^2$

$\Rightarrow BC^2 = AC^2 - AB^2$

$\Rightarrow BC = \sqrt{AC^2 - AB^2}$

$= \sqrt{(u^2+v^2)^2 - (u^2-v^2)^2}$

$= \sqrt{4u^2v^2} = 2uv$

Now, area of $\Delta ABC = \dfrac{1}{2} \times \text{base} \times \text{height}$

$= \dfrac{1}{2} \times (u^2-v^2) \times (2uv)$

$= uv(u^2-v^2)$

$\Rightarrow uv(u+v)(u-v) = 2016$

$\Rightarrow 9 \times 7 \times 16 \times 2 = 2016$

$\therefore u = 9$ and $v = 7$

$\therefore$ Perimeter of triangle

$= (u^2+v^2) + (u^2-v^2) + 2uv$

$= 2u^2 + 2uv = 2u(u+v)$

$= 2 \times 9 \times (9+7) = 288$ unit

72. A circle is inscribed in an equilateral triangle of side of length. The area of any square inscribed in the circle is

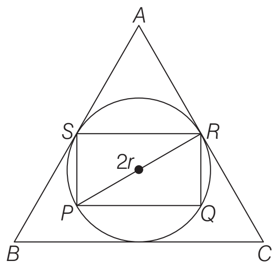
(a) $\dfrac{l^2}{2}$ (b) $\dfrac{\sqrt{3}l^2}{4}$ (c) $\dfrac{l^2}{4}$ (d) $\dfrac{l^2}{6}$

(d) Side of equilateral $\Delta ABC = l$

Radius of circle inscribed in ΔABC

$r = \dfrac{\text{Area of } \Delta ABC}{\text{Semi-perimeter of } \Delta ABC}$

$\Rightarrow r = \dfrac{\frac{\sqrt{3}}{4}l^2}{3\frac{l}{2}}$



$\Rightarrow r = \dfrac{l}{2\sqrt{3}}$

Area of square $PQRS$

$= PQ^2 = \dfrac{PR^2}{2} = \dfrac{4r^2}{2} = 2r^2$

$= 2\left(\dfrac{l}{2\sqrt{3}}\right)^2 = \dfrac{2l^2}{4\times 3} = \dfrac{l^2}{6}$

73. Walls (excluding roofs and floors) of 5 identical rooms having length, breadth and height 6 m, 4 m and 2.5 m respectively are to be painted. Out of five rooms, two rooms have one square window each having a side of 2.5 m. Paints are available only in cans of 1 L; and 1 L of paint can be used for painting 20 m^2. The number of cans required for painting is

(a) 10 (b) 12 (c) 13 (d) 14

(b) Length, breadth and height of walls are 6m, 4m and 2.5 m respectively

$\therefore$ Area of 4 walls

$= 2h(l+b) = 2 \times 2.5(6+4) = 50\ m^2$

Area of 4 walls of 5 rooms

$= 5 \times 50 = 250\ m^2$

Area of windows $= (2.5)^2 = 6.25\ m^2$

Area of two windows

$= 2 \times 6.25$

$= 13.50\ m^2$

Total area required for painting

$= 250 - 13.50$

$= 236.50$

Now, one cans paints $= 20\ m^2$

$\therefore$ Number of cans required for painting

$= \dfrac{236.50}{20} = 11.825 = 12$ cans

74. Let S be the parallelogram obtained by joining the mid-points of the parallelogram T. Consider the following statements :

1. The ratio of area of T to that S is 2 : 1
2. The perimeter of S is half of the sum of diagonals of T.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

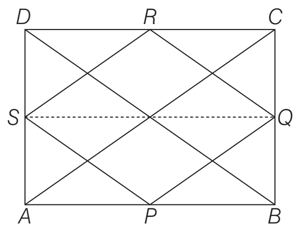
(a) Given,

Area of parallelogram $ABCD = T$

Area of parallelogram $PQRS = S$

Area of $\Delta PQS = \dfrac{1}{2} \times$ area of $||^{gm}$ $ABQS$

Area of $\Delta RSQ = \dfrac{1}{2} \times$ area of $||^{gm}$ $SQCD$



$\therefore$ Area of parallelogram $PQRS = \dfrac{1}{2}$ (area of parallelogram $ABCD$)

$S = \dfrac{1}{2}T \Rightarrow S : T = 1 : 2$

$\Rightarrow T : S = 2 : 1$

$\therefore$ Statement 1 is correct

2. Perimeter of parallelogram $PQRS$

$= PQ + RQ + RS + PS$

$PQ = \frac{1}{2} AC = RS$

$RQ = PS = \frac{1}{2} BD$

$\therefore PQ + RQ + RS + PS$

$= (AC + BD)$

Perimeter of S is equal to sum of diagonals T

$\therefore$ Statement 2 is incorrect

75. The sides of a triangle are 5 cm, 6 cm and 7 cm. The area of the triangle is approximately

(a) 14.9 cm^2 (b) 14.7 cm^2

(c) 14.5 cm^2 (d) 14.3 cm^2

(b) The sides of triangles are 5cm , 6 cm and 7cm respectively

Area of $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

Where $s = \frac{a+b+c}{2}$

$\therefore \quad s = \frac{5+6+7}{2} = 9$

$\Delta = \sqrt{9(9-5)(9-6)(9-7)}$

$\Delta = \sqrt{9 \times 4 \times 3 \times 2}$

$\Delta = 6\sqrt{6}$

$= 6 \times 2.44 = 14.69 = 14.7$

76. There is a path of width 5 m around a circular plot of land whose area is $144\pi \text{ m}^2$. The total area of the circular plot including the path surrounding it is

(a) $349\pi \text{ m}^2$ (b) $289\pi \text{ m}^2$

(c) $209\pi \text{ m}^2$ (d) $149\pi \text{ m}^2$

(b) Area of circular

Plot of land $= 144\pi \text{ m}^2$

Let r = radius of circular plot

$\therefore \; 144\pi = \pi r^2$

$\Rightarrow \quad r^2 = 144$

$\Rightarrow \quad r = 12$

Area of circular plot including the path

$= \pi(r+5)^2 = \pi(12+5)^2$

$= \pi(17)^2 = 289\pi \text{ m}^2$

77. The lateral surface area of a cone is 462 cm^2. Its slant height is 35 cm. The radius of the base of the cone is

(a) 8.4 cm (b) 6.5 cm

(c) 4.2 cm (d) 3.2 cm

(c) Lateral surface area of cone $= \pi rl$

$\therefore 462 = \pi r(35) \quad (\because l = 35)$

$\Rightarrow \quad r = \frac{462}{\pi \times 35}$

$\Rightarrow \quad r = \frac{462 \times 7}{22 \times 35} = 4.2$ cm

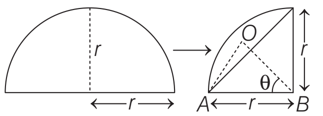
78. A semi-circular plate is rolled up to form a conical surface. The angle between the generator and the axis of the cone is

(a) 60° (b) 45°

(c) 30° (d) 15°

(c) Clearly, the radius of semi-circle = slant height of cone = r

And the circumference of semi-circle = circumference of the base of cone



$\Rightarrow \quad \pi r = 2\pi R \quad \Rightarrow \quad R = \frac{r}{2}$

Now, In ΔAOB, we have

$\sin\theta = \frac{\frac{r}{2}}{r} = \frac{1}{2} \quad \Rightarrow \theta = 30^\circ$

79. A solid right cylinder is of height π cm. If its lateral surface area is half its total surface area, then the radius of its base is

(a) $\pi / 2$ cm (b) π cm

(c) $1/\pi$ cm (d) $2/\pi$ cm

(b) Height of cylinder = π cm

Lateral surface area $= \frac{1}{2}$ total surface area

$\therefore \quad 2\pi rh = \frac{1}{2}(2\pi r^2 + 2\pi rh)$

$\Rightarrow \; 2\pi rh = \frac{1}{2} 2\pi r(r+h)$

$\Rightarrow \quad 2h = r + h$

$\Rightarrow \quad r = h$

$\therefore \quad r = \pi$ cm

80. A rectangular block of length 20 cm, breadth 15 cm and height 10 cm is cut up into exact number of equal cubes. The least possible number of cubes will be

(a) 12 (b) 16

(c) 20 (d) 24

(d) Volume of rectangular block

$= 20 \times 15 \times 10 = 3000 \text{ cm}^3$

Let n is the number of cube formed

Side of cube is 5 cm possible

$\therefore n = \frac{20 \times 15 \times 10}{5 \times 5 \times 5} = 24$

81. If the diagonal of a cube is of length l, then the total surface area of the cube is

(a) $3l^2$ (b) $\sqrt{3}l^2$ (c) $\sqrt{2}l^2$ (d) $2l^2$

(d) Diagonal of cube $= \sqrt{3} \times$ sides of cube

$\therefore \quad l = \sqrt{3} \times$ side of cube

Side of cube $= \frac{l}{\sqrt{3}}$

Surface area of cube = $6(\text{side})^2$

$= 6 \times \left(\frac{l}{\sqrt{3}}\right)^2 = \frac{6l^2}{3} = 2l^2$

82. An equilateral triangle, a square and a circle have equal perimeter. If T, S and C denote the area of the triangle, area of the square and area of the circle respectively, then which one of the following is correct?

(a) $T < S < C$ (b) $S < T < C$

(c) $C < S < T$ (d) $T < C < S$

(a) An equilateral triangle, a square and a circle of perimeter are equal

Let side of triangle = l cm

Side of square = x cm

Radius of circle = r cm

$\therefore 3l = 4x = 2\pi r = k$

$\Rightarrow \; l = k/3, \; x = k/4, \; r = \frac{k}{2\pi}$

Area of equilateral Δ

$= \frac{\sqrt{3}}{4} l^2 = \frac{\sqrt{3}}{4}\left(\frac{k^2}{9}\right) = \frac{k^2}{12\sqrt{3}}$

Area of square $= x^2 = \frac{k^2}{16}$

Area of circle $= \pi r^2 = \pi \frac{k^2}{4\pi^2} = \frac{k^2}{4\pi}$

$\frac{k^2}{4\pi} > \frac{k^2}{16} > \frac{k^2}{12\sqrt{3}}$

$C > S > T$

$\Rightarrow \quad T < S < C$

83. The areas of two similar triangles are $(7 - 4\sqrt{3}) \text{ cm}^2$ and $(7 + 4\sqrt{3}) \text{ cm}^2$ respectively. The ratio of their corresponding sides is

(a) $7 - 4\sqrt{3}$ (b) $7 - 3\sqrt{3}$

(c) $5 - \sqrt{3}$ (d) $5 + \sqrt{3}$

(a) Ratio of area of two similar triangle = Ratio of square of their corresponding sides

$\therefore \sqrt{\frac{7 - 4\sqrt{3}}{7 + 4\sqrt{3}}}$ = ratio of their corresponding sides $= 7 - 4\sqrt{3}$

84. The chord of a circle is $\sqrt{3}$ times its radius. The angle subtended by this chord at the minor arc is k times the angle subtended at the major arc. What is the value of k ?

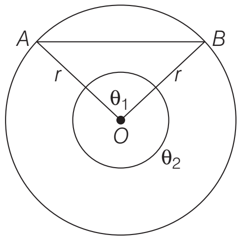
(a) 5 (b) 2
(c) 1/2 (d) 1/5

(c) Given, $AB = \sqrt{3}r$

Now, $\cos\theta_1 = \dfrac{OA^2 + OB^2 - AB^2}{2OA\cdot OB}$

$\cos\theta_1 = \dfrac{r^2 + r^2 - 3r^2}{2r^2}$

$\cos\theta_1 = -\dfrac{1}{2}$



$\theta_1 = 120° \Rightarrow \theta_1 = k\theta_2$

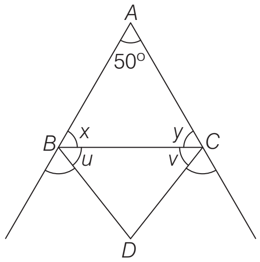
$\Rightarrow \quad k = \dfrac{120}{360 - 120}$

$= \dfrac{120}{240} = \dfrac{1}{2}$

85. In a triangle ABC, the sides AB, AC are produced and the bisectors of exterior angles of $\angle ABC$ and $\angle ACB$ intersect at D. If $\angle BAC = 50°$, then $\angle BDC$ is equal to

(a) 115°
(b) 65
(c) 55°
(d) 40°

(b) $\angle BDC = 180° - (u + v)$



$= 180° - \left(\dfrac{50 + y}{2} + \dfrac{50 + x}{2}\right)$

$= 180° - \left(50 + \dfrac{x + y}{2}\right)$

$= 180° - \left(\dfrac{50}{2} + 90°\right)$

$= 180° - (25 + 90)$

$= 180° - 115° = 65°$

86. Two cones have their heights in the ratio 1 : 3. If the radii of their bases are in the ratio 3 : 1, then the ratio of their volumes will be

(a) 1 : 1 (b) 2 : 1 (c) 3 : 1 (d) 9 : 1

(c) Let the height of cones are h and $3h$ and their radii of bases are $3r$ and r

$\therefore$ Volumes of cones are $\dfrac{1}{3}\pi(3r)^2(h)$

and $\dfrac{1}{3}\pi(r^2)(3h)$

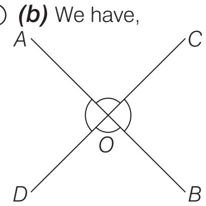
$\therefore$ Ratio of volumes will be

$= \dfrac{\frac{1}{3}\pi\, 9r^2h}{1/3\ \pi r^2 3h} = \dfrac{3}{1} = 3 : 1$

87. If two lines AB and CD intersect at O such that $\angle AOC = 5\angle AOD$, then the four angles at O are

(a) 40°, 40°, 140°, 140°
(b) 30°, 30°, 150°, 150°
(c) 30°, 45°, 75°, 210°
(d) 60°, 60°, 120°, 120°

(b) We have,



$\angle AOC = 5\angle AOD$

$\therefore \quad \angle AOD + \angle AOC = 180°$

$\Rightarrow 5\angle AOD + \angle AOD = 180°$

$\Rightarrow \quad 6\angle AOD = 180°$

$\Rightarrow \quad \angle AOD = 30°$

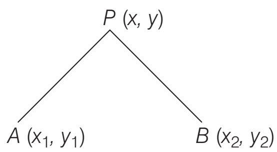
$\therefore$ Other four angles are

30°, 30°, 150°, 150°

88. If a point P moves such that the sum of the squares of its distances from two fixed points A and B is a constant, then the locus of the point P is

(a) A straight line
(b) A circle
(c) Perpendicular bisector of AB
(d) An arbitrary curve

(b) Given, $PA^2 + PB^2 =$ constant



$\therefore (x - x_1)^2 + (y - y_1)^2 + (x - x_2)^2 + (y - y_2)^2 = \text{constant}$

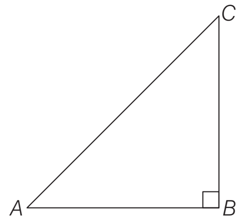
$\Rightarrow 2(x^2 + y^2) - 2x(x_1 + x_2) - 2y(y_1 + y_2) + x_1^2 + x_2^2 + y_1^2 + y_2^2 = \text{constant}$

Which represents the locus circle.

89. If ABC is a right-angled triangle with AC as its hypotenuse, then which one of the following is correct ?

(a) $AC^3 < AB^3 + BC^3$
(b) $AC^3 > AB^3 + BC^3$
(c) $AC^3 \le AB^3 + BC^3$
(d) $AC^3 \ge AB^3 + BC^3$

(b) Here, ABC is a right angle triangle with AC as its hypotenuse



Let $AC = 5$
$AB = 3$
$BC = 4$

$\therefore AC^3 = (5)^3 = 125$

$AB^3 = (3)^3 = 27$

$BC^3 = (4)^3 = 64$

Now, $AC^3 > AB^3 + BC^3$

90. The area of the region bounded externally by a square of side $2a$ cm and internally by the circle touching the four sides of the square is

(a) $(4 - \pi)a^2$ (b) $(\pi - 2)a^2$
(c) $\dfrac{(8 - \pi)a^2}{2}$ (d) $\dfrac{(\pi - 2)a^2}{2}$

(a) Side of square $ABCD$ is 2a cm

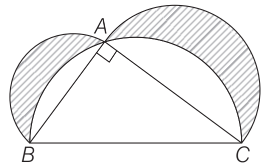
$\therefore$ Radius of circle = a cm

Area of shaded region

= area of square – area of circle

$= (2a)^2 - \pi a^2 = 4a^2 - \pi a^2 = (4 - \pi)a^2$

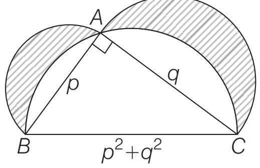
91. In the figure given below, ABC is a right-angled triangle where $\angle A = 90°$, $AB = p$ cm and $AC = q$ cm. On the three sides as diameters semi-circles are drawn as shown in the figure. The area of the shaded portion, in square cm, is



(a) pq (b) $\dfrac{\pi(p^2+q^2)}{2}$

(c) $\pi(p^2+q^2)$ (d) $\dfrac{pq}{2}$

(d)



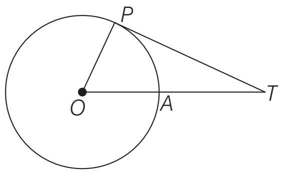
Area of shaded region =

area of semi-circle on AB + area of semi-circle on AC + area of ΔABC − area of semi-circle on BC

$$= \frac{\pi}{2}\left(\frac{p^2}{4}\right) + \frac{\pi}{2}\left(\frac{q^2}{4}\right) + \frac{1}{2}pq - \frac{\pi}{2}\left(\frac{p^2+q^2}{4}\right)$$

$$= \frac{1}{2}pq$$

92. In the figure given below, the radius of the circle is 6 cm and $AT = 4$ cm. The length of tangent PT is



(a) 6 cm

(b) 8 cm

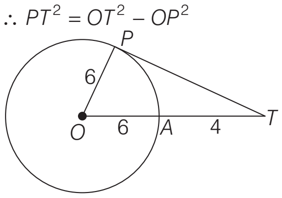
(c) 9 cm

(d) 10 cm

(b) Here, $OP = 6$

$OT = OA + AT = 10$

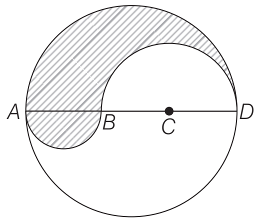
$\therefore PT^2 = OT^2 - OP^2$



$\Rightarrow \quad PT = \sqrt{(10)^2 - (6)^2}$

$\Rightarrow \sqrt{100-36} = \sqrt{64} = 8$

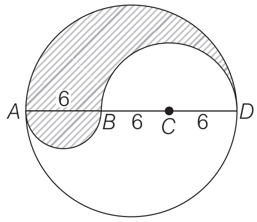
93. In the figure given below, $ABCD$ is the diameter of a circle of radius 9 cm. The lengths AB, BC and CD are equal. Semi-circles are drawn on AB and BD as diameters as shown in the figure. What is the area of the shaded region?



(a) 9π (b) 27π (c) 36π (d) 81π

(b) Area of shaded regions =

Area of semi-circle on AD + area of semicircle on AB − area of semi-circle on BD



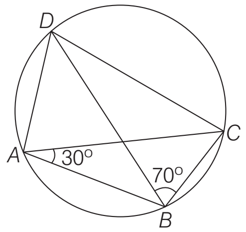
$$= \frac{\pi}{2}(9)^2 + \frac{\pi}{2}(3)^2 - \frac{\pi}{2}(6)^2$$

$$= \frac{\pi}{2}(9^2 + 3^2 - 6^2)$$

$$= \frac{\pi}{2}(81 + 9 - 36)$$

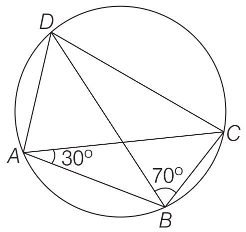
$$= \frac{\pi}{2}(54) = 27\pi$$

94. In the figure given below, what is $\angle BCD$ equal to?



(a) 70° (b) 75° (c) 80° (d) 90°

(c) $\angle CAD = \angle CBD$



$\angle CAD = \angle CBD$

(angle on same segment)

$\therefore \angle CAD = 70°$

$\therefore \angle BAD = \angle BAC + \angle CAD$

$= 30° + 70° = 100°$

$\angle BAD + \angle BCD = 180°$

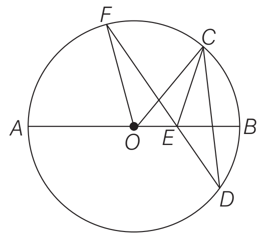
($ABCD$ is cycle quadilateral)

$\therefore \angle BCD = 180° - \angle BAD$

$= 180° - 100°$

$= 80°$

95. In the figure given below, AB is the diameter of the circle whose centre is at O. Given that $\angle ECD = \angle EDC = 32°$, then $\angle CEF$ and $\angle COF$ respectively are



(a) 32°, 64° (b) 64°, 64°

(c) 32°, 32° (d) 64°, 32°

(b) Given, $\angle ECD = \angle EDC = 32°$

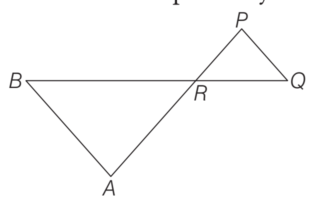
$\angle CEF = \angle ECD + \angle EDC$

(exterior angle of Δ is sum of interior opposite angle)

$\therefore \angle CEF = 32° + 32° = 64°$

$\angle COF = 2\angle CDF = 2(32°) = 64°$

96. In the figure given below, $\Delta ABR \sim \Delta PQR$. If $PQ = 3$ cm, $AB = 6$ cm, $BR = 8.2$ cm and $PR = 5.2$ cm, then QR and AR are respectively



(a) 8.2 cm, 10.4 cm

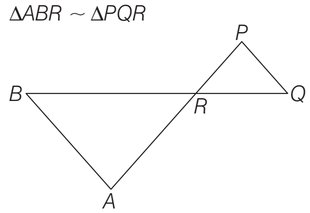
(b) 4.1 cm, 6 cm

(c) 2.6 cm, 5.2 cm

(d) 4.1 cm, 10.4 cm

(d) Given,

$\Delta ABR \sim \Delta PQR$



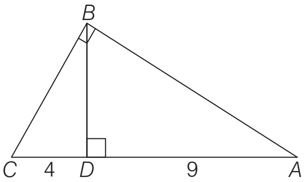
$$\therefore \quad \frac{AB}{PQ} = \frac{AR}{PR} = \frac{BR}{QR}$$

$$\frac{6}{3} = \frac{AR}{5.2} = \frac{8.2}{QR}$$

Solving and, we get

$QR = 4.1$ cm and $AR = 10.4$ cm

97. In the figure given below ABC is a triangle with AB perpendicular to BC. Further BD is perpendicular to AC. If $AD = 9$ cm and $DC = 4$ cm, then what is the length of BD?



(a) 13/36 cm (b) 36/13 cm

(c) 13/2 cm (d) 6 cm

(d) In ΔABD,

$$BD^2 = AB^2 - AD^2 \quad \ldots\text{(i)}$$

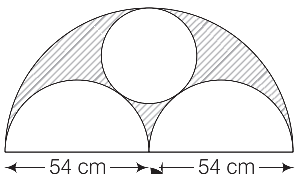
In ΔBCD,

$$BD^2 = BC^2 - CD^2 \quad \ldots\text{(ii)}$$

Adding Eqs. (i) and (ii), we get

$$2BD^2 = (AB^2 + BC^2) - (AD^2 + CD^2)$$
$$= (AC^2) - (AD^2 + CD^2)$$
$$= (13)^2 - (4^2 + 9^2)$$
$$2BD^2 = 169 - 97 = 72$$
$$BD = \sqrt{\frac{72}{2}} = 6$$

98. In the figure given below, the diameter of bigger semi-circle is 108 cm. What is the area of the shaded region?



(a) 201π cm^2

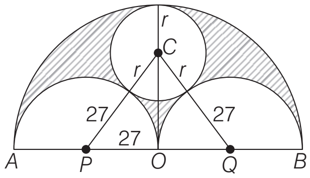
(b) 186.3π cm^2

(c) 405π cm^2

(d) 769.5π cm^2

(c) Given, $AO = OB = 54$ cm

$\therefore \quad OP = OQ = 27$ cm



In ΔOPC, $PC^2 = OP^2 + OC^2$

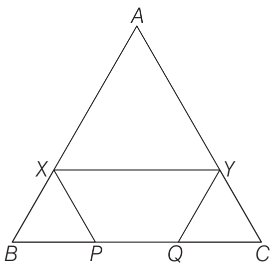
$$(27 + r)^2 = 27^2 + (54 - r)^2$$
$$\Rightarrow (27)^2 + 54r + r^2 = 27^2 + (54)^2 - 108r + r^2$$
$$\Rightarrow \quad r = 18$$

$\therefore$ Area of shaded region =

Area of bigger semi-circle – 2(area of smaller semi-circle) – area of smallest circle

$$= \frac{\pi}{2}(54)^2 - 2\left(\frac{\pi}{2}(27)^2\right) - \pi(18)^2$$
$$= \frac{\pi}{2}[54^2 - 2(27)^2 - 2(18)^2]$$
$$= \frac{\pi}{2}[2916 - 1458 - 648]$$
$$= \frac{\pi}{2} \times 810 = 405\pi \text{ cm}^2$$

99. In the figure given below, ABC is an equilateral triangle with each side of length 30 cm. XY is parallel to BC, XP is parallel to AC and YQ is parallel to AB. If $XY + XP + YQ$ is 40 cm, then the value of PQ is



(a) 5 cm (b) 12 cm

(c) 15 cm (d) 10 cm

(d) Given,

$$AB = BC = AC = 30$$

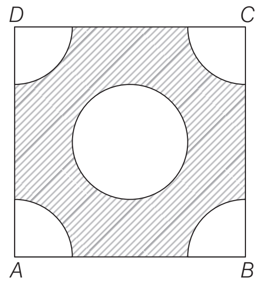
XY is parallel to BC

$\therefore AXY$ is also equilateral triangle

Similarly, XBP and CYQ is an equilateral triangle

$$\therefore \quad XY = AX,\ XP = BX,$$
$$QY = CY = CQ$$
$$XY + XP + CY = 40$$
$$AX + BX + CY = 40$$
$$AB + CQ = 40$$
$$\Rightarrow CQ = 40 - 30 = 10 = BP$$
$$PQ = BC - (BP + CQ)$$
$$= 30 - 20 = 10$$

100. In the figure given below, $ABCD$ is a square of side 4 cm. Quadrants of a circle of diameter 2 cm are removed form the four corners and a circle of diameter 2 cm is also removed. What is the area of the shaded region?



(a) $5\frac{7}{9}$ cm^2 (b) $7\frac{7}{9}$ cm^2

(c) $9\frac{5}{7}$ cm^2 (d) $9\frac{5}{6}$ cm^2

(c) Area of shaded region =

= Area of square – 4 (area of quadrant)– area of circle

$$= (4)^2 - 4\left(\frac{\pi}{4}(1)^2\right) - \pi(1)^2$$
$$= 4^2 - 2\pi$$
$$= 16 - 2 \times \frac{22}{7}$$
$$= 16 - \frac{44}{7}$$
$$= \frac{112 - 44}{7}$$
$$= \frac{68}{7} = 9\frac{5}{7} \text{ cm}^2$$

PAPER II English

Directions (Q. Nos. 1-10) Each of the following sentence in this section has a blank space and four words or group of words given after the sentences. Select the word or group of words you consider most appropriate for the blank space. and indicate your response on the answer sheet accordingly.

1. My teacher was .... us for being late.
(a) annoyed at (b) annoyed with
(c) annoyed about (d) annoys

✓ *(b)* 'annoyed with' is the most appropriate option as when we are angry with people we use 'annoyed with'.

2. Sandhya .... me from the top of the house.
(a) shouted to (b) shouted at
(c) shouted on (d) shouted

✓ *(b)* With verb 'shouted', preposition to be used is 'at'. So option 'b' 'shouted at' is correct here as we shout at some or the other person. When we 'shout to' some person, we want to be heard. Here, option (b) should be used.

3. Ravi has the habit of .... a headache.
(a) complaining (b) complain
(c) complaining to (d) complaining of

✓ *(d)* 'complaining of' is the most suitable option. When we speak about illnesses, we use complaining of.

4. I always want to go alone for a ride, but my mother .... going with my brother.
(a) insists (b) insists on
(c) insists in (d) insisted

✓ *(a)* 'insists' is appropriate option. 'Insist' means to say firmly or demand forcefully. No need to use any preposition.

5. The new student found it difficult to .... with his classmates.
(a) get along (b) get among
(c) get well (d) get up

✓ *(a)* 'get along' is the phrasal verb which should be used here in the blank. 'Get along' means to have a friendly relationship with others.

6. The visiting Diplomat .... the Prime Minister.
(a) called in
(b) called at
(c) called on
(d) called upon

✓ *(c)* 'called on' is the appropriate phrasal verb to fill in the blank. 'Called on' means to come to see or visit someone.

7. .... sincere, he would have got the prize.
(a) Had he been
(b) Has he been
(c) Would he have been
(d) He is

✓ *(a)* From the given option 'had he been' is most suitable. The sentence is 'conditional' type, here, use of past perfect is required.

8. Ten years .... for me to live in a foreign country.
(a) are a long time
(b) is a long time
(c) has a long time
(d) of time

✓ *(b)* Ten years is singular noun, so we will use 'is a long time' to fill in the blank.

9. If I .... you, I .... love to accept the offer.
(a) was …… will
(b) was …… would
(c) were …… would
(d) were …… will

✓ *(c)* Use of were and would is required in this conditional type of sentence.

10. My sister asked me .... willing to go abroad for my studies.
(a) if I were
(b) if I could be
(c) whether I should be
(d) whether I will

✓ *(a)* 'if I were' seems to be the only correct option.

Directions (Q.Nos. 11-20) Each item in this section consists of a sentence with an underlined word/words followed by four words. Select the option that is nearer to meaning to the underlined word/words.

11. Rahul is always <u>thrifty</u>.
(a) reckless (b) economical
(c) naive (d) extravagant

✓ *(b)* economical is synonym of the underlined word thrifty which means using money and other resources carefully. Other options are antonyms and do not give similar meaning. An economic person is one who is careful with spending money.

12. His <u>salubrious</u> words calmed the students.
(a) provoking (b) pleasant
(c) ridiculous (d) thanking

✓ *(b)* pleasant is nearer to the meaning of underlined word 'salubrious'. Other options are not relevant here.

13. He felt <u>desolated</u> after he lost his business.
(a) deserted (b) joyful
(c) strong (d) annoyed

✓ *(a)* 'deserted' is closest in meaning to word desolated. The word means to leave someone in a situation when they have no one to support. The word desolated also means to feel miserable or gloomy.

14. Don't <u>condone</u> such acts which lead to unrest in the country.
(a) regard (b) punish
(c) aware of (d) overlook

✓ *(d)* 'overlook' is closest in meaning to the word 'condone'. The word condone means to accept or allow behaviour that is wrong.

15. A good work place shall not encourage <u>ineptitude</u> even in a hidden manner.
(a) incompetence (b) courage
(c) gossip (d) radical thinking

✓ *(a)* 'incompetence' is closest in meaning to the word ineptitude. The word means lack of skill or ability and incompetence also means the same.

16. Learning of foreign language should not <u>impede</u> one's mother tongue learning.
(a) facilitate (b) acts for
(c) hinder (d) accept

✓ *(c)* 'hinder' is the most suitable meaning of the word 'impede'. Both words mean to delay or prevent someone by obstructing them.

17. <u>Extradition</u> of the leader of the group was debated for hours in the meeting.
(a) acceptance (b) sentence
(c) extension (d) deportation

✓ *(d)* 'deportation', is closest in meaning to the word 'extradition'. The word means the action of deporting a person accused of a crime.

18. It was felt that the decision to remove the group from the exercise would be <u>detrimental</u> to the organisation.
(a) beneficial (b) harsh
(c) disadvantageous (d) demanding

✓ *(c)* 'disadvantageous' is closest in meaning to the word 'detrimental'. The word means adverse, undesirable or harmful.

19. His <u>derisive</u> behaviour has led to the situation we face now.
(a) mockery (b) conducive
(c) encouraging (d) contemptuous

✓ *(d)* The word 'derisive' means expressing contempt or ridicule. So, option (d) 'contemptuous' is the most suitable option to select.

20. Any classroom should provide an <u>engaging</u> environment for learners.

(a) carefree (b) appealing
(c) thinking (d) dreaming

✓ ***(b)*** appealing is closest in meaning to the word 'engaging' which means delightful or pleasing.

Directions (21-25) In this section, you have few short passages. After each passage, you will find some items based on the passage. First, read a passage and answer the items based on it. You are required to select your answer based on the contents of the passage and opinion of the author only.

PASSAGE 1

Daily consumption of a certain form of curcumin improved memory and mood in people with mild, age-related memory loss. The research examined the effects of an easily absorbed curcumin supplement on memory performance in people without dementia, as well as curcumin's potential impact on the microscopic plaques and tangled in the brains of people with Alzheimer's disease.

Found in turmeric, curcumin has previously been shown to have anti-inflammatory and antioxidant properties in laboratory studies. It has also been suggested as a possible reason that senior citizens in India, where curcumin is a dietary staple, have a lower prevalence of Alzheimer's disease and better cognitive performance.

21. Which of the following statements are true?

1. Senior citizens in India have high level of Alzheimer's disease because of consumption of turmeric.
2. Senior citizens in India do not, have high prevalence of Alzheimer's because of consumption of turmeric.
3. Consumption of turmeric enhances cognitive performance.
4. Curcumin is an antioxidant.

Select the correct answer using the codes given below

(a) 2, 3 and 4 (b) 1, 3 and 4
(c) 1 and 4 (d) 1 and 3

✓ ***(a)*** Statement 2, 3 and 4 are true as per the passage.

22. Curcumin has positive effect on people

(a) without dementia
(b) with Alzheimer's disease
(c) without dementia and with Alzheimer's disease
(d) with dementia and with Alzheimer's disease

✓ ***(c)*** As stated in the passage curcumin has positive effect on people without dementia and also on people suffering from Alzheimer's disease.

23. Which word in the passage means 'earlier'?

(a) Performance (b) Absorbed
(c) Properties (d) Previously

✓ ***(d)*** Previously means earlier or in the time gone by.

24. Eating turmeric

(a) will reduce the chance of getting Alzheimer's disease
(b) will increase curcumin
(c) will enhance dementia
(d) will reduce chance of getting cancer

✓ ***(a)*** As inferred from the passage, consumption of turmeric can reduce the chance of getting Alzheimer's disease as it contains as antioxidant called curcumin.

25. .... of a disease in a region depends on the food habits too.

(a) Dominance (b) Prevalence
(c) Affection (d) Death

✓ ***(b)*** Prevalence is the best option for this sentence. The word means widespread presence or a condition which is quite common.

PASSAGE 2

Mr. Rowland Hill, when a young man was walking through the Lake district, when he one day saw the postman deliver a letter to a woman at a cottage door. The woman turned it over and examined it and then returned it, saying she could not pay the postage, which was a shilling. Hearing that the letter was from her brother, Mr. Hill paid the postage, in spite of the manifest unwillingness of the woman. As soon as the postman was out of sight, she showed Mr. Hill how his money had been wasted, as far as she was concerned. The sheet was blank. There was an agreement between her brother and herself that as long as all went well with him, he should send a blank sheet in this way once a quarter and she thus had tidings of him without expense of postage.

26. The story uses irony as a technique because

(a) the woman returned her own brother's letter without opening it
(b) the woman broke the agreement of receiving blank letters to convey well being of her brother
(c) Mr. Hill accepted the letter addressed to the woman
(d) in the modern times a brother has no time to write a letter to his own sister

✓ ***(a)*** Woman returned her own brother's letter without reading it.

27. The woman returned the letter to the postman because

(a) she could not pay the postage
(b) the letter was not addressed to her
(c) she already knew the contents of the letter
(d) she hated the person who wrote the letter

✓ ***(c)*** The woman returned the letter to the postman because she already knew the contents of the letter.

28. Mr. Hill paid the postage because

(a) the letter was from her brother
(b) the woman was his relative
(c) the letter was addressed to him
(d) he wanted to be kind to her

✓ ***(d)*** Mr. Hill paid the postage because he wanted to be kind to the woman as letter was from her brother.

29. The envelope contained

(a) a currency note (b) two written sheets
(c) no sheet at all (d) a blank sheet

✓ ***(d)*** As mentioned in the passage, the envelope contained a blank sheet only.

30. The woman and her brother had agreed that

(a) the letter with no postage meant good news
(b) the blank sheet meant being well
(c) the blank sheet meant bad news
(d) the letter with no postage meant unimportant news

✓ ***(b)*** There was an agreement between the woman and her brother that the blank sheet meant all was well with him.

PASSAGE 3

In good many cases unnecessary timidity makes the trouble worse than it needs to be. Public opinion is always more tyrannical towards those who obviously fear it than towards those who feel indifferent to it. A dog will bark more loudly and bite more easily when people are afraid of it than when they treat him with contempt and the human herd has something of this same characteristic. If you show that you are afraid of them, you give promise of good hunting, whereas if you show indifference, they begin to doubt their own power and therefore, tend to let you alone.

31. If we are afraid of public opinion, the attitude of the people towards us is

(a) sympathetic
(b) indifferent
(c) admiration
(d) ruthless

✓ ***(d)*** According to the passage if we are afraid of public opinion, the altitude of the people towards us is ruthless attitude or tyrannical.

32. The statement, ''A dog will bark more loudly and bite more easily when people are afraid of him, than when they treat him with contempt ......'' implies that

(a) barking dogs seldom bite
(b) we should not be afraid of dogs
(c) if we are afraid of others, they will leave us alone
(d) if we are afraid of people, they will try to scare us more

✓ ***(d)*** This statement implies that if we are afraid of people, they will try to scare us more.

33. The author compares men with dogs in respect of

(a) attacking others without any reason
(b) attacking others when they are weak
(c) barking and biting
(d) faithfulness to the master

✓ ***(b)*** People as well as dogs try to attack others when they are weak or afraid of them.

34. 'You give promise of good hunting' means

(a) you are vulnerable
(b) you are challenging
(c) you are indomitable
(d) you are confused

✓ ***(a)*** The statement 'you give promise of good hunting' means that you are vulnerable which means exposed to the possibility of being attacked or harmed either physically or emotionally.

PASSAGE 4

We live in a curious age. We are offered glimpses of a world civilisation slowly emerging, e.g., the U.N. special agencies dedicated to health and education. But along with these are sights and sounds that suggest that the whole civilisation is rapidly being destroyed.

Two official policies clash and instantly embassies are attacked by howling mobs of students, at once defying law, custom and usage. And that this may not be merely so many hot-headed lads escaping all control and may itself be part of the policy of the political parties, that is, mob antics as additional propaganda to deceive world opinion, makes the situation even worse.

Parties have always been dishonest, but now it seems as if power-mania is ready to destroy those civilities that make international relations possible.

There is something even worse. What inspires these students to burn cars and books is not their political enthusiasm but a frenzied delight in destruction, an urge towards violent demolition.

35. The author calls our age curious because

(a) it is an age of science and scientists are curious by nature
(b) it is witnessing the emergence of a world civilisation
(c) it is witnessing incidents that threaten to shake the very foundations of civilisation
(d) it is an age of contradictions consisting of constructive and destructive activities

✓ ***(d)*** The author says that this modern age is curious because it is an age of contradictions consisting of constructive and destructive activities.

36. It is deplorable to witness mob attacks on embassies following a clash of policies of two official policy makers because

(a) students should not take part in politics, but should concentrate on their studies
(b) they may result in the loss of lives of young and promising students
(c) they are overlooked by the policy planners themselves
(d) they are indicative of the complete failure of the government in controlling the rebellious students

✓ ***(c)*** As remarked or mentioned in the passage, the attacks by students are overlooked by the policy planners or policy makers themselves.

37. One aspect of the mob indulging in violence and arson is that they

(a) destroy very costly things like vehicles
(b) destroy very valuable artifacts and books
(c) get a mad delight in destruction for the sake of destruction only
(d) are motivated by certain political ideology to resort to destruction

✓ ***(c)*** A worrying aspect of mob attack is that they get a mad delight in destruction for the sake of destruction only.

38. In the passage, the word 'demolition' has the meaning as the word

(a) defying
(b) antics
(c) destruction
(d) urge

✓ ***(c)*** destruction or knocking down.

Directions (Q.Nos. 39-48) Each item in this section has a sentence with three underlined parts labelled (a), (b) and (c).

Read each sentence to find out whether there is any error in any underlined part and indicate your response on the answer sheet against the corresponding letter i.e., (a), (b), (c). If you find no error, your response should be indicated as (d).

39. The letter has been written (a) I insist on (b) it being sent at once. (c) No error (d)

✓ ***(b)*** Delete 'on' after insist. The word does not require any preposition. It means to say firmly or demand forcefully.

40. "I'm tired of my boys", said the mother, (a) "Both of them keep quarrelling all the time (b) right now also they are quarrelling with one another." (c) No error (d)

✓ ***(c)*** Use 'each other' in place of 'one another' as reference is for two as suggested by the use of word 'both'.

41. Sherly wants to know (a) whether you are going (b) to Delhi today night. (c) No error (d)

✓ ***(c)*** Today night is not correct. We must write 'tonight'. That means approaching evening or night.

42. The visitor's to the zoo are requested, (a) in the interest of all concerned, (b) not to carry sticks, stones or food inside and not to tease animals. (c) No error (d)

✓ ***(a)*** Use of apostrophe is incorrect here. Just write the 'visitors'.

43. The legendary hero (a) laid down his precious life (b) for our country . (c) No error (d)

✓ ***(c)*** Use of pronoun 'our' is incorrect here. Use 'his' in its place.

44. Our gardener, which is very lazy, (a) says that (b) there will be no apples this year. (c) No error (d)

✓ ***(a)*** Use of 'which' pronoun is incorrect. Use 'who' for gardener. Who is generally used to refer to people where as 'which' is used to refer to animals or things.

45. When I asked the guest (a) what she would like to drink (b) she replied that she preferred coffee much more than tea. (c) No error (d)

✓ ***(c)*** Use of 'much more than' is incorrect here. We should write she preferred coffee to tea. The word prefer means more liking to a particular thing.

46. No sooner did I reached there (a) the children left the place (b) with their parents. (c) No error (d)

✓ ***(a)*** Part 'A' has an error. With helping verb, 'did' we use first form of verb. So the use of reached is not correct, write 'did I reach' there.

47. I did not want to listen to him, (a) but he was adamant and (b) discussed about the matter. (c) No error (d)

✓ ***(c)*** Use of preposition 'about' after discussed is not correct, remove it. The word 'discussed' itself means to talk about something.

48. Please note (a) that the interview for the post (b) shall be held on 15th June, 2019 between 10.00 a.m. to 2.00 p.m. (c) No error (d)

✓ ***(c)*** Use of 'shall' is incorrect here. As interview is singular in form, it should be replaced by 'will be held'.

Directions (Q. Nos. 49-63) Each of the following sentences in this section has a blank space with four words or group of words given. Select whichever word or group of words you consider most appropriate for the blank space.

This cultural form .... 49. and indicate your response on the answer sheet accordingly. Japan has a name which means 'whimsical or impromptu pictures'. It .... 50. in existence since the 12th century when the first ...... 51. for this art form was seen. Since the language itself is read from right to left, the books with .... 52. art form follow the same pattern. .... 53. when english translations were made, they flipped the pictures and published it. This .... 54. the purists as it showed left-handed samurai, who did not exist in the original book. Hence, nowadays even English translations follow .... 55. right to left format.

The name of this art form is Manga. So the .... 56. is that in the present social structure, discipline has become an important factor because we want large numbers of children .... 57. together and .... 58. Educated to be what? To be bank clerks or super salesmen, capitalists or commissars. When you are a superman .... 59. as/or a super governor or a subtle parliamentary debater, what have you done? You are probably very clever, full of facts. Anybody can pick up facts; but we are human beings, not factual machines, not .... 60. routine automations. But again, sirs, you are not interested. You are listening to me and .... 61. each other, your are not going to do a thing about radically changing the education system; so it will drag on .... 62. a monstrous revolution, which will merely be another substitution-there will be much more control because the totalitarian government knows how to shape the minds and hearts of the people, they .... 63. the trick.

49. (a) originating (b) originates (c) originated (d) organising

✓ ***(a)*** originating. As whole passage is in present tense, use of 'originating' is correct which means 'having a specified beginning'.

50. (a) had been (b) has been (c) was (d) is

✓ ***(b)*** has been, use of present perfect is required here.

51. (a) instance (b) incident (c) accident (d) events

✓ ***(a)*** Instance

52. (a) that (b) this (c) these (d) which

✓ ***(b)*** this

53. (a) For (b) Beginning (c) During (d) Initially

✓ ***(d)*** Initially

54. (a) enrage (b) enlarged (c) engraved (d) enraged

✓ ***(d)*** Enraged means to make people angry.

55. (a) the (b) a (c) some (d) same

✓ ***(d)*** Same. The word 'the' should come before same.

56. (a) difficulty (b) difficult (c) difference different

✓ ***(a)*** difficulty

57. (a) educated (b) to be educated (c) to be educating (d) to educate

✓ ***(b)*** to be educated (use of passive voice is required here.

58. (a) as quick as possible (b) as quickly as possible (c) as possible as (d) quickly

✓ ***(b)*** as quickly as possible

59. (a) of some kind (b) of same kind (c) of some (d) of same

✓ ***(a)*** of some kind

60. (a) beast (b) bear (c) beastly (d) bare

✓ ***(d)*** Use of 'bare' which means here basic and simple.

61. (a) smiling for (b) smiling to (c) smiling with (d) smiling at

✓ ***(d)*** smiling at

62. (a) until there are (b) still there is (c) till there was (d) till there is

✓ ***(d)*** till there is

63. (a) had learnt (b) learnt (c) have learnt (d) had been learnt

✓ ***(c)*** have learnt

Directions (Q. Nos. 64-68) Given below are some idioms/phrases followed by four alternative meanings to each. Choose the response (a), (b), (c) or (d) which is the most appropriate expression.

64. He makes decision on the fly.
(a) He decides quickly without any seriousness
(b) He decides with all seriousness
(c) He decides non-chalantly
(d) He is unwilling to decide

✓ ***(a)*** The phrase 'on the fly' means to do something quickly without planning or thinking about it in advance. e.g. The new rules have been created 'on the fly'.

65. Follow suit
(a) Following someone's suit
(b) Suiting to someone
(c) Doing the same as someone else has just done
(d) Doing the same kind of mistake

✓ ***(c)*** 'Follow suit' means 'doing the same as someone else has just done. e.g. When one airline reduces ticket prices, the others usually follow suit.

66. Close shave
(a) Shaving very closely
(b) Miraculous escape
(c) Saving someone from danger
(d) Easy escape

✓ ***(b)*** The idiom 'close shave' means miraculous escape. A close shave means a situation where one avoids something dangerous. e.g. The car almost hit the child who ran out in front of it. It was a 'close shave'.

67. At the crossroads
(a) At important point of a decision
(b) At an important point of journey
(c) At the important road of a journey
(d) At an important stage or decision

✓ ***(a)*** The idiom 'At the cross roads' means at important point of a decision. When a choice must be made, it is called at the cross roads. e.g. He is at the cross roads in his career-either he stays in his current job and waits for promotion or accept new post abroad.

68. A pearl of wisdom
(a) A wise man
(b) An important piece of order
(c) An important piece of pearl
(d) An important piece of advice

✓ ***(d)*** 'A pearl of wisdom' means an important or valuable piece of advice. e.g. The old woman shared her 'pearls of wisdom' with the youngsters.

Directions (Q. Nos. 69-88) Each of the following items in this section consists of a sentence the parts of which have been jumbled. These parts have been labelled P, Q, R and S are given below each sentence in four sequences namely (a), (b), (c) and (d). You are required to rearrange the jumbled parts of the sentence and mark your response accordingly.

69. have become integral to most people's lives
(P)
debate for years as the devices
(Q)
have drawn intense interest and
(R)
safety questions about cell phones
(S)
(a) P Q R S (b) R S P Q
(c) S P Q R (d) S R Q P

✓ ***(d)*** SRQP is the correct order of sentences.

70. by means of education
(P)
civilisation to bring about
(Q)
it is difficult in modern
(R)
an integrated individual
(S)
(a) R Q S P (b) R S P Q
(c) S P Q R (d) P R Q S

✓ ***(a)*** RQSP

71. is that it is not professional enough
(P)
have not done their home work
(Q)
a valid criticism of the profession of politics in India
(R)
as the majority of its practitioners
(S)
(a) R S P Q (b) R P S Q
(c) S P Q R (d) P Q R S

✓ ***(b)*** RPSQ

72. that suit partisan political objectives
(P)
when great historical figures are appropriated
(Q)
we are living at a time
(R)
and reduced into stereotypes
(S)
(a) R Q P S (b) R Q S P
(c) S Q R P (d) P R Q S

✓ ***(b)*** RQSP

73. it is in this context that
(P)
and prosperity must be viewed
(Q)
the role of agriculture
(R)
as a provider of jobs
(S)
(a) P Q R S (b) R S P Q
(c) P R S Q (d) R S Q P

✓ ***(c)*** PRSQ

74. and they largely relied on agriculture, fishing and hunting
(P)
the people had a subsistence economy
(Q)
from excavation sites indicate that
(R)
rich materials found
(S)
(a) R Q P S (b) Q S P R
(c) S P Q R (d) S R Q P

✓ ***(d)*** SRQP

75. and that is 'To learn to say I am sorry'
(P)
something important enough that
(Q)
but surely there must be
(R)
everyone should learn it
(S)
(a) R Q S P (b) R Q P S
(c) S P Q R (d) P R Q S

✓ ***(a)*** RQSP

76. or an independent judiciary
(P)
a free press is
(Q)
as essential a limb of democracy as a Parliament
(R)
freely elected by the people
(S)
(a) R Q S P
(b) Q R P S
(c) S P Q R
(d) Q R S P

✓ ***(d)*** QRSP

77. the opinion that a human life
(P)
and that he would quite like to live that long
(Q)
could span 125 years
(R)
there was a time when Gandhi expressed
(S)
(a) S P R Q
(b) R Q P S
(c) S P Q R
(d) Q R S P

✓ ***(a)*** SPRQ

78. I must say what I feel
(P)
I am a votary of truth and
(Q)
to what I may have said before
(R)
and think at a given moment without regards
(S)

(a) R Q S P (b) Q R P S
(c) P S R Q (d) Q P S R

✓ *(d)* QPSR

79. The man in the competition
(P)
has been elected
(Q)
as the chairperson of the sports committee
(R)
in red who stood first
(S)

(a) S P Q R (b) S R P Q
(c) P S R Q (d) Q R S P

✓ *(a)* SPQR

80. One of the difficulties the whole of mankind
(P)
or affect the masses
(Q)
the day after tomorrow
(R)
is that we want to transform
(S)

(a) S P Q R (b) P R S Q
(c) S P R Q (d) Q R S P

✓ *(a)* SPQR

81. The speaker of their in action
(P)
has identified (Q) charging the opponents (R)
many issues besides
(S)

(a) P S Q R (b) Q S R P
(c) S P R Q (d) Q S P R

✓ *(b)* QSRP

82. The government and job markets
(P)
must offer convincing solutions
(Q)
to the crises in the rural economy
(R)
that are causing social ferment
(S)

(a) Q R P S (b) Q S R P
(c) S P R Q (d) P R S Q

✓ *(a)* QRPS

83. The best part of literary flourishes
(P)
and locates the story with the larger framework of our world
(Q)
long-formed journalism is that
(R)
it brings back the importance of writing skills
(S)

(a) Q R P S (b) R S P Q
(c) S P R Q (d) P R S Q

✓ *(d)* PRSQ

84. Children that grow into beautiful trees
(P)
of a warm home and supportive surroundings
(Q)
are like the tender samplings
(R)
with the sunshine and rain
(S)

(a) Q R P S (b) R P Q S
(c) R P S Q (d) P R S Q

✓ *(c)* RPSQ

85. We with real life experiences
(P)
tend to learn with interest
(Q)
when we see beauty in our work
(R)
and connect learning
(S)

(a) Q R S P (b) R P Q S
(c) S P R Q (d) P R S Q

✓ *(a)* QRSP

86. Elementary education ensuring the growth of a nation
(P)
is inevitable (Q) in developing the children (R)
to further education, thereby
(S)

(a) Q R P S (b) R P Q S
(c) S P R Q (d) Q R S P

✓ *(d)* QRSP

87. National Building Organisation besides conducting surveys on housing
(P)
and disseminates the statistical information
(Q)
collects, tabulates
(R)
on housing and building construction activities
(S)

(a) Q R P S (b) R P Q S
(c) S P R Q (d) R Q S P

✓ *(d)* RQSP

88. The Himalayan range sacred to the Gaddi people
(P)
is home
(Q)
to a chain of high altitude lakes
(R)
that towers over the Kangra valley
(S)

(a) Q R P S
(b) S P Q R
(c) S Q R P
(d) R Q S P

✓ *(c)* SQRP

Directions (Q. Nos. 89-98) In this section each item consists of six sentences of a passage. The first and the sixth sentences are given as S1 and S6. The middle four sentences in each have been jumbled up and labelled P, Q, R and S. You are required to find the proper sequences of the four sentences.

89. S1 : The giant wall of the Dhauladhar range in Himachal Pradesh is one of the most stunning sights in the Himalayas.

P : As the life line of the region it acts as a watershed ridge between Chamba's Ravi river system and Kangra's Beas river system.

Q : Although of modest altitude compared to other Himalayan ranges-the highest Dhauladhar peak is less than 5,000 m.

R : Thus, the Dhauladhar could be stated as the life line of the region.

S : Despite of that, the range sweeps up an astounding 12,000 ft. from the valley floor, creating a barrier wall in that is striking to look at.

S6 : Looming over the hill stations of Dharmsala and McLeodganj, the Dhauladhar is a popular trekking destination.

(a) Q R P S (b) S P Q R
(c) Q S R P (d) R Q S P

✓ *(d)* RQSP

90. S1 : Truth is far more important than the teacher.

P : Without self-knowledge, the airplane becomes the most destructive instrument in life; but with self-knowledge, it is a means of human help.

Q : Wisdom begins with self-knowledge; and without self-knowledge, mere information leads to destruction.

R : In other words, you have to be the perfect teacher to create a new society; and to bring the perfect teacher into being, you have to understand yourself.

S : Therefore you, who are the seeker of truth, have to be both the pupil and the teacher.

S6 : So, a teacher must obviously be one who is not within the clutches of society, who does not play power politics or seeks position or authority.

(a) Q R S P (b) S R Q P
(c) Q S R P (d) R Q S P

✓ *(b)* SRQP

91. S1 : Though most of us talk of discipline, what do we mean by that word?

P : The teacher would understand each child and help him in the way required.

Q : But if you have five or six in a class and an intelligent understanding teacher with a warm heart, I am sure there would be no need discipline.

R : When you have a hundred boys in a class, you will have to have discipline; otherwise there will be complete chaos.

S : Discipline in schools becomes necessary when there is one teacher to a hundred boys and girls.

S6 : And most of us are interested in mass movements, large schools with a great many boys and girls; we are not interested in creative intelligence, therefore we put up huge schools with enormous attendances.

(a) Q R S P (b) S R Q P
(c) Q S R P (d) R Q P S

✓ *(d)* RQPS

92. S1 : Tolstoy Farm was founded in 1910 by which time Gandhi had already conceptualised ideas that he would develop in India.

P : He was rich and used his money to buy the land and help set up the farm.

Q : A Jewish architect, Kallenbach was by his side through this period.

R : Tolstoy Farm became the subject of research for different kinds of cooperative communities across the world.

S : He first put in the social, moral, religious components of his doctrine.

S6 : Both he and Gandhi often referred to the time that they spent in Tolstoy Farm as among the happiest in their lives.

(a) Q R S P (b) S Q P R
(c) S Q R P (d) R Q P S

✓ *(b)* SQPR

93. S1 : Decentralised planning is a process of planning that begins from the grassroots level taking into confidence all the beneficiaries.

P : Under decentralised planning, the operation is from bottom to top.

Q : It can be said that it is more connected with the capitalistic economies.

R : It empowers the individuals and small groups to carry out their plans for their achievement of a common goal.

S : The decentralised planning is implemented through market mechanism.

S6 : But it cannot be described as undemocratic for most national states adopt such a planning now.

(a) Q R S P (b) S R Q P
(c) S Q R P (d) S R P Q

✓ *(c)* SQRP

94. S1 : It is doubtful if mankind, throughout his long history, has ever lived at all 'sustainably'.

P : But in general mankind has regarded the environment as an endless 'resource' to be exploited and plundered.

Q : May be a few isolated tribal groups found the necessary balance with nature lived without the desire for endless 'more'.

R : Now we have reached a point where we are on the verge of destroying ourselves and most of the life on Earth.

S : This process has accelerated greatly since the industrial revolution.

S6 : The concept of 'sustainable' is so far from reality that it is almost laughable.

(a) P Q R S (b) Q P S R
(c) P Q S R (d) S R Q P

✓ *(b)* QPSR

95. S1 : Measurement is an important concept in performance management.

P : It also indicates where things are not going so well, so that corrective action can be taken.

Q : It identifies where things are going well to provide the foundations for building further success.

R : It is the basis for providing and generating feedback.

S : Measuring performance is relatively easy for those who are responsible for achieving quantified targets for example sales.

S6 : It is more difficult in the case of knowledge workers e.g. scientists and teachers.

(a) R Q P S (b) Q P S R
(c) P S Q R (d) S P Q R

✓ *(b)* QPSR

96. S1 : Equity theory is concerned with the perception people have about how they are being treated compared with others.

P : To be dealt with equitably is to be treated fairly in comparison with another group of people or a relevant other person.

Q : Equity involves feelings and perceptions and is always a comparative process.

R : Equity theory states, in effect, that people will be better motivated if they are treated equitably and demotivated if they are treated inequitably.

S : It is not synonymous with equality, which means treating everyone the same, since this would be inequitable if they deserve to be treated differently.

S6 : This explains only one aspect of the process of motivation and job satisfaction, although it may be significant in terms of morale.

(a) P Q R S (b) P Q S R
(c) R S Q P (d) Q P R S

✓ *(c)* RSQP

97. S1 : We cannot understand the power of rumours and prophecies in history by checking whether they are factually correct or not.

P : The rumours in 1857 began to make sense when seen in the context of the policies the British pursued from the late 1820s.

Q : Rumours circulate only when they resonate with the deeper fears and suspicions of people.

R : Under the leadership of Governor General Lord William Bentinck, the British adopted policies aimed at reforming Indian society by introducing Western education, Western ideas and Western Institutions.

S : We need to see what they reflect about the minds of people who believed them-their fears and apprehensions, their faiths and convictions.

S6 : With the cooperation of sections of Indian society they set up English-medium schools, colleges and universities which taught Western sciences and liberal arts.

(a) S Q P R (b) Q S P R
(c) P R S Q (d) R S P Q

✓ *(b)* QSPR

98. S1 : The Constitution of India thus emerged through a process of intense debate and discussion.

P : This was an unprecedented act of faith, for in other democracies the vote had been granted slowly, and in stages.

Q : However, on one central feature of the Constitution there was substantial agreement.

R : Many of the provisions were arrived at through a process of give and take, by forgiving a middle ground between two opposed positions.

S : This was on the granting of the vote to every adult Indian.

S6 : In countries such as the United States and the United Kingdom, only men with education were allowed into the charmed circle.

(a) P R S Q
(b) R Q S P
(c) S P R Q
(d) Q S R P

✓ *(b)* RQSP

Directions (Q. Nos. 99-109) **Each item in this section consists of a sentence with an underlined word/words followed by four words.**
Select the option that is opposite in meaning to the underlined word/ words.

99. His ideas are <u>obscure</u>.
(a) New (b) Clear
(c) Infamous (d) Obscene

✓ *(b)* clear is antonym of word 'obscure', which means something uncertain or doubtful. 'Clear' means which is easy to understand or transparent.

100. Ravi is jovial and he makes the environment <u>sanguine</u>.
(a) 'Pessimistic' (b) Optimistic
(c) Humorous (d) Rebellious

✓ *(a)* 'Pessimistic', the word 'Sanguine' means optimistic or positive especially in a bad or difficult situation. So, its antonym is 'Pessimistic' which means gloomy or believer in that worst will happen.

101. There prevailed a <u>woebegone</u> feeling in the room.
(a) Sad (b) Cheerful
(c) Sleepy (d) Thoughtful

✓ *(b)* The word 'woebegone' means someone sad or miserable in appearance. Its antonym will be 'cheerful' which means noticeably happy and optimistic.

102. It appears that the whole group is <u>mutinous</u>.
(a) Arrogant (b) Lucky
(c) Obedient (d) Sincere

✓ *(c)* The word 'mutinuous' means a person or soldier refusing to obey the authority. Its antonym is 'obedient'. The word means a person who is willing to comply with an order or request.

103. They consider themselves as <u>foes</u> from birth.
(a) Protagonists (b) Opponents
(c) Friends (d) Soul mates

✓ *(c)* 'Foe' means enemy or opponent. Its antonym is friends which means a person with whom one has bond of affection.

104. This painting has a <u>distinctive</u> element which can be noticed well.
(a) Salient (b) Common
(c) Great (d) Unique

✓ (b) The word 'distinctive' means unique or remarkable or uncommon. So, from the options given 'common' is appropriate antonym which means something which is found or done very often or which is prevalent.

105. The entry was carried out <u>inadvertently</u>.
(a) Purposely
(b) Purposively
(c) Accidently
(d) Not noticing

✓ *(a)* The word 'inadvertently' means something done without intention or accidentally. Its antonym is 'purposely' which means something done deliberately or intentionally.

106. The whole audience showed a <u>disdainful</u> attitude during the match.
(a) Sneering
(b) Respectful
(c) Mocking
(d) Cheerful

✓ *(b)* The word 'disdainful' means showing lack of respect or derisive. Its antonym in 'respectful' which means showing respect or reverence.

107. <u>Efficacy</u> of the project needs an examination.
(a) Inefficiency (b) Efficiency
(c) Value (d) Effectiveness

✓ *(a)* The word 'efficacy' means the ability to produce a desired or intended result. Its antonym is 'inefficiency' suggesting a state of not achieving maximum productivity, or failure to make best use of time or resources.

108. Her <u>rebuttal</u> that she was not involved in the case was considered by the court.
(a) Refusal (b) Denial
(c) Acceptance (d) Kindness

✓ *(c)* The word 'rebuttal' means denial or negation. Its antonym from the given option is 'Acceptance' which means a general agreement, or the act the agreeing to an offer, plan or invitation.

109. The baby could not move as the place was <u>soggy</u>.
(a) Sodden (b) Dry
(c) Hot (d) Wet

✓ *(b)* dry. The word 'soggy' means wet and soft. Its antonym is 'dry' which means something free of moisture, not wet.

Directions (Q. Nos. 110-115) **Each item in this section has a direct statement followed by its reported form in indirect speech. Select the correct statement in indirect speech and mark it in the answer sheet accordingly.**

110. The captain said to his soldiers, "Move forward and face the target now.''
(a) The captain ordered his soldiers to move forward and face the target
(b) The captain informed his soldiers that they should move forward and face the target now
(c) The captain asked his soldiers to move forward and face the target then
(d) The captain told his soldiers that they move forward and face the target immediately

✓ *(a)* It is correct sentence (Imperative type) in Indirect speech. Here, principal verb is 'ordered' which is correct as a captain orders to his soldiers.

111. Vivek said to his friend, "Could you please turn off the switch ?"

(a) Vivek told his friend to turn off the switch
(b) Vivek asked his friend to please turn off the switch
(c) Vivek requested his friend to turn off the switch
(d) Vivek told his friend that he should turn off the switch

✓ ***(c)*** As the sentence in direct speech is a type of request, option (c) is correct statement in indirect speech.

112. The manager said to his colleagues, "We have received a serious threat to our business now and we need to act to face it."

(a) The manager told his colleagues that they had received a serious threat to our business then and they needed to act to face it
(b) The manager told his colleagues that they received a serious threat to their business then and they needed to act to face it
(c) The manager said his colleagues that they had received a serious threat to our business then and they needed to act to face it.
(d) The manager told his colleagues that they had received a serious threat to their business at that time and they needed to act to face it.

✓ ***(d)*** As the sentence is in past tense 'have received' will change to 'had received' and 'need' will be changed to needed.

113. Romila said to Rahim, "Where were your ideas when we faced the troubles last week ?"

(a) Romila asked Rahim where his ideas had been when they had faced the trouble the week before
(b) Romila asked Rahim where his ideas had been when they faced the trouble the last week
(c) Romila requested Rahim where his ideas had been when they faced the trouble the week before
(d) Romila told Rahim where his ideas were when they faced the trouble the week before

✓ ***(a)*** As sentence is interrogative type, the correct statement in Indirect speech with proper tense used in option (a).

114. The actor said to his co-star, Sarita, "Will you go with me for a cup of tea in the evening today?"

(a) The actor said to his co-star if she would go for a cup of tea with him in evening today
(b) The actor told his co-star, Sarita if she would go with him for a cup of tea in evening that day
(c) The actor requests his co-star, Sarita if she would go with him for a cup of tea in that evening that day
(d) The actor asked his co-star Sarita if she would go with him for a cup of tea in the evening that day

✓ ***(d)*** It is correct sentence in Indirect speech.

115. The preacher said to the crowd, "The Sun rises everyday for all of us without any expectations in return."

(a) The preacher told the crowd that the Sun rose everyday for all of them without any expectations in return
(b) The preacher told the crowd that the Sun rises everyday for all of us without any expectations in return
(c) The preacher told the crowd that the Sun has risen everyday for all of them without any expectations in return
(d) The preacher told the crowd that the Sun rises everyday for all of them without any expectations in return

✓ ***(d)*** It is correct sentence in Indirect speech.

Directions (Q. Nos. 116-120) Each item in this section has a sentence in active voice followed by four sentences one of which is the correct passive voice statement of the same. Select the correct one and mark it in the answer sheet accordingly.

116. The Members of the Parliament elect their group leader either by consensus or by voice vote.

(a) The group leader is elected by the Members of the Parliament either by consensus or by voice vote
(b) The group leader was elected by the Members of the Parliament either by consensus or by voice vote
(c) The group leader has been elected by the Members of the Parliament either by consensus or by voice vote
(d) The Members of the Parliament are elected by their group leader either by consensus or by voice vote

✓ ***(a)*** It is the correct sentence in Passive voice. Sentence is in simple present tense.

117. All the examinees have answered one particular question in the long answer writing section.

(a) One particular question is answered by all the examinees in the long answer writing section
(b) One particular question was answered by all the examinees in the long answer writing section
(c) All the examinees answered one particular question in the long answer writing section
(d) One particular question has been answered by all the examinees in the long answer writing section

✓ ***(d)*** It is the correct sentence in Passive voice. Sentence is in present perfect tense.

118. The writer who passed away recently has authored a dozen novels and a number of poetry collections.

(a) A dozen novels and a number of poetry collections have been authored by the writer who passed away recently
(b) A dozen novels and a number of poetry collections has been authored by the writer who passed away recently
(c) A dozen novels and a number of poetry collections were authored by the writer who passed away recently
(d) A dozen novels and a number of poetry collections had been authored by the writer who passed away recently

✓ ***(a)*** It is the correct sentence in Passive voice. Sentence is in present perfect tense.

119. Shut the door.

(a) Shut the door
(b) Let the door be shut
(c) The door be shut
(d) The door is shut

✓ ***(b)*** It is the correct sentence in Passive voice. An Imperative sentence in passive voice has the following structure. Let + object + be +past participle.

120. India won freedom with the blood and sweat of hundreds and thousands of Indians.

(a) India had won freedom with the blood and sweat of hundreds and thousands of Indians
(b) Freedom had been won by India with the blood and sweat of hundreds and thousands of Indians
(c) Freedom was won by India with the blood and sweat of hundreds and thousands of Indians
(d) Freedom was won by hundreds and thousands of Indians with their blood and sweat

✓ ***(c)*** It is the correct sentence in Passive voice. The sentence in active speech is in simple past tense.

PAPER III General Studies

1. Two reactants in a flask at room temperature are producing bubbles of a gas that turn limewater milky. The reactants could be

(a) zinc and hydrochloric acid
(b) magnesium carbonate and hydrochloric acid
(c) methane and oxygen
(d) copper and dilute hydrochloric acid

✓ ***(b)*** When magnesium carbonate ($MgCO_3$) reacts with hydrochloric acid (HCl), the product will be magnesium chloride ($MgCl_2$), water (H_2O) and carbon dioxide (CO_2).

$$MgCO_3(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)\uparrow$$

Above two reactants when take in a flask at room temperature produce bubbles of CO_2 (carbon dioxide) gas.

After that, CO_2 reacts with lime water to form calcium carbonate ($CaCO_3$), which is white and does not dissolve in water. Thus, causing the lime water turn milky.

[Lime water → Lime water is the common name for a diluted solution of calcium hydroxide ($Ca(OH)_2$). It is clear and colourless.]

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

2. How many moles of CO can be obtained by reacting 2.0 mole of CH_4 with 2.0 mole of O_2 according to the equation given below?

$$CH_4(g) + \frac{1}{2}O_2 \longrightarrow CO + 2H_2$$

(a) 2.0 (b) 0.5
(c) 2.5 (d) 4.0

✓ ***(a)*** **Step I** To find the limiting reagent,

$$CH_4(g) + \frac{1}{2}O_2 \longrightarrow CO + 2H_2$$

$$1 \quad : \quad \frac{1}{2} \quad : \quad 1 \quad : \quad 2$$

As per balanced equation,

For $\frac{1}{2}$ mole of O_2 we need = 1 mole of CH_4

∴ For 2 mole of O_2 (given) we need = 4 mole of CH_4

But we have given only 2 moles of CH_4, thus CH_4 is out limiting reagent.

Thus,

Step II Quantity of CO is obtained

∵ 1 mole of CH_4 give 1 mole of CO

∴ 2 moles of CH_4 give = 2 moles of CO

Hence, 2 moles of CO will be produced.

3. Reaction between which of the following two reactants will produce hydrogen gas?

(a) Magnesium and hydrochloric acid
(b) Copper and dilute nitric acid
(c) Calcium carbonate and hydrochloric acid
(d) Zinc and nitric acid

✓ ***(a)***

(i) Reaction between magnesium and hydrochloric acid

$$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)\uparrow$$

When magnesium reacts with hydrochloric acid, they produce hydrogen (H_2) gas.

(ii) Reaction between copper and dilute nitric acid

$$3Cu + 8HNO_3(\text{dilute}) \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

When copper reacts with dil. nitric acid, they cannot produce hydrogen (H_2) gas.

(iii) Reaction between calcium carbonate and hydrochloric acid

$$CaCO_3 + HCl \longrightarrow CaCl_2 + CO_2 + H_2O$$

Calcium carbonate reacts with hydrochloric acid which cannot produce hydrogen (H_2) gas.

(iv) Reaction between zinc and nitric acid

$$Zn + 4HNO_3 \longrightarrow Zn(NO_3)_2 + 2H_2O + 2NO_2$$

When zinc reacts with nitric acid, they also cannot produce hydrogen (H_2) gas.

Note Nitric acid is an oxidizing agent which produces oxides of nitrogen on reacting with metal. It do not produces hydrogen gas.

4. Which of the following characteristics is common to hydrogen, nitrogen, oxygen and carbon dioxide?

(a) They are all diatomic.
(b) They are all gases at room temperature.
(c) They are all coloured.
(d) They all have same reactivity.

✓ ***(b)*** H_2, N_2, O_2 and CO_2 are all gases at room temperature.

H_2,N_2 and O_2 are diatomic but CO_2 is triatomic. They all are colourless gases and they all have different reactivity.

5. The compound $C_7H_7NO_2$ has

(a) 17 atoms in a molecule of the compound
(b) equal molecules of C and H by mass
(c) twice the mass of oxygen atoms compared to nitrogen atoms
(d) twice the mass of nitrogen atoms compared to hydrogen atoms

✓ ***(a)*** $C_7H_7NO_2$

Number of carbon atoms = 7
Number of hydrogen atoms = 7
Number of nitrogen atoms = 1
Number of oxygen atoms = 2

An oxygen molecule O_2 contains 2 oxygen atoms. Total number of molecules = 7 + 7 + 1 + 2 = 17. So, 17 atoms are present in a molecule of the compound.

6. Which of the following is the general formula for saturated hydrocarbons?

(a) C_nH_{2n+2} (b) C_nH_{2n-2}
(c) C_nH_{2n+1} (d) C_nH_{2n-1}

✓ ***(a)*** Alkanes are saturated hydrocarbons. This means that they contain only carbon and hydrogen atoms bonded by single bonds only.

The general formula for an alkane is C_nH_{2n+2}.

In this formula, n = number of carbon.

7. A particle moves with uniform acceleration along a straight line from rest. The percentage increase in displacement during sixth second compared to that in fifth second is about

(a) 11% (b) 22% (c) 33% (d) 44%

✓ ***(b)*** The particle is moving with uniform acceleration along a straight line from rest.

Displacement during 6th second

$$S_{6th} = \vec{u} + \frac{1}{2}a(2n-1)$$

∵ $n = 6$

$$S_{6th} = \vec{u} + \frac{1}{2}a(12-1)$$

Given, $\vec{u} = 0$

So, $S_{6th} = \vec{a}\,\frac{11}{2} = \frac{11}{2}\vec{a}$

In 5th second $= \frac{1}{2}\vec{a}\,(10-1) = \frac{9\vec{a}}{2}$

$$= \frac{S_{6th} - S_{5th}}{S_{5th}}$$

$$= \frac{\frac{11\vec{a}}{2} - \frac{9\vec{a}}{2}}{\frac{9\vec{a}}{2}} = \frac{2}{9}$$

∴ $\% = \frac{2}{9} \times 100$

$= 22.22 \simeq 22\%$

8. If two miscible liquids of same volume but different densities P_1 and P_2 are mixed, then the density of the mixture is given by

(a) $\frac{P_1 + P_2}{2}$ (b) $\frac{2P_1P_2}{P_1 + P_2}$
(c) $\frac{2P_1P_2}{P_1 - P_2}$ (d) $\frac{P_1P_2}{P_1 + P_2}$

✓ ***(b)*** Density of liquid $A = P_1$

Density of liquid $B = P_2$

We assume the mass of liquid A or $B = m$

Volume of liquid $A = \frac{m}{P_1}$

$$\left[\text{Formula} \rightarrow \text{Volume} = \frac{\text{Mass}}{\text{Density}}\right]$$

Volume of liquid $B = \frac{m}{P_2}$

Total volume of the mixture $= \frac{m}{P_1} + \frac{m}{P_2}$

Total mass $= m + m = 2m$

Hence, density of mixture $= \frac{\text{Mass}}{\text{Volume}}$

$$= \frac{2m}{\left(\frac{m}{P_1} + \frac{m}{P_2}\right)} = \frac{2m}{m\left(\frac{1}{P_1} + \frac{1}{P_2}\right)}$$

$$= \frac{2}{\left(\frac{1}{P_1} + \frac{1}{P_2}\right)} \text{ (cancelling } m) = \frac{2}{\left(\frac{P_2 + P_1}{P_1 P_2}\right)}$$

Density of mixture $= \frac{2P_1P_2}{P_1 + P_2} = \frac{P_1 + P_2}{2}$

9. The position vector of a particle is $\vec{r} = 2t^2\hat{x} + 3t\hat{y} + 4\hat{z}$

Then the instantaneous velocity $\vec{v}$ and acceleration $\vec{a}$ respectively lie

(a) on xy -plane and along z-direction
(b) on yz-plane and along x-direction
(c) on yz-plane and along y-direction
(d) on xy -plane and along x-direction

✓ ***(d)*** $\vec{r} = 2t^2\hat{x} + 3t\ \hat{y} + 4\hat{z}$

Instaneous velocity $\vec{V}\frac{d\vec{v}}{dt} = 4t\hat{x} + 3\hat{y}$

Hence, istantaneous velocity lie on XY plane.

Instaneous acceleration $a = \frac{d\,\vec{v}}{dt} = 4\hat{x}$

Hence, acceleration lie along x – axis.

10. Two persons are holding a rope of negligible mass horizontally. A 20 kg mass is attached to the rope at the midpoint; as a result the rope deviates from the horizontal direction. The tension required to completely straighten the rope is ($g = 10\text{ m/s}^2$)

(a) 200 N
(b) 20 N
(c) 10 N
(d) infinitely large

✓ ***(a)*** Given, $m = 20$ kg

Tension required to completely straighten the rope.

From Newton's second law of motion, $F = ma$

$F = 20 \times 10$ ($\because$ acc n due to $g = 10$ m/s)

$= 20\,\text{kg} \times 10\,\text{m/s}^2 = 200$ N

11. Which one of the following does *not* convert electrical energy into light energy?

(a) A candle
(b) A light-emitting diode
(c) A laser
(d) A television set

✓ ***(a)*** Burning of a candle is a chemical as well as physical change in which chemical energy is converted into heat and light energy. A light-emitting diode, a laser, a television set convert electrical energy into light energy.

12. Which of the following is/are the main absorbing organ/organs of plants?

(a) Root only
(b) Leaf only
(c) Root and leaf only
(d) Root, leaf and bark

✓ ***(a)*** Roots are the main absorbing organ in plants. This underground part of plant absorbs water and soluble mineral salts from soil.

13. Which of the following is *not* a primary function of a green leaf?

(a) Manufacture of food
(b) Interchange of gases
(c) Evaporation of water
(d) Conduction of food and water

✓ ***(d)*** Conduction of food and water is not the primary function of green leaf. Manufacture of food by photosynthesis, gaseous interchange by stomatas and evaporation of water (Transpiration) are primary functions of leaves. Conduction of water and food is done by vascular tissues, i.e. conduction of water occurs *via* xylem and conduction of food occurs *via* phloem.

14. Which one of the following denotes a 'true' fruit?

(a) When only the thalamus of the flower grows and develops into a fruit
(b) When only the receptacle of the flower develops into a fruit
(c) When fruit originates only from the calyx of a flower
(d) When only the ovary of the flower grows into a fruit

✓ ***(d)*** True fruit develops from ovary of flower after fertilization, while false or pseudo fruit develops from other parts of flower, i.e. the edible part of apple is fleshy receptacle.

15. In which one of the following physiological processes, excess water escapes in the form of droplets from a plant?

(a) Transpiration
(b) Guttation
(c) Secretion
(d) Excretion

✓ ***(b)*** Guttation is special physiological process in which excess water escapes in the form of liquid droplets from the tips and margins of leaves. It occurs during night or early morning, when there is high atmospheric humidity and rate of transpiration is lower than the rate of water absorption.

16. If the xylem of a plant is mechanically blocked, which of the following functions of the plant will be affected?

(a) Transport of water only
(b) Transport of water and solutes
(c) Transport of solutes only
(d) Transport of gases

✓ ***(b)*** Xylem is responsible for transportation of water and solutes (soluble mineral ions). Thus its blocking will affect transportation of water and solutes.

17. Which one of the following agents does not contribute to propagation of plants through seed dispersal?

(a) Wind
(b) Fungus
(c) Animal
(d) Water

✓ ***(b)*** Except fungus, all of rest factors are helpful in seed dispersal. The dislocation of seeds from their parental plants by different mediums is known as seed dispersal. The seeds of calotropis, rafflesia and lotus are dispersed by wind, elephant and water respectively.

18. The visible portion of the electromagnetic spectrum is

(a) infrared
(b) radiowave
(c) microwave
(d) light

✓ ***(d)*** The visible portion of the electromagnetic spectrum is light. Visible rays are the most familiar form of electromagnetic waves. It is the part of the spectrum that is detected by human eyes. It runs from about 4×10^{14} Hz to about 7×10^{14} Hz or a wavelength of about 700-400 nm. Visible light emitted or reflected from objects around us provides us information about the world. Infrared, radiowaves, microwaves, etc are not visible portion of electromagnetic spectrum.

19. Which one of the following is the correct ascending sequence of states in terms of their population density as per Census 2011?

(a) Arunachal Pradesh–Sikkim–Mizoram–Himachal Pradesh
(b) Arunachal Pradesh–Mizoram–Sikkim–Himachal Pradesh
(c) Mizoram–Arunachal Pradesh–Himachal Pradesh–Sikkim
(d) Arunachal Pradesh–Himachal Pradesh–Sikkim–Mizoram

✓ ***(b)*** The correct ascending sequence of states in terms of their population density as per Census 2011 is

- Arunachal Pradesh–17
- Mizoram–52
- Sikkim–86
- Himachal Pradesh–550

20. The rate of population growth during 2001-2011 decade declined over the previous decade (1991-2001) in all of the following states, except

(a) Tamil Nadu
(b) Kerala
(c) Goa
(d) Andhra Pradesh

✓ ***(a)*** The rate of population growth during 2001 - 2011 decade and previous decade (1991-2001) in the given states is stated below

	State	State growth (2001-11)	Growth (1991-2001)
1.	Tamil Nadu	15.60	11.19
2.	Kerala	4.9	9.42
3.	Goa	8.2	14.89
4.	Andhra Pradesh	11.0	14.59

Thus, decadal growth rate of Tamil Nadu was increased from the decade 1991-2001 to 2001-2011.

21. Which one of the following statements with regard to growth of coral reefs is not correct?

(a) Coral can grow abundantly in fresh water.
(b) It requires warm water between 23°C-25°C.
(c) It requires shallow saltwater, not deeper than 50 metres.
(d) It requires plenty of sunlight to aid photosynthesis.

✓ ***(a)*** The formation of coral reefs only occurs in saltwater of oceans, where temperature is between 23°-25°C and for high productivity. They do not exist below 50 m due to requirement of plenty of sunlight.

22. As per Census 2011, the concentration of Scheduled Caste population (going by percentage of Scheduled Caste population to total population of the State) is the highest in the state of

(a) Uttar Pradesh
(b) Himachal Pradesh
(c) Punjab
(d) West Bengal

✓ ***(c)*** Punjab has the largest share of Schedule Caste people in its population at 31.9%. Himachal Pradesh and West Bengal follow Punjab with 25.2% and 23.5%. Uttar Pradesh has 20.7% Schedule Caste people in its population.

23. Which one of the following states has more than two major ports?

(a) Maharashtra (b) West Bengal
(c) Odisha (d) Tamil Nadu

✓ ***(d)*** Tamil Nadu has more than two major ports i.e., Chennai Port, Tuticorin Port and Ennore Port. Maharashtra has two major ports i.e., Jawaharlal Nehru Port Trust and Mumbai Port. Both West Bengal and Odisha have one major port.

24. The equivalent weight of $Ba(OH)_2$ is (given, atomic weight of Ba is 137.3)

(a) 85.7 (b) 137.3
(c) 154.3 (d) 171.3

✓ ***(a)*** The equivalent weight is that weight which will furnish 1 mole of H^+ or 1 mole of OH^- ions.

$$Ba(OH)_2 = Ba^{2+} + 2OH^-$$

One mole of $Ba(OH)_2$ furnishes 2 moles of OH^- (2 equivalents).

$Ba(OH)_2$ = molar mass

$1(137.3) + 2(16) + 2(1) = 171.3$ gm/mole

$Ba(OH)_2$ = equivalent weight $= \frac{171.3}{2}$

$= 85.7$ g/eq

25. Which one of the following nitrogen oxides has the highest oxidation state on nitrogen?

(a) NO (b) NO_2
(c) N_2O (d) N_2O_5

✓ ***(d)*** Oxidation state of oxygen = – 2
Oxidation state of nitrogen = x

(a) $NO = x + (-2) = 0;\ x - 2 = 0;\ x = +2$

Where, oxidation state of nitrogen is + 2.

(b) $NO_2;\ x + 2(-2) = 0;\ x - 4 = 0;\ x = +4$

In above compound, oxidation state of nitrogen is + 4.

(c) $N_2O;\ 2x - 2 = 0;\ 2x - 2,\ x = \frac{2}{2} = 1$

Where, oxidation state of nitrogen is + 1.

(d) $N_2O_5;\ 2x + 5(-2) = 0;\ 2x - 10 = 0$

$$x = \frac{10}{2} = 5$$

In this compound, oxidation state of nitrogen is + 5, which is the highest oxidation state of nitrogen.

26. Which one of the following is not true for the form of carbon known as diamond?

(a) It is harder than graphite.
(b) It contains the same percentage of carbon as graphite.
(c) It is a better electric conductor then graphite.
(d) It has different carbon to carbon distance in all directions.

✓ ***(c)*** (i) **Diamond** → It has a crystalline lattice.

- In diamond, each carbon atom undergoes sp^3 hybridisation and linked to four other carbon atoms.
- The structure extends in space and produces a rigid three-dimensional network of carbon atom.
- In this structure, directional covalent bonds are present throughout the lattice.
- There are no free electrons, in the structure of Diamond. Therefore, it is a very poor electric conductor.

(ii) **Graphite**

- Graphite has layered structure.
- Its layers are held by Van der Waals forces.
- Each layer is composed of planar hexagonal rings of carbon atoms.
- Electrons are mobile in graphite, therefore graphite conducts electricity.
- It is a good electric conductor than diamond.

27. In which one of the following reactions, the maximum quantity of H_2 gas is produced by the decomposition of 1 g of compound by H_2O/O_2?

(a) $CH_4 + H_2O \longrightarrow CO + 3H_2$
(b) $CO_4 + H_2O \longrightarrow CO_2 + H_2$
(c) $CH_4 + \frac{1}{2}O_2 \longrightarrow CO + 2H_2$
(d) $C_{12}H_{24} + 6O_2 \longrightarrow 12CO + 12H_2$

✓ ***(a)*** $n = \frac{w}{m}$ where, n = no. of moles

w = given mass
m = atomic mass

(i) $CH_4 + H_2O \longrightarrow CO + 3H_2$

$(12 + 4 = 16)$ gm + 1 gm $\longrightarrow$ 6 gm

∵ 16 gm will produce = 6 gm of H_2

∴ 1 gm will produce

$= \frac{6}{16} = 0.375$ gm of H_2

(ii) $CO + H_2O \longrightarrow CO_2 + H_2$

$(12 + 16 = 28)$ gm + 1 gm $\longrightarrow$ 2 gm

28 gm will produce = 2 gm of H_2

1 gm will produce $= \frac{2}{28} = 0.0714$ gm of H_2

(iii) $CH_4 + \frac{1}{2}O_2 \longrightarrow CO + 2H_2$

$(12 + 4 = 16)$ gm + 1 gm $\longrightarrow$ 4 gm of H_2

∴ 16 gm will produce = 4 gm of H_2

∴ 1 gm will produce

$= \frac{4}{16} = 0.25$ gm of H_2

(iv) $C_{12}H_{24} + 6O_2 \longrightarrow 12CO + 12H_2$

$(144 + 24 = 168)$ gm + 1 gm → 24gm of H_2

168 gm will be produced = 24 gm of H_2

∴ 1 gm will be produced $= \frac{24}{168}$

$= 0.142$ gm of H_2

28. When a convex lens produces a real image of an object, the minimum distance between the object and image is equal to

(a) the focal length of the convex lens
(b) twice the focal length of the convex lens

(c) four times the focal length of the convex lens
(d) one half of the focal length of the convex lens

✓ ***(c)*** Let s be the distance

$$\frac{1}{u}+\frac{1}{s-u}=\frac{1}{f}$$

$\Rightarrow \quad f(s-u)+fu-u(s-u)=0$

$\Rightarrow \quad fs-fu+fu-su+u^2=0$

$\Rightarrow \quad fs-su+u^2=0$

$\Rightarrow \quad u^2=(u-f)s$ or $s=\dfrac{u}{u-f}$

On differentiating, $\dfrac{ds}{du}=\dfrac{(u-f)(2u)-u^2}{(u-f)^2}$

$$=\frac{(u^2-2uf)}{(u-f)^2}=\frac{u(u-2f)}{(u-f)^2}=0$$

or $u(u-2f)=0 \Rightarrow u=0$, or $u=2f$

Thus, image and object is $2(2f)=4f$

Hence, the minimum distance between the object and image is equal to four times the focal length of the convex lens.

29. The direction of magnetic field at any location on the earth's surface is commonly specified in terms of
(a) field declination
(b) field inclination
(c) both field declination and field inclination
(d) horizontal component of the field

✓ ***(d)*** It is termed as horizontal component of the field. At a place, it is defined as the component of earth's magnetic field along the horizontal in the magnetic meridian.

Its value is different at different places. Field declination is the acute angle between magnetic meridian and geographical meridian at a place. Inclination is the angle which resultant earth's magnetic field at a place makes with the horizontal surface.

30. A circuit has a fuse having a rating of 5 A. What is the maximum number of 100 W-220 V bulbs that can be safely connected in parallel in the circuit?
(a) 20 (b) 15 (c) 11 (d) 10

✓ ***(c)*** Given,

$P=100$ W (power of 1 bulb)

$V=220$ Volt $\quad I=5$ A

Let, x be maximum number of bulbs that can be safely connected in parallel circuit.

Potential difference $(V)=220$ volt

$$P=VI$$

$$100\times x=220\times 5$$

$$x=\frac{220\times 5}{100}=11$$

31. Which one of the following can extinguish fire more quickly?
(a) Cold water (b) Boiling water
(c) Hot water (d) Ice

✓ ***(b)*** Boiling water can quickly entinguish fire more quickly because the process of fire extinguishing involves absorption of heat. Absorption of heat in converting hot water to steam is more than the heat absorbed in heating cold water to the boiling temperature. Hence, boiling water can extinguish fire more quickly than ice, cold water and hot water.

32. In which of the following, heat loss is primarily not due to convection?
(a) Boiling water
(b) Land and sea breeze
(c) Circulation of air around blast furnace
(d) Heating of glass surface of a bulb due to current in filament

✓ ***(d)*** Convection is the process of heat transfer of the bulk movement of molecules within fluids such as gases and liquids. In this process, the heat transfers is by the actual motion of matter. It happens in liquid and gases. Boiling water, land and sea breeze, circulation of air around blast furnace, etc are the example of convection. Whereas heating of glass surface of a bulb due to current in filament occurs due to conduction of heat.

33. Which one of the following features is an indication for modification of stem of a plant?
(a) Presence of 'eye' on potato
(b) 'Scale' found in onion
(c) 'Tendril' found in pea
(d) 'Hair' present in carrot

✓ ***(a)*** Potato is a modified stem, known as tuber. The eye of potato is a very small axillary bud, which contains small internodes and nodes. It can give rise to a new plant via asexual reproduction. The tendrils of pea and scales of onion are modification of leaves, while hairs on carrot are part of root.

34. Which of the following roles is/are played by epididymis, vas deferens, seminal vesicles and prostate in male reproductive system of human?
(a) Spermatogenesis and maturation of sperms
(b) Maturation and motility of sperms
(c) Spermatogenesis and motility of sperms
(d) Motility of sperms only

✓ ***(b)*** Epididymis, vas deferens, seminal vesicles and prostate in male reproductive system of human are involved in maturation and motility of sperms. Spermatogenesis occurs in testes.

35. Which one of the following is the special type of milk produced by a lactating mother, essential for the development of immune response of newborn baby in human?
(a) Breast milk produced after a month of childbirth
(b) Transitional milk
(c) Colostrum
(d) Mineralised milk

✓ ***(c)*** After the childbirth, the first yellow and thick milk produced by a lactating mother is known as colostrum. It contains IgA antibodies in abundant, which are essential for the development of immune response against many diseases in new born baby.

36. Which one of the following statements explains higher mutation rate and faster evolution found in RNA virus?
(a) RNA is relatively unstable compared to DNA.
(b) Virus can multiply only within the living cell of a host.
(c) Metabolic processes are absent in virus.
(d) Virus can remain latent for a long period.

✓ ***(a)*** RNA is relatively unstable compared to DNA, which explains higher mutation rate and faster evolution found in RNA virus. The main reasons of unstability of RNA are presence of ribose, sugar and single strand structure.

37. Which one of the following is the correct ascending sequence of states with regard to percentage of urban population (2011)?
(a) Tamil Nadu–Mizoram–Goa–Maharashtra
(b) Goa–Mizoram–Maharashtra–Kerala
(c) Maharashtra–Kerala–Mizoram–Goa
(d) Mizoram–Goa–Maharashtra–Kerala

✓ ***(c)*** Percentage of urban population in Goa is 62.17%, Mizoram is 51.51% , Maharashtra is 45.23% and Kerala is 47.72%.

38. Which one of the following places does not fall on leeward slope?
(a) Pune (b) Bengaluru
(c) Leh (d) Mangaluru

✓ ***(d)*** Leeward slope is the part of a mountain that does not face the wind i.e. the slope opposite the windward slope. Mangaluru does not fall on leeward slope. Mangaluru is situated on the West coast of India and is bounded by the Arabian Sea to its West and the Western Ghats to its East. Mangaluru experiences moderate to gusty winds during day time and gentle winds at night.

39. South Arcot and Ramanathapuram receive over 50 percent of their annual rainfall from which one of the following?
(a) South-West monsoon
(b) North-East monsoon
(c) Bay of Bengal branch of summer monsoon
(d) Western disturbances

✓ ***(b)*** Tamil Nadu receives rainfall in the winter season due to North-East trade winds. The normal annual rainfall of the state is about 945 mm (37.2 in) of which 48% is through the North-East monsoon and 32% through the South-West monsoon.

40. The Eight Degree Channel separates which of the following?
(a) India from Sri Lanka
(b) Lakshadweep from Maldives
(c) Andaman from Nicobar Islands
(d) Indira Point from Indonesia

✓ ***(b)*** Lakshadweep islands are separated from Maldives by Eight Degree Channel. Minicoy is separated from rest of the Lakshadweep by Nine Degree Channel.

41. Match List I with List II and select the correct answer using the code given below the Lists

List I (Classification of Town)		List II (Example)	
A.	Industrial Town	1.	Vishakhapatnam
B.	Transport Town	2.	Bhilai
C.	Mining Town	3.	Singrauli
D.	Garrison Cantonment Town	4.	Ambala

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	3	1	4
(c)	4	3	1	2	(d)	4	1	3	2

✓ ***(a)*** Industrial Town-Bhilai, Garrison Cantonment Town-Ambala, Transport Town-Vishakhapatnam, Mining Town-Singrauli

42. Which of the following statements with regard to the land use situation in India is/are correct?

1. There has been a tremendous decline in area under forest in recent years.
2. The rate of increase in land use in recent years is the highest in case of area under non-agricultural use.
3. Land use such as barren and wasteland, area under pastures and tree crops have experienced decline in recent year.

Select the correct answer using the code given below.
(a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

✓ ***(c)*** India's tree and forest cover has registered an increase of 1% or 8,021 sq. km in two years since 2015, according to the latest assessment by the government. Land devoted to non-agricultural use has increased three times since independence. It is set to increase further and faster.

43. Which one of the following was not a part of the strategies followed by the Government of India to increase food grain production in India immediately after independence?
(a) Intensification of cropping over already cultivated land
(b) Increasing cultivable area by bringing cultivable and fallow land under plough
(c) Using High Yielding Varieties (HYV) seeds
(d) Switching over from cash crops to food crops

✓ ***(c)*** Use of High Yielding Varieties (HYV) seeds were not a part of strategies followed by Government of India to increase food grain production in India immediately after independence. HYV seeds are not used in India till beginning of Green Revolution.

44. Which one of the following is a West-flowing river?
(a) Mahanadi (b) Godavari
(c) Krishna (d) Narmada

✓ ***(d)*** Narmada is the largest West-flowing river of the Peninsular India. Narmada flows Westwards through a rift valley between the Vindhyan Range on the North and the Satpura Range on the South.

45. Khasi language is included in
(a) Munda branch of Austro-Asiatic sub-family
(b) Mon-Khmer branch of Austro-Asiatic sub-family
(c) North Assam branch of Sino-Tibetan family
(d) Assam-Myanmari branch of Sino-Tibetan family

✓ ***(b)*** Khasi is an Austro-Asiatic language spoken primarily in Meghalaya state in India by the Khasi people. It is also spoken by a sizable population in Assam and Bangladesh. Khasi is part of the Austro-Asiatic language family and is related to Cambodian and Mon languages of Southeast Asia and the Munda branch of that family which is spoken in East-central India.

46. The headquarters of Metro Railway Zone is located in
(a) New Delhi (b) Mumbai
(c) Kolkata (d) Chennai

✓ ***(c)*** Apart from the headquarters of the Eastern and the South Eastern Railways, Kolkata also has the headquarters of the Kolkata Metro Railways which is now a zone of the Indian Railways.

47. Which one among the following is not a tributary of river Luni?
(a) Khari (b) Sukri
(c) Jawai (d) Banas

✓ ***(d)*** Major tributaries of Luni river are the Sukri, Mithri, Bandi, Khari, Jawai, Guhiya and Sagi from the left and the Jojari river from the right. The Luni river begins near Ajmer in the Pushkar valley of the Western Aravalli Range at an elevation of about 550 m.

The Banas is a river of Rajasthan state in Western India. It is a tributary of the Chambal river.

48. Match List I with List II and select the correct answer using the code given below the Lists

List I (Major Dam)		List II (State)	
A.	Cheruthoni Dam	1.	Madhya Pradesh
B.	Indira Sagar Dam	2.	Tamil Nadu
C.	Krishnaraja Sagar Dam	3.	Karnataka
D.	Mettur Dam	4.	Kerala

Codes

	A	B	C	D		A	B	C	D
(a)	2	1	3	4	(b)	2	3	1	4
(c)	4	3	1	2	(d)	4	1	3	2

✓ ***(d)*** The Mettur Dam is the largest in Tamil Nadu located across the river Cauvery.

The Cheruthoni Dam, located in Idukki district, Kerala, India is a 138m tall concrete gravity dam. The Indira Sagar Dam is a multipurpose project of Madhya Pradesh on the Narmada river.

Krishnaraja Sagar, also popularly known as KRS, is a lake and the dam that creates it. They are close to settlement of Krishnaraja Sagar in the Indian state of Karnataka.

49. Which one among the following Union Territories of India shares the shortest length of National Highways?
(a) Chandigarh (b) Delhi
(c) Daman and Diu
(d) Dadra and Nagar Haveli

✓ ***(a)*** Length of National Highway in Chandigarh is 15km, in Delhi is 80km, Daman and Diu is 22 km Dadar and Nagar Haveli is 31km.

50. Which one among the following passes links Lhasa with Ladakh?
(a) Lanak La
(b) Burzil
(c) Babusar
(d) Khyber

✓ ***(a)*** Lanak La is a well-established frontier point between Ladakh and Tibet. It is on the Southeastern boundary of the Aksai Chin region that is controlled by China.

51. According to the latest Reserve Bank of India study on State finances, capital spending is maximum on
(a) rural development
(b) water supply and sanitation
(c) urban development
(d) education

✓ ***(c)*** Capital outlay by all states is expected to touch a staggering ₹ 5.37 trillion in 2018-19 , up from ₹ 4.7 trillion in 2017-18 (RE), revealed by the latest Reserve Bank of India (RBI) study on state finances. The spending amounts to 2.9 per cent of Gross Domestic product (GDP). Under AMRUT scheme, states have to spend large amounts on urban development. Thus, capital spending is maximum on urban development.

52. According to the World Bank's Doing Business Report, 2018, India's ranking has improved in 2018 as compared to 2017 in which of the following areas?

1. Paying taxes
2. Resolving insolvency
3. Starting a business
4. Getting electricity

Select the correct answer using the code given below.

(a) 1 only (b) 1 and 2 only
(c) 1, 2 and 3 (d) 2, 3 and 4

✓ *(b)* India improved its ranking from 100th position in 2017 to 77th position in 2018 (a jump of 23 positions) among 190 countries.

Also India has improved its rank by 53 positions in the last two years and 65 positions in the last four years [2014 – 2018].

World Bank's Ease of Doing Business Index ranks 190 countries based on 10 parameters, including starting a business, construction permits, getting electricity, getting credit, paying taxes, trade across borders, enforcing contracts, and resolving insolvency.

53. The Fourteenth Finance Commission assigned different weights to the following parameters for distribution of tax proceeds to the states

1. Income distance
2. Population
3. Demographic changes
4. Area

Arrange the aforesaid parameters in descending order in terms of their weights.

(a) 1-2-3-4 (b) 1-2-4-3
(c) 1-3-2-4 (d) 4-3-2-1

✓ *(b)* The 14th Finance Commission of India is constituted in the Chairmanship of the former RBI Governor Mr. Y.V. Reddy for the period of April 1, 2015 to March 31, 2020. The 14th Finance Commission has recommended a record 10% increase in the states' share in the Union taxes to 42% as compared to the 13th Finance Commission. The total devolution to states during the five year (2015-20) period will be ₹ 39.48 lakh crore. Criteria and weights for the horizontal distribution of the tax is as follows:

Criteria	Weight (%)
1. Income Distance	50
2. Population (1971)	17.5
3. Area	15
4. Demographic Change (2011)	10
5. Forest Cover	7.5

54. The natural rate of unemployment hypothesis was advocated by

(a) Milton Friedman (b) A. W. Phillips
(c) J. M. Keynes (d) R. G. Lipsey

✓ *(a)* The natural rate of unemployment is the name that was given to a key concept in the study of economic activity by Milton Friedman. The natural rate of unemployment is the difference between those who would accept a job at the current wage rate and those who are able and willing to take a job.

55. The Harappan site at Kot Diji is close to which one of the following major sites of that civilization?

(a) Harappa (b) Mohenjo-daro
(c) Lothal (d) Kalibangan

✓ *(b)* Kot Diji is located in the vicinity of several other important historic sites. It sits to the East of Mohenjo-daro, a group of mounds that contain the remains of largest city of the Indus civilization.

56. The story *Gandatindu Jataka* was written in which language?

(a) Sanskrit (b) Telugu
(c) Tamil (d) Pali

✓ *(d)* The Jatakas were written in Pali around the middle of the first millennium CE. One story known as the Gandatindu Jataka describes the plight of the subjects of a wicked king. These included elderly women and men, cultivators, herders, village boys and even animals. When the king went in disguise to find out what his subjects thought about him, each one of them cursed him for their miseries. To escape from this situation, people abandoned their village and went to live in the forest.

57. According to the Tamil Sangam texts, who among the following were the large landowners?

(a) Gahapatis (b) Uzhavars
(c) Adimais (d) Vellalars

✓ *(d)* Vellalars were, originally an elite caste of Tamil agricultural landlords in Tamil Nadu , Kerala states in India and in neighbouring Sri Lanka.

58. According to the *Manusmriti*, women can acquire wealth through which of the following means?

(a) Purchase (b) Investment
(c) Token of affection (d) Inheritance

✓ *(d)* Manusmriti provides a woman with property rights to six types of property in verses 9.192-9.200. One of them is through inheritance.

59. The dialogue on Varna between King Avantiputta and Kachchana, a disciple of Buddha, appears in which one of the following Buddhist texts?

(a) Majjhima Nikaya (b) Samyutta Nikaya
(c) Anguttara Nikaya (d) Ambattha Sutta

✓ *(a)* The story, based on a Buddhist text in Pali known as the Majjhima Nikaya, is part of adialogue between a king named Avantiputta and a disciple of the Buddha named Kachchana. While it may not be literally true, it reveals Buddhist attitudes towards varna.

60. In the first century AD, which among the following was not a major item of Indian exports to Rome?

(a) Pepper (b) Spikenard
(c) Tortoise shell (d) Nutmeg

✓ *(d)* In the first century AD, periplus of Erythrean sea, provides vivid accounts of Indo-Roman trade relations . It was trade between the Indian subcontinent and the Roman Empire in Europe and the Mediterranean. In India, the ports of Barbaricum (modern Karachi), Barygaza, Muziris and Arikamedu on the Southern tip of India acted as the main centres of that trade. Items exported from these places are spikenard, costus, cotton cloth of all kinds, silk cloth, mallow cloth, yarn, long pepper etc. Tortoiseshell receives more mention in the Periplus than any other object of trade.

61. Which of the following statements relating to the Government of India Act, 1858 is/are correct?

1. The British Crown assumed sovereignty over India from the East India Company.
2. The British Parliament enacted the first statute for the governance of India under the direct rule of the British.
3. This Act was dominated by the principle of absolute imperial control without any population participation in the administration of the country.

Select the correct answer using the code given below.

(a) 1 and 2 only (b) 2 only
(c) 1, 2 and 3 (d) 1 and 3 only

✓ *(c)* The Government of India Act, 1858 was an Act of the British Parliament that transferred the government and territories of the East India Company to the British Crown. The Company's rule over British territories in India came to an end and it was passed directly to the British government. Queen Victoria, who was the monarch of Britain, also became the sovereign of British territories in India as a result of this Act.

This act was dominated by the principle of absolute imperial control without any popular participation in administration of the country. Under this act, all the powers of Crown were exercised by Secretary of State of India and was assisted by Council of India (Council of fifteen members).

62. Which of the following statements relating to the Indian Councils Act, 1861 is/are correct?

1. The Act introduced a grain of popular element by including non-official members in the Governor-General's Executive Council.
2. The members were nominated and their functions were confined

exclusively to consideration of legislative proposals placed before it by the Governor-General.

3. The Governor-General did not have effective legislative power.

Select the correct answer using the code given below.

(a) 1 and 2 only (b) 2 and 3 only
(c) 1, 2 and 3 (d) 1 only

✓ ***(a)*** The Indian Councils Act, 1861 was passed by British Parliament on 1st August, 1861 to make substantial changes in the composition of the Governor-General council.

- The executive council of Governor-General was added a fifth finance member. For legislative purposes, the Governor-General's Council was enlarged. Now, there were to be between 6 and 12 additional members (nominated by the Governor-General).
- There were appointed for a period of 2 years. Out of these, atleast half of the additional members were to be non-official (British or Indian).
- Their functions were confined to legislative measures.
- Any bill related to public revenue or debt, military, religion or foreign affairs could not be passed without the Governor-General's assent.
- The Viceroy had the power to overrule the council, if necessary.
- The Governor-General also had the power to promulgate ordinances without the council's concurrence during emergencies.

63. Which of the following statements relating to the historic objectives resolution, which was adopted by the constituent assembly, is/are correct?

1. The Objectives Resolution inspired the shaping of the Constitution through all its subsequent stages.
2. It was not just a resolution, but a declaration, a firm resolve and a pledge.
3. It provided the underlying philosophy of our Constitution.

Select the correct answer using the code given below.

(a) 1 and 2 only (b) 1 only
(c) 1, 2 and 3 (d) 2 and 3 only

✓ ***(c)*** Objectives Resolution' was the resolution moved by Jawaharlal Nehru on the 13th of December, 1946 in the 1st Session of India's Constituent Assembly, which was charged with the objective to frame India's Constitution. It was unanimously adopted on 22nd of January, 1947. It laid down the fundamentals and philosophy of the constitutional structure. The Constituent Assembly declared it's firm resolve to proclaim India an Independent Sovereign Republic and draw up a Constitution for her governance.

64. The 2+2 Bilateral Dialogue was held in September 2018 between

(a) External Affairs and Defence Ministers of India with their US counterparts
(b) Finance and Defence Ministers of India with their Russian counterparts
(c) Home and Defence Ministers of India and their counterparts in Pakistan
(d) External Affairs and Defence Ministers of India with their counterparts in Pakistan.

✓ ***(a)*** The first edition of 2+2 dialogue between India-US held in New Delhi. In the inaugural meeting, Minister of External Affairs (MEA) Sushma Swaraj and Minister of Defence Smt Nirmala Sitharaman met with their US counterparts Secretary of State Mr. Michael R. Pompeo and Secretary of Defence Mr. James N. Mattis. They welcomed the launch of the 2+2 Dialogue as a reflection of the shared commitment by Prime Minister Shri Narendra Modi and President Mr. Donald Trump to provide a positive, forward-looking vision for the India-US strategic partnership and to promote synergy in their diplomatic and security efforts. They resolved to continue meetings in this format on an annual basis.

65. Who is the Chairman of the Defence Planning Committee set up in April 2018?

(a) The Prime Minister
(b) The National Security Advisor
(c) The Defence Minister
(d) The Chief of the Army Staff

✓ ***(b)*** On April 18, 2018, the government has revamped the existing defence planning system by establishing a Defence Planning Committee(DPC) under the Chairmanship of the National Security Adviser (NSA). The committee will be a permanent body and it will prepare national security strategy besides undertaking strategic defence review and formulating international defence engagement strategy. It will consist of Chairman Chiefs of the Staff Committee (COSC), service chiefs, Defence Secretary, Foreign Secretary and Secretary (expenditure) in Finance Ministry.

66. 'Tejas' is the name of which one of the following?

(a) Main battle tank
(b) Nuclear submarine
(c) Light combat aircraft
(d) Aircraft carrier

✓ ***(c)*** Tejas is an Indian single-seat, single-jet engine, multirale light fighter designed by the Aeronautical Development Agency (ADA) and Hindustan Aeronautics Ltd (HAL) for the Indian Air Force and Indian Navy. It came from the Light Combat Aircraft (LCA) programme, which began in the 1980s to replace India's ageing MIG-21 fighters. In 2003, the LCA was officially named 'Tejas'.

67. As per the extant policy, Foreign Direct Ltd Investment is permitted in the defence sector under the automatic route upto which one of the following limits?

(a) 26 per cent (b) 74 per cent
(c) 51 per cent (d) 49 per cent

✓ ***(d)*** On 10th November, 2017 the government relaxed Foreign Direct Investment (FDI) norms in the defence sector by allowing FDI upto 49 per cent under automatic route and beyond that through the FIPB's approval. The government has also done away with the earlier requirement of mandatory permission from the Cabinet Committee on Security (CCS) beyond 49 per cent.

68. The policy on strategic partnership in defence was approved by the Ministry of Defence in May 2017. Which of the following is not among the four segments identified by the Ministry for acquisition through the strategic partnership route?

(a) Artillery guns
(b) Fighter aircraft and helicopters
(c) Submarines
(d) Armoured fighting vehicles and main battle tanks

✓ ***(a)*** The policy on Strategic Partnerships in Defence sector was approved by Defence Acquisition Council (DAC) in May, 2017.
The policy is intended to institutionalise a transparent, objective and functional mechanism to encourage broader participation of the private sector, in addition to DPSUs / OFB, in the manufacture of defence platforms and equipment such as aircraft, submarines, helicopters and armoured vehicles. It will serve to enhance competition, increase efficiencies, facilitate faster and more significant absorption of technology, create a tiered industrial ecosystem, ensure development of a wider skill base and trigger innovation, leading to reduction in dependence on imports and greater self-reliance in meeting national security objectives. The following four segments have been identified for acquisition under Strategic Partnership (SP) route:

- Fighter Aircraft
- Helicopters
- Submarines
- Armoured Fighting Vehicles (AFVs)/ Main Battle Tanks (MBTs).

69. The acronym 'CAATSA' refers to a piece of legislation enacted by which one of the following countries?

(a) United Kingdom
(b) United States of America
(c) Russia
(d) India

✓ ***(b)*** The Countering America's Adversaries Through Sanctions Act, CAATSA is a United States federal law that imposed sanctions on

Iran, North Korea and Russia. CAATSA is a specifically enacted legislation. Its "ultimate goal", in the words of a senior State Department official, "is to prevent revenue from flowing to the Russian Government."

70. Who among the following is the Convener of the 'Task Force' set up in November 2017 by the Government of India to review the Income-tax Act and draft a new direct tax law?

(a) Girish Ahuja (b) Mukesh Patel
(c) Arbind Modi (d) Mansi Kedia

✓ *(c)* The finance ministry sets up a six-member task force to draft a new direct tax law that will better serve the country's economic needs by widening the tax base, improving compliance and ease of doing business. Arbind Modi, member, Central Board of Direct Taxes, was named convener of the six-member panel that has been tasked to draft a new law. Modi was also a key contributor to the direct taxes code proposed by the previous United Progressive Alliance Government.

71. With regard to the cabinet decision in July 2018, the percentage increase in Minimum Support Price (MSP) is maximum in which one of the following crops?

(a) Jowar (Hybrid) (b) Bajra
(c) Maize (d) Soya bean

✓ *(a)* Giving a major boost for the farmers' income, the Cabinet Committee on Economic Affairs chaired by Prime Minister Shri Narendra Modi has approved the increase in the Minimum Support Prices (MSPs) for all kharif crops for 2018-19 season. Among these, the maximum increase in the MSP is MSP of jowar which is from 1700 to ₹ 2430.

Directions (Q. Nos.72-75) **The following four (4) items consists of two statements, Statement I and Statement II. Examine these two statements carefully and select the correct answer using the code given below.**

Code

(a) Both the statements are individually true and Statement II is the correct explanation of Statement I.
(b) Both the statements are individually true and Statement II is not the correct explanation of Statement I.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

72. **Statement I** The overall fiscal deficit of the States in India during 2017-18 stayed above the FRBM threshold level of 3 per cent for the third successive year.

Statement II Special Category States had run up a higher level of fiscal deficit in 2017-2018 compared to 2016-17.

✓ *(b)* State governments in aggregate were to revert to well below 3% fiscal deficit threshold in Financial Year 2017-18. This year saw a change from the previous few years with all deficit indicators worsening for special category states than that of non-special category states and most special category states recording GFDs above the 3% mark.

73. **Statement I** There has been a sharp decline in savings rate in Indian economy between 2007-2008 to 2015-2016.

Statement II There has been a fall in household and public savings.

✓ *(c)* Gross domestic savings have been falling as a percentage of GDP. However, there has been rise in household and public savings recorded.

74. **Statement I** Private investments in research have severely lagged public investments in India.

Statement II Universities play a relatively small role in the research activities of the country.

✓ *(c)* According to the Economic Survey 2017-18, India under-spends on R&D even relative to its level of development. In India, the government is not just the primary source of R&D funding but also its the primary user of these funds.

India's spending on R&D (about 0.6% of GDP) is well below that in major nations such as the US (2.8%), China (2.1%), Israel (4.3%) and Korea (4.2%). Private investments in research have severely lagged public investments in India. The Survey highlighted that universities play a relatively small role in the research activities of the country.

75. **Statement I** Agriculture in India still accounts for a substantial share in total employment.

Statement II There has been no decline in volatility of agricultural growth in India.

✓ *(c)* The Indian economy has moved decisively to a higher path of growth in Agriculture's share of total employment has declined. It is still a dominant source of employment, and presently it accounts for 52 per cent of the country's total labour force. The declining trend in agricultural growth has emerged as a major concern for researchers and policymakers but volatility in agriculture has declined significantly in India.

76. Who among the following European travellers never returned to Europe and settled down in India?

(a) Duarte Barbosa (b) Manucci
(c) Tavernier (d) Bernier

✓ *(b)* Niccolao Manucci (19 April, 1638–1717) was an Italian writer and traveller. He wrote a memoir about the Indian subcontinent during the Mughal era. His records have been a source of history about Shah Jahan, Aurangzeb, Dara Shikoh, Shah Alam, Raja Jai Singh and Kirat Singh. He spent almost his entire life in India. Italian doctor Manucci, never returned to Europe, and settled down in India.

77. The class of Amar Nayakas in Vijayanagara is a reference to which of the following?

(a) Village Chieftains
(b) Senior Civil Servants
(c) Tributary Chiefs
(d) Military Commanders

✓ *(d)* The Amara Nayaka system was a major political innovation of the Vijayanagara Empire. Nayakas were military chiefs usually maintained law and order in their areas of control. They maintained forests and kept armed supporters. They use to control and expand fertile land and agricultural settlements.

78. The important source for Akbar's reign, Tarikh-i-Akbari was written by which one of the following Persian language scholars?

(a) Arif Qandahari
(b) Bayazid Bayat
(c) Abdul Qadir Badauni
(d) Nizamuddin Ahmad

✓ *(a)* Tarikh-i-Akbari, also known as Tarikh-i-Qandahari, or Muzaffar Nama, the first chronicle of Emperor Akbar's reign is unaimously important source of information on the 16th century history of India; particularly the formative years of his career. It was written by Arif Qandahari.

Though the major pre-occupation of modern scholars has been with Abul Fazl's Akbar Nama and Badauni's Muntakhab-ut-Tawarikh, it supplements in many ways of the former's works while Farishta in his Gulshan-i-Ibrahimi and Nihawandi in his Maasir-i-Rahimi explicitly acknowledge their indebtedness to it.

79. The aristocrat Muqarrab Khan was a great favourite of which Mughal Emperor?

(a) Akbar (b) Jahangir
(c) Farrukhsiyar (d) Shah Alam

✓ *(b)* Muqarrab Khan was favourite of Mughal emperor Jahangir. Muqarrab Khan of Golconda (titled Khan-Zaman Fath Jang) was the most experienced commander in Golconda, during the reign of Abul Hasan Qutb Shah. Muqarrab Khan is known to have been an ally of Afzal khan and defended Golconda's Southern realms against Maratha raids. Aurangzeb sent the Golconda noble Muqarrab Khan to hunt down and kill Shambhaji. Later in 1687, Aurangzeb ordered Mu'azzam (Shah Alam) to march against the sultanate of Golconda. Meanwhile he came close to Muqarrab Khan.

80. Who was the first Nawab Wazir of Awadh in the 18th Century?

(a) Nawab Safdarjung
(b) Nawab Saadat Ali Khan
(c) Nawab Shuja-ud-Daula
(d) Nawab Saadat Khan

✓ *(b)* Saadat Khan Burhanul Mulk or Saadat Ali Khan was appointed Nawab in 1722 and established his court in Faizabad near Lucknow. He took advantage of a weakening Mughal Empire in Delhi to lay the foundation of the Awadh dynasty.

81. According to the French traveller Tavernier, the majority of houses in Varanasi during the 17th century were made of

(a) brick and mud
(b) stone and thatch
(c) wood and stone
(d) brick and stone

✓ *(a)* In 1665, the French traveller Jean Baptiste Tavernier described the architectural beauty of the Vindu Madhava temple on the side of the Ganges. According to him, the majority of houses in Varanasi during the 17th century were made of brick and mud.

82. Which of the following statements relating to the duties of the Governor is/are correct?

1. The duties of the Governor as a Constitutional Head of the State do not become the subject matter of questions or debate in the Parliament.
2. Where the Governor takes a decision independently of his Council of Ministers or where he acts as the Chief Executive of the State under President's rule, his actions are subject to scrutiny by the Parliament.

Select the correct answer using the code given below.

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

✓ *(b)* Since Governor is appointed by the President or Union Government, his duties as constitutional Head of the State can become the subject matter of questions or debate in the Parliament even in that condition when the Governor takes a decision independently of his Council of Ministers or where he acts as the Chief Executive of the State under Presidents rule, his actions are subject to scrutiny by the Parliament.

83. Which one of the following Articles of the Constitution of India deals with the special provision with respect to the State of Assam?

(a) Article 371A (b) Article 371B
(c) Article 371C (d) Article 371D

✓ *(b)* The Article 371B of Indian Constitution has special provision with respect to the State of Assam. It is mostly related to Sixth schedule and functioning of Legislative Assembly and Sixth schedule.

84. Provisions of which one of the following Articles of the Constitution of India apply to the State of Jammu and Kashmir?

(a) Article 238 (b) Article 370
(c) Article 371 (d) Article 371G

✓ *(b)* Article 370 of the Indian Constitution is an article that grants special autonomous status to the State of Jammu and Kashmir. The article is drafted in Part XXI of the Constitution, which relates to Temporary, Transitional and Special Provisions.

85. Which one of the following Schedules to the Constitution of India provides for setting up of Autonomous District Councils?

(a) Third Schedule (b) Fourth Schedule
(c) Fifth Schedule (d) Sixth Schedule

✓ *(d)* The Constitution of India makes special provisions for the administration of the tribal dominated areas in the four states viz. Assam, Meghalaya, Tripura and Mizoram. Though these areas fall within the executive authorities of the state, provision has been made for the creation of the district councils and regional councils for the excercise of the certain legislative and judicial powers.

Each district is an autonomous district and Governor can modify/divide the boundaries of the said Tribal areas by notification. Currently, there are ten such councils in the region.
The Constitution of India makes special provisions for the administration of the tribal dominated areas in four states viz. Assam, Meghalaya, Tripura and Mizoram. As per Article 244 and 6th Schedule, these areas are called "Tribal Areas".The Sixth Schedule envisages establishment of Autonomous District Councils (ADCs). These councils have been given Legislative, Administrative and Judicial powers under the Sixth schedule. No law of the Centre or the state in respect of the legislative powers conferred on the autonomous district councils could be extended to those areas without their prior approval. The district councils are also empowered to constitute village councils and also village courts.

86. In which one of the following States was 'DEFEXPO 2018' held in April 2018?

(a) Goa (b) Karnataka
(c) Tamil Nadu (d) Andhra Pradesh

✓ *(c)* The 10th edition of DEFEXPO has been held in Chennai from 11 to 14 April, 2018. The location of the event is Tiruvidanthal, Kancheepuram district on the East Coast Road near Chennai, Tamil Nadu. DEFEXPO 2018, for the first time, project India's Defence manufacturing capabilities to the world. This is reflected in the tagline for the Expo, 'India: The Emerging Defence Manufacturing Hub'.

87. The two defence industrial corridors announced by the Finance Minister in his 2018 Budget speech are coming up in which of the following states?

(a) Odisha and West Bengal
(b) Punjab and Haryana
(c) Gujarat and Maharashtra
(d) Uttar Pradesh and Tamil Nadu

✓ *(d)* The government will develop two defence industrial production corridors and bring out an industry-friendly military production policy to promote the defence industry in Uttar Pradesh and Tamil Nadu. The first of the two corridors will be constructed between Chennai and Bengaluru, linking Kattupalli port, Chennai, Tiruchi, Coimbatore and Hosur. The second industrial corridor will link Agra, Aligarh, Lucknow, Kanpur, Jhansi and Chitrakoot.

It will be constructed keeping in mind the development requirements of Bundelkhand region.

The government had earlier announced that it would develop two such corridors.

88. What is India's first Indigenous Aircraft Carrier (IAC) called?

(a) Vikrant (b) Virat
(c) Vaibhav (d) Varaha

✓ *(a)* INS Vikrant (IAC-I) is the first indigenous aircraft carrier built in India and the first Vikrant class aircraft carrier built by Cochin shipyard (CSL) in Kochi, Kerala for the Indian Navy. The motto of the ship is Jayema Sam Yudhi Sprdhah, which is taken from Rigveda 1.8.3 and can be translated as "I defeat those who fight against me". India's first domestically built aircraft carrier was built in India under the so-called Indigenous Aircraft Carrier (IAC) program.

89. Which one of the following manufacturers is engaged in upgradation of the Swedish 155-mm Bofors Howitzer under the project 'Dhanush'?

(a) Bharat Electronics Limited
(b) Ordnance Factory Board
(c) Bharat Dynamics Limited
(d) Mishra Dhatu Nigam

✓ *(b)* Dhanush is a 155mm x 45mm calibre artillery gun and is also called the Desi Bofors. Dhanush will be the first artillery gun to be acquired by the army since the purchase of Bofors guns from Sweden in 1987.
It is a modified version of the Bofors upgraded by Ordnance Factory Board (OFB). It is a conglomerate of 39 ordnance factories with another two new projects being set up at Nalanda in Bihar and Korwa in Uttar Pradesh.

90. Which one of the following is the official mascot of Tokyo 2020 Olympic Games?
(a) Soohorang
(b) Vinicius de Moraes
(c) The Hare, the Polar Bear and the Leopard
(d) Miraitowa

✓ ***(d)*** The mascot for the 2020 Tokyo Olympics is named Miraitowa. Miraitowa is a combination of the Japanese words for future and eternity.

91. 'Mission Satyanishtha', a programme on ethics in public governance, was launched recently by the
(a) Indian Railways
(b) Central Bureau of Investigation
(c) Supreme Court
(d) Enforcement Directorate

✓ ***(a)*** 'Mission Satyanishtha' launched on 27th July, 2018 aims at sensitizing all railway employees about the need to adhere to good ethics and to maintain high standards of integrity at work. Mission Satyanishtha launched by Railway Ministry aims to adhere to good ethics and to maintain high standards of integrity at work. This mission is a first of its kind event held by any government organisation.

92. The College of Fort William was established by which one of the following Governor-Generals?
(a) Warren Hastings
(b) Lord Cornwallis
(c) Richard Wellesley
(d) William Bentinck

✓ ***(c)*** Fort William College (also called the College of Fort William) was an academy and learning centre of Oriental studies established by Lord Wellesley in 1800, by the then Governor General of British India. Fort William College aimed at training British officials in Indian languages and, in the process, fostered the development of languages such as Bengali and Urdu.

93. The Economic historian, who has used the data collected by Buchanan-Hamilton to support the thesis of deindustrialization in the 19th century India, is
(a) Tirthankar Roy
(b) Amiya Kumar Bagchi
(c) Sabyasachi Bhattacharya
(d) Irfan Habib

✓ ***(a)*** Tirthankar Roy, a professor of London school of Economics has given his thesis on de-industrial through data of Buchanan-Hamilton data.

94. Tea growing in India in the 19th century was made possible by
(a) Joseph Banks
(b) James Cook
(c) Robert Fortune
(d) Robert Owen

✓ ***(c)*** Robert Fortune (1812-80) was a Scottish botanist, plant hunter and traveller. He is best known for stealing tea plants from China in the employment of the British East India Company.

95. Subhas Chandra Bose started the 'Azad Hind Radio' in which of the following countries?
(a) Japan (b) Austria
(c) Germany (d) Malaysia

✓ ***(c)*** Azad Hind Radio was a propaganda radio service that was started under the leadership of Netaji Subhas Chandra Bose in Germany in 1942 to encourage Indians to fight for freedom.

96. Which political party formally accepted the Cabinet Mission Plan on 6th June, 1946, which had rejected the demand for a sovereign Pakistan?
(a) The Hindu Mahasabha
(b) The Congress
(c) The Muslim League
(d) The Unionist Party

✓ ***(c)*** Cabinet Mission of 1946 came to India with an aim to discuss the transfer of power from the British Government to the Indian leadership, with the aim of preserving India's unity and granting it independence. The Cabinet Mission Plan of 1946 proposed that there shall be a Union of India which was to be empowered to deal with the defense, foreign affairs and communication. The Muslim League first approved the plan. But when Congress declared that it could change the scheme through its majority in the Constituent Assembly, they rejected the plan.

97. The elected President of the All India Kisan Sabha, which met in Vijayawada (1944), was
(a) Sahajananda Saraswati
(b) Vinoba Bhave
(c) Achyut Rao Patwardhan
(d) Narendra Dev

✓ ***(a)*** The eighth annual session of the All India Kisan Sabha was held, at Bezwada(Vijayawada), in Andhra, on March 14 and 15, 1944, under the presidentship of Swami Sahajanand Saraswati. The session was unique in many respects in the history of the Sabha. The total gathering at the open session numbered about a lakh including over 10,000 women—as big as at Gaya in 1939.

The presidential procession on 14th morning was a disciplined mass of nearly 10,000 people formed into a one-mile-long mobile column which marched for three hours through Bezwada town and its outskirts while about as many thousands were witnessing the arousing demonstration in amazement. The main political resolution was on the release of national leaders to end the deadlock. It was of paramount importance to our national life to-day because its main drive goes towards the achievement of Hindu-Muslim and Congress-League unity leading to the release of the leaders, ending the deadlock, and winning a National Government of National defence.

98. Which one of the following regarding the tenure of the elected members of the Autonomous District Council is correct?
(a) Five years from the date of election
(b) Five years from the date appointed for the first meeting of the council after the election
(c) Six years from the date of administration of oath
(d) Six years from the date of election

✓ ***(b)*** The Sixth schedule to the Constitution empowers the Governor to determine the administrative areas of the autonomous councils. Each district council or regional council provided under the Sixth schedule is a corporate body by name of District Council or Regional Council of (name of the district or name of the region) having perpetual succession and a common seal with the right to sue and be sued. The councils consists of 30 members, 26 are elected from the single member constituencies on the basis of adult franchise and not more than 4 persons are nominated by the Governor on the advice of the Chief Executive Member for a term of five years. The tenure of the elected members of the Autonomous District Council is five years from the date appointed for the first meeting of the Council after the election. They are known as MDC (Member of the District Council).

99. Who among the following shall cause the accounts of the Autonomous District and Regional Council Funds to be audited?
(a) The Comptroller and Auditor General of India
(b) The Chartered Accountant empanelled by the Government of India
(c) The State Government Auditors
(d) Any Chartered Accountant

✓ ***(a)*** The district and the regional councils are Responsible for framing rules for the management of finances with the approval of the Governor. They are also given mutually exclusive powers to collect land revenues, levy and collect taxes on lands.

The District Councils enjoy autonomy and the Acts of the Parliament and the State legislatures on the subject under them do not normally apply to the autonomous districts.

The Sixth schedule of the Constitution of India also envisages audit of accounts of district and regional councils of autonomous regions by the Comptroller and Auditor General of India (CAG).

100. Who has the power of annulment or suspension of acts and resolutions of the Autonomous District and Regional Councils?
(a) The Governor
(b) The President
(c) The Chief Minister of the State
(d) The Prime Minister

✓ ***(a)*** Governors of four states viz. Assam, Meghalaya, Tripura and Mizoram are empowered to declare some tribal dominated districts/areas of these states as autonomous districts and autonomous regions by order. No separate legislation is needed for this.

The Governor also has power to include any other area, exclude any area, increase, decrease, diminish these areas, unite two districts/regions, and alter the names and boundaries of these autonomous district councils. Governor has the power of annulment of suspension of Acts and Resolutions of the Autonomous District and regional council.

101. The audit reports of the Comptroller and Auditor General of India relating to the accounts of the Union shall be submitted to

(a) the President

(b) the Speaker of the Lok Sabha

(c) the Prime Minister

(d) the Vice President

✓ ***(a)*** Under Article 151 of Constitution, the reports of the Comptroller and Auditor-General of India relating to the accounts of the Union shall be submitted to the President, who shall cause them to be laid before each House of Parliament.

102. Which of the following is not related to the powers of the Governor?

(a) Diplomatic and Military powers

(b) Power of appoint Advocate General

(c) Summoning, proroguing and dissolving State Legislature

(d) Power to grant pardons, reprieves, respites or remission of punishments

✓ ***(a)*** Governor is the Head of the State, as President being the head of the country. He enjoys executive, legislative as well as judicial functions. Following are the powers and duties performed by Governor in states are

- They have power to appoint Advocate General.
- They can summoning, proroguing and dissolving State Legislature.
- They have power to grant pardons, reprieves, respites or remission of punishments. But in case of Military power, Governor don't have such power as like President.

103. Which one of the following regarding the procedure and conduct of business in the Parliament is ***not*** correct?

(a) To discuss State matters

(b) To discuss issues of the use of Police force in suppressing the Scheduled Caste and Scheduled Tribe communities

(c) To discuss issues in dealing with violent disturbances in an undertaking under the control of the Union Government.

(d) To discuss issues for putting down the demands of the industrial labour

✓ ***(a)*** To discuss state matters in general condition does not lies under procedure and conduct of business of Parliament. Under article 246 of Constitution, Parliament has right to make laws on Union list and Concurrent list of Schedule seven. Parliament can discuss the issue of Police action against the Schedule Castes and Tribes under the Scheduled Castes and Tribes (Prevention of Atrocities) Act, 1989. Issue of Industrial labours come under Concurrent list, so it can be discussed in Parliament. Parliament can also discuss issues in dealing with violent disturbances in an undertaking under the control of the Union Government.

104. Which of the following is not under the powers and functions of the Election Commission of India?

(a) Superintendence, direction and control of the preparation of electoral rolls

(b) Conduct of elections to the Parliament and to the Legislature of each State

(c) Conduct of election of the office of the President and the Vice President

(d) Appointment of the Regional Commissioners to assist the Election Commission in the performance of the functions conferred on the Commission

✓ ***(d)*** According to Article 324 of Indian Constitution, the Election Commission of India has superintendence, direction and control of the entire process for conduct of elections to Parliament and Legislature (state legislative assembly and state legislative council) of every State and to the offices of President and Vice-President of India. Appointment of the Regional Commission in the performance of the functions conferred on the Commission is not done by the Election Commission of India.

105. Which one of the following criteria is not required to be qualified for appointment as Judge of the Supreme Court?

(a) At least five years as a Judge of a High Court

(b) At least ten years as an Advocate of a High Court

(c) In the opinion of the President, a distinguished Jurist

(d) At least twenty years as a Sub-Judicial Magistrate

✓ ***(d)*** Article 124 (3) of the Constitution prescribes that for appointment as a Judge of the Supreme Court a person must be

- a citizen of India,
- has been a judge of any High Court for at least 5 years, or
- has been an advocate in a High Court for 10 years or is in the opinion of the President a distinguished jurist.

106. Which one of the following is not among the duties of the Chief Ministers?

(a) To communicate to the Governor of the State all decisions of the Council of Ministers relating to the administration of the affairs of the State and proposals for legislation

(b) To furnish information relating to the administration of the State and proposals for legislation as the Governor may call for

(c) To communicate to the President all decisions of the Council of Ministers relating to the administration of the State in the monthly report

(d) To submit for the consideration of the Council of Ministers any matter on which a decision has been taken by a Minister but has not been considered by the Council, if the governor so requires

✓ ***(c)*** To communicate to the President all decisions of the Council of Ministers relating to the administration of the State in the monthly report is the duty of Prime Minister under Article 78.

The Chief Minister is appointed by the Governor. Article. 164 of the Constitution provides that there shall be a Council of Ministers with the Chief Minister at its hand to aid and advise the Governor. Under Article 167 of the Constitution, it shall be the duty of the Chief Minister of each state

- to communicate to the Governor of the State all decisions of the Council of Ministers relating to the administration of the affairs of the State and proposals for legislation.
- to furnish such information relating to the administration of the affairs of the State and proposals for legislation as the Governor may call for.
- if the Governor so requires, to submit for the consideration of the Council of Ministers any matter on which a decision has been taken by a Minister but which has not been considered by the Council.

107. Which one of the following is not considered a part of the Legislature of States?

(a) The Governor

(b) The Legislative Assembly

(c) The Legislative Council

(d) The Chief Minister

✓ ***(d)*** Under Article 168 of the Constitution, for every State there shall be a Legislature which shall consist of the Governor, and where there are two Houses of the Legislature of a State, one shall be known as the Legislative Council and the other as the Legislative Assembly, and where there is only one House, it shall be known as the Legislative Assembly.

Hence, Chief Minister is not the part of Legislature.

108. Which one of the following regarding the ordinance-making power of the Governor is not correct?

(a) It is not a discretionary power.
(b) The Governor may withdraw the ordinance anytime.
(c) The ordinance power can be exercised when the Legislature is not in session.
(d) The aid and advice of Ministers is not required for declaring the ordinance.

✓ *(d)* The ordinance-making power of the Governor is exercised under Article 214. The Governor can issue ordinance only when two conditions are fulfilled ;

- the governor can only issue ordinances when the legislative assembly of a state both houses in session or where there are two houses in a state both houses are not in session.
- the governor must be satisfied that circumstance exist which render it necessary for him to take immediate action.

The Court cannot question the validity or the ordinance on the ground that there was no necessity or sufficient ground for issuing the ordinance by the Governor. The existence of such necessity is not a justiciable discretionary. The exercise of ordinance-making power is not discretionary. The Governor exercises this power on the advice of the Council of Minister. He may withdraw it at any time.

109. Which one of the following statements regarding the Universal Declaration of Human Rights is ***not*** correct?

(a) The UN General Assembly adopted the Human Rights Charter on 10th December, 1948.
(b) Some of the provisions of the Fundamental Rights enshrined in the Constitution of India are similar to the provisions of the Universal Declaration of Human Rights.
(c) The Rights to Property is not a part of the Universal Declaration of Human Rights.
(d) India is a signatory to the Universal Declaration of Human Rights.

✓ *(c)* The Universal Declaration of Human Rights (UDHR) was adopted by the United Nations General Assembly at its third session on 10th December, 1948. India is a signatory to the Universal Declaration of Human Rights. Our Indian Constitution was greatly influenced by the Universal Declaration of Human Rights specially the part three i.e. Fundamental rights. Right to Property (Housing) is also a human right under this declaration.

110. Which one of the following statements regarding the Human Rights Council is ***not*** correct?

(a) It is an inter-governmental body within the United Nations system made up of all members of the UN.
(b) It is responsible for the promotion and protection of all human rights around the globe.
(c) It replaced the former United Nations Commission on Human Rights.
(d) It is made up of 47 UN Member States which are elected by the UN General Assembly.

✓ *(a)* The Human Rights Council is an inter-governmental body within the United Nations system made up of 47 states responsible for the promotion and protection of all human rights around the globe. It replaced the former United Nations Commission on Human Rights. Hence, all UN members are not of its part.

111. Ace athlete Neerja Chopra is an accomplished player in

(a) Hammer throw (b) Javelin throw
(c) Shot put throw (d) Discus throw

✓ *(b)* Neeraj Chopra is an Indian track and field athlete, who competes in the Javelin Throw. He represented India at the 2018 Asian Games where he won a gold medal, setting the national record of 88.06 m, and at the 2018 Commonwealth Games where he also clinched the gold medal.

112. According to the updated World Bank data for 2017, India is the sixth biggest economy of the world (in terms of GDP). Which one of the following is ***not*** ahead of India?

(a) Japan
(b) UK
(c) France
(d) Germany

✓ *(c)* In 2017, India became the sixth largest economy with a gross domestic product of \$2.60 trillion, relegating France to the seventh position. As per the data, GDP of France stood at \$2.58 trillion.

113. Which one of the following is correct about 'Aaykar Setu'?

(a) It is a mechanism for achieving excellence in public sector delivery related to GST.
(b) With the use of a mobile app, it facilitates online payment of taxes.
(c) It is a communication strategy designed to collect information and build a database of tax defaulters.
(d) It enables electronic filing and processing of import and export declarations.

✓ *(b)* The Central Board of Direct Taxes has launched an app 'Aaykar Setu'. The app helps to file Income Tax Return (ITR) online, locate the nearest Tax Return Prepares (TRP), provides calculators and other tools, helps manage PAN and TDS.

114. SWAYAM is

(a) a network that aims to tap the talent pool of scientists and entrepreneurs towards global excellence
(b) a Massive Open Online Courses (MOOCs) initiative on a national platform
(c) an empowerment scheme for advancing the participation of girls in education.
(d) a scheme that supports differently abled children to pursue technical education.

✓ *(b)* Under SWAYAM or Study Webs of Active –Learning for young aspiring minds programme of Ministry of Human Resource Development, Government of India, professors and faculties of centrally funded institutions like IITs, IIMs, central universities will offer online courses to citizens of India.

115. Under the PRASAD Tourism Scheme, which one of the following has not been identified as a religious site for development?

(a) Ajmer (Rajasthan)
(b) Haridwar (Uttrakhand)
(c) Somnath (Gujarat)
(d) Velankanni (Tamil Nadu)

✓ *(b)* PRASAD (Pilgrimage Rejuvenation and Spiritual Augmentation Drive) scheme has been launched to identify and develop pilgrimage tourist destinations on the principles of high tourist visits, competitiveness and sustainability to enrich the religious tourism experience. Twelve cities namely Amaravati (Andhra Pradesh), Gaya (Bihar), Dwaraka (Gujarat), Amritsar (Punjab), Ajmer (Rajasthan), Kanchipuram (Tamil Nadu), Vellankani (Tamil Nadu), Puri (Odisha), Varanasi (Uttar Pradesh), Mathura (Uttar Pradesh), Kedarnath (Uttarakhand) and Kamakhya (Assam) have been identified for development under Pilgrimage Rejuvenation and Spirituality Augmentation Drive (PRASAD) by the Ministry of Tourism.

116. Name the Indian cricketer who is not inducted to the ICC Cricket Hall of Fame (till July 2018).

(a) Rahul Dravid
(b) Sunil Gavaskar
(c) Sachin Tendulkar
(d) Anil Kumble

✓ *(c)* The ICC Cricket 'Hall of Fame' recognises the achievements of the legends of the cricket. Since 1932 till date just five Indians have been included in this list. Rahul Dravid, the former wall of Indian cricket is the latest Indian player who is included in the list of ICC Hall of Fame in 2018. Other are Bishan Singh Bedi, Kapil Dev, Sunil Gavaskar in 2009 and Anil Kumble (2015).

117. The Central Water Commission has recently entered into a collaborative agreement with which one of the following entities for flood forecasting?

(a) Skymet
(b) Google
(c) MetService
(d) AccuWeather

✓ ***(b)*** The Central Water Commission (CWC) has entered into an agreement with Google to improve flood forecast systems and disseminate flood-related information by using technology developed by the tech giant. The initiative is likely to help crisis management agencies to deal extreme hydrological events in a better manner.

118. The tagline 'Invaluable Treasures of Incredible India' is associated with the logo for

(a) Archaelogical Survey of India
(b) India Tourism Development Corporation
(c) Geological Survey of India
(d) Geographical Indications (GI) of India

✓ ***(d)*** Department of Industrial Policy and Promotion (DIPP) under Ministry of Commerce and Industry has unveiled tricolour logo for Geographical Indication (GI) certified products.

The logo has tagline "Invaluable Treasures of Incredible India" printed below it.

119. Who took over the 'Eka Movement' started by the Congress in Awadh during 1921-1922?

(a) Bhagwan Ahir
(b) Madari Pasi
(c) Baba Ramchandra
(d) Shah Naeem Ata

✓ ***(b)*** Eka Movement or Unity Movement is a Peasant Movement which surfaced in Hardoi, Bahraich and Sitapur during the end of 1921 by Madari Pasi, an offshoot of Non-cooperation Movement. The initial thrust was given by the leaders of Congress and Khilafat movement.

120. Which organization was started at the Haridwar Kumbh Mela in 1915?

(a) Sanatan Dharma Sabha
(b) Dev Samaj
(c) Brahmin Sabha
(d) Hindu Mahasabha

✓ ***(d)*** Hindu Mahasabha was founded in 1915 by Madan Mohan Malviya. It worked with Arya Samaj and other Hindu organisations. It was directly linked with Rashtriya Swam Sevak Sangh founded in 1925 at Nagpur by K.B.Hegewar. The first All India Hindu Mahasabha Conference was organised at Haridwar in 1915 during Kumbh Mela.

MATHEMATICS

TREND ANALYSIS
(2016-2014)

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Number System	11	10	12	13
2	HCF and LCM of Numbers	4	3	4	4
3	Decimal Fractions	-	1	1	-
4	Square Roots and Cube Roots	2	1	2	1
5	Time and Distance	7	10	7	5
6	Time and Work	1	6	1	4
7	Percentage	2	4	2	2
8	Simple Interest	1	-	2	1
9	Compound Interest	-	1	2	1
10	Profit and Loss	1	1	-	1
11	Ratio and Proportion	2	3	2	1
12	Logarithm	1	-	-	-
13	Basic Operations and Factorisation	11	5	5	1
14	HCF and LCM of Polynomials	4	3	4	4
15	Rational Expressions	-	-	-	1
16	Linear Equations	1	1	1	1
17	Quadratic Equations and Inequations	7	4	1	9
18	Set Theory	-	1	1	-
19	Measurement of Angles and Trigonometric Ratio	8	11	12	11
20	Height and Distances	2	2	4	1
21	Lines Angles	3	-	1	-
22	Triangles	12	2	3	7
23	Quadrilateral and Polygon	3	5	4	4
24	Circle	7	4	5	6
25	Locus	-	-	-	-
26	Area and Perimeter of Plane Figures	4	6	11	4
27	Surface Area and Volume of Solids	-	11	8	11
28	Statistics	6	5	5	7
	Total	**100**	**100**	**100**	**100**

01

NUMBER SYSTEM

Generally (10-12) questions have been asked from this chapter. Questions, from this section usually test your basic knowledge of numbers and are mostly based on various properties of multiplication and division. A good number of statement based questions have been asked from this chapter.

NUMBER SYSTEM

Numbers are collection of certain symbols or figures called digits. The common number system in use is decimal number system. In this system, we use ten symbols each representing a digit. These are 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. A combination of these figures representing a number is called a numeral.

Types of Numbers

1. **Natural numbers** Numbers which are used for counting i.e. 1, 2, 3, 4, ... are called natural numbers. The set of natural numbers is denoted by 'N'. Smallest natural number is 1 but we cannot find the largest natural number as successor of every natural number is again a natural number.
2. **Whole numbers** Natural numbers including zero are known as whole numbers. The set of whole numbers is denoted by W.
   - Every natural number is a whole number.
   - Zero (0) is the only whole number which is not a natural number.
3. **Even numbers** The numbers which are divisible by 2 are called as even numbers. e.g 2, 4, 6, 8, 10, … . In general these are represented by $2m$, where $m \in N$.
4. **Odd numbers** The number which are not divisible by 2 are called as odd numbers. e.g. 1, 3, 5, 7, 9, … . In general, these are represented by $(2m-1)$, where $m \in N$.
5. **Prime numbers** Those numbers which are divisible by 1 and the number itself are known as prime numbers. e.g. 2, 3, 5, 7,..., etc. are prime numbers.
   - If a number is not divisible by any of the prime numbers upto square root of that number, then it is a prime number.
   - 2 is the only even number which is prime.
   - The prime numbers upto 100 are : 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89 and 97 i.e. there are 25 prime numbers upto 100.

6. **Coprime numbers** Two natural numbers x and y are said to be coprime, if they do not have any common divisor other than 1.

   e.g. (9, 2), (5, 6), (11, 15) are the pairs of coprime numbers.
   - If x and y are any two coprimes, a number p is divisible by x as well as by y, then the number is also divisible by xy.
   - Coprime are also called as relatively prime numbers.
   - **Twin primes** Twin primes are pair of primes which differ by 2. e.g. (3, 5), (7, 9), (11, 13) etc.
7. **Composite numbers** A composite number is any number greater than one that is not a prime number. e.g. 4, 6, 8, 9, … all are composite numbers.
   - '1' is neither prime nor composite.

INTEGERS

The collection of positive numbers, negative numbers and zero are called integers. The set of integers is denoted by Z or I.

Thus, Z or $I = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$ is the set of integers. Every natural number and whole number is a part of integer. So, $N \subset W \subset I$.

Types of Integers

Integers are of three types

(i) **Positive Integers** It is a set of all positive numbers. It is denoted by I^+, $I^+ = \{1, 2, 3, 4, \dots\}$

(ii) **Negative Integers** It is a set of all negative numbers. It is denoted by I^-, $I^- = \{\dots, -3, -2, -1\}$

(iii) **Non-negative Integers** An integer that is either 0 or positive is called non-negative integer. $\{0, 1, 2, 3, \dots\}$

Note '0' is neither positive nor negative.

EXAMPLE 1. The smallest 3 digit prime number is

a. 101 b. 103 c. 109 d. 113

Sol. a. The smallest 3 digit number is 100, which is divisible by 2.

$\therefore$ 100 is not a prime numbers.

$\sqrt{101} < 11$ and 101 is not divisible by 2, 3, 5 and 7.

$\therefore$ 101 is a prime number.

Hence, 101 is the smallest 3 digit prime number.

RATIONAL NUMBERS

The numbers which are expressed in the form of p/q where p and q are integers and are coprimes, $q \neq 0$ are called rational numbers.

The set of rational numbers is represented by 'Q'.

e.g. $\frac{3}{5}, \frac{-3}{7}, 7, -6$ are rational numbers.

- The decimal expansion of every rational number is either terminating or non-terminating repeating.

e.g. $\frac{1}{5} = 0.2, \frac{1}{3} = 0.333\dots, \frac{8}{44} = 0.181818\dots$ etc.

- The recurring decimal have been given a short notation as $0.3333 = 0.\overline{3}, 0.181818 = 0.\overline{18}$

Note
- Zero is a rational number, since we can write $0 = 0/1$.
- Every natural number, whole number and integer is a rational number.

IRRATIONAL NUMBERS

The number which cannot be expressed in the form p/q, where p and q both are integers and $q \neq 0$ are known as irrational numbers. The irrational numbers when expressed in decimal form are in non-terminating and non-repeating form. e.g. $\sqrt{2}, \sqrt{5}, \sqrt{7}$, 0.101005001, etc.

- Here, it is notable that exact value of π is not $\frac{22}{7}$ or 3.14, as $\frac{22}{7}$ is a rational number while π is irrational.

IMPORTANT FACTS

1. If $a + \sqrt{b} = x + \sqrt{y}$, where a and x are rational and $\sqrt{b}$ and $\sqrt{y}$ are irrational, then $a = x$ and $b = y$.
2. The sum or difference of a rational and an irrational number is irrational number.
3. The product of rational and irrational number is also an irrational number.
4. If we add, subtract, multiply or divide two irrational numbers, we may get an irrational number or rational number.

EXAMPLE 2. The rational number lying between $\sqrt{2}$ and $\sqrt{3}$ is

a. $\frac{49}{28}$ b. $\frac{56}{35}$

c. $\frac{63}{45}$ d. $\frac{85}{68}$

Sol. b. We have, $\sqrt{2} = 1.414\dots$ and $\sqrt{3} = 1.732\dots$

$\frac{49}{28} = \frac{7}{4} = 1.75, \quad \frac{63}{45} = \frac{7}{5} = 1.4$

$\frac{56}{35} = \frac{8}{5} = 1.6, \quad \frac{85}{68} = \frac{5}{4} = 1.25$

Clearly, 1.6 lies between $\sqrt{2}$ and $\sqrt{3}$.

Hence, $\frac{56}{35}$ lies between $\sqrt{2}$ and $\sqrt{3}$.

REAL NUMBERS

The collection of all rational and all irrational numbers together forms the set of real numbers and is denoted by 'R'. Thus, all natural numbers, whole numbers, integers, rational and irrational numbers are real numbers.

Properties of Real Numbers

Properties of real numbers are as follows

General Properties of R

1. If x and y are two real numbers, then either $x > y$, $y > x$ or $x = y$.
2. If x and y are two real numbers, then
   (i) $x > y \Rightarrow \frac{1}{x} < \frac{1}{y}$ (ii) $x > y \Rightarrow -x < -y$
   (iii) $x > y \Rightarrow x + a > y + a$
   (iv) $x > y \Rightarrow xa > ya$, when $a > 0$
3. If $xy = 0 \Rightarrow x = 0$ or $y = 0$

Properties of operations on R

Let '$*$' be any operation defined on R.

1. **Closure Property** If $a \in R$ and $b \in R$, then $a * b \in R$.
2. **Associative Property** If $a, b, c \in R$, then $a * (b * c) = (a * b) * c$.
3. **Commutative Property** If $a, b \in R$, then $a * b = b * a$.
4. **Identity** Let 'I' be the identity, then $I * a = a * I = a \; \forall \; a \in R$.
5. **Inverse** Let a' be the inverse of a, then $a' * a = a * a' = I \; \forall \; a \in R$.

- The above properties hold for addition and multiplication on R.
- The additive identity of R is O and additive inverse of $a \in R$ is $-a$.
- The multiplicative identity of R is 1 and multiplicative inverse of $a \in R$ is $1/a$.
- R is closed for subtraction, if $a, b \in R$, then $a - b \in R$.
- If $a, b, c \in R$, then $a \times (b + c) = (a \times b) + (a \times c)$, is the distributive property on R.

Absolute Value of a Real Number

The absolute value of a real number x is denoted by $|x|$.

Thus, $|3| = 3$ and $|-4| = 4$.

If x is any real number, then $|x| = \begin{cases} x, \text{ when } & x > 0 \\ 0, \text{ when } & x = 0 \\ -x, \text{ when } & x < 0 \end{cases}$

e.g.

1. If $x = 5, |5| = 5$, if $x = 0, |0| = 0$
   If $x = -5, |-5| = -(-5) = 5$
2. $|x - 2| = x - 2$, if $x > 2$, $|x - 2| = 0$, if $x = 2$
   $|x - 2| = 2 - x$, if $x < 2$

Some Properties of Absolute Values

1. $|x| \geq 0$ for all real x.
2. $|x| = a$ means $x = a$ or $x = -a$
3. $|x| > a$ means $x > a$ or $x < -a$
4. If $n = x^2$; $\sqrt{n} = \sqrt{x^2} = |x| = x$, if $x > 0 = -x$, if $x < 0$

EXAMPLE 3. Find the value of x which satisfy the inequalities $|x| \geq x$ and $2x - 1 > 3$.

a. All positive number
b. All positive number greater than 2
c. All negative number less than -2
d. All negative number

Sol. b. $|x| \geq x$ is true for all real values of x. Now, consider $2x - 1 > 3$ or $2x > 4$ or $x > 2$.

So, the solution set is all positive number greater than 2.

FACTORS IN SET OF INTEGERS

Let $a, b \in I$, we say that a is a factor of b, if there exists an integer p such that $b = ap$; in short we write a / b read it 'a' divides 'b'.

If a is factor of b, then b is called a multiple of a.

Properties of factor and multiples

For real numbers a and b, if $b = ac$ then 'a' is a factor 'b' and we write a / b.

(i) $a / b, b / c \Rightarrow a / c$ (This law is known as transitivity)
(ii) $a / a, \forall a \in R$ (This law is known as reflexivity)
(iii) a / b and $a / c \Rightarrow a / b + c$ and $a / b - c$
(iv) If p is prime number and p divides ab where a,b are integers, then p divides 'a' or p divides 'b'. Thus $p / ab \Rightarrow p / a$ or p / b.

EXAMPLE 4. How many factors of $2^5 \times 3^6$ are perfect squares?

a. 9 b. 12
c. 18 d. 4

Sol. b. Any factor of this number should be of the form $2^a \times 3^b$.

For the factor to be a perfect square a, b have to be even

a can take values 0, 2, 4 and b can take values 0, 2, 4, 6.

$\therefore$ Total number of perfect squares $= 3 \times 4 = 12$

To find the Unit's Place Digit of a given Expression

When Number is in the form of Product

To find the unit digit in the product of two or more number we take unit digit of every number and then multiply them. Then, the unit digit of the resultant product is the unit digit of the product of original numbers.

When Number is in the form of Index

Let the exponential number of the form be a^n and $n \in I$.

1. In case, if a is any of (0, 1, 5, 6), then the unit's place digit is 0, 1, 5 and 6, respectively.
2. In case, if a is any of (4 and 9)
   (i) and if power is odd, then the unit's place digit is 4 and 9, respectively.
   (ii) and if power is even, then the unit's place digit is 6 and 1, respectively.
3. In case, if a is any of (2, 3, 7, 8), then see the following steps

Step I First, divide the exponent of a by 4.

Step II If any remainder comes on division. Put it as the power of a and get the result.

Step III If any remainder does not come on division. Put 4 as the power of a and get the result.

EXAMPLE 5. Find the unit digit of $207 \times 781 \times 39 \times 94$.

a. 4 b. 2 c. 1 d. 5

Sol. b. Taking unit digit of every number and then multiplying them $= 7 \times 1 \times 9 \times 4 = 7 \times 36$

Again taking unit digit and then multiplying $= 7 \times 6 = 42$

$\therefore$ Required unit digit $= 2$

EXAMPLE 6. What is the last digit in $7^{402} + 3^{402}$?

a. 0 b. 4 c. 8 d. None of these

Sol. c. On division of 402 by 4, we get 2 as remainder.

$\therefore$ Last digit of 7^{402} = Last digit of 7^2 = Last digit of $49 = 9$

Last digit of 3^{402} = Last digit of 3^2

= Last digit of $9 = 9$

$\therefore$ Last digit of $7^{402} + 3^{402}$ = Last digit of $(9 + 9) = 8$

DIVISION ON NUMBERS
(DIVISION ALGORITHM)

Let 'a' and 'b' be two integers such that $b \neq 0$. On dividing 'a' by 'b', 'q' will be the quotient and 'r' will be the remainder, then the relationship between a, b, q and r is $a = bq + r$, where $0 \le r < b$. Or in general, we have

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

EXAMPLE 7. When a positive integer n is divided by 5, the remainder is 2. What is the remainder when the number $3n$ is divided by 5?

a. 1 b. 2 c. 3 d. 4

Sol. a. Let $n = 5q + 2$ and $3n = 3(5q + 2)$

$\Rightarrow \quad 3n = 15q + 6 = 15q + (5 + 1) = 5(3q + 1) + 1$

when $3n$ is divided by 5, then remainder is 1.

Divisibility Test

Divisor	Condition with Example
2	If the unit place of a number is '0' or divisible by 2 i.e. unit place is even. e.g. 17980, 314782, 6148, 316 etc.
3	If the sum of the digits of the given number is divisible by 3. e.g. 24375, here $2 + 4 + 3 + 7 + 5 = 21 \div 3 = 7$ Hence, 24375 is divisible by 3.
4	If last two digits of number is divisible by 4. e.g. 589372, 72 is divisible by 4. Hence, 589372 is also divisible by 4.
5	If digit at unit place is 5 or 0. e.g. 895, 700 etc.
6	If given number is divisibly by both 2 and 3. e.g. 759312. Here the last digit is divisible by 2. And $(7 + 5 + 9 + 3 + 1 + 2) = 27$, is divisible by 3. Hence, 759312 is divisible by 6.
7	If twice the number at units place is subtracted from rest of the digits and the remainder is divisible by 7. e.g. (a) $875 = 87 - 2(5) = 87 - 10 = 77 \div 7 = 11$ Hence, 875 is divisible by 7. (b) $5103 = 510 - 2(3) = 510 - 6 = 504 \div 7 = 72$ Hence, 5103 is divisible by 7. **Note** Trick is applicable for number greater than 99.
8	If last three digits are divisible by 8. e.g. (a) $96432 \rightarrow 432 \div 8 = 54$ (b) $16000 \rightarrow 000 \div 8 = 0$
9	If sum of all the digits are divisible by 9. e.g. $317349 \Rightarrow (3 + 1 + 7 + 3 + 4 + 9) = 27 \div 9 = 3$ Hence, 317349 is divisible by 9.
10	If last digit of a number is '0'. e.g. 130, 36980, etc.
11	If the difference between sum of digits at even places and sum of digits at odd places is divisible by 11. e.g. 10615. Sum of digits at odd place $= 1 + 6 + 5 = 12$ Sum of digits at even place $= 0 + 1 = 1$ Difference $= 12 - 1 = 11 \div 11 = 1$ So, 10615 is divisible by 11.

EXAMPLE 8. What is the remainder when 4^{1000} is divided by 7? **2014 I**

a. 1 b. 2 c. 4 d. 5

Sol. c. On division of $4^1, 4^2, 4^3, 4^4, 4^5, 4^6$, and 4^7 by 7,

we get remainders 4, 2, 1, 4, 2, 1 and 4 respectively.

Now, 4^4 gives us same remainder as 4^1, so the cyclicity is of 3.

So, any power of 3 or a multiple of 3 will give a remainder of 1.

So, 4^{999} will give a remainder of 1.

$\therefore$ Remainder of $\frac{4^{1000}}{7} = 4$

Theorem of Divisibility

1. If N is a composite number of the form $N = a^p \cdot b^q \cdot c^r \ldots$, where a, b and c are primes, then the number of divisors of N, represented by m is given by $m = (p+1)(q+1)(r+1)\ldots$
2. The sum of the divisors of N, represented by S is given by

$$S = \frac{(a^{p+1}-1)}{(a-1)} \cdot \frac{(b^{q+1}-1)}{(b-1)} \cdot \frac{(c^{r+1}-1)}{(c-1)}$$

SOME IMPORTANT RESULTS ON DIVISION

1. If p divides q and r, then p also divides their sum and difference also.
2. For any natural number n, $(n^3 - n)$ is divisible by 6.
3. The product of three consecutive natural numbers is always divisible by 6.
4. $(x^m - a^m)$ is divisible by $(x+a)$ for even values of m.
5. $(x^m + a^m)$ is divisible by $(x+a)$ for odd values of m.
6. $(x^m - a^m)$ is divisible by $(x-a)$ for all values of m.

EXAMPLE 9. $19^5 + 21^5$ is divisible by

a. Only 10 b. Only 20
c. Both 10 and 20 d. Neither 10 nor 20

Sol. c. We can check divisibility of $19^5 + 21^5$ by 10 by adding the unit digits of 9^5 and 1^5 which is equal to $9+1=10$. So, it must be divisible by 10.

Now, for divisibility by 20 we add 19 and 21 which is equal to 40.

So, it is clear that it is also divisible by 20.

So, $19^5 + 21^5$ is divisible by both 10 and 20.

Shortcut Methods for Multiplication

1. Multiplication of a Given Number by 9, 99 etc.

Method Place as many zeros at the right of the multiplicand as is the number of nines in the multiplier and subtract from the number, so formed from the multiplicand, to get the result. e.g.

(i) Multiply 8886 by 9999

So, $8886 \times 9999 = 88860000 - 8886 = 88851114$

(ii) Multiply 56985 by 999

So, $56985 \times 999 = 56985000 - 56985 = 56928015$

2. Multiplication of a Given Number by a Power of 5

Method Put as many zeros to the right of the multiplicand as is the power of 5 in the multiplier. Divide the number so formed by 2 raised to the same power as is the power of 5.

e.g. Multiply 6798 by 125

Here, $6798 \times 125 = 6798 \times 5^3 = \frac{6798000}{2^3}$

$$= \frac{6798000}{8} = 849750$$

Important Identities

The identities given below are very useful for quick multiplication.

Here, if a, b and c are real numbers, then

1. (i) $(a+b)^2 = (a+b)(a+b) = a^2 + b^2 + 2ab$
   (ii) $(a+b)^3 = a^3 + b^3 + 3ab\,(a+b)$
2. (i) $(a-b)^2 = (a-b)(a-b) = a^2 + b^2 - 2ab$
   (ii) $(a-b)^3 = a^3 - b^3 - 3ab\,(a-b)$
3. (i) $(a+b) = \sqrt{(a-b)^2 + 4ab}$
   (ii) $(a-b) = \sqrt{(a+b)^2 - 4ab}$
4. (i) $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
   (ii) $(a+b)^2 - (a-b)^2 = 4ab$
5. $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$
6. $a^2 - b^2 = (a-b)(a+b)$

EXAMPLE 10. The value of $\frac{1.073 \times 1.073 - 0.927 \times 0.927}{1.073 - 0.927} + \frac{(3^4)^4 \times 9^6}{(27)^7 \times 3^9}$ is

a. $2\frac{1}{3}$ b. $2\frac{1}{5}$
c. $2\frac{1}{9}$ d. 3

Sol. c. $\frac{1.073 \times 1.073 - 0.927 \times 0.927}{1.073 - 0.927} + \frac{(3^4)^4 + (9)^6}{(27)^7 \times (3)^9}$

$$= \frac{(1.073)^2 - (0.927)^2}{1.073 - 0.927} + \frac{(3^4)^4 \times (3^2)^6}{(3^3)^7 \times (3)^9}$$

$$= \frac{(1.073 + 0.927)\,(1.073 - 0.927)}{1.073 - 0.927} + \frac{3^{28}}{3^{30}}$$

$$= 2 + \frac{1}{3^2} = 2 + \frac{1}{9} = 2\frac{1}{9}$$

SIMPLIFICATION

The word simplification refers to a procedure of converting a complex arithmetical expression into a simple expression.

For simplifying an expression, follow the rule 'VBODMAS'. The letter of word means

V — Under line portion i.e. bar
B — Brackets
O — Orders or powers
D — Division
M — Multiplication
A — Addition
S — Subtraction

- The operations have to be carried out in the order, in which they appear in the word VBODMAS.
- The order of brackets to be simplied is (), {}, [ ].

EXAMPLE 11. Simply the following expression $52-4$ of $(17-12)+4\times7$.

a. 38 b. 85 c. 60 d. 72

Sol. c. $52-4$ of $(17-12)+4\times7$

$= 52-4$ of $5+4\times7$ (simplifying parenthesis)
$= 52-4\times5+4\times7$ (simplifying of)
$= 52-20+4\times7$ (simplifying multiplication)
$= 52-20+28$ (simplifying multiplication)
$= 32+28$ (simplifying subtraction)
$= 60$ (simplifying addition)

PRACTICE EXERCISE

1. Let 'a' and 'b' be natural number, not necessarily distinct. For all values of 'a' and 'b' the natural number would be
(a) $(a+b)$ (b) a/b (c) $a-b$ (d) $\log(ab)$

2. x and y are two natural numbers such that x is less than y, q is the quotient and r is the remainder when y is divided by x. Therefore,
(a) $r=0$ (b) $r<0$ (c) $r>x$ (d) $0\le r<x$

3. If p is a prime number and p divides ab i.e. p/ab, where 'a' and 'b' are integers, then
(a) p/a or p/b (b) p/a and p/b
(c) $p/a-b$ (d) None of these

4. Which one of the following is a prime number?
(a) 161 (b) 171
(c) 173 (d) 221

5. If n is a natural number, then $\sqrt{n}$ is
(a) always a whole number
(b) always a natural number
(c) sometimes a natural number and sometimes an irrational number
(d) always an irrational number

6. The product of a rational number and an irrational number is
(a) natural number (b) an irrational number
(c) a composite number (d) a rational number

7. What is the value of x for which x, $x+1$, $x+3$ are all prime numbers?
(a) 0 (b) 1
(c) 2 (d) 101

8. In a division operation, the divisor is 5 times the quotient and twice the remainder. If the remainder is 15, then what is the dividend?
(a) 175 (b) 185 (c) 195 (d) 250

9. If $42*8$ is a multiple of 9, then the digit represented by $*$ is
(a) 0 (b) 1 (c) 2 (d) 4

10. The value at $*$ in $\frac{16}{7}\times\frac{16}{7}-\frac{*}{7}\times\frac{9}{7}+\frac{9}{7}\times\frac{9}{7}=1$ is
(a) 33 (b) -33 (c) -11 (d) 32

11. The value of $10\div\overline{4+6}\times4$ is
(a) 4 (b) 1/4 (c) 5 (d) None of these

12. If $\frac{a}{4}=\frac{b}{5}=\frac{c}{6}$, then the value of $\frac{a+b+c}{b}$ is
(a) 3 (b) 2 (c) 6 (d) 4

13. The value of $[a-b+(b-a)]-[2a-2b+(b-2a)]$ is
(a) $b-2a$ (b) $a-2b$ (c) $b+2a$ (d) b

14. If $3.325\times10^k=0.0003325$, the value of '$k$' is
(a) 4 (b) -4 (c) -3 (d) -2

15. If $\sqrt{[0.04\times0.4\times x]}=0.4\times0.04\times\sqrt{y}$, then the value of x/y is
(a) 0.0016 (b) 0.16 (c) 0.016 (d) 0.160

16. The unit digit in the product $(127)^{170}$ is
(a) 3 (b) 9 (c) 7 (d) 3

17. The unit digit in the product $(7^{71}\times6^{59}\times3^{65})$ is
(a) 6 (b) 2 (c) 4 (d) 1

18. When n is divided by 4, the remainder is 3. What is the remainder when $2n$ is divided by 4?
(a) 0 (b) 2 (c) 6 (d) 3

19. If $\dfrac{\frac{1}{4}-\frac{1}{6}-\frac{1}{48}}{\frac{1}{4}-\left(\frac{1}{6}-\frac{1}{48}\right)} \div \dfrac{\frac{1}{4}\times\frac{1}{6}-\frac{1}{48}}{\frac{1}{4}\times\left(\frac{1}{6}-\frac{1}{48}\right)} = x$, the value of x is
(a) $\frac{20}{21}$ (b) $-\frac{21}{20}$ (c) $\frac{21}{20}$ (d) $-\frac{20}{21}$

20. If x is negative real number, then
(a) $|x| = x$ (b) $|x| = -x$ (c) $|x| = \frac{1}{x}$ (d) $|x| = -\frac{1}{x}$

21. If $\frac{x}{y} = \frac{3}{5}$, then the value of $\frac{x-y}{x+y}$ is
(a) $-\frac{1}{4}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) -6

22. If x is real, then $\left|\frac{7-x}{3}\right| < 2$, if and only if
(a) $1 < x < 13$ (b) $-1 < x < 13$ (c) $x < 13$ (d) $x > 13$

23. A number is divisible by 25 only, if
(a) the last digit of the number is zero
(b) the last digit of the number is 5
(c) the last two digit of the number is divisible by 15
(d) the last two digit of the number is divisible by 25

24. The value of $\frac{2.48\times 2.48 - 1.52\times 1.52}{0.96}$ is
(a) 4 (b) 0.96 (c) 16 (d) 15.04

25. Which one of the following is correct regarding the number 222222?
(a) It is divisible by 3 but not divisible by 7
(b) It is divisible by 3 and 7 but not divisible by 11
(c) It is divisible by 2 and 7 but not divisible by 11
(d) It is divisible by 3, 7 and 11

26. What is the sum of positive integers less than 100 which leave a remainder 1 when divided by 3 and leave a remainder 2 when divided by 4?
(a) 416 (b) 620 (c) 1250 (d) 1314

27. What is the last digit in the expansion of $(2457)^{754}$?
(a) 3 (b) 7 (c) 8 (d) 9

28. A three-digit number is divisible by 11 and has its digit in the unit's place equal to 1. The number is 297 more than the number obtained by reversing the digits. What is the number?
(a) 121 (b) 231 (c) 561 (d) 451

29. The remainder on dividing given integers a and b by 7 are, respectively 5 and 4. What is the remainder when ab is divided by 7?
(a) 3 (b) 4 (c) 5 (d) 6

30. If k is any even positive integer, then $(k^2 + 2k)$ is
(a) divisible by 24
(b) divisible by 8 but may not be divisible by 24
(c) divisible by 4 but may not be divisible by 8
(d) divisible by 2 but may not be divisible by 4

31. The number 2784936 is divisible by which one of the following numbers?
(a) 86 (b) 87 (c) 88 (d) 89

32. Which among the following is the largest four-digit number that is divisible by 88?
(a) 9988 (b) 9966 (c) 9944 (d) 8888

33. The pair of rational numbers that lies between $\frac{1}{4}$ and $\frac{3}{4}$ is
(a) $\frac{262}{1000}, \frac{752}{1000}$ (b) $\frac{24}{100}, \frac{74}{100}$
(c) $\frac{9}{40}, \frac{31}{40}$ (d) $\frac{252}{1000}, \frac{748}{1000}$

34. By adding x to 1254934, the resulting number becomes divisible by 11, while adding y to 1254934 makes the resulting number divisible by 3. Which one of the following is the set of values for x and y?
(a) $x = 1, y = 1$ (b) $x = 1, y = -1$
(c) $x = -1, y = 1$ (d) $x = -1, y = -1$

35. What is the last digit in the expansion of 3^{4798} ?
(a) 1 (b) 3 (c) 7 (d) 9

36. What least value must be given to ⊗, so that the number 84705 ⊗ 2 is divisible by 9?
(a) 0 (b) 1 (c) 2 (d) 3

37. What is the total number of three digit numbers with unit digit 7 and divisible by 11?
(a) 6 (b) 7 (c) 8 (d) 9

38. Find the value of 'a' and 'b' $3\frac{7}{a} \times b\frac{3}{15} = 8$.
(a) 2, 11 (b) 11, 2 (c) 1, 1 (d) 2, 1

39. If a, b and c are real numbers such that $a < b$ and $c < 0$, then which of the statements is true?
(a) $(a/c) < (b/c)$ (b) $ac < bc$
(c) $(c/a) > (c/b)$ (d) $ac > bc$

40. If we divide a positive integer by another positive integer, what is the resulting number?
(a) It is always a natural number
(b) It is always an integer
(c) It is a rational number
(d) It is an irrational number

41. What can be said about the expansion of $2^{12n} - 6^{4n}$, where n is a positive integer?
(a) Last digit is 4 (b) Last digit is 8
(c) Last digit is 2 (d) Last two digits are zero

42. If p is an integer, then every square integer is of the form

(a) $2p$ or $(4p-1)$ (b) $4p$ or $(4p-1)$
(c) $3p$ or $(3p+1)$ (d) $4p$ or $(4p+1)$

43. If r and s are any real numbers such that $0 \le s \le 1$ and $r + s = 1$, then what is the maximum value of the product?

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

44. A number when divided by 2, 3 or 5 gives remainder 1. The number is

(a) 31 (b) 47 (c) 49 (d) 53

45. The largest integer that divides product of any four consecutive integers is

(a) 4 (b) 6 (c) 12 (d) 24

46. Which one of the following three digit numbers divides 9238 and 7091 with the same remainder in each case?

(a) 113 (b) 209 (c) 317 (d) 191

47. For a positive integer n, define $d(n) =$ The number of positive divisors of n. What is the value of $d(d(d(12)))$?

(a) 1 (b) 2 (c) 4 (d) None of these

48. If three sides of a right angled triangle are integers in their lowest form, then one of its sides is always divisible by

(a) 6 (b) 5 (c) 7 (d) None of these

49. If $-1 \le x \le 3$ and $1 \le y \le 3$, then the maximum value of $(3y - 4x)$ is

(a) 18 (b) 13 (c) 5 (d) –6

50. If 'p' is an integer greater than 3, then on dividing $p^{11} + 1$ by $p - 1$, we would get the remainder as

(a) 2 (b) 0 (c) –2 (d) –1

51. If A is the set of squares of natural numbers and x and y are any two element of A, then the correct statement is

(a) $x + y$ belongs to A (b) $x - y$ belongs to A
(c) $\frac{x}{y}$ belongs to A (d) $x\,y$ belongs to A

52. A ten-digit number is divisible by 4 as well as by 5. What could be the possible digit at the ten's place in the given number?

(a) 0, 1, 2, 4, or 6 (b) 1, 2, 4, 6 or 8
(c) 2, 3, 4, 6 or 8 (d) 0, 2, 4, 6 or 8

53. If $x < 0 < y$, then which one of the following relations is correct?

(a) $\frac{1}{x^2} < \frac{1}{xy} < \frac{1}{y^2}$ (b) $\frac{1}{x^2} > \frac{1}{xy} < \frac{1}{y^2}$
(c) $\frac{1}{x} < \frac{1}{y}$ (d) $\frac{1}{x} > \frac{1}{y}$

54. What will be the remainder when 19^{100} is divided by 20?

(a) 19 (b) 20 (c) 3 (d) 1

55. $7^{12} - 4^{12}$ is exactly divisible by which of the following number?

(a) 34 (b) 33 (c) 36 (d) 35

56. If $2^p + 3^q = 17$ and $2^{p+2} - 3^{q+1} = 5$, then find the value of p and q.

(a) –2, 3 (b) 2, – 3 (c) 3, 2 (d) 2, 3

57. If $\frac{49}{15} = 3 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$, where x, y and z are natural numbers, then what is z equal to?

(a) 1 (b) 2 (c) 3
(d) Cannot be determined due to insufficient data

58. The least number which is a perfect square and has 540 as a factor is

(a) 8100 (b) 6400 (c) 4900 (d) 3600

59. If A is real and $1 + A + A^2 + A^3 = 40$, then A is equal to

(a) –3 (b) –1 (c) 1 (d) 3

60. How many factors of 1080 are perfect squares?

(a) 4 (b) 6 (c) 8 (d) 5

61. Consider the following statements:

I. In a given whole number, if the sum of the odd numbered digit is equal to the sum of even numbered digits, then the number is divisible by 11.
II. In a given whole number, if the difference of sum of odd numbered digits and even numbered digits is divisible by 11, then the number is divisible by 11.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Both I and II
(c) Only II (d) None of these

62. Consider the following statements:

A number $a_1a_2a_3a_4a_5$ is divisible by 9, if

I. $a_1 + a_2 + a_3 + a_4 + a_5$ is divisible by 9.
II. $a_1 - a_2 + a_3 - a_4 + a_5$ is divisible by 9.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

63. Consider the following statements:

If p is a prime such that $p + 2$ is also a prime, then

I. $p(p + 2) + 1$ is a perfect square.
II. 12 is a divisor of $p + (p + 2)$, if $p > 3$

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

64. Consider the following statements:

I. If x and y are composite integers, so also is $x + y$.

II. If x and y are composite integers and $x > y$, then $x - y$ is also a composite integer.

III. If x and y are composite integers, so also in xy.

Which of the statement(s) given above is/are correct?

(a) I, II and III (b) I and II
(c) Only III (d) None of these

65. Consider the following statements:

I. The product of any three consecutive integers is divisible by 6.

II. Any integer can be expressed in one of the three forms $3k$, $3k+1$, $3k+2$, where k is an integer.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

66. Consider the following statements:

I. A natural number is divisible by 2, if its last digit is divisible by 2.

II. A natural number is divisible by 2, if its last digit is either zero or 2.

III. A natural number is divisible by 2, if its last digit is even.

Which of the statements given above are correct?

(a) I and II (b) I and III (c) II and III (d) All of these

67. Consider the following statements for natural numbers a, b and c :

I. If 'a' is divisible by 'b' and 'b' is divisible by 'c', then a must be divisible by 'c'.

II. If 'a' is a factor of both 'b' and 'c', then 'a' must be a factor of '$b + c$'.

III. If 'a' is a factor of both 'b' and 'c', then 'a' must be a factor of '$b - c$'.

Which of the statements given above are correct?

(a) I, II and III (b) I and II
(c) II and III (d) None of these

68. Consider the following statements:

I. Set of positive powers of 2 is closed under multiplication.

II. The set $\{1, 0, -1\}$ is closed under multiplication.

III. The number 35 has exactly four divisors.

IV. The set $\{1, 0, -1\}$ is closed under addition.

Which of the statements given above are correct?

(a) I, II and III are true (b) Only III
(c) Only IV (d) All of these

PREVIOUS YEARS QUESTIONS

69. What number should be added to 231228 to make it exactly divisible by 33? **2012 I**

(a) 1 (b) 2 (c) 3 (d) 4

70. If a positive integer leaves remainder 28 when divided by 143, then what is the remainder obtained on dividing the same number by 13? **2012 I**

(a) 0 (b) 2 (c) 9 (d) 10

71. Which one of the following is neither prime number nor composite number? **2012 II**

(a) 1 (b) 2 (c) 3 (d) None of these

72. Which one of the following has least number of divisors? **2012 II**

(a) 88 (b) 91 (c) 96 (d) 99

73. How many numbers between −11 and 11 are multiples of 2 or 3? **2012 II**

(a) 11 (b) 14 (c) 15 (d) None of these

74. Consider the following statements

I. If n is a prime number greater than 5, then $n^4 - 1$ is divisible by 2400.

II. Every square number is of the form $5n$ or $(5n-1)$ or $(5n+1)$, where n is a whole number.

Which of the statement(s) given above is/are correct? **2012 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

75. If N, $(N+2)$ and $(N+4)$ are prime numbers, then the number of possible solutions for N are **2013 I**

(a) 1 (b) 2
(c) 3 (d) None of these

76. The two-digit number, which when divided by sum of the digits and product of the digits, respectively leaves the same remainder the number is and the difference of quotients is one.

(a) 14 (b) 23 (c) 32 (d) 41 **2013 I**

77. If x is positive even integer and y is negative odd integer, then x^y is **2013 I**

(a) odd integer (b) even integer
(c) rational number (d) None of these

78. Consider the following statements

I. There is a finite number of rational numbers between any two rational numbers.

II. There is an infinite number of rational numbers between any two rational numbers.

III. There is a finite number of irrational number between any two rational numbers.

Which of the statement(s) given above is/are correct? **2013 I**

(a) Only I (b) Only II
(c) Only III (d) Both II and III

79. If k is a positive integer, then every square integer is of the form 2013 II

(a) $4k$ (b) $4k$ or $4k+3$
(c) $4k+1$ or $4k+3$ (d) $4k$ or $4k+1$

80. If b is the largest square divisor of c and a^2 divides c, then which one of the following is correct (where, a, b and c are integers)? 2013 II

(a) b divides a (b) a does not divide b
(c) a divides b (d) a and b are coprime

81. Every prime number of the form $3k+1$ can be represented in the form $6m+1$ (where, k and m are integers), when 2013 II

(a) k is odd
(b) k is even
(c) k can be both odd and even
(d) No such form is possible

82. Consider the following statements

I. 7710312401 is divisible by 11.
II. 173 is a prime number.

Which of the statement(s) given above is/are correct? 2013 II

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

83. Consider the following statements

I. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7.
II. Every composite number less than 121 is divisible by a prime number less than 11.

Which of the statement(s) given above is/are correct? 2013 II

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

84. Consider the following statements

I. No integer of the form $4k+3$, where k an integer, can be expressed as the sum of two squares.
II. Square of an odd integer can expressed in the form $8k+1$, where k is an integer.

Which of the statement(s) given above is/are correct? 2014 I

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

85. What is 26^2+97^2 equal to? 2014 I

(a) 27^2+93^2 (b) 34^2+93^2 (c) 82^2+41^2 (d) 79^2+62^2

86. If n is a whole number greater than 1, then $n^2(n^2-1)$ is always divisible by 2014 I

(a) 12 (b) 24 (c) 48 (d) 60

87. What is the remainder when $(17^{23}+23^{23}+29^{23})$ is divided by 23? 2014 II

(a) 0 (b) 1 (c) 2 (d) 3

88. What is the remainder when $(1235\times 4523\times 2451)$ is divided by 12? 2014 II

(a) 1 (b) 3 (c) 5 (d) 7

89. What is the number of divisors of 360? 2014 II

(a) 12 (b) 18
(c) 24 (d) None of these

90. The multiplication of a three-digit number $XY5$ with digit Z yields $X215$. What is $X+Y+Z$ equal to? 2014 II

(a) 13 (b) 15 (c) 17 (d) 18

91. How many pairs of X and Y are possible in the number $763X4Y2$, if the number is divisible by 9? 2014 II

(a) 8 (b) 9 (c) 10 (d) 11

92. If $a_n = 3-4n$, then what is $a_1+a_2+a_3+\ldots+a_n$ equal to? 2014 II

(a) $-n(4n-3)$ (b) $-n(2n-1)$ (c) $-n^2$ (d) $-n(2n+1)$

93. Consider all those two-digit positive integers less than 50, which when divided by 4 yield unity as remainder. What is their sum? 2014 II

(a) 310 (b) 314 (c) 218 (d) 323

94. p, q and r are prime numbers such that $p<q<r<13$. In how many cases would $(p+q+r)$ also be a prime number? 2014 II

(a) 1 (b) 2 (c) 3 (d) None of these

95. The digit in the units place of the product $81\times 82\times 83\times 84\times\ldots\times 99$ is 2015 I

(a) 0 (b) 4 (c) 6 (d) 8

96. What is the remainder obtained when $1421\times 1423\times 1425$ is divided by 12? 2015 I

(a) 1 (b) 2 (c) 3 (d) 4

97. What is the remainder when 4^{96} is divided by 6? 2015 I

(a) 4 (b) 3
(c) 2 (d) 1

98. What is the maximum value of m, if the number $N = 35\times 45\times 55\times 60\times 124\times 75$ is divisible by 5^m? 2015 I

(a) 4 (b) 5
(c) 6 (d) 7

99. If $a-b=4$ and $a^2+b^2=40$, where a and b are positive integers, then a^3+b^6 is equal to 2015 I

(a) 264 (b) 280
(c) 300 (d) 324

100. If n is a natural number and $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$, where p_1, p_2, p_3 are distinct prime factors, then the number of prime factors for n is 2015 I

(a) $x_1+x_2+x_3$ (b) $x_1x_2x_3$
(c) $(x_1+1)(x_2+1)(x_3+1)$ (d) None of these

101. If $\frac{37}{13} = 2 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$, where x, y and z are natural numbers, then what is z equal to? **2015 I**

(a) 1 (b) 2 (c) 3
(d) Cannot be determined due to insufficient data

102. A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **2015 I**

(a) 12 (b) 15 (c) 25 (d) 32

103. The number of pairs (x, y), where x, y are integers satisfying the equation $21x + 48y = 5$, is **2015 II**

(a) Zero (b) One (c) Two (d) Infinity

104. The largest natural number which divides every natural number of the form $(n^3 - n)(n - 2)$, where n is a natural number greater than 2, is **2015 II**

(a) 6 (b) 12
(c) 24 (d) 48

105. The digit in the units place of the resulting number of the expression $(234)^{100} + (234)^{101}$, is **2015 II**

(a) 6 (b) 4
(c) 2 (d) 0

106. A number when divided by 7 leaves a remainder 3 and the resulting quotient, when divided by 11 leaves a remainder 6. If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n. What are the values of m and n, respectively? **2015 II**

(a) 1 and 4 (b) 4 and 1 (c) 3 and 6 (d) 6 and 3

107. The seven digit number $876p37q$ is divisible by 225. The values of p and q can be respectively **2015 II**

(a) 9, 0 (b) 0, 0
(c) 0, 5 (d) 5,9

108. Let x and y be positive integers such that $x > y$. The expressions $3x + 2y$ and $2x + 3y$, when divided by 5 leave remainders 2 and 3, respectively. What is the remainder when $(x - y)$, is divided by 5? **2015 II**

(a) 4 (b) 2 (c) 1 (d) 0

109. The sum of first 47 terms of the series

$\frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \dots$, is **2015 II**

(a) 0 (b) $-\frac{1}{6}$ (c) $\frac{1}{6}$ (d) $\frac{9}{20}$

110. The value of the expression $\frac{(243 + 647)^2 + (243 - 647)^2}{243 \times 243 + 647 \times 647}$ is equal to **2016 I**

(a) 0 (b) 1 (c) 2 (d) 3

111. What is the maximum value of m, if the number $N = 90 \times 42 \times 324 \times 55$ is divisible by 3^m? **2016 I**

(a) 8 (b) 7 (c) 6 (d) 5

112. $7^{10} - 5^{10}$ is divisible by **2016 I**

(a) 5 (b) 7 (c) 10 (d) 11

113. If $a^3 = 117 + b^3$ and $a = 3 + b$, then the value of $a + b$ is (given that $a > 0$ and $b > 0$) **2016 I**

(a) 7 (b) 9 (c) 11 (d) 13

114. Let m be a non-zero integer and n be a positive integer. Let R be the remainder obtained on dividing the polynomial $x^n + m^n$ by $(x - m)$. Then, **2016 I**

(a) R is a non-zero even integer
(b) R is odd, if m is odd
(c) $R = s^2$ for some integer s, if n is even
(d) $R = t^3$ for some integer t, if 3 divides n

115. If $\frac{61}{19} = 3 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{x}}}$ where x, y and z are natural numbers, then what z is equal to? **2016 I**

(a) 1 (b) 2 (c) 3 (d) 4

116. Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct?

I. $(y - x)$ can be an even integer.
II. xy can be an even integer.
III. $0.5\ (x + y)$ can be an even integer.

Select the correct answer using the code given below **2016 I**

(a) I and II (b) II and III
(c) I and III (d) All of these

117. Consider the following statements:

I. Every natural number is a real number.
II. Every real number is a rational number.
III. Every integer is real number.
IV. Every rational number is a real number.

Which of the above statements are correct? **2016 I**

(a) I, II and III (b) I, III and IV
(c) II and III (d) III and IV

118. Consider the following statements in respect of two different non-zero integers p and q.

I. For $(p + q)$ to be less than $(p - q)$, q must be negative.
II. For $(p + q)$ to be greater than $(p - q)$, both p and q must be positive.

Which of the above statements is/are correct? **2016 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

119. If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?

I. $a - b < a - c$

II. If $a < b$, then $\dfrac{a}{c} < \dfrac{b}{c}$

III. $\dfrac{1}{b} < \dfrac{1}{c}$

Select the correct answer using the code given below **2016 I**

(a) Only I (b) Only II
(c) Only III (d) II and III

120. If m and n are distinct natural numbers, then which of the following is/are integer/integers?

I. $\dfrac{m}{n} + \dfrac{n}{m}$ II. $mn\left(\dfrac{m}{n} + \dfrac{n}{m}\right)(m^2 + n^2)^{-1}$

III. $\dfrac{mn}{m^2 + n^2}$

Select the correct answer using the code given below **2016 (I)**

(a) I and II (b) Only II
(c) II and III (d) Only III

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42		43		44		45		46		47		48		49		50	
51		52		53		54		55		56		57		58		59		60	
61		62		63		64		65		66		67		68		69		70	
71		72		73		74		75		76		77		78		79		80	
81		82		83		84		85		86		87		88		89		90	
91		92		93		94		95		96		97		98		99		100	
101		102		103		104		105		106		107		108		109		110	
111		112		113		114		115		116		117		118		119		120	

HINTS AND SOLUTIONS

1. *(a)* $(a + b)$ always represent a natural number $\forall\, a, b \in N$.

2. *(d)* As, $y = qx + r$, so $0 \le r < x$.

3. *(a)* As p is prime, so p/a or p/b.

4. *(c)* Consider the given options

(a) 161 is divisible by 7. Hence, 161 is not a prime number.

(b) Since, $1 + 7 + 1 = 9$. Therefore, 171 is divisible by 3. Hence, 171 is not a prime number.

(c) $173 < (14)^2$

173 is not divisible by any of the numbers 2, 3, 5, 7, 11, 13. Hence, 173 is a prime number.

(d) 221 is divisible by 13. Hence, 221 is not a prime number.

5. *(c)* Consider, $2, 4 \in N$

So, $\sqrt{4} = 2$, a natural number

and $\sqrt{2}$ = irrational number.

6. *(b)* We know that, the product of a rational number and an irrational number is an irrational number.

7. *(c)* If $x = 2$, then, x, $x + 1$ and $x + 3$ are all prime numbers.

8. *(c)* $\because$ Dividend $= D \times Q + R$

Given, $D = 5Q$ and $D = 2R$

When $R = 15, D = 2 \times 15 = 30$

$\therefore \quad Q = \dfrac{D}{5} = \dfrac{30}{5} = 6$

$\therefore$ Dividend $= 30 \times 6 + 15 = 195$

9. *(d)* As, $4 + 2 + * + 8 = 14 + *$; $42*8$ is divisible by 9, if $14 + *$ divisible by 9.

So, $14 + * = 18$ (nearest multiple of 9)

$\Rightarrow \quad * = 18 - 14 = 4$

10. *(d)* Here, $\dfrac{16}{7} \times \dfrac{16}{7} - \dfrac{*}{7} \times \dfrac{9}{7} + \dfrac{9}{7} \times \dfrac{9}{7} = 1$

Put $* = x$

$\Rightarrow \quad \dfrac{256}{49} - \dfrac{9x}{49} + \dfrac{81}{49} = 1$

$\Rightarrow \quad 256 + 81 - 9x = 49$

$\Rightarrow \quad 9x = 288 \Rightarrow x = 32$

$\Rightarrow \quad 9x = 288 \Rightarrow x = 32$

11. *(a)* By BODMAS, $10 \div \overline{4 + 6} \times 4$

$= 10 \div (4 + 6) \times 4$

$= 10 \div 10 \times 4 = 1 \times 4 = 4$

12. *(a)* $\dfrac{a}{4} = \dfrac{b}{5} = \dfrac{c}{6} = k$ (say)

$\Rightarrow \quad a = 4k,\ b = 5k$ and $c = 6k$

So, $\dfrac{a + b + c}{b} = \dfrac{4k + 5k + 6k}{5k} = \dfrac{15k}{5k} = 3$

13. *(d)* $[a - b + (b - a)]$

$- [2a - 2b + (b - 2a)]$

$\Rightarrow [a - b + b - a] - [2a - 2b + b - 2a]$

$\Rightarrow \quad 0 - [2a - 2a - b] = 0 + b = b$

14. *(b)* $10^k = \frac{0.0003325}{3.325}$ (given)

$= \frac{3.325 \times 10^{-4}}{3.325} = 10^{-4}$

$\Rightarrow \quad 10^k = 10^{-4}$

$\therefore \quad k = -4$

15. *(c)* $\sqrt{0.04 \times 0.4 \times x} = 0.4 \times 0.04 \times \sqrt{y}$ (given)

$\therefore \quad \frac{\sqrt{x}}{\sqrt{y}} = \frac{0.4 \times 0.04}{\sqrt{0.04 \times 0.4}}$

$\frac{x}{y} = \frac{0.4 \times 0.4 \times 0.04 \times 0.04}{0.04 \times 0.4} = 0.016$

(squaring both sides)

16. *(b)* Since, unit digit in 7^4 is 1.

$\therefore \quad 7^{168} = (7^4)^{42}$ give unit digit 1.

$\therefore \quad 7^{170} = 7^{168} \times 7^2$ gives the unit digit as $1 \times 7 \times 7 = 9$

17. *(c)* Unit place in $7^4 = 1$, so unit place in 7^{68} is 1.

$\therefore$ Unit place in $7^{68} \times 7^3 = 3$. Similarly, unit place in 6^{59} is 6 and unit place in 3^4 is 1, so unit place in $3^{65} = 3^{64} \times 3 = 3$.

$\therefore$ Unit place in $7^{71} \times 6^{59} \times 3^{65}$ is the unit place of $3 \times 6 \times 3 = 4$

18. *(b)* As, n is divided by 4 the remainder is 3, so

$n = 4q + 3$, where q is quotient.

$\Rightarrow \quad 2n = 8q + 6$

$\Rightarrow \quad 2n = (8k + 4) + 2 = 4(2k + 1) + 2$

So, if $2n$ is divided by 4 the quotient is $2k + 1$ and remainder is 2.

19. *(c)* Given, $\dfrac{\frac{1}{4} - \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} - \left(\frac{1}{6} - \frac{1}{48}\right)} \div \dfrac{\frac{1}{4} \times \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} \times \left(\frac{1}{6} - \frac{1}{48}\right)} = x$

$\Rightarrow \quad x = \dfrac{\frac{12 - 8 - 1}{48}}{\frac{12 - (8 - 1)}{48}} \div \dfrac{\frac{1}{24} - \frac{1}{48}}{\frac{1}{4} \times \left(\frac{8 - 1}{48}\right)}$

$\Rightarrow \quad x = \dfrac{\frac{3}{48}}{\frac{5}{48}} \div \dfrac{\frac{2 - 1}{48}}{\frac{7}{4 \times 48}}$

$\Rightarrow \quad x = \frac{3}{5} \div \frac{4}{7} \Rightarrow x = \frac{21}{20}$

20. *(b)* Clearly, absolute value is defined by $|x| = -x$.

21. *(a)* $\frac{x - y}{x + y} = \frac{x/y - 1}{x/y + 1} = \frac{3/5 - 1}{3/5 + 1}$

$= \frac{(3 - 5)/5}{(3 + 5)/5} = \frac{-2}{8} = -\frac{1}{4}$

22. *(a)* As, $\left|\frac{7 - x}{3}\right| < 2$

$\Rightarrow \quad \frac{7 - x}{3} < 2$ or $-\left(\frac{7 - x}{3}\right) < 2$

$[\because |x| < a \Rightarrow x < a \text{ or } -x < a]$

$\Rightarrow \quad 7 - x < 6$ or $\frac{x - 7}{3} < 2$

$\Rightarrow \quad -x < -1$ or $x - 7 < 6$

$\Rightarrow \quad x > 1$ or $x < 13$

$\Rightarrow \quad 1 < x < 13$

23. *(d)* A number is divisible by 25 when its last 2 digits are either zero or divisible by 25.

24. *(a)* $\frac{(2.48)^2 - (1.52)^2}{0.96}$

$= \frac{(2.48 - 1.52)(2.48 + 1.52)}{0.96}$

$[\because a^2 - b^2 = (a - b)(a + b)]$

$= \frac{0.96 \times 4}{0.96} = 4$

25. *(d)* Given, number is 222222.

Here, sum of digits

$= 2 + 2 + 2 + 2 + 2 + 2 = 12,$

which is divisible by 3. So, given number is divisible by 3.

Now, sum of odd terms of digits − Sum of even terms of digits = 6 − 6 = 0,

it is divisible by 11.

Also, in a number if a digit is repeated six times, then the number is divisible by 7, 11 and 13.

Hence, the given number is divisible by 3, 7 and 11.

26. *(a)* Required numbers are of the form of $12q - 2$

i.e. 10, 22, 34, 46, 58, 70, 82, 94

$\therefore$ Total sum $= 10 + 22 + 34 + 46 + 58 + 70 + 82 + 94 = 416$

27. *(d)* The last digit in the expansion of $(2457)^{754}$ is equal to last digit of $(7)^{754}$.

$(7)^{754} = (7^4)^{188} \times 7^2 = 1 \times 7^2$

$= (7)^2 = 49$

Hence, last digit is 9.

28. *(d)* On taking option (d),

The reverse digit of 451 is 154.

Now, $154 + 297 = 451$ is equal to the original number.

29. *(d)* Let $a = 7p + 5$ and $b = 7q + 4$

where, p and q are natural numbers.

$\therefore \quad ab = (7p + 5)(7q + 4)$

$ab = 49pq + (4p + 5q)\,7 + 20$

$= 7(7pq + 4p + 5q) + 7 \times 2 + 6$

when ab is divided by 7, we get the remainder 6.

30. *(b)* If k is any even positive integer, then $k^2 + 2k$ is divisible by 8 but may not be divisible by 24.

Let $k = 2m, m \in N$, then

$k^2 + k \cdot 2 = 4m^2 + 4m = 4m(m + 1)$

which is divisible by 8.

31. *(c)* Given number is 2784936.

Sum of digits at odd places= 25 and sum of digits at even places = 14

$\therefore$ Difference $= 25 - 14 = 11$

So, number is divisisble by 11.

Also last three digits of 2784936 i.e. 936 is divisible by 8.

Hence, 2784936 is divisible by both 8 and 11 i.e. 88.

32. *(c)* A number divisible by 88, if it is divisible by 8 and 11.

In the given options 9944 and 8888 are divisible by 88. Hence, maximum number is 9944.

33. *(d)* $\because \frac{1}{4} = 0.25$ and $\frac{3}{4} = 0.75$

Only option (d) lies between 0.25 and 0.75. Since $\frac{252}{1000} = 0.252$

and $\frac{748}{1000} = 0.748$

34. *(b)* Difference of sums of even and odd places digit of 1254934

$= (1 + 5 + 9 + 4) - (2 + 4 + 3)$

$= 19 - 9 = 10$

This number will be divisible by 11, after adding x, if $x = 1$.

Also, the sum of digits of 1254934

$= 1 + 2 + 5 + 4 + 9 + 3 + 4 = 28$

1254934 will be divisible by 3, after adding y, if $y = -1$

35. *(d)* Last digit in the expansion of 3^{4798}

= Last digit in the expansion of $(3^4)^{1199} \cdot 3^2$

= Last digit in the expansion of $3^2 = 9$

36. *(b)* The given number is divisible by 9, if sum of the digits is divisible by 9.

Here, sum of digits

$= 8 + 4 + 7 + 0 + 5 + 2 + \otimes = 26 + \otimes$

If $\otimes = 1$, then $26 + 1 = 27$ is divisible by 9.

37. *(c)* The total number of three-digit numbers with unit digit 7 and divisible by 11 are 187, 297, 407, 517, 627, 737, 847, 957.

$\therefore$ Total numbers = 8

38. *(b)* Apply hit and trial method from the given option. As, here when $a = 11$, $b = 2$, then

$$3\frac{7}{a} \times b\frac{3}{15} = 3\frac{7}{11} \times 2\frac{3}{15}$$
$$= \frac{40}{11} \times \frac{33}{15} = 8$$

39. *(d)* Since, $a < b \Rightarrow a - b < 0$. Also, $c < 0$

$\therefore (a - b)\, c > 0 \Rightarrow ac - bc > 0 \Rightarrow ac > bc$

40. *(c)* When we divide a positive integer by another positive integer, the resultant will be a rational number i.e. in the form of p/q, where p and q are positive integers and $q \neq 0$.

41. *(d)* $2^{12n} - 6^{4n} = (2^{12})^n - (6^4)^n$
$= (4096)^n - (1296)^n$

$\therefore \quad 2^{12n} - 6^{4n} = (4096 - 1296)\, k$

As $x^n - y^m$ is always divisible by $(x - m)$
$= 2800\,(k)$

Hence, last two digits are always be zero.

42. *(c)* We know that

$4 = 3p + 1$, for $p = 1$, $9 = 3p$, for $p = 3$
$16 = 3p + 1$, for $p = 5$,
$25 = 3p + 1$, for $p = 8$
$36 = 3p$, for $p = 12$

Hence, every square integer is of form either $3p$ or $3p + 1$.

43. *(d)* Given, $r + s = 1$

For maximum product, $r = s = \frac{1}{2}$

$\therefore \qquad rs = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

44. *(a)* Required number
$= [\text{LCM of 2, 3 and 5}] + 1$
$= 30 + 1 = 31$

45. *(d)* The largest integer that divides product of any four consecutive integers is 24.
e.g. 1, 2, 3, 4 are four consecutive integers.

Multiplication $= 1 \times 2 \times 3 \times 4 = 24$

which divided by 24.

46. *(a)* When we divide the number 9238 and 7091 by 113, we get the same remainder 85.

47. *(d)* $d(d(d(12))) = d(d(6))$
[$\because$ positive integer divisor of $12 = 1, 2, 3, 4, 6, 12$]
$d(d(6)) = d(4)$
[$\because$ positive integer divisor of $6 = 1, 2, 3, 6$]
and $\quad d(4) = 3$
[$\because$ positive integer divisor of $4 = 1, 2, 4$]

48. *(b)* Given, by pythagoras theorem,
$(3)^2 + (4)^2 = (5)^2$

Let the sides of a right triangle be 3, 4, 5.

Hence, one of its sides is always divisible by 5.

49. *(b)* $\because 1 \leq y \leq 3 \Rightarrow 3 \leq 3y \leq 9 \qquad \ldots(\text{i})$

and $-1 \leq x \leq 3 \Rightarrow -4 \leq 4x \leq 12$

$\Rightarrow -12 \leq -4x \leq 4 \qquad \ldots(\text{ii})$

From Eqs. (i) and (ii), we get
$-9 \leq 3y - 4x \leq 13$

Maximum value is 13.

50. *(a)* When $p^{11} + 1$ is divided by $(p - 1)$, then the remainder is

$(1)^{11} + 1 = 1 + 1 = 2$

51. *(d)* Let $x = a^2$ and $y = b^2$ for some $a, b \in N$, then $xy = a^2 b^2 = (ab)^2$, where $ab \in N$.

So, $xy \in A$.

But $\frac{x}{y} = \frac{a^2}{b^2} = \left(\frac{a}{b}\right)^2 \notin A$ as $\frac{a}{b} \notin N$.

52. *(d)* Since, a ten-digit number is divisible by 4 as well as by 5, then this number must be divisible by 20.

We known that any number is divisible by 20, if last two digits is divisible by 20. It means unit place will be zero and ten's place may be 0, 2, 4, 6 or 8.

53. *(c)* As, $x < 0 < y$ [given]

$\Rightarrow \quad x < 0$ and $y > 0$

$\therefore \quad \frac{1}{x} < 0$ and $\frac{1}{y} > 0 \;\therefore\; \frac{1}{x} < \frac{1}{y}$

54. *(d)* On division of $(19)^n$ by 20, we get remainder either 19 or 1.

Since, last digit of $(19)^{100}$ is 1.

$\therefore$ Remainder of $\frac{(19)^{100}}{20}$ is 1.

55. *(b)* We know that $(x^n - y^n)$ is divisible by $(x - y)$ for all 'n' and is divisible by $(x + y)$ for even 'n'.

$\therefore (7^{12} - 4^{12})$ is divisible by $(7 + 4) = 11$ and $(7 - 4) = 3$

Thus, $(7^{12} - 4^{12})$ is divisible by 33.

56. *(c)* Here, $2^p + 3^q = 17 \qquad \ldots(\text{i})$

$2^{p+2} - 3^{q+1} = 5$

or $\quad 4 \cdot 2^p - 3 \cdot 3^q = 5 \qquad \ldots(\text{ii})$

On multiplying Eq. (i), by 3 and adding it with Eq. (ii), we get

$7 \cdot 2^p = 56,\; 2^p = 8 = 2^3, \Rightarrow p = 3$

Put $p = 3$ in Eq. (i), we get

$2^3 + 3^q = 17,\; 3^q = 17 - 8 = 9 = 3^2$

$\therefore \qquad q = 2$

57. *(c)* $\frac{49}{15} = 3 + \cfrac{1}{x + \cfrac{1}{y + \cfrac{1}{z}}}$

$\Rightarrow \quad \cfrac{1}{x + \cfrac{1}{\cfrac{yz + 1}{z}}} = \frac{49}{15} - 3 = \frac{4}{15}$

$\Rightarrow \quad \cfrac{1}{x + \cfrac{z}{yz + 1}} = \frac{4}{15}$

$\Rightarrow \quad \frac{yz + 1}{xyz + x + z} = \frac{4}{15}$

$\Rightarrow \quad 15(yz + 1) = 4(xyz + x + z)$

$\Rightarrow \quad 15yz + 15 = 4xyz + 4x + 4z$

$\Rightarrow \quad 15 - 4x = (4xy + 4 - 15y)z$

$z = \frac{15 - 4x}{(4xy + 4 - 15y)}$

As x, y and z are natural numbers.

Again, let $x = 3$ and $y = 1$

Then, $z = \frac{15 - 4(3)}{4(3)(1) + 4 - 15(1)}$
$= \frac{15 - 12}{12 + 4 - 15} = \frac{3}{1}$

$\Rightarrow \quad z = 3$, which is a natural number.

58. *(a)* Required number is

$x^2 = 540 \times q = 3 \times 3 \times 3 \times 2 \times 2 \times 5 \times q$

In order to make x^2 a perfect square, the least number we must have to put is $q = 3 \times 5 = 15$

$\therefore \quad x^2 = 540 \times 15 = 8100$

59. *(d)* We have, $1 + A + A^2 + A^3 = 40$

$\Rightarrow \qquad (1 + A) + A^2(1 + A) = 40$

$\Rightarrow \qquad (1 + A)(1 + A^2) = 40$

Only $A = 3$ satisfies the above equation, so, $A = 3$.

60. *(a)* We have, $1080 = 2^3 \times 3^3 \times 5$

For any perfect square, all the powers of the primes have to be even numbers.

So, if the factor is of the form

$2^a \times 3^b \times 5^c$

The values a can be 0 and 2, b can be 0 and 2 and c can take the value 0.

Totally, there are 4 possibilities.

61. *(b)* Clearly, both statements satisfies divisibility rule of 11.

62. *(a)* As, we know that a number $a_1 a_2 a_3 a_4 a_5$ is divisible by 9, if sum of the digits, i.e. $a_1 + a_2 + a_3 + a_4 + a_5$ is divisible by 9. Hence, only statement I is true.

63. *(c)* Taking $p = 11$

$p + 2 = 13$ [a prime number]

I. $11 \times 13 + 1 = 144$ [a square number]

II. $11 + 13 = 24$ [12 is a divisor of 24]

Hence, both statements I and II are correct.

64. *(c)* I. If $x = 15$ and $y = 14$, then $x + y = 15 + 14 = 29$, which is a prime number. So, if x and y are composite, then $x + y$ is not always composite.

II. If $x = 15$ and $y = 14$, then $x - y = 15 - 14 = 1$ which is neither prime nor composite, hence again $x - y$ is not always composite.

III. Third condition is satisfied for all measure.

Hence, only III is correct.

65. *(c)* I. The product of any three consecutive integers is divisible by 6.

II. Here, $3k = \{\ldots -6, -3, 0, 3, 6, \ldots\}$

$3k + 1 = \{\ldots -5, -2, 1, 4, 7, \ldots\}$

and $3k + 2 = \{\ldots -4, -1, 2, 5, 8, \ldots\}$

$\therefore \quad \{3k, 3k + 1, 3k + 2\}$

$= \{\ldots -6, -5, -4, -3, -2, -1$

$0, 1, 2, 3, 4, 5, 6 \ldots\}$

Hence, it is true.

66. *(d)* All statements are true.

67. *(a)* (i) If b / a and c / b, then $a = bx$ and

$b = cy$ for $x, y \in N$

$\Rightarrow \quad a = bx = cy\,(x) = cxy \Rightarrow c / a$

(ii) If a / b and a / c, then $b = ax$ and $c = ay$ for $xy \in N$,

$\Rightarrow \quad (b + c) = ax + ay = a\,(x + y)$

So, $a / (b + c)$

(iii) From (ii), we have $(b - c) = a(x - y)$

$\Rightarrow a / (b - c)$

68. *(d)* All are true.

69. *(c)* Given,

$$\begin{array}{l} 33\,)\,231228\,(\,7006 \\ \quad 231 \\ \quad \overline{228} \\ \quad 198 \\ \quad \overline{30} \end{array}$$

Now, $33 - 30 = 3$

So, on adding 3 to 231228, it will be completely divisible by 33.

70. *(b)* Given, $N = 143k + 28$

$\Rightarrow N = 143k + 26 + 2$

$\Rightarrow N = 13\,(11k + 2) + 2$

$\therefore$ When the number is divided by 13 the remainder is 2.

71. *(a)* 1 is neither prime number nor composite number.

72. *(b)* Here, $88 = 2 \times 2 \times 2 \times 11 = (2)^3 \times (11)^1$

$91 = (7)^1 \times (13)^1$

$96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$

$= (2)^5 \times (3)^1$

and $\quad 99 = 3 \times 3 \times 11 = (3)^2 \times (11)^1$

So, 91 has least number of divisors.

73. *(c)* Following are the numbers between -11 and 11 which are multiples of 2 or 3.

$-10, -9, -8, -6, -4, -3, -2,$

$0, 2, 3, 4, 6, 8, 9, 10$

So, the numbers of multiples 2 or 3, between -11 and 11 are 15.

74. *(b)* I. Given, n is a prime number greater than 5.

Now, $n^4 - 1 = (n^2 - 1)(n^2 + 1)$

$= (n - 1)(n + 1)(n^2 + 1)$

Put $n = 11$,

$n^4 - 1 = (11 - 1)(11 + 1)(121 + 1)$

[prime number greater than 5]

$= 10 \times 12 \times 122$

$= 14640$ which is not divisible by 2400.

So, statement I is not true.

II. Every square number can be of the form $5n$ or $(5n \pm 1)$ or $(5n \pm 4)$.

So, statement II is true.

75. *(a)* When N is a natural number, then there is only one possible case that N, $(N + 2)$, $(N + 4)$ are prime numbers.

When $N = 3$, then N, $(N + 2)$, $(N + 4)$

$= 3, 5, 7$ all are primes numbers.

76. *(c)* From options,

(c) $\dfrac{32}{(3 + 2)} = \dfrac{32}{5} = 2$ (remainder)

and $\dfrac{32}{(3 \times 2)} = \dfrac{32}{6} = 2$ (remainder)

$\therefore$ Remainder are same.

and difference of quotients $= 6 - 5 = 1$

77. *(c)* If x is a positive even integer and y is negative odd integer, then x^y is a rational number.

78. *(b)* We know that, between any two rational numbers, there are an infinite number of rational and irrational numbers.

Hence, only statement II is correct.

79. *(d)* If k is a positive integer, then every square integer is of the form $4k$ or $4k + 1$, as every square number is either a multiple of 4 or exceeds multiple of 4 by unity.

80. *(c)* Since, b is largest square divisor of c.

So, $\quad c = bx$

[where, x is not a whole square number]

Also, a^2 divides c.

So, a^2 will divide bx or a will divide b.

[since, it cannot divide x as it is not a whole square]

81. *(b)* Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ only, when k is even.

82. *(c)* I. In 7710312401, difference between sum of even place digits and the sum of odd place digits 0. So, it is divisible by 11.

II. Since, $173 < (14)^2$ and it is not divisible by 2, 3, 5, 7, 11 and 13. So, it is a prime number.

Hence, both statements I and II are correct.

83. *(c)* Both the statements given are correct. As 121 is the square of 11. So, to obtain prime numbers less than 121, we reject all the multiples of prime numbers less than 11 i.e. 2, 3, 5 and 7. Similarly, every composite number less than 121 is divisible by a prime number less than 11 i.e. 2, 3, 5 or 7.

84. *(a)* I. $f(k) = 4k + 3$

For $k = 1$, $f(1) = 4 \times 1 + 3 = 7$

For $k = 2$, $f(2) = 4 \times 2 + 3 = 11$

For $k = 3$, $f(3) = 4 \times 3 + 3 = 15$

Values of $f(k)$ for $k = 1, 2, \ldots$ cannot be expressed as sum of two squares, since $1^2 + 2^2 = 5$, $1^2 + 3^2 = 10$,

$2^2 + 3^2 = 13$.

II. $f(k) = 8k + 1$

For $k = 1$, $f(1) = (8 \times 1) + 1 = 9$

For $k = 2$, $f(2) = (8 \times 2) + 1 = 17$

For $k = 3$, $f(3) = (8 \times 3) + 1 = 25$

For $k = 4$, $f(4) = (8 \times 4) + 1 = 33$

For $k = 5$, $f(5) = (8 \times 5) + 1 = 41$

$f(k) = 8k + 1$ is square of an odd integer only for some values of k.

So, only statement I is correct.

85. *(d)* To check which option is equal to $26^2 + 97^2$, we take the sum of unit digit's square of both number of the question as well as the answer options. Whichever answer option shows the same result will be the answer.

Here, in $26^2 + 97^2$, $6^2 = 36$

and $7^2 = 49$

So, $\quad 36 + 49 = 85$

For option (a), $7^2 + 3^2 = 49 + 9 = 58$

For option (b), $4^2 + 3^2 = 16 + 9 = 25$

For option (c), $2^2 + 1^2 = 4 + 1 = 5$

For option (d), $9^2 + 2^2 = 81 + 4 = 85$

Only option (d) satisfies the condition.

86. *(a)* If n is greater than 1, then $n^2(n^2 - 1)$ is always divisible by 12.

Illustration 1 Put $n = 2$, then

$n^2(n^2-1)=(2)^2(2^2-1)=4\times3=12$

Illustration 2 Put $n=3$, then

$n^2(n^2-1)=(3)^2(3^2-1)=9\times8=72$

87. *(a)* We have, $\frac{17^{23}+29^{23}}{23}+\frac{23^{23}}{23}$

$(17^{23}+29^{23})$ is divisible by $(17+29)$ i.e. 46

So, it is divisible by 23.

Also, 23^{23} is always divisible by 23.

$\therefore$ Remainder $=0$

88. *(b)* Let $E=(1235\times4523\times2451)$

$=(12\times102+11)(12\times376+11)\times(12\times204+3)$

When we divide E by 12, then

Remainder = Remainder when

$11\times11\times3$ or 363 is divided by $12=3$

89. *(c)* $\because$ $360=2^3\times3^2\times5^1$

$\therefore$ Number of divisors

$=(3+1)(2+1)(1+1)$
$=4\times3\times2=24$

90. *(a)* Given, three-digit number $=XY5$

and $\frac{XY5}{X215}=Z$

Here, Z can take values 1, 3, 5, 7 and 9.

But only 9 satisfies it,

$\because$ $\begin{array}{r}135\\ \times9\\ \hline 1215\end{array}$

then $X=1, Y=3$ and $Z=9$

Now, $X+Y+Z=1+3+9=13$

91. *(d)* Given number is $763X4Y2$.

Since, given number is divisible by 9.

$\therefore$ $7+6+3+X+4+Y+2=9k$

$\Rightarrow$ $22+X+Y=9k$

It is clear that LHS is divisible by 9, if $X+Y=5,14$

When sum of X and Y is 5, then possible pairs are (1, 4), (4, 1), (2, 3), (3, 2), (0, 5), (5, 0). When sum of X and Y is 14, then possible pairs are (5, 9), (9, 5), (6, 8), (8, 6) and (7, 7).

Hence, the possible pairs are 11.

92. *(b)* Given, $a_n=3-4n$

$\therefore$ $\Sigma a_n=\Sigma(3-4n)$

$=3n-4\frac{[n\times(n+1)]}{2}$

$=3n-2n^2-2n$

$=n-2n^2=-n(2n-1)$

93. *(a)* Let the two-digit numbers less than, 50 which when divided by 4 yield unity as remainder be 13, 17, ... , 49.

Here, first term, $a=13$, common difference, $d=4$ and $n=10$

$\therefore$ Required sum $=\frac{n}{2}[2a+(n-1)d]$

$=\frac{10}{2}[2\times13+(10-1)4]$

$=\frac{10}{2}[26+36]=\frac{10\times62}{2}=310$

94. *(b)* The prime numbers less than 13 are 2, 3, 5, 7, 11.

Also, $p<q<r<13$

and $p+q+r$ is a prime number.

Hence, only two possible pairs exist i.e. (3, 5, 11) and (5, 7, 11).

95. *(a)* Product of unit digits

$=1\times2\times3\times4\times5\times6\times7\times8\times9\times1\times2\times3\times........\times8\times9$

$=0$

$\therefore$ Required digit in the unit place is 0.

96. *(c)* See question 88.

97. *(d)* See example 8.

98. *(c)* $N=35\times45\times55\times60\times124\times75$

$=5\times7\times5\times9\times5\times11\times5\times12\times124\times5^2\times3$

$=5^6\times7\times9\times11\times12\times124\times3$

Hence, the maximum value of m is 6.

99. *(b)* Given $a-b=4$ ...(i)

On squaring both sides, we get

$(a-b)^2=(4)^2\Rightarrow a^2+b^2-2ab=16$

$\Rightarrow 40-2ab=16$ $[\because a^2+b^2=40]$

$\Rightarrow$ $2ab=24$ $\Rightarrow$ $ab=12$

$\because$ $a+b=\sqrt{a^2+b^2+2ab}$

$=\sqrt{40+2\times12}=8$

$\therefore$ $a+b=8$ ...(ii)

On solving Eq. (i) and Eq. (ii), we get $a=6$ and $b=2$

Now, $a^3+b^6=6^3+2^6=2^3\times3^3+2^6$

$=2^3(3^3+2^3)$

$=8(27+8)=8\times35=280$

Hence, the value of a^3+b^6 is 280.

100. *(c)* If factor of given number is of the form of $p_1^{\alpha_1}, p_2^{\alpha_2}\dots p_n^{\alpha_n}$, then number of prime factors are

$(\alpha_1+1)(\alpha_2+1)\dots(\alpha_n+1)$

Hence, the prime factor of n are

$(x_1+1)(x_2+1)$ and (x_3+1)

101. *(b)* See question 57.

102. *(a)* Let x be the required number.

$\therefore$ $52x-25x=324\Rightarrow 27x=324$

$\Rightarrow$ $x=\frac{324}{27}=12$

Hence, the required number is 12.

103. *(a)* $21x+48y=5\Rightarrow 3(7x+16y)=5$

If x, y are integer, then LHS of the above equation is multiple of 3, but the RHS of above equation is not multiple of 3.

$\therefore$ There is no any integral values of x and y exist.

104. *(c)* Let $x=(n^3-n)(n-2)$, where $n>2$

Take $n=3$, we get

$x=(3^3-3)(3-2)=(27-3)(1)=24$

[which is divisible by 6, 12 and 24]

Take $n=4$, we get $x=(4^3-4)(4-2)$

$=(64-4)\times2=120$

[which is again divisible by 6, 12 and 24]

Now, take $n=5$, we get

$x=(5^3-5)(5-2)=(125-5)\times3$

$=120\times3=360$

[which is again divisible by 6, 12 and 24]

Hence, 24 is the largest natural number.

105. *(d)* We have, $(234)^{100}+(234)^{101}$

$\Rightarrow$ $(234)^{100}(1+234)=235(234)^{100}$

We know that square of any number having 4 at unit place is a number in which 6 at unit place.

Any exponent of a number 6 at unit place is always 6 at unit place.

$\because$ $(235)(\dots6)=\dots30$

Resulting number have 0 at unit place.

106. *(a)* Let the number be y.

$\therefore$ $y=7q+3$ and $q=11p+6$

$\therefore$ $y=7(11p+6)+3$

$\Rightarrow$ $y=77p+45$

When divided by 11 remainder is 1 and quotient is $7p+4$ and when the quotient is divided by 7 remainder is 4.

107. *(c)* Seven digits number $876p37q$ is divisible by 225, if this number is divisible by 9 and 5.

If this number is divisible by 9.

Then, sum of its digits is divisible by 9.

Now, sum of digits

$=8+7+6+p+3+7+q$

$=31+p+q$

$\therefore p+q=5$ or $p+q=14$

$\therefore$ Given number is divisible by 225 when $q=5$

if $q=5$, $p=0$ or 9.

108. *(a)* We have, when $3x+2y$ is divided by 5, remainder is 2.

$\therefore$ $3x+2y=5q+2$ ...(i)

and when $2x+3y$ is divided by 5, the remainder is 3.

$\therefore \quad 2x + 3y = 5m + 3 \quad \ldots(ii)$

Subtract Eq. (ii) from Eq.(i), we get

$$x - y = 5(q - m) - 1$$
$$x - y = 5(q - m) - 5 + 4$$
$$x - y = 5(q - m - 1) + 4$$

$\therefore$ $x - y$ is divided by 5 remainder is 4.

109. *(b)* The sum of first 47 terms of the series

$$\frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} \ldots 47 \text{ term}$$

It is clear that sum of first 6 term is zero.

Similarly, sum of first 42 terms is zero.

Sum of last 5 terms

$$= \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} = \frac{-1}{6}$$

110. *(c)* $\dfrac{(243 + 647)^2 + (243 - 647)^2}{243 \times 243 + 647 \times 647}$

$$= \frac{2[(243)^2 + (647)^2]}{(243)^2 + (647)^2} = 2$$

$$[\because (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)]$$

111. *(b)* We have, $N = 90 \times 42 \times 324 \times 55$

$$= 3^2 \times 10 \times 3 \times 14 \times 3^4 \times 4 \times 55$$
$$= 3^7 \times 10 \times 14 \times 4 \times 55$$

Hence, N is divisible by 3^7.

So, the maximum value of m is 7 when N is divisible by 3^m.

112. *(d)* $7^{10} - 5^{10} = (7^5)^2 - (5^5)^2$

$$= (7^5 + 5^5)(7^5 - 5^5)$$
$$= (16807 + 3125)(7^5 - 5^5)$$
$$= 19932 \times (7^5 - 5^5)$$

Since, 19932 is divisible by 11.

Hence, $7^{10} - 5^{10}$ is divisible by 11.

113. *(a)* We have, $a - b = 3$

Squaring both sides, we get

$$a^2 + b^2 - 2ab = 9 \quad \ldots(i)$$

Also, $\quad a^3 - b^3 = 117$

$\Rightarrow \quad (a-b)(a^2 + b^2 + ab) = 117$

$\Rightarrow \quad a^2 + b^2 + ab = \dfrac{117}{3} = 39 \quad \ldots(ii)$

On subtracting Eq. (i) from Eq. (ii)

$$3ab = 30 \quad \Rightarrow ab = 10$$

Now, $a + b = \sqrt{(a-b)^2 + 4ab}$

$$= \sqrt{9 + 40} = \sqrt{49} = 7$$

114. *(a)* If R is the remainder obtained by dividing the polynomial $x^n + m^n$ by $(x - m)$, then $(x^n + m^n - R)$ is divisible by $(x - m)$.

Let $f(x) = x^n + m^n - R$

$\because$ $f(x)$ is divisible by $(x - m)$.

$\therefore \quad f(m) = 0$

$\Rightarrow \quad (m)^n + (m)^n - R = 0$

$$R = 2(m)^n$$

As, m is a non-zero integer and n is a positive integer, then R is a non-zero even integer.

115. *(c)* See question 57.

116. *(d)* x is prime and y is composite number.

Since, x is prime number.

$\therefore$ x may be {2, 3, 5, 7, 11, …} and y may be {4, 6, 8, 9, 10, 12, 14, 15, …}

Here, y may be even or odd number and x is odd number except 2.

$\therefore$ $(y - x)$ can be even number.

Also, xy can be even number.

If y is odd integer and $x \geq 3$.

Then, $0.5(x + y)$ is an even integer.

Hence, statements I, II and III are correct.

117. *(b)* A real number is collection of all rational and irrational numbers.

So, statement II is false.

118. *(a)* Given, p and q are non-zero integers.

I. $\because \quad p + q < p - q$

$\Rightarrow \quad q + q < 0$

$\Rightarrow \quad 2q < 0$

$\Rightarrow \quad q < 0$

$\therefore$ q must be negative.

Hence, statement I is correct.

II. $\because \quad p + q > p - q$

$\Rightarrow \quad q + q > 0$

$\therefore \quad 2q > 0$

$\therefore$ q must be positive irrespective of p.

Hence, statement II is incorrect.

119. *(d)* I. $a - b < a - c$

$\Rightarrow b - c > 0 \quad \ldots(i)$

As, b is negative real number and c is positive real number, then Eq. (i) is not true.

II. If $a < b$, when a and b are negative real numbers and c is a positive real number, then $\dfrac{a}{c} < \dfrac{b}{c}$ is always true for $b > a$.

III. $\dfrac{1}{b} < \dfrac{1}{c}$ is always true, as c is a positive real number and b is a negative real number.

120. *(d)* I. If m and n are distinct natural numbers, then $\dfrac{m}{n} + \dfrac{n}{m}$ is integer if and only if $m = n$. Hence statement I is incorrect.

II. $mn\left(\dfrac{m}{n} + \dfrac{n}{m}\right)(m^2 + n^2)^{-1}$

$$= mn\left(\frac{m^2 + n^2}{mn}\right)\left(\frac{1}{m^2 + n^2}\right) = 1$$

Hence, statement II is correct for all values of m and n.

III. Now, $\dfrac{mn}{m^2 + n^2}$ is a fraction.

So, statement III is incorrect.

02

SEQUENCE AND SERIES

Usually (1-2) questions have been asked from this chapter. Questions are mostly based on relation between arithmatic geometric and harmonic mean.

SEQUENCE

A set of numbers arranged in a definite order according to some definite rule is called a sequence.
e.g. 2, 4, 6, 8, ... is a sequence.

SERIES

If $a_1, a_2, a_3, \ldots a_n$ is a sequence, then the expression $a_1 + a_2 + a_3 + \ldots + a_n$ is called the series.
The series is said to be finite or infinite depending upon the last term is given or not. e.g.

(i) $a_1, a_2, a_3, \ldots, a_n$ is a finite sequence and is denoted as $\{a_k\}_{k=1}^{n}$.

(ii) $a_1, a_2, a_3, a_4, \ldots$ is an infinite sequence and is denoted by $\{a_n\}_{n=1}^{\infty}$ or simply $\{a_n\}$.

Here, a_1 is called the first term and in case of finite sequence a_n is called the last term.
e.g. 2, 4, 6, 8, … , 100 is a finite sequence.
1, 2, 3, 4, … is an infinite sequence.

PROGRESSION

Sequence following certain patterns are called progressions.
e.g. 2, 3, 4, 5, … is a progression, here each term is increasing by 1.

Arithmetical Progression (AP)

An arithmetic progression is a sequence in which the difference between any term and its preceding term is constant throughout. The constant '*d*' is called the common difference. The first term of an AP is represented by '*a*'.

If an AP has first term = a and common difference = d, then the general form of an AP is

$$a, a+d, a+2d, a+3d, \ldots, a+(n-1)d$$

- nth term of an AP is $T_n = a+(n-1)d$.
- Sum of the first n terms of an AP is

$$S_n = \frac{n}{2}[2a+(n-1)d] \text{ or } S_n = \frac{n}{2}[a+l]$$

where, l = last term

EXAMPLE 1. Find the sum of 11 terms of $-7, -2, 3, 8 \ldots$.

a. 200 b. 198 c. 326 d. 137

Sol. d. Here, $a=-7$, $l=-7+(10\times 5)=43$

$$\therefore \quad S_{11} = n\left(\frac{a+l}{2}\right) = 11\times\left(\frac{-7+43}{2}\right)$$

$$= 11\times\frac{36}{2} = 198$$

Arithmetic Mean (AM)

When three terms are in AP, then the middle term is called arithmetic mean of the other two.

If a, b and c are in AP, then b is AM of a and c.

Let a and b be two numbers and M be the AM between them. Then, a, M, b are in AP, $M = \frac{a+b}{2}$.

EXAMPLE 2. Find the arithmetic mean (AM) between 3 and 9.

a. 4 b. 6
c. 8 d. None of these

Sol. b. The arithmetic mean is $\frac{3+9}{2} = \frac{12}{2} = 6$.

So, 3, 6, 9 are in AP.

Geometric Progression

A geometric progression is a progression of numbers, whose first term is non-zero and each of the term is obtained by multiplying its preceding term by a constant quantity. This constant quantity is called the common ratio of the GP.

Thus, if t_1, t_2 and t_3 are in GP, then common ratio

$$r = \frac{\text{Second term}}{\text{First term}} = \frac{t_2}{t_1}$$

If 'a' is the first term and 'r' is the common ratio, then GP can be written as $a, ar, ar^2, ar^3, \ldots, ar^{n-1} = (a \neq 0)$

- n th term of a GP is $T_n = ar^{n-1} = l$.

[where, l = last term]

- Sum of the n terms of a GP is $S_n = \frac{a(r^n-1)}{r-1}$, if $r>1$ and $S_n = \frac{a(1-r^n)}{1-r}$, if $r<1$.
- Sum of infinite terms of a GP is $S_n = \frac{a}{1-r}$.

Geometric Mean (GM)

If three terms are in GP, then the middle term is called the geometric mean of the other two. If a, b and c are in GP, then b is the GM of a and c.

Let a and b be two numbers and G be the GM between them. Then, a, G, b are in GP, $G=\sqrt{ab}$, $a>0$, $b>0$.

EXAMPLE 3. If the 4th, 10th and 16th terms of a GP are x, y and z respectively, then x, y, z are in

a. AP b. GP
c. AGP d. HP

Sol. b. Let the first term and common ratio be a and r, respectively.

Given, $T_4 = x \Rightarrow ar^{4-1} = x \Rightarrow ar^3 = x$ …(i)

$T_{10} = y \Rightarrow ar^{10-1} = y \Rightarrow ar^9 = y$ …(ii)

$T_{16} = z \Rightarrow ar^{16-1} = z \Rightarrow ar^{15} = z$ …(iii)

On multiplying Eq. (i), by Eq. (iii), we get

$$ar^3 \times ar^{15} = xz \Rightarrow a^2r^{18} = xz$$

$$\Rightarrow \quad (ar^9)^2 = xz \Rightarrow y^2 = xz \quad \text{[from Eq. (i)]}$$

Hence, x, y and z are in GP.

Harmonic Progression (HP)

A sequence is said to be harmonic progression (HP). If the reciprocals of its terms are in arithmetic progression (AP).

e.g. The sequence, $1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7} \ldots$ is an HP because the sequence 1, 3, 5, 7 … is an AP.

If $a_1, a_2, a_3, \ldots, a_n$ are in HP, then $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots, \frac{1}{a_n}$ are in AP.

- nth term of an HP is $T_n = \dfrac{1}{\frac{1}{a_1}+(n-1)\left(\frac{1}{a_2}-\frac{1}{a_1}\right)}$

Harmonic Mean (HM)

If three terms are in HP, then the middle term is called the harmonic mean of the other two. If a, b and c are in HM, then b is the HM of a and c.

Let a and b be two numbers and H be the HM between them.

Then, a, b and c are in HP

$\therefore \quad H = \dfrac{2ab}{a+b}$

Relation between Arithmetic, Geometric and Harmonic Mean

Let A, G and H be the arithmetic, geometric and harmonic means between a and b, then

(i) $A \geq G \geq H$ (ii) $G^2 = AH$

EXAMPLE 4. If a, b, c are in AP, p, q, r are in HP and ap, bq, cr are in GP, then $\dfrac{p}{r} + \dfrac{r}{p}$ is equal to

a. $\dfrac{a}{c} - \dfrac{c}{a}$ b. $\dfrac{a}{c} + \dfrac{c}{a}$
d. $\dfrac{b}{q} + \dfrac{q}{b}$ d. $\dfrac{b}{q} - \dfrac{q}{b}$

Sol. **b.** Since, a, b, c are in AP $\Rightarrow \quad b = \dfrac{a+c}{2}$ ...(i)

p, q, r are in HP $\Rightarrow \quad q = \dfrac{2pr}{p+r}$ ...(ii)

and ap, bq, cr are in GP, $b^2q^2 = apcr$

$\Rightarrow \quad \left(\dfrac{a+c}{2}\right)^2 \left(\dfrac{4p^2r^2}{(p+r)^2}\right) = apcr$ [from Eqs. (i) and (ii)]

$\Rightarrow \quad \dfrac{(a+c)^2 pr}{(p+r)^2} = ac$

$\Rightarrow \quad \dfrac{(a+c)^2}{ac} = \dfrac{(p+r)^2}{pr}$

$\Rightarrow \quad \dfrac{p^2}{pr} + \dfrac{r^2}{pr} + 2 = \dfrac{a^2}{ac} + \dfrac{c^2}{ac} + 2 \Rightarrow \dfrac{p}{r} + \dfrac{r}{p} = \dfrac{a}{c} + \dfrac{c}{a}$

Sum to n Terms of Special Series

The sum of first n terms of some special series is given below

1. The sum of first n natural numbers
$$= \sum n = 1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$$
2. The sum of square of the first n natural numbers
$$= \sum n^2 = 1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$$
3. The sum of cubes of the first n natural numbers
$$= \sum n^3 = 1^3 + 2^3 + \ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

PRACTICE EXERCISE

1. If $\dfrac{3+5+7+\cdots+n}{5+8+11+\cdots+10 \text{ terms}} = 7$, then the value of n is

(a) 35 (b) 36 (c) 37 (d) 40

2. If sum of n terms of an AP is $3n^2 + 5n$ and $T_m = 164$, then m is equal to

(a) 26 (b) 27 (c) 28 (d) None of these

3. In a GP, if the $(m+n)$th term be p and $(m-n)$th term be q, then its mth term is

(a) $\sqrt{pq}$ (b) $\sqrt{p/q}$ (c) $\sqrt{q/p}$ (d) $\sqrt{p+q}$

4. The sum of the first 'n' terms of the series $\dfrac{1}{2} + \dfrac{3}{4} + \dfrac{7}{8} + \dfrac{15}{16} + \cdots$ is

(a) $2^n - n - 1$ (b) $1 - 2^{-n}$
(c) $n + 2^{-n} - 1$ (d) $2^n - 1$

5. An AP consists of n (odd terms) and its middle term is m. Then, the sum of the AP is

(a) $2mn$ (b) $\dfrac{1}{2}mn$ (c) mn (d) mn^2

6. The sum of $1 + \dfrac{2}{5} + \dfrac{3}{5^2} + \dfrac{4}{5^3} + \cdots \infty$ upto n terms is

(a) $\dfrac{25}{16}$ (b) $\dfrac{15}{16}$
(c) $\dfrac{5}{16}$ (d) $\dfrac{3}{2}$

7. The sum of n terms of an AP is $an(n-1)$. The sum of the squares of these terms is equal to

(a) $a^2n^2(n-1)^2$ (b) $\dfrac{a^2}{6}n(n-1)(2n-1)$
(c) $\dfrac{2a^2}{3}n(n-1)(2n-1)$ (d) $\dfrac{2a^2}{3}n(n+1)(2n+1)$

8. If S be the sum to infinity of a GP, whose first term is a, then the sum of first n terms is

(a) $S\left(1-\frac{a}{S}\right)^n$

(b) $S\left[1-\left(1-\frac{a}{S}\right)^n\right]$

(c) $a\left[1-\left(1-\frac{a}{S}\right)^n\right]$

(d) None of the above

9. If the non-zero numbers a, b, c are in AP and $\tan^{-1} a, \tan^{-1} b, \tan^{-1} c$ are also in AP, then

(a) $a=b=c$ (b) $b^2=2ac$
(c) $a^2=bc$ (d) $c^2=ab$

10. $\frac{1}{b-a}+\frac{1}{b-c}=\frac{1}{a}+\frac{1}{c}$, then a, b, c are in

(a) AP (b) GP
(c) HP (d) None of these

11. The value of $x+y+z$ is 15, if a, x, y, z, b are in AP while the value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is $\frac{5}{3}$, if a, x, y, z, b are in HP. Then, a and b are

(a) 1, 9 (b) 3, 7
(c) 7, 3 (d) None of these

12. If the mth and nth term of a HP are n and m respectively, then the mnth term is

(a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$

13. If $a, 2a+2, 3a+3$ are in GP, then what is the fourth term of the GP?

(a) -13.5 (b) 13.5 (c) -27 (d) 27

14. If the sum of first 'n' natural numbers is $\frac{n(n+1)}{2}$. Then, what will be the sum of first 'n' terms of the series of alternate positive and negative numbers when 'n' is even?

$$1^2-2^2+3^2-4^2+5^2-\ldots$$

I. $\frac{n(n+1)}{2}$ II. $\frac{n^2(n+1)}{2}$ III. $\frac{-n(n+1)}{2}$

Which of the above statement(s) is/are correct?

(a) Only I (b) Only III (c) Only II (d) None of these

PREVIOUS YEARS QUESTIONS

15. If A, G and H are the arithmetic, geometric and harmonic means between a and b respectively, then which one of the following relations is correct? **2015 I**

(a) G is the geometric mean between A and H
(b) A is the arithmetic mean between G and H
(c) H is the harmonic mean between A and G
(d) None of the above

16. Consider the following statements in respect of the expression $S_n=\frac{n(n+1)}{2}$, where 'n' is an integer.

I. There are exactly two values of n for which $S_n=861$.

II. $S_n=S_{-(n+1)}$ and hence for any integer m we have two values of n for which $S_n=m$.

Which of these statement(s) is/are correct? **2016 (I)**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16									

HINTS AND SOLUTIONS

1. (a) $\because \dfrac{\frac{n}{2}[2\times3+(n-1)2]}{\frac{10}{2}[2\times5+(10-1)\times3]}=7$

$\Rightarrow \dfrac{n(n+2)}{5\times37}=7$

$\Rightarrow n^2+2n-1295=0$

$\Rightarrow n^2+37n-35n-1295=0$

$\Rightarrow (n+37)(n-35)=0$

$\therefore n=35$

2. (b) $\because T_m=S_m-S_{m-1}$

$\Rightarrow 164=3(2m-1)+5\cdot1$

$\Rightarrow 6m=162$

$\therefore m=27$

3. (a) $T_{m+n}=ar^{m+n-1}=p$,

$T_{m-n}=ar^{m-n-1}=q$

On multiplying, we get $a^2r^{2m-2}=pq$

$\therefore T_m=ar^{m-1}=\sqrt{pq}$

4. (c) $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\frac{15}{16}+\ldots+n$ terms

$=\left(1-\frac{1}{2}\right)+\left(1-\frac{1}{4}\right)+\left(1-\frac{1}{8}\right)+\ldots+\left(1-\frac{1}{2^n}\right)$

$=n-\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\ldots+\frac{1}{2^n}\right)$

$=n-\frac{1}{2}\left(\frac{1-(1/2)^n}{1-1/2}\right)=n+2^{-n}-1$

5. (*c*) Middle term $= T_{\frac{n+1}{2}}$

$\therefore \quad a + \left(\dfrac{n+1}{2} - 1\right) d = m$ [given]

$2a + (n-1)\,d = 2m$ ...(i)

Now, $S_n = \dfrac{n}{2}[2a + (n-1)\,d] = nm$

6. (*a*) The given sequence is arithmetic geometric series, where $r = \dfrac{1}{5}$ and $d = 1$

$$S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

$$= \frac{1}{1-\frac{1}{5}} + \frac{1 \times \frac{1}{5}}{\left(1-\frac{1}{5}\right)^2}$$

$$= \frac{5}{4} + \frac{5}{16} = \frac{25}{16}$$

7. (*c*) Let $S_n = an(n-1)$, then

$S_{n-1} = a(n-1)(n-2)$

$\therefore \quad T_n = S_n - S_{n-1} = 2a(n-1)$

$T_n^2 = 4a^2(n-1)^2$

$\therefore$ Sum $= \Sigma T_n^2 = 4a^2 \dfrac{(n-1)(n)(2n-1)}{6}$

$= \dfrac{2a^2 n(n-1)(2n-1)}{3}$

8. (*b*) Let r be the common ratio of GP, then

$S = \dfrac{a}{1-r}, \quad r = 1 - \dfrac{a}{S}$

$\therefore \quad S_n = \dfrac{a(1-r^n)}{1-r} = \dfrac{a}{1-r}(1-r^n)$

$= S\left[1 - \left(1 - \dfrac{a}{S}\right)^n\right]$

9. (*a*) Since, $2b = a + c$ ...(i)

and $2\tan^{-1} b = \tan^{-1} a + \tan^{-1} c$

$\Rightarrow \quad \dfrac{2b}{1-b^2} = \dfrac{a+c}{1-ac}$

$\Rightarrow \quad b^2 = ac$ [from Eq. (i)]

$\Rightarrow \quad 4b^2 = 4ac$

$\Rightarrow (a+c)^2 - 4ac = 0$ [from Eq. (i)]

$\Rightarrow \quad (a-c)^2 = 0$

$\Rightarrow \quad a = c = b$

10. (*c*) Given, $\dfrac{1}{b-a} + \dfrac{1}{b-c} = \dfrac{1}{a} + \dfrac{1}{c}$

$\Rightarrow \quad \dfrac{1}{b-a} - \dfrac{1}{c} = \dfrac{1}{a} - \dfrac{1}{b-c}$

$\Rightarrow \quad \dfrac{(c-b+a)}{c(b-a)} = \dfrac{(b-c-a)}{a(b-c)}$

$\Rightarrow \quad \dfrac{1}{c(b-a)} = -\dfrac{1}{a(b-c)}$

$\Rightarrow \quad ba - ca = -cb + ac$

$\Rightarrow \quad ab + bc = 2ac$

$\therefore \quad b = \dfrac{2ac}{a+c}$

Hence, a, b, c are in HP.

11. (*a*) $\because a, x, y, z, b$ are in AP.

$\therefore x + y + z = 3\left(\dfrac{a+b}{2}\right)$

$\Rightarrow \quad 15 = 3\left(\dfrac{a+b}{2}\right)$

$\Rightarrow \quad a + b = 10$ ...(i)

Also, a, x, y, z, b are in HP.

$\Rightarrow \quad \dfrac{1}{a}, \dfrac{1}{x}, \dfrac{1}{y}, \dfrac{1}{z}, \dfrac{1}{b}$ are in AP.

$\Rightarrow \quad \dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} = 3\left(\dfrac{a+b}{2ab}\right)$

$\Rightarrow$
$\dfrac{5}{3} = \dfrac{3 \times 10}{2ab}$ [$\because a + b = 10$]

$\Rightarrow \quad ab = 9$

On solving Eqs. (i) and (ii), we get

$a = 1, b = 9$ or $b = 1, a = 9$

12. *(b)* Let a be the first term and d the common difference of corresponding AP.

So, m th and nth term of AP are $\dfrac{1}{n}$ and $\dfrac{1}{m}$.

$\therefore \quad \dfrac{1}{n} = a + (m-1)\,d$ ...(i)

and $\quad \dfrac{1}{m} = a + (n-1)\,d$ ...(ii)

On solving Eqs. (i) and (ii), we get,

$d = \dfrac{1}{mn}$ and $a = \dfrac{1}{mn}$

$\therefore mn$th term of AP

$= \dfrac{1}{mn} + (mn-1) \times \dfrac{1}{mn}$

$= \dfrac{1 + mn - 1}{mn} = 1$

$\therefore mn$th term of HP = 1

13. *(c)* $a, 2a+2, 3a+3$ are in GP.

$\Rightarrow \quad (2a+2)^2 = (3a+3)a$

$\Rightarrow \quad 4a^2 + 4 + 8a = 3a^2 + 3a$

$\Rightarrow \quad a^2 + 5a + 4 = 0$

$\Rightarrow \quad a = -1, -4$

Now, $a = -1$ does not satisfy the given series.

$\therefore \; -4, -6, -9$ are in GP.

$\therefore \quad t_4 = -4\left(\dfrac{3}{2}\right)^3 = -13.5$

14. (*b*) When 'n' is even.

Let $n = 2m$, then

$= 1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \dots$

$= (1^2 - 2^2) + (3^2 - 4^2) + (5^2 - 6^2)$
$+ \dots + (2m-1)^2 - (2m)^2$

$= (1+2)(1-1) + (3+4)(-1)$
$+ (5+6)(-1) + \dots$
$+ (2m - 1 + 2m)(-1)$

$= -(1 + 2 + 3 + 4 + \dots + 2m)$

$= \dfrac{-2m(2m+1)}{2} = \dfrac{-n(n+1)}{2}$

15. *(a)* Given, A, G and H are the arithmetic, geometric and harmonic means between a and b, respectively.

$\therefore A = \dfrac{a+b}{2}$, $G = \sqrt{ab}$ and $H = \dfrac{2ab}{a+b}$
(i) (ii) (iii)

On multiplying Eq. (i) and (iii), we get

$\therefore \quad AH = \dfrac{a+b}{2} \times \dfrac{2ab}{a+b} = ab = (\sqrt{ab})^2$

[from Eq. (ii)]

$AH = G^2$

Hence, the option (a) is correct.

16. *(a)* I. $\quad S_n = \dfrac{n(n+1)}{2} = 861$

$\Rightarrow \; n^2 + n - 861 \times 2 = 0$

$\Rightarrow (n+42)(n-41) = 0$

$\Rightarrow \; n = -42, 41$

Hence, statement I is correct.

II. Given, $S_n = S_{-(n+1)}$

If $S_n = m$, then we have two values of n if and only if m is positive integer.

Hence, statement II is incorrect.

03

HCF AND LCM OF NUMBERS

Usually (2-3) questions have been asked from this chapter. Generally questions based on LCM and HCF are related to traffic lights, racitracks, largest size of tile etc. This concept is also useful in the chapters of time and distance, time and work, pipes and cristerns etc.

FACTORS

If number 'a' divides the number b without leaving a remainder, then 'a' is said to be the factor of 'b'.

e.g. (i) 4 is a factor of 16 as $16 = 4 \times 4$ (ii) 9 is a factor of 729 as $729 = 9 \times 9 \times 9$

Prime Factors

The factors that cannot be again factorized i.e. the factors which are prime numbers are called prime factors.

e.g. (i) 9 is a factor of 729, but 3 is a prime factor of 729.

(ii) 8 and 9 are factors of 72, but 2 and 3 are prime factors of 72.

Least Common Multiple (LCM)

1. **Common Multiple** A common multiple of two or more numbers is a number which is exactly divisible (without leaving remainder) by each of them.
   e.g. 30 is a common multiple of 2, 3, 5, 6, 10 and 15 because 30 is exactly divisible by each number. Similarly, 72 is a common multiple of 2, 3, 4, 6, 9, 12 and 18.
2. **Least Common Multiple** The least common multiple of two or more given numbers is the least number which is exactly divisible by all the given numbers.
   e.g. 84, 162, 252 are the common multiples of 2, 3, 4, 7. But, 84 is the LCM of 2, 3, 4, 7.

Methods of Finding LCM

1. **Prime Factorization Method** Write down the given numbers as the product of prime factors. Then, the LCM is the product of the highest powers of all the prime factors.

EXAMPLE 1. The LCM of 30, 250, 490 is

a. 46750 b. 36750
c. 26750 d. None of these

Sol. b. Here, $30 = 2 \times 3 \times 5$

$250 = 5 \times 5 \times 2 \times 5 = 2 \times 5^3$

and $490 = 7 \times 7 \times 2 \times 5 = 2 \times 5 \times 7^2$

$\therefore$ LCM of 30, 250 and 490 $= 2 \times 5^3 \times 7^2 \times 3 = 36750$

2. **Division Method** Write the given number in a row and divide them with the common prime divisor. On division. Write the quotient in each case below the number. If any number is not divisible by the respective divisor, then write it as such in the next row. Keep on dividing the quotients until you get 1. Multiply all the divisors to get the required LCM.

EXAMPLE 2. What is the LCM of 120, 144, 160 and 180.

a. 1450 b. 1620 c. 1440 d. 1380

Sol. c.

2	120,	144,	160,	180
2	60,	72,	80,	90
2	30,	36,	40,	45
3	15,	18,	20,	45
3	5,	3,	10,	15
5	5,	1,	10,	5
2	1,	1,	2,	1
	1,	1,	1	1

$\therefore$ LCM of 120, 144, 160 and 180 $= 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 2$
$= 1440$

IMPORTANT POINTS

1. The least number which when divided by x, y and z leaving the remainders a, b and c, respectively is given by [LCM of $(x, y, z) - p$], where $p = (x - a) = (y - b) = (z - c)$.
2. The least number which when divided by x, y and z leaving the same remainder R in each case is given by [LCM of $(x, y, z) + R$].

EXAMPLE 3. What is the least number which when divided by 42, 72 and 84 leaves the remainders 25, 55 and 67, respectively?

a. 521 b. 512 c. 504 d. 487

Sol. d. Here, difference $= (42 - 25) = (72 - 55) = (84 - 67) = 17$

Now, LCM (42, 72, 84) = 504

$\therefore$ Required number = 504 − 17 = 487

EXAMPLE 4. Find the least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 but when divided by 9, leaves no remainder.

a. 1620 b. 1683
c. 1635 d. 1672

Sol. b. LCM (5, 6, 7, 8) = 840.

Here, $R = 3 \Rightarrow$ Number is of the form $840k + 3$.

Least value of k for which $(840k + 3)$ when divided by 9 leaves no remainder is 2.

$\therefore$ Required number $= 840 \times 2 + 3 = 1683$

Highest Common Factor (HCF)

1. **Common Factor** A common factor of two or more numbers is a number which divides each of them exactly.
e.g. 2 is common factor of 2, 10, 20.
2. **Highest Common Factor** The Highest Common Factor (HCF) of two or more numbers is the largest number that divides all the given numbers exactly. It is also known as Greatest Common Divisor (GCD). HCF is always a factor of LCM.
e.g. HCF of the numbers 18 and 24 is 6.

Methods of Finding HCF

1. **HCF by Prime Factorization** Write the given number as product of prime factors and then find the product of least powers of common prime factors. This product is the required HCF of given numbers.

EXAMPLE 5. The HCF of 65, 75 and 105 is

a. 4 b. 5 c. 6 d. 8

Sol. b. Here, $65 = 13 \times 5$, $75 = 5 \times 5 \times 3$

and $105 = 7 \times 3 \times 5$

$\therefore$ HCF of 65, 75 and 105 = 5

2. **HCF by Division Method** Suppose we have to find the HCF of two given numbers, divide the large number by the smaller one. Now, you will get the remainder. Divide the divisor by the remainder. Repeat this process until no remainder is left, the last divisor used in this process is the desired greatest common divisor i.e. HCF. In order to find the HCF of three numbers, then, HCF of [(HCF of any two) and (third number)] gives the HCF of given three numbers.

EXAMPLE 6. The HCF of 204, 1190 and 1445 is

a. 85 b. 15 c. 17 d. 75

Sol. c. Here,

```
1190)1445 (1
     1190
     ----
      255)1190 (4
          1020
          ----
           170) 255 (1
                170
                ---
                 85)170 (2
                    170
                    ---
                     ×
```

So, HCF of 1190 and 1445 is 85.

Now,

$$
\begin{array}{l}
85)\ 204\ (2 \\
\underline{170} \\
\quad 34)\ 85\ (2 \\
\quad \underline{68} \\
\qquad 17)\ 34\ (2 \\
\qquad \underline{34} \\
\qquad \underline{\times}
\end{array}
$$

$\therefore$ HCF of 85 and 204 is 17.

Hence, HCF of 204,1190 and 1145 is 17.

IMPORTANT POINTS

1. For integers x, y and z, if HCF $(x, y) = 1$ and HCF $(x, z) = 1$, then HCF of (x, y, z) is always 1.
2. The greatest number that will divide x, y and z leaving remainders a, b and c, respectively is given by HCF of $(x - a), (y - b), (z - c)$.
3. The greatest number that will divide x, y and z leaving the same remainder in each case is given by [HCF of $|x - y|, |y - z|, |z - x|$]

EXAMPLE 7. Find the greatest number which will divide 400, 435 and 541 leaving 9,10 and 14 as remainders respectively.

a. 19 b. 17 c. 13 d. 9 **2014 I**

Sol. b Required number = HCF of (400-9, 435-10, 541-14)
= HCF of (391, 425, 527) = 17

EXAMPLE 8. For any integers 'a' and 'b' with HCF $(a, b) = 1$, what is HCF $(a+b, a-b)$ equal to?

a. It is always 1 b. It is always 2 **2014 I**
c. Either 1 or 2 d. None of these

Sol. c. Put arbitrary values of a and b.

Illustration 1 Let $a = 9$ and $b = 8$.

$\therefore$ HCF $(8 + 9, 9 - 8) \Rightarrow$ HCF $(17, 1) = 1$

Illustration 2 Let $a = 23$ and $b = 17$.

$\therefore$ HCF $(17 + 23, 23 - 17) \Rightarrow$ HCF $(40, 6) = 2$

Hence, HCF $(a + b, a - b)$ can either be 1 or 2.

How to Calculate LCM and HCF of Fractions

The LCM and HCF of fractions can be obtained from the following formula

1. HCF of fractions $= \dfrac{\text{HCF of numerators}}{\text{LCM of denominators}}$
2. LCM of fractions $= \dfrac{\text{LCM of numerators}}{\text{HCF of denominators}}$

EXAMPLE 9. The HCF of $\frac{14}{3}, \frac{21}{9}, \frac{7}{15}$ is

a. $\frac{7}{45}$ b. $3\frac{2}{5}$ c. $\frac{7}{30}$ d. None of these

Sol. a. HCF of numerators i.e. 14, 21 and 7 is 7 and LCM of denominators i.e. 3, 9 and 15 is 45. So, HCF of given fractions $= \frac{7}{45}$.

Relation between LCM and HCF of Two Numbers

Product of two numbers = (Their HCF) $\times$ (Their LCM)

EXAMPLE 10. The LCM of two numbers is 90 times their HCF. The sum of LCM and HCF is 1456. If one of the numbers is 160, then what is the other number? **2014 II**

a. 120 b. 136 c. 144 d. 184

Sol. c. Let the HCF of two numbers be x.
LCM of two numbers be $90x$.
According to the question,

$$\text{LCM} + \text{HCF} = 1456$$
$$\Rightarrow \quad 90x + x = 1456$$
$$\Rightarrow \quad 91x = 1456$$
$$\Rightarrow \quad x = 16$$

$\therefore$ HCF of two numbers $= 16$
and LCM of two numbers $= 90 \times 16 = 1440$
We know that,

$$\text{LCM} \times \text{HCF} = \text{Product of two numbers}$$
$$\Rightarrow \quad 1440 \times 16 = 160 \times \text{Second number}$$
$$\therefore \text{Second number} = \frac{1440 \times 16}{160} = 144$$

EXAMPLE 11. What is the greatest number that divides 13850 and 17030 and leaves a remainder 17?

a. 477 b. 159 c. 107 d. 87 **2012 II**

Sol. b. Required number = HCF of $(13850 - 17), (17030 - 17)$
$=$ HCF of $(13833, 17013) = 159$

EXAMPLE 12. There are three drums with 1653 litre 2261 litre and 2527 litre of petrol. The greatest possible size of the measuring vessel with which we can measure the petrol of any drum while every time the vessel must be completely filled is

a. 31 b. 27 c. 19 d. 41

Sol. c. The maximum capacity of the vessel = HCF of 1653, 2261 and 2527 = 19

EXAMPLE 13. John, Kate and smith at same time, same point and in same direction to run around a circular in 150 seconds. Find after what time will they meet again?

a. 30 min b. 25 min c. 20 min d. 15 min

Sol. b. LCM of 250, 300 and 150 = 1500 sec
= 25 min.
Hence, john, kate and smith meet after 25 min.

PRACTICE EXERCISE

1. If $x = 2^3 \times 3^2 \times 5^4$ and $y = 2^2 \times 3^2 \times 5 \times 7$, then HCF of x and y is
(a) 180 (b) 360 (c) 540 (d) 35

2. LCM of $2^3 \times 3 \times 5$ and $2^4 \times 5 \times 7$ is
(a) $2^{12} \times 3 \times 5^2 \times 7$ (b) $2^4 \times 5 \times 7 \times 9$
(c) $2^4 \times 3 \times 5 \times 7$ (d) $2^3 \times 3 \times 5 \times 7$

3. LCM of $\frac{4}{5}, \frac{3}{10}$ and $\frac{7}{15}$ is
(a) $8\frac{2}{3}$ (b) $\frac{8}{15}$ (c) 20 (d) $16\frac{4}{5}$

4. What is the HCF of $\frac{3}{2}, \frac{9}{7}$ and $\frac{15}{14}$?
(a) $\frac{3}{7}$ (b) $\frac{3}{14}$ (c) $\frac{3}{2}$ (d) 3

5. The least number divisible by 12, 15, 20 and is a perfect square is
(a) 900 (b) 400 (c) 36 (d) 256

6. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3 is
(a) 423 (b) 843 (c) 1683 (d) 2523

7. The HCF of two numbers is 1/5th of their LCM. If the product of the two numbers is 720, then the HCF of the numbers is
(a) 13 (b) 12 (c) 14 (d) 18

8. The LCM of two numbers is 39780 and their ratio is 13 : 15. Then, the numbers are
(a) 273, 315 (b) 2652, 3060
(c) 516, 685 (d) None of these

9. If the highest common factor of two positive integers is 24, then their least common multiple cannot be
(a) 72 (b) 216 (c) 372 (d) 600

10. If the HCF of three numbers 144, x and 192 is 12, then the number x cannot be
(a) 180 (b) 84 (c) 60 (d) 48

11. Consider those numbers between 300 and 400 such that when each number is divided by 6, 9 and 12, it leaves 4 as remainder in each case. What is the sum of the numbers?
(a) 692 (b) 764 (c) 1080 (d) 1092

12. What is the smallest positive integer which when divided by 4, 5, 8 and 9 leaves remainder 3, 4, 7 and 8, respectively?
(a) 119 (b) 319 (c) 359 (d) 719

13. What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$?
(a) ab (b) a^2b^2 (c) a^2b^3 (d) a^3b^2

14. If a number is exactly divisible by 11 and 13, which of the following types the number must be?
(a) Divisible by (11+13) (b) Divisible by (13−11)
(c) Divisible by (11×13) (d) Divisible by (13÷11)

15. What is the sum of the digits of the least number which when divided by 52, leaves 33 as remainder, when divided by 78 leaves 59 and when divided by 117, leaves 98 as remainder ?
(a) 17 (b) 18 (c) 19 (d) 21

16. For any integer n, what is the HCF of integers $m = 2n + 1$ and $k = 9n + 4$?
(a) 3 (b) 1 (c) 2 (d) 4

17. For any three natural numbers a, b and c, if HCF $(a, b) = c$, then HCF $\left(\frac{a}{c}, \frac{b}{c}\right)$ is
(a) a/c (b) b/c (c) c (d) Always 1

18. Raj, Rachit and Asha begin to tag around a circular stadium. They complete their revolutions in 42 s, 56 s and 63 s, respectively. After how many seconds will they be together at the starting point?
(a) 366 (b) 252 (c) 504 (d) 605

19. Find the side of the largest possible square slabs which can be paved on the floor of a room 2m 50cm long and 1m 50cm broad. Also, find the number of such slabs to pave the floor.
(a) 25,20 (b) 30,15 (c) 50,15 (d) 55,10

20. Four bells begin to toll together and toll, respectively at intervals of 5, 6, 8 and 12 s. How many times will they toll together in an hour excluding the one at the start?
(a) 10 (b) 19 (c) 13 (d) 9

21. 21 mango trees, 42 apple trees and 56 orange trees have to be planted in rows such that each row contains the same number of trees of one variety only. What is the minimum number of rows in which the above trees may be planted?
(a) 3 (b) 15 (c) 17 (d) 20

22. A person has four iron bars whose lengths are 24 m, 36 m, 48 m and 72 m, respectively. This person wants to cut pieces of same length from each of four bars. What is the least number of total pieces, if he is to cut without any wastage?
(a) 10 b) 15 (c) 20 (d) 25

23. For two natural numbers m and n, let g_{mn} denote the greatest common factor of m and n. Consider the following in respect of three natural numbers k, m and n.

I. $g_{m(nk)} = g_{(mn)k}$ II. $g_{mn}g_{nk} = g_{mk}$

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

24. Consider the following in respect of integers a and b

I. HCF $(a,b) =$ HCF $(a+b,b)$
II. HCF $(a,b) =$ HCF $(a,b-a)$ for $b > a$

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) None of these

PREVIOUS YEARS' QUESTIONS

25. The HCF and LCM of two natural numbers are 12 and 72, respectively. What is the difference between the two numbers, if one of the numbers is 24? **2012 I**

(a) 12 (b) 18 (c) 21 (d) 24

26. The sum of two numbers is 232 and their HCF is 29. What is the number of such pairs of numbers satisfying the above condition? **2012 I**

(a) One (b) Two (c) Four (d) None of these

27. The product of HCF and LCM of 18 and 15 is **2012 II**

(a) 120 (b) 150 (c) 175 (d) 270

28. Three planets revolve round the Sun once in 200, 250 and 300 days, respectively in their own orbits. When do they all come relatively to the same position at a certain point of time in their orbits? **2012 II**

(a) After 3000 days (b) After 2000 days
(c) After 1500 days (d) After 1200 days

29. The LCM of two numbers is 2376 while their HCF is 33. If one of the numbers is 297, then the other number is **2013 I**

(a) 216 (b) 264 (c) 642 (d) 792

30. The HCF of two numbers is 98 and their LCM is 2352. The sum of the numbers may be **2013 II**

(a) 1372 (b) 1398 (c) 1426 (d) 1484

31. If for integers a, b and c, HCF $(a,b)=1$ and HCF $(a,c)=1$, then which one of the following is correct? **2013 II**

(a) HCF $(a, bc) = 1$ (b) HCF $(a, bc) = a$
(c) HCF $(a, bc) = b$ (d) None of these

32. What is the number of integral solutions of the equations HCF $(a, b) = 5$ and $a + b = 65$? **2014 I**

(a) Less than 65 (b) Infinitely many
(c) Exactly one (d) None of these

33. For any integer n, HCF of $(22n+7, 33n+10)$ is equal to **2014 I**

(a) n (b) 1 (c) 11 (d) None of these

34. In a fire range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth shooter hit the target once in every 5 s, 6 s, 7 s and 8 s, respectively. If all of them hit their target at 9 : 00 am, when will they hit their target together again? **2014 I**

(a) 9 : 04 am (b) 9 : 08 am
(c) 9 : 14 am (d) None of these

35. If a and b be positive integers, then HCF of $\left(\frac{a}{\text{HCF}(a,b)}, \frac{b}{\text{HCF}(a,b)}\right)$ equal to? **2014 I**

(a) a (b) b (c) 1 (d) $\frac{a}{\text{HCF}(a,b)}$

36. The HCF of two natural numbers m and n is 24 and their product is 552. How many sets of values of m and n are possible? **2014 II**

(a) 1 (b) 2 (c) 4
(d) No set of m and n is possible satisfying the given conditions

37. The LCM of two integers is 1237. What is their HCF? **2014 II**

(a) 37 (b) 19
(c) 1 (d) Cannot be determined

38. There are 48 cricket balls, 72 hockey balls and 84 tennis balls and they have to be arranged in several rows in such a way that every row contains the same number of balls of one type. What is the minimum number of rows required for this to happen? **2014 II**

(a) 12 (b) 16 (c) 17 (d) 19

39. Consider all positive two digit numbers each of which when divided by 7 leaves a remainder 3. What is their sum? **2015 II**

(a) 661 (b) 666 (c) 676 (d) 777

40. What is the sum of digits of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11, respectively, as the remainders? **2015 II**

(a) 5 (b) 6 (c) 7 (d) 8

41. The LCM of two numbers is 12 times their HCF. The sum of HCF and LCM is 403. If one of the numbers is 93, then the other number is **2015 II**

(a) 124 (b) 128 (c) 134 (d) 138

42. Consider the following in respect of natural numbers a, b and c **2016 I**

I. LCM $(ab, ac) = a$ LCM (b,c)
II. HCF $(ab, ac) = a$ HCF (b,c)
III. HCF $(a,b) <$ LCM (a,b)
IV. HCF (a,b) divides LCM (a,b).

Which of the above statement(s) is/are correct?

(a) III and II (b) III and IV
(c) I, II and IV (d) All of these

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42																	

HINTS AND SOLUTIONS

1. *(a)* Given, $x = 2^3 \times 3^2 \times 5^4$
and $y = 2^2 \times 3^2 \times 5 \times 7$
$\therefore$ HCF $= 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$

2. *(c)* Here, say $a = 2^3 \times 3 \times 5$ and
$b = 2^4 \times 5 \times 7$, then
LCM $= 2^4 \times 3 \times 5 \times 7$

3. *(d)* Here, LCM
$$= \frac{\text{LCM of } 4, 3, 7}{\text{HCF of } 5, 10, 15} = \frac{84}{5} = 16\frac{4}{5}$$

4. *(b)* HCF $\left(\frac{3}{2}, \frac{9}{7}, \frac{15}{14}\right) = \frac{\text{HCF }(3, 9, 15)}{\text{LCM }(2, 7, 14)}$
$$= \frac{3}{14}$$

5. *(a)*

2	12, 15, 20
2	6, 15, 10
3	3, 15, 5
5	1, 5, 5
	1, 1, 1

LCM of 12, 15 and 20 $= 2 \times 2 \times 3 \times 5$
$\therefore$ Required perfect square
$= 2 \times 2 \times 3 \times 3 \times 5 \times 5 = 900$

6. *(b)* To find out the least number, firstly we find out the LCM of given numbers.
$\therefore$ LCM (5, 6, 7, 8)
$=$ LCM $(5, 2 \times 3, 7, 2^3) = 840$
$\therefore$ Required number $= 840 + 3 = 843$

7. *(b)* Let LCM $= 5x$, then HCF $= x$
Now, product of numbers $= 720$
So, $5x \times x = 720$
$\Rightarrow 5x^2 = 720 \Rightarrow x = 12$

8. *(b)* The numbers are $13x$ and $15x$.
So, x is the HCF. Now,
HCF $\times$ LCM $=$ Product of numbers
$x \times 39780 = 13x \times 15x$
$\Rightarrow x \times 39780 = 13 \times 15 \times x^2$
$\Rightarrow x = \frac{39780}{13 \times 15} = 204.$
$\therefore$ Numbers are $13 \times 204 = 2652$ and $15 \times 204 = 3060$

9. *(c)* In the given options, only 372 is not divisible by 24. Therefore, LCM of numbers cannot be 372.

10. *(d)* Here, we know that
$144 = 12 \times 2 \times 2 \times 3$
and $192 = 12 \times 2 \times 2 \times 2 \times 2$
By taking option (d), $48 = 12 \times 2 \times 2$
Hence, the value of x will not be 48 otherwise the HCF of given numbers becomes 48.

11. *(a)* LCM of 6, 9 and 12 = 36
So, number is the form of $36p + 4$.
Since, the required numbers are between 300 and 400.
$\therefore$ $p = 9$ and 10
$\therefore$ Required sum $= 328 + 364 = 692$

12. *(c)* Here,
$4 - 3 = 5 - 4 = 8 - 7 = 9 - 8 = 1$
Now, $4 = 2 \times 2, 5 = 5$
$8 = 2 \times 2 \times 2, 9 = 3 \times 3$
$\therefore$ LCM $= 5 \times 2 \times 2 \times 2 \times 3 \times 3 = 360$
Required number $= 360 - 1 = 359$

13. *(b)* $a^2b^4 + 2a^2b^2 = a^2b^2(b^2 + 2)$ ...(i)
and $(ab)^7 - 4a^2b^9 = a^7b^7 - 4a^2b^9$
$= a^2b^2(a^5b^5 - 4b^7)$ ...(ii)
From Eqs. (i) and (ii), we get
HCF $= a^2b^2$

14. *(c)* LCM of 11 and 13 will be (11×13). Hence, if a number is exactly divisible by 11 and 13, then the same number must be exactly divisible by their LCM i.e. (11×13).

15. *(a)* Here, $52 - 33 = 78 - 59$
$= 117 - 98 = 19$
Now, $52 = 13 \times 2 \times 2$
$\Rightarrow 78 = 13 \times 2 \times 3$
$\Rightarrow 117 = 13 \times 3 \times 3$
$\therefore$ LCM $= 13 \times 2 \times 2 \times 3 \times 3 = 468$
$\therefore$ Required number $= 468 - 19 = 449$
Hence, the sum of the digits is 17.

16. *(b)* Since, $m = 2n + 1$ is an odd integer, so its factors may be 1 or 3 and $k = 9n + 4$ its factors may be 1, 2 and 4.
Hence, HCF of (m, k) is 1.

17. *(d)* It is always 1
Illustrations Let $a = 21$ and $b = 35$
Then, HCF (21, 35) = 7
$\therefore$ HCF $\left(\frac{21}{7}, \frac{35}{7}\right) =$ HCF $(3, 5) = 1$

18. *(c)* Required time = LCM of 42,56 and 63 s LCM of 42, 56 and 63 is

2	42, 56, 63
3	21, 28, 63
7	7, 28, 21
	1, 4, 3

$\therefore$ Required time
$= 2 \times 3 \times 7 \times 4 \times 3 = 504$ s.

19. *(c)* Side of largest possible square slab is the HCF of 250 cm and 150 cm
$250 = 5 \times 5 \times 5 \times 2$
$150 = 5 \times 5 \times 3 \times 2$
$\therefore$ HCF is 50. Then, number of slabs
$$= \frac{\text{Area of floor}}{\text{Area of slab}} = \frac{250 \times 150}{50 \times 50} = 15$$

20. *(a)* Here, LCM of 5, 6, 8 and 12 is 360, so the bells will toll after 360 s.
So, in an hour they will toll together
$$= \frac{60 \times 60}{360} = 10 \text{ times}$$

21. *(c)* The HCF of (21, 42, 56) is 7.
$\therefore$ The minimum number of rows
$$= \frac{21}{7} + \frac{42}{7} + \frac{56}{7}$$
$= 3 + 6 + 8 = 17$

22. *(b)* Here, $24 = 12 \times 2$, $36 = 12 \times 3$
$48 = 12 \times 4$, $72 = 12 \times 6$
$\therefore$ HCF (24, 36, 48, 72) = 12
Total piece $= \frac{24}{12} + \frac{36}{12} + \frac{48}{12} + \frac{72}{12}$
$= 2 + 3 + 4 + 6 = 15$

23. *(d)* I. Let three natural numbers are $m = 16, n = 15, k = 20$

$\therefore \quad g_{m\ (nk)} = g_{16\ (300)} = 4$

and $\quad g_{(m\ n)\ k} = g_{240\ (20)} = 20$

$\therefore \quad g_{m\ (nk)} \neq g_{(mn)\ k}$

II. $\quad g_{mn}\ g_{nk} = g_{16,\ 15}\ g_{15,\ 20} = 1 \times 5 = 5$

and $\quad g_{mk} = g_{16,\ 20} = 4$

$\therefore g_{mn}\ g_{nk} \neq g_{mk}$

So, neither I nor II is correct.

24. *(c)* I. Let $a = 4$, and $b = 10$

$\therefore \quad a + b = 14$

HCF $(4, 10) = 2$

and HCF $(14, 10) = 2$

$\therefore$ HCF (a, b) = HCF $(a + b, b)$

II. Let $a = 6$ and $b = 15$

$\therefore b - a = 15 - 6 = 9$

HCF $(6, 15) = 3$

HCF $(6, 9) = 3$

$\therefore$ HCF (a, b) = HCF $(a, b - a)$

25. *(a)* Second number $= \dfrac{\text{LCM} \times \text{HCF}}{\text{First number}}$

$= \dfrac{72 \times 12}{24} = 36$

$\therefore$ Difference between two numbers

$= 36 - 24 = 12$

26. *(b)* Let two numbers be $29x$ and $29y$.

$29x + 29y = 232 \Rightarrow x + y = 8$

$\therefore \quad (x, y) = (1, 7), (3, 5)$

Hence, one such pair is 87 and 145.

and the other pair is 203 and 29.

27. *(d)* Here, $18 = 2 \times 3 \times 3$ and $15 = 3 \times 5$

HCF of 18 and 15 = 3

LCM of 18 and 15 $= 2 \times 3 \times 3 \times 5 = 90$

$\therefore$ Product of HCF and LCM of both numbers $= 3 \times 90 = 270$

28. *(a)* Given that, three planets revolves the Sun once in 200, 250 and 300 days, respectively in their own orbits.

$\therefore$ Required time = LCM of 200, 250 and 300 = 3000 days

Hence, after 3000 days they all come relatively to the same position at a certain point of time in their orbits.

29. *(b)* Given, LCM of two numbers = 2376

HCF of two numbers = 33

One of the number = 297

$\because$ (HCF of two numbers) $\times$ (LCM of two numbers)

= (First number) $\times$ (Second number)

$\therefore$ Second number $= \dfrac{33 \times 2376}{297} = 264$

30. *(a)* It is given that, the HCF of two numbers is 98. This means that both the numbers are multiples of 98. Therefore, the sum of these two numbers must also be a multiple of 98. Among all the four options given, only option (a) satisfies this condition.

31. *(a)* For integers a, b and c, if HCF (a, b) = 1 and HCF (a, c) = 1 then, HCF $(a, b\ c)$ = 1

32. *(a)* $\because$ HCF $(a, b) = 5$

Let $\quad a = 5x$ and $b = 5y$

$\therefore \quad 5x + 5y = 65 \Rightarrow x + y = 13$

$\therefore$ Number of pairs of $(x, y) = (1, 12), (2, 11), (3, 10), (4, 9), (5, 8), (6, 7)$

Hence, number of solutions is less than 65.

33. *(b)* HCF of $(22n + 7, 33n + 10)$ is always 1.

Illustration For $n = 1$, HCF (29, 43) = 1

For $n = 2$, HCF (51, 76) = 1

For $n = 3$, HCF (73, 109) = 1

34. *(c)* Time after which they will hit the target together again

= LCM of 5, 6, 7 and 8

$= 5 \times 3 \times 7 \times 2 \times 2 \times 2 = 840$ s

Duration after which they will hit target together

$= \dfrac{840}{60} = 14$ min.

So, they will hit the target together after 14 min.

Hence, they will hit together again at 9 : 14 am.

35. *(c)* HCF $\left(\dfrac{a}{\text{HCF}\ (a, b)}, \dfrac{b}{\text{HCF}\ (a, b)}\right)$ is always equal to 1.

Illustration Let the two positive integers be $a = 24$ and $b = 36$.

$\therefore$ HCF $\left(\dfrac{24}{\text{HCF}\ (24, 36)}, \dfrac{36}{\text{HCF}\ (24, 36)}\right)$

HCF $\left(\dfrac{24}{12}, \dfrac{36}{12}\right) \Rightarrow$ HCF (2, 3) = 1

36. *(d)* LCM of two natural numbers

$= \dfrac{\text{Product of } m \text{ and } n}{\text{HCF of } m \text{ and } n} = \dfrac{552}{24} = 23$

Here, no set of m and n is possible satisfying the given conditions as LCM is always a multiple of HCF.

37. *(c)* Given, LCM of two integers is 1237, which is a prime number.

So, their HCF is 1.

38. *(c)* Given, number of cricket balls

$= 48 = 2^4 \times 3$

Number of hockey balls $= 72 = 2^3 \times 3^2$

and number of tennis balls

$= 84 = 2^2 \times 3 \times 7$

$\therefore$ HCF of 48, 72 and 84 $= 2^2 \times 3 = 12$

Now, minimum number of rows

$= \dfrac{48}{12} + \dfrac{72}{12} + \dfrac{84}{12}$

$= 4 + 6 + 7 = 17$

39. *(c)* The required numbers are 10, 17, 24,…, 94.

Total number of numbers is 13.

$\because$ Sum of these numbers

$= \dfrac{13}{2}[10 + 94] = \dfrac{13}{2} \times 104$

$= 13 \times 52 = 676$

40. *(d)* Here, $6 - 5 = 1$, $8 - 7 = 1$

$12 - 11 = 1$

LCM of 6,8 and 12 = 24

Required number

$= 24k - 1$, k is any natural number

For $k = 6$, the number $= 144 - 1 = 143$

which is multiple of 13

So, sum of digits $= 1 + 4 + 3 = 8$

41. *(a)* Let other number be b and HCF be x.

$\Rightarrow$ LCM $= 12x$

We have, $x + 12x = 403 \Rightarrow 13x = 403$

$\therefore \quad x = 31$

$\because$ Product of two numbers

= LCM $\times$ HCF

$\therefore \quad 93 \times b = x \times 12x$

$\Rightarrow \quad 93 \times b = 12 \times 31 \times 31$

$\therefore \quad b = 124$

42. *(d)* a, b and c are natural numbers.

I. LCM of $(ab, ac) = abc$

$a \times$ LCM of $(b, c) = abc$

Hence, statement I is correct.

II. HCF $(ab, ac) = a$ HCF (b, c)

HCF of (ab, ac) = Common factor of (ab, ac)

and $a \times$ HCF $(b, c) = a \times$ common factor of (b, c)

Hence, statement II is correct.

III. We know that HCF is always less than LCM.

Hence, statement III is correct.

IV. HCF (a, b) divides LCM (a, b) because a common factor between a, b always divides $(a \times b)$.

Hence, statement IV is correct.

04

DECIMAL FRACTIONS

Regularly (1-2) questions have been asked from this chapter. It is one of the most common chapters which we are studying from the very starting age.

FRACTION

If any unit is divided into some parts, then ...... each of these parts is called fraction of that unit. A fraction is represented as p/q, where $q \neq 0$ and here q is called as denominator and p is called as numerator.

e.g. 1/2, 3/4, 6/7 etc. are fractions.

Simple fraction

The fraction which has denominator other than power of 10, is called simple fraction. e.g. $\frac{6}{5}, \frac{2}{9}$ etc.

Note Simple fraction is also known as vulgar fraction.

DECIMAL FRACTION

Fraction that has powers of 10 in the denominator are called decimal fraction. e.g.

(i) $\frac{1}{10}$ is the tenth part of 1 is written as 0.1. (ii) $\frac{1}{100}$ is the hundredth part of 1 is written as 0.01.

Types of Decimals

1. **Recurring Decimals** A decimal number in which a digit or set of digits repeats regularly is called non-terminating repeating decimals or recurring decimals. To represent these fractions, a line is drawn on repeating digits.

e.g. $$\frac{1}{7} = 0.1428571428571\ldots = 0.\overline{142857}$$

$$\frac{2}{3} = 0.66666\ldots = 0.\overline{6}$$

$$\frac{4}{3} = 1.33333\ldots = 1.\overline{3}$$

2. **Pure recurring decimal** A decimal fraction in which all the digits after the decimal points are repeated, is called a pure recurring decimal.
   e.g. $0.786786786\ldots = 0.\overline{786}$
   $0.77777\ldots = 0.\overline{7}$

EXAMPLE 1. Which one of the following is a non-terminating repeating decimal?

a. $\frac{13}{8}$ b. $\frac{3}{16}$ c. $\frac{3}{11}$ d. $\frac{137}{25}$

Sol. c. $\because \frac{3}{11} = 0\cdot 272727\ldots$

$= 0\cdot\overline{27}$

$\frac{13}{8} = \frac{13\times125}{8\times125} = \frac{1625}{1000} = 1.625$

$\frac{3}{16} = \frac{3\times625}{16\times625} = \frac{1875}{10000} = 0.1875$

$\frac{137}{25} = \frac{137\times4}{25\times4} = \frac{548}{100} = 5.48$

From above it is clear that all of these are terminating decimals. Hence, $\frac{3}{11}$ is a non-terminating repeating decimal.

3. **Mixed recurring decimal** A decimal fraction in which some digits after the decimal point are not repeated while some are repeated, is called a mixed recurring decimal.
   e.g. $0.179999\ldots = 0.17\overline{9}$
   $0.053939\ldots = 0.05\overline{39}$

Note The decimal expansion of a rational number is either terminating or non-terminating recurring. In other words, a number whose decimal expansion is terminating or non-terminating recurring is a rational number.

Operations on Decimal Fractions, Addition and Subtraction

While making addition or subtraction of decimal fractions the numbers are placed in such a way that the decimal point lie in one column. Then, the numbers can be added or subtracted as usual.

EXAMPLE 2. $7.093 - 3.57 = ?$

a. 3.523 b. 3.513
c. 3.143 d. 3.532

Sol. a. Here,
$$\begin{array}{r} 7.093 \\ -\ 3.570 \\ \hline 3.523 \end{array}$$

EXAMPLE 3. $51.3 + 7.078 + 1.38 + 0.9 = ?$

a. 61.668 b. 59.238
c. 60.658 d. None of these

Sol. c. Here,
$$\begin{array}{r} 51.300 \\ 7.078 \\ 1.380 \\ +\ 0.900 \\ \hline 60.658 \\ \hline \end{array}$$

Note Adding/Annexing zeroes to the extreme right of a decimal fraction does not change its value.

Multiplication

While multiplying two or more decimal fractions, consider them without decimal point and multiply them as usual. In the product so obtained the decimal point is marked off as many places from the right as in the sum of the decimal places in the given numbers.

EXAMPLE 4. $5\times 0.25\times 6.301\times 0.00394 = ?$

a. 0.0310324220 b. 0.0310324210
c. 0.0310324250 d. None of these

Sol. c. For the product of $5\times0.25\times6.301\times0.00394$

Here, $5\times25\times6301\times394 = 310324250$

Total decimal places $(2+3+5) = 10$

$\therefore\quad 5\times0.25\times6.301\times0.00394 = 0.031032425$

Division

While dividing the given decimal fraction by a natural number, divide it without the decimal point to get quotient. Here, in the quotient so obtained, place the decimal point after as many places from the right as are there in the dividend.

For dividing a decimal fraction by a decimal fraction, multiply the dividend and the divisor by a suitable power of 10 to make the divisor a whole number and then proceed as in the previous rule.

EXAMPLE 5. $0.01834 \div 13 = ?$

a. 141.077 b. 1.41077
c. 0.00141077 d. None of these

Sol. c. Here, for $0.01834\div13$

$\Rightarrow\quad \frac{1834}{13} = 141.077$

So, $0.01834\div13 = 0.00141077$

EXAMPLE 6. $0.0066 \div 0.22 = ?$

a. 0.03 b. 0.3 c. 3 d. None of these

Sol. a. $0.0066\div0.22$

So, $\frac{0.0066\times10000}{0.22\times10000} = \frac{66}{22\times100} = \frac{3}{100} = 0.03$

To Convert a Pure Recurring Decimal into a Simple Fraction

In order to convert a pure recurring decimal fraction into simple (vulgar) fraction, we write the repeated figures only once in the numerator without decimal point and write as many nines in the denominator as the number of repeating figure.

e.g. $0.\overline{5} = \frac{5}{9}$ or $0.\overline{45} = \frac{45}{99}$

EXAMPLE 7. $1.\overline{27}$ in the form p/q is equal to

a. $\frac{127}{100}$ b. $\frac{73}{100}$ c. $\frac{14}{11}$ d. $\frac{11}{14}$

Sol. c. $1.\overline{27} = 1 + 0.\overline{27} = 1 + 27/99 = \frac{99+27}{99} = \frac{126}{99} = \frac{14}{11}$

To Convert Mixed Recurring Decimal into a Simple Fraction

The numerator is obtained by taking the difference between the number formed by all the digits after decimal point. (Here, repeated digits are taken only once) and the number formed by non-repeating digits. The denominator is obtained by placing, as many nines as in repeating digits followed by as many zeroes as the number of non-repeating digits.

EXAMPLE 8. The vulgar fraction of $0.12\overline{36}$ is

a. $\frac{120}{823}$ b. $\frac{102}{825}$ c. $\frac{93}{825}$ d. None of these

Sol. b. $0.12\overline{36} = \frac{1236-12}{9900} = \frac{1224}{9900} = \frac{102}{825}$

EXAMPLE 9. What is the value of $1.\overline{34} + 4.1\overline{2}$?

a. $\frac{133}{90}$ b. $\frac{371}{90}$ c. $5\frac{219}{990}$ d. $5\frac{461}{990}$

Sol. d. $\because 1.\overline{34} = \frac{134-1}{99} = \frac{133}{99}$ and $4.1\overline{2} = \frac{412-41}{90} = \frac{371}{90}$

$\therefore 1.\overline{34} + 4.1\overline{2} = \frac{133}{99} + \frac{371}{90} = \frac{1330+4081}{990} = \frac{5411}{990} = 5\frac{461}{990}$

Comparison of Fractions By Converting in Decimal Form

To compare the fraction using decimal, convert each one of the given fractions in decimal form and now arrange them in ascending or descending order as per the requirement.

By Equating Denominators

For comparison of fractions, take LCM of the denominator of all fractions. So, that their denominators are same. Now, the fractions having largest numerator is the largest fraction.

EXAMPLE 10. Arrange $\frac{2}{3}, \frac{5}{6}, \frac{16}{25}, \frac{3}{7}$ in ascending order.

a. $\frac{2}{3}, \frac{5}{6}, \frac{16}{25}$ and $\frac{3}{7}$ b. $\frac{16}{25}, \frac{5}{6}, \frac{3}{7}$ and $\frac{2}{3}$

c. $\frac{3}{7}, \frac{16}{25}, \frac{2}{3}$ and $\frac{5}{6}$ d. None of these

Sol. c. Firstly, we convert each of the fraction in decimal form.

$\frac{2}{3} = 0.\overline{666}, \frac{5}{6} = 0.8\overline{33}, \frac{16}{25} = 0.64, \frac{3}{7} = 0.\overline{428571}$

Here, $0.\overline{428571} < 0.64 < 0.\overline{666} < 0.8\overline{333}$

Hence, $\frac{3}{7} < \frac{16}{25} < \frac{2}{3} < \frac{5}{6}$ and so $\frac{3}{7}, \frac{16}{25}, \frac{2}{3}$ and $\frac{5}{6}$ are in ascending order.

EXAMPLE 11. Arrange $\frac{7}{8}, \frac{5}{6}, \frac{6}{7}$ in descending order

a. 6/7, 5/6, 7/8 b. 7/8, 5/6, 6/7
c. 7/8, 6/7, 5/6 d. 6/7, 7/8, 5/6

Sol. c. LCM of 8, 6, 7 = 168

$\therefore \quad \frac{7}{8} = \frac{7\times 21}{8\times 21} = \frac{147}{168} \quad \Rightarrow \quad \frac{5}{6} = \frac{5\times 28}{6\times 28} = \frac{140}{168}$

and $\frac{6}{7} = \frac{6\times 24}{7\times 24} = \frac{144}{168}$

$\because \quad \frac{140}{168} < \frac{144}{168} < \frac{147}{168}$

So, descending order of fractions is $\frac{7}{8}, \frac{6}{7}, \frac{5}{6}$.

HCF and LCM of Decimal Fractions

For finding the HCF or LCM of decimal fractions, first make the given fractions to have same number of decimal places by annexing zeroes, if needed. Now, find the HCF or LCM of the numbers without considering decimal. Finally, in the result mark as many decimal places as are there in each of the given numbers.

EXAMPLE 12. The HCF and LCM of 1.600, 8 and 2.4 is

a. 0.800 and 24 b. 0.6000 and 38
c. 0.900 and 42 d. None of these

Sol. a. We can write the given numbers as 1.600, 8.000, 2.400.

So, without decimal, the numbers are 1600, 8000 and 2400.

Now, HCF of 1600, 8000 and 2400 is 800.

$\therefore$ HCF of 1.600, 8.000 and 2.400 is 0.800.

And LCM of 1600, 8000, 2400 is 24000.

$\therefore$ LCM of 1.600, 8.000 and 2.400 is 24.000.

PRACTICE EXERCISE

1. If $(15.9273 - x) = 11.0049$, then the value of x is

(a) 4.9224 (b) 0.4922 (c) 0.4294 (d) 6.932

2. If $(15.39 + 0.236 + 5.290 + 0.0002) = x$, then the value of x is

(a) 0.20916 (b) 2.0916 (c) 209.16 (d) 20.9162

3. If $175 \times 1.24 = 2.17$, then the value of 1.75×124 is

(a) 217 (b) 0.0217

(c) 2.17 (d) None of these

4. If $111.744 \div 28.8 = 3.88$, then the value of $1117.44 \div 288$ is

(a) 3.88 (b) 0.388 (c) 388.0 (d) 38.8

5. The value of $\dfrac{(0.5)^4 - (0.4)^4}{(0.5)^2 + (0.4)^2}$ is equal to

(a) 0.9 (b) 0.09 (c) 9 (d) 0.009

6. The value of $\dfrac{0.004 \times 0.0008}{0.02}$ equals to

(a) 0.000016 (b) 0.00016

(c) 0.0016 (d) None of these

7. Which of the following sets of the fractions is in ascending order?

(a) $\frac{6}{8}, \frac{7}{9}, \frac{5}{6}, \frac{11}{13}$ (b) $\frac{5}{6}, \frac{6}{8}, \frac{7}{9}, \frac{11}{13}$

(c) $\frac{11}{13}, \frac{5}{6}, \frac{7}{9}, \frac{6}{8}$ (d) $\frac{11}{13}, \frac{7}{9}, \frac{6}{8}, \frac{5}{6}$

8. When 0.232323...is converted into a fraction, then the result is

(a) $\frac{1}{5}$ (b) $\frac{2}{9}$ (c) $\frac{23}{99}$ (d) $\frac{23}{100}$

9. If $3.245 \times 10^k = 0.0003245$, then the value of k is

(a) 4 (b) –4 (c) 3 (d) 5

10. The value of $(0.\overline{6} + 0.\overline{8} + 0.\overline{7})$ is

(a) $2\frac{1}{8}$ (b) $2\frac{1}{9}$ (c) $2\frac{1}{3}$ (d) $1\frac{1}{9}$

11. The value of $(6.\overline{88} - 2.\overline{58})$ is

(a) $4.\overline{30}$ (b) $4.\overline{29}$ (c) $3.\overline{22}$ (d) $4.\overline{38}$

12. The expression $(11.98 \times 11.98 + 11.98 \times x + 0.02 \times 0.02)$ will be a perfect square for x equal to

(a) 0.02 (b) 0.2 (c) 0.04 (d) 0.4

13. If $2.5x = 0.5y$, then the value of $\dfrac{x+y}{x-y}$ is

(a) –1.3 (b) –1.5 (c) 1.5 (d) 1.3

14. 7.2 exceeds its one-tenth by

(a) 8.48 (b) 5.48 (c) 6.48 (d) 5.28

15. If $2.5252525\ldots = \dfrac{p}{q}$ (in the lowest form), then what is the value of $\dfrac{q}{p}$?

(a) 0.4 (b) 0.42525 (c) 0.0396 (d) 0.396

16. What decimal of an hour is a second?

(a) 0.25 (b) 0.0256 (c) 0.00027 (d) 0.0125

17. If $\sqrt{5} = 2.24$, then the value of $\dfrac{3\sqrt{5}}{2\sqrt{5} - 0.48}$ is

(a) 1.68 (b) 16.8 (c) 168 (d) 0.168

18. The value of $\left\{\dfrac{(0.1)^2 - (0.01)^2}{0.0001} + 1\right\}$ is equal to

(a) 1001 (b) 11 (c) 101 (d) 100

19. The value of $\dfrac{3}{3 + \dfrac{0.3 - 3.03}{3 \times 0.91}}$ is equal to

(a) 0.75 (b) 1.5 (c) 15 (d) 0.15

20. What should be subtracted from the multiplication of 0.527 and 2.013 to get 1?

(a) 0.939085 (b) 0.060851 (c) 1.91984 (d) 2.16085

21. The value of $0.34\overline{67} + 0.13\overline{33}$ is equal to

(a) 0.48 (b) $0.48\overline{01}$

(c) $0.\overline{48}$ (d) 0.48

22. The greatest fraction out of $\frac{2}{5}, \frac{5}{6}, \frac{11}{12}$ and $\frac{7}{8}$ is

(a) $\frac{7}{8}$ (b) $\frac{5}{6}$ (c) $\frac{11}{12}$ (d) $\frac{2}{5}$

23. Which of the following fractions is greater than $\frac{3}{4}$ and less than $\frac{5}{6}$.

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{4}{5}$ (d) $\frac{9}{10}$

24. Consider the following statements:

I. $\frac{1}{22}$ cannot be written as a terminating decimal.

II. $\frac{2}{15}$ can be written as a terminating decimal.

III. $\frac{1}{16}$ can be written as a terminating decimal.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II

(c) I and III (d) II and III

25. Consider the following decimal numbers:

I. 1.16666666… II. 1.181181118…
III. 2.010010001… IV. 1.454545…

Which of the above numbers represent(s) rational number(s)?

(a) Only IV (b) II and III
(c) I and IV (d) None of these

PREVIOUS YEARS' QUESTIONS

26. What is the value of $0.00\overline{7} + 17.\overline{83} + 310.020\overline{2}$? **2012 I**

(a) $327.86\overline{638}$ (b) $327.866\overline{38}$
(c) $327.866\overline{83}$ (d) $327.866\overline{8}$

27. What is the value of 0.242424…? **2012 II**

(a) $\frac{23}{99}$ (b) $\frac{8}{33}$ (c) $\frac{7}{33}$ (d) $\frac{47}{198}$

28. Representation of $0.2\overline{341}$ in the form $\frac{p}{q}$, where p and q are integers, $q \neq 0$, is **2013 I**

(a) $\frac{781}{3330}$ (b) $\frac{1171}{4995}$ (c) $\frac{2341}{9990}$ (d) $\frac{2339}{9990}$

29. Let p be a prime number other than 2 or 5. One would like to express the vulgar fraction $1/p$ in the form of a recurring decimal. Then, the decimal will be **2015 I**

(a) a pure recurring decimal and its period will be necessarily $(p-1)$
(b) a mixed recurring decimal and its period will be necessarily $(p-1)$
(c) a pure recurring decimal and its period will be some factor of $(p-1)$
(d) a mixed recurring decimal and its period will be some factor of $(p-1)$

30. The value of $(0.\overline{63} + 0.\overline{37})$ is **2015 II**

(a) 1 (b) $\frac{100}{91}$ (c) $\frac{100}{99}$ (d) $\frac{1000}{999}$

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	

HINTS AND SOLUTIONS

1. *(a)* $x = 15.9273 - 11.0049 = 4.9224$

2. *(d)*
$$\begin{array}{r} 15.3900 \\ 0.2360 \\ 5.2900 \\ +\ 0.0002 \\ \hline 20.9162 \end{array}$$
$x = 20.9162$

3. *(a)* $1.75 \times 1.24 = 2.17$

$\Rightarrow\ 1.75 \times 124 = 217$

4. *(a)* Since, $\frac{111.744}{28.8} = 3.88$

$\Rightarrow\ \frac{111744}{28800} = 3.88$ [Rule 6]

Now, $\frac{1117.44}{288} = \frac{11744}{28800} = 3.88$ [Rule 4]

5. *(b)* $\frac{[(0.5)^2]^2 - [(0.4)^2]^2}{(0.5)^2 + (0.4)^2}$

$[(a^2 - b^2) = (a+b)(a-b)]$

$= \frac{[(0.5)^2 - (0.4)^2][(0.5)^2 + (0.4)^2]}{[(0.5)^2 + (0.4)^2]}$

$= (0.5)^2 - (0.4)^2 = (0.5 - 0.4)(0.5 + 0.4)$ [Rule 1]

$= 0.1 \times 0.9 = 0.09$ [Rule 2]

6. *(b)* $\frac{0.004 \times 0.0008}{0.002} = \frac{0.0000032}{0.02}$

$= 0.00016$

7. *(a)* Here, $\frac{6}{8} = 0.75$, $\frac{7}{9} = 0.\overline{7}$, $\frac{5}{6} = 0.8\overline{3}$, $\frac{11}{13} = 0.846$

So, $0.75 < 0.\overline{7} < 0.8\overline{3} < 0.846$

8. *(c)* Given, $0.232323\ldots = 0.\overline{23}$

(which is a recurring decimal) $= \frac{23}{99}$

9. *(b)* Here, $10^k = \frac{0.0003245}{3.245}$

$= \frac{3.245 \times 10^{-4}}{3.245}$

$10^k = 10^{-4}$

So, $k = -4$

10. *(c)* $0.\overline{6} + 0.\overline{8} + 0.\overline{7} = \frac{6}{9} + \frac{8}{9} + \frac{7}{9}$

$= \frac{21}{9} = \frac{7}{3} = 2\frac{1}{3}$

11. *(a)* Here, $6.\overline{88} - 2.\overline{58}$

$= \left(6 + \frac{88}{99}\right) - \left(2 + \frac{58}{99}\right)$

$= (6 - 2) + \left(\frac{88}{99} - \frac{58}{99}\right)$

$= 4 + \frac{88 - 58}{99} = 4 + \frac{30}{99} = 4.\overline{30}$

12. *(c)* Given expression

$= (11.98)^2 + (0.02)^2 + 11.98 \times x$

For the given expression to be a perfect square, we must have

$11.98 \times x = 2 \times 11.98 \times 0.02$

$\Rightarrow\ x = 0.04$

[by using $(a+b)^2 = a^2 + b^2 + 2ab$]

13. *(b)* $2.5x = 0.5y$

$\Rightarrow \quad \frac{x}{y} = \frac{0.5}{2\cdot 5} = 0.20$

Now, the expression is [dividing numerator and denominator by y]

$\frac{x+y}{x-y} = \frac{x/y+1}{x/y-1} = \frac{0.20+1}{0.20-1} = \frac{1.20}{-0.80}$

$= -1.5$

Shortcut Method

$\frac{x}{y} = \frac{0.5}{2.5} \Rightarrow \frac{x+y}{x-y} = \frac{0.5+2.5}{0.5-2.5}$

[using componendo and dividendo]

$\Rightarrow \quad \frac{x+y}{x-y} = \frac{3}{-2} = -1.5$

14. *(c)* $7.2 - \frac{7.2}{10}$

$\Rightarrow \quad 7.2 - 0.72 = 6.48$

15. *(d)* Given, $\frac{p}{q} = 2.\overline{52}$ …(i)

Now, 100 multiply both sides, we get

$\frac{100p}{q} = 252.\overline{52}$ …(ii)

On subtracting Eq. (i) from Eq. (ii), we get

$\frac{99p}{q} = 250 \Rightarrow \frac{q}{p} = \frac{99}{250} = 0.396$

16. *(c)* Required decimal fraction $= \frac{1}{60\times 60}$

$= \frac{1}{3600} = 0.00027$

17. *(a)* $\frac{3\sqrt{5}}{2\sqrt{5}-0.48} = \frac{3\times 2.24}{2\times 2.24 - 0.48}$

$[\because \sqrt{5} = 2.24]$

$= \frac{6.72}{4.48-0.48} = \frac{6.72}{4} = 1.68$

18. *(d)* Here, $\left[\frac{(0.1)^2-(0.01)^2}{0.0001} + 1\right]$

$= \frac{0.01-0.0001}{0.0001} + 1 = \left(\frac{0.0099}{0.0001} + 1\right)$

$= (99+1) = 100$

19. *(b)* $\dfrac{\dfrac{3}{3+0.3-3.03}}{3\times 0.91} = \dfrac{\dfrac{3}{3-273}}{3\times 91}$

$= \frac{3}{3-1} = \frac{3}{2} = 1.5$

20. *(b)* Let x should be subtracted

$(0.527\times 2.013) - x > 1$

$1.060851 - x > 1$

$x = 0.060851$

21. *(b)* $0.34\overline{67} + 0.13\overline{33}$

$= \frac{3467-34}{9900} + \frac{1333-13}{9900}$

$= \frac{3433+1320}{9900} = \frac{4753}{9900}$

$= \frac{4801-48}{9900} = 0.48\overline{01}$

22. *(c)* $\frac{2}{5} = 0.4$, $\frac{5}{6} = 0.8\overline{3}$, $\frac{11}{12} = 0.91\overline{6}$

and $\frac{7}{8} = 0.875$

Clearly, the greatest fraction is $0.91\overline{6}$ i.e. $\frac{11}{12}$.

23. *(c)* $\frac{3}{4} = 0.75$, $\frac{5}{6} = 0.833$

$\frac{1}{2} = 0.5$, $\frac{2}{3} = 0.66$, $\frac{4}{5} = 0.8$ and $\frac{9}{10} = 0.9$

Clearly, 0.8 lies between 0.75 and 0.8333.

$\therefore \quad \frac{4}{5}$ lies between $\frac{3}{4}$ and $\frac{5}{6}$.

24. *(c)* In $\frac{1}{22}$ and $\frac{2}{15}$, 22 and 15 are not in the form of $2^m \times 5^n$ but in $\frac{1}{16}$.

16 in the form of $2^4 \times 5^0$. So, $\frac{1}{16}$ can be written as a terminating decimal.

Hence, statements I and III are correct.

25. *(c)* Since, 1.16666… and 1.454545… are recurring numbers and we know that, recurring numbers represent rational numbers.

Hence, statements I and IV are rational numbers.

26. *(b)* $0.00\overline{7} = 0.007777777$

$\Rightarrow \quad 17.\overline{83} = 17.83838383$

$310.020\overline{2} = 310.020222222$

$= 327.8663838$ on adding

$= 327.866\overline{38}$

27. *(b)* Given, $0.242424... = 0.\overline{24}$

[which is a recurring decimal number]

$= \frac{24}{99} = \frac{8}{33}$

28. *(d)* $0.2\overline{341} = \frac{2341-2}{9990} = \frac{2339}{9990}$

29. *(c)* Value p may be 3, 7, 11, 13.

$\frac{1}{3} = 0.\overline{3}$, Period = 1

Here, $p - 1 = 2$ and 1 is a factor of 2.

$\frac{1}{7} = 0.\overline{142857}$, Period = 6

Here, $p - 1 = 6$ and 6 is a factor of 6.

$\frac{1}{11} = 0.\overline{09}$, Period = 2

Here, $p - 1 = 10$ and 2 is a factor of 10.

30. *(c)* We have, $0.\overline{63} + 0.\overline{37}$

Let $x = 0.63\ 6363\ 63\ \ldots$

$100x = 63.636363\ \ldots$, $99x = 63$

and $x = \frac{63}{99}$

Similarly, $y = 0.\overline{37} = 0.373737$

$\because \quad y = \frac{37}{99}$

$\therefore \quad x + y = \frac{63}{99} + \frac{37}{99} = \frac{100}{99}$

05

SQUARE ROOTS AND CUBE ROOTS

Generally (1-3) questions have been asked from this chapter. This section will be useful in solving simplification questions and will save lots of time while doing fuzzy calculations.

SQUARE

The square of a number is obtained by multiplying the number by itself.

e.g. Square of 5 $= 5 \times 5 = 25$, square of 6 $= 6 \times 6 = 36$

Perfect square A natural number n is called a perfect square or a square number, if there exists a natural number m such that $n = m^2$.

e.g. The numbers 1, 4, 9, 16, 25, ... are perfect square.

Properties of Perfect Square

The properties of perfect squares are discussed below.

(i) A number having 2, 3, 7 or 8 at unit's place is never a perfect square.
e.g. 172, 2783

(ii) The squares of odd numbers are odd and the squares of even numbers are even.

(iii) The difference of squares of two consecutive natural numbers is equal to their sum.
e.g. $7^2 - 6^2 = 7 + 6 = 13$

(iv) The product of four consecutive natural numbers plus one is a always square.

(v) A number ending in an odd number of zeroes is never a perfect square.

Square Root

The square root of a number x is that number which when multiplied by itself gives x as the product. It is denoted by $\sqrt{\ }$.

In general, $y^2 = x \Rightarrow y = \sqrt{x}$

Here, y is the square root of x, if and only if x is the square of y.

e.g. $\sqrt{4} = \sqrt{2 \times 2} = 2$

Thus, 2 is the square root of 4.

EXAMPLE 1. If $\sqrt{\frac{x}{64}} = \frac{4}{8}$, then the value of x is

a. 16 b. 12 c. 8 d. 4

Sol. a. We have, $\sqrt{\frac{x}{64}} = \frac{4}{8}$,

On squaring both sides, we get

$$\left(\sqrt{\frac{x}{64}}\right)^2 = \left(\frac{4}{8}\right)^2 \Rightarrow \frac{x}{64} = \frac{16}{64} \Rightarrow x = 16$$

Methods to Find Square Root

The square root of a given number can be determined by using any of the following two methods

1. Prime Factorization Method
2. General Method/Division Method

Prime Factorization

In this method, we express the given number as the product of prime factors. Now, for finding square root, we take the product of these prime factors choosing one out of every pair of same prime factors.

EXAMPLE 2. The square root of 213444 is

a. 332 b. 368 c. 432 d. 462

Sol. d. The prime factorisation of 213444 is

$$213444 = 2 \times 2 \times 3 \times 3 \times 7 \times 7 \times 11 \times 11$$

$\therefore \sqrt{213444} = \sqrt{\overline{2\times 2}\times\overline{3\times 3}\times\overline{7\times 7}\times\overline{11\times 11}}$

Now, taking one number from each pair and multiplying them, we get $\sqrt{213444} = 2 \times 3 \times 7 \times 11 = 462$

EXAMPLE 3. What is the square root of $\frac{0.324 \times 0.64 \times 129.6}{0.729 \times 1.024 \times 36}$?

a. 4 b. 3

c. 2 d. 1

Sol. d. $\sqrt{\frac{0.324 \times 0.64 \times 129.6}{0.729 \times 1.024 \times 36}} = \sqrt{\frac{324 \times 64 \times 1296}{729 \times 1024 \times 36}}$

$$= \sqrt{\frac{(2^2 \times 3^4) \times (2^6) \times (2^4 \times 3^4)}{(3^6) \times (2^{10}) \times (2^2 \times 3^2)}} = \frac{(2 \times 3^2) \times 2^3 \times (2^2 \times 3^2)}{3^3 \times 2^5 \times (2 \times 3)}$$

$$= \frac{18 \times 8 \times 36}{27 \times 32 \times 6} = 1$$

Division Method

To find the square root of a given number using this method, the following steps are to be followed.

Step I In the given number, place bars over every pair of digits starting with the unit's digit. Each pair and the remaining one digit (if any) on the extreme left is called a period.

Step II Think of the largest number whose square is less than or equal to the first period.

Step III Put the quotient above the period and write the product of divisor and quotient just below the first period.

Step IV Subtract the product of divisor and quotient from the first period and bring down the next period to the right of the remainder. This becomes the new dividend.

Step V Double the quotient and enter it with a blank on its right.

Step VI Guess a largest possible digit to fill the blank which will also become the next digit in the quotient, such that when the new divisor is multiplied to the new quotient the product is less than or equal to the dividend.

Step VII Repeat the above steps till all the periods have been taken up.

Now, the quotient so obtained is the required square root of the given number.

EXAMPLE 4. The square root of 1522756 is

a. 1182 b. 1222

c. 1234 d. 1334

Sol. c

$$\begin{array}{r|l|l}
1 & \overline{1}\ \overline{52}\ \overline{27}\ \overline{56} & 1234 \\
 & 1 & \\ \hline
22 & \overline{52} & \\
\times 2 & 44 & \\ \hline
243 & 08\overline{27} & \\
\times 3 & 729 & \\ \hline
2464 & 9856 & \\
\times 4 & 9856 & \\ \hline
 & \times &
\end{array}$$

Hence, the square root of 1522756 is 1234.

EXAMPLE 5. The least number of four digits which is a perfect square is 2012 II

a. 1204 b. 1024

c. 1402 d. 1420

Sol. b. Here, the greatest three-digit number $= 999$

$$\begin{array}{r|l}
 & 31 \\ \hline
3 & 9\ 99 \\
\times 3 & 9 \\ \hline
61 & 99 \\
 & 61 \\ \hline
 & 38
\end{array}$$

Here, the greatest number of three digits which is perfect square $= 999 - 38 = 961 = (31)^2$

So, the smallest four-digit number which is perfect square $= (31 + 1)^2 = (32)^2 = 1024$

CUBE

If a number is multiplied three times with itself, then the result of this multiplication is called the cube of that number.
e.g. $8 = 2\times 2\times 2 = 2^3$

Properties of Cube

1. Cubes of all even natural numbers are even and all odd natural numbers are odd.
2. Cubes of the numbers ending in digit 0, 1, 4, 5, 6 and 9 are the numbers ending in the same digit.
3. Cubes of negative integers are negative.
4. Cubes of numbers ending in digits 3 and 7 ends in digit 7 and 3, respectively.

CUBE ROOT

The cube root of a given number is the number whose cube is the given number.
In general, we can say that $a^3 = b$, then cube root of 'b' is 'a'. Cube root of a number 'a' is denoted as $\sqrt[3]{a}$.
e.g. $\sqrt[3]{64} = \sqrt{4\times 4\times 4} = 4$. Thus, 4 is the cube root of 64.

EXAMPLE 6. If $\sqrt[3]{\frac{x}{27}} = \frac{5}{3}$, then the value of x is

a. 125 b. 25 c. 27 d. 9

Sol. a. We have $\sqrt[3]{\frac{x}{27}} = \frac{5}{3}$

On cubing both sides, we get

$$\left(\sqrt[3]{\frac{x}{27}}\right)^3 = \left(\frac{5}{3}\right)^3 \Rightarrow \frac{x}{27} = \frac{125}{27}$$

$\therefore \quad x = 125$

Method to Find Cube Root

Method to calculate the cube root of a number is as follow

Prime Factorization Method

In this method, we express the given number as the product of prime factors.

Now, for finding cube root, we take the product of these prime factors choosing one out of every three, from the factors.

EXAMPLE 7. Find the cube root of 373248.

a. 42 b. 52 c. 62 d. 72

Sol. d. Here, $373248 = 8\times 8\times 8\times 9\times 9\times 9$

$$373248 = \overline{2\times 2\times 2}\times\overline{2\times 2\times 2}\times\overline{2\times 2\times 2}\times\overline{3\times 3\times 3}\times\overline{3\times 3\times 3}$$

$\Rightarrow \quad \sqrt[3]{373248} = 2\times 2\times 2\times 3\times 3 = 72$

Hence, the cube root of 373248 is 72.

POWERS OR EXPONENTS

An expression that represents repeated multiplication of the same factor is called a power. It is written as 'x^n' and is read as 'x' raised to the power n.

Here, x is called the 'base'.

and n is called the 'exponent'.

e.g. $6^5 = 6\times 6\times 6\times 6\times 6 = 7776$.

Laws of Exponents

Let $\frac{x}{y}$ be any rational number and m, n be any integers. Then,

(i) $\left(\frac{x}{y}\right)^m \times \left(\frac{x}{y}\right)^n = \left(\frac{x}{y}\right)^{m+n}$

(ii) $\left(\frac{x}{y}\right)^m \div \left(\frac{x}{y}\right)^n = \left(\frac{x}{y}\right)^{m-n}$

(iii) $\left[\left(\frac{x}{y}\right)^m\right]^n = \left(\frac{x}{y}\right)^{mn}$

(iv) $\left(\frac{x}{y}\right)^{-n} = \left(\frac{y}{x}\right)^n$

(v) $\left(\frac{x}{y}\right)^0 = 1$

EXAMPLE 8. If $2^m + 2^{1+m} = 24$, then the value of m is

a. 1 b. 2 c. 3 d. 5

Sol. c. Given, $2^m + 2^{1+m} = 24 \Rightarrow 2^m + 2^1\times 2^m = 24$

$\Rightarrow \quad 2^m(1+2) = 24 \Rightarrow 2^m\times 3 = 24 \Rightarrow 2^m = 8 = 2^3$

Now, on comparing both sides, we get

$\therefore \quad m = 3$

Hence, the value of m is 3.

SURDS OR RADICALS

Let x be a rational number and n be a positive integer, such that $\sqrt[n]{x}$ is irrational, then $\sqrt[n]{x}$ is called a radical or surd of order n and here x is called the radicand.

- A surd of order 2 is called a quadratic surd i.e. $\sqrt{2}, \sqrt{3}$.
- A surd of order 3 is called a cubic surd. i.e. $\sqrt[3]{2}, \sqrt[3]{3}$.
- A surd of order 4 is called a biquadratic surd i.e. $\sqrt[4]{5}, \sqrt[4]{7}$.

Laws of Radicals

The laws of exponents which are applicable to the surds are

(i) $(\sqrt[n]{x})^n = x$

(ii) $\sqrt[n]{xy} = \sqrt[n]{x}\cdot\sqrt[n]{y}$

(iii) $\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$

(iv) $(\sqrt[n]{x})^m = (\sqrt[n]{x^m})$

(v) $\sqrt[m]{\sqrt[n]{x}} = \sqrt[mn]{x} = \sqrt[n]{\sqrt[m]{x}}$

Types of Surds

1. **Pure surds** A surd which has only 1 as a rational factor is called a pure surd.
   e.g. $\sqrt{2}, \sqrt[3]{2}, \sqrt[4]{3}$ all are pure surds.
2. **Mixed surds** A surd which is not a pure surd or has factor other than unity is called a mixed surd.
   e.g. $2\sqrt{3}, 5\sqrt[3]{12}, \frac{4}{3}\sqrt[3]{35}$ are mixed surds.
3. **Like and Unlike surds** When the radicands of two surds are same, then those are known as like surds.
   e.g. $6\sqrt{3}$ and $2^2 \cdot \sqrt{3}$.
   When radicands of two surds are different, then they are called unlike surds. e.g. $4\sqrt{7}$ and $6\sqrt{5}$.

Comparison of Two Surds

1. If two surds are of the same order, then the one whose radicand is larger, is the larger of the two.
   e.g. $\sqrt{17} > \sqrt{13}$, $\sqrt[3]{21} > \sqrt[3]{16}$
2. If any two surds have different orders are to be compared, then we will first reduce them to the same but smallest order and then compare them.

Note Let $\sqrt[n]{x}$ be a surd of order n.
Then, $\sqrt[n]{x} = \sqrt[m]{x^{m/n}}$ i.e. $x^{1/n} = (x^{m/n})^{1/m}$ is a surd of order m.

EXAMPLE 9. Convert $\sqrt{2}$ into a surd of order 4.

a. $\sqrt[4]{2}$ b. $\sqrt[4]{8}$ c. $\sqrt[4]{4}$ d. None of these

Sol. c. $\sqrt{2} = 2^{1/2} = (2^{4/2})^{1/4} = (2^2)^{1/4} = \sqrt[4]{4}$

EXAMPLE 10. Consider the following in respect of the numbers $\sqrt{2}, \sqrt[3]{3}$, and $\sqrt[6]{6}$. **2014 (I)**

I. $\sqrt[6]{6}$ is the greatest number.
II. $\sqrt{2}$ is the smallest number.

Which of these statements is/are correct?

a. Only I b. Only II c. Both I and II d. Neither I nor II

Sol. d. Taking LCM of 2, 3 and 6 = 12

Now, $\sqrt{2} = 2^{1/2} = 2^{6/12} = \sqrt[12]{2^6} = \sqrt[12]{64}$

$$\sqrt[6]{6} = (6)^{\frac{1}{6}} = (6)^{\frac{2}{12}} = \sqrt[12]{6^2} = \sqrt[12]{36}$$

$$\sqrt[3]{3} = 3^{1/3} = 3^{4/12} = \sqrt[12]{81}$$

So, neither I nor II is correct.

Operations on Surds

Addition and Subtraction of Surds

Addition or subtraction takes place only between like surds, using laws of numbers. e.g.

(i) $2\sqrt{3} + 4\sqrt{3} + \sqrt{5} = 6\sqrt{3} + \sqrt{5}$

(ii) $6\sqrt{5} - 3\sqrt{2} + 7\sqrt{5} + 4\sqrt{2} = (6\sqrt{5} + 7\sqrt{5}) + (4\sqrt{2} - 3\sqrt{2})$
$= 13\sqrt{5} + \sqrt{2}$

Multiplication and Division of Surds

Suppose given surds are $\sqrt[p]{a}, \sqrt[q]{b}$ and $\sqrt[r]{c}$, then first find the LCM of p, q and r to make the radial powers of the surds same and then their multiplication and division can be done using following rules.

(i) $\sqrt[n]{x} \times \sqrt[n]{y} = \sqrt[n]{xy}$ (ii) $\frac{\sqrt[n]{x}}{\sqrt[n]{y}} = \sqrt[n]{\frac{x}{y}}$

EXAMPLE 11. Simplify $\sqrt[6]{12} \div \sqrt{3} \cdot \sqrt[3]{2}$.

a. $\sqrt[3]{3}$ b. $\sqrt{3}$
c. $\sqrt[3]{2}$ d. $\sqrt[3]{\frac{1}{3}}$

Sol. d. Here, the order of $\sqrt{3}$, $\sqrt[3]{2}$ is 2 and 3, respectively.

So, LCM 2, 3 and 6 = 6

$$\sqrt{3} = 3^{1/2} = (3^{6/2})^{1/6} = (3^3)^{1/6} = (27)^{1/6} = \sqrt[6]{27}$$

$$\sqrt[3]{2} = (2)^{1/3} = (2^{6/3})^{1/6} = (2^2)^{1/6} = \sqrt[6]{4}$$

So, $\frac{\sqrt[6]{12}}{\sqrt{3} \cdot \sqrt[3]{2}} = \frac{\sqrt[6]{12}}{\sqrt[6]{27} \cdot \sqrt[6]{4}} = \frac{\sqrt[6]{12}}{\sqrt[6]{27 \times 4}} = \sqrt[6]{\frac{12}{27 \times 4}} = \sqrt[6]{\frac{1}{9}}$

$$= \left[\left(\frac{1}{3}\right)^2\right]^{1/6} = \left[\frac{1}{3}\right]^{2/6} = \left[\frac{1}{3}\right]^{1/3} = \sqrt[3]{\frac{1}{3}}$$

Rationalisation of Surds

Process of converting surd into rational number is called as rationalisation of surds.

When the product of two surds is a rational number, then each surd is called a rationalising factor of each other.

e.g. $\sqrt[3]{25} \times \sqrt[3]{5} = \sqrt[3]{25 \times 5} = \sqrt[3]{125} = \sqrt[3]{5^3} = 5$

So, $\sqrt[3]{25}$ and $\sqrt[3]{5}$ are rationalising factor of each other.

Rationalising factors of a given surd

- Rationalising factor of $\frac{1}{\sqrt{a}} = \sqrt{a}$
- Rationalising factor of $\frac{1}{\sqrt{a} \pm \sqrt{b}} = \sqrt{a} \mp \sqrt{b}$

EXAMPLE 12. $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ is equal to

a. 16 b. 8 c. 4 d. $\sqrt{15}$

Sol. b. $\frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$

$$= \frac{(\sqrt{5}+\sqrt{3})(\sqrt{5}+\sqrt{3})}{(\sqrt{5}-\sqrt{3})(\sqrt{5}+\sqrt{3})} + \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}+\sqrt{3})} \times \frac{(\sqrt{5}-\sqrt{3})}{(\sqrt{5}-\sqrt{3})}$$

$$= \frac{(\sqrt{5}+\sqrt{3})^2}{\sqrt{5^2}-\sqrt{3^2}} + \frac{(\sqrt{5}-\sqrt{3})^2}{\sqrt{5^2}-\sqrt{3^2}} = \frac{(\sqrt{5}+\sqrt{3})^2 + (\sqrt{5}-\sqrt{3})^2}{(\sqrt{5})^2 - (\sqrt{3})^2}$$

$$= \frac{2[(\sqrt{5})^2 + (\sqrt{3})^2]}{5-3} = \frac{2(5+3)}{2} = 8$$

PRACTICE EXERCISE

1. If $\sqrt{\frac{x}{49}} = \frac{4}{7}$, then the value of x is

(a) 9 (b) 25 (c) 16 (d) 8

2. Which of the following cannot be a digit in the unit place of a perfect square?

(a) 1 (b) 5 (c) 7 (d) 0

3. The number $\sqrt{0.0001}$ is

(a) a rational number less than 0.01
(b) a rational number
(c) an irrational number
(d) Neither a rational number nor an irrational number

4. If m and n are natural numbers, then $\sqrt[m]{n}$ is

(a) always irrational
(b) irrational unless n is the mth power of an integer
(c) irrational unless m is the nth power of an integer
(d) irrational unless m and n are coprime

5. What is that fraction which when multiplied by itself gives 227.798649?

(a) 15.093 (b) 15.099 (c) 14.093 (d) 9.0019

6. If $x = \sqrt{3018 + \sqrt{36 + \sqrt{169}}}$, then the value of x is

(a) 55 (b) 44 (c) 63 (d) 42

7. If $\sqrt[3]{\sqrt{x}} = x/5$, then x is equal to

(a) $\sqrt[5]{5}$ (b) $\sqrt[6]{5^5}$ (c) $\sqrt[5]{5^6}$ (d) $\sqrt{5}$

8. What is the value of

$\sqrt{29.16} + \sqrt{0.2916} + \sqrt{0.002916} + \sqrt{0.00002916}$?

(a) 5.9949 (b) 5.9894 (c) 5.9984 (d) 5.9994

9. $(\sqrt{5})^{-5/2} \times (\sqrt{5})^{-3/2}$

(a) $\frac{1}{625}$ (b) $\frac{1}{25}$ (c) $\frac{1}{125}$ (d) None of these

10. If $a = 3$, $b = 9$ and $c = 10$, then the value of $\sqrt{13 + a} + \sqrt{112 + b} + \sqrt{c - 1}$ is

(a) 15 (b) 18 (c) 16 (d) 10

11. If $\sqrt{0.9 \times 0.09 \times x} = 0.9 \times 0.09 \times \sqrt{z}$, then the value of $\frac{x}{z}$ is

(a) 0.081 (b) 0.810 (c) 0.81 (d) 8.09

12. $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$ is equal to

(a) 0 (b) 2 (c) 1 (d) $2^{31/32}$

13. If $\sqrt{35} = 5.9160$, then the value of $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$ is

(a) 9.1060 (b) 10.9160 (c) 11.9160 (d) 12

14. Find the value of

$\sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + \sqrt{225}}}}}$

(a) 18 (b) 19 (c) $\sqrt{19}$ (d) $18 + \sqrt{3}$

15. What is $\frac{1}{\sqrt{9} - \sqrt{8}} - \frac{1}{\sqrt{8} - \sqrt{7}} + \frac{1}{\sqrt{7} - \sqrt{6}} - \frac{1}{\sqrt{6} - \sqrt{5}} + \frac{1}{\sqrt{5} - \sqrt{4}}$ equal to?

(a) 0 (b) 1 (c) 5 (d) $\frac{1}{3}$

16. What is one of the square roots of $9 - 2\sqrt{14}$?

(a) $\sqrt{7} - \sqrt{3}$ (b) $\sqrt{6} - \sqrt{3}$ (c) $\sqrt{7} - \sqrt{5}$ (d) $\sqrt{7} - \sqrt{2}$

17. If $p^x = r^y = m$ and $r^w = p^z = n$, then which one of the following is correct?

(a) $xw = yz$ (b) $xz = yw$
(c) $x + y = w + z$ (d) $x - y = w - z$

18. If $a^x = b^y = c^z$ and $abc = 1$, then what is $xy + yz + zx$ equal to

(a) xyz (b) $x + y + z$ (c) 0 (d) 1

19. If $a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ and $b = \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$, then the value of $\left(\frac{a^2 + ab + b^2}{a^2 - ab + b^2}\right)$ is

(a) 3/4 (b) 4/3 (c) 3/5 (d) 5/3

20. If $3\sqrt{5} + \sqrt{125} = 17.88$, then what will be the value of $\sqrt{80} + 6\sqrt{5}$

(a) 13.41 (b) 20.46 (c) 21.66 (d) 22.35

21. If $N = 2^{0.15}$ and $N^b = 16$, then b is equal to

(a) 80/3 (b) 5/3 (c) 4 (d) 3/5

22. If $6^A = 2$, $6^B = 5$ and $6^Q = 15$, then Q equals to

(a) $B + 3$ (b) $5A + B$ (c) $B^2 - 2B$ (d) $B - A + 1$

23. Arrange $\sqrt[4]{3}, \sqrt[6]{10}, \sqrt[12]{25}$ in descending order.

(a) $\sqrt[6]{10} > \sqrt[4]{3} > \sqrt[12]{25}$ (b) $\sqrt[12]{25} > \sqrt[4]{3} > \sqrt[6]{10}$
(c) $\sqrt[6]{10} > \sqrt[12]{25} > \sqrt[4]{3}$ (d) $\sqrt[4]{3} > \sqrt[12]{25} > \sqrt[6]{10}$

24. If $(3.7)^x = (0.037)^y = 10000$, then what is the value of $\frac{1}{x} - \frac{1}{y}$?

(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

25. The greatest six digit number which is a perfect square is

(a) 998004 (b) 998006 (c) 998049 (d) 998001

26. What is the smallest number by which 26244 must be divided to get a perfect cube?
(a) 4 (b) 6 (c) 36 (d) 16

27. What is the smallest number that must be added to 1780 to make it a perfect square?
(a) 39 (b) 49 (c) 59 (d) 69

28. A gardener plants 17956 trees in such a way that there are as many rows as there are trees in a row. The number of trees in a row are
(a) 136 (b) 164 (c) 134 (d) 166

29. A group of student decided to collect as many paise from each member of the group as is the number of members. If the total collection amounts to ₹ 32.49 the number of members in the group, is
(a) 37 (b) 47 (c) 57 (d) 27

30. A general arranges his soldiers in rows to form a perfect square. He find that in doing, so 60 soldiers are left out. If the total number of soldiers be 8160. Then, the number of soldiers in each row is
(a) 90 (b) 91 (c) 92 (d) 80

31. A ball is dropped from a height 64 m above the ground and every time it hits the ground it rises to a height equal to half of the previous. What is the height attained after it hits the ground for the 16th time?
(a) 2^{-12} m (b) 2^{-11} m (c) 2^{-10} m (d) 2^{-9} m

32. If $x = \dfrac{\sqrt{a+3b}+\sqrt{a-3b}}{\sqrt{a+3b}-\sqrt{a-3b}}$, then $3bx^2 - 2ax + 3b$ is equal to (given that, $b \neq 0$).
(a) 1 (b) 0 (c) ab (d) $2ab$

PREVIOUS YEARS' QUESTIONS

33. If $a^x = b, b^y = c$ and $xyz = 1$, then what is the value of c^z? **2012 I**
(a) a (b) b (c) ab (d) a/b

34. If $196\,x^4 = x^6$, then x^3 is equal to which one of the following? **2012 I**
(a) $\dfrac{x^6}{14}$ (b) $14x^4$ (c) $\dfrac{x^2}{14}$ (d) $14x^2$

35. If $\sqrt{10+\sqrt[3]{x}} = 4$, then what is the value of x? **2012 I**
(a) 150 (b) 216 (c) 316 (d) 450

36. The expression $[(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}}$ gives **2013 I**
(a) a natural number
(b) an integer and not a natural number
(c) a rational number but not an integer
(d) a real number but not a rational number

37. If $16 \times 8^{n+2} = 2^m$, then m is equal to **2013 I**
(a) $n + 8$ (b) $2n + 10$
(c) $3n + 2$ (d) $3n + 10$

38. When a ball bounces, it rises to $\dfrac{2}{3}$ of the height from which it fell. If the ball is dropped from a height of 36 m, how high will it rise at the third bounce? **2013 I**
(a) $10\dfrac{1}{3}$ m (b) $10\dfrac{2}{3}$ m (c) $10\dfrac{1}{3}$ m (d) $12\dfrac{2}{3}$ m

39. The product of four consecutive natural numbers plus one is **2014 I**
(a) a non-square
(b) always sum of two square numbers
(c) a square
(d) None of the above

40. The difference of two consecutive cubes **2014 I**
(a) is odd or even (b) is never divisible by 2
(c) is always even (d) None of these

41. The square root of $\dfrac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$ is **2015 I**
(a) 1 (b) 2 (c) 3 (d) 4

42. What is $\dfrac{5+\sqrt{10}}{5\sqrt{5}-2\sqrt{20}-\sqrt{32}+\sqrt{50}}$ equal to? **2015 I**
(a) 5 (b) $5\sqrt{2}$ (c) $5\sqrt{5}$ (d) $\sqrt{5}$

43. If $x = \dfrac{91}{216}$, then the value of $3 - \dfrac{1}{(1-x)^{1/3}}$ is **2015 II**
(a) $\dfrac{9}{5}$ (b) $\dfrac{5}{9}$ (c) $\dfrac{4}{9}$ (d) $\dfrac{4}{5}$

44. Which one of the following is correct? **2015 II**
(a) $\sqrt{2} < \sqrt[4]{6} < \sqrt[3]{4}$ (b) $\sqrt{2} > \sqrt[4]{6} > \sqrt[3]{4}$
(c) $\sqrt[4]{6} < \sqrt{2} < \sqrt[3]{4}$ (d) $\sqrt[4]{6} > \sqrt{2} > \sqrt[3]{4}$

45. If $x = \sqrt{3}+\sqrt{2}$, then the value of $x^3 + x + \dfrac{1}{x} + \dfrac{1}{x^3}$, is **2015 II**
(a) $10\sqrt{3}$ (b) $20\sqrt{3}$ (c) $10\sqrt{2}$ (d) $20\sqrt{2}$

46. If $x = 2^{1/3} + 2^{-1/3}$, then the value of $2x^3 - 6x - 5$ is equal to **2016 I**
(a) 0 (b) 1 (c) 2 (d) 3

47. If $4^x 2^y = 128$ and $3^{3x}\,3^{2y} - 9^{xy} = 0$, then the value of $x + y$ can be equal to **2016 I**
(a) 7 (b) 5 (c) 3 (d) 1

48. If $x = \dfrac{\sqrt{a+2b}+\sqrt{a-2b}}{\sqrt{a+2b}-\sqrt{a-2b}}$, then $bx^2 - ax + b$ is equal to (given that $b \neq 0$) **2016 I**
(a) 0 (b) 1 (c) ab (d) $2ab$

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42		43		44		45		46		47		48					

HINTS AND SOLUTIONS

1. *(c)* Given, $\sqrt{\dfrac{x}{49}} = \dfrac{4}{7}$ or $\dfrac{\sqrt{x}}{7} = \dfrac{4}{7}$

Now, on squaring both sides, we get

$$(\sqrt{x})^2 = (4)^2 \Rightarrow x = 16$$

Hence, the value of x is 16.

2. *(c)* The digits 2, 3, 7 and 8 cannot be at the unit place of a perfect square number.

3. *(b)* $\sqrt{0.0001} = 0.01$ which is a rational number.

4. *(b)* If m and n are natural numbers, then $\sqrt[m]{n}$ is irrational unless n is mth power of an integer.

5. *(a)* Let fraction be x, then

$$x^2 = 227.798649$$

$$\Rightarrow \quad x = \sqrt{227.798649} = 15.093$$

6. *(a)* Given, $x = \sqrt{3018 + \sqrt{36 + \sqrt{169}}}$

$$= \sqrt{3018 + \sqrt{36 + 13}}$$

$$= \sqrt{3018 + \sqrt{49}} = \sqrt{3018 + 7}$$

$$= \sqrt{3025} = 55$$

Hence, the value of x is 55.

7. *(c)* $\sqrt[3]{\sqrt{x}} = x/5$

On cubing both sides, we get

$$\sqrt{x} = x^3/5^3$$

On squaring both sides, we get

$$x = x^6/5^6 \Rightarrow 5^6 = \frac{x^6}{x} = x^5$$

$$\Rightarrow x^5 = 5^6 \qquad \Rightarrow \quad x = \sqrt[5]{5^6}$$

8. *(d)* $\sqrt{29.16} + \sqrt{0.2916} + \sqrt{0.002916} + \sqrt{0.00002916}$

$$= 5.4 + 0.54 + 0.054 + 0.0054$$

$$= 5.9994$$

9. *(b)* $(\sqrt{5})^{-5/2} \times (\sqrt{5})^{-3/2}$

$$= (\sqrt{5})^{-5/2 - 3/2} = (\sqrt{5})^{-8/2}$$

$$= (\sqrt{5})^{-4} = \left(\frac{1}{\sqrt{5}}\right)^4$$

$$= \left(\frac{1}{5}\right)^{(1/2)\times 4} = \left(\frac{1}{5}\right)^2 = \frac{1}{25}$$

10. *(b)* Given, $a = 3, b = 9$ and $c = 10$

$$\therefore \quad \sqrt{13 + a} + \sqrt{112 + b} + \sqrt{c - 1}$$

$$= \sqrt{13 + 3} + \sqrt{112 + 9} + \sqrt{10 - 1}$$

$$= \sqrt{16} + \sqrt{121} + \sqrt{9} = 4 + 11 + 3 = 18$$

11. *(a)* Here, $\sqrt{\dfrac{x}{z}} = \dfrac{0.9 \times 0.09}{\sqrt{0.9 \times 0.09}}$

$$= \frac{\sqrt{0.9 \times 0.09} \times \sqrt{0.9 \times 0.09}}{\sqrt{0.9 \times 0.09}}$$

$$= \sqrt{0.9 \times 0.09}$$

$$\Rightarrow \left(\sqrt{\frac{x}{z}}\right)^2 = (\sqrt{0.9 \times 0.09})^2$$

$$\Rightarrow \quad \frac{x}{z} = 0.081$$

12. *(d)* Here, $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$

$$= \sqrt{2\sqrt{2\sqrt{2\sqrt{2 \times 2^{1/2}}}}} = \sqrt{2\sqrt{2\sqrt{2 \times 2^{3/4}}}}$$

$$= \sqrt{2\sqrt{2 \times 2^{7/8}}} = \sqrt{2 \times 2^{15/16}} = 2^{31/32}$$

13. *(c)* Here, $\dfrac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$

$$= \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} + \sqrt{5}} = \frac{(\sqrt{7} + \sqrt{5})^2}{7 - 5}$$

$$= \frac{7 + 5 + 2\sqrt{35}}{2} = \frac{12 + 2\sqrt{35}}{2}$$

$$[\because \ (a + b)^2 = a^2 + b^2 + 2ab]$$

$$= 6 + \sqrt{35} = 6 + 5.9160 = 11.9160$$

14. *(b)* $\sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + \sqrt{225}}}}}$

$$= \sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + 15}}}}$$

$$= \sqrt{343 + \sqrt{307 + \sqrt{273 + 16}}}$$

$$= \sqrt{343 + \sqrt{307 + 17}}$$

$$= \sqrt{343 + 18} = \sqrt{361} = 19$$

15. *(c)* $\dfrac{1}{\sqrt{9} - \sqrt{8}} - \dfrac{1}{\sqrt{8} - \sqrt{7}} + \dfrac{1}{\sqrt{7} - \sqrt{6}} - \dfrac{1}{\sqrt{6} - \sqrt{5}} + \dfrac{1}{\sqrt{5} - \sqrt{4}}$

$$= (\sqrt{9} + \sqrt{8}) - (\sqrt{8} + \sqrt{7}) + (\sqrt{7} + \sqrt{6}) - (\sqrt{6} + \sqrt{5}) + (\sqrt{5} + \sqrt{4})$$

[on rationalisation]

$$= \sqrt{9} + \sqrt{4} = 3 + 2 = 5$$

16. *(d)* $\sqrt{9 - 2\sqrt{14}} = \sqrt{7 + 2 - 2 \times \sqrt{7} \times \sqrt{2}}$

$$= \sqrt{(\sqrt{7} - \sqrt{2})^2} = \sqrt{7} - \sqrt{2}$$

17. *(a)* Given, $p^x = r^y = m$ and

$$r^w = p^z = n$$

Now, $\quad p^x = r^y$

On multiply power with w on both sides, we get

$$(p^x)^w = (r^y)^w \Rightarrow p^{xw} = r^{yw}$$

$$\Rightarrow \quad p^{xw} = (r^w)^y \qquad \text{...(i)}$$

On putting $r^w = p^z$ in Eq. (i), we get

$$p^{xw} = (p^z)^y \quad \Rightarrow \quad p^{xw} = p^{zy}$$

On comparing both sides, we get

$$\therefore \quad xw = zy$$

18. *(c)* Given, $a^x = b^y = c^z = k$ [Let]

$$\Rightarrow \quad a = k^{1/x}, \ b = k^{1/y}$$

and $\quad c = k^{1/z}$

$$\therefore \quad abc = k^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$$

$$\Rightarrow \quad 1 = k^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} = k^0$$

$$[\because \ abc = 1, \text{given}]$$

On comparing both sides, we get

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$\Rightarrow \quad xy + yz + zx = 0$$

19. *(b)* $a = \dfrac{\sqrt{5}+1}{\sqrt{5}-1} \times \dfrac{\sqrt{5}+1}{\sqrt{5}+1}$

$= \dfrac{(\sqrt{5}+1)^2}{5-1} = \dfrac{3+\sqrt{5}}{2}$

$b = \dfrac{\sqrt{5}-1}{\sqrt{5}+1} \times \dfrac{\sqrt{5}-1}{\sqrt{5}-1}$

$= \dfrac{(\sqrt{5}-1)^2}{5-1} = \dfrac{3-\sqrt{5}}{2}$

$\therefore a^2 + b^2 = \left(\dfrac{3+\sqrt{5}}{2}\right)^2 + \left(\dfrac{3-\sqrt{5}}{2}\right)^2$

$= \dfrac{2(9+5)}{4} = 7$

Also, $ab = \dfrac{\sqrt{5}+1}{\sqrt{5}-1} \times \dfrac{\sqrt{5}-1}{\sqrt{5}+1} = 1$

$\therefore \dfrac{a^2+ab+b^2}{a^2-ab+b^2} = \dfrac{(a^2+b^2)+ab}{(a^2+b^2)-ab}$

$= \dfrac{7+1}{7-1} = \dfrac{4}{3}$

20. *(d)* $3\sqrt{5} + \sqrt{125} = 17.88$

$\Rightarrow 3\sqrt{5} + 5\sqrt{5} = 17.88$

$\Rightarrow 8\sqrt{5} = 17.88$

$\Rightarrow \sqrt{5} = \dfrac{17.88}{8} = 2.235$

$\therefore \sqrt{80} + 6\sqrt{5} = 4\sqrt{5} + 6\sqrt{5}$

$= 10\sqrt{5} = 10 \times 2.235 = 22.35$

21. *(a)* $N = 2^{0.15}$

$\Rightarrow N = (2)^{3/20} \Rightarrow (N)^b = (2)^{3b/20}$

But, $N^b = 16$

$\therefore 16 = (2)^{3b/20} \Rightarrow 2^4 = (2)^{3b/20}$

$\Rightarrow 4 = \dfrac{3b}{20} \Rightarrow b = \dfrac{80}{3}$

22. *(d)* Since $15 = 3 \times 5 = 3 \times 6^B$

As, $3 = \dfrac{6}{2} = \dfrac{6^1}{6^A} = 6^{1-A}$

$\therefore 15 = 3 \times 6^B = 6^{1-A} \times 6^B$

$\Rightarrow 6^Q = 6^{1-A} \times 6^B \Rightarrow 6^Q = 6^{1-A+B}$

$\Rightarrow Q = 1 - A + B$

23. *(a)* LCM of 4, 6, 12 = 12

$\sqrt[4]{3} = \sqrt[12]{3^3} = \sqrt[12]{27}$

$\sqrt[6]{10} = \sqrt[12]{10^2} = \sqrt[12]{100}$

$\sqrt[12]{25} = \sqrt[12]{25}$

Clearly, $\sqrt[12]{100} > \sqrt[12]{27} > \sqrt[12]{25}$

Thus, $\sqrt[6]{10} > \sqrt[4]{3} > \sqrt[12]{25}$

24. *(c)* Given, $(3.7)^x = (0.037)^y = 10000$

$\Rightarrow (3.7)^x = 10^4$ and $(0.037)^y = 10^4$

$\Rightarrow 37 = 10^{4/x+1}$ and $37 = 10^{4/y+3}$

$\Rightarrow 10^{4/x+1} = 10^{4/y+3}$

On comparing both sides, we get

$\dfrac{4}{x} + 1 = \dfrac{4}{y} + 3$

$\Rightarrow \dfrac{4}{x} - \dfrac{4}{y} = 3 - 1 \Rightarrow \dfrac{1}{x} - \dfrac{1}{y} = \dfrac{1}{2}$

Hence, the value of $\dfrac{1}{x} - \dfrac{1}{y}$ is $\dfrac{1}{2}$.

25. *(d)* Here, the greatest six-digit number = 999999

	999
9	$\overline{99}\ \overline{99}\ \overline{99}$
	81
189	1899
×9	1701
1989	19899
×9	17901
	1998

Here, the greatest number of six digit which is perfect square

$= 999999 - 1998 = 998001$

26. *(c)* We have,

$26244 = 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3$

So, to get a perfect cube,

26244 must be divided by $2 \times 2 \times 3 \times 3$ i.e., 36.

27. *(d)* We know that, $(42)^2 = 1764$ and $(43)^2 = 1849$

Since, $1764 < 1780 < 1849$

Hence, the smallest number that must be added to 1780 is $(1849 - 1780)$, i.e. 69.

28. *(c)* Given, total number of tress = 17956

$\therefore$ Number of trees in each row

$= \sqrt{17956} = 134$

29. *(c)* Total rupees collected = ₹ 32.49

$= 32.49 \times 100$ paise = 3249 paise

$\therefore$ Number of member in the group

$= \sqrt{3249} = 57$

30. *(a)* Here, number of soldier arranged

$= 8160 - 60 = 8100$

As number of soldier in each row is equal to number of row.

So, number of soldier in each row

$= \sqrt{8100} = 90$

31. *(c)* After Ist hit, height of the ball

$= \dfrac{1}{2}(64)$

After IInd hit, height of the ball

$= \left(\dfrac{1}{2}\right)^2 (64)$

After 16th hit, height of the ball

$= \left(\dfrac{1}{2}\right)^{16} (64)$

$= \dfrac{1}{2^{16}}(2^6) = 2^{-10}$ m

32. *(b)* Given, $x = \dfrac{\sqrt{a+3b} + \sqrt{a-3b}}{\sqrt{a+3b} - \sqrt{a-3b}}$

By rationalising, we get

$x = \dfrac{(\sqrt{a+3b})^2 + (\sqrt{a-3b})^2}{(\sqrt{a+3b})^2 - (\sqrt{a-3b})^2}$

$+ \dfrac{2\sqrt{(a+3b)(a-3b)}}{(\sqrt{a+3b})^2 - (\sqrt{a-3b})^2}$

$x = \dfrac{a+3b+a-3b+2\sqrt{a^2-9b^2}}{a+3b-a+3b}$

$x = \dfrac{2a+2\sqrt{a^2-9b^2}}{a+3b-a+3b}$

$= \dfrac{(a+\sqrt{a^2-9b^2})}{(3b)}$

$3bx^2 - 2ax + 3b = 0$

33. *(a)* Given, $a^x = b$, $b^y = c$ and $xyz = 1$

Now, $a^x = b$

On multiplying both sides by y in power, we get

$(a^x)^y = (b)^y \Rightarrow a^{xy} = b^y$

Here, $b^y = c \Rightarrow a^{xy} = c$

Again, multiply with z in power, we get

$(a^{xy})^z = c^z \Rightarrow a^{xyz} = c^z$

Given, $xyz = 1 \Rightarrow a = c^z$

$\therefore c^z = a$

Hence, the value of c^z is a.

34. *(d)* $196x^4 = x^6$, $196 = \dfrac{x^6}{x^4} = x^2$

$\Rightarrow x = \sqrt{196} = 14$

$\Rightarrow x^3 = 14x^2$

35. *(b)* Given, $\sqrt{10 + \sqrt[3]{x}} = 4$

On squaring both sides, we get

$10 + \sqrt[3]{x} = 16$

$\Rightarrow \sqrt[3]{x} = 16 - 10 = 6$

Now, on cubing both sides, we get

$x = (6)^3 = 216$

Hence, the value of x is 216.

36. *(d)* Expression $= [(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}}$

$= (\sqrt{2})^{(2)^{\frac{\sqrt{2}}{2}}} = (\sqrt{2})^{(2)^{\frac{1}{\sqrt{2}}}}$

$= (2)^{\frac{1}{2} \times 2^{\frac{1}{\sqrt{2}}}} = (2)^{(2)^{\left(\frac{1}{\sqrt{2}} - 1\right)}}$

which denotes a real number but not a rational number.

37. *(d)* Given, $16 \times 8^{n+2} = 2^m$

$\Rightarrow (2)^4 \times 2^{3(n+2)} = 2^m$

$\Rightarrow 2^{(4+3n+6)} = 2^m \Rightarrow 2^{(3n+10)} = 2^m$

On comparing, we get

$3n + 10 = m$

$\Rightarrow m = 3n + 10$

38. *(b)* After first bounce, height of ball

$$= \left(\frac{2}{3}\right)^1 \times 36$$

and after third bounce, height of ball

$$= \left(\frac{2}{3}\right)^3 \times 36$$

$$= \frac{8}{27} \times 36 = \frac{8 \times 4}{3} = \frac{32}{3} \text{ m} = 10\frac{2}{3} \text{ m}$$

Hence, the required height at third bounce is $10\frac{2}{3}$ m.

39. *(c)* Product of four consecutive numbers plus one is always a square.

Illustration 1 Let four consecutive numbers be 3, 4, 5 and 6.

$\therefore \quad (3 \times 4 \times 5 \times 6) + 1 = 361 = (19)^2$

Illustration 2 Let four consecutive numbers be 9, 10, 11 and 12.

$\therefore \ (9 \times 10 \times 11 \times 12) + 1 = 11881 = (109)^2$

40. *(b)* The difference of two consecutive cubes is never divisible by 2.

Illustration 1 Let the two consecutive numbers be 4 and 5.

$\therefore \quad (5)^3 - (4)^3 = 125 - 64 = 61$

Illustration 2 Let the two consecutive numbers be 9 and 10.

$\therefore \ (10)^3 - (9)^3 = 1000 - 729 = 271$

41. *(b)* $\sqrt{\frac{(0.75)^3}{1 - 0.75} + [0.75 + (0.75)^2 + 1]}$

$$= \sqrt{\frac{(0.75)^3}{0.25} + [1.75 + (0.75)^2]}$$

$$= \sqrt{3 \times (0.75)^2 + 1.75 + (0.75)^2}$$

$$= \sqrt{4 \times (0.75)^2 + 1.75} = \sqrt{4} = 2$$

Hence, the required square root is 2.

42. *(d)* $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$

$$= \frac{5 + \sqrt{10}}{5\sqrt{5} - 4\sqrt{5} - 4\sqrt{2} + 5\sqrt{2}}$$

$$= \frac{(5 + \sqrt{10})}{(\sqrt{5} + \sqrt{2})} \times \frac{(\sqrt{5} - \sqrt{2})}{(\sqrt{5} - \sqrt{2})}$$

[by rationalisation]

$$= \frac{5\sqrt{5} - 5\sqrt{2} + \sqrt{50} - \sqrt{20}}{\sqrt{5^2} - \sqrt{2^2}}$$

$[\because (a - b)(a + b) = a^2 - b^2]$

$$= \frac{5\sqrt{5} - 5\sqrt{2} + 5\sqrt{2} - 2\sqrt{5}}{5 - 2} = \frac{3\sqrt{5}}{3} = \sqrt{5}$$

43. *(a)* We have, $x = \frac{91}{216}$

$$\Rightarrow \quad 1 - x = 1 - \frac{91}{216} = \frac{125}{216} = \left(\frac{5}{6}\right)^3$$

$$\Rightarrow (1 - x)^{\frac{1}{3}} = \frac{5}{6} \Rightarrow \frac{1}{(1 - x)^{\frac{1}{3}}} = \frac{6}{5}$$

$$\Rightarrow \ 3 - \frac{1}{(1 - x)^{\frac{1}{3}}} = 3 - \frac{6}{5} = \frac{9}{5}$$

44. *(b)* We have, $\sqrt{2}, 6^{1/4}, 4^{1/3}$

The LCM of 2, 4, 3 are 12.

$$(2^6)^{\frac{1}{12}}; (6^3)^{\frac{1}{12}}; (4^4)^{\frac{1}{12}}$$

$$(64)^{\frac{1}{12}}; (216)^{\frac{1}{12}}; (256)^{\frac{1}{12}}$$

$$\sqrt{2} < 6^{\frac{1}{4}} < 4^{\frac{1}{3}} \text{ or } \sqrt{2} > \sqrt[4]{6} > \sqrt[3]{4}$$

45. *(b)* We have, $x = \sqrt{3} + \sqrt{2}$

$$\because \quad \frac{1}{x} = \frac{1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}}$$

$$= \frac{\sqrt{3} - \sqrt{2}}{3 - 2} = \sqrt{3} - \sqrt{2}$$

$$\therefore \quad x + \frac{1}{x} = 2\sqrt{3}$$

On cubing both sides, we get

$$x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) = 24\sqrt{3}$$

$$\because x^3 + \frac{1}{x^3} + x + \frac{1}{x} = 24\sqrt{3} - 2\left(x + \frac{1}{x}\right)$$

$$= 24\sqrt{3} - 2 \times 2\sqrt{3}$$

$$= 24\sqrt{3} - 4\sqrt{3} = 20\sqrt{3}$$

46. *(a)* Given, $x = 2^{1/3} + 2^{-1/3}$

$$\therefore \quad 2x^3 - 6x - 5 = 2(2^{1/3} + 2^{-1/3})^3 - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 2[2 + 2^{-1} + 3(2^{1/3} + 2^{-1/3})] - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 4 + 2 \times \frac{1}{2} + 6(2^{1/3} + 2^{-1/3}) - 6(2^{1/3} + 2^{-1/3}) - 5$$

$$= 4 + 1 - 5 = 0$$

47. *(b)* We have, $4^x 2^y = 128$

$$\Rightarrow \quad 2^{2x} 2^y = (2)^7$$

$$\Rightarrow \quad 2^{2x + y} = (2)^7$$

$$\Rightarrow \quad 2x + y = 7$$

$$\Rightarrow \quad y = 7 - 2x \quad \ldots(\text{i})$$

Now, $\quad 3^{3x} \cdot 3^{2y} = 9^{xy}$

$$\Rightarrow \quad 3^{3x + 2y} = 3^{2xy}$$

$$\Rightarrow \quad 3x + 2y = 2xy \quad \ldots(\text{ii})$$

$$\Rightarrow \ 3x + 2(7 - 2x) = 2x(7 - 2x)$$

[from Eq. (i)]

$$\Rightarrow \quad 3x + 14 - 4x = 14x - 4x^2$$

$$\Rightarrow \quad 4x^2 - 15x + 14 = 0$$

$$\Rightarrow \quad 4x^2 - 8x - 7x + 14 = 0$$

$$\Rightarrow \quad 4x(x - 2) - 7(x - 2) = 0$$

$$\Rightarrow \quad (4x - 7)(x - 2) = 0$$

$$\Rightarrow \quad x = \frac{7}{4} \text{ or } x = 2$$

$$\therefore \quad y = \frac{7}{2} \text{ or } y = 3$$

$$\therefore \quad x + y = \frac{21}{4} \text{ or } x + y = 5$$

48. *(a)* Refer to question 32.

06

TIME AND DISTANCE

Regularly (4-7) questions have been asked from this chapter. Generally the type of questions vary from simple to quite complex. But if you know the basics, then this section becomes easy to score area.

Distance

The length of the path travelled by any object or a person between two places is known as distance. The unit of distance is m or km.

Time

The duration in hours, minutes or seconds spent to cover a certain distance is called time.

Speed

The distance travelled by any object or a person per unit time is known as speed of that object or a person. The unit of speed is m/s or km/h.

So, $$\text{speed} = \frac{\text{distance}}{\text{time}}$$

EXAMPLE 1. There are 20 poles with a constant distance between each pole. A car takes 24 s to reach the 12th pole. How much time will it take to reach the last pole?

a. 25.25 s b. 17.45 s c. 35.75 s d. 41.45 s

Sol. d. Suppose distance between each pole is 1m.

$\therefore$ Total distance = 19 m

It takes 24 s to cover 11 m.

$\therefore$ To cover 19 m, it will take $\frac{24}{11} \times 19 = 41.45$.

Conversion of Unit

(i) To convert speed of an object from km/h to m/s, multiply it by $\frac{5}{18}$.

e.g. 36 km/h $= 36 \times \frac{5}{18}$ m/s = 10 m/s

(ii) To convert speed of an object from m/s to km/h, multiply it by $\frac{18}{5}$.

e.g. $5 \text{ m/s} = 5 \times \frac{18}{5} \text{ km/h} = 18 \text{ km/h}$

Average Speed

The average speed of an object is defined as total distance travelled divided by total time taken.

i.e. Average speed $= \frac{\text{Total distance}}{\text{Total time}}$

EXAMPLE 2. A man completes 30 km of a journey at 6 km/h and the remaining 40 km of the journey in 5 h. Then, his average speed for the whole journey is

a. 5 km/h b. 7 km/h
c. 7.5 km/h d. None of these

Sol. b. Average speed $= \frac{\text{Total distance travelled}}{\text{Total time taken}} = \frac{70}{10} = 7$ km/h

Relative Speed

The speed of an object with respect to other is called relative speed. Suppose two bodies are moving with speeds of x km/h and y km/h respectively, then their relative speed will be

(i) $(x + y)$ km/h, if both the bodies are moving in opposite directions.

(ii) $(x - y)$ km/h, if both the bodies are moving in same direction.

EXAMPLE 3. Two persons 27 km apart setting out at the same time are together in 9 h, if they walk in the same direction but in 3 h, if they walk in opposite directions. Then, their rates of walking (speeds) are

a. 2 km/h and 4 km/h b. 3 km/h and 5 km/h
c. 4 km/h and 8 km/h d. None of these

Sol. d. Let the first person be walking faster with speed x km/h and second walking with speed y km/h.

Case I Both walking in same directions.

∴ Distance travelled by first person in 9 h = $9x$ km

and distance travelled by second person in 9 h = $9y$ km

As both are 27 km apart

∴ $9x - 9y = 27 \Rightarrow x - y = 3$ ...(i)

Case II Both walking in opposite directions.

∴ Distance travelled by first person in 3 h = $3x$

and distance travelled by second person in 3 h = $3y$

So, by condition, $3x + 3y = 27 \Rightarrow x + y = 9$ ...(ii)

On adding Eqs. (i) and (ii), we get

$\Rightarrow 2x = 12 \Rightarrow x = 6$ km/h

Put the value of x in Eq. (ii), we get

$6 + y = 9 \Rightarrow y = 3$ km/h

So, their speeds are 6 km/h and 3 km/h.

Important Rules and Formulae

Rule 1 If A travels with speed x km/h for t_1 h and with speed y km/h for t_2 h, then average speed $= \frac{xt_1 + yt_2}{t_1 + t_2}$

EXAMPLE 4. A man walks at the rate of 5 km/h for 6 h and at 4 km/h for 12 h. Find out the average speed (in km/h) of the man.

a. $2\frac{1}{3}$ b. $5\frac{1}{3}$ c. $7\frac{1}{3}$ d. None of these

Sol. d. Here, $x = 5$ km/h, $y = 4$ km/h, $t_1 = 6$ h and $t_2 = 12$ h

∴ Average speed $= \frac{xt_1 + yt_2}{t_1 + t_2} = \frac{5 \times 6 + 4 \times 12}{6 + 12} = \frac{30 + 48}{18}$

$= \frac{78}{18} = 4\frac{1}{3}$ km/h

Rule 2 If a man/vehicle covers two equal distances with the speed of x km/h and y km/h respectively, then the average speed of the man/vehicle for complete journey will be $\frac{2xy}{x + y}$.

EXAMPLE 5. A man covers half of his journey at 6 km/h and the remaining half at 3 km/h. Then, his average speed is

a. 1 km/h b. 2 km/h c. 3 km/h d. 4 km/h

Sol. d. Here, $x = 6$ km/h and $y = 3$ km/h

∴ Average speed $= \frac{2xy}{x + y} = \frac{2 \times 6 \times 3}{6 + 3} = \frac{36}{9} = 4$ km/h

Rule 3 If a man changes his speed to $\left(\frac{x}{y}\right)$ of his usual speed and gets late by t min, then the usual time taken by him $= \frac{t \times x}{(y - x)}$.

Note If a person changes his speed to $\left(\frac{x}{y}\right)$ of his usual and reaches early by t min, then usual time taken by him $= \frac{t \times x}{(x - y)}$.

EXAMPLE 6. If a man travels with a speed of 2/5 times of his original speed and he reached his office 15 min late to fixed time, then the time taken with his original speed, is

a. 10 min b. 15 min c. 20 min d. 25 min

Sol. a. Here, $x = 2$, $y = 5$ and $t = 15$ min

∴ Required time $= \frac{x \times t}{y - x} = \frac{2 \times 15}{5 - 2}$

$= \frac{2 \times 15}{3} = 2 \times 5 = 10$ min

Rule 4 When a man travels from A to B with a speed x km/h and reaches t_1 h later after the fixed time and when he travels with a speed y km/h from A to B, he reaches his destination t_2 h before the fixed time, then distance between A and $B = \frac{xy(t_1 + t_2)}{y - x}$ km.

EXAMPLE 7. Walking at 3 km/h, Rajeev reaches his school 5 min late, if he walks at 4 km/h, he will be 5 min early. The distance of Rajeev's school from his house is

a. $1\frac{1}{2}$ km b. 2 km c. $2\frac{1}{2}$ km d. 5 km

Sol. b. Here, $x = 3$ km/h, $y = 4$ km/h, $t_1 = \frac{5}{60}$ h and $t_2 = \frac{5}{60}$ h

$$\therefore \text{ Required distance} = \frac{xy(t_1 + t_2)}{y - x} = \frac{3 \times 4\left(\frac{5}{60} + \frac{5}{60}\right)}{4 - 3}$$

$$= 3 \times 4 \times \frac{10}{60} = 2 \text{ km}$$

Rule 5 A person travels from A to B with a speed x km/h and reaches t_1 h late. Later he increases his speed by y km/h to cover the same distance and he still gets late by t_2 h, then the distance between A and $B = (t_1 - t_2)(x + y)\frac{x}{y}$.

EXAMPLE 8. A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increased his speed by 5 km/h but still he is late by 10 min. What is the distance of the school from his home?

a. $\frac{100}{3}$ km b. $\frac{50}{3}$ km c. $\frac{100}{7}$ km d. $\frac{2}{5}$ km

Sol. a. Given, $x = 20$ km/h, $y = 5$ km/h, $t_1 = 30$ min $= \frac{1}{2}$ h

and $t_2 = 10$ min $= \frac{1}{6}$ h

$$\therefore \text{ Required distance} = (t_1 - t_2)(x + y)\frac{x}{y}$$

$$= \left(\frac{1}{2} - \frac{1}{6}\right)(20 + 5) \times \frac{20}{5} = \frac{100}{3} \text{ km}$$

Problem Based on Trains

Problems based on trains are same as the problems related to speed, Time and Distance. The only difference is that the length of the moving object (train) is taken into consideration in these types of problems.

Rule 6 (i) Time taken by a train x m long in passing a single post or pole or standing man = Time taken by the train to cover x m.

(ii) Time taken by a train x m long in passing an object of length y m = Time taken by the train to cover $(x + y)$ m.

EXAMPLE 9. A 100 m long train is moving at a speed of 60 km/h. In what time will it cross a signal pole?

a. 6 s b. 12 s c. 15 s d. None of these

Sol. a. Here, speed of train $= 60$ km/h $= 60 \times \frac{5}{18} = \frac{50}{3}$ m/s

and distance covered by train in passing a pole

= Length of the train

$\because$ Length of train = 100 m

$$\therefore \text{Time taken to pass the pole} = \frac{\text{Distance travelled by train}}{\text{Speed of train}} = 6 \text{ s}$$

Rule 7 Time taken by a train of length l_1 metre moving at a speed of s_1 m/s to cross another train of length l_2 moving at a speed of s_2 m/s in same/opposite direction is $\frac{l_1 + l_2}{s_1 \pm s_2}$

Use '+' for opposite direction.

Use '–' for same direction.

EXAMPLE 10. Two trains of lengths 110 m and 130 m travel on parallel track. If they move in the same direction, the first one which is faster takes one minute to pass the other one completely. If they move in opposite directions then they pass each other in 3s, then the speed of the trains is

a. 41 m/s and 39 m/s b. 32 m/s and 43 m/s
c. 42 m/s and 38 m/s d. None of these

Sol. c. Let v_1 be the velocity of faster train and v_2 be the velocity of slower train.

Case I If they move in the same direction, then

$$\frac{110 + 130}{v_1 - v_2} = 60 \text{ s} \Rightarrow v_1 - v_2 = 4 \quad \ldots\text{(i)}$$

Case II If they move in opposite directions,

$$\frac{110 + 130}{v_1 + v_2} = 3 \Rightarrow v_1 + v_2 = 80 \quad \ldots\text{(ii)}$$

On adding Eqs. (i) and (ii), we get $2v_1 = 84 \Rightarrow v_1 = 42$ m/s

Now, putting the value of v_1 in Eq. (ii), we get

$$42 + v_2 = 80,\ v_2 = 38 \text{ m/s} = 3 \times 4 \times \frac{10}{60} = 2 \text{ km}$$

Rule 8 If two trains start at the same time from points A and B towards each other and after crossing each other, they take t_1 and t_2 time in reaching points B and A respectively, then ratio of their speed $= \sqrt{\frac{t_2}{t_1}}$

or $$\frac{S_1}{S_2} = \sqrt{\frac{t_2}{t_1}}$$

EXAMPLE 11. Two trains, one from Howrah to Patna and the other from Patna to Howrah, start simultaneously. After they meet, the trains reach their destinations after 9 h and 16 h, respectively. What is the ratio of their speeds?

a. 4:3
b. 3:1
c. 4:5
d. 3:2

Sol. a. Let us name the trains as A and B. Then, A's speed : B's speed $= \sqrt{t_2} : \sqrt{t_1} = \sqrt{16} : \sqrt{9} = 4 : 3$.

Problems Based on Boat and Stream

Boats and streams is an application of speed, time and distance. Some of the important terms are explained below

- **Still water** If the speed of water is zero, then water is considered to be still water.
- **Stream water** If the water of a river is moving at a certain speed, then it is called stream water.
- **Downstream motion** In water, the direction along the stream is called downstream.
- **Upstream motion** In water, the direction against the stream is called upstream.

Some rules and problems related to boat and stream are given below

Rule 9 If the speed of a boat in still water be x km/h and speed of stream be y km/h, then

(i) Speed of boat downstream $= (x + y)$ km/h
(ii) Speed of boat upstream $= (x - y)$ km/h

EXAMPLE 12. A sailor goes 8 km downstream in 40 min and returns back in 1 h. Then, the speed of the sailor in still water and the speed of the current is

a. 5 km/h and 3 km/h
b. 10 km/h and 2 km/h
c. 7 km/h and 10 km/h
d. None of these

Sol. b. Let the speed of the sailor in still water $= x$ km/h

and speed of the current (stream) $= y$ km/h

Then, speed of the sailor downstream $= (x + y)$ km/h

and speed of the sailor upstream $= (x - y)$ km/h

$\therefore$ Time to travel 8 km downstream = 40 min $= \frac{40}{60}$ h $= \frac{2}{3}$ h

$$\Rightarrow \quad \frac{8}{x+y} = \frac{2}{3} \quad \Rightarrow x + y = \frac{24}{2} \qquad \left[\because \frac{\text{distance}}{\text{speed}} = \text{time}\right]$$

$$\Rightarrow \quad x + y = 12 \qquad \ldots\text{(i)}$$

and time to return = 1 h $\Rightarrow \frac{8}{x-y} = 1$

$$\Rightarrow \quad x - y = 8 \qquad \ldots\text{(ii)}$$

On adding Eqs. (i) and (ii), we get $x = 10$ and $y = 2$

Hence, speed of sailor in still water = 10 km/h

Speed of current (stream) = 2 km/h

Rule 10 Let the speed of boat in downstream $= u$ km/h and speed of boat in upstream $= v$ km/h, then

(i) Speed of boat in still water $= \frac{1}{2}(u + v)$ km/h

(ii) Speed of stream (current) $= \frac{1}{2}(u - v)$ km/h

EXAMPLE 13. The speed of boat upstream and speed of boat downstream are 7 km/h and 13 km/h, respectively. Then, the speed of stream and speed of boat in still water is

a. 10 km/h and 3 km/h
b. 15 km/h and 9 km/h
c. 20 km/h and 6 km/h
d. 40 km/h and 12 km/h

Sol. a. Let the speed of boat in downstream and upstream be u km/h and v km/h, respectively.

Given, $v = 7$ km/h and $u = 13$ km/h

Speed of boat in still water $= \frac{1}{2}(7 + 13) = \frac{1}{2}(20) = 10$ km/h

Speed of the stream $= \frac{1}{2}(13 - 7) = \frac{1}{2}(6) = 3$ km/h

RACES

A contest of speed in running, driving, riding, sailing or rowing over a specified distance is called a race.

Suppose A and B are two contestants in a race. If A beats B by x m and length of the track is dm.

Then, Distance travelled by B when A finishes the race

$$= d - x \text{ and } \frac{\text{Speed of } A}{\text{Speed of } B} = \frac{d}{d - x}$$

Note (i) A gives B a start of x m means when A starts at starting point, B starts x m ahead from starting point.

(ii) A gives B a start of t s means A starts t s after B starts from the same point.

EXAMPLE 14. In a 200 m race, A can beat B by 50m and B can beat C by 8m. In the same race, A can beat C by what distance?

a. 60 m
b. 72 m
c. 56 m
d. 66 m

Sol. c. A can beat B by 50 m.

$$\therefore \quad \frac{\text{speed of } A}{\text{speed of } B} = \frac{200}{200 - 50} = \frac{200}{150}$$

Also, B can beat C by 8m,

$$\therefore \quad \frac{\text{speed of } B}{\text{speed of } C} = \frac{200}{200 - 8} = \frac{200}{192}$$

$$\text{Now, } \frac{\text{speed of } A}{\text{speed of } C} = \frac{\text{speed of } B}{\text{speed of } C} \times \frac{\text{speed of } A}{\text{speed of } B} = \frac{200}{150} \times \frac{200}{192} = \frac{200}{144}$$

So, A beats C by $(200 - 144) = 56$ m

PRACTICE EXERCISE

1. A car completes a journey in 6 h with a speed of 50 km/h. At what speed must it travel to complete the journey in 5 h?

(a) 60 km/h (b) 55 km/h (c) 45 km/h (d) 61 km/h

2. Kiran covers a certain distance at 80 km/h and returns back to the same point at 20 km/h. Then, the average speed during the whole journey is

(a) 35 km/h (b) 32 km/h (c) 30 km/h (d) 28 km/h

3. Normally Sarita takes 3 h to travel between two stations with a constant speed. One day her speed was reduced by 12 km/h and she took 45 min more to complete the journey. Then, the distance between the two stations is

(a) 60 km (b) 120 km (c) 180 km (d) 95 km

4. Rani goes to school at 10 km/h and reaches the school 6 min late. Next day, she covers this distance at 12 km/h and reaches the school 9 min earlier than the scheduled time. What is the distance of her school from her house?

(a) 16 km (b) 12 km (c) 10 km (d) 15 km

5. A man travels first 50 km at 25 km/h next 40 km at 20 km/h and then 90 km at 15 km/h. His average speed (in m/s) for the whole journey is

(a) 18 (b) 5 (c) 10 (d) 36

6. A boy is running at a speed of p km/h to cover a distance of 1 km. But due to the slippery ground, his speed is reduced by q km/h ($p > q$). If he takes r h to cover the distance, then

(a) $\frac{1}{r} = \frac{pq}{p+q}$ (b) $\frac{1}{r} = p + q$ (c) $r = p - q$ (d) $\frac{1}{r} = p - q$

7. A train passes telegraph post in 40 s moving at a rate of 36 km/h. Then, the length of the train is

(a) 400 m (b) 500 m (c) 450 m (d) 395 m

8. A person can run around a circular path of radius 21 m in 44 s. In what time will the same person run a distance of 3 km?

(a) 18 min 40 s (b) 16 min 30 s
(c) 18 min 30 s (d) 16 min 40 s

9. A car is ahead of a scooter by 30 km. Car goes at the rate of 50 km/h and the scooter goes at the rate of 60 km/h. The scooter overtakes the car after

(a) 3 h (b) 3.5 h (c) 4 h (d) $3\frac{1}{4}$ h

10. Two towns A and B are 250 km apart. A bus starts from A to B at 6 : 00 am at a speed of 40 km/h. At the same time another bus starts from B to A at a speed of 60 km/h. The time of their meeting is

(a) 8 : 30 am (b) 8 : 00 am (c) 9 : 00 am (d) 9 : 15 am

11. A train T_1 leaves a place P at 5:00 am and reaches another place Q at 9 : 00 am another train T_2 leaves the place Q at 7 : 00 am and reaches the place P at 10 : 30 am. The time at which the two trains cross each other is

(a) 8 : 26 am (b) 7 : 56 am (c) 8 : 15 am (d) 8 : 00 am

12. A certain distance is covered at a certain speed. If half of the distance is covered in double time, then the ratio of the two speeds is

(a) 4 : 1 (b) 1 : 4 (c) 2 : 1 (d) 1 : 2

13. Two trains start running at the same time from two stations which are 210 km apart and going in opposite directions cross each other at a distance of 100 km from one of the station. The ratio of their speed is

(a) 11 : 9 (b) 10 : 11 (c) 11 : 10 (d) 9 : 11

14. A police car is ordered to chase a speeding car that is 5 km ahead. The thief car is travelling at an average speed of 80 km/h and the police car pursues it at an average speed of 100 km/h. How long does it take for the police car to overtake the other car?

(a) 17 min (b) 19 min (c) 13 min (d) 15 min

15. Points A and B are 70 km apart on a highway. A car starts from A and another car starts from B at the same time. If they travel in the same direction they meet in 7 h, but they travel towards each other they meet in 1 h. What are the speeds of the cars?

(a) 30 km/h, 40 km/h (b) 36 km/h, 40 km/h
(c) 19 km/h, 20 km/h (d) 40 km/h, 50 km/h

16. Assume that the distance that a car runs on 1 L of petrol varies inversely as the square of the speed at which it is driven. It gives a run of 25 km/L at a speed of 30 km/h. At what speed should it be driven to get a run of 36 km/L?

(a) 12.5 km/h (b) 25 km/h (c) 30 km/h (d) 40 km/h

17. A man standing on a railway platform observes that a train going in one direction takes 4 s to pass him. Another train of same length going in the opposite direction takes 5 s to pass him. The time taken (in seconds) by the two trains to cross each other will be

(a) $\frac{32}{9}$ (b) $\frac{33}{7}$ (c) $\frac{40}{9}$ (d) $\frac{49}{9}$

18. A bullock cart has to cover a distance of 80 km in 10 h. If it covers half of the journey in 3/5th time, what should be its speed to cover the remaining distance in the time left?
(a) 5 km/h (b) 10 km/h (c) 15 km/h (d) 18 km/h

19. By walking at $\frac{3}{4}$ of his usual speed, a man reaches his office 25 min later than usual. His usual time is
(a) 60 min (b) 70 min (c) 75 min (d) 80 min

20. A train covers a distance in 50 min, if it runs at a speed of 48 km/h on an average. The speed at which the train must run to reduce the time of journey to 40 min, will be
(a) 10 km/h (b) 20 km/h (c) 40 km/h (d) 60 km/h

21. Two trains travel in the same direction at 50 km/h and 32 km/h, respectively. A man in the slower train observes that 15 s elapse before the faster train completely passes him. What is the length of the faster train?
(a) 75 m (b) 125 m (c) 150 m (d) $\frac{625}{3}$ m

22. A father and his son start at a point A with speeds of 12 km/h and 18 km/h, respectively and reach another point B. If his son starts 60 min after his father at A and reaches B, 60 min before his father, what is the distance between A and B?
(a) 90 km (b) 72 km (c) 36 km (d) None of these

23. Two trains of lengths 100 m and 150 m are travelling in opposite directions at speeds of 75 km/h and 50 km/h, respectively. What is the time taken by them to cross each other?
(a) 7.4 s (b) 7.2 s (c) 7 s (d) 6.8 s

24. A man can walk uphill at the rate of 2.5 km/h and downhill at the rate of 3.25 km/h. If the total time required to walk a certain distance up the hill and return to the starting position is 4 h 36 min, what is the distance he walked up the hill?
(a) 3.5 km (b) 4.5 km (c) 5.5 km (d) 6.5 km

25. A boat goes 30 km upstream and 44 km downstream in 10 h. In 13 h it can go 40 km upstream and 55 km downstream. The speed of the boat in still water is
(a) 9 km/h (b) 8 km/h (c) 4 km/h (d) 3 km/h

26. A motorboat takes 2 h to travel a distance of 9 km down the current and it takes 6 h to travel the same distance against the current. What is the speed of the boat in still water (in km/h)?
(a) 3 (b) 2 (c) 1.5 (d) 10

27. Ram travels from P to Q at 10 km/h and return at 15 km/h. Shyam travels from P to Q and return at 12.5 km/h. If he takes 12 min less than Ram, then what is the distance between P and Q?
(a) 60 km (b) 45 km (c) 36 km (d) 30 km

28. A person x started 5 min earlier at 60 km/h from a place P, then another person y followed him at 48 km/h, started his journey at 3:05 pm. Which of the following is/are correct.
I. At 3 : 15 pm, x and y are 7 km apart.
II. At 3 : 25 pm, y will overtake x
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

29. A boy walking at a speed of 20 km/h reaches his school 30 min late. Next time he increases his speed by 4 km/h but still he is late by 10 min.
Which of the following statement is/are correct?
I. The distance of the school from his home is 50 km/h.
II. If he increases his speed by 10 km/h but still late by 10 min, then distance of the school will be 20 km.
(a) Both I and II are correct (b) Only I is correct
(c) Only II is correct (d) Neither I nor II are correct.

Directions (Q. Nos.30-31) Ramesh and Prateek start running at same time in opposite directions from two points and after passing each other they complete their journeys in 'x' and 'y' h, respectively.
Then, Speed of Ramesh : Speed of Prateek = $\sqrt{y} : \sqrt{x}$

30. Find the ratio of the speeds of Ramesh and Prateek, if Ramesh and Prateek completed their journeys in 16 h and 25 h after passing each other.
(a) 5 : 4 (b) 5 : 3 (c) 4 : 5 (d) 3 : 5

31. What if two persons completed their journey with speeds in the ratio 7 : 9, then how much time they should have taken to complete their journey after they meet each other.
(a) 81 : 49 (b) $\sqrt{7} : \sqrt{9}$ (c) 49 : 81 (d) 14 : 18

PREVIOUS YEARS' QUESTIONS

32. A train 280 m long is moving at a speed of 60 km/h. What is the time taken by the train to cross a platform 220 m long? **2012 I**
(a) 45 s (b) 40 s (c) 35 s (d) 30 s

33. A car travels along the four sides of a square at speeds v, $2v$, $3v$ and $4v$, respectively. If u is the average speed of the car in its travel around the square, then which one of the following is correct? **2012 I**
(a) $u = 2.25\,v$ (b) $u = 3\,v$
(c) $v < u < 2\,v$ (d) $3\,v < u < 4\,v$

34. A car is travelling at a constant rate of 45 km/h. The distance travelled by car from 10 : 40 am to 1 : 00 pm is **2012 II**
(a) 165 km (b) 150 km (c) 120 km (d) 105 km

35. A person travels a certain distance at 3 km/h and reaches 15 min late. If he travels at 4 km/h, he reaches 15 min earlier. The distance he has to travel is **2013 I**

(a) 4.5 km (b) 6 km (c) 7.2 km (d) 12 km

36. If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then the average speed (in km/h) is **2013 I**

(a) $\frac{x+y}{2}$ (b) $\sqrt{xy}$ (c) $\frac{2xy}{x+y}$ (d) $\frac{x+y}{xy}$

37. A sailor sails a distance of 48 km along the flow of a river in 8 h. If it takes 12 h to return the same distance, then the speed of the flow of the river is **2013 I**

(a) 0.5 km/h (b) 1 km/h (c) 1.5 km/h (d) 2 km/h

38. A train running at the speed of 72 km/h goes past a pole in 15 s. What is the length of the train? **2013 II**

(a) 150 m (b) 200 m (c) 300 m (d) 350 m

39. Two cars A and B start simultaneously from a certain place at the speed of 30 km/h and 45 km/h, respectively. The car B reaches the destination 2 h earlier than A. What is the distance between the starting point and destination? **2013 II**

(a) 90 km (b) 180 km (c) 270 km (d) 360 km

40. A man cycles with a speed of 10 km/h and reaches his office at 1 : 00 pm. However, when he cycles with a speed of 15 km/h, he reaches his office at 11 : 00 am. At what speed should he cycle, so that he reaches his office at 12 noon? **2013 II**

(a) 12.5 km/h (b) 12 km/h (c) 13 km/h (d) 13.5 km/h

41. A train takes 9 s to cross a pole. If the speed of the train is 48 km/h, the length of the train is **2014 I**

(a) 150 m (b) 120 m (c) 90 m (d) 80 m

42. A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of the train? **2014 II**

(a) 50 m (b) 100 m (c) 150 m (d) 200 m

43. A train travels at a speed of 40 km/h and another train at a speed of 20 m/s. What is the ratio of speed of the first train to that of the second train? **2014 II**

(a) 2 : 1 (b) 5 : 9 (c) 5 : 3 (d) 9 : 5

44. The distance between two points (A and B) is 110 km. X starts running from point A at a speed of 60 km/h and Y starts running from point B at a speed of 40 km/h at the same time. They meet at a point C, somewhere on the line AB. What is the ratio of AC to BC? **2014 II**

(a) 3 : 2 (b) 2 : 3 (c) 3 : 4 (d) 4 : 3

45. A man rides one-third of the distance from A to B at the rate of x km/h and the remainder at the rate of $2y$ km/h. If he had travelled at a uniform rate of $6z$ km/h, then he could have ridden from A to B and back again in the same time. Which one of the following is correct? **2014 II**

(a) $z = x + y$ (b) $3z = x + y$

(c) $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$ (d) $\frac{1}{2z} = \frac{1}{x} + \frac{1}{y}$

46. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight increased by 30 min. The duration of the flight is **2015 I**

(a) 1 h (b) 2 h (c) 3 h (d) 4 h

47. With a uniform speed, a car covers a distance in 8 h. Had the speed been increased by 4 km/h, the same distance could have been covered in 7 h and 30 min. What is the distance covered? **2015 I**

(a) 420 km (b) 480 km (c) 520 km (d) 640 km

48. A car travels the first one-third of a certain distance with a speed of 10 km/h, the next one-third distance with a speed of 20 km/h and the last one-third distance with a speed of 60 km/h. The average speed of the car for the whole journey is **2015 I**

(a) 18 km/h (b) 24 km/h (c) 30 km/h (d) 36 km/h

49. A man rows 32 km downstream and 14 km upstream, and he takes 6 h to cover each distance. What is the speed of the current? **2015 I**

(a) 0.5 km/h (b) 1 km/h (c) 1.5 km/h (d) 2 km/h

50. A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80 m, how far must the winning post from the starting point be so that A and B might reach it at the same time? **2015 I**

(a) 200 m (b) 300 m (c) 270 m (d) 160 m

51. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the policeman chases him. The thief and the policeman run at the speed of 10 km/h and 11 km/h, respectively. What is the distance between them after 6 min? **2015 I**

(a) 100 m (b) 120 m (c) 150 m (d) 160 m

52. Two persons A and B start simultaneously from two places c km apart and walk in the same direction. If A travels at the rate of p km/h and B travels at the rate of q km/h, then A has travelled before he overtakes B a distance of **2015 I**

(a) $\frac{qc}{p+q}$ km (b) $\frac{pc}{p-q}$ km (c) $\frac{qc}{p-q}$ km (d) $\frac{pc}{p+q}$ km

53. By increasing the speed of his car by 15 km/h, a person covers 300 km distance by taking an hour less than before. The original speed of the car was **2015 II**

(a) 45 km/h (b) 50 km/h (c) 60 km/h (d) 75 km/h

54. Two trains, one is of 121 m in length at the speed of 40 km/h and the other is of 99 m in length at the speed of 32 km/h are running in opposite directions. In how much time will they be completely clear from each other from the moment they meet ? **2015 II**

(a) 10 s (b) 11 s (c) 16 s (d) 21 s

55. The speeds of three buses are in the ratio 2 : 3 : 4. The time taken by these buses to travel the same distance will be in the ratio. **2015 II**

(a) 2 : 3 : 4 (b) 4 : 3 : 2
(c) 4 : 3 : 6 (d) 6 : 4 : 3

56. Two trains are moving in the same direction at 1.5 km/min and 60 km/h, respectively. A man in the faster train observes that it takes 27 s to cross the slower train. The length of the slower train is **2015 II**

(a) 225 m (b) 230 m (c) 240 m (d) 250 m

57. In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race? **2015 II**

(a) 2 (b) 2.04 (c) 2.08 (d) 3.2

58. Three athletes run a 4 km race. Their speeds are in the ratio 16 : 15 : 11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third position is **2015 II**

(a) 1000 m (b) 800 m (c) 750 m (d) 600 m

59. A motorboat, whose speed is 15 km/h in still water goes 30 km downstream and comes back in a total of 4h and 30 min. The speed of the stream is **2015 II**

(a) 4 km/h (b) 5 km/h
(c) 6 km/h (d) 10 km/h

60. A bike consumes 20 mL of petrol per kilometre, if it is driven at a speed in the range of 25-50 km/h and consumes 40 mL of petrol per kilometre at any other speed. How much petrol is consumed by the bike in travelling a distance of 50 km, if the bike is driven at a speed of 40 km/h for the first 10 km, at a speed of 60 km/h for the next 30 km and at a speed of 30 km/h for the last 10 km? **2015 II**

(a) 1 L (b) 1.2 L (c) 1.4 L (d) 1.6 L

61. A passenger train takes 1 h less for a journey of 120 km, if its speed is increased by 10 km/h from its usual speed. What is its usual speed? **2016 I**

(a) 50 km/h (b) 40 km/h (c) 35 km/h (d) 30 km/h

62. A man walking at 5 km/h noticed that a 225 m long train coming in the opposite direction crossed him in 9 s. The speed of the train is **2016 I**

(a) 75 km/h (b) 80 km/h (c) 85 km/h (d) 90 km/h

63. A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by 1 km/h, then he takes 20 min more to cover the same distance. The original average speed of the cyclist is **2016 I**

(a) 5 km/h (b) 6 km/h
(c) 7 km/h (d) None of these

64. In a race of 1000 m, A beats B by 100 m or 10 s. If they start a race of 1000 m simultaneously from the same point and if B gets injured after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B? **2016 I**

(a) 65 s (b) 60 s (c) 50 s (d) 45 s

65. In a race A, B and C take part. A beats B by 30 m, B beats C by 20 m and A beats C by 48 m.

Which of the following is/are correct?

I. The length of the race is 300 m.

II. The speeds of A, B and C are in the ratio 50 : 45 : 42.

Select the correct answer using codes given below.

(a) Only I (b) Only II **2016 I**
(c) Both I and II (d) Neither I nor II

ANSWERS

1	a	2	b	3	c	4	d	5	b	6	d	7	a	8	d	9	a	10	a
11	b	12	a	13	c	14	d	15	a	16	b	17	c	18	b	19	c	20	d
21	a	22	b	23	b	24	d	25	b	26	a	27	d	28	a	29	c	30	a
31	a	32	d	33	c	34	d	35	b	36	c	37	b	38	c	39	b	40	b
41	b	42	d	43	b	44	a	45	c	46	a	47	b	48	a	49	c	50	a
51	a	52	b	53	c	54	b	55	d	56	a	57	b	58	a	59	b	60	d

HINTS AND SOLUTIONS

1. *(a)* Let speed be v km/h.

We know that,

Distance = Speed × Time

$6 \times 50 = 5 \times v$

$\Rightarrow \quad v = \dfrac{6 \times 50}{5} = 60$ km/h

2. *(b)* Average speed $= \dfrac{2xy}{x+y}$ [Rule 2]

$= \dfrac{2 \times 80 \times 20}{80+20} = 32$ km/h

3. *(c)* Let Sarita's speed be u and distance is constant.

Then, $\quad u \times 3 = (u-12)\dfrac{15}{4}$

$\left[\because 3\text{ h} + 45\text{ min} = \dfrac{15}{4}\text{ h}\right]$

$\Rightarrow \quad u = \dfrac{45 \times 4}{3} = 60$ km/h

$\therefore$ Distance = Speed × Time

$= 60 \times 3 = 180$ km

4. *(d)* Here, $x = 10$, $y = 12$, $t_1 = \dfrac{6}{60}$ h and $t_2 = \dfrac{9}{60}$ h

$\therefore$ Required distance $= \dfrac{xy(t_1 + t_2)}{y - x}$

[Rule 4]

$= \dfrac{10 \times 12 \times 15}{60 \times 2}$

$= 15$ km

5. *(b)* Total distance covered

$= 50 + 40 + 90 = 180$ km

Total time taken

$= \left(\dfrac{50}{25} + \dfrac{40}{20} + \dfrac{90}{15}\right) = 10$ h

$\therefore$ Average speed for the whole journey

$= \dfrac{\text{Total distance travelled}}{\text{Total time taken}}$

$= \dfrac{180}{10} = 18$ km/h

$\therefore$ 18 km/h $= \dfrac{18 \times 5}{18}$ m/s $= 5$ m/s

6. *(d)* Actual speed of boy $= (p - q)$ km/h

Time taken to cover 1 km $= \dfrac{1}{p-q}$

$\therefore \quad \dfrac{1}{p-q} = r \Rightarrow \dfrac{1}{r} = p - q$

7. *(a)* Length of train = Distance covered in 40 s at the rate of 36 km/h.

[Rule 6 (i)]

$\therefore$ Length of train

$= 40 \times 36 \times \dfrac{5}{18} = 400$ m

8. *(d)* Distance travelled in 44 s $= 2\pi r$

$= 2 \times \dfrac{22}{7} \times 21 = 132$ m

$\therefore$ Speed $= \dfrac{132}{44} = 3$ m/s

$\left[\because \text{speed} = \dfrac{\text{distance}}{\text{time}}\right]$

Time taken to travel 3 km $= \dfrac{3000}{3}$

$= 1000\text{ s} = \dfrac{1000}{60}$ min $= 16$ min 40 s

9. *(a)* Distance between car and scooter

$= 30$ km

Relative speed

$y = 60 - 50 = 10$ km/h

So, the time taken by scooter to overtake the car $= \dfrac{30}{10} = 3$ h

10. *(a)* Let the buses meet x h after 6:00 am.

Then, the distance covered by the two buses is 250 km.

$\therefore \quad 40x + 60x = 250$

$\Rightarrow \quad x = \dfrac{250}{100} = 2.5$ h = 2h and 30 min

So, they will meet at 8 : 30 am.

11. *(b)* Let the distance between P and Q be x km and let the two trains meet y h after 7 am.

Then, T_1 covers x km in 4h and T_2 covers x km in $3\dfrac{1}{2}$ h.

$\therefore$ Speed of train $T_1 = \dfrac{x}{4}$ km/h

and speed of train $T_2 = \dfrac{2x}{7}$ km/h

According to the question,

$\dfrac{x(y+2)}{4} + \dfrac{2xy}{7} = x$

$\Rightarrow \quad \dfrac{(y+2)}{4} + \dfrac{2y}{7} = 1$

$\Rightarrow \quad \dfrac{7(y+2)+8y}{28} = 1$

$\Rightarrow \quad 7y + 14 + 8y = 28$

$\therefore \quad y = \dfrac{14}{15}\text{ h} = \dfrac{14}{15} \times 60\text{ min} = 56$ min

So, trains will meet at 7 : 56 am.

12. *(a)* Let x km distance be covered in y h.

Then, speed of object in first case

$= \dfrac{x}{y}$ km/h

As, half of this distance is covered in double time.

Then, speed of object in second case

$= \dfrac{x}{2} \div 2y = \dfrac{x}{2} \times \dfrac{1}{2y} = \dfrac{x}{4y}$ km/h

$\therefore$ Ratio of first and second speeds

$= \dfrac{x}{y} : \dfrac{x}{4y} = 1 : \dfrac{1}{4} = 4 : 1$

13. *(c)* Let the speed of two trains be x km/h and y km/h, respectively. Then, the time taken by first train to cover 110 km = Time taken by second train to cover 100 km .

Thus, $\quad \dfrac{110}{x} = \dfrac{100}{y} \Rightarrow \dfrac{x}{y} = \dfrac{110}{100}$

$\therefore \quad x : y = 11 : 10$

14. *(d)* Distance travelled by thief car in one hour = 80 km

Distance travelled in one hour by police car = 100 km

So, police travels extra 20 km in 1 h.

So, to overtake thief, police car has to travel 5 km extra.

$\therefore$ Time $= \dfrac{5}{20} = \dfrac{1}{4}\text{ h} = \dfrac{60}{4}$ min $= 15$ min

15. *(a)* Let the speed of car $A = x$ km/h

Let the speed of car $B = y$ km/h

Total distance covered by b other the cars in 1h = 70 km

$x(1) + y(1) = 70$

$x + y = 70$ ...(i)

[in opposite direction]

Distance covered by both the cars in

7 h = 70 km

But in same direction

$7x - 7y = 70 \Rightarrow x - y = 10$ ...(ii)

added Eq. (i) and (ii), we get

$2x = 80$, $x = 40$ km/h, $y = 30$ km/h

16. *(b)* Let the speed of car be v and distance covered by car in one litre be A.

$\therefore \quad A \propto \dfrac{1}{v^2} \Rightarrow A = \dfrac{K}{v^2}$

Then, $A = 25$ km/h and $v = 30$ km/h

So, $\quad 25 = \dfrac{K}{(30)^2} \Rightarrow K = 900 \times 25$

$\Rightarrow \quad K = 22500 \Rightarrow A = \dfrac{22500}{v^2}$

When $A = 36$ km/L

$$v^2 = \frac{22500}{36} \Rightarrow v = \sqrt{\frac{22500}{36}} = \frac{150}{6}$$

Hence, the speed of a car is 25 km/h.

17. *(c)* Let the length of each train be I m.

$\Rightarrow$ Speed of first train $= \left(\frac{I}{4}\right)$ m/s

and speed of second train $= \left(\frac{I}{5}\right)$ m/s

As, both trains are moving in opposite direction.

Time taken to cross each other

$$= \frac{I + I}{\frac{I}{4} + \frac{I}{5}} \quad \text{[Rule 7]}$$

$$= \left(\frac{2I}{\frac{9I}{20}}\right) \text{s} = \left(\frac{20 \times 2}{9}\right) = \frac{40}{9} \text{ s}$$

18. *(b)* Given, total distance to cover in 10 h $= 80$ km

If it covers 40 km in $\frac{3}{5}$ th of time i.e. 40 km in 6 h.

$\therefore$ Remaining time $= 10 - 6 = 4$ h

Remaining distance $= 40$ km

$\therefore$ Required speed $= (40 \div 4)$ km/h $= \frac{40}{4} = 10$ km/h

19. *(c)* Here, $x = 3$ m/s, $y = 4$ m/s and $t = 25$ min

$\therefore$ Required time $= \frac{x \times t}{y - x}$ [Rule 3]

$$= \frac{3 \times 25}{4 - 3} = 75 \text{ min}$$

20. *(d)* Distance travelled in 1h $= 48$ km

$\therefore$ Distance travelled in 50 min

$$= \frac{48}{60} \times 50 = 40 \text{ km}$$

Time to be reduced $= \frac{40}{60}$ h

$\therefore$ Required speed

$$= \frac{40}{40/60} = \frac{40 \times 60}{40} = 60 \text{ km/h}$$

21. *(a)* Let the length of faster train be x km.

$\because$ Trains travel in the same direction.

$\therefore$ Relative speed $= (50 - 32) = 18$ km/h

Elapsed time $= 15$ s $= \frac{15}{3600}$ h

Now, time $= \frac{\text{distance}}{\text{speed}}$

$$\Rightarrow \frac{15}{3600} = \frac{x}{18} \Rightarrow x \times 3600 = 18 \times 15$$

$$\therefore x = \frac{18 \times 15}{3600} = 0.075 \text{km} = 75 \text{ m}$$

22. *(b)* Let distance between A and B be x km.

By given condition,

$$\frac{x}{12} - \frac{x}{18} = 2 \Rightarrow 6x = 2 \times 18 \times 12$$

$$\therefore \quad x = \frac{2 \times 18 \times 12}{6} = 72 \text{ km}$$

Hence, the required distance is 72 km.

23. *(b)* Relative speed when trains are in opposite direction

$$= V_1 + V_2$$
$$= 75 + 50 = 125 \text{ km/h}$$
$$= \frac{125 \times 5}{18} \text{ m/s}$$

and total distance covered

$$= (100 + 150) = 250 \text{ m}$$

$\therefore$ Time taken to cross each other

$$= \frac{\text{Total covered distance}}{\text{Relative speed}}$$

$$= \frac{250 \times 18}{125 \times 5} = 7.2 \text{ s}$$

24. *(d)* Let he walked up the hill at a distance of x km.

By given condition,

$$\frac{x}{2.5} + \frac{x}{3.25} = 4\frac{36}{60}$$

$$\Rightarrow x\left(\frac{1}{2.5} + \frac{1}{3.25}\right) = \frac{276}{60}$$

$$\Rightarrow x\left(\frac{2}{5} + \frac{4}{13}\right) = \frac{276}{60}$$

$$\therefore x = \frac{276}{60} \times \frac{65}{46} = 6.5 \text{ km}$$

25. *(b)* Let speed of boat in still water $= x$ km/h

and speed of boat in current water $= y$ km/h

$\therefore$ Downstream speed $= (x + y)$ km/h

Upstream speed $= (x - y)$ km/h [Rule 9]

According to the question,

$$\frac{30}{x - y} + \frac{44}{x + y} = 10 \quad \text{...(i)}$$

and

$$\frac{40}{x - y} + \frac{55}{x + y} = 13 \quad \text{...(ii)}$$

Let $\frac{1}{x - y} = u$ and $\frac{1}{x + y} = v$ ...(iii)

Then, $30u + 44v = 10$

$40u + 55v = 13$

On solving the above equations, we get

$$u = \frac{1}{5}, v = \frac{1}{11}$$

From Eq. (iii), we get

$$x - y = 5 \text{ and } x + y = 11$$

On solving the above equations, we get

$$x = 8, y = 3$$

Thus, the speed of boat in still water $= 8$ km/h

26. *(a)* Let the speed of motorboat be x km/h and the speed of water be y km/h.

Now, speed of boat downstream

$= (x + y)$ km/h [Rule 9]

and speed of boat upstream

$= (x - y)$ km/h

By given condition,

$$\frac{9}{x + y} = 2 \Rightarrow 2x + 2y = 9 \quad \text{...(i)}$$

and

$$\frac{9}{x - y} = 6 \Rightarrow 6x - 6y = 9$$

$$\Rightarrow 2x - 2y = 3 \quad \text{...(ii)}$$

On solving Eqs. (i) and (ii), we get

$x = 3$ km/h and $y = 1.5$ km/h

$\therefore$ Speed of boat $= 3$ km/h

27. *(d)* Let the distance between P and $Q = d$ km. Total time taken by Ram

$$= \frac{d}{10} + \frac{d}{15} = \frac{25d}{150}$$

Total time taken by Shyam $= \frac{2d}{12.5} = \frac{4d}{25}$

According to question,

$$\frac{25d}{150} - \frac{4d}{25} = \frac{12}{60}$$

$$\Rightarrow d\left[\frac{25 - 24}{150}\right] = \frac{1}{5}$$

$$\Rightarrow \frac{d}{150} = \frac{1}{5} \Rightarrow d = 30 \text{ km}$$

28. *(a)* Distance travelled by x in 15 min

$$= 60 \times \frac{15}{60} = 15 \text{ km}$$

Distance travelled by y in 10 min

$$= 48 \times \frac{10}{60} = 8 \text{ km}$$

Difference $= (15 - 8)$ km $= 7$ km

Hence, at 3:15 pm they are 7 km apart.

So, statement I is true. As speed of x is greater than y. So, y will never overtake x. Thus, statement II is false.

29. *(c)* I. Here, $x = 20$ km/h, $y = 4$ km/h, $t_1 = 30$ min, $t_2 = 10$ min

According to formula,

$\therefore$ Required distance

$$= (t_1 - t_2)(x + y)\frac{x}{y} \quad \text{[Rule 5]}$$

$$= \frac{(30 - 10)}{60}(20 + 4)\left(\frac{20}{4}\right)$$

$$= \frac{20}{60} \times 24 \times \frac{20}{4} = 5 \times 8 = 40 \text{ km}$$

So, I is incorrect.

II. Here, $x = 20$ km/h, $y = 10$ km/h

$t_1 = 30$ min, $t_2 = 10$ min

According to formula,

$\therefore$ Required distance

$$= \left(\frac{30 - 10}{60}\right)(20 + 10)\left(\frac{20}{10}\right)$$

[Rule 5]

$$= \frac{20}{60} \times 30 \times \frac{20}{10} = 20 \text{ km}$$

So, II is correct.

30. *(a)* $x = 16$, $y = 25$

According to formula,

$$\therefore \frac{\text{Ramesh's Speed}}{\text{Prateek's Speed}} = \sqrt{\frac{y}{x}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

or 5:4

31. *(a)* Here, we know the relation

$$\frac{\text{Speed of Ramesh}}{\text{Speed of Prateek}} = \sqrt{\frac{y}{x}} \Rightarrow \frac{7}{9} = \sqrt{\frac{y}{x}}$$

On squaring both sides, $\frac{49}{81} = \frac{y}{x} = 81 : 49$

32. *(d)* To cover a distance by train

$= 280 + 220 = 500$ m

Speed of train $= 60$ km/h

$$= 60 \times \frac{5}{18} \text{ m/s}$$

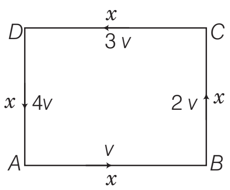
$$= \frac{50}{3} \text{ m/s}$$

$\therefore$ Time taken by train

$$= \frac{\text{total distance}}{\text{speed}}$$

$$= \frac{500}{50/3} = 30 \text{ s}$$

33. *(c)* Let side of a square be x.



$$\because \text{ Average speed } (u) = \frac{\text{total distance}}{\text{total time}}$$

$$= \frac{(x + x + x + x)}{\frac{x}{v} + \frac{x}{2v} + \frac{x}{3v} + \frac{x}{4v}}$$

$$= \frac{4x}{x\left(\frac{1}{v} + \frac{1}{2v} + \frac{1}{3v} + \frac{1}{4v}\right)}$$

$$= \frac{4 \times v}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{48v}{25} = 1.92\,v$$

which lies in the interval $v < u < 2v$.

34. *(d)* Given, speed of a car $= 45$ km/h

Time taken by the car

$$= 140 \text{ min} = \frac{140}{60} \text{ h}$$

So, required distance travelled by car

$=$ speed $\times$ time

$$= 45 \times \frac{140}{60} = 105 \text{ km}$$

35. *(b)* Given, $x = 3$ km/h, $y = 4$ km/h,

$$t_1 = 15 \text{ min} = \frac{15}{60} \text{ h}$$

and $t_2 = 15 \text{ min} = \frac{15}{60}$ h

$$\therefore \text{Required distance} = \frac{xy(t_1 + t_2)}{y - x}$$

[Rule 4]

$$= \frac{3 \times 4\left(\frac{15}{60} + \frac{15}{60}\right)}{4 - 3}$$

$$= 12 \times \frac{30}{60} = 12 \times \frac{1}{2} = 6 \text{ km}$$

36. *(c)* If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then $= \frac{2xy}{x + y}$

[Rule 2]

37. *(b)* Let speed of the flow of water be v km/h and rate of sailing of sailer be u km/h.

Speed of sailor downstream

$= (u + v)$ km/h [Rule 9]

Speed of sailor upstream $= (u - v)$ km/h

Condition I $u + v = \frac{48}{8}$

$\Rightarrow$ $u + v = 6$ ...(i)

Condition II $u - v = \frac{48}{12}$

$\Rightarrow$ $u - v = 4$ ...(ii)

On subtracting Eqs. (ii) from (i), we get

$v = 1$ km/h

38. *(c)* Speed of train $= 72$ km/h

$$= 72 \times \frac{5}{18}$$

$= 20$ m/s

Length of train $=$ distance covered

$x = 20 \times 15$

$x = 300$ m

Length of the train is 300 m.

39. *(b)* Let the distance between two points be x km

Time taken by A – Time taken by $B = 2$

$$\Rightarrow \quad \frac{x}{30} - \frac{x}{45} = 2$$

$$\Rightarrow \quad x\left[\frac{3 - 2}{90}\right] = 2$$

$\Rightarrow$ $x = 90 \times 2 = 180$ km

40. *(b)* Let t h be the time taken by the man to reach his office at speed of 15 km/h.

Then, time taken to reach office at the speed of 10 km/h $= (t + 2)$ h

Now, $s_1 \times t_1 = s_2 \times t_2$

$\Rightarrow$ $10 \times (t + 2) = 15 \times t$

$\Rightarrow$ $10t + 20 = 15t \Rightarrow 5t = 20$

$\therefore$ $t = 4$ h

Now, distance covered to reach office

$= s_1 \times t_1$

$= 10 \times (4 + 2) = 10 \times 6 = 60$ km

Now, speed required to reach office at 12 noon in 5 h.

$\therefore$ Speed $= \frac{60}{5} = 12$ km/h

41. *(b)* Speed of train $= 48$ km/h

$$= \left(48 \times \frac{5}{18}\right) \text{ m/s}$$

Let the length of train be x m

$$x = 48 \times \frac{5}{18} \times 9$$

$x = 120$ m

Length of the train is 120 m.

42. *(d)* Let the speed of a train be x m/s and length be y m.

Condition I When $t = 10$ s

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$\Rightarrow$ $10 = \frac{y}{x}$ $\therefore$ $y = 10x$ ...(i)

Condition II When $t = 20$ s

and total distance $= y + 200$ m

$$\therefore \text{ Time} = \frac{\text{Total distance}}{\text{Speed}}$$

$$20 = \frac{y + 200}{x} \Rightarrow 20x = y + 200$$

$$20 \times \frac{y}{10} = y + 200$$

$\Rightarrow 2y - y = 200$ $\Rightarrow$ $y = 200$

Hence, the length of train is 200 m.

43. *(b)* Given, speed of a train $= 40$ km/h

$$= 40 \times \frac{5}{18} \text{ m/s}$$

Speed of another train $= 20$ m/s

$\therefore$ Required ratio

$$= \frac{\text{Speed of first train}}{\text{Speed of second train}}$$

$$= \frac{40 \times \frac{5}{18}}{20} = \frac{40 \times 5}{20 \times 18} = \frac{2 \times 5}{18}$$

$$= \frac{10}{18} = \frac{5}{9} \text{ or } 5 : 9$$

44. *(a)* Given, distance between two points (A and B) = 110 km

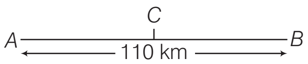
Their relative speed

$= 60 + 40 = 100$ km/h

Time after which they meet

$= \dfrac{\text{Total distance}}{\text{Relative speed}}$

$= \dfrac{110}{100}$



$= 1.10$ h

Distance covered by A in 1.1 h

$= AC = 60 \times 1.1 = 66$ km

Remaining distance

$= BC = 110 - 66 = 44$ km

$\therefore$ Required ratio $= AC : BC$

$= 66 : 44 = 3 : 2$

45. *(c)* Let the total distance be d.

Time taken to cover $\frac{1}{3}$rd distance,

$$t_1 = \frac{\frac{1}{3}d}{x} = \frac{d}{3x} \quad \left[\because \text{time} = \frac{\text{distance}}{\text{speed}}\right]$$

Remaining distance $= d - \frac{1}{3}d = \frac{2}{3}d$

Time taken to cover $\frac{2}{3}$rd distance,

$$t_2 = \frac{\frac{2}{3}d}{2y} = \frac{2d}{6y} = \frac{d}{3y}$$

Time taken to cover distance from A to B and B to A,

$$\therefore \quad t = \frac{2d}{6z} = \frac{d}{3z}$$

According to the question,

$$t_1 + t_2 = t \Rightarrow \frac{d}{3x} + \frac{d}{3y} = \frac{d}{3z}$$

$$\Rightarrow \frac{1}{3x} + \frac{1}{3y} = \frac{1}{3z} \quad \therefore \quad \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

46. *(a)* Let the original speed of an aircraft be x km/h and its reduced speed $= (x - 200)$ km/h.

Condition I

Time taken by aircraft to cover 600 km $= \dfrac{600}{x}$ h.

Condition II Time taken by aircraft to cover

600 km $= \dfrac{600}{(x - 200)}$ h

According to the question,

$$\frac{600}{x - 200} - \frac{600}{x} = \frac{1}{2} \quad \left[\because 30 \text{ min} = \frac{1}{2} \text{ h}\right]$$

On dividing both sides by 600, we get

$$\frac{1}{x - 200} - \frac{1}{x} = \frac{1}{600 \times 2}$$

$$\Rightarrow \quad \frac{x - (x - 200)}{x(x - 200)} = \frac{1}{1200}$$

$\Rightarrow \ x^2 - 200x - 240000 = 0$

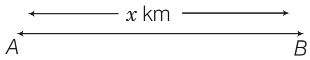
$\Rightarrow \ (x - 600)(x + 400) = 0$

$\Rightarrow \quad x = 600$ km/h

or $\quad x \neq -400$ km/h

$\therefore$ Required time $= \dfrac{600}{600} = 1$ h.

47. *(b)* Let the distance between A and B be x km and speed be V km/h.



Case I Given, distance $= x$ km, speed $= V$ km/h and time = 8 h.

$\therefore$ Speed $= \dfrac{\text{Distance}}{\text{Time}} \Rightarrow V = \dfrac{x}{8}$ ... (i)

Case II If speed $= (V + 4)$ km/h

and time $= 7\frac{1}{2}$ h $= \frac{15}{2}$ h

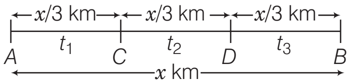
then, $\quad V + 4 = \dfrac{x}{15/2} \Rightarrow V + 4 = \dfrac{2x}{15}$

$\Rightarrow \ \dfrac{x}{8} + 4 = \dfrac{2x}{15}$ [from Eq .(i)]

$\Rightarrow \ \dfrac{2x}{15} - \dfrac{x}{8} = 4 \Rightarrow \dfrac{x}{120} = 4$

$\Rightarrow \quad x = 480$ km

48. *(a)*



Let total distance of AB be x km.

For distance AC, $\quad t_1 = \dfrac{x/3}{10} = \dfrac{x}{30}$ h

For distance CD, $t_2 = \dfrac{x/3}{20} = \dfrac{x}{60}$ h.

For distance BD, $t_3 = \dfrac{x/3}{60} = \dfrac{x}{180}$ h.

$\therefore$ Total time taken

$$= t_1 + t_2 + t_3 = \frac{x}{30} + \frac{x}{60} + \frac{x}{180}$$

$$= \frac{10x}{180} = \frac{x}{18}$$

$\therefore$ Average speed $= \dfrac{\text{Total distance}}{\text{Total time taken}}$

$$= \frac{x}{\frac{x}{18}} = 18 \text{ km/h}$$

49. *(c)* Let speed of a boat in still water be x km/h and speed of the stream be y km/h.

Then, speed of the boat downstream

$= (x + y)$ km/h [Rule 9]

and speed of the boat upstream

$= (x - y)$ km/h

According to the question,

$$\frac{32}{x + y} = 6 \quad \left[\because \text{speed} = \frac{\text{distance}}{\text{time}}\right]$$

and $\dfrac{14}{x - y} = 6$

$\Rightarrow \quad 6x + 6y = 32$ ... (i)

and $\quad 6x - 6y = 14$ ... (ii)

On subtracting Eq. (ii) from Eq. (i), we get

$12y = 18$

$y = \dfrac{18}{12} = 1.5$ km/h

50. *(a)* Let the speed of B be x m/s.

$\therefore$ Speed of $A = 1\frac{2}{3}x = \frac{5x}{3}$ m/s

Ratio of speed of rates of A and B

$= \dfrac{5x}{3} : x = 5 : 3$

$\because$ 2 m are gained in a race of 5 m.

$\therefore$ 1 m are gained in a race of $\frac{5}{2}$ m.

So, 80 m are gained in a race of

$\frac{5}{2} \times 80$ m $= 200$ m

51. *(a)* Given, speed of thief = 10 km/h

$= \dfrac{10 \times 1000}{60}$ m/min $= \dfrac{500}{3}$ m/min

and speed of policeman = 11 km/h

$= \dfrac{11 \times 1000}{60}$ m/min $= \dfrac{550}{3}$ m/min

Now, distance travelled by thief in 6 min

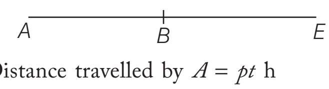
$= \dfrac{500}{3} \times 6 = 1000$ m

and distance travelled by policeman in 6 min $= \dfrac{550}{3} \times 6 = 1100$ m

$\therefore$ Difference $= (1100 - 1000) = 100$ m

Hence, the distance between them after 6 min is 100 m.

52. *(b)* Let A and B will meet after t h at point E.



Distance travelled by $A = pt$ h

and distance travelled by $B = qt$ h

According to the question,

$pt = qt + c \Rightarrow pt - qt = c$

$\Rightarrow \ t(p - q) = c \Rightarrow t = \dfrac{c}{p - q}$ ...(i)

$\because$ Distance travelled by $A = pt$

$= \dfrac{pc}{p - q}$ [from Eq. (i)]

53. *(c)* Let the original speed of car be x km/h.

Time taken to cover 300 km = $\frac{300}{x}$ h

If the speed of car is increased by 15 km/h.

Then, $\frac{300}{x+15} = \frac{300}{x} - 1$

$\Rightarrow \quad 1 = \frac{300}{x} - \frac{300}{x+15}$

$= 300\left[\frac{x+15-x}{x(x+15)}\right]$

$\Rightarrow \quad 1 = \frac{4500}{x^2 + 15x}$

$\Rightarrow \quad x^2 + 15x - 4500 = 0$

$\Rightarrow \quad (x+75)(x-60) = 0$

$\therefore \quad x = 60$ km/h

54. *(b)* Total length of trains

$= 121 + 99 = 220$ m

Relative speed of trains $= (40 + 32)$ km/h

$= 72$ km/h

$= 72 \times \frac{5}{18}$ m/s $= 20$ m/s

$\therefore \quad \text{Time} = \frac{\text{distance}}{\text{speed}} = \frac{220}{20} = 11$ s

55. *(d)* Let the distance be x.

Now, ratio of time taken to travel the distance by each bus is

$\frac{x}{2} : \frac{x}{3} : \frac{x}{4} \Rightarrow \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$

$= 6 : 4 : 3$

56. *(a)* Speed of faster train is 1.5 km/min.

i.e. 90 km/h or 25 m/s

and speed of slower train = 60 km/h

$= \frac{50}{3}$ m/s

Since, these trains are moving in same direction.

So, relative speed of train

$= \left(25 - \frac{50}{3}\right)$ m/s $= \frac{25}{3}$ m/s

Time taken by crossing slower from

= 27 s

$\therefore$ Distance = speed × time

$= \frac{25}{3} \times 27 = 225$ m

57. *(b)* When A covers 100 m, then B covers 96 m and C covers 98 m.

i.e. when C covers 98 m, then B covers 96 m.

$\therefore$ when C covers 100 m, then B covers $\frac{100 \times 96}{98}$ m, i.e. 97.96

So, C beats B by approximately 2.04 m.

58. *(a)* Ratio of speeds are 16 : 15 : 11.

Let speed of winner athletic be $16x$ km/h.

Similarly, speed of athletic in second and third position be $15x$ and $11x$, respectively.

Total distance travelled by winner is 4 km.

$\therefore \quad \text{Time} = \frac{4}{16x}$ h

Distance travelled by second athletic in

$\frac{4}{16x}$ h $= \frac{4}{16x} \times 15x = \frac{15}{4}$ km

Similarly, distance travelled by third athletic in

$\frac{4}{16x}$ h $= \frac{4}{16x} \times 11x = \frac{11}{4}$ km

$\therefore$ Difference between their distances

$= \left(\frac{15}{4} - \frac{11}{4}\right)$ km

$= 1$ km $= 1000$ m

59. *(b)* Refer to example 12.

60. *(d)* Petrol consumed by the bike

$= (10 \times 20) + (30 \times 40) + (10 \times 20)$ mL

$= (200 + 1200 + 200)$ mL

$= 1600$ mL $= 1.6$ L

61. *(d)* Refer to question 53.

62. *(c)* Let the speed of the train $= x$ km/h

Then, relative speed of train

$= (x + 5)$ km/h

and length of the train

$= 225$ m $= 0.225$ km [given]

Time taken by train to cross the man

$= \frac{0.225}{x+5}$ h

According to the question,

$\frac{0.225}{x+5} = \frac{9}{3600}$

$\Rightarrow \quad x + 5 = 90$

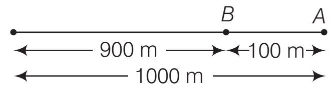
$\Rightarrow \quad x = 85$ km/h

Hence, the speed of the train is 85 km/h.

63. *(c)* Refer to question 46.

64. *(a)* Since, either A beats B by 100 m or 10 s. It means that B runs 100 m in 10 s.

$\therefore$ Speed of $B = \frac{100}{10} = 10$ m/s



$\because$ B gets injured at a distance of 450 m and his speed gets halved.

So, time taken by B to cover 1000 m

$= \frac{450}{10} + \frac{550}{5} = 155$

$\because$ Ratio of speed of A and B is equal to ratio of distance covered by A and B.

$A : B = 1000 : 900 = 10 : 9$

Now, speed of $A = \frac{10}{9} \times 10 = \frac{100}{9}$ m/s

Time taken by A to cover

1000 m $= \frac{1000}{100} \times 9 = 90$ s

Hence, A beat B by length of time

$= (155 - 90) = 65$ s

65. *(c)* I. Let the length of race be x m.

Then, distance covered by $A = x$ m.

Distance covered by B when A reaches the destination $= x - 30$

Distance covered by C when A reaches the destination $= x - 48$

Ratio of speed of B to $A = \frac{x-30}{x}$

Ratio of speed of C to $A = \frac{x-48}{x}$

and distance covered by C when B reaches the destination $= x - 20$

Ratio of speed B to $C = \frac{x}{x-20}$

$\frac{x-30}{x} = \frac{x-48}{x-20} \quad \left[\frac{B}{A} = \frac{C}{A} \times \frac{B}{C}\right]$

$\Rightarrow \quad x^2 - 50x + 600 = x^2 - 48x$

$\therefore \quad x = 300$ m

II. The speeds of A, B and C are in the ratio

300 : 270 : 252 = 50 : 45 : 42

07

TIME AND WORK

Usually (2-4) questions have been asked from this chapter. Generally questions are asked on Man-hours formulae, total wages paid for a work, number of days to complete a certain work and pipes and cisterns.

In this chapter, we will study the relationship among the quantity of work given, wages given, stipulated time, number of persons etc., and after it, we will be able to calculate the work in stipulated time by arranging some persons according to the work but before that lets discuss some basic rules.

IMPORTANT RULES AND FORMULAE

Rule 1 If a person can do a piece of work in 'n' days, then he will do $\frac{1}{n}$ of the work in one day and if a person can do $\frac{1}{n}$th of work in one day, then he will complete the work in n days.

e.g. If Raj can do a piece of work in 20 days, then he will do 1/20th of the work in one day.

Note In the problems related to time and work it is always considered that a man/woman works at uniform rate.

Rule 2 A and B can do a piece of work in x days and y days, respectively. Then, time taken by $(A + B)$ to complete the work is equal to reciprocal of $(A + B)$'s one day's work, i.e. $\frac{xy}{x+y}$.

Rule 3 If A and B can complete a work in x days and A alone can finish that work in y days, then

number of days required to complete the work by $B = \frac{xy}{y-x}$ days

Rule 4 If A and B can do a piece of work in x days. B and C can do same work in y days, C and A can do same work in z days. Then, they will complete the same work in $\frac{2xyz}{xy+yz+zx}$ days by working together.

EXAMPLE 1. Raj can do a piece of work in 20 days and Rohan can do it in 12 days. How long will they take if both work together?

a. $5\frac{1}{2}$ days b. $7\frac{1}{2}$ days c. $3\frac{1}{2}$ days d. $9\frac{1}{2}$ days

Sol. b. Raj's one day's work $= \frac{1}{20}$

Rohan's one day's work $= \frac{1}{12}$

$\therefore$ (Raj + Rohan)'s one day's work $= \frac{1}{20} + \frac{1}{12} = \frac{3+5}{60} = \frac{8}{60}$

$\therefore$ Number of day's taken by Raj and Rohan together to complete the work $= \frac{60}{8}$ days or $7\frac{1}{2}$ days

Here, $x = 20$ days and $y = 12$ days

$\therefore$ Required time $= \frac{xy}{x+y}$ [Rule 2]

$$= \frac{20\times 12}{20+12} = \frac{20\times 12}{32} = \frac{15}{2} = 7\frac{1}{2} \text{ days}$$

EXAMPLE 2. *A* and *B* together can do a piece of work in 12 days and *A* alone can do it 18 days. In how many days can *B* alone do it?

a. 14 days b. 24 days c. 36 days d. 28 days

Sol. c. Here, $x = 12$ and $y = 18$

$\therefore$ Time taken by $B = \frac{xy}{y-x} = \frac{12\times 18}{18-12} = 36$ days [Rule 3]

EXAMPLE 3. *A* and *B* can do a piece of work in 3 days. *B* and *C* in 9 days and *A* and *C* in 12 days. Find the time in which *A*, *B* and *C* can finish the work, working together.

a. 3 days b. $3\frac{1}{4}$ days

c. $3\frac{15}{19}$ days d. None of these

Sol. c. Here, $x = 3$, $y = 9$ and $z = 12$

According to rule,

Required time taken by, A, B and $C = \frac{2xyz}{xy+yz+zx}$ [Rule 4]

$$= \frac{2\times 3\times 9\times 12}{3\times 9 + 9\times 12 + 3\times 12} = \frac{6\times 12}{19} = \frac{72}{19} = 3\frac{15}{19} \text{ days}$$

Rule 5 If two groups, M_1 persons of the first group can do 'W_1', work in 'D_1' days working T_1 h in a day earning a sum of ₹ R_1 and M_2 persons of the second group can do W_2 work in D_2 days working T_2h in a day earning a sum of ₹ R_2. If each person of both group has the same efficiency of work, then

$$\frac{M_1 D_1 T_1}{W_1 R_1} = \frac{M_2 D_2 T_2}{W_2 R_2}$$

EXAMPLE 4. 15 men complete a work in 16 days. If 24 men are employed, then the time required to complete that work will be **2014 I**

a. 7 days b. 8 days c. 10 days d. 12 days

Sol. c. Let the work done be 1.

Here, $M_1 = 15$, $D_1 = 16$, $W_1 = W_2 = 1$, $M_2 = 24$ and $D_2 = ?$

Now, according to the formula,

$$M_1 D_1 W_2 = M_2 D_2 W_1 \Rightarrow 15\times 16\times 1 = 24\times D_2 \times 1$$

$$\Rightarrow D_2 = \frac{15\times 16}{24} \Rightarrow = \frac{240}{24} = 10 \text{ days}$$

Therefore, 10 days are required to complete the work.

Rule 6 If '*m*' men or '*n*' women can do a piece of work in '*a*' days, then *x* men and *y* women can do the same work in $\frac{1}{\frac{x}{m\times a} + \frac{y}{n\times a}}$ days.

EXAMPLE 5. If 3 men or 4 women can reap a field in 43 days. How long will 7 men and 5 women take to reap it?

a. 3 days b. 7 days c. 12 days d. 15 days

Sol. c. Here, $m = 3$, $n = 4$, $a = 43$, $x = 7$ and $y = 5$

$\therefore$ Required days $= \frac{1}{\frac{x}{m\times a} + \frac{y}{n\times a}} = \frac{1}{\frac{7}{3\times 43} + \frac{5}{4\times 43}}$ [Rule 6]

$$= \frac{1}{\frac{1}{43}\left(\frac{7}{3} + \frac{5}{4}\right)} = \frac{1}{\frac{1}{43}\times\frac{43}{12}} = 12 \text{ days}$$

Rule 7 If *A* can do a work in *x* days and *B* can do *y*% fast than *A*, then *B* will complete the work in $\frac{100\,x}{(100+y)}$ days.

EXAMPLE 6. *x* can do a work in 16 days. In how many days will the work be completed by *y*, if the efficiency of *y* is 60% more than that of *x*? **2013 II**

a. 10 days b. 12 days c. 25 days d. 30 days

Sol. a. Here, $x = 16$ days and $y = 60\%$ faster

$\therefore$ Required days $= \frac{100\,x}{100+y} = \frac{100\times 16}{100+60} = \frac{1600}{160} = 10$ days

[Rule 7]

Rule 8 If *A*, *B* and *C* can do a piece of work in *x*, *y* and *z* days, respectively and they received ₹ *k* as wages by working together, then

$$\text{share of } A = ₹\,\frac{yz}{xy+yz+zx}\times k$$

$$\text{share of } B = ₹\,\frac{xz}{xy+yz+zx}\times k$$

and $$\text{share of } C = ₹\,\frac{xy}{xy+yz+zx}\times k$$

Note If A and B can do a piece of work in x days and y days, respectively and they received ₹ k as wages by working together, then share of $A = ₹\frac{yk}{x+y}$ and share of $B = ₹\frac{xk}{x+y}$.

Wages are directly proportional to the work done and indirectly proportional to the time taken by the individual.

EXAMPLE 7. X completes a job in 2 days and Y completes it in 3 days and Z takes 4 days to complete it. If they work together and get ₹ 3900 for the job, then how much amount does Y get?

a. ₹ 1800 b. ₹ 1200 c. ₹ 900 d. ₹ 800

Sol. b. Here, $a = 2$ days, $b = 3$ days and $c = 4$ days and $k = ₹\,3900$

Now, amount of $y = \frac{ac}{ab+bc+ca} \times k$ [Rule 8]

$$= \frac{2\times 4}{2\times 3+3\times 4+4\times 2}\times 3900$$

$$= \frac{8}{6+12+8}\times 3900 = \frac{8\times 3900}{26} = ₹\,1200$$

SOME OTHER FORMULAS

1. If A and B can do a piece of work in x and y days, respectively. A and B started working together but A left the work t days before completing the work, then time taken to complete the work will be $\frac{(x+t)\,y}{(x+y)}$ days.
2. A and B do a piece of work in 'a' and 'b' days, respectively. Both begin together but after some days, A leaves the work and the remaining work is completed by B in x days. Then, the time after which A left is given by $T = \frac{(b-x)\,a}{a+b}$
3. If A and B can do a piece of work in x and y days, respectively. They start working together and after t days B leaves the work, then time taken to finish the whole work will be $\frac{x}{y}\times(y-t)$ days.

EXAMPLE 8. Akshu can do a piece of work in 10 days and Harshal can do same work in 12 days. They started working together but Akshu left the work 2 days before completion of work, then time taken to complete the work?

a. $6\frac{6}{11}$ days b. $5\frac{3}{10}$ days c. $4\frac{3}{2}$ days d. $7\frac{2}{5}$ days

Sol. a. $6\frac{6}{11}$ days $= \frac{12\times(10+2)}{10+12} = \frac{12\times 12}{22} = 6\frac{6}{11}$ days

EXAMPLE 9. A can do a piece of work in 10 days and B can do same work in 15 days. They started working together but after 2 days, A left work. The remaining work was completed by B alone, then time taken to complete the work?

a. $8\frac{3}{11}$ days b. $8\frac{2}{3}$ days c. $5\frac{3}{4}$ days d. $2\frac{5}{3}$ days

Sol. b. $8\frac{2}{3}$ days $= \frac{10}{15}(15-2) = \frac{26}{3} = 8\frac{2}{3}$ days

PIPES AND CISTERNS

Problems on pipes and cisterns are based on the basic concept of time and work. Pipes are connected to a tank or cistern and are used t4o fill or empty the tank or cistern.

Important Rules and Formulae

Rule 9 If there are two pipes A and B takes 'a' and 'b' h respectively to fill a tanker, then the two pipes together fill $\left(\frac{1}{a}+\frac{1}{b}\right)$ part of the tank in 1h and time taken to fill the tank will be $\left(\frac{ab}{a+b}\right)$ h. If a pipe fills a tank in 'a' h, then in 1 h only $\frac{1}{a}$ of the tank is filled.

EXAMPLE 10. Pipe A can fill a tank in 45 h and pipe B can fill it in 36 h. If both the pipes are opened in the empty tank. In how many hours will it be full?

a. 10 h b. 15 h c. 20 h d. 28 h

Sol. c. Here, $a = 45$ h and $b = 36$ h.

$\therefore$ Required time $= \frac{ab}{a+b} = \frac{45\times 36}{45+36} = \frac{1620}{81} = 20$ h

Rule10 (i) If a pipe can fill a tank in 'a' h and another can fill the tank in 'b' h but a third pipe empties the filled tank in 'c' h, then in one hour $\left(\frac{1}{a}+\frac{1}{b}-\frac{1}{c}\right)$ part of the tank will be filled (when all the three pipes are open) and time taken to fill the tank will be $\frac{abc}{bc+ac-ab}$ h.

(ii) If a pipe can fill a tank in a_1 h and another pipe can empty the filled tank in a_2 h, then in one hour $\left(\frac{1}{a_1}-\frac{1}{a_2}\right)$ part of the tank will be filled. (when both pipes are open) and time taken to fill the tank will be $\left(\frac{a_1\,a_2}{a_2-a_1}\right)$ h.

EXAMPLE 11. Two pipes 'A' and 'B' can fill a tank in 36 min and 45 min, respectively. A waste pipe 'C' can empty the tank in 30 min. In how much time the tank is full if all three pipes are opened?

a. 60 min b. 90 min c. 115 min d. None of these

Sol. a. Here, $a = 36, b = 45$ and $c = 30$.

So, time taken to fill the tank will be

$$= \frac{abc}{ac+bc-ab} \quad \text{[Rule 8]}$$

$$= \frac{36\times 45\times 30}{36\times 30+45\times 30-36\times 45} = 60 \text{ min}$$

PRACTICE EXERCISE

1. X can do $\frac{3}{4}$ of a work in 12 days. In how many days X can finish the 1/2 work?

(a) 8 days (b) 16 days (c) 12 days (d) 24 days

2. A can do a piece of work in 10 days and B can do the same work in 12 days. How long will they take to finish the work, if both work together?

(a) $5\frac{5}{11}$ days (b) $6\frac{5}{11}$ days

(c) $5\frac{1}{5}$ days (d) None of these

3. A, B and C working together take 30 min to address a pile of envelopes. A and B together would take 40 min, A and C together would take 45 min. How long would each take working alone?

(a) A : 72 min, B : 90 min, C : 120 min

(b) A : 42 min, B : 90 min, C : 120 min

(c) A : 72 min, B : 90 min, C : 100 min

(d) A : 72 min, B : 80 min, C : 120 min

4. A and B together can do a piece of work in 12 days and A alone can do it in 36 days. In how many days can B alone do it?

(a) 18 days (b) 12 days

(c) 15 days (d) 20 days

5. Ram can do a piece of work in 6 days and Shyam can finish the same work in 12 days. How much work will be finished, if both work together for 2 days?

(a) One-fourth of the work (b) One-third of the work

(c) Half of the work (d) Whole of the work

6. A and B can do given work in 8 days; B and C can do the same work in 12 days and A, B, C complete it in 6 days. In how many days can A and C finish it?

(a) 12 (b) 8 (c) 14 (d) 16

7. A can finish a work in 8 days and B can do it in 12 days. After A had worked for 3 days, B also joins A to finish the remaining work. In how many days will the remaining work be finished?

(a) 2 days (b) 3 days

(c) 4 days (d) 5 days

8. Two taps can fill a tub in 5 min and 7 min, respectively. A pipe can empty it in 3 min. If all the three are kept open simultaneously, when will the tub be full?

(a) 60 min (b) 85 min (c) 90 min (d) 105 min

9. A and B can do a piece of work in 40 days and 50 days, respectively. Both begin together but after a certain time, A leaves off. In this case B finishes the remaining work in 20 days. After how may days did A leave?

(a) 14 days (b) $13\frac{1}{3}$ days (c) 13 days (d) 15 days

10. P and Q can do a job in 2 days, Q and R can do it in 4 days and P and R in 12/5 days. What is the number of days required for P alone to do the job?

(a) 5/2 (b) 3 (c) 14/5 (d) 6

11. $\frac{1}{48}$ of a work is completed in half a day by 5 persons. Then, $\frac{1}{40}$ of the work can be completed by 6 persons in how many days?

(a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$

12. A garrison of 'n' men had enough food to last for 30 days. After 10 days, 50 more men joined them. If the food now lasted for 16 days, what is the value of n?

(a) 200 (b) 240 (c) 280 (d) 320

13. Two pipes A and B can fill a tank in 12 and 16 min, respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 9 min?

(a) 3 min (b) 5 min (c) 4 min (d) 2 min

14. 2 men undertake to do a job for ₹ 1400. One can do it alone in 7 days and the other in 8 days. With the assistance of a boy they finish the work in 3 days. How should the money be divided?

(a) ₹ 600, ₹ 525, ₹ 275 (b) ₹ 550, ₹ 500, ₹ 350

(c) ₹ 650, ₹ 470, ₹ 280 (d) None of these

15. A, B and C can do a piece of work individually in 8, 10 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed?

(a) 53/9 days (b) 34/7 days

(c) 85/13 days (d) 53/10 days

16. A can do a piece of work in 'x' days and B can do the same work $3x$ days. To finish the work together they take 12 days. What is the value of 'x'?

(a) 8 (b) 10 (c) 12 (d) 16

17. If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, what is the time taken by 15 men and 20 boys in doing the same type of work?

(a) 4 days (b) 5 days (c) 6 days (d) 7 days

18. Four taps can individually fill a cistern of water in 1h, 2h, 3h and 6h, respectively. If all the four taps are opened simultaneously, the cistern can be filled in how many minutes?

(a) 20 (b) 30 (c) 35 (d) 40

19. 76 ladies complete a job in 33 days. Due to some reason some ladies did not join the work and therefore, it was completed in 44 days. The number of ladies who did not report for the work is

(a) 17 (b) 18 (c) 19 (d) 20

20. 9 men finish one-third work in 10 days. The number of additional men required for finishing the remaining work in 2 more days will be

(a) 78 (b) 81 (c) 55 (d) 30

21. Ravi and Sneha working separately can finish a job in 8 and 12 h, respectively. If they work for an hour alternately, Ravi beginning at 9:00 am. When will the job be finished?

(a) 7 : 30 pm (b) 7 : 00 pm (c) 6 : 30 pm (d) 6 : 00 pm

22. Consider the following statements:

I. If 18 men can earn ₹ 1440 in 5 days, then 10 men can earn ₹ 1280 in 6 days.

II. If 16 men can earn ₹ 1120 in 7 days, then 21 men can earn ₹ 800 in 4 days.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 23-24) If a_1 men and b_1 boys can complete a work in x days, while a_2 men and b_2 boys can complete the same work in y days, then

$$\frac{\text{One day work of 1 man}}{\text{One day work of 1 boy}} = \frac{(yb_2 - xb_1)}{(xa_1 - ya_2)}$$

23. If 14 men and 12 boys can finish a work in 4 days, while 8 men and 16 boys can finish the same work is 5 days. Compare the 1 day work of 1 man and 1 boy.

(a) 2 (b) $1\frac{1}{2}$ (c) $\frac{1}{2}$ (d) 3

24. 28 men and '*m*' boys can finish a work in 4 days, while 20 men and 6 boys can finish the same work in 5 days. Now, if the ratio of 1 day work of 1 man and 1 boy is 1 : 2, what will be the value of '*m*'?

(a) 7 (b) 6 (c) 8 (d) 12

PREVIOUS YEARS' QUESTIONS

25. 45 people take 18 days to dig a pond. If the pond would have to be dig in 15 days, then the number of people to be employed will be **2012 I**

(a) 50 (b) 54 (c) 60 (d) 72

26. *A* and *B* can do a piece of work in 10 h. *B* and *C* can do it in 15 h, while *A* and *C* take 12 h to complete the work. *B* independently can complete the work in **2012 I**

(a) 12 h (b) 16 h (c) 20 h (d) 24 h

27. *X* can do a piece of work in 25 days. *Y* is 25% more efficient than *X*. The number of days taken by *Y* is **2012 II**

(a) 15 days (b) 20 days (c) 21 days (d) 30 days

28. A mason can build a tank in 12 h. After working for 6 h, he took the help of a boy and finished the work in another 5 h. The time that the boy will take alone to complete the work is **2013 I**

(a) 30 h (b) 45 h (c) 60 h (d) 64 h

29. *X* can complete a job in 12 days. If *X* and *Y* work together, they can complete the job in $6\frac{2}{3}$ days. *Y* alone can complete the job in **2013 I**

(a) 10 days (b) 12 days (c) 15 days (d) 18 days

30. Pipe *A* can fill a tank in 10 min and pipe *B* can empty it in 15 min. If both the pipes are opened in an empty tank, the time taken to make it full is **2013 I**

(a) 20 min (b) 25 min

(c) 30 min (d) None of these

31. 4 goats and 6 sheeps can graze a field in 50 days. 2 goats and 9 sheeps can graze the field in **2013 II**

(a) 100 days (b) 75 days (c) 50 days (d) 25 days

32. *A* can finish a work in 15 days, *B* in 20 days and *C* in 25 days. All these three worked together and earned ₹ 4700. The share of *C* is **2013 II**

(a) ₹ 1200 (b) ₹ 1500 (c) ₹ 1800 (d) ₹ 2000

33. 20 workers working for 5 h per day complete a work in 10 days. If 25 workers are employed to work 10 h per day, what is the time required to complete the work? **2013 II**

(a) 4 days (b) 5 days (c) 6 days (d) 8 days

34. 18 men can earn ₹ 360 in 5 days. How much money will 15 men earn in 9 days? **2013 II**

(a) ₹ 600 (b) ₹ 540 (c) ₹ 480 (d) ₹ 360

35. 2 men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets ₹ 90 per day, what should be the wages per day of a woman? **2013 II**

(a) ₹ 48 (b) ₹ 60 (c) ₹ 72 (d) ₹ 135

36. A can do a piece of work in 4 days and B can complete the same work in 12 days. What is the number of days required to do the same work together? **2013 II**

(a) 2 days (b) 3 days (c) 4 days (d) 5 days

37. A, B and C can do a piece of work individually in 8, 12 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed? **2014 II**

(a) $5\frac{8}{9}$ days (b) $4\frac{6}{7}$ days (c) $6\frac{7}{13}$ days (d) $3\frac{3}{4}$ days

38. A is thrice as efficient as B and hence completes a work in 40 days less than the number of days taken by B. What will be the number of days taken by both of them when working together? **2014 II**

(a) 22.5 days (b) 15 days (c) 20 days (d) 18 days

39. The efficiency of P is twice that of Q, whereas the efficiency of P and Q together is three times that of R. If P, Q and R work together on a job, in what ratio should they share their earnings? **2015 I**

(a) 2 : 1 : 1 (b) 4 : 2 : 1 (c) 4 : 3 : 2 (d) 4 : 2 : 3

40. A and B are two taps which can fill a tank individually in 10 min and 20 min, respectively. However, there is a leakage at the bottom, which can empty a filled tank in 40 min. If the tank is empty initially, then how much time will both the taps take to fill the tank with leakage? **2015 II**

(a) 2 min (b) 4 min (c) 5 min (d) 8 min

41. If 4 men working 4 h per day for 4 days complete 4 units of work, then how many units of work will be completed by 2 men working for 2 h per day in 2 days? **2015 II**

(a) 2 (b) 1 (c) $\frac{1}{2}$ (d) $\frac{1}{8}$

42. If m persons can paint a house in d days, then how many days will it take for $(m + 2)$ persons to paint the same house? **2015 II**

(a) $md + 2$ (b) $md - 2$ (c) $\frac{m+2}{md}$ (d) $\frac{md}{m+2}$

43. Two pipes A and B can fill a tank in 60 min and 75 min, respectively. There is also an outlet C. If A, B and C are opened together, then the tank is full in 50 min. How much time will be taken by C to empty the full tank? **2016 I**

(a) 100 min (b) 110 min (c) 120 min (d) 125 min

ANSWERS

1	a	2	a	3	a	4	a	5	c	6	b	7	b	8	d	9	b	10	b
11	d	12	a	13	c	14	a	15	d	16	d	17	a	18	b	19	c	20	b
21	c	22	d	23	a	24	b	25	b	26	d	27	b	28	c	29	c	30	c
31	d	32	a	33	a	34	b	35	b	36	b	37	a	38	b	39	a	40	d
41	c	42	d	43	a														

HINTS AND SOLUTIONS

1. *(a)* Since, X can do $\frac{3}{4}$ of work in 12 days. [Rule 1]

So, X can do 1 work in $\frac{12 \times 4}{3}$ days.

$\therefore X$ can do $\frac{1}{2}$ work in $\frac{12 \times 4 \times 1}{3 \times 2} = 8$ days

2. *(a)* Here, $x = 10$ and $y = 12$

$\therefore$ Number of days taken by A and B

$$= \frac{xy}{x+y} = \frac{12 \times 10}{12 + 10} \text{ [Rule 2]}$$

$$= 5\frac{5}{11} \text{ days}$$

3. *(a)* B's one min's work

$= (A + B + C)$'s one min's work $-(A + C)$'s one min's work

$$= \frac{1}{30} - \frac{1}{45} = \frac{6-4}{180} = \frac{2}{180} = \frac{1}{90}$$

C's one min's work $= (A + B + C)$'s one min work $- (A + B)$'s one min work

$$= \frac{1}{30} - \frac{1}{40} = \frac{4-3}{120} = \frac{1}{120}$$

A' one min's work $= (A + B)$'s one min work $-$ B's one min work

$$= \frac{1}{40} - \frac{1}{90} = \frac{9-4}{360}$$

$$= \frac{5}{360} = \frac{1}{72}$$

Hence, A, B and C alone can finish the work in 72 min, 90 min and 120 min, respectively.

4. *(a)* Here, $x = 12$ and $y = 36$

$\therefore$ Time taken by $B = \frac{xy}{y-x} = \frac{12\times36}{36-12}$ [Rule 3]

$= \frac{12\times36}{24} = 18$ days

5. *(c)* $\therefore$ One day work of Ram $= \frac{1}{6}$

$\therefore$ One day work of Shyam $= \frac{1}{12}$

Hence, one day work of, Ram and Shyam

$= \frac{1}{6} + \frac{1}{12} = \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}$

$\therefore$ Two day's work $= \frac{1}{2}$

Thus, if they work together for 2 day's, then half of the work will be complete.

6. *(b)* Here, $x = 8$, $y = 12$, $z = 12$

Now, time taken by $(A + B + C)$

$= \frac{2xyz}{xy + yz + zx}$ [Rule 4]

$\Rightarrow 6 = \frac{2\times8\times12\times z}{8\times12 + 12\times z + z\times8}$

$\Rightarrow 96 + 20z = 32z$

$\Rightarrow 96 = 12z$

$\Rightarrow z = 8$

So, A and C can do the work in 8 days.

7. *(b)* One day's work of $A = \frac{1}{8}$

One day's work of $B = \frac{1}{12}$

3 day's work of $A = \frac{3}{8}$

Remaining work of $A = 1 - \frac{3}{8}$

$= \frac{5}{8}$

One day's work of A and B together

$= \frac{1}{8} + \frac{1}{12} = \frac{3+2}{24} = \frac{5}{24}$

Number of days to finish the work

$= \frac{5}{8} \div \frac{5}{24} = 3$ days

8. *(d)* 1 min work of first tap $= \frac{1}{5}$

1 min work of second tap $= \frac{1}{7}$

and 1 min work of third $= -\frac{1}{3}$

$\therefore$ 1 min work of all taps $= \frac{1}{5} + \frac{1}{7} - \frac{1}{3}$

$= \frac{21+15-35}{105} = \frac{1}{105}$

Hence, tap will be filled in 105 min, if they work together.

9. *(b)* Here, $a = 40$ days, $b = 50$ days, $x = 20$ and $T = ?$

$\therefore$ Required time $= \frac{(b-x)a}{a+b}$

$= \frac{(50-20)\times40}{(40+50)} = \frac{30\times40}{90}$

$= \frac{40}{3} = 13\frac{1}{3}$ days

10. *(b)* Here, $x = 2$, $y = 4$ and $z = \frac{12}{5}$

$\therefore$ Time taken by $(P + Q + R)$

$= \frac{2xyz}{xy + yz + zx}$ [Rule 4]

$= \frac{2\times2\times4\times\frac{12}{5}}{2\times4 + 4\times\frac{12}{5} + \frac{12}{5}\times2}$

$= \frac{192}{40 + 48 + 24}$

$= \frac{192}{112} = \frac{12}{7}$ days

Now, P one days work

$= \frac{7}{12} - \frac{1}{4} = \frac{7-3}{12}$

$= \frac{4}{12} = \frac{1}{3}$

Hence, P alone can do in 3 days.

11. *(d)* Using the formula $\frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$

12. *(a)* By given condition,

$n\times30 = n\times10 + (n+50)\times16$

$\Rightarrow 20n = 16n + 800$

$\therefore n = \frac{800}{4} = 200$

13. *(c)* Here, $x = 12$, $y = 16$ and $t = 9$

Two pipes A and B can fill a tank in x h and y h, respectively. If both the pipes ax opened simultaneously, then the time after which B should be closed, so that the tank is filled in

$t\text{h} = [y(1 - t/x)]^n$

Required time after which B should be closed $= y\left(1-\frac{t}{x}\right) = 16\left(1-\frac{9}{12}\right)$

$= 16\times\frac{3}{12} = 4$ min

14. *(a)* Let the boy completes the work in x days.

According to the condition,

$\frac{1}{7} + \frac{1}{8} + \frac{1}{x} = \frac{1}{3} \Rightarrow \frac{1}{x} = \frac{1}{3} - \frac{1}{8} - \frac{1}{7}$

$\Rightarrow \frac{1}{x} = \frac{56-21-24}{168} = \frac{11}{168}$

$\therefore x = \frac{168}{11}$ days

So, money is to be shared in the ratio

$\frac{1}{7} : \frac{1}{8} : \frac{11}{168}$ or $24 : 21 : 11$

Thus,

A's amount $= \frac{24}{56}\times1400 =$ ₹ 600

B's amount $= \frac{21}{56}\times1400 =$ ₹ 525

Boy's amount $= \frac{11}{56}\times1400 =$ ₹ 275

15. *(d)* A's work in one day $= \frac{1}{8}$, B's work in one day $= \frac{1}{10}$

C' work in one day $= \frac{1}{15}$

$(A+B)$'s work in one day

$= \frac{1}{8} + \frac{1}{10} = \frac{5+4}{40} = \frac{9}{40}$

$(A+B)$'s work in two days $= \frac{2\times9}{40} = \frac{9}{20}$

Remaining work $= 1 - \frac{9}{20} = \frac{11}{20}$

$(B+C)$'s work in one day

$= \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$

Since, $(B+C)$ complete the work in 6 days.

$\therefore \frac{11}{20}$ work will be completed in

$6\times\frac{11}{20} = \frac{11\times3}{10} = \frac{33}{10}$ days

$\therefore$ Total number of days

$= 2 + \frac{33}{10} = \frac{20+33}{10} = \frac{53}{10}$ days

16. *(d)* 1 day work of $A = \frac{1}{x}$

1 day work of $B = \frac{1}{3x}$

$\therefore$ 1 day work of both A and B

$= \frac{1}{x} + \frac{1}{3x} = \frac{4}{3x}$

given, one day work of both A and B $= \frac{1}{12}$

$\Rightarrow \quad \frac{4}{3x} = \frac{1}{12} \quad \Rightarrow \quad 3x = 48 \Rightarrow \quad x = 16$

Hence, the value of x is 16.

17. *(a)* Given, $6\ M + 8\ B = 10$ days ...(i)

and $26\ M + 48\ B = 2$ days ... (ii)

$15\ M + 20\ B = ?$

Here, $M_1 D_1 = M_2 D_2$

$\Rightarrow (6M + 8B) \times 10 = (26M + 48B) \times 2$

$\Rightarrow \quad 60M + 80\ B = 52\ M + 96\ B$

$\Rightarrow \quad 8M = 16B$

$\therefore \quad M = 2B$

Then, $15\ M + 20\ B = 15 \times 2\ B + 20\ B$
$= 30\ B + 20\ B = 50\ B$

On putting the value of M in Eq. (i), we get

$\Rightarrow \quad 6 \times 2B + 8\ B = 10$ days

$\Rightarrow \quad 12\ B + 8\ B = 10$ days

$\therefore$ 20 boys finish the work in 10 days

One boy finish the work in

10×20 days $= 200$ days

and 50 boys finish the work in $\frac{200}{50}$ days

$= 4$ days

18. *(b)* Part filled by first tap in 1h = 1

Part filled by second tap in 1h = $\frac{1}{2}$

Part filled by third tap in 1h = $\frac{1}{3}$

Part filled by fourth tap in 1h = $\frac{1}{6}$

Total tank filled by all taps in 1h

$= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$

$\therefore$ Required time $= \frac{1}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{6}}$

$= \frac{6}{6 + 3 + 2 + 1} = \frac{6}{12}$ h $= \frac{6}{12} \times 60$ min

$= 30$ min

19. *(c)* Given, $M_1 = 76, D_1 = 33$

Let number of ladies who did not report for the work = x

By given condition,

$M_2 = 76 - x$ and $D_2 = 44$

We know that,

$M_1 D_1 = M_2 D_2$ [Rule 5]

$\therefore \quad 76 \times 33 = (76 - x) \times 44$

$\Rightarrow \quad 76 - x = \frac{76 \times 3}{4} = 19 \times 3 \Rightarrow x = 19$

Hence, the number of ladies is 19.

20. *(b)* Refer to Question No 11

[Using Rule 5]

21. *(c)* Work done by Ravi in one hour = $\frac{1}{8}$

Work done by Sneha in one hour = $\frac{1}{12}$

Total work done by them in 2 h

$= \frac{1}{8} + \frac{1}{12} = \frac{5}{24}$

Work done is 4 pairs of hours

$= \frac{5}{24} \times 4 = \frac{5}{6}$

Remaining work after 8 hours

$= 1 - \frac{5}{6} = \frac{1}{6}$

Now, its Ravi's turn

So, work left after 9 hours = $\frac{1}{6} - \frac{1}{8} = \frac{1}{24}$

$\frac{1}{24}$ work will be done by sneha in

$12 \times \frac{1}{24} h = 30$ min

Total time taken = 9 h and 30 min

Required time = 6 : 30 pm

22. *(d)* I. 18 men can earn in 5 days = ₹ 1440

1 man can earn in 1 day = ₹ $\frac{1440}{18 \times 5}$

$\therefore$ 10 men can earn in 6 days

$= \frac{1440}{18 \times 5} \times 6 \times 10$

$=$ ₹ 960 $\neq$ ₹ 1280

II. 16 men can earn in 7 days = ₹ 1120

1 man can earn in 1 day = $\frac{1120}{16 \times 7}$

$\therefore$ 21 men can earn in 4 days

$= \frac{1120}{16 \times 7} \times 21 \times 4 =$ ₹ 840 $\neq$ ₹ 800

So, neither statement I nor II is correct.

23. *(a)* Here, $a_1 = 14, b_1 = 12, x = 4,$

$a_2 = 8, b_2 = 16$ and $y = 5$

$\frac{\text{One day work of 1 man}}{\text{One day work of 1 boy}} = \frac{(yb_2 - xb_1)}{(xa_1 - ya_2)}$

$= \frac{5 \times 16 - 4 \times 12}{4 \times 14 - 5 \times 8} = 2$

24. *(b)* Refer to Question No 23.

25. *(b)* Using the formula, $M_1 D_1 = M_2 D_2$

26. *(d)* Here, $x = 10,\ y = 15,\ z = 12.$

$\therefore$ Time taken by $(A + B + C)$

$= \frac{2xyz}{xy + yz + zx}$ [Rule 4]

$= \frac{2 \times 10 \times 15 \times 12}{10 \times 15 + 15 \times 12 + 12 \times 10}$

$= \frac{3600}{450} = 8$ h

So, 1 h work of $B = \frac{1}{8} - \frac{1}{12}$

$= \frac{3 - 2}{24} = \frac{1}{24}$

Hence, B alone can do in 24 h.

27. *(b)* Here, $x = 25$ days and $y = 25$

$\therefore$ Required days $= \frac{100x}{100 + y} = \frac{100 \times 25}{100 + 25}$

$= \frac{2500}{125} = 20$ days

28. *(c)* Mason work for 1 h = $\frac{1}{12}$

Mason Work for 6 h = $\frac{6}{12} = \frac{1}{2}$

Work left = $1 - \frac{1}{2} = \frac{1}{2}$

Now, let the boy can finish the work in x h.

Then, their 1 h work $= \frac{1}{12} + \frac{1}{x} = \frac{x + 12}{12x}$

$\therefore \quad \frac{x + 12}{12x} \times 5 = \frac{1}{2} \quad \Rightarrow \quad \frac{5x + 60}{12x} = \frac{1}{2}$

$\Rightarrow \quad 10x + 120 = 12x \ \Rightarrow \ 120 = 2x$

$\therefore \quad x = 60$ h

29. *(c)* X's one day work $= \frac{1}{12}$ and $(X + Y)$'s one day work $= \frac{3}{20}$

$\therefore$ Y's one day work

$= \frac{3}{20} - \frac{1}{12} = \frac{4}{60} = \frac{1}{15}$

$\therefore$ Number of day's taken by Y to complete the work = 15 days

30. *(c)* Part filled by pipe A in 1 min = $\frac{1}{10}$

and part emptied by pipe B in 1 min $= \frac{1}{15}$

$\therefore$ Total tank filled in minutes

$= \frac{1}{10} - \frac{1}{15} = \frac{3 - 2}{30} = \frac{1}{30}$

Hence, the tank will be filled in 30 min.

31. *(d)* Here, $m = 4$, $n = 6$, $a = 50$ and $x = 2$, $y = 9$

$\therefore$ Required days $= \frac{1}{\frac{x}{m \times a} + \frac{y}{n \times a}}$

$= \frac{1}{\frac{2}{4 \times 50} + \frac{9}{6 \times 50}}$ [Rule 6]

$= \frac{1}{\frac{1}{100} + \frac{3}{100}} = \frac{1}{\frac{4}{100}} = \frac{100}{4} = 25$ days

32. *(a)* Here, $x = 15$ days, $y = 20$ days and $z = 25$ days and $K = ₹4700$

Share of $C = ₹\left(\frac{kxy}{xy + yz + zx}\right)$

$= ₹\ \frac{4700 \times 15 \times 20}{15 \times 20 + 20 \times 25 + 25 \times 15}$

$= ₹\ \frac{4700 \times 15 \times 20}{1175} = ₹\ 1200$

33. *(a)* Refer to Question No 11 [Using Rule 5]

34. *(b)* Refer to Question No 22

35. *(b)* $\because$ 14 days work of 2 men and 1 women

= 8 days work of 4 women and 2 men

$\Rightarrow$ 1 day work of 28 men and 14 women

= 1 day work of 32 women and 16 men

$\Rightarrow$ 28 men + 14 women = 32 women + 16 men

$\Rightarrow \quad 28\,m - 16\,m = 32\,w - 14\,w$

$\Rightarrow \quad 12\,m = 18\,w \quad \Rightarrow \quad \frac{m}{w} = \frac{18}{12} = \frac{3}{2}$

So, efficiency of 1 man and 1 woman is 3 : 2.

So, their wages must be in the same ratio

$\therefore \quad \frac{90}{x} = \frac{3}{2}$

[where, x = wages per day of a woman]

$\therefore \quad x = \frac{90 \times 2}{3} = ₹\ 60$

36. *(b)* A's one day work $= \frac{1}{4}$

and B's one day work $= \frac{1}{12}$

One day work of A and B together

$= \frac{1}{4} + \frac{1}{12} = \frac{3+1}{12} = \frac{4}{12} = \frac{1}{3}$

Days required by A and B together to do the work

$= \frac{1}{\text{One day work of } A \text{ and } B \text{ together}}$

= 3 days

37. *(a)* Work done by A in one day $= \frac{1}{8}$

Work done by B in one day $= \frac{1}{12}$

and work done by C in one day $= \frac{1}{15}$

Let the work will be completed in x days.

Then, $\quad \frac{2}{8} + \frac{x}{12} + \frac{x-2}{15} = 1$

$\Rightarrow \quad \frac{1}{4} + \frac{x}{12} + \frac{x-2}{15} = 1$

$\Rightarrow \quad \frac{15 + 5x + 4(x-2)}{60} = 1$

$\Rightarrow 15 + 5x + 4x - 8 = 60 \Rightarrow 9x + 7 = 60$

$\Rightarrow \quad 9x = 53 \Rightarrow x = \frac{53}{9} = 5\frac{8}{9}$

Hence, the work will completed in $5\frac{8}{9}$ days.

38. *(b)* Let efficiency of A be x days.

Then, efficiency of $B = 3x$ days.

$\therefore$ B's one day work $= \frac{1}{3x}$

and $\quad A$'s one day work $= \frac{1}{x}$

Again, let time taken by $B = t$ days

Then, time taken by $A = (t - 40)$ days

Now, $\frac{t}{3x} = \frac{t-40}{x} \quad \Rightarrow \quad t = 3(t-40)$

$\Rightarrow \ 3t - t = 120, \ \ t = \frac{120}{2} = 60$ days

So, B complete the work in 60 days and A complete the work in 20 days.

$(A + B)$'s 1 day work

$= \frac{1}{60} + \frac{1}{20} = \frac{4}{60} = \frac{1}{15}$

Hence, they will complete the work in 15 days when they work together.

39. *(a)* Let efficiency of P be x days.

Then, efficiency of $Q = 2x$ days

$\therefore \quad Q$'s one day work $= \frac{1}{2x}$

and $\quad P$'s one day work $= \frac{1}{x}$

Now, $(P + Q)$'s one day work

$= \frac{1}{x} + \frac{1}{2x} = \frac{3}{2x}$

$\therefore (P + Q)$ will complete the whole work in $\frac{2x}{3}$ days.

According to the question,

R will complete this work $= 2x$ days

$\therefore \quad R$'s one day work $= \frac{1}{2x}$

Required ratio $= \frac{1}{x} : \frac{1}{2x} : \frac{1}{2x} = 1 : \frac{1}{2} : \frac{1}{2}$

$= 2 : 1 : 1$

40. *(d)* Part of tank filled by A in 1 min $= \frac{1}{10}$

Part of tank filled by B in 1 min $= \frac{1}{20}$

Part of tank emptied by leakage in 1 min $= \frac{1}{40}$

Part of tank filled by both taps and leakage is $\left(\frac{1}{10} + \frac{1}{20} - \frac{1}{40}\right) = \frac{1}{8}$

Hence, total time taken to fill the tank is 8 min.

41. *(c)* Here, $M_1 = 4, T_1 = 4, W_1 = 4, D_1 = 4$ and $M_2 = 2, T_2 = 2, D_2 = 2, W_2 = ?$

Using the formula,

$\frac{M_1 \times T_1 \times D_1}{W_1} = \frac{M_2 \times T_2 \times D_2}{W_2}$ [Rule 5]

$\Rightarrow \quad \frac{4 \times 4 \times 4}{4} = \frac{2 \times 2 \times 2}{W_2}$

$\Rightarrow \quad W_2 = \frac{8}{16} = \frac{1}{2}$

42. *(d)* m persons paint a house in d days.

$\therefore$ 1 person paints a house in $(m \times d)$ days.

and $m + 2$ persons paint a house in $\left(\frac{md}{m+2}\right)$ days.

43. *(a)* Since, two pipes A and B fill a tank in 60 min and 75 min, respectively.

$\therefore$ Part of tank filled by pipe A in 1 min $= \frac{1}{60}$

and part of tank filled by pipe B in 1 min $= \frac{1}{75}$

Now, part of tank filled by A and B together in 1 min

$= \frac{1}{60} + \frac{1}{75} = \frac{9}{300} = \frac{3}{100}$

Let part of tank emptied by pipe C in 1 min $= \frac{1}{C}$.

So, net part of tank filled by pipes A, B and C together in 1 min

$= \frac{3}{100} - \frac{1}{C} = \frac{1}{50}$ [Given]

$\Rightarrow \quad \frac{1}{C} = \frac{3}{100} - \frac{1}{50} = \frac{1}{100}$

$\therefore \quad C = 100$

Hence, the time taken by pipe C to empty the tank is 100 min.

08

PERCENTAGE

Generally (2-3) questions have been asked from this chapter. Generally questions which are asked from this chapter are tricky but you can easily solve them by using the short-tricks formulae in very less time.

PER CENT

Per cent is a fraction whose denominator is 100 and numerator of the fraction is called the rate per cent. Per cent is denoted by the symbol '%'.

Important Rules and Formulae

Rule 1 For expressing $a\%$ as a fraction, we can write $a\% = \frac{a}{100}$, i.e. divide 'a' by 100 and reduce it to the lowest form, e.g. $25\% = \frac{25}{100} = \frac{1}{4}$

Rule 2 For expressing a fraction $\frac{a}{b}$ as a per cent, we can write $\frac{a}{b} = \left(\frac{a}{b} \times 100\right)\%$

e.g. $\frac{2}{3} = \frac{2}{3} \times 100\% = \frac{200}{3}\%$

Rule 3 To find how much per cent one quantity is of another quantity, we can write

$$\text{Required percentage} = \frac{\text{The quantity to be expressed in per cent}}{2^{\text{nd}}\text{ quantity (in respect of which the per cent has to be obtained)}} \times 100\%$$

EXAMPLE 1. 20g is what per cent of 1000g?

a. 2% b. 5% c. 7% d. 9%

Sol. a. Required percentage $= \frac{20}{1000} \times 100\% = 2\%$

Rule 4 Percentage increase or decrease in a quantity when it increases or decreases by some value.

$$\text{Percentage} = \frac{\text{Increase or decrease in the value}}{\text{Initial value}} \times 100\ \%$$

EXAMPLE 2. A person's salary has increased from ₹ 7200 to ₹ 8100. What is the percentage increase in his salary? 2013 II

a. 25% b. 18% c. $16\frac{2}{3}\%$ d. $12\frac{1}{2}\%$

Sol. d. Percentage increase in salary $= \frac{8100-7200}{7200} \times 100\%$

$$= \frac{900}{7200} \times 100\% = 12.5\% = 12\frac{1}{2}\%$$

Rule 5 If A is x% more than 'B', then B would be $\left[\frac{x}{(100+x)} \times 100\right]$% less than '$A$'.

Rule 6 If A is x% less than 'B', then B would be $\left[\frac{x}{(100-x)} \times 100\right]$% more than A.

EXAMPLE 3. Raj get 10% less marks than Rohit in an examination. What percentage of marks does Rohit gets more than Raj?

a. $11\frac{1}{6}\%$ b. $11\frac{1}{9}\%$ c. $12\frac{1}{9}\%$ d. None of these

Sol. b. Given, $x = 10\%$

$\therefore$ Rohit more percentage than Raj $= \frac{10}{100-10} \times 100$

$$= \frac{10 \times 100}{90} = 11\frac{1}{9}\%$$

Rule 7 If the price of a commodity is increased (or decreased) by x%, then the decrease (or increase) in consumption, so as not to increase or (decrease) the expenditure is $\left(\frac{x}{100 \pm x} \times 100\right)$%.

EXAMPLE 4. If the price of the cooking gas rises by 15%, by what per cent should a family reduce its consumption so as not to exceed the budget on cooking gas?

a. $12\frac{1}{23}\%$ b. $13\frac{1}{23}\%$ c. $14\frac{1}{23}\%$ d. None of these

Sol. b. Let initial price of cooking gas be ₹ 100.

Price after increase = ₹ 115

On ₹ 115 he should reduce ₹ 15 on ₹ 100, he should reduce

$$= \frac{15}{115} \times 100 = 13\frac{1}{23}\%$$

Shortcut Method

Here, $x = 15\%$

$\therefore$ Reduction in consumption $= \left(\frac{x}{100+x} \times 100\right)\%$

$$= \frac{15}{115} \times 100\% = 13\frac{1}{23}\%$$

Rule 8 If the value of an object is first changed (increased or decreased) by x% and then changed (increased or decreased) by y%, then

Net effect $= \left(\pm x \pm y + \frac{(\pm x)(\pm y)}{100}\right)\%$

Here, + sign is used in case of increment and – sign is used in case of decrement.

EXAMPLE 5 The price of an article is first increased by 20% and later on the price were decreased by 25% due to reduction in sales. Find the net percentage change in final price of article.

a. 20% b. 10% c. 30% d. 15%

Sol. b. Here, $a = 20\%$, $b = 25\%$

So, required percentage $= \frac{10}{100} \times 100 = 10\%$

$$\text{Required change} = \left[(\pm a) + (\pm b) + \frac{(\pm a)(\pm b)}{100}\right]\%$$

$$= \left[20 - 25 + \frac{20 \times (-25)}{100}\right]\% \left[\begin{matrix}\text{positive sign for increase}\\ \text{negative sign for decrease}\end{matrix}\right]$$

$$= [-5-5]\%$$

$$= -10\%$$

$\therefore$ Net percentage change is a decrease of 10 % because final result is negative.

Rule 9 If the population of a town is P and it increases (or decreases) at the rate of R% per annum, then

(i) Population after n yr $= P\left(1 \pm \frac{R}{100}\right)^n$

(ii) Population, n yr ago $= \frac{P}{\left(1 \pm \frac{R}{100}\right)^n}$

EXAMPLE 6 The population of a town is 352800. If it increases at the rate of 5% per annum, then what will be its population 2 yr hence, also find the population 2 yr ago.

a. 315000 b. 316500 c. 200045 d. 320000

Sol d. Given that, $p = 352800$, $R = 5\%$ and $n = 2$

According to the formula,

Population after 2 yr $= P\left(1 + \frac{R}{100}\right)^n = 352800 \times \left(1 + \frac{5}{100}\right)^2$

$$= 352800 \times \left(\frac{100+5}{100}\right)^2 = 352800 \times \left(\frac{21}{20} \times \frac{21}{20}\right) = 388962$$

Population 2 yr ago

$$= \frac{P}{\left(1 + \frac{R}{100}\right)^n} = \frac{352800}{\left(1 + \frac{5}{100}\right)^2} = 352800 \times \frac{20}{21} \times \frac{20}{21} = 320000$$

Rule 10 If the present population of a city is P and there is a increment or decrement of $R_1\%$, $R_2\%$ and $R_3\%$ in first, second and third year respectively, then

Population of city after 3 yr

$$= P\left(1 \pm \frac{R_1}{100}\right)\left(1 \pm \frac{R_2}{100}\right)\left(1 \pm \frac{R_3}{100}\right)$$

Note Use '+' sign for increases and '–' sign for decreases.

EXAMPLE 7 Population of a city in 2004 was 1000000. If in 2005 there is an increment of 15%, in 2006 there is a decrement of 35% and in 2007 there is an increment of 45%, then find the population of city at the end of year 2007.

a. 1083875 b. 1083000 c. 1089000 d. 1135000

Sol a. Given that, $P = 1000000$, $R_1 = 15\%$, $R_2 = 35\%$ (decrease) and $R_3 = 45\%$

Population of city at the end of year 2007.

$$= P\left(1 + \frac{R_1}{100}\right)\left(1 - \frac{R_2}{100}\right)\left(1 + \frac{R_3}{100}\right)$$

$$= 1000000\left(1 + \frac{5}{100}\right)\left(1 - \frac{35}{100}\right)\left(1 + \frac{45}{100}\right)$$

$$= 1000000 \times \frac{115}{100} \times \frac{65}{100} \times \frac{145}{100}$$

$$= 1083875$$

PRACTICE EXERCISE

1. If 90% of A = 30% of B and B = x% of A, the value of x is

(a) 700 (b) 600 (c) 300 (d) 1100

2. When 40% of a number is added to 42, the result is the number itself. The number is

(a) 70 (b) 90 (c) 82 (d) 72

3. If x% of y is $13x$, then the value of y is

(a) 880 (b) 1300 (c) 1200 (d) 700

4. What is the number when 20% of number is 30% of 40?

(a) 90 (b) 80 (c) 60 (d) 50

5. What is the number when increased by 20% becomes 300?

(a) 250 (b) 200 (c) 180 (d) 280

6. If 50% of $(x - y)$ = 40% of $(x + y)$, then what percent of x is y?

(a) $10\frac{1}{9}\%$ (b) $11\frac{1}{9}\%$ (c) $13\frac{1}{9}\%$ (d) $21\frac{1}{9}\%$

7. Two candidates fought an election one get 65% of the votes and won by 300 votes. The total number of votes polled in the election is

(a) 700 (b) 950 (c) 1000 (d) 900

8. A man spends ₹ 3500 and saves $12\frac{1}{2}\%$ of his income. His monthly income (in ₹) is

(a) 4000 (b) 3800 (c) 4200 (d) 4500

9. In an examination 52% of the candidates failed in English, 42% in Mathematics and 17% in both. The number of those who passed in both the subjects is

(a) 23% (b) 40% (c) 53% (d) 33%

10. To pass an examination, a candidate needs 40% marks. All questions carry equal marks. A candidate just passed by getting 10 answers correct by attempting 15 of the total questions. How many questions are there in the examination?

(a) 25 (b) 30
(c) 40 (d) 45

11. If salary of X is 20% more than salary of Y, then by how much percentage is salary of Y less than X?

(a) 25 (b) 20
(c) $\frac{50}{3}$ (d) $\frac{65}{4}$

12. Water contains $14\frac{2}{7}\%$ of hydrogen and the rest is oxygen. In 350 g of water, oxygen will be

(a) 300 g (b) 250 g
(c) 200 g (d) None of these

13. The income of 'A' is 20% higher than that of 'B'. The income of 'B' is 25% less than of C. What per cent less is A's income from 'C's income?

(a) 7% (b) 8%
(c) 10% (d) 12.5%

14. 38 L of milk was poured into a tub and the tub was found to be 5% empty. To completely fill the tub, what amount of additional milk must be poured?

(a) 1 L (b) 2 L (c) 3 L (d) 4 L

15. 10% of the inhabitants of a certain city left that city. Later on 10% of the remaining inhabitants of that city again left the city. What is the remaining percentage of population of that city?

(a) 80% (b) 80.4% (c) 80.4% (d) 81%

16. If after 24% of wastage the net output of a coal-mine is 68400 quintals. Then, the total output of the coal-mine in quintals is
(a) 70000 (b) 90000 (c) 80000 (d) 89000

17. A rise of 25% in the price of grapes compels a person to buy 1.5 kg of grapes less for ₹ 240. Then,the original price of grapes per kg is
(a) ₹ 40 (b) ₹ 32 (c) ₹ 30 (d) ₹ 28

18. The price of an item is increased by 20% and then decreased by 20% the final price as compared to original price is
(a) 4% less (b) 4% more (c) 20% less (d) 20% more

19. Sohan saves 14% of his salary while George saves 22%. If both gets the same salary and George saves ₹ 1540. Then, the salary of each of them is
(a) ₹ 9500 (b) ₹ 17000 (c) ₹ 7000 (d) ₹ 7500

20. A's salary is half that of B. If A got a 50% rise in his salary and B got a 25% rise in his salary, then the percentage increase in combined salaries of both is
(a) 13% (b) $33\frac{1}{3}\%$ (c) 33% (d) 45%

21. A man donated 4% of his salary to a charity and deposited 10% of the rest in the bank. If now he has ₹ 10800, then his income was
(a) ₹ 13500 (b) ₹ 14500
(c) ₹ 40000 (d) ₹ 12500

22. Rohit saves 30% of his salary. When his expenses increased by 30%, he is able to save ₹ 1215 per month. His monthly salary is
(a) ₹ 13500 (b) ₹ 14500
(c) ₹ 30000 (d) ₹ 12500

23. A man spends 75% of his income. If his income is increased by 20% and he increased his expenditure by 10%. His savings percentage is increased by
(a) 25% (b) 50% (c) 75% (d) 10%

24. The price of wheat has increased by 60%. In order to restore to the original price, the new price must be reduced by
(a) 37.5% (b) 33% (c) 34% (d) 40%

25. If the numerator of a fraction increased by 20% and its denominator be diminished by 10%. The value of the fraction is $\frac{16}{27}$, then the fraction is
(a) 4/9 (b) 3/2 (c) 3/8 (d) 9/4

26. 1 L of water is evaporated from 6 L of a solution having 4% of sugar. The percentage of sugar in the remaining solution is
(a) 4% (b) 5% (c) $4\frac{4}{5}\%$ (d) $\frac{4}{5}\%$

27. Two numbers are less than a third number by 30% and 40%, respectively. How much per cent is the second number less than the first?
(a) 35% (b) 36% (c) 14% (d) $14\frac{2}{7}\%$

28. A sample of 5 L of glycerine is formed to be adulterated to the extent of 20%. Find how glycerine should be added to bring down percentage of impurity to 5%?
(a) 10 L (b) 25 L (c) 15 L (d) 20 L

29. The daily wages of a worker increase by 20% but the number of hours worked by him also dropped by 20%. If originally he was getting ₹ 200 per week, his wages per week now is
(a) ₹ 160 (b) ₹ 192 (c) ₹ 210 (d) ₹ 198

30. 140 L of a liquid contains 90% of acid and the rest water. How much water must be added to make the water 12.5% of the resulting mixture?
(a) 4 L (b) 10 L (c) 12 L (d) 3 L

31. A person spends 30% of monthly salary on rent, 25% on food, 20% on children's education and 12% on electricity and the balance of ₹ 1040 on the remaining items. What is the monthly salary of the person?
(a) ₹ 8000 (b) ₹ 9000 (c) ₹ 9600 (d) ₹ 10600

32. In an office, 40% of the staff is female and rest is male. 60% of the male and 40% of female voted for Ramesh. The percentage of votes Ramesh got was
(a) 24% (b) 42% (c) 50% (d) 52%

33. In a class-X of 30 students, 24 passed in first class; in another class-Y of 35 students, 28 passed in first class. In which class was the percentage of students passed first class more?
(a) Class-X (b) Class-Y
(c) Both X and Y (d) None of these

Directions (Q. Nos. 34-36) *The population of the town is 126800. It increases by 15% in the 1st year due to increase crime in the city.*

34. What is the population of the town at the end of 2nd year if the population decreases by 20% in the second year?
(a) 174984 (b) 135996 (c) 116656 (d) 145820

35. What is the population of the town, if the population decrease by 15% at the end of 2nd year?
(a) 123749 (b) 123479
(c) 123947 (d) None of these

36. What is the population of the town, if the population increases by 20% at the end of 2nd year?
(a) 194784 (b) 174984 (c) 179484 (d) 178494

PREVIOUS YEARS' QUESTIONS

37. A man losses 20% of his money. After spending 25% of the remainder, he has ₹ 480 left. What is the amount of money he originally had? **2012 I**

(a) ₹ 600 (b) ₹ 720 (c) ₹ 800 (d) 840

38. The price of an article is ₹ 25. After two successive cuts by the same percentage, the price becomes ₹ 20.25. If each time the cut was x%, then **2012 II**

(a) $x = 9$ (b) $x = 10$ (c) $x = 11$ (d) $x = 11.5$

39. What is 5% of 50% of 500? **2012 II**

(a) 12.5 (b) 25 (c) 1.25 (d) 6.25

40. X, Y and Z had taken a dinner together. The cost of the meal of Z was 20% more than that of Y and the cost of the meal of X was 5/6 as much as the cost of the meal of Z. If Y paid ₹ 100, then what was the total amount that all the three of them had paid? **2013 II**

(a) ₹ 285 (b) ₹ 300
(c) ₹ 355 (d) None of these

41. A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece? **2013 II**

(a) 140% (b) $\frac{400}{3}$%
(c) 40% (d) None of these

42. If m% of $m + n$% of n = 2% of $(m \times n)$, then what percentage of m is n? **2014 II**

(a) 50% (b) 75%
(c) 100% (d) Cannot be determined

43. The price of a commodity increased by 5% from 2010 to 2011, 8% from 2011 to 2012 and 77% from 2012 to 2013. What is the average price increase (approximate) from 2010 to 2013? **2014 II**

(a) 26% (b) 32% (c) 24% (d) 30%

44. A person could save 10% of his income. But 2 yr later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased? **2015 I**

(a) $22\frac{2}{9}$% (b) $23\frac{1}{3}$%
(c) $24\frac{2}{9}$% (d) $25\frac{2}{9}$%

45. 20% of a number when added to 20 becomes the number itself, then the number is **2015 II**

(a) 20 (b) 25 (c) 50 (d) 80

46. A's salary was increased by 40% and then decreased by 20%. On the whole A's salary is increased by **2015 II**

(a) 60% (b) 40% (c) 20% (d) 12%

47. In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 votes more than his rival. The number of voters on the voter list is **2015 II**

(a) 3600 (b) 6200 (c) 6028 (d) 6400

48. The salary of a person is increased by 10% of his original salary. But he received the same amount even after increment. What is the percentage of his salary he did not receive? **2016 I**

(a) 11% (b) 10% (c) $\frac{100}{11}$% (d) $\frac{90}{11}$%

49. The expenditure of a household for a certain month is ₹ 20000, out of which ₹ 8000 is spent on education, ₹ 5900 on food, ₹ 2800 on shopping and the rest on personal care. What percentage of expenditure is spent on personal care? **2016 I**

(a) 12% (b) 16.5% (c) 18% (d) 21.8%

ANSWERS

1	c	2	a	3	b	4	c	5	a	6	b	7	c	8	a	9	a	10	a
11	c	12	a	13	c	14	b	15	d	16	b	17	b	18	a	19	c	20	b
21	d	22	a	23	b	24	a	25	a	26	c	27	d	28	c	29	b	30	a
31	a	32	d	33	c	34	c	35	c	36	b	37	c	38	b	39	a	40	d
41	b	42	c	43	d	44	a	45	b	46	d	47	b	48	c	49	b		

HINTS AND SOLUTIONS

1. *(c)* Given, 90% of A = 30% of B

$$\frac{90A}{100} = \frac{30B}{100} \quad \text{[Rule 1]}$$

$\Rightarrow \frac{A}{B} = \frac{3}{9} \Rightarrow B = 3A$

Now, $B = x\%$ of A, $3A = \frac{xA}{100}$ [Rule 1]

$\therefore \quad x = 300$

Hence, the value of x is 300.

2. *(a)* Let the number be x,

then, $42 + 40\%$ of $x = x$

$\Rightarrow \quad x = 42 + \frac{40}{100}x$ [Rule 1]

$\Rightarrow \frac{3}{5}x = 42$

$\therefore \quad x = \frac{42 \times 5}{3} = 70$

Hence, the required number is 70.

3. *(b)* Given, $x\%$ of $y = 13x$

$\Rightarrow \quad \frac{x}{100}y = 13x$ [Rule 1]

$\therefore \quad y = 13 \times 100 = 1300$

4. *(c)* Let the number be x.

By given condition, 20% of x

= 30% of 40

$\Rightarrow \frac{20x}{100} = \frac{30 \times 40}{100} \Rightarrow 20x = 30 \times 40$

$\therefore \quad x = 60$

Hence, the required number is 60.

5. *(a)* Let the number be x.

$\therefore \; x + \frac{20x}{100} = 300$

$\Rightarrow \quad x = \frac{300 \times 100}{120} = 250$

6. *(b)* Given, 50% of $(x - y)$ = 40% of $(x + y)$

$\frac{50}{100} \times (x - y) = \frac{40}{100} \times (x + y)$

$\Rightarrow \quad 5x - 5y = 4x + 4y$ [Rule 1]

$\Rightarrow \quad x = 9y$ ... (i)

Let $\quad r\%$ of $x = y \Rightarrow \frac{r}{100} \times x = y$

$\Rightarrow \quad \frac{r}{100} \times 9y = y$ [from Eq. (i)]

$\therefore \quad r = \frac{100}{9} = 11\frac{1}{9}\%$

7. *(c)* Let total number of votes polled be x.

$\therefore$ 65% of x – 35% of x = 300

$\frac{65}{100}x - \frac{35}{100}x = 300$

$\Rightarrow \frac{30}{100}x = 300 \Rightarrow x = \frac{300 \times 100}{30}$

$\therefore \quad x = 1000$ votes

Hence, the total number of votes is 1000.

8. *(a)* Let monthly income be ₹ x.

$\Rightarrow \quad 87\frac{1}{2}\%$ of x = ₹ 3500

$\Rightarrow \quad \frac{175}{2 \times 100}x$ = ₹ 3500

$\therefore \quad x = \frac{3500 \times 2 \times 100}{175}$ = ₹ 4000

9. *(a)* Number of candidates who passed in both the subject $= 100 - (x + y - z)\%$

$= 100 - (52 + 42 - 17)\%$

$= 23\%$

10. *(a)* Let the total number of questions in examination be x.

By given condition, 40% of x = 10

$\Rightarrow \quad \frac{x \times 40}{100} = 10 \Rightarrow x = \frac{1000}{40} = 25$

11. *(c)* Let the salary of y be ₹ x.

$\therefore$ Salary of x = ₹ $\frac{120x}{100}$ = ₹ $\frac{6x}{5}$

Difference of their salaries

= ₹ $\left(\frac{6x}{5} - x\right)$ = ₹ $\frac{x}{5}$

$\therefore$ Required percentage

$= \frac{x/5}{6x/5} \times 100\% = \frac{50}{3}\%$

[Rule 3]

Shortcut Method

Here, $x = 20\%$

$\therefore$ Required percentage

$= \left(\frac{x}{100 + x} \times 100\right)\%$ [Rule 5]

$= \left(\frac{20}{100 + 20} \times 100\right)\%$

$= \frac{2000}{120}\% = \frac{50}{3}\%$

12. *(a)* Percentage of oxygen in water

$= 100 - 14\frac{2}{7} = 100 - \frac{100}{7} = 85\frac{5}{7}\%$

Percentage of oxygen in 350 g of water

$= 85\frac{5}{7}\%$ of $350 = \frac{600}{7} \times \frac{350}{100} = 300$ g

13. *(c)* Let the income of C be ₹ x.

$\therefore$ Income of $B = x - x \times 25\% = x - \frac{25x}{100}$

$= x - \frac{x}{4} = \frac{3x}{4}$

and, the income of $A = \frac{3x}{4} + \frac{3x}{4} \times 20\%$

$= \frac{3x}{4} + \frac{3x}{4} \times \frac{1}{5} = \frac{9x}{10}$

Now, difference between C's salary and A's salary

$= x - \frac{9x}{10} = \frac{x}{10}$

$\therefore$ The required percentage

$= \frac{x}{10 \times x} \times 100\% = 10\%$

14. *(b)* Amount of milk in tub = 38 L

$\because$ Tub found to be 5% empty

$\therefore$ Total quantity of milk in per cent = 95%

Now, completely fill the tub, total amount of additional milk

$= \frac{38 \times 5}{95} = 2$ L

15. *(d)* Remaining percentage

$= \left(1 - \frac{10}{100}\right)\left(1 - \frac{10}{100}\right)$

$= \frac{90}{100} \times \frac{90}{100} = 81\%$

16. *(b)* Let total output be x quintal.

$\therefore$ Useful output = 100 – 24 = 76%

$\Rightarrow \quad \frac{76}{100}x = 68400$

$\Rightarrow \quad x = \frac{68400 \times 100}{76}$

$\therefore \quad x = 90000$ quintal

17. *(b)* Let original price of grapes be ₹ x.

$\therefore$ Increased price $= x + \frac{25}{100}x$

= ₹ $\frac{125}{100}x$ ...(i)

Increased price of 1.5 kg of grapes

$= \frac{25}{100} \times 240$ = ₹ 60

$\therefore$ Increased price of 1 kg $= \frac{60}{1.5}$

= ₹ 40 ...(ii)

So, from Eqs. (i) and (ii), we have

$\frac{125}{100}x$ = ₹ 40

$\therefore \quad x = \frac{40 \times 100}{125}$ = ₹ 32

18. *(a)* The price of item first increased by 20% and then decreased by 20%.

$\therefore$ Net effect $= \left(20 - 20 + \frac{20 \times (-20)}{100}\right)$

$= \frac{-400}{100} = -4\%$ [Rule 8]

19. *(c)* Let salary of each of them be ₹ x.
George saves 22% of x and his saving amount is ₹ 1540.

$\Rightarrow \quad \frac{22}{100}x = 1540$

$\Rightarrow \quad x = \frac{1540 \times 100}{22} = ₹\ 7000$

20. *(b)* Let A's salary = x. Then,
B's salary = $2x$

New salary of A = 150% of $x = \frac{3}{2}x$

Total salary of B = 125% of $2x = \frac{5}{2}x$

Total combined salary

$= \left(\frac{3}{2}x + \frac{5}{2}x\right) = 4x$

∴ Required increment in salary

$= \frac{4x - 3x}{3x} = \frac{x}{3x} \times 100 = \frac{100}{3} = 33\frac{1}{3}\%$

[Rule 4]

21. *(d)* Let total income be ₹ x.

Income deposited = 10% of $\left[x - \frac{4}{100}x\right]$

$= \frac{10}{100}\left(x - \frac{4}{100}x\right) = \frac{96}{1000}x$

Remaining income = ₹ 10800

$\therefore \quad \frac{4}{100}x + \frac{96x}{1000} + 10800 = x$

$\Rightarrow \quad 10800 = x - \frac{136x}{1000}$

$\Rightarrow \quad 10800 = \frac{864x}{1000} \Rightarrow x = \frac{10800 \times 1000}{864}$

$\therefore \quad x = ₹\ 12500$

22. *(a)* Let the salary of Rohit be ₹ 100, then saving = ₹ 30

Expenses = ₹ 70

New expenses = (100 + 30)% of ₹ 70
= ₹ 91

New saving = ₹ (100 – 91) = ₹ 9

He saves ₹ 9, his salary = ₹ 100

If he saves ₹ 1215.

Then, his salary = ₹ $\left(\frac{100}{9} \times 1215\right)$

= ₹ 13500

23. *(b)* Let income of man be ₹ 100.
Then, his expenditure = ₹ 75
and savings = ₹ 25
New income = ₹ (100 + 20) = ₹ 120
New expenditure = ₹ (75 + 7.5)
= ₹ 82.50
New saving = ₹ (120 – 82.5) = ₹ 37.50
Saving difference = 37.50 – 25.0 = 12.5
∴ Percentage increase saving

$= \frac{12.5}{25} \times 100 = 50\%$

24. *(a)* New price reduced by

$= \left(\frac{x}{100 + x}\right) \times 100\%$ [Rule 7]

$= \left(\frac{60}{100 + 60}\right) \times 100\% = 37.5\%$

25. *(a)* Let fraction be $\frac{x}{y}$.

New fraction $= \frac{120\% \text{ of} x}{90\% \text{ of } y} = \frac{4x}{3y}$

According to question,

[by given condition]

$\frac{4x}{3y} = \frac{16}{27} \Rightarrow \frac{x}{y} = \frac{16}{27} \times \frac{3}{4} = \frac{4}{9}$

26. *(c)* Amount of sugar in 6 L of solution

$= \frac{4}{100} \times 6 = 0.24$ L

After evaporation, sugar in 5 L = 0.24 L

∴ Percentage of sugar

$= \left(\frac{0.24}{5} \times 100\right) = 4\frac{4}{5}\%$

27. *(d)* Let the third number be z.

∴ First number $x = \frac{(100 - 30)}{100} \times z = \frac{7z}{10}$

Second number

$y = \frac{(100 - 40)}{100} \times z = \frac{6z}{10}$

Difference between first and second number

$= (x - y) = \frac{7z}{10} - \frac{6z}{10} = \frac{z}{10}$

Hence, the required percent [Rule 4]

$= \frac{\frac{z}{10}}{\frac{7z}{10}} \times 100\% = \frac{100}{7}\% = 14\frac{2}{7}\%$

28. *(c)* Glycerine in the given sample = 80% of 5 L

$= \frac{80}{100} \times 5 = 4$ L

Let x L of glycerine be added, then

$\frac{4 + x}{(5 + x)} \times 100 = 95$

$\Rightarrow \quad 80 + 20x = 95 + 19x$

$\therefore \quad x = 15$ L

29. *(b)* Increased wages of the worker = 200 + 20% of 200 = ₹ 240

Also, let he worked for x h.

∴ Reduced working hours

$= x - \frac{20}{100}x = 0.80x$

Required wages $= \frac{240}{x} \times 0.8x = ₹\ 192$

30. *(a)* Water in the mixture = 10% of 140 L

$= \frac{10}{100} \times 140 = 14$ L

Let x L of water added in the mixture, then $\left(\frac{14 + x}{140 + x}\right) \times 100 = 12.5$

$\Rightarrow \quad 1400 + 100x = 1750 + 12.5x$

$\Rightarrow \quad 87.5x = 350$

$\Rightarrow \quad x = \frac{350}{87.5} = 4$L

31. *(a)* Let the monthly salary of the person be ₹ x.

By given condition,

$\frac{(100 - 30 - 25 - 20 - 12) \times x}{100} = 1040$

$\Rightarrow \quad \frac{13x}{100} = ₹\ 1040 \Rightarrow x = \frac{1040 \times 100}{13}$

$\Rightarrow \quad x = ₹\ 8000$

Hence, the monthly salary of the person is ₹ 8000.

32. *(d)* Let total number of staff be 100.

Female staff = 40

Male staff = (100 – 40) = 60

Votes casted by females

$= \frac{40}{100} \times 40 = 16$

Votes casted by males $= \frac{60}{100} \times 60 = 36$

Votes casted by both males and females

= 16 + 36 = 52

∴ Percentage votes obtained = 52%.

33. *(c)* For class-X, let the student passed in first class = a%.

Then, by condition given in question,

$a\%$ of $30 = 24 \Rightarrow \frac{a \times 30}{100} = 24$

$\therefore \quad a = 80\%$

Now, for class-Y let the student passed in first class = b%.

According to the question,

$b\%$ of $35 = 28$

$\Rightarrow \quad \frac{b}{100} \times 35 = 28$

$\therefore \quad b = 80\%$

Hence, both classes have equal percentage of students getting first class.

34. *(c)* Given, $R_1 = 15\%$ and $R_2 = 20\%$.

∴ Required population

$= P\left(1 + \frac{R_1}{100}\right)\left(1 - \frac{R_2}{100}\right)$ [Rule 10]

$= 126800\left(1 + \frac{15}{100}\right)\left(1 - \frac{20}{100}\right)$

$= 126800\left(1+\frac{3}{20}\right)\left(1-\frac{1}{5}\right)$

$= 126800\left(\frac{23}{20}\right)\left(\frac{4}{5}\right) = 116656$

35. *(c)* Given, $R_1 = 15\%$, $R_2 = 15\%$

$\therefore$ Population $= P\left(1+\frac{R_1}{100}\right)\left(1-\frac{R_2}{100}\right)$

$= 126800\left(1+\frac{15}{100}\right)\left(1-\frac{15}{100}\right)$

$= 126800\times\frac{115}{100}\times\frac{85}{100} = 123947$

36. *(b)* Given, $R_1 = 15\%$ and $R_2 = 20\%$

$\therefore$ Required population

$= P\left(1+\frac{15}{100}\right)\left(1+\frac{20}{100}\right)$

$= 126800\left(\frac{23}{20}\right)\left(\frac{6}{5}\right) = 174984$

37. *(c)* Refer to question 21.

38. *(b)* Refer to question 34.

39. *(a)* 5% of 50% of 500

$= \frac{5}{100}\times\frac{50}{100}\times 500 = 12.5$ [Rule 1]

40. *(d)* Given, the cost of meal of Y = ₹ 100

Now, according to the question,

Cost of the meal of Z = 20% more than that of Y

$= \left(100+\frac{20}{100}\times 100\right)$

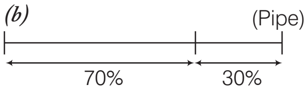
$= (100+20) =$ ₹ 120

and, the cost of the meal of $X = \frac{5}{6}$ as much as the cost of the meal of $Z = \frac{5}{6}\times 120 =$ ₹ 100

$\therefore$ Total amount that all the three of them has paid

$= 100 + 100 + 120 =$ ₹ 320

41. *(b)*



$\therefore$ Increase in percentage of longer piece compared to shorter piece

$= \frac{70-30}{30}\times 100\%$ [Rule 4]

$= \frac{40}{30}\times 100\% = \frac{400}{3}\%$

42. *(c)* Refer to question 6.

43. *(d)* Average price increase

$= \left(\frac{5+8+77}{3}\right)\% = \frac{90}{3}\% = 30\%$

44. *(a)* Refer to question 23.

45. *(b)* Refer to question 2.

46. *(d)* Refer to question 18.

47. *(b)* Let the number of voters on the voter list be x.

Total cast vote = 90% of $x - 60$

Winner was supported by 47% of total voter. i.e. 47% of x.

Hence, rival got vote = (90% of $x - 60$) $-$ 47% of x.

= 43% of $x - 60$

It is given that difference between their vote is 308.

Then, 47% of $x -$ 43% of $x + 60 = 308$

$\Rightarrow$ 4% of $x = 308 - 60$

$\Rightarrow \frac{4}{100}x = 248$

$\therefore x = \frac{248\times 100}{4} = 6200$

48. *(c)* Let the original salary be x.

Then, increased salary

$= \left(\frac{110}{100}\right)x =$ ₹ $\frac{11x}{10}$

$\because$ He received the same salary even after increment.

Amount of salary he did not receive

$= \frac{11x}{10} - x =$ ₹ $\frac{x}{10}$

$\therefore$ Amount of salary in percentage

$= \frac{\left(\frac{x}{10}\right)}{\left(\frac{11x}{10}\right)}\times 100\%$

$= \frac{x}{11x}\times 100\% = \frac{100}{11}\%$

49. *(b)* Refer to question 31.

09

SIMPLE INTEREST

Regularly (1-2) questions have been asked from this chapter. Generally direct formula based questions are asked from this chapter and hence making it easy to score area.

INTEREST

When money is borrowed by a person, then customarily the money lender used to charge some extra money in lieu of the money lent by him. This extra money earned by the money lender is called interest.

Some terms related to interest are given below

Principal (P) The money which is borrowed from a money lender, is called Principal.

Amount (A) The sum of the principal and the interest is called Amount i.e. Amount (A) = Principal (P) + Interest (I).

Rate (R) It is the rate at which the interest is charged on Principal.

Time (T) The time period, for which the money is lent or deposited or borrowed, is called Time.

SIMPLE INTEREST

If the interest is calculated on the original Principal for any length of time, then it is called simple interest.

$$\text{Simple Interest (SI)} = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100} \quad \text{or} \quad \text{SI} = \frac{P \times R \times T}{100}$$

$$\text{Amount } (A) = \text{Principal } (P) + \text{Interest } (I) \text{ and } P = \frac{100 \times A}{100 + RT}$$

Where, $A \to$ Amount

$R \to$ Rate of simple Interest

$T \to$ Time

EXAMPLE 1. The sum required to earn a monthly interest of ₹400 at 10% per annum at simple interest is

a. ₹ 2000 b. ₹ 12000 c. ₹ 24000 d. ₹ 48000

Sol. **d.** Given, SI = ₹ 400, $R = 10\%$ and $T = 1$ month $= \frac{1}{12}$ yr

$$SI = \frac{P \times R \times T}{100} \quad \text{or} \quad P = \frac{100 \times SI}{R \times T} = \frac{100 \times 400 \times 12}{10 \times 1} = ₹\ 48000$$

EXAMPLE 2. In what time will the simple interest on ₹ 400 at 10% per annum be the same as the simple interest on ₹ 1000 for 4 yr at 4% per annum?

a. 2 yr **b.** 3 yr **c.** 4 yr **d.** 6 yr

Sol. **c.** Here, $P =$ ₹1000, $T = 4$ yr, and $R = 4\%$

$\therefore$ Simple interest $= \frac{P \times R \times T}{100} = \frac{1000 \times 4 \times 4}{100} =$ ₹ 160

Now, simple interest = ₹ 160, $P =$ ₹ 400, $R = 10\%$,

then, $T = \frac{100 \times SI}{P \times R} = \frac{100 \times 160}{400 \times 10} = 4$ yr

EXAMPLE 3. A sum at simple interest of 4% per annum amounts to ₹ 3120 in 5 yr. Find the sum.

a. ₹ 2500 **b.** ₹ 1300 **c.** ₹ 4000 **d.** ₹ 2600

Sol. **d.** Here, $T = 5$ yr, $R = 4\%$, $A =$ ₹ 3120

We know that, $P = \frac{100 \times A}{100 + RT} = \frac{100 \times 3120}{100 + 4 \times 5} = \frac{100 \times 3120}{120}$
= ₹ 2600

Important Rules and Formulae

Rule 1 If a sum of money becomes n times in 'T' yr at simple interest, then rate of interest will be, $R = \left(\frac{100(n-1)}{T}\right)\%$

EXAMPLE 4. At what rate per cent per annum will a sum of money double in 8 yr?

a. 12% **b.** $12\frac{1}{2}\%$

c. 13% **d.** 15%

Sol. **b.** Here, $n = 2$ and $T = 8$ yr

$\therefore$ Rate $= \frac{100(n-1)}{T} = \frac{100 \times (2-1)}{8} = \frac{100}{8} = 12\frac{1}{2}\%$

Rule 2 If a certain Principal Amounts to ₹ A_1 in t_1 yr and to ₹ A_2 in t_2 yr, then the sum (P) is given by ₹ $\left(\frac{A_2 t_1 - A_1 t_2}{t_1 - t_2}\right)$ and the rate per cent (R) per annum is given by $\left[\frac{100 (A_2 - A_1)}{A_1 t_2 - A_2 t_1}\right]\%$.

EXAMPLE 5. A certain sum amounts to ₹ 1586 in 2 yr and ₹ 1729 in 3 yr. Find the rate and the sum.

a. 8%, ₹ 1200
b. 9%, ₹ 1300
c. 10%, ₹ 1000
d. 11%, ₹ 1300

Sol. **d.** Here, $A_1 =$ ₹1586, $t_1 = 2$ yr, $A_2 = 1729$ and $t_2 = 3$ yr

$\therefore$ Required principle $= \frac{A_2 t_1 - A_1 t_2}{t_1 - t_2} = \frac{1729 \times 2 - 1586 \times 3}{2 - 3}$

= ₹ 1300

and required rate $R = \left[\frac{(A_2 - A_1) \times 100}{A_1 t_2 - A_2 t_1}\right]\%$

$= \frac{(1729 - 1586) \times 100}{1586 \times 3 - 1729 \times 2} = \frac{143 \times 100}{1300} = 11\%$

Rule 3 At the same rate of simple interest, if a sum of money becomes n_1 times of itself in t_1 yr and n_2 times in t_2 years, then $t_2 = \frac{(n_2 - 1)}{(n_1 - 1)} t_1$ yr.

EXAMPLE 6. A sum of money becomes 3 times in 5 yr at simple interest. In how many years, will the same sum become 6 times at the same rate of simple interest?

a. 10 yr
b. 12 yr
c. 12.5 yr
d. 10.5 yr

Sol. **c.** Here, $n_1 = 3$, $t_1 = 5$yr and $n_2 = 6$, $t_2 = ?$

$\therefore$ Required time $(t_2) = \frac{(n_2 - 1)}{(n_1 - 1)} t_1 = \frac{(6-1)}{(3-1)} \times 5 = \frac{25}{2} = 12.5$ yr

PRACTICE EXERCISE

1. Find the amount on a sum of ₹ 400 for 3 yr at simple interest at 5% per annum.

(a) ₹ 460 (b) ₹ 415 (c) ₹ 435 (d) ₹ 412

2. Find what sum of money will amount to ₹ 900 in 4 yr at 5% per annum on simple interest?

(a) ₹ 750 (b) ₹ 650
(c) ₹ 500 (d) ₹ 550

3. If a certain sum is doubled in 8 yr on simple interest, in how many years will it be four times?

(a) 24 yr (b) 16 yr (c) 32 yr (d) 12 yr

4. A sum of money at simple interest amount to ₹ 1260 in 2 yr and ₹1350 in 5 yr, then the rate per cent per annum is

(a) 30% (b) 10% (c) 2.5% (d) 5%

5. The difference of 13% per annum and 12% of a sum in 1 yr is ₹ 110. Then, the sum is
(a) ₹ 12000 (b) ₹ 13000 (c) ₹ 11000 (d) ₹ 16000

6. The simple interest on a sum of money at 10% per annum for 6 yr is half the sum. Then, the sum is
(a) ₹ 5000 (b) not possible (c) ₹ 4000 (d) ₹ 6000

7. The sum which amounts to ₹ 840 in 5 yr at the rate of 8% per annum simple interest is
(a) ₹ $\frac{100 \times 8 \times 5 + 100}{840}$ (b) ₹ $\frac{100 + 840}{100 + 5 \times 8}$
(c) ₹ $\frac{100 \times 840}{100 + 5 \times 8}$ (d) ₹ $\frac{840 \times 5 \times 8}{100}$

8. A certain sum at simple interest amounts to ₹ 1040 in 3 yr and to ₹ 1360 in 7 yr. Then, the sum is
(a) ₹ 750 (b) ₹ 800 (c) ₹ 900 (d) ₹ 1000

9. A sum becomes 6 fold at 5% per annum. At what rate, the sum becomes 12 fold?
(a) 10% (b) 12% (c) 9% (d) 11%

10. A man borrowed ₹ 40000 at 8% simple interest per year. At the end of second year he paid back certain amount and at the end of fifth year he paid back ₹ 35960 and cleared the debt. What is the amount did he pay back after the second year?
(a) ₹ 16200 (b) ₹ 17400
(c) ₹ 18600 (d) None of these

11. A sum of ₹ 1550 was lent partly at 5% and partly at 8% simple interest. The total interest received after 3 yr was ₹ 300. The ratio of money lent at 5% to 8% is
(a) 11 : 12 (b) 16 : 15 (c) 12 : 21 (d) 11 : 13

12. Rahim buys a house and pays ₹ 8000 cash and ₹ 9600 at 5 yr credit at 4% per annum simple interest. Then, the cash price of the house
(a) ₹ 10000 (b) ₹ 9600 (c) ₹ 17000 (d) ₹ 16000

13. At what rate per cent per annum simple interest, will a sum of money triple itself in 25 yr?
(a) 8% (b) 9% (c) 10% (d) 12%

14. A man invested ₹ 1000 on a simple interest at a certain rate and ₹ 1500 at 2% higher rate. The total interest in 3 yr is ₹ 390. What is the rate of interest for ₹ 1000?
(a) 4% (b) 5% (c) 6% (d) 8%

15. In what time the simple interest on a sum of money be 3/ 8 of the principal with rate of interest 3(1/ 8)%?
(a) 9 yr (b) 6 yr (c) 12 yr (d) 15 yr

16. If the rate of simple interest is 12% per annum the amount that would fetch interest of ₹ 6000 per annum is
(a) ₹ 7200 (b) ₹ 72000 (c) ₹ 50000 (d) ₹ 48543.69

17. A lends a sum of money for 10 yr at 5% simple interest, B lends double that amount for 5 yr at the same rate of interest. Which of the following statement is true in this regard?
(a) A will get twice the amount of interest that B would get
(b) B will get twice the amount of interest that A would get.
(c) A and B will get the same amount as interest.
(d) B will get four times the amount of interest that A would get

18. Out of a sum of ₹ 625 a part was lent at 5% and the other at 10% simple interest. If the interest on the first part after 2 yr is equal to the interest on the second part after 4 yr, then the second sum (in ₹) is
(a) ₹ 125 (b) ₹ 200 (c) ₹ 250 (d) ₹ 300

19. A man invests an amount of ₹ 15860 in the names of his three sons A, B and C in such a way that they get the same interest after 2, 3 and 4 yr, respectively. If the rate of simple interest is 5%, then the ratio of the amounts invested among A, B and C will be
(a) 10 : 15 : 20 (b) $\frac{1}{10} : \frac{1}{15} : \frac{1}{20}$
(c) 110 : 115 : 120 (d) $\frac{1}{110} : \frac{1}{115} : \frac{1}{120}$

20. A sum was put at simple interest at a certain rate for 2 yr. Had it been put at 3% higher rate, it would have fetched ₹ 72 more. The sum is
(a) ₹ 1200 (b) ₹ 1600 (c) ₹ 1900 (d) ₹ 1400

21. Harsha makes a fixed deposit of ₹ 20000 in Bank of India for a period of 3 yr. If the rate of interest be 13% SI per annum charged half yearly, what amount will he get after 42 months?
(a) ₹ 27800 (b) ₹ 28100 (c) ₹ 29100 (d) ₹ 30000

22. A sum of money lent on simple interest triples itself in 15 yr and 6 months. In how many year still it be doubled?
(a) 6 yr and 3 months (b) 7 yr and 9 months
(c) 8 yr and 3 months (d) 9 yr and 6 months

23. Mr Pawan invests an amount of ₹ 24200 at the rate of 4% per annum for 6 yr to obtain a simple interest, later he invests the principal amount as well as the amount obtained as simple interest for another 4 yr at the same rate of interest. What amount of simple interest will be obtained at the end of the last 4 yr?
(a) ₹ 4800 (b) ₹ 4850.32 (c) ₹ 4801.28 (d) ₹ 4700

24. A person invested some amount at the rate of 12% simple interest and the remaining at 10%. He received yearly an interest of ₹ 130. Had he interchanged the amounts invested, he would have received an interest of ₹ 134. How much money did he invest at different rates?
(a) ₹ 500 at the rate of 10%, ₹ 800 at the rate of 12%
(b) ₹ 700 at the rate of 10%, ₹ 600 at the rate of 12%
(c) ₹ 800 at the rate of 10%, ₹ 400 at the rate of 12%
(d) ₹ 700 at the rate of 10%, ₹ 500 at the rate of 12%

Directions (Q. Nos. 25-27) *If SI for a certain sum P_1 for time T_1 and rate of interest R_1 is I_1 and SI for another sum P_2 for time T_2 and rate of interest R_2 is I_2, then difference of SI* $= I_2 - I_1 = \frac{P_2R_2T_2 - P_1R_1T_1}{100}$.

25. Simple interest for the sum of ₹ 1500 is ₹ 50 in 4 yr and ₹ 80 in 8 yr, the rate of SI is
(a) 0.6% (b) 5% (c) 0.05% (d) 0.5%

26. Simple interest for the sum of ₹ 1230 for 2 yr is ₹ 10 more than the simple interest for ₹ 1130 for the same duration. Find the rate of interest.
(a) 5% (b) 6% (c) 8% (d) 2%

27. If the annual payment oa ₹ A will discharge a debt of ₹ 1092 due in 2 yr at 12% simple interest, then
I. A will be ₹ 515 (approx.).
II. A will be ₹ 530 (approx.), if interest rate is 6%.
(a) Only I (b) Both I and II
(c) Only II (d) Neither I nor II

PREVIOUS YEARS' QUESTIONS

28. The principal on which a simple interest of ₹ 55 will be obtained after 9 months at the rate of 3(2/ 3)% per annum is **2013 I**
(a) ₹ 1000 (b) ₹ 1500
(c) ₹ 2000 (d) ₹ 2500

29. In how much time would the simple interest on a principal amount be 0.125 times the principal amount at 10% per annum? **2015 I**
(a) $1\frac{1}{4}$ yr (b) $1\frac{3}{4}$ yr (c) $2\frac{1}{4}$ yr (d) $2\frac{3}{4}$ yr

30. If a sum of money at a certain rate of simple interest per year doubles in 5 yr and at a different rate of simple interest per year becomes three times in 12 yr, then the difference in the two rates of simple interest per year is **2016 I**
(a) 2% (b) 3% (c) $3\frac{1}{3}$% (d) $4\frac{1}{3}$%

ANSWERS

1	a	2	a	3	a	4	c	5	c	6	b	7	c	8	b	9	d	10	b
11	b	12	d	13	a	14	a	15	c	16	c	17	c	18	a	19	b	20	a
21	c	22	b	23	c	24	d	25	d	26	a	27	b	28	c	29	a	30	c

HINTS AND SOLUTIONS

1. *(a)* Given, $P =$ ₹400, $R = 5\%$ and $T = 3\,yr$

Simple interest $= \frac{P \times R \times T}{100}$

$$SI = \frac{400 \times 3 \times 5}{100} = ₹\ 60$$

$\therefore$ Amount $= P + SI = 400 + 60 =$ ₹ 460

2. *(a)* Let the sum of money be ₹ x.

$\because$ Amount = Sum + SI

$\therefore$ Amount $= x + \frac{x \times 5 \times 4}{100}$ $\left[\because SI = \frac{P \times R \times T}{100}\right]$

But amount = ₹ 900

$\therefore$ $900 = x + \frac{20x}{100}$

$\Rightarrow$ $900 = \frac{6x}{5} \Rightarrow x = \frac{900 \times 5}{6} = ₹\ 750$

Shortcut Method

Here, $A =$ ₹900, $R = 5\%$ and $T = 4$yr

$\because$ Sum

$= \frac{100 \times A}{100 + RT} = \frac{100 \times 900}{100 + 5 \times 4} = ₹750$

3. *(a)* Let the sum be ₹ x, so amount = ₹ $2x$

$\therefore$ SI = ₹ x

Let R be rate of interest.

$\therefore$ $R = \frac{100 \times SI}{P \times T} = \frac{100 \times x}{x \times 8} = 12.5\%$

Now, the needed amount = ₹ $4x$

$\therefore$ SI = ₹ $(4x - x) =$ ₹ $3x$

$\therefore$ $T = \frac{100 \times SI}{P \times R} = \frac{100 \times 3x}{x \times 12.5} = 24$ yr

Shortcut Method

Here, $n_1 = 2$, $n_2 = 4$ and $t_1 = 8$ yr

$\therefore$ Required time $= \frac{(n_2 - 1)}{(n_1 - 1)} \times t_1$ [Rule 3]

$= \frac{(4-1)}{(2-1)} \times 8 = \frac{3}{1} \times 8 = 24$ yr

4. *(c)* Simple interest in 3 yr

= ₹ (1350 − 1260) = ₹ 90

$\therefore$ Simple interest for 2 yr

$= \frac{2}{3} \times 90 =$ ₹ 60

$\because$ Principal = ₹ (1260 − 60) = ₹ 1200

$\therefore$ Rate,

$R = \frac{100 \times SI}{P \times T} = \frac{100 \times 60}{1200 \times 2} = \frac{60}{24} = \frac{5}{2}$

$R = 2.5\%$

Shortcut Method

Here, $A_1 =$ ₹1260, $t_1 = 2$yr,
$A_2 =$ ₹1350 and $t_2 = 5$yr

$\therefore$ Required rate

$= \left[\frac{(A_2 - A_1)}{A_1t_2 - A_2t_1}\right] \times 100\%$ [Rule 2]

$= \frac{(1350 - 1260)}{1260 \times 5 - 1350 \times 2} \times 100\% = 2.5\%$

5. *(c)* Let the sum be ₹ x.

Then, $\frac{x \times 13 \times 1}{100} - \frac{x \times 12 \times 1}{100} = 110$

$\Rightarrow \frac{x}{100} = 110$

$\Rightarrow x = 110 \times 100 = ₹ 11000$

Shortcut Method

Here, $P = ₹ 110$, $r_1 = 13\%$, $r_2 = 12\%$ and $t_1 = t_2 = 1$yr

$\therefore$ Short trick

$= \frac{110 \times 100}{13 - 12}$

$= ₹ 11000$

6. *(b)* Let the sum (principal) be ₹ x.

$\therefore$ Simple interest = ₹ $\frac{x}{2}$

and $T = 6$ yr, $R = 10\%$ per annum

$\therefore$ $\text{SI} = \frac{P \times R \times T}{100}$

$\Rightarrow \frac{x}{2} = \frac{x \times 10 \times 6}{100} \Rightarrow \frac{1}{2} = \frac{6}{10}$

Which is not true, so it is not a possible case.

7. *(c)* Let the sum be ₹ 100.

Then, amount = (Sum + SI)

$= \left(100 + \frac{100 \times 8 \times 5}{100}\right) = (100 + 8 \times 5)$

So, when the amount is $(100 + 8 \times 5)$, then sum = 100

When the amount is ₹ 840, then sum

$= ₹ \left(\frac{100 \times 840}{100 + 8 \times 5}\right)$

8. *(b)* Simple interest for 4 yr

$= ₹ (1360 - 1040) = ₹ 320$

Simple interest for 3 yr

$= ₹ \left(\frac{320}{4} \times 3\right) = ₹ 240$

$\therefore$ Sum or principle

$= ₹ (1040 - 240) = ₹ 800$

Shortcut Method

Here, $A_1 = ₹ 1040$, $t_1 = 3$yr,

$A_2 = ₹ 1360$ and $t_2 = 7$yr

$\therefore$ Required principle (sum)

$= ₹ \left(\frac{A_2 t_1 - A_1 t_2}{t_1 - t_2}\right)$ [Rule 2]

$= \frac{1360 \times 3 - 1040 \times 7}{3 - 7} = ₹ 800$

9. *(d)* SI at 5% $= 6P - P = 5P$

$\therefore$ $5P = \frac{P \times 5 \times T}{100} \Rightarrow T = 100$ yr

Now, for new rate (R),

$11P = \frac{P \times R \times 100}{100}$

$\therefore$ $R = 11\%$

Shortcut Method

As, $R_1 = 5\%$, $n_1 = 6$, $n_2 = 12$.

According to the formula,

$R_2 = \frac{n_2 - 1}{n_1 - 1} \times R_1 = \frac{12 - 1}{6 - 1} \times 5$ [Rule 3]

$= \frac{11}{5} \times 5 = 11\%$

10. *(b)* Total borrowed money = ₹ 40000

and rate of interest = 8%

The interest for 2 yr $= \frac{40000 \times 8 \times 2}{100}$

$= ₹ 6400$

Let he paid ₹ x at the end of second year.

$\therefore$ Interest will be calculated on

₹ $(40000 - x + 6400)$

Interest for 3 yr $= \frac{(46400 - x) \times 3 \times 8}{100}$

$= ₹ \frac{6}{25}(46400 - x)$

$\therefore \frac{6}{25}(46400 - x) + 46400 - x = 35960$

$\Rightarrow x = \frac{21576 \times 25}{31} = ₹ 17400$

11. *(b)* Let sum lent at rate 5% be ₹ x.

Then, sum lent at rate 8%

$= ₹ (1550 - x)$

$\therefore$ Simple interest at rate 5% $= \frac{x \times 5 \times 3}{100}$

Simple interest at rate 8%

$= \frac{(1550 - x) \times 8 \times 3}{100}$

$\therefore \frac{x \times 15}{100} + \frac{(1550 - x) \times 24}{100} = 300$

$\Rightarrow \frac{15x}{100} + \frac{37200}{100} - \frac{24x}{100} = 300$

$\Rightarrow x = ₹ 800$

Amount lent at rate 8%

$= ₹ (1550 - 800) = ₹ 750$

$\therefore$ Required ratio $= \frac{800}{750} = \frac{16}{15} = 16 : 15$

12. *(d)* Let the amount remaining to pay be ₹ x.

$\therefore$ Price of house = ₹ $(x + 8000)$

$\Rightarrow 9600 - \frac{x \times 4 \times 5}{100} = x$

$\Rightarrow 9600 - \frac{x}{5} = x$

$\Rightarrow 9600 = \frac{6x}{5} \Rightarrow \frac{9600 \times 5}{6} = x$

$\Rightarrow x = ₹ 8000$

$\therefore$ Cash price of the house

$= ₹ (8000 + 8000) = ₹ 16000$

13. *(a)* Let the principal be ₹ P

As, amount $= 3P$ and $T = 25$ yr

$\therefore$ $\text{SI} = 3P - P = 2P$

$\because$ $\text{Rate} = \frac{100 \times \text{SI}}{\text{principal} \times T}$

$= \frac{100 \times 2P}{P \times 25} = 8\%$

14. *(a)* Let a man invest ₹ 1000 at a R%.

Now, rate is increased by 2%.

$\therefore$ New rate $= (R + 2)\%$

By given condition,

$\frac{1000 \times R \times 3}{100} + \frac{1500 \times (R + 2) \times 3}{100} = 390$

$\Rightarrow 30R + 45R + 90 = 390$

$\Rightarrow 75R = 300 \Rightarrow R = 4\%$

15. *(c)* Here, rate of interest $= 3\frac{1}{8}\% = \frac{25}{8}\%$

Let principal be ₹ x.

and simple interest = ₹ $\frac{3}{8}x$

$\therefore$ $\frac{3}{8}x = \frac{x \times \frac{25}{8} \times T}{100}$

$\Rightarrow \frac{300}{25} = T \Rightarrow T = 12$ yr

16. *(c)* Given, rate of interest = 12% per annum

Simple interest = ₹ 6000 per annum

Let principal is ₹ P.

$\therefore$ $6000 = \frac{P \times 1 \times 12}{100}$

$\because$ $P = \frac{6000 \times 100}{12} = ₹ 50000$

Hence, the required Principal amount is ₹ 50000.

17. *(c)* For A Let the amount be ₹ x.

Rate of interest = 5% and Time = 10 yr

$\therefore$ Simple interest $= \frac{x \times 5 \times 10}{100} = \frac{x}{2}$

For B The amount be ₹ $2x$.

Rate of interest = 5% and Time = 5 yr

$\text{SI} = \frac{2x \times 5 \times 5}{100} = \frac{x}{2}$

So, A and B both will get the same amount as interest.

18. *(a)* Let the first part and the second part be ₹ x and ₹ $(625 - x)$, respectively.

$\because$ The simple interest on both part in same.

So, $\frac{x \times 5 \times 2}{100} = \frac{(625 - x) \times 10 \times 4}{100}$

$\Rightarrow 10x = (625 - x)40$

$\Rightarrow x = (625 - x)4$

$\therefore$ $x = ₹ 500$

$\therefore$ Second part

$= ₹ (625 - 500) = ₹ 125$

19. *(b)* Let the amount of $A = ₹\ a$, time = 2 yr and rate = 5%

$\therefore$ Simple Interest of

$$A = \frac{a \times 2 \times 5}{100} = \frac{10a}{100}$$

Let the amount of $B = ₹\ b$, rate = 5% and time = 3 yr.

$\therefore$ Simple interest of

$$B = \frac{b \times 3 \times 5}{100} = \frac{15b}{100}$$

Let the amount of $C = ₹\ c$, time = 4 yr and rate = 5%

$\therefore$ Simple interest of $C = \frac{c \times 4 \times 5}{100} = \frac{20c}{100}$

But $\frac{a \times 10}{100} = \frac{b \times 15}{100} = \frac{c \times 20}{100}$

$\Rightarrow \quad 10a = 15b = 20c = k$

So, $\quad a = \frac{k}{10},\ b = \frac{k}{15},\ c = \frac{k}{20}$

$\therefore \quad a : b : c = \frac{1}{10} : \frac{1}{15} : \frac{1}{20}$

20. *(a)* Let the sum be ₹ x and the original rate r%, then

Simple interest $= \frac{x \times r \times 2}{100}$

Now, rate is increased by 3%.

$\therefore$ New rate $= (r + 3)\%$

$\because$ Simple interest $= \frac{x \times (r+3) \times 2}{100}$

$$\therefore \frac{x \times (r+3) \times 2}{100} - \frac{x \times r \times 2}{100} = 72$$

$$\Rightarrow \frac{(xr + 3x)2}{100} - \frac{2xr}{100} = 72$$

$$\Rightarrow \frac{2xr + 6x - 2xr}{100} = 72$$

$\therefore \quad x = ₹\ 1200$

21. *(c)* As rate of interest is charged half yearly,

So, $\quad$ rate $= \frac{13}{2}\%$ half yearly

time $\left(\frac{42}{12} \times 2\right)$ half yearly = 7 half yearly

$$\text{SI} = \frac{20000 \times 13 \times 7}{100 \times 2} = ₹\ 9100$$

$\therefore$ Amount $(A) = 20000 + 9100$
$= ₹\ 29100$

22. *(b)* Let initial amount be ₹ P, then $A = ₹\ 3P$ and

$T = 15$ yr and 6 months $= \frac{31}{2}$ yr

$\because$ SI $= A - P = 3P - P = ₹\ 2P$

$$\Rightarrow P \times \frac{31}{2} \times \frac{R}{100} = 2P$$

$$\Rightarrow R = \frac{2 \times 2 \times 100}{31}\%$$

$$= \frac{400}{31}\%$$

Let amount doubled be t_1 yr.

Now, SI $= 2P - P = ₹\ P$

$$\therefore \quad t_1 = \frac{SI \times 100}{P \times R}$$

$$\Rightarrow \quad t_1 = \frac{P \times 100 \times 31}{P \times 400}$$

$$= \frac{31}{4} = 7 \text{ yr and } 9 \text{ months}$$

23. *(c)* **Case I** $\quad \text{SI} = \frac{P \times R \times T}{100}$

$$R = \frac{24200 \times 4 \times 6}{100} = ₹\ 5808$$

$\therefore$ Amount = Principal + SI

$$\text{SI} = 24200 + 5808 = 30008$$

Case II $\quad \text{SI} = \frac{30008 \times 4 \times 4}{100} = ₹\ 4801.28$

24. *(d)* Let the person invest amount x and y into two different rates of interest.

$$\therefore \quad \frac{x \times 12 \times 1}{100} + \frac{y \times 10 \times 1}{100} = 130 \quad \left[\because \text{SI} = \frac{\text{PRT}}{100}\right]$$

$\Rightarrow \quad 12x + 10y = 13000 \quad \ldots$(i)

and $\quad \frac{y \times 12 \times 1}{100} + \frac{x \times 10 \times 1}{100} = 134$

$\Rightarrow \quad 12y + 10x = 13400 \quad \ldots$(ii)

On solving Eqs. (i) and (ii), we get

$x = ₹\ 500$ and $y = ₹\ 700$

25. *(d)* Here, $I_1 = 50$, $I_2 = 80$, $T_1 = 4$ yr, $T_2 = 8$ yr, and $P = ₹\ 1500$

According to formula,

$$I_2 - I_1 = \frac{P \times R \times (T_2 - T_1)}{100}$$

$$\Rightarrow \quad 80 - 50 = \frac{1500 \times R \times (8 - 4)}{100}$$

$\Rightarrow \quad 30 = 15 \times R \times 4$

$\Rightarrow \quad 2R = 1$

$\Rightarrow \quad R = \frac{1}{2} = 0.5\%$

26. *(a)* Here, $I_1 - I_2 = ₹\ 10$, $P_1 = ₹\ 1230$, $P_2 = ₹\ 1130$ and $T = 2$ yr

According to formula,

$$I_2 - I_1 = \frac{T \times R \times (P_2 - P_1)}{100}$$

$$-10 = \frac{R \times 2(1130 - 1230)}{100}$$

$$-10 = \frac{-200R}{100}$$

$\therefore \quad R = 5\%$

27. *(b)* **Case I** Annual payment

$$= \frac{100P}{100T + \frac{RT(T-1)}{2}}$$

$$= \frac{1092 \times 100}{100 \times 2 + \frac{24(2-1)}{2}}$$

$$= \frac{1092 \times 100}{212} = ₹\ 515.09 \approx ₹\ 515$$

Case II $\quad$ Annual payment

$$= \frac{1092 \times 100}{100 \times 2 + \frac{12(2-1)}{2}}$$

$$= \frac{1092 \times 100}{206} = ₹\ 530.09 \approx ₹\ 530$$

28. *(c)* Let P be the principal amount.

SI = ₹ 55, time, $t = 9$ months $= \frac{9}{12}$ yr

and $\quad$ rate, $r = 3\frac{2}{3}\% = \frac{11}{3}\%$

$$\because \quad \text{SI} = \frac{P \times R \times T}{100}$$

$$\Rightarrow \quad P = \frac{\text{SI} \times 100}{R \times T} = \frac{55 \times 100}{11 \times 9} \times 3 \times 12$$

$$= 5 \times 100 \times 4 = ₹\ 2000$$

$\therefore$ Principal $(P) = ₹\ 2000$

29. *(a)* Let the principal be ₹ x and the time be t yr.

Rate = 10%

$\because$ Simple interest $= \frac{P \times R \times T}{100}$

According to the question,

$0.125 \times$ Principal = Simple interest

$$\therefore \quad 0.125\ P = \frac{P \times 10 \times T}{100}$$

$$\Rightarrow \quad \frac{125P}{1000} = \frac{10 \times P \times T}{100}$$

$$\Rightarrow \quad \frac{125}{100} = T$$

$$\Rightarrow \quad T = \frac{5}{4} = 1\frac{1}{4} \text{ yr}$$

30. *(c)* Let principal = ₹ P, then amount of money = ₹ $2P$

$\therefore \quad$ SI $= 2P - P = ₹P$

Now, $\quad P = \frac{P \times r \times 5}{100} \Rightarrow r = 20\%$

Amount of money after 12 yr = ₹$3P$

$\therefore \quad$ SI $= 3P - P = 2P$

Now, $\quad 2P = \frac{P \times R \times 12}{100} \Rightarrow R = \frac{50}{3}\%$

$\therefore$ Difference between two interest rates

$$= \left(20 - \frac{50}{3}\right)\% = \frac{10}{3}\% = 3\frac{1}{3}\%$$

10

COMPOUND INTEREST

Usually (1-2) questions have been asked from this chapter. Questions from this chapter are based on the direct application of compound interest formula.

COMPOUND INTEREST

When the interest is calculated on the amount of previous year, then it is known as compound interest. It will make a deposit or loan grow at a faster rate than simple interest.

Important Rules and Formulae

Consider $P =$ Principal (amount borrowed), $A =$ Amount (Principal + Interest)
$R =$ Rate of interest per annum and $n =$ Number of years

Rule 1 When interest is compounded annually, then

(i) Amount $= P\left(1+\dfrac{R}{100}\right)^n$ (ii) $\text{CI} = A - P = P\left[\left(1+\dfrac{R}{100}\right)^n - 1\right]$

EXAMPLE 1. The amount and the compound interest on ₹ 24000 compounded annually for 2 yr at the rate of 10% per annum is

a. ₹ 39040 and ₹ 4040 b. ₹ 29040 and ₹ 5040 c. ₹ 19040 and ₹ 3040 d. None of these

Sol. **b.** Here, $P =$ ₹ 24000, $R = 10\%$ per annum and $n = 2$ yr

$$\because A = P\left(1+\frac{R}{100}\right)^n = 24000\left(1+\frac{10}{100}\right)^2 = 24000\left(\frac{110}{100}\right)^2 = 24000 \times \frac{110}{100} \times \frac{110}{100} = ₹\ 29040$$

$\because$ Compound interest = Amount − Principal = ₹ 29040 − ₹ 24000 = ₹ 5040

So, amount = ₹ 29040 and compound interest = ₹ 5040

Rule 2 When interest is compounded half-yearly (every 6 months), then

(i) Amount $= P\left(1+\dfrac{r}{100\times 2}\right)^{2n}$ (ii) $\text{CI} = \left[\left(1+\dfrac{R}{100\times 2}\right)^{2n} - 1\right]$

EXAMPLE 2. The compound interest on ₹ 24000 compounded semi-annually for $1\frac{1}{2}$ yr at the rate of 10% per annum is

a. ₹ 3783 b. ₹ 2783 c. ₹ 2763 d. ₹ 3763

Sol. a. Given, $P =$ ₹ 24000

Rate of interest = 10% per annum

and time $= 1\frac{1}{2}$ yr $= \frac{3}{2}$ yr

Since, the interest is compounded half-yearly.

Then, rate $(R) = \frac{10}{2}$ % half-yearly

and time $(n) = 2 \times \frac{3}{2} = 3$ half-yearly

$$\therefore \quad A = 24000\left(1+\frac{5}{100}\right)^3 \qquad \left[\because A = P\left(1+\frac{R}{100}\right)^n\right]$$

$$= 24000\left(1+\frac{1}{20}\right)^3 = 24000\left(\frac{21}{20}\right)^3 \qquad \text{[Rule 1]}$$

$$= 24000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} = ₹\ 27783$$

Now, compound interest $= A - P$

$=$ ₹ 27783 − ₹ 24000 = ₹ 3783

Rule 3 When interest is compounded quarterly (every 3 months), then

(i) $\text{Amount} = P\left(1+\frac{R}{100\times 4}\right)^{4n}$

(ii) $\text{CI} = P\left[\left(1+\frac{R}{100\times 4}\right)^{4n} - 1\right]$

EXAMPLE 3. The amount and the compound interest on ₹ 100000 compounded quarterly for 9 months at the rate of 4% per annum is

a. ₹ 106060 and ₹ 6060.10 b. ₹ 103030 and ₹ 3030.10
c. ₹ 103030.10 and ₹ 3030.10 d. ₹ 106060.10 and ₹ 6060.10

Sol. c. Given, $P =$ ₹ 100000, rate of interest = 4% per annum and time = 9 months

Since, the interest is compounded quarterly.

Then, rate $(R) = \frac{4}{4} = 1\%$ quarterly

and time $(n) = 9$ months $= \frac{9}{12} \times 4 = 3$ quarter yearly

$\because$ Amount, $A = P\left(1+\frac{R}{100}\right)^n$

$$\therefore \quad A = 100000\left(1+\frac{1}{100}\right)^3 = 100000\left(\frac{101}{100}\right)^3$$

$$= 100000 \times \frac{101}{100} \times \frac{101}{100} \times \frac{101}{100} = ₹\ 103030.10$$

$\therefore$ Compound interest $= A - P =$ ₹ 103030.10 − ₹ 100000

$=$ ₹ 3030.10

Rule 4 When interest is compounded annually but time is given in fraction (say, $t\frac{a}{b}$ yr), then

$$\text{Amount} = P\left(1+\frac{R}{100}\right)^t \times \left(\frac{1+(a/b)R}{100}\right)$$

EXAMPLE 4. The amount and compound interest on ₹ 5000 compounded annually for 2 yr 6 months at the rate of 10% per annum is

a. ₹ 6252.5 and ₹ 1342.5 b. ₹ 6000 and ₹ 1300
c. ₹ 6250 and ₹ 1340 d. ₹ 6352.5 and ₹ 1352.5

Sol. d. Here, time = 2 yr and 6 months

$\therefore \quad t = 2$ yr and $\frac{a}{b} = \frac{1}{2}$ yr

$$\because \quad \text{Amount} = P\left(1+\frac{R}{100}\right)^t \times \left(\frac{1+\frac{a}{b}R}{100}\right)$$

$$\therefore \quad A = 5000\left(1+\frac{10}{100}\right)^2 \left(1+\frac{\frac{1}{2}\cdot 10}{100}\right)$$

$$= 5000\left(\frac{110}{100}\right)^2 \left(\frac{105}{100}\right)$$

$$= 5000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{21}{20} = ₹\ 6352.50$$

$\therefore$ Compound interest $= A - P$

CI = 6352.50 − 5000 = ₹ 1352.50

Rule 5 When rate of interest for n_1, n_2 and n_3 yr are R_1, R_2 and R_3 respectively, then

$$\text{Amount} = P\left(1+\frac{R_1}{100}\right)^{n_1}\left(1+\frac{R_2}{100}\right)^{n_2}\left(1+\frac{R_3}{100}\right)^{n_3}$$

EXAMPLE 5. The compound interest on ₹ 5000 for 4 yr if the rate of interest is 10% per annum for the first two years and 15% for the next two years is

a. ₹ 3000 b. ₹ 3001 c. ₹ 3002 d. None of these

Sol. b. Here, $R_1 = 10\%$, $n = n_1 + n_2 = 4$ yr

$n_1 = 2$ yr, $R_2 = 15\%$, $n_2 = 2$ yr

and principal = ₹ 5000

$$\because \quad \text{Amount} = P\left(1+\frac{R_1}{100}\right)^{n_1}\left(1+\frac{R_2}{100}\right)^{n_2}$$

$$\therefore \quad A = 5000\left(1+\frac{10}{100}\right)^2\left(1+\frac{15}{100}\right)^2$$

$$= 5000 \times \left(\frac{110}{100}\right)^2\left(\frac{115}{100}\right)^2$$

$$= 5000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{23}{20} \times \frac{23}{20} = ₹\ 8001.125$$

$\therefore$ Compound interest

$= A - P =$ ₹ 8001.125 − ₹ 5000 $\simeq$ ₹ 3001

Rule 6 The difference D between simple and compound interest accrued on ₹ P at the rate of interest of R% is given by

(i) For 2 yr, $D = \frac{PR^2}{(100)^2}$

(ii) For 3 yr, $D = \frac{PR^2(300+r)}{(100)^3}$

EXAMPLE 6. The difference between the compound interest and the simple interest on a certain sum at 12% per annum for 2 yr is ₹ 90. Then, sum is

a. ₹ 6250 b. ₹ 6350 c. ₹ 6520 d. ₹ 6950

Sol. a. Let the sum be ₹ 100.

Case I Then, simple interest on ₹ 100 for 2 yr

$$= \left(\frac{100 \times 12 \times 2}{100}\right) = ₹\ 24$$

Case II Amount when ₹ 100 is borrowed for 2 yr

$$= 100 \times \left(1 + \frac{12}{100}\right)^2 \quad \left[A = P\left(1+\frac{R}{100}\right)^n, \text{Rule 1 (i)}\right]$$

$$= 100 \times \frac{28}{25} \times \frac{28}{25} = ₹\left(\frac{3136}{25}\right)$$

$\therefore$ Compound interest for 2yr $= \left(\frac{3136}{25} - 100\right) = ₹\left(\frac{636}{25}\right)$

Difference between CI and SI $= \left(\frac{636}{25} - 24\right) = ₹\ \frac{36}{25}$

If the difference between CI and SI is ₹ $\frac{36}{25}$, then the sum = ₹ 100

If the difference between CI and SI is ₹ 1,

then the sum = ₹ $\left(\frac{100 \times 25}{36}\right)$

If the difference between CI and SI is ₹ 90, then the sum

$$= \left(100 \times \frac{25}{36} \times 90\right) = ₹\ 6250$$

Here, the sum is ₹ 6250.

Shortcut Method

Let the required sum be ₹ P.

Here, difference = ₹ 90 and $R = 12\%$

$\therefore$ Difference between CI and SI

$$90 = \frac{PR^2}{(100)^2} = \frac{(P)(12)^2}{(100)^2} \Rightarrow 90 = \frac{P \times 144}{100 \times 100}$$

$\therefore$ $P = ₹\ 6250$

Required sum = ₹ 6250

Rule 7 If a certain sum at compound interest becomes A_1, in n yr and A_2 in $(n+1)$ yr, then

(i) Rate of compound interest $= \frac{(A_2 - A_1)}{A_1} \times 100\%$

(ii) Sum $= A_1\left(\frac{A_1}{A_2}\right)^n$

EXAMPLE 7. A sum of money on compound interest amount to ₹ 9680 in 2 yr and to ₹ 10648 in 3 yr. What is the rate of interest per annum?

a. 5% b. 10% c. 15% d. 20%

Sol. b. Given, $A_1 = ₹\ 10648$ and $A_2 = ₹\ 9680$

$\therefore$ Rate of compound interest $= \frac{(A_2 - A_1)}{A_1} \times 100\%$

$$= \frac{10648 - 9680}{9680} \times 100\% = 10\%$$

DEPRECIATION

The value of certain things like the machine, vehicle etc., decreases over a period of time. The decrement in the value things is called depreciation. The depreciation per unit time is called the rate of depreciation.

Rule 8 If the present value of a article is P and it depreciate at the rate of R% per annum. Then,

(i) Value of article after n yr $= P \times \left(1 - \frac{R}{100}\right)^n$

(ii) Value of article n yr ago $= \frac{P}{\left(1 - \frac{R}{100}\right)^n}$

EXAMPLE 8. Given that carbon – 14 (C_{14}) decays at a constant rate in such a way that it reduces to 50% in 5568 yr. Then, the age of an old wooden piece in which the carbon is only 12.5% of the original is

a. 16000 yr b. 16244 yr c. 16702 yr d. None of these

Sol. d. Let the rate of decay be R% and the age of the wooden piece be n yr.

Then, $\left(\frac{50}{100}\right) = \left(1 - \frac{R}{100}\right)^{5568} \Rightarrow \left(1 - \frac{R}{100}\right) = (0.5)^{1/5568}$ ...(i)

$\Rightarrow \frac{12.5}{100} = \left(1 - \frac{R}{100}\right)^n \Rightarrow (0.125) = \left(1 - \frac{R}{100}\right)^n$

$\Rightarrow (0.125)^{1/n} = \left(1 - \frac{R}{100}\right) \Rightarrow (0.5)^{3/n} = \left(1 - \frac{R}{100}\right)$ ...(ii)

From Eqs. (i) and (ii), $(0.5)^{3/n} = (0.5)^{\frac{1}{5568}}$

On comparing both sides, we get

$$\frac{3}{n} = \frac{1}{5568} \Rightarrow n = 3 \times 5568 = 16704 \text{ yr}$$

INSTALMENTS

When a borrower pays the sum in parts, then we say that he/she is paying in instalments.

$$\therefore\ P = \frac{x}{\left(1 + \frac{R}{100}\right)} + \frac{x}{\left(1 + \frac{R}{100}\right)^2} + \ldots + \frac{x}{\left(1 + \frac{R}{100}\right)^n}$$

x = value of each instalment

Total amount paid in instalments, $A = P\left(1+\frac{R}{100}\right)^n$

n = number of instalments

EXAMPLE 9. Sapna borrowed some money on compound interest and returned it in 3 yr in equal annual instalments. If rate of interest is 15% per annum and annual instalment is ₹ 486680, then the sum borrowedes.

a. 111300
b. 122400
c. 1111200
d. 154320

Sol. c. Given that, rate of interest, $R = 15\%$ pa

Annual instalment, $x =$ ₹ 486680

Total number of instalments, $n = 3$

$$\therefore\ P = \left[\frac{x}{\left(1+\frac{R}{100}\right)} + \frac{x}{\left(1+\frac{R}{100}\right)^2} + \frac{x}{\left(1+\frac{R}{100}\right)^3}\right]$$

$$\Rightarrow P = x\left[\frac{100}{(100+R)} + \frac{(100)^2}{(100+R)^2} + \frac{(100)^3}{(100+R)^3}\right]$$

$$= 486680\left[\left(\frac{100}{100+15}\right) + \left(\frac{100}{100+15}\right)^2 + \left(\frac{100}{100+15}\right)^3\right]$$

$$= 486680 \times \left[\frac{20}{23} + \left(\frac{20}{23}\right)^2 + \left(\frac{20}{23}\right)^3\right]$$

$$= 486680 \times \frac{20}{23}\left(1 + \frac{20}{23} + \frac{400}{529}\right)$$

$$= 1111200$$

$\therefore$ Principal borrowed = ₹ 1111200

SOME IMPORTANT RESULTS

1. Simple interest and compound interest are equal for first year period on the same sum and at the same rate.
2. If A and B are the amounts of a certain sum for two consecutive years, then simple interest for 1 yr = $B - A$
3. If certain sum at compound interest becomes x times in n_1 yr and y times in n_2 yr, then $x^{1/n_1} = y^{1/n_2}$

PRACTICE EXERCISE

1. Kiran purchased a scooter for ₹ 24000. The value of scooter is depreciating at the rate of 5% per annum. Then, its value after 3 yr is
(a) ₹ 20577 (b) ₹ 20977 (c) ₹ 20677 (d) ₹ 20877

2. If P be the principal amount and the rate of interest be r% per annum and the compound interest is calculated k times in a year, then what is the amount at the end of n yr?
(a) $P\left(1+\frac{r}{100k}\right)^{nk}$ (b) $P\left(1+\frac{kr}{100}\right)^{nk}$
(c) $P\left(1+\frac{kr}{100}\right)^{n/k}$ (d) $P\left(1+\frac{kr}{100k}\right)^{n/k}$

3. The amount of a certain sum at compound interest for 2 yr at 5% is ₹ 4410. The sum is
(a) ₹ 4000 (b) ₹ 4200 (c) ₹ 3900 (d) ₹ 3800

4. A person borrowed ₹ 7500 at 16% compound interest. How much does he have to pay at the end of 2 yr to clear the loan?
(a) ₹ 9900 (b) ₹ 10092 (c) ₹ 11000 (d) ₹ 11052

5. If the rate of interest is 10% per annum and is compounded half-yearly, then the principal of ₹ 400 in 3/2 yr will amount to
(a) ₹ 463.00 (b) ₹ 463.05
(c) ₹ 463.15 (d) ₹ 463.20

6. At compound interest, if a certain sum of money doubles in n yr, then the amount will be four fold in
(a) $2n^2$ yr (b) n^2 yr (c) $2n$ yr (d) $4n$ yr

7. The simple interest on a certain sum of money for 3 yr at 8% per annum is half the compound interest on ₹ 4000 for 2 yr at 10% per annum. What is the sum placed on simple interest?
(a) ₹ 1550 (b) ₹ 1650 (c) ₹ 1750 (d) ₹ 2000

8. What is the least number of complete years in which a sum of money at 20% compound interest will be more than doubled?
(a) 7 (b) 6 (c) 5 (d) 4

9. The difference between the simple interest and the compound interest (compounded annually) on ₹ 1250 for 2 yr at 8% per annum will be
(a) ₹ 18 (b) ₹ 2 (c) ₹ 8 (d) ₹ 4

10. The compound interest on a sum for 2 yr is ₹ 832 and the simple interest on the same sum at the same rate for the same period is ₹ 800. What is the rate of interest?
(a) 6% (b) 8% (c) 10% (d) 12%

11. A saving bank gives interest which compounds annually. Raju deposited ₹ 100 and received ₹ 121 at the end of second year. Rate of compound interest per annum is
(a) 10% (b) 15% (c) 11.5% (d) 20.5%

12. The amount of a certain sum at compound interest for 4 yr at 10% in ₹ 4410. The sum is
(a) 3012.08 (b) 3015 (c) 3020.16 (d) 3016.9

13. The compound interest on ₹ 5000 for 3 yr at 8% for first year, 10% for second year and 12% for third year will be
(a) ₹ 1560.40 (b) ₹ 1500 (c) ₹ 1565.60 (d) ₹ 1652.80

14. An amount of ₹ x at compound interest at 20% per annum for 3 yr becomes y. What is $y:x$?
(a) 3 : 1 (b) 36 : 25
(c) 216 : 125 (d) 125 : 216

15. The compound interest on ₹ 2000 for 1 yr at the rate of 8% per annum, when the interest is compounded semi-annually
(a) ₹ 163.20 (b) ₹ 2163.20 (c) ₹ 3153.20 (d) ₹ 1163

16. ₹ 16000 invested at 10% per annum compounded semi-annually amounts to ₹ 18522. Then, the period of investment is
(a) $1\frac{1}{2}$ yr (b) 3 yr (c) 2 yr (d) $\frac{5}{2}$ yr

17. A sum compounded annually becomes $\frac{25}{16}$ times of itself in 2 yr. Then, the rate of interest per annum is
(a) 25% (b) 20% (c) 15% (d) $7\frac{1}{2}$%

18. A sum of ₹ 3200 invested at 10% per annum compounded quarterly amounts to ₹ 3362, then the time period is
(a) $1\frac{1}{2}$ yr (b) $\frac{1}{2}$ yr (c) 2 yr (d) 1 yr

19. A sum amount to ₹ 9680 in 2 yr and to ₹ 10648 in 3 yr compounded annually. Then, the sum and rate of interest, respectively are
(a) ₹ 8000, 10% (b) ₹ 8500, 10%
(c) ₹ 8500, 9% (d) ₹ 8000, 9%

20. If the value of a machine depreciates by 10% of its value at the beginning of the year and its present value is estimated as ₹ 10935, then what was its value three years back?
(a) ₹ 15000 (b) ₹ 7000
(c) ₹ 8050 (d) None of these

21. A sum of money doubles itself at compound interest in 15 yr. It will becomes 8 times in

(a) 30 yr (b) 40 yr
(c) 60 yr (d) 45 yr

22. A man borrows ₹ 4000 at 8% per annum on compound interest. At the end of every year he pays ₹ 1500 as part payment of loan and interest. How much does he still owe to the bank after three such annual payments?

(a) ₹ 1799 (b) ₹ 2000
(c) ₹ 169.25 (d) None of these

23. A sum of ₹ 10000 deposited at compound interest becomes double after 5 yr. After 20 yr, the amount will be

(a) ₹ 160000 (b) ₹ 40000
(c) ₹ 50000 (d) ₹ 60000

24. Vinod and Karan each invested ₹ 15000 for 3 yr at the same rate of interest but Vinod's investment is compounded annually while Karan's investment is charged on simple interest.

I. Vinod receive more interest than karan.
II. Data is insufficient to calculate the interest.

Which one is correct?

(a) Only I (b) Only II
(c) Neither I nor II (d) Both I and II

25. An amount is invested in a bank at compound rate of interest. The total amount, including interest, after first and third year is ₹ 1200 and ₹ 1587, respectively.

I. The principal amount will be ₹ 1403.
II. The rate of interest is 16%

Which statement(s) is/are correct?

(a) Both I and II (b) Only I
(c) Only II (d) Neither I nor II

26. From the given statements, identify which of the following or both are correct and then select the appropriate option.

I. The Present worth of ₹ 169 due in 2 yr at 4% *Pa* compound interest is 156.25.
II. If the simple interest on a certain sum for 2 yr is ₹ 120 and compound interest is ₹ 129, then the rate of interest must be 15%.

Which statement(s) is/are correct?

(a) Only II (b) Only I
(c) Neither I nor II (d) Both I and II

PREVIOUS YEARS' QUESTIONS

27. What is the compound interest on ₹ 1600 at 25% per annum of 2 yr compound annually? **2012 II**

(a) ₹ 700 (b) ₹ 750 (c) ₹ 800 (d) ₹ 900

28. The difference between compound interest and simple interest for 2 yr at the rate of 10% over principal amount of ₹ X is ₹ 10. What is the value of X? **2014 II**

(a) ₹ 100 (b) ₹ 1000 (c) ₹ 500 (d) ₹ 5000

29. The difference between compound interest and simple interest at the same rate of interest R per cent per annum on an amount of ₹ 15000 for 2 yr is ₹ 96. What is the value of R? **2015 I**

(a) 8% (b) 10%
(c) 12% (d) Cannot be determined

30. There is 60% increase in an amount in 6 yr at simple interest. What will be the compound interest on ₹ 12000 after 3 yr at the same rate of interest? **2015 I**

(a) ₹ 2160 (b) ₹ 3120 (c) ₹ 3972 (d) ₹ 6240

31. A sum of ₹ 10000, is deposited for 1 yr at the rate of interest 10% compounded half yearly. What will be the interest at the end of one year? **2015 II**

(a) ₹ 1000 (b) ₹ 1025 (c) ₹ 1050 (d) ₹ 1100

ANSWERS

1	a	2	a	3	a	4	b	5	b	6	c	7	c	8	d	9	c	10	b
11	a	12	a	13	d	14	c	15	a	16	a	17	a	18	b	19	a	20	a
21	d	22	c	23	a	24	b	25	d	26	d	27	d	28	b	29	a	30	c
31	b																		

HINTS AND SOLUTIONS

1. *(a)* Given, $P = ₹\ 24000$, $R = 5\%$ per annum and $n = 3$ yr

$\therefore \quad A = P\left(1 - \frac{R}{100}\right)^n$ [Rule 8 (i)]

$= 24000\left(1 - \frac{5}{100}\right)^3$

$= 24000 \times \left(\frac{95}{100}\right)^3$

$= ₹\ 20577$

2. *(a)* Given, principal amount $= ₹\ P$

Rate of interest, $R = \frac{r}{k}\%$ per annum and

Time, $T = nk$

$\because \quad A = P\left(1 + \frac{R}{100}\right)^T$ [Rule 1. (i)]

$\therefore \quad A = P\left(1 + \frac{r}{100k}\right)^{nk}$

3. *(a)* Let the principal be $= ₹\ x$.

Given, $n = 2$ yr, $R = 5\%$, $A = ₹\ 4410$

$\because \quad A = P\left(1 + \frac{R}{100}\right)^n$ [Rule 1. (i)]

$\therefore \quad 4410 = x\left(1 + \frac{5}{100}\right)^2$

$\Rightarrow \quad 4410 = x\left(\frac{21}{20}\right)^2$

$\Rightarrow \quad x = \frac{4410 \times 400}{441} = ₹\ 4000$

4. *(b)* Given, $P = ₹\ 7500$, $R = 16\%$ and $n = 2$ yr

$\because \ A = P\left(1 + \frac{R}{100}\right)^n$ [Rule 1. (i)]

$= 7500\left(1 + \frac{16}{100}\right)^2$

$= 7500\left(\frac{116}{100}\right)^2$

$= 7500 \times \frac{29}{25} \times \frac{29}{25}$

$= 12 \times 841 = ₹\ 10092$

5. *(b)* Refer to example 2.

6. *(c)* Let the principal be $₹\ x$.

$\therefore$ Rate $= ₹\ R$, Amount $= ₹\ 2x$

and Time $= n$ yr

$\because \quad A = P\left(1 + \frac{R}{100}\right)^n$ [Rule 1. (i)]

$\therefore \quad x\left(1 + \frac{R}{100}\right)^n = 2x$

$\Rightarrow \quad \left(1 + \frac{R}{100}\right)^n = 2$ ...(i)

Let it becomes four fold in N yr.

Then, $x\left(1 + \frac{R}{100}\right)^N = 4x$

$\Rightarrow \quad \left(1 + \frac{R}{100}\right)^N = 4$

$\Rightarrow \quad 2^2 = \left(1 + \frac{R}{100}\right)^N$

$\Rightarrow \left(1 + \frac{R}{100}\right)^{2n} = \left(1 + \frac{R}{100}\right)^N$

[from Eq. (i)]

$\therefore \quad N = 2n$ yr

7. *(c)* Let the principal amount be $₹\ P$.

By given condition, $\text{SI} = \frac{1}{2}\ \text{CI}$

$\Rightarrow \quad \frac{P \times 8 \times 3}{100} = \frac{1}{2}\left[4000\left(1 + \frac{10}{100}\right)^2 - 4000\right]$

[Rule 1 (ii)]

$\Rightarrow \frac{24P}{100} = \frac{1}{2}\left[4000 \times \frac{121}{100} - 4000\right]$

$= \frac{1}{2}[4840 - 4000]$

$\Rightarrow \frac{24P}{100} = 420$

$\therefore \quad P = \frac{420 \times 100}{24} = ₹\ 1750$

8. *(d)* Let the sum of money be $₹\ P$.

$\therefore$ Amount $= ₹\ 2P$

$\Rightarrow \quad A = P\left(1 + \frac{R}{100}\right)^n$ [Rule 1. (i)]

$\Rightarrow \quad 2P = P\left(1 + \frac{20}{100}\right)^n$

$\Rightarrow \quad \frac{2P}{P} = \left(\frac{6}{5}\right)^n$

$\Rightarrow \quad 2 = \left(\frac{6}{5}\right)^n$

On putting $n = 4$, we get

$\left(\frac{6}{5}\right)^4 = \frac{1296}{625} = 2$ (approx)

$\therefore$ Least number of years $= 4$

9. *(c)* Given, $P = ₹1250$, $R = 8\%$ per annum and $n = 2$yr

$\because \quad$ Simple interest $= \frac{P \times R \times T}{100}$

$\therefore \quad$ Simple interest $= \frac{1250 \times 8 \times 2}{100}$

$= ₹\ 200$

$\because \quad$ Compound interest

$= P\left[\left(1 + \frac{R}{100}\right)^n - 1\right]$ [Rule 1. (ii)]

$\because \quad \text{CI} = 1250\left(1 + \frac{8}{100}\right)^2 - 1250$

$= 1250 \times \left(\frac{108}{100}\right)^2 - 1250$

$= 1458 - 1250 = ₹\ 208$

$\therefore \quad$ Difference in SI and CI

$= 208 - 200 = ₹\ 8$

Shortcut Method

Here, $p = ₹\ 1250$ and $R = 8\%$

$\therefore$ Required difference $= \frac{PR^2}{(100)^2}$

[Rule 6. (i)]

$= \frac{1250 \times 64}{100 \times 100} = ₹\ 8$

10. *(b)* Given, $\text{CI} = ₹\ 832$, $\text{SI} = ₹\ 800$, $n = 2$ yr

$\text{CI} = P\left\{\left(1 + \frac{R}{100}\right)^n - 1\right\}$

[Rule 1. (ii)]

$\therefore \quad 832 = P\left\{\left(1 + \frac{R}{100}\right)^2 - 1\right\}$ ...(i)

Also, $\quad \text{SI} = \frac{P \times R \times T}{100}$

$\Rightarrow \quad 800 = \frac{P \times R \times 2}{100}$

$\Rightarrow \quad P = \frac{40000}{R}$

From Eq. (i),

$832 = \frac{40000}{R}\left(\frac{R^2}{10000} + \frac{2R}{100}\right)$

$\Rightarrow \quad 832 = \frac{40000}{100}\left(\frac{R}{100} + 2\right)$

$\Rightarrow \quad 832 = 4R + 800$

$\therefore \quad R = \frac{32}{4} = 8\%$

11. *(a)* Given, principal (P) = ₹ 100

Amount (A) received after 2 yr = ₹ 121

Let rate of interest = R% per annum

$\because A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

$\therefore \quad 121 = 100\left(1+\frac{R}{100}\right)^2$

$\Rightarrow \quad \frac{121}{100} = \left(\frac{100+R}{100}\right)^2$

$\Rightarrow \quad \left(\frac{11}{10}\right)^2 = \left(\frac{100+R}{100}\right)^2$

$\Rightarrow \quad \frac{11}{10} = \frac{100+R}{100}$

$\Rightarrow 100 + R = \frac{11\times 100}{10}$

$\Rightarrow 100 + R = 110$

$\therefore \quad R = 110 - 100 = 10$

$\therefore$ Rate of interest = 10%

12. *(a)* Refer to example 3.

13. *(d)* Given, P = ₹ 5000,

$R_1 = 8\%, R_2 = 10\%, R_3 = 12\%$

and $n_1 = n_2 = n_3 = 1$ yr

$\because$ Amount

$= P\left(1+\frac{R_1}{100}\right)^{n_1}\left(1+\frac{R_2}{100}\right)^{n_2}\left(1+\frac{R_3}{100}\right)^{n_3}$

[Rule 5]

$= \left[5000\times\left(1+\frac{8}{100}\right)\left(1+\frac{10}{100}\right)\left(1+\frac{12}{100}\right)\right]$

$= \left(5000\times\frac{27}{25}\times\frac{11}{10}\times\frac{28}{25}\right)$ = ₹ 6652.80

$\therefore$ Compound interest = 6652.80 − 5000

= ₹ 1652.80

14. *(c)* Given, P = ₹ x, R = 20% per annum, $n = 3$ yr

and A = ₹ y

$\because \quad A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

$\therefore \quad y = x\left(1+\frac{20}{100}\right)^3$

$\Rightarrow \quad y = x\left(\frac{6}{5}\right)^3 \Rightarrow \frac{y}{x} = \frac{216}{125}$

$\Rightarrow \quad y : x = 216 : 125$

15. *(a)* Given, principal (P) = ₹ 2000

Time = 1 yr

Rate of interest = 8% per annum

Since, the interest is compounded semi-annually.

Then, rate $(R) = \frac{8}{2}\% = 4$ half-yearly

and time $(n) = 1\times 2 = 2$ half-yearly

$\because \quad \text{CI} = P\left[\left(1+\frac{R}{100}\right)^n - 1\right]$

[Rule 1. (ii)]

$\text{CI} = 2000\left(1+\frac{4}{100}\right)^2 - 2000$

$= 2000\times\left(\frac{26}{25}\right)^2 - 2000$

= 2163.20 − 2000 = ₹ 163.20

16. *(a)* Given, principal (P) = ₹ 16000

Amount received at the end of period (A) = ₹ 18522

Let time = t yr

$\therefore \quad n = 2t$

Rate = 10% per annum i.e 5% half-yearly

$\because \quad A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

$\therefore \quad 18522 = 16000\left(1+\frac{5}{100}\right)^n$

$\Rightarrow \left(\frac{18522}{16000}\right) = \left(\frac{21}{20}\right)^n$

$\Rightarrow \left(\frac{9261}{8000}\right) = \left(\frac{21}{20}\right)^n \Rightarrow \left(\frac{21}{20}\right)^3 = \left(\frac{21}{20}\right)^n$

$\Rightarrow \quad n = 3$

$\because \quad n = 2t \Rightarrow 2t = 3$

$\therefore \quad t = \frac{3}{2}\text{ yr} = 1\frac{1}{2}\text{ yr}$

17. *(a)* Let sum/ principal be ₹ x and the rate be R% per annum.

$\therefore$ Amount, $A = \frac{25}{16}x$ and $n = 2$ yr

$\because \quad A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

$\therefore \quad \frac{25}{16}x = x\left(1+\frac{R}{100}\right)^2$

$\Rightarrow \quad \frac{25}{16} = \left(1+\frac{R}{100}\right)^2$

$\Rightarrow \quad \left(\frac{5}{4}\right)^2 = \left(1+\frac{R}{100}\right)^2$

$\Rightarrow \quad \frac{5}{4} = \left(1+\frac{R}{100}\right) \Rightarrow \frac{R}{100} = \frac{1}{4}$

$\therefore \quad R = 25\%$

18. *(b)* Here, P = ₹ 3200,

A = ₹ 3362

[since, amount is payable quarterly]

$\therefore R = 10\%$ per annum $= \frac{10}{4}\%$ quarterly

and let time = t

$\Rightarrow \quad n = 4t$

$\because \quad A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

$\Rightarrow \quad 3362 = 3200\left(1+\frac{10}{4\times 100}\right)^n$

$\Rightarrow \quad \frac{3362}{3200} = \left(\frac{410}{400}\right)^n \Rightarrow \left(\frac{41}{40}\right)^2 = \left(\frac{41}{40}\right)^n$

$\Rightarrow \quad n = 2 \Rightarrow 4t = 2$ yr [as $n = 4t$]

$\therefore \quad t = \frac{1}{2}$ yr

19. *(a)* Given, A_1 = ₹ 9680, A_2 = ₹ 10648

and $n = 2$ yr

$\therefore$ Rate of compound interest

$= \frac{A_2 - A_1}{A_1}\times 100\%$ [Rule 7 (i)]

$= \frac{10648 - 9680}{9680}\times 100\% = 10\%$

$\therefore$ Sum $= A_1\left(\frac{A_1}{A_2}\right)^n$ [Rule 7 (ii)]

$= 9680\left(\frac{9680}{10648}\right)^2$

$= 9680\left(\frac{110}{121}\right)^2 = 9680\left(\frac{10}{11}\right)^2$

$= \frac{9680\times 100}{121}$ = ₹ 8000

20. *(a)* Let the value of machine 3 yr ago be ₹ x.

and given, P = ₹ 10935, $R = 10\%$ and $n = 3$ yr

$\because \quad x = P\left(1-\frac{R}{100}\right)^n$ [Rule 8 (i)]

$\therefore x\left(1-\frac{10}{100}\right)^3 = 10935$

$\Rightarrow \quad x\left(\frac{90}{100}\right)^3 = 10935$

$\therefore \quad x = \frac{10935\times 10\times 10\times 10}{9\times 9\times 9}$ = ₹ 15000

21. *(d)* Let the sum be ₹ x and rate be R% per annum.

$\therefore$ Amount, A = ₹ $2x$

and time, $n = 15$ yr

$\because \quad A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

Then, $\quad x\left(1+\frac{R}{100}\right)^{15} = 2x$

$\Rightarrow \quad \left(1+\frac{R}{100}\right)^{15} = 2 \quad \ldots\text{(i)}$

Suppose the sum becomes eight times in 'n' yr, then

$x\left(1+\frac{R}{100}\right)^n = 8x$ [by given condition]

$\Rightarrow \left(1+\frac{R}{100}\right)^n = 8 = 2^3 \quad \ldots\text{(ii)}$

[but $2^3 = \left[\left(1+\frac{R}{100}\right)^{15}\right]^3$ from Eq. (i)]

$\Rightarrow \quad \left(1+\frac{R}{100}\right)^n = 2^3 = \left(1+\frac{R}{100}\right)^{45}$

[on comparing]

$\therefore \quad n = 45$ yr

Shortcut Method

Since, a sum of money at compound interest doubles itself in 15 yr.

Hence, it will becomes eight times (2^3 times) in $= 15 \times 3 = 45$ yr

22. *(c)* Amount due at the end of 1 yr

$= 4000\left(1+\frac{8}{100}\right) = ₹\ 4320$

$\therefore$ Amount due after 1 yr

$= (4320 - 1500) = ₹\ 2820$

Amount due at the end of second year

$= 2820\left(1+\frac{8}{100}\right) = ₹\ 3045.60$

$\therefore$ Amount due after second year

$= (3045.60 - 1500) = ₹\ 1545.60$

Amount due at the end of third year

$= 1545.60\left(1+\frac{8}{100}\right) = ₹\ 1669.25$

Amount due after the payment of third instalment

$= (1669.25 - 1500) = ₹\ 169.25$

23. *(a)* Let the sum be ₹ x, then

$x\left(1+\frac{R}{100}\right)^5 = 2x \Rightarrow \left(1+\frac{R}{100}\right)^5 = 2$ ...(i)

The amount after 20 yr,

$x\left(1+\frac{R}{100}\right)^{20} = x\left[\left(1+\frac{R}{100}\right)^5\right]^4$

$= 2^4 x = 16x$ [from Eq. (i)]

$= 16 \times 10000 = ₹\ 160000$

[put $x = ₹\ 10000$]

24. *(b)* Data is insufficient as the rate of interest is not given to calculate further.

25. *(d)* Let amount be ₹ x and rate of interest is R % annually.

According to the questions,

Amount after 1st yr = ₹ 1200

$x\left(1+\frac{R}{100}\right) = 1200$ ...(i)

Amount after 3rd yr = ₹ 1587

$x\left(1+\frac{R}{100}\right)^3 = ₹\ 1587$ ...(ii)

On dividing Eq. (ii) by Eq. (i), we get

$\left(1+\frac{R}{100}\right)^2 = \frac{1587}{1200} = \frac{529}{400}$

$1+\frac{R}{100} = \frac{23}{20} \quad \Rightarrow \quad \frac{R}{100} = \frac{3}{20}$

$\therefore \quad R = 15\%$

Put $R = 15\%$ in Eq. (i),

$x\left(1+\frac{15}{100}\right) = ₹\ 1200 \Rightarrow x = \frac{1200 \times 100}{115}$

$\therefore \quad x = ₹\ 1043.478$

26. *(d)* I. Given, $R = 4\%$, $n = 2$ yr and $A = ₹\ 169$, $P = ?$

$A = P\left(1+\frac{R}{100}\right)^n$ [Rule 1. (i)]

$169 = P\left(1+\frac{4}{100}\right)^2$

$\Rightarrow 169 = P\left(\frac{26}{25}\right)^2$

$P = \frac{169 \times 25 \times 25}{26 \times 26} = ₹\ 156.25$

II. Given, SI = ₹ 120 , $n = 2$ yr

and CI = ₹ 129.

$SI = \frac{P \times R \times T}{100}$

$120 = \frac{P \times R \times 2}{100} \quad \Rightarrow \quad PR = ₹\ 6000$

$\therefore \quad P = \frac{6000}{R}$ ...(i)

$CI = P\left[\left(1+\frac{R}{100}\right)^n - 1\right]$ [Rule 1. (ii)]

$129 = P\left[\left(1+\frac{R}{100}\right)^2 - 1\right]$.

$1290000 = P\,[(100+R)^2 - 100^2]$.

$= P\,[R^2 + R \times 200]$.

$= \frac{6000}{R}[R^2 + R \times 200]$

[from Eq. (i)]

$1290000 = 6000R + 1200000$

$R = \frac{90000}{6000} = 15\%$

Hence, both statement are correct.

27. *(d)* $P = ₹\ 1600$, $r = 25\%$ and $n = 2$ yr

$\therefore \quad A = P\left[1+\frac{R}{100}\right]^n$ [Rule 1. (i)]

$= 1600\left[1+\frac{25}{100}\right]^2$

$= 1600 \times \frac{5}{4} \times \frac{5}{4} = ₹\ 2500$

$\therefore$ Compound interest

$= ₹\ 2500 - ₹1600 = ₹\ 900$

28. *(b)* Refer to example 6.

29. *(a)* Given, principal = ₹ 15000

Rate, $R = ?$

Difference, $D = ₹\ 96$

For 2 yr Difference, $D = \frac{PR^2}{(100)^2}$

[Rule 6 (i)]

$\Rightarrow \quad 96 = \frac{15000 \times R^2}{100 \times 100}$

$\Rightarrow \quad R^2 = 64$

$\therefore \quad R = 8\%$

30. *(c)* $\therefore$ Let the principal be ₹x.

Then, SI = ₹ $\frac{60x}{100}$

$\because \quad SI = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$

$\Rightarrow \quad \frac{60x}{100} = \frac{x \times \text{Rate} \times 6}{100}$ [$\because$ Time = 6 yr]

$60 = \text{Rate} \times 6$

$\therefore \quad$ Rate = 10%

Again principal = ₹ 12000, Time = 3 yr

$\text{Amount} = \text{Principal}\left(1+\frac{\text{Rate}}{100}\right)^{\text{Time}}$

$= 12000\left(1+\frac{10}{100}\right)^3$

$= 12000\left(\frac{11}{10}\right)^3$

$= 12000 \times \frac{11 \times 11 \times 11}{1000}$

$= 12 \times 121 \times 11 = ₹\ 15972$

$\therefore$ CI = Amount − Principal

$= 15972 - 12000 = ₹\ 3972$

31. *(b)* Refer to example 2.

11

PROFIT AND LOSS

Generally (2-4) questions have been asked from this chapter. Generally questions are based on basic formulae and tricks explained in this chapter which are helpful to solve the problems more easily and quickly.

Cost Price (CP)

The price, at which an article is bought is called its cost price. All the overhead expenses in the transection like freight, damage etc., are added to the cost price.

Selling Price (SP)

The price at which an article sold is called its selling price.

Profit (SP>CP) When an article is sold at a price more than its cost price, then profit is earned.

- Profit = Selling price – Cost price
- Loss = Cost price – Selling price

Loss (CP>SP) When an article is sold at a price lower than its cost price, then loss is incurred.

- $\text{Profit } \% = \frac{\text{Profit}}{\text{CP}} \times 100\% = \frac{(\text{SP} - \text{CP})}{\text{CP}} \times 100\%$
- $\text{Loss } \% = \frac{\text{Loss}}{\text{CP}} \times 100\ \% = \frac{(\text{CP} - \text{SP})}{\text{CP}} \times 100\%$

Note Profit and loss percentage is always calculated as the percentage of CP unless otherwise specified.

List Price/Marked Price

The price of an article which is displayed on the price tag is known as marked price. It is the normal price of the thing/products without a discount. Sometimes the shopkeeper increases or decreases the cost price, then this price is taken as the list price of the article.

Important Rules and Formulae

Rule 1 If there is a profit of $r\%$, then

$$\text{(i) } SP = \frac{(100+r)\times CP}{100} \qquad \text{(ii) } CP = \frac{100\times SP}{(100+r)}$$

EXAMPLE 1. If cost price of a fan is ₹ 720. If there is a profit of $16\frac{2}{3}\%$. Then, selling price is

a. ₹ 840 b. ₹ 940
c. ₹ 1050 d. None of these

Sol. a. Given, $r = 16\frac{2}{3}\%$ and cost price = ₹ 720

$$\therefore \text{Selling price (SP)} = \frac{CP\times(100+r)}{100} = \frac{720\times\left(100+16\frac{2}{3}\right)}{100}$$

$$= \frac{720\times\left(100+\frac{50}{3}\right)}{100} = \frac{720\times 350}{300} = ₹\ 840$$

Rule 2 If there is a loss of $r\%$, then

$$\text{(i) } SP = \frac{(100-r)}{100}\times CP \qquad \text{(ii) } CP = \frac{100\times SP}{(100-r)}$$

EXAMPLE 2. When Selling price is ₹ 75 and there is a loss of 12%. Then, cost price is

a. ₹ 22 b. ₹ 44
c. ₹ 55 d. ₹ 85.22

Sol. d. Given, $r = 12\%$ and selling price = ₹ 75

$$\therefore \text{ Cost price} = \frac{100\times SP}{(100-r)} = \frac{100\times 75}{(100-12)} = \frac{100\times 75}{88} = 85.22$$

Rule 3 (i) When there are two successive profits or losses of $x\%$ and $y\%$, then the resultant profit or loss percent is given by $\left(\pm x \pm y + \frac{(\pm x)(\pm y)}{100}\right)\%$.

Note
- Take positive for profit and negative for loss.
- If the sign of result is positive, then there is a total gain.
- If the sign of result is negative, then there is a total loss.

EXAMPLE 3. By selling a watch for ₹ 132 a trademan got two successive profits of 10% and 20%, then the resultant profit percentage is

a. 22% b. 30%
c. 32% d. 34%

Sol. c. Given, the successive profits are 10%, 20%.

$$\text{So, resultant profit percentage} = \left(10+20+\frac{20\times 10}{100}\right)\%$$

$$= 32\%$$

Rule 4 When two different articles are sold at the same selling price, getting gain/loss of $x\%$ on the first and gain/loss of $y\%$ on the second, then the overall % gain or % loss in the transection is given by

$\left[\frac{100(x+y)+2xy}{(100+x)+(100+x)}\right]\%$. The above expression represent overall gain or loss accordingly as its sign is positive or negative.

When two different articles are sold at the same selling price getting a gain of $x\%$ on the first and loss of $x\%$ on the second, then the overall % loss in the transection is given by $\left(\frac{x}{10}\right)^2\%$.

Note That in such questions there is always a loss.

EXAMPLE 4. A trader sells two cycles at ₹ 1188 each and gains 10% on the first and loses 12% on the second. What is the profit or loss percent on the whole?

a. 1% loss b. 1% gain c. No loss no gain d. 3.2% loss

Sol. d. When there is a profit of $x\%$ and loss of $y\%$, then the resultant

$$\text{Profit/Loss percentage} = \left(x-y-\frac{xy}{100}\right)\%$$

Here, $x = 10\%$ and $y = 12$

$$\therefore \text{Profit/Loss percentage} = \left(10-12-\frac{10\times 12}{100}\right)\% = -3.2\%$$

EXAMPLE 5. Harish sold two scooters, each for ₹ 24000. If he makes 20% profit on the first and 15% loss on the second. What is his gain or loss percent in the transection?

a. loss $\frac{20}{41}\%$ b. gain $\frac{20}{41}\%$ c. loss $\frac{32}{73}\%$ d. gain $\frac{32}{73}\%$

Sol. a. Here, $x = 20$ and $y = -15$

$$\therefore \text{Overall gain/loss }\% = \left[\frac{100(x+y)+2xy}{(100+x)+(100+y)}\right]\%$$

$$= \left[\frac{100(20-15)+2\times 20\times -15}{(100+20)+(100-15)}\right]\%$$

$$= \frac{-100}{205}\% = \frac{-20}{41}\%$$

Which represents loss.

EXAMPLE 6. Vikram sold two horses for ₹ 990 each gaining 10% on the one and lossing 10% on the other. Find his gain or loss percent.

a. 1% gain b. 1% loss c. 4 % gain d. 4 % loss

Sol. b. Here, $x = 10$

$$\therefore \quad \text{Overall loss } \% = \left(\frac{x}{10}\right)^2 \%$$
$$= \left(\frac{10}{10}\right)^2 \% = 1\%$$

Rule 5 A person buys two items for ₹ S. One is sold at a loss of r% and the other at a gain of R%, if each item was sold at the same price, then

$$\text{Cost price of item sold at loss} = \frac{S \times (100 + R)}{(100 - r) + (100 + R)}$$

$$\text{Cost price of item sold at gain} = \frac{S \times (100 - r)}{(100 - r) + (100 + R)}$$

EXAMPLE 7. Sudhir bought two boxes for ₹ 1300. He sold one box at a profit of 20% and the other at a loss of 12%. If the selling price of both boxes is the same, then the cost price of each box is

a. ₹ 700 and ₹ 500 b. ₹ 750 and ₹ 550
c. ₹ 800 and ₹ 600 d. None of these

Sol. b. Given, cost of two boxes $S =$ ₹ 1300, profit on first box $R = 20\%$

and loss on second box $r = 12\%$

Cost price of box sold at loss

$$= \frac{1300 \times (100 + 20)}{(100 - 12) + (100 + 20)} = \frac{1300 \times 120}{88 + 120} = ₹\,750$$

Cost price of box sold at gain

$$= \frac{1300 \times (100 - 12)}{(100 - 12) + (100 + 20)} = \frac{1300 \times 88}{88 + 120} = ₹\,550$$

Rule 6 If a trader professer to sell his goods at cost price, but uses false weights, then gain %

$$= \frac{\text{true weight} - \text{false weight}}{\text{false weight}} \times 100$$

EXAMPLE 8. A dishonest dealer sell his goods at cost price, but he uses a weight of 960 gm for the kg weight, then the percentage of gain is

a. $2\frac{1}{3}\%$ b. $4\frac{1}{6}\%$ c. $6\frac{1}{9}\%$ d. $3\frac{1}{2}\%$

Sol. b. $\text{Gain } \% = \dfrac{\text{true weight} - \text{false weight}}{\text{false weight}} \times 100$

$$= \frac{1000 - 960}{960} \times 100 = \frac{40}{960} \times 100 = 4\frac{1}{6}\%$$

Rule 7 If cost price of 'n' articles is equal to the selling price of 'm' articles, then

$$\text{Profit percentage} = \left(\frac{n - m}{m}\right) \times 100\% \ (n > m)$$

$$\text{and loss percentage} = \left(\frac{m - n}{m}\right) \times 100\% \ (m > n)$$

EXAMPLE 9. If the cost price of 18 chairs be equal to selling price of 16 chairs. The gain percent is

a. 12% b. 12.5%
c. 14% d. 15.5%

Sol. b. Given, $n = 18$, $m = 16$ $[\because n > m]$

$$\therefore \text{ Profit percentage} = \left(\frac{18 - 16}{16}\right) \times 100 \quad \left[\because \frac{n - m}{m} \times 100\right]$$
$$= \frac{2}{16} \times 100 = \frac{100}{8} = 12.5\%$$

DISCOUNT

The reduction allowed on the marked price of an article is called as discount. Discount is always reckoned on the marked price.

Selling price = Marked price − Discount

Rule 8 If discount allowed is r%, then

$$\text{Selling price} = \frac{(100 - r)}{100} \times \text{Marked price}$$

EXAMPLE 10. If the marked price of a fan is ₹ 700 and a discount of 10% is given on it, then what is the selling price of the fan?

a. ₹ 500 b. ₹ 575
c. ₹ 610 d. ₹ 630

Sol. d. Given, $r = 10\%$

$$\text{Selling price} = \frac{(100 - 10)}{100} \times 700$$
$$= \frac{90 \times 700}{100} = ₹\,630$$

Rule 9 If a shopkeeper marks his items at x% above the cost price and allows customers a discount of y% then there is $\left(x - y - \dfrac{xy}{100}\right)\%$ profit or loss according to positive or negative sign, respectively.

EXAMPLE 11. A dealer marked his goods 20% above the cost price and allows a discount of 10%. Then, his gain percent is

a. 2% b. 4%
c. 6% d. 8%

Sol. d. Given, $x = 20\%$ and $y = 10\%$

$$\text{Profit/Loss percentage} = \left(x - y - \frac{xy}{100}\right)\%$$
$$= \left(20 - 10 - \frac{20 \times 10}{100}\right)\% = 8\%$$

Hence, positive sign shows profit i.e. profit of 8%.

Successive Discounts
(Discount Series)

Rule 10 Single discount equivalent to three successive discounts $r_1\%, r_2\%$ and $r_3\%$

$$=\left[1-\left(1-\frac{r_1}{100}\right)\left(1-\frac{r_2}{100}\right)\left(1-\frac{r_3}{100}\right)\right]\times 100\%$$

Note This formula also can be apply for more than three successive discounts.

EXAMPLE 12. A single discount which is equivalent to two successive discounts of 20% and 5% is

a. 20% b. 22% c. 24% d. 25%

Sol. c. Given, $r_1 = 20\%$, $r_2 = 5\%$

∴ Single discount equivalent to two succesive discount 20% and 5%

$$=\left[1-\left(1-\frac{20}{100}\right)\left(1-\frac{5}{100}\right)\right]\times 100\%$$

$$=\left(1-\frac{4}{5}\times\frac{19}{20}\right)\times 100\% = \left(1-\frac{19}{25}\right)\times 100\%$$

$$=\frac{6}{25}\times 100\% = 24\%$$

Rule 11 If marked price of an item is ₹ x and the successive discount rates are $r_1\%, r_2\%, r_3\%$ and so on, then selling price of the item

$$SP = x\times\left(1-\frac{r_1}{100}\right)\left(1-\frac{r_2}{100}\right)\left(1-\frac{r_3}{100}\right)$$

EXAMPLE 13. The marked price of a watch is ₹ 1600. The shopkeeper gives successive discount of 10%, r% to the customer. If the customer pays ₹ 1224 for the watch, then the value of r is

a. 20% b. 35% c. 37% d. 15%

Sol. d. Given, cost price $x =$ ₹ 1600, $r_1 = 10\%$

and $r_2 = r\%$

∴ Selling price = Cost price $\left(1-\frac{r_1}{100}\right)\left(1-\frac{r_2}{100}\right)$

$$1224 = 1600\left(1-\frac{10}{100}\right)\left(1-\frac{r}{100}\right)$$

$$\Rightarrow \quad 1-\frac{r}{100} = \frac{1224}{160\times 9} \quad\Rightarrow\quad \frac{r}{100} = 1-\frac{1224}{1440}$$

$$\therefore \quad r = 15\%$$

PRACTICE EXERCISE

1. By selling an article for ₹ 247.50, Sonu get a profit of 12.5%. The cost of the article is
(a) ₹ 220 (b) ₹ 205 (c) ₹ 210 (d) ₹ 200

2. If cost price of a fan is ₹ 720 and its SP is ₹ 840. Find the gain percent.
(a) 16% (b) $16\frac{2}{3}\%$ (c) $16\frac{1}{3}\%$ (d) $16\frac{7}{3}\%$

3. By selling an article for ₹ 110, a man losses 12%. For how much should he sell it to gain 8%?
(a) ₹ 120 (b) ₹ 125 (c) ₹ 135 (d) ₹ 140

4. A man buys 4 tables and 5 chairs for ₹ 1000. If he sells the tables at 10% profit and chairs 20% profit, he earns a profit of ₹ 120. Then, what is the cost of one table?
(a) ₹ 200 (b) ₹ 220 (c) ₹ 240 (d) ₹ 260

5. If selling price of 8 articles is equal to the cost price of 10 articles, then per cent gain or loss is
(a) 20% (b) 25% (c) 30% (d) 35%

6. A man sells fans at the same price on one he gain 20% and losses 20% on the other. His gain or loss is
(a) 4% loss (b) 4% gain
(c) Neither gain nor loss (d) 1% loss

7. Sneha gains 10% on selling a pen. If she sells it at double the price, the profit per cent is
(a) 120% (b) 60% (c) 100% (d) 200%

8. On selling an article for ₹ 240, a trader loses 4%. In order to gain 10%, he must sell the article for
(a) ₹ 275 (b) ₹ 280 (c) ₹ 285 (d) ₹ 300

9. By selling 8 dozen pencils, a shopkeeper gains the selling price of 1 dozen pencils. What is the gain?
(a) $12\frac{1}{2}\%$ (b) $13\frac{1}{7}\%$ (c) $14\frac{2}{7}\%$ (d) $87\frac{1}{2}\%$

10. A man purchased a watch for ₹ 400 and sold it at a gain of 20% of the selling price. The selling price of the watch is
(a) ₹ 300 (b) ₹ 320 (c) ₹ 440 (d) ₹ 500

11. A person A sells a table costing ₹ 2000 to a person B and earns a profit of 6%. The person B sells it to another person C at a loss of 5%. At what price did B sell the table?
(a) ₹ 2054 (b) ₹ 2050 (c) ₹ 2024 (d) ₹ 2014

12. What price did the seller mark at the printed price of a watch purchased at ₹ 380, so that after giving 5% discount, there is 25% profit?
(a) ₹ 400 (b) ₹ 450 (c) ₹ 500 (d) ₹ 600

13. A discount series of 10%, 20% and 40% is equal to a single discount of
(a) 50% (b) 60% (c) 56.8% (d) 70.28%

14. Successive discounts of $12\frac{1}{2}\%$ and $7\frac{1}{2}\%$ are given on the marked price of a cupboard. If the customer pays ₹ 2590, then what is the marked price?
(a) ₹ 3108 (b) ₹ 3148 (c) ₹ 3200 (d) ₹ 3600

15. The difference between a discount of 40% on ₹ 1000 and two successive discounts of 35% and 5% on the same amount is
(a) ₹ 15.50 (b) ₹ 16.50 (c) ₹ 17.50 (d) ₹ 18.00

16. A dealer buys an article listed at ₹ 100 and gets two successive discounts of 10% and 20%. He spends 10% of the cost price on transport etc. At what price should he sell the article to earn a profit of 15%?
(a) ₹ 90 (b) ₹ 91 (c) ₹ 91.08 (d) ₹ 91.10

17. An item costing ₹ 200 is being sold at 10% loss. If the price is further reduced by 5%, then the selling price will be
(a) ₹ 170 (b) ₹ 171 (c) ₹ 175 (d) ₹ 179

18. A man bought a number of oranges at 3 for a rupee and an equal number at 2 for a rupee. At what price per dozen should he sell them to make a profit of 20%?
(a) ₹ 4 (b) ₹ 5 (c) ₹ 6 (d) ₹ 7

19. A milk vendor bought 28 L of milk at the cost of ₹ 8.50 per L. After adding some water, he sold the mixture at the same price. If he gains 12.5%, then how much water did he add?
(a) 5.5 L (b) 4.5 L
(c) 3.5 L (d) 2.5 L

20. A man sold two watches, each for ₹ 495. If he gained 10% on one watch and suffered a loss of 10% on the other, then what is the loss or gain percentage in the transaction?
(a) 1% gain (b) 1% loss
(c) 100/99% loss (d) No gain no loss

21. Jyoti bought a computer system for ₹ 40000. She sold it to Brajesh at a loss of 4%. If Brajesh sells it for ₹ 40320 to Yash, then the profit per cent earned by Brajesh is
(a) 3% (b) 5%
(c) 7% (d) 10%

22. List price of a video cassette is ₹ 100. A dealer sells three video cassettes for ₹ 274.50 after allowing discount at certain rate. The rate of discount allowed is
(a) 7% (b) 7.5%
(c) 8% (d) 8.5%

23. A fruit-seller buys lemons at 2 for a rupee and sells them at 5 for three rupees. What is his gain per cent?
(a) 10% (b) 15% (c) 20% (d) 25%

24. A trader marks 10% higher than the cost price. He gives a discount of 10% on the marked price. In this kind of sales how much per cent does the trader gain or loss?
(a) 5% gain (b) 2% gain (c) 1% loss (d) 3% loss

25. One saree was purchased for ₹ 564 after getting a discount of 6% and another saree was purchased for ₹ 396 after getting a discount of 1%. Taking both the items as a single transaction, what is the percentage of discount?
(a) 3.5% (b) 4% (c) 7% (d) 7.5%

26. A dishonest dealer professes to sell his good at cost price but uses a false weight and thus gains 25%. For a kilogram he uses a weight of
(a) 700 g (b) 750 g (c) 800 g (d) 850 g

27. A person bought two old scooters for ₹ 9000. By selling one at a profit of 25% and the other at a loss of 20%, to neither gain nor loses. The cost of each scooter is
(a) ₹ 3500, ₹ 500 (b) ₹ 4500, ₹ 4500
(c) ₹ 4000, ₹ 5000 (d) ₹ 5300, ₹ 3700

28. The manufacturer of a certain item can sell all he can produce at the selling price of ₹ 60 each. It costs him ₹ 40 in materials and labour to produce each item and he has overhead expenses of ₹ 3000 per week in order to operate the plant. The number of items he should produce and sell in order to make a profit of atleast ₹ 1000 per week is
(a) 400 (b) 300 (c) 250 (d) 200

29. A person sold a table at a profit of $6\frac{1}{2}\%$. If he had sold it for ₹ 1250 more, he would have gained 19%
I. The CP of the table is ₹ 10000.
II. CP will ₹ 19000, if he had sold it for ₹ 2400 more and gained same profit
Which one is correct?
(a) Only II (b) Only I
(c) Neither I nor II (d) Both I and II

30. A bookseller sells a book at a gain of 10%. If he had bought it at 4% less and sold it for ₹ 6 more, he would have gained $18\frac{3}{4}\%$
I. The SP of the book is ₹ 165.
II. The CP of the book is ₹ 150.
Which one is correct?
(a) Neither I nor II (b) Both I and II
(c) Only I (d) Only II

PREVIOUS YEARS' QUESTIONS

31. A person bought 8 quintal of rice for certain rupees. After a week, he sold 3 quintal of rice at 10% profit, 3 quintal of rice with neither profit nor loss and 2 quintal at 5% loss. In this transaction, what is the profit? **2012 I**

(a) 10% (b) 20% (c) 25% (d) None of these

32. The cost of two articles are in the ratio 3 : 5. If there is 30% loss on the first article and 20% gain on the second article, then what is overall percentage of loss or gain? **2012 I**

(a) 2.25% gain (b) 5.25% loss
(c) 2% loss (d) None of these

33. A cloth store is offering Buy 3, get 1 free. What is the net percentage discount being offered by the store? **2012 II**

(a) 20% (b) 25% (c) 30% (d) $33\frac{1}{3}\%$

34. A person sold an article for ₹ 136 and got 15% loss. Had he sold it for ₹ x, he would have got a profit of 15%. Which one of the following is correct? **2012 II**

(a) $190 < x < 200$ (b) $180 < x < 190$
(c) $170 < x < 180$ (d) $160 < x < 170$

35. A man buys a television set which lists for ₹ 5000 at 10% discount. He gets an additional 2% discount (after the first discount) for paying cash. What does he actually pay for the set? **2012 II**

(a) ₹ 4410 (b) ₹ 4400 (c) ₹ 4000 (d) ₹ 4500

36. A merchant earns a profit of 20% by selling a basket containing 80 apples which costs ₹ 240 but he gives one-fourth of it to his friend at cost price and sells the remaining apples. In order to earn the same profit, at what price must he sell each apple? **2012 II**

(a) ₹ 3.00 (b) ₹ 3.60 (c) ₹ 3.80 (d) ₹ 4.80

37. A person sold an article for ₹ 3600 and got a profit of 20%. Had he sold the article for ₹ 3150, how much profit would he have got? **2013 II**

(a) 4% (b) 5% (c) 6% (d) 10%

38. Two lots of onions with equal quantity, one costing ₹ 10 per kg and the other costing ₹ 15 per kg, are mixed together and whole lot is sold at ₹ 15 per kg. What is the profit or loss? **2013 II**

(a) 10% loss (b) 10% profit
(c) 20% profit (d) 20% loss

39. On a 20% discount sale, an article costs ₹ 596. What was the original price of the article? **2014 I**

(a) ₹ 720 (b) ₹ 735 (c) ₹ 745 (d) ₹ 775

40. A man buys 200 oranges for ₹ 1000. How many oranges for ₹ 100 can be sold, so that his profit percentage is 25%? **2014 II**

(a) 10 (b) 14 (c) 16 (d) 20

41. When an article is sold at 20% discount, the selling price is ₹ 24. What will be the selling price when the discount is 30%. **2014 II**

(a) ₹ 25 (b) ₹ 23 (c) ₹ 21 (d) ₹ 20

42. A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000 g for 800 g. What is his actual profit percentage? **2014 II**

(a) 25% (b) 20%
(c) 40% (d) 30%

43. A person selling an article for ₹ 96 finds that his loss per cent is one-fourth of the amount of rupees that he had paid for the article. What can be the cost price? **2014 II**

(a) Only ₹ 160 (b) Only ₹ 240
(c) Either ₹ 160 or ₹ 240 (d) Neither ₹ 160 nor ₹ 240

44. A milkman claims to sell milk at its cost price but he is making a profit of 20% since, he has mixed some amount of water in the milk. What is the percentage of milk in the mixture? **2015 I**

(a) 80% (b) $\frac{250}{3}\%$ (c) 75% (d) $\frac{200}{3}\%$

45. The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and 40% on the same amount, is equal to **2015 II**

(a) 43.2% (b) 50% (c) 56.8% (d) 70%

46. A cloth merchant buys cloth from a weaver and cheats him by using a scale which is 10 cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but while selling uses a scale which is 10 cm shorter than a normal metre scale. What is his gain? **2016 I**

(a) 20% (b) 21% (c) $22\frac{2}{9}\%$ (d) $23\frac{1}{3}\%$

ANSWERS

1	a	2	b	3	c	4	a	5	b	6	a	7	a	8	a	9	c	10	d
11	d	12	c	13	c	14	c	15	c	16	c	17	b	18	c	19	c	20	b
21	b	22	d	23	c	24	c	25	b	26	c	27	c	28	d	29	b	30	b
31	d	32	d	33	b	34	b	35	a	36	c	37	b	38	c	39	c	40	c
41	c	42	a	43	c	44	b	45	c	46	a								

HINTS AND SOLUTIONS

1. *(a)* Given, selling price of article

$= ₹\ 247.50$ and gain $= \frac{25}{2}\%$

$\therefore$ Cost price

$= ₹ \left\{ \frac{100}{\left(100 + \frac{25}{2}\right)} \times 247.50 \right\}$ [Rule 1]

$= ₹ \frac{100 \times 2 \times 247.50}{225} = ₹\ 220$

2. *(b)* Total gain = SP − CP

$= (840 - 720) = ₹\ 120$

$\therefore$ Gain percent $= \frac{120}{720} \times 100 = 16\frac{2}{3}\%$

3. *(c)* Given, SP = ₹ 110 and loss = 12%

$\therefore$ Costprice $= ₹ \left(\frac{100}{88} \times 110\right) = ₹\ 125$

Now, CP = ₹ 125, gain required = 8%

$\therefore$ SP $= ₹ \left(\frac{(100+8)}{100} \times 125\right) = ₹\ 135$

4. *(a)* Let cost of one table and one chair be ₹ x and ₹ y, respectively.

$\therefore$ Cost of 4 tables and 5 chairs

$= ₹\ 4x + 5y$

$\therefore 4x + 5y = 1000$ ...(i)

$\therefore$ Now, profit on 4 tables

$= 4x \times 10\% = ₹ \frac{4x}{10}$

Profit on 5 chairs $= 5y \times 20\% = ₹\ y$

$\therefore \frac{4x}{10} + y = 120$ ...(ii)

Now, on solving Eqs. (i) and (ii), we get

$x = ₹\ 200$

5. *(b)* Refer to example 9.

6. *(a)* Refer to example 4.

7. *(a)* Let the selling price be ₹ 100.

$\therefore$ Cost price $= \frac{\text{SP} \times 100}{100 + 10}$ [Rule 1]

$= \frac{100 \times 100}{110} = ₹ \frac{1000}{11}$

Now, if SP is ₹ 200.

$\therefore$ Gain $= ₹ \left(200 - \frac{1000}{11}\right) = ₹ \frac{1200}{11}$

Gain percent $= \frac{1200/11}{1000/11} \times 100 = 120\%$

8. *(a)* Given, selling price of an article

$= ₹\ 240$

Loss = 4%

$\therefore$ Cost price of an article for the loss of 4%

$= ₹ \frac{240 \times 100}{96}$ [Rule 2]

Selling price of an article for a profit of 10%

$= \frac{240 \times 100}{96} \times \frac{110}{100} = \frac{240 \times 110}{96}$

$= ₹\ 275$

9. *(c)* Let selling price of 1 dozen pencil be ₹ x.

$\therefore$ Selling price of 8 dozen pencils = ₹ $8x$ and profit = ₹x

$\because$ Cost price of 8 dozen pencils

$= 8x - x = ₹\ 7x$

$\therefore$ Gain, percent $= \frac{x}{7x} \times 100\% = 14\frac{2}{7}\%$

10. *(d)* Given, CP = ₹ 400, let selling price be ₹ x.

Then, 400 + 20% of $x = x$

$\Rightarrow 400 + \frac{x}{5} = x \Rightarrow \frac{4x}{5} = 400$

$\Rightarrow x = \frac{400 \times 5}{4} \Rightarrow x = ₹\ 500$

$\therefore$ Selling price = ₹ 500

11. *(d)* The cost price of table for person B

$= 2000 + 6 \times \frac{2000}{100}$

$= 2000 + 120 = ₹\ 2120$

Selling price for person B

$= 2120 - \frac{2120 \times 5}{100}$

$= 2120 - 106 = ₹\ 2014$

12. *(c)* Let marked price be ₹ x.

Selling price after 5% discount

$= x - \frac{5}{100}x = \frac{19}{20}x$

Profit = SP − CP $= \frac{19}{20}x - 380$

Profit % $= \frac{\frac{19}{20}x - 380}{380} \times 100$

$25 = \frac{\frac{19}{20}x - 380}{380} \times 100$

$x = 475 \times \frac{20}{19} = ₹\ 500$

13. *(c)* Refer to example 12.

14. *(c)* Refer to example 13.

15. *(c)* **Case I** Discount = 40 %

$\Rightarrow$ Selling price = 60% of 1000

$= \frac{60}{100} \times 1000 = ₹\ 600$

Case II Two successive discounts are of 35% and 5%.

$\Rightarrow$ Selling price

$= 1000 \times \left(1 - \frac{35}{100}\right)\left(1 - \frac{5}{100}\right)$

[Rule 11]

$= \frac{65}{100} \times \frac{95}{100} \times 1000 = ₹\ 617.50$

$\therefore$ Difference $= (617.50 - 600) = ₹\ 17.50$

16. *(c)* List price of article be ₹ 100.

Cost price for dealer

$= \frac{(100-20)}{100} \times \frac{(100-10)}{100} \times 100$

$= \frac{80 \times 90 \times 100}{100 \times 100} = ₹\ 72$

Money spent on transport

$= \frac{10}{100} \times 72 = ₹\ 7.20$

$\therefore$ Total cost price = 72 + 7. 20 = ₹ 79.20

$\therefore$ Selling price

$= \frac{(100+15)}{100} \times 79.20$

$= \frac{115 \times 79.20}{100} = ₹\ 91.08$

17. *(b)* Given, cost of article = ₹ 200

$\therefore$ Selling price of article = 95% of (90% of 200)

$= \frac{95}{100} \times \frac{90}{100} \times 200 = ₹\ 171$

18. *(c)* Given, CP of 3 oranges of Ist variety

= ₹ 1

CP of 1 orange of Ist variety $= ₹ \frac{1}{3}$

CP of 2 oranges of IInd variety = ₹ 1

CP of 1 orange of IInd variety $= ₹ \frac{1}{2}$

Total CP of 2 oranges of different variety $= \frac{1}{3} + \frac{1}{2} = ₹ \frac{5}{6}$

Profit on 2 oranges of different variety

$= 20\%$ of $\frac{5}{6} = \frac{20}{100} \times \frac{5}{6} = ₹ \frac{1}{6}$

$\therefore$ SP of 2 oranges of different variety

$= \frac{5}{6} + \frac{1}{6} = ₹\ 1$

Hence, SP of 12 oranges is ₹ 6.

19. *(c)* Given, cost price of 1L = ₹ 8.50

$\therefore$ Total CP of milk $= 28 \times 8.50 = ₹\ 238$

$\therefore$ Profit = 12.5% of 238

$= \frac{12.5}{100} \times 238 = ₹\ 29.75$

Let he added x L of water.

$\therefore$ Profit $= x \times 8.5 \Rightarrow 29.75 = x \times 8.5$

$\therefore$ $x = 3.5$ L

20. *(b)* Refer to example 6.

21. *(b)* Cost price of computer for Brajesh

$= 40000 - \frac{4}{100}(40000) =$ ₹ 38400

Selling price of computer for Brajesh

= ₹ 40320

$\therefore$ Profit amount $= (40320 - 38400)$

= ₹ 1920

$\therefore$ Profit percentage earned by Brajesh

$= \frac{1920}{38400} \times 100 = 5\%$

22. *(d)* Given, list price of a video cassette

= ₹ 100

Let the rate of discount be r%.

Selling price of 3 video cassette

= ₹ 274.50

$\therefore$ Selling price of 1 video cassette

= ₹ $\frac{274.50}{3}$ = ₹ 91.50

$\therefore$ $100 - \frac{r}{100} \times 100 =$ ₹ 91.50

$\Rightarrow$ $100 - 91.50 = r \Rightarrow 8.50 = r$

$\therefore$ Rate of discount = 8.5%

23. *(c)* CP of 1 lemon $= \frac{1}{2}$

SP of 1 lemon $= \frac{3}{5}$

$\therefore$ Gain %

$= \frac{\frac{3}{5} - \frac{1}{2}}{\frac{1}{2}} \times 100 = \frac{6-5}{10} \times 2 \times 100 = 20\%$

24. *(c)* Let the cost price be ₹ x.

Marked price $= \frac{x \times 110}{100} =$ ₹ $\frac{11x}{10}$

$\therefore$ SP $= \frac{11x}{10} \times \frac{90}{100} = \frac{99x}{100}$

$\therefore$ Required gain/loss per cent

$= \frac{\frac{99x}{100} - x}{x} \times 100 = -1\%$ [Rule 8]

[negative sign i.e. loss]

25. *(b)* Let marked price of two sarees be ₹ x and ₹ y, respectively.

$\therefore$ $x - \frac{6x}{100} = 564 \Rightarrow \frac{94x}{100} = 564$

$\Rightarrow$ $x =$ ₹ 600 and $y - \frac{y}{100} = 396$

$\Rightarrow$ $\frac{99y}{100} = 396 \Rightarrow y =$ ₹ 400

$\therefore$ Total MP amount = 600 + 400

= ₹ 1000

Total amount after discount

= 564 + 396 = ₹ 960

$\therefore$ Discount per cent

$= \frac{1000 - 960}{1000} \times 100 = \frac{40}{10}\% = 4\%$

26. *(c)* Here, true weight = 1000

and gain = 25%

$\Rightarrow$ $25 = \frac{1000 - \text{false weight}}{\text{false weight}} \times 100$

$\Rightarrow$ $\frac{\text{false weight}}{4} = 1000 - \text{false weight}$

$\Rightarrow$ false weight $= 1000 \times \frac{4}{5} = 800$

Hence, he uses the weight 800 g.

27. *(c)* Let cost price of one scooter be ₹ x.

$\therefore$ Cost price of other scooter

= ₹ $(9000 - x)$

$\therefore$ Selling price of Ist scooter

$= x + \frac{25x}{100} = \frac{125x}{100}$

Also, selling price of 2nd scooter

$= (9000 - x)\left(1 - \frac{20}{100}\right)$

$= (9000 - x)\left(\frac{80}{100}\right)$

Total selling price of both scooters

$= \frac{125x}{100} + (9000 - x)\frac{80}{100}$

$\therefore$ $\frac{125x}{100} + (9000 - x)\frac{80}{100} = 9000$

[given]

$\Rightarrow$ $\frac{45x}{100} + 7200 = 9000$

$\therefore$ $x = 4000$

$\therefore$ Cost price of 1st scooter = ₹ 4000

Hence, cost price of 2nd scooter

$= (9000 - x) =$ ₹ 5000

28. *(d)* Let the number of items be x.

Then, selling price of items $= 60x$

Cost of material of items $= 40x$

Overhead expenses = ₹ 3000

$\therefore$ $60x - (40x + 3000) = 1000$

$\Rightarrow$ $20x = 4000$

$\therefore$ $x = \frac{4000}{20} = 200$

29. *(b)* I. Let CP $= x$

Then, SP $= \frac{213}{200}x$.

If SP $= \frac{213x}{200} + 1250$, then gain = 19%

$\Rightarrow$ $19 = \frac{\frac{213x}{200} + 1250 - x}{x} \times 100$

$\Rightarrow$ $\frac{19x}{100} - \frac{213x}{200} = 1250$

$\Rightarrow$ $x = 1250 \times 8 = 10000$

II. If SP $= \frac{213x}{200} + 2400$ and gain

= 19%, then

$19 = \frac{\frac{213x}{200} + 2400 - x}{x} \times 100$

$\Rightarrow$ $\frac{19x}{100} - \frac{13x}{200} = 2400$

$\Rightarrow$ $x = 2400 \times 8 = 19200$

Hence, I is correct but II is incorrect.

30. *(b)* Let CP $= x$

then SP $= \frac{110x}{100} = \frac{11x}{10}$

Now, CP = 96% of $x = \frac{96x}{100} =$ ₹ $\frac{24x}{25}$

According to the questions

SP = ₹ $\left(\frac{11x}{10} + 6\right)$

$\therefore$ $\left(\frac{11x}{10} + 6\right) = 118\frac{3}{4}\%$ of $\frac{24x}{25}$

$\Rightarrow$ $\frac{11x + 60}{10} = \frac{475}{400} \times \frac{24x}{25} = \frac{57x}{50}$

$\Rightarrow$ $570x = 550x + 3000$

$\Rightarrow$ $x = \frac{3000}{20} = 150$

$\therefore$ CP = ₹ 150

$\therefore$ SP $= \frac{11}{10} \times 150 =$ ₹ 165

Hence, I and II both are correct.

31. *(d)* Let CP of 8 quintal be ₹ x.

$\therefore$ CP of 1 quintal = ₹ $\frac{x}{8}$

$\therefore$ SP of 3 quintal rice at 10% profit

$= \frac{3x}{8} + \frac{3x}{8} \times \frac{1}{10} = \frac{3x}{8} + \frac{3x}{80} = \frac{33x}{80}$

SP of 3 quintal rice without profit or loss = ₹ $\frac{3x}{8}$

SP of 2 quintal rice at 5% loss

$= \frac{2x}{8} - \frac{2x}{8} \times \frac{5}{100} = \frac{x}{4} - \frac{x}{4 \times 20}$

$= \frac{19x}{4 \times 20} = \frac{19x}{80}$

$\therefore$ Total SP $= \frac{33x}{80} + \frac{3x}{8} + \frac{19x}{80}$

$= \frac{33x + 30x + 19x}{80} = \frac{82x}{80}$

$\therefore$ Profit percentage $= \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100\%$

$= \frac{\frac{82x}{80} - x}{x} \times 100\% = \frac{(82 - 80)x}{80x} \times 100\%$

$= \frac{2}{80} \times 100 = 2.5\%$

32. *(d)* Let the CP of two articles be $3x$ and $5x$, respectively.

$\therefore$ SP of first article $= \frac{3x \times 70}{100} = \frac{21x}{10}$ [Rule 2]

and SP of second article

$= \frac{5x \times 120}{100} = 6x$

$\therefore$ Total SP

$= 6x + \frac{21x}{10} = \frac{60x + 21x}{10} = \frac{81x}{10}$

$\therefore$ Total CP $= 3x + 5x = 8x$

$\therefore$ Profit $= \frac{81x}{10} - 8x = \frac{81x - 80x}{10} = \frac{x}{10}$

$\therefore$ Overall percentage of gain percentage

$= \frac{\frac{x}{10} \times 100}{8x} = \frac{x \times 100}{10 \times 8x} = 1.25\%$

33. *(b)* We know that,

Net percentage discount

$= \frac{\text{Discount}}{\text{Cost price}} \times 100\% = \frac{1}{4} \times 100\% = 25\%$

34. *(b)* We know that,

Cost price $= \left(\frac{100}{100 - \text{Loss}\%}\right) \times \text{SP}$ [Rule 2]

$= \left(\frac{100}{100 - 15}\right) \times 136$

$= \frac{136 \times 100}{85} = ₹\ 160$

$\because$ Given, profit percentage = 15%

$\therefore$ Selling price (x)

$= \frac{160 \times (100 + 15)}{100} = \frac{160 \times 115}{100} = ₹ 184$

So, option (b) is correct because $180 < x < 190$.

35. *(a)* Given, list price of television = ₹ 5000

and discount = 10%

$\therefore$ After discount television cost

$= 5000 - 5000 \times 10\%$

$= 5000 - \frac{5000 \times 10}{100} = ₹\ 4500$

Additional discount = 2%

$\therefore$ Actually price of television

$= 4500 - \frac{4500 \times 2}{100} = ₹\ 4410$

36. *(c)* $\because$ CP of 80 apples = ₹ 240

$\therefore$ CP of 1 apple $= \frac{240}{80} = ₹\ 3$

$\therefore$ CP of 20 apples $= ₹ 3 \times 20 = ₹\ 60$

To earn a profit of 20%, SP = 120% of 240 = ₹ 288

But he sold $\frac{1}{4}$ of his apples i.e. 20 apples for ₹ 60.

$\therefore$ SP of remaining 60 apples

$= (288 - 60) = ₹\ 228$

$\therefore$ SP of 1 apple $= \frac{228}{60} = ₹\ 3.80$

37. *(b)* Let the cost price of the article be ₹ x.

Given, profit percentage = 20%

Now, $x + \frac{20x}{100} = 3600 \Rightarrow \frac{120x}{100} = 3600$

$\therefore \quad x = ₹\ 3000$

Now, profit percentage when the article is sold for ₹ 3150

$\frac{3150 - 3000}{3000} \times 100 = \frac{150}{3000} \times 100 = 5\%$

38. *(c)* Let each lot of onion contains x kg. Then, total cost price of these two lots together $= 10x + 15x = 25x$

Selling price of whole lot

$= 15 \times (x + x) = 15 \times 2x = 30x$

$\therefore$ Profit percentage $= \frac{30x - 25x}{25x} \times 100$

$= \frac{5x}{25x} \times 100 = 20\%$

Hence, the profit is 20%.

39. *(c)* Let the original price be ₹ x.

Since, at discount of 20% article costs ₹ 596.

Then, $596 = \frac{80}{100} \times x \Rightarrow x = \frac{596 \times 100}{80}$

$= ₹\ 745$

Hence, the original price of an article is ₹ 745.

40. *(c)* Given, cost price of 200 oranges = ₹ 1000

$\therefore$ Cost price of 1 orange $= \frac{1000}{200} = ₹\ 5$

Selling price of 1 orange $= 5\left(\frac{100 + 25}{100}\right)$

$= 5 \times \frac{125}{100} = ₹\ 6.25$

Now, in ₹ 6.25 number of oranges can be sold = 1

In ₹ 100, number of oranges can be sold

$= \frac{1}{6.25} \times 100 = 16$

Hence, 16 oranges can be sold in ₹ 100 for profit 25%.

41. *(c)* Let the cost price of article be ₹ x.

Discount = 20% and selling price = 24 [given]

$\therefore \quad x = 24 + 20\%$ of x

$\Rightarrow \quad x = 24 + \frac{20}{100}x \Rightarrow x = 24 + \frac{x}{5}$

$\Rightarrow \quad x - \frac{x}{5} = 24 \Rightarrow \frac{4x}{5} = 24$

$\Rightarrow \quad x = \frac{24 \times 5}{4} = 30$

$\therefore$ Cost price of an article = ₹ 30

Now, selling price after 30% discount

$= 30 - 30\%$ of $30 = 30 - \frac{30}{100} \times 30$

$= 30 - 9 = ₹\ 21$

42. *(a)* $\because$ Actual profit percentage

$= \frac{\text{Fair weight} - \text{Unfair weight}}{\text{Unfair weight}} \times 100\ \%$

$= \frac{1000 - 800}{800} \times 100\% = 25\%$

43. *(c)* Let the cost price of an article be ₹ x.

and selling price of an article = ₹ 96 [given]

$\frac{x - 96}{x} \times 100 = \frac{1}{4}x$

$\Rightarrow \quad x^2 - 400x + 38400 = 0$

$\Rightarrow \quad x^2 - 160x - 240x + 38400 = 0$

$\Rightarrow \quad (x - 160)(x - 240) = 0$

$\therefore \quad x = 160$ or 240

Hence, the cost price of the article is either

₹ 160 or ₹ 240.

44. *(b)* Let CP of 1 L of milk be ₹ x.

$\therefore$ SP of 1 L of milk

$= x \times 120\% = ₹\ 1.2x$

Now, as in ₹ $1.2x$, the quantity of milk sold = 1 L

$\therefore$ In ₹ x, quantity of milk sold

$= \frac{1}{1.2x} \times x = \frac{5}{6}$ L

According to the question,

CP of milk and SP of mixture are same, therefore in mixture, quantity of milk must be $\frac{5}{6}$ L.

Hence, the required percentage

$= \frac{5}{6} \times 100\% = \frac{250}{3}\%$

45. *(c)* Refer to example 12.

46. *(a)* Let the amount of cloth = 100 cm

Then, amount of cloth purchased

= 110 cm

and amount of cloth sold = 90 cm

$\therefore$ Gain % $\frac{110 - 90}{100} \times 100\%$

$= \frac{20}{100} \times 100\% = 20\%$

12

RATIO AND PROPORTION

Regularly (1-2) questions have been asked from this chapter. Generally both the topics i.e. ratio and proportion have same importance but more emphasis is given on proportion. The concepts of this chapter are extremely used in other chapters like-age, average. etc.

RATIO

A ratio is the comparison of two or more quantities of the same type (or kind) by division. i.e. if a and b are two quantities of same kind (or same unit), then the fraction a/b is called the ratio a to b and we write it as $a : b$. Here, a is called antecedent and b is called the consequent.

e.g. If a fruit box contains 8 oranges and 7 lemons then the ratio of oranges to lemons is 8 to 7 or 8:7.

Note If both terms a and b of a ratio are multiplied or divided by the same quantity, then ratio remains unchanged.

i.e. $a:b$ is same as $na : nb \Rightarrow \frac{a}{b} = \frac{na}{nb}$ and $a : b$ is same as $\frac{a}{n} : \frac{b}{n} \Rightarrow \frac{a}{b} = \frac{a/n}{b/n}$.

EXAMPLE 1. If the ratio of 90 cm to 1.5 m is same as 3 : x, then x is

a. 4 **b.** 5 **c.** 2 **d.** 1

Sol. **b.** 1.5 m = 150 cm (units must be same)

Ratio of 90 cm to 1.5 m $= \frac{90}{150} = \frac{90/30}{150/30} = \frac{3}{5}$.

As this ratio is same as 3 : x

$\therefore \quad \frac{3}{x} = \frac{3}{5}$

$\Rightarrow \quad 3x = 5 \times 3$

$\Rightarrow \quad x = 5$

Types of Ratios

1. **Compound Ratio** When two or more ratios are multiplied together, they are said to be compound ratio.
   If $\frac{a}{b}, \frac{c}{d}, \frac{e}{f}$ and $\frac{g}{h}$ are all ratios, then their compound ratio is $aceg : bdfh$.
   e.g. Compound ratio of $\frac{2}{3}, \frac{4}{7}$ and $\frac{1}{3} = \frac{2\times4\times1}{3\times7\times3} = \frac{8}{63}$.
2. **Duplicate Ratio** When a ratio is compounded with itself, the resulting ratio is called the duplicate ratio.
   So, $a^2 : b^2$ is duplicate ratio of $a : b$.

EXAMPLE 2. The duplicate ratio of the ratio $2\sqrt{2} : 3\sqrt{5}$ is

a. 4 : 9 b. 8 : 45 c. 2 : 3 d. 6 : 45

Sol. b. The duplicate ratio of $2\sqrt{2} : 3\sqrt{5}$ is

$$\frac{2\sqrt{2}}{3\sqrt{5}} \times \frac{2\sqrt{2}}{3\sqrt{5}} = \frac{4\sqrt{4}}{9\sqrt{25}} = \frac{8}{45} = 8 : 45$$

3. **Triplicate Ratio** If a ratio is compounded three times with itself, then resulting ratio is called triplicate ratio.
   So, $a^3 : b^3$ is the triplicate ratio of $a : b$.

EXAMPLE 3. If $4x + 3 : 9x + 10$ is the triplicate ratio of 3 : 4. Then, the value of x is

a. 2 b. 6 c. 8 d. 12

Sol. b. Since, $4x + 3 : 9x + 10$ is the triplicate ratio of 3 : 4.

Therefore, $\frac{4x+3}{9x+10} = \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \Rightarrow \frac{4x+3}{9x+10} = \frac{27}{64}$

$\Rightarrow \quad 64(4x + 3) = 27(9x + 10)$

$\Rightarrow \quad 256x + 192 = 243x + 270$

$\Rightarrow \quad 256x - 243x = 270 - 192 \Rightarrow 13x = 78$

$\therefore \quad x = \frac{78}{13} = 6$

Hence, the value of x is 6.

4. **Subduplicate Ratio** If two numbers are in ratio, then the ratio of their square roots is called sub duplicate ratio.
   If $a : b$ is ratio, then the subduplicate ratio is $\sqrt{a} : \sqrt{b}$.
   e.g. Subduplicate ratio of 9 : 16 is $\sqrt{9} : \sqrt{16} = 3 : 4$.
5. **Subtriplicate Ratio** If two numbers are in ratio, then the ratio of their cube roots is called subtriplicate ratio.
   If $a : b$ is ratio, then the subtriplicate ratio is $\sqrt[3]{a} : \sqrt[3]{b}$.
   e.g. Subtriplicate ratio of 64 : 27 is $\sqrt[3]{64} : \sqrt[3]{27} = 4 : 3$
6. **Reciprocal Ratio** If $a : b$ is a ratio, then $\frac{1}{a} : \frac{1}{b}$ is its reciprocal ratio, i.e. $b : a$.
   e.g. Reciprocal ratio of 3 : 7 is $\frac{1}{3} : \frac{1}{7}$ i.e. 7 : 3.

EXAMPLE 4. Compound ratio of the duplicate ratio of 5 : 6, the reciprocal ratio of 25:42 and the subduplicate ratio of 36:49 is

a. 1 : 2 b. 2 : 3 c. 1 : 1 d. 4 : 9

Sol. c. The duplicate ratio of 5 : 6 is $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

The reciprocal ratio of 25 : 42 is $\frac{1}{25} : \frac{1}{42}$ i.e. $\frac{42}{25}$

and subduplicate ratio of $\frac{36}{49}$ is $\frac{\sqrt{36}}{\sqrt{49}} = \frac{6}{7}$

$\therefore$ Compound ratio is $\frac{25}{36} \times \frac{42}{25} \times \frac{6}{7} = \frac{1}{1} = 1 : 1$

IMPORTANT POINTS

- Usually, the ratio is expressed in its lowest terms.
- Ratio exists only between quantities of the same kind.
- Ratio is a fraction, so it has no units.
- Ratio is taken only between positive quantities.

PROPORTION

The equality of two ratios is called proportion.

': :' is the sign of proportion.

So, if $a : b = c : d$, we write it as

$a : b :: c : d$ and say that a, b, c and d are in proportion.

The quantities a, b, c and d are called the first, second, third and fourth terms of proportion respectively. First and fourth terms are called **extreme terms** and second and third terms are called **means** or **middle terms.**

If quantities a, b, c and d are in proportion, then

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

i.e. product of extreme terms = product of middle terms.

This is also called cross-product rule.

Types of Proportion

1. **Continued Proportion** The non-zero quantities of same kind, $a, b, c, d, e, f, \ldots$ are said to be continued proportion, if $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{e}{f} = \ldots$

EXAMPLE 5. If a, b, c, d and e are in continued proportion, then a/e is equal to

a. a^3/b^3 b. a^4/b^4 c. b^3/a^3 d. b^4/a^4

Sol. b. Since, a, b, c, d and e are in continued proportion.

$$\because \quad \frac{a}{b}=\frac{b}{c}=\frac{c}{d}=\frac{d}{e} \Rightarrow \frac{e}{d}=\frac{d}{c}=\frac{c}{b}=\frac{b}{a} \Rightarrow c=\frac{b^2}{a}$$

$$d=\frac{c^2}{b}=\frac{b^4}{a^2}\cdot\frac{1}{b}=\frac{b^3}{a^2}\ , \quad e=\frac{d^2}{c}=\frac{b^6}{a^4}\cdot\frac{a}{b^2}=\frac{b^4}{a^3}$$

$$\therefore \quad \frac{a}{e}=\frac{a}{(b^4/a^3)}=\frac{a^4}{b^4}$$

2. **Mean Proportional** If a, b and c are in continued proportion, then b is called the mean proportional of a and c.

i.e. $$\frac{a}{b}=\frac{b}{c} \Rightarrow b^2=ac \Rightarrow b=\sqrt{ac}$$

EXAMPLE 6. What is the mean proportional between $(15+\sqrt{200})$ and $(27-\sqrt{648})$? 2012 I

a. 4 b. $14\sqrt{7}$ c. $3\sqrt{5}$ d. $5\sqrt{3}$

Sol. c. Let mean proportional between $(15+\sqrt{200})$ and $(27-\sqrt{648})$ is x

Then, $x=\sqrt{(15+\sqrt{200})(27-\sqrt{648})}$

$$=\sqrt{(15+10\sqrt{2})(27-18\sqrt{2})}$$

$$=\sqrt{5(3+2\sqrt{2}).9(3-2\sqrt{2})}=\sqrt{45[(3)^2-(2\sqrt{2})^2]}$$

$$=\sqrt{45(9-8)}=\sqrt{45}=3\sqrt{5}$$

3. **Third Proportional** If $a:b::b:c$, then c is called the 3rd proportional to a and b. Now, c will be calculated as below:

$$a:b::b:c \Rightarrow a:b=b:c \Rightarrow c=\frac{b^2}{a}$$

EXAMPLE 7. The two numbers such that their mean proportional is 24 and the third proportional is 1536, are

a. 3 and 98 b. 6 and 96 c. 4 and 92 d. None of these

Sol. b. Let a and b be the required numbers, then mean proportional between a and b is 24.

$$\therefore \quad a:24=24:b \Rightarrow ab=24^2,\ ab=576 \quad \ldots\text{(i)}$$

and third proportional to a and b is 1536.

$$\Rightarrow \quad a:b=b:1536$$

$$\Rightarrow \quad b^2=1536a \Rightarrow a=\frac{b^2}{1536} \quad \ldots\text{(ii)}$$

On putting the value of a in Eq. (i), we get

$$\Rightarrow \quad \frac{b^2}{1536}\cdot b=576 \Rightarrow b^3=576\times1536$$

$$\Rightarrow \quad b^3=(96\times6)\times(96\times16)=96^3 \Rightarrow b=96$$

$$\therefore \quad a=\frac{576}{b}=\frac{576}{96}=6 \quad \text{[from Eq. (i)]}$$

Hence, the required numbers are 6 and 96.

4. **Fourth Proportional** If four numbers or quantities a, b, c and d are in proportion, then d is known as fourth proportional.

i.e. $a:b::c:d$

EXAMPLE 8. The fourth proportional to p^2-pq+q^2, p^3+q^3, $p-q$ is

a. $p+q$ b. $p-q$ c. p^2+q^2 d. p^2-q^2

Sol. d. Let the fourth proportional be x.

Then, $p^2-pq+q^2 : p^3+q^3 = p-q : x$

$$\Rightarrow \quad \frac{p^2-pq+q^2}{p^3+q^3}=\frac{p-q}{x} \Rightarrow x=\frac{(p-q)(p^3+q^3)}{p^2-pq+q^2}$$

$$\therefore \quad x=\frac{(p-q)(p^3+q^3)}{p^2-pq+q^2}=(p-q)(p+q)=(p^2-q^2)$$

Properties of Proportion

If four non zero quantities a, b, c and d are in proportion, then properties of proportion are as follows.

1. **Invertendo** If $a:b::c:d$, then $b:a::d:c$

i.e. $$\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{b}{a}=\frac{d}{c}$$

2. **Alternendo** If $a:b::c:d$, then $a:c::b:d$

i.e. $$\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a}{c}=\frac{b}{d}$$

3. **Componendo** (Adding the Denominator)

If $a:b::c:d$, then $(a+b):b::(c+d):d$

i.e. $$\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a+b}{b}=\frac{c+d}{d}$$

4. **Dividendo** (Subtracting the Denominator)

If $a:b::c:d$, then $a:(a-b)::c:(c-d)$

i.e. $$\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a}{a-b}=\frac{c}{c-d}$$

5. **Componendo and Dividendo**

If $a:b::c:d$, then $(a+b):(a-b)::(c+d):(c-d)$

i.e. $$\frac{a}{b}=\frac{c}{d} \Rightarrow \frac{a+b}{a-b}=\frac{c+d}{c-d}$$

EXAMPLE 9. If $(a-b):(a+b)=1:5$, then what is $(a^2-b^2):(a^2+b^2)$ equal to?

a. $\frac{13}{5}$ b. $\frac{4}{13}$ c. $\frac{5}{13}$ d. None of these

Sol. c. Given, $\frac{a-b}{a+b}=\frac{1}{5} \Rightarrow \frac{a+b}{a-b}=\frac{5}{1} \Rightarrow \frac{a+b+a-b}{a+b-a+b}=\frac{5+1}{5-1}$

(using componendo and dividendo rule)

$$\Rightarrow \quad \frac{2a}{2b}=\frac{6}{4}=\frac{3}{2} \Rightarrow \frac{a}{b}=\frac{3}{2} \Rightarrow \frac{a^2}{b^2}=\frac{9}{4}$$

Again, using componendo and dividendo rule

$$\Rightarrow \quad \frac{a^2+b^2}{a^2-b^2}=\frac{9+4}{9-4}=\frac{13}{5}$$

$$\Rightarrow \quad \frac{a^2-b^2}{a^2+b^2}=\frac{5}{13} \quad \therefore (a^2-b^2):(a^2+b^2)=5:13$$

MIXTURE OR ALLIGATION

If two or more quantities are mixed together in a certain ratio, then the product is called mixture.

Mean Price

Cost price of a unit quantity of the mixture is called the mean price.

Rule of Mixture or Alligation

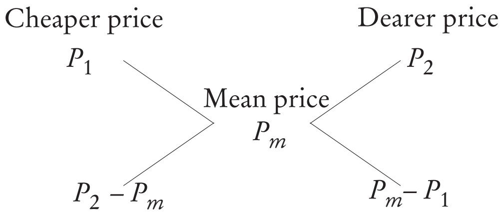
It is the rule to find the ratio in which two or more ingredients are mixed or to be mixed.

If two quantities are mixed in the ratio $n_1 : n_2$, then

$$\frac{n_1}{n_2} = \frac{P_2 - P_m}{P_m - P_1}$$

where P_1, P_2 and P_m are cheaper price, dearer price and mean price respectively.

It can also be expressed as



$$\therefore \quad \frac{\text{Cheaper quantity}}{\text{Dearer quantity}} = \frac{n_1}{n_2} = \frac{P_2 - P_m}{P_m - P_1}$$

Note $P_1 < P_m < P_2$

EXAMPLE 10. In what proportion, must wheat at ₹ 6.20 per kg be mixed with wheat at ₹ 7.20 per kg, so that the mixture be worth ₹ 6.50 per kg?

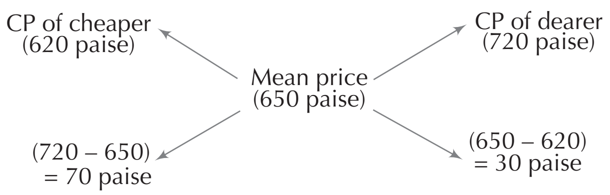
a. 6 : 4 b. 7 : 4 c. 7 : 3 d. 5 : 2

Sol. c. Given, cost price of cheaper quantity = ₹ 6.20 per kg

Cost price of dearer quantity = ₹ 7.20 per kg

Mean price = ₹ 6.50 per kg

According to the rule of alligation,



$\therefore$ Required ratio $= 70 : 30 = 7 : 3$

VARIATIONS

When a change in a certain quantity leads to a certain change in another quantity then it is called variations.

Direct Variation

If x varies directly with y, then as x increses (decrease) y also increases (decreases)

i.e. $x \propto y$

$\Rightarrow x = ky$. where, k is the constant of variation.

Inverse Variation

If x varies inversely with y, then an increase is the value of x causes a decreases in the value of y.

i.e. $x \propto \dfrac{1}{y}$

$\Rightarrow \quad x = \dfrac{k'}{y}$

where, k' is the constant of variation.

EXAMPLE 11. If a quantity y varies as the sum of three quantities of which the first varies as x, the second varies as $-x + x^2$, the third varies as $x^3 - x^2$, then what is y equal to?

a. kx^3, where k is a constant
b. $kx + lx^2 + mx^3$, where k, l and m are constants
c. kx^2, where k is a constant
d. kx, where k is a constant

Sol. b. Since, first term $\propto x$

$\Rightarrow$ First term $= c_1 x$,

$\Rightarrow$ Second term $\propto (-x + x^2)$

$\Rightarrow$ Second term $= c_2(-x + x^2)$

and third term $\propto (x^3 - x^2) \Rightarrow$ Third term $= c_3 (x^3 - x^2)$

Also, $y \propto [c_1 x + c_2 (-x + x^2) + c_3 (x^3 - x^2)]$ (given)

$\Rightarrow \quad y = c_4[(c_1 - c_2)x + (c_2 - c_3)x^2 + c_3 x^3]$

$= c_4 (c_1 - c_2) x + (c_2 - c_3) c_4 x^2 + c_3 c_4 x^3$

$= kx + l x^2 + mx^3$

where, $k = c_4 (c_1 - c_2)$,

$l = (c_2 - c_3) c_4$,

and $m = c_3 c_4$

PRACTICE EXERCISE

1. The compounded ratio of $2a:6b, 7a:49b$ and $3a:12b$ is
(a) $a:84b$ (b) $a^3:84b^3$ (c) $a:24b$ (d) $2a:3b$

2. The subduplicate ratio of $16x^4:625y^6$ is
(a) $4x^2:25y^3$ (b) $4x^2:25y^2$ (c) $4x:25y$ (d) $8x^2:125y^3$

3. If $x+5:3x+4$ is the duplicate ratio of $5:8$, then the value of x is
(a) 16 (b) 18 (c) 20 (d) 22

4. If $x:6::5:3$, then the value of x is
(a) 8 (b) 10 (c) 12 (d) 13

5. The third proportional to 9 and 12 is
(a) 12 (b) 14 (c) 16 (d) 18

6. The mean proportional between $(2+\sqrt{3})$ and $(8-\sqrt{48})$ is
(a) 2 (b) 3 (c) 4 (d) 5

7. The fourth proportional to 7, 11, 14 is
(a) 16 (b) 18 (c) 20 (d) 22

8. In a ratio which is equal to 7 : 8, if the antecedent is 35, then the consequent is
(a) 30 (b) 32 (c) 36 (d) 40

9. If $x:y=1:3$, $y:z=5:k$, $z:t=2:5$ and $t:x=3:4$, then what is the value of k?
(a) 1/2 (b) 1/3 (c) 2 (d) 3

10. If the ratio of x to y is 25 times the ratio of y to x, then what is the ratio of x to y?
(a) 1 : 5 (b) 5 : 1 (c) 25 : 1 (d) 1 : 25

11. If $p\%$ of ₹ x is equal to t times $q\%$ of ₹ y, then what is the ratio of x to y?
(a) $pt:q$ (b) $p:qt$ (c) $qt:p$ (d) $q:pt$

12. If $a:b=2:3$ and $x:y=3:4$, then the value of $\dfrac{4ay-3bx}{5ax-2by}$ is
(a) $\frac{5}{3}$ (b) $\frac{5}{6}$ (c) $\frac{4}{5}$ (d) $\frac{5}{4}$

13. If $\dfrac{x^3+3x}{3x^2+1}=\dfrac{341}{91}$, then the value of x is
(a) 16 (b) 14 (c) 13 (d) 11

14. What is the number which has to be added to each term of the ratio 49 : 68, so that it becomes 3 : 4?
(a) 3 (b) 5 (c) 8 (d) 9

15. Divide 1870 into three parts in such a way that half of the first part, one-third of the second part and one-sixth of the third part are equal. Then, the third part is
(a) 340 (b) 510 (c) 1020 (d) 1320

16. Out of the ratios 7 : 20, 13 : 25, 17 : 30 and 11 : 15, the smallest one is
(a) 11 : 15 (b) 7 : 20 (c) 11 : 16 (d) 17 : 30

17. What is the ratio of the numbers of 0.5 of a number is equal to 0.07 of another?
(a) 50 : 7 (b) 5 : 7 (c) 1 : 14 (d) 7 : 50

18. What is the ratio whose terms differ by 40 and the measure of which is $\frac{2}{7}$?
(a) 14 : 56 (b) 15 : 56 (c) 16 : 56 (d) 16 : 72

19. If $P:Q=\frac{3}{5}:\frac{5}{7}$ and $Q:R=\frac{3}{4}:\frac{2}{5}$, then what is $P:Q:R$ equal to?
(a) $\frac{3}{5}:\frac{5}{7}:\frac{2}{5}$ (b) $\frac{9}{20}:\frac{15}{28}:\frac{2}{7}$ (c) $\frac{3}{5}:\frac{3}{4}:\frac{2}{5}$ (d) $\frac{3}{5}:\frac{5}{7}:\frac{3}{4}$

20. Two numbers are in the ratio 3 : 5. If 9 is subtracted from each number, then they are in the ratio of 12 : 23. What is the second number?
(a) 44 (b) 55 (c) 66 (d) 77

21. The mean proportional between two numbers is 28 and their third proportional to them is 224. The two numbers are
(a) 7 and 112 (b) 14 and 56 (c) 28 and 28 (d) 21 and 36

22. ₹ 770 have been divided among A, B, C in such a way that A receives $\frac{2}{9}$th of what B and C together receive. What is A's share?
(a) ₹ 154 (b) ₹ 140 (c) ₹ 250 (d) ₹ 254

23. x varies directly as y and inversely as square of z. When $y=4$ and $z=14$, then $x=10$. If $y=16$ and $z=7$, what is x?
(a) 180 (b) 160 (c) 154 (d) 140

24. If $x=\dfrac{\sqrt{m+3n}+\sqrt{m-3n}}{\sqrt{m+3n}-\sqrt{m-3n}}$, then
(a) $3nx^2-2mx+3n=0$ (b) $2nx^2-2mx+3n=0$
(c) $3nx^2-2mx-3n=0$ (d) $3nx^2+3mx+3n=0$

25. If $p+r=2q$ and $\frac{1}{q}+\frac{1}{s}=\frac{2}{r}$, then
(a) $p:s=r:q$ (b) $p^2:r^2=q:s$
(c) $p:q=r:s$ (d) $p^2=2r+s$

26. If q is the mean proportional between p and r, then $\dfrac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}}$ is equal to

(a) p^2q^3 (b) q^3 (c) q^4 (d) $p^2r^2q^4$

27. The monthly incomes of A and B are in the ratio 4 : 3. Each of them saves ₹ 600. If the ratio of the expenditure is 3 : 2, then what is the monthly income of A?

(a) ₹ 2400 (b) ₹ 1800 (c) ₹ 2000 (d) ₹ 3600

28. Three numbers are in the ratio 3 : 2 : 5 and the sum of their squares is 1862. What are the three numbers?

(a) 18, 12, 30 (b) 24, 16, 40
(c) 15, 10, 25 (d) 21, 14, 35

29. The speeds of three cars are in the ratio 4 : 3 : 2. What is the ratio between the times taken by the cars to cover the same distance?

(a) 2 : 3 : 4 (b) 3 : 4 : 6
(c) 1 : 2 : 3 (d) 4 : 3 : 2

30. The ratio between the ages of A and B is 2 : 5. After 8 yr, their ages will be in the ratio 1 : 2. What is the difference between their present ages?

(a) 20 yr (b) 22 yr (c) 24 yr (d) 25 yr

31. Let y is equal to the sum of two quantities of which one varies directly as x and the other inversely as x. If $y = 6$ when $x = 4$ and $y = 10/3$, when $x = 3$, then what is the relation between x and y?

(a) $y = x + (4/x)$ (b) $y = -2x + (4/x)$
(c) $y = 2x + (8/x)$ (d) $y = 2x - (8/x)$

32. If ₹ 8400 is divided among A, B and C in the ratio $\frac{1}{5} : \frac{1}{6} : \frac{1}{10}$, what is the share of A?

(a) ₹ 1800 (b) ₹ 3000 (c) ₹ 3600 (d) ₹ 4000

33. A certain amount of money has to be divided between two persons P and Q in the ratio 3 : 5. But it was divided in the ratio of 2 : 3 and there by Q loses ₹ 10. What was the amount?

(a) ₹ 250 (b) ₹ 300 (c) ₹ 350 (d) ₹ 400

34. 20 L of a mixture contains milk and water in the ratio 4 : 3. If 6 L of this mixture are replaced by 6 L of milk, the ratio of milk to water in the new mixture will become

(a) 7 : 3 (b) 8 : 3 (c) 9 : 7 (d) 4 : 6

35. Two vessels are full with milk and water in the ratios 1 : 3 and 3 : 5, respectively. If both are mixed in the ratio 3 : 2, what is the ratio of milk and water in the new mixture?

(a) 4 : 15 (b) 3 : 7
(c) 6 : 7 (d) None of these

36. A mixture contains milk and water in the ratio 5 : 1. On adding 5 L of water, the ratio of milk and water becomes 5 : 2. What is the quantity of milk in the original mixture?

(a) 5 L (b) 25 L
(c) 27.5 L (d) 32.5 L

37. If x varies as the mth power of y, y varies as the nth power of z and x varies as the p th power of z, then which one of the following is correct?

(a) $p = m + n$ (b) $p = m - n$
(c) $p = mn$ (d) None of these

38. The wages of labourers in a factory has increased in the ratio 22 : 25 and their number decreased in the ratio 3 : 2. What was the original wage bill of the factory, if the present bill is ₹ 5000?

(a) ₹ 4000 (b) ₹ 6000
(c) ₹ 8000 (d) None of these

39. The ratio of A to B is $x : 8$ and the ratio of B to C is 12 : z. If the ratio of A to C is 2 : 1, then what is the ratio of $x : z$?

(a) 2 : 3 (b) 3 : 2 (c) 4 : 3 (d) 3 : 4

40. A bag contains ₹ 112 in the form of ₹ 1, 50 paise and 10 paise coins in the ratio 3 : 8 : 10. What is the number of 50 paise coins?

(a) 112 (b) 108 (c) 96 (d) 84

41. In a class, the number of boys is more than the number of girls by 12% of the total students. What is the ratio of number of boys to that of girls?

(a) 11 : 14 (b) 14 : 11
(c) 28 : 25 (d) 25 : 28

42. A sum of ₹ 53 is divided among A, B, C in such a way that A gets ₹ 7 more than what B gets and B gets ₹ 8 more than what C gets. The ratio of their shares is

(a) 25 : 18 : 10 (b) 6 : 7 : 8
(c) 12 : 14 : 9 (d) 15 : 8 : 30

43. A person P started a business with a capital of ₹ 2525 and another person Q joined P after some months with a capital of ₹ 1200. Out of the total annual profit of ₹ 1644, P's share was ₹ 1212. When did Q join as partner?

(a) After 2 months (b) After 3 months
(c) After 4 months (d) After 5 months

44. Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75%, respectively What will be the ratio of increased seats?

(a) 2 : 3 : 4 (b) 6 : 7: 8
(c) 6 : 8: 9 (d) 5 : 7 : 12

45. The ratio between the number of passengers travelling by I and II class between the two railway stations is 1:50, whereas the ratio of I and II class fares between the same station is 3:1, If on a particular day, ₹ 1325 were collected from the passengers travelling between these stations, then what was the amount collected from the II class passengers?

(a) ₹ 750 (b) ₹ 1000
(c) ₹ 850 (d) ₹ 1250

46. If x varies inversely as the square of y in such a way that, if $x=1$, then $y=6$.

I. If $y=3$, then $x=4$ II. If $y=6$, then $x=1$

Which of the following options is correct?

(a) Only II (b) Both I and II
(c) Only I (d) Neither I nor II

47. If $a:b=c:d=e:f=1:2$, then

I. $\dfrac{3a+5c+7e}{3b+5d+7f}=\dfrac{1}{3}$ II. $\dfrac{a^2+c^2+e^2}{b^2+d^2+f^2}=\dfrac{1}{2}$

Which of the following option is correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

48. A cat takes 5 leaps for every 4 leaps of a dog but 3 leaps of the dog are equal to 4 leaps of the cat. Now, the

I. Ratio of the speeds of the cat to that of the dog is 15 : 16.

II. Ratio of the distance of the cat to that of the dog is 15:16, covered in 30 min.

Which of the following option is correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

49. Determine the ratio of the number of people having characteristic X to the number of people having characteristic Y in a population of 100 subjects from the following table:

Having X and Y	10
Having X but not Y	30
Having Y but not X	20
Having neither X nor Y	40

(a) 4:3 (b) 3:2 (c) 1:2 (d) 2:3

50. Fresh grapes contain 90 percent water by weight while dried grapes contain 20 percent water by weight. What is the weight of dry grapes contain 20 percent water by weight. What is the weight of dry grapes available from 20 kg of fresh grapes?

(a) 2 kg (b) 2.4 kg
(c) 2.5 kg (d) None of these

PREVIOUS YEARS' QUESTIONS

51. If $x:y=7:5$, then what is the value of $(5x-2y):(3x+2y)$? **2012 I**

(a) 5/4 (b) 6/5 (c) 25/31 (d) 31/42

52. Two numbers are in the ratio 2 : 3. If 9 is added to each number, they will be in the ratio 3 : 4. What is the product of the two numbers? **2012 I**

(a) 360 (b) 480 (c) 486 (d) 512

53. A milkman bought 15 kg of milk and mixed 3 kg of water in it. If the price per kg of the mixture becomes ₹ 22, what is cost price of the milk per kg? **2012 II**

(a) ₹ 28.00 (b) ₹ 26.40 (c) ₹ 24.00 (d) ₹ 22.60

54. Sex ratio is defined as the number of females per 1000 males. In a place, the total inhabitants are 1935000, out of which 935000 are females. What is the sex ratio for the place? **2012 II**

(a) 935 (b) 1000
(c) 1935 (d) 9350

55. In a certain school, the ratio of boys to girls is 7 : 5. If there are 2400 students in the school, then how many girls are there? **2013 I**

(a) 500 (b) 700 (c) 800 (d) 1000

56. If $A:B=2:3, B:C=5:7$ and $C:D=3:10$, then what is $A:D$ equal to? **2014 I**

(a) 1 : 7 (b) 2 : 7 (c) 1 : 5 (d) 5 : 1

57. $(x+y):(x-y)=3:5$ and $xy=$ positive imply that **2014 II**

(a) x and y are both positive
(b) x and y are both negative
(c) one of them is positive and one of them is negative
(d) no real solution for x and y exists

58. The ratio of ages of A and B is $2:5$ and the ratio of ages of B and C is $3:4$. What is the ratio of ages of A, B and C? **2014 II**

(a) 6 : 15 : 20 (b) 8 : 5 : 3
(c) 6 : 5 : 4 (d) 2 : 15 : 4

59. The height of a tree varies as the square root of its age (between 5 to 17 yr). When the age of the tree is 9 yr, its height is 4 ft. What will be the height of the tree at the age of 16 yr? **2014 II**

(a) 5 ft 4 inch (b) 5 ft 5 inch
(c) 4 ft 4 inch (d) 4 ft 5 inch

60. 16 L of a mixture contains milk and water in the ratio $5:3$. If 4 L of milk is added to this mixture, the ratio of milk to water in the new mixture would be **2015 I**

(a) 2 : 1 (b) 7 : 3 (c) 4 : 3 (d) 8 : 3

61. If $a:b=3:5$ and $b:c=7:8$, then $2a:3b:7c$ is equal to 2015 II
(a) 42 : 105 : 320 (b) 15 : 21 : 40
(c) 6 : 15 : 40 (d) 30 : 21 : 350

62. The speeds of three buses are in the ratio 2 : 3 : 4. The time taken by these buses to travel the same distance will be in the ratio. 2015 II
(a) 2 : 3 : 4 (b) 4 : 3 : 2
(c) 4 : 3 : 6 (d) 6 : 4 : 3

63. In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7 : 3. The quantity of water to be added to the mixture to make the ratio of milk and water 1 : 2 is 2015 II
(a) 30 (b) 32
(c) 33 (d) 35

64. The annual incomes of two persons are in the ratio 9 : 7 and their expenses are in the ratio 4 : 3. If each of them saves ₹ 2000 per year, what is the difference in their annual incomes? 2016 I
(a) ₹ 4000 (b) ₹ 4500 (c) ₹ 5000 (d) ₹ 5500

65. If $\frac{a}{b}=\frac{b}{c}=\frac{c}{d}$, then which of the following is/are correct?
I. $\frac{b^3+c^3+d^3}{a^3+b^3+c^3}=\frac{d}{a}$ II. $\frac{a^2+b^2+c^2}{b^2+c^2+d^2}=\frac{a}{d}$
Select the correct answer using the code given below. 2016 I
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWERS

1	b	2	a	3	c	4	b	5	c	6	a	7	d	8	d	9	a	10	b
11	c	12	b	13	d	14	c	15	c	16	b	17	d	18	c	19	b	20	b
21	b	22	b	23	b	24	a	25	c	26	c	27	a	28	d	29	b	30	c
31	d	32	c	33	d	34	a	35	b	36	b	37	c	38	d	39	c	40	a
41	b	42	a	43	b	44	a	45	d	46	b	47	d	48	c	49	a	50	c
51	c	52	c	53	b	54	a	55	d	56	a	57	d	58	a	59	a	60	b
61	c	62	d	63	c	64	a	65	a										

HINTS AND SOLUTIONS

1. *(b)* The required compounded ratio is
$$\frac{2a}{6b}\times\frac{7a}{49b}\times\frac{3a}{12b}=\frac{a^3}{84b^3}=a^3:84b^3$$

2. *(a)* The subduplicate ratio of $16x^4:625y^6$ is
$$\frac{\sqrt{16x^4}}{\sqrt{625y^6}}=\frac{4x^2}{25y^3}=4x^2:25y^3$$

3. *(c)* Duplicate ratio of $5:8=\frac{5^2}{8^2}$
$$\Rightarrow \frac{5^2}{8^2}=\frac{x+5}{3x+4}\Rightarrow\frac{25}{64}=\frac{x+5}{3x+4}$$
$$\Rightarrow 25(3x+4)=64(x+5)$$
$$\Rightarrow 75x+100=64x+320$$
$$\Rightarrow 75x-64x=320-100$$
$$\Rightarrow 11x=220$$
$$\therefore x=\frac{220}{11}=20$$

4. *(b)* Given, $x:6::5:3$
$$\therefore \frac{x}{6}=\frac{5}{3}\Rightarrow 3x=30\Rightarrow x=\frac{30}{3}=10$$

5. *(c)* Let the third proportional to 9 and 12 be x.
$$\therefore 9:12::12:x$$
$$\Rightarrow \frac{9}{12}=\frac{12}{x}\Rightarrow 9x=144$$
$$\therefore x=\frac{144}{9}=16$$

6. *(a)* Mean proportional between $(2+\sqrt{3})$ and $(8-\sqrt{48})$
$$=\sqrt{(2+\sqrt{3})(8-\sqrt{48})}$$
$$=\sqrt{(2+\sqrt{3})\,4(2-\sqrt{3})}$$
$$=2\sqrt{(2+\sqrt{3})(2-\sqrt{3})}$$
$$=2\sqrt{(4-3)}=2\times1=2$$
$$[\because (a-b)(a+b)=a^2-b^2]$$

7. *(d)* Let fourth proportional be x, then
$$7:11::14:x$$
$$\Rightarrow \frac{7}{11}=\frac{14}{x}\Rightarrow 7x=11\times14$$
$$\Rightarrow x=2\times11$$
$$\therefore x=22$$

8. *(d)* Let the consequent be x.
$$\therefore 35:x=7:8$$
Then, $\frac{35}{x}=\frac{7}{8}$ [by given condition]
$$\Rightarrow 7x=35\times8 \;\therefore\; x=\frac{35\times8}{7}=40$$

9. *(a)* Given, $x:y=1:3$, $y:z=5:k$, $z:t=2:5$ and $t:x=3:4$
$$\frac{x}{y}\times\frac{y}{z}\times\frac{z}{t}\times\frac{t}{x}=1$$
$$\Rightarrow \frac{1}{3}\times\frac{5}{k}\times\frac{2}{5}\times\frac{3}{4}=1\Rightarrow k=\frac{1}{2}$$

10. *(b)* Given, $\frac{x}{y}=25\left(\frac{y}{x}\right)$
$$\Rightarrow \frac{x^2}{y^2}=\frac{25}{1}\Rightarrow\frac{x}{y}=\frac{5}{1}\text{ or },5:1$$

11. *(c)* According to the question,
$$p\%\text{ of }x=t\,(q\%\text{ of }y)$$
$$\Rightarrow \frac{xp}{100}=\frac{yq}{100}\times t$$
$$\therefore x:y=qt:p$$

12. *(b)* $\dfrac{4ay-3bx}{5ax-2by}$ [dividing numerator and denominator by by]

$$=\dfrac{\frac{4ay}{by}-\frac{3bx}{by}}{\frac{5ax}{by}-\frac{2by}{by}}=\dfrac{4\left(\frac{a}{b}\right)-3\left(\frac{x}{y}\right)}{5\left(\frac{a}{b}\right)\left(\frac{x}{y}\right)-2}$$

$$=\dfrac{4\left(\frac{2}{3}\right)-3\left(\frac{3}{4}\right)}{5\left(\frac{2}{3}\right)\left(\frac{3}{4}\right)-2}=\dfrac{\frac{8}{3}-\frac{9}{4}}{\frac{5}{2}-2}$$

$$=\dfrac{\frac{32-27}{12}}{\frac{5-4}{2}}=\dfrac{\frac{5}{12}}{\frac{1}{2}}=\frac{5}{12}\times\frac{2}{1}=\frac{5}{6}$$

13. *(d)* Given, $\dfrac{x^3+3x}{3x^2+1}=\dfrac{341}{91}$

By componendo and dividendo rule,

$$\frac{x^3+3x+3x^2+1}{x^3+3x-3x^2-1}=\frac{341+91}{341-91}=\frac{432}{250}$$

$$\Rightarrow\quad \frac{(x+1)^3}{(x-1)^3}=\frac{216}{125}$$

$[\because (a+b)^3=a^3+b^3+3ab(a+b)]$

On cube roots both sides, we get

$$\Rightarrow\quad \frac{x+1}{x-1}=\frac{6}{5}\Rightarrow 5(x+1)=6(x-1)$$

$\therefore\quad x=11$

14. *(c)* Let the number be x

$\therefore\quad \dfrac{49+x}{68+x}=\dfrac{3}{4}$

$\Rightarrow 196+4x=204+3x$

$\therefore\quad x=8$

15. *(c)* Let the three parts be A, B and C. By given condition,

$\frac{1}{2}A=\frac{1}{3}B=\frac{1}{6}C=x$ [say]

$\therefore\quad \frac{1}{2}A=x\Rightarrow A=2x$

$\frac{1}{3}B=x\Rightarrow B=3x$

$\frac{1}{6}C=x\Rightarrow C=6x$

As, $A+B+C=1870$ [by given condition]

$\therefore\quad 2x+3x+6x=1870$

$\Rightarrow\quad 11x=1870$

$\therefore\quad x=\dfrac{1870}{11}=170$

$\therefore$ Third part i.e. $C=6\times170=1020$

16. *(b)* LCM of 20, 25, 30, 15 is 300.

So, $\dfrac{7}{20}=\dfrac{7}{20}\times\dfrac{15}{15}=\dfrac{105}{300}$

$\Rightarrow\quad \dfrac{13}{25}=\dfrac{13}{25}\times\dfrac{12}{12}=\dfrac{156}{300}$

$\Rightarrow\quad \dfrac{17}{30}=\dfrac{17}{30}\times\dfrac{10}{10}=\dfrac{170}{300}$

and $\dfrac{11}{15}=\dfrac{11}{15}\times\dfrac{20}{20}=\dfrac{220}{300}$

As, $\dfrac{105}{300}$ is the smallest, so $\dfrac{7}{20}$ or, 7 : 20 is the smallest ratio.

17. *(d)* Let the numbers be A and B.

$\therefore\quad 0.5A=0.07B$ [by given condition]

$\Rightarrow\quad \dfrac{A}{B}=\dfrac{0.07}{0.5}=\dfrac{7}{50}$ i.e. $7:50$

18. *(c)* Let the ratio be x and $(x+40)$.

Then, by given condition, $\dfrac{x}{x+40}=\dfrac{2}{7}$

$\Rightarrow\quad 7x=2x+80\Rightarrow\quad x=16$

So, the required ratio is $16:56$.

19. *(b)* Given, $P:Q=\dfrac{3}{5}:\dfrac{5}{7}$ …(i)

and $\quad Q:R=\dfrac{3}{4}:\dfrac{2}{5}$ …(ii)

From Eq. (i), $\quad P:Q=\dfrac{3}{5}\times\dfrac{3}{4}:\dfrac{5}{7}\times\dfrac{3}{4}$

$=\dfrac{9}{20}:\dfrac{15}{28}$ …(iii)

From Eq. (ii), $\quad Q:R=\dfrac{3}{4}\times\dfrac{5}{7}:\dfrac{2}{5}\times\dfrac{5}{7}$

$=\dfrac{15}{28}:\dfrac{2}{7}$ …(iv)

From Eqs. (iii) and (iv),

$P:Q:R=\dfrac{9}{20}:\dfrac{15}{28}:\dfrac{2}{7}$

20. *(b)* Let two numbers be $3x$ and $5x$, respectively.

According to the question,

$(3x-9):(5x-9)::12:23$

$\Rightarrow\quad \dfrac{3x-9}{5x-9}=\dfrac{12}{23}$

$\Rightarrow 69x-207=60x-108$

$69x-60x=207-108\ \Rightarrow 9x=99$

$\therefore\quad x=11$

$\therefore$ Second number $=5x=5\times11=55$

21. *(b)* Let the numbers are a and b and its mean proportional is x

$\therefore\quad a:x::x:b$

$a:28::28:b\ (\because x=28)$

$\Rightarrow\quad 28^2=ab$ …(i)

Again, $a:b::b:224$

$\Rightarrow b^2=a\times224\Rightarrow a=\dfrac{b^2}{224}$ …(ii)

From Eq. (i) say,

$28^2=\dfrac{b^2}{224}\times b$

$\Rightarrow b^3=28\times28\times224$

$\Rightarrow\quad b=56$

Again from Eq. (i) say,

$a=\dfrac{28\times28}{56}=14$

22. *(b)* As, $A:(B+C)=2:9$ [given]

Sum of ratios $=2+9=11$

So, A's part $=\ 770\times\dfrac{2}{11}=$ ₹ 140

23. *(b)* Given, $x\propto y$ and $x\propto\dfrac{1}{z^2}$

Now, $\ x\propto\dfrac{y}{z^2}\Rightarrow x=\dfrac{ky}{z^2}$

$\because\quad x=10$, $y=4$ and $z=14$

$\therefore\ 10=\dfrac{k\cdot4}{196}\Rightarrow k=\dfrac{1960}{4}=490$

Now, $\ z=7$ and $y=16$, then

$x=\dfrac{490\times16}{7\times7}=160$

24. *(a)* $\dfrac{x}{1}=\dfrac{\sqrt{m+3n}+\sqrt{m-3n}}{\sqrt{m+3n}-\sqrt{m-3n}}$

By componendo and dividendo rule,

$$\frac{x+1}{x-1}=\frac{\sqrt{m+3n}+\sqrt{m-3n}+\sqrt{m+3n}-\sqrt{m-3n}}{\sqrt{m+3n}+\sqrt{m-3n}-\sqrt{m+3n}+\sqrt{m-3n}}$$

$$\Rightarrow\quad \frac{x+1}{x-1}=\frac{\sqrt{m+3n}}{\sqrt{m-3n}}$$

On squaring both sides, we get

$$\frac{(x+1)^2}{(x-1)^2}=\frac{m+3n}{m-3n}$$

Again applying componendo and dividendo rule,

$$\frac{(x+1)^2+(x-1)^2}{(x+1)^2-(x-1)^2}=\frac{m+3n+m-3n}{m+3n-m+3n}$$

$$\Rightarrow\quad \frac{2(x^2+1)}{2(2x)}=\frac{2m}{2(3n)}$$

$$\Rightarrow\quad \frac{x^2+1}{2x}=\frac{m}{3n}\Rightarrow 3n(x^2+1)=2mx$$

$\therefore\quad 3nx^2-2mx+3n=0$

25. *(c)* Given, $p+r=2q\Rightarrow\ 2=\dfrac{p+r}{q}$

and $\quad \dfrac{1}{q}+\dfrac{1}{s}=\dfrac{2}{r}$

$\Rightarrow\quad \dfrac{1}{q}+\dfrac{1}{s}=\dfrac{p+r}{qr}\Rightarrow\dfrac{s+q}{sq}=\dfrac{p+r}{qr}$

$\Rightarrow\quad r(s+q)=s(p+r)$

$\Rightarrow\quad rq=sp\ \Rightarrow\ \dfrac{p}{q}=\dfrac{r}{s}$

$\therefore\quad p:q=r:s$

26. *(c)* As, given $q^2 = pr$

$$\frac{p^2 - q^2 + r^2}{p^{-2} - q^{-2} + r^{-2}} = \frac{p^2 - q^2 + r^2}{\frac{1}{p^2} - \frac{1}{pr} + \frac{1}{r^2}}$$

$$= \frac{p^2 - q^2 + r^2}{\frac{r^2 - pr + p^2}{p^2 r^2}} = \frac{(p^2r^2)(p^2 - q^2 + r^2)}{(r^2 - pr + p^2)}$$

$$= \frac{(p^2r^2)(p^2 - pr + r^2)}{(r^2 - pr + p^2)}$$

$$= p^2r^2 = (pr)^2 = (q^2)^2 = q^4$$

27. *(a)* Let monthly incomes of A's and B's be $4x$ and $3x$ respectively

Each saving = ₹ 600

$\because$ Income − saving = Expenditure

$\therefore \frac{4x - 600}{3x - 600} = \frac{3}{2}$

$\Rightarrow \quad 2(4x - 600) = 3(3x - 600)$

$8x - 1200 = 9x - 1800$

$\Rightarrow \quad 1800 - 1200 = 9x - 8x \Rightarrow x = 600$

$\therefore \quad A$'s income = $4x = 4 \times 600$ = ₹ 2400

28. *(d)* Let three numbers be $3x$, $2x$ and $5x$, respectively.

According to the question,

$$(3x)^2 + (2x)^2 + (5x)^2 = 1862$$

$\Rightarrow \quad 9x^2 + 4x^2 + 25x^2 = 1862$

$\Rightarrow \quad x^2 = \frac{1862}{38} = 49 \quad \therefore \ x = 7$

Hence, the numbers are 3×7, 2×7 and 5×7 i.e. 21, 14 and 35.

29. *(b)* Speed $\propto \frac{1}{\text{Time}}$

$\therefore$ Required ratio $= \frac{1}{4} : \frac{1}{3} : \frac{1}{2} = 3 : 4 : 6$

30. *(c)* Let the ages of A and B be $2x$ yr and $5x$ yr, respectively.

After 8 yr age of $A = (2x + 8)$ yr

and age of $B = (5x + 8)$ yr

According to the question, $\frac{2x + 8}{5x + 8} = \frac{1}{2}$

$\Rightarrow \quad 4x + 16 = 5x + 8$

$\Rightarrow \quad x = 8$

$\therefore$ Difference between their present ages

$= 5x - 2x = 3x = 3 \times 8 = 24$ yr

31. *(d)* By given condition,

$$y = lx + \frac{m}{x} \qquad \text{...(i)}$$

where, l and m are proportionality constants.

when $y = 6$ and $x = 4$,

then $6 = 4l + \frac{m}{4}$

$\Rightarrow \quad 16l + m = 24 \qquad$ ...(ii)

when $y = \frac{10}{3}$ and $x = 3$,

then $\frac{10}{3} = 3l + \frac{m}{3} \Rightarrow 9l + m = 10$ ...(iii)

On solving Eqs. (ii) and (iii),

$l = 2$ and $m = -8$

From Eq. (i), $\quad y = 2x - \frac{8}{x}$

32. *(c)* Given, $\quad A : B : C = \frac{1}{5} : \frac{1}{6} : \frac{1}{10}$

$= 6 : 5 : 3$

$\therefore$ Share of $A = \frac{6}{6 + 5 + 3} \times 8400$

$= \frac{6}{14} \times 8400$ = ₹ 3600

33. *(d)* Let the amount be ₹ x.

In first condition, Q's part $= \frac{5x}{5 + 3} = \frac{5}{8}x$

In second condition, Q's part

$= \frac{3x}{2 + 3} = \frac{3}{5}x$

By given condition, $\frac{5}{8}x - \frac{3}{5}x = 10$

$\Rightarrow \quad \frac{x}{40} = 10 \Rightarrow x$ = ₹ 400

34. *(a)* Remaining mixture = 20 − 6 = 14 L

Quantity of milk $= 14 \times \frac{4}{7} = 8$ L

$\therefore$ Water contents = 14 − 8 = 6 L

$\because$ Milk in new mixture

= 8 + 6 = 14 L and water = 6 L

$\therefore$ Ratio of milk and water in new mixture $= \frac{14}{6} = \frac{7}{3}$

Hence, the required ratio 7 : 3.

35. *(b)* Let x be the part of milk in new mixture.

By law of mixture,

$\therefore \quad \frac{\frac{3}{8} - x}{x - \frac{1}{4}} = \frac{3}{2} \Rightarrow \frac{3}{4} - 2x = 3x - \frac{3}{4}$

$\Rightarrow \quad 5x = \frac{6}{4} = \frac{3}{2} \quad \therefore \quad x = \frac{3}{10}$

Required ratio $= \frac{3}{10 - 3} = \frac{3}{7}$ or, 3 : 7

36. *(b)* Let quantities of milk and water be $5x$ and x L, respectively.

According to the question, $\frac{5x}{x + 5} = \frac{5}{2}$

$\Rightarrow \quad 10x = 5x + 25$

$\therefore \quad x = 5$

Hence, the quantity of milk in the original mixture = 5 × 5 = 25 L

37. *(c)* Given, $x \propto y^m \qquad$ ...(i)

$y \propto z^n \qquad$ ...(ii)

$x \propto z^p \qquad$ ...(iii)

On putting the values of x and y from Eqs. (ii) and (iii) in Eq. (i), we get

$z^p \propto (z^n)^m \Rightarrow z^p \propto z^{mn}$

$\therefore \quad p = mn$

38. *(d)* Let the initial and final salary be ₹ $22x$ and ₹ $25x$, respectively.

and let initial and final number of employees be $3y$ and $2y$, respectively.

$\therefore$ Present bill = Final salary × Final number of employees

$5000 = 25x \times 2y$

$\Rightarrow \quad \frac{5000}{50} = xy \Rightarrow xy = 100$

$\therefore$ Original bill = Initial salary × Initial number of employees

$= 22x \times 3y$

$= 66\,xy = 66 \times 100$ = ₹ 6600

39. *(c)* Given,

$A : B = x : 8$ and $B : C = 12 : z$

$\Rightarrow A : C = \frac{A}{C} = \frac{A}{B} \times \frac{B}{C} = \frac{x}{8} \times \frac{12}{z} = \frac{3x}{2z}$

But $\quad A : C = 2 : 1$

$\Rightarrow \quad \frac{3x}{2z} = \frac{2}{1} \Rightarrow \quad x : z = 4 : 3$

40. *(a)* Let the number of coins of ₹ 1, 50 paise and 10 paise be $3x$, $8x$ and $10x$, respectively

According to the question,

$\frac{3x}{1} + \frac{8x}{2} + \frac{10x}{10} = 112$

$\Rightarrow \quad 3x + 4x + x = 112$

$\Rightarrow \quad x = \frac{112}{8} = 14$

$\therefore$ Number of 50 paise coins

$= 14 \times 8 = 112$

41. *(b)* Let the number of boys and girls be x and y, respectively.

$\therefore$ Total number of students $= (x + y)$

According to the question,

Number of boys = Number of girls + Total Number of students × 12%

$x = y + (x + y) \times 12\%$

$x - y = \frac{(x + y) \times 12}{100}$

$\Rightarrow \quad 25x - 25y = 3x + 3y$

$\Rightarrow \quad 22x = 28y$

$\therefore \quad x : y = 14 : 11$

42. *(a)* Let C gets ₹ x.

Now, by given conditions,

Amount of B = ₹ $(x + 8)$

Amount of A = ₹ $(x + 15)$

$\therefore \quad x + (x + 8) + (x + 15) = 53$

$\Rightarrow \quad x = 10$

$\therefore \quad A : B : C = (10 + 15) : (10 + 8) : 10$

$= 25 : 18 : 10$

43. *(b)* Let Q join for x months.

$\therefore$ Ratio of capital

$= 2525 \times 12 : 1200 \times x$

$= 2525 : 100x = 101 : 4x$

$\therefore \quad P$'s profit $= \dfrac{101}{101 + 4x} \times 1644$

$\Rightarrow 1212 = \dfrac{101 \times 1644}{101 + 4x}$

$\Rightarrow \dfrac{1}{137} = \dfrac{1}{101 + 4x}$

$\Rightarrow \quad 4x = 36 \quad \Rightarrow \quad x = 9$

Hence, Q joined for 9 months i.e. he joined after 3 months.

44. *(a)* Originally, let the numbers of seats for Mathematics, Physics and Biology be $5x$, $7x$ and $8x$, respectively.

Number of increased seats are (140% of $5x$) (150% of $7x$) and (175% of $8x$).

Required ratio can be obtained as

$\Rightarrow \left(\dfrac{140}{100} \times 5x\right) : \left(\dfrac{150}{100} \times 7x\right) : \left(\dfrac{175}{100} \times 8x\right)$

$\Rightarrow \left(7x : \dfrac{21}{2}x : 14x\right) \times 2$

$\Rightarrow \quad (14x : 21x : 28x) \div 7x$

$\therefore \quad 2 : 3 : 4$

45. *(d)* Let the number of passengers travelling by class I and class II be x and $50x$ respectively.

Then, amount collected from class I and II will be ₹ $3 \times x$ and ₹ $50x$ respectively.

Given, $\quad 3x + 50x = 1325$

$53x = 1325 \quad \Rightarrow \quad x = 25$

$\therefore$ Amount collected from class II

$= 50 \times 25 =$ ₹ 1250

46. *(b)* As $x \propto \dfrac{1}{y^2} \quad \Rightarrow \quad x = \dfrac{k}{y^2} \qquad \ldots$(i)

If $x = 1$

and $\quad y = 6$

$1 = \dfrac{k}{6^2} \qquad \Rightarrow \quad k = 36$

On putting the value of k in Eq. (i), we get

$x = \dfrac{36}{y^2} \qquad \ldots$(ii)

I. On putting $y = 3$ in Eq. (ii), $x = \dfrac{36}{9}$

$x = 4$

II. On putting $y = 6$ in Eq. (ii), we get

$x = \dfrac{36}{36} = 1$

$x = 1$

Both statements I and II are correct.

47. *(d)* Given that, $\dfrac{a}{b} = \dfrac{c}{d} = \dfrac{e}{f} = \dfrac{1}{2}$

$\Rightarrow \quad a = \dfrac{b}{2}, \quad c = \dfrac{d}{2}, \quad e = \dfrac{f}{2}$

I. $\dfrac{3a + 5c + 7e}{3b + 5d + 7f} = \left(\dfrac{\dfrac{3b}{2} + \dfrac{5d}{2} + \dfrac{7f}{2}}{3b + 5d + 7f}\right)$

$= \dfrac{\frac{1}{2}(3b + 5d + 7f)}{3b + 5d + 7f} = \dfrac{1}{2}$

II. $\dfrac{a^2 + c^2 + e^2}{b^2 + d^2 + f^2} = \dfrac{\dfrac{b^2}{4} + \dfrac{d^2}{4} + \dfrac{f^2}{4}}{b^2 + d^2 + f^2} = \dfrac{1}{4}$

$\therefore$ Neither I nor II correct.

48. *(c)* I. 4 leaps of cat = 3 leaps of dog

$\Rightarrow \quad$ 1 leap of cat $= \dfrac{3}{4}$ leap of dog

Cat takes 5 leap for every 4 leaps of dog.

$\therefore \quad$ Required Ratio

$= (5 \times$ cat's leap$) : (4 \times$ dog's leap$)$

$= \left(5 \times \dfrac{3}{4} \text{ dog's leap}\right) : (4 \times \text{dog's leap})$

$= 15 : 16$

Thus, $\dfrac{\text{Speed of cat}}{\text{Speed of dog}} = \dfrac{15}{16}$

II. $\dfrac{\text{Distance (cat)}}{\text{Distance (dog)}} = \dfrac{\text{s (cat)} \times \text{t}}{\text{s (dog)} \times \text{t}}$

$= \dfrac{\text{s (cat)} \times 30}{\text{s (dog)} \times 30} = \left(\dfrac{\text{s (cat)}}{\text{s (dog)}} = \dfrac{15}{16}\right)$

Thus, both statements I and II are correct.

49. *(a)* Number of people having characteristic X

$= 10 + 30 = 40$

Number of people having characteristic Y

$= 10 + 20 = 30$

Required ratio = 40:30 = 4:3.

50. *(c)* Fresh grapes contain 10% pulp.

$\therefore$ 20 kg fresh grapes contain 2 kg pulp.

Dry grapes contain 80% pulp.

2 kg pulp would contain

$\dfrac{2}{0.8} = \dfrac{20}{8} = 2.5$ kg dry grapes.

51. *(c)* Refer to question 12

52. *(c)* Refer to question 20

$\therefore$ Their product $= 18 \times 27 = 486$

53. *(b)* Let cost price of milk be ₹ x per kg.

Then, according to the rule of mixture,

$22 : (x - 22) = 15 : 3$

$\Rightarrow \quad \dfrac{22}{x - 22} = \dfrac{15}{3}$

$\Rightarrow \quad \dfrac{22}{x - 22} = 5$

$\therefore \quad x =$ ₹ 26.40

54. *(a)* Total number of inhabitants = 1935000

Total number of females = 935000

$\therefore \quad$ Total number of males

$= 1935000 - 935000 = 1000000$

$\therefore \quad$ sex ratio $= \dfrac{935000}{1000000} \times 1000 = 935$

55. *(d)* Given, ratio of boys to girls = 7 : 5

and total number of students = 2400

Sum of ratio = 7 + 5 = 12

$\therefore \quad$ Number of girls $= \dfrac{5}{12} \times 2400 = 1000$

56. *(a)* Given, $A : B = 2 : 3$, $B : C = 5 : 7$

and $\quad C : D = 3 : 10$

$\therefore \quad \dfrac{A}{D} = \dfrac{A}{B} \times \dfrac{B}{C} \times \dfrac{C}{D}$

$= \dfrac{2}{3} \times \dfrac{5}{7} \times \dfrac{3}{10} = \dfrac{1}{7}$

$\therefore \quad A : D = 1 : 7$

57. *(d)* Given that, $(x + y) : (x - y) = 3 : 5$

$\therefore \quad \dfrac{x + y}{x - y} = \dfrac{3}{5}$

On applying componendo and dividendo rule, we get

$\dfrac{(x + y) + (x - y)}{(x + y) - (x - y)} = \dfrac{3 + 5}{3 - 5}$

$\Rightarrow \quad \dfrac{2x}{2y} = \dfrac{8}{-2} \quad \Rightarrow \quad \dfrac{x}{y} = -4$

$\therefore \quad x = -4y$

But it is given , xy = positive

$\therefore \quad -4y \times y$ = positive

$\Rightarrow -4y^2$ = positive, which is not possible.

Hence, no real solution for x and y exists.

58. *(a)* Given, ratio of ages of A and B i.e. $A : B = 2 : 5$

Ratio of ages of B and C i.e. $B : C = 3 : 4$

$\therefore$ Ratio of ages of A, B and C = $\begin{matrix} 2 : 5 \\ \downarrow \swarrow \downarrow \\ 3 : 4 \end{matrix}$

$= 2 \times 3 : 3 \times 5 : 5 \times 4 = 6 : 15 : 20$

59. *(a)* Let height of tree be h ft and age be a yr.

Then, according to the question,

$h \propto \sqrt{a} \Rightarrow h = k\sqrt{a} \qquad$ ..(i)

where, k is a constant.

If age = 9 yr, then height = 4 ft

From Eq. (i), we get $4 = k\sqrt{9}$

$\Rightarrow \quad 4 = k \times 3 \quad \therefore \quad k = \frac{4}{3}$

If age = 16 yr, then $h = k\sqrt{16} = \frac{4}{3} \times 4$

$\left[\because k = \frac{4}{3}\right]$

$\therefore \quad h = \frac{16}{3}$ ft $= 5\frac{1}{3} = 5$ ft 4 inch

60. *(b)* Refer to question 34.

61. *(c)* We have, $\frac{a}{b} = \frac{3}{5}$

$\because \quad b = \frac{5a}{3}$ and $\frac{b}{c} = \frac{7}{8}$

$\Rightarrow \quad c = \frac{8}{7}b = \frac{8}{7} \times \frac{5a}{3} = \frac{40a}{21}$

$\because \quad a : b : c = a : \frac{5a}{3} : \frac{40a}{21}$

Now, $2a : 3b : 7c = 2a : 5a : \frac{40a}{3}$

$= 2 : 5 : \frac{40}{3} = 6 : 15 : 40$

62. *(d)* Let the distance be x.

Now, ratio of time taken to travel the distance by each bus is

$\frac{x}{2} : \frac{x}{3} : \frac{x}{4} \Rightarrow \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$

$\Rightarrow \quad 6 : 4 : 3$

63. *(c)* Since, milk $= 7x$ and water $= 3x$

Then, $7x + 3x = 30 \Rightarrow 10x = 30$

$\therefore \quad x = 3$

So, milk = 21 L and water = 9 L

Let y L of water to be added.

Then, $\frac{21}{9 + y} = \frac{1}{2}$

$\Rightarrow \quad 9 + y = 42$

$\therefore \quad y = 33$ L

64. *(a)* Refer to question 27.

65. *(a)* $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k$ (say)

$\Rightarrow a = bk, b = ck, c = dk$

$\therefore \quad a = dk^3, b = dk^2, c = dk$

I. $\frac{b^3 + c^3 + d^3}{a^3 + b^3 + c^3}$

$= \frac{b^3 + c^3 + d^3}{(bk)^3 + (ck)^3 + (dk)^3}$

$= \frac{1}{k^3}\frac{(b^3 + c^3 + d^3)}{(b^3 + c^3 + d^3)} = \frac{1}{k^3} = \frac{d}{a}$

Hence, statement I is correct.

II. Similarly $\frac{a^2 + b^2 + c^2}{b^2 + c^2 + d^2} = k^2 = \frac{b}{d}$

Hence, statement II is incorrect.

13

LOGARITHM

Usually (2-3) questions have been asked from this chapter. Generally questions are based on fundamental rules of logarithm. It is especially used to solve complicated mathematical calculations. Applications of square, cube, surds and indices are extremely used here.

DEFINITION

If 'a' is a positive real number, other than 1 and 'b' is a rational number such that $a^b = N$, then we say that logarithm of N to base 'a' is b or 'b' is the logarithm of N to base 'a', written as $\log_a N = b$

So, $a^b = N \Leftrightarrow \log_a N = b$, e.g. $7^0 = 1 \Leftrightarrow \log_7 1 = 0$, $81^{1/4} = 3 \Leftrightarrow \log_{81} 3 = 1/4$

EXAMPLE 1. Which one of the following has the value of $x = 4$?

a. $\log_2 x = 3$ b. $\log_5 x = 3$ c. $\log_{81} x = \frac{3}{2}$ d. $\log_{\sqrt{2}} x = 4$

Sol. **d.** a. $\log_2 x = 3 \Leftrightarrow x = 2^3 = 8$ b. $\log_5 x = 3 \Leftrightarrow x = 5^3 = 125$ [by definition]

c. $\log_{81} x = 3/2 \Leftrightarrow x = (81)^{3/2} = (3^4)^{3/2}, \Leftrightarrow x = 3^6 \quad \Leftrightarrow x = 729$

d. $\log_{\sqrt{2}} x = 4 \Leftrightarrow x = (\sqrt{2})^4,\ x = (2^{1/2})^4 = 2^2 = 4$

EXAMPLE 2. If $\log_5 y = x$ and $\log_2 z = x$, then the value of 20^x in terms of y and z is

a. zy b. z^2y c. xy^2 d. z/y

Sol. **b.** Given, $\log_5 y = x$ and $\log_2 z = x \Rightarrow y = 5^x$ and $z = 2^x$ ...(i)

Now, $20^x = (2^2 \times 5)^x = 2^{2x} \times 5^x = (2^x)^2 \cdot 5^x = z^2 \cdot y$ [from Eq. (i)]

$\therefore$ $20^x = z^2 \cdot y$

Fundamental Rules of Logarithms

Rule 1 If m and n are positive rational number, then $\log_a (mn) = \log_a m + \log_a n$

If $m_1, m_2, m_3, \ldots, m_n$ are positive rational numbers, then

$\log_a (m_1 m_2 m_3 \ldots m_n) = \log_a m_1 + \log_a m_2 + \ldots + \log_a m_n$

EXAMPLE 3. The value of $\log_{10}(15)$ is

a. $\log_{10} 2 - \log_{10} 5$ b. $\log_{10} 5$
c. $\log_{10} 3 + \log_{10} 5$ d. $\log_{10} 2 + \log_{10} 5$

Sol. c. $\log_{10} 15 = \log_{10}(3 \times 5) = \log_{10} 3 + \log_{10} 5$

Rule 2 If m and n are positive rational numbers, then

$$\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$$

EXAMPLE 4. The value of x in $\log x - \log(x-1) = \log 3$ is

a. 2/3 b. 3/2 c. 1/2 d. 1/4

Sol. b. $\log x - \log(x-1) = \log 3$

$\Rightarrow \quad \log \dfrac{x}{x-1} = \log 3 \Rightarrow \dfrac{x}{x-1} = 3$

$\Rightarrow \quad x = 3(x-1) \Rightarrow x = 3x - 3 \Rightarrow -2x = -3$

$\therefore \quad x = 3/2$

Rule 3 If m and n are positive rational numbers, then, $\log_a(m^n) = n \log_a m$

EXAMPLE 5. The value of x in
$\log_x 4 + \log_x 16 + \log_x 64 = 12$ is

a. 1 b. 2 c. 3 d. 4

Sol. b. $\log_x 4 + \log_x 16 + \log_x 64 = 12$

$\Rightarrow \quad \log_x 2^2 + \log_x 2^4 + \log_x 2^6 = 12$

$\Rightarrow \quad 2\log_x 2 + 4\log_x 2 + 6\log_x 2 = 12$

$\Rightarrow \quad 12\log_x 2 = 12 \Rightarrow \log_x 2 = 1$

$\therefore \quad x = 2$

EXAMPLE 6. The value of x in $\dfrac{\log 256}{\log 16} = \log x$ is

a. 1 b. 10 c. 100 d. 0

Sol. c. $\dfrac{\log 256}{\log 16} = \dfrac{\log 16^2}{\log 16} = \dfrac{2\log 16}{\log 16} = 2$

$\Rightarrow \quad \log x = 2 \Rightarrow \log_{10} x = 2 \quad [\because \log x = \log_{10} x]$

$\therefore \quad x = 10^2 = 100$

Rule 4 If m is a positive rational number and a, b are positive real numbers i.e. $a \neq 1$ and $b \neq 1$, then

$$\log_a m = \frac{\log_b m}{\log_b a}$$

EXAMPLE 7. The value of
$\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$ is

a. 1 b. 2 c. –1 d. None of these

Sol. b. $\log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9$

$$= \frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times \frac{\log 6}{\log 5} \times \frac{\log 7}{\log 6} \times \frac{\log 8}{\log 7} \times \frac{\log 9}{\log 8}$$

$$= \frac{\log 9}{\log 3} = \frac{\log 3^2}{\log 3} = \frac{2\log 3}{\log 3} = 2$$

SOME USEFUL RESULTS

- logarithm of 1 with any base is always zero i.e. $\log_a 1 = 0$
- logarithm of base to itself is always 1 i.e. $\log_a a = 1$
- $x > y \Rightarrow \log_a x > \log_a y$ for $a > 1$
- $x > y \Rightarrow \log_a x < \log_a y$ for $0 < a < 1$
- for $x > 0, 1 \neq a > 0$, $\log_a n(x) = \dfrac{1}{n}\log_a x$
- for $1 \neq a, b > 0$, $x > 0$, $a^{\log_b x} = x^{\log_b a}$
- If 'a' is a positive real number and n is a positive rational number, then

(i) $\log_{a^y} n^x = \dfrac{x}{y}\log_a n$.

(ii) $a^{\log_a n} = n$

(iii) $\log_a n = \log_{a^2} n^2 = \log_{a^3} n^3 = \log_{a^m} n^m$

Common Logarithm and Natural Logarithm

Logarithm can have any positive base other than 0. But there are two most important base which are generally used.

1. **Common Logarithm** Logarithm to the base '10' is called common logarithm. It is also called Brigg's logarithm.

   e.g. $\log 100 = \log_{10} 100 = 2$
   $\log 1000 = \log_{10} 1000 = 3$

2. **Natural Logarithm** Logarithm to the base e is called natural logarithm. It is also called Napier logarithm. $\log_e x$ is usually denoted by $\ln x$. e is irrational number between 2 and 3.

   $\ln x = y \Rightarrow e^y = x$

ANTILOGARITHM

The positive number 'a' is called the antilogarithm of a number b, if $\log a = b$. If 'a' is antilogarithm of b, it is written as $a = \text{antilog } b$.

So, $a = \text{antilog } b \Leftrightarrow \log a = b$

Characteristic and Mantissa of a Logarithm

The logarithm of positive real number 'n' consists of two parts

1. The integral part is known as the characteristic. It is always an integer (positive, negative or zero).
2. The decimal part is called as the mantissa. The mantissa is never negative and is always less than one.

To Find the Characteristic

Case I The characteristic of the log of a number greater than 1 is positive and numerically one less than the number of digits to the left of the decimal part.

e.g.

Value	Characteristic
log 3.257	1 – 1 = 0
log 32.57	2 – 1 = 1
log 325.7	3 – 1 = 2

Case II The characteristic of log of a number less than 1 is negative and numerically one more than the number of zeroes immediately after the decimal point. It is represented by bar over the digit.

The bar over the characteristic indicates that it is negative.

e.g.

Value	Characteristic
log 0.3257	$-(0+1)=-1$ i.e. $\bar{1}$
log 0.03257	$-(1+1)=-2$ i.e. $\bar{2}$
log 0.003257	$-(2+1)=-3$ i.e. $\bar{3}$

Rules for Inserting Decimal Point

Two rules are used for inserting a decimal point.

Rule 5 When the characteristic of the logarithm is positive, we insert the decimal point after the $(n+1)$ th digit, where n is the characteristic.

Rule 6 When the characteristic of the logarithm is negative, we insert the decimal point such that the first significant figures is at nth place, where $\bar{n}$ is the characteristic.

EXAMPLE 8. The characteristic of the logarithm of the number 0.00000014 is

a. 1
b. 7
c. –7
d. None of these

Sol. c. $0.00000014 = 1.4 \times \dfrac{1}{10000000} = 1.4 \times 10^{-7}$

$\therefore$ The characteristic of log 0.00000014 = –7

EXAMPLE 9. The characteristic of the logarithm of the number 8.188 is

a. 2
b. 1
c. –1
d. 0

Sol. d. $8.188 = 8.188 \times 1 = 8.188 \times 10^0$

$\therefore$ The characteristic of log 8.188 = 0

EXAMPLE 10. The characteristic of the logarithm of the number 566.37 is

a. 2
b. –2
c. 4
d. None of these

Sol. a. $566.37 = 5.6637 \times 10^2$

$\therefore$ The characteristic of log 566. 37 = 2

EXAMPLE 11. The characteristic of the logarithm of the number 313 is

a. 3
b. 2
c. 1
d. 0

Sol. b. $313 = 3.13 \times 10^2$

$\therefore$ The characteristic of log 313 = 2

SOME IMPORTANT POINTS

1. The base of a logarithm is never taken as zero and negative.
2. Log of negative integers are not defined and also $\log_e 0$ is not defined.
3. Logarithmic function is positive as well as negative but exponential function is always positive.

PRACTICE EXERCISE

1. What is the value of $\log_{100} 0.1$?
(a) 1/2 (b) –1/2 (c) –2 (d) 2

2. The value of $3 \log 3 + 2 \log 2$ is
(a) log 108 (b) log 106 (c) log 109 (d) None of these

3. If $\log_a \sqrt{2} = \frac{1}{6}$, then the value of a is
(a) $(\sqrt{2})^6$ (b) $(6)^{1/2}$ (c) 3 (d) –6

4. If $\log_3 x = -2$, then the value of x is
(a) $\frac{1}{9}$ (b) $-\frac{1}{9}$ (c) $\frac{1}{8}$ (d) $-\frac{1}{8}$

5. Find the logarithm of 1728 to the base $2\sqrt{3}$.
(a) 3.124 (b) 3.1732 (c) 6 (d) 5

6. What is the value of
$$(\log_{1/2} 2)(\log_{1/3} 3)(\log_{1/4} 4)\ldots(\log_{1/1000} 1000)?$$
(a) 1 (b) –1 (c) 1 or –1 (d) 0

7. What is the value of
$\frac{1}{2}\log_{10} 25 - 2\log_{10} 3 + \log_{10} 18$?
(a) 2 (b) 3 (c) 1 (d) 0

8. What is the value of $[\log_{10}(5 \log_{10} 100)]^2$?
(a) 4 (b) 3 (c) 2 (d) 1

9. The value of $\log_y x \cdot \log_z y \cdot \log_x z$ is
(a) log *xyz* (b) *xyz* (c) 1 (d) 0

10. The value of $\log_3 (27 \times \sqrt[4]{9} \times \sqrt[3]{9})$ is
(a) 4 (b) $4\frac{1}{3}$ (c) $8\frac{1}{3}$ (d) $4\frac{1}{6}$

11. The value of $\log_2 [\log_2 \log_2 \log_2 (65536)]$ is
(a) 8 (b) 16 (c) 4 (d) 1

12. What is the value of $[\log_{13}(10)]/[\log_{169}(10)]$?
(a) $\frac{1}{2}$ (b) 2 (c) 1 (d) $\log_{10} 13$

13. What is the value of
$\left(\frac{1}{3}\log_{10} 125 - 2\log_{10} 4 + \log_{10} 32 + \log_{10} 1\right)$?
(a) 0 (b) $\frac{1}{5}$ (c) 1 (d) $\frac{2}{5}$

14. If $\log_r 6 = m$ and $\log_r 3 = n$, then what is $\log_r (r/2)$ equal to?
(a) $m - n + 1$ (b) $m + n - 1$ (c) $1 - m - n$ (d) $1 - m + n$

15. What is $\log_{10}\left(\frac{3}{2}\right) + \log_{10}\left(\frac{4}{3}\right) + \log_{10}\left(\frac{5}{4}\right) + \cdots$ upto 8 terms equal to?
(a) 0 (b) 1 (c) $\log_{10} 5$ (d) None of these

16. The value of $\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)}$ is
(a) *xyz* (b) 2 (c) 0 (d) 1

17. The value of
$\frac{1}{1 + \log_x (yz)} + \frac{1}{1 + \log_y (xz)} + \frac{1}{1 + \log_z (xy)}$ is
(a) 1 (b) $\frac{1}{xy^2}$ (c) $x = yz$ (d) 0

18. If $\log_4 (x^2 + x) - \log_4 (x + 1) = 2$, then the value of x is
(a) 4 (b) 8 (c) 16 (d) 1

19. If $\log_4 x + \log_2 x = 6$, then the value of x is
(a) 16 (b) 4 (c) 2 (d) 1

20. Given $\log_{10} 2 = 0.3010$, the value of $\log_{10} 5$ is
(a) 0.6990 (b) 0.6919 (c) 0.6119 (d) 0.7525

21. If $\log \frac{x}{y} + \log \frac{y}{x} = \log (x + y)$, then
(a) $x + y = 1$ (b) $x - y = 0$ (c) $x - y = 1$ (d) $x = y$

22. The characteristic in $\log 6.7482 \times 10^{-5}$ is
(a) 6 (b) – 4 (c) 5 (d) – 5

23. If $10^x = 1.73$ and $\log_{10} 1730 = 3.2380$, then x equals to
(a) 2.380 (b) 0.2380 (c) 2.2380 (d) 1.380

24. If $2^{2x+3} = 6^{x-1}$, then x equals
(a) $\frac{4 \log 2 + \log 3}{\log 3 - \log 2}$ (b) $\frac{3 \log 2 + 2 \log 3}{\log 3 - 2 \log 2}$
(c) $\frac{\log 48}{\log 7}$ (d) None of these

25. The value of $10^{\log_{10} m + 2\log_{10} n + 3\log_{10} p}$ is
(a) m^2np^3 (b) mn^2p^3 (c) m^3np^2 (d) None of these

26. Given that $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$ and $\log_{10} 7 = 0.8491$, then $\log_{10} \frac{108}{\sqrt{7}}$ is equal to
(a) 2.6123 (b) 1.6088 (c) 1.6320 (d) 2.4558

27. If a, b and c are three consecutive integers, then $\log (ac + 1)$ is equal to
(a) $\log (2b)$ (b) $(\log b)^2$ (c) $2 \log b$ (d) None of these

28. If $\log_r p = 2, \log_r q = 3$, then the value of $\log_p q$ is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{2}$ (d) 6

29. If $\log x^2y^2 = a$ and $\log \frac{x}{y} = b$, then $\frac{\log x}{\log y}$ is equal to
(a) $\frac{a - 3b}{a + 2b}$ (b) $\frac{a + 3b}{a - 2b}$ (c) $\frac{a + 2b}{a - 2b}$ (d) $\frac{a - 2b}{a + 3b}$

30. If $\log_{10} 5 = 0.70$, then $\log_5 10$ is equal to

(a) 1.35 (b) 1.40 (c) 1.43143 (d) 1.56

31. The value of

$$\log_3\left(1+\frac{1}{3}\right)+\log_3\left(1+\frac{1}{4}\right)+\log_3\left(1+\frac{1}{5}\right)+\ldots+\log_3\left(1+\frac{1}{24}\right)$$ is

(a) $-1 + 2\log_3 5$ (b) 2
(c) 3 (d) 4

32. If $\log(x+y) = \log x + \log y$ and $x = 1.1568$, then y is equal to

(a) 7.3776 (b) $\overline{7}.3776$ (c) 5.3776 (d) 5.3116

33. If $\log_8 x + \log_4 x + \log_2 x = 11$. Then, the value of x is

(a) 128 (b) 16 (c) 32 (d) 64

34. What is the value of $\log_y x^5 \log_x y^2 \log_z z^3$?

(a) 10 (b) 30 (c) 20 (d) 60

35. If $(\log_3 x)(\log x^{2x})(\log_{2x} y) = \log_x x^2$, then what is the value of y?

(a) $\frac{9}{2}$ (b) 9 (c) 18 (d) 27

36. If $\log_{10} 2, \log_{10}(2^x - 1), \log_{10}(2^x + 3)$ are three consecutive terms of AP, then which one of the following is correct?

I. $x = 1$ II. $x = \log_2 5$

(a) Both I and II (b) Only II
(c) Only I (d) None of these

37. Consider the following statements

I. $(\log_{10} 0.1)^2 + \log_{10} 10 \cdot \log_{10} 100 = 3$

II. $\log_{10}\log_{10} 10 = 1$ III. $\log_{10}\sqrt{10} + \log_{10}\sqrt{10} = 1$

Which of the statements given above are correct?

(a) I and III (b) II and III
(c) I and II (d) All are correct

38. If $y = (a^x)^{(a^x)\cdots\infty}$, then which one of the following is correct?

(a) $\log y = xy \log a$ (b) $\log y = x + y\log a$
(c) $\log y = y + x\log a$ (d) $\log y = (y + x)\log a$

PREVIOUS YEARS' QUESTIONS

39. What is the logarithm of 0.0001 with respect to base 10? **2012 II**

(a) 4 (b) 3 (c) – 4 (d) – 3

40. If $\log_{10} a = p$ and $\log_{10} b = q$, then what is the value of $\log_{10}(a^p b^q)$? **2012 II**

(a) $p^2 + q^2$ (b) $p^2 - q^2$ (c) $p^2 q^2$ (d) $\frac{p^2}{q^2}$

41. What are the possible solutions for x of the equation $x^{\sqrt{x}} = \sqrt[n]{x^x}$, where x and n are positive integers? **2015 I**

(a) 0, n (b) 1, n (c) n, n^2 (d) 1, n^2

42. The value of $\frac{1}{5}\log_{10} 3125 - 4\log_{10} 2 + \log_{10} 32$ is **2016 I**

(a) 0 (b) 1 (c) 2 (d) 3

ANSWERS

1	b	2	a	3	a	4	a	5	c	6	b	7	c	8	d	9	c	10	d
11	d	12	b	13	c	14	d	15	c	16	b	17	a	18	c	19	a	20	a
21	a	22	d	23	b	24	a	25	b	26	b	27	c	28	c	29	c	30	c
31	a	32	a	33	d	34	b	35	b	36	b	37	a	38	a	39	c	40	a
41	a	42	b																

HINTS AND SOLUTIONS

1. *(b)* $\log_{100} 0.1 = \log_{10^2}\left(\frac{1}{10}\right)$

$= \frac{1}{2}\log_{10}\left(\frac{1}{10}\right) = \frac{1}{2}\log_{10}(10)^{-1}$

$= -\frac{1}{2}\log_{10} 10 = -\frac{1}{2}$ [Rule 3]

2. *(a)* $3\log 3 + 2\log 2 = \log 3^3 + \log 2^2$

$= \log 27 + \log 4$ [Rule 3]

$= \log(27 \times 4) = \log 108$ [Rule 1]

3. *(a)* $\because \log_a \sqrt{2} = \frac{1}{6} \Rightarrow a^{1/6} = \sqrt{2}$

$\therefore a = (\sqrt{2})^6$

4. *(a)* $\log_3 x = -2 \Rightarrow x = 3^{-2} = \frac{1}{3^2} = \frac{1}{9}$

5. *(c)* Let $\log_{2\sqrt{3}} 1728 = x$

$\Rightarrow (2\sqrt{3})^x = 1728$

$\because 1728 = 2^6(\sqrt{3})^6 = (2\sqrt{3})^6$

$(2\sqrt{3})^x = (2\sqrt{3})^6$

On comparing both sides, we get $x = 6$

6. *(b)* $(\log_{1/2} 2)(\log_{1/3} 3)(\log_{1/4} 4)\ldots(\log_{1/1000} 1000)$

$= \left(\frac{\log 2}{\log 1/2}\right)\left(\frac{\log 3}{\log 1/3}\right)\left(\frac{\log 4}{\log 1/4}\right)\ldots\left(\frac{\log 1000}{\log 1/1000}\right)$ $\left[\because \log_b a = \frac{\log a}{\log b}\right]$

[Rule 4]

$$= \left(\frac{\log 2}{-\log 2}\right)\left(\frac{\log 3}{-\log 3}\right)\left(\frac{\log 4}{-\log 4}\right) \cdots \left(\frac{\log 1000}{-\log 1000}\right)$$

$= (-1)\times(-1)\times(-1) \times \ldots \times (-1)$

[$\because$ number of terms is odd]

$= -1$

7. *(c)* $\frac{1}{2}\log_{10} 25 - 2\log_{10} 3 + \log_{10} 18$

$= \log_{10} 25^{1/2} - \log_{10} 3^2 + \log_{10} 18$ [Rule 3]

$= \log_{10} 5 - \log_{10} 9 + \log_{10} 18$

$= \log_{10} \frac{5 \times 18}{9} = \log_{10} \frac{90}{9}$

$= \log_{10} 10 = 1$

8. *(d)* $[\log_{10}(5\log_{10} 100)]^2$

$= [\log_{10}(5\log_{10} 10^2)]^2$

$= [\log_{10}(10\log_{10} 10)]^2$

$= [\log_{10} 10]^2$ [$\because \log_{10} 10 = 1$]

$= 1^2 = 1$

9. *(c)* $\log_y x \cdot \log_z y \cdot \log_x z$

$= \frac{\log x}{\log y} \times \frac{\log y}{\log z} \times \frac{\log z}{\log x} = 1$

$\left[\because \log_a b = \frac{\log b}{\log a}\right]$

10. *(d)* Let $\log_3 (27 \times \sqrt[4]{9} \times \sqrt[3]{9}) = x$

$\Rightarrow 3^x = 27 \times \sqrt[4]{9} \times \sqrt[3]{9}$

$\Rightarrow 3^x = 3^3 \times 3^{2/4} \times 3^{2/3}$

$3^x = 3^{25/6}$

On comparing both sides, we get

$\Rightarrow x = \frac{25}{6} = 4\frac{1}{6}$

11. *(d)* $\log_2 \log_2 \log_2 \log_2 2^{16}$

$= \log_2 \log_2 \log_2 (16)$ [$\because \log_2 2 = 1$]

$= \log_2 \log_2 \log_2 (2^4) = \log_2 \log_2 (4)$

$= \log_2 \log_2 (2^2) = \log_2 (2) = 1$

12. *(b)* $\frac{\log_{13}(10)}{\log_{169}(10)} = \frac{\log_{13}(10)}{\log_{13^2}(10)}$

$= \frac{\log_{13} 10}{\frac{1}{2}\log_{13} 10}$ $\left[\because \log_{a^b} c = \frac{1}{b}\log_a c\right]$

$= \frac{1}{1/2} = 2$

13. *(c)* We have, $\frac{1}{3}\log_{10} 125 - 2\log_{10} 4 + \log_{10} 32 + \log_{10} 1$

$= \log_{10}(125)^{1/3} - 2\log_{10}(2)^2 + \log_{10}(2)^5 + 0$ [$\because \log_{10} 1 = 0$]

$= \log_{10} 5 - 4\log_{10} 2 + 5\log_{10} 2$

$= \log_{10} 5 + \log_{10} 2 = \log_{10} 5 \times 2$

$= \log_{10} 10 = 1$ [Rule 1]

14. *(d)* Given, $\log_r 6 = m$ and $\log_r 3 = n$

$\because \log_r 6 = \log_r (2\times 3) = \log_r 2 + \log_r 3$

$\therefore \log_r 3 + \log_r 2 = m$

$\Rightarrow n + \log_r 2 = m$

$\Rightarrow \log_r 2 = m - n$

$\therefore \log_r \left(\frac{r}{2}\right) = \log_r r - \log_r 2$ [Rule 2]

$= 1 - m + n$

15. *(c)* $\log_{10}\left(\frac{3}{2}\right) + \log_{10}\left(\frac{4}{3}\right) + \log_{10}\left(\frac{5}{4}\right) + \cdots +$ 8th term

$\because T_n = \log_{10}\left(\frac{n+2}{n+1}\right)$

$\Rightarrow T_8 = \log_{10}\left(\frac{10}{9}\right)$

$\therefore \log_{10}\left(\frac{3}{2}\right) + \log_{10}\left(\frac{4}{3}\right) + \log_{10}\left(\frac{5}{4}\right) + \cdots + \log_{10}\left(\frac{10}{9}\right)$

$= \log_{10}\left(\frac{3}{2} \times \frac{4}{3} \times \frac{5}{4} \times \cdots \times \frac{10}{9}\right)$ [Rule 1]

$= \log_{10}\left(\frac{10}{2}\right) = \log_{10} 5$

16. *(b)* The given expression is

$= \log_{xyz}(xy) + \log_{xyz}(yz) + \log_{xyz}(zx)$

$= \log_{xyz}(xy \cdot yz \cdot zx)$ [Rule 1]

$= \log_{xyz}(xyz)^2 = 2\log_{xyz} xyz = 2$

17. *(a)* The given expression is

$$= \frac{1}{\log_x(yz) + \log_x x} + \frac{1}{\log_y(xz) + \log_y y} + \frac{1}{\log_z(xy) + \log_z z}$$

$$= \frac{1}{\log_x xyz} + \frac{1}{\log_y xyz} + \frac{1}{\log_z xyz}$$

$= \log_{xyz} x + \log_{xyz} y + \log_{xyz} z$

$= \log_{xyz} xyz = 1$ [Rule 1]

18. *(c)* $\log_4(x^2 + x) - \log_4(x+1) = 2$

$\Rightarrow \log_4\left(\frac{x^2+x}{x+1}\right) = 2 \Rightarrow 4^2 = \frac{x^2+x}{x+1}$

$\Rightarrow 16x + 16 = x^2 + x$

$\Rightarrow x^2 - 15x - 16 = 0$

$\therefore x = 16$ or $x = -1$ [not possible]

19. *(a)* Given, $\log_4 x + \log_2 x = 6$

$\Rightarrow \frac{\log x}{\log 4} + \frac{\log x}{\log 2} = 6$

$\Rightarrow \frac{\log x}{2\log 2} + \frac{\log x}{\log 2} = 6$

$\Rightarrow 3\log x = 12\log 2$

$\Rightarrow \log x = 4\log 2 \Rightarrow \log x = \log 2^4$

$\Rightarrow \log x = \log 16$

On comparing both sides, we get

$\therefore x = 16$

20. *(a)* $\log_{10} 5 = \log_{10}\frac{10}{2}$

$= \log_{10} 10 - \log_{10} 2 = 1 - 0.3010$

$= 0.6990$

21. *(a)* Given, $\log\frac{x}{y} + \log\frac{y}{x} = \log(x+y)$

$\Rightarrow \log\frac{x}{y}\cdot\frac{y}{x} = \log(x+y)$

$\Rightarrow \log 1 = \log(x+y)$

$\therefore x + y = 1$

22. *(d)* The characteristic in $\log 6.7482 \times 10^{-5}$ is -5.

23. *(b)* Given, $10^x = 1.73$, $x = \log_{10} 1.73$

$= \log_{10} 1730 - \log_{10} 1000$

$= \log_{10} 1730 - \log_{10} 10^3$

$= 3.2380 - 3 = 0.2380$

24. *(a)* $2^{2x+3} = 6^{x-1}$

Taking log on both sides, we get

$(2x+3)\log 2 = (x-1)\log 6$

$\Rightarrow 2x\log 2 + 3\log 2 = (x-1)(\log 2 + \log 3)$

$\Rightarrow 2x\log 2 + 3\log 2 = x(\log 2 + \log 3) - \log 2 - \log 3$

$\Rightarrow x(\log 2 - \log 3) = -4\log 2 - \log 3$

$\therefore x = \frac{4\log 2 + \log 3}{\log 3 - \log 2}$

25. *(b)* $10^{\log_{10} m + 2\log_{10} n + 3\log_{10} p}$

$= 10^{\log_{10} m + \log_{10} n^2 + \log_{10} p^3}$

$\Rightarrow 10^{\log_{10} mn^2p^3} = mn^2p^3$ [$\because a^{\log_a p} = p$]

26. *(b)* $\log_{10}\frac{108}{\sqrt{7}} = \log_{10} 108 - \log_{10}\sqrt{7}$

$= \log_{10} 2^2 \times 3^3 - \log_{10} 7^{1/2}$

$= 2\log_{10} 2 + 3\log_{10} 3 - \frac{1}{2}\log_{10} 7$

$= 2\times(0.3010) + 3(0.4771) - \frac{1}{2}(0.8491)$

$= 0.6020 + 1.4313 - 0.4245 = 1.6088$

27. *(c)* Let first integer be a.

Then, $b = a + 1$ and $c = a + 2$

$\therefore \quad ac + 1 = a(a+2) + 1$

$= a^2 + 2a + 1 = (a+1)^2$

$ac + 1 = b^2$

So, $\log(ac+1) = \log b^2 = 2\log b$

28. *(c)* Given, $\log_r p = 2$ and $\log_r q = 3$

By relation, $\log_p q = \dfrac{\log_r q}{\log_r p} = \dfrac{3}{2}$

29. *(c)* Given, $\log x^2 y^2 = a$

$[\because \log(mn) = \log m + \log n]$

$\therefore \quad \log x^2 + \log y^2 = a$

$\Rightarrow 2\log x + 2\log y = a \quad \ldots(i)$

Also, $\log x - \log y = b \quad \ldots(ii)$

On solving Eqs. (i) and (ii), we get

$\log x = \dfrac{a+2b}{4}$ and $\log y = \dfrac{a-2b}{4}$

$\therefore \dfrac{\log x}{\log y} = \dfrac{a+2b}{a-2b}$

30. *(c)* $\log_5 10 = \dfrac{\log_{10} 10}{\log_{10} 5} = \dfrac{1}{0.70} = 1.43143$

$[\because \log_{10} 10 = 1]$

31. *(a)* $\log_3\left(1+\dfrac{1}{3}\right) + \log_3\left(1+\dfrac{1}{4}\right)$

$+ \log_3\left(1+\dfrac{1}{5}\right) + \ldots + \log_3\left(1+\dfrac{1}{24}\right)$

$= \log_3 \dfrac{4}{3} + \log_3 \dfrac{5}{4} + \log_3 \dfrac{6}{5} + \ldots + \log_3 \dfrac{25}{24}$

$= \log_3 4 - \log_3 3 + \log_3 5 - \log_3 4 + \log_3 6 - \log_3 5 + \ldots + \log_3 25 - \log_3 24$

$= -\log_3 3 + \log_3 25 = -1 + 2\log_3 5$

32. *(a)* Given, $\log(x+y) = \log x + \log y$

$\Rightarrow \log(x+y) = \log xy$

$\Rightarrow x + y = xy$

$\Rightarrow y = \dfrac{x}{x-1} = \dfrac{1.1568}{1.1568-1} = \dfrac{1.1568}{0.1568}$

$= 7.37755 = 7.3776$

33. *(d)* Given, $\log_8 x + \log_4 x + \log_2 x = 11$

$\Rightarrow \dfrac{\log x}{\log 8} + \dfrac{\log x}{\log 4} + \dfrac{\log x}{\log 2} = 11$

$\Rightarrow \dfrac{\log x}{\log 2^3} + \dfrac{\log x}{\log 2^2} + \dfrac{\log x}{\log 2} = 11$

$\Rightarrow \dfrac{\log x}{3\log 2} + \dfrac{\log x}{2\log 2} + \dfrac{\log x}{\log 2} = 11$

$\Rightarrow \dfrac{11\log x}{6\log 2} = 11 \Leftrightarrow \log x = 6\log 2$

$\Rightarrow x = 2^6, \therefore x = 64$

34. *(b)* $\therefore \log_y x^5 \log_x y^2 \log_z z^3$

$= 5\log_y x \cdot 2\log_x y \cdot 3\log_z z$

$[\because \log_a b^n = n\log_a b]$

$= 5\log_y x \cdot 2\log_x y \cdot 3 \times 1$

$= \dfrac{5\log x}{\log y} \times 2 \cdot \dfrac{\log y}{\log x} \times 3 \left[\because \log_a b = \dfrac{\log b}{\log a}\right]$

$= 5 \times 2 \times 3 = 30$

35. *(b)* $(\log_3 x)(\log_x 2x)(\log_{2x} y) = \log_x x^2$

$\Rightarrow \dfrac{\log x}{\log 3} \times \dfrac{\log 2x}{\log x} \times \dfrac{\log y}{\log 2x} = \dfrac{\log x^2}{\log x}$

$\left[\because \log_b a = \dfrac{\log a}{\log b}\right]$

$\Rightarrow \dfrac{\log y}{\log 3} = \dfrac{2\log x}{\log x} [\because \log a^b = b\log a]$

$\Rightarrow \log y = 2\log 3 \Rightarrow \log y = \log 3^2$

$[\because \log m = \log n \Rightarrow m = n]$

$\Rightarrow \log y = \log 9$

$\therefore y = 9$

36. *(b)* Given that, $\log_{10} 2$, $\log_{10}(2^x - 1)$ and $\log_{10}(2^x + 3)$ are in AP.

then,

$2\log_{10}(2^x - 1) = \log_{10} 2 + \log_{10}(2^x + 3)$

$\Rightarrow \log_{10}(2^x - 1)^2 = \log_{10}(2^{x+1} + 6)$

$\Rightarrow (2^x - 1)^2 = (2^{x+1} + 6)$

$\Rightarrow 2^{2x} + 1 - 2.2^x = 2 \cdot 2^x + 6$

$\Rightarrow 2^{2x} - 4.2^x - 5 = 0$

$\Rightarrow (2^x)^2 - 4(2^x) - 5 = 0$

$\Rightarrow y^2 - 4y - 5 = 0$

$\therefore y = 5, -1$

Hence, $2^x = -1$ or $2^x = 5$

$\therefore x = \log_2(-1)$ is not possible

or $x = \log_2(5)$

Then, $x = \log_2(5)$ is answer.

Hence, II is correct.

37. *(a)* I. $[\log_{10}(0.1)]^2 + \log_{10} 10 \cdot \log_{10} 100$

$= [-\log_{10} 10]^2 + \log_{10} 10 \cdot \log_{10} 10^2$

$= (-1)^2 + (1) \cdot 2\log_{10} 10$

$= +1 + (1) \cdot 2 = 1 + 2 = 3$

Hence, statement I is correct.

II. $\log_{10} \log_{10} 10 = \log_{10}(1) = 0 \neq 1$

Hence, statement II is incorrect.

III. $\log_{10}\sqrt{10} + \log_{10}\sqrt{10}$

$= \dfrac{1}{2}\log_{10} 10 + \dfrac{1}{2}\log_{10} 10$

$= \dfrac{1}{2} + \dfrac{1}{2} = 1$

Hence, statement III is correct.

38. *(a)* Given, $y = (a^x)^{(a^x)^{\cdots\infty}}$

$\therefore y = (a^x)^y$

On taking log both sides, we get

$\log y = y\log a^x$

$\therefore \log y = xy\log a$

39. *(c)* Let, $\log_{10} 0.0001 = x$

Then, $x = \log_{10} \dfrac{1}{(10)^4}$

$= \log_{10} 1 - \log_{10}(10)^4$

$= 0 - 4 = -4$

40. *(a)* Given, $\log_{10} a = p$ and $\log_{10} b = q$

$\log_{10}(a^p b^q) = \log_{10} a^p + \log_{10} b^q$

$\Rightarrow \log_{10}(a^p b^q) = p\log_{10} a + q\log_{10} b$

$\therefore \log_{10}(a^p b^q) = p \times p + q \times q$

$= p^2 + q^2$

41. *(a)* Given, $x^{\sqrt{x}} = \sqrt[n]{x^x}$, where x and n are positive integers.

On taking log both sides, we get

$\log x^{\sqrt{x}} = \log[\sqrt[n]{x^x}]$

$\Rightarrow \sqrt{x}\log x = \log(x^x)^{1/n} = \log x^{x/n}$

$\Rightarrow \sqrt{x}\log x = \dfrac{x}{n}\log x$

$\Rightarrow \sqrt{x}\log x - \dfrac{x}{n}\log x = 0$

$\Rightarrow \log x\left[\sqrt{x} - \dfrac{x}{n}\right] = 0$

$\because \log x \neq 0$

$\therefore \sqrt{x} - \dfrac{x}{n} = 0$

$\Rightarrow \sqrt{x} = \dfrac{x}{n}$

$\therefore \dfrac{x}{\sqrt{x}} = n$

On squaring both sides, we get

$\dfrac{x^2}{x} = n$

$\Rightarrow x^2 - nx = 0$

$\Rightarrow x(x-n) = 0$

$\therefore x = 0, x = n$

Hence, the possible solution of x is 0, n.

42. *(b)* $\dfrac{1}{5}\log_{10} 3125 - 4\log_{10} 2 + \log_{10} 32$

$= \log_{10}[(5)^5]^{1/5} - \log_{10}(2)^4 + \log_{10} 32$

$= \log_{10} 5 - \log_{10} 16 + \log_{10} 32$

$= \log_{10}\left(\dfrac{5 \times 32}{16}\right) = \log_{10} = 1$

14

ALGEBRAIC OPERATIONS

Generally (3-4) questions have been asked from this chapter. Questions which are asked from this chapter are mostly based on direct identities and factor theorem.

Algebraic Expressions

A combination of constants and variables connected by the four fundamental operations +, –, × and ÷ is called an algebraic expression.

POLYNOMIALS

A polynomial is an algebraic expression consisting of variables and coefficients, having non-negative integral powers. e.g. $x^3 + 5x^2 - 1$, $3x^3 + 4x^2 y + 17$ etc.

(i) $3x^2 + 9x - 1 + 7/x$ is not a polynomial as it contains a term, namely $7/x$, having negative integral power of variable x.

(ii) $5x^2 - 7x^{7/2} + x - 1$ is not a polynomial as the term $-7x^{7/2}$ contains rational power of variable x.

Degree of a Polynomial

The exponent of the highest degree term in a polynomial, is known as degree of polynomial.

e.g. (i) $4x^3 - 9x^2 + 7x + 9$ is a polynomial with variable x of degree 3.

(ii) $4x^3 y - 3x^2 + 2xy + 1$ is a polynomial of degree 4.

Various Types of Polynomials

Polynomials are classified on the basis of degree of polynomial and number of terms which are given below.

Based on Degree of Polynomial

(i) **Constant polynomial** A polynomial of degree zero, is called constant polynomial. e.g. $f(x) = 8$.

Note The polynomial $f(x) = 0$ is called the zero polynomial. The degree of zero polynomial is not defined.

(ii) **Linear polynomial** A polynomial of degree 1, is called linear polynomial. e.g. $9x+7$, $x-9$, $x+2$ etc.

(iii) **Quadratic polynomial** A polynomial of degree 2, is called quadratic polynomial.

The general form of quadratic polynomial in x is ax^2+bx+c, where a, b and c are real and $a \neq 0$.

e.g. $x^2-7x+12, 5y^2-7, z^2-4$ etc., are all quadratic polynomials.

(iv) **Cubic polynomial** A polynomial of degree 3, is called cubic polynomial.

The general form of a cubic polynomial in x is ax^3+bx^2+cx+d, where a, b, c and d are real and $a \neq 0$.

e.g. $3x^3+2x^2+7x-1$, $8y^3+5y+2$, x^3-8 etc., are all cubic polynomials.

(v) **Biquadratic polynomial** A polynomial of degree 4, is called a biquadratic polynomial.

The general form of a biquadratic polynomial in x is $ax^4+bx^3+cx^2+dx+e$, where a, b, c, d and e are real numbers and $a \neq 0$.

e.g. $6x^4+3x^3+2x^2+x+2$, x^4-8 etc.,

are all biquadratic polynomials.

Based on Number of Terms

(i) **Monomial** A polynomial containing only one term is called a monomial. e.g. $6x^2$

(ii) **Binomial** A polynomial containing two terms is called a binomial. e.g. $5x^3+7xy$

(iii) **Trinomial** A polynomial containing three terms is called a trinomial. e.g. $7x^2y+5x+6$

FACTORISATION

To express a polynomial as the product of other polynomials of degree less than that of the given polynomial is called as factorisation.

e.g. $\quad x^2-49 = x^2-7^2 = (x-7)(x+7)$

Important Identities

- $(a^2-b^2) = (a+b)(a-b)$
- $(a+b)^2 = a^2+b^2+2ab$ and $(a-b)^2 = a^2+b^2-2ab$
- $(a+b)^2-(a-b)^2 = 4ab$
- $(a+b)^2+(a-b)^2 = 2(a^2+b^2)$
- $(a+b)^3 = a^3+b^3+3ab(a+b)$
- $(a-b)^3 = a^3-b^3-3ab(a-b)$
- $(a^3+b^3) = (a+b)(a^2+b^2-ab)$
- $(a^3-b^3) = (a-b)(a^2+b^2+ab)$
- $(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ac)$
- $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-ac-bc)$
- If $a+b+c=0$, then $a^3+b^3+c^3=3abc$
- $a^4+a^2b^2+b^4 = (a^2+ab+b^2)(a^2-ab+b^2)$

Methods of Factorisation

These methods are given below

Factorisation by Taking Out the Common Factor

If each term of an expression has a common factor, take out the common factors.

EXAMPLE 1. Factorise value of $4(3a-2b)^2-5(3a-2b)$ is

a. $(3a-2b)(12a-8b-5)$ b. $(2a-3b)(12a-8b)$
c. $(2a-3b)(8a-12b-5)$ d. None of these

Sol. a. $4(3a-2b)^2-5(3a-2b) = (3a-2b)[4(3a-2b)-5]$
$= (3a-2b)[12a-8b-5]$

Factorisation by Grouping

Sometimes in a given polynomial, it is not possible to take out a common factor directly. However, on rearranging the terms of the polynomial and grouping them such that all the terms have a common factor, the polynomial can be easily factorised.

EXAMPLE 2. Factorise $x^2+y-xy-x$

a. $(x-y)(x+1)$ b. $(x-y)(x^2-1)$
c. $(x+y)(x^2+1)$ d. $(x-y)(x-1)$

Sol. d. **Method I** $x^2+y-xy-x = x^2-x+y-xy$
$= x(x-1)+y(1-x) = x(x-1)-y(x-1) = (x-1)(x-y)$

Method II $x^2+y-xy-x = x^2-xy+y-x$
$= x(x-y)-1(x-y) = (x-y)(x-1)$

EXAMPLE 3. Factorise $x^2+\dfrac{4}{x^2}+4-2x-\dfrac{4}{x}$

a. $\left(x-\frac{2}{x}\right)\left(x+\frac{2}{x}+2\right)$ b. $\left(x-\frac{2}{x}\right)\left(x+\frac{2}{x}-2\right)$
c. $\left(x+\frac{2}{x}\right)\left(x+\frac{2}{x}-2\right)$ d. None of these

Sol. c. $\left(x^2+\frac{4}{x^2}+4-2x-\frac{4}{x}\right) = \left(x^2+\frac{4}{x^2}+4\right)-2\left(x+\frac{2}{x}\right)$
$= \left(x+\frac{2}{x}\right)^2-2\left(x+\frac{2}{x}\right) = \left(x+\frac{2}{x}\right)\left(x+\frac{2}{x}-2\right)$

Factorisation of Perfect Square Polynomials

If the polynomial is given as the perfect square quadratic polynomial, then use the following Identities to factorise it.

- $a^2 + 2ab + b^2 = (a+b)^2$
- $a^2 - 2ab + b^2 = (a-b)^2$

EXAMPLE 4. Factorise $16x^2 - 8x + 1$

a. $(2x-1)^2$ b. $(x-4)^2$ c. $(x-2)^2$ d. $(4x-1)^2$

Sol. d. $16x^2 - 8x + 1 = (4x)^2 - 2(4x) + 1^2 = (4x-1)^2$

$= (4x-1)(4x-1)$

EXAMPLE 5. Factorise $x^2 - 2\sqrt{5}x + 5$

a. $(x\sqrt{5}-1)^2$ b. $(x-5)^2$ c. $(x+\sqrt{5})(x-\sqrt{5})$ d. $(x-\sqrt{5})^2$

Sol. d. $x^2 - 2\sqrt{5}x + 5 = x^2 - 2\sqrt{5}x + (\sqrt{5})^2 = (x-\sqrt{5})^2$

$= (x-\sqrt{5})(x-\sqrt{5})$

Factorising the Difference of Two Squares

In case, the polynomial is given in the form of $a^2 - b^2$, to evaluate it, following identity is applied.

- $a^2 - b^2 = (a-b)(a+b)$

EXAMPLE 6. Factorise $2a^5 - 32a$

a. $(a^2+4)(a-2)(a+2)$ b. $2a(a^2+4)(a-2)(a+2)$
c. $2a(a^2+4)$ d. None of these

Sol. b. $2a^5 - 32a = 2a(a^4 - 16)$

$= 2a[(a^2)^2 - (4)^2] = 2a(a^2+4)(a^2-4)$

$= 2a(a^2+4)(a^2-2^2) = 2a(a^2+4)(a-2)(a+2)$

Factorisation of Quadratic Polynomials

Quadratic polynomials of the type $ax^2 + bx + c$, where $a \neq 0$, a and b are coefficients of x^2 and x respectively and c is constant, can be factorised by splitting the middle term. We find two numbers p and q such that

$p + q = b$ and $pq = ac$, then

$$ax^2 + bx + c = ax^2 + (p+q)x + c = ax^2 + px + qx + c$$

EXAMPLE 7. Factorise $x^2 + 9x + 14$

a. $(x+2)(x+7)$ b. $(x-2)(x-7)$
c. $(x+2)(x-7)$ d. $(x-2)(x+7)$

Sol. a. $x^2 + 9x + 14 = x^2 + (7+2)x + 14 = x^2 + 7x + 2x + 14$

$= x(x+7) + 2(x+7) = (x+7)(x+2)$

EXAMPLE 8. Factorise $4\sqrt{3}x^2 + 5x - 2\sqrt{3}$

a. $(2x+\sqrt{3})(4x-\sqrt{3})$ b. $(2x-\sqrt{3})(4x+\sqrt{3})$
c. $(\sqrt{3}x+2)(4x-\sqrt{3})$ d. None of these

Sol. c. $4\sqrt{3}x^2 + 5x - 2\sqrt{3} = 4\sqrt{3}x^2 + 8x - 3x - 2\sqrt{3}$

$[\because 4\sqrt{3} \times (-2\sqrt{3}) = -24 = 8 \times (-3)]$

$= 4x(\sqrt{3}x + 2) - \sqrt{3}(\sqrt{3}x + 2)$

$= (\sqrt{3}x + 2)(4x - \sqrt{3})$

Factorisation of Sum and Difference of Cubes

In case, the polynomial is given in the form of $a^3 + b^3$ or $a^3 - b^3$. To factorise it, following identities can be applied.

- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

EXAMPLE 9. Factorise $(2x+3y)^3 - (2x-3y)^3$

a. $18(4x^2+3y^2)$ b. $18y(3x^2+4y^2)$
c. $xy(4x^2+3y^2)$ d. $18y(4x^2+3y^2)$

Sol. d. $(2x+3y)^3 - (2x-3y)^3$

Put $2x + 3y = a$ and $2x - 3y = b$

$\Rightarrow$ $a^3 - b^3 = (a-b)(a^2 + ab + b^2) = (2x + 3y - 2x + 3y)$

$[(2x+3y)^2 + (2x+3y)(2x-3y) + (2x-3y)^2]$

$= 6y[4x^2 + 9y^2 + 12xy + 4x^2 - 9y^2 + 4x^2 + 9y^2 - 12xy]$

$= 6y[12x^2 + 9y^2] = 18y[4x^2 + 3y^2]$

Hence, $(2x+3y)^3 - (2x-3y)^3 = 18y[4x^2 + 3y^2]$

Factorisation of Polynomial of form $a^3 + b^3 + c^3 - 3abc$

Here, it is easy to use.

- $a^3 + b^3 + c^3 - 3abc$
  $= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ac)$
- If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

EXAMPLE 10. Factorise $x^3 + 27y^3 + 8x^3 - 18xyz$

a. $(x+3y+2z)(x^2+9y^2+4z^2+3xy+6yz+2xz)$
b. $(x+3y+2z)(x^2+9y^2+4z^2-3xy-6yz-2xz)$
c. $(x-3y-2z)(x^2+9y^2-4z^2-3xy+6yz+2xz)$
d. None of the above

Sol. b. $x^3 + 27y^3 + 8z^3 - 18xyz$

$= x^3 + (3y)^3 + (2z)^3 - 3(x)(3y)(2z)$

$= (x + 3y + 2z)[x^2 + 9y^2 + 4z^2 - x(3y) - 3y(2z) - x(2z)]$

$= (x + 3y + 2z)([x^2 + 9y^2 + 4z^2 - 3xy - 6yz - 2xz])$

EXAMPLE 11. Factorise
$(2x-3y)^3 + (3y-5z)^3 + (5z-2x)^3$

a. $3(2x-3y)(3y-5z)(5z-2x)$
b. $(2x-3y)(3y-5z)(5z-2x)$
c. $3(2x-3y)(3y+5z)(5z-2x)$
d. None of the above

Sol. **a.** Here, $2x-3y=a$, $3y-5z=b$ and $5z-2x=c$

$\because \quad a+b+c=0 \Rightarrow a^3+b^3+c^3=3abc$

$\therefore \quad (2x-3y)^3+(3y-5z)^3+(5z-2x)^3$

$= 3(2x-3y)(3y-5z)(5z-2x)$

Remainder Theorem

Let $p(x)$ be a polynomial of degree n greater than or equal to 1 (i.e. $n \geq 1$) and a be any real number. If $p(x)$ is divided by the linear polynomial $(x-a)$, then the remainder is $p(a)$. Remainder can be evaluated by substituting $x-a=0$, i.e. $x=a$ in $p(x)$.

EXAMPLE 12. The remainder when $12x^3-13x^2-5x+9$ is divided by $(3x+2)$ is

a. 1 **b.** 2 **c.** 3 **d.** 0

Sol. **c.** Let $p(x)=12x^3-13x^2-5x+9$ and $q(x)=3x+2$

When $p(x)$ is divided by $(3x+2)$, then remainder is $p\left(\frac{-2}{3}\right)$.

Now, $p\left(\frac{-2}{3}\right)=12\left(\frac{-2}{3}\right)^3-13\left(\frac{-2}{3}\right)^2-5\left(\frac{-2}{3}\right)+9$

$=12\times\left(\frac{-8}{27}\right)-13\times\frac{4}{9}+\frac{10}{3}+9=3$

Hence, the required remainder is 3.

Factor Theorem

Let $p(x)$ be a polynomial, if degree $n \geq 1$ and a be any real number. Then,

(i) If $p(a)=0$, then $(x-a)$ is a factor of $p(x)$.

(ii) If $(x-a)$ is a factor of $p(x)$, then $p(a)=0$.

EXAMPLE 13. For what values of k will $4x^5+9x^4-7x^3-5x^2-4kx+3k^2$ contain $x-1$ as a factor?

a. $3, -\frac{1}{2}$ **b.** $3, -1$ **c.** $0, \frac{1}{3}$ **d.** $1, \frac{1}{3}$

Sol. **d.** Given, $P(x)=4x^5+9x^4-7x^3-5x^2-4kx+3k^2$

Since, $(x-1)$ is a factor of $P(x)$. $\therefore \quad P(1)=0$

$\Rightarrow 4\times(1)^5+9\times(1)^4-7\times(1)^3-5\times(1)^2-4\times k\times(1)+3\times k^2=0$

$\Rightarrow \quad 3k^2-4k+1=0 \Rightarrow 3k^2-3k-k+1=0$

$\Rightarrow \quad (3k-1)(k-1)=0 \quad \Rightarrow k=\frac{1}{3}, 1$

Division Synthetic Algorithm

If $p(x)$ and $g(x)$ any two polynomials with $g(x)\neq 0$, then we can find polynomials $q(x)$ and $r(x)$ such that

$$p(x)=g(x)\times q(x)+r(x)$$

i.e. Dividend = (Divisor × Quotient) + Remainder

where, $r(x)=0$ or degree of $r(x) <$ degree of $g(x)$, then we say that $p(x)$ divided by $g(x)$, gives $q(x)$ as quotient and $r(x)$ as remainder.

If the remainder $r(x)$ is zero, we say that divisor $g(x)$ is a factor of $p(x)$.

EXAMPLE 14. If two factors of $a^4-2a^3-9a^2+2a+8$ are $(a+1)$ and $(a-1)$, then what are the other two factors?

a. $(a-2)$ and $(a+4)$ **b.** $(a+2)$ and $(a+4)$

c. $(a+2)$ and $(a-4)$ **d.** $(a-2)$ and $(a-4)$

Sol. **c.** Let $f(a)=a^4-2a^3-9a^2+2a+8$

and $(a+1)$ and $(a-1)$ are two factors of $f(a)$.

Now, by division method,

$$\begin{array}{l} a^2-1)\ a^4-2a^3-9a^2+2a+8\ (a^2-2a-8 \\ \quad a^4 \qquad\quad - a^2 \\ \hline \quad -2a^3-8a^2+2a+8 \\ \quad -2a^3 \qquad\quad 2a \\ \hline \qquad -8a^2 \quad +8 \\ \qquad -8a^2 \quad +8 \\ \hline \qquad\quad \times \end{array}$$

So, the required factor of $f(a)$ is (a^2-2a-8).

Further factorise it by splitting the middle term, i.e.

$a^2-2a-8=a^2-4a+2a-8$

$=a(a-4)+2(a-4)=(a+2)(a-4)$

Hence, $(a+2)$ and $(a-4)$ are other two factors of $f(a)$.

Factorisation of Polynomials Using Factor Theorem

To factorise the polynomial, follow the following steps

(i) Let $p(x)$ be a given polynomial.

(ii) find all the possible factors of constant term in $p(x)$.

(iii) Take any one of the factors, say (α) and find $p(\alpha)$. If $p(\alpha)=0$ the $(x-\alpha)$ is a factor of $p(x)$.

(iv) Divide the polynomial $p(x)$ by $(x-\alpha)$ to find its all other factors.

EXAMPLE 15. Factorise $2x^3-5x^2-19x+42$.

a. $(x-2)(2x-7)(x+3)$ **b.** $(x+2)(2x+7)(x-3)$

c. $(x-1)(3x-7)(x-4)$ **d.** $(2x+3)(4x-7)(x-1)$

Sol. **a.** Let $p(x)=2x^3-5x^2-19x+42$

Hence, constant term is 42 and its all factors are $=\pm 1$, ± 2, ± 3, ± 6, ± 7, ± 14, ± 21 and ± 42.

At $x=1$, $\quad p(1)=2(1)^3-5(1)^2-19+42$

$=2-5-19+42=20\neq 0$

So, $(x-1)$ is not a factor of $p(x)$.

At $x=2$, $p(2)=2(2)^3-5(2)^2-19(2)+42=58-58=0$

So, $(x-2)$ is a factor of $p(x)$.

Now, on dividing $p(x)$ by $(x-2)$, we get $(2x^2-x-2)$ as quotient i.e. other factor of $p(x)$.

So, $\quad p(x)=(x-2)[2x^2-x-21]$

$=(x-2)(2x^2-7x+6x-21)$

$=(x-2)[x(2x-7)+3(2x-7)]$

$\therefore \quad p(x)=(x-2)(2x-7)(x+3)$

PRACTICE EXERCISE

1. If $x = 5, y = 3$ and $z = 2$, then the value of $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$ is

(a) 125 (b) 0 (c) –25 (d) 10

2. The factors of $x^2 - 2\sqrt{3}x + 3$ are

(a) $(x + \sqrt{3})^2$ (b) $(x - \sqrt{3})^2$

(c) $(x + \sqrt{3})(x - \sqrt{3})$ (d) $(x + 2)(x + \sqrt{3})$

3. The factors of $(a^4b^4 - 16c^4)$ are

(a) $4(a^2b^2 + c^2)(ab - 2c)(ab + 2c)$

(b) $(a^2b^2 - 4c^2)(ab + 2c)^2$

(c) $(a^2b^2 + 4c^2)(ab + 2c)(ab - 2c)$

(d) $(a^2b^2 - 4c^2)^2(ab + 2c)(ab + 4c)$

4. The factors of $(x^8 - y^8)$ are

(a) $(x^4 + y^4)(x^2 + y^2)(x + y)(x - y)$

(b) $(x^2 + y^2)^2(x + y)(x - y)$

(c) $(x^4 + y^4)(x^2 + y^2)^2$

(d) $(x^2 + y^2)(x - y)^2$

5. The factors of $\left(a^2 + a + \frac{1}{4}\right)$ are

(a) $\left(a + \frac{1}{2}\right)^2$ (b) $\left(a + \frac{1}{2}\right)(a + 2)$

(c) $\left(a + \frac{1}{2}\right)\left(a - \frac{1}{2}\right)$ (d) $\left(a - \frac{1}{2}\right)^2$

6. The factors of $8 - 4x - 2x^3 + x^4$ are

(a) $(2 - x)(4 - x^3)$ (b) $(2 + x)(4 - x^3)$

(c) $(2 + x)(3 - x^3)$ (d) $(2 - x)(x^3 - 4)$

7. The factors of $(a^2 - b^2 - 4ac + 4c^2)$ are

(a) $(a + 2c + b)(a - 2c - b)$ (b) $(a - 2c + b)(a - 2c - b)$

(c) $(a - 2b + c)(a + b + 2c)$ (d) $(a - 2b)(a + 2b + 2c)$

8. What are the factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$?

(a) $(x - 2y - 4)$ and $(x - 2y + 2)$

(b) $(x - y + 2)$ and $(x - 4y + 4)$

(c) $(x - y + 2)$ and $(x - 4y - 4)$

(d) $(x + 2y - 4)$ and $(x + 2y + 2)$

9. Factorise $a^3 - \frac{1}{a^3} - 2a + \frac{2}{a}$

(a) $\left(a + \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2}\right) - 2\left(a - \frac{1}{a}\right)$

(b) $\left(a - \frac{1}{a}\right)\left(a^2 + \frac{1}{a^2} + 1\right) + 2\left(a - \frac{1}{a}\right)$

(c) $\left(a + \frac{1}{a}\right)\left(a - \frac{1}{a}\right)$

(d) None of the above

10. What are the factors of $\left(\frac{1}{3}x^2 - 2x - 9\right)$?

(a) $\frac{1}{3}(x - 9)(x + 3)$ (b) $\frac{1}{3}(x - 9)(x - 3)$

(c) $\frac{1}{3}(x + 9)(x + 3)$ (d) $\frac{1}{3}(x + 9)(x - 3)$

11. What are the factors of $(6\sqrt{3}x^2 - 47x + 5\sqrt{3})$?

(a) $(3\sqrt{3}x + 5\sqrt{3})(2x - 5\sqrt{3})$

(b) $(3\sqrt{3}x - 5\sqrt{3})(x\sqrt{3}x + 1)$

(c) $(2x + 5\sqrt{3})(3\sqrt{3}x + 1)$

(d) $(2x - 5\sqrt{3})(3\sqrt{3}x - 1)$

12. The factors of $(8a^3 + 125b^3 - 64c^3 + 120abc)$ are

(a) $(2a + 5b - 4c)(2a + 5b + 4c)$

(b) $(2a - 5b - 4c)(2a + 5b + 4c)$

(c) $(2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$

(d) $(2a + 5b + 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$

13. If $x^{1/3} + y^{1/3} + z^{1/3} = 0$, then

(a) $x + y + z = 0$ (b) $(x + y + z)^3 = 27xyz$

(c) $x + y + z = 3xyz$ (d) $x^3 + y^3 + z^3 = 0$

14. If $x + \frac{1}{x} = \sqrt{5}$, then the value of $x^3 + \frac{1}{x^3}$ is

(a) $8\sqrt{5}$ (b) $2\sqrt{5}$ (c) $5\sqrt{5}$ (d) $7\sqrt{5}$

15. If $\left(x + \frac{1}{x}\right) = 6$, then $\left(x^2 + \frac{1}{x^2}\right)$ is equal to

(a) 32 (b) 38 (c) 34 (d) 44

16. If $x - \frac{1}{x} = \frac{1}{3}$, then what is $9x^2 + \frac{9}{x^2}$ equal to?

(a) 18 (b) 19 (c) 20 (d) 21

17. If $a + b + c = 6$ and $a^2 + b^2 + c^2 = 26$, then what is $ab + bc + ca$ equal to?

(a) 0 (b) 2 (c) 4 (d) 5

18. If $(x^4 + x^{-4}) = 322$, then what is one of the value of $(x - x^{-1})$?

(a) 18 (b) 16 (c) 8 (d) 4

19. If $a + b + c = 0$, then what is the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$?

(a) – 3 (b) 0 (c) 1 (d) 3

20. If $a + b + c = 10$ and $ab + bc + ca = 31$, then the value of $a^2 + b^2 + c^2$ is

(a) 48 (b) 38 (c) 28 (d) 18

21. The factors of $x^2 + \frac{1}{4x^2} + 1 - 2x - \frac{1}{x}$ are

(a) $\left(x - \frac{1}{2x}\right)\left(x - \frac{1}{2x} + 2\right)$ (b) $\left(x + \frac{1}{2x}\right)\left(x + \frac{1}{2x} - 2\right)$

(c) $\left(x - \frac{1}{2x}\right)\left(x - \frac{1}{2x} - 2\right)$ (d) $\left(x + \frac{1}{2x}\right)\left(x - \frac{1}{2x} - 2\right)$

22. The factors of $(a^2 - b^2)(c^2 - d^2) - 4abcd$ are

(a) $(ac + bd + bc + ad)(ac - bd - bc - ad)$
(b) $(ac - bd - bc + ad)(ac + bd + dc + ab)$
(c) $(ac - bd + bc + ad)(ac - bd - bc - ad)$
(d) $(ac - bd - bc - ad)(ac + bd + dc + ab)$

23. If 'a' is an integer such that $a + \frac{1}{a} = \frac{17}{4}$, then the value of $\left(a - \frac{1}{a}\right)$ is

(a) 4 (b) $\frac{13}{4}$ (c) $\frac{17}{4}$ (d) $\frac{15}{4}$

24. $x^4 + 4y^4$ is divisible by which one of the following?

(a) $(x^2 + 2xy + 2y^2)$ (b) $(x^2 + 2y^2)$
(c) $(x^2 - 2y^2)$ (d) None of these

25. If $x(x + y + z) = 9$, $y(x + y + z) = 16$ and $z(x + y + z) = 144$, then x equal to

(a) $\frac{9}{5}$ (b) $\frac{9}{7}$ (c) $\frac{9}{13}$ (d) $\frac{16}{13}$

26. If $x = (b - c)(a - d)$, $y = (c - a)(b - d)$ and $z = (a - b)(c - d)$, then what is $x^3 + y^3 + z^3$ equal to?

(a) xyz (b) $2xyz$ (c) $3xyz$ (d) $-3xyz$

27. If the expression $(px^3 + x^2 - 2x - q)$ is divisible by $(x - 1)$ and $(x + 1)$, what are the values of p and q respectively?

(a) 2, – 1 (b) – 2, 1 (c) – 2, – 1 (d) 2, 1

28. If the expression $px^3 + 3x^2 - 3$ and $2x^3 - 5x + p$ when divided by $x - 4$ leave the same remainder, then what is the value of p?

(a) – 1 (b) 1 (c) – 2 (d) 2

29. Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n$, where $a_0, a_1, a_2 \ldots a_n$ are constants. If $f(x)$ is divided by $ax - b$, the remainder is

(a) $f\left(\frac{b}{a}\right)$ (b) $f\left(\frac{-b}{a}\right)$ (c) $f\left(\frac{a}{b}\right)$ (d) $f\left(\frac{-a}{b}\right)$

30. Which one of the following statements is correct?

(a) Remainder theorem is a special case of factor theorem
(b) Factor theorem is a special case of remainder theorem
(c) Factor theorem and remainder theorem are two independent results
(d) None of the above

31. What is $x(y - z)(y + z) + y(z - x)(z + x) + z(x - y)(x + y)$ equal to?

(a) $(x + y)(y + z)(z + x)$ (b) $(x - y)(x - z)(z - y)$
(c) $(x + y)(z - y)(x - z)$ (d) $(y - x)(z - y)(x - z)$

32. If the remainder of the polynomial $a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ when divided by $(x - 1)$ is 1, then which one of the following is correct?

(a) $a_0 + a_2 + \cdots = a_1 + a_3 + \cdots$
(b) $a_0 + a_2 + \cdots = 1 + a_1 + a_3 + \cdots$
(c) $1 + a_0 + a_2 + \cdots = -(a_1 + a_3 + \cdots)$
(d) $1 - a_0 - a_2 - \cdots = a_1 + a_3 + \cdots$

33. If u, v and w are real numbers such that $u^3 - 8v^3 - 27w^3 = 18uvw$, then which one of the following is correct?

(a) $u - v + w = 0$ (b) $u = -v = -w$
(c) $u - 2v = 3w$ (d) $u + 2v = -3w$

34. The value of $(a^{1/8} + a^{-1/8})(a^{1/8} - a^{-1/8})(a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})$ is

(a) $(a + a^{-1})$ (b) $(a - a^{-1})$ (c) $(a^2 + a^{-2})$ (d) $(a^2 - a^{-2})$

35. The polynomial $f(x) = x^4 - 2x^3 + 3x^2 - ax + b$, when divided by $(x - 1)$ and $(x + 1)$ leaves the remainders 5 and 19, respectively. The values of a and b are

(a) $a = 4, b = 3$ (b) $a = 3, b = 4$
(c) $a = 5, b = 8$ (d) $a = 9, b = 7$

36. If the expression $ax^2 + bx + c$ is equal to 4 when $x = 0$, leaves a remainder 4 when divided by $x + 1$ and a remainder 6 when divided by $x + 2$., then the values of a, b and c are respectively:

(a) 1, 1, 4 (b) 2, 2, 4 (c) 3, 3, 4 (d) 4, 4, 4

37. If $(x^{3/2} - xy^{1/2} + x^{1/2}y - y^{3/2})$ is divided by $x^{1/2} - y^{1/2}$ the quotient is

(a) $x - y$ (b) $x + y$ (c) $x^{1/2} + y^{1/2}$ (d) $x^2 - y^2$

38. If, $9x^2 + 3px + 6q$ when divided by $3x + 1$ leaves a remainder $-\frac{3}{4}$ and $qx^2 + 4px + 7$ is exactly divisible by $x + 1$, then the values of p and q respectively are

(a) $0, 7/4$ (b) $-7/4, 0$
(c) $\frac{7}{4}, 0$ (d) None of these

39. If $a + b + c = 1$, then which of the following statements are true?

I. $(a + b)(b + c)(c + a) = bc + ac + ab - abc$

II. $a^2 + b^2 - c^2 + 2ab = a + b - c$

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

40. If $x + \frac{1}{x} = 4$, then the value of expression

I. $x^3 + \frac{1}{x^3} = 52$ II. $x = 2 + \sqrt{3}$

Which of the following is correct?

(a) Only I (b) Only II
(c) Neither I nor II (d) Both I and II

PREVIOUS YEARS' QUESTIONS

41. What is the value of k that $(2x-1)$ may be a factor of $4x^4-(k-1)x^3+kx^2-6x+1$? 2012 I
(a) 8 (b) 9 (c) 12 (d) 13

42. $x^4+xy^3+x^3y+xz^3+y^4+yz^3$ is divisible by 2012 I
(a) Only $(x-y)$
(b) Only $(x^3+y^3+z^3)$
(c) Both $(x+y)$ and $(x^3+y^3+z^3)$
(d) None of the above

43. If x^3+5x^2+10k leaves remainder $-2x$ when divided by x^2+2, then what is the value of k? 2012 I
(a) -2 (b) -1 (c) 1 (d) 2

44. If $x+\frac{1}{x}=a$, then what is the value of $x^3+x^2+\frac{1}{x^3}+\frac{1}{x^2}$? 2012 I
(a) a^3+a^2 (b) a^3+a^2-5a
(c) a^3+a^2-3a-2 (d) a^3+a^2-4a-2

45. If $p(x)$ is a common multiple of degree 6 of the polynomials $f(x)=x^3+x^2-x-1$ and $g(x)=x^3-x^2+x-1$, then which one of the following is correct? 2012 I
(a) $p(x)=(x-1)^2(x+1)^2(x^2+1)$
(b) $p(x)=(x-1)(x+1)(x^2+1)^2$
(c) $p(x)=(x-1)^3(x+1)(x^2+1)$
(d) $p(x)=(x-1)^2(x^4+1)$

46. What is the value of k which will make the expression $4k^2+12k+k$ a perfect square? 2012 II
(a) 5 (b) 7 (c) 8 (d) 9

47. The factors of $5px-10qy+2rpx-4qry$ is/are 2013 I
(a) Only $(5+2r)$
(b) Only $(px-2qy)$
(c) Both $(5+2r)$ and $(px-2qy)$
(d) Neither $(5+2r)$ nor $(px-2qy)$

48. $(a+1)^4-a^4$ is divisible by 2013 I
(a) $-2a^2+2a-1$ (b) $2a^3-2a-1$
(c) $2a^3-2a+1$ (d) $2a^2+2a+1$

49. One of the factors of the polynomial $x^4-7x^3+5x^2-6x+81$ is 2013 I
(a) $x+2$ (b) $x-2$ (c) $x+3$ (d) $x-3$

50. If the expression x^3+3x^2+4x+k has a factor $x+5$, then what is the value of k? 2013 II
(a) -70 (b) 70 (c) 48 (d) -48

51. The quantity which must be added to $(1-x)(1+x^2)$ to obtain x^3 is 2013 II
(a) $2x^3+3x^2+x+1$ (b) $2x^3+x^2+x-1$
(c) $2x^3-x^2+x-1$ (d) $-x^3+x-1$

52. $x(y^2-z^2)+y(z^2-x^2)+z(x^2-y^2)$ is divisible by 2013 II
(a) Only $(y-z)$
(b) Only $(z-x)$
(c) Both $(y-z)$ and $(z-x)$
(d) Neither $(y-z)$ nor $(z-x)$

53. Consider the following statements
I. $x+3$ is the factor of x^3+2x^2+3x+8.
II. $x-2$ is the factor of x^3+2x^2+3x+8.
Which of the statement(s) given above is/are correct? 2013 II
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

54. $x^3+6x^2+11x+6$ is divisible by 2014 I
(a) Only $(x+1)$ (b) Only $(x+2)$
(c) Only $(x+3)$ (d) All of these

55. What should be added to the $x(x+a)(x+2a)(x+3a)$, so that the sum be a perfect square? 2014 I
(a) $9a^2$ (b) $4a^2$
(c) a^4 (d) None of these

56. If $\left(x^2+\frac{1}{x^2}\right)=\frac{17}{4}$, then what is $\left(x^3-\frac{1}{x^3}\right)$ equal to? 2014 I
(a) $\frac{75}{16}$ (b) $\frac{63}{8}$ (c) $\frac{95}{8}$ (d) None of these

57. What is the remainder when x^5-5x^2+125 is divided by $x+5$? 2014 II
(a) 0 (b) 125 (c) -3125 (d) 3125

58. If $ax+by-2=0$ and $axby=1$, where $a\neq 0, b\neq 0$, then what is (a^2x+b^2y) equal to? 2014 II
(a) $a+b$ (b) $2ab$ (c) a^3+b^3 (d) a^4+b^4

59. If $(x+k)$ is the common factor of x^2+ax+b and x^2+cx+d, then what is k equal to? 2014 II
(a) $(d-b)/(c-a)$ (b) $(d-b)/(a-c)$
(c) $(d+b)/(c+a)$ (d) $(d-b)/(c+a)$

60. Consider the following statements:
I. $(a-b-c)$ is one of the factors of $3abc+b^3+c^3-a^3$.
II. $(b+c-1)$ is one of the factors of $3bc+b^3+c^3-1$.
Which of the statement(s) given above is/are correct? 2014 II
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

61. For what value of k is $(x-5)$ a factor of $x^3-3x^2+kx-10$? 2015 I
(a) -8 (b) 4 (c) 2 (d) 1

62. If $x+y+z=0$, then $x^3+y^3+z^3+3xyz$ is equal to 2015 I
(a) 0 (b) $6xyz$ (c) $12xyz$ (d) xyz

63. The expression $x^3q^2 - x^3pt + 4x^2pt - 4x^2q^2 + 3xq^2 - 3xpt$ is divisible by **2015 I**

(a) Only $(x-1)$ (b) Only $(x-3)$
(c) Both $(x-1)$ and $(x-3)$ (d) Neither $(x-1)$ nor $(x-3)$

64. Which one of the following is correct? **2015 II**

(a) $(x+2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
(b) $(x+2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$
(c) $(x-2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
(d) $(x-2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$

65. For what value of k $(x+1)$ is a factor of $x^3 + kx^2 - x + 2$? **2016 I**

(a) 4 (b) 3 (c) 1 (d) -2

66. If the polynomial $x^6 + px^5 + qx^4 - x^2 - x - 3$ is divisible by $(x^4 - 1)$, then the value of $p^2 + q^2$ is **2016 I**

(a) 1 (b) 9 (c) 10 (d) 13

67. If the linear factors of $ax^2 - (a^2+1)x + a$ are p and q, then $p + q$ is equal to **2016 I**

(a) $(x-1)(a+1)$ (b) $(x+1)(a+1)$
(c) $(x-1)(a-1)$ (d) $(x+1)(a-1)$

68. If $\sqrt{\dfrac{x}{y}} = \dfrac{10}{3} - \sqrt{\dfrac{y}{x}}$ and $x - y = 8$, then the value of xy is equal to **2016 I**

(a) 36 (b) 24 (c) 16 (d) 9

ANSWERS

1	b	2	b	3	c	4	a	5	a	6	a	7	b	8	a	9	d	10	a
11	d	12	c	13	b	14	b	15	c	16	b	17	d	18	d	19	d	20	b
21	b	22	c	23	d	24	a	25	c	26	c	27	d	28	b	29	a	30	b
31	b	32	d	33	c	34	b	35	c	36	a	37	b	38	c	39	c	40	d
41	d	42	c	43	c	44	c	45	a	46	d	47	c	48	d	49	d	50	b
51	c	52	c	53	d	54	d	55	c	56	b	57	c	58	a	59	a	60	c
61	a	62	b	63	c	64	c	65	d	66	c	67	a	68	d				

HINTS AND SOLUTIONS

1. *(b)* $x^2 + y^2 + z^2 - 2xy + 2yz - 2zx$
$= (x - y - z)^2 = (5 - 3 - 2)^2 = 0$
$[\because x = 5, y = 3 \text{ and } z = 2]$

2. *(b)* $x^2 - 2\sqrt{3}\,x + 3$
$= x^2 - \sqrt{3}\,x - \sqrt{3}\,x + 3$
$= x(x - \sqrt{3}) - \sqrt{3}\,(x - \sqrt{3})$
$= (x - \sqrt{3})(x - \sqrt{3}) = (x - \sqrt{3})^2$

3. *(c)* $(a^4b^4 - 16c^4) = [(a^2b^2)^2 - (4c^2)^2]$
$= [a^2b^2 + 4c^2][a^2b^2 - 4c^2]$
$[\because a^2 - b^2 = (a-b)(a+b)]$
$= [a^2b^2 + 4c^2][(ab)^2 - (2c)^2]$
$= [a^2b^2 + 4c^2][(ab + 2c)(ab - 2c)]$

4. *(a)* $x^8 - y^8 = [(x^4)^2 - (y^4)^2]$
$= [x^4 + y^4][x^4 - y^4]$
$= [x^4 + y^4][(x^2)^2 - (y^2)^2]$
$= (x^4 + y^4)(x^2 + y^2)(x^2 - y^2)$
$= (x^4 + y^4)(x^2 + y^2)(x + y)(x - y)$

5. *(a)* $\left(a^2 + a + \dfrac{1}{4}\right) = a^2 + \dfrac{1}{2}a + \dfrac{1}{2}a + \dfrac{1}{4}$
$= a\left(a + \dfrac{1}{2}\right) + \dfrac{1}{2}\left(a + \dfrac{1}{2}\right)$
$= \left(a + \dfrac{1}{2}\right)\left(a + \dfrac{1}{2}\right) = \left(a + \dfrac{1}{2}\right)^2$

6. *(a)* $8 - 4x - 2x^3 + x^4$
$= 4(2 - x) - x^3(2 - x)$
$= (2 - x)(4 - x^3)$

7. *(b)* $(a^2 - b^2 - 4ac + 4c^2)$
$= (a^2 - 4ac + 4c^2) - b^2$
$[(a)^2 - 2(2c)(a) + (2c)^2] - b^2$
$= (a - 2c)^2 - b^2$
$= (a - 2c - b)(a - 2c + b)$

8. *(a)* $x^2 + 4y^2 + 4y - 4xy - 2x - 8$
$= x^2 - 4xy + 4y^2 - 2x + 4y - 8$
$= (x - 2y)^2 - 2(x - 2y) - 8$
Let $x - 2y = z$
$\therefore$ $z^2 - 2z - 8 = z^2 - 4z + 2z - 8$
$= z(z - 4) + 2(z - 4)$
$= (z + 2)(z - 4)$
Now, put $z = x - 2y$
$\therefore$ $(x - 2y + 2)$
and $(x - 2y - 4)$ are required factors.

9. *(d)* $a^3 - \dfrac{1}{a^3} - 2a + \dfrac{2}{a}$
$= a^3 - \left(\dfrac{1}{a}\right)^3 - 2\left(a - \dfrac{1}{a}\right)$
$= \left(a - \dfrac{1}{a}\right)\left(a^2 + \dfrac{1}{a^2} + 1\right) - 2\left(a - \dfrac{1}{a}\right)$
$= \left(a - \dfrac{1}{a}\right)\left(a^2 + \dfrac{1}{a^2} - 1\right)$

10. *(a)* $\left(\dfrac{1}{3}x^2 - 2x - 9\right) = \dfrac{(x^2 - 6x - 27)}{3}$
$= \dfrac{1}{3}[x^2 - 9x + 3x - 27]$
$= \dfrac{1}{3}[x(x - 9) + 3(x - 9)]$
$= \dfrac{1}{3}(x + 3)(x - 9)$

11. *(d)* $6\sqrt{3}\,x^2 - 47x + 5\sqrt{3}$
$= 6\sqrt{3}\,x^2 - 45x - 2x + 5\sqrt{3}$
$= 3\sqrt{3}\,x(2x - 5\sqrt{3}) - 1(2x - 5\sqrt{3})$
$= (2x - 5\sqrt{3})(3\sqrt{3}\,x - 1)$

12. *(c)* $8a^3 + 125b^3 - 64c^3 + 120abc$
$= (2a)^3 + (5b)^3 - (4c)^3 + 120abc$
$= (2a)^3 + (5b)^3 + (-4c)^3 - 3(2a)(5b)(-4c)$
$= (2a + 5b - 4c)(4a^2 + 25b^2 + 16c^2 - 10ab + 20bc + 8ac)$
$[\because a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)]$

13. *(b)* Since, $x^{1/3} + y^{1/3} + z^{1/3} = 0$
$\therefore (x^{1/3})^3 + (y^{1/3})^3 + (z^{1/3})^3 - 3x^{1/3}y^{1/3}z^{1/3} = 0$
$\Rightarrow \quad x + y + z - 3(xyz)^{1/3} = 0$
$\Rightarrow \quad x + y + z = 3(xyz)^{1/3}$
$\Rightarrow \quad (x + y + z)^3 = 27xyz$

14. *(b)* $\because \left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$
$(\sqrt{5})^3 = x^3 + \frac{1}{x^3} + 3(\sqrt{5})$
$\left[\because x + \frac{1}{x} = \sqrt{5}\right]$
$\Rightarrow \quad x^3 + \frac{1}{x^3} = 5\sqrt{5} - 3\sqrt{5} = 2\sqrt{5}$

15. *(c)* Given, $\left(x + \frac{1}{x}\right) = 6$
On squaring both sides, we get
$\left(x + \frac{1}{x}\right)^2 = 6^2$
$\Rightarrow \quad x^2 + \frac{1}{x^2} + 2 = 36$
$\Rightarrow \quad x^2 + \frac{1}{x^2} = 34$

16. *(b)* Given, $x - \frac{1}{x} = \frac{1}{3}$
$\Rightarrow \quad 3x - \frac{3}{x} = 1$
On squaring both sides, we get
$\left(3x - \frac{3}{x}\right)^2 = (1)^2$
$9x^2 + \frac{9}{x^2} - 2 \times 9 = 1$
$\Rightarrow \quad 9x^2 + \frac{9}{x^2} = 19$

17. *(d)* $\because (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
$\therefore (6)^2 = 26 + 2(ab + bc + ca)$
$\Rightarrow \quad 2(ab + bc + ca) = 10$
$\Rightarrow \quad ab + bc + ca = 5$

18. *(d)* Given, $x^4 + \frac{1}{x^4} = 322$
$\Rightarrow \left(x^2 + \frac{1}{x^2}\right)^2 - 2 = 322$
$\Rightarrow \quad \left(x^2 + \frac{1}{x^2}\right)^2 = 324 = 18^2$
$\Rightarrow \quad x^2 + \frac{1}{x^2} = 18$
$\Rightarrow \quad \left(x - \frac{1}{x}\right)^2 + 2 = 18 \Rightarrow x - \frac{1}{x} = 4$

19. *(d)* Given, $a + b + c = 0$
$\Rightarrow \quad a^3 + b^3 + c^3 = 3\,abc$
On dividing both sides by abc, we get
$\Rightarrow \quad \frac{a^3}{abc} + \frac{b^3}{abc} + \frac{c^3}{abc} = 3$
$\Rightarrow \quad \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = 3$

20. *(b)* $\because \quad (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ac$
$(10)^2 = (a^2 + b^2 + c^2) + 2(31)$
$a^2 + b^2 + c^2 = 100 - 62$
$\Rightarrow \quad a^2 + b^2 + c^2 = 38$

21. *(b)* $x^2 + \frac{1}{4x^2} + 1 - 2x - \frac{1}{x}$
$= \left(x^2 + \frac{1}{4x^2} + 1\right) - 2\left(x + \frac{1}{2x}\right)$
$= \left(x + \frac{1}{2x}\right)^2 - 2\left(x + \frac{1}{2x}\right)$
$= \left(x + \frac{1}{2x}\right)\left(x + \frac{1}{2x} - 2\right)$

22. *(c)* $(a^2 - b^2)(c^2 - d^2) - 4abcd$
$= a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2 - 4abcd$
$= (a^2c^2 + b^2d^2 - 2abcd) - (b^2c^2 + a^2d^2 + 2abcd)$
$= (ac - bd)^2 - (bc + ad)^2$
$= (ac - bd + bc + ad)(ac - bd - bc - ad)$

23. *(d)* $\because \left(a - \frac{1}{a}\right) = \sqrt{\left(a + \frac{1}{a}\right)^2 - 4}$
$= \sqrt{\left(\frac{17}{4}\right)^2 - 4}$
$\left[\because \text{given } \left(a + \frac{1}{a}\right) = \frac{17}{4}\right]$
$= \sqrt{\frac{289 - 64}{16}} = \sqrt{\frac{225}{16}} = \frac{15}{4}$

24. *(a)* $x^4 + 4y^4$
$= x^4 + 4y^4 + 4x^2y^2 - 4x^2y^2$
$= (x^2 + 2y^2)^2 - (2xy)^2$
$= (x^2 + 2y^2 - 2xy)(x^2 + 2y^2 + 2xy)$
From above it is clear that $x^4 + 4y^4$ is divisible by $x^2 + 2y^2 + 2xy$

25. *(c)* Given, $x(x + y + z) = 9$,
$y(x + y + z) = 16$
and $z(x + y + z) = 144$
On adding all three equations, we get
$x(x + y + z) + y(x + y + z) + z(x + y + z) = 9 + 16 + 144$
$\Rightarrow (x + y + z)(x + y + z) = 9 + 16 + 144$
$\Rightarrow \quad (x + y + z)^2 = 169$
$\Rightarrow \quad x + y + z = 13$
$\because \quad x(x + y + z) = 9$
$\Rightarrow x(13) = 9 \quad \Rightarrow \quad x = \frac{9}{13}$

26. *(c)* Given, $x = (b - c)(a - d)$,
$y = (c - a)(b - d)$
and $\quad z = (a - b)(c - d)$
$\because x + y + z = (b - c)(a - d) + (c - a)(b - d) + (a - b)(c - d)$
$= 0$
Hence, $x^3 + y^3 + z^3 = 3xyz$

27. *(d)* Here, $px^3 + x^2 - 2x - q$ is divisible by $(x - 1)$ and $(x + 1)$.
$\therefore \quad p(1)^3 + (1)^2 - 2(1) - q = 0$
$\Rightarrow \quad p - q = 1 \quad \ldots(\text{i})$
and $p(-1)^3 + (-1)^2 - 2(-1) - q = 0$
$\Rightarrow \quad p + q = 3 \quad \ldots(\text{ii})$
On solving Eqs. (i) and (ii), we get
$p = 2$ and $q = 1$

28. *(b)* Let $f(x) = px^3 + 3x^2 - 3$
and $g(x) = 2x^3 - 5x + p$.
On dividing $f(x)$ by $(x - 4)$, remainder is
$f(4) = p(4)^3 + 3(4)^2 - 3$
$= 64p + 48 - 3 = 64p + 45$
Similarly $g(4) = 2(4)^3 - 5(4) + p$
$= 128 - 20 + p = 108 + p$
But $\quad f(4) = g(4) \quad$ [given]
$\therefore \quad 64p + 45 = 108 + p$
$\Rightarrow \quad 63p = 63 \quad \Rightarrow \quad p = 1$

29. *(a)* $ax - b = 0 \;\Rightarrow x = b/a$
By remainder theorem, if $f(x)$ is divisible by $ax - b$, the remainder is $f(b/a)$.

30. *(b)* Factor theorem is a special case of remainder theorem.

31. *(b)* $x(y - z)(y + z) + y(z - x)(z + x) + z(x - y)(x + y)$
$= x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)$
$= x(y^2 - z^2) + yz^2 - yx^2 + zx^2 - zy^2$
$= (y - z)(xy + xz - x^2 - yz)$
$= (y - z)[y(x - z) + x(z - x)]$
$= (y - z)(z - x)(x - y)$
$= (x - y)(x - z)(z - y)$

32. *(d)* Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$

Since, $(x-1)$ is a factor of $f(x)$.

Put $x = 1$ in $f(x)$, then

$f(1) = a_0 + a_1 + a_2 + \cdots + a_n$

$\Rightarrow\ 1 = a_0 + a_1 + a_2 + \cdots + a_n$

$\therefore\ 1 - a_0 - a_2 - \cdots = a_1 + a_3 + \cdots$

33. *(c)* Given, $(u)^3 + (-2v)^3 + (-3w)^3 = 3 \times (-2)(-3)\,uvw$

$\therefore\ u + (-2v) + (-3w) = 0$

$[\because a^3 + b^3 + c^3 = 3abc$, then $a + b + c = 0]$

$\Rightarrow\ u - 2v - 3w = 0$

$\therefore\ u - 2v = 3w$

34. *(b)* We know that, $(a+b)(a-b) = a^2 - b^2$

So, $[(a^{1/8} + a^{-1/8})(a^{1/8} - a^{-1/8})(a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})]$

$= [(a^{1/8})^2 - (a^{-1/8})^2(a^{1/4} + a^{-1/4})]$

$= (a^{1/4} - a^{-1/4})(a^{1/4} + a^{-1/4})(a^{1/2} + a^{-1/2})$

$= [a^{1/2} - a^{-1/2}][a^{1/2} + a^{-1/2}]$

$= (a - a^{-1})$

35. *(c)* $f(x) = x^4 - 2x^3 + 3x^2 - ax + b$

Put $x = 1$ as $f(1) = 5$, $f(-1) = 19$

$f(1) = 1^4 - 2(1)^3 + 3(1)^2 - a + b$

$\Rightarrow 1 - 2 + 3 - a + b = 5$

$\Rightarrow\ -a + b = 3$ ...(i)

and $f(-1) = (-1)^4 - 2(-1)^3 + 3(-1)^2 - a(-1) + b$

$\Rightarrow\ 1 + 2 + 3 + a + b = 19$

$\Rightarrow\ a + b = 13$ ...(ii)

On solving Eqs. (i) and (ii), we get

$a = 5, b = 8$

36. *(a)* Let $f(x) = ax^2 + bx + c$

Given $f(0) = 4$

$\Rightarrow 0 + 0 + c = 4 \Rightarrow a - b + 4 = 4 \Rightarrow c = 4$

On dividing $f(x)$ by $x + 1$, remainder is 4

$\therefore\ f(-1) = 4$

$\Rightarrow\ a - b + c = 4 \Rightarrow a - b + 4 = 4$

$\Rightarrow\ a - b = 0$ ...(i)

On dividing $f(x)$ by $x + 2$, remainder is 6

$\therefore\ f(-2) = 6$

$\Rightarrow 4a - 2b + c = 6$

$\Rightarrow\ 4a - 2b = 2$ ...(ii)

$\Rightarrow\ a - b + 4 = 4$

On solving Eq. (i) and eq. (ii), we get

$a = 1, b = 1$

Hence,

$a = 1, b = 1, c = 4.$

37. *(b)* $x^{3/2} - xy^{1/2} + x^{1/2}y - y^{3/2}$

$= x(x^{1/2} - y^{1/2}) + y(x^{1/2} - y^{1/2})$

$= (x^{1/2} - y^{1/2})(x + y)$

$\Rightarrow$ Quotient

$$= \frac{(x^{1/2} - y^{1/2})(x + y)}{(x^{1/2} - y^{1/2})} = x + y$$

38. *(c)* Let $f(x) = 9x^2 + 3px + 6q$

Given, $f(-1/3) = -3/4$

$\Rightarrow 9\left(-\frac{1}{3}\right)^2 + 3p\left(-\frac{1}{3}\right) + 6q = -3/4$

$\Rightarrow\ 1 - p + 6q = -3/4$

$\Rightarrow 24q - 4p + 7 = 0$ ...(i)

Let $g(x) = qx^2 + 4px + 7$

Since, $(x + 1)$ is a factor of $g(x)$

$\therefore\ g(-1) = 0 \Rightarrow q - 4p + 7 = 0$...(ii)

On solving Eq. (i) and Eq. (ii), we get $q = 0$ and $p = 7/4$

39. *(c)* I. $(a + b)(b + c)(c + a)$

$= (1 - c)(1 - a)(1 - b)$ $[\because a + b + c = 1]$

$= (1 - c)[1 - b - a + ab]$

$= 1 - b - a + ab - c + bc + ac - abc$

$= 1 - (a + b + c) + ab + bc + ac - abc$

$= ab + bc + ac - abc$

II. $a^2 + b^2 - c^2 + 2ab$

$= a^2 + b^2 + 2ab - c^2$

$= (a + b)^2 - c^2$

$= (a + b + c)(a + b - c) = a + b - c$ $[\because a + b + c = 1]$

So, both statements are correct.

40. *(d)* We have, $x + \frac{1}{x} = 4$

I. $\left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3(x) \times \left(\frac{1}{x}\right)\left(x + \frac{1}{x}\right)$

$(4)^3 = x^3 + \frac{1}{x^3} + 3(4)$

$64 - 12 = x^3 + \frac{1}{x^3} \Rightarrow x^3 + \frac{1}{x^3} = 52$

II. $x + \frac{1}{x} = 4 \Rightarrow x^2 - 4x + 1 = 0$

$$x = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 1}}{2 \times 1}$$

$$= \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$$

Hence, both statements are correct.

41. *(d)* Let $f(x) = 4x^4 - (k - 1)x^3 + kx^2 - 6x + 1$ ...(i)

Since, $(2x - 1)$ is a factor of $f(x)$

$\therefore\ f\left(\frac{1}{2}\right) = 0$

$\Rightarrow\ 4 \times \left(\frac{1}{2}\right)^4 - (k - 1) \times \left(\frac{1}{2}\right)^3 + k \times \left(\frac{1}{2}\right)^2 - 6 \times \frac{1}{2} + 1 = 0$

$\Rightarrow\ \frac{1}{4} - \frac{(k - 1)}{8} + \frac{k}{4} - 2 = 0$

$\Rightarrow\ \frac{k}{4} - \frac{(k - 1)}{8} = 2 - \frac{1}{4}$

$\Rightarrow\ \frac{2k - (k - 1)}{8} = \frac{8 - 1}{4} = \frac{7}{4}$

$\Rightarrow\ 8k - 4k + 4 = 56$

$\Rightarrow\ 4k = 52 \Rightarrow k = 13$

Hence, the value of k is 13.

42. *(c)* Given, $x^4 + xy^3 + x^3y + xz^3 + y^4 + yz^3$

$= (x^4 + xy^3 + xz^3) + (x^3y + y^4 + yz^3)$

$= x(x^3 + y^3 + z^3) + y(x^3 + y^3 + z^3)$

$= (x + y)(x^3 + y^3 + z^3)$

Hence, given polynomial is divisible by both $(x + y)$ and $(x^3 + y^3 + z^3)$.

43. *(c)* $x^2 + 2\ )\ x^3 + 5x^2 + 10k\ (\ x + 5$

$\underline{-x^3 \pm 2x}$

$5x^2 - 2x + 10k$

$\underline{-5x^2 \pm 10}$

$-2x - 10 + 10k$ = Remainder

But given that, remainder $= -2x$

$\therefore\ -2x - 10 + 10k = -2x$

$\Rightarrow\ -10 + 10k = 0$

$\Rightarrow\ 10k = 10 \Rightarrow k = 1$

44. *(c)* Given that, $x + \frac{1}{x} = a$ ...(i)

Then, $x^3 + x^2 + \frac{1}{x^3} + \frac{1}{x^2}$

$= \left(x^3 + \frac{1}{x^3}\right) + \left(x^2 + \frac{1}{x^2}\right)$

$= \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) + \left(x + \frac{1}{x}\right)^2 - 2$

$= a^3 - 3a + a^2 - 2 = a^3 + a^2 - 3a - 2$

45. *(a)* Given, $f(x) = x^3 + x^2 - x - 1$

and $g(x) = x^3 - x^2 + x - 1$

$p(x) = f(x) \cdot g(x)$

$= (x^3 + x^2 - x - 1)(x^3 - x^2 + x - 1)$

$= [x^2(x + 1) - 1(x + 1)][x^2(x - 1) + 1(x - 1)]$

$= [(x + 1)(x^2 - 1)][(x - 1)(x^2 + 1)]$

$= (x + 1)(x + 1)(x - 1)(x - 1)(x^2 + 1)$

$= (x + 1)^2(x - 1)^2(x^2 + 1)$

46. *(d)* Put the value of all options in

$4k^2 + 12k + k$

when, $k = 9$,

then, $4(9)^2 + 12(9) + 9 = 441 = (21)^2$

47. *(c)* Given expression

$= 5px - 10qy + 2rpx - 4qry$
$= (5px + 2rpx) - (10qy + 4qry)$
$= px(5 + 2r) - 2qy(5 + 2r)$
$= (5 + 2r)(px - 2qy)$

So, factors of given expression are both $(5 + 2r)$ and $(px - 2qy)$.

48. *(d)* Given, $(a + 1)^4 - a^4$

$= \{(a + 1)^2 - a^2\}\{(a + 1)^2 + a^2\}$
$= \{(a + 1) + a\}\{(a + 1) - a\}\{a^2 + 1 + 2a + a^2\}$
$= (2a + 1)(1)(2a^2 + 2a + 1)$
$= (2a + 1)(2a^2 + 2a + 1)$

So, $(a + 1)^4 - a^4$ is divisible by $(2a^2 + 2a + 1)$

49. *(d)* Let $f(x) = x^4 - 7x^3 + 5x^2 - 6x + 81$

$f(3) = (3)^4 - 7(3)^3 + 5(3)^2 - 6(3) + 81$
$= 81 - 189 + 45 - 18 + 81 = 0$

$\therefore (x - 3)$ is a factor of $f(x)$

50. *(b)* Let $f(x) = x^3 + 3x^2 + 4x + k$

$\because$ $(x + 5)$ is a factor of $f(x)$

$\therefore$ $f(-5) = 0$

$\Rightarrow$ $(-5)^3 + 3(-5)^2 + 4(-5) + k = 0$

$\Rightarrow$ $-125 + 75 - 20 + k = 0$

$\Rightarrow$ $-70 + k = 0 \Rightarrow k = 70$

Hence, the value of k is 70.

51. *(c)* Let $p(x)$ be added to obtain x^3.

Then, $(1 - x)(1 + x^2) + p(x) = x^3$

$\Rightarrow$ $1 + x^2 - x - x^3 + p(x) = x^3$

$\Rightarrow$ $p(x) = x^3 - 1 - x^2 + x + x^3$

$\therefore$ $p(x) = 2x^3 - x^2 + x - 1$

So, $2x^3 - x^2 + x - 1$ is added to $(1 - x)(1 + x^2)$ to obtain x^3.

52. *(c)* Let $f(x) = x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)$

If it is divisible by $(y - z)$, then $y - z$ is a factor of $f(x)$.

$\therefore$ $y - z = 0 \Rightarrow y = z$

On putting $y = z$, we get

$x(z^2 - z^2) + z(z^2 - x^2) + z(x^2 - z^2)$
$= z^3 - zx^2 + zx^2 - z^3 = 0$

Hence, $y - z$ is a factor, so it is divisible by $(y - z)$.

On putting $z = x$, we get

$x(y^2 - x^2) + y(x^2 - x^2) + x(x^2 - y^2)$
$= xy^2 - x^3 + x^3 - xy^2 = 0$

So, $(z - x)$ is also a factor, so it is also divisible by $(z - x)$.

53. *(d)* Put $x = -3$ in $p(x)$, we get

Let $p(x) = x^3 + 2x^2 + 3x + 8$

$p(-3) = (-3)^3 + 2(-3)^2 + 3(-3) + 8$
$= -27 + 18 - 9 + 8$
$= -10 \neq 0$

So, $(x + 3)$ is not the factor of $x^3 + 2x^2 + 3x + 8$.

Similarly, put $x = 2$ in $p(x)$, we get

$p(2) = (2)^3 + 2(2)^2 + 3(2) + 8$
$= 30 \neq 0$ So, $x - 2$ is also not the factor of $x^3 + 2x^2 + 3x + 8$.

54. *(d)* Let $f(x) = x^3 + 6x^2 + 11x + 6$

Put $x = -1$ in $f(x)$, we get

$f(-1) = (-1)^3 + 6(-1)^2 + 11(-1) + 6$
$= -1 + 6 - 11 + 6$
$= -12 + 12 = 0$

Hence, $(x + 1)$ is a factor of $f(x)$.

$\therefore$ $f(x) = (x + 1)(x^2 + 5x + 6)$
$= (x + 1)(x^2 + 3x + 2x + 6)$
$= (x + 1)(x + 2)(x + 3)$

Hence, $(x + 1), (x + 2)$ and $(x + 3)$ are the factors of $f(x)$.

55. *(c)* $x(x + a)(x + 2a)(x + 3a)$

$= (x + a)(x + 2a)x(x + 3a)$
$= (x^2 + 3ax + 2a^2)(x^2 + 3ax)$
$= (x^2 + 3ax + a^2 + a^2)(x^2 + 3ax + a^2 - a^2)$
$= (x^2 + 3ax + a^2)^2 - a^4$

So, a^4 must be added to $x(x + a)(x + 2a)(x + 3a)$ to make it a perfect square.

56. *(b)* Given, $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$

$\Rightarrow$ $x^2 + \frac{1}{x^2} - 2 + 2 = \frac{17}{4}$

$\Rightarrow$ $\left(x - \frac{1}{x}\right)^2 + 2 = \frac{17}{4}$

$\Rightarrow$ $\left(x - \frac{1}{x}\right)^2 = \frac{17}{4} - 2$

$\Rightarrow$ $\left(x - \frac{1}{x}\right)^2 = \frac{9}{4}$

$\Rightarrow$ $\left(x - \frac{1}{x}\right) = \frac{3}{2}$ ...(i)

On cubing both sides, we get

$\left(x - \frac{1}{x}\right)^3 = \left(\frac{3}{2}\right)^3$

$\Rightarrow$ $x^3 - \frac{1}{x^3} - 3 \times \frac{1}{x} \cdot x\left(x - \frac{1}{x}\right) = \frac{27}{8}$

$\Rightarrow$ $x^3 - \frac{1}{x^3} = \frac{27}{8} + 3 \times \left(\frac{3}{2}\right)$ [From Eq. (i)]

$\Rightarrow$ $x^3 - \frac{1}{x^3} = \frac{27}{8} + \frac{9}{2}$

$\therefore$ $\left(x^3 - \frac{1}{x^3}\right) = \frac{63}{8}$

57. *(c)* Let $f(x) = x^5 - 5x^2 + 125$

Now, put $x = -5$,

$\therefore$ Required remainder $= f(-5)$
$= (-5)^5 - 5(-5^2) + 125$
$= -3125 - 125 + 125 = -3125$

58. *(a)* Given, $ax + by - 2 = 0$

$\Rightarrow$ $ax + by = 2$

Squaring on both sides, we get

$(ax + by)^2 = (2)^2$

$\Rightarrow$ $a^2x^2 + b^2y^2 + 2axby = 4$

$\Rightarrow$ $a^2x^2 + b^2y^2 + 2 = 4$ [$\because$ $axby = 1$]

$\Rightarrow$ $a^2x^2 + b^2y^2 = 2$

$\Rightarrow$ $a^2x^2 = 1$ and $b^2y^2 = 1$

$\Rightarrow$ $ax = 1$ and $by = 1$

$\Rightarrow$ $x = \frac{1}{a}$ and $y = \frac{1}{b}$

$\therefore$ $a^2x + b^2y = a^2 \cdot \frac{1}{a} + b^2 \cdot \frac{1}{b} = a + b$

59. *(a)* Given, $x + k$ is the common factor of $x^2 + ax + b$ and $x^2 + cx + d$. Then, put $x = -k$ in given polynomial.

$\therefore$ $k^2 - ka + b = 0$

and $k^2 - kc + d = 0$

Then, $k^2 - ka + b = k^2 - kc + d$

$\Rightarrow$ $k(c - a) = d - b \Rightarrow k = \frac{d - b}{c - a}$

60. *(c)* I. We have, $3abc + b^3 + c^3 - a^3$

$= -(a^3 - b^3 - c^3 - 3abc)$
$= -[a^3 + (-b)^3 + (-c)^3 - 3(a)(-b)(-c)]$
$= -(a - b - c)(a^2 + b^2 + c^2 + ab - bc + ac)$

So, $(a - b - c)$ is a factor of $3abc + b^3 + c^3 - a^3$.

Hence, statement I is correct.

II. Now, $3bc + b^3 + c^3 - 1$

$= b^3 + c^3 - (1)^3 - 3bc(-1)$
$= (b + c - 1)[b^2 + c^2 + 1^2 - bc + c + b]$

So, $(b + c - 1)$ is a factor of $3bc + b^3 + c^3 - 1$.

Hence, statement II is also correct.

61. *(a)* Let $p(x) = x^3 - 3x^2 + kx - 10$

Then, $(x-5)$ is a factor of $p(x)$.

$\therefore \quad p(5) = 0$

$\Rightarrow \; 5^3 - 3(5)^2 + k \times 5 - 10 = 0$

$\Rightarrow \quad 125 - 75 + 5k - 10 = 0$

$\Rightarrow \quad 40 + 5k = 0 \;\Rightarrow k = -8.$

62. *(b)* $\because \; x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$

Given, $\quad x + y + z = 0$

$\therefore \quad x^3 + y^3 + z^3 - 3xyz = 0$

$\Rightarrow \; x^3 + y^3 + z^3 + 3xyz = 6xyz$

63. *(c)* $x^3q^2 - x^3pt + 4x^2pt - 4x^2q^2 + 3xq^2 - 3xpt$

$= x^3q^2 - x^3pt - 4x^2q^2 + 4x^2pt + 3xq^2 - 3xpt$

$= x^3(q^2 - pt) - 4x^2(q^2 - pt) + 3x(q^2 - pt)$

$= (x^3 - 4x^2 + 3x)(q^2 - pt)$

$= x(x^2 - 4x + 3)(q^2 - pt)$

$= x(x^2 - 3x - x + 3)(q^2 - pt)$

$= x[x(x-3) - 1(x-3)](q^2 - pt)$

$= x(x-3)(x-1)(q^2 - pt)$

64. *(c)* In option *(c)*, we have

$(2)^4 - 6(2)^3 + 12(2)^2 - 24 \times 2 + 32$

$= 16 - 6(8) + 12(4) - 48 + 32$

$= 16 - 48 + 48 - 48 + 32 = 0$

Hence, $(x-2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$

65. *(d)* Let $f(x) = x^3 + kx^2 - x + 2$

If $(x+1)$ is a factor of $f(x)$.

Then, $f(-1) = 0$

$\Rightarrow \; (-1)^3 + k(-1)^2 - (-1) + 2 = 0$

$\Rightarrow \; -1 + k + 1 + 2 = 0 \;\Rightarrow\; k = -2$

66. *(c)* Let $f(x) = x^6 + px^5 + qx^4 - x^2 - x - 3$

$\because f(x)$ is divisible by $(x^4 - 1)$.

Then, $f(1) = f(-1) = 0$

Now, $f(1) = 1 + p + q - 1 - 1 - 3 = 0$

$\Rightarrow \; p + q = 4 \quad \ldots\text{(i)}$

and $\quad f(-1) = 1 - p + q - 1 + 1 - 3$

$\Rightarrow \quad q - p = 2 \quad \ldots\text{(ii)}$

On solving Eqs. (i) and (ii), we get $q = 3$ and $p = 1$

$\therefore \quad p^2 + q^2 = 3^2 + 1^2 = 10$

67. *(a)* Let $f(x) = ax^2 - (a^2 + 1)x + a$

$= ax^2 - x - a^2x + a$

$= x(ax - 1) - a(ax - 1)$

$= (x - a)(ax - 1)$

If p and q are linear factors of $f(x)$, then $p = (x - a)$ and $q = (ax - 1)$

$\therefore \quad p + q = (x - a) + (ax - 1)$

$= (x - 1)(a + 1)$

68. *(d)* We have, $x - y = 8 \quad \ldots\text{(i)}$

and $\quad \sqrt{\dfrac{x}{y}} = \dfrac{10}{3} - \sqrt{\dfrac{y}{x}}$

$$\Rightarrow \quad \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{10}{3}$$

$$\Rightarrow \quad \frac{(\sqrt{x})^2 + (\sqrt{y})^2}{\sqrt{xy}} = \frac{10}{3}$$

$$\Rightarrow \quad x + y = \frac{10}{3}\sqrt{xy}$$

On squaring both sides, we have

$$(x + y)^2 = \frac{100}{9}xy$$

$$\Rightarrow \quad (x - y)^2 + 4xy = \frac{100}{9}xy$$

$[\because (x + y)^2 = (x - y)^2 + 4xy]$

$$\Rightarrow (8)^2 = \left(\frac{100}{9} - 4\right)xy \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad 64 = \frac{64}{9}xy \quad \Rightarrow \quad xy = 9$$

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HCF AND LCM OF POLYNOMIALS

Regularly (1-2) questions have been asked from this chapter. Generally questions are based on your prior knowledge of GCD and LCM of numbers and expressions.

DIVISOR

A polynomial $d(x)$ is said to be a divisor of polynomial $p(x)$ if $d(x)$ is a factor of $p(x)$ i.e., $p(x)$ can be written as $p(x) = d(x) \cdot q(x)$, where $q(x)$ is a quotient polynomial e.g. $(x-3)$ is a divisor of $(x-3)^2(x+1)$.

HCF (GCD) of Polynomials

A polynomial $h(x)$ is called the HCF or GCD of two or more given polynomials, if $h(x)$ is a polynomial of highest degree dividing each one of the given polynomial without leaving any remainder.

Note The coefficient of highest degree term is always taken as positive.

EXAMPLE 1. The HCF of polynomials $(x+2)^2 (x-3)^2$ and $(x+1)(x+2)^2(x-3)$ is

a. $(x+2)(x-3)$ b. $(x+2)^2 (x-3)$ c. $(x+1)(x+2)(x-3)$ d. $(x+2)^2 (x-3)^2$

Sol. b. Let $p(x) = (x+2)^2 (x-3)^2$ and $q(x) = (x+1)(x+2)^2(x-3)$

The common factor between two polynomials is $(x+2)^2 (x-3)$, $\therefore$ HCF $= (x+2)^2 (x-3)$

HCF by Factorisation Method

Following are the steps for calculating HCF through factorisation method

Step I Resolve the given polynomials in the complete factored form.

Step II Find the HCF of the numerical factors (if any) of given polynomial.

Step III Find the factors of highest degree which is common to all given polynomials.

Step IV The product of all such common factors and HCF of the numerical factors is the HCF of given polynomials.

EXAMPLE 2. The HCF of $p(x) = 24(6x^4 - x^3 - 2x^2)$ and $q(x) = 20(2x^6 + 3x^5 + x^4)$ is

a. $4(2x+1)$ b. $x^2(2x+1)$ c. $4x^2$ d. None of these

Sol. d. Here, $P(x) = 24(6x^4 - x^3 - 2x^2) = 2^3 \cdot 3 \cdot x^2 \cdot (6x^2 - x - 2)$

$= 2^3 \cdot 3 \cdot x^2 \cdot (6x^2 - 4x + 3x - 2) = 2^3 \cdot 3 \cdot x^2 (2x+1) \cdot (3x-2)$

and $q(x) = 20(2x^6 + 3x^5 + x^4) = 2^2 \cdot 5 \cdot x^4 (2x^2 + 3x + 1)$

$= 2^2 \cdot 5 \cdot x^4 \cdot (2x^2 + 2x + x + 1) = 2^2 . 5 . x^4 (2x+1)(x+1)$

HCF of numerical factor $= 2^2 = 4$, Highest degree common factor $= x^2(2x+1)$

$\therefore$ Required HCF $= 4x^2(2x+1)$

HCF by Division Method

To find the HCF of polynomials which cannot be factorised easily, we use successive division method.

Step I Arrange the given polynomials in descending order of powers of its variables.

Step II If any common factor is present in the terms of each polynomial, it should be taken out.

Step III Divide the polynomial of highest degree by the polynomial of lowest degree.

or If both the polynomials are of the same degree then any one of them can be taken as divisor or dividend.

Step IV After the first division take the remainder as the new divisor and first divisor as new dividend.

Step V Continue this process of dividing the last divisor by the last remainder until the remainder becomes zero.

Step VI The product of common factors obtained from step II and last divisor is the HCF of given polynomials.

Note
- If the first term of a remainder is negative at any stage, the sign of all of its term must be changed.
- If at any stage, the remainder contains common factor it should be taken out.

EXAMPLE 3. The HCF of $22x^6 - 78x^5 - 16x^2$ and $2x^5 - 78x^2 - 44x$ is

a. $(x^2 - 3x - 2)$ b. $2x(x^2 - 3x - 2)$

c. $22x(x^2 - 3x - 2)$ d. None of these

Sol. b. Let $p(x) = 22x^6 - 78x^5 - 16x^2 = 2x^2(11x^4 - 39x^3 - 8)$

$q(x) = 2x^5 - 78x^2 - 44x = 2x(x^4 - 39x - 22)$

Let us divide $(11x^4 - 39x^3 - 8)$ by $(x^4 - 39x - 22)$

$x^4 - 39x - 22)\ 11x^4 - 39x^3 - 8\ (11$

$11x^4 - 242 - 429x$

$- \quad + \quad +$

$-39x^3 + 429x + 234$

-39 is taken out as common factor from remainder.

$x^3 - 11x - 6)\ x^4 - 39x - 22\ (x$

$x^4 - 11x^2 - 6x$

$- \quad + \quad +$

$11x^2 - 33x - 22$

Again 11 is taken out as common factor from remainder

$x^2 - 3x - 2)\ x^3 - 11x - 6\ (x + 3$

$x^3 - 3x^2 - 2x$

$- \quad + \quad +$

$3x^2 - 9x - 6$

$3x^2 - 9x - 6$

$- \quad + \quad +$

$\times$

$\therefore$ HCF of $11x^4 - 39x^3 - 8$

and $x^4 - 39x - 22$ is $x^2 - 3x - 2$.

Also, HCF of $2x^2$ and $2x$ is $2x$.

$\therefore$ Required HCF $= 2x(x^2 - 3x - 2)$

LCM of Polynomials

A polynomial $h(x)$ is called the LCM of two or more polynomials, if it is a polynomial of smallest degree which is divided by each one of the given polynomials without leaving any remainder.

EXAMPLE 4. The LCM of $12x^2y^3z^2$ and $18x^4y^2z^3$ is

a. $x^4y^3z^3$

b. $9x^2y^2z^3$

c. $36x^3y^3z^4$

d. $36x^4y^3z^3$

Sol. d. Here, $12x^2y^3z^2 = 2^2 \times 3^1 \times x^2 \times y^3 \times z^2$

$18x^4y^2z^3 = 2^1 \times 3^2 \times x^4 \times y^2 \times z^3$

$\therefore$ Required LCM $= 2^2 \times 3^2 \times x^4 \times y^3 \times z^3$

$= 36x^4y^3z^3$

LCM by Factorisation Method

Following are the steps for calculating LCM through factorisation method

Step I Resolve the given polynomials in the complete factored form

Step II Find the LCM of the numerical factors (if any) of given polynomials

Step III The required LCM is the product of LCM of numerical factors and each factor raised to the highest power.

EXAMPLE 5. The LCM of $x^3 - 2x^2 - x + 2$ and $x^3 - x^2 - 4x + 4$ is

a. $(x^2 - 1)(x^2 - 4)$ b. $(x^2 + 2)(x^2 - 4)$
c. $(x^2 + 1)(x^2 + 2)$ d. None of these

Sol. a. Here, $x^3 - 2x^2 - x + 2 = x^2(x - 2) - 1(x - 2)$

$$\Rightarrow \quad (x-2)(x^2-1) = (x-2)(x-1)(x+1) = (x-1)(x+1)(x-2)$$

and $x^3 - x^2 - 4x + 4 = x^2(x-1) - 4(x-1)$

$$\Rightarrow \quad (x-1)(x^2-4) = (x-1)(x-2)(x+2)$$

$\therefore$ Required LCM $= (x-1)(x+1)(x-2)(x+2)$

$= (x^2 - 1)(x^2 - 2^2)$ $[\because (a-b)(a+b) = a^2 - b^2]$

$= (x^2 - 1)(x^2 - 4)$

> For any two polynomials $p(x)$ and $q(x)$
> $p(x) \times q(x) =$ (Their HCF) $\times$ (Their LCM)
> i.e. Product of two polynomials
> = Product of their HCF and LCM

EXAMPLE 6. The HCF of two polynomials is $x + 3$ and their LCM is $x^3 - 7x + 6$. If one of the polynomials is $x^2 + 2x - 3$. Then, the other is

a. $x^2 - x - 6$ b. $x^2 - x + 6$
c. $x^2 + x + 6$ d. None of these

Sol. d. Here, HCF $= x + 3$ and LCM $= x^3 - 7x + 6$

Also, one polynomial $= x^2 + 2x - 3$

As, LCM $\times$ HCF = Product of two polynomials

$$\text{Other polynomial} = \frac{\text{LCM} \times \text{HCF}}{\text{One polynomial}}$$

$$= \frac{(x+3) \times (x^3 - 7x + 6)}{x^2 + 2x - 3}$$

$$= \frac{(x+3)(x^3 - 7x + 6)}{(x+3)(x-1)}$$

$$= \frac{(x+3)(x-1)(x^2 + x - 6)}{(x+3)(x-1)} = x^2 + x - 6$$

So, other polynomial is $x^2 + x - 6$.

PRACTICE EXERCISE

1. The LCM of $(x-1)(x-2)$ and $x^2(x-2)(x+3)$ is
(a) $(x-1)$ (b) $(x-1)(x-2)(x+3)$
(c) $x^2(x-1)(x-2)(x+3)$ (d) None of these

2. The LCM of $2(a^2 - b^2)$, $3(a^3 - b^3)$, $4(a^4 - b^4)$ is
(a) $6(a-b)(a+b)(a^2+b^2)$ (b) $12(a^4 - b^4)(a^2 + ab + b^2)$
(c) $a^3 - b^3$ (d) $12(a^4 - b^4)$

3. The HCF of two expressions a and b is 1. Their LCM is
(a) $(a + b)$ (b) $a - b$ (c) ab (d) $\frac{1}{ab}$

4. The LCM of the polynomials $(x+3)^2(x-2)(x+1)^2$; $(x+1)^3(x+3)(x+4)$ is
(a) $(x-2)(x+1)^3(x+3)^2(x+4)$
(b) $(x-2)(x+1)^3(x+3)(x+4)$
(c) $(x-2)(x+3)(x+4)$
(d) $(x-2)^2(x+1)(x+3)^2(x+4)$

5. HCF of $4y^4x - 9y^2x^3$ and $4y^2x^2 + 6yx^3$ is
(a) $y^2x(2y + 3x)$ (b) $yx(3x + 2y)$
(c) $yx^2(x + 3)$ (d) None of these

6. What is the HCF of $(x^4 - x^2 - 6)$ and $(x^4 - 4x^2 + 3)$?
(a) $x^2 - 3$ (b) $x + 2$ (c) $x + 3$ (d) $x^2 + 3$

7. The HCF of the polynomial A and B where
$A = (x+3)^2(x-2)(x+1)^2$
and $B = (x+1)^2(x+3)(x+4)$ is given by
(a) $(x+1)^2(x+3)$ (b) $(x+1)(x+3)^2$
(c) $(x+1)(x+3)$ (d) $(x+3)^2(x+1)^2$

8. The HCF of $22x(x+1)^2$ and $36x^2(2x^2 + 3x + 1)$ is
(a) $2x(x+1)$ (b) $x(x+1)$ (c) $2(x+1)$ (d) $2(x+1)^2$

9. The LCM of $a^2 - b^2 - c^2 - 2bc$, $b^2 - c^2 - a^2 - 2ac$ and $c^2 - a^2 - b^2 - 2ab$ is
(a) $(a+b+c)$
(b) $(a-b-c)(a+b+c)$
(c) $(a+b+c)(c-a-b)$
(d) $(a+b+c)(a-b-c)(b-c-a)(c-a-b)$

10. LCM of $[(x+3)(x-2)]^2$ and $[(x-2)(x-6)]$ is
(a) $(x+3)(x-2)^3(x-6)$ (b) $(x+3)(x-2)^2(x-6)$
(c) $(x+3)(x-2)(x-6)$ (d) $(x+3)(x-6)$

11. If $(z-1)$ is the HCF of (z^2-1) and $pz^2-q(z+1)$, then
(a) $2p=q$ (b) $p=2q$ (c) $3p=2q$ (d) $3p=2p$

12. The HCF of two expressions is $3x^2+4x-4$ and their LCM is $3x^4+4x^3-7x^2-4x+4$. The expressions are
(a) $(x-1)(3x^2+4x-4)$ and $(3x^2+4x-4)$
(b) $(x+1)(3x^2+4x-4)$ and $(x+2)(3x^2+4x-4)$
(c) $(x+2)(3x^2+4x+4)$ and $(x-1)$
(d) $(x+1)(3x^2+4x-4)$ and $(x-1)(3x^2+4x-4)$

13. The LCM and HCF of two polynomials $p(x)$ and $q(x)$ are $36x^2(x+a)(x^3-a^3)$ and $x^2(x-a)$, respectively. If $p(x)=(4x^2)(x^2-a^2)$, then $q(x)$
(a) $4x^3(x^3-a^3)$ (b) $12x^3(x^2-a^2)$
(c) $9x^3(x^3-a^3)$ (d) $36x^3(x^3-a^3)$

14. The LCM of two polynomials $p(x)$ and $q(x)$ is x^3-7x+6. If $p(x)=x^2+2x-3$ and $q(x)=x^2+x-6$, then the HCF is
(a) $(x+3)$ (b) $(x-3)$
(c) $(x+3)(x-2)$ (d) $(x-1)$

15. If $(x+k)$ is the HCF of (x^2+ax+b) and (x^2+px+q), then the value of k is
(a) $\left(\frac{b+q}{a+p}\right)$ (b) $\left(\frac{b-q}{a-p}\right)$ (c) $\left(\frac{a+b}{p+q}\right)$ (d) $\left(\frac{a-b}{p-q}\right)$

16. What is the value of k for which the HCF of $2x^2+kx-12$ and $x^2+x-2k-2$ is $(x+4)$?
(a) 5 (b) 7 (c) 10 (d) -4

17. If the HCF of (x^2+x-12) and $(2x^2-kx-9)$ is $(x-k)$, then what is the value of k?
(a) -3 (b) 3 (c) -4 (d) 4

18. If GCD of the polynomials (x^3-2x^2+px+6) and (x^2-5x+q) is $(x-3)$. Then, the value of $5q+6p$ is
(a) -1 (b) 1
(c) 0 (d) None of these

19. The sum and the difference of two expressions is $5x^2-x-4$ and $x^2+9x-10$ respectively, then their LCM would be equal to
(a) $(x-1)$ (b) $(2x+3)(3x+7)$
(c) $(2x-3)(3x+7)$ (d) $(x-1)(2x-3)(3x+7)$

20. Find the values of a and b so that the polynomials $p(x)$ and $q(x)$ have $(x+1)(x+3)$ as their HCF

$$p(x)=(x^2+3x+2)(x^2+2x+a)$$

and $q(x)=(x^2+7x+12)(x^2+7x+b)$
(a) $-3, 6$ (b) $3, -6$
(c) $6, -3$ (d) None of these

21. If $f(x)$ and $g(x)$ are two polynomials with integral coefficients which vanish at $x=1/2$, then what is the factor of HCF of $f(x)$ and $g(x)$?
(a) $x-1$ (b) $x-2$ (c) $2x-1$ (d) $2x+1$

22. The HCF of two polynomials $p(x)$ and $q(x)$ is $2x(x+2)$ and LCM is $24x(x+2)^2(x-2)$. If $p(x)=8x^3+32x^2+32x$, then what is $q(x)$ equal to?
(a) $4x^3-16x$ (b) $6x^3-24x$ (c) $12x^3+24x$ (d) $12x^3-24x$

23. If $(x-2)$ is the HCF of (ax^2+bx+c) and (bx^2+ax+c), then value of c is
(a) $2(a+b)$ (b) $(a+b)$ (c) $-3(a+b)$ (d) $-(a+b)$

24. We have three polynomials $A=8p+p^2+12$, $B=p^2+2p-24$ and $C=p^2+15p+54$
I. Their LCM is $(p+6)(p-4)(p+2)(p+9)$
II. Their HCF is $(p+6)(p-2)$
Then, which of the following codes is/are correct
(a) Only I (b) Only II
(c) Neither I nor II (d) Both I and II

25. Which of the following statements are true?
I. HCF of x^2-6x+9 and x^3-27 is $(x-3)$.
II. LCM of $10x^2yz, 15xyz, 20xy^2z^2$ is $120x^2y^2z^2$.
III. HCF of $(6x^2-7x-3)$ and $(2x^2+11x-21)$ is $(2x-3)$.
Select the correct answer using the codes given below
(a) I and III (b) I, II and III
(c) II and III (d) None of these

26. Consider the following statements :
I. The HCF of $x+y$ and $x^{10}-y^{10}$ is $x+y$.
II. The HCF of $x+y$ and $x^{10}+y^{10}$ is $x+y$.
III. The HCF of $x-y$ and $x^{10}+y^{10}$ is $x-y$.
IV. The HCF of $x-y$ and $x^{10}-y^{10}$ is $x-y$.
Which of the statement(s) given above is/are correct?
(a) I and II (b) II and III (c) I and IV (d) II and IV

Directions (Q. Nos. 27-29) *A student wrote five polynomials such as $A=pq-np$, $B=pq-mq$, $C=q^2-3nq+2n^2$, $D=pq-2pn-mq+2mn$, $E=pq-np-mq+mn$. Now, he divide the polynomials into groups and calculate the HCF and LCM.*

27. Calculate the HCF of A, C and E
(a) $(q-2n)$ (b) $(p-n)$ (c) $(q-n)$ (d) $(q-n)(q-2n)$

28. Calculate the LCM of D and E
(a) $(p-m)(q-2n)$ (b) $(p-m)(q-n)(q-2n)$
(c) $(q-n)(q-2n)(m-p)$ (d) $(q-n)(p-m)$

29. The HCF of all five polynomials together is
(a) $(q-n)$ (b) $(q-2n)$ (c) 1 (d) $(p-m)$

PREVIOUS YEARS' QUESTIONS

30. What is the LCM of $a^3b - ab^3$, $a^3b^2 + a^2b^3$ and $ab(a+b)$? 2012 I

(a) $a^2b^2(a^2 - b^2)$ (b) $ab(a^2 - b^2)$
(c) $a^2b^2 + ab^3$ (d) $a^3b^3(a^2 - b^2)$

31. What is the HCF of $36(3x^4 + 5x^3 - 2x^2)$, $9(6x^3 + 4x^2 - 2x)$ and $54(27x^4 - x)$? 2012 I

(a) $9x(x+1)$ (b) $9x(3x-1)$ (c) $18x(3x-1)$ (d) $18x(x+1)$

32. What is the HCF of the polynomials $x^3 + 8$, $x^2 + 5x + 6$ and $x^3 + 2x^2 + 4x + 8$? 2013 II

(a) $x+2$ (b) $x+3$ (c) $(x+2)^2$ (d) None of these

33. The LCM of $(x^3 - x^2 - 2x)$ and $(x^3 + x^2)$ is 2013 II

(a) $x^3 - x^2 - 2x$ (b) $x^2 + x$ (c) $x^4 - x^3 - 2x^2$ (d) $x - 2$

34. The HCF of $(x^4 - y^4)$ and $(x^6 - y^6)$ is 2013 II

(a) $x^2 - y^2$ (b) $x - y$ (c) $x^3 - y^3$ (d) $x^4 - y^4$

35. What is the LCM of $x^2 + 2x - 8$, $x^3 - 4x^2 + 4x$ and $x^2 + 4x$? 2013 II

(a) $x(x+4)(x-2)^2$ (b) $x(x+4)(x-2)$
(c) $x(x+4)(x+2)^2$ (d) $x(x+4)^2(x-2)$

36. What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7 - 4a^2b^9$? 2013 II

(a) ab (b) a^2b^3 (c) a^2b^2 (d) a^3b^2

37. What is the HCF of $8(x^5 - x^3 + x)$ and $28(x^6 + 1)$? 2014 I

(a) $4(x^4 - x^2 + 1)$ (b) $2(x^4 - x^2 + 1)$
(c) $(x^4 - x^2 + 1)$ (d) None of these

38. What is the highest common factor of $2x^3 + x^2 - x - 2$ and $3x^3 - 2x^2 + x - 2$? 2014 II

(a) $x-1$ (b) $x+1$ (c) $2x+1$ (d) $2x-1$

39. The HCF and LCM of two polynomials are $(x + y)$ and $(3x^5 + 5x^4y + 2x^3y^2 - 3x^2y^3 - 5xy^4 - 2y^5)$, respectively. If one of the polynomials is $(x^2 - y^2)$, then the other polynomial is 2015 I

(a) $3x^4 - 8x^3y + 10x^2y^2 + 7xy^3 - 2y^4$
(b) $3x^4 - 8x^3y - 10x^2y^2 + 7xy^3 + 2y^4$
(c) $3x^4 + 8x^3y + 10x^2y^2 + 7xy^3 + 2y^4$
(d) $3x^4 + 8x^3y - 10x^2y^2 + 7xy^3 + 2y^4$

40. If $(x+1)$ is the HCF of $Ax^2 + Bx + C$ and $Bx^2 + Ax + C$ where $A \neq B$, then the value of C is 2015 II

(a) A (b) B (c) $A - B$ (d) 0

41. The sum and difference of two expressions are $5x^2 - x - 4$ and $x^2 + 9x - 10$ respectively. The HCF of the two expressions will be 2016 I

(a) $x+1$ (b) $(x-1)$ (c) $(3x+7)$ (d) $(2x-3)$

ANSWERS

1	c	2	b	3	c	4	a	5	b	6	a	7	a	8	a	9	d	10	b
11	b	12	d	13	c	14	a	15	b	16	a	17	b	18	c	19	d	20	a
21	c	22	b	23	c	24	a	25	a	26	c	27	c	28	b	29	c	30	a
31	c	32	a	33	c	34	a	35	a	36	c	37	a	38	a	39	c	40	d
41	b																		

HINTS AND SOLUTIONS

1. *(c)* LCM = product of the largest power of each factor

$= x^2(x-1)(x-2)(x+3)$

2. *(b)* Here, $2(a^2 - b^2) = 2(a+b)(a-b)$

$3(a^3 - b^3) = 3(a-b)(a^2 + ab + b^2)$

and $4(a^4 - b^4) = 4(a+b)(a-b)(a^2 + b^2)$

LCM of numerical coefficients = 12 and LCM of algebraic expressions

$= (a-b)(a+b)(a^2+b^2)(a^2+ab+b^2)$

$= (a^4 - b^4)(a^2 + ab + b^2)$

$\therefore$ LCM of polynomials

$= 12(a^4 - b^4)(a^2 + ab + b^2)$

3. *(c)* $\text{LCM} = \dfrac{\text{Product of expressions}}{\text{HCF}}$

$= \dfrac{a \times b}{1} = ab$

4. *(a)* Given, $(x+3)^2(x-2)(x+1)^2$ and $(x+1)^3(x+3)(x+4)$

LCM

$= (x-2)(x+1)^3(x+3)^2(x+4)$

5. *(b)* $4y^4x - 9y^2x^3 = y^2x(4y^2 - 9x^2)$

$= y^2x(2y - 3x)(2y + 3x)$

$4y^2x^2 + 6yx^3 = 2yx^2(2y + 3x)$

$\therefore$ Required HCF $= xy(2y + 3x)$

6. *(a)* Let $p(x) = x^4 - x^2 - 6$

$= x^4 - 3x^2 + 2x^2 - 6$

$= x^2(x^2 - 3) + 2(x^2 - 3)$

$= (x^2 + 2)(x^2 - 3)$

$q(x) = x^4 - 4x^2 + 3$

$= x^4 - 3x^2 - x^2 + 3$

$= x^2(x^2 - 3) - 1(x^2 - 3)$

$= (x^2 - 3)(x^2 - 1)$

HCF of $p(x), q(x) = x^2 - 3$

7. *(a)* $A = (x+3)^2(x-2)(x+1)^2$ and

$B = (x+1)^2(x+3)(x+4)$

$\therefore$ HCF of polynomials

$= (x+3)(x+1)^2$

8. *(a)* Here, HCF of 22 and 36 is 2.

Now, $x(x+1)^2 = x(x+1)(x+1)$

$x^2(2x^2+3x+1) = x^2(2x+1)(x+1)$

Common factors of $x(x+1)^2$

and $x^2(2x^2+3x+1)$ are $x(x+1)$.

Hence, required HCF $= 2x(x+1)$

9. *(d)* $a^2-b^2-c^2-2bc = a^2-(b^2+c^2+2bc)$

$= a^2-(b+c)^2 = (a+b+c)(a-b-c)$

$b^2-c^2-a^2-2ac = b^2-(c^2+a^2+2ac)$

$= b^2-(c+a)^2$

$= (b-c-a)(b+c+a)$

$= (a+b+c)(b-c-a)$

and $c^2-a^2-b^2-2ab$

$= c^2-(a^2+b^2+2ab) = c^2-(a+b)^2$

$= (c-a-b)(c+a+b)$

$= (a+b+c)(c-a-b)$

$\therefore$ Required LCM

$= (a+b+c)(a-b-c)$

$(b-c-a)(c-a-b)$

10. *(b)* Given, $[(x+3)(x-2)^2]$

and $[(x-2)(x-6)]$

$\therefore$ LCM $= (x+3)(x-2)^2(x-6)$

11. *(b)* Since, $(z-1)$ is the HCF, so it will divide each one of the given polynomials. So, $z=1$ will make each one zero.

$\therefore \quad p(1)^2 - q(1+1) = 0 \Rightarrow p = 2q$

12. *(d)* Let $p(x)$ and $q(x)$ be two polynomials and $p(x) \div \text{HCF} = a$

and $q(x) \div \text{HCF} = b$

$\therefore \quad p(x) \times q(x) = \text{LCM} \times \text{HCF}$

$\Rightarrow a \times \text{HCF} \times b \times \text{HCF} = \text{LCM} \times \text{HCF}$

$\Rightarrow \quad a \times b \times \text{HCF} = \text{LCM}$

$$ab = \frac{\text{LCM}}{\text{HCF}}$$

$$= \frac{3x^4+4x^3-7x^2-4x+4}{3x^2+4x-4}$$

$= (x+1)(x-1)$

Let $a = (x+1)$ and $b = (x-1)$,

then the required expression are

$(x+1)(3x^2+4x-4)$

and $(x-1)(3x^2+4x-4)$.

13. *(c)* Here, LCM $= 36x^3(x+a)(x^3-a^3)$ and HCF $= x^2(x-a)$

$p(x) = 4x^2(x^2-a^2)$

But $p(x) \times q(x) = \text{HCF} \times \text{LCM}$

$$q(x) = \frac{\text{HCF} \times \text{LCM}}{p(x)}$$

$$= \frac{x^2(x-a)\,36x^3(x+a)(x^3-a^3)}{4x^2(x^2-a^2)}$$

$= 9x^3(x^3-a^3)$

14. *(a)* Given,

$p(x) = x^2+2x-3 = (x+3)(x-1)$

$q(x) = x^2+x-6 = (x+3)(x-2)$

and LCM

$= x^3-7x+6 = (x-1)(x^2+x-6)$

$= (x-1)(x+3)(x-2)$

$$\therefore \text{HCF} = \frac{p(x)\times q(x)}{\text{LCM}}$$

$$= \frac{(x+3)(x-1)\times(x+3)(x-2)}{(x-1)(x+3)(x-2)}$$

$= (x+3)$

15. *(b)* Since, $(x+k)$ is the HCF, it will divide both the polynomials without leaving any remainder, thus $x = -k$ will make both of them zero.

$\therefore k^2 - pk + q = k^2 - ak + b$

or $\quad -ak + b = -pk + q$

$\Rightarrow \quad ak - pk = b - q$

$$\therefore \quad k = \frac{b-q}{a-p}$$

16. *(a)* Since, $(x+4)$ is HCF, so it will divide both the expressions i.e. $x=-4$ will make each one zero.

$\therefore \quad 2(-4)^2 + k(-4) - 12 = 0$

$\Rightarrow \quad 32 - 12 = 4k$

$\therefore \quad 20 = 4k \quad \Rightarrow \quad k = 5$

17. *(b)* Since, HCF of x^2+x-12 and $2x^2-kx-9$ is $(x-k)$, then $(x-k)$ will be the factor of $2x^2-kx-9$.

$\therefore \quad 2k^2 - k^2 - 9 = 0$

$\Rightarrow \quad k^2 - 9 = 0$

$\Rightarrow \quad k = \pm 3$

and factor of x^2+x-12 are $(x+4)(x-3)$.

Hence, value of k is 3.

18. *(c)* Here, $(x-3)$ is GCD, so is a factor of both of them.

$\therefore$ Putting $x=3$, in both makes each polynomial zero.

$3^3 - 2(3)^2 + p(3) + 6 = 0 \Rightarrow p = -5$

$3^2 - 5(3) + q = 0 \Rightarrow q = 6$

$\therefore \quad 5q + 6p = 5(6) + (-5)(6)$

$= 30 - 30 = 0$

19. *(d)* Let the expressions be $p(x)$ and $q(x)$, then

$p(x) + q(x) = 5x^2 - x - 4 \quad \ldots\text{(i)}$

$p(x) - q(x) = x^2 + 9x - 10 \quad \ldots\text{(ii)}$

On adding Eqs. (i) and (ii), we have

$2p(x) = 6x^2 + 8x - 14$

$\Rightarrow \quad p(x) = 3x^2 + 4x - 7$

$= 3x^2 + 7x - 3x - 7$

$= x(3x+7) - 1(3x+7)$

$\therefore \quad p(x) = (3x+7)(x-1)$

Subtracting Eqs. (i) and (ii), we have

$2q(x) = 4x^2 - 10x + 6$

$\Rightarrow \quad q(x) = 2x^2 - 5x + 3$

$= 2x^2 - 3x - 2x + 3$

$= x(2x-3) - 1(2x-3)$

$= (x-1)(2x-3)$

$\therefore$ LCM of $p(x)$ and $q(x)$

$= (x-1)(2x-3)(3x+7)$

20. *(a)* $p(x) = (x+2)(x+1)(x^2+2x+a)$

$q(x) = (x+3)(x+4)(x^2+7x+b)$

As HCF is $(x+1)(x+3)$, then both $(x+1)$ and $(x+3)$ must be factors of $p(x)$ and $q(x)$.

For $p(x)$, $\quad (x+1)$ is already a factor, so $(x+3)$ must be a factor of x^2+2x+a.

So, $\quad (-3)^2 + 2(-3) + a = 0$

$\Rightarrow \quad 9 - 6 + a = 0$

$\therefore \quad a = -3$

For $q(x)$, $\quad (x+3)$ is already factor.

$\therefore (x+1)$ must be a factor of

$x^2 + 7x + b$.

$\therefore (-1)^2 + 7(-1) + b = 0$

$\therefore \quad b = 6$

So, $\quad a = -3$ and $b = 6$ is solution.

21. *(c)* Given, $f(x)$ and $g(x)$ vanish at $x = \frac{1}{2}$

So, $(2x-1)$ is a factor of $f(x)$ and $g(x)$ both.

Hence, HCF of $f(x)$ and $g(x) = 2x - 1$.

22. *(b)* Refer to question 13.

23. *(c)* As $(x-2)$ is the HCF of (ax^2+bx+c) and (bx^2+ax+c)

So, it will divide both the expressions,

$\therefore \quad a(2)^2 + b(2) + c = 0$

$\Rightarrow \quad 4a + 2b + c = 0 \quad \ldots\text{(i)}$

and $b(2)^2 + a(2) + c = 0$

$\Rightarrow \quad 4b + 2a + c = 0 \quad \ldots\text{(ii)}$

adding Eqs. (i) and (ii), we get

$\Rightarrow 6a + 6b + 2c = 0$

$\Rightarrow \quad 2c = -6a - 6b$

$\Rightarrow \quad c = -3(a+b)$

24. *(a)* $A = p^2 + 8p + 12 = (p+2)(p+6)$

$B = p^2 + 2p - 24 = (p-4)(p+6)$

$C = p^2 + 15p + 54 = (p+9)(p+6)$

I. LCM of A, B

and $C = (p+2)(p+6)(p-4)(p+9)$

Thus, I is correct

II. HCF of A, B and $C = (p+6)$

Hence, II is incorrect.

25. *(a)* I. $x^2 - 6x + 9 = (x-3)(x-3)$

and $x^3 - 27 = x^3 - (3)^3$

$= (x-3)(x^2 + 3x + 9)$

$\therefore$ HCF $= x - 3$.

Hence, it is true.

II. LCM of $10x^2yz$, $15xyz$ and $20xy^2z^2$ is $60x^2y^2z^2$.

Hence, it is false.

III. $6x^2 - 7x - 3 = (2x-3)(3x+1)$

and $2x^2 + 11x - 21$

$= (x+7)(2x-3)$

Hence, HCF $= (2x-3)$, it is also true.

Hence, the statement I and III are correct.

26. *(c)* We know that, $(x+y)$ and $(x-y)$ are the factors of $(x^{10} - y^{10})$.

Hence, statements I and IV are true.

27. *(c)* We have, $A = pq - np = p(q-n)$ ...(i)

$C = q^2 - 3nq + 2n^2$

$= q^2 - 2nq - nq + 2n^2$

$= q(q-2n) - n(q-2n)$

$= (q-n)(q-2n)$ ...(ii)

$E = pq - np - mq + mn$

$= p(q-n) - m(q-n)$

$= (p-m)(q-n)$ ...(iii)

$\therefore$ HCF of A, C and $E = (q-n)$

28. *(b)* We have, $E = (p-m)(q-n)$ ... (i)

$D = pq - 2np - mq + 2mn$

$= p(q-2n) - m(q-2n)$

$= (p-m)(q-2n)$ ...(ii)

$\therefore$ LCM of D and E

$= (p-m)(q-n)(q-2n)$

29. *(c)* The factors of given polynomials are as follows

$A = pq - np = p(q-n)$ ...(i)

$B = pq - mq = q(p-m)$ ...(ii)

$C = (q-n)(q-2n)$ ...(iii)

$D = (p-m)(q-2n)$ ...(iv)

$E = (p-m)(q-n)$ ...(v)

There is no such factor which is common to all given five polynomials. Thus, HCF $(A, B, C, D, E) = 1$

30. *(a)* Here, $a^3b - ab^3 = ab(a^2 - b^2)$

$= ab(a-b)(a+b)$

$a^3b^2 + a^2b^3 = a^2b^2(a+b)$

$\therefore$ LCM $[(a^3b - ab^3), (a^3b^2 + a^2b^3), ab(a+b)]$

$= a^2b^2(a+b)(a-b)$

$= a^2b^2(a^2 - b^2)$

31. *(c)* Let $P(x) = 36(3x^4 + 5x^3 - 2x^2)$

$= 36x^2(3x^2 + 5x - 2)$

$= 36x^2(x+2)(3x-1)$

$Q(x) = 9(6x^3 + 4x^2 - 2x)$

$= 18x(3x^2 + 2x - 1)$

$= 18x(3x-1)(x+1)$

$R(x) = 54(27x^4 - x) = 54x(27x^3 - 1)$

$= 54x(3x-1)(9x^2 + 1 + 3x)$

HCF of $[36, 18, 54] = 18$

$\therefore$ HCF of $[P(x), Q(x), R(x)]$

$= 18x(3x-1)$

32. *(a)* Let $f(x) = x^3 + 8 = x^3 + 2^3$

$= (x+2)(x^2 - 2x + 4)$

$= (x+2)(x-2)(x-2)$

$g(x) = x^2 + 5x + 6$

$= x^2 + 3x + 2x + 6$

$= (x+3)(x+2)$

and $h(x) = x^3 + 2x^2 + 4x + 8$

$= (x+2)(x^2 + 4)$

$\therefore$ HCF of $\{f(x), g(x), h(x)\} = (x+2)$

33. *(c)* Let $f(x) = x^3 - x^2 - 2x$

$= x(x^2 - x - 2)$

$= x(x+1)(x-2)$

and $g(x) = x^3 + x^2$

$= x^2(x+1) = x \cdot x(x+1)$

$\therefore$ LCM of $[f(x), g(x)]$

$= x^2(x+1)(x-2)$

$= x^2(x^2 - x - 2)$

$= x^4 - x^3 - 2x^2$

34. *(a)* Let $f(x) = (x^4 - y^4)$

$= (x^2 - y^2)(x^2 + y^2)$

$= (x-y)(x+y)(x^2 + y^2)$

and $g(x) = (x^6 - y^6)$

$= (x^3 + y^3)(x^3 - y^3)$

$= (x+y)(x^2 - xy + y^2)(x-y)(x^2 + xy + y^2)$

$= (x-y)(x+y)(x^2 - xy + y^2)(x^2 + xy + y^2)$

$\therefore$ HCF of

$[f(x), g(x)] = (x-y)(x+y)$

$= x^2 - y^2$

35. *(a)* $x^2 + 2x - 8 = (x-2)(x+4)$

$x^3 - 4x^2 + 4x = x[x^2 - 4x + 4]$

$= x(x-2)^2$

$x^2 + 4x = x(x+4)$

Now, LCM of $(x^2 + 2x - 8)$, $(x^3 - 4x^2 + 4x)$ and $(x^2 + 4x)$

$= x(x-2)^2(x+4)$

36. *(c)* $a^2b^4 + 2a^2b^2 = a^2b^2(b^2 + 2)$ ...(i)

and $(ab)^7 - 4a^2b^9 = a^7b^7 - 4a^2b^9$

$= a^2b^2(a^5b^5 - 4b^7)$ ...(ii)

From Eqs. (i) and (ii), HCF $= a^2b^2$

37. *(a)* Let $p(x) = 8(x^5 - x^3 + x)$

$= 4 \times 2 \times x(x^4 - x^2 + 1)$

and $q(x) = 28(x^6 + 1)$

$= 7 \times 4[(x^2)^3 + (1)^3]$

$= 4 \times 7 \times (x^2 + 1)(x^4 - x^2 + 1)$

$\therefore$ HCF of $p(x)$ and $q(x)$

$= 4(x^4 - x^2 + 1)$

38. *(a)* Let $f(x) = 2x^3 + x^2 - x - 2$

$= (x-1)(2x^2 + 3x + 2)$

and $g(x) = 3x^3 - 2x^2 + x - 2$

$= (x-1)(3x^2 + x + 2)$

Hence, the highest factor of $f(x)$ and $g(x)$ is $(x-1)$.

39. *(c)* Given, HCF $= (x+y)$ and

LCM $= 3x^5 + 5x^4y + 2x^3y^2 - 3x^2y^3 - 5xy^4 - 2y^5$

$= x^3(3x^2 + 5xy + 2y^2) - y^3(3x^2 + 5xy + 2y^2)$

$= (3x^2 + 5xy + 2y^2)(x^3 - y^3)$

We know that,

Product of two polynomials

= HCF × LCM

$\therefore$ Required polynomial

$$= \frac{(x+y)(x^3 - y^3)(3x^2 + 5xy + 2y^2)}{(x-y)(x+y)}$$

$$= \frac{(x-y)(x^2 + y^2 + xy)(3x^2 + 5xy + 2y^2)}{(x-y)}$$

$= (x^2 + y^2 + xy)(3x^2 + 5xy + 2y^2)$

$= 3x^4 + 8x^3y + 10x^2y^2 + 7xy^3 + 2y^4$

40. *(d)* $(x+1)$ is the HCF of $Ax^2 + Bx + C$ and $Bx^2 + Ax + C$

$\therefore \quad A(-1)^2 + B(-1) + C = 0$

$\Rightarrow \quad A - B + C = 0$

$\Rightarrow \quad C = B - A$

and $B(-1)^2 + A(-1) + C = 0$

$\Rightarrow \quad B - A + C = 0$

$\Rightarrow \quad C = A - B$

$\therefore \quad C = 0$

41. *(b)* Refer to question 19.

16

RATIONAL EXPRESSIONS

Usually (1-2) questions have been asked from this chapter. Generally questions are asked from this chapter are based on simplification of rational expressions.

RATIONAL EXPRESSIONS

An expression in the form of $\frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials and $q(x) \neq 0$ is called a rational expression.

- In the rational expression $\frac{p(x)}{q(x)}$, $p(x)$ is called the numerator and $q(x)$ is called the denominator of the rational expression.
- Every polynomial is a rational expression. Since, $p(x)$ can always be written as $\frac{p(x)}{1}$ and a constant function 1 is a polynomial of degree 0.
- Every rational expression need not be a polynomial.

Working Rule to Reduce the Given Rational Expression in its Lowest Term

1. Firstly, factorize both the polynomials $p(x)$ and $q(x)$.
2. Find the HCF of $p(x)$ and $q(x)$. If HCF of $p(x)$ and $q(x)$ is one, then rational expression $\frac{p(x)}{q(x)}$ is in its lowest terms.
3. If HCF is not equal to 1. Then, divide both $p(x)$ and $q(x)$ by their HCF and the rational expression is obtained in the lowest terms.

EXAMPLE 1. The lowest term of an expression $\frac{a^3 - b^3}{a^2 + ab + b^2}$ is

a. $a+b$ b. $a-b$ c. ab d. $\frac{a}{b}$

Sol. b. Rational expression

$$= \frac{a^3 - b^3}{a^2 + ab + b^2} = \frac{(a-b)(a^2 + ab + b^2)}{(a^2 + ab + b^2)} = a - b$$

$$[\because (a^3 - b^3) = (a-b)(a^2 + ab + b^2)]$$

EXAMPLE 2. The lowest term of an expression $\frac{3x^2 - 11x - 4}{6x^2 - 7x - 3}$ is

a. $\frac{x+4}{2x+3}$ b. $\frac{x+4}{2x-3}$ c. $\frac{x-4}{2x+3}$ d. $\frac{x-4}{2x-3}$

Sol. d. $\frac{3x^2 - 11x - 4}{6x^2 - 7x - 3} = \frac{3x^2 - 12x + x - 4}{6x^2 - 9x + 2x - 3}$

$$= \frac{3x(x-4) + 1(x-4)}{3x(2x-3) + 1(2x-3)} = \frac{(3x+1)(x-4)}{(3x+1)(2x-3)} = \frac{x-4}{2x-3}$$

EXAMPLE 3. The lowest term of an expression $\frac{12x^3y^5z^4}{18x^2y^6z^5}$ is

a. $\frac{2x}{yz}$ b. $\frac{xy}{z}$ c. $\frac{2x}{3yz}$ d. $\frac{3x}{2yz}$

Sol. c. $\frac{12x^3y^5z^4}{18x^2y^6z^5} = \frac{6 \times 2 \times x^2 \times x \times y^5 \times z^4}{6 \times 3 \times x^2 \times y \times y^5 \times z \times z^4} = \frac{2x}{3yz}$

Operations on Rational Expressions

1. **Addition** or **Subtraction of Rational Expressions with like denominators**

   It $\dfrac{P(x)}{q(x)}$ and $\dfrac{h(x)}{q(x)}$ are two rational expressions then

   $\dfrac{p(x)}{q(x)} \pm \dfrac{h(x)}{q(x)} = \dfrac{p(x) \pm h(x)}{q(x)}$

2. **Addition** or **subtraction of Rational Expressions with unlike denominators**

To add or subtract rational expressions with unlike denominators, follow the steps given below

Step I Write each denominator in the factor form

Step II Find the LCM of the denominators

Step III Rewrite each rational expression with LCM as the denominator

Step IV Add or subtract the numerators.

EXAMPLE 4. The sum of $\dfrac{x+1}{x-1}$ and $\dfrac{x-1}{x+1}$ is

a. $\dfrac{x^2+1}{x^2-1}$ b. $\dfrac{1}{x^2+1}$ c. $\dfrac{2x^2+2}{x^2-1}$ d. $\dfrac{x^2+2}{x^2-1}$

Sol. c. $\dfrac{x+1}{x-1} + \dfrac{x-1}{x+1} = \dfrac{(x+1)}{(x-1)} \times \dfrac{(x+1)}{(x+1)} + \dfrac{(x-1)}{(x+1)} \times \dfrac{(x-1)}{(x-1)}$

$$= \frac{(x+1)^2 + (x-1)^2}{(x^2-1)}$$

$$= \frac{x^2+1+2x+x^2+1-2x}{x^2-1} = \frac{2x^2+2}{x^2-1}$$

$[\because (a+b)^2 = a^2+b^2+2ab \text{ and } (a-b)^2 = a^2+b^2-2ab]$

EXAMPLE 5. What should be subtracted from $\dfrac{7x}{x^2+x-12}$ to get $\dfrac{4}{x+4}$?

a. $\dfrac{1}{x-3}$ b. $\dfrac{3}{x-3}$ c. $\dfrac{1}{x+3}$ d. $\dfrac{3}{x+3}$

Sol. b. Let $p(x)$ is to be subtracted, then

$$\frac{7x}{x^2+x-12} - p(x) = \frac{4}{x+4}$$

$$\Rightarrow \quad p(x) = \frac{7x}{x^2+x-12} - \frac{4}{x+4} = \frac{7x}{(x+4)(x-3)} - \frac{4}{(x+4)}$$

$$= \frac{7x}{(x+4)(x-3)} - \frac{4(x-3)}{(x+4)(x-3)} = \frac{7x-4(x-3)}{(x+4)(x-3)}$$

$$= \frac{3x+12}{(x+4)(x-3)} = \frac{3(x+4)}{(x+4)(x-3)} = \frac{3}{x-3}$$

3. **Multiplication of Rational Expressions If** $\dfrac{p(x)}{q(x)}$ and $\dfrac{g(x)}{h(x)}$ are two rational expressions, then their product is given by $\dfrac{p(x)}{q(x)} \times \dfrac{g(x)}{h(x)} = \dfrac{p(x)\cdot g(x)}{q(x)\cdot h(x)}$

   - Multiplicative inverse of $\dfrac{p(x)}{q(x)}$ is $\dfrac{q(x)}{p(x)}$.
   - 1 is the multiplicative identity.

EXAMPLE 6. The product of $\dfrac{x^2-1}{x^2+1}$ and $\dfrac{x+2}{x+1}$ is

a. $\dfrac{(x-1)}{(x^2+1)}$ b. $\dfrac{(x-2)}{(x^2+1)}$

c. $\dfrac{(x-1)(x+2)}{(x^2+1)}$ d. $\dfrac{x+2}{(x^2-1)}$

Sol. c. Here, product $= \dfrac{x^2-1}{x^2+1} \times \dfrac{(x+2)}{(x+1)} = \dfrac{(x-1)(x+1)(x+2)}{(x^2+1)(x+1)}$

$[\because a^2-b^2 = (a-b)(a+b)]$

$$= \frac{(x-1)(x+2)}{(x^2+1)}$$

4. **Division of Rational Expressions** If $\dfrac{p(x)}{q(x)}$ and $\dfrac{g(x)}{h(x)}$ are two rational expressions, then their division is given by $\dfrac{p(x)}{q(x)} \div \dfrac{g(x)}{h(x)} = \dfrac{p(x)}{q(x)} \times \dfrac{h(x)}{g(x)}$

   = Product of $\dfrac{p(x)}{q(x)}$ and the reciprocal of $\dfrac{g(x)}{h(x)}$ i.e. $\dfrac{h(x)}{g(x)}$.

EXAMPLE 7. The lowest term of an expression $\dfrac{x^2+8x+12}{x^2-7x+12} \div \dfrac{x^2+4x-12}{x-4}$ is

a. $\dfrac{(x-2)}{(x-3)(x+2)}$ b. $\dfrac{x-2}{x+2}$

c. $\dfrac{(x+2)}{(x-3)(x-2)}$ d. $\dfrac{x+3}{x-2}$

Sol. c. Here, $\dfrac{x^2+8x+12}{x^2-7x+12} \div \dfrac{x^2+4x-12}{x-4}$

$$= \frac{x^2+6x+2x+12}{x^2-4x-3x+12} \times \frac{x-4}{(x^2+6x-2x-12)}$$

$$= \frac{(x+6)(x+2)}{(x-4)(x-3)} \times \frac{x-4}{(x+6)(x-2)} = \frac{x+2}{(x-3)(x-2)}$$

PRACTICE EXERCISE

1. Which of the following are rational expressions?

I. $\dfrac{x^3-3x^2+2}{x^2+1}$ II. $\dfrac{z^3-3z^2}{2z+3}$

III. $\dfrac{x^2-x+2}{x+3}$ IV. $\dfrac{x^3+3x^2-1}{x^2+\sqrt{x-1}}$

Select the correct answer using the codes given below

(a) I, II and III (b) II, III and IV
(c) All of these (d) None of these

2. The simplified form of $\dfrac{a+2}{a+3}-\dfrac{(a+1)}{(a+2)}$.

(a) $\dfrac{1}{a^2+5a+6}$ (b) $\dfrac{a+2}{a^2+5a+6}$
(c) 0 (d) $\dfrac{(a+2)^2}{a^2+5a+6}$

3. $\left(\dfrac{x+1}{x^2-1}-\dfrac{2}{x}\right)$ expressed as a rational expression is

(a) $\dfrac{x^2-2}{x(x^2-1)}$ (b) $\dfrac{-(x-2)}{x(x-1)}$
(c) $\dfrac{2(x+1)}{(x^3-1)}$ (d) $\dfrac{1}{x(x+1)}$

4. The expression $\dfrac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$ in lowest terms is

(a) $\dfrac{1}{x-7}$ (b) $(x-7)$ (c) $(x-2)$ (d) $\dfrac{1}{x-2}$

5. $\dfrac{a-c}{(a-b)(x-a)}+\dfrac{b-c}{(b-a)(x-b)}$ is equal to

(a) $\dfrac{b-a}{(x-b)(x-a)}$ (b) $\dfrac{x-c}{(x-a)(x-b)}$
(c) $\dfrac{b-a}{(x-b)(x-c)}$ (d) None of these

6. The sum of the rational expression $\dfrac{x-3}{x^2+1}$ and its reciprocal is

(a) $\dfrac{x^3-3x^2+x-3}{x^4+3x^2-6x+10}$ (b) $\dfrac{x^4+3x^2-6x+10}{x^3-3x^2+x-3}$
(c) $\dfrac{x^4-3x^2+6x+10}{x^3-3x^2+x-3}$ (d) $\dfrac{(x-1)^3}{(x+3)^2}$

7. The value of

$$\frac{a}{(a-b)(a-c)}+\frac{b}{(b-c)(b-a)}+\frac{c}{(c-a)(c-b)}$$

(a) $2a$ (b) 0 (c) $2b$ (d) $b-c$

8. If $A=\left[\dfrac{x-1}{x+1}\right]$ and $B=\left[\dfrac{x+1}{x-1}\right]$, then $(A+B)^2$ is

(a) $\dfrac{4x^4+8x^2-4}{x^4-2x^2+1}$ (b) $\dfrac{4x^4+8x^2+4}{x^4-2x^2+1}$
(c) $\dfrac{4x^4+8x^2+4}{x^4+2x+1}$ (d) None of these

9. The value of

$$\frac{1}{(1-a)(1-b)}+\frac{a^2}{(1-a)(b-a)}-\frac{b^2}{(b-1)(a-b)} \text{ is}$$

(a) $1+a$ (b) $1-a^2$
(c) $1-b^2$ (d) None of these

10. The value of $\dfrac{\left[\dfrac{x}{y}-\dfrac{y}{x}\right]\left[\dfrac{y}{z}-\dfrac{z}{y}\right]\left[\dfrac{z}{x}-\dfrac{x}{z}\right]}{\left[\dfrac{1}{x^2}-\dfrac{1}{y^2}\right]\left[\dfrac{1}{y^2}-\dfrac{1}{z^2}\right]\left[\dfrac{1}{z^2}-\dfrac{1}{x^2}\right]}$ is

(a) $x^2y^2z^2$ (b) $-x^2y^2z^2$
(c) 1 (d) None of these

11. What is the simplified form of

$$\left(\frac{x^2-3x+2}{x^3-8}\right)\div\left(\frac{x^2-9}{x^2+7x+12}\right)\times\left(\frac{x^3+2x^2+4x}{x^2+3x-4}\right)?$$

(a) $\dfrac{x}{x-3}$ (b) $\dfrac{x-2}{x-3}$
(c) $\dfrac{x}{x+3}$ (d) $\dfrac{x+3}{x+4}$

12. If $pq+qr+rp=0$, then what is the value of

$$\frac{p^2}{p^2-qr}+\frac{q^2}{q^2-rp}+\frac{r^2}{r^2-pq}?$$

(a) 0 (b) 1
(c) –1 (d) 3

13. If $x+y+z=0$, then what is the value of

$$\frac{1}{x^2+y^2-z^2}+\frac{1}{y^2+z^2-x^2}+\frac{1}{z^2+x^2-y^2}?$$

(a) $\dfrac{1}{x^2+y^2+z^2}$ (b) 1
(c) –1 (d) 0

14. If $a=\dfrac{1+x}{2-x}$, then what is $\dfrac{1}{a+1}+\dfrac{2a+1}{a^2-1}$ equal to?

(a) $\dfrac{(1+x)(2+x)}{2x-1}$ (b) $\dfrac{(1-x)(2-x)}{x-2}$
(c) $\dfrac{(1+x)(2-x)}{2x-1}$ (d) $\dfrac{(1-x)(2-x)}{2x+1}$

15. If $\frac{1}{x+1}+\frac{2}{y+2}+\frac{1009}{z+1009}=1$, then what is the value of $\frac{x}{x+1}+\frac{y}{y+2}+\frac{z}{z+1009}$?

(a) 0 (b) 2 (c) 3 (d) 4

16. What is $\frac{x^2-3x+2}{x^2-5x+6} \div \frac{x^2-5x+4}{x^2-7x+12}$ equal to?

(a) $\frac{x+3}{x-3}$ (b) 1 (c) $\frac{x+1}{x+3}$ (d) 2

17. If $x+y+z=0$, then what is $\frac{xyz}{(x+y)(y+z)(z+x)}$ equal to (where, $x \neq -y, y \neq -z, z \neq -x$) ?

(a) -1 (b) 1
(c) $xy+yz+zx$ (d) None of these

PREVIOUS YEARS' QUESTIONS

18. What is $\frac{(x^2+y^2)(x-y)-(x-y)^3}{x^2y-xy^2}$ equal to? **2013 II**

(a) 1 (b) 2 (c) 4 (d) -2

19. What is $\frac{1}{a-b}-\frac{1}{a+b}-\frac{2b}{a^2+b^2}-\frac{4b^3}{a^4+b^4}-\frac{8b^7}{a^8-b^8}$ equal to? **2014 II**

(a) $a+b$ (b) $a-b$ (c) 1 (d) 0

20. If $a^2-by-cz=0$, $ax-b^2+cz=0$ and $ax+by-c^2=0$, then the value of $\frac{x}{a+x}+\frac{y}{b+y}+\frac{z}{c+z}$ will be **2016 I**

(a) $a+b+c$ (b) 3 (c) 1 (d) 0

ANSWERS

1	a	2	a	3	b	4	c	5	b	6	b	7	b	8	b	9	d	10	b
11	a	12	b	13	d	14	c	15	b	16	b	17	a	18	b	19	d	20	c

HINTS AND SOLUTIONS

1. *(a)* Here, $\frac{x^3+3x^2-1}{x^2+\sqrt{x-1}}$ is not a rational expression, since the denominator is not a polynomial.

2. *(a)* We know $\frac{a+2}{a+3}-\frac{(a+1)}{(a+2)}$

$$=\frac{(a+2)^2-(a+1)(a+3)}{(a+3)(a+2)}$$

$$=\frac{a^2+4+4a-(a^2+4a+3)}{a^2+5a+6}$$

$$=\frac{1}{a^2+5a+6}$$

3. *(b)* Here, $\left(\frac{x+1}{x^2-1}-\frac{2}{x}\right)=\frac{1}{x-1}-\frac{2}{x}$

$[\because x^2-1=(x+1)(x-1)]$

$$=\frac{x-2x+2}{x(x-1)}=\frac{-(x-2)}{x(x-1)}$$

4. *(c)* We know that,

$$\frac{(x-1)(x-2)(x^2-9x+14)}{(x-7)(x^2-3x+2)}$$

$$=\frac{(x-1)(x-2)(x^2-7x-2x+14)}{(x-7)(x^2-2x-x+2)}$$

$$=\frac{(x-1)(x-2)(x-7)(x-2)}{(x-7)(x-2)(x-1)}=x-2$$

5. *(b)* Here, $\frac{a-c}{(a-b)(x-a)}+\frac{b-c}{(b-a)(x-b)}$

$$=\frac{(a-c)(x-b)-(b-c)(x-a)}{(a-b)(x-a)(x-b)}$$

$$=\frac{ax-ab-xc+bc-(bx-ab-cx+ac)}{(a-b)(x-a)(x-b)}$$

$$=\frac{ax+bc-bx-ac}{(a-b)(x-a)(x-b)}$$

$$=\frac{x(a-b)-c(-b+a)}{(a-b)(x-a)(x-b)}$$

$$=\frac{(x-c)(a-b)}{(a-b)(x-a)(x-b)}=\frac{(x-c)}{(x-a)(x-b)}$$

6. *(b)* Here, reciprocal of $\frac{x-3}{x^2+1}$ is $\frac{x^2+1}{(x-3)}$

So,

$$\frac{x-3}{x^2+1}+\frac{x^2+1}{x-3}=\frac{(x-3)^2+(x^2+1)^2}{(x^2+1)(x-3)}$$

$$=\frac{x^2+9-6x+x^4+1+2x^2}{x^3-3x^2+x-3}$$

$$=\frac{x^4+3x^2-6x+10}{x^3-3x^2+x-3}$$

7. *(b)* We know that,

$$\frac{a}{(a-b)(a-c)}+\frac{b}{(b-c)(b-a)}+\frac{c}{(c-a)(c-b)}$$

$$=\frac{-a}{(a-b)(c-a)}-\frac{b}{(b-c)(a-b)}-\frac{c}{(c-a)(b-c)}$$

$$=\frac{(-a)(b-c)-b(c-a)-c(a-b)}{(a-b)(b-c)(c-a)}$$

$$=\frac{-ab+ac-bc+ab-ac+bc}{(a-b)(b-c)(c-a)}$$

$$=\frac{0}{(a-b)(b-c)(c-a)}=0$$

8. *(b)* Here,

$$(A+B)^2=\left[\left(\frac{x-1}{x+1}\right)+\left(\frac{x+1}{x-1}\right)\right]^2$$

$$=\left[\frac{(x-1)^2+(x+1)^2}{(x+1)(x-1)}\right]^2=\left[\frac{2(x^2+1)}{x^2-1}\right]^2$$

$[\because (a+b)^2+(a-b)^2=2(a^2+b^2)]$

$$=\frac{4(x^4+2x^2+1)}{x^4-2x^2+1}=\frac{4x^4+8x^2+4}{x^4-2x^2+1}$$

9. *(d)* Given,

$$\frac{1}{(1-a)(1-b)}+\frac{a^2}{(1-a)(b-a)}-\frac{b^2}{(b-1)(a-b)}$$

$$=\frac{(b-a)+a^2(1-b)-b^2(1-a)}{(1-a)(1-b)(b-a)}$$

$$= \frac{b - a + a^2 - a^2b - b^2 + ab^2}{(1-a)(1-b)(b-a)}$$

$$= \frac{(b-a) + (a^2 - b^2) + ab(b-a)}{(1-a)(1-b)(b-a)}$$

$$= \frac{(b-a) - (a+b)(b-a) + ab(b-a)}{(1-a)(1-b)(b-a)} = \frac{1 - a - b + ab}{1 - b - a + ab} = 1$$

10. *(b)* Given, $\dfrac{\left[\dfrac{x}{y} - \dfrac{y}{x}\right]\left[\dfrac{y}{z} - \dfrac{z}{y}\right]\left[\dfrac{z}{x} - \dfrac{x}{z}\right]}{\left[\dfrac{1}{x^2} - \dfrac{1}{y^2}\right]\left[\dfrac{1}{y^2} - \dfrac{1}{z^2}\right]\left[\dfrac{1}{z^2} - \dfrac{1}{x^2}\right]}$

$$= \frac{\left[\dfrac{x^2 - y^2}{xy}\right]\left[\dfrac{y^2 - z^2}{zy}\right]\left[\dfrac{z^2 - x^2}{xz}\right]}{\left[\dfrac{y^2 - x^2}{x^2y^2}\right]\left[\dfrac{z^2 - y^2}{z^2y^2}\right]\left[\dfrac{x^2 - z^2}{x^2z^2}\right]}$$

$$= -\frac{\dfrac{(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)}{x^2\,y^2\,z^2}}{\dfrac{(x^2 - y^2)(y^2 - z^2)(z^2 - x^2)}{x^4y^4z^4}} = -\frac{x^4y^4z^4}{x^2y^2z^2} = -\,x^2y^2z^2$$

11. *(a)* Given, $\left(\dfrac{x^2 - 3x + 2}{x^3 - 8}\right) \div \left(\dfrac{x^2 - 9}{x^2 + 7x + 12}\right) \times \left(\dfrac{x^3 + 2x^2 + 4x}{x^2 + 3x - 4}\right)$

$$= \left(\frac{x^2 - 3x + 2}{x^3 - 8} \times \frac{x^2 + 7x + 12}{x^2 - 9}\right) \times \left(\frac{x^3 + 2x^2 + 4x}{x^2 + 3x - 4}\right)$$

$$= \frac{(x-1)(x-2)}{(x-2)(x^2 + 4 + 2x)} \times \frac{(x+4)(x+3)}{(x-3)(x+3)} \times \frac{x(x^2 + 2x + 4)}{(x-1)(x+4)}$$

$$= \frac{x}{x-3} \qquad [\because a^3 - b^3 = (a-b)(a^2 + b^2 + ab)]$$

12. *(b)* Given, $pq + qr + rp = 0$

$$\therefore \quad \frac{p^2}{p^2 - qr} + \frac{q^2}{q^2 - rp} + \frac{r^2}{r^2 - pq}$$

$$= \frac{p^2}{p^2 + rp + pq} + \frac{q^2}{q^2 + pq + qr} + \frac{r^2}{r^2 + qr + rp}$$

$$= \frac{p^2}{p\,(p + r + q)} + \frac{q^2}{q\,(p + q + r)} + \frac{r^2}{r\,(p + q + r)} = \frac{p + q + r}{p + q + r} = 1$$

13. *(d)* Given, $x + y + z = 0 \Rightarrow x + y = -z.$

On squaring both sides, we get $x^2 + y^2 + 2xy = z^2$

$$x^2 + y^2 - z^2 = -2xy$$

Similarly, $y^2 + z^2 - x^2 = -2yz$ and $z^2 + x^2 - y^2 = -2zx$

$$\therefore \quad \frac{1}{x^2 + y^2 - z^2} + \frac{1}{y^2 + z^2 - x^2} + \frac{1}{z^2 + x^2 - y^2}$$

$$= \frac{1}{-2xy} + \frac{1}{-2yz} + \frac{1}{-2zx} = \frac{1}{2}\left(\frac{z + x + y}{xyz}\right) = 0$$

14. *(c)* Given, $a = \dfrac{1 + x}{2 - x}$ So, $\dfrac{1}{a+1} + \dfrac{2a + 1}{a^2 - 1} = \dfrac{3a}{a^2 - 1} = \dfrac{3\left(\dfrac{1+x}{2-x}\right)}{\left(\dfrac{1+x}{2-x}\right)^2 - 1}$

$$= \frac{3(1+x)(2-x)}{1 + x^2 + 2x - (4 + x^2 - 4x)} = \frac{3(1+x)(2-x)}{6x - 3} = \frac{(1+x)(2-x)}{(2x - 1)}$$

15. *(b)* Given, $\dfrac{1}{x+1} + \dfrac{2}{y+2} + \dfrac{1009}{z + 1009} = 1$

$$\Rightarrow \quad \frac{1}{x+1} - 1 + \frac{2}{y+2} - 1 + \frac{1009}{z + 1009} - 1 = 1 - 3$$

$$\Rightarrow \quad -\frac{x}{x+1} - \frac{y}{y+2} - \frac{z}{z + 1009} = -2$$

$$\therefore \quad \frac{x}{x+1} + \frac{y}{y+2} + \frac{z}{z + 1009} = 2$$

16. *(b)* Given, $\dfrac{x^2 - 3x + 2}{x^2 - 5x + 6} \div \dfrac{x^2 - 5x + 4}{x^2 - 7x + 12}$

$$= \frac{x^2 - 3x + 2}{x^2 - 5x + 6} \times \frac{(x^2 - 7x + 12)}{(x^2 - 5x + 4)}$$

$$= \frac{(x-1)(x-2)}{(x-3)(x-2)} \times \frac{(x-4)(x-3)}{(x-4)(x-1)} = 1$$

17. *(a)* Given, $x + y + z = 0$

$$\Rightarrow \quad x + y = -z,\; y + z = -x \text{ and } z + x = -y$$

$$\therefore \quad \frac{xyz}{(x+y)(y+z)(z+x)} = \frac{xyz}{(-z)(-x)(-y)} = \frac{xyz}{-xyz} = -1$$

18. *(b)* Given, $\dfrac{(x^2 + y^2)(x - y) - (x - y)^3}{x^2y - xy^2}$

$$= \frac{x^3 + xy^2 - x^2y - y^3 - (x^3 - y^3 - 3x^2y + 3xy^2)}{x^2y - xy^2}$$

$$= \frac{x^3 + xy^2 - x^2y - y^3 - x^3 + y^3 + 3x^2y - 3xy^2}{x^2y - xy^2}$$

$$= \frac{2x^2y - 2xy^2}{x^2y - xy^2} = \frac{2(x^2y - xy^2)}{x^2y - xy^2} = 2$$

19. *(d)* $\dfrac{1}{a-b} - \dfrac{1}{a+b} - \dfrac{2b}{a^2 + b^2} - \dfrac{4b^3}{a^4 + b^4} - \dfrac{8b^7}{a^8 - b^8}$

$$= \frac{(a+b) - (a-b)}{(a-b)(a+b)} - \frac{2b}{a^2 + b^2} - \frac{4b^3}{a^4 + b^4} - \frac{8b^7}{a^8 - b^8}$$

$$= \frac{2b}{a^2 - b^2} - \frac{2b}{a^2 + b^2} - \frac{4b^3}{a^4 + b^4} - \frac{8b^7}{a^8 - b^8}$$

$$= \frac{2b(a^2 + b^2) - 2b(a^2 - b^2)}{(a^2 - b^2)(a^2 + b^2)} - \frac{4b^3}{a^4 + b^4} - \frac{8b^7}{a^8 - b^8}$$

$$= \frac{4b^3}{a^4 - b^4} - \frac{4b^3}{a^4 + b^4} - \frac{8b^7}{a^8 - b^8}$$

$$= \frac{4b^3(a^4 + b^4) - 4b^3(a^4 - b^4)}{(a^4 - b^4)(a^4 + b^4)} - \frac{8b^7}{a^8 - b^8} = 0$$

20. *(c)* Given,

$$a^2 - by - cz = 0 \qquad \text{...(i)}$$

$$ax - b^2 + cz = 0 \qquad \text{...(ii)}$$

and

$$ax + by - c^2 = 0 \qquad \text{...(iii)}$$

On adding Eqs. (i), (ii) and (iii), we get

$$x = \frac{b^2 - a^2 + c^2}{2a} \Rightarrow \frac{x}{a + x} = \frac{b^2 - a^2 + c^2}{a^2 + b^2 + c^2}$$

Similarly, $\dfrac{y}{b + y} = \dfrac{a^2 - b^2 + c^2}{a^2 + b^2 + c^2}$ and $\dfrac{z}{c + z} = \dfrac{a^2 + b^2 - c^2}{a^2 + b^2 + c^2}$

$$\therefore \frac{x}{a + x} + \frac{y}{y + b} + \frac{z}{c + z} = \frac{(b^2 - a^2 + c^2) + (a^2 - b^2 + c^2) + (a^2 + b^2 - c^2)}{a^2 + b^2 + c^2} = 1$$

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LINEAR EQUATIONS

Generally (3-5) questions have been definitely asked from this chapter in CDS examination. Beside this, the linear equations are also applicable in solving various word problems related to number system, mensuration, time and work, etc.

A linear equation is an equation for a straight line.

So, the equation which has degree 1, i.e., which has linear power of the variables, is called a linear equation.

LINEAR EQUATIONS IN ONE VARIABLE

A linear equation in one variable is an equation which can be written in the form of $ax+b=0$, or $ax=c$, where a, b, c are real numbers with $a \neq 0$.

e.g. $5x+8=9-x$, $\frac{2}{3}y+7=\frac{y}{2}$ and $t+3t=9-t$ are linear equations in one variable.

Solving a linear Equation in One Variable

1. Simplify both sides of the equation.
2. Use the addition and subtraction properites to get all variables of terms on the LHS and all constant terms on the RHS.
3. Simplify and divide both sides of the equation by the coefficient of the variable.

EXAMPLE 1. Solve $2(x-3)-(5-3x)=3(x+1)-4(2+x)$.

a. 1 b. −1 c. 0 d. 3

Sol. a. $2(x-3)-(5-3x)=3(x+1)-4(2+x) \Rightarrow 2x-6-5+3x=3x+3-8-4x$

$\Rightarrow 5x-11=-x-5 \Rightarrow 6x=6 \Rightarrow x=1$, Hence, the value of x is 1.

LINEAR EQUATION IN TWO VARIABLES

An equation which can be put in the form $ax+by+c=0$, where a, b and c are real numbers and a, b both are not zero, is called a linear equation in two variables.

e.g. $2x+3y=5$, $\sqrt{2}x+\sqrt{3}y=0$ and $2a+3b=0$ are linear equations in two variables.

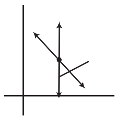
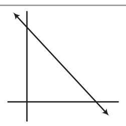
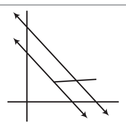
Pair of Linear Equations

Two linear equations in the two same variables are called pair of linear equations in two variables. The general form of pair of linear equations in two variables x and y is $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$
where, a_1, a_2, b_1, b_2 and c_1, c_2 are all real numbers and $a_1^2+b_1^2 \neq 0,\ a_2^2+b_2^2 \neq 0$.

Consistency Pair of linear Equations

A pair of linear equations which has no solution is called an **inconsistent pair** of linear equations. A pair of linear equations which has atleast one solution is called a **consistent pair** of linear equations. Let us consider two linear equations as $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$.

Pair of linear equations in two variables will represent two straight lines, both to be considered together. The table given below repersents the three possibilities.

Ratios	Consistency	Graphical Representation	Algebraic interpretation
$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Consistent	 Intersecting lines.	Exactly one solution (unique).
$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Dependent (consistent)	 Coincident lines.	Infinitely many solutions.
$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Inconsistent	 Parallel lines.	No solution.

HOMOGENEOUS PAIR OF LINEAR EQUATIONS

The pair of linear equations $a_1x+b_1y=0,\ a_2x+b_2y=0$ has

1. A unique solution, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ and the solution is $x=0,\ y=0$.
2. An infinite number of solutions, if $\frac{a_1}{a_2} = \frac{b_1}{b_2}$.

   This pair of linear equations is always consistent.

EXAMPLE 2. The values of k for which the system $kx+2y=5$ and $3x+y=1$ has

(*i*) unique solution (*ii*) no solution are

a. $k \neq 6$ and $k=6$ **b.** $k \neq 3$ and $k=3$
c. $k \neq 2$ and $k=2$ **d.** None of these

Sol. **a.** Given equations are, $kx+2y-5=0$ and $3x+y-1=0$

On comparing the given equations, with standard form of equations

$$a_1x+b_1y+c_1=0 \text{ and } a_2x+b_2y+c_2=0$$

We have, $a_1=k,\ b_1=2,\ c_1=-5$
and $a_2=3,\ b_2=1,\ c_2=-1$

(i) Here, the equations have a unique solution, if

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

i.e. if $\frac{k}{3} \neq \frac{2}{1} \Rightarrow k \neq 6$

(ii) The equations have no solution, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$\therefore$ $\frac{k}{3} = \frac{2}{1} \Rightarrow k=6$

EXAMPLE 3. The value of k for which the system of equations $kx-y=2,\ 6x-2y=3$ has infinitely many solutions is

a. 3 **b.** 4
c. 12 **d.** No such value of k exists.

Sol. **d.** Given equations are,

$$kx-y-2=0 \quad \text{and} \quad 6x-2y-3=0$$

On comparing the given equations, with standard form of equations

$$a_1x+b_1y+c_1=0 \text{ and } a_2x+b_2y+c_2=0$$

We have, $a_1=k,\ b_1=-1,\ c_1=-2$
and $a_2=6,\ b_2=-2,\ c_2=-3$

Here, the equations have infinite number of solutions, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

But, $\frac{a_1}{a_2} = \frac{k}{6},\ \frac{b_1}{b_2} = \frac{-1}{-2} = \frac{1}{2}$ and $\frac{c_1}{c_2} = \frac{-2}{-3} = \frac{2}{3}$

So, $\frac{b_1}{b_2} \neq \frac{c_1}{c_2}$.

Thus, the system has no such value of k, for which the given system has infinitely many solutions.

Algebraic Methods of Solutions

Different algebraic methods for solving pair of linear equations are discussed below in detail.

1. Substitution Method

In this method, we substitute the value of one variable in terms of other variable to solve the pair of linear equations, so this method is called substitution method.

Steps used in this method are given below.

Step I Find the value of one variable x (or y) in terms of other variable i.e. y (or x) from an equation.

Step II Substitute this value of x (or y) in other equation, and reduce it to a linear equation in one variable i.e. in terms of y (or x) which can be solved easily.

Step III Substitute the value of y (or x) obtained in Step II in the equation which is used to obtain the value of the other variable in Step I.

EXAMPLE 4. The solutions of the system of equations $x+y=14$ and $x-y=4$, is

a. $x=5$ and $y=9$ b. $x=9$ and $y=5$

c. $x=5$ and $y=3$ d. $x=3$ and $y=5$

Sol. b. Given equations are, $x+y=14$ ...(i)

and $x-y=4$ ...(ii)

from eq. (ii), $y=x-4$ ...(iii)

On substituting the value of y from Eq. (iii) in Eq. (i), we get

$$x+x-4=14 \Rightarrow 2x=18 \Rightarrow x=9$$

On substituting $x=9$ in Eq. (iii), we get

$$y=9-4=5 \Rightarrow y=5$$

Hence, $x=9$ and $y=5$

2. Elimination Method

In this method, one variable is eliminated from both equations to have an equation in one variable, so it is called elimination method.

Steps used in this method are given below.

Step I Make the coefficient of one variable (x *or* y) numerically equal by multiplying both equations by some suitable non zero constant.

Step II Now, add or subtract both equations, so that one variable is eliminated and the remaining equation is in one variable only.

Step III Solve the equation in one variable to get the value of this variable (x or y).

Step IV Substitute this value (x or y) in either of the given equations to get the value of other variable.

EXAMPLE 5. The solution of the given system of equations $2x+5y=11$ and $3x+4y=13$ is

a. (4, 2) b. (3, 1) c. (5, 2) d. (1, 1)

Sol. b. Given equations are, $2x+5y=11$ ...(i)

and $3x+4y=13$ ...(ii)

On multiplying Eq. (i) by 3 and Eq. (ii) by 2, we get

$$6x+15y=33 \quad \text{...(iii)}$$

$$6x+8y=26 \quad \text{...(iv)}$$

On subtracting Eq. (iv) from Eq. (iii), we get

$$(6x+15y)-(6x+8y)=33-26 \Rightarrow 7y=7 \Rightarrow y=1$$

Put $y=1$ in Eq. (i), $2x+5\times1=11$

$\Rightarrow$ $2x=11-5 \Rightarrow 2x=6 \Rightarrow x=3$

Hence, solution of system of equations is (3, 1).

3. Cross-multiplication Method

Let us consider a general system of two simultaneous linear equations

$$a_1x+b_1y+c_1=0 \quad \text{...(i)}$$

$$a_2x+b_2y+c_2=0 \quad \text{...(ii)}$$

where, $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ *then the solution of the system is given by,*

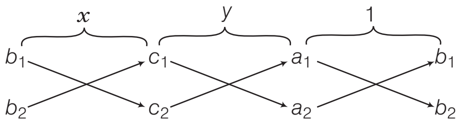
$$x=\frac{b_1c_2-b_2c_1}{a_1b_2-a_2b_1} \text{ and } y=\frac{c_1a_2-c_2a_1}{a_1b_2-a_2b_1}$$

$$\therefore \frac{x}{b_1c_2-b_2c_1}=\frac{y}{c_1a_2-c_2a_1}=\frac{1}{a_1b_2-a_2b_1}$$

This method is called cross multiplication method.

Remembering Technique (Cross-multiplication)

The following diagram helps in remembering the above solution.



The arrows between the two numbers indicate that they are to be multiplied. The downward arrows indicate first product while upward arrows indicate the second product. The second product is to be subtracted from the first.

EXAMPLE 6. The system of equations $ax-by=a^2-b^2$ and $x+y=a+b$ has the solution.

a. $x=a$ and $y=b$ b. $x=-a$ and $y=-b$

c. $x=b$ and $y=a$ d. $x=-b$ and $y=-a$

Sol. a. Given, pairs of linear equations are

$$ax-by-(a^2-b^2)=0 \quad \text{...(i)}$$

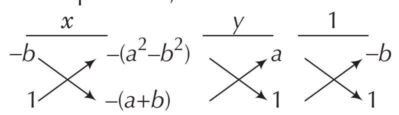
and $x+y-(a+b)=0$ ...(ii)

On comparing Eqs. (i) and (ii) with $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$, we get

$$a_1=a,\ b_1=-b,\ c_1=-(a^2-b^2)$$

$$a_2=1,\ b_2=1,\ c_2=-(a+b)$$

By cross multiplication, we have



$$\Rightarrow \frac{x}{b(a+b)+(a^2-b^2)}=\frac{y}{-(a^2-b^2+a(a+b))}=\frac{1}{a+b}$$

$$\Rightarrow \frac{x}{ab+a^2}=\frac{y}{b^2+ab}=\frac{1}{a+b} \Rightarrow x=\frac{ab+a^2}{a+b} \text{ and } y=\frac{b^2+ab}{a+b}$$

$$\Rightarrow x=\frac{a(a+b)}{a+b}=a \text{ and } y=\frac{b(b+a)}{a+b}=b$$

So, $x=a$ and $y=b$ is the required solution.

EQUATIONS REDUCIBLE TO A PAIR OF LINEAR EQUATIONS

We have several situations when there are two equations that are not linear, but can be reduced to a pair of linear equations. By making suitable substitutions.

EXAMPLE 7. If $\frac{2}{x}+\frac{3}{y}=13$ and $\frac{5}{x}-\frac{4}{y}=-2$, then what is the value of $\frac{x}{y}$?

a. $\frac{2}{3}$ b. $\frac{3}{2}$ c. $\frac{1}{3}$ d. $\frac{1}{2}$

Sol. **b.** Let $\frac{1}{x}=p$ and $\frac{1}{y}=q$

So, we have, $2p+3q=13$ ...(i)

and $5p-4q=-2$ ...(ii)

On multiplying Eq. (i) by 4 and Eq. (ii) by 3 and then adding, we get

$$8p+15p=52-6 \Rightarrow 23p=46 \Rightarrow p=2$$

On putting $p=2$ in Eq. (i), we get

$$2\times 2+3q=13 \Rightarrow 3q=13-4=9 \Rightarrow q=\frac{9}{3}=3$$

Now, $x=\frac{1}{p} \Rightarrow x=\frac{1}{2}$ and $y=\frac{1}{q} \Rightarrow y=\frac{1}{3}$.

Hence, the value of $\frac{x}{y}$ is $\frac{3}{2}$.

EXAMPLE 8. Find the solution of the system of equations $4x+3y=18xy$ and $2x-5y=-4xy$.

a. $x=2, y=3$ b. $x=3, y=2$ c. $x=\frac{1}{3}, y=\frac{1}{2}$ d. $x=\frac{1}{2}, y=\frac{1}{3}$

Sol. **d.** Given equations are, $4x+3y=18xy$ ...(i)

and $2x-5y=-4xy$ ...(ii)

On dividing both the equations by xy, we get

$$\frac{4x}{xy}+\frac{3y}{xy}=\frac{18xy}{xy} \text{ and } \frac{2x}{xy}-\frac{5y}{xy}=\frac{-4xy}{xy}$$

or $$\frac{4}{y}+\frac{3}{x}=18 \text{ and } \frac{2}{y}-\frac{5}{x}=-4$$

On putting $\frac{1}{y}=A$ and $\frac{1}{x}=B$, we get

$4A+3B=18$ ...(iii)

$2A-5B=-4$ ...(iv)

On multiplying Eq. (iv) by 2 and subtracting from Eq. (iii), we get

$$\begin{array}{r} 4A+3B=18 \\ 4A-10B=-8 \\ -\quad +\quad\quad + \\ \hline 13B=26 \end{array}$$

$\Rightarrow$ $B=2$

On putting, the value of B in Eq. (iii), we get

$\Rightarrow$ $4A+6=18 \Rightarrow 4A=12 \Rightarrow A=3$

Now $\frac{1}{x}=B=2 \Rightarrow x=\frac{1}{2}$ and $\frac{1}{y}=A=3 \Rightarrow y=\frac{1}{3}$

So, $x=\frac{1}{2}$ and $y=\frac{1}{3}$ is the required solution.

EXAMPLE 9. If $\frac{3}{x+y}+\frac{2}{x-y}=2$ and $\frac{9}{x+y}-\frac{4}{x-y}=1$, then what is value of $\frac{x}{y}$?

a. $\frac{3}{2}$ b. 5 c. $\frac{2}{3}$ d. $\frac{1}{5}$

Sol. **b.** Given equations are,

$\frac{3}{x+y}+\frac{2}{x-y}=2$ ...(i)

and $\frac{9}{x+y}-\frac{4}{x-y}=1$ ...(ii)

Let $\frac{1}{x+y}=a$ and $\frac{1}{x-y}=b$

$\therefore$ $3a+2b=2$ ...(iii)

$9a-4b=1$ ...(iv)

On multiplying Eq. (iii) by 2 and adding with Eq. (iv), we get

$\Rightarrow$ $6a+9a=5 \Rightarrow 15a=5$

$\therefore$ $a=\frac{1}{3}$

$\therefore$ $\frac{1}{x+y}=\frac{1}{3} \Rightarrow x+y=3$ ...(v)

On putting the value of a in Eq. (iii), we get

$$3\times\frac{1}{3}+2b=2 \Rightarrow b=\frac{1}{2}$$

$\Rightarrow$ $\frac{1}{x-y}=\frac{1}{2} \Rightarrow x-y=2$ ...(vi)

On adding Eqs. (v) and (vi), we get, $2x=5 \Rightarrow x=\frac{5}{2}$

From Eq. (v), $x+y=3 \Rightarrow y=3-\frac{5}{2}=\frac{1}{2}$

$\therefore$ $$\frac{x}{y}=\frac{\frac{5}{2}}{\frac{1}{2}}=5$$

LINEAR EQUATION IN THREE VARIABLES

An equation in the form of $ax+by+cz=r$, where a, b, c and r are real numbers and a, b and c are not all zeros is called a linear equation in three variables

e.g. $3x+4y-7z=2$, $-2x+y-z=6$ and $x+y+z=2$ are all linear equation in three variables.

Use substitution and elimination method to solve the system of three equations in three variables.

EXAMPLE 10. Solve the system $x+y+z=5$, $2x-y+z=9$ and $x-2y+3z=16$.

a. (2, – 1, 4) b. (3, 5, 6) c. (1, – 2, 5) d. (6, 1, – 3)

Sol. a. Given equations are, $x+y+z=5$ ...(i)

$2x-y+z=9$ ...(ii)

$x-2y+3z=16$ ...(iii)

First we eliminate y by adding Eqs. (i) and (ii), we get

$$\begin{array}{r} x+y+z=5 \\ \underline{2x-y+z=9} \\ 3x+2z=14 \end{array} \quad \text{...(iv)}$$

Next we eliminate y by multiplying Eq. (i) by 2 and then adding it to Eq. (iii), we get

$$\begin{array}{r} 2x+2y+2z=10 \\ \underline{x-2y+3z=16} \\ 3x+5z=26 \end{array} \quad \text{...(v)}$$

Now subtract Eq. (iv) from Eq. (v), we get

$3z=12 \Rightarrow z=4$

Substitute $z=4$ in Eq. (iv), $3x+2\times4=14 \Rightarrow 3x=6 \Rightarrow x=2$.

Finally put $x=2$ and $z=4$ in Eq. (i), we get

$2+y+4=5 \Rightarrow y=-1$

$\therefore$ (2, –1, 4) is the required solution.

APPLICATION OF LINEAR EQUATIONS

The application of linear equations are discussed below.

Problem Based on Ages

If the problem involves finding out the ages of two persons, then take the present age of one person as x and that of the other as y.

Then, 'a' years ago, age of Ist person was $(x-a)$ years and that of 2nd person was $(y-a)$ years.

After b years, age of Ist person will be $(x+b)$ years and that of 2nd person will be $(y+b)$ years.

Formulate the equations and then solve them.

EXAMPLE 11. Age of X is six times that of Y. After 4 yr, X is 4 times elder to Y. What is the present age of Y?

a. 4 yr b. 5 yr c. 6 yr d. 7 yr

Sol. c. Let the present age of X and Y be x and y yr, respectively.

Condition I Age of X = Age of $Y \times 6$, $x=6y$ ...(i)

After 4 yr, Age of $X=(x+4)$ yr, Age of $Y=(y+4)$ yr

Condition II $(x+4)=(y+4)\times4$, $x+4=4y+16$ ...(ii)

Put the value of $x=6y$ in Eq. (ii), we get

$\Rightarrow$ $6y+4=4y+16 \Rightarrow 2y=12$

$\therefore$ $y=6$

So, the present age of Y is 6 yr.

Problem Based on Numbers

Let the digit in unit's place be x and that in ten's place be y. Then, the two-digits number is given by $10y+x$. On interchanging the positions of the digits, the digit in unit place becomes y and that in ten's place becomes x, and the number becomes $10x+y$. Formulate the equations and then solve them.

EXAMPLE 12. A number consists of two digits, whose sum is 8. If 18 is added to the number, the digits are reversed. The number is equal to

a. 26 b. 35 c. 53 d. 62

Sol. b. Let the unit's place digit and ten's place digit be x and y, respectively.

$\therefore$ Original number $=10y+x$

Number obtained by reversing the order of digit $=10x+y$

Condition I Sum of digits = 8

$\Rightarrow$ $x+y=8$ ...(i)

Condition II $10x+y=10y+x+18$

$\Rightarrow$ $10x-10y+y-x=18 \Rightarrow 9x-9y=18$

$\Rightarrow$ $x-y=2$ ...(ii)

On adding Eqs. (i) and (ii), we get

$x+y+x-y=8+2 \Rightarrow 2x=10$

$\therefore$ $x=5$

On putting $x=5$ in Eq. (i), we get

$5+y=8 \Rightarrow y=3$

Hence, Required number $=10y+x=10\times3+5=35$

Problem Based on Fractions

Let the numerator of the fraction be x and denominator be y, then the fraction is x/y.

Formulate the linear equations on the basis of conditions given and solve them.

EXAMPLE 13. If we add 1 to the numerator and subtract 1 from the denominator, a fraction becomes 1. It also becomes $\frac{1}{2}$, if we only add 1 to the denominator. The fraction is

a. 4/5 b. 3/5 c. 2/5 d. 1/5

Sol. b. Let the numerator and the denominator of the fraction be x and y, respectively.

$\therefore$ Fraction $=\dfrac{x}{y}$

Condition I When 1 is added to the numerator and 1 is subtract from the denominator, then new fraction $=\dfrac{x+1}{y-1}$

According to the question,

$\dfrac{x+1}{y-1}=1 \Rightarrow x+1=y-1 \Rightarrow x-y=-2$ ...(i)

Condition II When 1 is added to denominator, then new fraction $= \dfrac{x}{y+1}$

According to the question, $\dfrac{x}{y+1} = \dfrac{1}{2} \Rightarrow 2x = y + 1$

$\Rightarrow \quad 2x - y = 1 \quad \ldots(ii)$

On subtracting Eq. (ii) from Eq. (i), we get

$-x = -3 \Rightarrow x = 3$

Put the value of x in Eq. (i), we get

$3 - y = -2 \Rightarrow y = 5$

Hence, the fractions is $\dfrac{x}{y} = \dfrac{3}{5}$

Problem Based on Distance, Speed and Time

Speed is the distance covered by an object per unit time.

i.e. $\quad \text{Speed} = \dfrac{\textit{Distance}}{\textit{Time}}$

If the speed of boat in still water be x km/h and speed of stream be y km/h. Then, the speed of boat downstream $= (x + y)$ km/h
and speed of boat upstream $= (x - y)$ km/h

EXAMPLE 14. A motorboat takes 2h to travel a distance of 9 km down the current and it takes 6h to travel the same distance against the current. What is the speed of the boat in still water ?

a. 3 km/h b. 2 km/h c. 1.5 km/h d. 1 km/h

Sol. **a.** Let speed of the motorboat in still water be x km/h and speed of the stream be y km/h.

Then, speed of the motorboat downstream $= (x + y)$ km/h
and speed of the boat upstream $= (x - y)$ km/h

Condition I When motorboat goes 9 km downstream, then

$$2 = \frac{9}{x+y} \qquad \left[\because \text{Time} = \frac{\text{Distance}}{\text{Speed}}\right]$$

$\Rightarrow \quad 2x + 2y = 9 \quad \ldots(i)$

Condition II When motorboat goes 9 km upstream, then

$6 = \dfrac{9}{x-y} \Rightarrow 6x - 6y = 9 \quad \ldots(ii)$

On multiplying Eq. (i) by 3, we get

$6x + 6y = 27 \quad \ldots(iii)$

Now, adding Eqs. (ii) and (iii), we get

$12x = 36 \Rightarrow x = 3$ km/h

Problem Based on Fixed and running Charges

Let the fixed charge of any commodity be ₹ x and rate of running charges be ₹ y, then total cost $= x +$ total running charges.
Formulate the equations and then solve them.

EXAMPLE 15. A railway ticket for a child costs half the full fare but the reservation charge is the same on half tickets as much as on full ticket. One reserved first class ticket for a journey between two stations is ₹ 362, one full and one half reserved first class tickets cost ₹ 554. What is the reservation charge?

a. ₹ 18 b. ₹ 22 c. ₹ 38 d. ₹ 46

Sol. **b.** Let full fare and reservation charge be ₹ x and ₹ y, respectively. Then, full ticket $= x + y$

half ticket $= \dfrac{x}{2} + y$

According to question, $x + y = 362 \quad \ldots(i)$

and $\quad (x + y) + \left(\dfrac{x}{2} + y\right) = 554 = \dfrac{3}{2}x + 2y = 554 \quad \ldots(ii)$

On multiplying Eq. (i) by 2, we get

$2x + 2y = 724 \quad \ldots(iii)$

On subtracting Eq. (ii) from Eq. (iii), we get

$\dfrac{x}{2} = 170 \Rightarrow x = 340$

On putting value of x in Eq. (i), we get

$340 + y = 362 \Rightarrow y = 22$

Hence, the reservation charge is ₹ 22.

Problems Based on Mensuration

Make use of given formulae to formulate the equation

1. Area of rectangle = length × breadth
   Perimeter of rectangle = 2 (length+Breadth)
2. The sum of angles of a triangle is 180°
3. The sum of opposite angles of cyclic quadrilateral is 180°
4. In parallelogram, opposite angles are equal.

EXAMPLE 16. The length of a rectangle is 8 cm more than its breadth. If the perimeter of the rectangle of 68 cm, then its length and breadth are

a. 21 cm, 13 cm b. 14 cm, 23 cm
c. 19 cm, 20 cm d. 9 cm, 15 cm

Sol. **a.** Let the length and breadth of a rectangle be x and y, respectively.

Condition I $\quad x = y + 8 \Rightarrow x - y = 8 \quad \ldots(i)$

Condition II Perimeter of rectangle = 68 cm

$\Rightarrow \quad 2[x + y] = 68, \quad x + y = 34 \quad \ldots(ii)$

On adding Eq. (i) and (ii), we get

$2x = 42 \Rightarrow x = 21$

On putting $x = 21$ in Eq. (ii), we get

$21 + y = 34 \Rightarrow y = 34 - 21 = 13$

Hence, length of rectangle = 21 cm
and breadth of rectangle = 13 cm

EXAMPLE 17. The area of a rectangle is decreased by $28\ m^2$, if the length is increased by 2 m and the breadth is decreased by 2 m. The area of a rectangle is increased by $33\ m^2$, if the length is decreased by 1m and the breadth is increased by 2 m, then the area of rectangle is

a. $253\ m^2$ b. $235\ m^2$ c. $532\ m^2$ d. $352\ m^2$

Sol. a. Let the length and breadth of a rectangle be x and y, respectively.

$\therefore$ Original area of rectangle $= xy\ m^2$

Condition I If length $= (x+2)$ m and breadth $= (y-2)$ m

$\therefore$ Area of rectangle $= (x+2)(y-2)\ m^2$

According to the question,

$$(x+2)(y-2) = xy - 28$$

$$\Rightarrow \quad xy - 2x + 2y - 4 = xy - 28$$

$$\Rightarrow \quad 2x - 2y = 24 \Rightarrow x - y = 12 \quad \ldots(i)$$

Condition II If, length $= (x-1)$ m and breadth $= (y+2)$ m

$\therefore$ Now the area of rectangle $= (x-1)(y+2)\ m^2$

According to the question,

$$(x-1)(y+2) = xy + 33$$

$$\Rightarrow \quad xy + 2x - y - 2 = xy + 33$$

$$\Rightarrow \quad 2x - y = 35 \quad \ldots(ii)$$

On subtracting Eq. (i) from (ii), we get, $x = 23$

On putting $x = 23$ in Eq. (i), we get,

$$23 - y = 12$$

$$\therefore \quad y = 23 - 12 = 11$$

So, original area of rectangle $= xy = 23 \times 11 = 253\ m^2$

PRACTICE EXERCISE

1. Find x, if $25x - 19 - [3 - \{4x - 5\}] = 3x - (6x - 5)$.

(a) $x = 1$ (b) $x = -1$ (c) $x = \frac{1}{2}$ (d) $x = 2$

2. If a number is subtracted from three-fourth of itself, the value so obtained is -130. Then, what is the number?

(a) 540 (b) 560 (c) 420 (d) 520

3. Sum of two numbers is 21 and their difference is 11, then the greatest number is

(a) 5 (b) 16 (c) 9 (d) 10

4. Which of the following equations have $x = 2$ and $y = 1$ as a solution?

I. $2x + 5y = 9$ II. $5x + 3y = 14$

III. $2x + 3y = 7$ IV. $2x - 3y = 1$

Select the correct answer using the codes given below

(a) I and IV (b) II and III (c) Only I (d) I, III and IV

5. The solution of the pair of equation $\frac{x}{2} + y = 0.8$ and $\frac{7}{x + \frac{y}{2}} = 10$ is

(a) $x = \frac{2}{5}, y = \frac{3}{5}$ (b) $x = \frac{2}{3}, y = 5$

(c) $x = \frac{2}{5}, y = \frac{5}{3}$ (d) $x = \frac{3}{5}, y = \frac{2}{5}$

6. A system of two simultaneous linear equations in two variables has a unique solution if their graphs

(a) are coincident (b) are parallel

(c) intersect in one point (d) None of these

7. The system of equations $6x + 5y = 11$ and $9x + \frac{15}{2}y = 21$ has

(a) a unique solution (b) many solution

(c) no solution (d) None of these

8. The sum of two numbers is 2490 and if 6.5% of one number is equal to 8.5% of the other, then numbers are

(a) 1414, 1076 (b) 1411, 1079

(c) 1412, 1078 (d) None of these

9. Given two linear equations $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$, if $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the graph is

(a) parallel (b) intersection at one point

(c) coincident (d) None of these

10. For what value of k, the following equations will be inconsistent? $4x + 6y = 11$ and $2x + ky = 7$

(a) $k = -3$ (b) $k = \frac{12}{5}$ (c) $k = 12$ (d) $k = 3$

11. For what value of k, the system of equations has infinitely many solutions $2x - ky = 4$ and $3x + 2y = 6$?

(a) $\frac{4}{3}$ (b) $\frac{-4}{3}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

12. For what value of α, the system of equation $\alpha x + 3y = \alpha - 3$, $12x + \alpha y = \alpha$ will have a unique solution?

(a) $\alpha = \pm 6$ (b) $\alpha = 6$ (c) $\alpha \neq \pm 6$ (d) $\alpha = -6$

13. Consider the following statements:

I. The solution of equation $2x+3y=425$ and $3x+2y=350$ is a positive integral pair.

II. for $k=-2$, the equations $x-ky=2$ and $3x+6y=-5$ have infinitely many solutions.

III. (2, 5) and (–1, 3) are solutions of $2x+5y=13$

Select the correct answer using the codes given below:

(a) Only I (b) Both I and II
(c) Both II and III (d) All of these

14. A horse and two cows together cost ₹ 680. If a horse costs ₹ 80 more than a cow, then the cost of horse is

(a) ₹ 170 (b) ₹ 280 (c) ₹ 200 (d) ₹ 220

15. Sunita has 10 paise and 50 paise coins in her purse. If the total number of coins is 17 and their total value is ₹ 4.50, then number of 10 paise coins is

(a) 9 (b) 7 (c) 10 (d) 5

16. The solution of the equation $\frac{3x-y+1}{3}=\frac{2x+y+2}{5}=\frac{3x+2y+1}{6}$ is given by which one of the following?

(a) $x=2, y=1$ (b) $x=1, y=1$
(c) $x=-1, y=-1$ (d) $x=1, y=2$

17. The line $3x-5y=-10$ cuts Y-axis at

(a) (0, 2) (b) (0, 1) (c) (0, 3) (0, 4)

18. The equations $px+q=0$ and $rx+s=0$ are consistent, if

(a) $ps=qr$ (b) $ps+qr=0$ (c) $pq-rs=0$ (d) $pq+rs=0$

19. A streamer goes downstream and covers the distance between two ports in 4 h while it covers the same distance upstream in 5 h. If the speed of the stream is 2 km/h, then the speed of the streamer in still water is

(a) 20 km/h (b) 19 km/h (c) 18 km/h (d) 19.5 km/h

20. The solution of the system of linear equations $0.4x+0.3y=1.7$ and $0.7x-0.2y=0.8$ is

(a) $x=3, y=2$ (b) $x=2, y=-3$
(c) $x=2, y=3$ (d) None of these

21. If $\sqrt{2}x-\sqrt{3}y=0$ and $\sqrt{7}x+\sqrt{2}y=0$, then find the value of $x+y$.

(a) 1 (b) 2 (c) 3 (d) 0

22. A man starts his job with a certain monthly salary and earns a fixed increment every year. If his salary was ₹ 1500 after 4 yr of services and ₹ 1800 after 10 yr of service. What was his starting salary?

(a) ₹ 1300 (b) ₹ 1200 (c) ₹ 50 (d) ₹ 1100

23. If $\frac{44}{x+y}+\frac{30}{x-y}=10, \frac{55}{x+y}+\frac{40}{x-y}=13$, then find the value of x and y.

(a) $x=2, y=8$ (b) $x=8, y=3$
(c) $x=8, y=2$ (d) $x=3, y=8$

24. Consider the following sets of equations

I. $2x-y=0$ and $6x-3y=0$

II. $3x-4y=0$ and $12x-20y=0$ Then,

(a) Both sets I and II possess unique solution
(b) Set I possesses unique solution and set II has infinitely many solutions
(c) Set I possesses infinitely many solutions and set II possess unique solution
(d) None of the sets I and II possesses a unique solution

25. A fraction becomes $\frac{4}{5}$, if 1 is added to both numerator and denominator. If however 5 is subtracted from both numerator and denominator the fraction becomes $\frac{1}{2}$. Then, the fractions is

(a) $\frac{7}{9}$ (b) $\frac{9}{7}$ (c) $\frac{3}{5}$ (d) $\frac{4}{3}$

26. For what value of k, the following system of equations $3x+4y=6$ and $6x+8y=k$ represents coincident lines?

(a) 12 (b) 11 (c) 13 (d) 10

27. The area of a rectangle remains the same if the length is increased by 7 m and the breadth is decreased by 3 m. The area remains unaffected if the length is decreased by 7 m and the breadth is increased by 5 m, then area of rectangle is

(a) 280 m^2 (b) 320 m^2 (c) 420 m^2 (d) 400 m^2

28. The value of x in the solution of the equation $2^{x+y}=2^{x-y}=\sqrt{8}$ is

(a) 0 (b) $\frac{3}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{8}$

29. A and B each have a certain number of mangoes. A says to B : "If you give 30 of your mangoes, I will have twice as many as left with you" B replies "If you give me 10, I will have thrice as many as left with you". How many mangoes did A has?

(a) 41 (b) 62 (c) 34 (d) 32

30. There are two examination rooms A and B. If 10 candidates are sent from room A to room B, the number of students in each room is the same. If 20 candidates are sent from B to A, the number of students in A is double the number of students in B. Then, number of students in room B is

(a) 40 (b) 100 (c) 80 (d) 60

31. A train started from a station with a certain number of passengers. At the first halt, 1/3rd of its passengers got down and 120 passengers got in. At the second halt, half of the passengers got down and 100 persons got in. Then, the train left for its destination with 240 passengers. How many passengers were there in the train when it started ?

(a) 540 (b) 480 (c) 360 (d) 240

32. The system of equations $x + 2y = 3$ and $3x + 6y = 9$ has

(a) unique solutions (b) no solution
(c) infinitely many solutions (d) finite number of solutions

33. The sum of digits of a two-digit number is 8 and the difference between the number and that formed by reversing the digits is 18. What is the difference between the digits of the number?

(a) 1 (b) 2 (c) 3 (d) 4

34. If $(x, y) = (4, 1)$ is the solution of the pair of linear equations $mx + y = 2x + ny = 5$, then what is $m + n$ equal to?

(a) –2 (b) –1 (c) 1 (d) 1

35. If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, then

(a) $a = 2, b = 3$ *(b)* $a = -2, b = 3$
(c) $a = 2, b = -3$ *(d)* $a = 3, b = 2$

36. The sum of two numbers is 80. If the larger number exceeds four times the smaller by 5, what is the smaller number ?

(a) 5 (b) 15 (c) 20 (d) 25

37. If $\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}$ and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy}$, where $x, y \neq 0$ and $y \neq 0$, then what is the value of $x + y$?

(a) 2 (b) 3 (c) 4 (d) 8

38. Under what condition do the equation $kx - y = 2$ and $6x - 2y = 3$ have a unique solution?

(a) $k = 3$ (b) $k \neq 3$ (c) $k = 0$ (d) $k \neq 0$

39. Let there be three simultaneous linear equations in two unknowns, which are non-parallel and non-collinear. What can be the number of solutions (if they do exist)?

(a) One or infinite (b) Only one
(c) Exactly two (d) Exactly three

40. A number consists of two digits, whose sum is 10. If 18 is subtracted from the number, digits of the number are reversed. What is the product?

(a) 15 (b) 18
(c) 24 (d) 32

41. What is the value of k for which the system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution?

(a) $-\frac{3}{14}$ (b) $-\frac{14}{3}$ (c) $\frac{1}{10}$ (d) 10

42. What is the solution of the equation $x - y = 0.9$ and $11(x + y)^{-1} = 2$?

(a) $x = 3.2$ and $y = 2.3$ (b) $x = 1$ and $y = 0.1$
(c) $x = 2$ and $y = 1.1$ (d) $x = 1.2$ and $y = 0.3$

43. Pooja started her job with certain monthly salary and gets a fixed increment every year. If her salary was ₹ 4200 after 3 yr and ₹ 6800 after 8 yr of service, then what are her initial salary and the annual increment, respectively ?

(a) ₹ 2640, ₹ 320 (b) ₹ 2460, ₹ 320
(c) ₹ 2460, ₹ 520 (d) ₹ 2640, ₹ 520

44. A person bought 5 tickets from the station P to a station Q and 10 tickets from the station P to a station R. He paid ₹ 350. If the sum of fare of a ticket from P to Q and a ticket from P to R is ₹ 42, then what is the fare from P to Q?

(a) ₹ 12 (b) ₹ 14
(c) ₹ 16 (d) ₹ 18

45. The Community Relief fund receives a large donation of \$ 2800. The foundation agrees to spend the money on \$ 20 school bags, \$ 25 sweaters, \$ 5 books. They want to buy 200 items and send them to schools in earthquake-hit areas. They must order as many books as school bages and sweaters combined.

How many of each item should they order?

(a) (40, 60, 100) (b) (20, 30, 80)
(c) (50, 100, 60) (d) 40, 80, 25)

Directions (Q. Nos. 46-47) *Some money is divided among Rajesh, Sonal and Chetan in such a way that 2 times share of Rajesh, 3 times share of Sonal and 5 times share of Chetan are all equal.*

Now, answer the following questions based on above information.

46. If the sum of 6 times the share with Rajesh and 6 times the share with Sonal is ₹ 150, then find the share of Chetan?

(a) ₹ 5 (b) ₹ 6 (c) ₹ 10 (d) ₹ 12

47. If the sum of shares of Rajesh, Chetan and Sonal is ₹ 155, then find the share of Rajesh.

(a) ₹ 75 (b) ₹ 35 (c) ₹ 50 (d) ₹ 25

PREVIOUS YEARS' QUESTIONS

48. The graphs of $ax + by = c$, $dx + ey = f$ will be

I. Parallel, if the system has no solution.
II. Coincident, if the system has finite number of solutions.
III. Intersecting, if the system has only one solutions

Which of the statement(s) given above is/are correct? **2012 I**

(a) Both I and II (b) Both II and III
(c) Both I and III (d) All of these

49. The sum of two numbers is 20 and their product is 75. What is the sum of their reciprocals? **2012 I**

(a) $\frac{1}{15}$ (b) $\frac{1}{5}$ (c) $\frac{4}{15}$ (d) $\frac{7}{15}$

50. If $3^{x+y} = 81$ and $81^{x-y} = 3$, then what is the value of x? **2012 I**

(a) $\frac{17}{16}$ (b) $\frac{17}{8}$ (c) $\frac{17}{4}$ (d) $\frac{15}{4}$

51. If $\frac{a}{b} - \frac{b}{a} = \frac{x}{y}$ and $\frac{a}{b} + \frac{b}{a} = x - y$, then what is the value of x? **2013 I**

(a) $\frac{a+b}{a}$ (b) $\frac{a+b}{b}$ (c) $\frac{a-b}{a}$ (d) None of these

52. The system of equations $3x + y - 4 = 0$ and $6x + 2y - 8 = 0$ has **2013 I**

(a) a unique solution $x = 1$, $y = 1$
(b) a unique solution $x = 0$, $y = 4$
(c) no solution
(d) infinite solutions

53. The sum of two numbers is 7 and the sum of their squares is 25. The product of the two numbers is **2013 I**

(a) 6 (b) 10 (c) 12 (d) 15

Directions (Q. Nos. 54-55) *A number consists of two digits whose sum is 10. If the digits of the number are reversed, then the number is decreased by 36.*

54. Which of the following is/are correct?

I. The number is divisible by a composite number.
II. The number is a multiple of a prime number.

Select the correct answer using the codes given below **2013 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

55. What is the products of the two digits? **2013 I**

(a) 21 (b) 24 (c) 36 (d) 42

56. Ten chairs and six tables together cost ₹ 6200, three chairs and two tables together cost ₹ 1900. The cost of 4 chairs and 5 tables is **2013 I**

(a) ₹ 3000 (b) ₹ 3300 (c) ₹ 3500 (d) ₹ 3800

57. A bus starts with some passengers. At the first stop, one-fifth of the passengers gets down and 40 passengers get in. At the second stop, half of the passengers gets down and 30 get in. The number of passengers now is 70. The number of passengers with which the bus started was **2013 II**

(a) 40 (b) 50 (c) 60 (d) 70

58. If $x + y - 7 = 0$ and $3x + y - 13 = 0$, then what is $4x^2 + y^2 + 4xy$ equal to? **2013 II**

(a) 75 (b) 85 (c) 91 (d) 100

59. If $\frac{x}{2} + \frac{y}{3} = 4$ and $\frac{2}{x} + \frac{3}{y} = 1$, then what is $x + y$ equal to? **2013 II**

(a) 11 (b) 10 (c) 9 (d) 8

60. If $x + y = 5$, $y + z = 10$ and $z + x = 15$, then which one of the following is correct? **2014 I**

(a) $z > x > y$ (b) $z > y > x$ (c) $x > y > z$ (d) $x > z > y$

61. The present age of Ravi's father is 4 times Ravi's present age. 5 yr back, Ravi's father was seven times as old as Ravi was at that time. What is the present age of Ravi's father? **2014 I**

(a) 84 yr (b) 70 yr (c) 40 yr (d) 35 yr

62. The sum of two positive number x and y is 2.5 times their difference. If the product of numbers is 84, then what is the sum of those two numbers? **2014 I**

(a) 26 (b) 24 (c) 22 (d) 20

63. Two chairs and one table cost ₹ 700 and one chair and Two tables cost ₹ 800. If cost m tables and m chairs is ₹ 30000, then what is m equal to? **2014 I**

(a) 60 (b) 55 (c) 50 (d) 45

64. A certain number of two digits is three times the sum of its digits. If 45 is added to the number, then the digits will be reversed. What is the sum of the squares of the two digits of the number?

(a) 41 (b) 45 **2014 II**
(c) 53 (d) 64

65. A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **2015 I**

(a) 12 (b) 15 (c) 25 (d) 32

66. A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was 3/4 full. Oil of how many bottles can the tin contain? (All bottles are of equal volume) **2015 II**

(a) 35 (b) 40 (c) 45 (d) 50

67. A number consists of two digits, whose sum is 7. If the digits are reversed, the number is increased by 27. The product of digits of the number is **2015 II**

(a) 6 (b) 8 (c) 10 (d) 12

68. If $\frac{p}{x}+\frac{q}{y}=m$ and $\frac{q}{x}+\frac{p}{y}=n$, then what is $\frac{x}{y}$ equal to? **2016 I**

(a) $\frac{np+mq}{mp+nq}$ (b) $\frac{np+mq}{mp-nq}$

(c) $\frac{np-mq}{mp-nq}$ (d) $\frac{np-mq}{mp+nq}$

69. The value of k, for which the system of equations $3x-ky-20=0$ and $6x-10y+40=0$ has no solution, is **2016 I**

(a) 10 (b) 6 (c) 5 (d) 3

70. There are three brothers. The sums of ages of two of them at a time are 4 yr, 6 yr and 8 yr. The age difference between the eldest and the youngest is **2016 I**

(a) 3 yr (b) 4 yr (c) 5 yr (d) 6 yr

71. The annual incomes of two persons are in the ratio 9 : 7 and their expenses are in the ratio 4 : 3. If each of them saves ₹ 2000 per year, then what is the difference in their annual incomes? **2016 I**

(a) ₹ 4000 (b) ₹ 4500 (c) ₹ 5000 (d) ₹ 5500

72. Let a two digits number be k times the sum of its digits. If the number formed by interchanging the digits is m times the sum of the digits, then the value of m is **2016 I**

(a) $9-k$ (b) $10-k$ (c) $11-k$ (d) $k-1$

ANSWERS

1	*a*	2	*d*	3	*b*	4	*d*	5	*a*	6	*c*	7	*c*	8	*b*	9	*b*	10	*d*
11	*b*	12	*c*	13	*a*	14	*b*	15	*c*	16	*b*	17	*a*	18	*a*	19	*c*	20	*c*
21	*d*	22	*a*	23	*b*	24	*c*	25	*a*	26	*a*	27	*c*	28	*b*	29	*c*	30	*c*
31	*d*	32	*c*	33	*b*	34	*a*	35	*d*	36	*b*	37	*c*	38	*b*	39	*b*	40	*c*
41	*d*	42	*a*	43	*d*	44	*b*	45	*a*	46	*b*	47	*a*	48	*c*	49	*c*	50	*b*
51	*d*	52	*d*	53	*c*	54	*b*	55	*a*	56	*a*	57	*b*	58	*d*	59	*b*	60	*a*
61	*c*	62	*d*	63	*a*	64	*c*	65	*a*	66	*b*	67	*c*	68	*c*	69	*c*	70	*b*
71	*a*	72	*c*																

HINTS AND SOLUTIONS

1. *(a)* We have, $25x-19-[3-\{4x-5\}]$

$$=3x-(6x-5)$$

$$\Rightarrow \quad 25x-19-[3-4x+5]=3x-6x+5$$

$$\Rightarrow \quad 25x-19+4x-8=-3x+5$$

$$\Rightarrow \quad 29x+3x=5+27$$

$$\Rightarrow \quad 32x=32 \Rightarrow x=\frac{32}{32}=1 \Rightarrow \quad x=1$$

Hence, the value of x is 1.

2. *(d)* Let the number be x.

According to the question,

$$\frac{3x}{4}-x=-130 \Rightarrow \frac{3x-4x}{4}=-130$$

$$\Rightarrow \quad \frac{-x}{4}=-130 \quad \Rightarrow \quad x=520$$

Hence, the number is 520.

3. *(b)* Let numbers be x and y, respectively.

Then by given condition,

$$x+y=21 \qquad \ldots\text{(i)}$$

$$x-y=11 \qquad \ldots\text{(ii)}$$

On adding Eqs. (i) and (ii), we get $2x=32$

$$x=\frac{32}{2}=16$$

On putting $y=16$ in Eq. (i), we get

$$x+16=21$$

$$\therefore \quad x=5$$

Hence, the greatest number is 16.

4. *(d)* Put $x=2$ and $y=1$ in each equation

I. $2x+5y=9 \quad \Rightarrow 2(2)+5(1)=9$

$9=9$, it is true.

II. $5x+3y=14$

$\Rightarrow \quad 5(2)+3(1)=14$

$\Rightarrow \quad 13=14$, it is false.

III. $2x+3y=7 \Rightarrow 2(2)+3(1)=7$

$7=7$, it is true.

IV. $2x-3y=1 \Rightarrow 2(2)-3(1)=1$

$1=1$, it is true.

So, $x=2$ and $y=1$ is a solution of I, III and IV.

5. *(a)* Given equations are,

$$x+2y=1.6 \qquad \ldots\text{(i)}$$

and $$x+\frac{y}{2}=\frac{7}{10} \qquad \ldots\text{(ii)}$$

On multiplying Eq. (i) by 10 and Eq. (ii) by 10, we get

or $$10x+20y=16 \qquad \ldots\text{(iii)}$$

and $$10x+5y=7 \qquad \ldots\text{(iv)}$$

On subtracting, Eq. (iv) from Eq. (iii), we get

$$(10x + 20y) - (10x + 5y) = 16 - 7$$

$$\Rightarrow \quad 15y = 9 \Rightarrow y = \frac{9}{15} \Rightarrow y = \frac{3}{5}$$

On putting the value of y in Eq (i), we get

$$x + 2y = \frac{8}{5}, \quad x + \frac{6}{5} = \frac{8}{5}$$

$$\Rightarrow \quad x = \frac{8}{5} - \frac{6}{5} = \frac{2}{5}$$

6. *(c)* A pair of linear equations in two variables has a unique solution if their graphs intersect in one point.

7. *(c)* Here, $\frac{a_1}{a_2} = \frac{6}{9} = \frac{2}{3}, \frac{b_1}{b_2} = \frac{5}{15/2} = \frac{2}{3}$

and $\quad \frac{c_1}{c_2} = \frac{11}{21} \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

So, the system has no solution.

8. *(b)* Let the numbers be x and $2490 - x$.

6.5% of $x = \frac{6.5}{100} \times x = \frac{13x}{200}$

8.5% of $(2490 - x) = \frac{8.5}{100}(2490 - x)$

$$= \frac{17}{200}(2490 - x)$$

By given condition, $\frac{13x}{200} = \frac{17(2490 - x)}{200}$

$$\Rightarrow \quad 13x = 17(2490 - x)$$

$$\Rightarrow \quad 13x + 17x = 42330$$

$$\Rightarrow \quad x = \frac{42330}{30} = 1411$$

$\therefore$ Second number $= (2490 - x)$

$$= 2490 - 1411 = 1079$$

9. *(b)* As this is the case of unique solution $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, so the graph of the equations will intersect at one point.

10. *(d)* Given equations are,

$4x + 6y = 11$ and $2x + ky = 7$

Here, $\quad a_1 = 4, b_1 = 6, c_1 = 11$

and $\quad a_2 = 2, b_2 = k, c_2 = 7$

Here, $\quad \frac{a_1}{a_2} = \frac{4}{2}, \frac{b_1}{b_2} = \frac{6}{k}$ and $\frac{c_1}{c_2} = \frac{11}{7}$

Since, system is inconsistent, so

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{4}{2} = \frac{6}{k}$$

$\Rightarrow 4k = 12 \Rightarrow k = 3$

Hence, the value of k is 3.

11. *(b)* Given equations are,

$2x - ky = 4$ and $3x + 2y = 6$

Here, $a_1 = 2, b_1 = -k, c_1 = 4$

and $a_2 = 3, b_2 = 2, c_2 = 6$

Now, $\frac{a_1}{a_2} = \frac{2}{3}, \frac{b_1}{b_2} = \frac{-k}{2}$ and $\frac{c_1}{c_2} = \frac{4}{6}$

Since, the system has infinitely many solutions

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}, \quad \frac{2}{3} = \frac{-k}{2} = \frac{2}{3}$$

$$\Rightarrow \quad -3k = 4 \Rightarrow k = \frac{-4}{3}$$

12. *(c)* Given equations are,

$\alpha x + 3y = \alpha - 3$ and $12x + \alpha y = \alpha$

Here, $\quad a_1 = \alpha, b_1 = 3, c_1 = \alpha - 3$

$a_2 = 12, b_2 = \alpha, c_1 = \alpha$

Since, system has unique solution,

So, $\quad \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow \quad \frac{\alpha}{12} \neq \frac{3}{\alpha}$

$$\Rightarrow \quad \alpha^2 \neq 36 \Rightarrow \alpha \neq \pm 6$$

13. *(a)* I. $2x + 3y = 425, 3x + 2y = 350$

Solving both the equations, we get $x = 40$ and $y = 115$.

II. When $k = -2$

$$\frac{a_1}{a_2} = \frac{1}{3}, \frac{b_1}{b_2} = \frac{2}{6} = \frac{1}{3}, \frac{c_1}{c_2} = \frac{-2}{5}$$

$$\because \quad \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

Hence, the given equations have no solution.

III. (2, 5) is not a solution of $2x + 5y = 13$ as

$$2(2) + 5(5) = 4 + 25 = 29 \neq 13.$$

Hence, only I is correct.

14. *(b)* Let cost of one horse and one cow be ₹x and ₹y, respectively.

Condition I $\quad x + 2y = 680 \quad$ ...(i)

Condition II $\quad x = y + 80$

$\Rightarrow \quad x - y = 80 \quad$ ...(ii)

On subtracting Eq. (ii) from Eq. (i), we get

$$(x + 2y) - (x - y) = 680 - 80$$

$$\Rightarrow \quad 3y = 600$$

$$\therefore \quad y = 200$$

On putting $y = 200$ in Eq. (i), we get

$$x + 2 \times 200 = 680$$

$$\Rightarrow \quad x + 400 = 680 \Rightarrow x = 680 - 400$$

$$\therefore \quad x = 280$$

Hence, the cost of one horse is ₹ 280.

15. *(c)* Let number of 10 paise coins be x and number of 50 paise coins be y.

According to the questions,

Then, $\quad x + y = 17 \quad$ ...(i)

and $\quad 10x + 50y = 450 \quad$ ...(ii)

From Eq. (ii), $\quad x + 5y = 45 \quad$ ...(iii)

On subtracting Eq. (i) from Eq. (iii), we get

$$(x + 5y) - (x + y) = 45 - 17$$

$$\Rightarrow \quad 4y = 28 \Rightarrow y = \frac{28}{4} = 7$$

$\therefore$ Number of 10 paise coins

$$= x = 17 - y = 17 - 7 = 10$$

16. *(b)* Given,

$$\underset{(\text{I})}{\frac{3x - y + 1}{3}} = \underset{(\text{II})}{\frac{2x + y + 2}{5}} = \underset{(\text{III})}{\frac{3x + 2y + 1}{6}}$$

Taking Ist and IInd terms,

$$5(3x - y + 1) = 3(2x + y + 2)$$

$\Rightarrow \quad 9x - 8y = 1 \quad$ ...(i)

Taking IInd and IIIrd terms

$$6(2x + y + 2) = 5(3x + 2y + 1)$$

$\Rightarrow \quad 3x + 4y = 7 \quad$ ...(ii)

On solving Eqs. (i) and (ii), we get $y = 1$ and $x = 1$

17. *(a)* Given, $3x - 5y = -10$

At Y-axis, $x = 0$

$$\Rightarrow \quad 0 - 5y = -10 \Rightarrow y = \frac{10}{5} = 2$$

So, the point on Y-axis is (0, 2).

18. *(a)* Given, $px + q = 0$ and $rx + s = 0$

$$\Rightarrow \quad x = \frac{-q}{p} \text{ and } x = \frac{-s}{r}$$

So, $\quad \frac{-q}{p} = \frac{-s}{r} \Rightarrow ps = qr$

19. *(c)* Let the speed of the streamer in still water $= x$ km/h

then speed of streamer downstream

$= (x + 2)$ km/h

speed of streamer upstream

$= (x - 2)$ km/h

Distance travelled by streamer downstream in 4h

$= 4(x + 2)$ km

[$\because$ distance = time $\times$ speed]

and distance travelled by streamer upstream in 5 h $= 5(x - 2)$

So, $\quad 4(x + 2) = 5(x - 2)$

$$\Rightarrow \quad 4x + 8 = 5x - 10$$

Hence, $x = 18$ km/h is the speed of streamer in still water.

20. *(c)* Given, $0.4x + 0.3y = 1.7$ and $0.7x - 0.2y = 0.8$

On multiplying above equation by 10, we get

$4x + 3y = 17 \quad$ ...(i)

and $\quad 7x - 2y = 8 \quad$ ...(ii)

On multiplying Eq. (i) by 2 and Eq. (ii) by 3, we get

$$8x + 6y = 34 \quad \text{...(iii)}$$
and $$21x - 6y = 24 \quad \text{...(iv)}$$
On adding Eqs. (iii) and (iv), we get
$$(8x + 6y) + (21x - 6y) = 34 + 24$$
$\Rightarrow \quad 29x = 58$
$\therefore \quad x = 2$
On putting $x = 2$ in Eq. (i), we get
$$8 + 3y = 17 \Rightarrow y = 3$$
Hence, $x = 2$ and $y = 3$ is the required solution.

21. *(d)* Given, $\sqrt{2}x - \sqrt{3}y = 0$
and $\sqrt{7}x + \sqrt{2}y = 0$
Here, $a_1 = \sqrt{2},\ b_1 = -\sqrt{3}$
$a_2 = \sqrt{7}$ and $b_2 = \sqrt{2}$
As the equations are homogeneous equations and also
$\frac{a_1}{a_2} \ne \frac{b_1}{b_2}$. So, equation has one solution,
$\therefore \quad x = y = 0 \Rightarrow x + y = 0$
Hence, the value of $x + y$ is zero.

22. *(a)* Let the starting salary be ₹ x and annual increment be ₹ y.
By given condition, $x + 4y = 1500$ ...(i)
and $x + 10y = 1800$ ...(ii)
On subtructing Eq. (i) from Eq. (ii), we get
$$6y = 300 \Rightarrow y = 50$$
On putting $y = 50$ in Eq. (i), we get
$$x + 200 = 1500$$
$\therefore \quad x =$ ₹ 1300

23. *(b)* Put $\frac{1}{x+y} = A$ and $\frac{1}{x-y} = B$
$\Rightarrow \quad 44A + 30B = 10 \quad$ ...(i)
and $\quad 55A + 40B = 13 \quad$ ...(ii)
On multiplying Eq. (i) by 4 and Eq. (ii) by 3 and subtracting, we get
$$\begin{aligned} 176A + 120B &= 40 \\ 165A + 120B &= 39 \\ - \quad\quad - \quad &\quad - \\ \hline 11A \quad\quad &= 1 \end{aligned}$$
$\Rightarrow \quad A = 1/11 \Rightarrow x + y = 11 \quad$ ...(iii)
On putting the value of A in Eq. (i), we get
$$4 + 30B = 10 \Rightarrow 30B = 6$$
$\therefore \quad B = \frac{1}{5},\quad x - y = 5 \quad$ ...(iv)
On adding Eqs. (iii) and (iv), we get
$$2x = 16$$
$\therefore \quad x = 8$
On putting $x = 8$ in Eq. (iii), we get
$$8 + y = 11 \Rightarrow y = 3$$
Hence, $x = 8$ and $y = 3$.

24. *(c)* **Set I** $\quad 2x - y = 0$ and $6x - 3y = 0$
Here, $\quad a_1 = 2,\ b_1 = -1$
and $\quad a_2 = 6,\ b_2 = -3$
$\Rightarrow \quad \frac{a_1}{a_2} = \frac{2}{6} = \frac{1}{3} \Rightarrow \frac{b_1}{b_2} = \frac{1}{3}$
$\therefore \quad \frac{a_1}{a_2} = \frac{b_1}{b_2}$
So, system of equations have infinitely many solutions.

Set II $\quad 3x - 4y = 0$ and $12x - 20y = 0$
Here, $a_1 = 3,\ b_1 = -4$
and $\quad a_2 = 12,\ b_2 = -20$
$\therefore \quad \frac{a_1}{a_2} = \frac{3}{12} = \frac{1}{4} \Rightarrow \frac{b_1}{b_2} = \frac{-4}{-20} = \frac{1}{5}$
Here $\quad \frac{1}{4} \ne \frac{1}{5}$
So, system of equatons has unique solution.

25. *(a)* Let the fraction be $\frac{x}{y}$.
Condition I $\quad \frac{x+1}{y+1} = \frac{4}{5}$
$\Rightarrow \quad 5(x + 1) = 4(y + 1)$
$\Rightarrow \quad 5x - 4y = -1 \quad$ ...(i)
Condition II $\quad \frac{x-5}{y-5} = \frac{1}{2}$
$\Rightarrow \quad 2(x - 5) = (y - 5)$
$\Rightarrow \quad 2x - y = 5 \quad$ ...(ii)
On multiplying Eq. (ii) by 4 and subtracting from Eq. (i), we get
$$\begin{aligned} 5x - 4y &= -1 \\ 8x - 4y &= 20 \\ - \quad + \quad &\quad - \\ \hline -3x \quad\quad &= -21 \end{aligned}$$
$\therefore \quad x = 7$
Put the value of x in Eq. (i),
$\Rightarrow \quad 35 - 4y = -1 \Rightarrow y = 9$
Hence, the fraction is $\frac{7}{9}$.

26. *(a)* Given equations are,
$$3x + 4y = 6 \quad \text{...(i)}$$
and $$6x + 8y = k \quad \text{...(ii)}$$
Here, $\quad a_1 = 3,\ b_1 = 4,\ c_1 = b$
$\quad a_2 = 6,\ b_2 = 8,\ c_2 = k$
Since the system of equations represents concident line
$\therefore \quad \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, i.e. $\frac{3}{6} = \frac{4}{8} = \frac{6}{k}$
$\Rightarrow \quad 4k = 6 \times 8 \Rightarrow k = \frac{48}{4} = 12$
Hence, the value of k is 12.

27. *(c)* Let the length of rectangle be x m and breadth be y m.
$\therefore$ Original area of rectangle $= xy$
Case I Length $= (x + 7)$ m and breadth $= (y - 3)$ m
$\therefore \quad$ New area of rectangle
$= (x + 7)(y - 3)$
Given, $\quad xy = xy - 3x + 7y - 21$
$\Rightarrow \quad 3x - 7y = -21 \quad$ ...(i)
Case II Length $= (x - 7)$
and Breadth $= (y + 5)$
$\therefore$ New area of rectangle
$= (x - 7)(y + 5)$
Given, $\quad xy = xy + 5x - 7y - 35$
$\Rightarrow \quad 5x - 7y = 35 \quad$ ...(ii)
On subtracting Eq. (ii) from Eq. (i), we get
$$\begin{aligned} 3x - 7y &= -21 \\ 5x - 7y &= 35 \\ - \quad + \quad &\quad - \\ \hline -2x \quad\quad &= -56 \end{aligned}$$
$\therefore \quad x = 28$ m
On putting $x = 28$ in Eq. (ii), we get
$$5 \times 28 - 7y = 35 \Rightarrow -7y = 35 - 140$$
$\therefore \quad y = 15$
$\therefore$ Area of rectangle
$= xy = 15 \times 28 = 420\ \text{m}^2$

28. *(b)* Given, $2^{x+y} = \sqrt{8},\ 2^{x-y} = \sqrt{8}$
or $\quad 2^{x+y} = 2\sqrt{2},\ 2^{x-y} = 2\sqrt{2}$
or $\quad 2^{x+y} = 2^{3/2},\ 2^{x-y} = 2^{3/2}$
On comparing both sides, we get
$$x + y = \frac{3}{2} \quad \text{...(i)}$$
and $x - y = \frac{3}{2} \quad$ ...(ii)
On adding Eqs. (i) and (ii), we get $2x = 3$
$\Rightarrow \quad x = \frac{3}{2}$
Hence, the value of x is $\frac{3}{2}$.

29. *(c)* Let A has x mangoes and B has y mangoes.
Case I $\quad x + 30 = 2(y - 30)$
$\Rightarrow \quad x + 30 = 2y - 60$
$\quad x - 2y = -90 \quad$ ...(i)
Case II $(y + 10) = 3(x - 10)$
$\quad y = 3x - 30 - 10$
$\quad 3x - y = 40 \quad$ ...(ii)
On solving Eqs. (i) and (ii), we get
$$x = 34,\ y = 62$$
So, A has 34 mangoes.

30. *(c)* Let number of students in room A be x and in room B be y.

So, by given condition

$x - 10 = y + 10 \Rightarrow x - y = 20$ ...(i)

and $x + 20 = 2(y - 20)$

$x - 2y = -60$ ...(ii)

On subtracting Eq. (ii) from Eq. (i), we get

$(x - y) - (x - 2y) = 20 + 60 \Rightarrow y = 80$

Hence, number of students in room B is 80.

31. *(d)* Let the number of passengers in the starting be x.

Number of passengers after first halt

$$= \left[x - \frac{x}{3}\right] + 120 = \frac{2}{3}x + 120$$

and number of passengers after second halt

$$= \frac{1}{2}\left[\frac{2}{3}x + 120\right] + 100$$

But number of passengers after second halt = 240

$\therefore \quad \frac{1}{2}\left[\frac{2}{3}x + 120\right] + 100 = 240$

$\Rightarrow \frac{2}{3}x + 120 = 280 \Rightarrow \frac{2}{3}x = 160$

$\Rightarrow \quad x = 240$

32. *(c)* Given system of equations are

$x + 2y = 3$ and $3x + 6y = 9$

Here, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{3}.$

Hence, given system of the equation has infinitely many solutions.

33. *(b)* Let x be the ten's digit and y be the unit's digit of two-digit number.

By given condition, $x + y = 8$

and $(10x + y) - (10y + x) = 18$ ...(i)

$\Rightarrow \quad 9x - 9y = 18 \Rightarrow x - y = 2$ ...(ii)

On adding Eqs. (i) and (ii), we get

$2x = 10 \Rightarrow x = 5$

On putting $x = 5$ in Eq. (i), we get

$\Rightarrow \quad y = 8 - 5 = 3 \Rightarrow x = 5$ and $y = 3$

$\therefore$ Required difference of digits,

$x - y = 5 - 3 = 2$

34. *(a)* Given, $(x, y) = (4, 1)$ and

$mx + y = 2x + ny = 5$

On putting $x = 4$ and $y = 1$, we get

$\therefore \quad \underset{\text{I}}{m(4) + 1} = \underset{\text{II}}{2 \times 4 + n} = \underset{\text{III}}{5}$

$\therefore \quad 4m + 1 = 5$ and $8 + n = 5$

$\Rightarrow \quad n = -3$ and $m = 1$

$\therefore \quad m + n = 1 - 3 = -2$

35. *(d)* Given, $2^a + 3^b = 17$

and $2^{a+2} - 3^{b+1} = 5$

$\Rightarrow \quad 2^a \times 2^2 - 3^b \times 3^1 = 5$

$\Rightarrow \quad 4 \cdot 2^a - 3 \cdot 3^b = 5$

Let $2^a = x$ and $3^b = y$

Then, $x + y = 17$ ...(i)

$4x - 3y = 5$ ...(ii)

On multiplying Eq. (i) by 3 and adding to Eq. (ii), we get

$$\begin{array}{r} 3x + 3y = 51 \\ 4x - 3y = 5 \\ \hline 7x = 56 \end{array}$$

$\therefore \quad x = 8$

On putting the value of x in Eq. (i), we get

$8 + y = 17 \Rightarrow y = 9$

Now, $2^a = x \Rightarrow 2^a = 8 = (2)^3$

$\therefore \quad a = 3$ and $3^b = y = 9 \Rightarrow 3^b = 3^2$

$\therefore \quad b = 2$

Hence, $a = 3$ and $b = 2$

36. *(b)* Let smaller number be x.

and larger number $= 80 - x$

By given condition, $80 - x = 4x + 5$

$\Rightarrow \quad 5x = 75 \Rightarrow x = 15$

Hence, the smaller number is 15.

37. *(c)* Given equations are,

$\frac{2}{x} + \frac{3}{y} = \frac{9}{xy} \Rightarrow 2y + 3x = 9$ ...(i)

and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy} \Rightarrow 4y + 9x = 21$ ...(ii)

On solving Eqs. (i) and (ii), we get

$x = 1$ and $y = 3$

$\therefore \quad x + y = 1 + 3 = 4$

38. *(b)* Since, the equation $kx - y = 2$ and $6x - 2y = 3$ have a unique solution

$\therefore \quad \frac{k}{6} \neq \frac{1}{2} \Rightarrow k \neq 3$

39. *(b)* Three lines $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$

and $a_3x + b_3y + c_3 = 0$

which are non-parallel and non-collinear they have only one solution if they meet in a common point in this case these lines are called concurrent lines.

40. *(c)* Let the unit's digit be x and ten's digit's be y. Then the two-digit number

$= 10y + x$

By given condition, $x + y = 10$ ...(i)

and $10y + x - 18 = 10x + y$

$\Rightarrow 9x - 9y = -18 \Rightarrow x - y = -2$

On solving Eqs. (i) and (ii), we get

$x = 4$ and $y = 6$

$\therefore$ Required product $= xy = 4 \times 6 = 24$

41. *(d)* Since, given system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution.

Then, $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$

$\therefore \quad \frac{1}{5} = \frac{2}{k} \neq \frac{-3}{7} \Rightarrow k = 10$

42. *(a)* Given, $x - y = 0.9$ ...(i)

and $11(x + y)^{-1} = 2$ ...(ii)

$\Rightarrow \quad 2x + 2y = 11$

On multiplying Eq. (i) by 2 and adding Eqs. (i) and (ii), we get

$4x = 12.8 \Rightarrow x = 3.2$

From Eq. (i) $\quad y = 3.2 - 0.9 = 2.3$

43. *(d)* Let Pooja's initial salary be ₹ x and fixed increment every year be ₹ y.

By given condition, $x + 3y = 4200$ ...(i)

and $x + 8y = 6800$ ...(ii)

On subtracting Eq. (ii) from Eq. (i), we get

$(x + 3y) - (x + 8y) = 4200 - 6800$

$\Rightarrow \quad -5y = -2600 \Rightarrow y =$ ₹520

On putting $y = 520$ in Eq. (i), we get

$x =$ ₹ 2640

44. *(b)* Let the fare of ticket from station P to station Q is ₹ x and that from station P to station R is ₹ y.

By given condition, $x + y = 42$

and $5x + 10y = 350$

On solving Eqs. (i) and (ii), we get

$x = 14$ and $y = 28$

Hence, fare from station P to station Q is ₹ 14.

45. *(a)* Let x represent school bags, y represent sweaters, z represent books. System of equations:

$x + y + z = 200$ ...(i)

$z = x + y$ ...(ii)

$20x + 25y + 5z = 2800$ ...(iii)

putting $z = x + y$ in Eq. (i)

$(x + y) + x + y = 200$, $2x + 2y = 200$

$x + y = 100$, $x = 100 - y$

putting $x = 100 - y$ in Eq. (iii)

$20(100 - y) + 25y + 5$

$[(100 - y) + y] = 2800$

$2000 - 20y + 25y + 500 = 2800$

$5y = 300 \Rightarrow y = 60$

$x = 100 - 60 \Rightarrow x = 40$

$z = 40 + 60 \Rightarrow z = 100$

40 school bags, 60 sweaters, 100 books

46. *(b)* Let the share of Rajesh be R, Sonal be S and Chetan be C, respectively.

According to question, $2R = 3S = 5C$

also, $6R + 6S = 150$

$\Rightarrow \frac{3}{2} \times 6S + 6S = 150$

$\Rightarrow 9S + 6S = 150 \Rightarrow S = 10$

$\therefore C = \frac{3}{5}S = \frac{3}{5} \times 10 = ₹\ 6.$

47. *(a)* $R + C + S = 155$

$R + \frac{2}{5}R + \frac{2}{3}R = 155$

$\frac{15R + 6R + 10R}{15} = 155 \Rightarrow R = ₹\ 75$

48. *(c)* The graph of $ax + by = c$, $dx + ey = f$ will be coincident, if the system has infinite number of solutions. So, statement II is false.

Thus, statements I and III are correct.

49. *(c)* Let the two numbers be x and y, respectively. According to the question,

$x + y = 20$ ...(i)

and $xy = 75$ ...(ii)

$\Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{y + x}{xy} = \frac{20}{75} = \frac{4}{15}$

[from Eqs. (i) and (ii)]

50. *(b)* Given equations are,

$3^{x+y} = 81 \Rightarrow 3^{x+y} = 3^4$

$\Rightarrow x + y = 4$ ...(i)

and $81^{x-y} = 3 \Rightarrow (3^4)^{x-y} = 3^1$

$\Rightarrow x - y = \frac{1}{4}$ ...(ii)

On adding Eqs. (i) and (ii), we get

$\Rightarrow 2x = \frac{17}{4} \Rightarrow x = \frac{17}{8}$

51. *(d)* Given equations are,

$\frac{a}{b} - \frac{b}{a} = \frac{x}{y}$ ...(i)

and $\frac{a}{b} + \frac{b}{a} = x - y$ ...(ii)

From Eqs. (i) and (ii),

$$\frac{a}{b} + \frac{b}{a} = x - \frac{x}{\left(\frac{a}{b} - \frac{b}{a}\right)} = \frac{\left(\frac{a}{b} - \frac{b}{a}\right)x - x}{\frac{a}{b} - \frac{b}{a}}$$

$$\Rightarrow \left(\frac{a}{b} + \frac{b}{a}\right)\left(\frac{a}{b} - \frac{b}{a}\right) = x\left(\frac{a}{b} - \frac{b}{a} - 1\right)$$

$$\Rightarrow \left(\frac{a^2}{b^2} - \frac{b^2}{a^2}\right) = x\left(\frac{a^2 - b^2 - ab}{ab}\right)$$

$$\Rightarrow x = \frac{ab}{(a^2 - b^2 - ab)} \times \left(\frac{a^4 - b^4}{a^2 b^2}\right)$$

$$\Rightarrow x = \frac{(a^4 - b^4)}{a^2 - b^2 - ab} \cdot \frac{1}{ab}$$

$$= \frac{(a - b)(a + b)(a^2 + b^2)}{ab(a^2 - b^2 - ab)}$$

52. *(d)* Given equations are,

$3x + y = 4$ ...(i)

$6x + 2y = 8$ ...(ii)

Here, $a_1 = 3$, $b_1 = 1$ and $c_1 = 4$

and $a_2 = 6$, $b_2 = 2$ and $c_2 = 8$

$\because \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2}$

So, the system of equations has infinitely many solutions.

53. *(c)* Let the two numbers be x and y, respectively.

Given, sum of two numbers = 7

$\Rightarrow x + y = 7$ ...(i)

and sum of their squares = 25

$\Rightarrow x^2 + y^2 = 25$ ...(ii)

Now, we have

$(x + y)^2 = (x^2 + y^2) + 2xy$

$\Rightarrow (x + y)^2 - (x^2 + y^2) = 2xy$

$\Rightarrow xy = \frac{1}{2}[(x + y)^2 - (x^2 + y^2)]$

$= \frac{1}{2}[(7)^2 - (25)]$

[from Eqs. (i) and (ii)]

$= \frac{1}{2}(49 - 25) = \frac{1}{2} \times 24 = 12$

Hence, the product of the two numbers is 12.

Solutions (Q. Nos. 54-55) Let the digit in unit's place by y and that in ten's place be x.

Then, the two-digit number is given by $10x + y$.

Number obtain by reversing the order of digit $= 10y + x$

Now, by given condition,

$x + y = 10$ ...(i)

and $(x + 10y) + 36 = (y + 10x)$

$\Rightarrow -9y + 9x = 36$

$\Rightarrow x - y = 4$ ...(ii)

On adding Eqs. (i) and (ii), we get

$2x = 14, \ x = 7$

On putting the value of x in Eq. (i), we get

$7 + y = 10 \Rightarrow y = 3$

54. *(b)* The required number is 73.

So, the number is a multiple of a prime number.

55. *(a)* Required product of two digits

$= 3 \times 7 = 21$

56. *(a)* Let the cost of one chair and one table be ₹ x and ₹ y, respectively.

By given condition,

$10x + 6y = 6200$ ...(i)

and $3x + 2y = 1900$

$\Rightarrow 9x + 6y = 5700$ ...(ii)

On subtracting Eq. (ii) from Eq. (i), we get

$(10x + 6y) - (9x + 6y) = 6200 - 5700$

$\Rightarrow x = 500$

$\therefore x = ₹500$

From Eq. (i), $5000 + 6y = 6200$

$\Rightarrow 6y = 1200 \Rightarrow y = ₹200$

$\therefore$ Cost of 4 chairs and 5 tables

$= 4x + 5y = 4 \times 500 + 5 \times 200$

$= 2000 + 1000 = ₹3000$

57. *(b)* Let bus starts with x number of passengers.

After 1st stoppage, number of passengers

$= x - \frac{x}{5} + 40 = \frac{5x - x + 200}{5}$

$= \frac{4x + 200}{5}$

After 2nd stoppage, number of passengers

$\frac{4x + 200}{5} - \frac{4x + 200}{5 \times 2} + 30$

According to question,

$\Rightarrow \frac{4x + 200}{5} - \frac{4x + 200}{10} + 30 = 70$

$\frac{4x + 200}{5}\left(1 - \frac{1}{2}\right) + 30 = 70$

$\Rightarrow \frac{4x + 200}{5} \times \frac{1}{2} = 40$

$\Rightarrow 4x + 200 = 400 \Rightarrow 4x = 200$

$\therefore x = \frac{200}{4} = 50$

Hence, the number of passengers is 50.

58. *(d)* We have, $x + y - 7 = 0$

$\Rightarrow x + y = 7$ ...(i)

and $3x + y - 13 = 0$

$\Rightarrow 3x + y = 13$ ...(ii)

On subtracting Eq. (i) from Eq. (ii), we get

$$\begin{array}{r} 3x + y = 13 \\ x + y = 7 \\ - \quad - \quad - \\ \hline 2x = 6 \end{array}$$

$\therefore x = 3$

On putting the value of x in Eq. (i), we get

$3 + y = 7 \Rightarrow y = 4$

Now, $4x^2 + y^2 + 4xy = (2x + y)^2$

$= (2 \times 3 + 4)^2$

$= (6 + 4)^2$

$= 10^2 = 100$

59. *(b)* Given, $\frac{x}{2} + \frac{y}{3} = 4 \Rightarrow \frac{3x + 2y}{6} = 4$

$\Rightarrow 3x + 2y = 24$ ...(i)

and $\frac{2}{x} + \frac{3}{y} = 1 \Rightarrow \frac{2y + 3x}{xy} = 1$

$\Rightarrow 2y + 3x = xy$ ...(ii)

On comparing Eqs. (i) and (ii), we get $xy = 24$

There are 8 possibilities for x and y, respectively.

$$1 \times 24 = 24 \qquad 6 \times 4 = 24$$
$$2 \times 12 = 24, \qquad 8 \times 3 = 24$$
$$3 \times 8 = 24, \qquad 12 \times 2 = 24$$
$$4 \times 6 = 24, \qquad 24 \times 1 = 24$$

On putting all the values of x and y in given equations we get that only $x = 4$ and $y = 6$ satisfy the equations.

$\therefore \quad x + y = 4 + 6 = 10$

60. *(a)* Given equations are,

$$x + y = 5 \qquad \ldots(i)$$
$$y + z = 10 \qquad \ldots(ii)$$
$$z + x = 15 \qquad \ldots(iii)$$

from Eq. (i), we have $y = 5 - x$

put the value of x in Eq. (ii) ,

$$z + 5 - x = 10$$
$$\Rightarrow \quad z - x = 5 \qquad \ldots(iv)$$

Add Eq. (iv) and Eq. (iii)

$$(z + x) + (z - x) = 15 + 5$$
$$\Rightarrow \quad 2z = 20 \Rightarrow z = 10$$

from Eq. (iii), we have $10 + x = 15$

$\Rightarrow \quad x = 5$

and hence $y = 5 - x = 5 - 5 = 0$

$\therefore$ correct sequence is $z > x > y$

61. *(c)* Let present age of Ravi be x yr.

$\therefore$ Present age of Ravi's father $= 4x$ yr

Now, 5 yr ago, age of Ravi's father

$$= (4x - 5) \text{ yr}$$

and age of Ravi $= (x - 5)$ yr

According to the question,

$$4x - 5 = 7(x - 5)$$
$$\Rightarrow \quad 4x - 5 = 7x - 35 \Rightarrow 3x = 30$$
$$\therefore \quad x = 10$$

$\therefore$ Present age of Ravi $= x = 10$ yr

Present age of Ravi's father

$$= 4x = 4 \times 10 = 40 \text{ yr}$$

62. *(d)* Let the two positive numbers be x and y, respectively. According to the question,

$$(x + y) = 2.5(x - y)$$
$$\Rightarrow \quad x + y = 2.5x - 2.5y$$
$$\Rightarrow \quad 3.5y = 1.5x \Rightarrow \frac{x}{y} = \frac{7}{3} \text{ or } x = \frac{7}{3}y$$

Now, product of numbers $xy = 84$

$$\Rightarrow \quad \frac{7}{3}y \times y = 84 \quad \Rightarrow \quad y^2 = \frac{84 \times 3}{7}$$
$$\Rightarrow \quad y^2 = 12 \times 3$$
$$\therefore \quad y = 6 \Rightarrow x = \frac{7}{3} \times 6 = 14$$

$\therefore$ Sum of numbers $= x + y$

$$= 14 + 6 = 20$$

63. *(a)* Let the cost of one chair be ₹ x and that of one table be ₹ y.

Then, $\quad 2x + y = 700 \qquad \ldots(i)$

and $\quad x + 2y = 800 \qquad \ldots(ii)$

On multiplying Eq. (ii) by 2 and subtracting from Eq. (i), we get

$$\begin{array}{r} 2x + y = 700 \\ 2x + 4y = 1600 \\ - \quad - \quad - \\ \hline -3y = -900 \end{array} \quad \Rightarrow \quad y = 300$$

On putting $y = 300$ in Eq. (ii), we get

$$x + 2 \times 300 = 800$$
$$x = 800 - 600 = 200$$

Since, m chairs and m tables are to be purchased for ₹ 30000.

$\therefore$ Cost of m table and m chairs $= 30000$

$\therefore \quad 200m + 300m = 30000$

$$\Rightarrow \quad m = \frac{30000}{500} \quad \Rightarrow \quad m = 60$$

64. *(c)* Let the two-digit number be $10x + y$,

Condition I

According to the question,

$$10x + y = 3(x + y)$$
$$\Rightarrow \quad 10x + y - 3x - 3y = 0$$
$$\Rightarrow \quad 7x - 2y = 0 \qquad \ldots(i)$$

Condition II

$$\Rightarrow \quad 10x + y + 45 - 10y - x = 0$$
$$\Rightarrow \quad 9x - 9y + 45 = 0$$
$$\Rightarrow \quad x - y = -5 \qquad \ldots(ii)$$

On solving Eqs. (i) and (ii), we get

$$x = 2 \text{ and } y = 7$$

$\therefore$ Sum of the squares of digits

$$= (2)^2 + (7)^2 = 4 + 49 = 53$$

65. *(a)* Let x be the required number.

$$\therefore \quad 52x - 25x = 324$$
$$\Rightarrow \quad 27x = 324 \Rightarrow x = \frac{324}{27} = 12$$

Hence, the required number is 12.

66. *(b)* Let a tin of oil contain x bottles of oil.

According to the question,

$$\frac{4}{5} - \frac{6}{x} + \frac{4}{x} = \frac{3}{4} \Rightarrow \frac{4}{5} - \frac{3}{4} = \frac{6}{x} - \frac{4}{x}$$
$$\Rightarrow \quad \frac{1}{20} = \frac{2}{x} \quad \Rightarrow \quad x = 40$$

$\therefore$ A tin contains 40 bottles of oil.

67. *(c)* Let the number be $10x + y$.

We have, $\quad x + y = 7 \qquad \ldots(i)$

and $10y + x = 10x + y + 27$

$\Rightarrow \quad 9y - 9x = 27 \Rightarrow y - x = 3 \qquad \ldots(ii)$

Solving Eqs. (i) and (ii), we get $y = 5$ and $x = 2$

$\therefore$ Number is 25.

Product of number $= 2 \times 5 = 10$

68. *(c)* Given, $\frac{p}{x} + \frac{q}{y} = m$ and $\frac{q}{x} + \frac{p}{y} = n$

Let $\frac{1}{x} = u$ and $\frac{1}{y} = v$

Then, $pu + qv = m \qquad \ldots(i)$

and $qu + pv = n \qquad \ldots(ii)$

On solving Eqs. (i) and (ii), we get

$$u = \frac{mp - nq}{p^2 - q^2} \text{ and } v = \frac{mq - np}{q^2 - p^2}$$
$$\therefore \quad \frac{x}{y} = \frac{1/u}{1/v} = \frac{v}{u} = \frac{-(mq - np)}{mp - nq}$$
$$= \frac{np - mq}{mp - nq}$$

69. *(c)* Given equations are,

$$3x - ky - 20 = 0 \quad \ldots(i)$$
$$6x - 10y + 40 = 0 \ldots(ii)$$

since, the given system has no solution

$$\therefore \quad \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \Rightarrow \frac{3}{6} = \frac{k}{10} \Rightarrow k = 5$$

70. *(b)* Let the ages of three brothers be a, b and c.

Then, $a + b = 4, \; b + c = 6$

and $\quad c + a = 8$

On solving these three equations, we get

$$a = 3, b = 1 \text{ and } c = 5$$

$\therefore$ Age difference between eldest and youngest $= 5 - 1 = 4$ yr

71. *(a)* Let annual incomes of two persons be $9x$ and $7x$ and expenses be $4y$ and $3y$, respectively.

Then, according to the question,

$$9x - 4y = 2000 \qquad \ldots(i)$$

and $\quad 7x - 3y = 2000 \qquad \ldots(ii)$

From Eqs. (i) and (ii), we get

$$9x - 4y = 7x - 3y \Rightarrow y = 2x$$

On putting the value of y in Eq. (i), we get

$$9x - 8x = 2000 \Rightarrow x = ₹\ 2000$$

$\therefore$ Difference between their annual incomes $= 9x - 7x = 2x = ₹\ 4000$

72. *(c)* Let the unit's place number be y and ten's place number be x.

Then, number $= 10x + y$

Now, after interchanging the digits,

New number $= 10y + x$ and sum of digits $= x + y$

According to the question,

$$10x + y = k(x + y) \qquad \ldots(i)$$

and $\quad 10y + x = m(x + y) \qquad \ldots(ii)$

On adding Eqs. (i) and (ii), we get

$$11(x + y) = (k + m)(x + y)$$
$$\Rightarrow \quad k + m = 11 \Rightarrow m = 11 - k$$

18

QUADRATIC EQUATIONS AND INEQUALITIES

Regularly (9-10) questions have been asked from this chapter. Questions from this section generally focus on finding roots of quadratic equation and their factorisation.

QUADRATIC EQUATION

A quadratic equation is an equation whose degree is 2, meaning that the highest exponent of variable is 2. The general form of a quadratic equation is $ax^2 + bx + c = 0$, where a, b and c are real numbers and $a \neq 0$.
e.g. $2x^2 + 3x + 5 = 0$, $x^2 + x + 1 = 0$ and $5x^2 + 6x + 8 = 0$ are all quadratic equations.

Roots of a Quadratic Equation

A real number α is said to be a root of the quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$, if it satisfies the equation i.e. $a\alpha^2 + b\alpha + c = 0$. We can also say that $x = \alpha$ is a solution of the quadratic equation.
e.g. 2 is the root of the quadratic equation, $x^2 - 6x + 8 = 0$ since $(2)^2 - 6(2) + 8 = 0$.

Note A quadratic equation has exactly two roots.
If α be the root of the quadratic equation $ax^2 + bx + c = 0$, then $(x - \alpha)$ is the factor of $ax^2 + bx + c$.

Solutions of Quadratic Equations

Solutions of quadratic equations can be determined by using any of the following two methods

1. By Factorisation

In this method, the middle term of the quadratic equation is splitted and the given quadratic equation is converted into a product of two linear factors. Then, the roots of the equation are obtained by equating each factor equal to zero.
Let the quadratic equation be $ax^2 + bx + c = 0$ and its linear factors are $(px + q)$ and $(rx + s)$, then

$$ax^2 + bx + c = (px + q)(rx + s)$$

Now, $ax^2 + bx + c = 0 \Rightarrow (px + q)(rx + s) = 0 \Rightarrow px + q = 0$ or $rx + s = 0 \Rightarrow x = \frac{-q}{p}$ or $x = \frac{-s}{r}$

Thus, $\frac{-q}{p}$ and $\frac{-s}{r}$ are the roots of the equation, $ax^2 + bx + c = 0$.

EXAMPLE 1. Solve the equation $x^2-7x+12=0$, and find the value of x.

a. $-3, -4$ b. $-3, 4$ c. $3, -4$ d. $3, 4$

Sol. d. Given, $x^2-7x+12=0$

$x^2-3x-4x+12=0$ [splitting the middle term]

$\Rightarrow x(x-3)-4(x-3)=0$

$\Rightarrow (x-3)(x-4)=0$

$\Rightarrow (x-3)=0$ or $(x-4)=0 \Rightarrow x=3$ or $x=4$

So, $x=3, 4$ are roots of the given equation.

EXAMPLE 2. Solve the equation $3a^2x^2+8abx+4b^2=0$, $a\neq 0$, and find the value of x.

a. $\frac{2b}{3a}, \frac{2b}{a}$ b. $\frac{-2b}{3a}, \frac{2b}{a}$ c. $\frac{-2b}{3a}, \frac{-2b}{a}$ d. $\frac{2b}{3a}, \frac{-2b}{a}$

Sol. c. The equation is $3a^2x^2+8abx+4b^2=0$

$\Rightarrow 3a^2x^2+2abx+6abx+4b^2=0$

$\Rightarrow ax(3ax+2b)+2b(3ax+2b)=0$

$\therefore (3ax+2b)(ax+2b)=0$

$\Rightarrow 3ax+2b=0$ or $ax+2b=0$

$\therefore x=\frac{-2b}{3a}$ or $\frac{-2b}{a}$

2. By Using the Quadratic Formula

Let the quadratic equation be $ax^2+bx+c=0$, where $a\neq 0$. Then, solution of this equation can be find using the formula.

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

- If α and β be considered as roots of the quadratic equation, then

$$\alpha=\frac{-b+\sqrt{b^2-4ac}}{2a} \text{ and } \beta=\frac{-b-\sqrt{b^2-4ac}}{2a}$$

- The quantity b^2-4ac is called the discriminant of the quadratic equation $ax^2+bx+c=0$ and is denoted by D. So, $D=b^2-4ac$.

EXAMPLE 3. Solve the equation $x^2-9x+18=0$ and find the value of x.

a. $3, 6$ b. $-3, -6$ c. $-3, 6$ d. $3, -6$

Sol. a. We have, $x^2-9x+18=0$

Here, $a=1$, $b=-9$ and $c=18$

$\therefore x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-(-9)\pm\sqrt{(-9)^2-4(1)(18)}}{2(1)}$

$=\frac{9\pm\sqrt{81-72}}{2}=\frac{9\pm\sqrt{9}}{2}=\frac{9\pm 3}{2}$

$\therefore x=\frac{9+3}{2}$ or $x=\frac{9-3}{2} \Rightarrow x=6$ or 3

Hence, the values of x are 6 and 3.

Nature of Roots of Quadratic Equation

Let $D=b^2-4ac$ be the discriminant of the quadratic equation, $ax^2+bx+c=0$, where $a\neq 0$. Then, the following cases arise

(i) If $D>0$, then the two roots are real and distinct.

(ii) If $D=0$, then the two roots are real and equal and are given by $\alpha=\beta=\frac{-b}{2a}$.

(iii) If $D<0$, then there are no real roots, i.e. given equation has imaginary roots.

IMPORTANT POINTS

1. If $a, b, c\in Q$ and D is a perfect square, then equation has rational roots.
2. If one roots of quadration equation is $p+\sqrt{q}$, then other root will be $p-\sqrt{q}$.

EXAMPLE 4. If the equation $x^2+2(1+k)x+k^2=0$ has equal roots, then what is the value of k?

a. $\frac{1}{2}$ b. $-\frac{1}{2}$

c. 1 d. -1

Sol. b. Given equation, $x^2+2(1+k)x+k^2=0$.

If it has equal roots, then $D=0$

$\Rightarrow \{2(1+k)\}^2-4k^2=0$

$\Rightarrow 4(1+k^2+2k)-4k^2=0$

$\Rightarrow 4+4k^2+8k-4k^2=0$

$\Rightarrow 4+8k=0 \Rightarrow k=-\frac{4}{8}$

$\therefore k=-\frac{1}{2}$

Hence, the value of k is $-\frac{1}{2}$.

Roots Under Particular Conditions

Consider the quadratic equation $ax^2+bx+c=0$.

(i) Both roots are positive, if a and b are opposite in sign and a and c are same in sign, i.e. $\frac{-b}{a}>0$ and $\frac{c}{a}>0$.

(ii) Both roots are negative, if a, b and c are of same sign. i.e. $\frac{b}{a}>0$ and $\frac{c}{a}>0$.

(iii) Roots are of opposite signs, if a and c are of opposite sign, i.e. $\frac{-a}{c}>0$.

(iv) Roots are equal but opposite in signs, if $b=0$.

(v) Roots are reciprocal to each other, if $a=c$.

EXAMPLE 5. What is the least integral value of k for which the equation $x^2 - 2(k-1)x + (2k+1) = 0$ has both the roots positive?

a. 1 b. $-\frac{1}{2}$ c. 4 d. 0

Sol. a. Both the roots of the equation $ax^2 + bx + c = 0$ are positive, if $-\frac{b}{a} > 0$ and $\frac{c}{a} > 0$

Given equation is $x^2 - 2(k-1)x + (2k+1) = 0$.

Now, $-\frac{b}{a} = \frac{2(k-1)}{1} \Rightarrow 2(k-1) > 0$, if $k > 1$

and $\frac{c}{a} = \frac{2k+1}{1} \Rightarrow (2k+1) > 0$, if $k > -\frac{1}{2}$

$\therefore \quad k > 1$

Hence, the least value k as per given in options is 1.

Relation between Roots and Coefficients

1. Quadratic Equation

Let α, β be the roots of the equation $ax^2 + bx + c = 0$ $(a \neq 0)$, then

Sum of the roots $= \alpha + \beta = -\frac{b}{a} = \frac{-\textit{Coefficient of } x}{\textit{Coefficient of } x^2}$

Product of the roots $= \alpha\beta = \frac{c}{a} = \frac{\textit{Constant term}}{\textit{Coefficient of } x^2}$

2. Cubic Equation

If α, β, γ are the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$, then

Sum of roots $= \alpha + \beta + \gamma = -b/a$

Product of roots two at a time $= \alpha\beta + \beta\gamma + \gamma\alpha = c/a$

Product of three roots $= \alpha\beta\gamma = -\frac{d}{a}$

EXAMPLE 6. What are the roots of the quadratic equation $a^2b^2x^2 - (a^2+b^2)x + 1 = 0$?

a. $\frac{1}{a^2}, \frac{1}{b^2}$ b. $-\frac{1}{a^2}, -\frac{1}{b^2}$ c. $\frac{1}{a^2}, -\frac{1}{b^2}$ d. $-\frac{1}{a^2}, \frac{1}{b^2}$

Sol. a. Let roots of the equation $a^2b^2x^2 - (a^2+b^2)x + 1 = 0$ be α and β.

$\therefore$ Sum of roots $= \alpha + \beta = \frac{a^2+b^2}{a^2b^2}$ ...(i)

and product of roots $= \alpha\beta = \frac{1}{a^2b^2}$

Now, $\alpha - \beta = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta} = \sqrt{\left(\frac{a^2+b^2}{a^2b^2}\right)^2 - \frac{4}{a^2b^2}}$

$\Rightarrow \quad \alpha - \beta = \sqrt{\frac{(a^2-b^2)^2}{(a^2b^2)^2}} = \frac{a^2-b^2}{a^2b^2}$ ...(ii)

On solving Eqs. (i) and (ii), we get

$\alpha = \frac{1}{b^2}$ and $\beta = \frac{1}{a^2}$

EXAMPLE 7. If one root of the equation $\frac{x^2}{a} + \frac{x}{b} + \frac{1}{c} = 0$ is reciprocal to the other, then which one of the following is correct?

a. $a = b$ b. $b = c$ c. $ac = 1$ d. $a = c$

Sol. d. Given, $\frac{x^2}{a} + \frac{x}{b} + \frac{1}{c} = 0$ ...(i)

Now, let one root be α, then the other root is $1/\alpha$.
We know that,

Product of roots $= \frac{\text{Constant term}}{\text{Coefficient of } x^2} = \frac{1/c}{1/a}$

$\Rightarrow \quad \alpha \cdot \frac{1}{\alpha} = \frac{a}{c} \Rightarrow c = a$

which is the required relation.

Symmetric Functions of the Roots

Let α and β be the roots of a quadratic equation. An expression in α and β which remains same when α and β are interchanged is known as a symmetric function in α and β. To evaluate the value of symmetric function of the roots, express the given function in terms of $(\alpha+\beta)$ and $\alpha\beta$.

You can use the following results.

- $(\alpha^2 + \beta^2) = [(\alpha+\beta)^2 - 2\alpha\beta]$
- $(\alpha^3 + \beta^3) = [(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)]$
- $(\alpha - \beta)^2 = [(\alpha+\beta)^2 - 4\alpha\beta]$
- $(\alpha^3 - \beta^3) = [(\alpha-\beta)^3 + 3\alpha\beta(\alpha-\beta)]$

EXAMPLE 8. If α and β are the roots of the equation $x^2 - 6x + 6 = 0$, what is $\alpha^3 + \beta^3 + \alpha^2 + \beta^2 + \alpha + \beta$ equal to?

a. 150 b. 138 c. 128 d. 124

Sol. b. Here, $\alpha + \beta = \frac{-\text{Coefficient of } x}{\text{Coefficient of } x^2} = 6$

and $\alpha\beta = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = 6$

$\therefore \quad (\alpha+\beta)^2 = 6^2 \Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = 36$

$\Rightarrow \quad \alpha^2 + \beta^2 = 36 - 2(6) = 24$

$\therefore (\alpha^3 + \beta^3) + (\alpha^2 + \beta^2) + (\alpha + \beta)$

$= (\alpha+\beta)(\alpha^2 + \beta^2 - \alpha\beta) + (\alpha^2 + \beta^2) + (\alpha + \beta)$

$= 6(24 - 6) + (24) + (6)$

$= 6(18) + 30 = 108 + 30 = 138$

Formation of a Quadratic Equation

If α, β are the roots of a quadratic equation, then the equation is $(x-\alpha)(x-\beta)=0 \Rightarrow x^2-(\alpha+\beta)x+\alpha\beta=0$

i.e. x^2 – (sum of the roots) x + product of the roots $=0$

EXAMPLE 9. The quadratic equation whose roots are 3 and –1, is

a. $x^2-4x+3=0$ b. $x^2-2x-3=0$
c. $x^2+2x-3=0$ d. $x^2+4x+3=0$

Sol. b. Given that, the roots of the quadratic equation are 3 and – 1.

Let $\alpha=3$ and $\beta=-1$

$\therefore$ Sum of roots $=\alpha+\beta=3-1=2$

and product of roots $=\alpha\beta=(3)(-1)=-3$

So, required quadratic equation is

$$x^2-(\alpha+\beta)x+\alpha\beta=0$$
$$\Rightarrow \quad x^2-(2)x+(-3)=0$$
$$\Rightarrow \quad x^2-2x-3=0$$

Equations Reducible to Quadratic Equations

The equations which at the outset are not quadratic equations, but can be reduced to quadratic equations by suitable substitutions or simplifications are called equations reducible to quadratic equations.

Rule 1 $ax^{2n}+bx^n+c=0$, where $n\geq 2$

Use substitution $x^n=y$

$\Rightarrow x^{2n}=y^2$ and equation reduces to $ay^2+by+c=0$.

EXAMPLE 10. Solve the quadratic equation $x^4-26x^2+25=0$, and find the value of x.

a. $\pm 1, \pm 25$ b. $\pm 1, \pm 5$
c. 1, 5 d. 1, 25

Sol. b. Put $x^2=z \Rightarrow x^4=z^2$

$x^4-26x^2+25=0$ can be written as

$$z^2-26z+25=0 \Rightarrow z^2-25z-z+25=0$$
$$\Rightarrow \quad z(z-25)-1(z-25)=0$$
$$\Rightarrow \quad (z-1)(z-25)=0 \Rightarrow z=1 \text{ or } z=25$$

when $z=1 \Rightarrow x^2=1 \Rightarrow x=\pm 1$

when $z=25 \Rightarrow x^2=25 \Rightarrow x=\pm 5$

Hence, the solutions are $\pm$ 1 and $\pm$ 5.

Rule 2 $Px+\dfrac{Q}{x}=R$

Reduce the given equation to a quadratic equation by multiplying both sides by 'x'. So, $Px^2+Q=Rx$

or $Px^2-Rx+Q=0$

EXAMPLE 11. Solve $2x-\dfrac{3}{x}=5$. Then, the value of x is

a. 1/3, –3 b. 1/2, –3 c. –1/2, 3 d. 1/2, 4

Sol. c. Here, $2x-\dfrac{3}{x}=5$ ...(i)

On multiplying Eq. (i) by x $\Rightarrow 2x^2-3=5x$

$\Rightarrow \quad 2x^2-5x-3=0 \Rightarrow 2x^2-6x+x-3=0$

$\Rightarrow (2x+1)(x-3)=0 \Rightarrow x=-1/2$ or $x=3$

Hence, the solution is $x=\dfrac{-1}{2}, 3$.

Rule 3 $\sqrt{x+b}+x=c$ or $\sqrt{a-x^2}=bx+c$

- Square both sides to get a quadratic equation without the radical.
- Solve the equation by factorisation or by quadratic formula.

EXAMPLE 12. Solve $\sqrt{2x+9}+x=13$, then the value of x is

a. 4, 10 b. 8, 20 c. 3, 15 d. –4, –10

Sol. b. Given, $\sqrt{2x+9}+x=13 \Rightarrow \sqrt{2x+9}=13-x$

On squaring both sides, we get $(\sqrt{2x+9})^2=(13-x)^2$

$$\Rightarrow \quad 2x+9=169+x^2-26x$$
$$\Rightarrow \quad x^2-28x+160=0 \Rightarrow x^2-20x-8x+160=0$$
$$\Rightarrow \quad (x-8)(x-20)=0 \Rightarrow x=8 \text{ or } x=20$$

Hence, the solution is $x=8, 20$.

Rule 4 $\sqrt{ax+b}\pm\sqrt{cx+d}=e$

or $\sqrt{ax+b}\pm\sqrt{cx+d}\pm\sqrt{ex+f}=0$

To solve such type of equations, follow the steps given below.

- Square both sides once, so that only one term containing radical is obtained.
- Keep the term containing radical on one side and all other terms on the other side.
- Square again to get, the quadratic equation and solve it.

EXAMPLE 13. Solve the following equation $\sqrt{4-x}+\sqrt{x+9}=5$. Then, the value of x is

a. 1, 5 b. –1, 3 c. 0, – 5 d. 1, 4

Sol. c. Given, $\sqrt{4-x}+\sqrt{x+9}=5$

On squaring both sides, we get $(\sqrt{4-x}+\sqrt{x+9})^2=25$

$$\Rightarrow \quad (4-x)+(x+9)+2(\sqrt{4-x}\sqrt{x+9})=25$$
$$\Rightarrow \quad 2(\sqrt{4-x})(\sqrt{x+9})=12 \Rightarrow (\sqrt{4-x})(\sqrt{x+9})=6$$

Again, squaring both sides, we get

$$\Rightarrow \quad (4-x)(x+9)=36 \Rightarrow -x^2+36-5x=36$$
$$\Rightarrow \quad x^2+5x=0 \Rightarrow x(x+5)=0 \Rightarrow x=0 \text{ or } x=-5$$

$\therefore$ $x=0, -5$ is the required solution.

EXAMPLE 14. Solve $\sqrt{11y-6}+\sqrt{y-1}-\sqrt{4y+5}=0$. Then, the value of y is

a. {5, 6/5} b. {1, 2}
c. {3, 5/2} d. None of these

Sol. a. Given, $\sqrt{11y-6}+\sqrt{y-1}-\sqrt{4y+5}=0$

$$\sqrt{11y-6}+\sqrt{y-1}=\sqrt{4y+5}$$

On squaring both sides, we get

$$[\sqrt{11y-6}+\sqrt{y-1}]^2=[\sqrt{4y+5}]^2$$

$$\Rightarrow (11y-6)+(y-1)+2\sqrt{(11y-6)(y-1)}=(4y+5)$$

$$\Rightarrow 12y-7+2\sqrt{11y^2-17y+6}=4y+5$$

$$\Rightarrow 2\sqrt{11y^2-17y+6}=12-8y$$

$$\Rightarrow \sqrt{11y^2-17y+6}=6-4y$$

Again, squaring both sides, we get

$$11y^2-17y+6=(6-4y)^2$$

$$\Rightarrow 11y^2-17y+6=36+16y^2-48y$$

$$\Rightarrow 5y^2-31y+30=0$$

Here, $a=5$, $b=-31$ and $c=30$

$$\Rightarrow y=\frac{31\pm\sqrt{(-31)^2-4(5)(30)}}{2\times5}\quad\left[\because y=\frac{-b\pm\sqrt{b^2-4ac}}{2a}\right]$$

$$\Rightarrow y=\frac{31\pm\sqrt{361}}{10}\Rightarrow y=\frac{31\pm19}{10}$$

$$\Rightarrow y=\frac{31+19}{10}\text{ and }y=\frac{31-19}{10}$$

$\therefore$ $y=\left\{5,\frac{6}{5}\right\}$ is the required solution.

Rule 5 $(x+a)(x+b)(x+c)(x+d)+k=0$, where $a+b=c+d$ and k may or may not be zero.
To solve such type of equations, put $x^2+(a+b)x=T$.

EXAMPLE 15. Solve the equation $(x+1)(x+2)(x+3)(x+4)-8=0$, and find the number of real roots of this equation.

a. 1 b. 2 c. 4 d. No real roots

Sol. b. $(x+1)(x+2)(x+3)(x+4)-8=0$

Here, $1+4=2+3$,

so, $[(x+1)(x+4)][(x+2)(x+3)]-8=0$

$$[x^2+5x+4][x^2+5x+6]-8=0$$

Put $x^2+5x=t$, so the equation becomes

$$(t+4)(t+6)-8=0$$

$$\Rightarrow t^2+10t+24-8=0\Rightarrow t^2+10t+16=0$$

$$\Rightarrow t^2+8t+2t+16=0$$

$$\Rightarrow t(t+8)+2(t+8)=0\Rightarrow (t+8)(t+2)=0$$

$$\Rightarrow t=-8\text{ or }-2$$

Now,

$x^2+5x=-8$	$x^2+5x=-2$
$\Rightarrow x^2+5x+8=0$	$\Rightarrow x^2+5x+2=0$
$\Rightarrow x=\frac{-5\pm\sqrt{25-32}}{2}$	$\Rightarrow x=\frac{-5\pm\sqrt{25-8}}{2}$
$\therefore x=\frac{-5\pm\sqrt{-7}}{2}$	$\therefore x=\frac{-5\pm\sqrt{17}}{2}$

Hence, the number of real roots of the given equation is 2.

Rule 6 (i) $a\left(x^2+\frac{1}{x^2}\right)+b\left(x+\frac{1}{x}\right)+c=0$

(ii) $a\left(x^2+\frac{1}{x^2}\right)+b\left(x-\frac{1}{x}\right)+c=0$

To solve an equation of the form (i) . put $\left(x+\frac{1}{x}\right)=y$ and to solve an equation of the form (ii), put $\left(x-\frac{1}{x}\right)=y$

EXAMPLE 16. Solve $4\left(x^2+\frac{1}{x^2}\right)-4\left(x+\frac{1}{x}\right)-7=0;$ $x\neq0$ and find the real roots of x.

a. {–2, –1/2} b. {0, 2}
c. {2, 1/2} d. None of these

Sol. c. $4\left(x^2+\frac{1}{x^2}\right)-4\left(x+\frac{1}{x}\right)-7=0$ ...(i)

Put $x+\frac{1}{x}=y$

On squaring both sides, we get

$$\left(x+\frac{1}{x}\right)^2=y^2\Rightarrow x^2+\frac{1}{x^2}+2=y^2\Rightarrow x^2+\frac{1}{x^2}=y^2-2$$

$$\Rightarrow 4(y^2-2)-4y-7=0\quad\text{[from Eq. (i)]}$$

$$\Rightarrow 4y^2-8-4y-7=0\Rightarrow 4y^2-4y-15=0$$

$$\Rightarrow 4y^2-10y+6y-15=0\Rightarrow 2y(2y-5)+3(2y-5)=0$$

$$\Rightarrow (2y-5)(2y+3)=0\Rightarrow y=\frac{5}{2}\text{ or }-\frac{3}{2}$$

Now,

For $y=\frac{5}{2}$	For $y=-\frac{3}{2}$
$\Rightarrow x+\frac{1}{x}=\frac{5}{2}$	$\Rightarrow x+\frac{1}{x}=-\frac{3}{2}$
$\Rightarrow \frac{x^2+1}{x}=\frac{5}{2}$	$\Rightarrow \frac{x^2+1}{x}=-\frac{3}{2}$
$\Rightarrow 2x^2+2=5x$	$\Rightarrow 2x^2+2=-3x$
$\Rightarrow 2x^2-5x+2=0$	$\Rightarrow 2x^2+3x+2=0$
$\Rightarrow (x-2)(2x-1)=0$	Here, $D=9-16=-7<0$
$\therefore x=2$ or $x=\frac{1}{2}$	So, there are no real roots.
	$[\because D=b^2-4ac]$

Hence, the values of x are 2 and $\frac{1}{2}$.

Rule 7 $ax^4 + bx^3 + cx^2 + bx + a = 0$

- To solve such type of equations, we divide the equation by x^2, to obtain $a(x^2 + 1/x^2) + b(x + 1/x) + c = 0$, and, then put $x + \frac{1}{x} = y$

EXAMPLE 17. If $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$, $x \neq 0$, then the roots of the equation is

a. $\left\{\pm 2, \frac{-3 \pm \sqrt{13}}{2}\right\}$ b. $\left\{\pm 1, \frac{3 \pm \sqrt{13}}{2}\right\}$

c. $\left\{\pm 3, \frac{3 \pm \sqrt{11}}{2}\right\}$ d. None of these

Sol. b. $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$

On dividing throughout by x^2 and rearranging the terms, we get

$$\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 = 0 \quad \ldots(i)$$

Put $x - \frac{1}{x} = y$

On squaring both sides, we get

$$x^2 + \frac{1}{x^2} - 2 = y^2 \Rightarrow x^2 + \frac{1}{x^2} = y^2 + 2$$

$\Rightarrow (y^2 + 2) - 3y - 2 = 0$ [from Eq. (i)]

$\Rightarrow y^2 - 3y = 0 \Rightarrow y(y - 3) = 0$

$\therefore y = 0$ or 3

For $y = 0$	For $y = 3$
$\Rightarrow x - \frac{1}{x} = 0$	$\Rightarrow x - \frac{1}{x} = 3$
$\Rightarrow \frac{x^2 - 1}{x} = 0$	$\Rightarrow \frac{x^2 - 1}{x} = 3$
$\Rightarrow x^2 = 1$	$\Rightarrow x^2 - 3x - 1 = 0$
$\therefore x = \pm 1$	$\Rightarrow x = \frac{3 \pm \sqrt{9 + 4}}{2}$
	$\therefore x = \frac{3 \pm \sqrt{13}}{2}$

So, $x = \left\{\pm 1, \frac{3 \pm \sqrt{13}}{2}\right\}$ is the required solution.

Note An equation of degree n has n roots.

Word Problems Involving Quadratic Equation

In this section, we will discuss some problems based on practical applications of quadratic equation.

In such type of problems,

(i) We formulate a quadratic equation in variable x with the help of the given conditions.

(ii) We then solve the quadratic equation to find the answer to the given problem.

(iii) Sometimes it may happen that, out of the roots of the quadratic equation only one satisfies the given condition in the problem. So, the root which does not satisfy the condition of the problem will be rejected.

EXAMPLE 18. A positive number, when increased by 10 equals 200 times its reciprocal. What is number?

a. 100 b. 10 c. 20 d. 200

Sol. b. Let the positive number be x.

Then, according to the question,

$$x + 10 = \frac{200}{x} \Rightarrow x^2 + 10x = 200$$

$\Rightarrow x^2 + 10x - 200 = 0 \Rightarrow x^2 + 20x - 10x - 200 = 0$

$\Rightarrow x(x + 20) - 10(x + 20) = 0 \Rightarrow (x - 10)(x + 20) = 0$

$\therefore x = 10, -20$

But $x \neq -20$, since x is a positive number.

So, the required number is 10.

EXAMPLE 19. Out of group of swans, 7/2 times the square root of the number are swimming in the pool. While the two remaining are playing outside the pool. What is the total number of swans?

a. 4 b. 8 c. 12 d. 16

Sol. d. Let total number of swans be x.

Number of swans swimming in the pool $= \frac{7}{2}\sqrt{x}$

Remaining swans $= 2$

By given condition, $\frac{7}{2}\sqrt{x} + 2 = x \Rightarrow \frac{7}{2}\sqrt{x} = x - 2$

On squaring both sides, we get $\frac{49}{4}x = x^2 + 4 - 4x$

$\Rightarrow 4x^2 - 65x + 16 = 0,\ 4x^2 - 64x - x + 16 = 0$

$\Rightarrow 4x(x - 16) - 1(x - 16) = 0 \Rightarrow (x - 16)(4x - 1) = 0$

$\therefore x = 16 \quad \left[\because x \neq \frac{1}{4}\right]$

Thus, total number of swans $= 16$

INEQUALITIES

Two real numbers or two algebraic expressions related by the symbols >, <, ≤ or ≥ form inequality. Here, the symbols < (less than), > (greater than), ≤ (less than or equal) and ≥ (greater than or equal) are known as inequality signs.

e.g. $5 < 7$, $x \leq 2$, $x + y \geq 11$ are all inequalities.

Linear Inequalities

A linear inequality is an inequality which involves a linear function i.e. each variable occurs in first degree only and there is no term involving the product of the variables.

e.g. $ax + b \leq 0$, $ax + by + c > 0$, $ax \leq 4$.

Linear inequality in one variable

A linear inequality which has only one variable is called linear inequality in one variable.

e.g. $ax+b<0$, where $a \neq \left(\frac{-b}{a}\right)$, $4x+7 \geq 0$

Linear inequality in two variables

A linear inequality which has two variables, is called linear inequality in two variables.

e.g. $3x+11y \leq 0$, $4t+3y>0$

General Rules to Solve Linear Inequalities

- Equal numbers may be added or subtracted from both sides of an inequality without changing the sign of in equality.
  e.g. If $a>b$, then for any number c,
  $a+c>b+c$ or $a-c>b-c$.
- If both sides of an inequality are multiplied or divided by the same positive number, then the sign of inequality remains the same. e.g.
  If $a>b$ and $c>0$, then $\frac{a}{c}>\frac{b}{c}$ and $ac>bc$.
- If both sides of an inequality are multiplied or divided by the same negative number, then the sign of inequality is reversed.
  e.g. If $a>b$ and $c<0$, then $\frac{a}{c}<\frac{b}{c}$ and $ac<bc$.

Graphical Solution of Linear Inequalities in Two Variables

Let the inequality be $ax+by+c \geq 0$.

Step I Consider it as $ax+by+c=0$ by changing the inequality sign to equality sign, Draw the graph of this equation.

Step II Choose any point not on the line, [usually $(0, 0)$].
Find whether this point satisfy the inequality or not. If it does, then shade the half-plane containing that point. Otherwise shade the other half-plane.

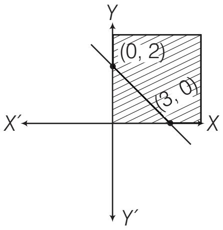
EXAMPLE 20. The solution set of inequality $2x+3y \geq 6$, $x \geq 0$ and $y \geq 0$ is

a. $x=1, y=1$ b. $x=-1, y=-1$
c. $x=3, y=1$ d. None of these

Sol. d. Draw the graph of $2x+3y=6$.

The values of (x, y) satisfying $2x+3y=6$ are

x	0	3
y	2	0



Clearly, here (0, 0) does not satisfies the inequality, so we shade the plane which does not contain (0, 0).

Also, since $x \geq 0$ and $y \geq 0$, so the solution set must be in Ist quadrant only.

Hence, the shaded region represents the solution set of the given system of inequalities.

Note (i) If the inequality is strict (> or <), graph a dashed line. If the inequality is not strict (≥ or ≤), graph a solid line.
(ii) In order to solve a system of two or more linear inequalities with the same variables, graph each inequality on the same set of axes. The intersection of the region of all the inequalities is the required solution set.

EXAMPLE 21. Find the solution set of $-2x+3y>6$ and $4x-6y>12$.

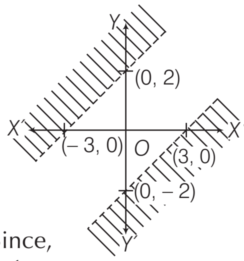
a. $x<-1, y>-3$ b. $x<1, y>3$
c. $x \geq 3, y \geq 2$ d. No solution

Sol. d. Consider, $-2x+3y=6$

x	0	-3
y	2	0

Again, $4x-6y=12$

x	0	3
y	-2	0



Now, graph each equation. Since, there is no intersection of these two shaded regions, therefore there is no solution.

Quadratic Inequalities

An equation of the form

$$ax^2+bx+c \geq 0 \quad \text{or} \quad ax^2+bx+c \leq 0$$
$$\text{or} \quad ax^2+bx+c>0 \quad \text{or} \quad ax^2+bx+c<0$$

where, $a \neq 0$ is called a quadratic inequality in one variable x. It is same as quadratic equations except for the inequality sign.

Solution of Quadratic Inequalities

Solving the inequality means finding the values of x that make the inequality true. Solution sets of quadratic inequalities are expressed in the form of intervals. Here are the steps:

- Replace the inequality symbol with an equal sign.
- Solve the quadratic equation by factorisation or by quadratic formula and find the real roots of the equation.
- Plot the two real roots on the numbers line. They divide the line into three sections or three intervals.

- Pick a number from each interval and test it in the original inequality.
- If the inequality holds true for the choosen point. then that interval is the solution of the given quadratic inequality.

Note (i) If the quadratic equation does not have real roots, then the quadratic expression is always positive (or always negative) depending on the sign of a meaning that the solution set will either be empty or the entire real number line.

e.g. $15x^2 - 18x + 7 > 0$ has no real roots as $D = b^2 - 4ac = 64 - 420 < 0$. Since $a > 0$, so the expression is always positive and hence the solution set will be the entire real number line.

(ii) When the inequality has an additional '=' sign ($\geq$ or $\leq$), use closed intervals like $[a, b]$ to indicate that the two end points are also included in the solution set. If the inequality is strict ($>$or$<$), use open intervals like (a, b) as end points are not included.

EXAMPLE 22. The solution of the inequality $x^2 + 4x + 3 \geq 0$ is

a. $x < -3$ and $x > -1$ b. $x \leq -3$ and $x \geq -1$
c. $x < -3$ and $x \geq -1$ d. $x \leq -3$ or $x > -1$

Sol. b. We have, $x^2 + 4x + 3 \geq 0$

Change the inequality to equality sign and solve

$$x^2 + 4x + 3 = 0 \Rightarrow x^2 + 3x + x + 3 = 0$$

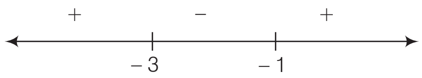
$\Rightarrow x(x+3) + (x+3) = 0 \Rightarrow (x+1)(x+3) = 0 \Rightarrow x = -1, -3.$

Place the value of x on the number line to create intervals.

Take $x = -4$, Take $x = -2$, Take $x = 0$

$\Rightarrow (-4)^2 + 4(-4) + 3 > 0,$

$(-2)^2 + 4(-2) + 3 < 0,\ 0^2 + 4(0) + 3 > 0$



So, the intervals which satisfy the given equation are $(-\infty, -3]$ and $[-1, \infty)$

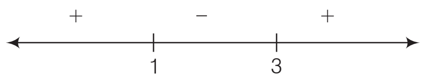
i.e. $x \leq -3$ and $x \geq -1$

EXAMPLE 23. The real values of x which satisfy $x^2 - 4x + 3 \geq 0$ and $x^2 - 3x - 4 \leq 0$ is

a. $(-1, 1) \cup (3, 4)$ b. $[-1, 1] \cap [3, 4]$
c. $[-1, 1] \cup [3, 4]$ d. $(-1, 1) \cup [3, 4]$

Sol. c. Given, $x^2 - 4x + 3 \geq 0$

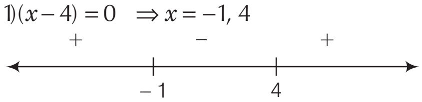
$\Rightarrow \quad x^2 - 4x + 3 = 0 \Rightarrow (x-1)(x-3) = 0 \Rightarrow x = 1, 3$



$\Rightarrow x \in (-\infty, 1]$ or $x \in [3, \infty)$. $\Rightarrow x \in (-\infty\ 1] \cup [3, \infty)$.

Again, $x^2 - 3x - 4 \leq 0 \Rightarrow x^2 - 3x - 4 = 0$

$\Rightarrow \quad (x+1)(x-4) = 0 \Rightarrow x = -1, 4$



$\Rightarrow x \in [-1, 4]$

$\therefore$ Solution set is

$x = (-\infty, 1] \cup [3, \infty) \cap [-1, 4] = [-1, 1] \cup [3, 4]$

PRACTICE EXERCISE

1. The quadratic equation has maximum
(a) one root (b) two roots
(c) four roots (d) three roots

2. The values of x in the equation $a^2b^2x^2 - (a^2 + b^2)x + 1 = 0,\ a \neq 0,\ b \neq 0$ is
(a) $1/a^2$ (b) $1/b^2$
(c) $1/a^2, 1/b^2$ (d) None of these

3. The value of 'a' for which the equation $ax^2 - 2\sqrt{5}\,x + 4 = 0$ has equal roots is
(a) 5/4 (b) 4/5
(c) – 5/4 (d) – 5/3

4. If one root of $3x^2 = 8x + (2k+1)$ is seven times the other, then the value of k is
(a) 5/3 (b) – 5/3
(c) 2/3 (d) – 3/2

5. The quadratic equation whose roots are $\frac{4+\sqrt{7}}{2}$ and $\frac{4-\sqrt{7}}{2}$ is
(a) $4x^2 + 16x + 9 = 0$ (b) $4x^2 - 16x - 9 = 0$
(c) $4x^2 - 16x + 9 = 0$ (d) $4x^2 + 16x - 9 = 0$

6. If α and β are the roots of the equation $x^2 - 8x + p = 0$ and $\alpha^2 + \beta^2 = 40$, then p is equal to
(a) 12 (b) 10 (c) 9 (d) 11

7. If α and β are the roots of the equation $x^2 - 5x + 6 = 0$, then the value of $\alpha^2 - \beta^2$
(a) 5 (b) – 5 (c) ± 5 (d) ± 4

8. If α, β are the roots of the equation $ax^2 + bx + c = 0$, then an equation whose roots are $1/\alpha$ and $1/\beta$ is
(a) $bx^2 + ax + c = 0$ (b) $ax^2 - bx + c = 0$
(c) $cx^2 + ax + b = 0$ (d) $cx^2 + bx + a = 0$

9. If α, β are the roots of a quadratic equation such that $\alpha + \beta = 24$ and $\alpha - \beta = 8$, then the equation is

(a) $x^2 - 24x - 128 = 0$ (b) $x^2 + 24x + 128 = 0$

(c) $x^2 + 24x - 128 = 0$ (d) None of these

10. Which one of the following is the equation whose roots are respectively three times the roots of the equation $ax^2 + bx + c = 0$?

(a) $ax^2 + bx + c = 0$ (b) $ax^2 + 3bx + 9c = 0$

(c) $ax^2 - 3bx + 9c = 0$ (d) $ax^2 + bx + 3c = 0$

11. How many real values of x satisfy the equation $x^{2/3} + x^{1/3} - 2 = 0$?

(a) only one value (b) two values

(c) three values (d) No value

12. If α, β are the roots of the quadratic equation $2x^2 - 4x + 1 = 0$. Then, the value of $\frac{1}{\alpha + 2\beta} + \frac{1}{\beta + 2\alpha}$ is equal to

(a) $\frac{12}{17}$ (b) $\frac{17}{12}$ (c) $\frac{11}{17}$ (d) $\frac{13}{17}$

13. Solve the equation $2^{x+2} + 2^{-x} = 5$ and find the roots of the equation.

(a) {0, 2} (b) {−2, 0} (c) {0, 1} (d) {−1, 0}

14. If α and β are roots of the equation $x^2 + p = 0$ where p is a prime, then which equation has the roots $1/\alpha$ and $1/\beta$?

(a) $\frac{1}{x^2} - \frac{1}{p} = 0$ (b) $px^2 + 1 = 0$ (c) $px^2 - 1 = 0$ (d) $\frac{1}{x^2} + \frac{1}{p} = 0$

15. The roots of the equation $x^2 + px + q = 0$ are 1 and 2. The roots of the equation $qx^2 - px + 1 = 0$ must be

(a) $\frac{-1}{2}$ and 1 (b) $\frac{1}{2}$ and 1

(c) $\frac{-1}{2}$ and -1 (d) None of these

16. Which one of the following is the quadratic equation whose roots are reciprocal to the roots of the quadratic equation $2x^2 - 3x - 4 = 0$?

(a) $3x^2 - 2x - 4 = 0$ (b) $4x^2 + 3x - 2 = 0$

(c) $3x^2 - 4x - 2 = 0$ (d) $4x^2 - 2x - 3 = 0$

17. The value of x satisfying the equation $\sqrt{x + 4} = x - 2$ is

(a) 0, 5 (b) 0, 4

(c) 5 (d) None of these

18. If the roots of the quadratic equation $px^2 + qx + r = 0$ are reciprocal to each other, then

(a) $q = r$ (b) $p = r$

(c) q divides r (d) p divides q

19. Sum of roots is − 1 and sum of their reciprocals is 1/6, then equation is

(a) $x^2 - 6x + 1 = 0$ (b) $x^2 - x + 6 = 0$

(c) $6x^2 + x + 1 = 0$ (d) $x^2 + x - 6 = 0$

20. If sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of their squares, then which one of the following is correct?

(a) $a^2 + b^2 = c^2$ (b) $a^2 + b^2 = a + b$

(c) $2ac = ab + b^2$ (d) $2c + b = 0$

21. If the roots of $x^2 - lx + m = 0$ differ by 1, then

(a) $l^2 = 4m - 1$ (b) $l^2 = 4m + 2$

(c) $l = 4m^2 + 1$ (d) $l^2 = 4m + 1$

22. If α, β are the roots of the equation $x^2 - (1 + a^2)\,x + \frac{1}{2}(1 + a^2 + a^4) = 0$, then $\alpha^2 + \beta^2$ is equal to

(a) $a^4 + a^2$ (b) a^2

(c) $(a^2 + a^4)^2$ (d) None of these

23. If the roots of $\frac{1}{x + p} + \frac{1}{x + q} = \frac{1}{r}$ are equal in magnitude and opposite in sign, then product of roots is

(a) $-\frac{1}{2}(p^2 + q^2)$ (b) $\frac{p^2 + q^2}{2}$

(c) $\frac{p + q}{2}$ (d) $\frac{1}{2}(p + q)^2$

24. If α, β are the roots of $3x^2 + 2x + 1 = 0$, then the equation whose roots are $\frac{1 - \alpha}{1 + \alpha}$ and $\frac{1 - \beta}{1 + \beta}$ is

(a) $x^2 + 2x + 3 = 0$ (b) $x^2 - 2x + 3 = 0$

(c) $x^2 + 2x - 3 = 0$ (d) $x^2 - 2x - 3 = 0$

25. If one root of $px^2 + qx + r = 0$ is double of the other root, then which one of the following is correct?

(a) $2q^2 = 9pr$ (b) $2q^2 = 9p$ (c) $4q^2 = 9r$ (d) $9q^2 = 2pr$

26. If the roots of the equation $x^2 + x + 1 = 0$ are in the ratio $m : n$, then

(a) $\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + 1 = 0$ (b) $\sqrt{m} + \sqrt{n} + 1 = 0$

(c) $\frac{m}{n} + \frac{n}{m} + 1 = 0$ (d) $m + n + 1 = 0$

27. What is one of the value of x in the equation $\sqrt{\frac{x}{1 - x}} + \sqrt{\frac{1 - x}{x}} = \frac{13}{6}$?

(a) $\frac{5}{13}$ (b) $\frac{7}{13}$ (c) $\frac{9}{13}$ (d) $\frac{11}{3}$

28. What are the roots of the equation

$$(a+b+x)^{-1}=a^{-1}+b^{-1}+x^{-1}\,?$$

(a) a, b (b) $-a, b$ (c) $a, -b$ (d) $-a, -b$

29. The number of roots of the quadratic equation $8\sec^2\phi-6\sec\phi+1=0$ is

(a) n (b) 2 (c) 0 (d) No solution

30. The number of real roots of $3^{2x^2-7x+7}=9$ is

(a) 3 (b) 1 (c) 2 (d) 4

31. If $x=\sqrt{2}+2$, then

(a) $x^2+4x+2=0$ (b) $x^2-2x-2=0$
(c) $x^2-4x+2=0$ (d) $x^2-4x-2=0$

32. For what value of k will the equation $\dfrac{x^2-bx}{ax-c}=\dfrac{k-1}{k+1}$ have roots reciprocal to each other?

(a) $\dfrac{1+c}{1-c}$ (b) $\dfrac{c+1}{c-1}$ (c) $a-c$ (d) $\dfrac{b+c}{c-1}$

33. If 'α' and 'β' be the roots of $ax^2-bx+b=0$, the value of $\sqrt{\dfrac{\alpha}{\beta}}+\sqrt{\dfrac{\beta}{\alpha}}$ is

(a) a/b (b) $\sqrt{b/a}$ (c) $\sqrt{a/b}$ (d) $-a/b$

34. If the equations $2x^2-7x+3=0$ and $4x^2+ax-3=0$ have a common root, then what is the value of a?

(a) -11 or 4 (b) -11 or -4 (c) 11 or -4 (d) 11 or 4

35. If the roots of $x^2+bx+c=0$ be α and β and those of $x^2+px+q=0$ be $k\alpha$ and $k\beta$, then

(a) $cb^2=qp^2$ (b) $qc^2=b^2p^2$
(c) $qb^2=cp^2$ (d) None of these

36. If -4 is a root of the equation $x^2+px-4=0$ and the equation $x^2+px+q=0$ has equal roots, then the values of p and q are, respectively

(a) -3 and $\dfrac{9}{4}$ (b) 3 and $\dfrac{9}{4}$ (c) $\dfrac{9}{4}$ and 3 (d) 4 and 3

37. If α,β are the roots of $2x^2-6x+3=0$, then the value of $\left(\dfrac{\alpha}{\beta}+\dfrac{\beta}{\alpha}\right)+3\left(\dfrac{1}{\alpha}+\dfrac{1}{\beta}\right)+2\alpha\beta$ is

(a) 12 (b) 23 (c) 13 (d) -13

38. If the roots of the equation $x^2+2ax+b=0$, are real and distinct and they differ by atmost 2m, then b lies in the interval

(a) (a^2-m^2, a^2) (b) $[a^2-m^2, a^2)$
(c) (a^2, a^2+m^2) (d) None of these

39. The solution of the equation

$$(x+2)(x-5)(x-6)(x+1)=144 \text{ is}$$

(a) $\{7,-3\}$ (b) $\{7,-3,2\}$ (c) $\{7,-3,2,1\}$ (d) $\{7,2\}$

40. The solution of the equation $\sqrt{x^2-16}-(x-4)=\sqrt{x^2-5x+4}$ is

(a) $\left\{4, 5, -\dfrac{13}{3}\right\}$ (b) $\{4, 5\}$ (c) $\{4\}$ (d) $\left\{5, -\dfrac{13}{3}\right\}$

41. If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2-bx+c=0$, then which one of the following is correct?

(a) $a^2+b^2+2ac=0$ (b) $a^2-b^2+2ac=0$
(c) $a^2+c^2+2ab=0$ (d) $a^2-b^2-2ac=0$

42. The equation $(1+n^2)x^2+2ncx+(c^2-a^2)=0$ will have equal roots, if

(a) $c^2=1+a^2$ (b) $c^2=1-a^2$
(c) $c^2=1+n^2+a^2$ (d) $c^2=(1+n^2)a^2$

43. What is the condition that the equation $ax^2+bx+c=0$, where $a\neq 0$ has both the roots positive?

(a) a, b and c are of same sign
(b) a and b are of same sign
(c) b and c have the same sign opposite to that of a
(d) a and c have the same sign opposite to that of b

44. The equation whose roots are twice the roots of the equation $x^2-2x+4=0$ is

(a) $x^2-2x+4=0$ (b) $x^2-2x+16=0$
(c) $x^2-4x+8=0$ (d) $x^2-4x+16=0$

45. If α and β are the roots of the equation $x^2+px+q=0$, then $-\alpha^{-1},-\beta^{-1}$ are the roots of which one of the following equations?

(a) $qx^2-px+1=0$ (b) $q^2+px+1=0$
(c) $x^2+px-q=0$ (d) $x^2-px+q=0$

46. If one root of the equation $ax^2+x-3=0$ is -1, then what is the other root?

(a) $\dfrac{1}{4}$ (b) $\dfrac{1}{2}$ (c) $\dfrac{3}{4}$ (d) 1

47. If the equation $(a^2+b^2)x^2-2(ac+bd)x+(c^2+d^2)=0$ has equal roots, then which one of the following is correct?

(a) $ab=cd$ (b) $ad=bc$
(c) $a^2+c^2=b^2+d^2$ (d) $ac=bd$

48. What is the solution of the equation

$$\sqrt{\frac{x}{x+3}}-\sqrt{\frac{x+3}{x}}=-\frac{3}{2}\,?$$

(a) 1 (b) 2 (c) 4 (d) None of these

49. What are the roots of the equation $4^x-3\cdot 2^{x+2}+32=0$?

(a) 1, 2 (b) 3, 4 (c) 2, 3 (d) 1, 3

50. If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is the value of $(\alpha^4 + \beta^4)$?
(a) 7 (b) 0
(c) 2 (d) None of these

51. When the roots of the quadratic equation $ax^2 + bx + c = 0$ are negative and reciprocals of each other, then which one of the following is correct?
(a) $b = 0$ (b) $c = 0$ (c) $a = c$ (d) $a = -c$

52. If sum as well as product of roots of a quadratic equation is 9, then what is the equation?
(a) $x^2 + 9x - 18 = 0$ (b) $x^2 - 18x + 9 = 0$
(c) $x^2 + 9x + 9 = 0$ (d) $x^2 - 9x + 9 = 0$

53. What are the roots of the equation $\log_{10}(x^2 - 6x + 45) = 2$?
(a) 9, – 5 (b) – 9, 5 (c) 11, – 5 (d) – 11, 5

54. The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. What is the product of the roots of the equation?
(a) $-\frac{(a+b)}{2}$ (b) $\frac{(a+b)}{2}$
(c) $-\frac{(a^2+b^2)}{2}$ (d) $\frac{(a^2+b^2)}{2}$

55. For what value of k, will the roots of the equation $kx^2 - 5x + 6 = 0$ be in the ratio of 2 : 3?
(a) 0 (b) 1 (c) – 1 (d) 2

56. What is the ratio of sum of squares of roots to the product of the roots of the equation $7x^2 + 12x + 18 = 0$?
(a) 6 : 1 (b) 1 : 6 (c) – 6 : 1 (d) – 6 : 7

57. If the roots of the equation $\frac{x(x-1)-(m+1)}{(x-1)(m-1)} = \frac{x}{m}$ are equal, then what is the value of m?
(a) 1 (b) 1/2 (c) 0 (d) – 1/2

58. If $3^x + 27(3^{-x}) = 12$, then what is the value of x?
(a) 1 (b) 2 (c) 1, 2 (d) 0, 1

59. What is the magnitude of difference of the roots of $x^2 - ax + b = 0$?
(a) $\sqrt{a^2 - 4b}$ (b) $\sqrt{b^2 - 4a}$
(c) $2\sqrt{a^2 - 4b}$ (d) $\sqrt{b^2 - 4ab}$

60. What can be said about the roots of the equation $x^2 - x - 2 = 0$?
(a) both of them are integers
(b) both of them are natural numbers
(c) the latter of the two is negative
(d) None of the above

61. An equation equivalent to the quadratic equation $x^2 - 6x + 5 = 0$ is
(a) $x^2 - 5x + 6 = 0$ (b) $5x^2 - 6x + 1 = 0$
(c) $|x - 3| = 2$ (d) $6x^2 - 5x + 1 = 0$

62. If the sum of a number and its reciprocal is $\frac{10}{3}$, then the numbers are
(a) $3, \frac{1}{3}$ (b) $3, -\frac{1}{3}$ (c) $-3, \frac{1}{3}$ (d) $-3, -\frac{1}{3}$

63. Divide 16 into two parts such that the twice of the square of the greater part exceeds, the square of the smaller part by 164. Then, the greater part is
(a) 58 (b) 10 (c) 6 (d) 15

64. The number of straight lines that can connect 'x' points is given by the equation $y = \frac{x(x-1)}{2}$. How many points does a figure have if only 15 lines can be drawn connecting them?
(a) 15 (b) 10 (c) 6 (d) 5

65. The two successive natural numbers whose squares have sum 221 are
(a) 10 and 11 (b) 11 and 12
(c) – 10 and – 11 (d) None of these

66. The two consecutive positive odd integers, the sum of whose squares is 130.
(a) – 7 and – 9 (b) 7 and 9
(c) 7 and 5 (d) 3 and – 5

67. The solution set of inequation $2x + 1 \geq 7$ is
(a) $x \geq 8$ (b) $x \geq 3$
(c) $x \geq 6$ (d) None of these

68. The values of x satisfying inequation $\frac{x-1}{3} \geq 4$ is
(a) $x \leq 13$ (b) $x \geq 12$ (c) $x \geq 13$ (d) $x = 13$

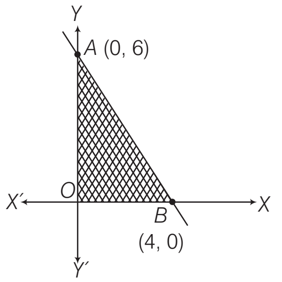
69. The values of x satisfying $3x + 2 \leq 5x - (4 - x)$ is
(a) $x \leq 2$ (b) $x \geq 2$
(c) $x = 2$ (d) None of these

70. The solution set of x for the inequations $2x + 3 \geq 8$ and $3x + 1 \leq 12$ is
(a) $\frac{5}{2} < x \leq \frac{11}{3}$ (b) $\frac{5}{2} < x < \frac{11}{3}$
(c) $\frac{5}{2} \leq x \leq \frac{11}{3}$ (d) $\frac{5}{2} \geq x \geq \frac{11}{3}$

71. The values of x satisfying $\frac{1}{2}\left(\frac{3}{5}x + 4\right) \geq \frac{1}{3}(x - 6)$ are
(a) $x \geq 120$ (b) $x \leq 120$ (c) $x \leq 12$ (d) $x \geq 12$

72. $4x^2 - 1 \leq 0$, then the solution set is
(a) $\frac{-1}{2} \leq x \leq \frac{1}{2}$ (b) $\frac{-1}{2} < x < \frac{1}{2}$
(c) $\frac{-1}{2} \geq x \geq \frac{1}{2}$ (d) None of these

73. The shaded region, including the boundary in the given graph, is exactly represented by



(a) $3x + 2y \leq 12, x < 0, y \geq 0$ (b) $3x + 2y \leq 12, x \geq 0, y \geq 0$
(c) $3x + 2y < 12, x \geq 0, y \geq 0$ (d) $3x + 2y > 12, x \geq 0, y \geq 0$

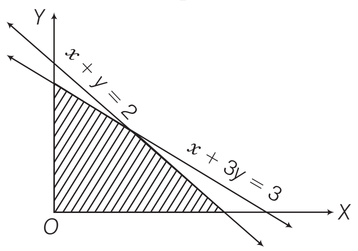
74. When is the expression $x^2 + 3x - 10$ positive only?
(a) $x \leq -5$ (b) $x \geq 2$ (c) $-5 < x < 2$ (d) $x < -5$ or $x > 2$

75. The solution set for the quadratic inequation $x^2 - 5x + 6 \geq 0$ is
(a) $[-\infty, 2] \cup [3, \infty]$ (b) $[-\infty, 2] \cup [3, \infty]$
(c) $[-\infty, 2] \cup [3, \infty]$ (d) $[2, 3]$

76. All real values of x for which $\dfrac{x-2}{3x+1} < \dfrac{x-3}{3x-2}$ are
(a) $\left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$ (b) $\left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$ (c) $\left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$ (d) R

77. If $x + 2y \leq 3, x > 0$ and $y > 0$, then one of the solution is
(a) $x = -1, y = 2$ (b) $x = 2, y = 1$
(c) $x = 1, y = 1$ (d) $x = 0, y = 0$

78. The shaded region in the given figure is the solution set of the inequalities



(a) $x + y \leq 2, x + 3y \geq 3, x \geq 0, y \geq 0$
(b) $x + y \geq 2, x + 3y \geq 3, x \geq 0, y \geq 0$
(c) $x + y \geq 2, x + 3y \leq 3, x \geq 0, y \geq 0$
(d) $x + y \leq 2, x + 3y \leq 3, x \geq 0, y \geq 0$

79. If $a + b = 2m^2$, $b + c = 6m$, $a + c = 2$, where m is a real number and $a \leq b \leq c$, then which one of the following is correct?
(a) $0 \leq m \leq 1/2$ (b) $-1 \leq m \leq 0$
(c) $1/3 \leq m \leq 1$ (d) $1 < m \leq 2$

80. If α and β are the roots of the equation $(x^2 - 3x + 2 = 0)$, then which of the following equation has the roots $(\alpha + 1)$ and $(\beta + 1)$?
I. $x^2 + 5x + 6 = 0$ II. $x^2 - 5x - 6 = 0$

Select the answer using the codes given below.
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

81. Consider the following statements
I. Every quadratic equation has atleast one real root.
II. A quadratic equation with integral coefficients has integral roots.
III. If the coefficient of x^2 and the constant term of a quadratic equation have opposite signs, then the quadratic equation has real roots.

Which of the above statements is/are correct?
(a) Only I and II (b) Only II and III
(c) Only III (d) All of these

82. Which of the following are quadratic equations?
I. $x^2 + \dfrac{1}{x^2} = 2$ II. $x + \dfrac{3}{x} = x^2$
III. $2x^2 - x + 2 = x^2 + 4x - 4$ IV. $x^3 + 6x^2 + 2x - 1 = 0$

Select the correct answer using the codes given below.
(a) I, III are quadratic (b) II, III and IV are quadratic
(c) Only III is quadratic (d) None of these

83. Consider the following statements
I. $x = 1$ is a root of $3x^2 - 2x - 1 = 0$.
II. $x = -2\sqrt{2}$ is a root of $x^2 + \sqrt{2}x - 4 = 0$.
III. $x^2 + x + 1 = 0$; $x = 1, x = -1$.
IV. $9x^2 - 3x - 2 = 0$; $x = -\dfrac{1}{3}, x = \dfrac{2}{3}$.

Which of the equation(s) given above is/are correct?
(a) I and II (b) I, II and IV
(c) III and IV (d) Only IV

Directions (Q. Nos. 84-86) *A ball is thrown upwards from a rooftop, 80 m above the ground. It will reach a maximum vertical height and then fall back to the ground. The height of the ball from the ground at time t is h which is given by, $h = -16t^2 + 64t + 80$ Now, answer the following questions based on the above information.*

84. What is the height reached by the ball after 1 sec?
(a) 150 m (b) 128 m
(c) 64 m (d) None of these

85. What is the maximum height reached by the ball?
(a) 110 m (b) 132 m
(c) 144 m (d) cannot be determined

86. How long will it take before hitting the ground?
(a) 5 sec (b) 6 sec
(c) 3 sec (d) cannot say

PREVIOUS YEARS' QUESTIONS

87. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then which one of the following is correct? **2012 I**
(a) $a < 2$ (b) $2 < a < 3$ (c) $3 < a < 4$ (d) $a > 4$

88. If one of the roots of quadratic equation $7x^2 - 50x + k = 0$ is 7, then what is the value of k? **2012 I**
(a) 7 (b) 1 (c) 50/7 (d) 7/50

89. Two students A and B solve an equation of the form $x^2 + px + q = 0$. A starts with a wrong value of p and obtains the roots as 2 and 6. B starts with a wrong value of q and gets the roots as 2 and -9. What are the correct roots of the equation? **2012 II**
(a) 3 and -4 (b) -3 and -4 (c) -3 and 4 (d) 3 and 4

90. If one of the roots of the equation $x^2 - bx + c = 0$ is the square of the other, then which of the following option is correct? **2013 I**
(a) $b^3 = 3bc + c^2 + c$ (b) $c^3 = 3bc + b^2 + b$
(c) $3bc = c^3 + b^2 + b$ (d) $3bc = c^3 + b^3 + b^2$

91. The difference of the roots of the equation $2x^2 - 11x + 5 = 0$ is **2013 I**
(a) 4.5 (b) 4 (c) 3.5 (d) 3

92. The sum of the squares of two numbers is 97 and the squares of their difference is 25. The product of the two numbers is **2013 I**
(a) 45 (b) 36 (c) 54 (d) 63

93. If $x + \frac{1}{x} = 2$, then what is value of $x - \frac{1}{x}$? **2013 I**
(a) 0 (b) 1 (c) 2 (d) -2

94. There are some benches in a classroom having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class of 158 students. The number of rows is **2013 I**
(a) 4 (b) 8 (c) 6 (d) 10

95. Which one of the following is factor of
$$x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14?$$
2013 II
(a) $x + \frac{1}{x} + 1$ (b) $x + \frac{1}{x} + 3$
(c) $x + \frac{1}{x} + 6$ (d) $x + \frac{1}{x} + 7$

96. If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is $\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)}$ equal to? **2013 II**
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) None of these

97. If $x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots \infty}}}$, then what is one of the values of x equal to? **2013 II**
(a) 6 (b) 5 (c) 4 (d) 3

98. Consider the following statements in respect of the quadratic equation $ax^2 + bx + b = 0$, where $a \neq 0$.
I. The product of the roots is equal to the sum of the roots.
II. The roots of the equation are always unequal and real.

Which of the above statements is/are correct? **2014 I**
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

99. If the roots of the equation $Ax^2 + Bx + C = 0$ are -1 and 1, then which one of the following is correct? **2014 I**
(a) A and C are both zero
(b) A and B are both positive
(c) A and C are both negative
(d) A and C are of opposite sign

100. If the roots of the equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$ are equal, where $b \neq 0$, then which one of the following is correct? **2014 I**
(a) $a + b + c = abc$ (b) $a^2 + b^2 + c^2 = 0$
(c) $a^3 + b^3 + c^3 = 0$ (d) $a^3 + b^3 + c^3 = 3abc$

101. If m and n are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $(m^2 + 1)/m$ and $(n^2 + 1)/n$ is **2014 I**
(a) $acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0$
(b) $acx^2 + (ab - bc)x + b^2 + (a - c)^2 = 0$
(c) $acx^2 + (ab - bc)x + b^2 - (a - c)^2 = 0$
(d) $acx^2 + (ab + bc)x + b^2 - (a - c)^2 = 0$

102. In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtains -9 and -1 for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was **2014 I**
(a) $x^2 + 10x + 9 = 0$ (b) $x^2 - 10x + 16 = 0$
(c) $x^2 - 10x + 9 = 0$ (d) None of these

103. If m and $n\,(m > n)$ are the roots of the equation $7(x + 2a)^2 + 3a^2 = 5a(7x + 23a)$, where $a > 0$, then what is $3m - n$ equal to? **2014 II**
(a) $12a$ (b) $14a$ (c) $15a$ (d) $18a$

104. If one of the roots of the equation $px^2 + qx + r = 0$ is three times the other, then which one of the following relations is correct? **2014 II**
(a) $3q^2 = 16pr$ (b) $q^2 = 24pr$
(c) $p = q + r$ (d) $p + q + r = 1$

105. If m and n are the roots of the equation $x^2 + ax + b = 0$ and m^2, n^2 are the roots of the equation $x^2 - cx + d = 0$, then which of the following is/are correct?

I. $2b - a^2 = c$ II. $b^2 = d$

Select the correct answer using the codes given below. **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

106. The sign of the quadratic polynomial $ax^2 + bx + c$ is always positive, if **2015 I**

(a) a is positive and $b^2 - 4ac \le 0$
(b) a is positive and $b^2 - 4ac \ge 0$
(c) a can be any real number and $b^2 - 4ac \le 0$
(d) a can be any real number and $b^2 - 4ac \ge 0$

107. The number of values of x satisfying $x + \frac{100}{x} > 50$, where x is a natural number less than or equal to 100, is **2015 II**

(a) 51 (b) 53 (c) 55 (d) 57

108. If the roots of the quadratic equation $x^2 - 4x - \log_{10} N = 0$ are all real, then the minimum value of N is **2015 II**

(a) $\frac{1}{100}$ (b) $\frac{1}{1000}$ (c) $\frac{1}{10000}$ (d) 10000

109. What is $\sqrt{4 + \sqrt{4 - \sqrt{4 + \ldots}}}$ equal to? **2015 II**

(a) 3 (b) $\frac{\sqrt{13} - 1}{2}$ (c) $\frac{\sqrt{13} + 1}{2}$ (d) 0

110. If $k = x - y + 2z$, where $-2 \le x \le 1$ and $-1 \le y \le 2$ and $3 \le z \le 6$, then which one of the following is correct? **2015 II**

(a) $0 \le k \le 9$ (b) $5 \le k \le 11$ (c) $2 \le k \le 14$ (d) $2 \le k \le 11$

111. If the sum of the roots of $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then which one of the following relations is correct? **2016 I**

(a) $ab^2 + bc^2 = 2a^2c$ (b) $ac^2 + bc^2 = 2b^2a$
(c) $ab^2 + bc^2 = a^2c$ (d) $a^2 + b^2 + c^2 = 1$

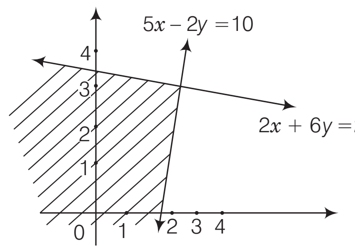
112. If the equations $x^2 - px + q = 0$ and $x^2 + qx - p = 0$ have a common root, then which one of the following is correct? **2016 I**

(a) $p - q = 0$ (b) $p + q - 2 = 0$
(c) $p + q - 1 = 0$ (d) $p - q - 1 = 0$

113. Under what condition on p and q, one of the roots of the equation $x^2 + px + q = 0$ is the square of the other? **2016 I**

(a) $1 + q + q^2 = 3pq$ (b) $1 + p + p^2 = 3pq$
(c) $p^3 + q + q^2 = 3pq$ (d) $q^3 + p + p^2 = 3pq$

114.



The linear inequations, for which the shaded area in the figure given above is the solution set, are **2016 I**

(a) $2x + 6y \le 21,\ 5x - 2y \le 10$
(b) $2x + 6y \le 21,\ 5x - 2y \ge 10$
(c) $2x + 6y \ge 21,\ 5x - 2y \le 10$
(d) $2x + 6y \ge 21,\ 5x - 2y \ge 10$

115. Let p and q be non-zero integers. Consider the polynomial $A(x) = x^2 + px + q$. It is given that $(x - m)$ and $(x - km)$ are simple factors of $A(x)$, where m is a non-zero integer and k is a positive integer, $k \ge 2$. Which one of the following is correct? **2016 I**

(a) $(k+1)^2 p^2 = kq$ (b) $(k+1)^2 q = kp^2$
(c) $k^2 q = (k+1)p^2$ (d) $k^2 p^2 = (k+1)^2 q$

116. Which of the points $P(5, -1), Q(3, -2)$ and $R(1, 1)$ lie in the solution of the system of inequations $x + y \le 4$ and $x - y \ge 2$? **2016 I**

(a) Q and R (b) P and R
(c) P and Q (d) P, Q and R

117. If $x = \frac{\sqrt{a + 2b} + \sqrt{a - 2b}}{\sqrt{a + 2b} - \sqrt{a - 2b}}$, then $bx^2 - ax + b$ is equal to (given that, $b \ne 0$) **2016 I**

(a) 0 (b) 1 (c) ab (d) $2ab$

118. If $\sqrt{3x^2 - 7x - 30} - \sqrt{2x^2 - 7x - 5} = x - 5$ has α and β as its roots, then the value of $\alpha\beta$ is **2016 I**

(a) -15 (b) -5 (c) 0 (d) 5

119. If the roots of the equation $lx^2 + mx + m = 0$ are in the ratio $p : q$, then $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}}$ is equal to **2016 I**

(a) 0 (b) 1 (c) 2 (d) 3

120. The solution of the inequation $1 + \frac{1}{x} - \frac{1}{x^2} \ge 0$ is (given that, $x \ne 0$) **2016 I**

(a) $x > 0$ (b) $x < 0$
(c) $\frac{-1 - \sqrt{5}}{5} \le x \le \frac{-1 + \sqrt{5}}{2}$ (d) $x \le \frac{-1 - \sqrt{5}}{2}$ or $x \ge \frac{-1 + \sqrt{5}}{2}$

ANSWERS

1	*b*	2	*c*	3	*a*	4	*b*	5	*c*	6	*a*	7	*c*	8	*d*	9	*d*	10	*b*
11	*b*	12	*a*	13	*b*	14	*b*	15	*c*	16	*b*	17	*c*	18	*b*	19	*d*	20	*c*
21	*d*	22	*b*	23	*a*	24	*b*	25	*a*	26	*a*	27	*c*	28	*d*	29	*d*	30	*c*
31	*c*	32	*b*	33	*b*	34	*a*	35	*c*	36	*b*	37	*c*	38	*b*	39	*b*	40	*b*
41	*b*	42	*d*	43	*d*	44	*d*	45	*a*	46	*c*	47	*b*	48	*a*	49	*c*	50	*a*
51	*c*	52	*d*	53	*c*	54	*c*	55	*b*	56	*d*	57	*d*	58	*c*	59	*a*	60	*a*
61	*c*	62	*a*	63	*b*	64	*c*	65	*a*	66	*b*	67	*b*	68	*c*	69	*b*	70	*c*
71	*b*	72	*a*	73	*b*	74	*d*	75	*a*	76	*b*	77	*c*	78	*d*	79	*c*	80	*d*
81	*c*	82	*c*	83	*b*	84	*b*	85	*c*	86	*a*	87	*a*	88	*a*	89	*b*	90	*a*
91	*a*	92	*b*	93	*a*	94	*b*	95	*c*	96	*b*	97	*d*	98	*d*	99	*d*	100	*d*
101	*a*	102	*c*	103	*c*	104	*a*	105	*b*	106	*a*	107	*c*	108	*c*	109	*c*	110	*c*
111	*a*	112	*d*	113	*c*	114	*a*	115	*b*	116	*c*	117	*a*	118	*a*	119	*a*	120	*d*

HINTS AND SOLUTIONS

1. *(b)* An equation of n degree has n roots. So, quadratic equation has two roots.

2. *(c)* $a^2b^2x^2 - a^2x - b^2x + 1 = 0$

$\Rightarrow \quad a^2x(b^2x - 1) - 1(b^2x - 1) = 0$

$\Rightarrow \quad (b^2x - 1)(a^2x - 1) = 0$

So, either $a^2x - 1 = 0 \Rightarrow a^2x = 1$

or $\quad b^2x - 1 = 0$ or $b^2x = 1$

$\Rightarrow \quad x = 1/a^2$ or $x = 1/b^2$

3. *(a)* Given, $ax^2 - 2\sqrt{5}x + 4 = 0$ has equal roots.

$\therefore$ Discriminant

$= (-2\sqrt{5})^2 - 4(a)4 = 0$

$[\because D = B^2 - 4AC]$

$\Rightarrow \quad 20 - 16a = 0 \Rightarrow a = 5/4$

4. *(b)* Given, $3x^2 = 8x + (2k + 1)$

or $\quad 3x^2 - 8x - (2k + 1) = 0$

Let first and second root be α and 7α, respectively.

$\therefore$ Sum of roots $= \alpha + 7\alpha = \frac{8}{3}$

$\Rightarrow \quad 8\alpha = \frac{8}{3} \Rightarrow \alpha = \frac{1}{3}$

So, roots are $\frac{1}{3}$ and $\frac{7}{3}$.

Also, product of roots $= \frac{1}{3} \times \frac{7}{3} = \frac{7}{9}$

$\Rightarrow \quad \frac{7}{9} = \frac{-(2k+1)}{3}$

$\Rightarrow \quad \frac{7}{3} = -(2k + 1)$

$\Rightarrow \quad 7 = -6k - 3 \Rightarrow 10 = -6k$

$\Rightarrow \quad k = \frac{-5}{3}$

5. *(c)* Here, sum of roots,

$$S = \frac{4+\sqrt{7}}{2} + \frac{4-\sqrt{7}}{2} = 4$$

Product of roots,

$$P = \left(\frac{4+\sqrt{7}}{2}\right)\left(\frac{4-\sqrt{7}}{2}\right) = \frac{16-7}{4} = \frac{9}{4}$$

The required equation is

$x^2 - Sx + P = 0$

$x^2 - 4x + \frac{9}{4} = 0$

$\Rightarrow \quad 4x^2 - 16x + 9 = 0$

6. *(a)* Given, $x^2 - 8x + p = 0$

Sum of roots $\alpha + \beta = 8$ and product of roots $\alpha\beta = p$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad$ ...(i)

$\Rightarrow \quad 40 = (8)^2 - 2(p) [\because \alpha^2 + \beta^2 = 40]$

$\Rightarrow \quad 40 - 64 = -2p$

$\Rightarrow \quad -24 = -2p \Rightarrow p = 12$

7. *(c)* Given, $x^2 - 5x + 6 = 0$

Sum of roots $= -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}$

$\Rightarrow \quad \alpha + \beta = 5$

Product of roots $= \frac{\text{Constant term}}{\text{Coefficient of } x^2}$

$\Rightarrow \quad \alpha\beta = 6$

Now, $\quad (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$= (5)^2 - 4 \times 6 = 1$

$\Rightarrow \quad \alpha - \beta = \pm 1$ and $\alpha + \beta = 5$

$\Rightarrow \quad \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$

$= 5(\pm 1)$

$\Rightarrow \quad \alpha^2 - \beta^2 = \pm 5$

8. *(d)* Here, $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$

Now, roots of required equation are $\frac{1}{\alpha}, \frac{1}{\beta}$.

$$\Rightarrow S = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha+\beta}{\alpha\beta} = \frac{-b/a}{c/a} = \frac{-b}{c}$$

$$P = \frac{1}{\alpha}\cdot\frac{1}{\beta} = \frac{1}{c/a} = \frac{a}{c}$$

$\therefore$ Required quadratic equation is

$x^2 - Sx + P = 0$

$\Rightarrow \quad x^2 - \left(\frac{-b}{c}\right)x + \frac{a}{c} = 0$

$\Rightarrow \quad cx^2 + bx + a = 0$

9. *(d)* Given, $\alpha + \beta = 24$ and $\alpha - \beta = 8$

On solving, we get $\alpha = 16$ and $\beta = 8$

$\therefore$ Sum of roots $= \alpha + \beta = 24$

and product of roots $= 16 \times 8 = 128$

So, required equation is

$x^2 - 24x + 128 = 0.$

10. *(b)* Let α and β be the roots of the equation $ax^2 + bx + c = 0$.

Then, $\quad \alpha + \beta = \frac{-b}{a}$ and $\quad \alpha\beta = \frac{c}{a}$

Now, $3\alpha + 3\beta = \frac{-3b}{a}$ and $3\alpha \cdot 3\beta = \frac{9c}{a}$

$\therefore$ Required equation $= x^2 -$ (Sum of roots) x + Product of roots

$\Rightarrow \quad x^2 - \left(\frac{-3b}{a}\right)x + \frac{9c}{a} = 0$

$\Rightarrow \quad ax^2 + 3bx + 9c = 0$

11. *(b)* Given equation is
$$x^{2/3}+x^{1/3}-2=0$$
$\Rightarrow \quad (x^{1/3})^2+x^{1/3}-2=0$

Let $\quad x^{1/3}=x$

$\Rightarrow \quad x^2+x-2=0,$

It is a quadratic equation in x.

$\therefore$ Discriminant of $x^2+x-2=0$ is

$B^2-4\,AC=1^2-4(1)(-2)=9>0$

Hence, two real values of x satisfy the given equation.

12. *(a)* Here, $\alpha+\beta=\frac{4}{2}=2, \alpha\beta=\frac{1}{2}$

Now, $\frac{1}{\alpha+2\beta}+\frac{1}{\beta+2\alpha}$

$=\frac{\beta+2\alpha+\alpha+2\beta}{(\alpha+2\beta)(\beta+2\alpha)}$

$=\frac{3\alpha+3\beta}{\alpha\beta+2\alpha^2+2\beta^2+4\alpha\beta}$

$=\frac{3(\alpha+\beta)}{2(\alpha+\beta)^2+\alpha\beta}=\frac{3(2)}{2(2)^2+\frac{1}{2}}=\frac{12}{17}$

13. *(b)* Given equation is $2^{x+2}+2^{-x}=5$

$\Rightarrow \quad 2^x2^2+2^{-x}=5 \Rightarrow 4\cdot 2^x+\frac{1}{2^x}=5$

Put $\quad 2^x=y, 4y+\frac{1}{y}=5$

$\Rightarrow \quad 4y^2+1=5y \Rightarrow 4y^2-5y+1=0$

$\Rightarrow \quad (y-1)(4y-1)=0$

$\Rightarrow \quad y-1=0 \quad \text{or} \quad 4y-1=0$

$\Rightarrow \quad y=1 \quad \text{or} \quad 4y=1$

$\Rightarrow \quad y=1, \frac{1}{4}$

By condition,

$2^x=y$	$2^x=y$
$\Rightarrow \quad 2^x=1$	$\Rightarrow \quad 2^x=\frac{1}{4}$
$\Rightarrow \quad 2^x=2^0$	$\Rightarrow \quad 2^x=\frac{1}{2^2}=2^{-2}$
$\therefore \quad x=0$	$\therefore \quad x=-2$

Hence, the roots of the equation are 0 and -2.

14. *(b)* Since, α and β are roots of the equation $x^2+p=0$.

$\therefore \quad \alpha+\beta=0 \quad \text{and} \quad \alpha\beta=p$

So, $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}=0$

and $\frac{1}{\alpha}\cdot\frac{1}{\beta}=\frac{1}{\alpha\beta}=\frac{1}{p}$

$\therefore$ Required equation $x^2-0\cdot x+\frac{1}{p}=0$

$\Rightarrow \quad px^2+1=0$

15. *(c)* The equation is $x^2+px+q=0$.
Sum of roots $=-p=1+2$

$\Rightarrow \quad p=-3$

Product of roots $=q=1\times 2=2$

$\therefore$ Equation $qx^2-px+1=0$ becomes
$$2x^2-(-3)x+1=0$$
$\Rightarrow \quad 2x^2+3x+1=0$

$\Rightarrow \quad (2x+1)(x+1)=0$

$\therefore \quad x=-\frac{1}{2}$ or $x=-1$

16. *(b)* Given, $2x^2-3x-4=0$
For getting a reciprocal roots, we replace x by $\frac{1}{x}$, we get
$$2\left(\frac{1}{x}\right)^2-3\left(\frac{1}{x}\right)-4=0$$
$\Rightarrow \quad \frac{2}{x^2}-\frac{3}{x}-4=0$

$\Rightarrow \quad -4x^2-3x+2=0$

$\Rightarrow \quad 4x^2+3x-2=0$

17. *(c)* As, $\sqrt{x+4}=x-2$
On squaring both sides, we get
$$(x+4)=(x-2)^2$$
$\Rightarrow \quad x+4=x^2+4-4x$

$\Rightarrow \quad x^2-5x=0 \Rightarrow x=0, x=5$

But for $x=0, \quad \sqrt{0+4}=0-2$

$\sqrt{4}\neq -2$

So, $x=5$ is the only solution.

18. *(b)* Let roots of equation be α and $\frac{1}{\alpha}$.

$\therefore$ Product of roots

$=\alpha\times\frac{1}{\alpha}=\frac{\text{Constant term}}{\text{Coefficient of } x^2}=\frac{r}{p}$

$\Rightarrow \quad 1=\frac{r}{p} \quad \Rightarrow \quad r=p$

19. *(d)* Let roots be α and β, then $\alpha+\beta=-1$
$$\frac{1}{\alpha}+\frac{1}{\beta}=\frac{1}{6} \quad \Rightarrow \quad \frac{\beta+\alpha}{\alpha\beta}=\frac{1}{6}$$
$$\frac{-1}{\alpha\beta}=\frac{1}{6} \quad \Rightarrow \quad \alpha\beta=-6$$
$\therefore$ Required equation is,
$$x^2-(\alpha+\beta)x+\alpha\beta=0$$
$\Rightarrow \quad x^2-(-1)x+(-6)=0$

$\therefore \quad x^2+x-6=0$

20. *(c)* Let α and β be the roots of the equation
$ax^2+bx+c=0$

$\therefore$ Sum of roots $(\alpha+\beta)=-\frac{b}{a}$

and product of roots $(\alpha\beta)=\frac{c}{a}$

By given condition,
$$\alpha+\beta=\alpha^2+\beta^2$$
$\Rightarrow \quad \alpha+\beta=(\alpha+\beta)^2-2\alpha\beta$

$\Rightarrow \quad -\frac{b}{a}=\left(-\frac{b}{a}\right)^2-2\left(\frac{c}{a}\right)$

$\Rightarrow \quad \frac{-b}{a}=\frac{b^2}{a^2}-\frac{2c}{a} \Rightarrow -ba=b^2-2ca$

$\Rightarrow \quad 2ac=b^2+ab$

21. *(d)* Here, roots are α and $\alpha+1$.

$\therefore \quad \alpha+(\alpha+1)=l \qquad$ [sum of roots]

$\Rightarrow \quad 2\alpha=l-1 \quad \Rightarrow \quad \alpha=\frac{l-1}{2}$

Also, $\alpha(\alpha+1)=m$ or $\alpha^2+\alpha=m$

$\Rightarrow \quad \left(\frac{l-1}{2}\right)^2+\left(\frac{l-1}{2}\right)=m$

$\Rightarrow \quad (l-1)^2+2(l-1)=4m$

$\Rightarrow \quad l^2-1=4m \Rightarrow l^2=4m+1$

22. *(b)* Here, $\alpha+\beta=(1+a^2)$

and $\quad \alpha\beta=\frac{1}{2}(a^4+a^2+1)$

$\because \quad \alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

$=(1+a^2)^2-(a^4+a^2+1)$

$=1+a^4+2a^2-a^4-a^2-1$

$\therefore \quad \alpha^2+\beta^2=a^2$

23. *(a)* Here, $\frac{1}{x+p}+\frac{1}{x+q}=\frac{1}{r}$

$\Rightarrow r(x+p+x+q)=(x+p)(x+q)$

$\Rightarrow \quad x^2+(p+q-2r)x+pq-(p+q)r=0$

Let roots be α and $(-\alpha)$, then
$$\alpha+(-\alpha)=0$$
$\Rightarrow -(p+q-2r)=0 \Rightarrow r=\frac{p+q}{2}$

So, product of roots $=pq-(p+q)r$

$=pq-(p+q)\cdot\frac{(p+q)}{2}$

$=pq-\frac{(p+q)^2}{2}=\frac{-1}{2}(p^2+q^2)$

24. *(b)* Here, $\alpha+\beta=-\frac{2}{3}$ and $\alpha\beta=\frac{1}{3}$

$\Rightarrow S=\frac{1-\alpha}{1+\alpha}+\frac{1-\beta}{1+\beta}$

$=\frac{(1-\alpha)(1+\beta)+(1+\alpha)(1-\beta)}{(1+\alpha)(1+\beta)}$

$=\frac{2-2\alpha\beta}{1+(\alpha+\beta)+\alpha\beta}=\frac{2-2\times\frac{1}{3}}{1+(-2/3)+1/3}$

$=\frac{4}{3}\times\frac{3}{2}=2$

$P=\frac{1-\alpha}{1+\alpha}\times\frac{1-\beta}{1+\beta}=\frac{1-(\alpha+\beta)+\alpha\beta}{1+(\alpha+\beta)+\alpha\beta}$

$= \frac{1-(-2/3)+1/3}{1+(-2/3)+1/3} = \frac{2}{2/3} = 3$

$\therefore$ Required equation is

$x^2 - Sx + P = 0$ or $x^2 - 2x + 3 = 0$

25. *(a)* Given, $px^2 + qx + r = 0$

Let the roots be α and β.

By given condition, $\beta = 2\alpha$

Product of roots $(\alpha\beta) = \frac{r}{p} = 2\alpha^2$

$\Rightarrow \quad \alpha^2 = \frac{r}{2p} \quad \ldots(i)$

Sum of roots $(\alpha + \beta) = -\frac{q}{p} = 3\alpha \ldots(ii)$

On squaring Eq. (ii), we get $9\alpha^2 = \frac{q^2}{p^2}$

$\Rightarrow \quad 9\left(\frac{r}{2p}\right) = \frac{q^2}{p^2}$ [from Eq. (i)]

$\Rightarrow \quad 9rp = 2q^2$

26. *(a)* Let roots be α, β, then $\frac{\alpha}{\beta} = \frac{m}{n}$ [given]

$\alpha + \beta = -1, \alpha\beta = 1$

$\sqrt{\frac{m}{n}} + \sqrt{\frac{n}{m}} + 1 = \sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + 1$

$= \frac{\alpha+\beta}{\sqrt{\alpha\beta}} + 1 = \frac{-1}{\sqrt{1}} + 1 = -1 + 1 = 0$

27. *(c)* Let $\sqrt{\frac{x}{1-x}} = y \Rightarrow \sqrt{\frac{1-x}{x}} = \frac{1}{y}$

$\therefore \quad y + \frac{1}{y} = \frac{13}{6} \Rightarrow (y^2 + 1)6 = 13y$

$\Rightarrow \quad 6y^2 - 13y + 6 = 0$

$\Rightarrow \quad 6y^2 - 9y - 4y + 6 = 0$

$\Rightarrow \quad 3y(2y-3) - 2(2y-3) = 0$

$\Rightarrow \quad (3y-2)(2y-3) = 0$

$\therefore \quad y = \frac{2}{3}$ and $\frac{3}{2}$

When, $y = \frac{2}{3} \Rightarrow \frac{x}{1-x} = \frac{4}{9}$

$\Rightarrow \quad 9x = 4 - 4x \Rightarrow x = \frac{4}{13}$

When, $y = \frac{3}{2} \Rightarrow \frac{x}{1-x} = \frac{9}{4}$

$\Rightarrow \quad 4x = 9 - 9x \Rightarrow x = \frac{9}{13}$

28. *(d)* Given, $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$

$\therefore \quad \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$

$\Rightarrow \quad \frac{-(a+b)}{(a+b+x)x} = \frac{(a+b)}{ab}$

$\Rightarrow \quad x^2 + (a+b)x + ab = 0$

$\Rightarrow \quad (x+a)(x+b) = 0$

$\Rightarrow \quad x = -a, -b$

29. *(d)* $8\sec^2\phi - 6\sec\phi + 1 = 0$

$\Rightarrow 8\sec^2\phi - 4\sec\phi - 2\sec\phi + 1 = 0$

$\Rightarrow \quad (4\sec\phi - 1)(2\sec\phi - 1) = 0$

$\Rightarrow \quad \sec\phi = 1/4$ or $\sec\phi = 1/2$

But $\sec\phi \geq 1$ or $\sec\phi \leq -1$

Hence, the equation has no solution.

30. *(c)* Here, $3^{2x^2 - 7x + 7} = 9 = 3^2$

On comparing, we get

$2x^2 - 7x + 7 = 2$

$2x^2 - 7x + 5 = 0$

Here, $D = b^2 - 4ac = 49 - 4(2)(5) = 9$

Since, $D > 0$, so it has two real roots.

31. *(c)* Here, $x = \sqrt{2} + 2 \Rightarrow x - 2 = \sqrt{2}$

On squaring both sides, we get

$(x-2)^2 = 2 \Rightarrow x^2 - 4x + 4 = 2$

$\Rightarrow x^2 - 4x + 2 = 0$

32. *(b)* $\frac{x^2 - bx}{ax - c} = \frac{k-1}{k+1}$

$\Rightarrow (x^2 - bx)(k+1) = (k-1)(ax - c)$

$\Rightarrow \quad x^2k + x^2 - bxk - bx = kax - kc - ax + c$

$\Rightarrow \quad (k+1)x^2 - x(bk + b + ka - a) + kc - c = 0$

Since, roots are reciprocal to each other.

So, product = 1

i.e. $\frac{kc - c}{(k+1)} = 1 \Rightarrow kc - c = k + 1$

$\Rightarrow \quad k(c-1) = c + 1 \Rightarrow k = \frac{c+1}{c-1}$

33. *(b)* Here, $\alpha + \beta = b/a$ and $\alpha\beta = b/a$

So, $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} = \frac{\alpha+\beta}{\sqrt{\alpha\beta}} = \frac{b/a}{\sqrt{b/a}} = \sqrt{\frac{b}{a}}$

34. *(a)* Given, $2x^2 - 7x + 3 = 0$

$\therefore \quad 2x^2 - 6x - x + 3 = 0$

$\Rightarrow \quad 2x(x-3) - 1(x-3) = 0$

$\Rightarrow \quad (2x-1)(x-3) = 0$

When, $x = \frac{1}{2}$

$4\left(\frac{1}{2}\right)^2 + a\left(\frac{1}{2}\right) - 3 = 0$

$\Rightarrow 1 + \frac{a}{2} - 3 = 0 \Rightarrow \frac{a}{2} = 2 \Rightarrow a = 4$

When, $x = 3$, $4(3)^2 + a(3) - 3 = 0$

$\Rightarrow \quad 36 + 3a - 3 = 0 \Rightarrow a = -11$

$\therefore \quad a = -11$ or 4

35. *(c)* Here, for equation $x^2 + bx + c = 0$

$\alpha + \beta = -b$ and $\alpha\beta = c$

and for $x^2 + px + q = 0$

$k\alpha + k\beta = -p$ and $k^2\alpha\beta = q$

Now, $qb^2 = k^2\alpha\beta(\alpha+\beta)^2$

$= k^2(\alpha+\beta)^2\alpha\beta$

$= [k(\alpha+\beta)]^2\alpha\beta$

$\therefore \quad qb^2 = p^2c$

36. *(b)* Since, -4 is a root of

$x^2 + px - 4 = 0$

$(-4)^2 + p(-4) - 4 = 0$

$\Rightarrow \quad 16 - 4p - 4 = 0 \Rightarrow p = 3$

As roots of $x^2 + px + q = 0$ are equal.

$\therefore \quad D = B^2 - 4AC = 0$

$\Rightarrow p^2 - 4q = 0$ or $3^2 - 4q = 0 \Rightarrow q = \frac{9}{4}$

So, $p = 3, q = \frac{9}{4}$

37. *(c)* Here, $\alpha + \beta = 3$ and $\alpha\beta = 3/2$

$\Rightarrow \quad \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + 3\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) + 2\alpha\beta$

$= \frac{\alpha^2 + \beta^2}{\alpha\beta} + \frac{3(\alpha+\beta)}{\alpha\beta} + 2\alpha\beta$

$= \frac{(\alpha+\beta)^2 - 2\alpha\beta}{\alpha\beta} + \frac{3(\alpha+\beta)}{\alpha\beta} + 2\alpha\beta$

$= \frac{3^2 - 2\cdot 3/2}{3/2} + \frac{3\cdot 3}{3/2} + 2\left(\frac{3}{2}\right)$

$= \frac{6}{3/2} + \frac{9}{3/2} + 3 = 4 + 6 + 3 = 13$

38. *(b)* Let α, β be the roots of

$x^2 + 2ax + b = 0 \quad \ldots(i)$

$\Rightarrow \quad \alpha + \beta = -2a$ and $\alpha\beta = b$.

By hypothesis, $|\alpha - \beta| \leq 2m$

$\Rightarrow \quad (\alpha - \beta)^2 \leq 4m^2$

$\Rightarrow \quad (\alpha+\beta)^2 - 4\alpha\beta \leq 4m^2$

$\Rightarrow \quad 4a^2 - 4b \leq 4m^2$

$\Rightarrow \quad a^2 - b \leq m^2 \quad \ldots(ii)$

and discriminant of Eq. (i) is greater than 0.

$\Rightarrow \quad 4a^2 - 4b > 0 \Rightarrow b < a^2 \quad \ldots(iii)$

From Eqs. (ii) and (iii),

$b \in [a^2 - m^2, a^2)$

39. *(b)* Refer to example 15. [Rule 5]

40. *(b)* Refer to example 14. [Rule 4]

41. *(b)* Since, $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$.

$\therefore \quad \sin\theta + \cos\theta = \frac{b}{a} \quad \ldots(i)$

and $\quad \sin\theta\cos\theta = \frac{c}{a}$

On squaring both sides of Eq. (i), we get

$\Rightarrow \quad \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = \frac{b^2}{a^2}$

$\Rightarrow 1 + 2\left(\frac{c}{a}\right) = \frac{b^2}{a^2} \Rightarrow 2\left(\frac{c}{a}\right) = \frac{b^2 - a^2}{a^2}$

$\Rightarrow 2ac = b^2 - a^2 \Rightarrow a^2 - b^2 + 2ac = 0$

42. *(d)* The equation will have equal roots,

if $B^2 - 4AC = 0$

$\therefore \quad (2nc)^2 - 4(1+n^2)(c^2-a^2) = 0$

$\Rightarrow \quad 4n^2c^2 - 4(c^2 + n^2c^2 - a^2 - n^2a^2) = 0$

$\Rightarrow \quad -4c^2 + 4a^2 + 4n^2a^2 = 0$

$\Rightarrow \quad c^2 = a^2(1+n^2)$

43. *(d)* a and c have the same sign opposite to that of b.

44. *(d)* Let the roots of equation $x^2 - 2x + 4 = 0$ be α and β.

Then, $\alpha + \beta = 2$ and $\alpha\beta = 4$

On taking, $\alpha \to 2\alpha$ and $\beta \to 2\beta$

$2\alpha + 2\beta = 4$ and $2\alpha \cdot 2\beta = 4 \times 4 = 16$

So, the new equation becomes

$$x^2 - 4x + 16 = 0$$

45. *(a)* Since, α and β are the roots of the equation $x^2 + px + q = 0$

$\therefore \quad \alpha + \beta = -p$ and $\alpha\beta = q$

Now, $\quad -\alpha^{-1} - \beta^{-1} = -\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$

$= -\left(\frac{\alpha+\beta}{\alpha\beta}\right) = \frac{p}{q}$

and $\left(-\frac{1}{\alpha}\right)\left(-\frac{1}{\beta}\right) = \frac{1}{\alpha\beta} = \frac{1}{q}$

Hence, required equation is

$x^2 - (-\alpha^{-1} - \beta^{-1})x + (-\alpha^{-1})(-\beta^{-1}) = 0$

$\Rightarrow x^2 - \frac{p}{q}x + \frac{1}{q} = 0$

$\Rightarrow qx^2 - px + 1 = 0$

46. *(c)* Since, one root of the equation $ax^2 + x - 3 = 0$ is -1.

$\therefore a(-1)^2 + (-1) - 3 = 0 \quad \Rightarrow \quad a = 4$

$\therefore \quad 4x^2 + x - 3 = 0$

Let other root of this equation be α.

Now, product of roots $= \frac{c}{a}$

$\therefore \quad -1 \cdot \alpha = -\frac{3}{4} \quad \Rightarrow \quad \alpha = \frac{3}{4}$

47. *(b)* Since, the roots of equation

$(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal.

$\therefore \quad B^2 = 4AC$

$\Rightarrow 4(ac+bd)^2 = 4(a^2+b^2)(c^2+d^2)$

$\Rightarrow a^2c^2 + b^2d^2 + 2abcd = a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

$\Rightarrow \quad (ad - bc)^2 = 0$

$\therefore \quad ad = bc$

48. *(a)* Given, $\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2}$

Let, $y = \sqrt{\frac{x}{x+3}}$, then $\sqrt{\frac{x+3}{x}} = \frac{1}{y}$

$y - \frac{1}{y} = -\frac{3}{2} \quad \Rightarrow \quad 2y^2 - 2 = -3y$

$\Rightarrow \quad 2y^2 + 3y - 2 = 0$

$\Rightarrow \quad 2y^2 + 4y - y - 2 = 0$

$\Rightarrow \quad 2y(y+2) - 1(y+2) = 0$

$\Rightarrow \quad (2y-1)(y+2) = 0$

If, $\quad (2y-1) = 0$

$\Rightarrow y = \frac{1}{2} \Rightarrow y = \sqrt{\frac{x}{x+3}} \Rightarrow \sqrt{\frac{x}{x+3}} = \frac{1}{2}$

On squaring both sides, we get

$$\frac{x}{x+3} = \frac{1}{4}$$

$\Rightarrow \quad 4x = x + 3 \Rightarrow \quad x = 1$

or $\quad y + 2 = 0 \quad \Rightarrow \quad y = -2$

Since, y cannot be negative.

Hence, $x = 1$ is the required solution.

49. *(c)* Given, $\quad 4^x - 3 \cdot 2^{x+2} + 32 = 0$

$\Rightarrow \quad 2^{2x} - 3 \cdot 2^x \cdot 2^2 + 32 = 0$

$\Rightarrow \quad 2^{2x} - 12 \cdot 2^x + 32 = 0$

$\Rightarrow \quad 2^{2x} - 8 \cdot 2^x - 4 \cdot 2^x + 32 = 0$

$\Rightarrow \quad (2^x - 8)(2^x - 4) = 0$

Either, $\quad 2^x = 8 \quad \Rightarrow \quad x = 3$

or $\quad 2^x = 4 \quad \Rightarrow \quad x = 2$

50. *(a)* Since, α and β are the roots of the equation $x^2 - x - 1 = 0$.

$\therefore \quad \alpha + \beta = 1$ and $\alpha\beta = -1$

Now, $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2$

$= [(\alpha+\beta)^2 - 2\alpha\beta]^2 - 2(\alpha\beta)^2$

$= (1+2)^2 - 2 = 9 - 2 = 7$

51. *(c)* Let the roots of equation $ax^2 + bx + c = 0$ be $-\alpha$ and $-\frac{1}{\alpha}$.

$\therefore$ Product of roots $= \frac{c}{a}$

$(-\alpha)\left(-\frac{1}{\alpha}\right) = \frac{c}{a} \quad \Rightarrow \quad 1 = \frac{c}{a} \Rightarrow c = a$

52. *(d)* Let the roots of the quadratic equation be α and β.

Then, $\quad \alpha + \beta = 9$ and $\alpha\beta = 9$

Hence, equation is

$$x^2 - (\alpha+\beta)x + (\alpha\beta) = 0$$

$\Rightarrow \quad x^2 - 9x + 9 = 0$

53. *(c)* Given, $\log_{10}(x^2 - 6x + 45) = 2$

$\Rightarrow \quad (x^2 - 6x + 45) = 10^2 = 100$

$\Rightarrow \quad x^2 - 6x - 55 = 0$

$\Rightarrow \quad x^2 - 11x + 5x - 55 = 0$

$\Rightarrow \quad x(x-11) + 5(x-11) = 0$

$\Rightarrow \quad (x+5)(x-11) = 0$

$\therefore \quad x = 11, -5$

54. *(c)* Given, $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$

$\Rightarrow \quad \frac{(x+b)+(x+a)}{(x+a)(x+b)} = \frac{1}{c}$

$\Rightarrow 2cx + (a+b)c = x^2 + (a+b)x + ab$

$\Rightarrow x^2 + (a+b-2c)x + ab - ac - bc = 0$

Let the roots of above equation be α and β.

Given, $\quad \alpha + \beta = 0$

$\Rightarrow -(a+b-2c) = 0 \Rightarrow a + b = 2c$ ...(i)

Now, $\alpha\beta = ab - ac - bc$

$= ab - (a+b)c$

$= ab - (a+b)\frac{(a+b)}{2}$

[from Eq. (i)]

$= \frac{2ab - (a^2 + b^2 + 2ab)}{2} = -\frac{(a^2+b^2)}{2}$

55. *(b)* Let the roots of the equation $kx^2 - 5x + 6 = 0$ be α and β.

$\therefore \quad \alpha + \beta = \frac{5}{k}$ and $\alpha\beta = \frac{6}{k}$

Given, $\quad \frac{\alpha}{\beta} = \frac{2}{3} \quad \Rightarrow \quad \alpha = \frac{2}{3}\beta$

$\therefore \quad \frac{2}{3}\beta + \beta = \frac{5}{k}$ and $\frac{2}{3}\beta^2 = \frac{6}{k}$

$\Rightarrow \quad \frac{5}{3}\beta = \frac{5}{k}$ and $\beta^2 = \frac{9}{k}$

$\beta = \frac{3}{k}$ and $\beta^2 = \frac{9}{k} \Rightarrow \quad \frac{9}{k^2} = \frac{9}{k}$

$\Rightarrow \quad k = 1$ and $k \neq 0$

[$\because$ leading coefficient is never zero.]

56. *(d)* Let α and β be the roots of the equation $7x^2 + 12x + 18 = 0$.

$\therefore \quad \alpha + \beta = \frac{-12}{7}$ and $\alpha\beta = \frac{18}{7}$

$\Rightarrow \quad \alpha^2 + \beta^2 + 2\alpha\beta = \frac{144}{49}$

$\Rightarrow \quad \alpha^2 + \beta^2 = \frac{144}{49} - \frac{36}{7} = -\frac{108}{49}$

$\therefore \quad \frac{\alpha^2+\beta^2}{\alpha\beta} = \frac{-\frac{108}{49}}{\frac{18}{7}} = -\frac{6}{7} = -6:7$

57. *(d)* Given, $\frac{x(x-1)-(m+1)}{(x-1)(m-1)} = \frac{x}{m}$

$\Rightarrow \quad m(x^2 - x - m - 1) = x(mx - x - m + 1)$

$\Rightarrow \quad mx^2 - mx - m(m+1) = mx^2 - x^2 - mx + x$

$\Rightarrow \quad x^2 - x - m(m+1) = 0$

Let one root be α, then other root is also α.

$\therefore \quad \alpha + \alpha = 1, \Rightarrow \alpha = \frac{1}{2}$

Also, $\alpha^2 = -m(m+1)$

$\Rightarrow \left(\frac{1}{2}\right)^2 = -m(m+1)$

$\Rightarrow 4m^2 + 4m + 1 = 0$

$\Rightarrow (2m+1)^2 = 0$

$\therefore m = -\frac{1}{2}$

58. *(c)* Given, $3^x + 27(3^{-x}) = 12$

Let $3^x = y$

$\therefore y + \frac{27}{y} = 12$

$\Rightarrow y^2 - 12y + 27 = 0$

$\Rightarrow y^2 - 9y - 3y + 27 = 0$

$\Rightarrow (y-3)(y-9) = 0 \Rightarrow y = 3, 9$

$\Rightarrow 3^x = 3$ or $3^x = 9 = 3^2$

$\therefore x = 1$ or $x = 2$

59. *(a)* Let the roots of the given equation be α and β.

$\therefore \alpha + \beta = a$ and $\alpha\beta = b$

Now, $|\alpha - \beta| = \sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$

$= \sqrt{a^2 - 4b}$

60. *(a)* $x^2 - x - 2 = 0$

$(x-2)(x+1) = 0 \Rightarrow x = 2, -1$

So, both the roots are integers.

61. *(c)* $x^2 - 6x + 5 = 0 \Rightarrow (x-5)(x-1) = 0$

$\Rightarrow x = 5$ or 1

Also, $|x-3| = 2 \Leftrightarrow (x-3) = 2$

or $-(x-3) = 2 \Rightarrow x = 5$ or $x = 1$

$\therefore x^2 - 6x + 5 = 0$

and $|x-3| = 2$ are equivalent.

62. *(a)* Let the number are x and $\frac{1}{x}$.

Then, $x + \frac{1}{x} = \frac{10}{3}$, $\frac{x^2+1}{x} = \frac{10}{3}$

$\Rightarrow 3x^2 - 10x + 3 = 0$

$\Rightarrow 3x^2 - 9x - x + 3 = 0$

$\Rightarrow 3x(x-3) - 1(x-3) = 0$

$\Rightarrow (3x-1)(x-3) = 0$

$\therefore x = \frac{1}{3}, x = 3$

63. *(b)* Let the smaller part $= x$ and greater part $= 16 - x$

By given condition,

$2(16-x)^2 - x^2 = 164$

$\Rightarrow 2(256 + x^2 - 32x) - x^2 = 164$

$\Rightarrow x^2 - 64x + 348 = 0$

$\Rightarrow (x-58)(x-6) = 0$

$\Rightarrow x = 58, x = 6$

Here $x \neq 58$

$\therefore x = 6$

and, hence larger part

$= 16 - x = 16 - 6 = 10$

64. *(c)* Here, $y = 15$, then $15 = \frac{x(x-1)}{2}$

$\Rightarrow x^2 - x - 30 = 0$

$\Rightarrow x^2 - 6x + 5x - 30 = 0$

$\Rightarrow x(x-6) + 5(x-6) = 0$

$\Rightarrow (x+5)(x-6) = 0$

Either $x = 6$ or $x = -5$

But $x \neq -5$, so $x = 6$

Thus, figure has 6 points.

65. *(a)* Let the natural numbers be x and $x+1$, respectively.

Then, $x^2 + (x+1)^2 = 221$

$\Rightarrow 2x^2 + 2x + 1 = 221$

$\Rightarrow 2x^2 + 2x - 220 = 0$

$\Rightarrow x^2 + x - 110 = 0$

$\Rightarrow (x+11)(x-10) = 0$

$\Rightarrow x = -11$ and $x = 10$

$x \neq -11$ [$\because$ numbers are natural]

So, for $x = 10$, next consecutive natural number

$= x + 1 = 10 + 1 = 11$

66. *(b)* Let the consecutive positive odd integers be $2x+1$ and $2x+3$,

so $(2x+1)^2 + (2x+3)^2 = 130$

$\Rightarrow (4x^2 + 4x + 1) + (4x^2 + 12x + 9) = 130$

$\Rightarrow x^2 + 2x - 15 = 0$

$\Rightarrow (x+5)(x-3) = 0$

$\Rightarrow x = 3, x = -5$, but $x \neq -5$

[$\because$ integers are positive.]

$\therefore$ Two consecutive integers are

$2x + 1 = 7$ and $2x + 3 = 9$

i.e. 7 and 9.

67. *(b)* Here, $2x + 1 \geq 7 \Rightarrow 2x \geq 7 - 1$

$\Rightarrow 2x \geq 6 \Rightarrow x \geq 3$

68. *(c)* Here, $\frac{x-1}{3} \geq 4$

$\Rightarrow x - 1 \geq 12 \Rightarrow x \geq 13$

69. *(b)* Here, $3x + 2 \leq 5x - (4 - x)$

$\Rightarrow 3x + 2 \leq 5x - 4 + x$

$\Rightarrow 3x + 2 \leq 6x - 4$

$\Rightarrow 3x - 6x \leq -4 - 2 \Rightarrow -3x \leq -6$

$\Rightarrow -x \leq -2 \Rightarrow x \geq 2$

70. *(c)* Here, $2x + 3 \geq 8 \Rightarrow 2x \geq 8 - 3$

$\Rightarrow 2x \geq 5 \Rightarrow x \geq \frac{5}{2}$

Again, $3x + 1 \leq 12$

$\Rightarrow 3x \leq 11 \Rightarrow x \leq \frac{11}{3}$

By combining values, we get

$\frac{5}{2} \leq x \leq \frac{11}{3}$

71. *(b)* $\frac{1}{2}\left(\frac{3}{5}x + 4\right) \geq \frac{1}{3}(x - 6)$

$\Rightarrow \frac{3}{10}x + 2 \geq \frac{1}{3}x - 2$

$\Rightarrow 9x + 60 \geq 10x - 60$

$\Rightarrow -x \geq -120$

[multiplying both sides by -1]

$\Rightarrow x \leq 120$

Thus, all real numbers x which are less than or equal to 120 satisfies the inequality.

72. *(a)* We have, $4x^2 - 1 \leq 0$

$\Rightarrow (2x)^2 - 1 \leq 0$

$\Rightarrow (2x-1)(2x+1) \leq 0$

So, either $(2x-1) \geq 0$

and $(2x+1) \leq 0$ ...(i)

or $(2x-1) \leq 0$

and $(2x+1) \geq 0$ ...(ii)

From Eq. (i), $2x \geq 1$ and $2x \leq -1$

$\Rightarrow x \geq \frac{1}{2}$ and $x \leq \frac{-1}{2}$

which is not possible.

From Eq. (ii), $(2x-1) \leq 0$

and $(2x+1) \geq 0$

$2x \leq 1$ and $2x \geq -1$

$x \leq \frac{1}{2}$ and $x \geq \frac{-1}{2}$

$\therefore \frac{-1}{2} \leq x \leq \frac{1}{2}$ is the required solution set.

73. *(b)* Here, $3x + 2y = 12$ passes through the points $(0, 6)$ and $(4, 0)$. So, its graph is line AB. Also, $3x + 2y \leq 12$ as the shaded region is below AB.

$x = 0$ and $y = 0$ are the Y-axis and X-axis, respectively. $x \geq 0$ and $y \geq 0$ implies the region on the right hand side of Y-axis and region above X-axis, respectively.

The three portion $x \geq 0$, $y \geq 0$ and $3x + 2y \leq 12$ intersects to give the shaded portion OAB.

74. *(d)* Given, $x^2 + 3x - 10 > 0$

$\Rightarrow x^2 + 5x - 2x - 10 > 0$

$\Rightarrow (x+5)(x-2) > 0$

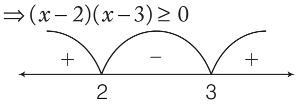
Either, $(x+5) > 0$ and $(x-2) > 0$

or $(x+5) < 0$ and $(x-2) < 0$

$\Rightarrow x > 2$ or $x < -5$

75. *(a)* $x^2 - 5x + 6 \geq 0$

$\Rightarrow (x-2)(x-3) \geq 0$



$\Rightarrow x \leq 2$ or $x \geq 3$

$\Rightarrow x \in [-\infty, 2]$ or $[3, \infty]$

$\Rightarrow x \in [-\infty, 2] \cup [3, \infty]$

76. *(b)* $\dfrac{x-2}{3x+1} < \dfrac{x-3}{3x-2}$

$\Leftrightarrow \dfrac{x-2}{3x+1} - \dfrac{x-3}{3x-2} < 0$

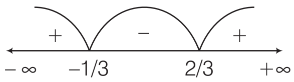
$\Leftrightarrow \dfrac{(x-2)(3x-2)-(x-3)(3x+1)}{(3x+1)(3x-2)} < 0$

$\Leftrightarrow \dfrac{7}{(3x+1)(3x-2)} < 0$

$\Leftrightarrow (3x+1)(3x-2) < 0$

$\Leftrightarrow 3\left(x+\dfrac{1}{3}\right)3\left(x-\dfrac{2}{3}\right) < 0$

$\Leftrightarrow \left(x+\dfrac{1}{3}\right)\left(x-\dfrac{2}{3}\right) < 0$



$\Leftrightarrow \dfrac{-1}{3} < x < \dfrac{2}{3}$

So, $x \in \left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$

77. *(c)* Here, as $x > 0$ and $y > 0$ so both are positive and satisfies $x + 2y \le 3$.

$\therefore$ When, $x = 3$ and $y = 1$, we get $x + 2y = 5$.

Clearly, this does not satisfy $x + 2y \le 3$

When $x = 1$ and $y = 1$,

then, $x + 2y \le 3$.

So, $x = 1$, $y = 1$ is one of the solutions.

78. *(d)* As, $(0, 0)$ satisfies $x + 3y \le 3$.

And $(0, 0)$ satisfies $x + y \le 2$.

Clearly, shaded region is the position common to the line $x + 3y = 3$ and below it and that on the line $x + y = 2$ and below it.

So, shaded region is the solution set

$x + y \le 2$, $x + 3y \le 3$, $x \ge 0$, $y \ge 0$.

79. *(c)* Given, $a + b = 2m^2$ …(i)

$b + c = 6m$ …(ii)

and $a + c = 2$ …(iii)

On adding Eqs. (i), (ii) and (iii), we get

$2(a + b + c) = 2m^2 + 6m + 2$

$\Rightarrow a + b + c = m^2 + 3m + 1$ …(iv)

On subtracting Eq. (ii) from Eq. (iv), we get

$a = m^2 - 3m + 1$

On subtracting Eq. (iii) from Eq. (iv), we get

$b = m^2 + 3m - 1$

On subtracting Eq. (i) from Eq. (iv), we get

$c = -m^2 + 3m + 1$

As, $a \le b$ and $b \le c$

$\Rightarrow m^2 - 3m + 1 \le m^2 + 3m - 1$

and $m^2 + 3m - 1 \le -m^2 + 3m + 1$

$\Rightarrow 6m \ge 2$ and $2m^2 \le 2$

$\Rightarrow m \ge \dfrac{1}{3}$ and $-1 \le m \le 1$

$\therefore \dfrac{1}{3} \le m \le 1$

80. *(d)* Given equation, $x^2 - 3x + 2 = 0$

$\Rightarrow x^2 - 2x - x + 2 = 0$

$\Rightarrow (x-2)(x-1) = 0 \Rightarrow x = 2, 1$

Let $\alpha = 1$ and $\beta = 2$

$\therefore \alpha + 1 = 2$ and $(\beta + 1) = 3$.

Now, sum of roots $= 2 + 3 = 5$

and product of roots $= 2 \times 3 = 6$

$\therefore$ Required equation

$= x^2 -$ (sum of roots) $+$ product of roots $= 0$

$\Rightarrow x^2 - 5x + 6 = 0$

Hence, the equation is neither I nor II.

81. *(c)* I. Every quadratic equation has two roots, which may or may not be real.

II. $x^2 - 4x + 2 = 0$ has integral coefficients but does not have integral roots.

III. Since, discriminant $= b^2 - 4ac$

If a and c have opposite sign, then $b^2 - 4ac \ge 0$

$\therefore$ The quadratic equation has real roots

82. *(c)* I. $x^2 + \dfrac{1}{x^2} = 2$

or $x^4 - 2x^2 + 1 = 0$

is not a quadratic equation.

II. $x + \dfrac{3}{x} = x^2$

$\Rightarrow \dfrac{x^2+3}{x} = x^2$

or $x^2 + 3 = x^3$

$\Rightarrow x^3 - x^2 - 3 = 0$

is not a quadratic equation.

III. $2x^2 - x + 2 = x^2 + 4x - 4$

or $2x^2 - x + 2 - x^2 - 4x + 4 = 0$

or $x^2 - 5x + 6 = 0$

is a quadratic equation.

IV. $x^3 + 6x^2 + 2x - 1 = 0$

is not a quadratic equation.

Hence, only III statement is correct.

83. *(b)* I. $3x^2 - 2x - 1 = 0$

Put $x = 1 \Rightarrow 3(1)^2 - 2(1) - 1 = 0$

$\therefore$ LHS = RHS

II. $x^2 + \sqrt{2x} - 4 = 0$

Put $x = -2\sqrt{2}$

$\Rightarrow (-2\sqrt{2})^2 + \sqrt{2}(-2\sqrt{2}) - 4 = 0$

$\therefore$ LHS = RHS

III. $x^2 + x + 1 = 0$; $x = 1, x = -1$

Put $x = 1$, we get $1^2 + 1 + 1 = 3 \ne 0$

$\therefore$ LHS $\ne$ RHS

$\therefore x = 1$ is not a solution.

Put $x = -1$, we get

$(-1)^2 + (-1) + 1 = 1 - 1 + 1$

$= 1 \ne 0$

$\therefore$ LHS $\ne$ RHS

$\therefore x = -1$ is not a solution.

IV. $9x^2 - 3x - 2 = 0$

Put $x = -1/3$

$\Rightarrow 9(-1/3)^2 - 3(-1/3) - 2 = 0$

$\therefore$ LHS = RHS

Now, put $x = 2/3$

$\Rightarrow 9(2/3)^2 - 3(2/3) - 2 = 0$

$\therefore$ LHS = RHS

84. *(b)* Given, $h = -16t^2 + 64t + 80$

put $t = 1$

$\Rightarrow h = -16(1)^2 + 64(1) + 80 = 128$ m

85. *(c)* By rearranging, we get

$h = -16(t^2 - 4t - 5)$

$\Rightarrow h = -16[(t-2)^2 - 9]$

$\Rightarrow h = -16(t-2)^2 + 144$

when the height is maximum, $t = 2$

$\therefore$ maximum height = 144 m

86. *(a)* When the ball hits the ground $h = 0$

$\Rightarrow -16t^2 + 64t + 80 = 0$

$\Rightarrow t^2 - 4t - 5 = 0$

$\Rightarrow t^2 - 5t + t - 5 = 0$

$\Rightarrow (t-5)(t+1) = 0 \Rightarrow t = 5, -1$

Since, the time cannot be negative.

So, $t = 5$ sec

87. *(a)* Let $f(x) = x^2 - 2ax + a^2 + a - 3$

$f(3) > 0 \Rightarrow 9 - 6a + a^2 + a - 3 > 0$ …(i)

Also, since roots are real

$\therefore B^2 - 4AC \ge 0$

$\Rightarrow 4a^2 - 4(a^2 + a - 3) \ge 0$ …(ii)

From Eqs. (i) and (ii), we have

$a^2 - 5a + 6 > 0$ and $a - 3 \le 0$

$\Rightarrow a^2 - 3a - 2a + 6 > 0$

$\Rightarrow (a-3)(a-2) > 0$

$\Rightarrow a \le 3$ and $a < 2$ or $a > 3$

Combining the above inequalities, we have $a < 2$

88. *(a)* Given, $7x^2 - 50x + k = 0$

Here, $a = 7$, $b = -50$ and $c = k$

Since, α and β are the roots of the given equation

$\therefore \alpha + \beta = \dfrac{-b}{a} \Rightarrow \alpha + \beta = \dfrac{50}{7}$

$\Rightarrow \beta = \dfrac{1}{7}$ [$\because \alpha = 7$, given]

and $\alpha\beta = \dfrac{c}{a} \Rightarrow 7 \times \dfrac{1}{7} = \dfrac{k}{7} \Rightarrow k = 7$

89. *(b)* Let α and β be the roots of the quadratic equation

$$x^2 + px + q = 0.$$

Given that, A starts with a wrong value of p and obtains the roots as 2 and 6. But this time q is correct.

i.e. Product of roots

$$= q = \alpha \cdot \beta = 6 \times 2 = 12 \quad ...(i)$$

and B starts with a wrong value of q and gets the roots as 2 and -9. But this time p is correct.

i.e. Sum of roots

$$= p = \alpha + \beta = -9 + 2 = -7 \quad ...(ii)$$

We know that,

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$
$$= (-7)^2 - 4 \times 12 = 49 - 48 = 1$$

[from Eqs. (i) and (ii)]

$$\Rightarrow \quad \alpha - \beta = 1 \quad ...(iii)$$

From Eqs. (ii) and (iii), $\alpha = -3$ and $\beta = -4$

which are correct roots.

90. *(a)* Let one root be α, then other roots is α^2

Given equation is $x^2 - bx + c = 0$

$\therefore$ Sum of roots $= \alpha + \alpha^2 = -\dfrac{(-b)}{1}$

$$\Rightarrow \quad \alpha(\alpha + 1) = b \quad ...(i)$$

and product of roots $= \alpha \cdot \alpha^2 = \dfrac{c}{1}$

$$\Rightarrow \quad \alpha^3 = c \Rightarrow \alpha = c^{1/3} \quad ...(ii)$$

From Eqs. (i) and (ii),

$$c^{1/3}(c^{1/3} + 1) = b \quad ...(iii)$$

On cubing both sides, we get

$$c(c^{1/3} + 1)^3 = b^3$$
$$\Rightarrow c\{c + 1 + 3c^{1/3}(c^{1/3} + 1)\} = b^3$$
$$\Rightarrow c(c + 1 + 3b) = b^3 \text{ [from Eq. (iii)]}$$
$$\Rightarrow b^3 = 3bc + c^2 + c$$

91. *(a)* Let α and β be the roots of the given quadratic equation $2x^2 - 11x + 5 = 0$.

$$\therefore \quad \alpha + \beta = -\frac{(-11)}{2} = \frac{11}{2} \quad ...(i)$$

$$\text{and} \quad \alpha \cdot \beta = \frac{5}{2} \quad ...(ii)$$

Now, $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$$= \left(\frac{11}{2}\right)^2 - 4\left(\frac{5}{2}\right)$$
$$= \frac{121}{4} - \frac{40}{4} = \frac{81}{4} = \left(\frac{9}{2}\right)^2$$

$\therefore$ Difference of roots $= (\alpha - \beta) = \dfrac{9}{2} = 4.5$

92. *(b)* Let the two numbers be x and y.

Now, by given condition,

$$x^2 + y^2 = 97 \quad ...(i)$$
$$\text{and} \quad (x - y)^2 = 25 \quad ...(ii)$$
$$\Rightarrow (x^2 + y^2) - 2xy = 25$$
$$\Rightarrow \quad 97 - 2xy = 25 \text{ [from Eq. (i)]}$$
$$\Rightarrow \quad 2xy = 72$$
$$\therefore \quad xy = 36$$

93. *(a)* Given that, $x + \dfrac{1}{x} = 2 \quad ...(i)$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = 4 \Rightarrow x^2 + \frac{1}{x^2} + 2 = 4$$
$$\Rightarrow \quad x^2 + \frac{1}{x^2} = 2 \quad ...(ii)$$

Now, $\left(x - \dfrac{1}{x}\right)^2 = \left(x^2 + \dfrac{1}{x^2}\right) - 2$

$$= 2 - 2 = 0 \text{ [from Eq. (ii)]}$$
$$\therefore \quad x - \frac{1}{x} = 0$$

94. *(b)* Let the number of columns be x.

Then, number of rows $= x + 4$

According to the question,

$$x(x + 4) \times 5 - 2 = 158$$
$$\Rightarrow \quad 5x(x + 4) = 160 \Rightarrow x(x + 4) = 32$$
$$\Rightarrow \quad x^2 + 4x - 32 = 0$$
$$\Rightarrow \quad x^2 + 8x - 4x - 32 = 0$$
$$\Rightarrow x(x + 8) - 4(x + 8) = 0$$
$$\Rightarrow \quad (x + 8)(x - 4) = 0$$

So, $x = 4$ as $x = -8$ is not possible.

$\therefore$ Number of rows

$$= x + 4 = 4 + 4 = 8$$

95. *(c)* Given, $x^2 + \dfrac{1}{x^2} + 8\left(x + \dfrac{1}{x}\right) + 14$

Let $x + \dfrac{1}{x} = y \quad ...(i)$

On squaring both sides, we get

$$\therefore \quad \left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = y^2$$
$$\Rightarrow \quad x^2 + \frac{1}{x^2} = y^2 - 2 \quad ...(ii)$$

Now, on putting values from Eqs. (i) and (ii) in the given equation, we get

$$y^2 - 2 + 8y + 14 = y^2 + 8y + 12$$
$$= y^2 + 2y + 6y + 12$$
$$= y(y + 2) + 6(y + 2)$$
$$= (y + 6)(y + 2)$$
$$= \left(x + \frac{1}{x} + 6\right)\left(x + \frac{1}{x} + 2\right)$$

So, one factor is $\left(x + \dfrac{1}{x} + 6\right)$.

96. *(b)* Given, $x^2 - x - 1 = 0$

Here, $a = 1$, $b = -1$ and $c = -1$

$\because$ α and β are the roots of the equation

$$x^2 - x - 1 = 0$$

$\therefore$ $\alpha + \beta = \dfrac{-b}{a} = 1$ and $\alpha\beta = \dfrac{c}{a} = -1$

$$\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha - \beta)^2(\alpha + \beta)}$$
$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{[(\alpha + \beta)^2 - 4\alpha\beta](\alpha + \beta)}$$
$$= \frac{(1)^2 - 2 \times -1}{[(1)^2 - 4 \times -1] \times 1} = \frac{3}{5}$$

97. *(d)* Given,

$$x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots \infty}}}$$

So, $x^2 = 6 + \sqrt{x^2} \Rightarrow x^2 = 6 + x$

$$\Rightarrow \quad x^2 - x - 6 = 0$$
$$\Rightarrow \quad x^2 + 2x - 3x - 6 = 0$$
$$\Rightarrow \quad x(x + 2) - 3(x + 2) = 0$$
$$\Rightarrow \quad (x - 3)(x + 2) = 0$$
$$\therefore \quad x = 3, -2$$

98. *(d)* Given, $ax^2 + bx + b = 0$

$$\Rightarrow \quad x^2 + \frac{b}{a}x + \frac{b}{a} = 0$$

$\therefore$ Sum of roots, $\alpha + \beta = \dfrac{-b}{a}$

and products of roots, $\alpha\beta = \dfrac{b}{a}$

Hence, product of roots is not equal to the sum of roots,

So, statement I is not correct.

Now, for roots to be real and unequal, $D > 0$

$$\Rightarrow \quad b^2 - 4ac > 0$$
$$\Rightarrow \quad b^2 - 4a(b) > 0 \quad [\because c = b]$$
$$\Rightarrow \quad b^2 - 4ab > 0 \Rightarrow b^2 > 4ab$$
$$\therefore \quad b > 4a$$

So, if $b > 4a$, then roots are unequal and real, so statement II is not always true, it depends on values of a and b.

99. *(d)* Given, $Ax^2 + Bx + C = 0 \quad ...(i)$

Since, the given roots are -1 and 1.

$\therefore$ Sum of roots $= -1 + 1 = 0$

and product of roots $= 1 \times (-1) = -1$

$\Rightarrow$ quadratic equation is $x^2 -$ (sum of roots) $x +$ product of roots $= 0$

$$\Rightarrow x^2 - 0 \cdot x - 1 = 0 \Rightarrow x^2 - 1 = 0$$

On comparing with above equation from Eq. (i), we get

$$A = 1 \text{ and } C = -1$$

So, A and C are of opposite sign.

100. *(d)* Given, $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$

Since, the given roots are equal.

$\therefore$ $D = 0$

i.e. $[2(b^2 - ac)]^2 - 4(a^2 - bc)(c^2 - ab) = 0$

$\Rightarrow\ 4(b^4 + a^2c^2 - 2ab^2c)$
$-4(a^2c^2 - bc^3 - a^3b + ab^2c) = 0$

$\Rightarrow\ 4b^4 + 4a^2c^2 - 8ab^2c - 4a^2c^2$
$+ 4bc^3 + 4a^3b - 4ab^2c = 0$

$\Rightarrow\ 4b^4 - 12ab^2c + 4bc^3 + 4a^3b = 0$

$\Rightarrow\ b^3 + c^3 + a^3 - 3abc = 0$

$\therefore\ a^3 + b^3 + c^3 = 3abc$

101. *(a)* Given m and n are the roots of the given equation $ax^2 + bx + c = 0$

$\therefore$ Sum of roots $= m + n = -b/a$ …(i)

and product of roots $= mn = \frac{c}{a}$ …(ii)

Now, $\frac{m^2+1}{m} + \frac{n^2+1}{n}$

$= \frac{m^2n + n + mn^2 + m}{mn}$

$= \frac{mn(m+n) + (m+n)}{mn}$

$= \frac{(m+n)(mn+1)}{mn}$

$= \frac{-\frac{b}{a}\left(\frac{c}{a}+1\right)}{\frac{c}{a}} = \frac{-b(a+c)}{ac}$

[using Eqs. (i) and (ii)]

and $\frac{m^2+1}{m} \times \frac{n^2+1}{n}$

$= \frac{(m^2+1)(n^2+1)}{mn}$

$= \frac{m^2n^2 + n^2 + m^2 + 1}{mn}$

$= \frac{(mn)^2 + (m+n)^2 - 2mn + 1}{mn}$

$= \frac{\left(\frac{c}{a}\right)^2 + \left(\frac{-b}{a}\right)^2 - 2\left(\frac{c}{a}\right) + 1}{\frac{c}{a}}$

$= \frac{c^2 + b^2 - 2ac + a^2}{ac}$

$= \frac{b^2 + (a-c)^2}{ac}$

We know that, quadratic equation can be written as

$x^2 -$ (Sum of roots) $x +$ Product of roots $= 0$

$\Rightarrow\ x^2 - \left(\frac{-b(a+c)}{ac}\right)x$
$+ \left(\frac{b^2 + (a-c)^2}{ac}\right) = 0$

$\Rightarrow\ acx^2 + b(a+c)x + b^2 + (a-c)^2 = 0$

$\Rightarrow\ acx^2 + (ab+bc)x + b^2 + (a-c)^2 = 0$

102. *(c)* When mistake is done in first degree term, then the roots of the equation are -9 and -1.

$\therefore$ Equation is $(x+1)(x+9)$

$= x^2 + 10x + 9$ ...(i)

When mistake is done in constant term, then the roots of equation are 8 and 2.

$\therefore$ Equation is $(x-2)(x-8)$

$= x^2 - 10x + 16$ ...(ii)

From Eqs. (i) and (ii), we get

$x^2 - 10x + 9 = 0$ is the correct equation.

103. *(c)* Given equation,

$7(x + 2a)^2 + 3a^2 = 5a\ (7x + 23a)$

$\Rightarrow\ 7(x^2 + 4a^2 + 4ax) + 3a^2$
$= 35ax + 115a^2$

$\Rightarrow\ 7x^2 - 7ax - 84a^2 = 0$

$\Rightarrow\ x^2 - ax - 12a^2 = 0$

$\Rightarrow\ (x+3a)(x-4a) = 0$

$\Rightarrow\ x = -3a$ and $x = 4a$

Since, m and n are the roots of the given equation.

Let $m = 4a$ and $n = -3a$ ($\because m > n$)

$\therefore\ 3m - n = 3(4a) - (-3a)$
$= 12a + 3a = 15a$

104. *(a)* Given equation is $px^2 + qx + r = 0$.

Let one root of the equation be α.

Then, other root $= 3\alpha$

$\therefore$ Sum of roots $= \alpha + 3\alpha = -\frac{q}{p}$

$\Rightarrow\ 4\alpha = \frac{-q}{p} \Rightarrow \alpha = \frac{-q}{4p}$ ...(i)

and product of roots $= (\alpha)\cdot(3\alpha) = \frac{r}{p}$

$\Rightarrow\ 3\alpha^2 = \frac{r}{p} \Rightarrow 3\left(\frac{-q}{4p}\right)^2 = \frac{r}{p}$

[from Eq. (i)]

$\Rightarrow\ \frac{3q^2}{16p^2} = \frac{r}{p} \Rightarrow 3q^2p = 16p^2r$

$\Rightarrow\ 3q^2 = 16pr$

105. *(b)* I. Given, m and n are the roots of the equation

$x^2 + ax + b = 0$.

$\therefore$ Sum of roots, $m + n = -a$ ...(i)

and product of roots, $mn = b$ ...(ii)

Also, given m^2 and n^2 are the roots of the equation

$x^2 - cx + d = 0$.

$\therefore\ m^2 + n^2 = c$ ...(iii)

and $m^2n^2 = d$ ...(iv)

On squaring both sides of Eq. (i), we get

$m^2 + n^2 + 2mn = a^2$

[from Eqs. (i) and (ii)]

$\Rightarrow\ c + 2b = a^2 \Rightarrow c = a^2 - 2b$

$\Rightarrow\ 2b - a^2 = -c$

Hence, statement I is not correct.

II. From Eq. (ii), $m^2n^2 = b^2$

$\Rightarrow\ b^2 = d$

Hence, statement II is correct.

106. *(a)* If a is positive and $b^2 - 4ac \le 0$, then the sign of quadratic polynomial $ax^2 + bx + c$ is always positive.

107. *(c)* We have, $x + \frac{100}{x} > 50$

and $1 \le x \le 100 \Rightarrow x^2 - 50x + 100 > 0$

$\Rightarrow\ x = \frac{50 \pm \sqrt{2500 - 400}}{2}$

$= \frac{50 \pm 10\sqrt{21}}{2}$

$x < 25 - 5\sqrt{21}$ or $x > 25 + 5\sqrt{21}$

$\Rightarrow\ x < 2.087$ or $x > 47.91$

$\Rightarrow x = 1$ and 2 or $x = 48, 49, \ldots, 100$

$\therefore$ Number of total values of

$x = 2 + 53 = 55$

108. *(c)* We have, $x^2 - 4x - \log_{10} N = 0$

We know that, roots are real, if $D \ge 0$

$\Rightarrow\ 16 - 4(-\log_{10} N) \ge 0$

$\Rightarrow\ 16 + 4\log_{10} N \ge 0$

$\Rightarrow\ \log_{10} N \ge -4$

$\Rightarrow\ N \ge 10^{-4} \Rightarrow N \ge \frac{1}{10000}$

Minimum value of $N = \frac{1}{10000}$

109. *(c)* Let $x = \sqrt{4 + \sqrt{4 - \sqrt{4 + \ldots}}}$

$\therefore\ x = \sqrt{4 + \sqrt{4 - x}}$

$\Rightarrow\ x^2 = 4 + \sqrt{4 - x}$

$\Rightarrow\ (x^2 - 4)^2 = 4 - x$

$\Rightarrow\ (x^2 - 4)^2 + x = 4$ …(i)

For option (c), $x = \frac{\sqrt{13}+1}{2}$

From Eq. (i),

$\text{LHS} = \left[\left(\frac{\sqrt{13}+1}{2}\right)^2 - 4\right]^2 + \frac{\sqrt{13}+1}{2}$

$= \left[\frac{13 + 1 + 2\sqrt{13} - 16}{4}\right]^2 + \frac{\sqrt{13}+1}{2}$

$= 4 = \text{RHS}$

110. *(c)* We have, $-2 \le x \le 1$ …(i)

$-1 \le y \le 2$

$\because\ -2 \le -y \le 1$ …(ii)

and $3 \le z \le 6$

$6 \le 2z \le 12$ …(iii)

On adding Eqs. (i), (ii) and (iii), we get

$-2 - 2 + 6 \le x - y + 2z \le 1 + 1 + 12$

$2 \le k \le 14$

111. *(a)* Let α and β be the roots of equation $ax^2 + bx + c = 0$.

Then, $\alpha + \beta = \frac{-b}{a}$

and $\alpha\beta = \frac{c}{a}$

Now, $\alpha + \beta = \left(\frac{1}{\alpha}\right)^2 + \left(\frac{1}{\beta}\right)^2$

$\Rightarrow \alpha + \beta = \frac{\beta^2 + \alpha^2}{(\alpha\beta)^2}$

$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$

$\Rightarrow \frac{-b}{a} = \frac{b^2 / a^2 - 2c / a}{c^2 / a^2}$

$\Rightarrow ab^2 - 2a^2c = -bc^2$

$\Rightarrow ab^2 + bc^2 = 2a^2c$

112. *(d)* Here, $a = 1, b = -p, c = q$

and $a' = 1, b' = q, c' = -p$

Applying common root condition,

$(a'c - ac')^2 = (bc' - b'c)(ab' - a'b)$

$\Rightarrow (q + p)^2 = (p^2 - q^2)(q + p)$

$\Rightarrow (p + q) = (p - q)(p + q)$

$\Rightarrow p - q = 1$

$\Rightarrow p - q - 1 = 0$

113. *(c)* Let α and α^2 be the roots of equation

$x^2 + px + q = 0$.

Then, $\alpha \cdot \alpha^2 = q$

$\therefore \alpha = (q)^{1/3}$ ...(i)

and $\alpha + \alpha^2 = -p$

$\therefore (q)^{1/3} + (q)^{2/3} = -p$ ...(ii)

On cubing both sides, we get

$q + q^2 + 3q(q^{1/3} + q^{2/3}) = -p^3$

$\Rightarrow q + q^2 + 3q(-p) = -p^3$

[from Eq. (ii)]

$\Rightarrow q + q^2 + p^3 = 3pq$

$\Rightarrow p^3 + q^2 + q = 3pq$

114. *(a)* Consider the line,

$2x + 6y = 21$ ...(i)

The point (0, 0) does not satisfy Eq. (i) but the point (0,0) satisfy the equation $2x + 6y \le 21$.

Now, consider the line,

$5x - 2y = 10$ ... (ii)

The point (0, 0) does not satisfy

Eq. (ii) but the point (0,0) satisfy the equation $5x - 2y \le 10$.

115. *(b)* We have, $A(x) = x^2 + px + q$

$\because (x - m)$ and $(x - km)$ are factors of $A(x)$, then m and km are roots of $A(x) = 0$

$\therefore m + km = -p \Rightarrow m(k + 1) = -p$

$\Rightarrow m = \frac{-p}{(k+1)}$ ...(i)

and $m \cdot km = q \Rightarrow m^2k = q$

$\Rightarrow \frac{p^2}{(k+1)^2} \cdot k = q$ [from Eq. (i)]

$\Rightarrow (k + 1)^2 q = kp^2$

116. *(c)* Given points are $P(5, -1)$, $Q(3, -2)$ and $R(1, 1)$.

If points P, Q and R lie in the solution of inequations $x + y \le 4$ and $x - y \ge 2$, then points P, Q and R satisfy these inequations.

For point $P(5, -1)$, $5 - 1 \le 4$ [true]

$5 + 1 \ge 2$ [true]

For point $Q(3, -2)$, $3 - 2 \le 4$ [true]

$3 + 2 \ge 2$ [true]

and for point $R(1, 1)$, $1 + 1 \le 4$ [true]

$1 - 1 \ge 2$ [false]

So, the point $R(1, 1)$ does not satisfy the inequations.

Hence, points P and Q lie in the solution of inequations.

117. *(a)* Given, $x = \frac{\sqrt{a + 2b} + \sqrt{a - 2b}}{\sqrt{a + 2b} - \sqrt{a - 2b}}$

By rationalising, we get

$$x = \frac{(\sqrt{a + 2b})^2 + (\sqrt{a - 2b})^2 + 2\sqrt{(a + 2b)(a - 2b)}}{(\sqrt{a + 2b})^2 - (\sqrt{a - 2b})^2}$$

$\Rightarrow x = \frac{a + 2b + a - 2b + 2\sqrt{a^2 - 4b^2}}{a + 2b - a + 2b}$

$\Rightarrow x = \frac{2a + 2\sqrt{a^2 - 4b^2}}{4b}$

$\Rightarrow x = \frac{a + \sqrt{a^2 - 4b^2}}{2b}$

$\Rightarrow bx^2 - ax + b = 0$

118. *(a)* Given, $\sqrt{3x^2 - 7x - 30}$

$= (x - 5) + \sqrt{2x^2 - 7x - 5}$

On squaring both sides, we get

$3x^2 - 7x - 30 = (x - 5)^2 + (2x^2 - 7x - 5) + 2(x - 5)\sqrt{2x^2 - 7x - 5}$

$\Rightarrow 3x^2 - 7x - 30 = x^2 + 25 - 10x + 2x^2 - 7x - 5 + (2x - 10)\sqrt{2x^2 - 7x - 5}$

$\Rightarrow 3x^2 - 7x - 30 - x^2 + 10x + 7x - 2x^2 - 20 = (2x - 10)\sqrt{2x^2 - 7x - 5}$

$\Rightarrow (10x - 50) = (2x - 10)\sqrt{2x^2 - 7x - 5}$

$\Rightarrow 10(x - 5) = 2(x - 5)\sqrt{2x^2 - 7x - 5}$

$\Rightarrow \sqrt{2x^2 - 7x - 5} = 5$ ...(i)

Again, on squaring both sides of Eq. (i), we get

$2x^2 - 7x - 5 = 25$

$\Rightarrow 2x^2 - 7x - 30 = 0$ ...(ii)

If α and β are roots of Eq. (ii), then

$\alpha\beta = \frac{-30}{2} = -15$

119. *(a)* Since, the roots of equation $lx^2 + mx + m = 0$ are in the ratio $p : q$.

Then, we can take roots as pk and qk.

$\therefore$ Sum of roots $= pk + qk = -\frac{m}{l}$

$\Rightarrow (p + q)k = -\frac{m}{l}$ ...(i)

and product of roots

$= (pq)k^2 = \frac{m}{l}$ ...(ii)

On dividing Eq. (i) by Eq. (ii), we get

$\frac{(p + q)k}{(pq)k^2} = -1$

$\Rightarrow \left(\frac{p + q}{pq}\right) = -k = \frac{m / l}{(p + q)}$

[using Eq. (i)]

$\Rightarrow \frac{(p + q)^2}{pq} = \frac{m}{l} \Rightarrow \frac{p + q}{\sqrt{pq}} = \pm\sqrt{\frac{m}{l}}$

$\Rightarrow \frac{p + q}{\sqrt{pq}} = -\sqrt{\frac{m}{l}}$

$\Rightarrow \sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}} = 0$

120. *(d)* We have, $1 + \frac{1}{x} - \frac{1}{x^2} \ge 0$

$\Rightarrow \frac{x^2 + x - 1}{x^2} \ge 0$

As, $x^2 \ge 0$, then $x^2 + x - 1 \ge 0$

If $x^2 + x - 1 = 0$,

then $x = \frac{-1 \pm \sqrt{5}}{2}$

$\therefore \left[x - \left(\frac{-1 + \sqrt{5}}{2}\right)\right]\left[x - \left(\frac{-1 - \sqrt{5}}{2}\right)\right] \ge 0$

$\Rightarrow x \le \frac{-1 - \sqrt{5}}{2}$ or $x \ge \frac{-1 + \sqrt{5}}{2}$

19

SET THEORY

Usually (1-2) questions have been asked from this chapter. Generally questions are asked from these topics more results on operations on set.

SET

A well defined collection of objects, is called a set. The objects in a set are called its members or elements. Sets are usually denoted by the capital letters A, B, C, X, Y and Z etc. And the elements of a set are denoted by small letters a, b, c etc. If x is an element of set A, we can write $x \in A$, which means that 'x belongs to A' or that x is an element of A.

If x does not belong to A, we can write, $x \notin A$. e.g.

- The collection of vowels in English alphabet is a set A containing five elements namely a, e, i, o and u, where $a \in A$ but $b \notin A$.
- The collection of first four prime numbers is a set A containing the elements 2, 3, 5 and 7, where $3 \in A$ but $1 \notin A$.

Representation of Sets

Sets are generally represented by following two ways:

1. **Roster or Tabular form or listing method** In this method, all the elements of a set are listed, within curly braces { } being separated by commas. e.g.
   (i) If A is a set of first eight prime numbers, then $A = \{2, 3, 5, 7, 11, 13, 17, 19\}$
   (ii) If B is a set of squares of first five natural numbers, then $B = \{1, 4, 9, 16, 25\}$
   (iii) If A is a set of vowels of English alphabets, then $A = \{a, e, i, o, u\}$

Note The order in which the elements are written in a set makes no difference and also the repeated elements are taken only once each.

2. **Set Builder form or Rule method** In this method, instead of listing all the elements of a set, we write the set by some special property or properties satisfied by all its elements and write it as.
   (i) The set B of all even natural numbers can be written as
   $B = \{x : x$ is a natural number and $x = 2n$ for $n \in N\}$ or $B = \{x : x \in N, x = 2n, n \in N\}$ and neads it as 'B' is the set consisting of all elements x such that x has the property of even natural number. The symbol ':' or '1' stands or such that
   (ii) The set $A = \{3, 5, 7, 9, 11\}$, then it is represented as
   $$A = \{x : x = 2n + 1 \text{ where } n \in N, n < 6\}$$

(iii) The set $A = \{0, 1, 4, 9, 16, \dots\}$ can be written as $A = \{x : x = n^2, n \in Z\}$, where Z is the set of integers.

Types of Sets

1. **Empty set** A set which does not contain any element is called an empty set or null set or void set. It is denoted by ϕ or { }.

   e.g. A = set of all odd numbers divisible by 2

   and $B = \{x : x \in N \text{ and } 5 < x < 6\}$

   The sets which have atleast one element are called non void or non-empty set.

2. **Singleton set** A set consisting of a single element is called a singleton set. e.g.
   (i) The set {5} is a singleton set.
   (ii) $\{x \mid x \in W \text{ and } x + 6 = 6\} = \{0\}$ which is a singleton set.

3. **Finite set** A set which consists of a definite number of elements, is called a finite set. Empty set is also *a* finite set. e.g.
   (i) The set {1, 2, 3, 4} is a finite set, because it contains a definite number of elements i.e. only 4 elements.
   (ii) B = Set of vowels in English Alphabets $= \{a, e, i, o, u\}$;
   (iii) A = Set of even prime natural numbers.
   $\Rightarrow \quad A = \{2\}$

Note The number of distinct elements contained in a set A is called the cardinal number of A and is denoted as $n(A)$. If $A = \{a, e, i, o, u\}$, then $n(A) = 5$.

4. **Infinite set** A set which consists of infinite number of elements is called an infinite set. It is represented by writing a few elements of the set followed by e.g.
   (i) Set of square of natural numbers is an infinite set, because natural numbers are infinite and it can be represented as {1, 4, 9, 16, 25, ...}.
   (ii) Set of all points in a plane.

5. **Equal sets** Two sets A and B are said to be equal, if they have exactly the same elements and we write $A = B$. Otherwise, two sets are said to be unequal and we write $A \neq B$.

   e.g. Let $A = \{1, 2, 3, 4\}$ and $B = \{4, 3, 1, 2\}$, then $A = B$ because each element of A is in B and *vice-versa*.

6. **Equivalent sets** Two sets A and B are equivalent, if their cardinal numbers are same i.e. $n(A) = n(B)$ and we write $A \leftrightarrow B$ or $A \sim B$

   e.g. Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$,
   then $n(A) = n(B)$
   $A \leftrightarrow B$ or $A \sim B$

7. **Subsets** Let A and B be two sets. If every element of A is an element of B, then A is called a subset of B.

   If A is subset of B, then we can write $A \subseteq B$ which is read as 'A is a subset of B' or 'A is contained in B'.
   (i) Every set is a subset of itself i.e. $A \subseteq A$, $B \subseteq B$.
   (ii) Empty set is a subset of every set i.e. $\phi \subseteq A$.

   e.g. Let $A = \{2, 4, 6\}$ and $B = \{6, 4, 2, 8\}$.

   Then, $A \subseteq B$ but $B \not\subseteq A$ i.e. A is a subset of B but B is not a subset of A.

Note The total number of subsets of a finite set containing n elements is 2^n.

8. **Super set** If A is a subset of B, then we say that B is superset of A and we write $B \supseteq A$.

   e.g. If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$,
   then, $B \supseteq A$.

9. **Comparability of sets** Two sets A and B are said to be comparable, if either $A \subset B$ or $B \subset A$ or $A = B$, otherwise A and B are said to be incomparable.

   e.g. Let $A = \{1, 2, 3\}$, $B = \{1, 2, 4, 6\}$ and $C = \{1, 2, 4\}$
   Since, $A \not\subseteq B$ and $B \not\subseteq A$.
   So, A and B are incomparable but $C \subset B$ and so B and C are comparable sets.

10. **Proper subset** If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B and we write $A \subset B$.

    e.g. If $A = \{1, 2, 3, 4, \dots\}$
    and $B = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$,
    then $A \subset B$.

Note The total number of proper subset of a finite set containing n elements is $(2^n - 1)$.

11. **Universal set** If there are some sets under consideration, then there happens to be a set which is a superset of each one of the given sets. Such a set is known as the universal set and it is denoted by U.

    e.g.
    - If $A = \{1, 2, 3\}$, $B = \{2, 3, 4, 5, 7\}$ and $C = \{2, 4, 6, 8\}$, then $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ is a universal set for A, B and C.
    - For the set of all integers, the universal set can be the set of rational numbers or the set of real numbers.

12. **Power set** The collection of all subsets of a set A, is called power set of A and it is denoted by $P(A)$. In $P(A)$, every element is a set.

    e.g. Let $A = \{1, 2, 3\}$

    Then, $P(A) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{3,1\}, \{1,2,3\}\}$

Properties of Power Sets

1. Each element of a power set is a set.
2. If set A has n elements, then $P(A)$ has 2^n elements.
3. If A is an empty set ϕ or $\{\ \}$, then $P(A)$ has just one element i.e. $P(A)=\{\phi\}$.
4. If $A \subseteq B$, then $P(A) \subseteq P(B)$.

EXAMPLE 1. The set $S=\{x \in N : x+3=3\}$ is a

a. null set
b. singleton set
c. infinite set
d. None of these

Sol. a. Given, $S=\{x \in N : x+3=3\}$

$\therefore \quad S=\{\}$

So, S is a null set.

EXAMPLE 2. If $A=\{x : x$ is an odd integer$\}$ and $B=\{x : x^2-8x+15=0\}$. Then, which one of the following is correct. **2013 I**

a. $A=B$
b. $A \subseteq B$
c. $B \subseteq A$
d. $A \subseteq B^C$

Sol. c. Given that, $A=\{x : x$ is an odd integer$\}$

and $B=\{x : x^2-8x+15=0\}=\{x : x^2-5x-3x+15=0\}$

$=\{x : x(x-5)-3(x-5)=0\}$

$=\{x : (x-5)(x-3)=0\}=\{3, 5\}$

Since, B has two odd elements.

$\therefore \quad B \subseteq A$

EXAMPLE 3. If $A=\{a, b\}$, then power set of A is

a. $\{\phi, a, b\}$
b. $\{\phi, a, b, A\}$
c. $\{\phi, a, b, ab\}$
d. $\{\phi, \{a\}, \{b\}, A\}$

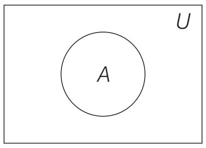
Sol. d. $A=\{a, b\}$

The subsets of A are ϕ, $\{a\}$, $\{b\}$ $\{a, b\}$.

$\therefore \quad P(A)=\{\phi, \{a\}, \{b\}, \{a, b\}\}=\{\phi, \{a\}, \{b\}, A\}$

VENN DIAGRAM

'Venn diagram' is the diagrammetic representation of various types of sets and operations on sets.

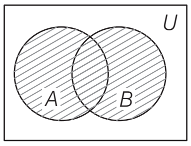


The universal set is usually represented by a rectangular region and other subsets of the universal set are represented by circles inscribed in it. Each inscribed circle represents a set (subset of universal set).

Operations on Sets

1. **Union of Sets** Let A and B be any two sets. The union of A and B is the set of all the elements of A and all the elements of B, the common elements being taken only once. It is denoted by $A \cup B$ and read as 'A union B.'

Thus, $A \cup B=\{x : x \in A$ or $x \in B\}$



In the given Venn diagram $A \cup B$ is denoted by the shaded region.

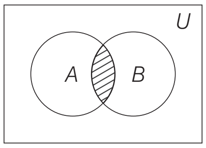
e.g. If $A=\{1, 2, 3, 4\}$

and $B=\{1, 2, 3, 5, 7\}$, then $A \cup B=\{1, 2, 3, 4, 5, 7\}$

Note • $x \in A \cup B \Leftrightarrow x \in A$ or $x \in B$
• $x \notin A \cup B \Leftrightarrow x \notin A$ and $x \notin B$ • $A \subseteq A \cup B, B \subseteq A \cup B$.

2. **Intersection of sets** Let A and B be any two sets. The intersection of A and B is the set of all those elements which belongs to both A and B. It is denoted by $A \cap B$ and read as A intersection B.

Thus, $A \cap B=\{x : x \in A$ and $x \in B\}$

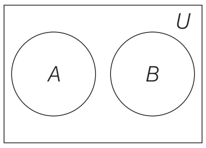


$A \cap B$ is represented by the shaded region in the given Venn diagram

e.g. If $A=\{1, 2, 3, 4\}$ and $B=\{1, 3, 7, 9\}$, then $A \cap B=\{1, 3\}$.

Note • $x \in A \cap B \Leftrightarrow x \in A$ and $x \in B$
• $x \notin A \cap B \Leftrightarrow x \notin A$ or $x \notin B$ • $A \cap B \subseteq A$ and $A \cap B \subseteq B$

3. **Disjoint sets** Two sets A and B are said to be disjoint, if they have no common element i.e. $A \cap B=\phi$

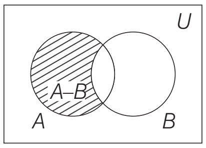


e.g. If $A=\{1, 2, 4\}$, $B=\{3, 5, 6\}$, then $A \cap B=\phi$

So, A and B are disjoint sets.

4. **Difference of sets** Let A and B be two sets. The difference of two sets is the set of all those elements of A which do not belong to B. It is denoted by $A-B$ and read as A minus B.

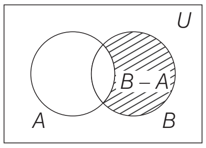
$\therefore \quad A-B=\{x : x \in A$ and $x \notin B\}$



The shaded part in the given Venn diagram represents $A-B$.

Similarly, the difference $B-A$ is the set of all those elements of B that do not belong to A.

$\therefore \quad B-A=\{x : x \in B$ and $x \notin A\}$

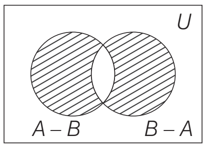


The shaded part in the given Venn diagram represents $B-A$.

e.g.

- If $A=\{2, 4, 6, 8, 10\}$ and $B=\{2, 4, 6, 12, 14\}$, then $A-B=\{8, 10\}$ and $B-A=\{12, 14\}$.
- If $A=\{1, 2, 3\}$ and $B=\{1, 2, 3, 4\}$, then $A-B=\phi$.

Symmetric Difference of Two Sets



Let A and B be two sets. The symmetric difference of sets A and B is the set $(A - B) \cup (B - A)$ and is denoted by $A \Delta B$ i.e. it is a set consisting of all those members of A which are not in B or all those which are in B but not in A.

Thus, $A \Delta B = \{x : x \in A \text{ but } x \notin B\} \cup \{x : x \in B \text{ but } x \notin A\}$

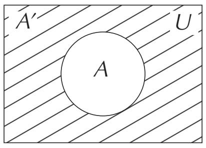
$A \Delta B = (A - B) \cup (B - A)$

The shaded part in given Venn diagram represents $A \Delta B$.

e.g. If $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $B = \{1, 3, 5, 6, 7, 8, 9\}$,

$\therefore$ $A - B = \{2, 4\}$ and $B - A = \{9\}$, then $A \Delta B = \{2, 4, 9\}$.

Complement of a Set



Let U be the universal set and A be any set such that $A \subset U$. Then, the complement of A with respect to U is the set of all those elements of U which are not in A. It is denoted by A' or A^C or $U - A$

Thus, $A' = \{x : x \in U \text{ and } x \notin A\}$

The shaded portion in the above Venn diagram shows the complement of set A. If A is a subset of the universal set U, then its complement A' is also a subset of U. e.g.

- Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, \dots\}$and $A = \{1, 3, 5, 7, \dots\}$.
  Then, the complement of A is
  $A' = U - A = \{2, 4, 6, 8, \dots\}$
- If $U = \{a, b, c, d, e, f\}$ and $A = \{a, e\}$,
  then $A' = \{b, c, d, f\}$.

EXAMPLE 4. If $A = \{0, 1 2, 3, 4\}$, $B = \{1, 2, 3\}$, $C = \{3, 5, 7, 9\}$, then $(A - B) \cap (B - C)$ is

a. $\{1, 2\}$ **b.** $\{0, 4\}$ **c.** $\{0, 1, 2, 4\}$ **d.** ϕ

Sol. **d.** Given, $A = \{0, 1, 2, 3, 4\}$, $B = \{1, 2, 3\}$ and $C = \{3, 5, 7, 9\}$

$\therefore$ $A - B = \{0, 4\}$ and $B - C = \{1, 2\}$

$\therefore$ $(A - B) \cap (B - C) = \phi$

Properties of Operations on Sets

1. $(A')' = A$
2. $A \cup A' = U$
3. $A \cap A' = \phi$
4. $\phi' = U$
5. $U' = \phi$
6. $A - B = A \cap B'$
7. $B - A = B \cap A'$
8. $A - B \subseteq A$
9. $B - A \subseteq B$
10. $A - B \neq B - A$
11. $A - \phi = A$ and $A - A = \phi$
12. $A \subseteq B \Rightarrow A - B = \phi$
13. $A - B = B - A \Leftrightarrow A = B$

Note The sets $A - B$, $A \cap B$ and $B - A$ are mutually disjoint sets i.e. the intersection of any of these two sets is an empty set.

Laws of Algebra of Sets

1. **Idempotent laws** For any set A, we have
   (i) $A \cup A = A$ (ii) $A \cap A = A$
2. **Identity Laws** For any set A,
   (i) $A \cup \phi = A$ (ii) $A \cap U = A$
   i.e. ϕ and U are identity elements for union and intersection, respectively.
3. **Commutative laws** For any two sets A and B, we have
   (i) $A \cup B = B \cup A$ (ii) $A \cap B = B \cap A$
4. **Associative laws** If A, B and C are any three sets, then
   (i) $(A \cup B) \cup C = A \cup (B \cup C)$
   (ii) $A \cap (B \cap C) = (A \cap B) \cap C$
5. **Distributive laws** If A, B and C are any three sets, then
   (i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
   (ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
6. **De-Morgan's laws** If A and B are any two sets, then
   (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$

More Results on Operations on Sets

1. If A, B and C are any three sets, then
   (i) $A - (B \cap C) = (A - B) \cup (A - C)$
   (ii) $A - (B \cup C) = (A - B) \cap (A - C)$
   (iii) $A \cap (B - C) = (A \cap B) - (A \cap C)$
   (iv) $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$
   (v) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

2. If A, B and C are finite sets and U be the finite universal set, then
   (i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
   (ii) $n(A \cup B) = n(A) + n(B) \Leftrightarrow A, B$ are disjoint sets.
   (iii) $n(A - B) = n(A) - n(A \cap B)$
   i.e. $n(A - B) + n(A \cap B) = n(A)$
   (iv) Number of elements which belong to exactly one of A or B
   $= n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$
   (v) If A, B and C are finite sets, then
   $$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$
   (vi) $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$
   (vii) $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

EXAMPLE 5. If a set A contains 60 elements and another set B contains 70 elements and there are 50 elements in common, then how many elements does $A \cup B$ contain?

a. 130 b. 100 c. 80 d. 70

Sol. c. Here, $n(A) = 60$, $n(B) = 70$ and $n(A \cap B) = 50$

Now, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 60 + 70 - 50 = 130 - 50 = 80$

EXAMPLE 6. A group of 40 people who speaks either english or Hindi, out of these 12 speak both English and Hindi and 22 speak Hindi. How many people speak only english not Hindi?

a. 30 b. 10 c. 18 d. 28

Sol. c. Let E be the set of students who speak English and H be the set of students who speak Hindi.

Given, $n(E \cup H) = 40$, $n(H) = 22$ and $n(E \cap H) = 12$

$\therefore \quad n(E \cup H) = n(E) + n(H) - n(E \cap H)$

$40 = n(E) + 22 - 12 \quad \Rightarrow n(E) = 30$

Number of people who speak English only
$= n(E) - n(E \cap H) = 30 - 12 = 18$

Ordered Pairs

An ordered pair consists of two objects or elements in a given fixed order.

If A and B are two sets and $a \in A$, $b \in B$, then the ordered pair of elements a and b is denoted by (a, b).

The natural numbers and their squares can be represented by ordered pair in the following way.

(1, 1), (2, 4), (3, 9), (4, 16), ...

Two ordered pairs (a, b) and (c, d) will be equal, if and only if $a = c$ and $b = d$.

Note $\{a, b\} = \{b, a\}$ but $(a, b) \neq (b, a)$.

PRACTICE EXERCISE

1. If $A = \{5, 6, 7\}$ and $B = \{7, 8, 9\}$, then $A \cup B$ is equal to

(a) {5, 6, 7, 8, 9} (b) {7, 8, 9} (c) {5, 6, 7} (d) ϕ

2. Given that $A = \{2, 6, 8, 9\}$, $B = \{7, 8, 9, 12\}$, then $B - A$ is equal to

(a) {7, 8, 9, 12} (b) {7, 12}
(c) {2, 6, 7, 8, 9, 12} (d) {2, 6, 8, 9, 12}

3. If U is the universal set of all natural numbers and $A = \{1, 2, 3, 4, 5\}$, then compute $A \cap U$.

(a) {1, 2, 3, 4} (b) ϕ (c) {1, 2, 3, 4, 5} (d) U

4. The set $\{2, 4, 16, 256, \ldots\}$ can be represented as which one of the following?

(a) $\{x \in N \mid x = 2^{2^n}, n \in N\}$
(b) $\{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \ldots\}$
(c) $\{x \in N \mid x = 2^{4n}, n = 0, 1, 2, \ldots\}$
(d) $\{x \in N \mid x = 2^{2n}, n = 0, 1, 2, \ldots\}$

5. If P and Q are any two sets and $P \subset Q$, then

(a) $P \cap Q = \phi$ (b) $P' \cap Q = P$ (c) $P \cap Q = P$ (d) $P \cap Q = Q$

6. Which one of the following is a true statement?

(a) $(A - B) \cap (B - A) = \phi$ (b) $(A - B) \cap (B - A) = A$
(c) $(A - B) \cap (B - A) = U$ (d) $(A - B) \cap (B - A) = B$

7. If P and Q are any two sets, then $P \cup Q = P \cap Q$, if

(a) P is the empty set (b) Q is the empty set
(c) Both P and Q are empty sets
(d) P and Q are non-empty sets

8. The number of non-empty proper subsets of set $A = \{2, 5, 7, 10\}$ is

(a) 16 (b) 15 (c) 14 (d) 8

9. Which of the following statements is false for the sets A, B and C, where

$A = \{x \mid x$ is letter of the word 'BOWL'$\}$
$B = \{x \mid x$ is a letter of the word 'ELBOW'$\}$
$C = \{x \mid x$ is a letter of the word 'BELLOW'$\}$

(a) $A \subset B$ (b) $B \supset C$
(c) $B \neq C$ (d) B is a proper subset of C

10. If A and B are two sets, then $A \cap (A \cup B)$ equals

(a) A (b) B
(c) ϕ (d) None of these

11. The smallest set B such that $B \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$ is

(a) {3, 5, 9} (b) {3, 5, 8}
(c) {1, 2, 3} (d) None of these

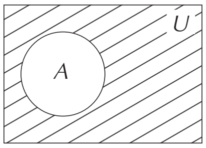
12. If $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $P = \{0, 1, 2, 3\}$, $Q = \{2, 3, 4, 5\}$, and $R = \{4, 5, 6\}$, then $Q' \cap (P \cup R)$ is equal to

(a) {0, 1, 6} (b) {0, 1, 2, 3, 4, 5}
(c) {6, 7, 8, 9} (d) {2, 3}

13. If P is a non-empty set, then $(P')'$ is equal to

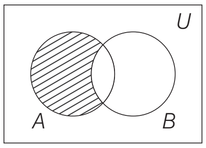
(a) ϕ (b) U (c) $U - P$ (d) P

14. The shaded region in the adjoining diagram is



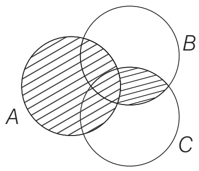
(a) $A \cup A'$ (b) U (c) A' (d) $A \cap A'$

15. The shaded region in the adjoining diagram represents



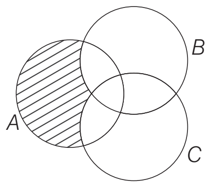
(a) $(A \cap B)'$ (b) $A \cup B$ (c) $A - B$ (d) $A \cap B$

16. The shaded portion in the following Venn diagram represents



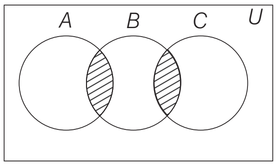
(a) $A \cup (B \cap C)$ (b) $A \cup (B \cup C)$
(c) $(A \cap B) \cap C$ (d) $(A \cap B) \cup C$

17. The shaded region in the adjoining diagram is



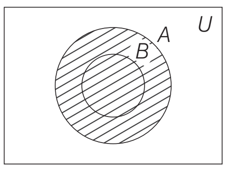
(a) $A \cap (B - C)$ (b) $A - (B \cup C)$
(c) $A \cap (B \cup C)$ (d) $A \cup (B \cap C)$

18. In the Venn diagram below, shaded portion represents

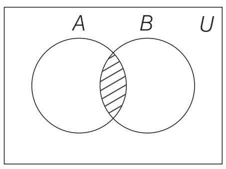


(a) $A \cup B \cup C$ (b) $A \cap B \cap C$
(c) $(A \cap B) \cup (B \cap C)$ (d) $A \cup B \cap C$

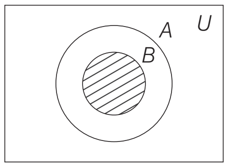
19. The Venn diagram for $A \cup B$ when $B \subset A$ is



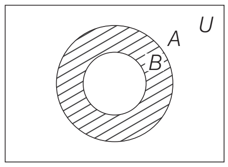
(a)



(b)



(c)



(d)

20. Which one of the following is a correct statement?

(a) $\phi \in \phi$ (b) $\phi \notin P(\phi)$ (c) $\phi = P(\phi)$ (d) $\phi \in P(\phi)$

21. Let $A = \left\{3, \pi, \sqrt{2}, \frac{2}{7}, -5, 3+\sqrt{7}\right\}$. The subset of A containing all the elements from it which are irrational numbers is

(a) $\left\{3, \frac{2}{7}, -5\right\}$ (b) $\left\{3, \pi, \frac{2}{7}, -5, 3+\sqrt{7}\right\}$
(c) $\{\pi, \sqrt{2}, 3+\sqrt{7}\}$ (d) $\{3, -5\}$

22. If $U = \{x : x \in N\}$, $A = \{x : x$ is an odd number$\}$, then A' is equal to

(a) $\{x : x$ is an even number$\}$ (b) $\{x : x$ is an odd number$\}$
(c) $\{x : x$ is a natural number$\}$ (d) $\{x : x$ is an integer$\}$

23. If two sets are disjoint, then their intersection is

(a) null set (b) singleton
(c) a infinite set (d) None of these

24. If A and B are two non-empty sets, then $A - (A - B)$ equals

(a) B (b) $A - B$ (c) $A \cap B$ (d) $A' \cap B'$

25. Let two sets A and B have $2n$ and $4n$ elements respectively, where n is a natural number. What can be the minimum number of elements in $A \cup B$?

(a) $2n$ (b) $3n$ (c) $4n$ (d) $6n$

26. If $P = \{x : x^2 - 3x + 2 = 0\}$, $Q = \{x : x^2 + 4x - 12 = 0\}$, then $P - Q$ is

(a) $\{1, 2\}$ (b) $\{2\}$ (c) $\{1\}$ (d) $\{4, 3\}$

27. If $A = \{(2^{2n} - 3n - 1) \mid n \in N\}$ and $B = \{9(n-1) \mid n \in N\}$, then which one of the following is correct?

(a) $A \subset B$ (b) $A \subset A$ (c) $A = B$
(d) Neither A is a subset of B nor B is a subset of A

28. If A, B and C are any three sets, then

(a) $A - (B \cup C) = (A - B) \cup (A - C)$
(b) $A - (B \cup C) = (A - B) - (A - C)$
(c) $A - (B \cup C) = (A - B) \cap (A - C)$
(d) $A - (B \cup C) = (A \cup B) - (A \cup C)$

29. If P and Q are two sets such that $n(P) = m$, $n(Q) = n$ and $n(P \cap Q) = p$, then $n(P \cup Q)$ is equal to

(a) $m + n$ (b) $m + n + p$ (c) $m + n - p$ (d) $m - n - p$

30. A and B are two sets such that $n(A) = 17$, $n(B) = 23$, $n(A \cup B) = 38$. Then, $n(A \cap B)$ is

(a) 40 (b) 78 (c) 2 (d) None of these

31. Let $A = \{x : x^2 - 6x + 8 = 0\}$ and $B = \{x : 2x^2 + 3x - 2 = 0\}$. Then, which one of the following is correct?
(a) $A \subseteq B$ (b) $B \subseteq A$
(c) Neither $A \subseteq B$ nor $B \subseteq A$ (d) $A = B$

32. If $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 4, 5\}$, $C = \{1, 3, 4, 5, 6, 7\}$, then $A \cap (B \cup C)$ is equal to
(a) $\{1, 2, 3, 4, 5, 6, 7\}$ (b) $\{1, 2, 3, 4\}$
(c) $\{1, 2, 3, 4, 5\}$ (d) ϕ

33. Let the set A and B be given by $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8, 10\}$ and the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, then $(A \cup B)'$ is
(a) $\{2, 4\}$ (b) U
(c) $\{1, 3, 5, 6, 7, 8, 9, 10\}$ (d) $\{5, 7, 9\}$

34. What is $\{[(A \cup B)' \cap A]\} - (A - B)$ equal to?
(a) ϕ (b) A (c) B (d) B'

35. Which one of the following is a correct statement?
I. $\{a\} \in \{\{a\}, \{b\}, \{c\}$ II. $\{a\} \subseteq \{\{a\}, b, c\}$
III. $\{a, b\} \subseteq \{\{a\}, b, c\}$ IV. $a \subseteq \{\{a\}, b, c\}$
(a) Only I (b) Only II (c) III and IV (d) Only IV

36. Which one of the following is an infinite set?
(a) $\{x : x$ is a whole number less than or equal to $1000\}$
(b) $\{x : x$ is a natural number less than $1000\}$
(c) $\{x : x$ is a positive integer less than or equal to $1000\}$
(d) $\{x : x$ is an integer and less than $1000\}$

37. Which one of the following is not correct in respect of the sets A and B?
(a) If $A \subseteq B$, then $B \cup A = B$
(b) If $A \subseteq B$, then $A \cap (A - B) = \phi$
(c) If $A \subseteq B$, then $B \cap A = A$
(d) If $A \cap B = \phi$, then either $A = \phi$ or $B = \phi$

38. If $R \in \overleftrightarrow{PQ}, S \notin \overleftrightarrow{PQ}$ and S does not lie on $\overleftrightarrow{PQ}$ extended, then
(a) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \phi$ (b) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{R\}$
(c) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{S\}$ (d) $\overleftrightarrow{PQ} \cap \overleftrightarrow{RS} = \{P\}$

39. In a group of 1000 people, there are 750 people who can speak Hindi and 400 who can speak English. How many can speak Hindi only?
(a) 600 (b) 150 (c) 300 (d) 500

40. In a committee 50 people speak French, 20 speak Spanish and 10 speak both Spanish and French. How many speak atleast one of these two languages?
(a) 60 (b) 50 (c) 30 (d) 70

41. Every student in a class of 42 students, studies atleast one of the subjects, Mathematics, English and Commerce, 14 students study Mathematics, 20 Commerce and 24 English. 3 students study Mathematics and Commerce, 2 English and Commerce and there is no student who studies all the three subjects. The number of students who study Mathematics but not Commerce is
(a) 4 (b) 3 (c) 12 (d) 11

42. In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% candidates failed in both English and Mathematics, what percentage of candidates passed in both the subjects?
(a) 18% (b) 21% (c) 23% (d) 25%

43. State which of the following statements about sets is/are true?
I. Every subset of a finite set is finite.
II. ϕ is a subset of $\{0\}$.
Select the correct answer using the codes given below
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

44. Which of the following is/are examples of empty set?
I. $A = \{x : x + 3 = 3, x \in I\}$
II. $B = \{x : x$ is a positive even integer and prime$\}$
III. $C = \{x : x^2 = 16, x$ is odd integer$\}$
Select the correct answer using the codes given below
(a) Only I (b) I and III (c) Only III (d) I, II and III

45. Consider the following statements :
I. $A' \cup B = (A \cap B)'$ II. $(\phi')' = \cup$
III. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Which of the statement(s) given above is/are correct?
(a) Only II (b) I and III (c) Only III (d) II and III

46. Let P = Set of all integral multiples of 3
Q = Set of all integral multiples of 4
R = Set of all integral multiples of 6
Consider the following relations
I. $P \cup Q = R$ II. $P \subset R$ III. $R \subset (P \cup Q)$
Which of the relation(s) given above is/are correct?
(a) Only I (b) Only II (c) Only III (d) II and III

47. Consider the following statements:
I. Set of all points of a given line is a infinite set.
II. The set of all birds in a zoo is a infinite set.
III. Good books in a school library is a set.
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II (c) Only III (d) All of these

48. Given that the set $A = \{0, 1, 2, 3\}$, which of the following statements about A are true?
I. A is a finite set.
II. A is a subset of the set of integers.
III. $\{1, 2\}$ is a proper subset of A.
IV. A is the null set.
Select the correct answer using the codes given below
(a) I, II and III (b) I and IV
(c) I and III (d) All of these

49. State which of the sets given below are infinite set?

I. Set of all concentric circles.

II. $\{x : x$ is a multiple of $2, x$ is an integer.$\}$

III. The set of lines which are parallel to X-axis.

IV. The set of positive integers greater than 100.

Select the correct answer using the codes given below

(a) I and II (b) II and III
(c) Only I (d) All of these

50. Which of the following sets are equivalent?

I. $A = \{1, 2, 3, 4, 5\}, B = \{7, 8, 9, 10, 11\}$

II. $A = \{x, y, z\}$ $B = \{p, q\}$

III. $P = \{2, 4, 6, 8\}, R = \{a, b, c, d\}$

IV. $A = \{36, 39, 42, 45\}$, $B = \{42, 39, 45, 36\}$

Select the correct answer using the codes given below

(a) I, II and IV (b) II, III and IV
(c) I, III and IV (d) None of these

Directions (Q. Nos. 51-52) *Answer the questions based on the following information.*

In a survey of 250 students, it was found that 150 play cricket, 100 play basketball and 120 play football, further, 30 of them play both basketball and football, 50 play both cricket and basketball and 60 play both cricket and football.

51. The maximum number of students who play all the three sports

(a) 20 (b) 10 (c) 15 (d) None of these

52. If 5 students play none of the three sports then numbers of students who play at least two sports

(a) 100 (b) 110 (c) 120 (d) 130

PREVIOUS YEARS' QUESTIONS

53. If $A = \{x : x$ is an even natural number$\}$, $B = \{x : x$ is a natural number and multiple of 5$\}$ and $C = \{x : x$ is a natural number and multiple of 10$\}$, then what is the value of $A \cap (B \cup C)$? **2012 I**

(a) $\{10, 20, 30, \ldots\}$ (b) $\{5, 10, 15, 20, \ldots\}$
(c) $\{2, 4, 6, \ldots\}$ (d) $\{20, 40, 60, \ldots\}$

54. Which one of the following is a null set? **2012 II**

(a) $A = \{x$ is a real number : $x > 1$ and $x < 1\}$
(b) $B = \{x : x + 3 = 3\}$
(c) $C = \{\phi\}$
(d) $D = \{x$ is a real number : $x \geq 1$ and $x \leq 1\}$

55. Let $x \in \{2, 3, 4\}$ and $y \in \{4, 6, 9, 10\}$. If A be the set of all order pairs (x, y) such that x is a factor of y. Then, how many elements does the set A contain? **2012 II**

(a) 12 (b) 10 (c) 7 (d) 6

56. Consider the following in respect of the sets A and B. **2013 I**

I. $(A \cap B) \subseteq A$ II. $(A \cap B) \subseteq B$ III. $A \subseteq (A \cup B)$

Which of the statement(s) given above is/are correct?

(a) I and II (b) II and III (c) I and III (d) All of these

57. In a school there are 30 teachers who teach Mathematics or Physics. Of these teachers, 20 teach Mathematics and 15 teach Physics, 5 teach both Mathematics and Physics. The number of teachers teaching only Mathematics is **2013 I**

(a) 5 (b) 10 (c) 15 (d) 20

58. In a class of 110 students, x students take both Mathematics and Statistics, $x + 20$ students take Mathematics and $x + 30$ students take Statistics. There are no students who take neither Mathematics nor Statistics. What is x equal to? **2013 II**

(a) 15 (b) 20 (c) 25 (d) 30

59. If A is a non-empty subset of a set E, then what is $E \cup (A \cap \phi) - (A - \phi)$ equal to? **2014 I**

(a) A (b) Complement of A
(c) ϕ (d) E

60. If A and B are any two non-empty subsets of a set E, then what is $A \cup (A \cap B)$ equal to? **2014 I**

(a) $A \cap B$ (b) $A \cup B$ (c) A (d) B

61. Out of 105 students taking an examination English and Mathematics, 80 students pass in English, 75 students pass in Mathematics, 10 students fail in both the subjects. How many students pass in only one subject? **2014 I**

(a) 26 (b) 30 (c) 35 (d) 45

62. Let A denotes the set of quadrilaterals having two diagonals equal and bisecting each other. Let B denotes the set of quadrilaterals having diagonals bisecting each other at 90°. Then, $A \cap B$ denotes **2015 II**

(a) the set of parallelograms (b) the set of rhombuses
(c) the set of squares (d) the set of rectangles

63. Let S be a set of first fourteen natural numbers. The possible number of pairs (a, b), where $a, b \in S$ and $a \neq b$ such that ab leaves remainder 1 when divided by 15, is **2016 I**

(a) 3 (b) 5 (c) 6 (d) None of these

64. In a gathering of 100 people, 70 of them can speak Hindi, 60 can speak English and 30 can speak French. Further, 30 of them can speak both Hindi and English, 20 can speak both Hindi and French. If x is the number of people who can speak both English and French, then which one of the following is correct? (Assume that everyone can speak atleast one of the three languages) **2016 I**

(a) $10 < x \leq 30$ (b) $0 \leq x < 8$ (c) $x = 9$ (d) $x = 8$

ANSWERS

1	a	2	b	3	c	4	b	5	c	6	a	7	c	8	c	9	a	10	a
11	a	12	a	13	d	14	c	15	c	16	a	17	b	18	c	19	a	20	d
21	c	22	a	23	a	24	c	25	c	26	c	27	a	28	c	29	c	30	c
31	c	32	b	33	d	34	a	35	a	36	d	37	d	38	b	39	a	40	a
41	d	42	c	43	c	44	c	45	c	46	c	47	a	48	a	49	d	50	c
51	a	52	b	53	a	54	a	55	d	56	d	57	c	58	b	59	b	60	c
61	c	62	c	63	d	64	a												

HINTS AND SOLUTIONS

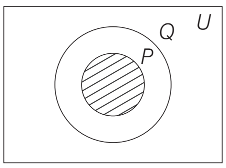
1. *(a)* $A \cup B = \{5, 6, 7\} \cup \{7, 8, 9\}$
$= \{5, 6, 7, 8, 9\}$

2. *(b)* $B - A = \{7, 8, 9, 12\} - \{2, 6, 8, 9\}$
$= \{7, 12\}$

3. *(c)* Given, $A = \{1, 2, 3, 4, 5\}$ and $A \subseteq U$.
$\Rightarrow A \cap U = \{1, 2, 3, 4, 5\} \cap \{U\}$
$= \{1, 2, 3, 4, 5\}$

4. *(b)* Let $A = \{2, 4, 16, 256, \ldots\}$
for $n = 0$, $2^{2^0} = 2^1 = 2$
for $n = 1$, $2^{2^1} = 2^2 = 4$
for $n = 2$, $2^{2^2} = 2^4 = 16$
Thus,
$A = \{x \in N \mid x = 2^{2^n}, n = 0, 1, 2, \ldots\}$

5. *(c)* $\because$ $P \subset Q$



So, $P \cap Q = P$

6. *(a)* Let $x \in A - B \Rightarrow x \in A$ and $x \notin B$
$\Rightarrow x \notin B - A$
Also, $x \in B - A \Rightarrow x \in B$
and $x \notin A \Rightarrow x \notin A - B$
So, $(A - B) \cap (B - A) = \phi$

7. *(c)* Only if P and Q both are empty set or P and Q have same elements, then $P \cup Q = P \cap Q$.

8. *(c)* Number of elements in $A = 4$
$\therefore$ Total number of non-empty Proper subsets of $A = 2^n - 2$
$= 2^4 - 2 = 14$

9. *(a)* Given, $A = \{B, O, W, L\}$
$B = \{B, O, W, L, E\}$
$C = \{B, O, W, L, E\}$
$\therefore$ $A \subset B$ and $B = C$

10. *(a)* Clearly, $A \cap (A \cup B) = A$

11. *(a)* $B \cup \{1, 2\} = \{1, 2, 3, 5, 9\}$
$\Rightarrow B = \{3, 5, 9\}$ is the required smallest set

12. *(a)* Here, $(P \cup R) = \{0, 1, 2, 3, 4, 5, 6\}$
$Q' = U - Q = \{0, 1, 6, 7, 8, 9\}$
So, $Q' \cap (P \cup R) = \{0, 1, 6\}$

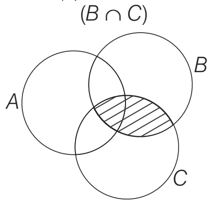
13. *(d)* Let $U = \{1, 2, 3, 4, 5, 6\}$ and $P = \{1, 3, 4\}$
Then, $P' = U - P = \{2, 5, 6\}$
and $(P')' = U - P' = \{1, 3, 4\} = P$

14. *(c)* The shaded region is $U - A = A'$.

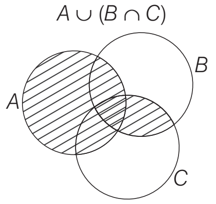
15. *(c)* The shaded region represents the elements in A and not in B, so it represents $A - B$.

16. *(a)* We have,

$(B \cap C)$



$A \cup (B \cap C)$



17. *(b)* The shaded region represents the elements in A and not in B or C.
$\therefore$ $A - (B \cup C)$

18. *(c)* Clearly, $(A \cap B) \cup (B \cap C)$

19. *(a)* Clearly, (a) is correct.

20. *(d)* Since, $P(\phi) = \{\phi\}$
Hence, $\phi \in P(\phi)$

21. *(c)* As, $\pi, \sqrt{2}, 3 + \sqrt{7}$ are irrational numbers.
So, required set is $\{\pi, \sqrt{2}, 3 + \sqrt{7}\}$.

22. *(a)* Here, $U = \{1, 2, 3, 4, 5, \ldots\}$
and $A = \{1, 3, 5, 7, 9, 11, \ldots\}$
So, $A' = U - A = \{2, 4, 6, 8, 10, \ldots\}$
$\therefore$ $A' = \{x : x$ is an even number$\}$

23. *(a)* Disjoint sets have no common element.
e.g. $A = \{1, 2, 3\}$, $B = \{4, 5, 6\}$
$\Rightarrow$ $A \cap B = \phi$

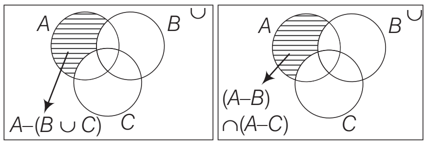
24. *(c)* $A - (A - B) = A - (A \cap B')$
$= A \cap (A \cap B')' = A \cap (A' \cup B)$
$= (A \cap A') \cup (A \cap B)$
$= \phi \cup (A \cap B) = A \cap B$

25. *(c)* $\because$ $n(A \cap B) = 2n$,
$\therefore$ $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 2n + 4n - 2n = 4n$
Hence, minimum number of elements in $A \cup B$ is $4n$.

26. *(c)* Here, $P = \{x: x^2 - 3x + 2 = 0\}$
$\Rightarrow$ $P = \{x : (x - 1)(x - 2) = 0\}$
$\Rightarrow$ $P = \{x : x = 1, x = 2\}$
$\therefore$ $P = \{1, 2\}$ and
$Q = \{x : x^2 + 4x - 12 = 0\}$
$\Rightarrow$ $Q = \{x : (x + 6)(x - 2) = 0\}$
$\Rightarrow$ $Q = \{-6, 2\}$
So, $P - Q = \{1, 2\} - \{-6, 2\} = \{1\}$

27. *(a)* Given, $A = \{(2^{2n} - 3n - 1) \mid n \in N\}$
$= \{0, 9, 54, 243, \ldots\}$
and $B = \{9(n - 1) \mid n \in N\}$
$= \{0, 9, 18, 27, \ldots\}$
From above, it is clear that $A \subset B$.

28. *(c)*



From the above two figures, shaded portion of

$A-(B\cup C)=(A-B)\cap(A-C)$

29. *(c)* $n(P\cup Q)=n(P)+n(Q)-n(P\cap Q)$

$\therefore\quad n(P\cup Q)=m+n-p$

30. *(c)* $n(A\cup B)=n(A)+n(B)-n(A\cap B)$

$\therefore n(A\cap B)=n(A)+n(B)-n(A\cup B)=17+23-38=2$

31. *(c)* Given, $A=\{x:x^2-6x+8=0\}$

$\Rightarrow\quad A=\{x:(x-4)(x-2)\}$

$\therefore\quad A=\{4,2\}$

And $\quad B=\{x:2x^2+3x-2=0\}$

$\Rightarrow\quad B=\{x:(2x-1)(x+2)\}$

$\therefore\quad B=\left\{\frac{1}{2},-2\right\}$

Hence, neither $A\subseteq B$ nor $B\subseteq A$.

32. *(b)* $(B\cup C)=\{1,2,3,4,5,6,7\}$

$\therefore\quad A\cap(B\cup C)=\{1,2,3,4\}\cap\{1,2,3,4,5,6,7\}=\{1,2,3,4\}$

33. *(d)* $(A\cup B)=\{1,2,3,4,6,8,10\}$

$(A\cup B)'=U-(A\cup B)$

$=\{1,2,3,4,5,6,7,8,9,10\}-\{1,2,3,4,6,8,10\}$

$=\{5,7,9\}$

34. *(a)* $\{(A\cup B)'\cap A\}-(A-B)$

$=\{(U-(A\cup B))\cap A\}-(A-B)$

$=\{(U\cap A)-\{(A\cup B)\cap A\}\}-(A-B)$

$=\{A-A\}-(A-B)=\phi-(A-B)=\phi$

35. *(a)* $\{a\}$ is an element of $\{\{a\},\{b\},\{c\}\}$, subsets of $\{\{a\},b,c\}$ are ϕ, $\{\{a\}$, $\{b\}$, $\{c\}$, $\{\{a\},b\}$, $\{b,c\}$, $\{\{a\},c\}$, $\{\{a\},b,c\}$

Hence, only statement I is correct.

36. *(d)* $\{x:x$ is an integer and less than $1000\}=[\ldots,998,999]$

i.e. $x\in(-\infty,1000)$ is an infinite set.

37. *(d)* If $A\cap B=\phi$, then it is not necessary that either.

$A=\phi$ or $B=\phi$.

38. *(b)* Since $R\in\overleftrightarrow{PQ}$ i.e R belongs to $\overleftrightarrow{PQ}$.

Again $S,\overleftrightarrow{PQ}$ and S does not lie on $\overleftrightarrow{PQ}$ extended

$\therefore\quad \overleftrightarrow{PQ}\cap\overleftrightarrow{RS}=\{R\}$

where R is point of intersection of the straight lines PQ and RS.

39. *(a)* Here, H = People, who can speak Hindi

E = People, who can speak English

Given, $n(H\cup E)=1000, n(H)=750,$

$n(E)=400$

$n(H\cup E)=n(H)+n(E)-n(H\cap E)$

$1000=750+400-n(H\cap E)$

$\Rightarrow n(H\cap E)=1150-1000=150$

Number of people who can speak Hindi only

$=n(H)-n(H\cap E)$

$=750-150=600$

40. *(a)* Let S be the set of people who speak spanish and F be the set of people who speak French

$\therefore\quad n(S)=20,\ n(F)=50,\ n(S\cap F)=10$

$\Rightarrow\quad n(S\cup F)=n(S)+n(F)-n(S\cap F)$

$=20+50-10=60$

41. *(d)* Here, M = Students who study Mathematics

E = Students who study English

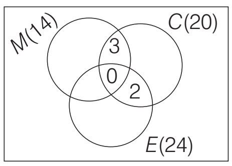
C = Students who study Commerce

$\therefore n(M\cup E\cup C)=42$

Also, $n(M)=14, n(E)=24, n(C)=20,$

$n(M\cap C)=3,\ n(E\cap C)=2$

and $n(M\cap E\cap C)=0$



From the Venn diagram, it is clear that number of students who study Mathematics but not Commerce

$=14-3=11$

42. *(c)* Let number of failed candidates in Mathematics and English be M and E, respectively.

Given, candidates failed in English, $n(E)=52\%$,

Candidates failed in Mathematics, $n(M)=42\%$

Candidates failed in both English and Mathematics,

$n(E\cap M)=17\%$

$\because$ Total candidates, failed

$n(E\cup M)=n(E)+n(M)-n(E\cap M)$

$=52+42-17=77\%$

$\therefore$ Total candidates passed $=(100-77)$

$=23\%$

43. *(c)* Clearly, both I and II are true.

44. *(c)* I. $A=\{0\}$ II. $B=\{2\}$

III. $C=\{\ \}$; ±4 is not an odd integer

Here, only III is empty set.

45. *(c)* By distributive law in sets

$A\cap(B\cup C)=(A\cap B)\cup(A\cap C)$

So, only III is correct.

46. *(c)* Here, $P=\{\ldots-6,-3,0,3,6,\ldots\}$,

$Q=\{\ldots-8,-4,0,4,8,\ldots\}$ and

$R=\{\ldots-12,-6,0,6,12,\ldots\}$

I. $P\cup Q=\{\ldots-8,-6,-4,-3,0,3,4,6,8,\ldots\}\neq R$

II. $P\not\subset R$ as $3\in P$ but $3\notin R$.

III. $R\subset(P\cup Q)$ is true.

Hence, only statement III is correct.

47. *(a)* I. There are infinite points lie on a line segment so, it is an infinite set.

II. Number of birds in zoo are countable so, it is a finite set.

III. It is not a well-defined set.

Hence, only I is correct.

48. *(a)* I. A is a finite set.

II. As all elements of A are integers, so A is subset of integers.

III. $\{1,2\}$ is a proper subset of A.

IV. $A\neq\phi$.

So, I, II and III are true.

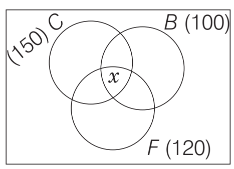
49. *(d)* All sets are infinite set.

50. *(c)* Equivalent sets have same cardinal numbers. Here, cardinal numbers of I, III, IV sets are same.

Solutions (Q. No. 51-52) *Let C, B and F denotes the number of students who play cricket, basketball and football respectively.*

Given, $n(C)=150,\ n(B)=100,\ n(F)=120,\ n(B\cap F)=30,\ n(C\cap B)=50$ and $n(C\cap F)=60$

Let x be the number of students who play all the three sports



51. *(a)* $n(C \cup B \cup F) = n(C) + n(B) + n(F) - n(C \cap F) - n(C \cap B) - n(B \cap F) + n(C \cap B \cap F)$

$\Rightarrow \quad n(C \cup B \cup F) = 150 + 100 + 120 - 60 - 50 - 30 + x$

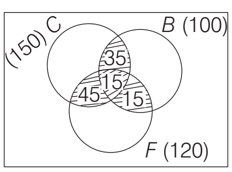
$\Rightarrow \quad n(C \cup B \cup F) = 230 + x$

Since, maximum value of $n(C \cup B \cup F)$ is 250

Hence, maximum value of $x = 250 - 230 = 20$

52. *(b)* Here, $n(C \cup B \cup F) = 250 - 5 = 245$

$\therefore \quad 245 = 230 + x \quad \Rightarrow \quad x = 15$



Number of students who play atleast two sports $= 15 + 35 + 45 + 15 = 110$

53. *(a)* Given, $A = \{2, 4, 6, 8, 10, 12, \ldots\}$,

$B = \{5, 10, 15, 20, 25, 30, \ldots\}$

and $C = \{10, 20, 30, 40, 50, 60, \ldots\}$

Now,

$(B \cup C) = \{5, 10, 15, 20, 25, 30, \ldots\}$

$\therefore \ A \cap (B \cup C) = \{2, 4, 6, 8, 10, 12, \ldots\} \cap \{5, 10, 15, 20, 25, 30, \ldots\}$

$= \{10, 20, 30, \ldots\}$

54. *(a)* From option (a), $A = \{x$ is a real number : $x > 1$ and $x < 1\}$.

Since, there is no such element which is greater than 1 and less than 1.

So, A is a null set.

From option (b), $B = \{x : x + 3 = 3\} = \{0\}$ = Singleton set

From option (c), $C = \{\phi\}$ = Singleton set

From option (d), $D = \{x$ is a real number : $x \geq 1$ and $x \leq 1\}$

$= \{1\}$ = Singleton set

55. *(d)* Given that, $x \in \{2, 3, 4\}$

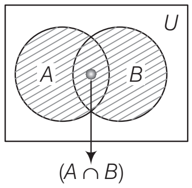
and $y \in \{4, 6, 9, 10\}$

and also $A = (x, y)$, such that x is a factor of y.

$\therefore \ A = \{(2, 4), (2, 6), (2, 10), (3, 6), (3, 9), (4, 4)\}$

Hence, A contain 6 elements.

56. *(d)* From figure,



I. $(A \cap B) \subseteq A$ [true]

II. $(A \cap B) \subseteq B$ [true]

III. $A \subseteq (A \cup B)$ [true]

So, all three statements are correct.

Shaded region $= (A \cup B)$

57. *(c)* Let M and P be the teachers who teach Mathematics and Physics, respectively.

Given,

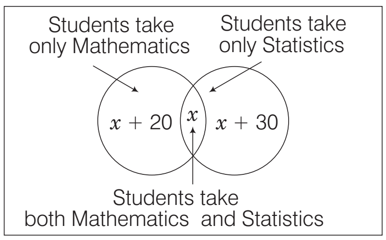
$n(M \cup P) = 30$, $n(M) = 20$, $n(P) = 15$ and $n(M \cap P) = 5$

$\therefore$ Number of teachers teaching only Mathematics is

$n(\text{only } M) = n(M) - n(M \cap P)$

$= 20 - 5 = 15$

58. *(b)* Venn diagram of given conditions is as shown in figure given below



Total students = Students those take only Mathematics + Student those take only Statistics + Student those take both Mathematics and Statistics

$\because \quad 110 = x + 20 + x + x + 30$

$\Rightarrow \quad 110 = 3x + 50 \quad \Rightarrow 3x = 60$

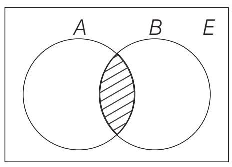
$\therefore \quad x = 20$

59. *(b)* $E \cup \{(A \cap \phi) - (A - \phi)\}$

$= E \cup \phi - A = E - A = A'$

$[\because (A \cap \phi) = \phi$ and $(A - \phi) = A]$

60. *(c)* Since, A and B are non-empty subsets of E.



$\therefore A \cup (A \cap B) = A \cup$ (Shaded portion)

$= A$

If A and B are two non-empty disjoint sets.

then, $A \cap B = \phi$

$\therefore \quad A \cup (A \cap B) = A \cup \phi = A$

61. *(c)* Let E and M represent the students pass in English and Mathematics.

Given, $n(U) = 105$, $n(E) = 80$,

$n(M) = 75$, and $n(E \cap M) = 10$

Now, $n(E \cup M) = n(U) - n(E \cap M)$

$= 105 - 10 = 95$

Also, $n(E \cup M) = n(E) + n(M) - n(E \cap M)$

$\Rightarrow \quad 95 = 80 + 75 - n(E \cap M)$

$\Rightarrow \quad n(E \cap M) = 60$

The number of students who pass in only one subject

$= n(E) + n(M) - 2\, n(E \cap M)$

$= 80 + 75 - 2 \times 60$

$= 155 - 120 = 35$

62. *(c)* A = diagonal equal and bisecting each other.

A is square or rectangle. and B diagonal bisecting each other at 90°.

So, $A \cap B$ = the set of squares.

63. *(d)* The possible set of pairs *(a, b)* such that *ab* leaves remainder 1 when divided by 15 are (2, 8), (8, 2) (7, 13) and (13, 7).

$\therefore$ Number of possible set of pairs = 4

64. *(a)* Let A, B and C be the number of people who can speak Hindi, English and French.

Then, $n(A) = 70$, $n(B) = 60$,

$n(C) = 30$, $n(A \cap B) = 30$,

$n(A \cap C) = 20$, and $n(B \cap C) = x$

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

$\Rightarrow \quad 100 = 70 + 60 + 30 - 30 - 20 - x + n(A \cap B \cap C)$

$\Rightarrow \quad x = 10 + n(A \cap B \cap C)$

Also, $n(A \cap B \cap C)$ can't be more than $n(A \cap C)$

$\Rightarrow \quad 10 \leq x \leq 30$

20

MEASUREMENTS OF ANGLES AND TRIGONOMETRIC RATIOS

Generally (13-14) questions have been asked from this chapter. Generally questions asked from this chapter are based on the trigonometric identities and formulae. So, detailed study of this chapter will help you to score in exam as good number of questions have been asked from this chapter.

Trigonometry is the branch of Mathematics which deals with the measurements of sides and angles of triangles and the problems based on these angles.

ANGLE

An angle is considered as the figure obtained by rotating a given ray about its endpoint. The original ray is called the initial side and the ray into which the initial side rotates is called terminal side. In figure, OA is initial side and OB is the terminal side of $\angle AOB$. The point O is called its vertex.

QUADRANTS

Let XOX' and YOY' be two lines at right angles in the plane of paper. These two perpendicular lines divide the plane of the paper into four equal parts, these four parts are known as four quadrants and lines XOX' and YOY' are known as X-axis and Y-axis, respectively.

The parts $XOY, YOX', X'OY'$ and $Y'OX$ are known as 1st, 2nd, 3rd and 4th quadrant, respectively.

Relation between Angles, Radius and Arc Length

If l' is the length of a circle of radius r. And this arc subtends an angle θ radians at the centre of the circle, then $l = r\,\theta \Rightarrow \theta = \frac{l}{r}$

Relation between Degree and Radian

A circle subtends an angle, at the centre whose radian measure is 2π and its degree measure is $360°$. It follows that

$$2\pi \text{ radian} = 360°$$
$$\pi \text{ radian} = 180°$$
$$\therefore \quad 1 \text{ radian} = \frac{180}{\pi} \text{ degree} = 57° \, 16' \, 22'' \quad \text{(approx)}$$
$$1 \text{ degree} = \frac{\pi}{180} \text{ radian} = 0.01746 \text{ rad} \quad \text{(approx)}$$

Note 1 radian is written as 1^c.

EXAMPLE 1. The radian measure corresponding to $-22° \, 30'$ is

a. $\frac{\pi^c}{8}$ b. $-\frac{\pi^c}{8}$ c. $\frac{\pi^c}{4}$ d. $-\frac{\pi^c}{4}$

Sol. b. As, $180° = \pi$

$$\therefore -22°30' = -\left(22 + \frac{1}{2}\right) = -\left(\frac{45}{2}\right)^{\circ} = -\left(\frac{45}{2} \times \frac{\pi}{180}\right)^{c} = -\frac{\pi^c}{8}$$

> **IMPORTANT POINTS**
>
> - The angle between two consecutive digits in a clock is $30° = \left(\frac{\pi}{6} \text{radian}\right)$.
> - The hour hand rotates through an angle of 30° in one hour i.e. $\left(\frac{1}{2}\right)^{\circ}$ in one minute.
> - The minute hand rotates through an angle of 6° in one minute.

EXAMPLE 2. If an arc 24π cm long of a circle subtends an angle of 72° at its centre, then the radius of the circle is

a. 30 cm b. 40 cm c. 50 cm d. 60 cm

Sol. d. Given, length of arc $(l) = (24\pi)$ cm

$$\text{Angle } (\theta) = 72° = \left(72 \times \frac{\pi}{180}\right)^c = \left(\frac{2\pi}{5}\right)^c$$

Let r be the radius of circle

$$\text{Since,} \quad \theta = \frac{1}{r} \Rightarrow r = \frac{1}{\theta} = \frac{24\pi}{\frac{2\pi}{5}} = 60 \text{ cm}$$

EXAMPLE 3. Consider the following statements:

I. The angular measure in radian of a circular arc of fixed length subtending at its centre decreases, if the radius of the arc increases.

II. 1800° is equal to 5π radian.

Which of the statements given above is/are correct? **2012 I**

a. Only I b. Only II
c. Both I and II d. Neither I nor II

Sol. a. I. We know that,

$$\text{Radius} = \frac{\text{arc}}{\text{angle}} \quad \text{(given, arc length is constant)}$$
$$\therefore \quad \text{Radius} \propto \frac{1}{\text{angle}}$$

So, angular measure in radian decreases, if the radius of the arc increases.

$$\text{II. } 1800° \times \frac{\pi}{180°} = 10\pi$$

Hence, only statement I is correct.

EXAMPLE 4. The angle between the hour hand and the minute hand of a clock at half-past three is

a. 54° b. 63° c. 75° d. 85°

Sol. c. Angle traced by hour hand in 1 hr = 30°

$$\text{Angle traced by hour hand in } 3\frac{1}{2} \text{ hr} = \left(30 \times \frac{7}{2}\right)^{\circ} = 105°$$

Angle traced by minute hand in 1 min = 6°

Angle traced by minute hand in 30 min = $6 \times 30° = 180°$

Angle between two hands = $180° - 105° = 75°$

TRIGONOMETRIC RATIOS

The ratio of the sides of a right angled triangle with respect to its angle are called trigonometric ratios.

The side opposite to the right angle is called the hypotenuse. Relative to the angle θ, the side opposite to angle θ is caused the prependicular side and the remaining side is called base.

(i) For angle θ, perpendicular = BC, base = AB and hypotenuse = AC

(ii) For angle α, perpendicular = AB, base = BC and hypotenuse = AC

Trigonometric ratios of θ in right angled ΔABC are defined below

$$\sin\theta = \frac{\textit{Perpendicular}}{\textit{Hypotenuse}} = \frac{BC}{AC} = \frac{P}{H}$$
$$\cos\theta = \frac{\textit{Base}}{\textit{Hypotenuse}} = \frac{AB}{AC} = \frac{B}{H}$$
$$\tan\theta = \frac{\textit{Perpendicular}}{\textit{Base}} = \frac{BC}{AB} = \frac{P}{B}$$
$$\sec\theta = \frac{\textit{Hypotenuse}}{\textit{Base}} = \frac{AC}{AB} = \frac{H}{B}$$
$$\cot\theta = \frac{\textit{Base}}{\textit{Perpendicular}} = \frac{AB}{BC} = \frac{B}{P}$$
$$\text{cosec}\,\theta = \frac{\textit{Hypotenuse}}{\textit{Perpendicular}} = \frac{AC}{BC} = \frac{H}{P}$$

Relation Between T-ratios

(i) $\sin\theta = \dfrac{1}{\text{cosec}\,\theta}$ (ii) $\cos\theta = \dfrac{1}{\sec\theta}$ (iii) $\tan\theta = \dfrac{1}{\cot\theta}$

(iv) $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ (v) $\cot\theta = \dfrac{\cos\theta}{\sin\theta}$

EXAMPLE 5. If $\sin\theta = \dfrac{8}{17}$, then the other trigonometric ratios $\cos\theta, \tan\theta, \text{cosec}\,\theta, \sec\theta, \cot\theta$ are

a. $\dfrac{15}{17}, \dfrac{8}{15}, \dfrac{17}{8}, \dfrac{17}{15}, \dfrac{15}{8}$
b. $-\dfrac{15}{17}, \dfrac{8}{15}, \dfrac{17}{8}, \dfrac{-17}{15}, \dfrac{8}{15}$
c. $\dfrac{17}{15}, \dfrac{15}{8}, \dfrac{8}{17}, \dfrac{15}{17}, \dfrac{8}{15}$
d. None of these

Sol. a. As, $\sin\theta = \dfrac{8}{17} = \dfrac{\text{perpendicular}}{\text{hypotenuse}}$

In right angled ΔPQR,

$\therefore PQ^2 + QR^2 = PR^2 \Rightarrow PQ^2 = PR^2 - QR^2$

$\Rightarrow \quad PQ = \sqrt{PR^2 - QR^2} = \sqrt{17^2 - 8^2}$

$= \sqrt{289 - 64} = \sqrt{225} = 15$

$\therefore \quad \cos\theta = \dfrac{PQ}{PR} = \dfrac{15}{17}, \quad \tan\theta = \dfrac{RQ}{PQ} = \dfrac{8}{15}$

$\text{cosec}\,\theta = \dfrac{PR}{RQ} = \dfrac{17}{8}, \quad \sec\theta = \dfrac{PR}{PQ} = \dfrac{17}{15}$

and $\quad \cot\theta = \dfrac{PQ}{RQ} = \dfrac{15}{8}$

EXAMPLE 6. If $\cot A = \dfrac{1}{(\sqrt{2} - 1)}$, then the value of $\sin A \cos A$ is

a. $\dfrac{1}{4}$ b. $\dfrac{1}{2}$ c. $\dfrac{1}{\sqrt{2}}$ d. $\dfrac{\sqrt{2}}{4}$

Sol. d. $\because \ \cot A = \dfrac{1}{\sqrt{2} - 1} = \dfrac{\text{base}}{\text{perpendicular}}$

$\Rightarrow \tan A = \dfrac{\sqrt{2} - 1}{1} \left(\because \tan A = \dfrac{1}{\cot A}\right)$

In right angled ΔABC,

$\therefore AC^2 = AB^2 + BC^2$

$\Rightarrow AC = \sqrt{AB^2 + BC^2}$

$= \sqrt{1^2 + (\sqrt{2} - 1)^2} = \sqrt{1 + 2 + 1 - 2\sqrt{2}} = \sqrt{4 - 2\sqrt{2}}$

$\therefore \sin A = \dfrac{BC}{AC} = \dfrac{\sqrt{2} - 1}{\sqrt{4 - 2\sqrt{2}}}$ and $\cos A = \dfrac{AB}{AC} = \dfrac{1}{\sqrt{4 - 2\sqrt{2}}}$

$\therefore \sin A \cos A = \dfrac{\sqrt{2} - 1}{\sqrt{4 - 2\sqrt{2}}} \cdot \dfrac{1}{\sqrt{4 - 2\sqrt{2}}} = \dfrac{\sqrt{2} - 1}{4 - 2\sqrt{2}}$

$= \dfrac{\sqrt{2} - 1}{2\sqrt{2}(\sqrt{2} - 1)} = \dfrac{1}{2\sqrt{2}} \times \dfrac{\sqrt{2}}{\sqrt{2}} = \dfrac{\sqrt{2}}{4}$

TRIGONOMETRIC IDENTITIES

A statement of equality involving trigonometric ratios of an angle is called a trigonometric identity. It is valid for all values of the angles. The three important identities are

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \text{cosec}^2\theta$

Note $\sin^2 A + \cos^2 B$ cannot be equal to 1 because the angles are different.

EXAMPLE 7. What is the value of $\dfrac{\tan A - \sin A}{\sin^3 A}$? **2013 I**

a. $\dfrac{\sec A}{1 - \cos A}$ b. $\dfrac{\sec A}{1 + \cos^2 A}$
c. $\dfrac{\sec A}{1 + \cos A}$ d. None of these

Sol. c. $\dfrac{\tan A - \sin A}{\sin^3 A} = \dfrac{\dfrac{\sin A}{\cos A} - \sin A}{\sin^3 A} = \dfrac{(1 - \cos A)}{\cos A \sin^2 A} \times \dfrac{(1 + \cos A)}{(1 + \cos A)}$

$= \dfrac{1 - \cos^2 A}{\cos A \sin^2 A (1 + \cos A)} = \dfrac{\sin^2 A}{\cos A \sin^2 A (1 + \cos A)}$

$= \dfrac{1}{\cos A} \cdot \dfrac{1}{1 + \cos A} = \dfrac{\sec A}{1 + \cos A}$

Values of Trigonometric Ratios for Some Specific Angles

Angles	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\cot\theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\text{cosec}\,\theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

EXAMPLE 8. What is the value of $\sin^3 60° \cot 30° - 2\sec^2 45° + 3\cos 60° \tan^2 45° - \tan^2 60°$?

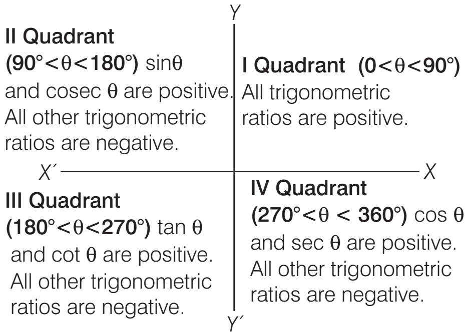
a. $\dfrac{35}{8}$ b. $\dfrac{-35}{8}$ c. $\dfrac{-11}{8}$ d. $\dfrac{11}{8}$

Sol. b. $\sin^3 60° \cot 30° - 2\sec^2 45° + 3\cos 60° \tan^2 45° - \tan^2 60°$

$= \left(\dfrac{\sqrt{3}}{2}\right)^3 \times \sqrt{3} - 2 \times (\sqrt{2})^2 + 3 \times \dfrac{1}{2} \times (1)^2 - (\sqrt{3})^2$

$= \dfrac{9}{8} - 4 + \dfrac{3}{2} - 3 = \dfrac{9 - 32 + 12 - 24}{8} = \dfrac{21 - 56}{8} = \dfrac{-35}{8}$

Sign of trigonometric ratio in different quadrants



EXAMPLE 9. If $0<\theta<\dfrac{\pi}{4}$, then what is $\sqrt{1-2\sin\theta\cos\theta}$ equal to? **2014 II**

a. $\cos\theta-\sin\theta$ b. $\sin\theta-\cos\theta$
c. $\pm(\cos\theta-\sin\theta)$ d. $\cos\theta\sin\theta$

Sol. a. Given, $0<\theta<\dfrac{\pi}{4}$, then $\sqrt{1-2\sin\theta\cos\theta}$

$$=\sqrt{\sin^2\theta+\cos^2\theta-2\sin\theta\cos\theta}$$

$[\because \sin^2\theta+\cos^2\theta=1]$

$$=\sqrt{(\cos\theta-\sin\theta)^2}$$

$\left[\because 0<\theta<\dfrac{\pi}{4}, \cos\theta>\sin\theta, \text{ so we take } (\cos\theta-\sin\theta)^2\right]$

$$=\cos\theta-\sin\theta$$

Domain and Range of Trigonometrical Functions

Trigonometric Ratio	Domain	Range
sin θ	R	[– 1, 1]
cos θ	R	[– 1, 1]
tan θ	$R\sim\left\{(2n+1)\dfrac{\pi}{2}, n\in I\right\}$	$(-\infty,\infty)=R$
cot θ	$R\sim\{n\pi, n\in I\}$	$(-\infty,\infty)=R$
sec θ	$R\sim\left\{(2n+1)\dfrac{\pi}{2}, n\in I\right\}$	$(-\infty,-1]\cup[1,\infty)$
cosec θ	$R\sim\{n\pi, n\in I\}$	$(-\infty,-1]\cup[1,\infty)$

The behaviour of the trigonometric function in different quadrants are defined in following table.

Trigonometric Ratio	Ist Quadrant (θ increase from 0 to $\pi/2$)	IInd Quadrant (θ increase from $\pi/2$ to π)	III Quadrant (θ increase from π to $3\pi/2$)	IVth Quadrant (θ increase from $\dfrac{3\pi}{2}$ to 2π)
sin θ	Increases from 0 to 1	Decreases from 1 to 0	Decreases from 0 to –1	Increases from –1 to 0
cos θ	Decreases from 1 to 0	Decreases from 0 to –1	Increases from –1 to 0	Increases from 0 to 1
tan θ	Increases from 0 to ∞	Increases from – ∞ to 0	Increases from 0 to ∞	Increases from – ∞ to 0
cot θ	Decreases from ∞ to 0	Decreases from 0 to – ∞	Decreases from ∞ to 0	Decreases from 0 to – ∞
sec θ	Increases from 1 to ∞	Increases from – ∞ to –1	Decreases from –1 to – ∞	Decreases from ∞ to 1
cosec θ	Decreases from ∞ to 1	Increases from 1 to ∞	Increases from – ∞ to –1	Decreases from –1 to – ∞

EXAMPLE 10. Which one of the following statements is true in respect of the expression $\sin 31°+\sin 32°$?

a. Its value is 0 b. Its value is 1
c. Its value is less than 1 d. Its value is greater than 1

Sol. d. We know that, $\sin 30°=\dfrac{1}{2}$

Value of sin increases 0° to 90°

$\therefore \quad \sin 31°>\sin 30°$ and $\sin 32°>\sin 30°$

$\Rightarrow \quad \sin 31°>\dfrac{1}{2}$ and $\sin 30°>\dfrac{1}{2}$

Adding both

$\therefore \quad \sin 31°+\sin 31°\Rightarrow\dfrac{1}{2}+\dfrac{1}{2}$

$\Rightarrow \quad \sin 31°+\sin 32°>1$

IMPORTANT POINTS

- $|\sin\theta|\le 1$
- $|\cos\theta|\le 1$
- $\sec\theta\ge 1$
- $\sec\theta\le -1$
- $\text{cosec}\,\theta\ge 1$
- $\text{cosec}\,\theta\le -1$
- tan θ and cot θ can take any value.

TRIGONOMETRIC RATIOS OF ALLIED ANGLES

Two angles are said to be allied when their sum or difference is either zero or a multiple of 90°.

The angles like $-\theta, 90°\pm\theta, 180°\pm\theta, 360°\pm\theta$ etc., are angles allied to the angle θ, if θ is measured in degrees.

Angle	sin θ	cos θ	tan θ	cot θ	sec θ	cosec θ
–θ	–sinθ	cosθ	– tanθ	–cotθ	secθ	–cosec θ
90°–θ	cosθ	sinθ	cotθ	tanθ	cosec θ	secθ
90°+θ	cosθ	–sinθ	–cotθ	– tanθ	–cosec θ	secθ
180°–θ	sinθ	–cosθ	– tanθ	–cotθ	–secθ	cosec θ
180°+θ	–sinθ	–cosθ	tanθ	cotθ	–secθ	–cosec θ
270°–θ	–cosθ	–sinθ	cotθ	tanθ	–cosec θ	–secθ
270°+θ	–cosθ	sinθ	–cotθ	– tanθ	cosec θ	–secθ
360°–θ	–sinθ	cosθ	– tanθ	–cotθ	secθ	–cosec θ
360°+θ	sinθ	cosθ	tanθ	cotθ	secθ	cosec θ

EXAMPLE 11. What is the value of $\sec(90°-\theta)\sin\theta\sec 45°$?

a. 1 b. $\frac{\sqrt{3}}{2}$ c. $\sqrt{2}$ d. $\sqrt{3}$

Sol. c. Given, $\sec(90°-\theta)\sin\theta\sec 45° = \text{cosec}\,\theta \times \sin\theta \times \sqrt{2}$

$$= \frac{1}{\sin\theta} \times \sin\theta \times \sqrt{2} = \sqrt{2}$$

EXAMPLE 12. The value of $\frac{\sin 135° - \cos 120°}{\sin 135° + \cos 120°}$ is

a. $2+3\sqrt{2}$ b. $2-3\sqrt{2}$ c. $3-2\sqrt{2}$ d. $3+2\sqrt{2}$

Sol. d. $\sin 135° = \sin(180° - 45°) = \sin 45° = \frac{1}{\sqrt{2}}$

$\cos 120° = \cos(180° - 60°) = -\cos 60° = -\frac{1}{2}$

$$\therefore \frac{\sin 135° - \cos 120°}{\sin 135° + \cos 120°} = \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{2}\right)}{\frac{1}{\sqrt{2}} + \left(-\frac{1}{2}\right)} = \frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \frac{(\sqrt{2}+1)^2}{2-1} = 2 + 1 + 2\sqrt{2} = 3 + 2\sqrt{2}$$

Sum, Difference and Product Formulae

- $\sin(A+B) = \sin A\cos B + \cos A \sin B$
- $\sin(A-B) = \sin A\cos B - \cos A \sin B$
- $\cos(A+B) = \cos A\cos B - \sin A \sin B$
- $\cos(A-B) = \cos A\cos B + \sin A \sin B$
- $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A+B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$
- $\cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$
- $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B$
  $= \cos^2 B - \cos^2 A$
- $\cos(A+B)\cos(A-B)$
  $= \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
- $2\sin A\cos B = \sin(A+B) + \sin(A-B)$
- $2\cos A\sin B = \sin(A+B) - \sin(A-B)$
- $2\cos A\cos B = \cos(A+B) + \cos(A-B)$
- $2\sin A\sin B = \cos(A-B) - \cos(A+B)$
- $\sin A + \sin B = 2\sin\frac{(A+B)}{2}\cdot\cos\frac{(A-B)}{2}$
- $\sin A - \sin B = 2\sin\frac{(A-B)}{2}\cdot\cos\frac{(A+B)}{2}$
- $\cos A + \cos B = 2\cos\frac{(A+B)}{2}\cdot\cos\frac{(A-B)}{2}$
- $\cos A - \cos B = -2\sin\frac{(A+B)}{2}\cdot\sin\frac{(A-B)}{2}$
  $= 2\sin\frac{(A+B)}{2}\cdot\sin\frac{(B-A)}{2}$
- $\tan A + \tan B = \frac{\sin(A+B)}{\cos A \cos B}$
- $\tan A - \tan B = \frac{\sin(A-B)}{\cos A \cos B}$
- $\cot A + \cot B = \frac{\sin(A+B)}{\sin A \sin B}$
- $\cot A - \cot B = \frac{\sin(B-A)}{\sin A \sin B}$

EXAMPLE 13. If $A - B = \frac{\pi}{3}$, then the value of $\cos A\cos B + \sin A \sin B$ is

a. 1/2 b. 1
c. 3/2 d. None of these

Sol. a. $\cos A\cos B + \sin A\sin B = \cos(A-B)$

$$= \cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \qquad \left(\because A - B = \frac{\pi}{3}\right)$$

EXAMPLE 14. The value of $\frac{\sqrt{3}\cos 23° - \sin 23°}{2}$ is

a. $\cos 7°$ b. $\sin 53°$ c. $\cos 53°$ d. $\sin 7°$

Sol. c. $\frac{\sqrt{3}\cos 23° - \sin 23°}{2} = \frac{\sqrt{3}}{2}\cos 23° - \frac{1}{2}\sin 23°$

$= \cos 30°\cos 23° - \sin 30°\sin 23°$

$= \cos(30° + 23°) = \cos 53°$

$[\because \cos(A+B) = \cos A\cos B - \sin A \sin B]$

EXAMPLE 15. The value of $\frac{\tan 47° + \tan 43°}{1 - \tan 47° \tan 43°}$ is

a. 1 b. ∞
c. 0 d. -1

Sol. b. $\because \frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(A+B) \Rightarrow \frac{\tan 47° + \tan 43°}{1 - \tan 47° \tan 43°}$

$$= \tan(47° + 43°) = \tan 90° = \infty$$

EXAMPLE 16. The value of $\sin 20°\sin 40°\sin 60°\sin 80°$ is

a. $\frac{1}{2}$ b. $\frac{3}{4}$ c. $\frac{1}{8}$ d. $\frac{3}{16}$

Sol. d. $\sin 20°\sin 40°\sin 80°\sin 60°$

$$= \sin 60°[\sin 80°\sin 40°]\sin 20°$$
$$= \frac{\sqrt{3}}{2}[\sin 80°\sin 40°]\sin 20°$$
$$= \frac{\sqrt{3}}{2}\times\frac{1}{2}[2\sin 80°\sin 40°]\sin 20°$$
$$= \frac{\sqrt{3}}{4}[\cos(80°-40°)-\cos(80°+40°)]\sin 20°$$
$$= \frac{\sqrt{3}}{4}[\cos 40°-\cos 120°]\sin 20°$$
$$= \frac{\sqrt{3}}{4}\left[\cos 40°-\left(-\frac{1}{2}\right)\right]\sin 20° \qquad \left(\because \cos 120° = -\frac{1}{2}\right)$$
$$= \frac{\sqrt{3}}{4}\left[\cos 40°+\frac{1}{2}\right]\sin 20°$$
$$= \frac{\sqrt{3}}{8}[\sin(40°+20°)-\sin(40°-20°)+\sin 20°]$$
$$= \frac{\sqrt{3}}{8}[\sin 60°-\sin 20°+\sin 20°]$$
$$= \frac{\sqrt{3}}{8}\times\sin 60° = \frac{\sqrt{3}}{8}\times\frac{\sqrt{3}}{2} = \frac{3}{16}$$

EXAMPLE 17. $\dfrac{\cos 4x+\cos 3x+\cos 2x}{\sin 4x+\sin 3x+\sin 2x}$ is equal to

a. $\cot 3x$ b. $\tan 3x$ c. $\cot x$ d. $\cot 2x$

Sol. a. $\dfrac{\cos 4x+\cos 3x+\cos 2x}{\sin 4x+\sin 3x+\sin 2x} = \dfrac{(\cos 4x+\cos 2x)+\cos 3x}{(\sin 4x+\sin 2x)+\sin 3x}$

$$= \frac{2\cos\left(\frac{4x+2x}{2}\right)\cos\left(\frac{4x-2x}{2}\right)+\cos 3x}{2\sin\left(\frac{4x+2x}{2}\right)\cos\left(\frac{4x-2x}{2}\right)+\sin 3x}$$
$$= \frac{2\cos 3x\cdot\cos x+\cos 3x}{2\sin 3x\cdot\cos x+\sin 3x} = \frac{\cos 3x(2\cos x+1)}{\sin 3x(2\cos x+1)} = \cot 3x$$

Trigonometric Ratios of Multiple Angles

- $\sin 2\theta = 2\sin\theta\cos\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$
- $\cos 2\theta = \cos^2\theta-\sin^2\theta = 2\cos^2\theta-1$
  $= 1-2\sin^2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$
- $\dfrac{1+\cos 2\theta}{2} = \cos^2\theta$ and $\dfrac{1-\cos 2\theta}{2} = \sin^2\theta$
- $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$
- $\sin 3\theta = 3\sin\theta-4\sin^3\theta$
- $\cos 3\theta = 4\cos^3\theta-3\cos\theta$
- $\tan 3\theta = \dfrac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}$

EXAMPLE 18. The value of $3\sin 15°-4\sin^3 15°$ is

a. 1 b. $\frac{1}{2}$ c. $\frac{1}{\sqrt{2}}$ d. 0

Sol. c. As, $3\sin\theta-4\sin^3\theta = \sin 3\theta$

So, $3\sin 15°-4\sin^3 15° = \sin 3(15°) = \sin 45° = \dfrac{1}{\sqrt{2}}$

Trigonometric Ratios of Submultiple Angles

- $\sin\theta = 2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}$
- $\cos\theta = \cos^2\dfrac{\theta}{2}-\sin^2\dfrac{\theta}{2} = 2\cos^2\dfrac{\theta}{2}-1 = 1-2\sin^2\dfrac{\theta}{2}$
- $\dfrac{1-\cos\theta}{\sin\theta} = \tan\dfrac{\theta}{2}$
- $\dfrac{1+\cos\theta}{\sin\theta} = \cot\dfrac{\theta}{2}$
- $\dfrac{1-\cos\theta}{1+\cos\theta} = \tan^2\dfrac{\theta}{2}$
- $\dfrac{1+\cos\theta}{1-\cos\theta} = \cot^2\dfrac{\theta}{2}$
- $\sin\theta = \dfrac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$
- $\cos\theta = \dfrac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$

Maximum and Minimum Values of Trigonometrical Functions

We know that, $-1\le\sin\theta\le 1$, $-1\le\cos\theta\le 1$

Maximum and minimum value of a trigonometrical function of the form $a\sin\theta\pm b\cos\theta$ are $\sqrt{a^2+b^2}$ and $-\sqrt{a^2+b^2}$, respectively.

EXAMPLE 19. The maximum value and minimum value of $8\sin\theta\cos\theta+4\cos 2\theta$ is

a. 4 and –4 b. 16 and –16
c. $4\sqrt{2}$ and $-4\sqrt{2}$ d. None of these

Sol. c. Here, $8\sin\theta\cos\theta+4\cos 2\theta = 4\sin 2\theta+4\cos 2\theta$

So, maximum value $= \sqrt{4^2+4^2} = 4\sqrt{2}$

and minimum value $= -\sqrt{4^2+4^2} = -4\sqrt{2}$

EXAMPLE 20. Solve $\cos\theta+\sin\theta = \sqrt{2}$, then the value of θ is

a. $\pi/3$ b. $\pi/4$ c. $\pi/6$ d. $\pi/2$

Sol. b. $\cos\theta+\sin\theta = \sqrt{2}$, $\dfrac{\cos\theta}{\sqrt{2}}+\dfrac{\sin\theta}{\sqrt{2}} = 1$

($\because$ dividing throughout by $\sqrt{2}$)

$\Rightarrow \quad \cos\theta\cos 45°+\sin\theta\sin 45° = 1$

$\cos(\theta-45°) = 1 = \cos 0°$

$\therefore \quad \theta-45° = 0 \Rightarrow \theta = 45° = \pi/4$

EXAMPLE 21. The value of ϕ for maximum value of $\sin 3\phi + \cos 3\phi$ is

a. $\sqrt{2}$ b. 90° c. 1 d. 15°

Sol. d. $\sin 3\phi + \cos 3\phi = \sqrt{2}\left(\frac{1}{\sqrt{2}}\cos 3\phi + \frac{1}{\sqrt{2}}\sin 3\phi\right)$

$= \sqrt{2}(\sin 45° \cos 3\phi + \cos 45° \sin 3\phi)$

$= \sqrt{2}\sin(45° + 3\phi)$

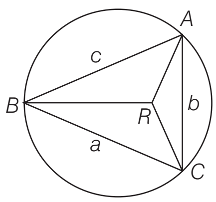
$\therefore$ The maximum value occurs when

$\sin(45° + 3\phi) = 1 = \sin 90°$

$\therefore \quad 45° + 3\phi = 90° \Rightarrow \phi = 15°$

Properties of Triangle

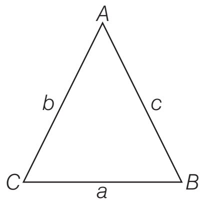
1. **Circumcircle of a triangle** The circle passing through the vertices of a ΔABC is called circumcircle. Its radius R is called the circumradius and its centre is known as circumcentre.



Here, $R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$

2. **Sine rule** In a ΔABC, if a, b, c be the three sides opposite to the angles A, B, C respectively, then

$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$



3. **Cosine rule** In a ΔABC, if a, b, c be the sides opposite to angles A, B and C respectively, then

$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, $\cos B = c^2 + a^2 - b^2$ and

$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

EXAMPLE 22. A triangle ABC is inscribed in a circle. If sum of the squares of sides of the triangle is equal to twice the square of the diameter, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

a. 2 b. 3 c. 4 d. None of these

Sol. a. Let the radius of inscribed circle be R.

$\therefore \quad R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$

$\Rightarrow \sin A = \frac{a}{2R}$, $\sin B = \frac{b}{2R}$ and $\sin C = \frac{c}{2R}$

$\sin^2 A + \sin^2 B + \sin^2 C = \frac{a^2 + b^2 + c^2}{4R^2}$

$\Rightarrow \sin^2 A = \frac{a^2}{4R^2}$, $\sin^2 B = \frac{b^2}{4R^2}$ and $\sin^2 C = \frac{c^2}{4R^2}$

According to the question,

$a^2 + b^2 + c^2 = 2 \times 2R^2 = 8R^2$

$\therefore \quad \sin^2 A + \sin^2 B + \sin^2 C = \frac{8R^2}{4R^2} = 2$

PRACTICE EXERCISE

1. A unit radian is approximately equal to

(a) 57° 17′ 43″ (b) 57° 16′22″
(c) 57° 17′ 47″ (d) 57° 17′ 49″

2. Find the radian measure corresponding to the degree – 47° 30′.

(a) $\frac{-19\pi}{72}$ rad (b) $\frac{17\pi}{72}$ rad
(c) $\frac{13\pi}{72}$ rad (d) $\frac{-15\pi}{72}$ rad

3. The length of a pendulum is 60 cm. The angle through which it swings when its tip describes an arc of length 16.5 cm is

(a) 15°30′ (b) 15°45′ (c) 16°15′ (d) 16°45′

4. A railway train is moving on a circular curve of radius 1500 m at a speed of 90 km/h. Through what angle has it turned in 11 seconds?

(a) 12° (b) 16°30′ (c) 10°30′ (d) 11°40′

5. The angle between the minute hand and the hour hand of a clock when the time is 7:35 pm is

(a) 16°45′ (b) 17°30′ (c) 18°15′ (d) 19°30′

6. When do the hands of a clock coincide between 5 pm and 6 pm?

(a) 5:30 pm (b) 5:27:16 pm
(c) 5:32:16 pm (d) 5:28:56 pm

7. If $\sin A = \frac{1}{3}$, then $\cos A \cdot \text{cosec } A + \tan A \cdot \sec A$ is equal to

(a) $\frac{16\sqrt{2} + 3}{8}$ (b) $\frac{4\sqrt{2} + 3}{8}$
(c) $\frac{\sqrt{3} + 2}{4}$ (d) $\frac{\sqrt{3} - 1}{8}$

8. The value of $\cos 15° - \sin 15°$ is equal to

(a) $\frac{1}{3}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$

9. If $3\tan\theta = 4$, then $\sqrt{\dfrac{1-\sin\theta}{1+\sin\theta}}$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) None of these

10. If $\tan A = 1$ and $\tan B = \sqrt{3}$, then $\cos A \cdot \cos B - \sin A \cdot \sin B$ is equal to

(a) $\frac{1+\sqrt{3}}{2\sqrt{2}}$ (b) $\frac{1-\sqrt{3}}{2\sqrt{2}}$ (c) $\frac{2\sqrt{2}}{3}$ (d) 1

11. The value of $\operatorname{cosec}^2\theta - 2 + \sin^2\theta$ is always

(a) less than zero (b) non-negative
(c) zero (d) 1

12. What is the value of

$$\sin^2 15° + \sin^2 20° + \sin^2 25° + \cdots + \sin^2 75°?$$

(a) 0 (b) $\frac{13}{2}$ (c) 6 (d) $\frac{11}{2}$

13. What is the value of

$$\frac{5\sin 75°\sin 77° + 2\cos 13°\cos 15°}{\cos 15°\sin 77°} - \frac{7\sin 81°}{\cos 9°}?$$

(a) -1 (b) 0 (c) 1 (d) 2

14. What is $\log(\tan 1°) + \log(\tan 2°) + \log(\tan 3°) + \cdots + \log(\tan 89°)$ equal to?

(a) 0 (b) 1 (c) 2 (d) -1

15. If $\tan\theta + \dfrac{1}{\tan\theta} = 2$, then the value of $\tan^2\theta + \dfrac{1}{\tan^2\theta}$ is equal to

(a) 6 (b) 4 (c) 2 (d) 3

16. If $\sec\theta = \dfrac{13}{5}$, then $\dfrac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$ is equal to

(a) $1/3$ (b) -3 (c) 3 (d) $\sqrt{3}$

17. $\sin(45° + A) - \cos(45° - A)$ is equal to

(a) $1/\sqrt{2}$ (b) 1 (c) 0 (d) $\sqrt{2}$

18. What is $\sqrt{\dfrac{1+\sin\theta}{1-\sin\theta}}$ equal to?

(a) $\sec\theta - \tan\theta$ (b) $\sec\theta + \tan\theta$
(c) $\operatorname{cosec}\theta + \cot\theta$ (d) $\operatorname{cosec}\theta - \cot\theta$

19. $\sin(n+1)A\sin(n-1)A + \cos(n+1)A\cos(n-1)A$ is equal to

(a) $\sin 2A$ (b) $\cos 2A$ (c) $\tan 2A$ (d) $\cot 2A$

20. The value of $\dfrac{\cos 11° + \sin 11°}{\cos 11° - \sin 11°}$ is

(a) $\tan 56°$ (b) $\tan 34°$ (c) $\cot 56°$ (d) $\tan 11°$

21. If $\sin\theta = \dfrac{a^2 - b^2}{a^2 + b^2}$, then $\tan\theta$ is equal to

(a) $\frac{a^2+b^2}{2ab}$ (b) $\frac{2ab}{a^2-b^2}$ (c) $\frac{a^2-b^2}{2ab}$ (d) $\frac{ab}{a^2+b^2}$

22. If $\cos\theta = 0.96$, then $\dfrac{1}{\sin\theta} + \dfrac{1}{\tan\theta}$ is equal to

(a) 0.98 (b) 3 (c) 4 (d) 7

23. If $0 < x < 45°$ and $45° < y < 90°$, then which one of the following is correct?

(a) $\sin x = \sin y$ (b) $\sin x < \sin y$
(c) $\sin x > \sin y$ (d) $\sin x \le \sin y$

24. For what value of θ is $(\sin\theta + \operatorname{cosec}\theta) = 2.5$, where $0 < \theta \le 90°$?

(a) 30° (b) 45° (c) 60° (d) 90°

25. If $\dfrac{\cos x}{\cos y} = n$ and $\dfrac{\sin x}{\sin y} = m$, then $(m^2 - n^2)\sin^2 y$ is equal to

(a) $1 - n^2$ (b) $1 + n^2$ (c) m^2 (d) n^2

26. If $p = \tan^2 x + \cot^2 x$, then which one of the following is correct?

(a) $p \le 2$ (b) $p \ge 2$ (c) $p < 2$ (d) $p > 2$

27. The difference of the two angles in degree measure is 1 and their sum in circular measure is also 1. What are the angles in circular measure?

(a) $\left(\frac{1}{2} - \frac{\pi}{360}\right), \left(\frac{1}{2} + \frac{\pi}{360}\right)$ (b) $\left(\frac{1}{2} - \frac{90}{\pi}\right), \left(\frac{1}{2} + \frac{90}{\pi}\right)$

(c) $\left(\frac{1}{2} - \frac{\pi}{180}\right), \left(\frac{1}{2} + \frac{\pi}{180}\right)$ (d) None of these

28. What is the value of $[(1 - \sin^2\theta)\sec^2\theta + \tan^2\theta](\cos^2\theta + 1)$ when $0 < \theta < 90°$?

(a) 2 (b) > 2 (c) ≥ 2 (d) < 2

29. If $7\cos^2\theta + 3\sin^2\theta = 4$ and $0 < \theta < \dfrac{\pi}{2}$, then what is the value of $\tan\theta$?

(a) $\sqrt{7}$ (b) $\frac{7}{3}$ (c) 3 (d) $\sqrt{3}$

30. If $\cos 1° = p$ and $\cos 89° = q$, then which one of the following is correct?

(a) p is close to 0 and q is close to 1
(b) $p < q$
(c) $p = q$
(d) p is close to 1 and q is close to 0

31. If $0 \le \theta < \dfrac{\pi}{2}$ and $p = \sec^2\theta$, then which one of the following is correct?

(a) $p < 1$ (b) $p = 1$ (c) $p > 1$ (d) $p \ge 1$

32. What is the value of $\cos 1°\cos 2°\cos 3° \ldots \cos 90°$?

(a) $\frac{1}{2}$ (b) 0 (c) 1 (d) 2

33. If $\tan A = \dfrac{1 - \cos B}{\sin B}$, then what is $\dfrac{2\tan A}{1 - \tan^2 A}$ equal to?

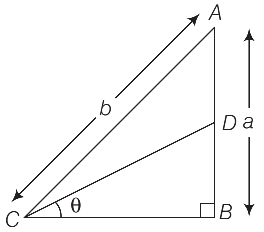
(a) $\frac{\tan B}{2}$ (b) $2\tan B$ (c) $\tan B$ (d) $4\tan B$

34. What is cot 15° cot 20° cot 70° cot 75° equal to?
(a) – 1 (b) 0 (c) 1 (d) 2

35. If $\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}=2$ with $0<\theta<90^\circ$, then what is θ equal to?
(a) 30° (b) 45° (c) 60° (d) 75°

36. In a right ΔABC, right angled at B, the ratio of AB to AC is $1:\sqrt{2}$, then $\frac{2\tan A}{1-\tan^2 A}$ is equal to
(a) 2 (b) 1 (c) 3 (d) undefined

37. In figure, $AD = DB$, $\angle B = 90^\circ$, then $\cos\theta$ is equal to



(a) $\frac{2\sqrt{b^2-a^2}}{\sqrt{4b^2-3a^2}}$ (b) $\frac{a}{\sqrt{4b^2-3a^2}}$
(c) $\frac{\sqrt{4b^2-3a^2}}{2}$ (d) None of these

38. If $0^\circ \le x \le 90^\circ$ and $\sin x+\sqrt{3}\cos x=1$, then what is the value of x?
(a) 30° (b) 45° (c) 60° (d) 90°

39. If $\tan A=\sqrt{2}-1$, then the value of $\text{cosec}\, A\cdot\sec A$ is equal to
(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{2}{\sqrt{2}}$ (c) $2\sqrt{2}$ (d) $\frac{\sqrt{3}}{2}$

40. $\tan\theta=3$ and θ lies in third quadrant, then the value of $\sin\theta$ is
(a) $\frac{1}{\sqrt{10}}$ (b) $-\frac{1}{\sqrt{10}}$ (c) $\frac{-3}{\sqrt{10}}$ (d) $\frac{3}{\sqrt{10}}$

41. The value of $(\sin 20^\circ\cos 70^\circ+\cos 20^\circ\sin 70^\circ)$ is
(a) 1 (b) 0 (c) – 1 (d) $\frac{1}{2}$

42. If ϕ and θ are supplementary angles, then
(a) $\sin\theta=\sin\phi$ (b) $\cos\theta=\cos\phi$
(c) $\tan\theta=\tan\phi$ (d) $\sec\phi=\text{cosec}\,\theta$

43. If $x+y=90^\circ$, then what is the value of $\sqrt{\cos x\ \text{cosec}\ y-\cos x\sin y}$?
(a) $\cos x$ (b) $\sin x$ (c) $\sqrt{\cos x}$ (d) $\sqrt{\sin x}$

44. The value of $\sqrt{2+\sqrt{2+2\cos 4\theta}}$ is
(a) $2\cos^2\theta$ (b) $2\cos 2\theta$ (c) $2\cos\theta$ (d) $2\cos\frac{\theta}{2}$

45. If $\tan A=\frac{5}{6}$ and $\tan B=\frac{1}{11}$, then $A+B$ is equal to
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{2}$ (d) π

46. The value of $\frac{1}{\sqrt{2}}(\cos A-\sin A)$ is
(a) $\cos\left(\frac{\pi}{3}+A\right)$ (b) $\cos\left(\frac{\pi}{2}+A\right)$
(c) $\cos\left(\frac{\pi}{4}+A\right)$ (d) $\sin\left(\frac{\pi}{4}+A\right)$

47. $\sin\frac{\pi}{4}\cos\frac{\pi}{12}-\cos\frac{\pi}{4}\sin\frac{\pi}{12}$ is equal to
(a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$

48. If $\tan\theta=\frac{x\sin\phi}{1-x\cos\phi}$ and $\tan\phi=\frac{y\sin\theta}{1-y\cos\theta}$, then $\frac{x}{y}$ is equal to
(a) $\frac{\sin\phi}{\sin\theta}$ (b) $\frac{\sin\phi}{\cos\theta}$ (c) $\frac{\sin\theta}{\sin\phi}$ (d) $\frac{\sin\theta}{1-\cos\phi}$

49. If $\sin(\theta+\phi)=2\sin(\theta-\phi)$, then
(a) $\cot\phi=3\tan\theta$ (b) $\tan\theta=3\tan\phi$
(c) $\sin\theta=3\sin\phi$ (d) $\sin\phi=\sin 2\theta$

50. If $\sin(\theta+\alpha)=\cos(\theta+\alpha)$, then $\tan\theta$ is equal to
(a) $\frac{1-\tan\alpha}{1+\tan\alpha}$ (b) $\frac{\tan\alpha+1}{\tan\alpha-1}$ (c) $\frac{1-\cot\alpha}{1+\cot\alpha}$ (d) $\frac{\sin 2(\theta+\alpha)}{\cos(\theta+\alpha)}$

51. $\cos^2\frac{\theta-\phi}{2}-\sin^2\frac{\theta+\phi}{2}$ is equal to
(a) $\cos\phi\sin\theta$ (b) $\cos 2\theta\sin\phi$
(c) $\cos\theta\cos\phi$ (d) $\sin\theta\sin\phi$

52. If $x=r\sin\theta\cos\phi$, $y=r\sin\theta\sin\phi$, $z=r\cos\theta$, then $x^2+y^2+z^2$ is equal to
(a) $r^2\cos^2\phi$ (b) $r^2\sin^2\theta+r^2\cos^2\phi$
(c) r^2 (d) $\frac{1}{r^2}$

53. If $\cot\theta=\frac{7}{24}$ and $\pi<\theta<\frac{3\pi}{2}$, then the value of $\cos\theta-\sin\theta$ is
(a) $\frac{19}{25}$ (b) $\frac{18}{25}$
(c) $\frac{17}{25}$ (d) $\frac{18}{25}$

54. If $\alpha+\beta=\frac{\pi}{2}$ and $\sin\alpha=\frac{1}{2}$, then β is
(a) 30° (b) 60° (c) 45° (d) $22\frac{1^\circ}{2}$

55. If $\sin x-\cos x=0$, then what is the value of $\sin^4 x+\cos^4 x$?
(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

56. If $3\cos x = 5\sin x$, then the value of $\dfrac{5\sin x - 2\sec^3 x + 2\cos x}{5\sin x + 2\sec^3 x - 2\cos x}$ is

(a) $\dfrac{361}{2397}$ (b) $\dfrac{271}{979}$ (c) $\dfrac{541}{979}$ (d) $\dfrac{127}{979}$

57. $\sqrt{\dfrac{1+\cos x}{1-\cos x}}$ is equal to

(a) $\sec x + \tan x$ (b) $\operatorname{cosec} x + \cot x$
(c) $\sec x - \tan x$ (d) $\operatorname{cosec} x - \cot x$

58. $\cos^4 x - \sin^4 x$ is equal to

(a) $2\cos^2 x - 1$ (b) $2\sin^2 x - 1$
(c) $\sin^2\theta - \cos^2\theta$ (d) None of these

59. $\dfrac{\sin\theta}{1+\cos\theta}$ is equal to

(a) $\dfrac{1-\cos\theta}{\sin\theta}$ (b) $\dfrac{\sin\theta}{\cos\theta}$ (c) $\dfrac{\cos\theta - 1}{\sin\theta}$ (d) $\dfrac{1+\sin\theta}{\cos\theta}$

60. If $0 \le \theta \le 90°$, then $\left(\dfrac{5\cos\theta - 4}{3 - 5\sin\theta} - \dfrac{3 + 5\sin\theta}{4 + 5\cos\theta}\right)$ is equal to

(a) 0 (b) 1 (c) $\dfrac{1}{4}$ (d) $\dfrac{1}{2}$

61. If $\cos(\alpha+\beta) = \dfrac{4}{5}$ and $\sin(\alpha-\beta) = \dfrac{5}{13}, \alpha, \beta$ lies between 0 and $\dfrac{\pi}{4}$, then the value of $\tan 2\alpha$ is

(a) $\dfrac{56}{33}$ (b) $\dfrac{56}{23}$ (c) $\dfrac{43}{33}$ (d) $\dfrac{34}{33}$

62. $\dfrac{\tan(45° + x)}{\tan(45° - x)}$ is equal to

(a) $\left(\dfrac{1+\tan x}{1-\tan x}\right)^2$ (b) $\left(\dfrac{1-\tan x}{1+\tan x}\right)^2$

(c) $\dfrac{2\sin^2\frac{x}{2}}{1+\cos^2\frac{x}{2}}$ (d) None of these

63. What is the simplest value of $\dfrac{(1-\sin A\cos A)(\sin^2 A - \cos^2 A)}{\cos A(\sec A - \operatorname{cosec} A)(\sin^3 A + \cos^3 A)}$?

(a) $\sin A$ (b) $\cos A$ (c) $\sec A$ (d) $\operatorname{cosec} A$

64. If $A + B = 45°$, then $(1+\tan A)(1+\tan B)$ is equal to

(a) 1 (b) -1 (c) 2 (d) -2

65. $\left(\dfrac{\cos 2B - \cos 2A}{\sin 2B + \sin 2A}\right)$ is equal to

(a) $\tan(A-B)$ (b) $\tan(A+B)$ (c) $\cot(A-B)$ (d) $\cot(A+B)$

66. $\sec^2 x + \tan^2 x = 7$, the value of x is

(a) 15° (b) 30° (c) 45° (d) 60°

67. $\cos 3\theta + \sin 3\theta$ is maximum, when θ is

(a) 15° (b) 30° (c) 45° (d) 60°

68. If $\sin^2 x + \sin^2 y + \sin^2 z = (\sin x + \sin y + \sin z)^2$, then which of the following expressions must necessarily vanish?

(a) $\tan x + \tan y + \tan z$ (b) $\cos x + \cos y + \cos z$
(c) $\dfrac{1}{\sin x} + \dfrac{1}{\sin y} + \dfrac{1}{\sin z}$ (d) $\dfrac{1}{\cos x} + \dfrac{1}{\cos y} + \dfrac{1}{\cos z}$

69. If $\sin x + \sin^2 x = 1$, then the value of $\cos^2 x + \cos^4 x$ is equal to

(a) -1 (b) 2 (c) -2 (d) 1

70. $\cos 35° + \cos 85° + \cos 155°$ is

(a) 0 (b) $\dfrac{1}{\sqrt{3}}$ (c) $\dfrac{1}{\sqrt{2}}$ (d) $\cos 275°$

71. $\sin 12° \sin 24° \sin 48° \sin 84°$ is

(a) $\dfrac{1}{16}$ (b) $\dfrac{3}{64}$ (c) $\dfrac{3}{15}$ (d) 0

72. If $\tan\theta = \dfrac{p}{q}$, then the value of $p\cos 2\theta + q\sin 2\theta$ is

(a) p (b) q
(c) $\dfrac{q(3q^2 - p^2)}{p^2 + q^2}$ (d) $\dfrac{p(3q^2 - p^2)}{p^2 + q^2}$

73. If $x = y\cos\dfrac{2\pi}{3} = z\cos\dfrac{4\pi}{3}$, then $xy + yz + zx$ is equal to

(a) 1 (b) -1 (c) 0 (d) 2

74. If $\sin\theta + \cos\theta = m$ and $\sec\theta + \operatorname{cosec}\theta = n$, then $n(m^2 - 1)$ is equal to

(a) $\dfrac{1}{2m}$ (b) m (c) $2m$ (d) $\dfrac{m}{2}$

75. If $\operatorname{cosec}\theta - \sin\theta = a^3$, $\sec\theta - \cos\theta = b^3$, then $a^2b^2(a^2 + b^2)$ is equal to

(a) -1 (b) 1 (c) 2 (d) -2

76. The maximum value and minimum value of $(1 + \cos 2x)$ are

(a) -1 and 1 (b) 1 and 2 (c) $\dfrac{-1}{2}$ and $\dfrac{1}{2}$ (d) 0 and 2

77. If $\tan\theta + \tan\phi = a$ and $\cot\theta + \cot\phi = b$, then $\cot(\theta + \phi)$ is equal to

(a) $\dfrac{1}{a} + \dfrac{1}{b}$ (b) $\dfrac{1}{a} - \dfrac{1}{b}$
(c) $a - b$ (d) $a + b$

78. $\tan 75° - \tan 30° - \tan 75° \tan 30°$ is equal to

(a) -1 (b) 1 (c) 0 (d) 2

79. The value of $\sin^2(15° + A) - \sin^2(15° - A)$ is equal to

(a) $\dfrac{1}{2}\cos 2A$ (b) $\dfrac{1}{2}\sin 2A$ (c) $\dfrac{1}{2}\tan 2A$ (d) $\cot 2A$

80. The value of $\frac{\sin 38° - \cos 68°}{\cos 68° + \sin 38°}$ is equal to

(a) $\sqrt{3} \tan 8°$ (b) $\sqrt{3} \cot 8°$
(c) $\sqrt{3} \sin 38°$ (d) $\sqrt{3} \sin 8°$

81. $2 \cos x - \cos 3x - \cos 5x$ is equal to

(a) $8 \cos^3 x \sin^2 x$ (b) $12 \cos^2 x \sin^3 x$
(c) $16 \cos^3 x \sin^2 x$ (d) $32 \cos^3 x \sin x$

82. $\cos 4x$ is equal to

(a) $1 + 2 \sin^2 2x$ (b) $2 \cos^2 2x$
(c) $1 - 8 \sin^2 x \cos^2 x$ (d) $1 + 8 \sin^2 x \cos^2 x$

83. $2 \sin A \cos^3 A - 2 \sin^3 A \cos A$ is equal to

(a) $\frac{1}{2} \sin 4A$ (b) $\frac{1}{2} \cos 4A$ (c) $\frac{1}{2} \tan 4A$ (d) $\frac{1}{2} \cot 4A$

84. If $a \cos \theta - b \sin \theta = c$, then what is the value of $a \sin \theta + b \cos \theta$?

(a) $\pm \sqrt{a^2 + b^2 + c^2}$ (b) $\pm \sqrt{a^2 - b^2 + c^2}$
(c) $\pm \sqrt{a^2 + b^2 - c^2}$ (d) $\pm \sqrt{a^2 - b^2 - c^2}$

85. If $\cos x + \cos^2 x = 1$, then the value of $\sin^8 x + 2 \sin^6 x + \sin^4 x$ is

(a) 0 (b) -1 (c) 2 (d) 1

86. If $m = \operatorname{cosec} x - \sin x$ and $n = \sec x - \cos x$, then $\tan x$ is equal to

(a) $\left(\frac{n}{m}\right)^{2/3}$ (b) $\left(\frac{n}{m}\right)^{1/3}$
(c) $\left(\frac{n}{m}\right)$ (d) $\left(\frac{n}{m}\right)^2$

87. If $2 \cos \theta = x + \frac{1}{x}$, then $2 \cos 3\theta$ is equal to

(a) $x^3 + \frac{1}{x^3}$ (b) $x^2 + \frac{1}{x^2}$ (c) $\frac{x^2}{1 + x^2}$ (d) None of these

88. If $\cos \theta \geq \frac{1}{2}$ in the first quadrant, then which one of the following is correct?

(a) $\theta \leq \frac{\pi}{3}$ (b) $\theta \geq \frac{\pi}{3}$ (c) $\theta \leq \frac{\pi}{6}$ (d) $\theta \geq \frac{\pi}{6}$

89. If $\sin A = \frac{3}{5}$ and $\cos B = \frac{12}{13}$, then the value of $\frac{\tan A - \tan B}{1 + \tan A \tan B}$ is equal to

(a) $\frac{23}{16}$ (b) $\frac{16}{63}$ (c) $\frac{1}{63}$ (d) $\frac{13}{63}$

90. If $B + C = 60°$, then which of the following is correct statement?

(a) $\sin (120° - B) = \sin (120° - C)$
(b) $\sin (120° + B) = \sin (180° + C)$
(c) $\cos (120° - B) = \sin (120° + C)$
(d) $\tan (B) = \tan (120° + C)$

91. What is the value of x in the equation
$x \frac{\operatorname{cosec}^2 30° \sec^2 45°}{8 \cos^2 45° \sin^2 60°} = \tan^2 60° - \tan^2 30°$?

(a) $x = 1$ (b) $x = 2$ (c) $x = \frac{1}{2}$ (d) $x = \frac{3}{2}$

92. Under which one of the following conditions is the trigonometrical identity $\sin x/(1 + \cos x) = (1 - \cos x)/\sin x$ true?

(a) x is not a multiple of 360°
(b) x is not an odd multiple of 180°
(c) x is not a multiple of 180°
(d) None of the above

93. If $\cos x = k \cos (x - 2y)$, then $\tan (x - y) \tan y$ is equal to

(a) $\frac{1+k}{1-k}$ (b) $\frac{1-k}{1+k}$ (c) $\frac{2k}{k+1}$ (d) $\frac{k-1}{2k+1}$

94. The value of $\frac{\sin (x + y) - 2 \sin x + \sin (x - y)}{\cos (x + y) - 2 \cos x + \cos (x - y)}$ is

(a) $\cot x$ (b) $\tan x$ (c) $\sin x$ (d) $\operatorname{cosec} x$

95. If $\sin x + \sin y = a$ and $\cos x + \cos y = b$, then $\tan \frac{x + y}{2}$ is

(a) $\frac{4}{a^2 + b^2}$ (b) $\frac{b}{a}$ (c) $\frac{a}{b}$ (d) $\frac{4}{a^2 - b^2}$

96. The largest hand of a clock is 42 cm long, then the distance covered by the extremity in 20 min is

(a) 88 cm (b) 80 cm (c) 82 cm (d) 84 cm

97. If $0° < x < 90°$ and $2 \sin x + 15 \cos^2 x = 7$, then what is the value of $\tan x$?

(a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) $\frac{3}{4}$ (d) $\frac{4}{3}$

98. If ABC is a triangle and $A + B + C = 180°$, then $\tan A + \tan B + \tan C$ is

(a) $\tan A \tan B \tan C$ (b) $\tan \frac{A + B}{2} + \tan \frac{B + C}{2}$
(c) $\frac{\tan A + \tan B}{\tan C}$ (d) $\cot A \cot B \cot C$

99. If $x = a \sec \theta \cos \phi$, $y = b \sec \theta \sin \phi$ and $z = c \tan \theta$, then the value of $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$ is

(a) 9 (b) 0 (c) 1 (d) 4

100. Which one of the following statements is correct?

(a) The squares of the tangents of the angles 30°, 45°, 60° are in G.P.
(b) The squares of the sines of the angles 30°, 45°, 60° are in G.P.
(c) The squares of the secants of the angles 30°, 45°, 60° are in A.P.
(d) The squares of the tangents of the angles 30°, 45°, 60° are in A.P.

101. If A, B, C and D are the successive angles of a cyclic quadrilateral, then what is $\cos A + \cos B + \cos C + \cos D$ are equal to?

(a) 4 (b) 2 (c) 1 (d) 0

102. Which one of the following is correct?

(a) There is only one θ with $0° < \theta < 90°$ such that $\sin\theta = a$, where a is a real number
(b) There is more than one θ with $0° < \theta < 90°$ such that $\sin\theta = a$, where a is a real number
(c) There is no θ with $0° < \theta < 90°$ such that $\sin\theta = a$, where a is a real number
(d) There are exactly two θ's with $0° < \theta$ $90°$ such that $\sin\theta = a$, where a is a real number

103. If $\sin(B + C - A) = \cos(C + A - B) = \tan(A + B - C) = 1$, then the angles A, B, C which are positive acute angles are, respectively

(a) 45°, 80°, 105° (b) $22\frac{1°}{2}$, $67\frac{1°}{2}$, 45°
(c) 20°, 70°, 90° (d) 30°, 60°, 90°

104. If A, B, C and D be the angles of a cyclic quadrilateral taken in order, then $\cos(180° + A) + \cos(180° + B) + \cos(180° + C) + \cos(180° + D)$ is equal to

(a) $2(\cos A + \cos B)$ (b) $2(\cos A + \cos D)$
(c) 0 (d) $4\cos A$

105. If ABC is a right angled triangle at C and having u units, v units and w units as the lengths of its sides opposite to be vertices A, B and C respectively, then what is $\tan A + \tan B$ equal to?

(a) $\frac{u^2}{vw}$ (b) 1 (c) $u + v$ (d) $\frac{w^2}{uv}$

106. The Earth takes 24 h to rotate about its own axis. Through what angle will it turn in 4 h and 12 min?

(a) 63° (b) 64° (c) 65° (d) 70°

107. Assume the Earth to be a sphere of radius R. What is the radius of the circle of latitude 40°S?

(a) $R\cos 40°$ (b) $R\sin 80°$
(c) $R\sin 40°$ (d) $R\tan 40°$

108. If $\sin 3\theta = \cos(\theta - 2°)$, where 3θ and $(\theta - 2°)$ are acute angles, then what is the value of θ?

(a) 22° (b) 23°
(c) 24° (d) 25°

109. Consider the following statements:

I. $\frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin^2\theta} = \cos^2\theta\,(1 + \tan\theta)(1 - \tan\theta)$

II. $\frac{1 + \sin\theta}{1 - \sin\theta} = (\tan\theta + \sec\theta)^2$

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

110. If $A = \frac{\pi}{6}$ and $B = \frac{\pi}{3}$, then which of the following is/are correct?

I. $\sin A + \sin B = \cos A + \cos B$
II. $\tan A + \tan B = \cot A + \cot B$

Select the correct answer using the codes given below

(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II

111. For any quadrilateral $ABCD$ which of the following statements are true?

I. $\sin(A + B) + \sin(C + D) = 0$
II. $\cos(A + B) + \cos(C + D) = 0$

Select the correct answer using the codes given below

(a) Only I (b) Only II
(c) Both I and II (d) None of these

112. Consider the following statements

I. 1° in radian measure is less than 0.03 radians.
II. 1 radian in degree measure is greater than 45°.

Which of the above statement is/are correct?

(a) Only II (b) Only I
(c) Neither I nor II (d) Both I of II

113. Consider the following statements

I. $\sin\theta \cdot \sin(60° + \theta) \cdot \sin(60° - \theta) = \frac{1}{4}\sin 3\theta$

II. $\cos\theta \cdot \sin(30° + \theta) \cdot \sin(30° - \theta) = \frac{1}{4}\cos 3\theta$

III. $\sin\theta \cdot \cos(30° + \theta) \cdot \cos(30° - \theta) = \frac{1}{4}\sin 3\theta$

Which of the above statement is/are correct?

(a) Only I
(b) Only II
(c) Only III
(d) All of the above

114. If ABC is a right angled triangle, then which of the following statements are correct?

I. $\sin(A + B) = \sin C$

II. $\sin\left(\frac{A + B}{2}\right) = \cos\frac{C}{2}$

III. $\tan\frac{(A + B - C)}{2} = \cot C$

IV. $\tan\frac{(A - B - C)}{2} = -\cot A$

Select the correct answer using the codes given below

(a) I and II (b) I, II and III
(c) I, II and IV (d) All of these

Directions. Q. Nos. (115-117) *Let* $\sin(A+B)=1$ *and* $\sin(A-B)=\frac{1}{2}$, *where* $A, B \in \left[0, \frac{\pi}{2}\right]$.

115. What is the value of A?

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{8}$

116. What is the value of $\tan(A+2B)\tan(2A+B)$?

(a) –1 (b) 0 (c) 1 (d) 2

117. What is the value of $\sin^2 A - \sin^2 B$?

(a) 0 (b) 1/2 (c) 1 (d) 2

Directions. Q. Nos. (118-120) *ABC is an obtuse triangle and tanA, tanB are the roots of the equation* $3x^2 - 2\sqrt{3}x + 1 = 0$.

118. The measure of angle C is

(a) $\pi/3$ (b) $\pi/2$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

119. Find the value of $\sin C + \cos C$.

(a) $\frac{2}{\sqrt{3}+1}$ (b) $\frac{\sqrt{3}+1}{2}$ (c) $\frac{\sqrt{3}-1}{2}$ (d) $\sqrt{3}+\frac{1}{2}$

120. If $2\sin B = \sqrt{\frac{1+\tan^2 C}{\operatorname{cosec}^2 C}}$, then the value of B is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) $\frac{\pi}{6}$

PREVIOUS YEARS' QUESTIONS

121. The expression $\sin^2 x + \cos^2 x - 1 = 0$ is satisfied by how many values of x? **2012 I**

(a) Only one value of x (b) Two values of x
(c) Infinite values of x (d) No value of x

122. If $3\sin x + 5\cos x = 5$, then what is the value of $(3\cos x - 5\sin x)$? **2012 I**

(a) 0 (b) 2 (c) 3 (d) 5

123. If $p = a\sin x + b\cos x$ and $q = a\cos x - b\sin x$, then what is the value of $p^2 + q^2$? **2012 I**

(a) $a+b$ (b) ab (c) a^2+b^2 (d) a^2-b^2

124. If α and β are complementary angles, then what is $\sqrt{\operatorname{cosec}\alpha \cdot \operatorname{cosec}\beta}\left(\frac{\sin\alpha}{\sin\beta}+\frac{\cos\alpha}{\cos\beta}\right)^{-\frac{1}{2}}$ equal to? **2012 I**

(a) 0 (b) 1
(c) 2 (d) None of these

125. Consider the following statements

I. $\frac{\cot 30^\circ + 1}{\cot 30^\circ - 1} = 2(\cos 30^\circ + 1)$

II. $2\sin 45^\circ \cos 45^\circ - \tan 45^\circ \cot 45^\circ = 0$

Which of the above statement(s) is/are correct? **2012 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

126. Consider the following statements

I. $\sin^2 1^\circ + \cos^2 1^\circ = 1$

II. $\sec^2 33^\circ - \cot^2 57^\circ = \operatorname{cosec}^2 37^\circ - \tan^2 53^\circ$

Which of the above statement(s) given above is/are correct? **2012 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

127. Consider the following statements:

I. There is only one value of x in the first quadrant that satisfies $\sin x + \cos x = 2$.

II. There is only one value of x in the first quadrant that satisfies $\sin x - \cos x = 0$.

Which of the above statement(s) given above is/are correct? **2012 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

128. If $\sin\theta\cos\theta = \sqrt{3}/4$, then $\sin^4\theta + \cos^4\theta$ is equal to **2012 II**

(a) 7/8 (b) 5/8 (c) 3/8 (d) 1/8

129. If the angle θ is in the first quadrant and $\tan\theta = 3$, then what is the value of $(\sin\theta + \cos\theta)$? **2012 II**

(a) $\frac{1}{\sqrt{10}}$ (b) $\frac{2}{\sqrt{10}}$ (c) $\frac{3}{\sqrt{10}}$ (d) $\frac{4}{\sqrt{10}}$

130. If $0^\circ < \theta < 90^\circ$, then all the trigonometric ratios can be obtained when **2012 II**

(a) only $\sin\theta$ is given
(b) only $\cos\theta$ is given
(c) only $\tan\theta$ is given
(d) any one of the six ratios is given

131. What is the value of $\sin A \cos A \tan A + \cos A \sin A \cot A$? **2012 II**

(a) $\sin^2 A + \cos^2 A$ (b) $\sin^2 A + \tan^2 A$
(c) $\sin^2 A + \cot^2 A$ (d) $\operatorname{cosec}^2 A + \cot^2 A$

132. What is the value of $\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}$? **2012 II**

(a) $2\operatorname{cosec}\theta$ (b) $2\sec\theta$
(c) $\sec\theta$ (d) $\operatorname{cosec}\theta$

133. What is the value of $\sec^2 D - \tan^2 D$? **2013 I**

(a) 1/2 (b) 2/3
(c) 1 (d) None of these

134. If $\cos A + \cos^2 A = 1$, then what is the value of $2(\sin^2 A + \sin^4 A)$? **2013 I**

(a) 4 (b) 2 (c) 1 (d) 1/2

135. $(1-\tan A)^2+(1+\tan A)^2+(1-\cot A)^2+(1+\cot A)^2$ is equal to **2013 I**

(a) $\sin^2 A\cos^2 A$ (b) $\sec^2 A\,\text{cosec}^2 A$
(c) $2\sec^2 A\,\text{cosec}^2 A$ (d) None of these

136. If $a^2=\dfrac{1+2\sin\theta\cos\theta}{1-2\sin\theta\cos\theta}$, then what is the value of $\dfrac{a+1}{a-1}$? **2013 I**

(a) $\sec\theta$ (b) 1 (c) 0 (d) $\tan\theta$

137. If $\sin\theta=\dfrac{x^2-y^2}{x^2+y^2}$, then which one of the following is correct? **2013 I**

(a) $\cos\theta=\dfrac{2xy}{x^2-y^2}$ (b) $\cos\theta=\dfrac{2xy}{x^2+y^2}$
(c) $\cos\theta=\dfrac{x-y}{x^2+y^2}$ (d) $\cos\theta=\dfrac{xy(x-y)}{x^2+y^2}$

138. Consider the following statements for $0\le\theta\le 90°$:

I. The value of $\sin\theta+\cos\theta$ is always greater than 1.
II. The value of $\tan\theta+\cot\theta$ is always greater than 1.

Which of the statement(s) given above is/are correct? **2013 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 139-141) *Read the following information carefully to answer the questions that follow.*

The angles A, B, C and D of a quadrilateral $ABCD$ are in the ratio 1 : 2 : 4 : 5.

139. What is the value of $\cos(A+B)$? **2013 I**

(a) 0 (b) 1/2
(c) 1 (d) None of these

140. What is the value of $\text{cosec}\,(C-D+B)$? **2013 I**

(a) 1 (b) 2 (c) 3 (d) 4

141. Consider the following statements:

I. $ABCD$ is a cyclic quadrilateral.
II. $\sin(B-A)=\cos(D-C)$

Which of the statement(s) given above is/are correct? **2013 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

142. If $\sin\theta+\cos\theta=\sqrt{3}$, then what is $\tan\theta+\cot\theta$ equal to? **2013 II**

(a) 1 (b) $\sqrt{2}$ (c) 2 (d) $\sqrt{3}$

143. What is $\text{cosec}\,(75°+\theta)-\sec(15°-\theta)$ equal to? **2013 II**

(a) 0 (b) 1 (c) $2\sin\theta$ (d) $2\cos\theta$

144. If $5\sin\theta+12\cos\theta=13$, then what is $5\cos\theta-12\sin\theta$ equal to? **2013 II**

(a) -2 (b) -1 (c) 0 (d) 1

145. If $\sin\theta-\cos\theta=0$, then what is $\sin^4\theta+\cos^4\theta$ equal to? **2013 II**

(a) 1 (b) $\frac{3}{4}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

146. What is $\dfrac{\cos^2(45°+\theta)+\cos^2(45°-\theta)}{\tan(60°+\theta)\tan(30°-\theta)}$ equal to? **2013 II**

(a) -1 (b) 0 (c) 1 (d) 2

147. If $\tan\theta+\sec\theta=m$, then what is $\sec\theta$ equal to? **2013 II**

(a) $\dfrac{m^2-1}{2m}$ (b) $\dfrac{m^2+1}{2m}$ (c) $\dfrac{m+1}{m}$ (d) $\dfrac{m^2+1}{m}$

148. If α,β and γ are acute angles such that $\sin\alpha=\dfrac{\sqrt{3}}{2}$, $\cos\beta=\dfrac{\sqrt{3}}{2}$ and $\tan\gamma=1$, then what is $\alpha+\beta+\gamma$ equal to? **2013 II**

(a) 105° (b) 120°
(c) 135° (d) 150°

149. What is $\dfrac{(1+\sec\theta-\tan\theta)\cos\theta}{(1+\sec\theta+\tan\theta)(1-\sin\theta)}$ equal to? **2013 II**

(a) 1 (b) 2 (c) $\tan\theta$ (d) $\cot\theta$

150. If ΔABC is right angled at C, then what is $\cos(A+B)+\sin(A+B)$ equal to? **2013 II**

(a) 0 (b) $\frac{1}{2}$ (c) 1 (d) 2

151. Consider the following statements:

I. $\tan\theta$ increases faster than $\sin\theta$ as θ increases.
II. The value of $\sin\theta+\cos\theta$ is always greater than 1.

Which of the statement(s) given above is/are correct? **2013 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

152. The value of $\cos 25°-\sin 25°$ is **2014 I**

(a) positive but less than 1
(b) positive but greater than 1
(c) negative
(d) 0

153. In a right angled ΔABC, right angle at B, if $\cos A=\dfrac{4}{5}$, then what is $\sin C$ is equal to? **2014 I**

(a) $\frac{3}{5}$ (b) $\frac{4}{5}$ (c) $\frac{3}{4}$ (d) $\frac{2}{5}$

154. If α and β are complementary angles, then what is $\sqrt{\cos\alpha \operatorname{cosec}\beta - \cos\alpha \sin\beta}$ equal to? **2014 I**
(a) $\sec\beta$ (b) $\cos\alpha$ (c) $\sin\alpha$ (d) $-\tan\beta$

155. If $\sec\theta + \tan\theta = 2$, then what is the value of $\sec\theta$? **2014 I**
(a) $\frac{3}{2}$ (b) $\sqrt{2}$ (c) $\frac{5}{2}$ (d) $\frac{5}{4}$

156. What is $\operatorname{cosec}(75° + \theta) - \sec(15° - \theta) - \tan(55° + \theta) + \cot(35° - \theta)$ equal to? **2014 I**
(a) -1 (b) 0 (c) 1 (d) $\frac{3}{2}$

157. What is $\sin 25° \sin 35° \sec 65° \sec 55°$ equal to? **2014 I**
(a) -1 (b) 0 (c) $\frac{1}{2}$ (d) 1

158. If $2\cot\theta = 3$, then what is $\frac{2\cos\theta - \sin\theta}{2\cos\theta + \sin\theta}$ equal to? **2014 I**
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

159. If $\sin\theta\cos\theta = \frac{1}{2}$, then what is $\sin^6\theta + \cos^6\theta$ equal to? **2014 I**
(a) 1 (b) 2 (c) 3 (d) $\frac{1}{4}$

160. If $\cos x + \sec x = 2$, then what $\cos^n x + \sec^n x$ equal to, where n is a positive integer? **2014 I**
(a) 2 (b) 2^{n-2} (c) 2^{n-1} (d) 2^n

161. If $\sin\theta + 2\cos\theta = 1$, where $0 < \theta < \pi/2$, then what is $2\sin\theta - \cos\theta$ equal to? **2014 I**
(a) -1 (b) $1/2$ (c) 2 (d) 1

162. If $\tan 8\theta = \cot 2\theta$, where $0 < 8\theta < \frac{\pi}{2}$, then what is the value of $\tan 5\theta$? **2014 I**
(a) $\frac{1}{\sqrt{3}}$ (b) 1 (c) $\sqrt{3}$ (d) 0

163. If $\sin(A + B) = 1$, where $0 < B < 45°$, then what is $\cos(A - B)$ equal to? **2014 I**
(a) $\sin 2B$ (b) $\sin B$ (c) $\cos 2B$ (d) $\cos B$

164. At what point of time after 3 O'clock, hour hand and the minute hand of a clock occur at right angles for the first time? **2014 I**
(a) 9 O'clock (b) 4 h $37\frac{1}{6}$ min
(c) 3 h $30\frac{8}{11}$ min (d) 3 h $32\frac{8}{11}$ min

165. If $\tan\theta + \cot\theta = 2$, then what is $\sin\theta + \cos\theta$ equal to? **2014 II**
(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\sqrt{2}$ (d) 1

166. What is $\frac{\sec x}{\cot x + \tan x}$ equal to? **2014 II**
(a) $\sin x$ (b) $\cos x$ (c) $\tan x$ (d) $\cot x$

167. What is $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$ equal to? **2014 II**
(a) 1 (b) 2 (c) $\sin x$ (d) $\cos x$

168. What is $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$ equal to? **2014 II**
(a) $\sin x + \cos x$ (b) $\sin x - \cos x$
(c) 2 (d) 1

169. What is $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$ equal to? **2014 II**
(a) $\frac{\sin x - 1}{\cos x}$ (b) $\frac{\sin x + 1}{\cos x}$ (c) $\frac{\sin x - 1}{\cos x + 1}$ (d) $\frac{\sin x + 1}{\cos x + 1}$

170. What is $(\sin^2 x - \cos^2 x)\ (1 - \sin^2 x \cos^2 x)$ equal to? **2014 II**
(a) $\sin^4 x - \cos^4 x$ (b) $\sin^6 x - \cos^6 x$
(c) $\cos^8 x - \sin^8 x$ (d) $\sin^8 x - \cos^8 x$

171. What is $(\sin x \cos y + \cos x \sin y)$ $(\sin x \cos y - \cos x \sin y)$ equal to? **2014 II**
(a) $\cos^2 x - \cos^2 y$ (b) $\cos^2 x - \sin^2 y$
(c) $\sin^2 x - \cos^2 y$ (d) $\sin^2 x - \sin^2 y$

172. If $A + B + C = 180°$, then $\cot A \cot B + \cot B \cot C + \cot C \cot A$ is equal to **2014 II**
(a) -1 (b) 2 (c) π (d) 1

173. If $\sin x + \operatorname{cosec} x = 2$, then what is $\sin^9 x + \operatorname{cosec}^9 x$ equal to? **2014 II**
(a) 2 (b) 18 (c) 512 (d) 1024

174. If $\sin x + \cos x = p$ and $\sin^3 x + \cos^3 x = q$, then what is $p^3 - 3p$ equal to **2014 II**
(a) 0 (b) $-2q$ (c) $2q$ (d) $4q$

175. Consider the following statements **2014 II**
I. $\sin 1° > \sin 1$ II. $\cos 1° < \cos 1$
Which of the statement(s) given above is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

176. The value of $\operatorname{cosec}^2 67° + \sec^2 57° - \cot^2 33° - \tan^2 23°$ is **2015 I**
(a) $2\sqrt{2}$ (b) 2 (c) $\sqrt{2}$ (d) 0

177. If $\tan(A + B) = \sqrt{3}$ and $\tan A = 1$, then $\tan(A - B)$ is equal to **2015 I**
(a) 0 (b) 1 (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{2}$

178. If $\tan A + \cot A = 4$, then $\tan^4 A + \cot^4 A$ is equal to **2015 I**
(a) 110 (b) 191 (c) 80 (d) 194

179. ABC is a triangle right angled at B and $AB:BC = 3:4$. What is $\sin A + \sin B + \sin C$ equal to? 2015 I

(a) 2 (b) $\frac{11}{5}$ (c) $\frac{12}{5}$ (d) 3

180. If $\sin x + \cos x = C$, then $\sin^6 x + \cos^6 x$ is equal to 2015 I

(a) $\frac{1+6C^2-3C^4}{16}$ (b) $\frac{1+6C^2-3C^4}{4}$

(c) $\frac{1+6C^2+3C^4}{16}$ (d) $\frac{1+6C^2+3C^4}{4}$

181. If $\frac{3-\tan^2 A}{1-3\tan^2 A} = k$, where k is a real number, then $\operatorname{cosec} A(3\sin A - 4\sin^3 A)$ is equal to 2015 I

(a) $\frac{2k}{k-1}$ (b) $\frac{2k}{k-1}$, where $\frac{1}{3} \le k \le 3$

(c) $\frac{2k}{k-1}$, where $k < \frac{1}{3}$ or $k > 3$ (d) $\frac{2k}{k+1}$

182. If $p = \sqrt{\frac{1-\sin x}{1+\sin x}}$, $q = \frac{1-\sin x}{\cos x}$, $r = \frac{\cos x}{1+\sin x}$, then which of the following is/are correct? 2015 I

I. $p = q = r$ II. $p^2 = qr$

Select the correct answer using the codes given below

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

183. Consider the following identity

I. $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \sin A + \cos A$

II. $(1-\sin A - \cos A)^2 = 2(1-\sin A)(1+\cos A)$

Which of the above identity/identities is/are correct?

(a) Only I (b) Only II 2015 I

(c) Both I and II (d) Neither I nor II

184. If a and b are positive, then the relation $\sin\theta = \frac{2a+3b}{3b}$ is 2015 II

(a) not possible (b) possible, if $a = b$

(c) possible, if $a > b$ (d) possible, if $a < b$

185. The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is 2015 II

(a) – 1 (b) 0

(c) 2 (d) 2

186. If $\tan\theta + \sec\theta = 2$, then $\tan\theta$ is equal to 2015 II

(a) $\frac{3}{4}$ (b) $\frac{5}{4}$ (c) $\frac{3}{2}$ (d) $\frac{5}{2}$

187. $\frac{\cos\theta}{1-\sin\theta}$ is equal to $\left(\text{where}, \theta \ne \frac{\pi}{2}\right)$ 2015 II

(a) $\frac{\tan\theta - 1}{\tan\theta + 1}$ (b) $\frac{1+\sin\theta}{\cos\theta}$ (c) $\frac{\tan\theta + 1}{\tan\theta - 1}$ (d) $\frac{1+\cos\theta}{\sin\theta}$

188. If $\tan(x+40)^\circ \tan(x+20)^\circ \tan(3x)^\circ \tan(70-x)^\circ \tan(50-x)^\circ = 1$, then the value of x is equal to 2015 II

(a) 30 (b) 20 (c) 15 (d) 10

189. The value of $32\cot^2\left(\frac{\pi}{4}\right) - 8\sec^2\left(\frac{\pi}{3}\right) + 8\cos^3\left(\frac{\pi}{6}\right)$ is equal to 2015 II

(a) $\sqrt{3}$ (b) $2\sqrt{3}$ (c) 3 (d) $3\sqrt{3}$

190. If $x = a\cos\theta$ and $y = b\cot\theta$, then $(ax^{-1} - by^{-1})(ax^{-1} + by^{-1})$ is equal to 2015 II

(a) 0 (b) 1 (c) $\tan^2\theta$ (d) $\sin^2\theta$

191. If θ is an acute angle and $\sin\theta\cos\theta = 2\cos^3\theta - 1.5\cos\theta$, then what is $\sin\theta$ equal to? 2015 II

(a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{1-\sqrt{5}}{4}$ (c) $\frac{\sqrt{5}+1}{4}$ (d) $-\frac{\sqrt{5}+1}{4}$

192. A clock is started at noon by 10 minutes past 5, through what angle, the hour hand moves? 2015 II

(a) 160° (b) 145° (c) 150° (d) 155°

193. Consider the following statements 2015 II

I. $\sin 66^\circ$ is less than $\cos 66^\circ$.

II. $\sin 26^\circ$ is less than $\cos 26^\circ$.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

194. Consider the following statements.

I. $\frac{1+\tan^2\theta}{1+\cot^2\theta} = \left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2$ is true for all $0 < \theta < \frac{\pi}{2}, \theta \ne \frac{\pi}{4}$

II. $\cot\theta = \frac{1}{\tan\theta}$ is true for $\theta = 45^\circ$ only.

Which of the above statements is/are correct? 2015 II

(a) Only I (b) Only II

(c) Both I and II (d) Neither I nor II

195. In a ΔABC, if $A - B = \frac{\pi}{2}$, then $C + 2B$ is equal to 2016 I

(a) $\frac{2\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) π (d) $\frac{\pi}{2}$

196. $\left(\frac{\sin 35^\circ}{\cos 55^\circ}\right)^2 - \left(\frac{\cos 55^\circ}{\sin 35^\circ}\right)^2 + 2\sin 30^\circ$ is equal to 2016 I

(a) –1 (b) 0 (c) 1 (d) 2

197. If $\tan\theta + \cot\theta = \frac{4}{\sqrt{3}}$, where $0 < \theta < \frac{\pi}{2}$, then $\sin\theta + \cos\theta$ is equal to 2016 I

(a) 1 (b) $\frac{\sqrt{3}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) $\sqrt{2}$

198. If $p = \cot\theta + \tan\theta$ and $q = \sec\theta - \cos\theta$, then $(p^2q)^{2/3} - (q^2p)^{2/3}$ is equal to **2016 I**

(a) 0 (b) 1 (c) 2 (d) 3

199. If $\frac{x}{a} - \frac{y}{b}\tan\theta = 1$ and $\frac{x}{a}\tan\theta + \frac{y}{b} = 1$, then the value of $\frac{x^2}{a^2} + \frac{y^2}{b^2}$ is **2016 I**

(a) $2\sec^2\theta$ (b) $\sec^2\theta$ (c) $\cos^2\theta$ (d) $2\cos^2\theta$

200. Which of the following is correct in respect of the equation $3 - \tan^2\theta = \alpha(1 - 3\tan^2\theta)$? (given that α is a real number.) **2016 I**

(a) $\alpha \in \left[\frac{1}{3}, 3\right]$ (b) $\alpha \in \left(-\infty, \frac{1}{3}\right] \cup [3, \infty)$

(c) $\alpha \in \left(-\infty, \frac{1}{3}\right) \cup [3, \infty)$ (d) None of these

201. A person goes to a market between 4 pm and 5 pm. When he comes back, he finds that the hour hand and the minute hand of the clock have interchanged their positions. For how much time (approximately) was he out of his house? **2016 I**

(a) 55.38 min (b) 55.48 min (c) 55.57 min (d) 55.67 min

202. Consider the following

I. $\sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} = \operatorname{cosec}\theta - \cot\theta$

II. $\sqrt{\frac{1 + \cos\theta}{1 - \cos\theta}} = \operatorname{cosec}\theta + \cot\theta$

Which of the above is/are identity/ identities? **2016 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

203. If $\frac{\cos^2\theta - 3\cos\theta + 2}{\sin^2\theta} = 1$, where $0 < \theta < \frac{\pi}{2}$, then which of the following statement(s) is/are correct?

I. There are two values of θ satisfying the above equation.

II. $\theta = 60°$ is satisfied by the above equation.

Select the correct answer using the codes given below **2016 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

204. Consider the following statements

I. There exists a positive real number m such that $\cos x = 2^{m+1}$.

II. $mn \geq m + n$, for all m, n belonging to set of natural numbers.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II **2016 I**
(c) Both I and II (d) Neither I nor II

ANSWERS

1	b	2	a	3	b	4	c	5	b	6	b	7	a	8	b	9	c	10	b
11	b	12	b	13	b	14	a	15	c	16	c	17	c	18	b	19	b	20	a
21	c	22	d	23	b	24	a	25	a	26	b	27	a	28	b	29	d	30	d
31	d	32	b	33	c	34	c	35	b	36	d	37	a	38	d	39	c	40	c
41	a	42	a	43	b	44	c	45	b	46	c	47	d	48	c	49	b	50	a
51	c	52	c	53	c	54	b	55	c	56	b	57	b	58	a	59	a	60	a
61	a	62	a	63	a	64	c	65	a	66	d	67	a	68	c	69	d	70	a
71	a	72	d	73	c	74	c	75	b	76	d	77	b	78	b	79	b	80	a
81	c	82	c	83	a	84	c	85	d	86	b	87	a	88	a	89	b	90	a
91	a	92	c	93	b	94	b	95	c	96	a	97	d	98	a	99	c	100	a
101	d	102	a	103	b	104	c	105	d	106	a	107	a	108	b	109	c	110	c
111	a	112	d	113	d	114	d	115	b	116	c	117	b	118	c	119	c	120	b
121	c	122	c	123	c	124	b	125	c	126	c	127	b	128	b	129	d	130	d
131	a	132	a	133	c	134	b	135	c	136	d	137	b	138	b	139	a	140	b
141	d	142	a	143	a	144	c	145	c	146	c	147	b	148	c	149	a	150	c
151	a	152	a	153	b	154	c	155	d	156	b	157	d	158	c	159	d	160	a
161	c	162	b	163	a	164	d	165	c	166	a	167	b	168	d	169	b	170	b
171	d	172	d	173	a	174	b	175	d	176	b	177	c	178	d	179	c	180	b
181	c	182	c	183	a	184	a	185	a	186	a	187	b	188	c	189	d	190	b
191	a	192	d	193	b	194	a	195	d	196	c	197	c	198	b	199	d	200	c
201	a	202	c	203	b	204	d												

HINTS AND SOLUTIONS

1. *(b)* We know that, π radian = 180°

$\Rightarrow 1 \text{ radian} = \dfrac{180°}{\pi} = \dfrac{180°}{22} \times 7$

$= \dfrac{630°}{11} = 57\dfrac{3°}{11} = 57° + \dfrac{3 \times 60}{11}$ min

$= 57° + 16' + \dfrac{4}{11}$ min

$= 57° + 16' + \dfrac{4}{11} \times 60$ s

$= 57° + 16' + 21.8''$

$= 57°\ 16'\ 218'' = 57°\ 16'\ 22''$

2. *(a)* $-47°\ 30' = -\left(47 + \dfrac{30}{60}\right)°$

$= -\left(47 + \dfrac{1}{2}\right)° \quad \left[\because 1' = \left(\dfrac{1}{60}\right)°\right]$

$= -\left(\dfrac{94+1}{2}\right)° = \dfrac{-95°}{2}$

$= \dfrac{-95}{2} \times \dfrac{\pi}{180} = \dfrac{-19\pi}{72}$ rad

3. *(b)* Here, $r = 60$ cm, $l = 16.5$ cm

$\therefore\ \theta = \dfrac{l}{r} = \left(\dfrac{16.5}{60}\right)^c$

$= \left(\dfrac{16.5}{60} \times \dfrac{180}{\pi}\right)° = \left(\dfrac{16.5}{60} \times \dfrac{180}{22} \times 7\right)°$

$= \left(\dfrac{63}{4}\right)° = 15°45'$

4. *(c)* Speed of train = 90 km/h

$= \left(90 \times \dfrac{5}{18}\right)$ m/sec = 25 m/sec

Distance moved in 11 sec

$= (25 \times 11)$ m = 275m

$\therefore\ l = 275$ m, $r = 1500$ m

$\because\ \theta = \dfrac{l}{r} \Rightarrow \theta = \left(\dfrac{275}{1500}\right)^c$

$= \left(\dfrac{275}{1500} \times \dfrac{180}{\pi}\right)° = \left(\dfrac{275}{1500} \times \dfrac{180}{22} \times 7\right)°$

$= \left(\dfrac{21}{2}\right)° = 10°30'$

5. *(b)* Angle traced by hour hand in 1 hr = 30°

Angle traced by hour hand in

$\dfrac{91}{12}$ hrs $= \left(30 \times \dfrac{91}{12}\right)° = 227°30'$

Angle traced by minute hand in 1min = 6°

Angle traced by minute hand in 35 min $= (6 \times 35)° = 210°$

$\therefore$ Required angle $= 227°30' - 210°$

$= 17°30'$

6. *(b)* Let the hands of a clock coincides after '*t*' min.

Angle traced by hour hand in

t min $= \dfrac{1}{2}t$

Angle traced by minute hand in t min $= 6t$

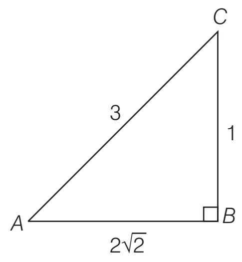
Angle between hour hand and minute hand at 5 pm $= 5 \times 30 = 150°$

$\therefore\ 6t - \dfrac{1}{2}t = 150° \Rightarrow t = \dfrac{150 \times 2}{11} = \dfrac{300}{11}$

$= 27 + \dfrac{3}{11} = 27$ min 16 sec

Therefore, required time = 5:27:16 pm

7. *(a)* Given, $\sin A = \dfrac{1}{3}$,



In right angled ΔABC,

$AB^2 = AC^2 - BC^2 = 3^2 - 1^2 = 8$

$\Rightarrow \quad AB = 2\sqrt{2}$

$\therefore \quad \cos A = \dfrac{AB}{AC} = \dfrac{2\sqrt{2}}{3}$

and $\quad \sec A = \dfrac{AC}{AB} = \dfrac{3}{2\sqrt{2}}$

$\tan A = \dfrac{BC}{AB} = \dfrac{1}{2\sqrt{2}}$

and $\operatorname{cosec} A = \dfrac{AC}{BC} = 3$

$\therefore\ \cos A \cdot \operatorname{cosec} A + \tan A \cdot \sec A$

$= \dfrac{2\sqrt{2}}{3} \cdot \dfrac{3}{1} + \dfrac{1}{2\sqrt{2}} \cdot \dfrac{3}{2\sqrt{2}}$

$= 2\sqrt{2} + \dfrac{3}{8} = \dfrac{16\sqrt{2} + 3}{8}$

8. *(b)* $\cos 15° - \sin 15° = \cos 15° - \sin(90° - 75°)$

$= \cos 15° - \cos 75°$

$= 2 \sin \dfrac{15° + 75°}{2} \cdot \sin \dfrac{75° - 15°}{2}$

$\left(\because \cos C - \cos D = 2 \sin \dfrac{C+D}{2} \sin \dfrac{D-C}{2}\right)$

$= 2 \sin 45° \cdot \sin 30° = 2\ \dfrac{1}{\sqrt{2}} \cdot \dfrac{1}{2} = \dfrac{1}{\sqrt{2}}$

9. *(c)* Given, $3 \tan\theta = 4 \Rightarrow \tan\theta = \dfrac{4}{3}$

$\sqrt{\dfrac{1 - \sin\theta}{1 + \sin\theta}} = \sqrt{\dfrac{1 - \sin\theta}{1 + \sin\theta} \times \dfrac{1 - \sin\theta}{1 - \sin\theta}}$

$= \dfrac{1 - \sin\theta}{\sqrt{1 - \sin^2\theta}}$

$= \dfrac{1 - \sin\theta}{\cos\theta} = \dfrac{1}{\cos\theta} - \dfrac{\sin\theta}{\cos\theta}$

$= \sec\theta - \tan\theta$

$= \sqrt{1 + \tan^2\theta} - \tan\theta$

$= \sqrt{1 + \left(\dfrac{4}{3}\right)^2} - \dfrac{4}{3}$

$= \sqrt{1 + \dfrac{16}{9}} - \dfrac{4}{3} = \dfrac{5}{3} - \dfrac{4}{3} = \dfrac{1}{3}$

10. *(b)* Given, $\tan A = 1 = \tan 45°$

$\Rightarrow \quad A = 45°$ and $\tan B = \sqrt{3} = \tan 60°$

$\therefore \quad B = 60°$

Now, $\cos A \cos B - \sin A \sin B$

$= \cos 45° \cos 60° - \sin 45° \sin 60°$

$= \dfrac{1}{\sqrt{2}} \times \dfrac{1}{2} - \dfrac{1}{\sqrt{2}} \times \dfrac{\sqrt{3}}{2} = \dfrac{1 - \sqrt{3}}{2\sqrt{2}}$

11. *(b)* $\operatorname{cosec}^2\theta - 2 + \sin^2\theta$

$= (\sin\theta - \operatorname{cosec}\theta)^2$

Hence, it is always non-negative.

12. *(b)* $\sin^2 15° + \sin^2 20° + \sin^2 25° + \ldots + \sin^2 75°$

$= (\sin^2 15° + \sin^2 75°)$

$+ (\sin^2 20° + \sin^2 70°) + \ldots + \sin^2 45°$

$= [\sin^2 15° + \sin^2(90° - 15°)]$

$+ [\sin^2 20° + \sin^2(90° - 20°)] + \ldots + \sin^2 45°$

$= (\sin^2 15° + \cos^2 15°)$

$+ (\sin^2 20° + \cos^2 20°) + \ldots + 1/2$

$= 6 + 1/2 = 13/2$

$(\because \sin^2\theta + \cos^2\theta = 1)$

13. *(b)* $\dfrac{5 \sin 75° \sin 77° + 2 \cos 13° \cos 15°}{\cos 15° \sin 77°} - \dfrac{7 \sin 81°}{\cos 9°}$

$= \dfrac{5 \cos 15° \sin 77° + 2 \sin 77° \cos 15°}{\cos 15° \sin 77°} - \dfrac{7 \cos 9°}{\cos 9°}$

$= \dfrac{7 \cos 15° \cdot \sin 77°}{\cos 15° \cdot \sin 77°} - \dfrac{7 \cos 9°}{\cos 9°}$

$= 7 - 7 = 0$

14. *(a)* log (tan 1°) + log (tan 2°) +…+ log (tan 89°)
= log (tan 1° tan 2°…tan 45° …tan 88° tan 89°)
= log [(tan 1° cot 1°) (tan 2° cot 2°) …tan 45°]
[$\because$ tan 89° = tan(90° − 1°) = cot 1°]
= log(1°·1° ...1°) = 0

15. *(c)* $\because$ Given, $\tan\theta + \dfrac{1}{\tan\theta} = 2$

On squaring both side, we get $\left(\tan\theta + \dfrac{1}{\tan\theta}\right)^2 = 2^2$

$\Rightarrow \tan^2\theta + \dfrac{1}{\tan^2\theta} + 2 = 4 \Rightarrow \tan^2\theta + \dfrac{1}{\tan^2\theta} = 2$

16. *(c)* Given, $\sec\theta = \dfrac{13}{5}$

$\therefore \tan\theta = \sqrt{\sec^2\theta - 1} = \sqrt{\dfrac{169}{25} - 1} = \sqrt{\dfrac{144}{25}} = \dfrac{12}{5}$

$\therefore \dfrac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta} = \dfrac{\dfrac{2\sin\theta}{\cos\theta} - 3}{4\dfrac{\sin\theta}{\cos\theta} - 9}$

(dividing numerator and denominator by cos θ)

$= \dfrac{2\tan\theta - 3}{4\tan\theta - 9} = \dfrac{2 \times \dfrac{12}{5} - 3}{4 \times \dfrac{12}{5} - 9} = \dfrac{9/5}{3/5} = \dfrac{9}{3} = 3$

17. *(c)* sin (45° + A) − cos (45° − A)
= sin(45°+A) − cos[(90° − (45°+A)] = sin(45°+A) − sin(45°+A)
= 0

18. *(b)* $\sqrt{\dfrac{1+\sin\theta}{1-\sin\theta}} = \sqrt{\dfrac{(1+\sin\theta)(1+\sin\theta)}{(1-\sin\theta)(1+\sin\theta)}}$

$= \sqrt{\dfrac{(1+\sin\theta)^2}{1-\sin^2\theta}} = \sqrt{\dfrac{(1+\sin\theta)^2}{\cos^2\theta}} = \dfrac{1+\sin\theta}{\cos\theta} = \dfrac{1}{\cos\theta} + \dfrac{\sin\theta}{\cos\theta}$

= sec θ + tan θ

19. *(b)* sin $(n+1)A$ sin $(n-1)A$ + cos $(n+1)A$ cos $(n-1)A$
= cos $[(n+1)A - (n-1)A]$
[$\because \cos A\cos B + \sin A\sin B = \cos(A - B)$]
= cos $[nA + A - nA + A]$ = cos $2A$

20. *(a)* $\dfrac{\cos 11° + \sin 11°}{\cos 11° - \sin 11°}$

Divide numerator and denominator by cos 11°

$= \dfrac{1 + \tan 11°}{1 - \tan 11°} = \dfrac{\tan 45° + \tan 11°}{1 - \tan 45° \tan 11°} = \tan(45° + 11°) = \tan 56°$

21. *(c)* $\tan\theta = \dfrac{\sin\theta}{\cos\theta} = \dfrac{\sin\theta}{\sqrt{1-\sin^2\theta}} = \dfrac{\dfrac{a^2-b^2}{a^2+b^2}}{\sqrt{1 - \dfrac{(a^2-b^2)^2}{(a^2+b^2)^2}}}$

$= \dfrac{\dfrac{a^2-b^2}{a^2+b^2}}{\dfrac{\sqrt{(a^2+b^2)^2 - (a^2-b^2)^2}}{a^2+b^2}} = \dfrac{a^2-b^2}{\sqrt{4a^2b^2}} = \dfrac{a^2-b^2}{2ab}$

22. *(d)* Given, $\cos\theta = 0.96$, $\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-(0.96)^2}$

$= \sqrt{1 - 0.9216} = 0.28$, $\tan\theta = \dfrac{\sin\theta}{\cos\theta} = \dfrac{0.28}{0.96} = \dfrac{7}{24}$

$\dfrac{1}{\sin\theta} + \dfrac{1}{\tan\theta} = \dfrac{100}{28} + \dfrac{24}{7} = \dfrac{196}{28} = 7$

23. *(b)* As we know, sin x is increasing from 0 to 90°.
$\therefore$ sin y > sin x

24. *(a)* Given, (sin θ + cosec θ) = 2.5

$\Rightarrow \left(\sin\theta + \dfrac{1}{\sin\theta}\right) = \dfrac{5}{2} \Rightarrow 2\sin^2\theta - 5\sin\theta + 2 = 0$

$\Rightarrow 2\sin^2\theta - 4\sin\theta - \sin\theta + 2 = 0$

$\Rightarrow 2\sin\theta(\sin\theta - 2) - 1(\sin\theta - 2) = 0$

$\Rightarrow (2\sin\theta - 1)(\sin\theta - 2) = 0 \Rightarrow \sin\theta = \dfrac{1}{2}$ $(\because \sin\theta \neq 2)$

$\therefore \theta = 30°$

25. *(a)* Given, $\dfrac{\cos x}{\cos y} = n$ and $\dfrac{\sin x}{\sin y} = m$ ...(i)

Now, $(m^2 - n^2)\sin^2 y = \left(\dfrac{\sin^2 x}{\sin^2 y} - \dfrac{\cos^2 x}{\cos^2 y}\right)\sin^2 y$

$= \dfrac{(1-\cos^2 x)\cos^2 y - \cos^2 x(1-\cos^2 y)}{\cos^2 y}$

$= \dfrac{\cos^2 y - \cos^2 x}{\cos^2 y} = 1 - n^2$ [from Eq. (i)]

26. *(b)* Given, $p = \tan^2 x + \cot^2 x = \tan^2 x + \dfrac{1}{\tan^2 x}$

$\because$ AM ≥ GM $\Rightarrow \tan^2 x + \dfrac{1}{\tan^2 x} \geq 2\left(\tan^2 x \cdot \dfrac{1}{\tan^2 x}\right)^{1/2}$

$\Rightarrow \tan^2 x + \dfrac{1}{\tan^2 x} \geq 2 \Rightarrow p \geq 2$

27. *(a)* Let angles in circular measure are A and B, then degree measures will be $A\left(\dfrac{180}{\pi}\right)$ and $B\left(\dfrac{180}{\pi}\right)$.

By given condition, $A + B = 1$ …(i)

$A\left(\dfrac{180}{\pi}\right) - B\left(\dfrac{180}{\pi}\right) = 1$ …(ii)

On multiplying Eq. (i) by $\dfrac{180}{\pi}$ and adding to Eq. (ii), we get

$2A\left(\dfrac{180}{\pi}\right) = \dfrac{180}{\pi} + 1 \Rightarrow A = \dfrac{1}{2} \times \dfrac{11}{180}\left[\dfrac{180}{\pi} + 1\right] = \dfrac{1}{2} + \dfrac{\pi}{360}$

$\therefore B = 1 - A = 1 - \left(\dfrac{1}{2} + \dfrac{\pi}{360}\right) = \dfrac{1}{2} - \dfrac{\pi}{360}$

28. *(b)* $[(1-\sin^2\theta)\sec^2\theta + \tan^2\theta](\cos^2\theta + 1)$
$= [\cos^2\theta \cdot \sec^2\theta + \tan^2\theta](\cos^2\theta + 1)$ $(\because \sin^2\theta + \cos^2\theta = 1)$
$= (1 + \tan^2\theta)(\cos^2\theta + 1) = \sec^2\theta(\cos^2\theta + 1)$
$(\because \sec^2\theta - \tan^2\theta = 1)$
$= \sec^2\theta \cdot \cos^2\theta + \sec^2\theta = 1 + \sec^2\theta > 2$
$(\because \sec^2\theta > 1$ for $0 < \theta < 90°)$

29. *(d)* Given, $7\cos^2\theta + 3\sin^2\theta = 4 \Rightarrow 7\cos^2\theta + 3(1-\cos^2\theta) = 4$

$\Rightarrow 3 - 4\cos^2\theta = 4 \Rightarrow \cos^2\theta = 1/4$

Now, $\tan^2\theta = \sec^2\theta - 1 = \dfrac{1}{\cos^2\theta} - 1 \Rightarrow \tan^2\theta = 4 - 1 = 3$

$\therefore \tan\theta = \sqrt{3}$ $(\because 0 < \theta < \pi/2)$

30. *(d)* We know that, the value of cos θ is decreasing in the interval $0 \leq \theta \leq 90°$

$\therefore$ cos 1° > cos 89° $\Rightarrow p > q$

Also, cos 1° is close to 1 and cos 89° is close to 0.
Hence, option *(d)* is correct.

31. *(d)* We know in the interval $\theta \in (0, \pi/2)$, $\sec^2\theta$ is increasing from 1 to ∞.

$\therefore \quad p \geq 1$

32. *(b)* $\because \cos 90° = 0$

$\therefore \quad \cos 1° \cos 2° \cos 3° \ldots \cos 90° = 0 \qquad (\because \cos 90° = 0)$

33. *(c)* Given, $\tan A = \dfrac{1-\cos B}{\sin B}$

$$\therefore \quad \frac{2\tan A}{1-\tan^2 A} = \frac{2\cdot\dfrac{(1-\cos B)}{\sin B}}{1-\left(\dfrac{1-\cos B}{\sin B}\right)^2}$$

$$= \frac{2(1-\cos B)\sin B}{\sin^2 B - 1 - \cos^2 B + 2\cos B} = \frac{2(1-\cos B)\sin B}{-2\cos^2 B + 2\cos B}$$

$$= \frac{2\sin B(1-\cos B)}{2\cos B(1-\cos B)} = \frac{\sin B}{\cos B} = \tan B$$

34. *(c)* $\cot 15° \cot 20° \cot 70° \cot 75°$

$= \tan(90° - 15°)\tan(90° - 20°)\cot 70° \cot 75°$

$= \tan 75° \tan 70° \dfrac{1}{\tan 70°}\cdot\dfrac{1}{\tan 75°} = 1$

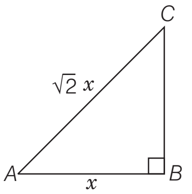
35. *(b)* Given, $\dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\sin\theta} = 2$

$\therefore \quad \sin^2\theta + \cos^2\theta = 2\sin\theta\cos\theta$

$\Rightarrow \quad \sin 2\theta = 1 = \sin 90° \Rightarrow 2\theta = 90° \Rightarrow \theta = 45°$

36. *(d)* Given, in right angled ΔABC, $\angle B = 90°$ and $AB : AC = 1 : \sqrt{2}$

Let $AB = x$, $AC = \sqrt{2}\,x$



$\Rightarrow \quad AB^2 + BC^2 = AC^2 \qquad$ (using pythagoras theorem)

$\Rightarrow \quad x^2 + BC^2 = (\sqrt{2}\,x)^2$

$\Rightarrow \quad BC^2 = 2x^2 - x^2 \Rightarrow BC = x$

$\therefore \quad \tan A = \dfrac{BC}{AB} = \dfrac{x}{x} = 1$

$\therefore \quad \dfrac{2\tan A}{1-\tan^2 A} = \dfrac{2(1)}{1-1^2} = \dfrac{2}{0} = \infty \qquad$ (undefined)

37. *(a)* Given, $AD = DB = \dfrac{a}{2}$ and $\angle B = 90°$

In right angled ΔABC, $BC^2 + AB^2 = AC^2$

$\Rightarrow BC^2 + a^2 = b^2 \Rightarrow BC = \sqrt{b^2 - a^2}$

In right angled ΔBCD, $CD^2 = BC^2 + BD^2$

$$CD^2 = (\sqrt{b^2-a^2})^2 + \left(\frac{a}{2}\right)^2 = \left(b^2 - a^2 + \frac{a^2}{4}\right)$$

$$\Rightarrow \quad CD^2 = \frac{4b^2 - 3a^2}{4} \Rightarrow CD = \frac{\sqrt{4b^2-3a^2}}{2}$$

$$\therefore \quad \cos\theta = \frac{BC}{CD} = \frac{\sqrt{b^2-a^2}}{\dfrac{\sqrt{4b^2-3a^2}}{2}} = \frac{2\sqrt{b^2-a^2}}{\sqrt{4b^2-3a^2}}$$

38. *(d)* Given, $\sin x + \sqrt{3}\cos x = 1$

On dividing both sides by 2, we get $\dfrac{1}{2}\sin x + \dfrac{\sqrt{3}}{2}\cos x = \dfrac{1}{2}$

$\Rightarrow \quad \sin x \sin 30° + \cos x \cos 30° = 1/2$

$\Rightarrow \cos(x - 30°) = \cos 60° \Rightarrow x - 30° = 60°$

$\therefore \quad x = 90°$

39. *(c)* $\operatorname{cosec} A \cdot \sec A = \dfrac{\sqrt{1+\tan^2 A}}{\tan A}\cdot\sqrt{1+\tan^2 A}$

$= \dfrac{1+\tan^2 A}{\tan A} = \dfrac{1+(\sqrt{2}-1)^2}{\sqrt{2}-1} \qquad (\because \tan A = \sqrt{2}-1)$

$= \dfrac{1+2+1-2\sqrt{2}}{\sqrt{2}-1} = \dfrac{4-2\sqrt{2}}{(\sqrt{2}-1)} = \dfrac{2\sqrt{2}(\sqrt{2}-1)}{(\sqrt{2}-1)} = 2\sqrt{2}$

40. *(c)* Given, $\tan\theta = \dfrac{3}{1} \Rightarrow \sin\theta = 3\cos\theta \Rightarrow \sin\theta = 3\sqrt{1-\sin^2\theta}$

Squaring both sides, we get $\sin^2\theta = 9(1-\sin^2\theta)$

$\Rightarrow \quad 10\sin^2\theta = 9 \Rightarrow \sin\theta = \sqrt{\dfrac{9}{10}} = -\dfrac{3}{\sqrt{10}}$

($\because \theta$ lies in IIIrd quadrant, so $\sin\theta$ will be negative)

41. *(a)* $\sin 20° \cos 70° + \cos 20° \sin 70°$

$= \sin(20° + 70°) = \sin 90° = 1$

$[\because \sin A \cos B + \cos A \sin B = \sin(A + B)]$

42. *(a)* $\phi + \theta = 180° \Rightarrow \theta = 180° - \phi \Rightarrow \sin\theta = \sin(180° - \phi)$

$\sin\theta = \sin\phi$

43. *(b)* $\sqrt{\cos x \operatorname{cosec} y - \cos x \sin y}$

$= \sqrt{\cos x \cdot \operatorname{cosec}(90° - x) - \cos x \cdot \sin(90° - x)}$

$(\because x + y = 90°$, given)

$= \sqrt{\cos x \cdot \sec x - \cos^2 x}$

$= \sqrt{1-\cos^2 x} = \sqrt{\sin^2 x} = \sin x$

44. *(c)* $\sqrt{2+\sqrt{2+2\cos 4\theta}} = \sqrt{2+\sqrt{2(1+\cos 4\theta)}}$

$= \sqrt{2+\sqrt{2(2\cos^2 2\theta)}} = \sqrt{2+\sqrt{4\cos^2 2\theta}} = \sqrt{2+2\cos 2\theta}$

$= \sqrt{2(1+\cos 2\theta)} = \sqrt{2(2\cos^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta$

45. *(b)* Given, $\tan A = \dfrac{5}{6}$ and $\tan B = \dfrac{1}{11}$

$$\because \quad \tan(A+B) = \frac{\tan A + \tan B}{1-\tan A \tan B} = \frac{\dfrac{5}{6}+\dfrac{1}{11}}{1-\dfrac{5}{6}\cdot\dfrac{1}{11}} = \frac{\dfrac{55+6}{66}}{\dfrac{66-5}{66}} = \frac{61}{61} = 1$$

$\Rightarrow \tan(A+B) = 1 \Rightarrow \tan(A+B) = \tan\dfrac{\pi}{4} \Rightarrow A + B = \dfrac{\pi}{4}$

46. *(c)* $\dfrac{1}{\sqrt{2}}(\cos A - \sin A) = \dfrac{\cos A}{\sqrt{2}} - \dfrac{\sin A}{\sqrt{2}}$

$= \cos\dfrac{\pi}{4}\cos A - \sin\dfrac{\pi}{4}\sin A = \cos\left(\dfrac{\pi}{4} + A\right)$

$[\because \cos A \cos B - \sin A \sin B = \cos(A + B)]$

47. *(d)* $\sin\dfrac{\pi}{4}\cdot\cos\dfrac{\pi}{12} - \cos\dfrac{\pi}{4}\cdot\sin\dfrac{\pi}{12} = \sin\left(\dfrac{\pi}{4} - \dfrac{\pi}{12}\right)$

$[\because \sin A \cos B - \cos A \sin B = \sin(A - B)]$

$= \sin\left(\dfrac{\pi}{6}\right) = \dfrac{1}{2}$

48. *(c)* Given, $\tan\theta = \dfrac{x\sin\phi}{1 - x\cos\phi} \Rightarrow x\sin\phi = \tan\theta - x\cos\phi\tan\theta$

$\Rightarrow \quad x = \dfrac{\tan\theta}{\sin\phi + \cos\phi\tan\theta} = \dfrac{\sin\theta}{\cos\theta\sin\phi + \cos\phi\sin\theta} = \dfrac{\sin\theta}{\sin(\theta+\phi)}$

Similarly, $y = \dfrac{\sin\phi}{\sin(\theta+\phi)}$, Hence, $\dfrac{x}{y} = \dfrac{\sin\theta}{\sin\phi}$

49. *(b)* Given, $\sin(\theta+\phi) = 2\sin(\theta-\phi)$

$\Rightarrow \quad \sin\theta\cos\phi + \cos\theta\sin\phi = 2(\sin\theta\cos\phi - \cos\theta\sin\phi)$

$\Rightarrow \quad 3\cos\theta\sin\phi = \sin\theta\cos\phi \Rightarrow \dfrac{3\sin\phi}{\cos\phi} = \dfrac{\sin\theta}{\cos\theta}$

$\Rightarrow \quad 3\tan\phi = \tan\theta$ or $\tan\theta = 3\tan\phi$

50. *(a)* Given, $\sin(\theta+\alpha) = \cos(\theta+\alpha) \Rightarrow \dfrac{\sin(\theta+\alpha)}{\cos(\theta+\alpha)} = 1$

$\Rightarrow \quad \tan(\theta+\alpha) = 1 \Rightarrow \dfrac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha} = 1$

$\Rightarrow \quad \tan\theta + \tan\alpha = 1 - \tan\theta\tan\alpha$

$\Rightarrow \quad \tan\theta(1 + \tan\alpha) = 1 - \tan\alpha \Rightarrow \tan\theta = \dfrac{1-\tan\alpha}{1+\tan\alpha}$

51. *(c)* $\cos^2\dfrac{\theta-\phi}{2} - \sin^2\dfrac{\theta+\phi}{2}$

$$= \cos\left(\frac{\theta-\phi}{2} + \frac{\theta+\phi}{2}\right)\cdot\cos\left(\frac{\theta-\phi}{2} - \frac{\theta+\phi}{2}\right)$$

$[\because \cos^2 A - \sin^2 B = \cos(A+B)\cos(A-B)]$

$$= \cos\theta\cdot\cos(-\phi) = \cos\theta\cdot\cos\phi$$

52. *(c)* $x^2 + y^2 + z^2 = r^2\sin^2\theta\cos^2\phi + r^2\sin^2\theta\sin^2\phi + r^2\cos^2\theta$

$$= r^2\sin^2\theta(\cos^2\phi + \sin^2\phi) + r^2\cos^2\theta$$

$$= r^2\sin^2\theta + r^2\cos^2\theta = r^2(\sin^2\theta + \cos^2\theta) = r^2$$

53. *(c)* Given, $\cot\theta = \dfrac{7}{24}$, $\operatorname{cosec}^2\theta = 1 + \cot^2\theta = 1 + \dfrac{49}{576} = \dfrac{625}{576}$

$\therefore \quad \operatorname{cosec}\theta = \pm\dfrac{25}{24}$

$\Rightarrow \quad \sin\theta = \pm\dfrac{24}{25}$, $\cos^2\theta = 1 - \sin^2\theta = 1 - \dfrac{576}{625} = \dfrac{49}{625}$

$\therefore \quad \cos\theta = \pm\dfrac{7}{25}$, As $\pi < \theta < \dfrac{3\pi}{2}$

$\therefore$ $\sin\theta$ and $\cos\theta$ both are negative.

$\therefore \quad \sin\theta = \dfrac{-24}{25}$, $\cos\theta = \dfrac{-7}{25}$

$\therefore \quad \cos\theta - \sin\theta = \dfrac{-7}{25} + \dfrac{24}{25} = \dfrac{17}{25}$

54. *(b)* Given, $\sin\alpha = \dfrac{1}{2} = \sin 30° \Rightarrow \alpha = 30°$

Also, $\quad \beta = \dfrac{\pi}{2} - \alpha = 90° - 30° = 60°$

55. *(c)* Given, $\sin x - \cos x = 0 \Rightarrow \sin x = \cos x \Rightarrow \tan x = 1$

$\Rightarrow \quad \tan x = \tan\dfrac{\pi}{4} \Rightarrow x = \dfrac{\pi}{4}$

$\therefore \sin^4 x + \cos^4 x = \sin^4\dfrac{\pi}{4} + \cos^4\dfrac{\pi}{4} = \left(\dfrac{1}{\sqrt2}\right)^4 + \left(\dfrac{1}{\sqrt2}\right)^4$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

56. *(b)* Given, $3\cos x = 5\sin x \Rightarrow \dfrac{\sin x}{\cos x} = \dfrac{3}{5} \Rightarrow \tan x = \dfrac{3}{5}$,

$$\frac{5\sin x - 2\sec^3 x + 2\cos x}{5\sin x + 2\sec^3 x - 2\cos x} = \frac{5\tan x - 2\sec^4 x + 2}{5\tan x + 2\sec^4 x - 2}$$

(dividing numerator and denominator by $\cos x$)

$$= \frac{5\tan x - 2(1+\tan^2 x)^2 + 2}{5\tan x + 2(1+\tan^2 x)^2 - 2}$$

$$= \frac{5\times\frac{3}{5} - 2\left(1+\frac{9}{25}\right)^2 + 2}{5\times\frac{3}{5} + 2\left(1+\frac{9}{25}\right)^2 - 2} = \frac{3 + 2 - 2\times\frac{1156}{625}}{3 - 2 + 2\times\frac{1156}{625}}$$

$$= \frac{3125 - 2312}{625 + 2312} = \frac{813}{2937} = \frac{271}{979}$$

57. *(b)* $\sqrt{\dfrac{1+\cos x}{1-\cos x}} = \sqrt{\dfrac{(1+\cos x)}{(1-\cos x)}\times\dfrac{1+\cos x}{1+\cos x}}$

$$= \frac{1+\cos x}{\sin x} = \frac{1}{\sin x} + \frac{\cos x}{\sin x} = \operatorname{cosec} x + \cot x$$

58. *(a)* $\cos^4 x - \sin^4 x = (\cos^2 x - \sin^2 x)(\sin^2 x + \cos^2 x)$

$= [(\cos^2 x) - (1 - \cos^2 x)] = 2\cos^2 x - 1 \quad (\because \sin^2 x + \cos^2 x = 1)$

59. *(a)* $\dfrac{\sin\theta}{1+\cos\theta} = \dfrac{\sin\theta}{1+\cos\theta}\times\dfrac{1-\cos\theta}{1-\cos\theta} = \dfrac{\sin\theta(1-\cos\theta)}{\sin^2\theta} = \dfrac{1-\cos\theta}{\sin\theta}$

60. *(a)* $\dfrac{5\cos\theta - 4}{3 - 5\sin\theta} - \dfrac{3 + 5\sin\theta}{4 + 5\cos\theta}$

$$= \frac{(5\cos\theta - 4)(4 + 5\cos\theta) - (3 + 5\sin\theta)(3 - 5\sin\theta)}{(3 - 5\sin\theta)(4 + 5\cos\theta)}$$

$$= \frac{(25\cos^2\theta - 16) - (9 - 25\sin^2\theta)}{(3 - 5\sin\theta)(4 + 5\cos\theta)}$$

$$= \frac{25(\sin^2\theta + \cos^2\theta) - 25}{(3 - 5\sin\theta)(4 + 5\cos\theta)} = \frac{25 - 25}{(3 - 5\sin\theta)(4 + 5\cos\theta)} = 0$$

61. *(a)* $\sin(\alpha+\beta) = \sqrt{1 - \cos^2(\alpha+\beta)} = \sqrt{1 - \dfrac{16}{25}} = \sqrt{\dfrac{9}{25}} = \dfrac{3}{5}$

$\cos(\alpha-\beta) = \sqrt{1 - \sin^2(\alpha-\beta)} = \sqrt{1 - \dfrac{25}{169}} = \sqrt{\dfrac{144}{169}} = \dfrac{12}{13}$

$\therefore \quad \tan(\alpha+\beta) = \dfrac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)} = \left(\dfrac{3}{5}\times\dfrac{5}{4}\right) = \dfrac{3}{4}$

$\tan(\alpha-\beta) = \dfrac{\sin(\alpha-\beta)}{\cos(\alpha-\beta)} = \dfrac{5}{13}\times\dfrac{13}{12} = \dfrac{5}{12}$

$\therefore \quad \tan(2\alpha) = \tan[(\alpha+\beta) + (\alpha-\beta)]$

$$= \frac{\tan(\alpha+\beta) + \tan(\alpha-\beta)}{1 - \tan(\alpha+\beta)\cdot\tan(\alpha-\beta)} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4}\times\frac{5}{12}} = \frac{56}{33}$$

62. *(a)* $\tan(45° + x) = \dfrac{\tan 45° + \tan x}{1 - \tan 45°\tan x} = \dfrac{1 + \tan x}{1 - \tan x}$

$\tan(45° - x) = \dfrac{\tan 45° - \tan x}{1 + \tan 45°\tan x} = \dfrac{1 - \tan x}{1 + \tan x}$

$$\therefore \quad \frac{\tan(45°+x)}{\tan(45°-x)} = \frac{\frac{1+\tan x}{1-\tan x}}{\frac{1-\tan x}{1+\tan x}} = \left[\frac{1+\tan x}{1-\tan x}\right]^2$$

63. *(a)* $\dfrac{(1-\sin A\cos A)(\sin^2 A-\cos^2 A)}{\cos A\,(\sec A-\operatorname{cosec} A)(\sin^3 A+\cos^3 A)}$

$$= \frac{(1-\sin A\cos A)(\sin^2 A-\cos^2 A)}{\left[\cos A\left(\frac{1}{\cos A}-\frac{1}{\sin A}\right)(\sin A+\cos A)\,(\sin^2 A+\cos^2 A-\sin A\cos A)\right]}$$

$$= \frac{[(1-\sin A\cos A)(\sin A\cdot\cos A)\times(\sin A+\cos A)(\sin A-\cos A)]}{[\cos A(\sin A-\cos A)(\sin A+\cos A)(1-\sin A\cos A)]} = \sin A$$

64. *(c)* $(1+\tan A)(1+\tan B)$

$= (1+\tan A)[1+\tan(45°-A)]$

$= (1+\tan A)\left[1+\dfrac{\tan 45°-\tan A}{1+\tan 45°\tan A}\right]$ $\quad (\because A+B=45°)$

$= (1+\tan A)\left[1+\dfrac{1-\tan A}{1+\tan A}\right]$

$= (1+\tan A)\left[\dfrac{1+\tan A+1-\tan A}{1+\tan A}\right]$

$= (1+\tan A)\left[\dfrac{2}{1+\tan A}\right] = 2$

65. *(a)* $\because \cos A-\cos B = 2\sin\dfrac{(A+B)}{2}\cdot\sin\dfrac{(B-A)}{2}$

and $\sin A+\sin B = 2\sin\dfrac{(A+B)}{2}\cdot\cos\dfrac{(A-B)}{2}$

$\therefore \dfrac{\cos 2B-\cos 2A}{\sin 2B+\sin 2A} = \dfrac{2\sin(A+B)\sin(A-B)}{2\sin(A+B)\cos(A-B)} = \tan(A-B)$

66. *(d)* $\sec^2 x+\tan^2 x = 7$ $\quad (\because \sec^2 x = 1+\tan^2 x)$

$1+\tan^2 x+\tan^2 x = 7 \Rightarrow 1+2\tan^2 x = 7$

$\Rightarrow \quad 2\tan^2 x = 6 \Rightarrow \tan x = \sqrt{3} = \tan 60°$

$\therefore \quad x = 60°$

67. *(a)* $\cos 3\theta+\sin 3\theta = \sqrt{2}\left(\dfrac{1}{\sqrt{2}}\cos 3\theta+\dfrac{1}{\sqrt{2}}\sin 3\theta\right)$

$= \sqrt{2}(\sin 45°\cos 3\theta+\cos 45°\sin 3\theta) = \sqrt{2}[\sin(45°+3\theta)]$

$\cos 3\theta+\sin 3\theta$ is maximum when $\sin(45°+3\theta)$ is maximum

The maximum value of $\sin(45°+3\theta) = 1 = \sin 90°$

$\Rightarrow \quad 45°+3\theta = 90° \Rightarrow \theta = 15°$

68. *(c)* $\sin^2 x+\sin^2 y+\sin^2 z = (\sin x+\sin y+\sin z)^2$

$\Rightarrow \sin x\sin y+\sin y\sin z+\sin z\sin x = 0$

$[\because (a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)]$

On dividing both sides by $\sin x\sin y\sin z$

$\Rightarrow \dfrac{1}{\sin x}+\dfrac{1}{\sin y}+\dfrac{1}{\sin z} = 0$

69. *(d)* Given, $\sin x+\sin^2 x = 1$ $\quad$ ...(i)

$\Rightarrow \quad \sin x = 1-\sin^2 x = \cos^2 x$ $\quad (\because \sin^2 x+\cos^2 x = 1)$

$\therefore \cos^2 x+\cos^4 x = (\sin x)+\sin^2 x = 1$ $\quad$ [from Eq. (i)]

70. *(a)* $(\cos 35°+\cos 85°)+\cos 155°$

$= 2\cos 60°\cos 25°+\cos(180°-25°)$

$= 2\times\dfrac{1}{2}\cos 25°-\cos 25° = \cos 25°-\cos 25° = 0$

71. *(a)* Refer to example 16.

72. *(d)* Given, $\tan\theta = \dfrac{p}{q}$,

$p\cos 2\theta+q\sin 2\theta = p\left(\dfrac{1-\tan^2\theta}{1+\tan^2\theta}\right)+q\left(\dfrac{2\tan\theta}{1+\tan^2\theta}\right)$

$= p\left(\dfrac{1-\frac{p^2}{q^2}}{1+\frac{p^2}{q^2}}\right)+q\left(\dfrac{\frac{2p}{q}}{1+\frac{p^2}{q^2}}\right) = p\left(\dfrac{q^2-p^2}{q^2+p^2}\right)+\left(\dfrac{2pq^2}{q^2+p^2}\right)$

$= \dfrac{p(q^2-p^2+2q^2)}{p^2+q^2} = \dfrac{p(3q^2-p^2)}{p^2+q^2}$

73. *(c)* Given, $x = y\cos\dfrac{2\pi}{3} = z\cos\dfrac{4\pi}{3} \Rightarrow x = \dfrac{-y}{2} = \dfrac{-z}{2} = k$ $\quad$ (let)

$x = k,\ y = -2k,\ z = -2k$

$\Rightarrow \quad xy+yz+zx = k(-2k)+(-2k)(-2k)+(-2k)k$

$= -2k^2+4k^2-2k^2 = 0$

74. *(c)* Given, $\sin\theta+\cos\theta = m$ $\quad$ ...(i)

and $\sec\theta+\operatorname{cosec}\theta = n \Rightarrow \dfrac{1}{\cos\theta}+\dfrac{1}{\sin\theta} = n$

$\Rightarrow \quad \dfrac{\sin\theta+\cos\theta}{\sin\theta\cos\theta} = n$

$\Rightarrow \quad \dfrac{m}{\sin\theta\cos\theta} = n \Rightarrow \sin\theta\cos\theta = \dfrac{m}{n}$ $\quad$ [from Eq. (i)]

On squaring Eq. (i), we get $\sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta = m^2$

$1+2\dfrac{m}{n} = m^2 \Rightarrow \dfrac{2m}{n} = m^2-1$ $\quad [\because \sin^2\theta+\cos^2\theta = 1]$

$\Rightarrow \quad 2m = (m^2-1)n$

75. *(b)* Given, $\operatorname{cosec}\theta-\sin\theta = a^3 \Rightarrow \dfrac{1}{\sin\theta}-\sin\theta = a^3$

$\Rightarrow \dfrac{1-\sin^2\theta}{\sin\theta} = a^3 \Rightarrow \dfrac{\cos^2\theta}{\sin\theta} = a^3 \Rightarrow \cos^2\theta = a^3\sin\theta$ $\quad$ ...(i)

and $\sec\theta-\cos\theta = b^3 \Rightarrow \dfrac{1}{\cos\theta}-\cos\theta = b^3 \Rightarrow \dfrac{1-\cos^2\theta}{\cos\theta} = b^3$

$\Rightarrow \quad \dfrac{\sin^2\theta}{\cos\theta} = b^3 \Rightarrow \sin^2\theta = b^3\cos\theta$ $\quad$ ...(ii)

On putting $\cos\theta = \dfrac{\sin^2\theta}{b^3}$ in Eq. (i), we get

$\dfrac{\sin^4\theta}{b^6} = a^3\sin\theta \Rightarrow \sin^3\theta = a^3 b^6$

$\therefore \quad \sin\theta = ab^2$ $\quad$ ...(iii)

Similarly, $\quad \cos\theta = a^2 b$ $\quad$ ...(iv)

On squaring and adding Eqs. (iii) and (iv), we get

$1 = a^2 b^4+a^4 b^2 \Rightarrow 1 = a^2 b^2(a^2+b^2)$

76. *(d)* As, we know that $\cos\theta \ge -1$ and $\cos\theta \le 1$

$\therefore \quad \cos 2x \ge -1 \Rightarrow 1+\cos 2x \ge 0$

$\cos 2x \le 1 \Rightarrow 1+\cos 2x \le 2$

So, minimum value of $1+\cos 2x$ is 0 and maximum value of $1+\cos 2x$ is 2.

77. *(b)* $\cot(\theta+\phi)=\dfrac{1}{\tan(\theta+\phi)}=\dfrac{1-\tan\theta\tan\phi}{\tan\theta+\tan\phi}$

$$=\frac{1}{\tan\theta+\tan\phi}-\frac{\tan\theta\tan\phi}{\tan\theta+\tan\phi}$$

$$=\frac{1}{\tan\theta+\tan\phi}-\frac{1}{\left(\dfrac{1}{\tan\theta}+\dfrac{1}{\tan\theta}\right)}$$

$$=\frac{1}{\tan\theta+\tan\phi}-\frac{1}{(\cot\theta+\cot\phi)}=\frac{1}{a}-\frac{1}{b}$$

[$\because$ $\tan\theta+\tan\phi=a$, or $\cot\theta+\cot\phi=b$]

78. *(b)* $\because$ $\tan 75°=\tan(45°+30°)$

$\Rightarrow$ $\tan 75°=\dfrac{\tan 45°+\tan 30°}{1-\tan 45°\tan 30°}$ $\Rightarrow$ $\tan 75°=\dfrac{1+\tan 30°}{1-\tan 30°}$

$\Rightarrow$ $\tan 75°-\tan 75°\cdot\tan 30°=1+\tan 30°$

$\Rightarrow$ $\tan 75°-\tan 30°-\tan 75°\cdot\tan 30°=1$

79. *(b)* $\sin^2(15°+A)-\sin^2(15°-A)=\sin(15°+A+15°-A)$

$\cdot\sin(15°+A-15°+A)$

[$\because$ $\sin^2 A-\sin^2 B=\sin(A+B)\cdot\sin(A-B)$]

$=\sin 30°\sin(2A)=\dfrac{1}{2}\sin 2A$

80. *(a)* $\dfrac{\sin 38°-\cos 68°}{\cos 68°+\sin 38°}=\dfrac{\cos(90°-38°)-\cos 68°}{\cos 68°+\sin(90°-52°)}$

$$=\frac{\cos 52°-\cos 68°}{\cos 68°+\cos 52°}=\frac{2\sin\left(\dfrac{52°+68°}{2}\right)\sin\left(\dfrac{68°-52°}{2}\right)}{2\cos\left(\dfrac{52°+68°}{2}\right)\cos\left(\dfrac{68°-52°}{2}\right)}$$

$$=\frac{\sin 60°\cdot\sin 8°}{\cos 60°\cdot\cos 8°}=\tan 60°\cdot\tan 8°=\sqrt{3}\tan 8°$$

81. *(c)* $2\cos x-(\cos 3x+\cos 5x)$

$=2\cos x-2\cos 4x\cdot\cos x=2\cos x(1-\cos 4x)$

$=2\cos x\cdot 2\sin^2 2x=4\cos x(2\sin x\cos x)^2=16\cos^3 x\sin^2 x$

82. *(c)* $\cos 4x=1-2\sin^2 2x=1-2(2\sin x\cos x)^2$

$=1-2(4\sin^2 x\cos^2 x)=1-8\sin^2 x\cos^2 x$

83. *(a)* $2\sin A\cos^3 A-2\sin^3 A\cos A$

$=2\sin A\cos A(\cos^2 A-\sin^2 A)$

$=\sin 2A(\cos 2A)=\dfrac{2\sin 2A\cos 2A}{2}=\dfrac{1}{2}\sin 4A$

84. *(c)* Given, $a\cos\theta-b\sin\theta=c$

On squaring both sides, we get

$a^2\cos^2\theta+b^2\sin^2\theta-2ab\cos\theta\sin\theta=c^2$

$\Rightarrow$ $a^2(1-\sin^2\theta)+b^2(1-\cos^2\theta)-2ab\sin\theta\cos\theta=c^2$

$\Rightarrow a^2+b^2-c^2=a^2\sin^2\theta+b^2\cos^2\theta+2ab\sin\theta\cos\theta$

$\Rightarrow (a\sin\theta+b\cos\theta)^2=a^2+b^2-c^2$

$\Rightarrow a\sin\theta+b\cos\theta=\pm\sqrt{a^2+b^2-c^2}$

85. *(d)* Given, $\cos x+\cos^2 x=1\Rightarrow\cos x=1-\cos^2 x=\sin^2 x$ ...(i)

$\sin^8 x+2\sin^6 x+\sin^4 x=(\cos x)^4+2\cos^3 x+\cos^2 x$

$=(\cos^2 x+\cos x)^2$ [from Eq. (i)]

$=(1)^2=1$

86. *(b)* $\dfrac{n}{m}=\dfrac{\sec x-\cos x}{\operatorname{cosec} x-\sin x}=\dfrac{\dfrac{1-\cos^2 x}{\cos x}}{\dfrac{1-\sin^2 x}{\sin x}}$

$=\dfrac{\sin^2 x}{\cos x}\cdot\dfrac{\sin x}{\cos^2 x}=\left(\dfrac{\sin x}{\cos x}\right)^3$

$\Rightarrow$ $(\tan x)^3=\left(\dfrac{n}{m}\right)$ $\Rightarrow$ $\tan x=\left(\dfrac{n}{m}\right)^{1/3}$

87. *(a)* $2\cos\theta=x+\dfrac{1}{x}$ ...(i)

On cubing both sides, we get $8\cos^3\theta=\left(x+\dfrac{1}{x}\right)^3$

$8\cos^3\theta=x^3+\dfrac{1}{x^3}+3x\cdot\dfrac{1}{x}\left(x+\dfrac{1}{x}\right)$

$8\cos^3\theta=x^3+\dfrac{1}{x^3}+3(2\cos\theta)$ [from Eq. (i)]

$\Rightarrow$ $8\cos^3\theta-6\cos\theta=x^3+\dfrac{1}{x^3}$

$\Rightarrow$ $2[4\cos^3\theta-3\cos\theta]=x^3+\dfrac{1}{x^3}$ $\Rightarrow$ $2[\cos 3\theta]=x^3+\dfrac{1}{x^3}$

88. *(a)* We know that, the value of $\cos\theta$ decreases in the interval $0\le\theta\le\pi/2$

$\because$ $\cos\theta\ge\dfrac{1}{2}\Rightarrow\cos\theta\ge\cos\dfrac{\pi}{3}$ $\Rightarrow$ $\theta\le\dfrac{\pi}{3}$

89. *(b)* Given, $\sin A=\dfrac{3}{5}$ and $\cos B=\dfrac{12}{13}$

$\therefore$ $\tan A=\dfrac{\sin A}{\cos A}=\dfrac{\sin A}{\sqrt{1-\sin^2 A}}=\dfrac{3/5}{\sqrt{1-(3/5)^2}}=\dfrac{3/5}{4/5}=\dfrac{3}{4}$

and $\tan B=\dfrac{\sin B}{\cos B}=\dfrac{\sqrt{1-\cos^2 B}}{\cos B}=\dfrac{\sqrt{1-\dfrac{144}{169}}}{\dfrac{12}{13}}=\dfrac{5/13}{12/13}=\dfrac{5}{12}$

$$\frac{\tan A-\tan B}{1+\tan A\tan B}=\frac{\dfrac{3}{4}-\dfrac{5}{12}}{1+\dfrac{3}{4}\cdot\dfrac{5}{12}}=\frac{\dfrac{9-5}{12}}{\dfrac{48+15}{4\times 12}}$$

$$=\frac{\dfrac{1}{3}}{\dfrac{63}{48}}=\frac{1}{3}\times\frac{48}{63}=\frac{16}{63}$$

90. *(a)* Given, $B+C=60°\Rightarrow B=60°-C$

$\sin(120°-B)=\sin[120°-(60°-C)]=\sin(60°+C)$

$=\sin[180°-(60°+C)]=\sin(120°-C)$

91. *(a)* Given, $\dfrac{x\cdot\operatorname{cosec}^2 30°\sec^2 45°}{8\cos^2 45°\sin^2 60°}=\tan^2 60°-\tan^2 30°$

$\Rightarrow$ $\dfrac{x\times(2)^2\times(\sqrt{2})^2}{8\times\left(\dfrac{1}{\sqrt{2}}\right)^2\times\left(\dfrac{\sqrt{3}}{2}\right)^2}=(\sqrt{3})^2-\left(\dfrac{1}{\sqrt{3}}\right)^2$

$\Rightarrow$ $\dfrac{x\times 4\times 2}{8\times\dfrac{1}{2}\times\dfrac{3}{4}}=3-\dfrac{1}{3}$ $\Rightarrow$ $\dfrac{8x}{3}=\dfrac{8}{3}\Rightarrow x=1$

92. *(c)* $\dfrac{\sin x}{1+\cos x}=\dfrac{1-\cos x}{\sin x}$

The above identity is possible for all values of x except multiples of 180°. Since, for $x=180°$, $\sin x=0$ and $1+\cos x=0$.

93. *(b)* Given, $\cos x = k\cos(x-2y) \Rightarrow \dfrac{1}{k} = \dfrac{\cos(x-2y)}{\cos x}$

Apply componendo and dividendo theorem

$$\frac{1-k}{1+k} = \frac{\cos(x-2y)-\cos x}{\cos(x-2y)+\cos x} = \frac{2\sin\dfrac{x-2y+x}{2}\sin\dfrac{x-x+2y}{2}}{2\cos\dfrac{x-2y+x}{2}\cos\dfrac{x-2y-x}{2}}$$

$$= \frac{2\sin(x-y)\sin y}{2\cos(x-y)\cos y} = \tan(x-y)\tan y$$

94. *(b)* $\dfrac{\sin(x+y)-2\sin x+\sin(x-y)}{\cos(x+y)-2\cos x+\cos(x-y)}$

$$= \frac{\sin(x+y)+\sin(x-y)-2\sin x}{\cos(x+y)+\cos(x-y)-2\cos x}$$

$$= \frac{2\sin\dfrac{x+y+x-y}{2}\cdot\cos\dfrac{x+y-x+y}{2}-2\sin x}{2\cos\dfrac{(x+y+x-y)}{2}\cdot\cos\dfrac{(x+y-x+y)}{2}-2\cos x}$$

$$= \frac{2\sin x\cos y-2\sin x}{2\cos x\cos y-2\cos x} = \frac{2\sin x(\cos y-1)}{2\cos x(\cos y-1)} = \frac{\sin x}{\cos x} = \tan x$$

95. *(c)* Given, $\sin x + \sin y = a \Rightarrow 2\sin\dfrac{x+y}{2}\cos\dfrac{x-y}{2} = a$ ...(i)

and $\cos x + \cos y = b \Rightarrow 2\cos\dfrac{x+y}{2}\cdot\cos\dfrac{(x-y)}{2} = b$ ...(ii)

On dividing Eq. (i) from Eq. (ii), we get

$$\therefore \quad \frac{2\sin\dfrac{x+y}{2}\cos\dfrac{x-y}{2}}{2\cos\dfrac{x+y}{2}\cos\dfrac{(x-y)}{2}} = \frac{a}{b} \Rightarrow \tan\frac{x+y}{2} = \frac{a}{b}$$

96. *(a)* Here, r = length of largest hand of clock

l = distance covered by largest hand of clock

Angle traced in 60 min = 360°

Angle traced in 20 min $= \dfrac{360°}{60}\times 20 = 120° \Rightarrow \theta = \dfrac{2\pi}{3}$

As, $\theta = \dfrac{l}{r} \Rightarrow l = r\theta, \Rightarrow l = 42\times\dfrac{2\pi}{3} = \dfrac{42\times2\times22}{3\times7} = 88$ cm

97. *(d)* Given, $2\sin x + 15\cos^2 x = 7 \Rightarrow 2\sin x + 15 - 15\sin^2 x = 7$

$\Rightarrow \quad 15\sin^2 x - 2\sin x - 8 = 0$

$\Rightarrow \quad 15\sin^2 x - 12\sin x + 10\sin x - 8 = 0$

$\Rightarrow \quad 3\sin x(5\sin x - 4) + 2(5\sin x - 4) = 0$

$\Rightarrow \quad (3\sin x + 2)(5\sin x - 4) = 0$

$\Rightarrow \quad \sin x = \dfrac{4}{5} \quad \left(\because \sin x \neq -\dfrac{2}{3}, 0° < x < 90°\right)$

$\therefore \quad \cos x = \sqrt{1-\sin^2 x} = \sqrt{1-\left(\dfrac{4}{5}\right)^2} = \sqrt{\left(\dfrac{3}{5}\right)^2} = \dfrac{3}{5}$

$\therefore \quad \tan x = \dfrac{\sin x}{\cos x} = \dfrac{4/5}{3/5} = \dfrac{4}{3}$

98. *(a)* In ΔABC, $A + B + C = 180°$

$\therefore \quad A + B = 180° - C \Rightarrow \tan(A+B) = \tan(180° - C)$

$\Rightarrow \quad \dfrac{\tan A + \tan B}{1-\tan A\tan B} = -\tan C$

$\Rightarrow \quad \tan A + \tan B = -\tan C + \tan A\tan B\tan C$

$\Rightarrow \quad \tan A + \tan B + \tan C = \tan A\tan B\tan C$

99. *(c)* Given, $x = a\sec\theta\cos\phi$, $y = b\sec\theta\sin\phi$ and $z = c\tan\theta$,

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2} = \frac{a^2\sec^2\theta\cos^2\phi}{a^2}+\frac{b^2\sec^2\theta\sin^2\phi}{b^2}-\frac{c^2\tan^2\theta}{c^2}$$

$$= \sec^2\theta[\cos^2\phi+\sin^2\phi]-\tan^2\theta = \sec^2\theta-\tan^2\theta = 1$$

100. *(a)* The squares of the tangents of the angles 30°, 45° and 60° are in G.P.

$\Rightarrow \tan^2 30°, \tan^2 45°, \tan^2 60°$ are in G.P. $\Rightarrow \left(\dfrac{1}{\sqrt{3}}\right)^2, (1)^2, (\sqrt{3})^2$

are in G.P. $\Rightarrow \dfrac{1}{3}, 1, 3$ are in G.P.

which is true as $1^2 = \dfrac{1}{3}\times 3 \Rightarrow 1 = 1$

101. *(d)* We know that, in a cyclic quadrilateral sum of opposite angle is 180°.

$\therefore \quad A + C = 180°$ ...(i)

and $\quad B + D = 180°$ ...(ii)

$\therefore \quad \cos A + \cos B + \cos C + \cos D$

$= \cos A + \cos B + \cos(180° - A) + \cos(180° - B)$

$= \cos A + \cos B - \cos A - \cos B = 0$

$[\therefore \cos(180° - \theta) = -\cos\theta]$

102. *(a)* We know that, for $0° < \theta < 90°$, there exist only one θ such that $\sin\theta = a$.

103. *(b)* $\because \sin(B + C - A) = 1 = \sin 90° \Rightarrow B + C - A = 90°$ ...(i)

$\because \quad \cos(C + A - B) = 1 = \cos 0°$

$\therefore \quad C + A - B = 0°$ ...(ii)

and $\tan(A + B - C) = 1 = \tan 45°$

$\therefore \quad A + B - C = 45°$ ...(iii)

On adding Eqs. (i), (ii) and (iii), we get

$A + B + C = 135°$ ...(iv)

On subtracting Eqs. (i), (ii), (iii) from Eq. (iv), we get

$2A = 45°, 2B = 135°, 2C = 90°$

$A = 22\dfrac{1°}{2}, B = 67\dfrac{1°}{2}, C = 45°$

104. *(c)* $\cos(180° + A) + \cos(180° + B) + \cos(180° + C) + \cos(180° + D)$

$= -\cos A - \cos B - \cos C - \cos D$

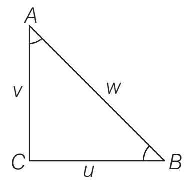
$= -\cos A - \cos B - \cos(180° - A) - \cos(180° - B)$

$(\because A + C = B + D = 180°)$

$= -\cos A - \cos B + \cos A + \cos B = 0$

105. *(d)* In ΔABC,

$$\tan A = \frac{BC}{AC} = \frac{u}{v} \text{ and } \tan B = \frac{v}{u}$$



Also, $u^2 + v^2 = w^2$ (by pythagoras theorem) ...(i)

$\therefore \tan A + \tan B = \dfrac{u}{v} + \dfrac{v}{u} = \dfrac{u^2+v^2}{uv} = \dfrac{w^2}{uv}$ [from Eq. (i)]

106. *(a)* $\because$ In 24 h, Earth rotate about its own axis = 360°

In 1 h Earth rotate about its own axis

$= \frac{360°}{24} = 15°$

In 4 h Earth rotate about its own axis

$= 15° \times 4 = 60°$

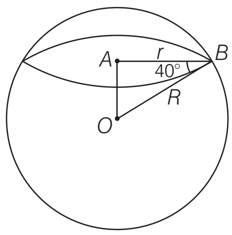
Since, in 60 min Earth rotate about its own axis = 15°

In 12 min Earth rotate about its own axis $= \frac{15° \times 12}{60°} = 3°$

$\therefore$ In 4 h 12 min Earth rotate about its own axis

$= 60° + 3° = 63°$

107. *(a)* In right angled ΔOAB,



$\cos 40° = \frac{AB}{OB} \Rightarrow \cos 40° = \frac{r}{R}$

$\Rightarrow \quad r = R \cos 40°$

So, the radius of the circle of latitude 40°S is $R \cos 40°$.

108. *(b)* Given, $\sin 3\theta = \cos(\theta - 2°)$

$\Rightarrow \quad \sin 3\theta = \sin[90° - (\theta - 2°)]$

$\Rightarrow \quad 3\theta = 90° - \theta + 2°$

$\Rightarrow \quad 4\theta = 92° \Rightarrow \theta = \frac{92°}{4} = 23°$

109. *(c)* I. RHS $= \cos^2\theta(1 + \tan\theta)(1 - \tan\theta)$

$= \cos^2\theta(1 - \tan^2\theta)$

$= \cos^2\theta\left(\frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta}\right)$

$= \frac{\cos^2\theta - \sin^2\theta}{\cos^2\theta + \sin^2\theta} =$ LHS

II. LHS $= \frac{1 + \sin\theta}{1 - \sin\theta} = \frac{(1 + \sin\theta)^2}{1 - \sin^2\theta}$

$= \left(\frac{1 + \sin\theta}{\cos\theta}\right)^2$

$= (\sec\theta + \tan\theta)^2$

Hence, both statements are correct.

110. *(c)* Given, $A = \frac{\pi}{6}$ and $B = \frac{\pi}{3}$

I. LHS $\sin A + \sin B = \sin\frac{\pi}{6} + \sin\frac{\pi}{3}$

$= \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1 + \sqrt{3}}{2}$

RHS $\cos A + \cos B = \cos\frac{\pi}{6} + \frac{\cos\pi}{3}$

$= \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3} + 1}{2}$

$\Rightarrow \quad \sin A + \sin B = \cos A + \cos B$

II. LHS $\tan A + \tan B = \tan\frac{\pi}{6} + \tan\frac{\pi}{3}$

$= \frac{1}{\sqrt{3}} + \sqrt{3} = \frac{4}{\sqrt{3}}$

RHS $\cot A + \cot B = \cot\frac{\pi}{6} + \cot\frac{\pi}{3}$

$= \sqrt{3} + \frac{1}{\sqrt{3}} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \tan A + \tan B = \cot A + \cot B$

Hence, both statements are true.

111. *(a)* $\because A + B + C + D = 360°$

$\therefore \quad A + B = 360° - (C + D)$

$\therefore \sin(A + B) = \sin[360° - (C + D)]$

$= -\sin(C + D)$

$\therefore \quad \sin(A + B) + \sin(C + D) = 0$

Also, $\cos(A + B) = \cos[360° - (C + D)]$

$\cos(A + B) = \cos(C + D)$

Hence, only statement I is correct.

112. *(d)* I. $1° = \frac{\pi}{180}$ radian $= \frac{3.14}{180} = 0.017$

$= 0.02$ radians (approx)

which is less than 0.03 radians.

II. 1 radian $= \frac{180}{\pi}$ degree

$= \frac{180}{3.14} = 57.32$ degree

which is greater than 45°.

Hence, both statements are true.

113. *(d)* I. $\sin\theta \cdot \sin(60° + \theta) \cdot \sin(60° - \theta)$

$= \sin\theta[\sin^2 60° - \sin^2\theta]$

$= \sin\theta\left[\frac{3}{4} - \sin^2\theta\right]$

$= \frac{1}{4}[3\sin\theta - 4\sin^3\theta] = \frac{1}{4}\sin 3\theta$

II. $\cos\theta \cdot \sin(30° + \theta) \cdot \sin(30° - \theta)$

$= \cos\theta[\sin^2 30° - \sin^2\theta]$

$= \cos\theta\left[\frac{1}{4} - (1 - \cos^2\theta)\right]$

$= \frac{1}{4}[4\cos^3\theta - 3\cos\theta] = \frac{1}{4}\cos 3\theta$

III. $\sin\theta \cdot \cos(30° + \theta) \cdot \cos(30° - \theta)$

$= \sin\theta[\cos^2 30° - \sin^2\theta]$

$= \sin\theta\left[\frac{3}{4} - \sin^2\theta\right] = \frac{1}{4}\sin 3\theta$

Hence, all three statements are correct.

114. *(d)* In right angled ΔABC,

$A + B + C = 180°$

I. $\sin(A + B) = \sin(180° - C) = \sin C$

II. $\sin\left(\frac{A + B}{2}\right) = \sin\left(\frac{180° - C}{2}\right)$

$= \sin\left(90° - \frac{C}{2}\right) = \cos\frac{C}{2}$

III. $\tan\left(\frac{A + B - C}{2}\right)$

$= \tan\frac{(180° - C) - C}{2}$

$= \tan\frac{(180° - 2C)}{2}$

$= \tan(90° - C) = \cot C$

IV. $\tan\frac{(A - B - C)}{2} = \tan\frac{A - (B + C)}{2}$

$= \tan\frac{A - (180° - A)}{2}$

$= \tan[-(90° - A)]$

$= -\tan(90° - A) = -\cot A$

Hence, all options are correct.

115. *(b)* Given, $\sin(A + B) = 1$

and $\quad \sin(A - B) = 1/2$

$\therefore \quad A + B = \frac{\pi}{2} \quad \ldots$(i)

$\therefore \quad A - B = \frac{\pi}{6} \quad \ldots$(ii)

On adding Eqs. (i) and (ii), we get

$2A = \frac{2\pi}{3} \Rightarrow A = \frac{\pi}{3}$ and $B = \frac{\pi}{6}$

116. *(c)* Now, $\tan(A + 2B) \cdot \tan(2A + B)$

$= \tan\left(\frac{\pi}{3} + \frac{\pi}{3}\right) \cdot \tan\left(\frac{2\pi}{3} + \frac{\pi}{6}\right)$

$= \tan\left(\frac{2\pi}{3}\right) \cdot \tan\left(\frac{5\pi}{6}\right)$

$= \tan\left(\frac{\pi}{2} + \frac{\pi}{6}\right) \cdot \tan\left(\frac{\pi}{2} + \frac{\pi}{3}\right)$

$= \left[-\cot\frac{\pi}{6}\right]\left[-\cot\frac{\pi}{3}\right] = (-\sqrt{3})\left(\frac{-1}{\sqrt{3}}\right) = 1$

117. *(b)* Now, $\sin^2 A - \sin^2 B$

$= \sin^2(\pi/3) - \sin^2(\pi/6)$

$= \left(\frac{\sqrt{3}}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$

118. *(c)* Since, $\tan A$, $\tan B$ are the roots of equation $3x^2 - 2\sqrt{3}x + 1 = 0$

$\therefore \quad \tan A + \tan B = \frac{2\sqrt{3}}{3}$

and $\quad \tan A \cdot \tan B = 1/3$

$\therefore \quad \tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$

$= \frac{\frac{2\sqrt{3}}{3}}{1 - 1/3} = \sqrt{3} = \tan\frac{\pi}{3}$

$\therefore \quad A + B = \pi/3$

Now, $A + B + C = \pi$

$\Rightarrow \quad C = \pi - (A + B) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

119. *(c)* Now, $\sin C + \cos C$
$= \sin(\pi - (A+B)) + \cos(\pi - (A+B))$
[as $A+B+C = \pi, C = \pi - (A+B)$]
$= \sin(A+B) - \cos(A+B)$
$= \sin(\pi/3) - \cos(\pi/3)$
$= \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2}$

120. *(b)* $2\sin B = \sqrt{\frac{1+\tan^2 C}{\text{cosec}^2 C}}$
$= \sqrt{\frac{1+\tan^2(2\pi/3)}{\text{cosec}^2(2\pi/3)}}$
$= \sqrt{\frac{1+\tan^2(\pi-\pi/3)}{\text{cosec}^2(\pi-\pi/3)}}$
$2\sin B = \sqrt{\frac{1+\tan^2 \pi/3}{\text{cosec}^2 \pi/3}}$
$= \sqrt{\frac{1+3}{4/3}} = \sqrt{\frac{4}{4}\times 3} = \sqrt{3}$
$\sin B = \frac{\sqrt{3}}{2} = \sin\frac{\pi}{3}, B = \frac{\pi}{3}$

121. *(c)* Given that, $\sin^2 x + \cos^2 x - 1 = 0$
$\Rightarrow \sin^2 x + \cos^2 x = 1$
which is an identity of trigonometric ratio and always true for every real value of x.
So, the equation has infinite solutions.

122. *(c)* Given that, $3\sin x + 5\cos x = 5$
On squaring both sides, we get
$9\sin^2 x + 25\cos^2 x + 30\sin x\cos x = 25$
$\Rightarrow 9(1-\cos^2 x) + 25(1-\sin^2 x) + 30\sin x\cos x = 25$
$\Rightarrow 9 + 25 - \{9\cos^2 x + 25\sin^2 x - 30\sin x\cos x\} = 25$
$\Rightarrow 9 = (3\cos x - 5\sin x)^2$
$\Rightarrow 3\cos x - 5\sin x = 3$

123. *(c)* Given, $p = a\sin x + b\cos x$ and $q = a\cos x - b\sin x$
$\Rightarrow p^2 = a^2\sin^2 x + b^2\cos^2 x + 2ab\sin x\cos x$
and $q^2 = a^2\cos^2 x + b^2\sin^2 x - 2ab\sin x\cos x$
$\therefore p^2 + q^2 = a^2(\sin^2 x + \cos^2 x) + b^2(\cos^2 x + \sin^2 x)$
$= a^2 + b^2$

124. *(b)* Given, $\alpha + \beta = 90°$ ...(i)
$\therefore \sqrt{(\text{cosec}\,\alpha \cdot \text{cosec}\,\beta)}\left(\frac{\sin\alpha}{\sin\beta} + \frac{\cos\alpha}{\cos\beta}\right)^{-1/2}$
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}}\left(\frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\sin\beta\cos\beta}\right)^{-1/2}$
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}}\left\{\frac{\sin(\alpha+\beta)}{\sin\beta\cos\beta}\right\}^{-1/2}$
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}}\left\{\frac{\sin 90°}{\cos(90°-\alpha)\sin\beta}\right\}^{-1/2}$ [from Eq. (i)]
$= \frac{1}{(\sin\alpha\sin\beta)^{1/2}} \times (\sin\alpha\sin\beta)^{1/2} = 1$

125. *(c)* I. $\frac{\cot 30°+1}{\cot 30°-1} = 2(\cos 30° + 1)$
$\Rightarrow \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2\left(\frac{\sqrt{3}}{2} + 1\right)$
$\Rightarrow \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = 2\left(\frac{\sqrt{3}+2}{2}\right)$
$\Rightarrow \frac{3+1+2\sqrt{3}}{3-1} = \sqrt{3} + 2$
$\Rightarrow \frac{4+2\sqrt{3}}{2} = \sqrt{3} + 2$
$\Rightarrow \frac{2(2+\sqrt{3})}{2} = \sqrt{3} + 2$
$\Rightarrow \sqrt{3} + 2 = \sqrt{3} + 2$
So, it is true.
II. $2\sin 45° \cos 45° - \tan 45° \cot 45° = 0$
$\Rightarrow 2 \times \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) - 1 \times 1 = 0$
or $2 \times \frac{1}{2} - 1 \times 1 = 0$
$\Rightarrow 1 - 1 = 0$, It is true.
Hence, both statements I and II are true.

126. *(c)* I. We know that, $\sin^2\theta + \cos^2\theta = 1$
$\therefore \sin^2 1° + \cos^2 1° = 1$
II. We have
$\sec^2 33° - \cot^2 57° = \text{cosec}^2 37° - \tan^2 53°$
LHS $= \sec^2 33° - \cot^2 57°$
$= \sec^2(90°-57°) - \cot^2 57°$
$= \text{cosec}^2 57° - \cot^2 57° = 1$
RHS $= \text{cosec}^2 37° - \tan^2 53°$
$= \text{cosec}^2 37° - \tan^2(90°-53°)$
$= \text{cosec}^2 37° - \cot^2 37° = 1$
LHS = RHS
Hence, both statements are correct.

127. *(b)* I. Given that, $\sin x + \cos x = 2$
$\Rightarrow (\sin x + \cos x)^2 = 4$
$\Rightarrow (\sin^2 x + \cos^2 x) + 2\sin x\cos x = 4$
$\Rightarrow 1 + \sin 2x = 4 \Rightarrow \sin 2x = 3$
As maximum value of $\sin\theta$ is 1.
Therefore, no value of x satisfy above equation. Thus, statement I is false.
II. $\sin x - \cos x = 0$
$\Rightarrow \tan x = 1 = \tan\pi/4$
It x lies in first quadrant, then $\tan x = \tan\pi/4 \Rightarrow x = \pi/4$
Thus, statement II is true.

128. *(b)* Given that, $\sin\theta \cdot \cos\theta = \frac{\sqrt{3}}{4}$ ...(i)
$\therefore \sin^4\theta + \cos^4\theta$
$= (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta \cdot \cos^2\theta$
$= (1)^2 - 2(\sin\theta \cdot \cos\theta)^2$
$= 1 - 2\left(\frac{\sqrt{3}}{4}\right)^2 = 1 - 2\cdot\frac{3}{16} = 1 - \frac{3}{8} = \frac{5}{8}$

129. *(d)* Given that, θ lies is in first quadrant and $\tan\theta = 3$
$\therefore \tan^2\theta = 9$
On adding both sides by 1, we get
$\Rightarrow 1 + \tan^2\theta = 10$
$\Rightarrow \sec^2\theta = 10 \Rightarrow \sec\theta = \sqrt{10}$
(since, θ lies in first quadrant)
$\Rightarrow \cos\theta = \frac{1}{\sqrt{10}}$ ...(i)
$\because \sin^2\theta = 1 - \cos^2\theta = 1 - \frac{1}{10} = \frac{9}{10}$
$\Rightarrow \sin\theta = \frac{3}{\sqrt{10}}$ ...(ii)
Now, $\sin\theta + \cos\theta = \frac{3}{\sqrt{10}} + \frac{1}{\sqrt{10}} = \frac{4}{\sqrt{10}}$

130. *(d)* If $0° < \theta < 90°$, then all the trigonometric ratios can be obtained when any one of the six ratios is given.

131. *(a)* $\sin A \cdot \cos A \cdot \tan A + \cos A \cdot \sin A \cdot \cot A$
$= \sin A \cdot \cos A \cdot \frac{\sin A}{\cos A} + \cos A \cdot \sin A \cdot \frac{\cos A}{\sin A}$
$= \sin^2 A + \cos^2 A$

132. *(a)* $\frac{\sin\theta}{1+\cos\theta} + \frac{1+\cos\theta}{\sin\theta}$
$= \frac{\sin^2\theta + (1+\cos\theta)^2}{\sin\theta(1+\cos\theta)}$
$= \frac{\sin^2\theta + 1 + \cos^2\theta + 2\cos\theta}{\sin\theta(1+\cos\theta)}$
$= \frac{\sin^2\theta + \cos^2\theta + 1 + 2\cos\theta}{\sin\theta(1+\cos\theta)}$
$= \frac{1+1+2\cos\theta}{\sin\theta(1+\cos\theta)}$
$= \frac{2(1+\cos\theta)}{\sin\theta(1+\cos\theta)} = 2\,\text{cosec}\theta$

133. *(c)* $\sec^2 D - \tan^2 D = 1$
Since, it is a trigonometric identity.

134. *(b)* Given, $\cos A + \cos^2 A = 1$
$\Rightarrow \cos A = 1 - \cos^2 A = \sin^2 A$ ...(i)
($\because \sin^2\theta + \cos^2\theta = 1$)
Now, $2(\sin^2 A + \sin^4 A)$
$= 2(\sin^2 A + \cos^2 A)$ [from Eq. (i)]
$= 2\cdot(1) = 2$

135. *(c)* $(1-\tan A)^2+(1+\tan A)^2+(1-\cot A)^2+(1+\cot A)^2$

$=2[1+\tan^2 A]+2[1+\cot^2 A]$

$[\because (a-b)^2+(a+b)^2=2(a^2+b^2)]$

$= 2\sec^2 A+2\operatorname{cosec}^2 A = 2\left(\frac{1}{\cos^2 A}+\frac{1}{\sin^2 A}\right)$

$=2\left(\frac{\sin^2 A+\cos^2 A}{\sin^2 A\cdot\cos^2 A}\right)=\frac{2\cdot(1)}{\sin^2 A\cdot\cos^2 A}=2\sec^2 A\cdot\operatorname{cosec}^2 A$

136. *(d)* Given, $a^2=\frac{1+2\sin\theta\cdot\cos\theta}{1-2\sin\theta\cdot\cos\theta}$

$\Rightarrow a^2=\frac{(\sin^2\theta+\cos^2\theta)+2\sin\theta\cdot\cos\theta}{(\sin^2\theta+\cos^2\theta)-2\sin\theta\cdot\cos\theta}$

$\Rightarrow a^2=\frac{(\sin\theta+\cos\theta)^2}{(\sin\theta-\cos\theta)^2}\Rightarrow\frac{a}{1}=\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta}$

$\Rightarrow \frac{a+1}{a-1}=\frac{(\sin\theta+\cos\theta)+(\sin\theta-\cos\theta)}{(\sin\theta+\cos\theta)-(\sin\theta-\cos\theta)}$

(applying componendo dividendo formula)

$\Rightarrow \frac{a+1}{a-1}=\frac{2\sin\theta}{2\cos\theta}=\tan\theta$

137. *(b)* Given, $\sin\theta=\frac{x^2-y^2}{x^2+y^2}\Rightarrow\cos^2\theta=1-\sin^2\theta=1-\left(\frac{x^2-y^2}{x^2+y^2}\right)^2$

$=\frac{(x^2+y^2)^2-(x^2-y^2)^2}{(x^2+y^2)^2}=\frac{4x^2y^2}{(x^2+y^2)^2}=\left(\frac{2xy}{x^2+y^2}\right)^2$

$\therefore \cos\theta=2xy/x^2+y^2$

138. *(b)* Let $f(\theta)=\sin\theta+\cos\theta$

Maximum and minimum value of $a\cos\theta+b\sin\theta$ is

$-\sqrt{a^2+b^2}\le a\cos\theta+b\sin\theta\le\sqrt{a^2+b^2}$

$\therefore -\sqrt{1+1}\le\cos\theta+\sin\theta\le\sqrt{1+1}$

$\Rightarrow -\sqrt{2}\le\cos\theta+\sin\theta\le\sqrt{2}$

for $0\le\theta\le 90°$, $1\le\cos\theta+\sin\theta\le\sqrt{2}$

Hence, statement I is false.

And let $g(\theta)=\tan\theta+\cot\theta=\tan\theta+\frac{1}{\tan\theta}$ $(\because \text{AM}\ge\text{GM})$

$\Rightarrow \frac{\tan\theta+\frac{1}{\tan\theta}}{2}\ge\left(\tan\theta\cdot\frac{1}{\tan\theta}\right)^{\frac{1}{2}}$

$\Rightarrow(\tan\theta+\cot\theta)\ge 2$

So, $(\tan\theta+\cot\theta)$ is always greater than 1.

Hence, statement II is true.

Solutions (Q. Nos. 139-141) Let the angle of A, B, C and D of a quadrilateral be x, $2x$, $4x$ and $5x$.

$\because x+2x+4x+5x=360°$

($\because$ sum of all interior in a quadrilateral is 360°)

$\Rightarrow 12x=360°,\ x=30°$

$\therefore \angle A=x=30°, \angle B=2x=60°$

$\angle C=4x=120°, \angle D=5x=150°$

139. *(a)* Now, $\cos(A+B)=\cos(30°+60°)=\cos 90°=0$

140. *(b)* Now, $\operatorname{cosec}(C-D+B)=\operatorname{cosec}(120°-150°+60°)$

$=\operatorname{cosec}(180°-150°)=\operatorname{cosec}30°=2$

141. *(d)* If $ABCD$ is a cyclic quadrilateral, then sum of opposite angles should be 180° but here

$\angle A+\angle C=30°+120°=150°\ne 180°$

and $\angle B+\angle D=60°+150°=210°\ne180°$

So, statement I is incorrect.

Now, $\sin(B-A)=\cos(D-C)$

$\Rightarrow \sin(60°-30°)=\cos(150°-120°)$

$\Rightarrow \sin 30°=\cos 30° \Rightarrow \frac{1}{2}\ne\frac{\sqrt{3}}{2}$

So, statement II is also incorrect.

142. *(a)* Given, $\sin\theta+\cos\theta=\sqrt{3}$

On squaring both sides, we get $(\sin\theta+\cos\theta)^2=(\sqrt{3})^2$

$\Rightarrow \sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta=3 \Rightarrow 1+2\sin\theta\cos\theta=3$

$\Rightarrow \sin\theta\cos\theta=\frac{3-1}{2}=1$ ...(i)

Now, $\tan\theta+\cot\theta=\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}=\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}=\frac{1}{\sin\theta\cos\theta}$

From Eq. (i), $\tan\theta+\cot\theta=\frac{1}{1}=1$

143. *(a)* $\operatorname{cosec}(75°+\theta)-\sec(15°-\theta)=\operatorname{cosec}(75°+\theta)-\sec[90°-(75°+\theta)]$

$=\operatorname{cosec}(75°+\theta)-\operatorname{cosec}(75°+\theta)=0$

144. *(c)* Given, $5\sin\theta+12\cos\theta=13$

On squaring both sides, we get

$25\sin^2\theta+144\cos^2\theta+120\sin\theta\cos\theta=169$

$\Rightarrow 25(1-\cos^2\theta)+144(1-\sin^2\theta)+120\sin\theta\cos\theta=169$

$\Rightarrow 25-25\cos^2\theta+144-144\sin^2\theta+120\sin\theta\cos\theta=169$

$\Rightarrow 25\cos^2\theta+144\sin^2\theta-120\sin\theta\cos\theta=169-169$

$\Rightarrow (5\cos\theta-12\sin\theta)^2=0 \Rightarrow 5\cos\theta-12\sin\theta=0$

145. *(c)* Given, $\sin\theta-\cos\theta=0$

$\therefore \sin\theta=\cos\theta$

Since, $\sin\theta$ and $\cos\theta$ are equal for $\theta=45°$.

$\therefore \sin^4\theta+\cos^4\theta=(\sin 45°)^4+(\cos 45°)^4$

$=\left(\frac{1}{\sqrt{2}}\right)^4+\left(\frac{1}{\sqrt{2}}\right)^4=\frac{1}{4}+\frac{1}{4}=\frac{1+1}{4}=\frac{2}{4}=\frac{1}{2}$

146. *(c)* $\frac{\cos^2(45°+\theta)+\cos^2(45°-\theta)}{\tan(60°+\theta)\tan(30°-\theta)}$

$=\frac{\cos^2(45°+\theta)+\cos^2[90°-(45°+\theta)]}{\tan(60°+\theta)\cdot\tan[90°-(60°+\theta]}$

$=\frac{\cos^2(45°+\theta)+\sin^2(45°+\theta)}{\tan(60°+\theta)\cdot\cot(60°+\theta)}=\frac{1}{1}=1$

147. *(b)* Given, $\tan\theta+\sec\theta=m \Rightarrow \tan\theta=m-\sec\theta$

On squaring both sides, we get

$\Rightarrow \tan^2\theta=m^2+\sec^2\theta-2m\sec\theta$

$\Rightarrow \tan^2\theta=m^2+1+\tan^2\theta-2m\sec\theta$

$\Rightarrow 0=m^2+1-2m\sec\theta \Rightarrow \sec\theta=\frac{m^2+1}{2m}$

148. *(c)* $\because\ \sin\alpha = \frac{\sqrt{3}}{2} \Rightarrow \sin\alpha = \sin 60°$

$\Rightarrow \alpha = 60°$ $\left(\because \sin 60° = \frac{\sqrt{3}}{2}\right)$

Now, $\cos\beta = \frac{\sqrt{3}}{2} \Rightarrow \beta = 30°$ $\left(\because \cos 30° = \frac{\sqrt{3}}{2}\right)$

and $\tan\gamma = 1 \Rightarrow \gamma = 45°$ $(\because \tan 45° = 1)$

$\therefore\ \alpha + \beta + \gamma = 60° + 30° + 45° = 135°$

149. *(a)* $$\frac{(1+\sec\theta - \tan\theta)\cos\theta}{(1+\sec\theta+\tan\theta)(1-\sin\theta)} = \frac{\left(1+\frac{1}{\cos\theta} - \frac{\sin\theta}{\cos\theta}\right)\cos\theta}{\left(1+\frac{1}{\cos\theta}+\frac{\sin\theta}{\cos\theta}\right)(1-\sin\theta)}$$

$$= \frac{\left(\frac{\cos\theta + 1 - \sin\theta}{\cos\theta}\right)\cos\theta}{\frac{(\cos\theta + 1 + \sin\theta)(1-\sin\theta)}{\cos\theta}}$$

$$= \frac{\cos\theta + 1 - \sin\theta}{\frac{\cos\theta + 1 + \sin\theta - \sin\theta\cos\theta - \sin\theta - \sin^2\theta}{\cos\theta}}$$

$$= \frac{\cos\theta + 1 - \sin\theta}{\frac{\cos\theta + 1 - \sin^2\theta - \sin\theta\cos\theta}{\cos\theta}} \quad (\because 1 - \sin^2\theta = \cos^2\theta)$$

$$= \frac{\cos\theta + 1 - \sin\theta}{\frac{\cos\theta + \cos^2\theta - \sin\theta\cos\theta}{\cos\theta}} = \frac{\cos\theta + 1 - \sin\theta}{\frac{\cos\theta\,(\cos\theta + 1 - \sin\theta)}{\cos\theta}} = 1$$

150. *(c)* In right angled ΔABC, if $\angle C$ is 90°, then

$\angle A + \angle B = 180° - 90° = 90°$

Now, $\cos(A+B) + \sin(A+B) = \cos 90° + \sin 90° = 0 + 1 = 1$

151. *(a)* Only statement I is correct as $\tan\theta$ increases faster than $\sin\theta$ as θ increases while statement II is wrong as the value of $\sin\theta + \cos\theta$ is not always greater than 1. It may also be equal to 1.

152. *(a)* Since, value of $\cos\theta$ decreases, from 0° to 90° and at 45° it is equal to the value of $\sin\theta$.

Similarly, value of $\sin\theta$ increases from 0 to 90° and at 45° it is equal to the value of $\cos\theta$.

For $0° < \theta < 45°$, $\cos\theta > \sin\theta$

So, value of $\cos 25° - \sin 25°$ is always positive but less than 1.

153. *(b)* In right angled ΔABC, $\angle B = 90°$

$\angle C = 180° - (\angle B + \angle A) = 180° - 90° - \angle A = 90° - \angle A$

$\therefore\ \sin C = \sin(90° - A) = \cos A = 4/5$

154. *(c)* Since, α and β are complementary angle.

$\therefore\ \alpha = 90° - \beta$

Now, $\sqrt{\cos\alpha\,\mathrm{cosec}\beta - \cos\alpha\sin\beta} = \sqrt{\frac{\cos\alpha}{\sin\beta} - \cos\alpha\sin\beta}$

$$= \sqrt{\frac{\cos\alpha}{\cos(90°-\beta)} - \cos\alpha\cos(90° - \beta)}$$

$$= \sqrt{\frac{\cos\alpha}{\cos\alpha} - \cos\alpha\cdot\cos\alpha} = \sqrt{1-\cos^2\alpha} = \sqrt{\sin^2\alpha} = \sin\alpha$$

155. *(d)* Given, $\sec\theta + \tan\theta = 2$ ...(i)

By trigonometric identity, $\sec^2\theta - \tan^2\theta = 1$

$\Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1 \Rightarrow \sec\theta - \tan\theta = 1/2$ ...(ii)

On adding Eqs. (i) and (ii), we get

$\Rightarrow 2\sec\theta = \frac{1}{2} + 2 \Rightarrow \sec\theta = \frac{5}{4}$

156. *(b)* $\mathrm{cosec}(75° + \theta) - \sec(15° - \theta) - \tan(55° + \theta) + \cot(35° - \theta)$

$= \mathrm{cosec}(75° + \theta) - \sec[90° - (75° + \theta)]$

$\quad - \tan(55° + \theta) + \cot[90° - (55° + \theta)]$

$= \mathrm{cosec}(75° + \theta) - \mathrm{cosec}(75° + \theta)$

$\quad - \tan(55° + \theta) + \tan(55° + \theta) = 0$

157. *(d)* $\sin 25° \sin 35° \sec 65° \sec 55°$

$= \sin 25° \cdot \sin 35° \cdot \frac{1}{\cos 65°} \cdot \frac{1}{\cos 55°}$

$= \sin 25° \cdot \sin 35° \cdot \frac{1}{\cos(90° - 25°)} \cdot \frac{1}{\cos(90° - 35°)}$

$= \sin 25° \cdot \sin 35° \cdot \frac{1}{\sin 25°} \cdot \frac{1}{\sin 35°} = 1$

158. *(c)* Given, $2\cot\theta = 3 \Rightarrow \cot\theta = \frac{3}{2}$

$$\therefore\ \frac{2\cos\theta - \sin\theta}{2\cos\theta + \sin\theta} = \frac{2\cot\theta - 1}{2\cot\theta + 1} = \frac{2\times\frac{3}{2} - 1}{2\times\frac{3}{2} + 1} = \frac{3-1}{3+1} = \frac{2}{4} = \frac{1}{2}$$

159. *(d)* $\sin^6\theta + \cos^6\theta = (\sin^2\theta)^3 + (\cos^2\theta)^3$

$= (\sin^2\theta + \cos^2\theta)(\sin^4\theta + \cos^4\theta - \sin^2\theta\cos^2\theta)$

$[\because a^3 + b^3 = (a+b)(a^2 + b^2 - ab)]$

$= (\sin^2\theta + \cos^2\theta)^2 - 2\sin^2\theta\cos^2\theta - \sin^2\theta\cos^2\theta$

$= (1 - 3\sin^2\theta\cos^2\theta) = 1 - 3\times\frac{1}{4} = 1 - \frac{3}{4} = \frac{1}{4}$

160. *(a)* Given, $\cos x + \sec x = 2$ ...(i)

$\Rightarrow \cos x + \frac{1}{\cos x} = 2 \Rightarrow \cos^2 x + 1 = 2\cos x$

$\Rightarrow \cos^2 x - 2\cos x + 1 = 0 \Rightarrow (\cos x - 1)^2 = 0 \Rightarrow \cos x = 1$

$\therefore\ \sec x = \frac{1}{\cos x} = 1$

$\therefore\ \cos^n x + \sec^n x = (1)^n + (1)^n = 1 + 1 = 2$

161. *(c)* Given, $\sin\theta + 2\cos\theta = 1$

On squaring both sides, we get $(\sin\theta + 2\cos\theta)^2 = 1$

$\Rightarrow \sin^2\theta + 4\cos^2\theta + 4\sin\theta\cos\theta = 1$

$\Rightarrow (1 - \cos^2\theta) + 4(1 - \sin^2\theta) + 4\sin\theta\cos\theta = 1$

$\Rightarrow -(\cos^2\theta + 4\sin^2\theta) + 4\sin\theta\cos\theta = 1 - 5$

$\Rightarrow \cos^2\theta + 4\sin^2\theta - 4\sin\theta\cos\theta = 4$

$\Rightarrow (2\sin\theta - \cos\theta)^2 = 4 \Rightarrow 2\sin\theta - \cos\theta = 2$

162. *(b)* Given, $\tan 8\theta = \cot 2\theta$

$\Rightarrow \tan 8\theta = \tan(90° - 2\theta) \Rightarrow 8\theta = 90° - 2\theta \Rightarrow \theta = 9°$

$\therefore\ \tan 5\theta \Rightarrow \tan 45° = 1$

163. *(a)* $\because\ \sin(A+B) = 1 = \sin 90° \Rightarrow (A+B) = 90°$

$\Rightarrow A = 90° - B$

Now, $\cos(A - B) = \cos A\cos B + \sin A\sin B$

$= \cos(90° - B)\cos B + \sin(90° - B)\sin B$

$= \sin B\cos B + \cos B\sin B = 2\sin B\cos B = \sin 2B$

164. *(d)* Clock will make right angle at $(5n + 15)\times\frac{12}{11}$ min past n.

Here, $n = 3$

$\therefore\ (5\times 3 + 15)\times\frac{12}{11}$ min past 3 $= 30\times\frac{12}{11}$ min past 3

$= 32\frac{8}{11}$ min past 3 i.e. 3 h $32\frac{8}{11}$ min

165. *(c)* Given, $\tan\theta + \cot\theta = 2$

$\Rightarrow \quad \dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\sin\theta} = 2 \Rightarrow \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} = 2$

$\Rightarrow \quad \dfrac{1}{\sin\theta\cos\theta} = 2 \Rightarrow \sin\theta\cos\theta = \dfrac{1}{2}$ ...(i)

Now, $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$

$= 1 + 2 \times \dfrac{1}{2} = 1 + 1$ [from Eq. (i)]

$\Rightarrow \quad (\sin\theta + \cos\theta)^2 = 2 \Rightarrow \sin\theta + \cos\theta = \sqrt{2}$

166. *(a)* $\dfrac{\sec x}{\cot x + \tan x} = \dfrac{1/\cos x}{\dfrac{\cos x}{\sin x} + \dfrac{\sin x}{\cos x}}$

$= \dfrac{1/\cos x}{\dfrac{\cos^2 x + \sin^2 x}{\sin x\cos x}} = \dfrac{1}{\cos x} \times \dfrac{\sin x\cos x}{1} = \sin x$

167. *(b)* $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$

$= (1 + \cot x - \operatorname{cosec} x)\left(1 + \dfrac{1}{\cot x} + \sec x\right)$

$= \dfrac{(1 + \cot x - \operatorname{cosec} x)(1 + \cot x + \operatorname{cosec} x)}{\cot x}$

$= \dfrac{(1 + \cot x)^2 - (\operatorname{cosec} x)^2}{\cot x}$

$= \dfrac{1^2 + \cot^2 x + 2\cot x - \operatorname{cosec}^2 x}{\cot x}$

$= \dfrac{1 + 2\cot x - (\operatorname{cosec}^2 x - \cot^2 x)}{\cot x} = \dfrac{1 + 2\cot x - 1}{\cot x} = 2$

168. *(d)* $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$

$= \left(\dfrac{1}{\sin x} - \sin x\right)\left(\dfrac{1}{\cos x} - \cos x\right)\left(\dfrac{\sin x}{\cos x} + \dfrac{\cos x}{\sin x}\right)$

$= \dfrac{(1 - \sin^2 x)(1 - \cos^2 x)(\sin^2 x + \cos^2 x)}{\sin x\cos x \cdot \sin x\cos x} = \dfrac{\cos^2 x\sin^2 x \times 1}{\sin^2 x\cos^2 x} = 1$

169. *(b)* $\dfrac{\sin x - \cos x + 1}{\sin x + \cos x - 1} = \dfrac{(\sin x - \cos x) + 1}{(\sin x + \cos x) - 1} \times \dfrac{(\sin x + \cos x) + 1}{(\sin x + \cos x) + 1}$

$= \dfrac{(\sin x - \cos x + 1)(\sin x + \cos x + 1)}{(\sin x + \cos x)^2 - 1}$

$[\because (a - b)(a + b) = a^2 - b^2]$

$= \dfrac{\sin^2 x + \sin x\cos x + \sin x - \cos x\sin x - \cos^2 x - \cos x + \sin x + \cos x + 1}{\sin^2 x + \cos^2 x + 2\sin x\cos x - 1}$

$= \dfrac{\sin^2 x + 2\sin x - \cos^2 x + 1}{1 + 2\sin x\cos x - 1}$

$= \dfrac{\sin^2 x + 2\sin x - (1 - \sin^2 x) + 1}{2\sin x\cos x}$ $[\because \cos^2 x = 1 - \sin^2 x]$

$= \dfrac{\sin^2 x + 2\sin x - 1 + \sin^2 x + 1}{2\sin x\cos x}$

$= \dfrac{2\sin^2 x + 2\sin x}{2\sin x\cos x} = \dfrac{2\sin x(\sin x + 1)}{2\sin x\cos x} = \dfrac{\sin x + 1}{\cos x}$

170. *(b)* $(\sin^2 x - \cos^2 x)(1 - \sin^2 x\cos^2 x)$

$= (\sin^2 x - \cos^2 x)[(\sin^2 x + \cos^2 x)^2 - \sin^2 x\cos^2 x]$

$(\because \sin^2 x + \cos^2 x = 1)$

$= (\sin^2 x - \cos^2 x)[(\sin^4 x + \cos^4 x + 2\sin^2 x\cos^2 x) - \sin^2 x\cos^2 x]$

$= (\sin^2 x - \cos^2 x)(\sin^4 x + \cos^4 x + \sin^2 x\cos^2 x)$

$= (\sin^2 x)^3 - (\cos^2 x)^3$ $[\because (a - b)(a^2 + b^2 + ab) = a^3 - b^3]$

$= \sin^6 x - \cos^6 x$

171. *(d)* $(\sin x \cdot \cos y + \cos x \cdot \sin y)(\sin x \cdot \cos y - \cos x \cdot \sin y)$

$= \sin(x + y) \cdot \sin(x - y) = \sin^2 x - \sin^2 y$

$[\because \sin^2 A - \sin^2 B = \sin(A + B)\sin(A - B)]$

172. *(d)* In ΔABC, $A + B + C = 180° \Rightarrow A + B = 180° - C$

$\tan(A + B) = \tan(180° - C) = -\tan C$

$\Rightarrow \quad \dfrac{\tan A + \tan B}{1 - \tan A\tan B} = -\tan C$

$\tan A + \tan B + \tan C = \tan A\tan B\tan C$

On dividing both sides by $\tan A\tan B\tan C$

$\dfrac{1}{\tan B\tan C} + \dfrac{1}{\tan A\tan C} + \dfrac{1}{\tan A\tan B} = 1$

$\Rightarrow \quad \cot B\cot C + \cot A\cot C + \cot A\cot B = 1$

173. *(a)* $\sin x + \operatorname{cosec} x = 2 \Rightarrow \sin x + \dfrac{1}{\sin x} = 2 \Rightarrow \sin^2 x + 1 = 2\sin x$

$\Rightarrow \quad \sin^2 x - 2\sin x + 1 = 0 \Rightarrow (\sin x - 1)^2 = 0 \Rightarrow \sin x = 1$

$\operatorname{cosec} x = \dfrac{1}{\sin x} = 1$

$\therefore \quad \sin^9 x + \operatorname{cosec}^9 x = (1)^9 + (1)^9 = 1 + 1 = 2$

174. *(b)* Given, $\sin x + \cos x = p$ ...(i)

and $\quad \sin^3 x + \cos^3 x = q$ ...(ii)

On cubing Eq. (i) on both sides, we get

$\sin^3 x + \cos^3 x + 3\sin x\cos x(\sin x + \cos x) = p^3$

$\Rightarrow \quad q + 3\sin x\cos x(p) = p^3$ [from Eqs. (i) and (ii)] ...(iii)

On squaring Eq. (i) on both sides, we get

$\sin^2 x + \cos^2 x + 2\sin x\cos x = p^2$

$\Rightarrow \quad \sin x\cos x = \dfrac{p^2 - 1}{2}$ $(\because \sin^2 x + \cos^2 x = 1)$

On putting this value in Eq. (iii), we get $q + \dfrac{3(p^2 - 1)p}{2} = p^3$

$\Rightarrow \quad 2q + 3p^3 - 3p = 2p^3 \Rightarrow p^3 - 3p = -2q$

175. *(d)* Since, 1 radian = 57°16′22″

So, sin 1° < sin 1 and cos 1° > cos 1

Hence, neither statement I nor II is correct.

176. *(b)* $\operatorname{cosec}^2 67° + \sec^2 57° - \cot^2 33° - \tan^2 23°$

$= \operatorname{cosec}^2(90° - 23°) + \sec^2(90° - 33°) - \cot^2 33° - \tan^2 23°$

$= \sec^2 23° + \operatorname{cosec}^2 33° - \cot^2 33° - \tan^2 23°$

$= 1 + \tan^2 23° + 1 + \cot^2 33° - \cot^2 33° - \tan^2 23° = 2$

$(\because 1 + \tan^2 q = \sec^2 q$ and $1 + \cot^2 q = \operatorname{cosec}^2 q)$

177. *(c)* Given, $\tan(A + B) = \sqrt{3} = \tan 60°$

$\therefore \quad A + B = 60°$ ...(i)

and $\quad \tan A = 1 \Rightarrow \tan A = \tan 45°$

$\therefore \quad A = 45°$

From Eq. (i), $A + B = 60° \Rightarrow 45° + B = 60° \Rightarrow B = 15°$

Now, $\quad \tan(A - B) = \tan(45° - 15°) = \tan 30° = 1/\sqrt{3}$

Hence, the value of $\tan(A - B)$ is $1/\sqrt{3}$.

178. *(d)* Given, $\tan A + \cot A = 4$

On squaring both sides, we get

$$(\tan A + \cot A)^2 = (4)^2$$

$\Rightarrow \tan^2 A + \cot^2 A + 2\tan A \cot A = 16$

$\Rightarrow \quad \tan^2 A + \cot^2 A + 2 = 16$

$\Rightarrow \quad \tan^2 A + \cot^2 A = 14$

Again, squaring both sides, we get

$$(\tan^2 A + \cot^2 A)^2 = (14)^2$$

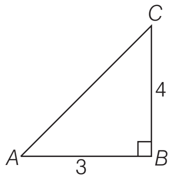
$\Rightarrow \quad \tan^4 A + \cot^4 A + 2\tan^2 A \cot^2 A = 196$

$\Rightarrow \quad \tan^4 A + \cot^4 A + 2 = 196$

$\Rightarrow \quad \tan^4 A + \cot^4 A = 194$

179. *(c)* In right angled ΔABC, $AB : BC = 3 : 4$

or $\dfrac{AB}{BC} = \dfrac{3}{4}$



Now, in ΔABC

$AC^2 = AB^2 + BC^2$

$= 3^2 + 4^2$

$= 9 + 16 = 25$

$\Rightarrow AC = 5 \Rightarrow \sin A = \dfrac{BC}{AC} = \dfrac{4}{5}$

$\sin B = 90° = \sin 90° = 1, \sin C = \dfrac{AB}{AC} = \dfrac{3}{5}$

Now, $\sin A + \sin B + \sin C = \dfrac{4}{5} + 1 + \dfrac{3}{5}$

$= \dfrac{4+5+3}{5} = \dfrac{12}{5}$

180. *(b)* Given, $\sin x + \cos x = C$

On squaring both sides, we get

$$(\sin x + \cos x)^2 = C^2$$

$\Rightarrow \quad \sin^2 x + \cos^2 x + 2\sin x \cos x = C^2$

$\Rightarrow \quad 1 + 2\sin x \cos x = C^2$

$\Rightarrow \quad \sin x \cos x = \dfrac{C^2 - 1}{2} \quad \ldots(\text{i})$

Now, $\sin^6 x + \cos^6 x = (\sin^2 x)^3 + (\cos^2 x)^3$

$= (\sin^2 x + \cos^2 x)[\sin^4 x + \cos^4 x - \sin^2 x \cos^2 x]$

$[\because a^3 + b^3 = (a+b)(a^2 + b^2 - ab)]$

$= 1[(\sin^2 x + \cos^2 x)^2 - 3\sin^2 x \cos^2 x]$

$= (1 - 3\sin^2 x \cos^2 x)$

$\therefore \sin^6 x + \cos^6 x = 1 - 3\sin^2 x \cos^2 x$

$= 1 - 3\left(\dfrac{C^2 - 1}{2}\right)^2 \quad$ [from Eq. (i)]

$= 1 - 3\left(\dfrac{C^4 + 1 - 2C^2}{4}\right)$

$= \dfrac{4 - 3C^4 - 3 + 6C^2}{4} = \dfrac{1 + 6C^2 - 3C^4}{4}$

181. *(c)* Given, $\dfrac{3 - \tan^2 A}{1 - 3\tan^2 A} = k$

$\Rightarrow \quad 3 - \tan^2 A = k(1 - 3\tan^2 A)$

$\Rightarrow \quad 3 - \tan^2 A = k - 3k\tan^2 A$

$\Rightarrow \quad 3k\tan^2 A - \tan^2 A = k - 3$

$\Rightarrow \quad \tan^2 A(3k - 1) = k - 3$

$\Rightarrow \quad \tan^2 A = \dfrac{k-3}{3k-1}$

where $\dfrac{k-3}{3k-1} > 0 \Rightarrow k < \dfrac{1}{3}$ or $k > 3$

Now, $\operatorname{cosec} A\,(3\sin A - 4\sin^3 A)$

$= \operatorname{cosec} A \times \sin A\,(3 - 4\sin^2 A)$

$= \dfrac{1}{\sin A} \times \sin A\,(3 - 4\sin^2 A)$

$\left(\because \operatorname{cosec} A = \dfrac{1}{\sin A}\right)$

$= 3 - 4\sin^2 A = 3 - 4\left[\dfrac{\tan^2 A}{1 + \tan^2 A}\right]$

$= \dfrac{3 - \tan^2 A}{1 + \tan^2 A} = \dfrac{3 - \dfrac{k-3}{3k-1}}{1 + \dfrac{k-3}{3k-1}}$

$= \dfrac{9k - 3 - k + 3}{3k - 1 + k - 3}$

$= \dfrac{8k}{4k - 4} = \dfrac{2k}{k-1}$,

where $k < \dfrac{1}{3}$ or $k > 3$

182. *(c)* Given, $p = \sqrt{\dfrac{1 - \sin x}{1 + \sin x}}$

$p = \sqrt{\dfrac{(1 - \sin x)(1 - \sin x)}{(1 + \sin x)(1 - \sin x)}}$

$= \dfrac{1 - \sin x}{\sqrt{1 - \sin^2 x}} = \dfrac{1 - \sin x}{\cos x}$

$r = \dfrac{\cos x}{1 + \sin x} \times \dfrac{(1 - \sin x)}{(1 - \sin x)} = \dfrac{\cos(1 - \sin x)}{1 - \sin^2 x}$

$r = \dfrac{\cos x\,(1 - \sin x)}{\cos^2 x} = \dfrac{1 - \sin x}{\cos x}$

$\Rightarrow \quad p = q = r$

Also, $q \times r = \dfrac{1 - \sin x}{\cos x} \times \dfrac{\cos x}{1 + \sin x}$

$= \dfrac{1 - \sin x}{1 + \sin x} = p^2$

Hence, both statements are correct.

183. *(a)* I. LHS $= \dfrac{\cos A}{1 - \tan A} + \dfrac{\sin A}{1 - \cot A}$

$= \dfrac{\cos A}{1 - \dfrac{\sin A}{\cos A}} + \dfrac{\sin A}{1 - \dfrac{\cos A}{\sin A}}$

$= \dfrac{\cos^2 A}{\cos A - \sin A} + \dfrac{\sin^2 A}{\sin A - \cos A}$

$= \dfrac{\cos^2 A - \sin^2 A}{(\cos A - \sin A)}$

$= \dfrac{(\cos A - \sin A)(\cos A + \sin A)}{(\cos A - \sin A)}$

$= (\cos A + \sin A) =$ RHS

II. $[(1 - \sin A) - \cos A]^2$

$= (1 - \sin A)^2 + \cos^2 A - 2\cos A(1 - \sin A)$

$= 1 + \sin^2 A - 2\sin A + \cos^2 A - 2\cos A(1 - \sin A)$

$= 1 + \sin^2 A + \cos^2 A - 2\sin A - 2\cos A(1 - \sin A)$

$= 2 - 2\sin A - 2\cos A(1 - \sin A)$

$= 2(1 - \sin A) - 2\cos A(1 - \sin A)$

$= 2(1 - \sin A)(1 - \cos A) \neq$ RHS

Hence, only statement I is correct.

184. *(a)* We have, $\sin\theta = \dfrac{2a + 3b}{3b} = 1 + \dfrac{2a}{3b}$

Since, $\sin\theta$ is always smaller or equal to 1 but $1 + \dfrac{2a}{3b} > 1$.

Hence, it is not possible.

185. *(a)* Since, $0 \le \cos^2 x \le 1$

$\because \; -1 \le \cos^2 x + \cos^2 y - \cos^2 z \le 2$

$\therefore$ Minimum value of the given expression is -1.

186. *(a)* We have, $\tan\theta + \sec\theta = 2$

$\Rightarrow \quad \sec\theta = 2 - \tan\theta$

On squaring both sides, we get

$$\sec^2\theta = 4 + \tan^2\theta - 4\tan\theta$$

$\Rightarrow \quad 1 + \tan^2\theta = 4 + \tan^2\theta - 4\tan\theta$

$\Rightarrow \quad 4\tan\theta = 3 \Rightarrow \tan\theta = \dfrac{3}{4}$

187. *(b)* We have, $\dfrac{\cos\theta}{1 - \sin\theta} \times \dfrac{1 + \sin\theta}{1 + \sin\theta}$

$= \dfrac{\cos\theta(1 + \sin\theta)}{1 - \sin^2\theta} = \dfrac{\cos\theta(1 + \sin\theta)}{\cos^2\theta}$

$= \dfrac{1 + \sin\theta}{\cos\theta}$

188. *(c)* We have,

$\tan(x + 40)° \tan(x + 20)° \tan(3x)° \tan(70 - x)° \tan(50 - x)° = 1$

$[\because \; \cot(90° - \theta) = \tan\theta]$

$\Rightarrow \quad \tan(x + 40)° \tan(x + 20)° \tan(3x)° \cot(90° - 70° + x)° \cot(90° - 50° + x)° = 1$

$\Rightarrow \quad \tan(x + 40)° \tan(x + 20)° \tan(3x)° \cot(20 + x)° \cot(40 + x)° = 1$

$\Rightarrow \quad \tan(3x)° = 1 \Rightarrow \tan(3x)° = \tan 45°$

$\Rightarrow \quad 3x = 45 \quad \Rightarrow \quad x = 15$

189. *(d)* We have, $32\cot^2\dfrac{\pi}{4} - 8\sec^2\left(\dfrac{\pi}{3}\right) + 8\cos^3\left(\dfrac{\pi}{6}\right)$

$= 32 \cdot (1) - 8 \cdot (2)^2 + 8 \cdot \left(\dfrac{\sqrt{3}}{2}\right)^3$

$= 32 - 8 \cdot (4) + 8 \cdot \dfrac{3\sqrt{3}}{8}$

$= 32 - 32 + 3\sqrt{3} = 3\sqrt{3}$

190. *(b)* We have, $x = a\cos\theta$ and $y = b\cot\theta$

$\therefore \left(\frac{a}{x}-\frac{b}{y}\right)\left(\frac{a}{x}+\frac{b}{y}\right)=\frac{a^2}{x^2}-\frac{b^2}{y^2}$

$= \sec^2\theta - \tan^2\theta = 1$

191. *(a)* We have, $\sin\theta\cos\theta = 2\cos^3\theta - \frac{3}{2}\cos\theta$

$\Rightarrow \quad 2\sin\theta = 4\cos^2\theta - 3$

$\Rightarrow \quad 2\sin\theta = 4 - 4\sin^2\theta - 3$

$\Rightarrow \quad 4\sin^2\theta + 2\sin\theta - 1 = 0,$

$\Rightarrow \quad \sin\theta = \frac{-2\pm\sqrt{4+16}}{8} = \frac{-2\pm 2\sqrt{5}}{8}$

$\Rightarrow \quad \sin\theta = \frac{-1\pm\sqrt{5}}{4}$

Since, θ is acute angle

$\therefore \quad \sin\theta > 0$

$\Rightarrow \quad \sin\theta = \frac{\sqrt{5}-1}{4}$

192. *(d)* Hours moved by hour hand = 5 hrs 10 min

$= 5 + \frac{10}{60} = \frac{31}{6}$ hrs

Angle traced by hour hand in 1 hr = 30°

$\therefore$ Angle traced by hour hand is

$\left(\frac{31}{6}\right)$ hrs $= 30\times\frac{31}{6} = 155°$

193. *(b)* I. If $45° < \theta < 90°$, then $\sin\theta > \cos\theta$

$\therefore \sin 66° > \cos 66°$

II. When $0° < \theta < 45°$, $\cos\theta > \sin\theta$

$\therefore \cos 26° > \sin 26°$

or $\quad \sin 26° < \cos 26°$.

Hence only II is correct.

194. *(a)* I. LHS $= \frac{1+\tan^2\theta}{1+\cot^2\theta} = \frac{\sec^2\theta}{\text{cosec}^2\theta}$

$= \tan^2\theta$

RHS $= \left(\frac{1-\tan\theta}{1-\cot\theta}\right)^2 = \left[\frac{\tan\theta[\cot\theta - 1]}{1-\cot\theta}\right]^2$

$= (-\tan\theta)^2 = \tan^2\theta$

$\therefore$ LHS = RHS

II. $\cot\theta = \frac{1}{\tan\theta} \Rightarrow \tan\theta\cdot\cot\theta = 1$

which true for all value of θ.

So, statement II is false.

Hence, only statement I is correct.

195. *(d)* We have, $A + B + C = \pi$

$\Rightarrow \quad \frac{\pi}{2} + B + B + C = \pi \; [A = B + \pi/2]$

$\Rightarrow \quad 2B + C = \pi - \frac{\pi}{2} = \frac{\pi}{2}$

196. *(c)* $\left(\frac{\sin 35°}{\cos 55°}\right)^2 - \left(\frac{\cos 55°}{\sin 35°}\right)^2 + 2\sin 30°$

$= \left[\frac{\sin(90°-55°)}{\cos 55°}\right]^2$

$- \left[\frac{\cos(90°-35°)}{\sin 35°}\right]^2 + 2\times\frac{1}{2}$

$= \left(\frac{\cos 55°}{\cos 55°}\right)^2 - \left(\frac{\sin 35°}{\sin 35°}\right)^2 + 1$

$= 1^2 - 1^2 + 1 = 1$

197. *(c)* We have, $\tan\theta + \cot\theta = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \frac{\sin^2\theta + \cos^2\theta}{\cos\theta\cdot\sin\theta} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \frac{1}{\frac{\sin 2\theta}{2}} = \frac{4}{\sqrt{3}}$

$\Rightarrow \quad \sin 2\theta = \frac{\sqrt{3}}{2}$

$\Rightarrow \quad 2\theta = \frac{\pi}{3} \Rightarrow \theta = \frac{\pi}{6}$

$\therefore \quad \sin\theta + \cos\theta = \sin\frac{\pi}{6} + \cos\frac{\pi}{6}$

$= \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{\sqrt{3}+1}{2}$

198. *(b)* We have,

$p = \cot\theta + \tan\theta = \text{cosec}\theta\cdot\sec\theta$

$q = \sec\theta - \cos\theta = \sin^2\theta\cdot\sec\theta$

$\therefore \quad (p^2q)^{2/3} - (q^2p)^{2/3}$

$= (\text{cosec}^2\theta\cdot\sec^2\theta\cdot\sin^2\theta\cdot\sec\theta)^{2/3}$

$- (\sin^4\theta\cdot\sec^2\theta\cdot\text{cosec}\theta\cdot\sec\theta)^{2/3}$

$= (\sec^3\theta)^{2/3} - (\sin^3\theta\cdot\sec^3\theta)^{2/3}$

$= \sec^2\theta - \sin^2\theta\cdot\sec^2\theta$

$= \sec^2\theta(1-\sin^2\theta) = 1$

199. *(d)* Given, $\frac{x}{a} - \frac{y}{b}\tan\theta = 1 \quad$ ...(i)

and $\quad \frac{x}{a}\tan\theta + \frac{y}{b} = 1 \quad$ ...(ii)

On solving Eqs. (i) and (ii), we get

$\frac{x}{a} = \frac{1+\tan\theta}{\sec^2\theta}$ and $\frac{y}{b} = \frac{1-\tan\theta}{\sec^2\theta}$

$\therefore \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{(1+\tan\theta)^2 + (1-\tan\theta)^2}{\sec^4\theta}$

$= \frac{2\sec^2\theta}{\sec^4\theta}$

$= 2\cos^2\theta$

200. *(c)* $\because \quad 3 - \tan^2\theta = \alpha(1 - 3\tan^2\theta)$

$\Rightarrow (3\alpha - 1)\tan^2\theta = \alpha - 3$

$\Rightarrow \quad \tan^2\theta = \frac{\alpha-3}{3\alpha-1}$

As, $\tan^2\theta \ge 0$, then $\frac{\alpha-3}{3\alpha-1} \ge 0 \Rightarrow \alpha \ge 3$

or, $\alpha < \frac{1}{3} \quad \Rightarrow \quad \alpha \in \left(-\infty, \frac{1}{3}\right) \cup [3, \infty)$

201. *(a)* To exchange the position, both hands has to cover 360° together.

Angle traced by hour hand in 1 min

$= (1/2)°$

Angle traced by minute hand in 1 min = 6°

Let the required time be t min. Then,

$6t + \frac{1}{2}t = 360° \Rightarrow t = \frac{360\times 2}{13}$

$= \frac{720}{13}$ min = 55.38 min

202. *(c)* I. $\sqrt{\frac{1-\cos\theta}{1+\cos\theta}} = \sqrt{\frac{1-\cos\theta}{1+\cos\theta}\times\frac{1-\cos\theta}{1-\cos\theta}}$

$= \sqrt{\frac{(1-\cos\theta)^2}{1-\cos^2\theta}} = \sqrt{\frac{(1-\cos\theta)^2}{\sin^2\theta}}$

$= \frac{1-\cos\theta}{\sin\theta} = \text{cosec}\,\theta - \cot\theta$

Hence, statement I is true.

II. $\sqrt{\frac{1+\cos\theta}{1-\cos\theta}} = \sqrt{\frac{1+\cos\theta}{1-\cos\theta}\times\frac{1+\cos\theta}{1+\cos\theta}}$

$= \sqrt{\frac{(1+\cos\theta)^2}{\sin^2\theta}} = \frac{1+\cos\theta}{\sin\theta}$

$= \text{cosec}\,\theta + \cot\theta$

Hence, statement II is true.

203. *(b)* $\because \frac{\cos^2\theta - 3\cos\theta + 2}{\sin^2\theta} = 1$

$\Rightarrow \cos^2\theta - 3\cos\theta + 1 + 1 - \sin^2\theta = 0$

$\Rightarrow \quad \cos^2\theta - 3\cos\theta + 1 + \cos^2\theta = 0$

$\Rightarrow \quad 2\cos^2\theta - 3\cos\theta + 1 = 0$

$\Rightarrow \quad (2\cos\theta - 1)(\cos\theta - 1) = 0$

$\Rightarrow \quad \cos\theta = \frac{1}{2}$ or, $\cos\theta = 1$

$\Rightarrow \quad \theta = \frac{\pi}{3} [\because \; 0 < \theta < \pi/2]$

Hence, only II is correct.

204. *(d)* I. As, $\cos x$ lies between -1 and 1, then $\cos x = 2^{m+1}$ does not exist for positive value of m.

II. The given relation $mn \ge m + n$ is true for $m, n \in N$ and $m > 1$ and $n > 1$.

Thus, statement II is not true.

Hence, neither I nor II is correct

21

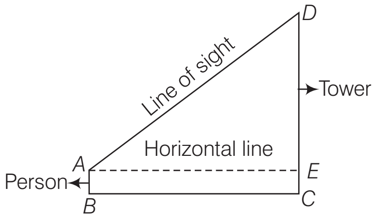
HEIGHT AND DISTANCE

Regularly (1-2) questions have been asked from this chapter. Generally this is the simplest application of trigonometry related to our real world and a little practice can get you command over this section.

It is an important application of trigonometry which helps us to find the height of any object and distance of that object from any point which are not directly measurable. If the angle of elevation/depression from a point is known.

Line of sight

A line of sight is the line drawn from the eye of an observer to the point, where the object is viewed by the observer.



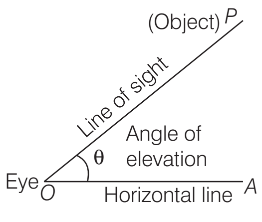
Horizontal Line

The line of sight which is parallel to ground level is known as horizontal line.

ANGLE OF ELEVATION

Angle of elevation is defined as an angle formed by the line of sight with the horizontal line when an object is viewed in the upward direction.

Let P be the position of the object above the horizontal line OA and O be the eye of the observer, then $\angle AOP$ is called angle of elevation.



IMPORTANT POINTS

- If the observer moves towards the perpendicular line (tower/building), then angle of elevation increases and if the observer moves away from the perpendicular line (tower/building), then the angle of elevation decreases.
- If the angle of elevation of Sun above a tower decreases, then the length of shadow of a tower increases.
- If height of tower is doubled and the distance between the observer and foot of the tower is also doubled, then the angle of elevation remains same.

EXAMPLE 1. The angle of elevation of the top of a tower of height $100\sqrt{3}$ m from a point at a distance of 100 m from the foot of the tower on a horizontal plane is

a. 30° **b.** 60° **c.** 75° **d.** 105°

Sol. **b.** Let ϕ be the angle of elevation.

Given, AC = Height of tower

$= 100\sqrt{3}$ m

and $AB = 100$ m

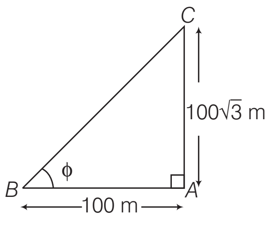
In right angled ΔABC,

$$\tan\phi = \frac{AC}{AB} = \frac{100\sqrt{3}}{100} = \sqrt{3}$$

$\Rightarrow \tan\phi = \tan 60°$

$\Rightarrow \phi = 60°$

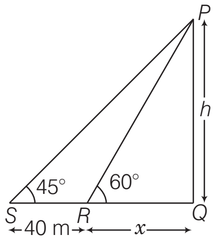
Hence, angle of elevation is 60°.



EXAMPLE 2. The angle of elevation of a tower at a point is 45°. After going 40 m towards the foot of the tower, the angle of elevation of the tower becomes 60°. Then, the height of the tower is

a. $20(\sqrt{3}+1)$ m
b. $\frac{20\sqrt{3}}{\sqrt{3}+1}$ m
c. $\frac{40\sqrt{3}}{\sqrt{3}+1}$ m
d. $\frac{40\sqrt{3}}{\sqrt{3}-1}$ m

Sol. d. Let $PQ(=h)$ be the height of the tower. Let S and R be the points where the angles subtended are 45° and 60°, respectively.



In right angled ΔPQR,

$$\tan 60° = \frac{PQ}{RQ}$$

$$\Rightarrow \quad \sqrt{3} = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \ldots(i)$$

In right angled ΔPQS, $\tan 45° = \frac{PQ}{SQ} = \frac{PQ}{SR+RQ} = \frac{h}{40+x}$

$$\Rightarrow \quad h = 40 + x \Rightarrow h = 40 + \frac{h}{\sqrt{3}} \quad \text{[from Eq. (i)]}$$

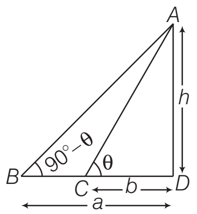
$$\Rightarrow h\left(1-\frac{1}{\sqrt{3}}\right) = 40 \Rightarrow h\left(\frac{\sqrt{3}-1}{\sqrt{3}}\right) = 40$$

$$\therefore \quad h = \frac{40\sqrt{3}}{\sqrt{3}-1} \text{ m}$$

EXAMPLE 3. The angles of elevation of the top of a tower from two points at distances 'a' and 'b' $(a > b)$ from the base and in the same straight line with it are complementary. Then, the height of the tower is

a. $\frac{1}{\sqrt{ab}}$
b. $\sqrt{ab}$
c. $\frac{a}{b}$
d. ab

Sol. b. Let $AD = h$ be the height of the tower and the angle of elevation of the top of tower at B and C be $(90° - \theta)$ and θ, respectively.



In right angled ΔADC,

$$\tan\theta = \frac{AD}{CD} = \frac{h}{b} \quad \left[\because \tan\theta = \frac{P}{B}\right]$$

$$h = b\tan\theta \quad \ldots(i)$$

In right angled ΔADB, $\tan(90° - \theta) = \frac{AD}{BD} \Rightarrow \cot\theta = \frac{h}{a}$

$$[\because \tan(90° - \theta) = \cot\theta]$$

$$\Rightarrow \quad \frac{h}{a} = \cot\theta \Rightarrow \frac{a}{h} = \tan\theta \quad \ldots(ii)$$

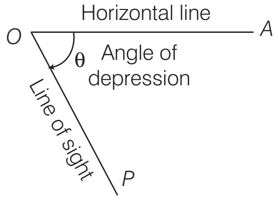
From Eqs. (i) and (ii), we get

$$h = b\left(\frac{a}{h}\right) \Rightarrow h^2 = ab$$

$$\therefore \quad h = \sqrt{ab}$$

ANGLE OF DEPRESSION

Angle of depression is defined as an angle formed by the line of sight with the horizontal line when an object is viewed in the downward direction.



Let P be the position of the object below the horizontal line OA and O be the eye of the observer, then $\angle AOP$ is called angle of depression.

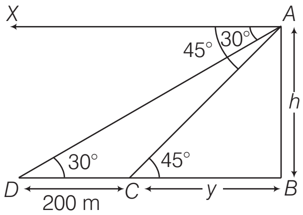
IMPORTANT POINTS

- The angle of depression of a point P as seen from a point O is numerically equal to the angle of elevation of O as seen from P.
- Angle of elevation and angle of depression are always acute angle.
- Unless mentioned, the height of the observer is not considered.

EXAMPLE 4. The angle of depression of two ships from the top of a light house are 45° and 30° towards East. If the ships are 200 m apart, the height of the light house is

a. $\frac{200}{\sqrt{3}-1}$ m
b. $\frac{200}{\sqrt{3}+1}$ m
c. $\frac{100}{\sqrt{3}-1}$ m
d. $\frac{100}{\sqrt{3}+1}$ m

Sol. a. Let $AB(=h)$ be the height of the light house and D, C be the ships.



Given, $DC = 200$ m

$$\angle ADC = \angle XAD = 30°$$

and $\angle ACB = \angle XAC = 45°$

Now, in ΔACB,

$$\tan 45° = \frac{AB}{CB} \Rightarrow 1 = \frac{h}{y} \Rightarrow \quad h = y \quad \ldots(i)$$

and in ΔADB, $\tan 30° = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{200+y}$

$$\Rightarrow \quad 200 + y = \sqrt{3}h \Rightarrow 200 + h = \sqrt{3}h \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad 200 = \sqrt{3}h - h \Rightarrow 200 = h(\sqrt{3}-1)$$

$$\therefore \quad h = \left(\frac{200}{\sqrt{3}-1}\right) \text{ m}$$

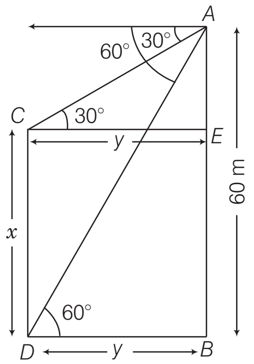
EXAMPLE 5. From the top of a building 60 m high the angles of depression of the top and bottom of a tower are observed to be 30° and 60°. Then, the height of the tower is

a. 10 m
b. 20 m
c. 30 m
d. 40 m

Sol. d. Let x be the height of the tower.

Here, $AB = 60$ m and $BD = y$ m

In right angled ΔABD,



$$\frac{AB}{BD} = \tan 60° \Rightarrow \frac{60}{y} = \sqrt{3}$$

$$\Rightarrow \quad y = \frac{60}{\sqrt{3}} \quad \ldots\text{(i)}$$

In right angled ΔAEC,

$$\frac{AE}{EC} = \tan 30° \Rightarrow \frac{AE}{BD} = \frac{1}{\sqrt{3}} \quad [\because CE = DB]$$

$$\Rightarrow \quad \frac{AE}{y} = \frac{1}{\sqrt{3}} \Rightarrow \frac{AE}{\frac{60}{\sqrt{3}}} = \frac{1}{\sqrt{3}} \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad AE = \frac{1}{\sqrt{3}} \times \frac{60}{\sqrt{3}} = 20 \text{ m} \quad [\because \sqrt{3} \times \sqrt{3} = 3]$$

$$\therefore \quad x = AB - AE = 60 - 20 = 40 \text{ m}$$

Hence, the height of tower is 40 m

PRACTICE EXERCISE

1. The length of shadow of a tree is 16 m when the angle of elevation of the Sun is 60°. What is the height of the tree?

(a) 8 m (b) 16 m (c) $16\sqrt{3}$ m (d) $\sqrt{3}$ m

2. From a light house the angles of depression of two ships on opposite sides of the light house are observed to 30° and 45°. If the height of light house is h, then what is the distance between the ships?

(a) $(\sqrt{3}+1)h$ (b) $(\sqrt{3}-1)h$ (c) $\sqrt{3}\,h$ (d) $\left(1+\frac{1}{\sqrt{3}}\right)h$

3. The angle of elevation of the top of a tower from the bottom of a building is twice that from its top. What is the height of the building, if the height of the tower is 75 m and the angle of elevation of the top of the tower from the bottom of the building is 60°?

(a) 25 m (b) 37.5 m (c) 50 m (d) 60 m

4. A radio transmitter antenna of height 100 m stands at the top of a tall building. At a point on the ground, the angle of elevation of bottom of the antenna is 45° and that of top of antenna is 60°. What is the height of the building?

(a) 100 m (b) 50 m (c) $50(\sqrt{3}+1)$ m (d) $25\sqrt{3}$ m

5. The shadow of a tower is 15 m when the Sun's altitude is 30°. What is the length of the shadow when the Sun's altitude is 60°?

(a) 3 m (b) 4 m (c) 5 m (d) 6 m

6. The angle of elevation of a moon when the length of the shadow of a pole is equal to its height is

(a) 60° (b) 45° (c) 90° (d) 30°

7. An aeroplane flying horizontally 1 km above the ground is observed at an elevation of 60°. After 10 s, its elevation is observed to be 30°, the uniform speed of the aeroplane in km/h is

(a) $\frac{240}{\sqrt{2}}$ (b) $240\sqrt{3}$

(c) 240 (d) $260\sqrt{3}$

8. A person of height 2m wants to get a fruit which is on a pole of height $(10/3)$ m. If he stands at a distance of $(4/\sqrt{3})$ m from the foot of the pole, then the angle at which he should throw the stone, so that it hits the fruit is

(a) 60° (b) 45° (c) 90° (d) 30°

9. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h. At a point on the plane, the angle of elevation of the bottom of the flagstaff is α and that of the top of the flagstaff is β. Then, the height of the tower is

(a) $\frac{h\tan\beta}{\tan\alpha - \tan\beta}$ (b) $\frac{h\tan\alpha}{\tan\alpha + \tan\beta}$

(c) $\frac{h\tan\beta}{\tan\alpha + \tan\beta}$ (d) $\frac{h\tan\alpha}{\tan\beta - \tan\alpha}$

10. From a window (h m high above the ground) of a house in a street, the angle of elevation and depression of the top and the foot of another house on the opposite side of the street are θ and ϕ, respectively. Then, the height of the opposite house is

(a) $h\tan\theta\cot\phi$ (b) $h[\tan\theta\cot\phi + 1]$

(c) $h[\cot\theta\tan\phi + 1]$ (d) $h\cot\theta\tan\phi$

11. The length of the shadow of a person s cm tall when the angle of elevation of the Sun is α is p cm. It is q cm when the angle of elevation of the Sun is β. Which one of the following is correct when $\beta = 3\alpha$?

(a) $p - q = s\left(\dfrac{\tan\alpha - \tan 3\alpha}{\tan 3\alpha \tan\alpha}\right)$ (b) $p - q = s\left(\dfrac{\tan 3\alpha - \tan\alpha}{3\tan 3\alpha \tan\alpha}\right)$

(c) $p - q = s\left(\dfrac{\tan 3\alpha - \tan\alpha}{\tan 3\alpha \tan\alpha}\right)$ (d) $p - q = s\left(\dfrac{\tan 2\alpha}{\tan 3\alpha \tan\alpha}\right)$

12. Two posts are 'K' m apart and the height of one is double that of the other. If from the middle point of the line joining their feet, an observer finds the angular elevations of their tops to be complementary, then the height (in metre) of the shorter post is

(a) $\sqrt{3}\,K$ (b) $\dfrac{K}{2\sqrt{2}}$ (c) $2\,K$ (d) $\dfrac{K}{\sqrt{2}}$

13. A flagstaff 5 m high stands on a building of 25 m height. At an observer at a height of 30 m, the flagstaff and the building subtend equal angles. The distance of the observer from the top of the flagstaff is

(a) $\dfrac{3}{2}$ m (b) $3\sqrt{\dfrac{5}{2}}$ m (c) $\dfrac{5\sqrt{3}}{2}$ m (d) $5\sqrt{\dfrac{3}{2}}$ m

14. A man is watching from the top of a tower a boat speeding away from the tower. The boat makes an angle of depression of 45° with the man's eye when at a distance of 60 m from the bottom of tower. After 5 s, the angle of depression becomes 30°. What is the approximate speed of the boat assuming that it is running in still water?

(a) 31.5 km/h (b) 36.5 km/h (c) 38.5 km/h (d) 40.5 km/h

15. Suppose the angle of elevation of the top of a tree at a point E due East of the tree is 60° and that at a point F due West of the tree is 30°. If the distance between the points E and F is 160 ft, then what is the height of the tree?

(a) $40\sqrt{3}$ ft (b) 60 ft (c) $\dfrac{40}{\sqrt{3}}$ ft (d) 23 ft

16. A telegraph post gets broken at a point against a storm and its top touches the ground at a distance 20 m from the base of the post making an angle 30° with the ground. What is the height of the post?

(a) $\dfrac{40}{\sqrt{3}}$ m (b) $20\sqrt{3}$ m (c) $40\sqrt{3}$ m (d) 30 m

17. If the angle of elevation of a cloud from a point h m above a lake is β and the angle of depression of its reflection in the lake is α, then the height of the cloud is

(a) $\dfrac{h\,\text{cosec}\,(\alpha - \beta)}{\text{cosec}\,(\alpha - \beta)}$ (b) $h\,\text{cosec}\,(\alpha - \beta)\sin(\alpha - \beta)$

(c) $h\sin(\alpha + \beta)\,\text{cosec}\,(\alpha - \beta)$ (d) $\dfrac{h\,\text{cosec}\,(\alpha + \beta)}{\sin(\alpha - \beta)}$

18. The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 100 m from its base is 45°. If the angle of elevation of the top of complete pillar at the same point is to be 60°, then the height of the incomplete pillar is to be increased by

(a) $50\sqrt{2}$ m (b) 100 m

(c) $100(\sqrt{3} - 1)$ m (d) $100(\sqrt{3} + 1)$ m

19. A man on the top of a vertical observation tower observes a car moving at a uniform speed coming directly towards it. If it takes 12 min for the angle of depression to change from 30° to 45°, how soon after this will the car reach the observation tower?

(a) 14 min 35 s (b) 15 min 49 s

(c) 16 min 23 s (d) 18 min 5 s

20. You are stationed at a radar base and you observe an unidentified plane at an altitude $h = 1000$ m flying towards your radar base at an angle of elevation $= 30°$. After exactly one minute, your radar sweep reveals that the plane is now at an angle of elevation = 60° maintaining the same altitude. What is the speed (in m/s) of the plane?

(a) 15.58 m/s (b) 19.25 m/s

(c) 18 m/s (d) 11.25 m/s

21. The angle of elevation of the top of the tower from a point on the ground is $\sin^{-1}(3/5)$. If the point of observation is 20 m away from the foot of the tower, what is the height of the tower?

(a) 9 m (b) 18 m (c) 15 m (d) 12 m

22. A balloon of radius r makes an angle α at the eye of an observer and the angle of elevation of its centre is β. The height of its centre from the ground level is given by

(a) $r\sin\beta\,\text{cosec}\,\alpha/2$ (b) $r\,\text{cosec}\,\alpha/2\sin\alpha$

(c) $r\,\text{cosec}\,\alpha\sin\beta$ (d) None of these

Directions (Q. Nos. 23-25) *The height of a tower is h and the angle of elevation of the top of the tower is α. When observer on moving a distance $h/2$ towards the tower, the angle of elevation becomes β.*

23. What is the value of $\cot\alpha - \cot\beta$?

(a) 1/2 (b) 2/3 (c) 1 (d) 2

24. If $\alpha = 30°$, then find the value of $\cot\beta$.

(a) $\sqrt{3} - 1/2$ (b) $\dfrac{2}{2\sqrt{3} - 2}$

(c) $\dfrac{1}{1 + \sqrt{3}}$ (d) $\dfrac{1}{1 - \sqrt{3}}$

25. If $\tan\beta = 4$, then distance between observer and tower when angle of elevation is β

(a) h (b) $h/4$ (c) $\dfrac{3h}{4}$ (d) $\dfrac{h}{3}$

PREVIOUS YEARS' QUESTIONS

26. What is the angle of elevation of the Sun when the shadow of a pole is $\sqrt{3}$ times the length of the pole? **2012 I**

(a) 30° (b) 45°
(c) 60° (d) None of these

27. From the top of a cliff 200 m high, the angles of depression of the top and bottom of a tower are observed to be 30° and 45°, respectively. What is the height of the tower? **2012 I**

(a) 400 m (b) $400\sqrt{3}$ m
(c) $400/\sqrt{3}$ m (d) None of these

28. The angles of elevation of the top of a tower from two points which are at distances of 10 m and 5 m from the base of the tower and in the same straight line with it are complementary. The height of the tower is **2012 II**

(a) 5 m (b) 15 m (c) $\sqrt{50}$ m (d) $\sqrt{75}$ m

29. A ladder 20 m long is placed against a wall, so that the foot of the ladder is 10 m from the wall. The angle of inclination of the ladder to the horizontal will be **2012 II**

(a) 30° (b) 45°
(c) 60° (d) 75°

Directions (Q. Nos. 30-33) *Read the following information carefully to answer the questions that follow.*

As seen from the top and bottom of a building of height h m, the angles of elevation of the top of a tower of height $\frac{(3+\sqrt{3})h}{2}$ m are α and β, respectively. **2013 I**

30. If $\beta = 30°$, then what is the value of $\tan\alpha$?

(a) 1/2 (b) 1/3
(c) 1/4 (d) None of these

31. If $\alpha = 30°$, then what is the value of $\tan\beta$?

(a) 1 (b) 1/2
(c) 1/3 (d) None of these

32. If $\alpha = 30°$ and $h = 30$ m, then what is the distance between the base of the building and the base of the tower?

(a) $(15+15\sqrt{3})$ m (b) $(30+15\sqrt{3})$ m
(c) $(45+15\sqrt{3})$ m (d) None of these

33. If $\beta = 30°$ and if θ is the angle of depression of the foot of the tower as seen from the top of the building, then what is the value of $\tan\theta$?

(a) $\frac{(3-\sqrt{3})}{3\sqrt{3}}$ (b) $\frac{(3+\sqrt{3})}{3\sqrt{3}}$
(c) $\frac{(2-\sqrt{3})}{3\sqrt{3}}$ (d) None of these

34. What is the angle of elevation of the Sun, when the shadow of a pole of height x m is $\frac{x}{\sqrt{3}}$ m? **2013 II**

(a) 30° (b) 45° (c) 60° (d) 75°

35. A spherical balloon of radius r subtends angle 60° at the eye of an observer. If the angle of elevation of its centre is 60° and h is the height of the centre of the balloon, then which one of the following is correct? **2013 II**

(a) $h = r$ (b) $h = \sqrt{2}r$
(c) $h = \sqrt{3}r$ (d) $h = 2r$

36. The angle of elevation of the top of a tower 30 m high from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30°. The distance between the two towers is m times the height of the shorter tower. What is m equal to? **2014 I**

(a) $\sqrt{2}$ (b) $\sqrt{3}$ (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

37. The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30°, then when it is 60°. What is the height of the tower? **2014 I**

(a) 25 m (b) $25\sqrt{3}$ m (c) $\frac{25}{\sqrt{3}}$ m (d) 30 m

38. From a certain point on a straight road, a person observe a tower in the West direction at a distance of 200 m. He walks some distance along the road and finds that the same tower is 300 m South of him. What is the shortest distance of the tower from the road? **2014 II**

(a) $\frac{300}{\sqrt{13}}$ m (b) $\frac{500}{\sqrt{13}}$ m (c) $\frac{600}{\sqrt{13}}$ m (d) $\frac{900}{\sqrt{13}}$ m

39. If from the top of a post a string twice the length of the post is stretched tight to a point on the ground, then what angle will the string make with the post? **2014 II**

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{5\pi}{12}$ (d) $\frac{\pi}{3}$

40. The angle of elevation of a cloud from a point 200 m above a lake is 30° and the angle of depression of its reflection in the lake is 60°. The height of the cloud is **2015 I**

(a) 200 m (b) 300 m (c) 400 m (d) 600 m

41. From the top of a tower, the angles of depression of two objects P and Q (situated on the ground on the same side of the tower) separated at a distance of $100(3-\sqrt{3})$ m are 45° and 60°, respectively. The height of the tower is **2015 I**

(a) 200 m (b) 250 m
(c) 300 m (d) None of these

42. The angles of elevation of the top of a tower from two points P and Q at distance m^2 and n^2 respectively, from the base and in the same straight line with it are complementary. The height of the tower is **2015 I**

(a) $(mn)^{1/2}$ (b) $mn^{1/2}$
(c) $m^{1/2}n$ (d) mn

43. A pole is standing erect on the ground which is horizontal, The tip of the pole is tied tight with a rope of length $\sqrt{12}$ m to a point on the ground. If the rope is making 30° with the horizontal, then the height of the pole is **2015 II**

(a) $2\sqrt{3}$ m (b) $3\sqrt{2}$ m
(c) 3 m (d) $\sqrt{3}$ m

44. An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant, when the angles of elevation of the two planes from some point on the ground are 60° and 45°, respectively. Then, the vertical distance between the two planes is **2015 II**

(a) $1000(\sqrt{3}-1)$ m (b) $1000\sqrt{3}$ m
(c) $1000(3-\sqrt{3})$ m (d) $3000\sqrt{3}$ m

45. Two observers are stationed due North of a tower (of height x m) at a distance y m from each other. The angles of elevation of the tower observed by them are 30° and 45°, respectively. Then, x/y is equal to **2016 I**

(a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{3}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) 1

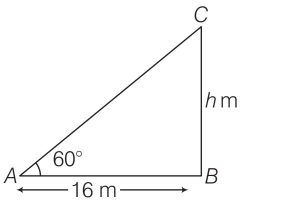
ANSWERS

1	c	2	a	3	c	4	c	5	c	6	b	7	b	8	d	9	d	10	b
11	c	12	b	13	d	14	a	15	a	16	b	17	c	18	c	19	c	20	b
21	c	22	a	23	a	24	a	25	b	26	a	27	d	28	c	29	c	30	b
31	a	32	c	33	a	34	c	35	c	36	b	37	b	38	c	39	d	40	c
41	c	42	d	43	d	44	c	45	c										

HINTS AND SOLUTIONS

1. *(c)* Let the height of the tree be h m.

$\therefore$ $BC = h$ m and $AB = 16$ m



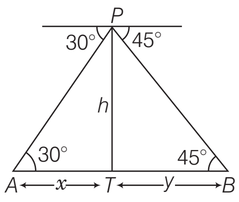
In right angled ΔABC,

$$\tan 60° = \frac{BC}{AB} = \frac{h}{16} \Rightarrow \sqrt{3} = \frac{h}{16}$$

$\Rightarrow$ $h = 16\sqrt{3}$ m

Hence, the height of the tree is $16\sqrt{3}$ m.

2. *(a)* In right angled ΔPBT, $\tan 45° = \frac{h}{y}$ $\Rightarrow$ $1 = \frac{h}{y}$



$\therefore$ $y = h$ ...(i)

and in right angled ΔPTA,

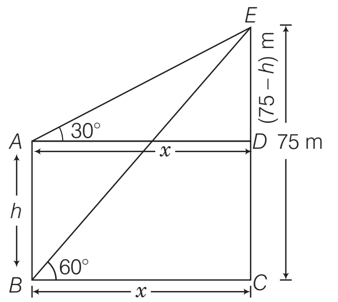
$$\tan 30° = \frac{h}{x} \Rightarrow x = \sqrt{3}h \quad \text{...(ii)}$$

$\therefore$ Required distance, $AB = x + y$

$$x + y = \sqrt{3}h + h = h(\sqrt{3}+1)\text{ m}$$

3. *(c)* Let height of the building be h m and distance between building and tower be x m.

$\therefore$ $AB = h$ m and $BC = x$ m



In right angled ΔADE,

$$\tan 30° = \frac{ED}{AD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{75-h}{x}$$

$\Rightarrow$ $x = 75\sqrt{3} - h\sqrt{3}$ ...(i)

and in right angled ΔBCE,

$$\tan 60° = \frac{CE}{BC} \Rightarrow \sqrt{3} = \frac{75}{x}$$

$\Rightarrow$ $x\sqrt{3} = 75$

$\Rightarrow (75\sqrt{3} - h\sqrt{3})\sqrt{3} = 75$ [from Eq. (i)]

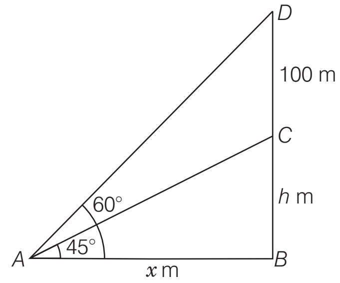
$\Rightarrow$ $75 \times 3 - 3h = 75$

$\Rightarrow$ $3h = 75 \times 3 - 75 \Rightarrow h = \frac{75 \times 2}{3}$

$\therefore$ $h = 50$ m

Hence, the height of the building is 50 m.

4. *(c)* Let BC be a building of height h m and CD be a height of antenna. and let distance between A and $B = x$ m



In right angled ΔABC,

$$\tan 45° = \frac{BC}{AB} = \frac{h}{x} \Rightarrow x = h \quad \ldots(i)$$

and in right angled ΔABD,

$$\tan 60° = \frac{DB}{BA} = \frac{DC + CB}{AB} = \frac{100 + h}{x}$$

$$\Rightarrow \quad \sqrt{3} = \frac{100 + h}{h} \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad \sqrt{3}h = 100 + h$$

$$\Rightarrow (\sqrt{3} - 1)\, h = 100 \Rightarrow h = \frac{100}{\sqrt{3} - 1}$$

$$\Rightarrow \quad h = \frac{100}{(\sqrt{3} - 1)} \times \frac{(\sqrt{3} + 1)}{(\sqrt{3} + 1)}$$

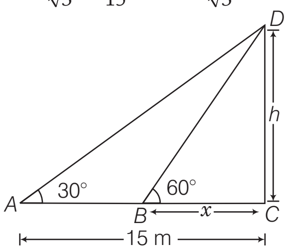
$$= 50\,(\sqrt{3} + 1)\text{ m}$$

Hence, the height of the building is $50\,(\sqrt{3} + 1)$ m.

5. *(c)* Let the height of the tower be h m and length of the shadow (BC) be x m.
In right angled ΔACD,

$$\tan 30° = \frac{CD}{AC}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{h}{15} \Rightarrow h = \frac{15}{\sqrt{3}}\text{ m} \quad \ldots(i)$$



and in right angled ΔBCD,

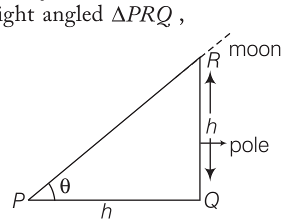
$$\tan 60° = \frac{CD}{BC} \Rightarrow \sqrt{3} = \frac{h}{x} \Rightarrow \frac{h}{\sqrt{3}} = x$$

$$\therefore \quad x = \frac{15}{3} = 5\text{ m}$$

[from Eq. (i)]

Hence, the length of shadow is 5 m.
When sun's altitude is 60°.

6. *(b)* Let the height of pole be h m, then $PQ = RQ = h$
In right angled ΔPRQ,

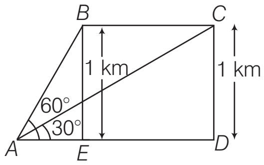


$$\tan\theta = \frac{RQ}{PQ} = \frac{h}{h} = 1$$

$\Rightarrow \tan\theta = 1 = \tan 45° \quad \therefore \quad \theta = 45°$
Hence, the angle of elevation of moon is 45°.

7. *(b)* Let A be the observer. When the aeroplane was at point B the angle of elevation was 60° and after 10 s when it was at point C the angle of elevation was 30°.
Here, in the figure,

$$BE = CD = 1\text{ km} = 1000\text{ m}$$



In right angled ΔBAE,

$$\tan 60° = \frac{BE}{AE} = \frac{1000}{AE}$$

$$\Rightarrow \quad AE = 1000 \cot 60°$$

$$= \frac{1000}{\sqrt{3}}\text{ m} \quad \ldots(i)$$

In right angled ΔACD,

$$\tan 30° = \frac{DC}{AD} = \frac{1000}{AD}$$

$$\Rightarrow AD = 1000 \cot 30° = 1000\sqrt{3}$$

$$\therefore \quad ED = AD - AE$$

$$= 1000\sqrt{3} - \frac{1000}{\sqrt{3}} = \frac{2000}{\sqrt{3}}\text{ m}$$

So, distance travelled by plane in 10 s, then

$$BC = DE = \frac{2000}{\sqrt{3}}\text{ m}$$

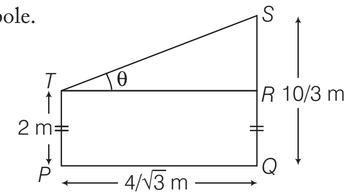
$$\therefore \text{ Speed of aeroplane} = \frac{\text{Distance}}{\text{Time}}$$

$$= \frac{\frac{2000}{\sqrt{3}}}{10}\text{ m/s} = \frac{2000 \times 60 \times 60}{10\sqrt{3} \times 1000}\text{ km/h}$$

$$= 240\sqrt{3}\text{ km/h}$$

Hence, the uniform speed of the aeroplane is $240\sqrt{3}$ km/h.

8. *(d)* Let TP be the man and SQ be the pole.



Let $\angle STR = \theta$

Now, $\quad SR = SQ - RQ$

$$= \frac{10}{3} - 2 = \frac{4}{3}\text{ m} \quad [\because RQ = TP]$$

In right angled ΔSTR,

$$\tan\theta = \frac{SR}{TR} = \frac{SR}{PQ} \quad [\because TR = PQ]$$

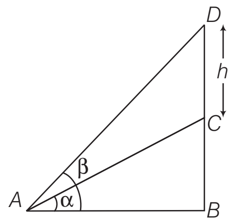
$$= \frac{4}{3} \times \frac{\sqrt{3}}{4} = \frac{1}{\sqrt{3}} = \tan 30°$$

$$\therefore \quad \theta = 30°$$

9. *(d)* Let BC be the tower and CD be the flagstaff.

$\therefore \quad \angle BAC = \alpha$ and $\angle BAD = \beta$

In right angled ΔABC,



$$\tan\alpha = \frac{BC}{AB} \quad \ldots(i)$$

and in right angled ΔABD,

$$\frac{BD}{AB} = \frac{BC + h}{AB} = \tan\beta \quad \ldots(ii)$$

On dividing Eq. (ii) by Eq. (i), we get

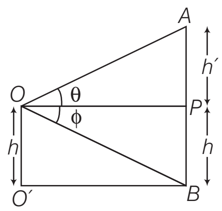
$$\frac{BC + h}{BC} = \frac{\tan\beta}{\tan\alpha}$$

$$\Rightarrow \quad (BC + h)\tan\alpha = BC\tan\beta$$

$$\Rightarrow \quad BC(\tan\beta - \tan\alpha) = h\tan\alpha$$

$$\therefore \quad BC = \frac{h\tan\alpha}{\tan\beta - \tan\alpha}$$

10. *(b)* Let O be the window and AB be the house on the opposite side of the street.



Let $\quad AP = h'$
and $\quad BP = OO' = h$

In right angled ΔAOP,

$$\tan\theta = \frac{AP}{OP} = \frac{h'}{OP} \quad \ldots(i)$$

and in right angled ΔBOP,

$$\tan\phi = \frac{PB}{OP} = \frac{h}{OP} \quad \ldots(ii)$$

$$\therefore \quad \frac{h'}{h} = \frac{\tan\theta}{\tan\phi} \quad \text{[by Eq. (i) ÷ Eq. (ii)]}$$

$$\Rightarrow \quad h' = h\tan\theta\cot\phi$$

$\therefore$ Height of the house

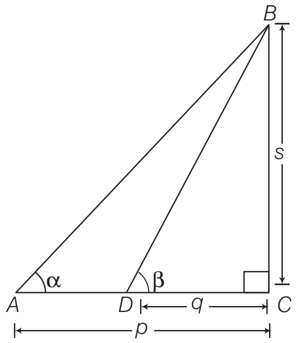
$$AB = AP + PB = h + h'$$

$$= h\tan\theta\cot\phi + h$$

$$= h\,[\tan\theta\cot\phi + 1]$$

11. *(c)* In right angled ΔBAC,

$$\tan\alpha = \frac{BC}{AC} = \frac{s}{p} \Rightarrow p = \frac{s}{\tan\alpha} \quad \ldots(i)$$



In right angled ΔBDC,

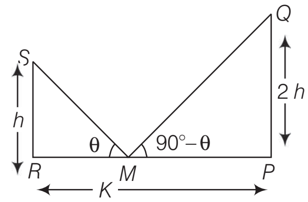
$$\tan\beta = \frac{BC}{DC} = \frac{s}{q} \Rightarrow q = \frac{s}{\tan 3\alpha} \quad \ldots(ii)$$

$[\because \beta = 3\alpha, \text{ given}]$

On subtracting Eq. (ii) from Eq. (i), we get

$$p - q = \frac{s}{\tan\alpha} - \frac{s}{\tan 3\alpha}$$
$$= s\left(\frac{\tan 3\alpha - \tan\alpha}{\tan\alpha \tan 3\alpha}\right)$$

12. *(b)* Let PQ and RS be the two posts such that $PQ = 2RS$ [given]



Given, M is the mid-point of RP.

$\therefore \quad RM = PM = \dfrac{K}{2}$ $[\because RP = K \text{ given}]$

Let $\angle RMS = \theta$

$\therefore \quad \angle QMP = 90° - \theta$

Let $RS = h$, then $PQ = 2h$

Now, in right angled ΔPMQ,

$$\tan(90° - \theta) = \frac{PQ}{MP}$$

$$\Rightarrow \quad \cot\theta = \frac{2h}{k/2} = \frac{4h}{k}$$

or $$\cot\theta = \frac{4h}{K} \quad \ldots(i)$$

In right angled ΔSRM,

$$\tan\theta = \frac{SR}{RM} = \frac{h}{K/2}$$

$$\Rightarrow \quad \frac{2h}{K} = \tan\theta \quad \ldots(ii)$$

On multiplying Eq. (i) by Eq. (ii), we get

$$\Rightarrow \quad \frac{4h}{K} \times \frac{2h}{K} = 1$$

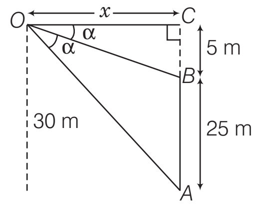
$$\Rightarrow \quad 8h^2 = K^2 \Rightarrow h^2 = \frac{K^2}{8}$$

$$\therefore \quad h = \frac{K}{2\sqrt{2}} \text{ m}$$

Hence, the height of shorter post is $\dfrac{K}{2\sqrt{2}}$ m.

13. *(d)* Let O be an observer at a height 30 m. Let x be the distance of observer from the top of the flagstaff.

i.e. $OC = x$



Here, AB be the tower of the flag staff = 25 m and BC be the flag staff = 5 m

In right angled ΔOBC,

$$\tan\alpha = \frac{BC}{OC} = \frac{5}{x}$$

$$\Rightarrow \quad x = 5\cot\alpha \quad \ldots(i)$$

In right angled ΔOCA,

$$\tan 2\alpha = \frac{AC}{OC} = \frac{30}{x} [\because AC = AB + BC]$$

$$\therefore \quad \tan 2\alpha = \frac{30}{5\cot\alpha} \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad \tan 2\alpha = 6\tan\alpha$$

$$\Rightarrow \frac{2\tan\alpha}{1-\tan^2\alpha} = 6\tan\alpha$$

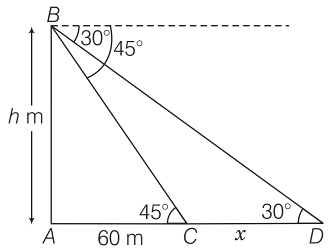
$$\left[\because \tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}\right]$$

$$\Rightarrow 3 - 3\tan^2\alpha = 1 \Rightarrow \tan^2\alpha = \frac{2}{3}$$

$$\Rightarrow \quad \tan\alpha = \sqrt{\frac{2}{3}} \text{ or } \cot\alpha = \sqrt{3/2}$$

or $$x = 5\sqrt{\frac{3}{2}} \text{ m} \quad \text{[from Eq. (i)]}$$

14. *(a)* Let height of tower AB be h m and distance between C and D be x m.



In right angled ΔACB, $\tan 45° = \dfrac{AB}{AC}$

$$\Rightarrow \quad 1 = \frac{h}{60} \Rightarrow h = 60 \text{ m} \quad \ldots(i)$$

Now, in right angled ΔADB,

$$\tan 30° = \frac{AB}{AD} = \frac{AB}{AC + CD} = \frac{60}{60 + x}$$

[from Eq. (i)]

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{60}{60 + x} \Rightarrow 60 + x = 60\sqrt{3}$$

$$\Rightarrow \quad x = 60(\sqrt{3} - 1) = 60(1.73 - 1)$$
$$= 60 \times 0.73 = 43.8 \text{ m}$$

Now, given time = 5 s

We know that, Speed = $\dfrac{\text{Distance}}{\text{Time}}$

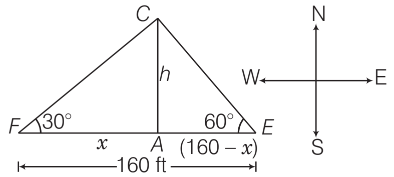
$$\therefore \quad \text{Speed of boat} = \frac{43.8}{5} \times \frac{18}{5} = \frac{788.4}{25}$$
$$= 31.5 \text{ km/h}$$

15. *(a)* Let $AC = h$ ft = Height of a tree and x ft = Distance between A and F then $AE = (160 - x)$ ft

In right angled ΔAFC,

$$\tan 30° = \frac{AC}{AF} = \frac{h}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x}$$

$$\Rightarrow \quad x = \sqrt{3}h \quad \ldots(i)$$



and in right angled ΔAEC,

$$\tan 60° = \frac{AC}{AE} = \frac{h}{160 - x}$$

$$\Rightarrow \quad \sqrt{3} = \frac{h}{160 - x} [\because \tan 60° = \sqrt{3}]$$

$$\Rightarrow \quad \sqrt{3}(160 - x) = h$$

$$\Rightarrow \quad \sqrt{3}(160 - \sqrt{3}h) = h \quad \text{[from Eq. (i)]}$$

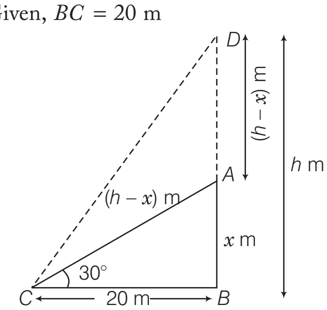
$$\Rightarrow \quad 160\sqrt{3} - 3h = h \Rightarrow 4h = 160\sqrt{3}$$

$$\therefore \quad h = 40\sqrt{3} \text{ ft}$$

Hence, the height of the tree is $40\sqrt{3}$ ft.

16. *(b)* Let the height of the post be h m, and $AB = x$ m.

Given, $BC = 20$ m



In right angled ΔABC,

$$\tan 30° = \frac{AB}{BC} \Rightarrow \frac{1}{\sqrt{3}} = \frac{x}{20}$$

$$\Rightarrow \quad x = \frac{20}{\sqrt{3}} \text{ m} \quad \ldots\text{(i)}$$

$$\text{and } \cos 30° = \frac{BC}{AC} = \frac{20}{h - x}$$

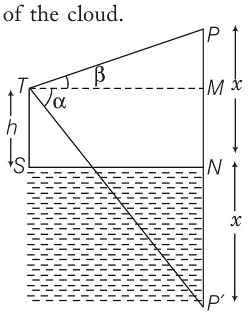
$$\Rightarrow \quad \frac{\sqrt{3}}{2} = \frac{20}{h - x} \Rightarrow h - x = \frac{40}{\sqrt{3}}$$

From Eq. (i), putting the value of x,

$$h = \frac{40}{\sqrt{3}} + \frac{20}{\sqrt{3}} = \frac{60}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 20\sqrt{3} \text{ m}$$

Hence, the height of the post is $20\sqrt{3}$ m.

17. *(c)* Let P be the cloud and P' its image in the lake. Let T be the point 'h' m above the surface of the lake and let x be the height of the cloud.



$\therefore \quad ST = h$

Also, $\quad NP = NP' = x \quad$ [say]

Then, $\quad PM = x - h$

$\quad P'M = x + h$

In right angled ΔPTM,

$$\frac{PM}{TM} = \tan \beta$$

$$\therefore \quad x - h = TM \tan \beta \quad \ldots\text{(i)}$$

In right angled $\Delta P'TM$,

$$\frac{P'M}{TM} = \tan \alpha$$

$$x + h = TM \tan \alpha \quad \ldots\text{(ii)}$$

From Eqs. (i) and (ii), we get

$$\frac{x - h}{x + h} = \frac{\tan\beta}{\tan\alpha}$$

Using componendo and dividendo rule, we get

$$\frac{x - h + x + h}{x - h - x - h} = \frac{\tan\beta + \tan\alpha}{\tan\beta - \tan\alpha}$$

$$\Rightarrow \quad \frac{2x}{-2h} = \frac{\tan\beta + \tan\alpha}{\tan\beta - \tan\alpha}$$

$$\Rightarrow \quad \frac{x}{h} = \frac{\tan\beta + \tan\alpha}{\tan\alpha - \tan\beta}$$

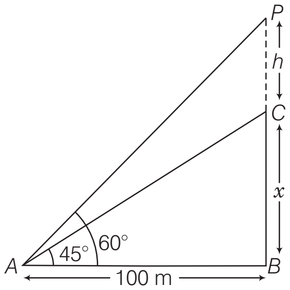
$$\Rightarrow \quad \frac{x}{h} = \frac{\sin\beta\cos\alpha + \cos\beta\sin\alpha}{\sin\alpha\cos\beta - \cos\alpha\sin\beta}$$

$$= \frac{\sin(\alpha + \beta)}{\sin(\alpha - \beta)}$$

$$\therefore \quad x = h \sin(\alpha + \beta) \operatorname{cosec}(\alpha - \beta)$$

18. *(c)* Let the height of the incomplete pillar be x m and the increasing height be $PC = h$ m.

Given, $AB = 100$ m



In right angled ΔABC,

$$\tan 45° = \frac{BC}{AB} = \frac{x}{100} \Rightarrow x = 100 \text{ m} \ldots\text{(i)}$$

and in right angled ΔAPB,

$$\tan 60° = \frac{PB}{AB} = \frac{PC + BC}{AB} = \frac{x + h}{100}$$

$$\Rightarrow \quad x + h = 100\sqrt{3}$$

$$\Rightarrow h = 100\sqrt{3} - x = 100\sqrt{3} - 100$$

[from Eq. (i)]

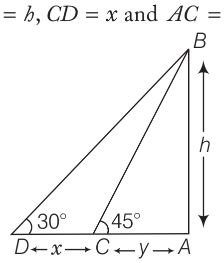
$\therefore \; h = 100(\sqrt{3} - 1)$ m

Hence, the height of the incomplete pillar is $100(\sqrt{3} - 1)$ m.

19. *(c)* Let AB be the tower and C, D be the two positions of the car.

Then, $\angle ACB = 45°$, $\angle ADB = 30°$.

Let $AB = h$, $CD = x$ and $AC = y$.



In right angled ΔABC, we get

$$\frac{AB}{AC} = \tan 45° = 1 \Rightarrow \frac{h}{y} = 1$$

$$\Rightarrow \quad y = h \quad \ldots\text{(i)}$$

In right angled ΔABD, we get

$$\frac{AB}{AD} = \tan 30° = \frac{1}{\sqrt{3}} \Rightarrow \frac{h}{x + y} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \quad x + y = \sqrt{3}h \quad \ldots\text{(ii)}$$

$$\therefore x = \sqrt{3}h - h = h(\sqrt{3} - 1)$$

[by Eqs. (i) and (ii)]

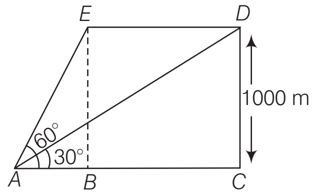
Now, $h(\sqrt{3} - 1)$ is covered in 12 min.

So, h will be covered in

$$\left[\frac{12}{h(\sqrt{3} - 1)} \times h\right] = \frac{12}{(\sqrt{3} - 1)} \text{min.}$$

$$= \left(\frac{1200}{73}\right) \text{min} \approx 16 \text{ min } 23 \text{ s}$$

20. *(b)* Let the radar base is at point A. The plane is at point D in the first sweep and at point E in the second sweep. The distance it covers in the one minute interval is DE.



From the figure,

In right angled ΔADC, we get

$$\tan 30° = \frac{DC}{AC} = \frac{1000}{AC} \Rightarrow AC = \frac{1000}{\tan 30°}$$

Similarly, in ΔEAB, we get

$$\tan 60° = \frac{EB}{AB} = \frac{1000}{AB} \Rightarrow AB = \frac{1000}{\tan 60°}$$

Total distance covered by plane in 1 min, then

$$DE = AC - AB$$

$$DE = \frac{1000}{\tan 30°} - \frac{1000}{\tan 60°}$$

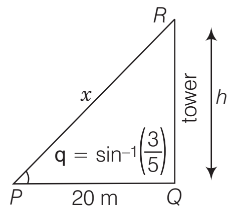
$$= 1000\sqrt{3} - \frac{1000}{\sqrt{3}} = 1154.70 \text{ m}$$

The speed of the plane is given by

s = distance covered/time taken

$= DE / 60 = 19.25$ m/s.

21. *(c)*



Let P be the point of observation and QR be the tower.

Given that, $\theta = \sin^{-1}\left(\frac{3}{5}\right)$, $PQ = 20$ m.

Let the height of the tower, $QR = h$ and $PR = x$

From the right angled ΔPQR,

$$\sin\theta = \frac{QR}{PR} \Rightarrow \sin\left[\sin^{-1}\left(\frac{3}{5}\right)\right] = h / x$$

$$\Rightarrow \quad \frac{3}{5} = \frac{h}{x} \Rightarrow x = \frac{5h}{3} \quad \ldots\text{(i)}$$

From pythagoras theorem, we get,

$$PQ^2 + QR^2 = PR^2$$

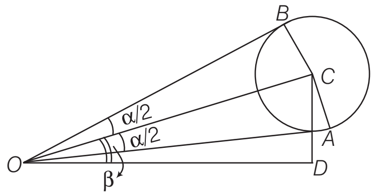
$$\Rightarrow \quad 20^2 + h^2 = x^2$$

$$\Rightarrow \quad 20^2 + h^2 = \left(\frac{5h}{3}\right)^2 \text{ [using Eq. (i)]}$$

$$\Rightarrow \quad 20^2 + h^2 = \frac{25h^2}{9} \Rightarrow \frac{16h^2}{9} = 20^2$$

$$\therefore \quad h = \frac{3 \times 20}{4} = 15 \text{ m}$$

22. *(a)* Let O be the position of the man's eye and C be the centre of the balloon.



Now, $\angle BOC = \angle COA = \dfrac{\alpha}{2}$

Given, $CA = BC = r$

In right angled ΔCOD, $\sin\beta = \dfrac{CD}{OC}$

$\therefore \quad CD = OC \sin\beta \quad \ldots\text{(i)}$

In right angled $\Delta\, COA$,

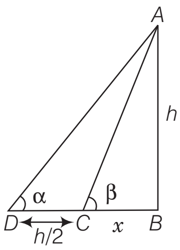
$$\sin\frac{\alpha}{2} = \frac{CA}{OC}$$

$$\therefore \quad OC = \frac{r}{\sin\frac{\alpha}{2}} = r\,\text{cosec}\,\frac{\alpha}{2}$$

From Eq. (i), $CD = r\,\text{cosec}\,\dfrac{\alpha}{2}\sin\beta$

Hence, the height of centre of the balloon is $r\sin\beta\,\text{cosec}\,\dfrac{\alpha}{2}$.

23. *(a)* Let AB be the tower of height h m, BC be x m and DC be $h/2$ m.



In right angled ΔABD,

$$\tan\alpha = \frac{h}{x + h/2} \Rightarrow \cot\alpha = \frac{x + h/2}{h}. \quad \ldots\text{(i)}$$

Now, in right angled ΔABC,

$$\tan\beta = \frac{h}{x} \Rightarrow \cot\beta = \frac{x}{h} \quad \ldots\text{(ii)}$$

Subtracting eq. (ii) from eq. (i), we get

$$\cot\alpha - \cot\beta = \frac{x + h/2}{h} - \frac{x}{h}$$

$$= \frac{x + h/2 - x}{h} = \frac{h/2}{h} = \frac{1}{2}$$

24. *(a)* We have, $\cot\alpha - \cot\beta = 1/2$

$\Rightarrow \quad \cot 30° - \cot\beta = 1/2$

$\Rightarrow \quad \cot\beta = \cot 30° - 1/2$

$= \sqrt{3} - 1/2$

25. *(b)* $\tan\beta = 4 \Rightarrow \cot\beta = 1/4$

we have, $\cot\alpha - \cot\beta = 1/2$

$\Rightarrow \cot\alpha = \cot\beta + 1/2 = 1/4 + 1/2 = \dfrac{3}{4}$

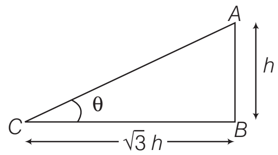
In ΔABD, $\cot\alpha = \dfrac{x + h/2}{h}$

$\therefore \quad \dfrac{3}{4} = \dfrac{x + h/2}{h}$

$\Rightarrow \quad 3h = 4x + 2h \Rightarrow x = \dfrac{h}{4}$

Hence, required distance is $\dfrac{h}{4}$

26. *(a)* Let θ be the angle of elevation of Sun and height of the pole be h m.



In right angled ΔABC,

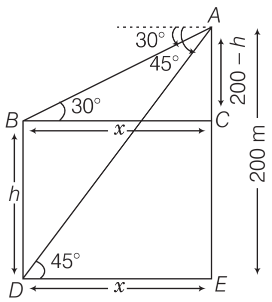
$$\tan\theta = \frac{AB}{BC} \Rightarrow \tan\theta = \frac{h}{\sqrt{3}h}$$

$\Rightarrow \quad \tan\theta = \dfrac{1}{\sqrt{3}} = \tan 30°$

$\therefore \quad \theta = 30°$

Hence, the angle of elevation of Sun is 30°.

27. *(d)* Let AE be the height of the cliff and h m (BD) be the height of the tower.



In right angled ΔABC,

$$\tan 30° = \frac{AC}{BC} = \frac{200 - h}{x}$$

$\Rightarrow \quad \dfrac{1}{\sqrt{3}} = \dfrac{200 - h}{x}$

$\Rightarrow \quad x = (200 - h)\sqrt{3} \quad \ldots\text{(i)}$

and in right angled ΔADE,

$$\tan 45° = \frac{AE}{DE} = \frac{200}{x}$$

$\Rightarrow \quad 1 = \dfrac{200}{x} \Rightarrow x = 200$ m

From Eq. (i), $200 = (200 - h)\sqrt{3}$

$\Rightarrow \quad 200 = 200\sqrt{3} - h\sqrt{3}$

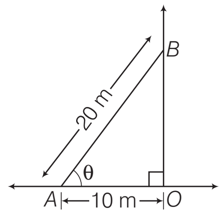
$\Rightarrow \quad h\sqrt{3} = 200(\sqrt{3} - 1)$

$\therefore \quad h = 200\left(\dfrac{\sqrt{3} - 1}{\sqrt{3}}\right)$ m

Hence, the height of tower is $\dfrac{200(\sqrt{3} - 1)}{\sqrt{3}}$ m.

28. *(c)* Refer to example 3.

29. *(c)* Let θ be the inclination of the ladder to the horizontal.



Now, in right angled ΔAOB,

$$\cos\theta = \frac{AO}{AB} = \frac{10}{20} = \frac{1}{2}$$

$\Rightarrow \quad \cos\theta = \cos 60°$

$\therefore \quad \theta = 60°$

Hence, the angle of inclination of the ladder is 60°.

Solutions (Q. Nos. 30-33)

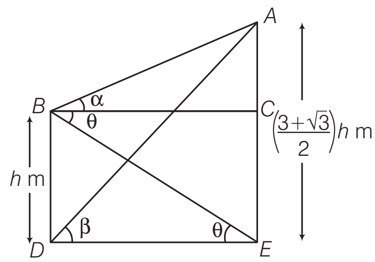
30. *(b)* Given, $\beta = 30°$

In right angled ΔADE, $\tan\beta = \dfrac{AE}{DE}$

$$\tan 30° = \frac{AE}{DE} \Rightarrow \frac{1}{\sqrt{3}} = \frac{AE}{DE}$$

$\Rightarrow \quad DE = \sqrt{3}\,AE = \sqrt{3}\left(\dfrac{3 + \sqrt{3}}{2}\right)h$

$\Rightarrow \quad BC = DE = \dfrac{3}{2}(1 + \sqrt{3})\,h \quad \ldots\text{(i)}$



[$\because BC = DE$]

Now, in right angled ΔABC,

$\Rightarrow \quad \tan\alpha = \dfrac{AC}{BC}$

$\Rightarrow \quad BC\tan\alpha = (AE - CE)$

$= (AE - BD)$ [$\because BD = CE$]

$\Rightarrow BC\tan\alpha = \left(\dfrac{3 + \sqrt{3}}{2}\right)h - h$

$= h\left(\dfrac{3 + \sqrt{3} - 2}{2}\right)$

$\Rightarrow \quad \dfrac{3}{2}(1 + \sqrt{3})\,h\tan\alpha = \left(\dfrac{1 + \sqrt{3}}{2}\right)h$

[from Eq. (i)]

$\therefore \quad \tan\alpha = \dfrac{1}{3}$

31. *(a)* Given, $\alpha = 30°$

In right angled ΔABC,

$$\tan\alpha = \tan 30° = \frac{AC}{BC}, \frac{1}{\sqrt{3}} = \frac{AC}{BC}$$

$$\Rightarrow BC = \sqrt{3}\, AC = \sqrt{3}\,(AE - CE)$$

$$= \sqrt{3}\,(AE - BD) \quad [\because BD = CE]$$

$$= \sqrt{3}\left(\frac{3+\sqrt{3}}{2} - 1\right)h$$

$$= \frac{\sqrt{3}}{2}(1+\sqrt{3})\, h \qquad \text{...(ii)}$$

Now, in right angled ΔADE,

$$\tan\beta = \frac{AE}{DE}$$

$$\Rightarrow \quad \tan\beta = \frac{AE}{BC} \qquad [\because DE = BC]$$

$$\Rightarrow \quad \frac{\left(\frac{3+\sqrt{3}}{2}\right)h}{\frac{\sqrt{3}}{2}(1+\sqrt{3})\,h} = \frac{\frac{\sqrt{3}(1+\sqrt{3})}{2}h}{\frac{\sqrt{3}\,(1+\sqrt{3})}{2}h}$$

$\therefore \qquad \tan\beta = 1$

32. *(c)* Given, $\alpha = 30°$ and $h = 30$ m

In right angled ΔABC,

$$\tan\alpha = \tan 30° = \frac{1}{\sqrt{3}} = \frac{AC}{BC}$$

$$\Rightarrow \frac{BC}{\sqrt{3}} = (AE - CE) = (AE - BD)$$

$$[\because BD = CE]$$

$$\Rightarrow BC = \sqrt{3}\left(\frac{3+\sqrt{3}}{2} - 1\right)h$$

$$\Rightarrow BC = \sqrt{3}\,\frac{(1+\sqrt{3})}{2}\cdot 30$$

$$= (\sqrt{3} + 3)\cdot 15$$

$$\therefore \quad DE = BC = (45 + 15\sqrt{3})\,\text{m}$$

$$[\because DE = BC]$$

33. *(a)* Given, $\beta = 30°$

In right angled ΔADE, $\tan\beta = \dfrac{AE}{DE}$

$$\Rightarrow \tan 30° = \frac{\left(\frac{3+\sqrt{3}}{2}\right)h}{DE}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{\sqrt{3}\left(\frac{1+\sqrt{3}}{2}\right)h}{DE}$$

$$\Rightarrow \quad DE = \frac{3}{2}(1+\sqrt{3})\, h \qquad \text{...(i)}$$

In right angled ΔBDE,

$$\tan\theta = \frac{BD}{DE} = \frac{h}{DE}$$

$$\Rightarrow \tan\ \theta = \frac{h}{\frac{3}{2}(1+\sqrt{3})\,h} \qquad [\text{from Eq. (i)}]$$

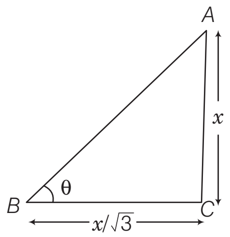
$$\Rightarrow \quad \tan\theta = \frac{2}{3}\,\frac{(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)}$$

$$\Rightarrow \quad \tan\theta = \frac{2(\sqrt{3}-1)}{3\cdot 2}$$

$$\Rightarrow \qquad \tan\theta = \frac{(\sqrt{3}-1)}{3}\cdot\frac{\sqrt{3}}{\sqrt{3}}$$

$$\therefore \qquad \tan\theta = \frac{(3-\sqrt{3})}{3\sqrt{3}}$$

34. *(c)* Let θ be the angle of elevation,



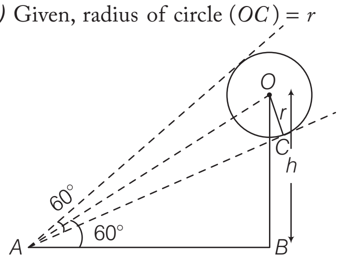
In right angled ΔABC,

$$\tan\theta = \frac{AC}{BC} = \frac{x}{\frac{x}{\sqrt{3}}} = \frac{\sqrt{3}x}{x} = \sqrt{3}$$

Here, $\tan\theta = \sqrt{3} = \tan 60°$

$\therefore \qquad \theta = 60° \qquad [\because \tan 60° = \sqrt{3}]$

35. *(c)* Given, radius of circle $(OC) = r$



In right angled ΔABO,

$$\sin 60° = \frac{OB}{AO} \Rightarrow AO = \frac{OB}{\sin 60°} \qquad \text{...(i)}$$

Now, in right angled ΔAOC,

$$\sin\frac{60°}{2} = \frac{OC}{AO} \Rightarrow AO = \frac{OC}{\sin 30°} \qquad \text{...(ii)}$$

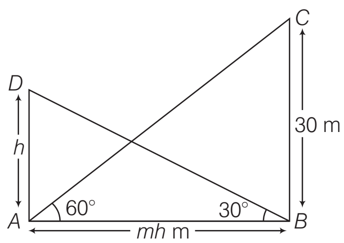
From Eqs. (i) and (ii),

$$\frac{OB}{\sin 60°} = \frac{OC}{\sin 30°} \Rightarrow \frac{h}{\frac{\sqrt{3}}{2}} = \frac{r}{\frac{1}{2}}$$

$\therefore \qquad h = \sqrt{3}\, r$

36. *(b)* Let h m be the height of shorter tower and the distance between the two towers is mh m.

Given, $\angle ABD = 30°$ and $\angle BAC = 60°$



In right angled ΔABD,

$$\tan\ 30° = \frac{h}{mh} \Rightarrow \frac{1}{\sqrt{3}} = \frac{1}{m}$$

$\Rightarrow \qquad m = \sqrt{3}$

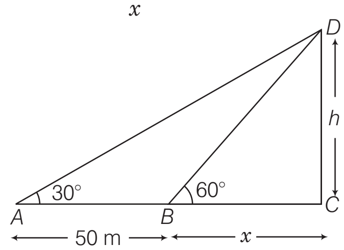
Hence, the value of m is $\sqrt{3}$.

37. *(b)* Let h m be the height of the tower and BC be x m.

In right angled ΔBCD,

$$\tan 60° = \frac{DC}{BC} = \frac{h}{x}$$

$$\Rightarrow \qquad \sqrt{3} = \frac{h}{x} \Rightarrow h = x\sqrt{3} \qquad \text{...(i)}$$



Now, in right angled ΔACD,

$$\tan 30° = \frac{DC}{AC}$$

$$= \frac{DC}{AB + BC} = \frac{h}{50 + x}$$

$$\Rightarrow \qquad \frac{1}{\sqrt{3}} = \frac{x\sqrt{3}}{50 + x} \qquad [\text{from Eq. (i)}]$$

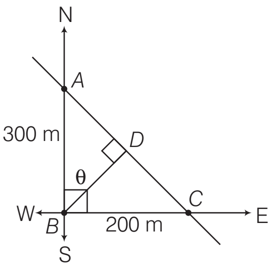
$$\Rightarrow \quad 50 + x = 3x \quad \Rightarrow \quad x = 25 \text{ m}$$

$\therefore \qquad h = 25\sqrt{3}$ m

Hence, the height of tower is $25\sqrt{3}$ m.

38. *(c)* Let person be at point C and observes a tower in West direction at B.

$\therefore \quad BC = 200$ m



He walks some distance and reach at A. Now, he observe tower in South direction at B.

$\therefore \quad AB = 300$ m

Let BD be the shortest distance of tower from the road, which is a perpendicular distance.

If $\angle ABD = \theta$, then

$\angle CBD = 90° - \theta$

[$\because$ angle between S and W = 90°]

In right angled ΔADB,

$$\cos\theta = \frac{BD}{AB} \Rightarrow \cos\theta = \frac{BD}{300} \qquad \text{...(i)}$$

In right angled ΔCDB,

$$\cos(90° - \theta) = \frac{BD}{BC}$$

$$\Rightarrow \qquad \sin\theta = \frac{BD}{200} \qquad \text{...(ii)}$$

We know that, $\cos^2\theta + \sin^2\theta = 1$

$$\Rightarrow \quad \left(\frac{BD}{300}\right)^2 + \left(\frac{BD}{200}\right)^2 = 1$$

[from Eqs. (i) and (ii)]

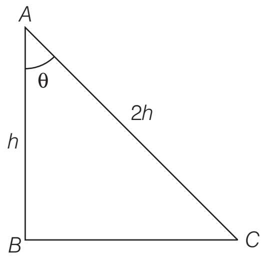
$$\Rightarrow \quad BD^2\left[\frac{40000+90000}{90000\times 40000}\right] = 1$$

$$\Rightarrow \quad BD^2\left[\frac{130000}{3600000000}\right] = 1$$

$$\Rightarrow \quad BD^2 = \frac{360000}{13}$$

$$\therefore \quad BD = \sqrt{\frac{360000}{13}} = \frac{600}{\sqrt{13}} \text{ m}$$

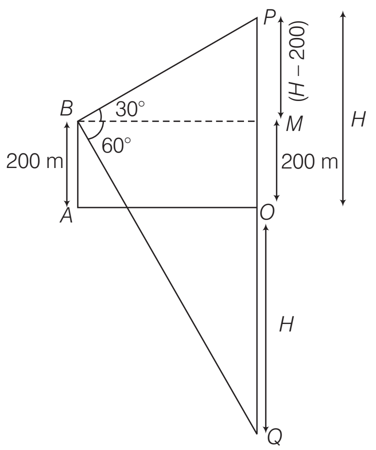
39. *(d)* Let AB be the height of the post, AC be the string and the angle made by string with the post be θ.



Now, $\cos\theta = \frac{AB}{AC} = \frac{h}{2h} \Rightarrow \frac{1}{2} = \cos 60°$

$$\therefore \quad \theta = \frac{\pi}{3}$$

40. *(c)* Let P be the cloud at a height Hm above the level of the water in the lake and Q be its image in the water.



$\therefore \quad OQ = OP = H$

Given, $\angle PBM = 30°$ and $\angle MBQ = 60°$

In right angled ΔPBM,

$$\tan 30° = \frac{PM}{BM} = \frac{H-200}{BM}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{H-200}{BM}$$

$$\Rightarrow \quad BM = \sqrt{3}(H-200) \qquad \text{...(i)}$$

In right angled ΔQMB,

$$\tan 60° = \frac{MQ}{BM} = \frac{H+200}{BM}$$

$$\Rightarrow \sqrt{3} = \frac{H+200}{\sqrt{3}(H-200)} \quad \text{[from Eq. (i)]}$$

$$\Rightarrow \quad H + 200 = 3(H-200)$$

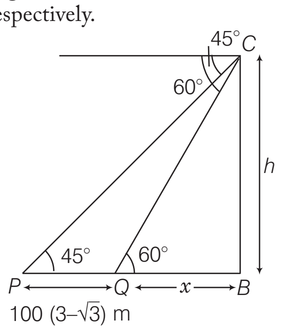
$$\Rightarrow \quad H + 200 = 3H - 600$$

$$\Rightarrow \quad 2H = 800$$

$$\therefore \quad H = 400 \text{ m}$$

Hence, the height of the cloud is 400 m.

41. *(c)* Let $BC = h$ m be height of tower. Let P and Q be the points, where the angle subtended are 45° and 60°, respectively.



In right angled ΔBCQ,

$$\tan 60° = \frac{BC}{BQ} \Rightarrow \sqrt{3} = \frac{h}{x}$$

$$\Rightarrow \quad x = \frac{h}{\sqrt{3}} \qquad \text{...(i)}$$

In right angled ΔPBC,

$$\tan 45° = \frac{BC}{PB} = \frac{BC}{PQ+QB}$$

$$\Rightarrow \quad 1 = \frac{h}{100(3-\sqrt{3})+x}$$

$$\Rightarrow \quad 100(3-\sqrt{3}) + x = h$$

$$\Rightarrow 100(3-\sqrt{3}) + \frac{h}{\sqrt{3}} = h \text{ [from Eq. (i)]}$$

$$\Rightarrow \quad h - \frac{h}{\sqrt{3}} = 100(3-\sqrt{3})$$

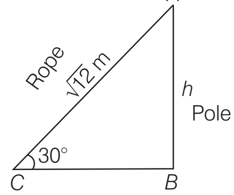
$$\Rightarrow \quad \frac{h(\sqrt{3}-1)}{\sqrt{3}} = 100\sqrt{3}(\sqrt{3}-1)$$

$$\Rightarrow \quad h = 100\sqrt{3}\times\sqrt{3}$$

$$\therefore \quad h = 300 \text{ m}$$

42. *(d)* Refer to example 3.

43. *(d)* AB is a pole and AC is rope.

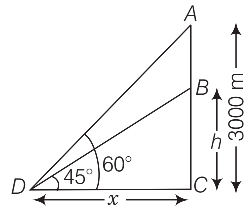


In ΔABC, $\sin 30° = \frac{AB}{AC} = \frac{h}{\sqrt{12}}$

$$\Rightarrow \quad \frac{h}{\sqrt{12}} = \frac{1}{2}$$

$$\therefore \quad h = \frac{\sqrt{12}}{2} = \frac{2\sqrt{3}}{2} = \sqrt{3} \text{ m}$$

44. *(c)* Let A and B be the position of two planes and D be a point.



In ΔBCD, $\tan 45° = \frac{h}{x}$

$$\Rightarrow \quad h = x$$

In ΔACD, $\tan 60° = \sqrt{3} = \frac{3000}{x}$

$$\because \quad x = \frac{3000}{\sqrt{3}}\times\frac{\sqrt{3}}{\sqrt{3}} = 1000\sqrt{3} \text{ m}$$

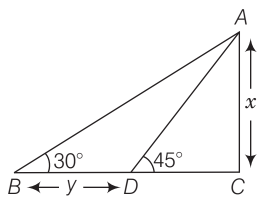
$$\therefore \quad AB = 3000 - h$$

$$= 3000 - 1000\sqrt{3}$$

$$= 1000(3-\sqrt{3}) \text{ m}$$

45. *(c)* In right angled ΔADC, $\tan 45° = \frac{AC}{DC}$

$$\Rightarrow \quad DC = \frac{AC}{\tan 45°} = x \qquad \text{...(i)}$$



In right angled ΔBCA, we get

$$\tan 30° = \frac{AC}{BC} \Rightarrow \frac{x}{DC+BD}$$

$$\Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{x}{y+x} \Rightarrow y + x = \sqrt{3}x$$

$$\Rightarrow \quad y = x(\sqrt{3}-1)$$

$$\Rightarrow \quad \frac{x}{y} = \frac{1}{(\sqrt{3}-1)}$$

$$\Rightarrow \quad \frac{x}{y} = \frac{1\times(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)}$$

$$\Rightarrow \quad \frac{x}{y} = \frac{(\sqrt{3}+1)}{(\sqrt{3})^2-(1)^2}$$

$$\therefore \quad \frac{x}{y} = \frac{(\sqrt{3}+1)}{2}$$

22

LINES AND ANGLES

Usually (2-3) questions have been asked from this chapter. And this is the easiest topic of geometry and so candidate can easily score marks in examinations.

In this chapter, we will study about the properties of the angles which formed, when two lines intersect each other and when a line intersect two or more parallel lines at distinct points.

Basic Terms and Definitions

Point

The figure of which length, breadth and height cannot be measured is called a point. It is infinitesind.

Line Segment

The straight path between two points P and Q is called a line segment.

This can be represented as $\overline{PQ}$.

- P and Q are called the end points of the line segment.
- The line segment has a definite length.
- Distance between P and Q is called the length of the line segment PQ.

Line

A line segment extended endlessly on both sides, is called a line. It is denoted by $\overleftrightarrow{PQ}$ or $\overleftrightarrow{QP}$.

- A line is a set of infinite number of points.
- A line have no end points and no definite length.

Ray

A ray extends indefinitely in one direction. This is exhibited by an arrow i.e. $\overrightarrow{PQ}$.

- P is called the initial point of the ray. The ray has no definite length.

Collinear Points

Three or more points are said to be collinear if a single straight line passes through them. Here, A, B and C are collinear.

Non-collinear Points

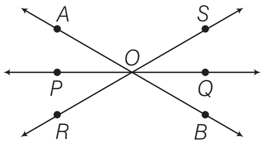
Three or more points not lying on a single straight line are called non-collinear points. Here, A, B and C are non-collinear points.

Intersecting Lines

Two lines having a common point are called intersecting lines. This common point is the point of intersection i.e. O.

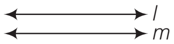
Concurrent Lines

Three or more lines intersecting at the same point are said to be concurrent. This common point is the point of concurrence i.e. O.



Non Intersecting lines/Parallel Lines

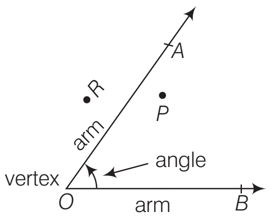
If two lines lie in the same plane and do not intersect when produced on either side, then such lines are said to be parallel to each other.



If l and m are two parallel lines, we write $l || m$ and read it as l is parallel to m.

ANGLES

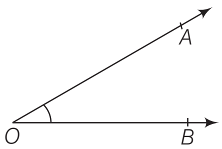
An angle is formed by two rays with a common initial point. Let 'O' is the initial point, then O is called the vertex.



Where, P is a point in the interior of $\angle AOB$ and R is a point in the exterior of $\angle AOB$.

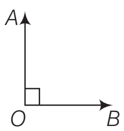
Types of Angle

1. **Acute angle** An angle whose measure is more than 0°, but less than 90°, is called an acute angle.

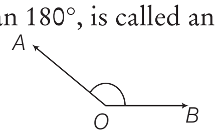


In the given figure, $0° < \angle AOB < 90°$

2. **Right angle** An angle whose measure is 90°, is called a right angle.
   In the given figure, $\angle AOB = 90°$.

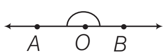


3. **Obtuse angle** An angle whose measure is more than 90° but less than 180°, is called an obtuse angle.

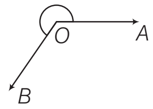


In the given figure, $90° < \angle AOB < 180°$.

4. **Straight angle** An angle whose measure is 180°, is called a straight angle.
   In the figure, $\angle AOB = 180°$

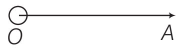


5. **Reflex angle** An angle whose measure is more than 180°, but less than 360°, is called a reflex angle.



In the given figure, $180° < \angle AOB < 360°$

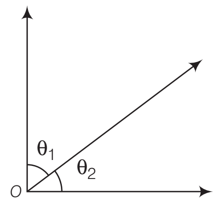
6. **Complete angle** An angle whose measure is 360°, is called a complete angle. In the given figure, $\angle AOA$ is a complete angle.



Pairs of Angle

1. Complementary Angles

Two angles are said to be complementary, if the sum of their measure is 90°. Thus, $\angle\theta_1$ and $\angle\theta_2$ are complementary, if $\angle\theta_1 + \angle\theta_2 = 90°$



- Complementary angles are complement of each other.
- Complement of x is $(90° - x)$.

EXAMPLE 1. The measure of an angle which is 28° more than its complement is

a. 23° b. 59° c. 77° d. None of these

Sol. **b.** Let the measure of the required angle be $x°$. Then, measure of its complement $= 90° - x$

$\therefore \quad x - (90° - x) = 28° \Leftrightarrow 2x = 118°$

$\Leftrightarrow \quad x = 59°$

Hence, the measure of the required angle is 59°.

EXAMPLE 2. The measure of the complement of an angle of 48° 36′ 24″ is

a. 41°23′36″ b. 42°23′36′ c. 41°24′36″ d. 42°24′36″

Sol. **a.** As, $90° = 89°\ 59'\ 60''$

$\therefore$ Complement of an angle of $(48°\ 36'\ 24'')$

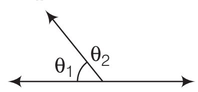
$= \text{Angle of } [90° - 48°\ 36'\ 24'']$

$= \text{Angle of } [89°\ 59'\ 60'' - 48°\ 36'\ 24'']$

$= \text{Angle of } (41°\ 23'\ 36'')$

2. Supplementary Angles

Two angles are said to be supplementary, if the sum of their measures is 180°. Thus, $\angle\theta_1$ and $\angle\theta_2$ are supplementary if $\theta_1 + \theta_2 = 180°$



- Supplementary angles are supplement of each other.
- Supplement of x is $(180° - x)$.

EXAMPLE 3. The measure of an angle, which is 32° less than its supplement is

a. 31° b. 64° c. 74° d. 148°

Sol. c. Let the measure of the required angle be x. Then, measure of its supplement $= (180° - x)$

$\therefore \quad (180° - x) - x = 32° \Leftrightarrow 2x = 148° \Leftrightarrow x = 74°$

EXAMPLE 4. Two supplementary angles are in the ratio 3 : 2. Then, the measurement of the smaller angle is

a. 36° b. 72° c. 108° d. 112°

Sol. b. Let the supplementary angles be $3x$ and $2x$, respectively. Then, according to the definition of supplementary angle.

$$3x + 2x = 180°$$

$$\Rightarrow \quad 5x = 180° \Rightarrow x = 36°$$

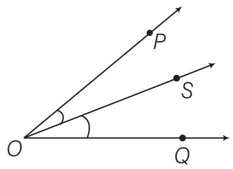
$\therefore$ Angles will be $3x = 3 \times 36 = 108°$

and $2x = 2 \times 36 = 72°$.

Thus, the smaller angle is 72°.

Bisector of an Angle

A ray which divides the angle into two equal parts, is called the bisector of an angle.



If a ray OS is the bisector of $\angle POQ$, then $\angle POS = \angle QOS$.

- $\angle POS = \angle QOS = \dfrac{1}{2} \angle POQ$

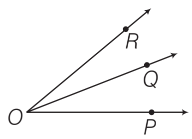
Adjacent Angles

Two angles are said to be adjacent, if

(i) they have a common vertex.

(ii) they have a common arm

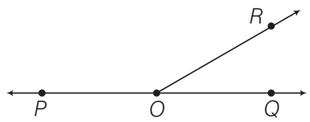
(iii) their non-common arms are on either side of the common arm.



Here, $\angle POQ$ and $\angle ROQ$ are adjacent angles.

Linear Pairs of Angles

If the non-common arms of two adjacent angles form a line, then these angles are called linear pair of angles. $\angle ROP$ and $\angle ROQ$ form a linear pair of angles.

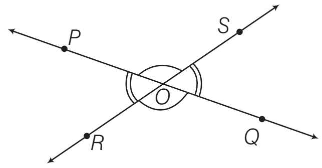


$$\therefore \quad \angle POQ = \angle ROP + \angle ROQ = 180°$$

Vertically Opposite Angles

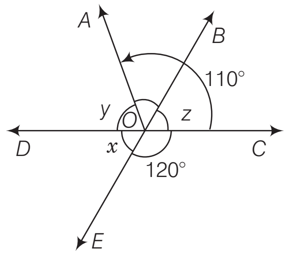
Two angles are called a pair of vertically opposite angles, if their arms form two pairs of opposite rays.

If two lines PQ and RS intersects at a point O, then the pair of $\angle POR$ and $\angle QOS$ or pair of $\angle POS$ and $\angle ROQ$ is said to be a pair of vertically opposite angles.



- Vertically opposite angles are always equal i.e. $\angle POS = \angle ROQ$ and $\angle POR = \angle SOQ$.
- Sum of the angles around a point is 360°.

EXAMPLE 5. In the given figure if DOC is a straight ray OB is bisector of $\angle AOC$, where $\angle AOC = 110°$ and $\angle COE = 120°$, then the sum of $\angle x$, $\angle y$ and $\angle z$ is



a. 160° b. 115° c. 170° d. 180°

Sol. d. Since, DOC is a straight line

$$\angle AOC + \angle DOA = 180° \quad \text{[Linear pair]}$$

$$110° + \angle DOA = 180° \quad [\because \angle AOC = 110°]$$

$$\angle DOA = 70°$$

$$\angle DOA = \angle y = 70°$$

Also, ray OB is angle bisector of $\angle AOC$

$$\angle z = \frac{\angle AOC}{2} = \frac{110°}{2} = 55°$$

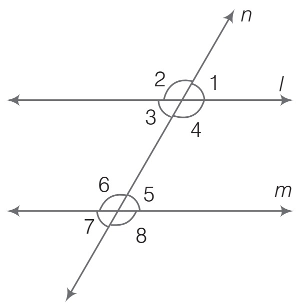
$\because$ Line BE and DC intersect each other at O.

$$\angle x = \angle z = 55° \quad [\because \text{vertically opposite angle}]$$

$$\therefore \angle x + \angle y + \angle z = 55° + 70° + 55° = 180°$$

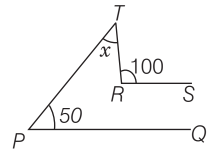
Some Other Angles

Let l and m are parallel lines and n is the transversal which cuts these parallel lines. The different angles formed are as follows.



1. **Corresponding angles** Corresponding angle pairs are
$\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 4$ and $\angle 8$, $\angle 3$ and $\angle 7$ and all the corresponding pair are equal
i.e., $\angle 1 = \angle 5$, $\angle 2 = \angle 6$, $\angle 4 = \angle 8$ and $\angle 3 = \angle 7$.
2. **Vertically opposite angles** Vertically opposite angles pairs are
$\angle 1$ and $\angle 3$, $\angle 4$ and $\angle 2$, $\angle 8$ and $\angle 6$, $\angle 5$ and $\angle 7$ and all vertically opposite angles are equal
i.e., $\angle 1 = \angle 3$, $\angle 4 = \angle 2$, $\angle 8 = \angle 6$ and $\angle 5 = \angle 7$.
3. **Alternate angles** Alternate angles pairs are $\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$ and they are equal
i.e., $\angle 3 = \angle 5$ and $\angle 4 = \angle 6$.
   - The sum of interior angles on the same side of transversal is equal to 180°.
     i.e., $\angle 3 + \angle 6 = 180^\circ$ and $\angle 4 + \angle 5 = 180^\circ$
   - The sum of exterior angles on the same side of transversal is equal to 180°.
     i.e., $\angle 2 + \angle 7 = 180^\circ$ and $\angle 1 + \angle 8 = 180^\circ$

EXAMPLE 6. In the given figure, $PQ \parallel RS$. The value of x is



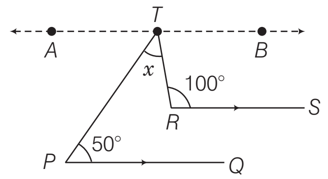
a. 50° **b.** 80° **c.** 75° **d.** 65°

Sol. **a.** Draw $AB \parallel PQ$

$\therefore$ $\angle ATP = \angle TPQ = 50^\circ$ [alternate angles]

and $\angle BTR + \angle TRS = 180^\circ$

[Angles on the same side of transversal]



$\Rightarrow$ $\angle BTR = 80^\circ$

and $\angle ATP + x + \angle BTR = 180^\circ$ [$\because$ ATB is a straight line]

$\therefore$ $x = 180^\circ - (50^\circ + 80^\circ) = 50^\circ$

PRACTICE EXERCISE

1. Three lines intersect each other in pairs. What is the total number of angles so formed?
(a) 3 (b) 6 (c) 9 (d) 12

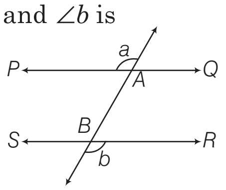
2. An angle is 14° more than its complement. Then, its measure is
(a) 166° (b) 86° (c) 76° (d) 52°

3. The measure of an angle is twice the measure of its supplementary angle. So, its measure is
(a) 120° (b) 60° (c) 100° (d) 90°

4. What is the least number of straight lines for a bounded plane figure?
(a) 1 (b) 2 (c) 3 (d) 4

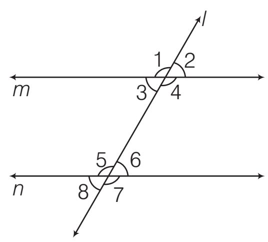
5. The supplement of $\frac{3}{5}$ of a right angle is
(a) 122° (b) 126° (c) 130° (d) 132°

6. In the given figure, if $PQ \parallel SR$, then the relation between $\angle a$ and $\angle b$ is



(a) $\angle a \neq \angle b$ (b) $\angle a < \angle b$ (c) $\angle a = \angle b$ (d) $\angle a > \angle b$

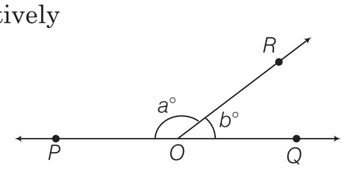
7. If $\angle 1 = (5x - 20)^\circ$ and $\angle 7 = (2x + 10)^\circ$, then $\angle 7$ is



(a) 38° (b) 10°
(c) 30° (d) 70°

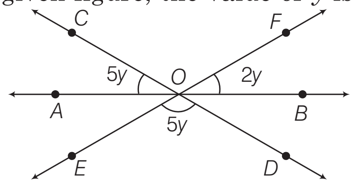
8. The measure of complementary angle of 12° 25′ 40″ is
(a) 77° 34′ 20″ (b) 77° 36′ 20″ (c) 77° 24′ 20″ (d) 77° 34′

9. $\angle POR$ and $\angle QOR$ form a linear pair and if $a - b = 80^\circ$, then the values of a and b are, respectively



(a) 95°, 85° (b) 108°, 72°
(c) 130°, 50° (d) 105°, 75°

10. In the given figure, the value of y is

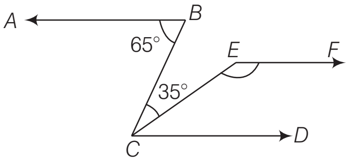


(a) 25° (b) 35° (c) 15° (d) 40°

11. AB and CD are two parallel lines. PQ cuts AB and CD at E and F, respectively. EL is the bisector of $\angle FEB$. If $\angle LEB = 35°$; then $\angle CFQ$ will be

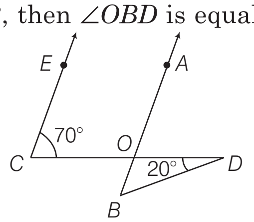
(a) 110° (b) 85° (c) 70° (d) 95°

12. AB and CD are two parallel lines. The points B and C are joined such that $\angle ABC = 65°$. A line CE is drawn making angle of 35° with the line CB, EF is drawn parallel to AB, as show in figure, then $\angle CEF$ is equal to



(a) 160° (b) 155° (c) 150° (d) 145°

13. In the given figure, if $EC \parallel AB$, $\angle ECD = 70°$, $\angle BDO = 20°$, then $\angle OBD$ is equal to

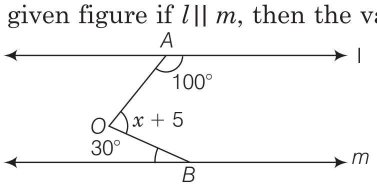


(a) 70° (b) 60° (c) 50° (d) 20°

14. Two parallel lines AB and CD are intersected by a transversal line EF at M and N, respectively. The lines MP and NP are the bisectors of the interior angles BMN and DNM on the same side of the transversal. Then, $\angle MPN$ is equal to

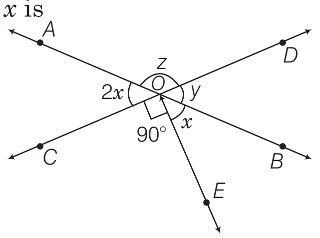
(a) 90° (b) 45° (c) 135° (d) 60°

15. In the given figure if $l \parallel m$, then the value of x is



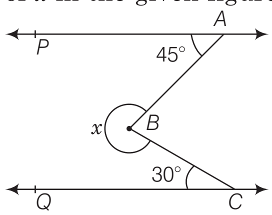
(a) 105° (b) 100° (c) 110° (d) 115°

16. In the given figure, if $\angle COE = 90°$, then the value of x is



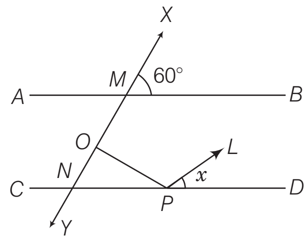
(a) 120° (b) 60° (c) 45° (d) 30°

17. The value of x in the given figure is



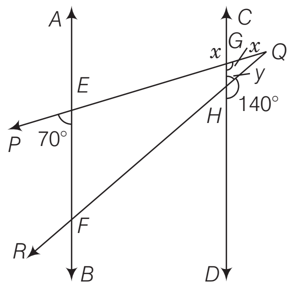
(a) 75° (b) 185°
(c) 285° (d) 245°

18. In the given figure, $AB \parallel CD$, $\angle DPL = \frac{1}{2}\angle NPO$, and $OP \perp XY$ the value of $x°$ is



(a) 30° (b) 40° (c) 15° (d) 25°

19. In the given figure, $AB \parallel CD$ and they cut PQ and QR at E, F and G, H respectively. Then find the value of $x + y$



(a) 120° (b) 130°
(c) 150° (d) 132°

20. Two parallel lines are cut by a transversal, then which of the following are true?

I. Pair of alternate interior angles are congruent.
II. Pair of corresponding angles are congruent.
III. Pair of interior angles on the same side of the transversal are supplementary.

Select the correct answer using the codes given below

(a) I, II and III are true (b) I and III are true
(c) I and II are true (d) II and III are true

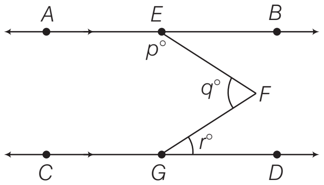
21. Consider the following statements related to three lines L_1, L_2 and L_3 in the same plane.

I. If L_2 and L_3 are both parallel to L_1, then they are parallel to each other.
II. If L_2 and L_3 are both perpendicular to L_1 then they are parallel to each other.
III. If there is acute angle between L_1 and L_3, then L_2 is parallel to L_3.

Which of the statement(s) given above is/are correct?

(a) I and II (b) II and III
(c) All of these (d) None of these

22. In the given figure, $AB \| CD$, then which one of the following is true?



(a) $p + q - r = 180°$ (b) $p + q + r = 180°$
(c) $p - q + r = 180°$ (d) $p + q - 2r = 180°$

23. LM is a straight line and O is a point on LM. Line ON is drawn not coinciding with OL or OM. If $\angle MON$ is one-third of $\angle LON$, then what is $\angle MON$ equal to?

(a) 45° (b) 60°
(c) 75° (d) 80°

24. Consider the following statements :

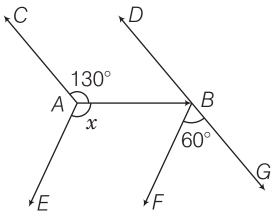
Two lines intersected by a transversal are parallel, if

I. The pairs of corresponding angles are equal.
II. The interior angles on the same side of the transversal are supplementary.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

25. In the given figure $AC \| BD$ and $AE \| BF$. What is the value of $\angle x$?



(a) 130° (b) 110°
(c) 70° (d) 50°

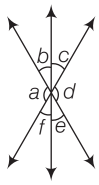
26. The necessary conditions for the line l and m to be parallel when these lines are intersected by a transversal line n is that

I. Interior angles on the same side are equal.
II. Corresponding angles are equal.
III. Vertically opposite angles are equal.
IV. Alternate interior angles are equal.

Select the correct answer using the codes given below

(a) II and III (b) I, II and III (c) II and IV (d) I, II and IV

27. In the given figure, which of the following statements must be true?

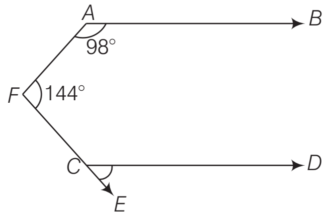


I. $a + b = d + c$ II. $a + c + e = 180°$
III. $b + f = c + e$ IV. $a + b + c = d + e + f$

Select the correct answer using the codes given below

(a) Only I (b) I, II and III
(c) II, III and IV (d) All of these

28. In the given figure, AB is parallel to CD. If $\angle BAF = 98°$ and $\angle AFC = 144°$, then $\angle ECD$ is equal to?



(a) 62° (b) 64°
(c) 82° (d) 84°

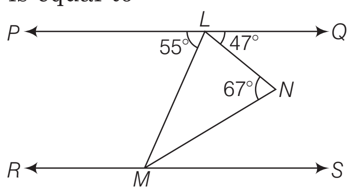
29. Consider the following statements :
If two parallel lines are intersected by a transversal, then

I. each pair of corresponding angles are equal.
II. each pair of alternate angle are unequal.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

30. In the given figure, PQ is parallel to RS. Then $\angle NMS$ is equal to

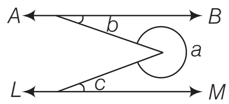


(a) 20° (b) 23° (c) 27° (d) 47°

31. The line segments AB and CD intersect at O. OF is the internal bisector of obtuse $\angle BOC$ and OE is the internal bisector of acute $\angle AOC$. If $\angle BOC = 130°$, then what is the measure of $\angle FOE$?

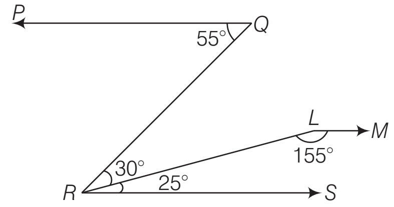
(a) 90° (b) 110° (c) 115° (d) 120°

32. In the given figure below, AB is parallel to LM. What is the angle a equal to?



(a) $\pi + b + c$ (b) $2\pi - b + c$
(c) $2\pi - b - c$ (d) $2\pi + b - c$

33. In the given figure, PQ is parallel to RS. What is the angle between the lines PQ and LM?

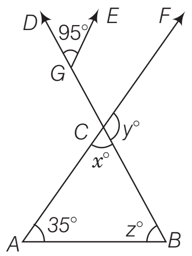


(a) 175° (b) 177° (c) 179° (d) 180°

34. The length of a line segment AB is 2 units. It is divided into two parts at the point C such that $AC^2 = AB \times CB$. What is the length of CB?

(a) $3+\sqrt{5}$ units (b) $3-\sqrt{5}$ unit
(c) $2-\sqrt{5}$ unit (d) $\sqrt{3}$ units

Directions (Q.Nos 35-36) *Read the information and answer the questions.'*
In the given figure, the lines CB and AC of a triangle ABC are extended to D and F, respectively and CF || GE.



35. The sum of the value of x, y and z.

(a) 250° (b) 180° (c) 230° (d) 110°

36. The value of the angle which equals one-fifth of the supplement of z ?

(a) 50° (b) 26° (c) 10° (d) 130°

PREVIOUS YEARS' QUESTIONS

37. AB is a straight line. C is a point whose distance from AB is 3 cm. What is the number of points which are at a distance of 1 cm from AB and 5 cm from C ? **2012 I**

(a) 1 (b) 2 (c) 3 (d) 4

38. The ratio of two complementary angle is 1 : 5. What is the difference between the two angles? **2012 I**

(a) 60° (b) 90°
(c) 120° (d) Cannot be determined

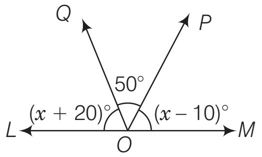
39. Consider the following statements :
If two straight lines intersect, then

I. vertically opposite angles are equal.
II. vertically opposite angles are supplementary.
III. adjacent angles are complementary.

Which of the statement(s) given above is/are correct? **2012 I**

(a) Only III (b) Only I (c) I and III (d) II and III

40. In the given figure, LOM is a straight line. What is the value of $x°$? **2012 II**



(a) 45° (b) 60° (c) 70° (d) 80°

41. If the arms of one angle are respectively parallel to the arms of another angle, then the two angles are **2013 I**

(a) Neither equal nor supplementary
(b) not equal but supplementary
(c) equal but not supplementary
(d) Either equal or supplementary

42. The complement angle of 80° is **2015 I**

(a) $\frac{18}{\pi}$ radian (b) $\frac{5\pi}{9}$ radian
(c) $\frac{\pi}{18}$ radian (d) $\frac{9}{5\pi}$ radian

43. Let OA, OB, OC and OD be rays in the anti-clockwise direction? Such that $\angle AOB = \angle COD = 100°$, $\angle BOC = 82°$ and $\angle AOD = 78°$.
Consider the following statements

I. AOC and BOD are lines.
II. $\angle BOC$ and $\angle AOD$ are supplementary. **2015 I**

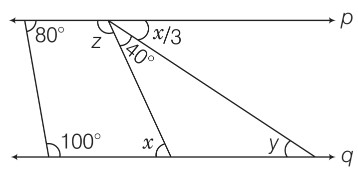
Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

44. If a transversal intersects four parallel straight lines, then the number of distinct values of the angles formed will be **2016 I**

(a) 2 (b) 4 (c) 8 (d) 16

45.



In the figure given above, p and q are parallel lines. What are the values of the angles x, y and z ? **2016 I**

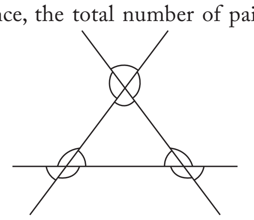
(a) $x = 80°$, $y = 40°$, $z = 100°$
(b) $x = 80°$, $y = 50°$, $z = 105°$
(c) $x = 70°$, $y = 40°$, $z = 110°$
(d) $x = 60°$, $y = 20°$, $z = 120°$

ANSWERS

1	d	2	d	3	a	4	c	5	b	6	c	7	c	8	a	9	c	10	c
11	a	12	c	13	c	14	a	15	a	16	d	17	c	18	c	19	c	20	a
21	a	22	a	23	a	24	c	25	b	26	c	27	c	28	a	29	a	30	a
31	a	32	c	33	d	34	b	35	c	36	b	37	d	38	a	39	b	40	b
41	b	42	c	43	d	44	a	45	d										

HINTS AND SOLUTIONS

1. *(d)* We know that, when two lines intersect each other, it makes 4 angles. Since, the total number of pairs = 3



$\therefore$ Total number of angles $= 3 \times 4 = 12$

2. *(d)* Let angle be x and its complement be $90° - x$.
According to the question,

$x = (90° - x) + 14$

$\Rightarrow 2x = 104°$

$\Rightarrow x = \dfrac{104°}{2} = 52°$

3. *(a)* Let angle be x and its supplementary $= 180° - x$

Then, $x = 2(180° - x)$

$\Rightarrow 3x = 360°$

$\Rightarrow x = 120°$

4. *(c)* The least number of straight lines for a bounded plane figure is 3.

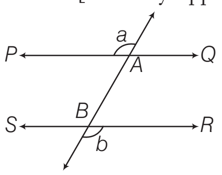
5. *(b)* Given, angle $= \dfrac{3}{5}$ of right angle

$= \dfrac{3}{5} \times 90° = 3 \times 18° = 54°$

Supplement of $54° = (180° - 54°)$

= An angle of measure 126°

6. *(c)* We have, $\angle QAB = \angle a$
[vertically opposite angles]



and $\angle QAB = \angle b$
[corresponding angles]

$\Rightarrow \angle a = \angle b$

7. *(c)* Here, $\angle 1 = \angle 5$
[corresponding angles]

$\angle 7 = \angle 5$ (vertical opposite)

$\Rightarrow \angle 1 = \angle 7$ (angle)

So, $5x - 20° = 2x + 10°$

$3x = 30° \Rightarrow x = 10°$

Hence,

$\angle 7 = 2x + 10° = 2(10°) + 10° = 30°$

8. *(a)* Complementary angle of $12° 25' 40''$

$= 90° - 12° 25' 40''$

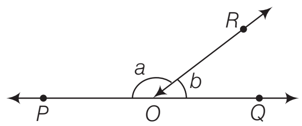
$= 89° 59' 60'' - 12° 25' 40''$

$= (89 - 12)° + (59' - 25') + (60 - 40)''$

$= 77° + 34' + 20'' = 77° 34' 20''$

9. *(c)* Since, $\angle POR$ and $\angle QOR$ is a linear pair.

$\therefore \angle POR + \angle ROQ = 180°$



$\Rightarrow a + b = 180°$ ... (i)

and $a - b = 80°$ ... (ii)

On solving Eqs. (i) and (ii), we get

$2a = 260° \Rightarrow a = 130°$

and $b = 180° - 130° = 50°$

So, $a = 130°, b = 50°$

10. *(c)* Since, AOB is a straight line.

$\therefore \angle AOB = 180°$

$\angle AOC + \angle COF + \angle FOB = 180°$
[straight line]

$5y + 5y + 2y = 180°$

[$\because \angle COF = \angle DOE$, vertically opposite angles]

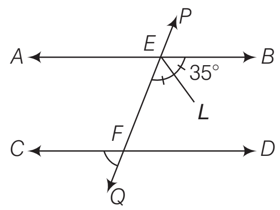
$\Rightarrow 12y = 180°$

$\Rightarrow y = \dfrac{180°}{12°} = 15°$

Hence, the value of y is 15°.

11. *(a)* Given, $\angle LEB = 35°$

$\angle FEB = 2 \times \angle LEB = 2 \times 35° = 70°$

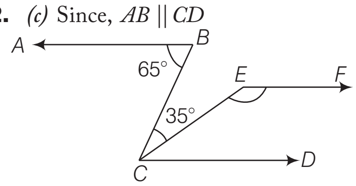


$\Rightarrow \angle AEB = \angle AEF + \angle BEF = 180°$
[straight line]

$\Rightarrow \angle AEF = 180° - 70° = 110°$

$\Rightarrow \angle CFQ = \angle AEF = 110°$
[corresponding angles]

12. *(c)* Since, $AB \parallel CD$



$\therefore \angle ABC = \angle BCD = 65°$
[alternate angles]

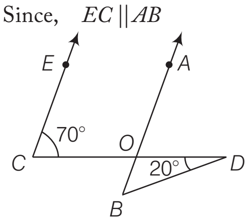
$\angle ECD = 65° - \angle BCE$

$= 65° - 35° = 30°$

$\angle CEF + \angle ECD = 180°$
[Angles on the same side of transversal]

$\Rightarrow \angle CEF = 180° - 30° = 150°$

13. *(c)* Since, $EC \parallel AB$



$\angle AOD = \angle ECO$
[$\because$ corresponding angles]

$\Rightarrow \angle AOD = 70°$

So, $\angle BOD = 180° - 70°$

$= 110°$ [linear point]

Now, in ΔBOD

$\angle OBD + \angle BOD + \angle ODB = 180°$

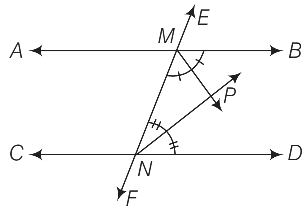
$\Rightarrow \angle OBD = 180° - (110 + 20)° = 50°$

14. *(a)* Given, $\angle PMN = \frac{1}{2}\angle BMN$

and $\angle PNM = \frac{1}{2}\angle DNM$

As, $\angle BMN + \angle DNM = 180°$

[angles on the same side of transversal]



$\therefore$ In ΔMPN, $\angle PMN + \angle PNM = 90°$

$\Rightarrow \angle MPN = 180° - (\angle PMN + \angle PNM)$

[angle sum property]

$\therefore \angle MPN = 180° - 90° = 90°$

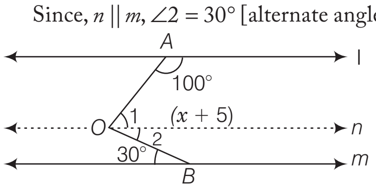
15. *(a)* Draw a line n passing through O and parallel to l and m

Since, $l \parallel n$, $\angle 1 + 100° = 180°$

[Sum of the interior angles on the same side of the transversal]

$\angle 1 = 80°$

Since, $n \parallel m$, $\angle 2 = 30°$ [alternate angles]



Now, $\angle AOB = \angle 1 + \angle 2$

$= (80° + 30°) = 110°$

But, $\angle AOB = (x + 5)° = 110°$

$x = (110 - 5)° = 105°$

16. *(d)* Here, $\angle BOD = \angle AOC$

$\therefore$ $2x = y$ [vertically opposite angles]

Now, COD is a straight line.

$\angle COD = 180°$

$\Rightarrow \angle COE + \angle EOB + \angle BOD = 180°$

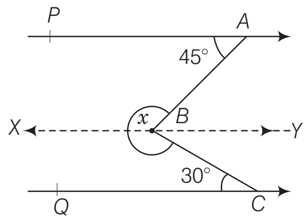
$\Rightarrow 90° + x + 2x = 180°$

$\Rightarrow 3x = 90° \Rightarrow x = 30°$

Hence, the value of x is $30°$.

17. *(c)* From figure, $PA \parallel QC$

Draw a line XY parallel to PA and QC.



$\because PA \parallel XY$,

$\Rightarrow \angle ABY = \angle PAB = 45°$

[alternate angles]

and $\because XY \parallel QC$, $\angle QCB = \angle CBY = 30°$

[alternate angles]

$\Rightarrow \angle ABC = \angle ABY + \angle CBY$

$\Rightarrow \angle ABC = 45° + 30° = 75°$

$\because x = 360° - \angle ABC = 360° - 75°$

$\therefore x = 285°$

18. *(c)* $\because$ $AB \parallel CD$, $\angle ONP = \angle XMB = 60°$

[corresponding angles]

$\angle OPN = 90° - \angle ONP$

$= 90° - 60° \Rightarrow \angle OPN = 30°$

But, $\angle DPL = \frac{1}{2}\angle NPO = \frac{1}{2} 30°$

$\Rightarrow \angle DPL = 15° \Rightarrow x = 15°$

19. *(c)* Since, $AB \parallel CD$ and PQ is transversal

$\angle PEF = \angle EGH$ [corresponding angles]

$\Rightarrow \angle EGH = 70°$ [$\because \angle PEF = 70°$]

Now, $\angle EGH + \angle HGQ = 180°$

[linear pair]

$\Rightarrow \angle HGQ = 180° - 70° = 110°$

Also, $\angle DHQ + \angle GHQ = 180°$

[linear pair]

$\angle GHQ = 180° - 140° = 40°$

$\therefore x + y = 110° + 40° = 150°$

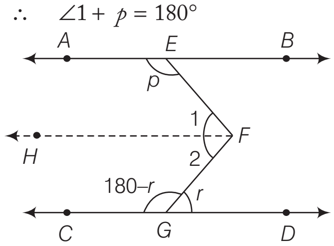
20. *(a)* All the three statements are true.

21. *(a)* Only statements I and II are true.

22. *(a)* Draw $FH \parallel AB \parallel CD$

(sum of interior angles)

$\therefore \angle 1 + p = 180°$ ...(i)



$\angle 2 + 180° - r = 180°$ ...(ii)

(sum of interior angle)

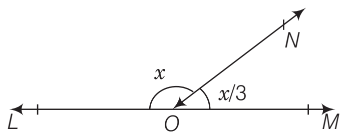
On adding Eqs. (i) and (ii), we get

$\angle 1 + \angle 2 + p + 180° - r = 360°$

$\Rightarrow p + q - r = 180°$ [$\because \angle 1 + \angle 2 = q°$]

23. *(a)* Given that, $\angle MON = \frac{1}{3}\angle LON$

Let $\angle LON = x$, then, $\angle MON = \frac{x}{3}$



We know that,

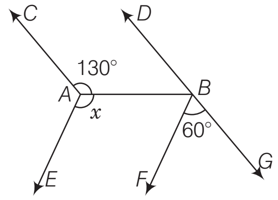
$\angle LON + \angle MON = 180°$ [linear pair]

$\Rightarrow x + \frac{x}{3} = 180° \Rightarrow x = \frac{180° \times 3}{4} = 135°$

Thus, $\angle MON = \frac{x}{3} = \frac{135°}{3} = 45°$

24. *(c)* Both the statements I and II are correct.

25. *(b)* Since, $AC \parallel BD$



$\therefore \angle DBA = 180° - 130° = 50°$

[interior angle]

[$\because BAC = 130°$]

DBG is straight line.

$\therefore \angle DBA + \angle ABF + \angle FBG = 180°$

[linear pairs]

$\Rightarrow 50° + \angle ABF + 60° = 180°$

$\Rightarrow \angle ABF = 70°$

Since, $AE \parallel BF$

$\therefore x = 180° - \angle ABF$

$= 180° - 70° = 110°$

26. *(c)* When two lines are parallel intersected by a transversal then corresponding as well as alternate interior angles are equal.

Hence, the statement II and IV are correct.

27. *(c)* We have, $a = d$, $b = e$ and $c = f$

[vertically opposite angles]

and $a + b + c = d + e + f = 180°$

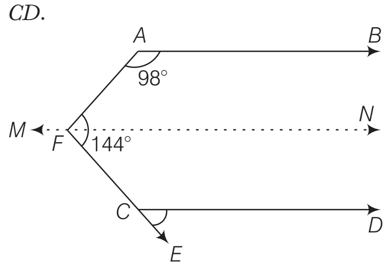
II. $a + b + c = 180° \Rightarrow a + c + e = 180°$

III. $b + f = c + e$

IV. $a + b + c = d + e + f$

Hence, statements II, III and IV are true.

28. *(a)* Draw a line MN parallel to AB and CD.



$\Rightarrow \angle AFN = 180° - 98° = 82°$

(sum of interior angles)

$\angle CFN = 144° - 82° = 62°$

and $\angle ECD = \angle CFN = 62°$

[corresponding angles]

29. *(a)* If two parallel lines are intersected by a transversal, then each pair of corresponding angles and of alternate angles are equal.

Therefore, statement I is correct.

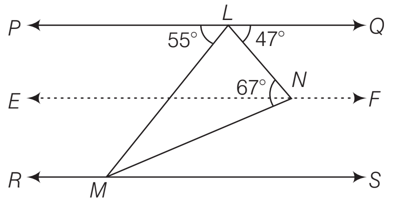
30. *(a)* Since, $PQ \parallel RS$

$\therefore \angle PLM = \angle LMS = 55°$ [alternate angle]

Draw a line EF which is parallel to PQ.

Then, $\angle QLN = \angle LNE = 47°$

$\because \angle ENL + \angle MNE = 67°$



$\Rightarrow 47° + \angle MNE = 67°$

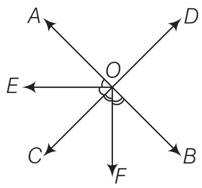
$\Rightarrow \angle MNE = 67° - 47°$

$\Rightarrow \angle MNE = 20°$

Similarly, $EF \parallel RS$, then

$\angle ENM = \angle NMS = 20°$ [alternate angle]

31. *(a)* Given, $\angle BOC = 130°$



$\because AOB$ is a straight line.

$\because \angle BOC + \angle AOC = 180°$ [linear pair]

$\Rightarrow 130° + \angle AOC = 180°$

$\Rightarrow \angle FOC + \angle FOC = 130°$

$\Rightarrow \angle AOC = 50°$

Now, $\angle BOC = 130°$

$\Rightarrow \angle BOF + \angle FOC = 130°$

[$\because OF$ is bisector of $\angle BOC$]

$\Rightarrow \angle FOC = 65°$

Now, $\angle AOC = 50°$

$\Rightarrow \angle AOE + \angle EOC = 50°$

$\Rightarrow \angle EOC + \angle EOC = 50°$

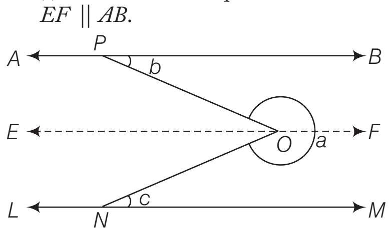
$\Rightarrow \angle EOC = 25°$

[$\because OE$ is bisector of $\angle AOC$]

$\therefore \angle EOF = \angle EOC + \angle FOC$

$= 25° + 65° = 90°$

32. *(c)* Draw a line parallel to AB i.e. $EF \parallel AB$.



$\therefore \angle EOP = \angle OPB = b$ [alternate angle]

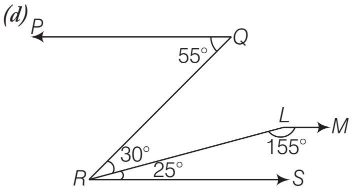
and $\angle EON = \angle ONM = c$ [alternate angle]

$\Rightarrow \angle PON = b + c$

$\because \angle PON + a = 2\pi$

$\therefore a = 2\pi - \angle PON = 2\pi - b - c$

33. *(d)*



$PQ \parallel RS$ ...(i)

Since, $\angle PQR = \angle QRS = 30° + 25° = 55°$ [alternate angle]

$PQ \parallel RS$ ...(i)

and $\angle SRL + \angle RLM = 180°$

$\Rightarrow RS \parallel LM$ ...(ii)

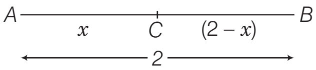
From Eqs. (i) and (ii), we get

$PQ \parallel LM$

So, the angle between the lines PQ and LM is 180°.

34. *(b)* Given, $AC^2 = AB \times CB$

$\Rightarrow x^2 = 2 \times (2 - x)$



$\Rightarrow x^2 = 4 - 2x$

$\Rightarrow x^2 + 2x - 4 = 0$

$\Rightarrow x = \dfrac{-2 \pm \sqrt{4+16}}{2 \times 1}$

$\Rightarrow x = -1 \pm \sqrt{5}$

Now, $BC = 2 - (-1 \pm \sqrt{5}) = 3 - \sqrt{5}$

(neglect $3 + \sqrt{5} \because 3 + \sqrt{5} > 2$)

35. *(c)* We have, $\angle DGE = \angle GCF = 95°$ [corresponding angles]

Also, $\angle GCF + \angle FCB = 180°$ [linear pair]

$\Rightarrow 95° + y = 180°$

$\Rightarrow y = 180° - 95° = 85°$

Again, $\angle ACB + y = 180°$

$\Rightarrow x = 180° - 85° = 95°$

Now, In ΔABC $\angle A + \angle B + \angle C = 180°$

$\Rightarrow 35° + z + 95° = 180°$

$\Rightarrow z + 130° = 180°$

$\Rightarrow z = 180° - 130° = 50°$

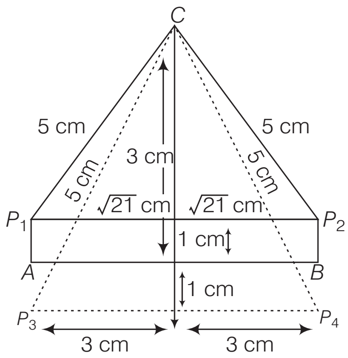
$\therefore x + y + z = 95° + 85° + 50° = 230°$

36. *(b)* Supplement of $z = 130°$

$\therefore \dfrac{1}{5}$th of $130° = \dfrac{130°}{5} = 26°$

37. *(d)* $\therefore$ Required number of points

$= 4\,(P_1, P_2, P_3, P_4)$



38. *(a)* Given that, $\dfrac{\alpha}{\beta} = \dfrac{1}{5}$

Let $\alpha = k$ and $\beta = 5k$

Sum of two complementary angles

$= 90°$

$\Rightarrow \alpha + \beta = 90°$,

$\alpha = 90° - \beta$

$\Rightarrow k = 90° - 5k$

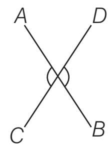
$\Rightarrow k = 15°$

$\therefore \alpha = 15°$ and $\beta = 75°$

$\therefore$ Difference between angles

$= 75° - 15° = 60°$

39. *(b)* Here, AB and CD are two lines.



If two straight lines intersect, then vertically opposite angles are equal.

40. *(b)* From the given figure,

$\angle LOQ + \angle QOP + \angle POM = 180°$ [straight line]

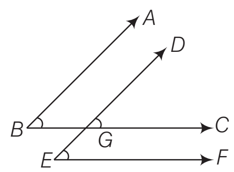
$\therefore (x° + 20°) + 50° + (x° - 10°) = 180°$

$\Rightarrow 2x° + 60° = 180°$

$\Rightarrow 2x° = 120°$

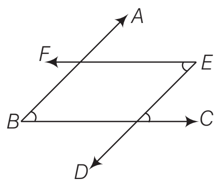
$\therefore x° = 60°$

41. *(b)* **Case I** When both pairs of arms are parallel same sense.



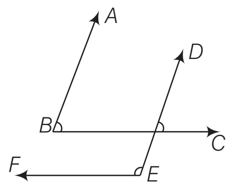
Here, $\angle ABC = \angle DEF$

Case II When both pairs of arms are parallel in opposite sense.



Here, $\angle ABC = \angle DEF$

Case III When one pair of arms is parallel in same direction and other pairs are parallel in opposite direction.



Here, $\angle ABC + \angle DEF = 180°$

So, the two angles are not equal but supplementary.

42. *(c)* Let the angle be θ.

$\therefore\ \theta = 80°$

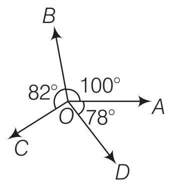
Complement angle

$$= 90° - \theta = 90° - 80° = 10°$$

[$\because$ sum of two complement angles is 90°]

$$= 10 \times \frac{\pi}{180°} = \frac{\pi}{18} \text{ radian.}$$

43. *(d)*



Given, $\angle AOB = \angle COD = 100°$,

$\angle BOC = 82°$

and $\angle AOD = 78°$

If, AOC is a straight line, then

$\angle AOB + \angle BOC = 180°$

$\Rightarrow\quad 100° + 82° = 180°$

$\Rightarrow\quad 182° \neq 180°$

If, BOD is a straight line, then

$\angle BOA + \angle AOD = 180°$

$\Rightarrow\quad 100° + 78° = 180°$

$\Rightarrow\quad 178° \neq 180°$

If, $\angle BOC$ and $\angle AOD$

are supplementary angles, then

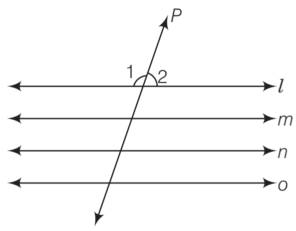
$\angle BOC + \angle AOD = 180°$

$\Rightarrow\quad 82° + 78° = 180°$

$\Rightarrow\quad 160° \neq 180°$

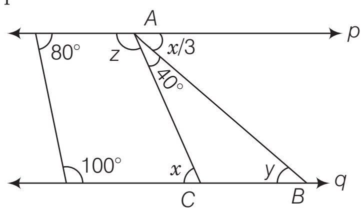
Hence, statements I and II are incorrect.

44. *(a)* If, line P intersect four parallel lines l, m, n and o, then 16 angles will be formed.



As these lines are parallel, hence distinct angle will be $\angle 1$ and $\angle 2$.

45. *(d)* In the given figure, lines p and q are parallel.



$\therefore\quad x = 40° + \dfrac{x}{3}$ [alternate angles]

$\Rightarrow\quad \dfrac{2x}{3} = 40° \Rightarrow x = 60°$

$y = x/3$ [alternate angle]

$y = \dfrac{60°}{3} = 20°,\ z + 40° + \dfrac{x}{3} = 180°$

$\Rightarrow\quad z + 20° = 180° - 40°$

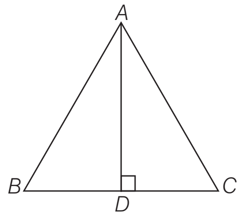
$\Rightarrow\quad z = 120°$

23

TRIANGLES

Generally (8-10) questions have been asked from this chapter. Generally questions are asked from the topic of pythagoras theorem, similarity of triangles and mid-point theorem.

TRIANGLE



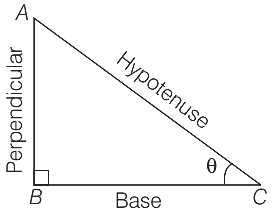
A plane (closed) figure bounded by three line segments is called a triangle. It is denoted by Δ.

ΔABC has

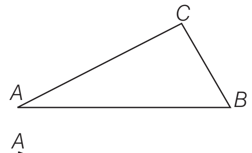
- three vertices, namely A, B and C.
- three sides, namely AB, BC and CA.
- three angles, namely $\angle A$, $\angle B$ and $\angle C$.
- Sum of three angles of a triangle is $180°$. i.e., $\angle A + \angle B + \angle C = 180°$
- Area of a triangle $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times BC \times AD$

Classification of Triangle

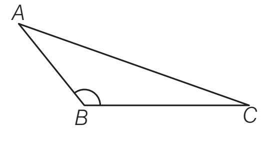
1. On the Basis of Angles



(i) **Right Angled Triangle** A triangle in which one of the angle measures $90°$ is called a right angled triangle. The side opposite to the right angle is called its hypotenuse and the remaining two sides are called as perpendicular and base depending upon conditions. Here, ΔABC is a right angled triangle in which $\angle B = 90°$ and AC is hypotenuse.



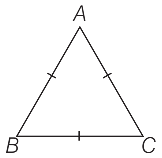
(ii) **Acute Angled Triangle** A triangle in which every angle measure more than $0°$ but less than $90°$ is called an acute angled triangle.
Here, ΔABC is an acute angled triangle.



(iii) **Obtuse Angled Triangle** A triangle in which one of the angle measures more than $90°$ but less than $180°$ is called an obtuse angled triangle.
Here, $\Delta\, ABC$ is an obtuse angled triangle and $\angle ABC$ is the obtuse angle.

2. On the Basis of Sides

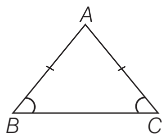
(i) **Equilateral Triangle** A triangle having all sides equal is called an equilateral triangle. Here ΔABC, is an equilateral triangle in which $AB = BC = AC$.



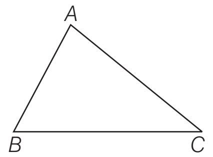
All angles are equal and are of measure $60°$.

(ii) **Isosceles Triangle** A triangle in which two sides are equal, is called an isosceles triangle. Here, ΔABC is an isosceles triangle as $AB = AC$.

- Angle opposite to equal sides are equal.
  i.e. $\angle B = \angle C$



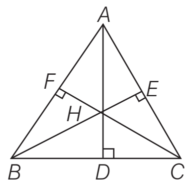
(iii) **Scalene Triangle** A triangle in which all the sides are of different lengths is called a scalene triangle. $\Delta\, ABC$ is a scalene triangle as $AB \neq BC \neq AC$.



Note The sum of the lengths of three sides of a triangle is called its perimeter.
So, perimeter of $\Delta ABC = AB + BC + AC$

Some Term Related to a Triangle

Altitudes and Orthocentre The altitude of a triangle is a line segment perpendicular drawn from vertex to the side opposite to it. The side on which the perpendicular is being drawn is called its base.

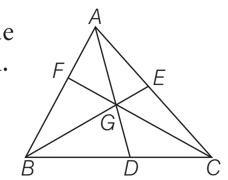


Here, In ΔABC, AD, BE and CF are altitudes drawn on BC, AC and AB, respectively.

- Altitudes of a triangle are concurrent.
- The point of intersection of all the three altitudes of a triangle is called its orthocentre.

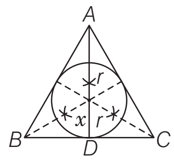
Medians and Centroid A line segment joining the mid-point of a side to the opposite vertex is called median.

Here, In ΔABC, AD, BE and CF are medians.



- The medians of a triangle are concurrent.
- The point of intersection of all the three medians of a triangle is called its centroid. It is denoted by G.
- Centroid divides the median in the ratio 2 : 1.
- A median bisects the area of the triangle i.e. $\text{Ar}(\Delta ABD) = \text{Ar}(\Delta ADC) = \frac{1}{2}\text{Ar}(\Delta ABC)$, etc.

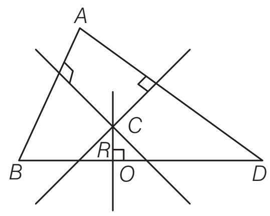
Incentre and Angle Bisector The point of intersection of all the three angle bisectors of a triangle is called its incentre. It is denoted by I.



- The circle with centre I and touches all the sides is called incircle and radius of this circle is called inradius denoted by 'r'.
- Inradius $= \frac{1}{3} \times$ Height $= \frac{\text{Side}}{2\sqrt{3}}$,

  $ID \to$ Inradius, $AD \to$ Height

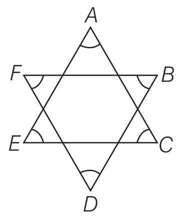
Perpendicular Bisector and Circumcentre The point of intersection of the perpendicular bisectors of the sides of a triangle is called its circumcentre. It is denoted by C.



The circle with centre C and passing through vertices A, B and C is called circumcircle. Radius of circumcircle is called circumradius denoted by R.

Circumradius $(CO) = \frac{2}{3} \times$ Height $= \frac{\text{Side}}{3}$

EXAMPLE 1. In the diagram given below what is the sum of all the angles $\angle A$, $\angle B$, $\angle C$, $\angle D$, $\angle E$ and $\angle F$?



a. $120°$ b. $180°$
c. $290°$ d. $360°$

Sol. d. Since, sum of the angles of a triangle is $180°$.

In $\Delta\, AEC$, $\angle A + \angle C + \angle E = 180°$ ...(i)

and In $\Delta\, BDF$, $\angle B + \angle D + \angle F = 180°$ ...(ii)

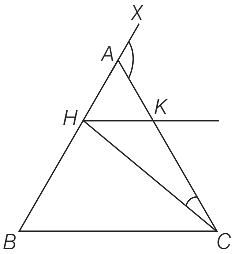
On adding the Eqs. (i) and (ii), we get

$$\angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 180° + 180° = 360°$$

EXAMPLE 2. ABC is an isosceles triangle in which $AB = AC$, $CH = CB$ and HK is parallel to BC. If the exterior $\angle CAX = 137°$, then what is the measure of $\angle HCK$?

a. $68\frac{1}{2}°$ b. $43°$
c. $25\frac{1}{2}°$ d. $137°$

Sol. **c.** $\because\ \angle CAX = 137°$



$\therefore \quad \angle ABC = \frac{1}{2}(137°) \; = 68\frac{1}{2}°$

$\therefore$ Again, $\quad BC = CH$ and $\angle ABC = 68\frac{1}{2}°$

Therefore, $\angle CHB = 68\frac{1}{2}°$, Therefore, $\angle HCB = 43°$

Hence, $\quad \angle HCK = 68\frac{1}{2}° - 43° = 25\frac{1}{2}°$

Some Important Results of a Triangle

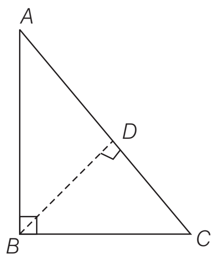
- If two angles of a triangle are equal, then the sides opposite to them are also equal.
- In a triangle, the side opposite to a greater angle is the longest side.
- The sum of all the three interior angles of a triangle is 180°.
- If a side of a triangle is produced, then the exterior angle so formed is equal to the sum of the two interior opposite angles.
- An exterior angle of a triangle is greater than either of the interior opposite angles.
- A triangle must have atleast two acute angles.
- If a perpendicular is drawn from the vertex of a right angled triangle to the hypotenuse, then the triangles on both sides of the perpendicular are similar to the original triangle and also to each other.
- In a right angle ΔABC, $\angle B = 90°$ and AC is hypotenuse. The perpendicular BD is dropped on hypotenuse AC from right angle vertex B, then

(i) $BD = \dfrac{AB \times BC}{AC}$

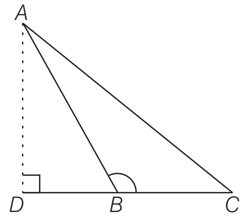
(ii) $AD = \dfrac{AB^2}{AC}$

(iii) $CD = \dfrac{BC^2}{AC}$

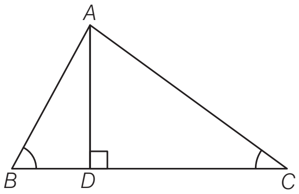
(iv) $\dfrac{1}{BD^2} = \dfrac{1}{AB^2} + \dfrac{1}{BC^2}$



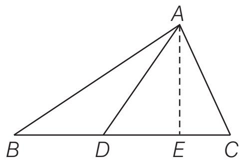
- The area of the equilateral triangle described on the side of a square is half the area of the equilateral triangle described on its diagonal.
- In ΔABC, if $\angle B > 90°$ and AD is perpendicular dropped on BC, then $AC^2 = AB^2 + BC^2 + 2BC \cdot BD$



- In ΔABC, if $\angle B < 90°$ and $AD \perp BC$, then $AC^2 = AB^2 + BC^2 - 2BC \cdot BD$



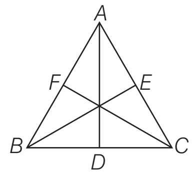
- In any triangle, the sum of the square of any two sides is equal to twice the square of half of the third side together with twice the square of the median which bisect the third side.



Here, AD is median, so

$$AB^2 + AC^2 = 2AD^2 + 2\left(\frac{1}{2}BC\right)^2$$

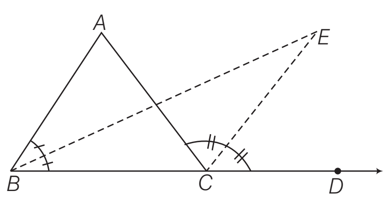
- In a $\Delta\, ABC$, three times the sum of the squares of the sides of a triangle is equal to four times the sum of the squares of the medians of the triangle.



So, in triangle, if AD, BE, FC are the medians, then

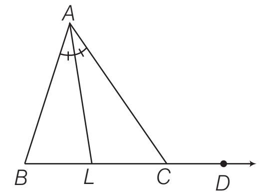
$$3(AB^2 + BC^2 + AC^2) = 4(AD^2 + BE^2 + CF^2)$$

- In an equilateral $\Delta\, ABC$, the side BC is trisected at D. Then, $\quad 9\,AD^2 = 7\,AC^2$
- When the bisector of one internal base angle and the bisector of external angle meet at a point then the formed angle is equal to one half of the vertical angle.

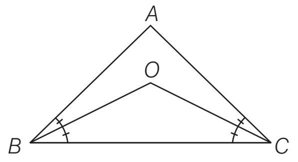


Here, $\quad \angle BEC = \frac{1}{2}\angle BAC$

- The side BC of $\Delta\, ABC$ is produced to D. The bisector of $\angle A$ meets BC in L. Then
  $\angle ABC + \angle ACD = 2\,\angle ALC$.

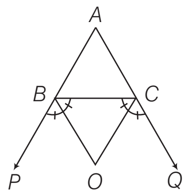


- In a $\Delta\, ABC$ the bisector of $\angle B$ and $\angle C$ intersect each other at a point O.



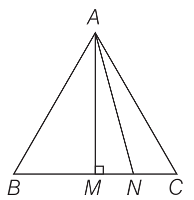
then, $\angle BOC = 90° + \dfrac{1}{2}\angle A$

- In a ΔABC, the side AB and AC are produced to P and Q, respectively. The bisectors of $\angle PBC$ and $\angle QCB$ intersect at a point O.



Then, $\angle BOC = 90° - \dfrac{1}{2}\angle A$

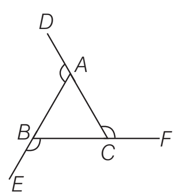
- In $\Delta\, ABC$, $\angle B > \angle C$. If AN is the bisector of $\angle BAC$ and $AM \perp BC$, then



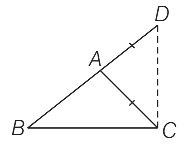
$$\angle MAN = \frac{1}{2}(\angle B - \angle C)$$

- The bisectors of the base angles of a triangle can never enclose a right angle.
- A triangle is an isosceles if and only if any two altitudes are equal.
- If the three sides of a triangle be produced in order, then the sum of all the exterior angles so formed is 360°.

  So, $\angle DAB + \angle EBC + \angle ACF = 360°$

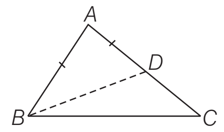


- Sum of any two sides of a triangle is greater than its third side.



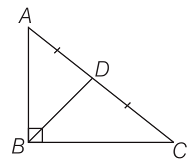
Here, in $\Delta\, ABC$, $AB + AC > BC$
similarly, $AB + BC > AC$, $BC + AC > AB$

- The difference between any two sides of a triangle is less than its third side.

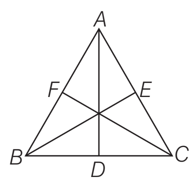


Here, in $\Delta\, ABC$, $AC - AB < BC$
Similarly, $BC - AC < AB$, $BC - AB < AC$

- If the bisector of the vertical angle of a triangle bisects the base, then that triangle is an isosceles.
- If the altitude from one vertex of a triangle bisects the opposite side, then the triangle is an isosceles.
- The perpendiculars drawn from the vertices of equal angles of an isosceles triangle to the opposite sides are equal.
- Medians of equilateral triangle are equal.
- If D is the mid-point of the hypotenuse AC of a right angled $\Delta\, ABC$. Then, $BD = \dfrac{1}{2}AC$

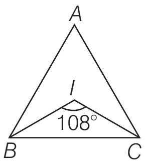


- The sum of any two sides of a triangle is greater than twice the median drawn to the third side.
- Perimeter of a triangle is greater than the sum of its three medians.



So, $AB + BC + AC > AD + BE + CF$.

EXAMPLE 3. In the given figure, I is the incentre of ΔABC. What is the measure of angle A .



a. 54° b. 18° c. 36° d. None of these

Sol. c. Here, I is the incentre of the ΔABC.

$\therefore$ BI and CI are the bisectors of $\angle B$ and $\angle C$, then

we know that, $\angle BIC = 90° + \frac{1}{2}\angle A$

$\Rightarrow \quad 108° = 90° + \frac{1}{2}\angle A$

$\Rightarrow \quad \frac{1}{2}\angle A = 108° - 90° = 18°$

$\therefore \quad \angle A = 36°$

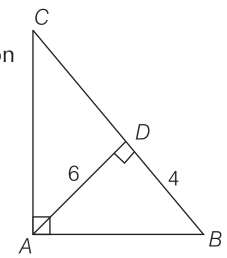
EXAMPLE 4. In a triangle ABC, $\angle BAC = 90°$ and AD is perpendicular to BC. If $AD = 6$ cm and $BD = 4$ cm, then the length of BC is

a. 8 cm b. 10 cm c. 9 cm d. 13 cm

Sol. d. In ΔABC, $\angle BAC = 90°$

$\therefore \quad AB = \sqrt{AD^2 + BD^2}$ [By pythagoras theorem]

$= \sqrt{36+16} = \sqrt{52}$



We know that, in a right angled triangle the perpendicular is drawn on hypotenuse from right angle vertex, then

$$BD = \frac{AB^2}{BC}$$

$$AB^2 = BD \times BC$$

$\Rightarrow \quad (\sqrt{52})^2 = BC \times 4$

$\Rightarrow \quad BC = \frac{52}{4} = 13\text{ cm}$

Congruent Figures

The geometrical figures having the same shape and size are known as congruent figures.

e.g. Two circles of the same radii and two squares of the same sides are congruent.

Congruent Triangles

Two triangles are said to be congruent, if both are exactly of same size i.e. all angles and sides are equal to corresponding angles and sides of other.

- Every triangle is congruent to itself $\Delta ABC \cong \Delta ABC$
- If $\Delta ABC \cong \Delta DEF$, then $\Delta DEF \cong \Delta ABC$
- If $\Delta ABC \cong \Delta DEF$ and $\Delta DEF \cong \Delta PQR$, then $\Delta ABC \cong \Delta PQR$

In congruent triangles, corresponding parts are equal and we write in short 'CPCT', i.e. corresponding part of congruent triangles.

Criteria for Congruence of Triangles

Theorem 1 Two triangles are congruent if two sides and the included angle of one triangle are equal to the corresponding sides and the included angle of other triangle. (SAS criteria)

Theorem 2 Two triangles are congruent if two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangles. (ASA criteria)

Theorem 3 Two triangles are congruent if three sides of one triangle are equal to the corresponding three sides of the other triangle (SSS).

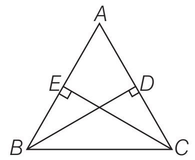
Theorem 4 Two triangles are congruent if the hypotenuse and other side of one triangle are equal to the hypotenuse and the corresponding side of the other triangle (RHS).

Theorem 5 Two triangles are congruent if two angles and a side other than the included side of one triangle are equal to the corresponding two angles and a side other than the included side of other triangle. (AAS Criteria)

EXAMPLE 5. In a ΔABC, the altitudes BD and CE are equal and $\angle A = 36°$. What is the value of the $\angle B$?

a. 72° b. 84° c. 18° d. 36°

Sol. a. For the ΔBDC and ΔBEC,

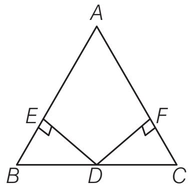


$BD = EC$, $BC = BC$ and $\angle BEC = \angle BDC = 90°$

Thus, $\Delta BEC \cong \Delta BDC$ [by RHS rule]

$\therefore \quad \angle B = \angle C = \frac{180° - 36°}{2} = 72°$ each

EXAMPLE 6. In the given figure, D is the midpoint of BC, $DE \perp AB$ and $DF \perp AC$ such that $DE = DF$, then which of the following is correct?



a. $AB = AC$ b. $AC = BC$
c. $AB = BC$ d. None of these

Sol. **a.** In right angled ΔBED and right angled ΔCFD

$DE = DF$ (given)

hypotenuse BD = hypotenuse CD

[$\because$ D is the mid-point of BC]

$\therefore \Delta BED \cong \Delta CFD$ [by RHS congruency]

$\Rightarrow \angle B = \angle C \quad \Rightarrow AC = AB$ [sides opposite to equal angles]

Similar Figures

The geometrical figures having the same shape, but different sizes are known as similar figures.

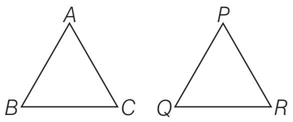
- The congruent figures are always similar, but two similar figures need not be congruent.
  e.g. Any two circles are similar. Any two rectangles are similar.

Similar Triangles

Two triangles are said to be similar to each other, if

(i) their corresponding sides are proportional.

(ii) their corresponding angles are equal.



Here, ΔABC and ΔPQR are similar triangles.

$\therefore \quad \angle A = \angle P, \ \angle B = \angle Q, \ \angle C = \angle R$

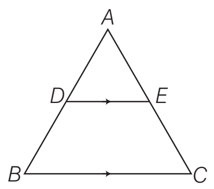
and $\quad \dfrac{AB}{PQ} = \dfrac{BC}{QR} = \dfrac{AC}{PR}$

Then, $\Delta ABC \sim \Delta PQR$

where, symbol ~ is read as, 'is similar to'.

Some Results on Similar Triangles

Theorem 1 If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct point, then it divides these sides in the same ratio. It is also called basic proportionality theorem.

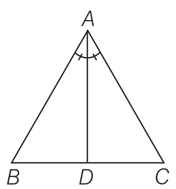


Here, $DE \parallel BC$, then $\dfrac{AD}{DB} = \dfrac{AE}{EC}$

or $\dfrac{AD}{AB} = \dfrac{AE}{AC}$

or $\dfrac{AB}{DB} = \dfrac{AC}{EC}$

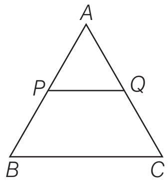
Theorem 2 The internal bisector of an angle of a triangle divides the opposite sides internally in the ratio of the sides containing the angle.



Here, AD is internal bisector of $\angle A$, then

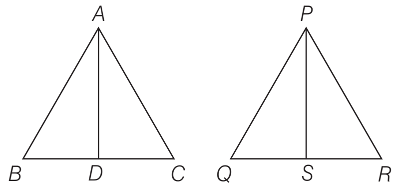
$\dfrac{AB}{AC} = \dfrac{BD}{DC}$

Theorem 3 The line joining the mid-points of any two sides of a triangle is parallel to the third side and is half of the third side.



Here, P and Q are mid-point of AB and AC. So, $PQ = \dfrac{1}{2} BC$.

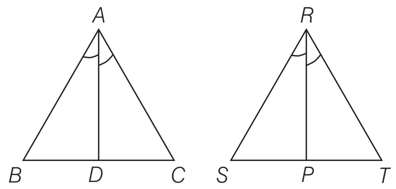
Theorem 4 If two triangles are equiangular, then the ratio of their corresponding sides is the same as the ratio of the corresponding altitudes.



Here, $\Delta ABC \sim \Delta PQR$ and AD and PS are altitude on BC and QR, respectively, then

$$\frac{BC}{QR} = \frac{AD}{PS}$$

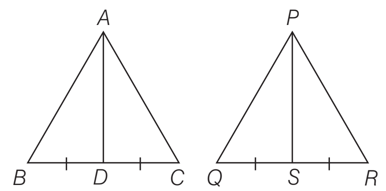
Theorem 5 If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding angle bisector segments.



Here, ΔABC and ΔRST are equiangular/similar and AD, RP are the angle bisectors of $\angle A$ and $\angle R$, then

$$\frac{BC}{ST} = \frac{AD}{RP}$$

Theorem 6 If two triangles are equiangular, then the ratio of the corresponding sides is the same as the ratio of the corresponding medians.



Here, ΔABC and ΔPQR are equiangular and AD, PS are the medians, then

$$\frac{BC}{QR} = \frac{AD}{PS}$$

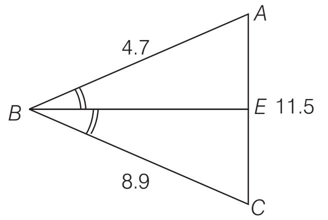
Note: If two triangles are similar, then ratio of their corresponding sides is same as ratio of their perimeters.

Criteria for Similarity of Two Triangles

1. **AAA similarity** If the corresponding angles of two triangles are equal then their corresponding sides are proportional and hence the two triangles are similar.

   Corollary AA similarity If two angles of one triangle are equal to the corresponding two angles of another triangle, then the two triangles are similar.
2. **SSS similarity** If the corresponding sides of two triangles are proportional, then their corresponding angles are equal, and hence the two triangles are similar.
3. **SAS similarity** If one angle of a triangle is equal to the corresponding angle of the other triangle and the sides including these angles are proportional, then the two triangles are similar.
4. **RHS similarity** If hypotenuse and one side of a right triangle are proportional to the hypotenuse and corresponding side of other right triangle then the two triangles are similar.

EXAMPLE 7. If $AB = 4.7$, $BC = 8.9$, $CA = 11.5$, then EA is



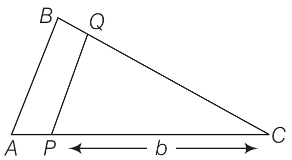
a. 6.07   b. 3.97
c. 2.37   d. None of these

Sol. b. Since, BE is the bisector of $\angle ABC$.

$$\therefore \quad \frac{AB}{BC} = \frac{AE}{EC} \Rightarrow \frac{4.7}{8.9} = \frac{AE}{11.5 - AE} \quad [\because EC = AC - AE]$$

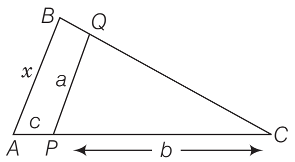
$$\Rightarrow \quad AE = 3.97$$

EXAMPLE 8. In the given triangle, AB is parallel to PQ. $AP = c$, $PC = b$, $PQ = a$, $AB = x$. What is the value of x?



a. $a + \frac{ab}{c}$ b. $a + \frac{bc}{a}$
c. $b + \frac{ca}{b}$ d. $a + \frac{ac}{b}$

Sol. d. In ΔABC and ΔPQC.



$$AB || PQ$$

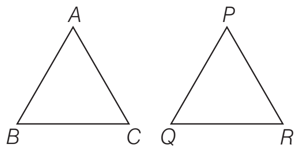
$$\therefore \quad \Delta ABC \sim \Delta PQC$$

$$\therefore \quad \frac{PC}{AC} = \frac{PQ}{AB} \Rightarrow \frac{b}{c+b} = \frac{a}{x}$$

$$\therefore \quad x = \frac{a(c+b)}{b} = \frac{ac}{b} + a$$

Areas of Similar Triangles

Theorem 1 The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

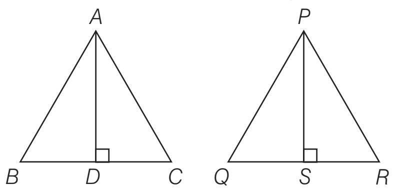


Here, $\Delta ABC \sim \Delta PQR$

$$\frac{Area\,(\Delta ABC)}{Area\,(\Delta PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{AC^2}{PR^2}$$

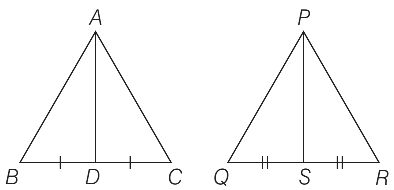
Theorem 2 The areas of two similar triangles is equal to the ratio of the squares of corresponding altitudes.

Here, $\quad \Delta ABC \sim \Delta PQR$



Then, $$\frac{Area\,(\Delta ABC)}{Area\,(\Delta PQR)} = \frac{AD^2}{PS^2}$$

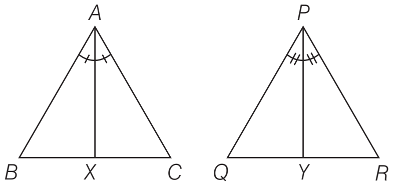
Theorem 3 The areas of two similar triangles is equal to the ratio of the squares of the corresponding medians.



Here, $\Delta ABC \sim \Delta PQR$,

then $$\frac{Area\,(\Delta ABC)}{Area\,(\Delta PQR)} = \frac{AD^2}{PS^2}$$

Theorem 4 The areas of two similar triangles is equal to the ratio of squares of the corresponding angle bisector segments.



Here, $\Delta\, ABC \sim \Delta\, PQR$

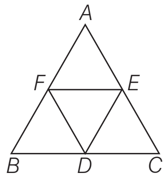
So, here $\dfrac{Area\,(\Delta\, ABC)}{Area\,(\Delta\, PQR)} = \dfrac{AX^2}{PY^2}$

Theorem 5 If the areas of two similar triangles are equal, then the triangles are congruent.

or

Equal and similar triangles are congruent.

Theorem 6 The line segments is joining the mid-points of the sides of a triangle form four triangles, each of which is similar to the original triangle.



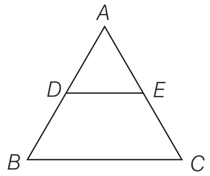
Here, D, E and F are mid-point of BC, AC and AB. Then, $\Delta\, AFE$, $\Delta\, FBD$, $\Delta\, EDC$ and $\Delta\, DEF$ is similar to $\Delta\, ABC$.

Here, also

$$\frac{Area\,(\Delta\, DEF)}{Area\,(\Delta\, ABC)} = \frac{DE^2}{AB^2} = \frac{\left(\frac{1}{2}AB\right)^2}{AB^2} = \frac{1}{4}$$

So, area $(\Delta\, DEF)$: area $(\Delta\, ABC) = 1 : 4$

EXAMPLE 9. In the figure given below, BC is parallel to DE and $DE : BC = 3 : 5$. What is the ratio of area of the ΔABC to that of ΔADE?



a. 3 : 1 b. 5 : 3 c. 9 : 2 d. 25 : 9

Sol. d. $\because$ Given, $DE : BC = 3 : 5$

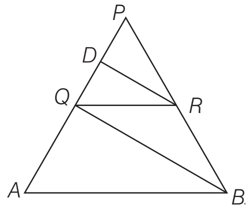
Since, $DE \parallel BC \Rightarrow \angle ADE = \angle ABC$

and $\angle AED = \angle ACB$

$\therefore \Delta ABC \sim \Delta ADE$ [by AA similarity]

$\therefore \dfrac{\text{Area of } \Delta ABC}{\text{Area of } \Delta ADE} = \left(\dfrac{BC}{DE}\right)^2 = \dfrac{25}{9}$ or, 25 : 9

EXAMPLE 10. In a given figure, QR is parallel to AB and DR is parallel to QB. What is the number of distinct pairs of similar triangles?



a. 1 b. 2 c. 3 d. 4

Sol. c. Since, QR is parallel to AB.

$\therefore \Delta PQR \sim \Delta PAB$

Also, DR is parallel to QB., $\Delta PQB \sim \Delta PDR$

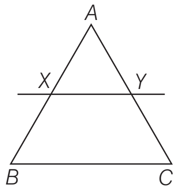
Again, $DR \parallel QB$ and $QR \parallel AB$,

$\therefore \Delta DQR \sim \Delta QAB$

EXAMPLE 11. In a triangle, a line XY is drawn parallel to BC meeting AB in X and AC in Y. The area of the $\Delta\, ABC$ is 2 times the area of the $\Delta\, AXY$. In what ratio X divides AB?

a. $1 : \sqrt{2}$ b. $\sqrt{2} : 1$ c. $(\sqrt{2}-1) : 1$ d. $1 : (\sqrt{2}-1)$

Sol. d. $\because \dfrac{\text{Area } (\Delta\, ABC)}{\text{Area } (\Delta\, AXY)} = \dfrac{AB^2}{AX^2}$



$\Rightarrow \dfrac{2 \text{ Area } (\Delta\, AXY)}{\text{Area } (\Delta\, AXY)} = \dfrac{AB^2}{AX^2} \Rightarrow \dfrac{2}{1} = \dfrac{AB^2}{AX^2}, \dfrac{AB}{AX} = \sqrt{2}$

$\Rightarrow AB = \sqrt{2}\, AX \Rightarrow AX + BX = \sqrt{2}\, AX$

$\Rightarrow BX = AX(\sqrt{2}-1)$

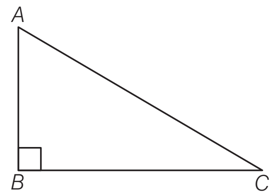
$\therefore \dfrac{AX}{BX} = \dfrac{1}{\sqrt{2}-1}$

So, X divides AB in $1 : (\sqrt{2}-1)$.

PYTHAGORAS THEOREM

In a right angled triangle, the square of the hypotenuse is equal to the sum of the square of the other two sides.

i.e. In ΔABC, if $\angle B = 90°$, then $AB^2 + BC^2 = AC^2$



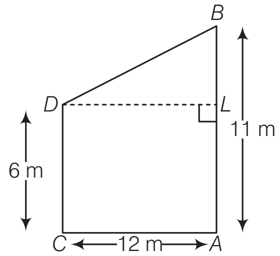
Converse of Pythagoras theorem

In a triangle, if the square of one side is equal to the sum of square of the other two sides then angle opposite to the first side is a right angle i.e. In ΔABC, if $AC^2 = AB^2 + BC^2$, then $\angle B = 90°$.

EXAMPLE 12. Two poles of heights 6m and 11m stand vertically on a plane ground at a distance of 12m from each other. The distance between their tops is

a. 9 m b. 10 m
c. 13 m d. 15 m

Sol. c. Let AB and CD be two poles such that $AB = 11\,m$, $CD = 6$ m and $AC = 12$ m.



Draw $DL \perp AB$

Then, $BL = AB - AL = AB - CD = (11-6)\,m = 5\,m$

and $DL = AC = 12\,m$

In right angled ΔBDL,

$$BD^2 = DL^2 + BL^2 = 5^2 + 12^2 = 169$$

$\Rightarrow$ $BD = \sqrt{169} = 13\,m$

PRACTICE EXERCISE

1. If the bisector of an angle of a triangle bisects the opposite side, then the triangle is
(a) equilateral (b) isosceles
(c) scalene (d) right angled triangle

2. The line segments joining the mid-points of the sides of a triangle form four triangles each of which is
(a) similar to the original triangle
(b) congruent to the original triangle
(c) an equilateral triangle (d) an isosceles triangle

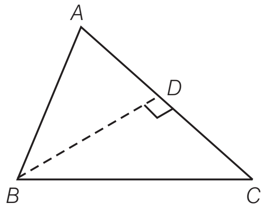
3. In a $\Delta\, ABC, BD$ and CE are perpendicular on AC and AB, respectively. If $BD = CE$, then the $\Delta\, ABC$ is
(a) equilateral (b) isosceles (c) right angled (d) scalene

4. If the length of hypotenuse of a right angled triangle is 5 cm and its area is 6 cm^2, then the length of the remaining sides are
(a) 1 cm and 3 cm (b) 3 cm and 2 cm
(c) 3 cm and 4 cm (d) 4 cm and 2 cm

5. $\Delta\, ABC$ is such that $AB = 3$ cm, $BC = 2$ cm and $AC = 2.5$ cm. $\Delta\, DEF$ is similar to $\Delta\, ABC$. If $EF = 4$ cm, then the perimeter of $\Delta\, DEF$ is
(a) 5 cm (b) 7.5 cm (c) 15 cm (d) 18 cm

6. A soldier goes to a warfield and runs in the following manner. From the starting point, he goes West 25 m, then due North 60 m, then due East 80 m, and finally due South 12 cm. The distance between the starting point and the finishing point is
(a) 177 m (b) 103 m (c) 83 m (d) 73 m

7. Let ABC be an isosceles triangle in which $AB = AC$ and $BD \perp AC$. Then, $BD^2 - CD^2$ is equal to



(a) $2DC \cdot AD$ (b) $2AD \cdot BC$ (c) $3DC \cdot AD$ (d) $\frac{1}{2} AD \cdot DC$

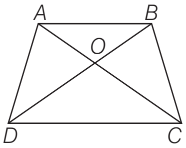
8. D and E are the points on the sides AB and AC respectively of a $\Delta\, ABC$ and $AD = 8$ cm, $DB = 12$ cm, $AE = 6$ cm and $EC = 9$ cm, then BC is equal to
(a) $\frac{2}{5} DE$ (b) $\frac{5}{2} DE$ (c) $\frac{3}{2} DE$ (d) $\frac{2}{3} DE$

9. A vertical stick 15m long casts a shadow 12m long on the ground. At the same time, a tower casts a shadow 50m long on the ground. The height of the tower is
(a) 60 m (b) 62 m (c) 62.5 m (d) 63 m

10. The areas of two similar triangles are $81\,cm^2$ and $49\,cm^2$, respectively. The ratio of their corresponding heights is
(a) 9 : 7 (b) 7 : 9 (c) 6 : 5 (d) 81 : 49

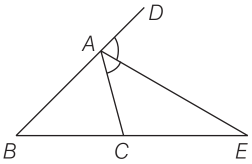
11. If D and E are points on the sides AB and AC, respectively of a $\Delta\, ABC$ such that $DE \| BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$. The value of x is
(a) 2.5 (b) 2 (c) 3 (d) 4

12. In the adjoining figure, $ABCD$ is a trapezium in which $AB || CD$ and its diagonals intersect at O. If $AO = (3x - 1), OC = (5x - 3)$, $BO = (2x + 1)$ and $OD = (6x - 5)$, then x is equal to



(a) 1 (b) 2 (c) 3 (d) 4

13. In the adjoining figure, AE is the bisector of exterior $\angle CAD$ meeting BC produced in E. If $AB = 10$ cm, $AC = 6$ cm and $BC = 12$ cm, then CE is equal to



(a) 6 cm (b) 12 cm (c) 18 cm (d) 20 cm

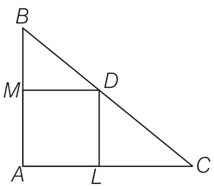
14. If D, E and F are respectively the mid-points of sides BC, AC and AB of a ΔABC. If $EF = 3$ cm, $FD = 4$ cm and $AB = 10$ cm, then DE, BC and CA, respectively will be equal to

(a) 6, 8 and 20 cm (b) $\frac{10}{3}$, 9 and 12 cm
(c) 4, 6 and 8 cm (d) 5, 6 and 8 cm

15. In $\Delta PQR, \angle Q = 3a, \angle P = a, \angle R = b$ and $3b - 5a = 30$, then the triangle is

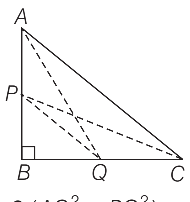
(a) scalene (b) isosceles
(c) equilateral (d) right angled

16. In ΔABC show in the figure $\angle A = 90°$. Let D be a point on BC such that $BD : DC = 1 : 3$. If DM and DL, respectively are perpendicular on AB and AC, then DM and LC are in the ratio of



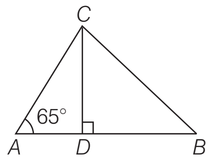
(a) 1 : 3 (b) 1 : 2 (c) 1 : 1 (d) 4 : 1

17. In a right angled ΔABC, right angled at B, if P and Q are points on the sides AB and AC respectively, then



(a) $AQ^2 + CP^2 = 2(AC^2 + PQ^2)$
(b) $2(AQ^2 + CP^2) = AC^2 + PQ^2$
(c) $AQ^2 + CP^2 = AC^2 + PQ^2$
(d) $AQ + CP = \frac{1}{2}(AC + PQ)$

18. In $\Delta ABC, \angle C = 90°$ and $CD \perp AB$, also $\angle A = 65°$, then $\angle CBA$ is equal to

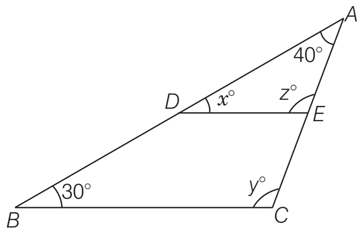


(a) 25° (b) 35° (c) 65° (d) 40°

19. The angles of a triangle are in the ratio 2 : 3 : 4. The angles of triangle are, respectively

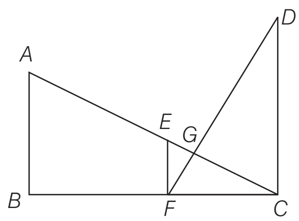
(a) 30°, 60°, 90° (b) 40°, 60°, 80°
(c) 60°, 40°, 80° (d) 20°, 60°, 80°

20. In figure, D and E are points on sides AB, AC of ΔABC such that $DE || BC$. If $\angle B = 30°$ and $\angle A = 40°$, then x, y and z are, respectively



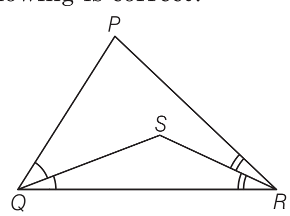
(a) 30°, 110°, 110° (b) 30°, 105°, 105°
(c) 30°, 85°, 85° (d) 30°, 95°, 95°

21. In figure, AB, EF and CD are parallel lines. Given that $GE = 5$ cm, $GC = 10$ cm and $DC = 18$ cm, then EF is equal to



(a) 11 cm (b) 5 cm (c) 6 cm (d) 9 cm

22. In the given figure, $PQ > PR$. QS and RS are the bisectors of $\angle Q$ and $\angle R$ respectively, then which of the following is correct?



(a) $SQ < SR$ (b) $SQ = SR$
(c) $SQ > SR$ (d) None of these

23. In a ΔABC, $AB = AC$ and $60° < A < 90°$. If $AB = c$, $AC = b$ and $BC = a$, then which of the following is correct?

(a) $b < a < 2b$ (b) $c/3 < a < 3c$
(c) $b < a < b\sqrt{3}$ (d) $c < a < c\sqrt{2}$

24. In ΔABC, $AD \perp BC$, then

(a) $AB^2 - BD^2 = AC^2 - CD^2$ (b) $AB^2 + BD^2 = AC^2 - CD^2$
(c) $AB^2 + BD^2 = AC^2 + CD^2$ (d) $AB^2 + AC^2 = BD^2 + DC^2$

25. ΔABC is a right angled at C and P is the length of the perpendicular from C to AB. If $BC = a$, $AC = b$, $AB = c$, then

(a) $\frac{a}{b} = \frac{p}{c}$ (b) $pc = ab$ (c) $\frac{1}{a} + \frac{1}{b} = \frac{1}{ab}$ (d) None of these

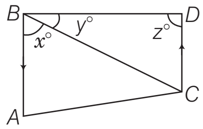
26. ABC is a triangle and the perpendicular drawn from A meets BC in D. If $AD^2 = BD \cdot DC$, then which one of the following is correct?

(a) ABC must be an obtuse angled triangle
(b) ABC must be an acute angled triangle
(c) Either $\angle B \geq 45°$ or $\angle C \geq 45°$ (d) $BC^2 = AB^2 + AC^2$

27. If Δ is the area of a right angled triangle and b is one of the sides containing the right angle, then what is the length of the altitude on the hypotenuse?

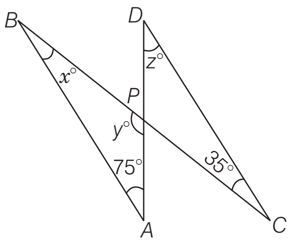
(a) $\frac{2\Delta b}{\sqrt{b^4 + 4\Delta^2}}$ (b) $\frac{2\Delta^2 b}{\sqrt{b^4 + 4\Delta^2}}$
(c) $\frac{2\Delta b^2}{\sqrt{b^4 + 4\Delta^2}}$ (d) $\frac{2\Delta^2 b^2}{\sqrt{b^4 + \Delta^2}}$

28. In figure, $AB || CD$. If $x = \frac{4}{3}y$ and $y = \frac{3}{8}z$, then the value of x is



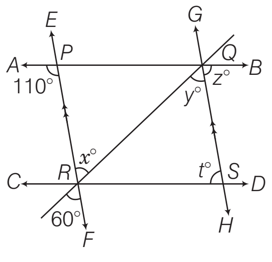
(a) 48° (b) 96° (c) 36° (d) None of these

29. In the given figure, $AB || CD$, then the values of x, y and z are, respectively



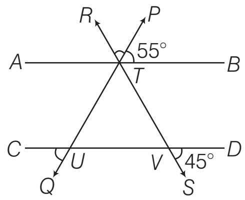
(a) 75°, 35°, 80° (b) 70°, 35°, 60°
(c) 35°, 70°, 75° (d) 70°, 35°, 80°

30. In the given figure, $AB || CD$, and $EF || GH$. The values of x, y, z and t are respectively



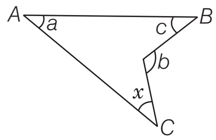
(a) 60°, 75°, 75°, 60° (b) 50°, 75°, 75°, 65°
(c) 60°, 70°, 60°, 70° (d) 60°, 60°, 70°, 70°

31. In the given figure, $AB || CD$, $\angle PTB = 55°$ and $\angle DVS = 45°$, then what is the sum of the measures of $\angle CUQ$ and $\angle RTP$?



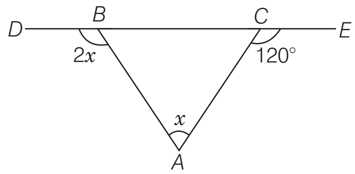
(a) 180° (b) 135° (c) 110° (d) 100°

32. What is the value of x in the figure given below?



(a) $b - a - c$ (b) $b - a + c$ (c) $b + a - c$ (d) $\pi - (a + b + c)$

33. In the given figure, what is the value of x?

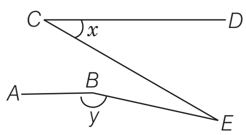


(a) 30° (b) 40° (c) 45° (d) 60°

34. PQR is a triangle right angled at Q. If X and Y are the mid-points of the sides PQ and QR respectively, then which one of the following is not correct?

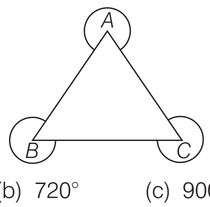
(a) $RX^2 + PY^2 = 5XY^2$ (b) $RX^2 + PY^2 = XY^2 + PR^2$
(c) $4(RX^2 + PY^2) = 5PR^2$ (d) $RX^2 + PY^2 = 3(PQ^2 + QR^2)$

35. In the given figure, AB is parallel to CD. If $\angle DCE = x$ and $\angle ABE = y$, then what is $\angle CEB$ equal to?



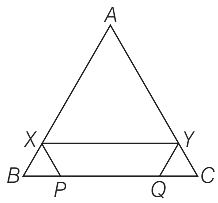
(a) $y - x$ (b) $(x + y)/2$ (c) $x + y - (\pi/2)$ (d) $x + y - \pi$

36. In the figure given below, what is the sum of the angles formed around A, B, C except the angles of the ΔABC?



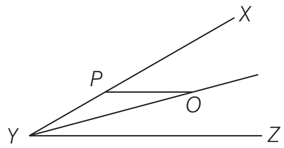
(a) 360° (b) 720° (c) 900° (d) 1000°

37. In the given figure, ABC is an equilateral triangle of side length 30 cm. XY is parallel to BC, XP is parallel to AC and YQ is parallel to AB. If $(XY + XP + YQ)$ is 40 cm, then what is PQ equal to?



(a) 5 cm (b) 12 cm
(c) 15 cm (d) None of these

38. O is any point on the bisector of the acute angle $\angle XYZ$. The line OP is parallel to ZY. Then, $\Delta\, YPO$ is



(a) scalene
(b) isosceles but not right angled
(c) equilateral
(d) right angled and isosceles

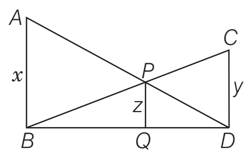
39. The median BD of the $\Delta\, ABC$ meets AC at D. If $BD = \frac{1}{2} AC$, then which one of the following is correct?

(a) $\angle ACB = 1$ right angle (b) $\angle BAC = 1$ right angle
(c) $\angle ABC = 1$ right angle (d) None of these

40. ABC is a triangle, X is a point outside the $\Delta\, ABC$ such that $CD = CX$, where D is the point of intersection of BC and AX and $\angle BAX = \angle XAC$. Which one of the following is correct?

(a) ΔABD and ΔACX are similar (b) $\angle ABD < \angle ACD$
(c) $AC = CX$ (d) $\angle ADB > \angle DXC$

41. In the figure given, $\angle ABD = \angle PQD = \angle CDQ = \frac{\pi}{2}$. If $AB = x$, $PQ = z$ and $CD = y$, then which one of the following is correct?



(a) $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$

(b) $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$

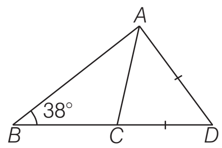
(c) $\frac{1}{z} + \frac{1}{y} = \frac{1}{x}$

(d) $\frac{1}{x} + \frac{1}{y} = \frac{2}{z}$

42. $\Delta\, PQR$ is right angled at Q, $PR = 5$ cm and $QR = 4$ cm. If the lengths of sides of another $\Delta\, ABC$ are 3 cm, 4 cm and 5 cm, then which one of the following is correct?

(a) Area of $\Delta\, PQR$ is double that of $\Delta\, ABC$
(b) Area of $\Delta\, ABC$ is double that of $\Delta\, PQR$
(c) $\angle B = \frac{\angle Q}{2}$
(d) Both triangles are congruent

43. In the figure, $\angle B = 38°$, $AC = BC$ and $AD = CD$. What is the value $\angle D$?



(a) 26° (b) 28°
(c) 38° (d) 52°

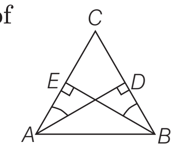
44. Consider the following statements in respect of any triangle.

I. The three medians of a triangle divide it into six triangles of equal area.
II. The perimeter of a triangle is greater than the sum of the lengths of its three medians.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

45. Consider the following in respect of the given figure :



I. $\Delta\, DAC \sim \Delta\, EBC$
II. $CA/CB = CD/CE$
III. $AD/BE = CD/CE$

Which of the statement(s) given above is/are correct?

(a) II and III (b) I and II (c) I and III (d) All of these

46. Consider the following statements :

I. Let PQR be a triangle in which $PQ = 3$ cm, $QR = 4$ cm and $RP = 5$ cm. If D is a point in the plane of the $\Delta\, PQR$ such that D is either outside it or inside it, then
$$DP + DQ + DR > 6 \text{ cm}$$
II. PQR is a right angled triangle.

Which one of the following is correct in respect of the above two statements?

(a) Both statements I and II are individually true and statement II is the correct explanation of statement I
(b) Both statements I and II are individually true but statement II is not the correct explanation of statement I
(c) Statement I is true and statement II is false
(d) Statement I is false and statement II is true

47. I. Let LMN be a triangle. Let P, Q be the mid-points of the sides LM, LN, respectively. If $PQ^2 = MP^2 + NQ^2$, then LMN is a right angled triangle at L.

II. If in a $\Delta\, ABC$, $AB^2 > BC^2 + CA^2$, then $\angle\, ACB$ is obtuse.

Which of the following is correct in the light of the above statements?

(a) Both statements I and II are individually true and statement II is the correct explanation of statement I
(b) Both statements I and II are individually true but statement II is not the correct explanation of statement I
(c) Statement I is true and statement II is false
(d) Statement I is false and statement II is true

PREVIOUS YEARS' QUESTIONS

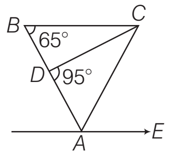
48. In a ΔABC, $\frac{1}{2}\angle A+\frac{1}{3}\angle C+\frac{1}{2}\angle B = 80°$, then what is the value of $\angle C$? **2012 II**

(a) 35° (b) 40° (c) 60° (d) 70°

49. In a ΔABC, side AB is extended beyond B, side BC beyond C and side CA beyond A. What is the sum of the three exterior angles? **2012 II**

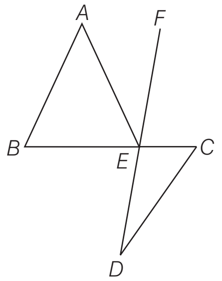
(a) 270° (b) 305° (c) 360° (d) 540°

50. In the given figure, ABC is a triangle, BC is parallel to AE. If $BC = AC$, then what is the value of $\angle CAE$? **2012 II**



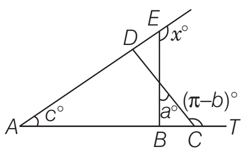
(a) 20° (b) 30° (c) 40° (d) 50°

51. In the given figure, AB is parallel to CD. $\angle ABC = 65°$, $\angle CDE = 15°$ and $AB = AE$. What is the value of $\angle AEF$? **2012 II**



(a) 30° (b) 35° (c) 40° (d) 45°

52. The angles $x°$, $a°$, $c°$ and $(\pi - b)°$ are indicated in the figure given below : **2012 II**



Which one of the following is correct ? **2012 II**

(a) $x° = a° + c° - b°$ (b) $x° = b° - a° - c°$
(c) $x° = a° + b° + c°$ (d) $x° = a° - b° - c°$

53. Consider the following statements:

I. If G is the centroid of ΔABC, then $GA = GB = GC$.

II. If H is the orthocentre of ΔABC, then $HA = HB = HC$.

Which of the statement(s) given above is are correct? **2013 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

54. I. Let the side DE of a ΔDEF be divided at S, so that $\frac{DS}{DE} = \frac{1}{\sqrt{2}}$. If a line through S parallel to EF meets DF at T, then the area of ΔDEF is twice the area of the ΔDST.

II. The areas of the similar triangles are proportional to the squares of the corresponding sides.

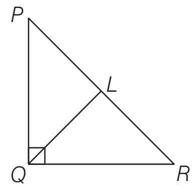
Which one of the following is correct in respect of the above statements? **2013 II**

(a) Both statements I and II are true and statement II is the correct explanation of statement I
(b) Both statements I and II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement II is true but statement I is false

55. Let ABC be a triangle with AB = 3 cm and AC = 5 cm. If AD is a median drawn from the vertex A to the side BC, then which one of the following is correct? **2013 II**

(a) AD is always greater than 4 cm but less than 5 cm
(b) AD is always greater than 5 cm
(c) AD is always less than 4 cm
(d) None of the above

56. In the figure given above, $\angle PQR = 90°$ and QL is a median, PQ = 5 cm and QR = 12 cm. Then, QL is equal to **2013 II**



(a) 5 cm (b) 5.5 cm (c) 6 cm (d) 6.5 cm

57. In a ΔABC, $\angle BCA = 90°$ and CD is perendicular to AB. If AD = 4 cm and BD = 9 cm, then the value of DC will be **2013 II**

(a) $\sqrt{18}$ cm (b) $\sqrt{20}$ cm (c) $\sqrt{65}$ cm (d) 6 cm

58. ABC is a triangle, where $BC = 2AB$, $\angle C = 30°$ and $\angle A = 90°$. The magnitude of the side AC is **2013 II**

(a) $\frac{2BC}{3}$ (b) $\frac{3BC}{4}$ (c) $\frac{BC}{\sqrt{3}}$ (d) $\frac{\sqrt{3}\,BC}{2}$

59. Let ABC be an equilateral triangle. If the side BC is produced to the point D so that $BC = 2\,CD$, then AD^2 is equal to **2013 II**

(a) $3\,CD^2$ (b) $4\,CD^2$ (c) $5\,CD^2$ (d) $7\,CD^2$

60. ABC is a right angled triangle such that $AB = a - b$, $BC = a$ and $CA = a + b$. D is a point on BC such that $BD = AB$. The ratio of $BD : DC$ for any value of a and b is given by **2013 II**

(a) 3 : 2 (b) 4 : 3 (c) 5 : 4 (d) 3 : 1

61. The side BC of a ΔABC is produced to D, bisectors of the $\angle ABC$ and $\angle ACD$ meet at P. If $\angle BPC = x°$ and $\angle BAC = y°$, then which one of the following option is correct? **2013 II**

(a) $x° = y°$ (b) $x° + y° = 90°$
(c) $x° + y° = 180°$ (d) $2x° = y°$

62. The side AC of a ΔABC is produced to D such that $BC = CD$. If $\angle ACB$ is 70°, then what is $\angle ADB$ equal to? **2013 II**

(a) 35° (b) 45° (c) 70° (d) 110°

63. The heights of two trees are x and y, where $x > y$. The tops of the trees are at a distance z a part. If s is the shortest distance between the trees, then what is s^2 equal to? **2013 II**

(a) $x^2 + y^2 - z^2 - 2xy$ (b) $x^2 + y^2 - z^2$
(c) $x^2 + y^2 + z^2 - 2xy$ (d) $z^2 - x^2 - y^2 + 2xy$

64. In a ΔABC, $\angle B = 90°$ and $\angle C = 2\angle A$, then what is AB^2 equal to? **2013 II**

(a) $2BC^2$ (b) $3BC^2$ (c) $4BC^2$ (d) $5BC^2$

65. PQR is an equilateral triangle. O is the point of intersection of altitudes PL, QM and RN. If $OP = 8$ cm, then what is the perimeter of the ΔPQR? **2013 II**

(a) $8\sqrt{3}$ cm (b) $12\sqrt{3}$ cm (c) $16\sqrt{3}$ cm (d) $24\sqrt{3}$ cm

66. ΔDEF is formed by joining the mid-points of the sides of ΔABC. Similarly, a ΔPQR is formed by joining the mid-points of the sides of the ΔDEF. If the sides of the ΔPQR are of lengths 1, 2 and 3 units, what is the perimeter of the ΔABC? **2013 II**

(a) 18 units (b) 24 units
(c) 48 units (d) Cannot be determined

67. E is the mid-point of the median AD of a ΔABC. If BE produced meets the side AC at F, then CF is equal to **2013 II**

(a) $AC/3$ (b) $2AC/3$ (c) $AC/2$ (d) None of these

68. If AD is the internal angular of ΔABC with $AB = 3$ cm and $AC = 1$ cm, then what is $BD : BC$ equal to? **2014 I**

(a) 1 : 3 (b) 1 : 4 (c) 2 : 3 (d) 3 : 4

69. In a ΔABC, AD is perpendicular to BC and BE is perpendicular to AC. Which of the following is correct? **2014 I**

(a) $CE \times CB = CA \times CD$ (b) $CE \times CA = CD \times CB$
(c) $AD \times BD = AE \times BE$ (d) $AB \times AC = AD \times BE$

70. The sides of a triangle are in geometric progression with common ratio $r < 1$. If the triangle is a right angled triangle, the square of common ratio is given by **2014 I**

(a) $\frac{\sqrt{5}+1}{2}$ (b) $\frac{\sqrt{5}-1}{2}$ (c) $\frac{\sqrt{3}+1}{2}$ (d) $\frac{\sqrt{3}-1}{2}$

71. If triangles ABC and DEF are similar such that $2AB = DE$ and $BC = 8$ cm, then what is EF equal to? **2014 I**

(a) 16 cm (b) 12 cm (c) 10 cm (d) 8 cm

72. The sides of a right angled triangle are equal to three consecutive numbers expressed in centimeters. What can be the area of such a triangle? **2014 I**

(a) 6 cm^2 (b) 8 cm^2 (c) 10 cm^2 (d) 12 cm^2

73. The three sides of a triangle are 15, 25, x units. Which one of the following is correct? **2014 I**

(a) $10 < x < 40$ (b) $10 \le x \le 40$
(c) $10 \le x < 40$ (d) $10 < x \le 40$

74. In a ΔABC, if $\angle B = 2$, $\angle C = 2\angle A$. Then, what is the ratio of AC to AB? **2014 II**

(a) $\sqrt{2} : 1$ (b) $\sqrt{3} : 1$ (c) $1 : 1$ (d) $1 : \sqrt{2}$

75. Three straight lines are drawn through the three vertices of a ΔABC, the line through each vertex being parallel to the opposite side. The ΔDEF is bounded by these parallel lines. **2014 II**

Consider the following statements in respect of the ΔDEF.

I. Each side of ΔDEF is double the side of ΔABC to which it is parallel.
II. Area of ΔDEF is four times the area of ΔABC.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

76. In a ΔABC, AD is the median through A and E is the mid-point of AD and BE produced meets AC at F. Then, AF is equal to **2014 II**

(a) $AC/5$ (b) $AC/4$ (c) $AC/3$ (d) $AC/2$

77. The point O is equidistant from the three sides of a ΔABC.
Consider the following statements: **2015 II**

I. $\angle OAC + \angle OCB + \angle OBA = 90°$
II. $\angle BOC = 2\angle BAC$
III. The perpendiculars drawn from any point on OA to AB and AC are always equal

Which of the above statements are correct?

(a) I and II (b) II and III (c) I and III (d) All of these

78. An equilateral ΔBOC is drawn inside a square $ABCD$. If $\angle AOD = 2\theta$, what is $\tan\theta$ equal to? **2015 II**

(a) $2 - \sqrt{3}$ (b) $1 + \sqrt{2}$ (c) $4 - \sqrt{3}$ (d) $2 + \sqrt{3}$

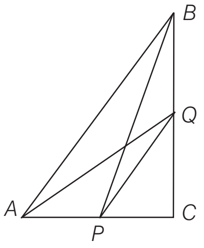
79. In a ΔPQR, point X is on PQ and point Y is on PR such that $XP = 1.5$ units, $XQ = 6$ units, $PY = 2$ units and $YR = 8$ units. Which of the following are correct?

I. $QR = 5XY$ II. QR is parallel to XY.
III. ΔPYX is similar to ΔPRQ.

Select the correct answer using the codes given below. **2016 I**

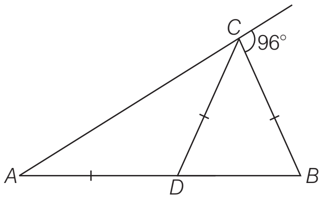
(a) I and II (b) II and III
(c) I and III (d) All of these

80. ABC is a triangle right angled at C as shown below. Which one of the following is correct? **2016 I**



(a) $AQ^2 + AB^2 = BP^2 + PQ^2$
(b) $AQ^2 + PQ^2 = AB^2 + BP^2$
(c) $AQ^2 + BP^2 = AB^2 + PQ^2$
(d) $AQ^2 + AP^2 = BK^2 + KQ^2$

81. In the given figure, $AD = CD = BC$. What is the value of $\angle CDB$? **2016 I**



(a) 32° (b) 64°
(c) 78°
(d) Cannot be determined due to insufficient data

82. ABC is an equilateral triangle and X, Y and Z are the points on BC, CA and AB respectively, such that $BX = CY = AZ$. Which of the following is/are correct? **2016 I**

I. XYZ is an equilateral triangle.
II. ΔXYZ is similar to ΔABC.

Select the correct answer using the codes given below.

(a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

83. Let ABC and $A'B'C'$ be two triangles in which $AB > A'B'$, $BC > B'C'$ and $CA > C'A'$. Let D, E and F be the mid-points of the sides BC, CA and AB, respectively. Let D', E' and F' be the mid-points of the sides $B'C'$, $C'A'$ and $A'B'$, respectively.

Consider the following statements **2016 I**

I. $AD > A'D'$, $BE > B'E'$ and $CF > C'F'$ are always true.

II. $\dfrac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \dfrac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2}$

Which one of the following is correct in respect of the above statements?

(a) Both statement I and statement II are true and statement II is the correct explanation of statement I
(b) Both statement I and statement II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

84. ABC and DEF are similar triangles. If the ratio of side AB to side DE is $(\sqrt{2}+1):\sqrt{3}$, then the ratio of area of ΔABC to that of ΔDEF is **2016 I**

(a) $(3-2\sqrt{2}):3$ (b) $(9-6\sqrt{2}):2$
(c) $1:(9-6\sqrt{2})$ (d) $(3+2\sqrt{2}):2$

85. Let ΔABC and ΔDEF be such that $\angle ABC = \angle DEF$, $\angle ACB = \angle DFE$ and $\angle BAC = \angle EDF$. Let L be the mid-point of BC and M be the mid-point of EF.

Consider the following statements **2016 I**

I. ΔABL and ΔDEM are similar.
II. ΔALC is congruent to ΔDMF even, if $AC \neq DF$.

Which one of the following is correct in respect of the above statements?

(a) Both statement I and statement II are true and statement II is the correct explanation of statement I
(b) Both statement I and statement II are true but statement II is not the correct explanation of statement I
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

86. ABC is a triangle in which D is the mid-point of BC and E is the mid-point of AD.

I. The area of ΔABC is equal to four times the area of ΔBED.
II. The area of ΔADC is twice the area of ΔBED.

Select the correct answer using the codes given below. **2016 I**

(a) Only I (b) Only II (c) Both I and II (d) Neither I nor II

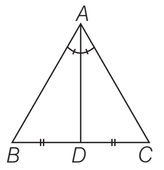
ANSWERS

1	b	2	a	3	b	4	c	5	c	6	d	7	a	8	b	9	c	10	a
11	d	12	b	13	c	14	d	15	d	16	a	17	c	18	a	19	b	20	a
21	d	22	c	23	d	24	a	25	b	26	d	27	a	28	a	29	c	30	d
31	b	32	a	33	d	34	d	35	d	36	c	37	d	38	b	39	c	40	a
41	a	42	d	43	b	44	c	45	d	46	b	47	b	48	c	49	c	50	d
51	b	52	c	53	d	54	a	55	c	56	d	57	d	58	d	59	d	60	d
61	d	62	a	63	d	64	b	65	d	66	b	67	b	68	d	69	b	70	b
71	a	72	a	73	a	74	a	75	c	76	c	77	c	78	d	79	d	80	c
81	b	82	c	83	b	84	c	85	c	86	c								

HINTS AND SOLUTIONS

1. *(b)* Let ABC is any triangle and AD is the bisector of angle A.

Also, $BD = DC$ (given)



$\because$ AD is the internal bisector of $\angle A$.

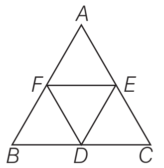
$\therefore \quad \dfrac{AB}{AC} = \dfrac{BD}{DC}$

$\Rightarrow \quad \dfrac{AB}{AC} = \dfrac{BD}{DC} = 1 \quad [\because BD = DC \text{ given}]$

$\Rightarrow \quad AB = AC$

Hence, triangle is an isosceles triangle.

2. *(a)* The line segments joining the mid-points of the sides of a triangle form four triangles each of which is similar to the original triangle.



Here, $\Delta\, BDF \sim \Delta\, ABC$

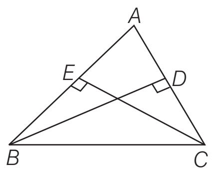
Also, $\Delta\, DEC, \Delta\, DEF$

$\Delta\, AFE \sim \Delta\, ABC$

3. *(b)* As $BD = EC$,

$\angle AEC = \angle BDA = 90°$each

$[\because BD \perp AC \text{ and } CE \perp AB]$



Also, $\angle A = \angle A$ [common]

$\therefore \quad \Delta BDA \cong \Delta\, AEC$

[by AAS congruency]

$\Rightarrow \quad AB = AC$ [by cpct]

So, triangle is an isosceles triangle.

4. *(c)* Let the other side be b and p.

$\therefore \quad \dfrac{1}{2} b \times p = 6 \quad \Rightarrow \quad b \times p = 12$

$\Rightarrow \quad b = \dfrac{12}{p}$

Also, by pythagoras theorem

$$h^2 = b^2 + p^2$$

$$5^2 = \left(\frac{12}{p}\right)^2 + p^2$$

$\Rightarrow \quad 25 = \dfrac{144}{p^2} + p^2$

$25p^2 = 144 + p^4$

$\Rightarrow \quad p^4 - 25p^2 + 144 = 0$

$\Rightarrow \quad p^4 - 16p^2 - 9p^2 + 144 = 0$

$\Rightarrow \quad p^2(p^2 - 16) - 9(p^2 - 16) = 0$

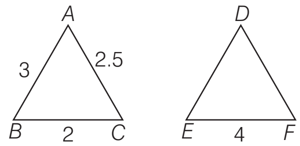
$\Rightarrow \quad (p^2 - 9)(p^2 - 16) = 0$

$\Rightarrow \quad p = 3 \text{ or } p = 4$

Hence, other sides are 3 cm and 4 cm.

5. *(c)* As $\Delta\, ABC \sim \Delta\, DEF$

$\Rightarrow \quad \dfrac{AB}{DE} = \dfrac{AC}{DF} = \dfrac{BC}{EF}$



$\because \quad \dfrac{BC}{EF} = \dfrac{2}{4} = \dfrac{1}{2}$

$\therefore \quad DE = 2AB = 2 \times 3 = 6$ cm

and $\quad DF = 2 \times AC = 2 \times 2.5 = 5$ cm

$\therefore$ Perimeter of $\Delta\, DEF = (6 + 5 + 4)$

$= 15$ cm

Shortcut Method

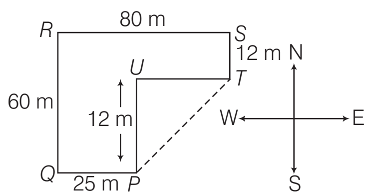
$\dfrac{\text{Perimeter of } \Delta ABC}{\text{Perimeter of } \Delta DEF}$

$=$ Ratio of corresponding sides

$\therefore \quad \dfrac{(3 + 2 + 2.5)}{\text{Perimeter of } \Delta DEF} = \dfrac{1}{2}$

$\therefore$Perimeter of $\Delta DEF = 2\,(7.5) = 15$ cm.

6. *(d)* Let P be the starting point of his run, then PT is the distance between the starting and the finishing point.



$\therefore PU = RQ - ST = 60 - 12 = 48$ m

and $\quad UT = RS - QP = 80 - 25 = 55$ m

$\therefore$ In $\Delta\, PUT,\ PT^2 = (PU)^2 + (TU)^2$

$\therefore PT = \sqrt{(48)^2 + (55)^2} = \sqrt{2304 + 3025}$

$= \sqrt{5329} = 73$ m

7. *(a)* As ADB is a right angled triangle.

So, $\ AB^2 = AD^2 + BD^2$

$\Rightarrow AC^2 = AD^2 + BD^2 \quad [\because AB = AC]$

$\Rightarrow (AD + DC)^2 = AD^2 + BD^2$

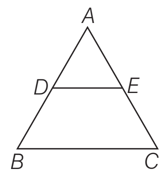
$\Rightarrow \quad AD^2 + DC^2 + 2AD \cdot DC$

$= AD^2 + BD^2$

$\Rightarrow \quad BD^2 - CD^2 = 2DC \cdot AD$

8. *(b)* As in $\Delta\, ADE$ and $\Delta\, ABC$,

$\dfrac{AD}{AB} = \dfrac{8}{20} = \dfrac{2}{5}$ and $\dfrac{AE}{AC} = \dfrac{6}{15} = \dfrac{2}{5}$



$\therefore \quad \dfrac{AD}{AB} = \dfrac{AE}{AC}$

and $\quad \angle A = \angle A$ [common]

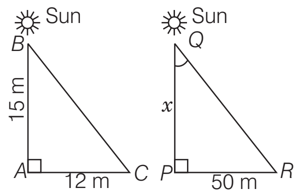
$\therefore \quad \Delta\, ADE \sim \Delta\, ABC$

[by SAS similarity]

$\therefore \quad \dfrac{DE}{BC} = \dfrac{AD}{AB} \quad \Rightarrow \quad \dfrac{DE}{BC} = \dfrac{2}{5}$

$\Rightarrow \quad BC = \dfrac{5}{2} DE.$

9. *(c)* Let AB be a vertical stick and AC be its shadow. Also, let PQ be a tower having shadow PR.



$\angle A = \angle P$

$\angle B = \angle Q$

[$\because$ Sun with tower and stick forms same angle]

As, $\Delta\, ABC \sim \Delta\, PQR$

[$\therefore$ By AA similarity]

$\therefore \quad \dfrac{AB}{PQ} = \dfrac{AC}{PR} \quad \Rightarrow \quad \dfrac{15}{x} = \dfrac{12}{50}$

$\Rightarrow \quad x = \dfrac{15 \times 50}{12} = 62.5$ m

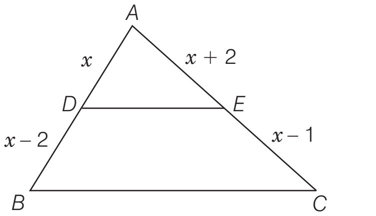
Hence, the height of the tower is 62.5 m.

10. *(a)* Let the ratio of their corresponding height be $h_1 : h_2$.

But the ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding heights.

$\therefore \quad \dfrac{h_1^2}{h_2^2} = \dfrac{81}{49} \quad \Rightarrow \quad h_1 : h_2 = 9 : 7$

11. *(d)* As $DE \,||\, BC$, by basic proportionality theorem,



$$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{x}{x-2} = \frac{x+2}{x-1}$$

$\Rightarrow \; x(x-1) = (x+2)(x-2)$

$\Rightarrow \; x^2 - x = x^2 - 2x + 2x - 4$

$\Rightarrow \; x = 4$

Hence, the value of x is 4.

12. *(b)* As $AB \,||\, CD$ and the diagonals of a trapezium divide each other proportionally.

So, $\dfrac{AO}{OC} = \dfrac{BO}{OD}$

$\Rightarrow \; \dfrac{3x-1}{5x-3} = \dfrac{2x+1}{6x-5}$

$(3x-1)(6x-5) = (5x-3)(2x+1)$

$\Rightarrow \; 18x^2 - 15x - 6x + 5$

$= 10x^2 + 5x - 6x - 3$

$\Rightarrow \; 8x^2 - 20x + 8 = 0$

$\Rightarrow \; 4x^2 - 10x + 4 = 0$

$\Rightarrow \; 2(2x^2 - 5x + 2) = 0$

$\Rightarrow \; 2(2x^2 - 4x - x + 2) = 0$

$\Rightarrow \; 2[2x(x-2) - 1(x-2)] = 0$

$\Rightarrow \; 2(2x-1)(x-2) = 0$

But as $x = \dfrac{1}{2}$ will make OC negative.

$\therefore \; x = 2$

13. *(c)* **Exterior angle bisector theorem** The exterior bisector of an angle of a triangle divides the opposite side externally in the ratio of the sides containing the angle.

$\because \dfrac{BE}{CE} = \dfrac{AB}{AC}$ as AE is an exterior angle bisector.

Let $CE = x$, $BE = BC + EC = 12 + x$

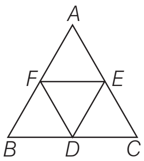
$\Rightarrow \; \dfrac{12+x}{x} = \dfrac{10}{6}$

$\Rightarrow \; (12 + x)\,6 = 10x$

$\Rightarrow \; 72 + 6x = 10x \quad \Rightarrow 4x = 72$

$\Rightarrow \; x = 18$ cm

14. *(d)* As the line joining the mid-points of any two sides or a triangle is parallel to the third side and is half of the third side.



$\therefore \; DE = \dfrac{1}{2} AB = \dfrac{1}{2} \times 10 = 5$ cm

$EF = \dfrac{1}{2} BC \Rightarrow BC = 2EF$

$= 2 \times 3 = 6$ cm

$DF = \dfrac{1}{2} AC \Rightarrow AC = 2 \times DF$

$= 2 \times 4 = 8$ cm

15. *(d)* In $\Delta PQR, \angle P + \angle Q + \angle R = 180°$

$\Rightarrow \; a + 3a + b = 180°$

$\Rightarrow \; 4a + b = 180° \quad \ldots$(i)

Given, $-5a + 3b = 30° \quad \ldots$(ii)

Solving Eqs.(i) and (ii), we get $a = 30°$ and $b = 60°$

$\therefore \angle P = 30°, \angle Q = 90°$ and $\angle R = 60°$

So, $\Delta\, PQR$ is right angled triangle.

16. *(a)* Consider ΔBMD and ΔDLC,

As $\angle BMD = \angle DLC = 90°$ [each]

Also, $\angle BDM = \angle DCL$ corresponding angle

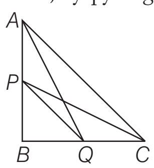
$\therefore \Delta BMD \sim \Delta DLC$ [by AA similarity]

$\therefore \; \dfrac{BD}{DC} = \dfrac{DM}{LC} = \dfrac{BM}{DL}$

$\Rightarrow \; \dfrac{BD}{DC} = \dfrac{DM}{LC} = \dfrac{1}{3}$

$\therefore \; DM : LC = 1 : 3$

17. *(c)* In ΔABC, by pythagoras theorem,



$AC^2 = AB^2 + BC^2 \quad \ldots$(i)

And in ΔPBQ,

$PQ^2 = PB^2 + BQ^2 \quad \ldots$(ii)

Adding Eqs. (i) and (ii),

$AC^2 + PQ^2 = (AB^2 + BC^2) + PB^2 + BQ^2$

$= (AB^2 + BQ^2) + (PB^2 + BC^2)$

$AC^2 + PQ^2 = AQ^2 + CP^2$

18. *(a)* In ΔBCA, $\angle CAB = 65°$ and $\angle ACB = 90°$ [$\because \angle C = 90°$ given]

$\angle CBA = 180° - (\angle BCA + \angle CAB)$

[by angle sum property of a triangle]

$= 180° - (65° + 90°)$

$= 180° - 155° = 25°$

19. *(b)* Let the angles of a triangle be $2x, 3x, 4x$, then

$2x + 3x + 4x = 180°$

[by angle sum property of a triangle]

$9x = 180° \Rightarrow x = 20°$

So, angles are $2x = 40°$, $3x = 60°$, $4x = 80°$.

20. *(a)* In ΔABC, $\angle A + \angle B + y = 180°$

$y = 180° - (40 + 30)° = 110°$

$\angle ADE = \angle ABC$ [corresponding angle]

$x = 30°$

Similarly, $y = z = 110°$

21. *(d)* In ΔGEF and ΔGCD, we have

$\angle EFG = \angle GDC$ [alternate angles]

$\angle EGF = \angle CGD$ [vertically opposite angles]

$\Delta GEF \sim \Delta GCD$ [by AA similarity]

$\therefore \; \dfrac{GE}{CG} = \dfrac{EF}{DC}$

$\Rightarrow \; \dfrac{5}{10} = \dfrac{EF}{18}$

$\Rightarrow \; EF = \dfrac{5 \times 18}{10} = 9$ cm

22. *(c)* Given, In ΔPQR, $PQ > PR$

$\Rightarrow \; \angle PRQ > \angle PQR$

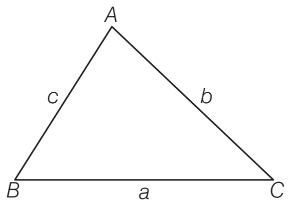
[angle opp to larger side is greater]

$\Rightarrow \; \dfrac{1}{2}\angle PRQ > \dfrac{1}{2}\angle PQR$

$\Rightarrow \; \angle SRQ > \angle SQR \Rightarrow SQ > SR$

[side opp to greater angle is larger]

23. *(d)* Given, $AB = AC$



$\Rightarrow \; \angle ABC = \angle ACB$

If $\angle A = 60°$, then

ΔABC is an equilateral triangle $a = b = c$

If $\angle A = 90°$, then

ΔABC is right angled triangle

$a^2 = b^2 + c^2 = c^2 + c^2 = 2c^2$ [$\because b = c$]

$\Rightarrow a = \sqrt{2}\,c$

$\therefore$ If $60° < A < 90°$, then $c < a < c\sqrt{2}$

24. *(a)* Here, in ΔABC

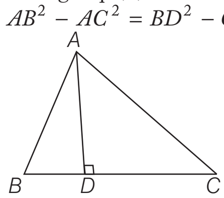
$AB^2 = AD^2 + BD^2 \quad \ldots$(i)

In right angled ΔADC, we have

$AC^2 = AD^2 + CD^2 \quad \ldots$(ii)

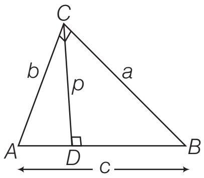
Subtracting Eq. (ii) from Eq. (i),

$AB^2 - AC^2 = BD^2 - CD^2$



$AB^2 - BD^2 = AC^2 - CD^2$

25. *(b)* Since, c is the base and p is the altitude of ΔABC.



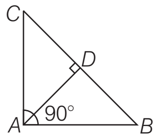
Here, area of $\Delta ABC = \frac{1}{2}pc$ …(i)

Also, area of $\Delta ABC = \frac{1}{2}ab$ …(ii)

From Eqs. (i) and (ii), we get

$$\frac{1}{2}pc = \frac{1}{2}ab \Rightarrow pc = ab$$

26. *(d)* Hence, $AD^2 = BD \cdot DC$



$$\Rightarrow \quad \frac{AD}{BD} = \frac{DC}{AD}$$

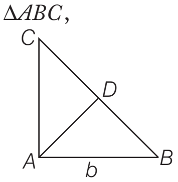
$\therefore \quad \Delta ADB \sim \Delta CDA \Rightarrow \angle BAD = \angle ACD$

Now, $\angle CAB = \angle CAD + \angle BAD$

$= \angle CAD + \angle ACD = 90°$

So, ΔABC must be right angled triangle.
Hence, $BC^2 = AC^2 + AB^2$

27. *(a)* In ΔABC,



Area of $\Delta ABC = \frac{1}{2} \times$ Base $\times$ Altitude

$$\Delta = \frac{1}{2}b \times AC,\ AC = \frac{2\Delta}{b}$$

In ΔABC, Using pythagoras theorem,

$AC^2 + AB^2 = BC^2$

$$\Rightarrow \quad BC = \sqrt{\frac{4\Delta^2}{b^2} + b^2}$$

Again in $\Delta\ ABC$, area of ΔABC

$$\Delta = \frac{1}{2} \times BC \times AD$$

$$\Rightarrow \quad AD = \frac{2\Delta}{\sqrt{\frac{4\Delta^2 + b^4}{b^2}}} = \frac{2\Delta b}{\sqrt{4\Delta^2 + b^4}}$$

28. *(a)* As, $AB \| CD$ and BC cuts them.

$\therefore \quad \angle BCD = \angle ABC = x$ [alternate $\angle S$]

$\therefore \quad$ In ΔBCD, $x + y + z = 180°$

$$\frac{4}{3}y + y + z = 180°$$

$$\Rightarrow \quad \frac{7y}{3} + z = 180°$$

$$\frac{7}{3}\left(\frac{3}{8}z\right) + z = 180° \Rightarrow \frac{7}{8}z + z = 180°$$

$$\left[\because\ y = \frac{3}{8}z\right]$$

$$\frac{15z}{8} = 180° \Rightarrow z = 96°$$

So, $\quad y = \frac{3}{8} \times 96° = 36°$

and $\quad x = \frac{4}{3} \times 36° = 48°$

29. *(c)* We have, $x = 35°$ [alternate angles]

$z° = 75°$ [alternate angles]

In ΔABP, $x + y + 75° = 180°$

$y = 180° - (75° + 35°) \Rightarrow y = 70°$

Hence the value of

$x = 35°, y = 70°, z = 75°$

30. *(d)* $x° = 60°$ [vertically opposite angles]

$y = 60°$ [corresponding angles]

$\angle PRS = 110°$ [alternate angles]

$\angle QRS + x° = 110°$ [Alternate angles]

$\angle QRS = 110° - 60° = 50°$

$\therefore \quad t = 180° - (y + \angle QRS)$

$= 180° - (60° + 50°)$

$t = 70°$

Also, $\quad t = z$ [alternate angles]

$\therefore \quad z = 70°$

31. *(b)* Since, $\angle PTB = 55°$ [given]

Then, $\quad \angle TUV = 55°$

[corresponding angle]

Also, $\quad \angle TUV = \angle CUQ = 55°$

[vertically opposite angle]

Also given, $\quad \angle DVS = 45°$

Then, $\quad \angle UVT = 45°$

[vertically opposite angle]

In ΔUTV,

$\angle T = 180° - (55° + 45°) = 80°$

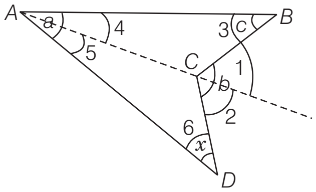
[angle sum property of a triangle]

$\Rightarrow \quad \angle T = \angle PTR = 80°$

[vertically opposite angle]

$\therefore \quad \angle CUQ + \angle RTP = 55° + 80° = 135°$

32. *(a)*



$\angle 1 = \angle 3 + \angle 4, \angle 2 = \angle 5 + \angle 6$

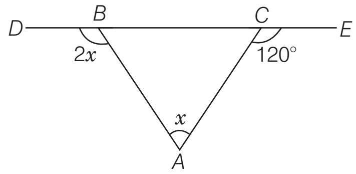
$\Rightarrow \angle 1 + \angle 2 = \angle 3 + \angle 4 + \angle 5 + \angle 6$

[since, exterior angle is equal to sum of two opposite interior angles]

$\Rightarrow \quad b = c + a + x$

$\therefore \quad x = b - c - a$

33. *(d)* $\because \angle ABC = 180° - \angle DBA = 180° - 2x$



and $\angle ACB = 180° - \angle ACE$

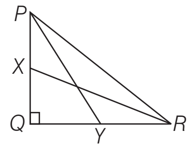
$= 180° - 120° = 60°$

In ΔABC, $\angle ABC + \angle ACB + \angle BAC = 180°$

$\Rightarrow 180° - 2x + 60° + x = 180° \Rightarrow x = 60°$

34. *(d)* In ΔPQY, by pythagoras theorem,

$PY^2 = PQ^2 + QY^2$



$$\Rightarrow \quad PY^2 = PQ^2 + \left(\frac{QR}{2}\right)^2$$

$$\left[QY = YR = \frac{QR}{2}\right] \quad \text{…(i)}$$

and in ΔXQR, $RX^2 = QX^2 + QR^2$

$$\Rightarrow \quad RX^2 = \left(\frac{PQ}{2}\right)^2 + QR^2$$

$$\left[\because PX = XQ = \frac{PQ}{2}\right] \quad \text{…(ii)}$$

On adding Eqs. (i) and (ii), we get

$$PY^2 + RX^2 = \frac{5PQ^2}{4} + \frac{5QR^2}{4}$$

$\Rightarrow \quad PY^2 + RX^2 = 5/4(PQ^2 + QR^2)$

$\Rightarrow 4(PY^2 + RX^2) = 5(PR^2)$

In ΔXQR, by pythagoras theorem,

$XR^2 = XQ^2 + QR^2$ …(iii)

and In ΔPQY, by pythagoras theorem,

$PY^2 = PQ^2 + QY^2$ …(iv)

On adding Eqs. (iii) and (iv),

$XR^2 + PY^2 = XQ^2 + QR^2 + PQ^2 + QY^2$

$\Rightarrow \quad XR^2 + PY^2 = XQ^2 + QY^2 + QR^2 + PQ^2$

$\Rightarrow \quad XR^2 + PY^2 = XY^2 + PR^2$

and again,

$XR^2 + PY^2 = XY^2 + PR^2$

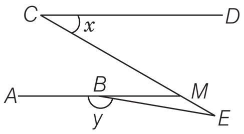
$= XY^2 + (2XY)^2$

$XR^2 + PY^2 = XY^2 + 4XY^2$

$\Rightarrow \quad XR^2 + PY^2 = 5XY^2$

Thus, option (d) is incorrect.

35. *(d)* We produced line AB to meet line CE at M.



Since, $AM \parallel CD$.

$\angle DCM = \angle BMC = x$ [alternate angles]

CME is a straight line

$\therefore \quad \angle BME = \pi - x$

Also, ABM is a straight line.

$\therefore \quad \angle EBM = \pi - y$

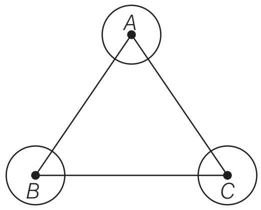
Now, in ΔBEM

$\angle EBM + \angle BME + \angle BEM = \pi$

$\Rightarrow \quad \pi - y + \pi - x + \angle E = \pi$

$\Rightarrow \quad \angle E = x + y - \pi$

36. *(c)* $\therefore \angle A = 360° - \text{Ext } \angle A$



Similarly, $\angle B = 360° - \text{Ext } \angle B$ and $\angle C = 360° - \text{Ext } \angle C$

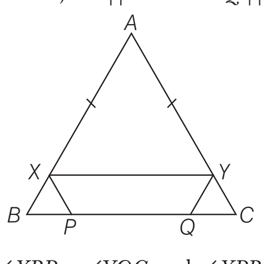
We know that, the sum of all the angles of a triangle is 180°.

$\therefore \quad \angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad 360° - \text{Ext } \angle A + 360° - \text{Ext } \angle B + 360° - \text{Ext } \angle C = 180°$

$\Rightarrow \quad \text{Ext } \angle A + \text{Ext } \angle B + \text{Ext } \angle C = 1080° - 180° = 900°$

37. *(d)* Since, $XP \parallel AC$ and $YQ \parallel AB$



$\therefore \angle XBP = \angle YQC$ and $\angle XPB = \angle YCQ$

Also, $XY \parallel BC$

$\therefore$ $XYQB$ and $XYCP$ are parallelogram

$\Rightarrow$ $XB = YQ$ and $XP = YC$

So, ΔXBP and ΔYCQ are congruent triangles.

Now, $XY \parallel BC$

$\therefore \quad \dfrac{AX}{AB} = \dfrac{XY}{BC} \Rightarrow AX = XY$

[$\because AB = BC = 30$ cm]

Also, $XY + XP + YQ = 40$ [given]

$\Rightarrow \quad AX + XB + YQ = 40$

[$\because XY = AX, XP = XB$]

$\Rightarrow \quad AB + YQ = 40$

$\Rightarrow \quad YQ = 40 - 30 = 10$ cm

$\therefore \quad YQ = XP = 10$ cm

$\therefore \quad BP = CQ = 10$ cm

$\therefore \quad PQ = 30 - BP - CQ = 30 - 10 - 10 = 10$ cm

38. *(b)* As, $OP \parallel YZ$

$\Rightarrow \angle POY = \angle OYZ$ [alternate angles]

$\Rightarrow \quad \angle PYO = \angle POY$

[since OY is angle bisector of $\angle Y$]

$\therefore \quad PY = PO$

As $\angle XYZ$ is an acute angle.

$\therefore \quad \dfrac{1}{2} \angle XYZ < 45°$

$\therefore \quad \angle POY = \angle PYO < 45°$

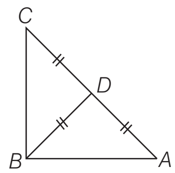
$\therefore \quad \angle YPO > 90°$

Hence, ΔPYO is an isosceles triangle but not a right angled triangle.

39. *(c)* Here, we see that

$CD = BD = DA$

This is possible only when ABC is right angled triangle.



40. *(a)* In $\Delta DCX, CD = CX$ [given]

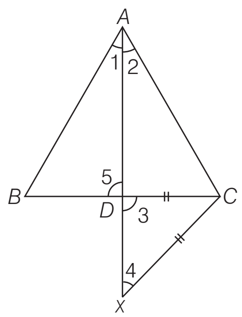
$\angle 3 = \angle 4$

[opposite angle of same sides]

But $\angle 3 = \angle 5$, So, $\angle 4 = \angle 5$

In Δ ABD and Δ ACX,

$\angle 1 = \angle 2$ [given]



$\angle 4 = \angle 5$

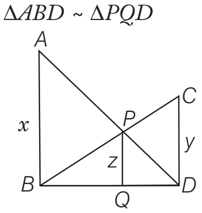
$\therefore \quad \Delta ABD \sim \Delta ACX$

[by AA similarity]

41. *(a)* Since, $AB \perp BD$ and $PQ \perp BD$

$\Rightarrow \quad AB \parallel PQ$

So, $\quad \Delta ABD \sim \Delta PQD$



Then, $\dfrac{x}{z} = \dfrac{BD}{QD}$ ...(i)

Since $CD \parallel PQ$

So, $\quad \Delta$ $BCD \sim \Delta$ BPQ,

$\therefore \quad \dfrac{z}{y} = \dfrac{BQ}{BD}$

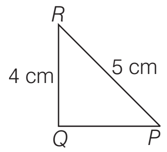
$\Rightarrow \quad \dfrac{z}{y} = \dfrac{BD - QD}{BD}$

$\Rightarrow \quad \dfrac{z}{y} = 1 - \dfrac{QD}{BD}$

$\Rightarrow \quad \dfrac{z}{y} = 1 - \dfrac{z}{x}$ [from Eq. (i)]

$\Rightarrow \quad \dfrac{z}{x} + \dfrac{z}{y} = 1 \Rightarrow \dfrac{1}{x} + \dfrac{1}{y} = \dfrac{1}{z}$

42. *(d)* In right angle Δ PQR, by pythagoras theorem,



$QP^2 = (5)^2 - (4)^2 = 9 \Rightarrow QP = 3$ cm

In second ΔABC whose sides are 3 cm, 4 cm and 5 cm. So, the sides of both triangle are same, hence they are congruent.

43. *(b)* Given, $AC = BC$

$\therefore \quad \angle ABC = \angle BAC$

[angles of an isosceles triangle]

$\Rightarrow \quad \angle BAC = \angle ABC = 38°$

In ΔABC,

$\angle ACB = 180° - (\angle BAC + \angle ABC)$

[by angle sum property of a triangle]

$= 180° - (38° + 38°)$

$= 180° - 76° = 104°$

In ΔACD, $\angle ACD = 180° - 104° = 76°$

[by linear pair]

and $\quad \angle CAD = \angle ACD = 76°$

[$\because CD = AD$]

$\therefore \quad \angle ADC = 180° - (\angle CAD + \angle ACD)$

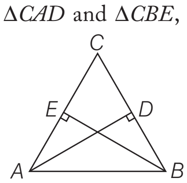
$= 180° - (76° + 76°) = 28°$

44. *(c)* I. It is true that the three medians of a triangle divide it into six triangles of equal area.

II. It is also true that, the perimeter of a triangle is greater than the sum of the lengths of its three medians.

Hence, I and II are correct.

45. *(d)* In ΔCAD and ΔCBE,



$\angle C = \angle C$ [common]

$\angle CEB = \angle ADC$ [each 90°]

$\therefore \quad \Delta CAD \sim \Delta CBE$

[by AA similarity]

Sides will be in same proportion.

$$\frac{CA}{CB} = \frac{CD}{CE} \text{ and } \frac{AD}{BE} = \frac{CD}{CE}$$

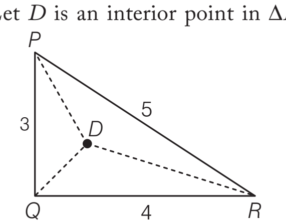
Hence, all three statements are correct.

46. *(b)* Given, $PQ = 3$ cm, $QR = 4$ cm and $RP = 5$ cm

Here, $RP^2 = PQ^2 + QR^2$

So, PQR is a right angled triangle.

Let D is an interior point in ΔPQR



In ΔDPQ, $DP + DQ > 3$ ...(i)

similarly, $DQ + DR > 4$ ...(ii)

and $DP + DR > 5$ ...(iii)

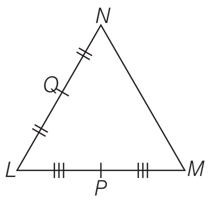
adding (i), (ii) and (iii), we get

$2(DP + DQ + DR) > 12$

$\Rightarrow \quad DP + DQ + DR > 6$

Hence, both statements I and II are individually true and statement II is not correct explanation of statement I.

47. *(b)* Given,



I. $PQ^2 = MP^2 + NQ^2$

$\Rightarrow \quad PQ^2 = LP^2 + LQ^2$

[$\because LP = MP$ and $NQ = LQ$]

$\Rightarrow \quad \angle QLP = 90°$

It means, ΔNLM is a right angled triangle.

II. It also true that if in a ΔABC. $AB^2 > BC^2 + CA^2$, then $\angle ACB$ is obtuse.

Hence, both statements are individually true but statement II is not the correct explanation of statement I.

48. *(c)* Given that,

$$\frac{1}{2}\angle A + \frac{1}{3}\angle C + \frac{1}{2}\angle B = 80°$$

$\Rightarrow \quad 3\angle A + 2\angle C + 3\angle B = 480°$

$\Rightarrow \quad 3(\angle A + \angle B) + 2\angle C = 480°$ ...(i)

Also, in ΔABC,

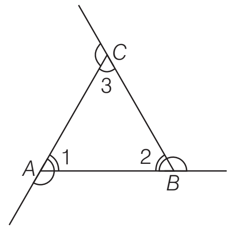
$\angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad 3(\angle A + \angle B) + 3\angle C = 540°$ ...(ii)

On subtracting Eq. (i) from Eq. (ii), we get

$\angle C = 60°$

49. *(c)*



By Exterior angle theorem,

ext. $\angle A = \angle 2 + \angle 3$

ext. $\angle B = \angle 1 + \angle 3$

ext. $\angle C = \angle 1 + \angle 2$

Sum of the three exterior angles

$= (\angle 1 + \angle 2) + (\angle 2 + \angle 3) + (\angle 3 + \angle 1)$

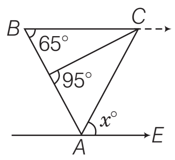
$= 2(\angle 1 + \angle 2 + \angle 3)$

$= 2 \times 180° = 360°$

[$\because \angle 1 + \angle 2 + \angle 3 = 180°$]

50. *(d)* Given that, $BC \parallel AE$

$\angle CBA + \angle EAB = 180°$



$\Rightarrow \quad \angle EAB = 180° - 65° = 115°$

$\because \quad BC = AC$

Hence, ΔABC is an isosceles triangle.

$\Rightarrow \quad \angle CBA = \angle CAB = 65°$

Now, $\angle EAB = \angle EAC + \angle CAB$

$\Rightarrow \quad 115° = x + 65° \Rightarrow x = 50°$

51. *(b)* Given that, $\angle ABC = 65°$

and $\angle CDE = 15°$

Since, $AB \parallel CD$

$\angle DCB = \angle ABC = 65°$

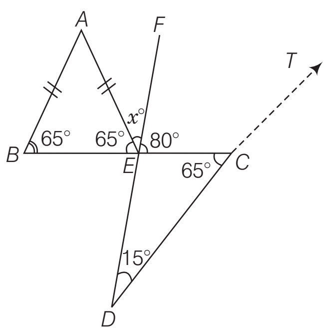
[alternate angles]

Also, $\angle DCE = \angle DCB$ [same angle]

Now, $\angle FEC = \angle CDE + \angle DCE$

[exterior angle]

$15° + 65° = 80°$



$\because \angle DEC + \angle FEC = 180°$ [linear pair]

Given that, $AB = AE$

$\therefore \quad \angle ABE = \angle AEB = 65°$

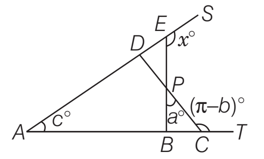
$\because \quad \angle AEB + \angle AEF + \angle FEC = 180°$

[straight line]

$\Rightarrow \quad 65° + x° + 80° = 180°$

$\therefore \quad x° = 180° - 145° = 35°$

52. *(c)* $\because \angle PCT + \angle PCB = \pi$ [linear pair]



$\angle PCB = \pi - (\pi - b°) = b°$ ...(i)

In ΔBPC, $\angle PCB + \angle BPC + \angle PBC = \pi$

[by angle sum property of a triangle]

$\Rightarrow \quad \angle PBC = \pi - \angle PCB - \angle BPC$

$= \pi - b° - a°$ ...(ii)

$\because \angle ABE + \angle EBC = \pi$ [linear pair]

$\Rightarrow \quad \angle ABE = \pi - \angle PBC$

[$\because \angle PBC = \angle EBC$]

$= \pi - (\pi - b° - a°) = a° + b°$...(iii)

Now, in ΔABE

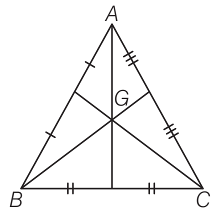
Sum of two interior angles = Exterior angle

$\angle EAB + \angle ABE = \angle BES$

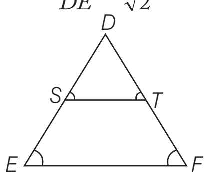
$c° + b° + a° = x°$

$\therefore \quad x° = a° + b° + c°$

53. *(d)* $GA = GB = GC$ is true only and only for equilateral triangle and here it is not given that ABC is an equilateral triangle. So, statement I is not correct. Similarly, statement II will also hold only for equilateral triangle.



54. *(a)* Given, $\dfrac{DS}{DE} = \dfrac{1}{\sqrt{2}}$



$\therefore \quad DE = \sqrt{2} \times DS$

Since, ST is parallel to EF.

$\therefore \quad \angle S = \angle E$ and $\angle T = \angle F$

So, ΔDTS and ΔDEF are similar.

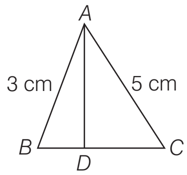
Ratio of areas = Ratio of squares of corresponding sides.

$$\Rightarrow \quad \frac{\Delta DST}{\Delta DEF} = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

$$\Rightarrow \quad \Delta\, DEF = 2\,\Delta\, DST$$

Hence, both statements I and II are true and statement II is the correct explanation of statement I.

55. *(c)* We know that, The sum of any two sides of a triangle is greater than twice the median drawn to the third side.

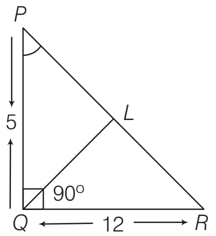


i.e. $(AB + AC) > 2\,AD$

$\Rightarrow \quad (3+5) > 2\,AD \quad \Rightarrow \quad AD < 4$

Hence, AD is always less than 4 cm.

56. *(d)* Given that, $PQ = 5$ cm, $QR = 12$ cm and QL is a median.



$\therefore \quad PL = LR = \dfrac{PR}{2}$ ...(i)

In ΔPQR, $(PR)^2 = (PQ)^2 + (QR)^2$

[by pythagoras theorem]

$(PR)^2 = (5)^2 + (12)^2$

$(PR)^2 = 25 + 144 \Rightarrow 169 = (13)^2$

$\Rightarrow \quad PR^2 = (13)^2 \quad \Rightarrow \quad PR = 13$

Now, by mid-point theorem, if L is the mid-point of the hypotenuse PR of a right angled ΔPQR, then

$$QL = \frac{1}{2}\,PR = \frac{1}{2}(13) = 6.5 \text{ cm}$$

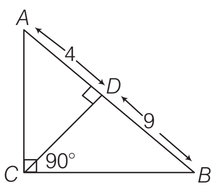
57. *(d)* In ΔABC and ΔACD,

$\because \quad \dfrac{AC}{AB} = \dfrac{4}{AC}$

$\therefore \quad AC^2 = 4 \times 13 = 52$ ... (i)

[$\because AB = 4 + 9 = 13$]

In ΔABC and ΔBCD, $\dfrac{BC}{AB} = \dfrac{9}{BC}$



$\Rightarrow \quad BC^2 = 9 \times 13 = 117$ ...(ii)

Now, $\dfrac{1}{CD^2} = \dfrac{1}{AC^2} + \dfrac{1}{BC^2} = \dfrac{1}{52} + \dfrac{1}{117}$

$$= \frac{9+4}{13 \times 4 \times 9} \quad \Rightarrow \quad \frac{1}{CD^2} = \frac{1}{36}$$

$\therefore \quad CD = 6$ cm

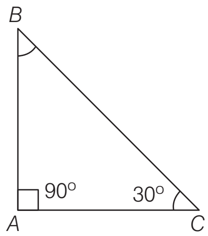
Shortcut Method

$CD^2 = AD \cdot DB$

$CD = \sqrt{AD \cdot DB} = \sqrt{4 \times 9}$

$= \sqrt{36} = 6$ cm

58. *(d)* Given that, $\angle A = 90°$ and $\angle C = 30°$



$\therefore \quad ABC$ is a right angled triangle.

Also, given that $BC = 2AB$

$\Rightarrow \quad AB = \dfrac{BC}{2}$ ...(i)

By pythagoras theorem,

$BC^2 = AC^2 + AB^2$

$\Rightarrow \quad (2AB)^2 = AC^2 + AB^2$

$\Rightarrow \quad AC^2 = 4AB^2 - AB^2 = 3AB^2$

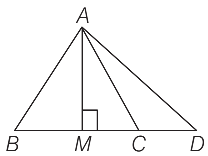
$\Rightarrow \quad AC = \sqrt{3} \cdot AB = \dfrac{\sqrt{3}}{2} \cdot (2AB)$

$\therefore \quad AC = \dfrac{\sqrt{3}}{2} \cdot BC$ [from Eq.(i)]

59. *(d)* Draw $AM \perp BC$

Let $AB = BC = AC = x$

Then, $\quad BM = MC = \dfrac{x}{2}$



Also, $\quad CD = \dfrac{x}{2}$ [$\because BC = 2CD$]

Now, in ΔAMC, by pythagoras theorem,

$AM^2 = AC^2 - MC^2$

$= x^2 - \dfrac{x^2}{4} = \dfrac{3x^2}{4}$

In ΔAMD, by pythagoras theorem,

$AD^2 = AM^2 + MD^2$

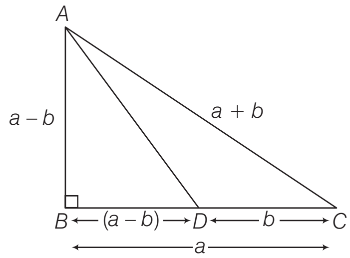
$= \dfrac{3x^2}{4} + x^2 = \dfrac{7x^2}{4}$

$= 7\left(\dfrac{x^2}{4}\right) = 7CD^2 \quad \left[\because CD = \dfrac{x}{2}\right]$

60. *(d)* In right angle ΔABC,

By pythagoras theorem,

$(a+b)^2 = (a-b)^2 + a^2$



$\Rightarrow a^2 + b^2 + 2ab = a^2 + b^2 - 2ab + a^2$

$\Rightarrow \quad 4ab = a^2 \quad \Rightarrow \quad 4b = a$

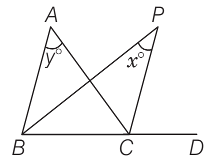
Now, $\quad \dfrac{BD}{DC} = \dfrac{a-b}{b}$

$= \dfrac{4b-b}{b} = \dfrac{3b}{b} = \dfrac{3}{1}$

or $3:1$

61. *(d)* Given that, $\angle BPC = x°$

and $\angle BAC = y°$



Since, BP and CP are the angle bisectors of $\angle ABC$ and $\angle ACD$, respectively.

$\therefore \quad \angle ABC = 2\angle PBC$ ...(i)

and $\quad \angle ACD = 2\angle PCD$ ...(ii)

since sum of two interior angles is equal to exterior angle

$\therefore \quad \angle PCD = x° + \angle PBC$ ...(iii)

and $\angle ACD = y° + \angle ABC$

$\Rightarrow \quad 2\angle PCD = y° + 2\angle PBC$

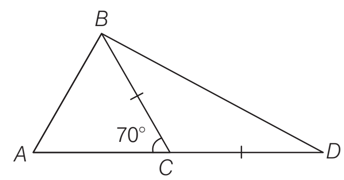
[using (i) and (ii)]

$\Rightarrow \quad 2\,[x° + \angle PBC] = y° + 2\angle PBC$

$\Rightarrow \quad 2x° = y°$

62. *(a)* $\because \angle ACB + \angle BCD = 180°$ [linear pair]

$\angle BCD = 180° - 70° = 110°$



In ΔBCD, $BC = CD$

$\Rightarrow \quad \angle CBD = \angle CDB$ ...(i)

[angles opposite to equal side]

Also, $\angle BCD + \angle CBD + \angle CDB = 180°$

[by angle sum property of a triangle]

$2\angle CDB = 180° - \angle BCD$

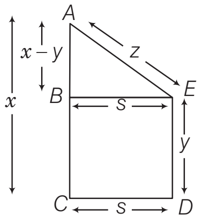
$= 180° - 110° = 70°$

$\therefore \angle CDB = \angle ADB = \dfrac{70°}{2} = 35°$

63. *(d)* Now, applying pythagoras theorem in ΔABE,

$$AE^2 = AB^2 + BE^2$$

$$\Rightarrow \quad z^2 = (x-y)^2 + s^2$$

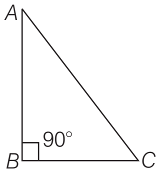


$$\Rightarrow \quad z^2 = x^2 + y^2 - 2xy + s^2$$

$$\therefore \quad s^2 = z^2 - x^2 - y^2 + 2xy$$

64. *(b)* Given, $\angle C = 2\angle A$ and $\angle B = 90°$

$\therefore \quad \angle A + \angle C = 90°$



$\Rightarrow \quad \angle A + 2\angle A = 90° \quad [\because \angle C = 2\angle A]$

$\Rightarrow \quad 3\angle A = 90° \Rightarrow \angle A = 30°$

So, $\quad \angle C = 90° - 30° = 60°$

Now, if the angles of a triangle are of measure 30°, 60° and 90°.
Then, side opposite to 30° i.e.

$$BC = \frac{1}{2} \times \text{hypotenuse}$$

$$\Rightarrow \quad BC = \frac{1}{2}AC \quad \Rightarrow \quad AC = 2\,BC$$

In right angled ΔABC,

by pythagoras theorem

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow 4BC^2 = AB^2 + BC^2 \quad [\because AC = 2BC]$$

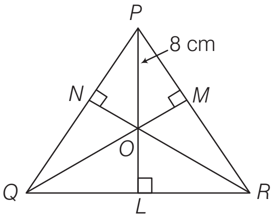
$$\Rightarrow \quad AB^2 = 4BC^2 - BC^2 = 3BC^2$$

65. *(d)* Since, PQR is an equilateral triangle. Then, PL is also the median of ΔPQR. Similarly, RN and QM are also the median of ΔPQR and O is the centroid.

So, $\dfrac{PO}{OL} = \dfrac{2}{1} \Rightarrow OL = \dfrac{PO}{2} = \dfrac{8}{2} = 4$ cm

Now, altitude of $\Delta PQR = \dfrac{\sqrt{3}a}{2}$

[where, a = length of side of an equilateral ΔPQR]



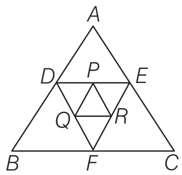
$$PO + OL = \frac{\sqrt{3}a}{2}, \; 8 + 4 = \frac{\sqrt{3}a}{2}$$

$$a = \frac{12 \times 2}{\sqrt{3}} = \frac{24}{\sqrt{3}} \text{ cm}$$

$\therefore$ Perimeter of $\Delta PQR = 3a$

$$= \frac{3 \times 24}{\sqrt{3}} = 24\sqrt{3} \text{ cm}$$

66. *(b)* Perimeter of $\Delta PQR = 1 + 2 + 3 = 6$ units



$\because$ P, Q, R are the mid-points of DE, DF and FE of ΔDEF.

$\therefore \qquad 2PQ = FE$

[by mid-point theorem]

Similarly, $\quad DF = 2PR$ and $DE = 2QR$

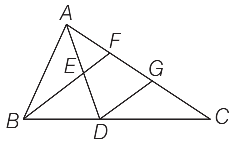
$\therefore$ Perimeter of $\Delta DEF = 2 \times 6 = 12$ units

Similarly, perimeter of ΔABC

$= 2 \times$ perimeter of ΔDEF

$= 2 \times 12 = 24$ units

67. *(b)* Draw a line segment DG parallel to BF. Then, in ΔADG,



$BF \parallel DG$

$\therefore \quad EF \parallel DG$

and $\quad AE = ED$

[since, E is mid-point of AD]

$\therefore \quad AF = FG \qquad$ ...(i)

Similarly, in ΔBCF

$DG \parallel BF \quad$ and $\quad BD = DC$

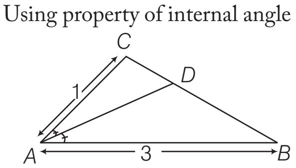
$\therefore \quad FG = GC \qquad$ ...(ii)

From Eqs. (i) and (ii), we get

$$CF = \frac{2}{3}AC$$

68. *(d)* In ΔABC, AD is the internal angle bisector of $\angle A$

Using property of internal angle bisector.



$$\frac{BD}{CD} = \frac{AB}{AC} \quad \Rightarrow \quad \frac{CD}{BD} = \frac{AC}{AB}$$

On adding both sides, we get

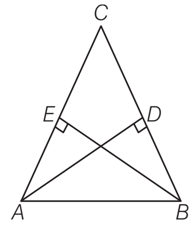
$$\Rightarrow \quad \frac{CD}{BD} + 1 = \frac{AC}{AB} + 1$$

$$\Rightarrow \quad \frac{CD + BD}{BD} = \frac{AC + AB}{AB}$$

$$\Rightarrow \quad \frac{BC}{BD} = \frac{3+1}{3} \quad \Rightarrow \quad \frac{BD}{BC} = \frac{3}{4}$$

$$\therefore \quad BD : BC = 3 : 4$$

69. *(b)* In ΔADC and ΔBEC



$\angle BEC = \angle ADC = 90°$

$\angle ACD = \angle BCE \qquad$ [common]

$\therefore \quad \Delta ADC \sim \Delta BEC$ [by AA similarity]

$$\frac{AC}{BC} = \frac{CD}{CE}$$

$$\Rightarrow \quad AC \times CE = BC \times CD$$

$$\Rightarrow \quad CE \times CA = CD \times CB$$

70. *(b)* Let the sides of triangle be $\dfrac{a}{r}$, a, ar and since $r < 1$.

$$\therefore \quad \frac{a}{r} > a > ar$$

Now, triangle is right angled.

Using pythagoras theorem,

$$\left(\frac{a}{r}\right)^2 = (a)^2 + (ar)^2$$

$$\Rightarrow \quad \frac{a^2}{r^2} = a^2 + a^2 r^2$$

$$\Rightarrow \quad \frac{a^2}{r^2} = a^2(1 + r^2) \Rightarrow r^2 + r^4 = 1$$

Put $\qquad r^2 = x,$

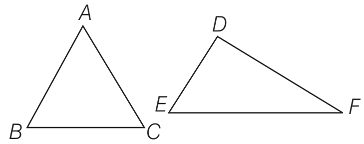
$$\Rightarrow x^2 + x - 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1 - 4(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

Since, sides of triangle cannot be negative

$$\therefore \quad r^2 = \frac{\sqrt{5} - 1}{2}$$

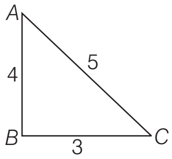
71. *(a)* $\Delta ABC \sim \Delta DEF$



$$\therefore \quad \frac{AB}{DE} = \frac{BC}{EF} \quad \Rightarrow \quad \frac{1}{2} = \frac{8}{EF}$$

$$EF = 16 \text{ cm}$$

72. *(a)* Since, the triangle is right angled. So, all the three consecutive sides must satisfy pythagoras theorem.



Hence, 3, 4 and 5 are the sides of triangle which satisfy pythagoras theorem.

$\therefore$ Area of triangle $= \frac{1}{2} \times 4 \times 3 = 6\text{ cm}^2$

73. *(a)* In a triangle,
Sum of two sides is always greater than the third side

i.e. $\quad x < 40 \quad$ ...(i)

Difference of two sides is always less than third side

i.e. $\quad 10 < x \quad$ ...(ii)

From Eqs. (i) and (ii), $10 < x < 40$

74. *(a)* Given, in $\Delta ABC, \angle B = 2\angle C = 2\angle A$

We know that, sum of all angles of a triangle = 180°

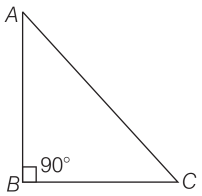
$\Rightarrow \quad \angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad \angle A + 2\angle A + \angle A = 180°$

$[\because 2\angle C = 2\angle A = \angle B]$

$\Rightarrow \quad 4\angle A = 180°$

$\Rightarrow \quad \angle A = \frac{180°}{4} = 45°$



$\therefore \quad \angle B = 90°$ and $\angle C = 45°$

Thus, ΔABC is a right angled triangle, right angle at B and $AB = BC$.

In ΔABC, by pythagoras theorem,

$AB^2 + BC^2 = AC^2$

$\Rightarrow \quad AB^2 + AB^2 = AC^2$

$[\because AB = BC]$

$\Rightarrow \quad 2AB^2 = AC^2$

$\Rightarrow \quad \sqrt{2}\,AB = AC$

[taking square root on both sides]

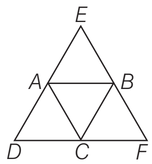
$\Rightarrow \quad \frac{AC}{AB} = \frac{\sqrt{2}}{1}$

$\therefore \quad AC : AB = \sqrt{2} : 1$

75. *(c)* I. On drawing the three straight lines through the three vertices of ΔABC, we get the following figure

Here, $AB \parallel DF$, $BC \parallel DE$ and $AC \parallel EF$

Obviously, A, B and C are the mid-points of DE, EF and DF, respectively.



By mid-point theorem,

$BC = \frac{1}{2}DE$ or $DE = 2BC$

Similarly, $DF = 2AB$

and $\quad EF = 2AC$

Hence, statement I is correct.

II. Also, area of

$\Delta ABC = \frac{1}{4}$ area of ΔDEF

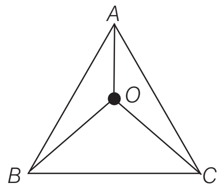
or area of $\Delta DEF = 4$ area of ΔABC

Hence, statement II is also correct.

76. *(c)* See solved answer of 67.

77. *(c)* Given, O is equidistant from AB, BC and AC

$\therefore$ O is the incentre of ΔABC



Join OA, OB and OC.

In ΔABC, OA, OB and OC are angle bisectors.

$\because \quad \angle OAB = \angle OAC = \frac{1}{2}\angle BAC$

$\therefore \quad \angle OBA = \angle OBC = \frac{1}{2}\angle ABC$

and $\angle OCB = \angle OCA = \frac{1}{2}\angle ACB$

Now, in ΔABC,

$\angle ABC + \angle BAC + \angle ACB = 180°$

[$\because$ angle sum property]

$\therefore \angle OAC + \angle OCB + \angle OBA = 90°$

[option]

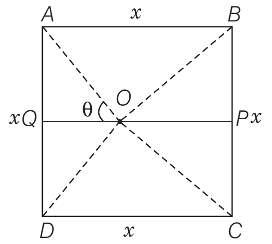
II. This statement is false as

$\angle BOC = 90° + \frac{1}{2}\angle BAC$

III. The statement perpendiculars drawn from any point on OA to AB and AC are always equal so it is true, because O is equidistant from AB and AC.

Hence, the statements I and III are correct.

78. *(d)*



Let side of square $ABCD$ be x and drawn equilateral ΔBOC inside $ABCD$.

Such that, $BO = OC = BC = x$

and $\quad BP = \frac{BC}{2} = \frac{x}{2}$

$\therefore \quad PO = \sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \sqrt{x^2 - \frac{x^2}{4}}$

$= \sqrt{\frac{3x^2}{4}} = \frac{\sqrt{3}}{2}x$

and $\quad OQ = AB - PO = x - \frac{\sqrt{3}}{2}x$

$= x\left(\frac{2-\sqrt{3}}{2}\right)$

Since, $\angle AOD = 2\theta$

$\Rightarrow \quad \angle AOQ = \theta$

In ΔAOQ,

$$\tan\theta = \frac{AQ}{OQ} = \frac{\frac{AD}{2}}{OQ} = \frac{\frac{x}{2}}{x\left(\frac{2-\sqrt{3}}{2}\right)}$$

$= \frac{x}{2} \times \frac{2}{x(2-\sqrt{3})}$

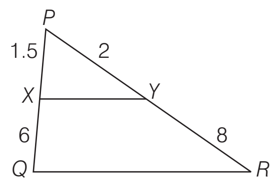
$= \frac{1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{2+\sqrt{3}}{4-3}$

$[\because (a+b)(a-b) = a^2 - b^2]$

$= 2 + \sqrt{3}$

79. *(d)* In ΔPQR,

$\frac{PX}{XQ} = \frac{1.5}{6} = \frac{1}{4}$ and $\frac{PY}{YR} = \frac{2}{8} = \frac{1}{4}$



So, $\quad \frac{PX}{XQ} = \frac{PY}{YR}$

I. $\frac{PX}{XY} = \frac{PQ}{QR} \Rightarrow \frac{1.5}{XY} = \frac{7.5}{QR}$

$\Rightarrow \quad QR = 5XY$

II. Also, $\quad \frac{PX}{PQ} = \frac{PY}{PR} = \frac{1}{5}$

$\Rightarrow$ QR is parallel of XY.

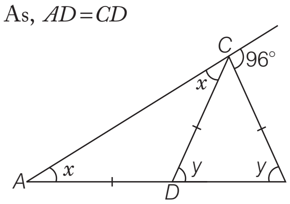
III. ΔPYX is similar to ΔPRQ.

Hence, all statements are correct.

80. *(c)* See solved answer 17.

81. *(b)* Let $\angle CDB = y$ and $\angle CAD = x$

As, $AD = CD$



$\Rightarrow\ \angle ACD = \angle CAD = x$ and $CD = CB$

$\Rightarrow\ \angle CBD = \angle CDB = y$

From exterior angle property,

$$\angle CDB = \angle CAD + \angle ACD$$

$$\Rightarrow \quad y = x + x \Rightarrow y = 2x$$

Now, $\angle ACD + \angle DCB + 96° = 180°$

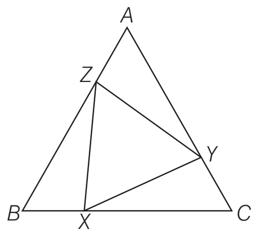
$$\Rightarrow \quad x + 180° - 2y + 96° = 180°$$

$$\Rightarrow \frac{y}{2} - 2y + 96° = 0 \Rightarrow \frac{3}{2}y = 96°$$

$$\therefore \quad y = 64°$$

82. *(c)* In an equilateral ΔABC,

$AB = BC = CA$ and $\angle B = \angle C = \angle A$



Given that, $BX = CY = AZ$

Now, in ΔXYC, ΔZYA and ΔXZB,

$$BX = CY = AZ \quad \ldots\text{(i)}$$

$$\Rightarrow (BC - XC) = (AC - AY) = AB - BZ$$

$$\Rightarrow \quad XC = AY = BZ \quad \ldots\text{(ii)}$$

and $\angle B = \angle C = \angle A$ ...(iii)

From Eqs. (i), (ii) and (iii), we get ΔXYC, ΔZYA and ΔXZB are congruent triangles.

$$\therefore \quad XY = YZ = XZ$$

So, ΔXYZ is an equilateral triangle and ΔXYZ is similar to ΔABC.

So, both statements are correct.

83. *(b)* I. This statement is always true.

II. In any triangle three times the sum of squares of the sides of a triangle is equal to four times the sum of squares of medians

i. e. $3(AB^2 + BC^2 + CA^2)$

$$= 4(AD^2 + BE^2 + CF^2)$$

Similarly $3(A'B'^2 + B'C'^2 + C'A'^2)$

$$= 4(A'D'^2 + B'E'^2 + C'F'^2)$$

$$\therefore \quad \frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2} = \frac{4}{3}$$

Hence, both statement are correct but statement II is not correct explanation of statement I.

84. *(c)* $\dfrac{\text{Area of } \Delta ABC}{\text{Area of } \Delta DEF} = \dfrac{AB^2}{DE^2}$

$$= \frac{(\sqrt{2}+1)^2}{(\sqrt{3})^2} = \frac{3 + 2\sqrt{2}}{3}$$

$$= \frac{(3+2\sqrt{2})(3-2\sqrt{2})}{3(3-2\sqrt{2})}$$

$$[\because a^2 - b^2 = (a-b)(a+b)]$$

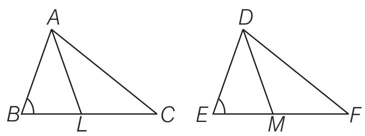
$$= \frac{1}{9 - 6\sqrt{2}}$$

Hence, the required ratio is $1 : (9 - 6\sqrt{2})$.

85. *(c)* Given that, $\angle ABC = \angle DEF$,

$$\angle ACB = \angle DFE$$

and $\quad \angle BAC = \angle EDF$



So, ΔABC and ΔDEF are similar.

$$\therefore \quad \frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

Now, L is the mid-point of BC, then

$$BL = \frac{1}{2}BC$$

Also, M is the mid-point of EF, then

$$EM = \frac{1}{2}EF \Rightarrow \frac{AB}{DE} = \frac{2BL}{2EM} = \frac{BL}{EM}$$

and $\angle ABL = \angle DEM$

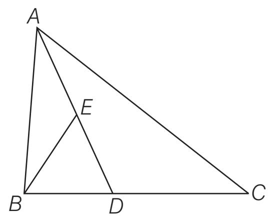
$\therefore \Delta ABL$ is similar to ΔDEM.

[By SAS Similarity]

Hence, statement I is true but statement II is false.

86. *(c)* In ΔABC, AD is the median and median AD bisects the area of ΔABC.

Area of $\Delta ABD = \frac{1}{2} \times$ Area of ΔABC



Since, E is the mid-point of AD. Then, BE is the median of ΔABD. So, BE bisects of area of ΔABD.

I. Area of $\Delta BED = \frac{1}{2} \times$ Area of ΔABD

$$= \frac{1}{2}\left[\frac{1}{2} \times \text{Area of } \Delta ABC\right]$$

$$= \frac{1}{4} \times \text{Area of } \Delta ABC$$

Hence, statement I is correct.

II. Area of ΔADC = Area of ΔABD

$$= 2 \times \text{Area of } \Delta BED$$

So, statement II is also correct.

Hence, the statements I and II are correct.

24

QUADRILATERAL AND POLYGON

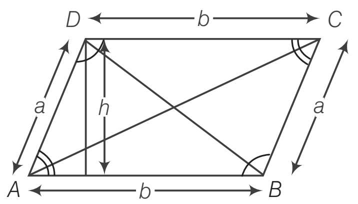
Regularly (2-3) questions have been asked from this chapter. Generally, questions from this chapter are tricky and mostly statement based. So, a clear concept of all the properties related to quadrilateral is necessary to do well.

QUADRILATERAL

A figure enclosed by four sides is called a quadrilateral. A quadrilateral has four angles and sum of these angles is equal to 360°.

Various types of quadrilateral are discussed below.

Parallelogram



A quadrilateral, in which opposite sides are parallel is called a parallelogram.

(i) Area = Base × Height = $b \times h$ (ii) Perimeter = $2(a + b)$

Properties of Parallelogram

(i) Diagonals of a parallelogram bisect each other.

(ii) Each diagonal of a parallelogram divides it into two congruent triangles.

(iii) Sum of any two adjacent angles is 180°.

(iv) A parallelogram inscribed in a circle is a rectangle.

(v) A parallelogram circumscribed about a circle is a rhombus.

(vi) Two parallelograms have equal areas if they are on the same base and between the same parallel lines.

(vii) Lines joining the midpoints of the sides of parallelogram is a parallelogram.

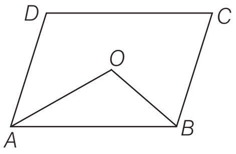
(viii) The opposite angles of parallelogram are equal. ($\angle A = \angle C$ and $\angle B = \angle D$) (from above figure)

(ix) The sum of the squares of the four sides is equal to the sum of squares of diagonal.

$AC^2 + BD^2 = AB^2 + BC^2 + CD^2 + AD^2$

EXAMPLE 1. In a parallelogram *ABCD*, the bisectors of $\angle A$ and $\angle B$ meet at *O*. Then, the value of $\angle AOB$ is

a. 55° **b.** 75° **c.** 90° **d.** 120°



Sol. **c.** As, *ABCD* is a parallelogram.

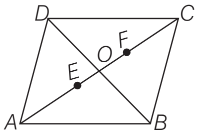
$\therefore \angle A + \angle B = 180°$ [cointerior angle]

$$\frac{1}{2}\angle A + \frac{1}{2}\angle B = 90°$$

$\Rightarrow \angle OAB + \angle OBA = 90°$

In ΔAOB, $\angle AOB = 180° - (\angle OAB + \angle OBA) = 180° - 90° = 90°$

EXAMPLE 2. In the adjoining figure, *ABCD* is a parallelogram and *E*, *F* are the centroids of ΔABD and ΔBCD, respectively, then the length of *EF* is



a. *AE* **b.** *OB* **c.** $\frac{1}{3}AE$ **d.** $\frac{1}{3}FC$

Sol. **a.** As *E* is the centroid of ΔABD and *AO* is one of its medians.

$\Rightarrow \quad OA : EO = 3 : 1 \Rightarrow EO = \frac{1}{3}OA$

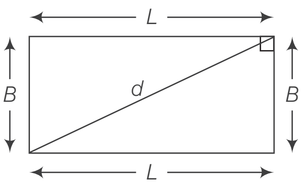
Similarly, $\quad FO = \frac{1}{3}OC$

$\therefore \quad EO + OF = \frac{1}{3}OA + \frac{1}{3}OC = \frac{1}{3}AC = AE$

$\therefore \quad EF = AE$

Rectangle

It is a parallelogram with opposite sides equal and each angle is equal to 90°.



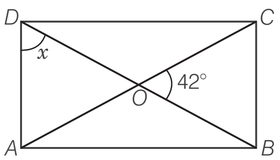
(i) Area = Length × Breadth $= L \times B$

(ii) Perimeter $= 2(L + B)$ (iii) Diagonal $(d) = \sqrt{L^2 + B^2}$

Properties of Rectangle

(i) The diagonals of a rectangle are of equal lengths and they bisect each other.
(ii) All rectangles are parallelograms but reverse is not true.
(iii) Diagonal of a rectangle inscribed in a circle is equal to the diameter of the circle.
(iv) The figure formed by joining the mid-points of the sides of a rectangle is a rhombus.
(v) A rectangle and a parallelogram have equal area if they are on the same base and between same parallel lines.

EXAMPLE 3. In adjoining figure, if *ABCD* is a rectangle and diagonals *AC* and *BD* intersect each other at *O*. Find the value of x



a. 75° **b.** 70° **c.** 69° **d.** 63°

Sol. **c.** Since, diagonals of rectangle bisect each other

$\therefore \quad OD = OA \Rightarrow \angle ODA = \angle OAD$

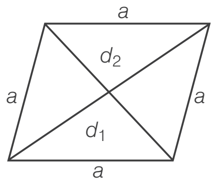
Also, $\quad \angle DOA = \angle COB = 42°$

In ΔDOA, $\angle DOA + \angle ODA + \angle OAD = 180°$

$\Rightarrow \quad 42° + x + x = 180° \Rightarrow 2x = 138° \Rightarrow x = 69°$

Rhombus

It is a parallelogram with all 4 sides equal. The opposite angles in a rhombus are equal but they are not right angle.



(i) Area $= \frac{1}{2} \times d_1 \times d_2$

(ii) Perimeter $= 4a$ (iii) Side $(a) = \frac{1}{2}\sqrt{d_1^2 + d_2^2}$

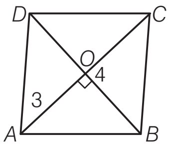
(iv) $4a^2 = d_1^2 + d_2^2$

where, a = side, d_1 and d_2 are diagonals.

Properties of Rhombus

(i) The diagonale of a rhombus bisect each other at right angles (90°). But they are not necessarily equal.
(ii) Diagonals bisect the vertex angles.
(iii) The figure formed by joining the mid-points of the sides of rhombus is a rectangle.
(iv) All rhombus are parallelogram but reverse is not true.
(v) A rhombus may or may not be a square but all squares are rhombus.

EXAMPLE 4. In a rhombus *ABCD*, diagonals intersect each other at *O*. If $AO = 3$ cm and $OB = 4$ cm, then find the perimeter of *ABCD*.



a. 14 cm **b.** 30 cm **c.** 20 cm **d.** 28 cm

Sol. **c.** The diagonals of a rhombus are perpendicular bisectors. Which means they form right angles at the point of their intersection.

$\therefore$ In ΔAOB, by pythagoras theorem,

$AO^2 + OB^2 = AB^2 \Rightarrow AB = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}$

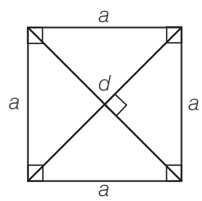
$\therefore \quad AB = 5$ cm

Hence, perimeter of rhombus

$ABCD = AB + BC + CD + DA = 5 + 5 + 5 + 5 = 20$ cm

Square

It is a parallelogram with all four sides equal and each angle is equal to 90°.



(i) Area $= (\text{side})^2 = a^2$ or $\frac{1}{2}d^2$

(ii) Perimeter $= 4 \times$ side $= 4a$

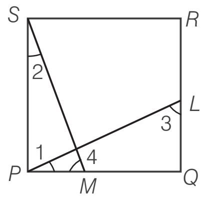
(iii) Diagonal $(d) = a\sqrt{2}$

where, a = side, d = diagonal

Properties of Square

(i) Diagonal of a square are equal and bisect each other at right angles (90°).
(ii) All square are rhombus but converse is not true.
(iii) Diagonal of square is the diameter of the circumscribing circle that circumscribes the square.
(iv) Side of a circumscribed square is equal to the diameter of the inscribed circle.
(v) The figure formed by joining the mid-points of the sides of square is a square.

EXAMPLE 5. *PQRS* is a square, *P* is joined to a point *L* on *QR* and *S* is joined to a point *M* on *PQ*. If $PL = SM$, then



a. $\angle POM > 90°$ b. $\angle POM = 90°$
c. $\angle POM < 90°$ d. None of these

Sol. **b.** In ΔSPM and ΔPQL, $\angle SPM = \angle PQL = 90°$

$SP = PQ$ and $SM = PL \Rightarrow \Delta SPM \cong \Delta PQL$ [By AAS]

$\Rightarrow \angle 1 = \angle 2$ and $\angle 3 = \angle 4$

In ΔSPM, $\angle 2 + 90° + \angle 4 = 180° \Rightarrow \angle 2 + \angle 4 = 90°$

In ΔOPM, $\angle 1 + \angle 4 + \angle POM = 180°$

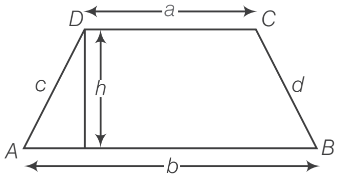
$\Rightarrow \angle 2 + \angle 4 + \angle POM = 180°$

$\Rightarrow 90° + \angle POM = 180°$ $[\because \angle 2 + \angle 4 = 90°]$

$\Rightarrow \angle POM = 90°$

Trapezium

It is a quadrilateral with any one pair of opposite sides parallel.



(i) Area $= \frac{1}{2}$ (Sum of the parallel sides × Height)

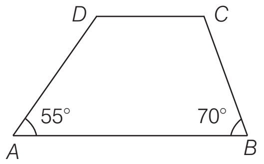
$= \frac{1}{2}(a+b)h$ where, a and b are parallel sides and h is the height or perpendicular distance between a and b.

(ii) Perimeter $= AB + BC + CD + AD$

Properties

(i) If the non-parallel sides are equal then the diagonals will also be equal to each other.
(ii) Diagonals intersect each other proportionally in the ratio of lengths of parallel sides.
(iii) If a trapezium is inscribed in a circle, then it is an isosceles trapezium with equal oblique sides.
(iv) The line joining the mid-points of non-parallel sides is half the sum of parallel side and is called median.
i.e. Median $= \frac{1}{2} \times$ sum of parallel sides.

EXAMPLE 6. In the adjoining figure, *ABCD* is a trapezium in which $AB \parallel DC$. If $\angle A = 55°$ and $\angle B = 70°$, then the value of $\angle C$ and $\angle D$ is



a. 75° and 85° b. 90° and 120°
c. 110° and 125° d. 115° and 120°

Sol. **c.** As, $AB \parallel CD$

So, $\angle A + \angle D = 180° \Rightarrow \angle D = 180° - 55° = 125°$

Also, $\angle B + \angle C = 180° \Rightarrow \angle C = 180° - \angle B$

$\Rightarrow \angle C = 180° - 70° = 110°$

So, $\angle C = 110°$ and $\angle D = 125°$

POLYGON

A polygon is a closed, plane figure bounded by '*n*' straight lines ($n \geq 3$). Each of the line segment forming the polygon is called its sides.

A polygon may be a triangle, quadrilateral, pentagon etc. Polygons are classified according to the number of sides as given below:

Number of sides	Name
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

Regular Polygon

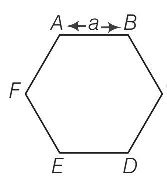
In regular polygons, all sides and all interior angles are equal. If each side of a regular polygon of n sides $= a$, then

(i) Area of regular pentagon $= 5a^2 \frac{\sqrt{3}}{4}$

(ii) Area of regular hexagon $= 6a^2 \frac{\sqrt{3}}{4}$

(iii) Area of regular octagon $= 2(\sqrt{2}+1)a^2$

(iv) Each exterior angle $= \frac{360°}{n}$



Regular Hexagon

(v) Each interior angle $= 180° - \text{Exterior angle}$

$$= \frac{(n-2)\times 180}{n}$$

(vi) Number of diagonals $= \left\{\frac{n(n-1)}{2} - n\right\}$

(vii) Sum of all interor angles $= (n-2)\times 180°$.

EXAMPLE 7. A polygon has 35 diagonals. Then, the number of sides of that polygon is

a. 7 b. 10 c. 11 d. 12

Sol. b. Let number of sides be n, then

$$\frac{n(n-1)}{2} - n = 35 \Rightarrow \frac{n^2 - n - 2n}{2} = 35$$

$\Rightarrow n^2 - 3n - 70 = 0 \Rightarrow n^2 - 10n + 7n - 70 = 0$

$\Rightarrow n(n-10) + 7(n-10) = 0 \Rightarrow (n-10)(n+7) = 0$

$\Rightarrow n = 10 \text{ and } n \neq -7$

Hence, the number of sides of the polygon is 10.

EXAMPLE 8. The angles of a hexagon are $x°$, $(x-5)°$, $(x-5)°$, $(2x-5)°$, $(2x-5)°$ and $(2x+20)°$. Then, the value of x is

a. 60° b. 75° c. 80° d. 90°

Sol. c. Sum of interior angles of a hexagon $= (n-2)\times 180°$

$= 4 \times 180° = 720°$

$\therefore x° + (x-5)° + (x-5)° + (2x-5)° + (2x-5)° + (2x+20)° = 720°$

$\therefore 9x = 720° = \frac{720°}{9} = 80°$

Hence, the value of x is 80°.

EXAMPLE 9. The difference between the interior and exterior angles of a regular polygon is 60°. Then, how many sides are there in that polygon?

a. 5 b. 6 c. 7 d. 8

Sol. b. Here, (interior angle) − (exterior angle) = 60°

$\Rightarrow \frac{(n-2)\times 180}{n} - \frac{360}{n} = 60$

$\Rightarrow \frac{1}{n}[(n-2)\times 180 - 360] = 60$

$\Rightarrow \frac{1}{n}[180n - 360 - 360] = 60 \Rightarrow \frac{1}{n}[180n - 720] = 60$

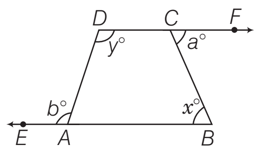
$\Rightarrow 180n - 720 = 60n \Rightarrow 180n - 60n = 720$

$\Rightarrow 120n = 720 \Rightarrow n = 6$

Therefore, the polygon contains 6 sides.

PRACTICE EXERCISE

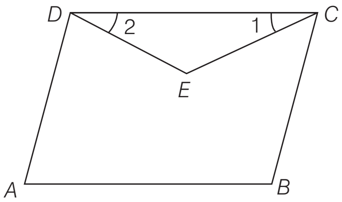
1. The sides BA and DC of quadrilateral $ABCD$ are produced as shown in figure. Then, which of the following statement is correct?



(a) $2x° + y° = a° + b°$ (b) $x° + \frac{1}{2}y° = \frac{a° + b°}{2}$

(c) $x° + y° = a° + b°$ (d) $x° + a° = y° + b°$

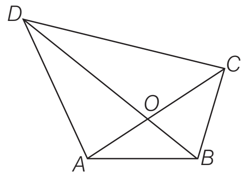
2. In the quadrilateral $ABCD$, the line segments bisecting $\angle C$ and $\angle D$ meet at E. Then, the correct statement is



(a) $\angle A + \angle B = \angle CED$ (b) $\angle A + \angle B = 2\angle CED$

(c) $\angle A + \angle B = 3\angle CED$ (d) None of these

3. If $ABCD$ is a quadrilateral whose diagonals AC and BD intersects of O, then



(a) $(AB + BC + CD + DA) < (AC + BD)$

(b) $(AB + BC + CD + DA) > 2(AC + BD)$

(c) $(AB + BC + CD + DA) > (AC + BD)$

(d) $(AB + BC + CD + DA) = 2(AC + BD)$

4. If area of a parallelogram with sides p and q is R and that of a rectangle with sides p and q is S, then

(a) $R > S$ (b) $R < S$

(c) $R = S$ (d) None of these

5. Two parallelogram stand on equal bases and between the same parallel. The ratio of their areas is

(a) 1 : 2 b) 2 : 1 (c) 1 : 3 (d) 1 : 1

6. If $ABCD$ is a rhombus, then

(a) $AC^2 + BD^2 = 4AB^2$ (b) $AC^2 + BD^2 = AB^2$
(c) $AC^2 + BD^2 = 2AB^2$ (d) $2(AC^2 + BD^2) = 3AB^2$

7. A point O in the interior of a rectangle $ABCD$ is joined with each of the vertices A, B, C and D. Then,

(a) $OB + OD = OC + OA$ (b) $OB^2 + OA^2 = OC^2 + OD^2$
(c) $OB \cdot OD = OC \cdot OA$ (d) $OB^2 + OD^2 = OC^2 + OA^2$

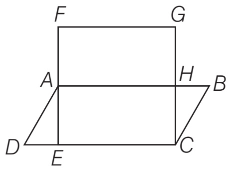
8. In a trapezium $ABCD$, if $AB || CD$, then $AC^2 + BD^2$ is equal to

(a) $BC^2 + AD^2 + 2AB \cdot CD$ (b) $AB^2 + CD^2 + 2AD \cdot BC$
(c) $AB^2 + CD^2 + 2AB \cdot CD$ (d) $BC^2 + AD^2 + 2BC \cdot AD$

9. $ABCD$ is a trapezium in which $AB || CD$ and $AB = 2CD$. Its diagonals intersects each other at O, then the ratio of the areas of the ΔAOB and ΔCOD is

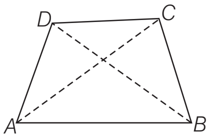
(a) 1 : 2 (b) 2 : 1 (c) 4 : 1 (d) 1 : 4

10. In the figure, $ABCD$ is a parallelogram with $AD = a$ units, $DC = 2a$ units and $DE : EC = 1 : 2$. $CEFG$ is a rectangle with $FE = 3\ AE$. What is the ratio of the area of a parallelogram and the rectangle?



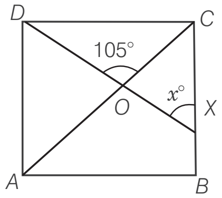
(a) 1 : 1 (b) 1 : 2 (c) $\frac{a^2}{5} : 1$ (d) 2 : 1

11. In the adjoining figure, $ABCD$ is a quadrilateral in which AB is the longest side and CD is the shortest side, then



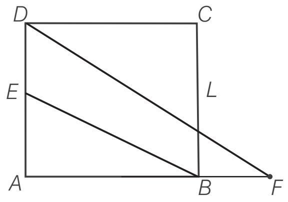
(a) $\angle C > \angle A$ and $\angle D > \angle B$ (b) $\angle C > \angle A$ and $\angle B > \angle D$
(c) $\angle C < \angle A$ and $\angle D < \angle B$ (d) $\angle C < \angle A$ and $\angle D = \angle B$

12. In the given figure, $ABCD$ is a square. A line segment DX cuts the side BC at X and the diagonal AC at O such that $\angle COD = 105°$ and $\angle OXC = x$. The value of x is



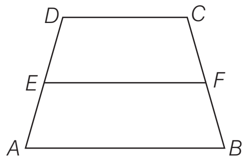
(a) 40° (b) 60° (c) 80° (d) 85°

13. In the given figure, $ABCD$ is a parallelogram and E is the mid-point of AD. A line through D, drawn parallel to EB, meets AB produced at F and BC at L. Then,



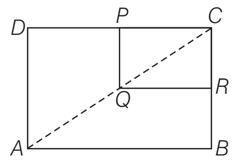
(a) $AF = 2DC$ (b) $2AF = 3DC$
(c) $2DF = 3DL$ (d) $AF + DF = 3DC + DL$

14. Let $ABCD$ be a trapezium in which $AB || DC$ and let E be the mid-point of AD. Let F be a point on BC such that $EF || AB$. Then, consider the following statements.



(a) F is the mid-point of BC (b) $EF = \frac{1}{2}(AB + DC)$
(c) Both (a) and (b) (d) None of these

15. In the figure given below, $ABCD$ and $PQRC$ are rectangles, where Q is the mid-point of AC, then



(a) $DP = PC$ (b) $\frac{1}{3} PR = \frac{3}{2} DB$
(c) $DP = PQ$ (d) $DP = \frac{1}{2} AC$

16. How many diagonals are there in a octagon?

(a) 8 (b) 16 (c) 18 (d) 20

17. The angles of a pentagon are in the ratio $1 : 2 : 3 : 5 : 9$, the largest angle is

(a) 81° (b) 135° (c) 243° (d) 249°

18. The ratio of an interior angle to the exterior angle of a regular polygon is 5 : 1. The number of sides of polygon is

(a) 10 (b) 11 (c) 12 (d) 14

19. The measure of each angle of a regular hexagon is

(a) 90° (b) 120° (c) 105° (d) 135°

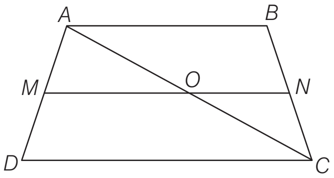
20. Each interior angle of a regular polygon is 150°. The number of sides of the polygon is

(a) 4 (b) 8 (c) 12 (d) 16

21. Let $LMNP$ be a parallelogram and NR be perpendicular to LP. If the area of the parallelogram is six times the area of ΔRNP and $RP = 6$ cm what is LR equal to?

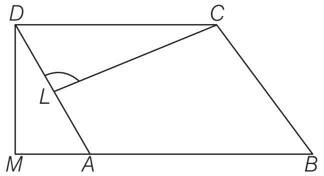
(a) 15 cm (b) 12 cm (c) 9 cm (d) 8 cm

22. $ABCD$ is a trapezium in which $AB || CD$. M and N are the mid-points of AD and BC, respectively. If $AB = 12$ cm and $MN = 14$ cm find CD.



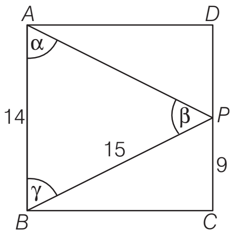
(a) 2 cm (b) 5 cm (c) 12 cm (d) 16 cm

23. $ABCD$ is a parallelogram. CL is perpendicular to AD and DM is perpendicular to BA produced. If $CD = 16$ units, $DM = 12$ units and $CL = 15$ units then $AD = ?$



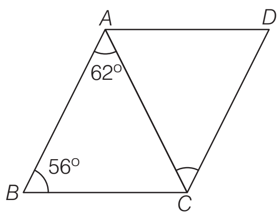
(a) 12.8 units (b) 13.6 units (c) 11.1 units (d) 12.4 units

24. $ABCD$ is a rectangle. $PC = 9$ cm, $BP = 15$ cm, $AB = 14$ cm. Then, the angles of ΔAPB are such that



(a) $\alpha > \beta > \gamma$ (b) $\alpha > \gamma > \beta$ (c) $\beta > \gamma > \alpha$ (d) $\alpha < \beta < \gamma$

25. $ABCD$ is a rhombus with $\angle ABC = 56°$ and $\angle BAC = 62°$, then $\angle ACD$ is equal to

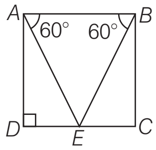


(a) 90° (b) 60° (c) 56° (d) 62°

26. One angle of a pentagon is 140°. If the remaining angles are in the ratio 1 : 2 : 3 : 4, then the size of the greatest angle is

(a) 150° (b) 180° (c) 160° (d) 170°

27. In the given figure $ABCD$ is a quadrilateral with AB parallel to DC and AD parallel to BC, $\angle ADC$ is a right angle. If the perimeter of the ΔABE is 6 units, what is the area of the quadrilateral?

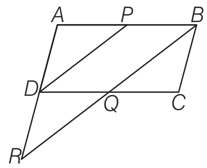


(a) $2\sqrt{3}$ sq units (b) 4 sq units
(c) 3 sq units (d) $4\sqrt{3}$ sq units

28. A square and a rhombus have the same base and the rhombus is inclined at 30°. What is the ratio of area of the square to the area of the rhombus?

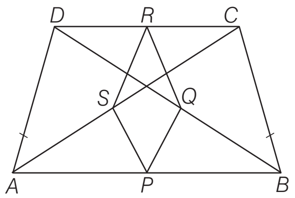
(a) $\sqrt{2} : 1$ (b) $2 : 1$ (c) $1 : 1$ (d) $2 : \sqrt{3}$

29. P is the mid-point of side AB of parallelogram $ABCD$. A line through B parallel to PD meets DC at Q and AD produced at R. Then BR is equal to



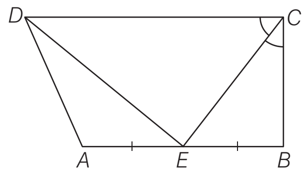
(a) BQ (b) $\frac{1}{2}$ (c) $2BQ$ (d) None of these

30. $ABCD$ is a trapezium in which $AB || DC$ and $AD = BC$. If P, Q, R, S be respectively the mid-point of BA, BD, CD and CA, then $PQRS$ is a



(a) rhombus (b) rectangle
(c) parallelogram (d) square

31. $ABCD$ is a parallelogram, E is the mid-point of AB and CE bisects $\angle BCD$. Then $\angle DEC$ is



(a) 60° (b) 90° (c) 100° (d) 120°

32. $ABCD$ is a square, P Q, R and S are points on the sides AB, BC, CD and DA, respectively such that $AP = BQ = CR = DS$. What is the value of $\angle SPQ$?

(a) 30° (b) 45° (c) 60° (d) 90°

33. The middle points of the parallel sides AB and CD of a parallelogram $ABCD$ are P and Q, respectively. If AQ and CP divide the diagonal BD into three parts BX, XY and YD, then which one of the following is correct?

(a) $BX \neq XY \neq YD$ (b) $BX = YD \neq XY$
(c) $BX = XY = YD$ (d) $XY = 2BX$

34. A parallelogram and a rectangle stand on the same base and on the same side of the base with the same height. If l_1, l_2 be the perimeters of the parallelogram and the rectangle respectively, then which one of the following is correct?

(a) $l_1 < l_2$ (b) $l_1 = l_2$
(c) $l_1 > l_2$ but $l_1 \neq 2l_2$ (d) $l_1 = 2l_2$

35. Two similar parallelograms have corresponding sides in the ratio $1:k$. What is the ratio of their areas?

(a) $1:3k^2$ (b) $1:4k^2$ (c) $1:k^2$ (d) $1:2k^2$

36. Let $WXYZ$ be a square. Let P, Q and R be the mid-points of WX, XY and ZW, respectively and K, L be the mid-points of PQ and PR, respectively. What is the value of

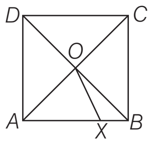
$$\frac{\text{area of } \Delta PKL}{\text{area of square } WXYZ}?$$

(a) $\frac{1}{32}$ (b) $\frac{1}{16}$
(c) $\frac{1}{8}$ (d) $\frac{1}{64}$

37. ABC is a triangle in which $AB = AC$. Let BC be produced to D. From a point E on the line AC let EF be a straight line such that EF is parallel to AB. Consider the quadrilateral $ECDF$ thus formed. If $\angle ABC = 65°$ and $\angle EFD = 80°$, then what is the value of $\angle FDC$?

(a) 43° (b) 41°
(c) 37° (d) 35°

38. In the given figure, $ABCD$ is a square in which $AO = AX$. What is $\angle XOB$?

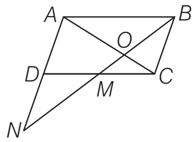


(a) 22.5° (b) 25°
(c) 30° (d) 45°

39. Two sides of a parallelogram are 10 cm and 15 cm. If the altitude corresponding to the side of length 15 cm is 5 cm, then what is the altitude to the side of length 10 cm ?

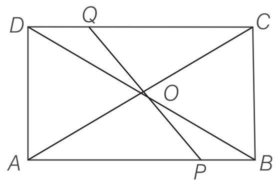
(a) 5 cm (b) 7.5 cm
(c) 10 cm (d) 15 cm

40. In the given figure, M is the mid-point of the side CD of the parallelogram $ABCD$. What is $ON:OB$?



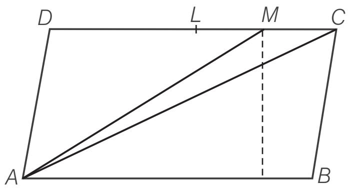
(a) 3 : 2 (b) 2 : 1 (c) 3 : 1 (d) 5 : 2

41. $ABCD$ is a parallelogram whose diagonals AC and BD intersect at O. A line segment PQ through O meets AB at P and DC at Q.



(a) ar (quadrilateral $APQD$) $= \frac{1}{2} ar$ (|| gm $ABCD$)
(b) ar (quadrilateral $APQD$) $= \frac{1}{4} ar$ (|| gm $ABCD$)
(c) ar (quadrilateral $APQD$) $= \frac{1}{3} ar$ (|| gm $ABCD$)
(d) None of the above

42. In the figure, $ABCD$ is a ||gm, L is the mid-point of DC and M is the mid-point of LC. If $ar\ (\Delta AMC) = 4\ \text{cm}^2$, then ar (|| gm $ABCD$) is



(a) 32 cm^2 (b) 16 cm^2
(c) 64 cm^2 (d) Data insufficient

43. Consider the following statements.

I. All squares are not parallelograms.
II. All parallelograms are trapezium.
III. All squares are rhombuses and also rectangles.
IV. All rhombuses are parallelograms.

Which of the above statements are correct?

(a) I and II (b) III and IV (c) I and III (d) II and IV

44. Consider the following statements.

I. If the side of a rhombus is 10 cm. Its diagonals should have values 16 cm and 12 cm.
II. The diagonals of a rhombus cut at right angles.
III. The diagonals do not bisect each other.

Which of the above statements are true?

(a) I and II (b) II and III (c) I and III (d) I, II and III

45. Consider the following statements in respect of a quadrilateral

I. The line segments joining the mid-points of the two pairs of opposite sides bisect each other at the point of intersection.
II. The area of the quadrilateral formed by joining the mid-points of the four adjacent sides is half of the total area of the quadrilateral.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

PREVIOUS YEARS' QUESTIONS

Directions (Q.Nos. 46-47) *Let ABCD be a quadrilateral, the diagonals AC and BD meet at O and the perpendicular drawn from A to CD, meet CD at E. Further,* $AO:OC = BO:OD$, $AB = 30$ cm, $CD = 40$ cm *and the area of the quadrilateral ABCD is* 1050 cm^2.

46. What is the value of BE? **2012 I**

(a) 30 cm (b) $30\sqrt{2}$ cm
(c) $30\sqrt{3}$ cm (d) None of these

47. What is the area of the ΔADC? **2012 I**

(a) 300 cm^2 (b) 450 cm^2
(c) 600 cm^2 (d) None of these

48. $ABCD$ is a rhombus with diagonals AC and BD. Then, which one among the following is correct? **2012 I**

(a) AC and BD bisect each other but not necessarily perpendicular to each other
(b) AC and BD are perpendicular to each other but not necessarily bisect each other
(c) AC and BD bisect each other and perpendicular to each other
(d) AC and BD neither bisect each other nor perpendicular to each other

49. The sides of a parallelogram are 12 cm and 8 cm long and one of the diagonals is 10 cm long. If d is the length of other diagonal, then which one of the following is correct? **2012 I**

(a) $d < 8$ cm (b) 8 cm $< d <$ 10 cm
(c) 10 cm $< d <$ 12 cm (d) $d >$12 cm

50. Let X be any point within a square $ABCD$. On AX a square $AXYZ$ is described such that D is within it. Which one of the following is correct? **2012 II**

(a) $AX = DZ$ (b) $\angle ADZ = \angle BAX$
(c) $AD = DZ$ (d) $BX = DZ$

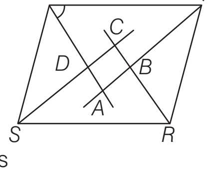
51. $ABCD$ is a parallelogram. If the bisectors of the $\angle A$ and $\angle C$ meet the diagonal BD at points P and Q respectively, then which one of the following is correct? **2012 II**

(a) $PCQA$ is a straight line (b) ΔAPQ is similar to ΔPCQ
(c) $AP = CP$ (d) $AP = AQ$

52. The locus of a point in rhombus $ABCD$ which is equidistant from A and C is **2012 II**

(a) a fixed point on diagonal BD
(b) diagonal BD (c) diagonal AC
(d) None of the above

53. In the figure given below, $PQRS$ is a parallelogram. If AP, AQ, CR and CS are bisectors of $\angle P, \angle Q, \angle R$ and $\angle S$ respectively, then $ABCD$ is a **2013 I**



(a) square (b) rhombus
(c) rectangle (d) None of these

54. $ABCD$ is a trapezium with parallel sides $AB = 2$ cm and $DC = 3$ cm. E and F are the mid-points of the non-parallel sides. The ratio of area of $ABFE$ to area of $EFCD$ is **2013 I**

(a) 9 : 10 (b) 8 : 9 (c) 9 : 11 (d) 11 : 9

55. If the diagonals of a rhombus are 4.8 cm and 1.4 cm, then what is the perimeter of the rhombus? **2013 II**

(a) 5 cm (b) 10 cm (c) 12 cm (d) 20 cm

56. $ABCD$ is a quadrilateral such that $BC = BA$ and $CD > AD$. Which one of the following is correct? **2013 II**

(a) $\angle BAD = \angle BCD$ (b) $\angle BAD < \angle BCD$
(c) $\angle BAD > \angle BCD$ (d) $2\angle BAD = \angle BCD$

57. Let $ABCD$ be a parallelogram. Let P, Q, R and S be the mid-points of sides AB, BC, CD and DA, respectively.
Consider the following statements.

I. Area of triangle APS < Area of triangle DSR, if $BD < AC$.
II. Area of triangle $ABC = 4$ (Area of triangle BPQ).

Select the correct answer using the codes given below **2014 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

58. In a trapezium, the two non-parallel sides are equal in length, each being of 5 cm. The parallel sides are at a distance of 3 cm apart. If the smaller side of the parallel sides is of length 2 cm, then the sum of the diagonals of the trapezium is **2014 I**

(a) $10\sqrt{5}$ cm (b) $6\sqrt{5}$ cm (c) $5\sqrt{5}$ cm (d) $3\sqrt{5}$ cm

59. Two light rods $AB = a + b$, $CD = a - b$ symmetrically lying on a horizontal plane. They are kept intact by two strings AC and BD. The perpendicular distance between rods is a. The length of AC is given by **2014 I**

(a) a (b) b (c) $\sqrt{a^2 - b^2}$ (d) $\sqrt{a^2 + b^2}$

60. Let $ABCD$ be a parallelogram. Let X and Y be the mid-points of the sides BC and AD, respectively. Let M and N be the mid-points of the sides AB and CD, respectively.
Consider the following statements.

I. The straight line MX cannot be parallel to YN.
II. The straight lines AC, BD, XY and MN meet at a point.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

61. If each interior angle of a regular polygon is 135°, then the number of diagonals of the polygon is equal to **2015 I**

(a) 54 (b) 48 (c) 20 (d) 18

62. If a star figure is formed by elongating the sides of a regular pentagon, then the measure of each angle at the angular points of the star figure is **2015 I**

(a) 36° (b) 35° (c) 32° (d) 30°

63. If X is any point within a square $ABCD$ and on AX a square $AXYZ$ is described, which of the following is/are correct?

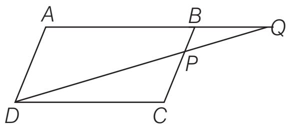
I. $BX = DZ$ or $BZ = DX$

II. $\angle ABX = \angle ADZ$ or $\angle ADX = \angle ABZ$

Select the correct answer using the code given below. **2015 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

64. In the adjoining figure, $ABCD$ is a parallelogram. P is a point on BC such that $PB : PC = 1 : 2$. DP and AB when both produced meet at Q. If area of ΔBPQ is 20 sq units, the area of ΔDCP is **2015 I**



(a) 20 sq units (b) 30 sq units
(c) 40 sq units (d) None of these

65. $ABCD$ is a parallelogram with AB and AD as adjacent sides. If $\angle A = 60°$ and $AB = 2AD$, then the diagonal BD will be equal to **2015 II**

(a) $\sqrt{2}AD$ (b) $\sqrt{3}AD$ (c) $2AD$ (d) $3AD$

66. If each interior angle of a regular polygon is 140° then number of vertices of the polygon is equal to **2016 I**

(a) 10 (b) 9 (c) 8 (d) 7

67. Consider the following statements.

I. There exists a regular polygon whose exterior angle is 70°.

II. Let $n \geq 5$, then the exterior angle of any regular polygon of n sides is acute.

Which of the above statements is/are correct? **2016 I**

(a) Only I (b) Only II
(b) Both I and II (d) Neither I nor II

68. Consider the following statements.

I. If $n \geq 3$ and $m \geq 3$ are distinct positive integers, then the sum of the exterior angles of a regular polygon of m sides is different from the sum of the exterior angles of a regular polygon of n sides.

II. Let m, n be integers such that $m > n \geq 3$. Then, the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides and their sum is $(m + n)\pi / 2$.

Which of the above statements is/are correct? **2016 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWERS

1	c	2	b	3	c	4	b	5	d	6	a	7	d	8	a	9	c	10	b
11	a	12	b	13	a	14	b	15	a	16	d	17	c	18	c	19	b	20	c
21	b	22	d	23	a	24	a	25	d	26	c	27	a	28	b	29	c	30	a
31	b	32	d	33	c	34	c	35	c	36	b	37	d	38	a	39	b	40	b
41	a	42	a	43	b	44	a	45	c	46	b	47	c	48	c	49	d	50	d
51	b	52	a	53	c	54	c	55	b	56	c	57	b	58	b	59	d	60	b
61	c	62	a	63	c	64	d	65	b	66	b	67	b	68	d				

HINTS AND SOLUTIONS

1. *(c)* As $\angle A + b° = 180°$ [linear pair]

$\Rightarrow \quad \angle A = 180° - b$

Also, $\angle C + a° = 180°$ [linear pair]

$\Rightarrow \quad \angle C = 180° - a°$

But $\angle A + \angle B + \angle C + \angle D = 360°$

[by angle sum property of a quadrilateral]

$\Rightarrow (180° - b°) + x° + (180° - a°) + y° = 360°$

$\Rightarrow \quad x° + y° = a° + b°$

2. *(b)* $\because \angle 1 = \frac{1}{2}\angle C$ and $\angle 2 = \frac{1}{2}\angle D$

In ΔDEC, $\angle 1 + \angle 2 + \angle CED = 180°$

[angle sum property of a triangle]

$\therefore \quad \angle CED = 180° - (\angle 1 + \angle 2)$

In quadrilateral $ABCD$,

$\angle A + \angle B + \angle C + \angle D = 360°$

$\angle A + \angle B + 2(\angle 1 + \angle 2) = 360°$

$\angle A + \angle B = 360° - 2(\angle 1 + \angle 2)$

$\angle A + \angle B = 2[180° - (\angle 1 + \angle 2)]$

$\angle A + \angle B = 2\angle CED$

3. *(c)* In ΔABC, ΔACD, ΔBCD and ΔABD

$AB + BC > AC$, $CD + DA > AC$

$BC + CD > BD$, $DA + AB > BD$

[Sum of any two sides is greater, than the third side.]

adding above inequalities,

$2(AB + BC + CD + DA) > 2(AC + BD)$

$(AB + BC + CD + DA) > (AC + BD)$

4. *(b)* Let h be the height of the parallelogram. Then, clearly $h < q$

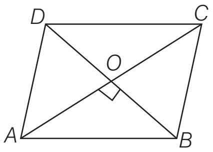
$\therefore \quad R = p \times h < p \times q = S$

$R < S$

5. *(d)* Two parallelograms on the same base and between the same parallels are equal in area. So, the ratio of their areas is $1:1$.

6. *(a)* As $ABCD$ is a rhombus.

So, $AO = OC = \frac{1}{2} AC$



$BO = OD = \frac{1}{2} BD$, $\angle AOB = 90°$

$\therefore \quad AB^2 = OA^2 + OB^2$

$AB^2 = \frac{AC^2}{4} + \frac{BD^2}{4}$

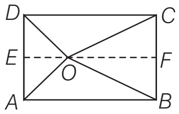
$\Rightarrow \; 4AB^2 = AC^2 + BD^2$

7. *(d)* Draw $EF \parallel AB$.

In right angled ΔEOA and ΔOCF,

$OA^2 = OE^2 + AE^2 \quad \ldots \text{(i)}$

and $\quad OC^2 = OF^2 + CF^2 \quad \ldots \text{(ii)}$



On adding Eqs. (i) and (ii), we get

$\therefore \; OA^2 + OC^2 = OE^2 + AE^2 + OF^2 + CF^2 \quad \ldots\text{(iii)}$

In right angled ΔDEO and ΔOBF,

$OD^2 = OE^2 + DE^2 \quad \ldots \text{(iv)}$

$OB^2 = OF^2 + BF^2 \quad \ldots\text{(v)}$

On adding Eqs. (iv) and (v), we get

$\Rightarrow$ Area of || gm $ABCD$

$\Rightarrow \; OD^2 + OB^2 = OE^2 + OF^2 + DE^2 + BF^2 \quad \ldots\text{(vi)}$

As $\; FB = EA \;$ and $\; DE = CF$

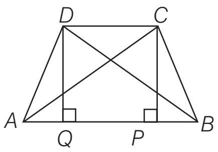
$\therefore \quad OD^2 + OB^2 = OE^2 + OF^2 + CF^2 + AE^2 \; \ldots\text{(vii)}$

Here, from Eqs. (iii) and (vii), we get

$OA^2 + OC^2 = OD^2 + OB^2$

8. *(a)* Draw $DQ \perp AB$ and $CP \perp AB$.

In ΔABD, $\angle A$ is acute.



So, $BD^2 = AD^2 + AB^2 - 2AB \cdot AQ \quad \ldots\text{(i)}$

In ΔABC, $\angle B$ is acute.

So, $AC^2 = BC^2 + AB^2 - 2AB \cdot BP \quad \ldots\text{(ii)}$

Adding Eqs. (i) and (ii), we get

$\therefore \quad AC^2 + BD^2 = (BC^2 + AD^2) + 2AB\,(AB - BP - AQ)$

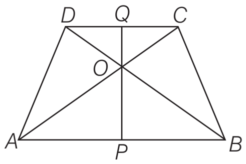
$= (BC^2 + AD^2) + 2AB \cdot PQ$

$[\because \; PQ = AB - AQ - BP]$

$= BC^2 + AD^2 + 2AB \cdot CD$

$[\because PQ = DC]$

9. *(c)* Let PQ be the perpendicular distance between the two parallel sides AB and CD. $\quad [\because PO = 2OQ]$



So, $\dfrac{\text{Area of } \Delta AOB}{\text{Area of } \Delta COD} = \dfrac{\frac{1}{2} AB \times OP}{\frac{1}{2} CD \times OQ}$

$= \dfrac{2CD \times 2OQ}{CD \times OQ} = \dfrac{4}{1}$ or $4:1$

10. *(b)* Area of parallelogram $ABCD$

$= 2$ (Area of ΔADE) + (Area of rectangle $AECH$)

$= 2 \times \frac{1}{2} DE \times AE + AE \times EC$

$= DE \times AE + AE \times EC$

$= AE\,(DE + EC) = AE \times [DC]$

$\because \quad \dfrac{DE}{EC} = \dfrac{1}{2} \Rightarrow \dfrac{DE}{EC} + 1 = \dfrac{1}{2} + 1$

$\Rightarrow \quad \dfrac{DE + EC}{EC} = \dfrac{3}{2} \quad \Rightarrow \quad \dfrac{DC}{EC} = \dfrac{3}{2}$

Area of parallelogram

$ABCD = \dfrac{FE}{3} \times \dfrac{3EC}{2} = \dfrac{FE \times EC}{2}$

$[\because FE = 3AE]$

$= \dfrac{\text{Area of rectangle } EFGC}{2}$

$\therefore$ Required ratio $= 1:2$

11. *(a)* Join AC and BD.

As $\quad AB > BC$

$\Rightarrow \quad \angle ACB > \angle BAC \quad \ldots\text{(i)}$

Also, $\quad AD > DC$

$\Rightarrow \quad \angle ACD > \angle CAD \quad \ldots\text{(ii)}$

On adding Eqs. (i) and (ii), we get

$\therefore \; \angle ACB + \angle ACD > \angle BAC + \angle CAD$

$\Rightarrow \angle C > \angle A$, similarly $\angle D > \angle B$

12. *(b)* $\angle COD = 105°$

$\therefore \angle BOA = 105°$

[vertically opposite angles]

$\therefore \angle AOD = \angle COX = \lambda$

$\Rightarrow 105° + 105° + x + x = 360°$

$\Rightarrow 2x = 150°$

$\Rightarrow x = 75°$

Also, $\angle OCX = \frac{1}{2} \times \angle C = \frac{1}{2} \times 90° = 45°$

In ΔOCX,

$\angle OCX + \angle COX + \angle OXC = 180°$

[by angle sum of property of a triangle]

$\Rightarrow \quad 45° + 75° + \angle OXC = 180°$

$\Rightarrow \angle OXC = 180° - 120° = 60°$

$\Rightarrow \quad x = 60°$

13. *(a)* $EB \parallel DL$ and $ED \parallel BL$

$\Rightarrow \quad EBLD$ is a parallelogram.

$\therefore \quad BL = ED = \frac{1}{2} AD = \frac{1}{2} BC = LC$

Also, in ΔDCL and ΔFBL,

$LC = BL$, $\angle DLC = \angle FLB$

$\because$ [vertically opposite angles]

and $\angle CDL = \angle BFL$ [alternate angles]

$\therefore \quad \Delta DCL \cong \Delta FBL \quad$ [By AAS rule]

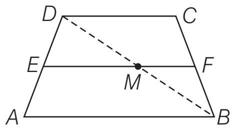
$\therefore \quad DC = BF \;$ and $DL = FL$

$\therefore \quad BF = DC = AB$

$\Rightarrow \quad 2AB = 2DC$

$\Rightarrow \quad AF = 2DC \quad [\because BF = AB]$

14. *(b)* Join BD, cutting EF at M.



So, M is mid-point of BD.

[$\because E$ is mid-point of AD and $EM \parallel AB$]

$\therefore \quad EM = \frac{1}{2} AB \quad \ldots\text{(i)}$

Similarly, F is mid-point of BC in ΔBCD.

$\therefore \quad FM = \frac{1}{2} DC \quad \ldots\text{(ii)}$

$\therefore \quad EF = EM + MF = \frac{1}{2}(AB + DC)$

[from Eqs. (i) and (ii)]

15. *(a)* $\angle CRQ = \angle CBA = 90° \Rightarrow QR \parallel AB$

In ΔABC, Q is the mid-point of AC and $QR \parallel AB$.

So, R is mid-point of BC.

Similarly, P is the mid-point of DC.

$\therefore \quad DP = PC$

16. *(d)* Number of diagonals of a polygon of 8 sides $= \dfrac{n\,(n-1)}{2} - n$

$= \dfrac{8\,(8-1)}{2} - 8 = 20$

17. *(c)* Sum of all angles of a pentagon

$= (5-2) \times 180° = 3 \times 180° = 540°$

Let the angle be $x, 2x, 3x, 5x$ and $9x$.

$\therefore \quad x + 2x + 3x + 5x + 9x = 540°$

$20x = 540° \quad \Rightarrow \quad x = 27°$

$\therefore \quad$ Largest angle $= 9x = 9 \times 27° = 243°$

18. *(c)* By condition,

$$\frac{\text{Interior angle of a regular polygon}}{\text{Exterior angle a regular polygon}} = \frac{5}{1}$$

$$\Rightarrow \frac{\frac{(n-2)}{n} \times 180°}{\frac{360°}{n}} = \frac{5}{1} \Rightarrow \frac{(n-2)}{2} = \frac{5}{1}$$

$$\Rightarrow \quad n - 2 = 10 \quad \Rightarrow \quad n = 12$$

19. *(b)* Here, $n = 6$

Sum of interior angles of a hexagon

$= (n-2)180° = (6-2) \times 180°$

$= 4 \times 180° = 720°$

Number of angles = 6

$\therefore$ Each angle of a regular hexagon

$$= \frac{720°}{6} = 120°$$

20. *(c)* Each interior angle of a regular polygon

$$= \frac{(n-2) \times 180°}{n}$$

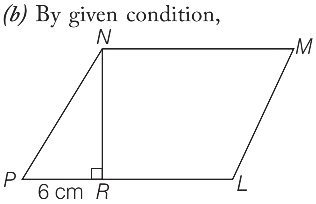
$$\therefore \frac{(n-2) \times 180°}{n} = 150° \quad \text{(given)}$$

$$(n-2)\ 180 = n \times 150$$

$$\Rightarrow \quad 30n = 360 \quad \Rightarrow \quad n = \frac{360}{30}$$

$$\Rightarrow \quad n = 12$$

21. *(b)* By given condition,



Area of parallelogram $LMNP$

$= 6 \times$ Area of ΔNPR

$$\therefore \; NR \times PL = 6 \times \frac{1}{2} \times NR \times PR$$

$$\Rightarrow \quad PL = 3PR$$

$$\Rightarrow PR + RL = 3PR \; [\because PL = PR + RL]$$

$$\Rightarrow \quad RL = 2PR = 2 \times 6 = 12 \text{ cm}$$

22. *(d)* $ABCD$ is a trapezium in which $AB || DC$ and M, N are the mid-points of AD and BC.

Hence, $MN || AB$ and $MN || DC$.

In ΔACB, ON passes through the mid-point N of BC and $ON || AB$

$$\therefore \quad ON = \frac{1}{2} AB = \frac{1}{2} \times 12 \text{ cm} = 6 \text{ cm}$$

And, $MO = MN - ON = (14 - 6)$ cm

$= 8$ cm

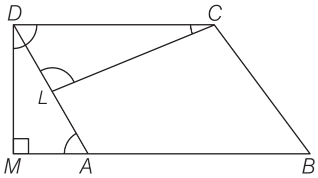
Again, MO passes through the mid-point M of AD and $MO || DC$

$$\therefore \quad MO = \frac{1}{2} DC = \frac{1}{2} CD$$

Hence, $CD = 2(MO) = 2(8) = 16$ cm

23. *(a)* $\Delta ADM \sim \Delta DCL$

Since, $\angle DMA = \angle CLD = 90°$ and $\angle CDL = \angle DAM$ as they are alternate angles



$(AD / CD) = (DM / CL)$

$$\Rightarrow \quad AD = \frac{DM \times CD}{CL} = \frac{12 \times 16}{15}$$

$= 12.8$ units

24. *(a)* Evidently $DP = 14 - 9 = 5$ cm

From ΔBPC,

$$BC^2 = 15^2 - 9^2 = 12^2$$

$$\Rightarrow \quad BC = 12 \text{ cm}$$

From $\Delta APD, AP^2 = AD^2 + DP^2$

$$= 12^2 + 5^2 = 169$$

$$\Rightarrow \quad AP = 13 \text{ cm}$$

In $\Delta ABP, AP < AB < BP$.

Therefore, $\gamma < \beta < \alpha$ (i.e.) $\alpha > \beta > \gamma$

25. *(d)* Since, $AB = BC$

$\therefore \angle BAC = \angle BCA = 62°$

Also as $AB || CD$ and AC transversal

So, $\angle BAC = 62° = \angle ACD$

(alternate interior angles)

$\therefore \quad \angle ACD = 62°$

26. *(c)* One angle of the pentagon is 140°.

Since, the remaining angles are in the ratio 1 : 2 : 3 : 4.

$\therefore$ Let the remaining angles be $x°$, $(2x)°$, $(3x)°$ and $(4x)°$.

But the sum of interior angles of a pentagon

$$(2 \times 5 - 4) \times 90 = 6 \times 90° = 540°$$

$$\therefore \quad 140 + x + 2x + 3x + 4x = 540$$

$$\Rightarrow \quad 10x = 400 \quad \Rightarrow \quad x = 40$$

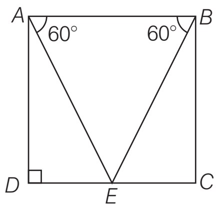
$\therefore$ The angles of the pentagon are 140°, 40°, 80°, 120° and 160°.

Hence, the size of the greatest angle

$= 160°$

27. *(a)* Given that, $AB || DC$ and $AD || BC$

In ΔABE,



$$\angle EAB = \angle ABE = 60°$$

$$\Rightarrow \quad \angle AEB = 60°$$

So, ΔABE is an equilateral triangle.

Now, perimeter of $\Delta ABE = 6$

$$\Rightarrow \quad AB + BE + EA = 6$$

$$\Rightarrow \quad AB = 2 \text{ units}$$

Now, In ΔADE and ΔBCE,

$\angle ADE = \angle BCE$ [each 90°]

$\angle DAC = \angle CBE$ [each 30°]

and $AE = BE$

$\therefore \Delta ADE \cong \Delta BCE$ [By AAS]

$\Rightarrow DE = CE$ [By CPCT]

In ΔADE by pythagoras theorem,

$$AE^2 = AD^2 + ED^2$$

$$\Rightarrow \quad 4 = AD^2 + 1$$

($\because E$ is mid-point of CD)

$$\Rightarrow \quad AD = \sqrt{3} \text{ units}$$

Hence, area of quadrilateral

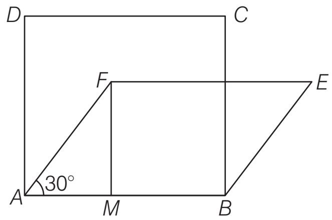
$ABCD = AB \times AD$

$= 2 \times \sqrt{3} = 2\sqrt{3}$ sq units

28. *(b)* $ABCD$ is square and $ABEF$ is a rhombus.

$$\frac{FM}{AF} = \sin 30° = \frac{1}{2}$$

$$\therefore FM = \frac{AF}{2}, \; AF = AB$$



Area of square $= a^2 \quad (AB = AD = a)$

Area of rhombus $= \dfrac{a \times a}{2} \quad \left(FM = \dfrac{a}{2}\right)$

(Area of rhombus = base × height)

$$\therefore \frac{\text{Area of square}}{\text{Area of rhombus}} = \frac{2}{1}$$

29. *(c)* In ΔARB, P is the mid-point of AB and $PD || BR$.

$\Rightarrow D$ is the mid-point of AR.

$\because ABCD$ is a parallelogram.

$\Rightarrow DC || AB \Rightarrow DQ || AB$

Thus, in ΔRAB, D is the mid-point of AR and $DQ || AB$.

$\therefore Q$ is the mid-point of $RB \Rightarrow BR = 2BQ$

30. *(a)* In ΔBDC, Q and R are the mid-points of BD and CD respectively.

$$\therefore QR || BC \text{ and } QR = \frac{1}{2} BC$$

Similarly, $PS || BC$ and $PS = \dfrac{1}{2} BC$

$\therefore \; PS || QR$ and $PS = QR$

$$\left[\text{each equal to } \frac{1}{2} BC\right]$$

Similarly, $PQ || SR$ and $PQ = SR$

[each equal to $1/2\ AD$]

$\therefore PS = QR = SR = PQ$ $\quad [\because AD = BC]$

Hence, $PQRS$ is a rhombus.

31. *(b)* $AB || DC$ and EC is a transversal

$\Rightarrow \angle BEC = \angle ECD$

$\Rightarrow \angle BEC = \angle ECB [\because \angle ECD = \angle ECB]$

$\Rightarrow \quad EB = BC \quad \Rightarrow \quad AE = AD$

Now, $AE = AD \Rightarrow \angle ADE = \angle AED$

$\Rightarrow \angle ADE = \angle EDC$

[$\because$ alternate Int. angles]

$\therefore DE$ bisects $\angle ADC$

Again, $\angle ADC + \angle BCD = 180°$

[Co. Int. angles]

$\Rightarrow \frac{1}{2}\angle ADC + \frac{1}{2}\angle BCD = 90°$

$\Rightarrow \quad \angle EDC + \angle DCE = 90°$

But, $\angle EDC + \angle DEC + \angle DCE = 180°$

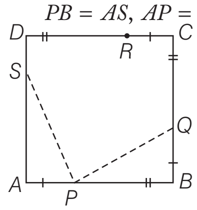
[$\because$ sum of the $\angle S$ of a Δ is 180°]

$\therefore \quad \angle DEC = 180° - 90°$

$\therefore \quad \angle DEC = 90°$

32. *(d)* In ΔAPS and ΔPBQ, [given]

$PB = AS,\ AP = BQ$



and $\quad \angle A = \angle B = 90°$

$\therefore \quad \Delta APS \cong \Delta BQP$ (by SAS rule)

$\therefore \quad SP = PQ,\ \angle SPA = \angle BQP$

and $\angle ASP = \angle BPQ$

$\angle SPQ = 180° - \angle APS - \angle BPQ$

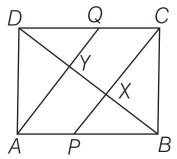
$= 180° - (\angle APS + \angle ASP)$

$= 180° - 90° = 90°$

33. *(c)* As $ABCD$ is a || gm

$\therefore AB || DC$ and $AB = DC$

and $\frac{1}{2}AB = \frac{1}{2}DC$



$\Rightarrow \quad AP = QC$

$\therefore APCQ$ is a || gm

$\Rightarrow \quad AQ || PC$

In $\Delta BAY, XP || AY$

and P is the mid-point of AB

$\therefore BX = YX$

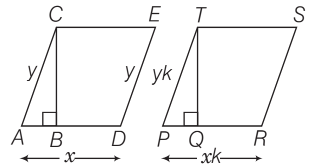
Similarly, in $\Delta DXC, DY = YX$

$\therefore BX = XY = DY$

34. *(c)* If a parallelogram and a rectangle stand on the same base and on the same side of the base with the same height, then perimeter of parallelogram is greater than perimeter of rectangle.

$\therefore \quad l_1 > l_2$

35. *(c)* Let the sides of two parallelograms are x, y and xk, yk, respectively.



Since, sides of two parallelogram are in the ratio $1 : k$.

$\therefore \quad \Delta ABC \sim \Delta PQT$

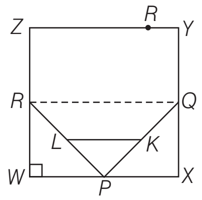
$\therefore \quad \frac{AC}{PT} = \frac{BC}{QT}$

$\Rightarrow \quad \frac{BC}{QT} = \frac{y}{yk} = \frac{1}{k}$

Let $\quad BC = z$ and $QT = zk$

$\therefore$ Ratio of areas of two similar parallelograms $= \frac{x \times z}{xk \times zk} = \frac{1}{k^2}$ or $1 : k^2$

36. *(b)* Area of $(\Delta PRQ) = \frac{1}{2}$ Area of $(WXQR)$



$= \frac{1}{2}\left[\frac{1}{2} \text{Area of } (WXYZ)\right]$

$= \frac{1}{4}$ Area of $(WXYZ)$ $\quad$ ...(i)

$\frac{\text{Area of } (\Delta PRQ)}{\text{Area of } (\Delta PLK)} = \frac{RP^2}{LP^2}$

(by properties of similar triangle)

$\Rightarrow \quad \frac{\text{Area of } (\Delta PRQ)}{\text{Area of } (\Delta PLK)} = \frac{(2LP)^2}{LP^2}$

[$\because L$ and K are the mid-point of PR and PQ]

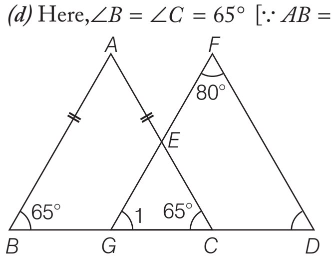
$\Rightarrow$ Area of $(\Delta PRQ) = 4$Area of (ΔPLK)

$\Rightarrow \frac{1}{4}$Area of $(WXYZ)$

$= 4$Area of (ΔPLK) [from Eq. (i)]

$\Rightarrow \quad \frac{\text{Area of } (\Delta PLK)}{\text{Area of } (WXYZ)} = \frac{1}{16}$

37. *(d)* Here, $\angle B = \angle C = 65°$ [$\because AB = AC$]



$\angle 1 = \angle B = 65°$ (corresponding angles)

In ΔFGD,

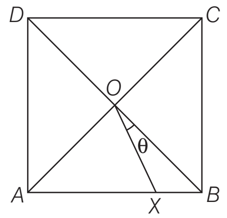
$\angle 1 + \angle F + \angle D = 180°$

(by angle sum of property of a triangle)

$\Rightarrow 65° + 80° + \angle D = 180°$

$\Rightarrow \quad \angle D = 35°$

38. *(a)* Let $\angle XOB = \theta$



In ΔOXB,

$\angle XOB + \angle OBX + \angle OXB = 180°$

(by angle sum of property of a triangle)

$\Rightarrow \quad \theta + 45° + \angle OXB = 180°$

$\Rightarrow \angle OXB = 180° - 45° - \theta = 135° - \theta$

Here, $\quad \angle OXA + \angle OXB = 180°$

(linear pair)

$\Rightarrow \quad \angle OXA + 135° - \theta = 180°$

$\Rightarrow \quad \angle OXA = 45° + \theta$

In ΔOXA, $\quad AO = AX$ $\quad$ (given)

$\therefore \angle OXA = \angle AOX = 45° + \theta$

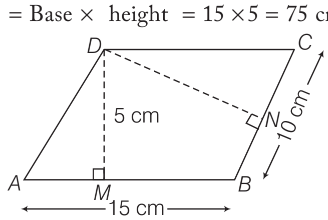
Since, $\angle AOX + \angle XOB = 90°$

$\Rightarrow \quad 45° + \theta + \theta = 90°$

$\Rightarrow \quad 2\theta = 45° \quad \Rightarrow \quad \theta = 22.5°$

39. *(b)* Area of parallelogram

$=$ Base $\times$ height $= 15 \times 5 = 75\ \text{cm}^2$



Area of parallelogram $=$ Base $\times$ Height

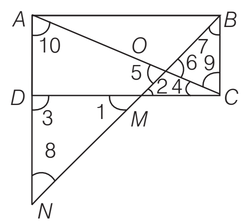
$= 10 \times DN$

$\therefore \quad 10 \times DN = 75$

$\Rightarrow \quad DN = \frac{75}{10} = 7.5$ cm

40. *(b)* In ΔDMN and ΔBMC,

$DM = MC$ $\quad$ (given)



$\angle 1 = \angle 2$ (vertically opposite angle)

$\angle 3 = \angle 4 + \angle 9$

(alternate interior angle)

$\Delta DMN \cong \Delta BMC$ $\quad$ (as ASA rule)

$DN = BC = AD$ (by CPCT)

So, $AN = 2BC \Rightarrow \frac{AN}{BC} = \frac{2}{1}$ ...(i)

In ΔOAN and ΔOBC,

$\angle 5 = \angle 6$ (vertically opposite angle)

$\angle 7 = \angle 8$ (alternate interior angle)

$\angle 9 = \angle 10$ (altenate angle)

$\therefore \Delta OAN \sim \Delta OBC$ (by AAA similarity)

So, the sides will be in same ratio

$\frac{AN}{BC} = \frac{ON}{OB} \Rightarrow \frac{2}{1} = \frac{ON}{OB}$ or 2 : 1

[from Eq. (i)]

41. *(a)* As $ABCD$ is a parallelogram.

$\therefore AB || CD$ and $AB = CD$

Draw $QR \perp AB$

In ΔDOQ and ΔPOB,

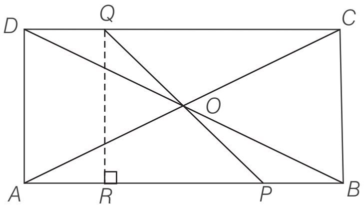
$\angle ODQ = \angle OPB$

$\angle OQD = \angle OBP$

and $OD = OB$

$\therefore \Delta OQD \cong \Delta OPB$ [By AAS]

$\Rightarrow PB = DQ$

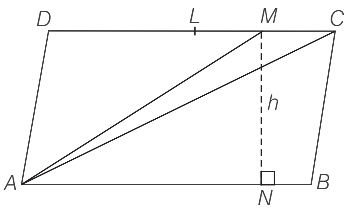


Now, ar (|| gm $ABCD$) $= QR \times AB$

$= QR \times (AP + PB) = QR \times (AP + DQ)$

$= 2 \times ar$ (quadrilateral $APDQ$)

42. *(a)* Draw $MN \perp AB$ and let $MN = h$



Now, ar (trapezium $ABCM$)

$= ar(\Delta\ AMC) + ar(\Delta ABC)$

$\Rightarrow \frac{1}{2}(MC + AB) \times h = 4 + \frac{1}{2}\ ar$

(parallelogram $ABCD$)

$\Rightarrow \frac{1}{2}\left(\frac{AB}{4} + AB\right) \times h = 4 + \frac{1}{2} \times AB \times h$

$\left[\because MC = \frac{AB}{4}\right]$

$\Rightarrow \frac{AB \times h}{8} = 4 \Rightarrow AB \times h = 32$

$\Rightarrow ar$ (||gm $ABCD$) $= 32$ cm^2

43. *(b)* I. All squares are parallelograms.

II. No parallelogram is a trapezium.

III. All squares are rhombuses and also rectangles.

IV. All rhombuses are parallelograms.

So, statements (III) and (IV) are correct.

44. *(a)* In a rhombus $ABCD$, if AC and BD are two diagonals then

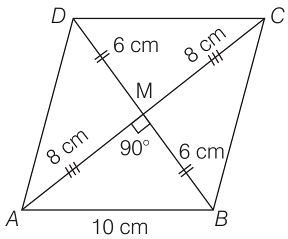
$AB^2 + BC^2 + CD^2 + AD^2$

$= AC^2 + BD^2$

$\Rightarrow (10)^2 + (10)^2 + (10)^2 + (10)^2$

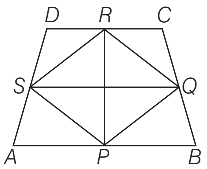
$= (16)^2 + (12)^2$

$\Rightarrow 400 = 400$



Hence, both I and II are true but III is false.

45. *(c)* $PQRS$ can be shown parallelogram, so the diagonal PR and SQ bisect each other.



II. Area (ΔRSQ)

$= \frac{1}{2}$ Area ($SQCD$) ...(i)

and area $\Delta(PSQ) = \frac{1}{2}$ Area ($ABQS$)...(ii)

On adding Eqs. (i) and (ii), we get

Area (RSQ) + Area (PSQ) $= \frac{1}{2}$ [Area ($SQCD$) + Area ($ABQS$)]

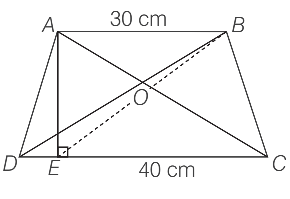
$\Rightarrow$ Area ($PQRS$) $= \frac{1}{2}$ Area ($ABCD$)

Hence, both statements are true.

Solutions (Q.Nos. 46-47)

Given, $AO : OC = BO : OD$

and $AB = 30$ cm and $CD = 40$ cm



$\therefore \Delta AOB \sim \Delta COD$

$\Rightarrow \frac{OC}{OA} = \frac{AB}{CD} = \frac{30}{40} = \frac{3}{4}$

$\therefore \angle OAB = \angle OCD$

and $\angle OBA = \angle ODC$

It means $DC || AB$.

So, it is a trapezium.

Area of trapezium

$ABCD = \frac{1}{2}(AB + CD) \times AE$

$\Rightarrow 1050 = \frac{1}{2}(30 + 40) \times AE$

$\Rightarrow AE = 30$ cm

46. *(b)* In right ΔEAB, $EB = \sqrt{AE^2 + AB^2}$

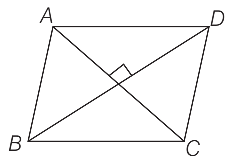
$= \sqrt{30^2 + 30^2} = 30\sqrt{2}$ cm

47. *(c)* Area of $\Delta\ ADC = \frac{1}{2} \times CD \times AE$

$= \frac{1}{2} \times 40 \times 30 = 600$ cm^2

48. *(c)* $ABCD$ is a rhombus.

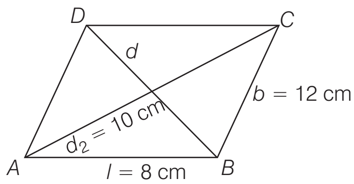
$\therefore AB = BC = CD = DA$



and diagonals bisect each other at right angles.

49. *(d)* In parallelogram,

$d^2 + d_2^2 = 2(l^2 + b^2)$



$\therefore d^2 + (10)^2 = 2(64 + 144)$

$\Rightarrow d^2 = 2 \times 208 - 100$

$\Rightarrow d^2 = 416 - 100 = 316$

$\Rightarrow d = \sqrt{316} \Rightarrow d = 17.76$ cm

$\therefore d > 12$

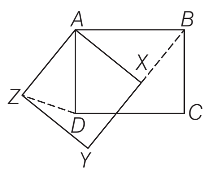
50. *(d)* In ΔABX and ΔADZ,

$AB = AD$ (side of a square $ABCD$)

and $AX = AZ$ (side of a square $AXYZ$)

Let $\angle BAX = \theta$

$\therefore \angle XAD = 90° - \theta$



Also, $AXYZ$ is a square.

$\therefore \quad \angle ZAX = 90°$

$\Rightarrow \quad \angle ZAD + \angle XAD = 90°$

$\Rightarrow \quad \angle ZAD = 90° - (90° - \theta) = \theta$

i.e. $\quad \angle BAX = \angle ZAD$

$\therefore \quad \Delta ABX \cong \Delta ADZ$

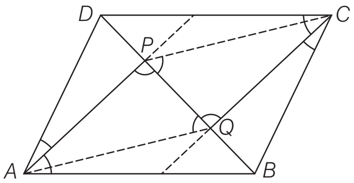
$\therefore \quad BX = DZ$ (by CPCT)

51. *(b)* Since, line segment AP and CQ bisects the $\angle A$ and $\angle C$, respectively.

Then, $\quad AP || CQ$

Now, in ΔAPQ and ΔCQP,

$\because \quad AP || QC$



$\because \quad \angle APQ = \angle PQC$ (alternate angle)

$PQ = PQ$ (common)

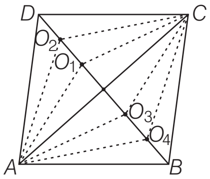
Also, $\quad PC || AQ$

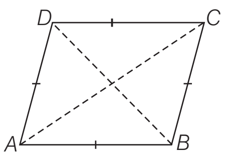
$\therefore \quad \angle CPQ = \angle PQA$ (alternate angle)

$\therefore \quad \Delta APQ \sim \Delta CQP$ (by ASA)

Hence, ΔAPQ is similar to ΔPCQ.

52. *(a)* We know that, in a rhombus $ABCD$ diagonals bisect each other at point O which means that the distance of O from four vertices of rhombus *i.e.*, A, B, C and D are equal. Even, if we take any fixed point on diagonal BD of rhombus $ABCD$ and join with vertices A and C, we get a point which is equidistant from A and C. (by property of congruent triangle). Hence, the locus of a point in rhombus $ABCD$ which is equidistant from A and C is a fixed point on diagonal BD.





53. *(c)* In parallelogram $PQRS$

$\angle P + \angle S = 180°$

$\Rightarrow \quad \frac{1}{2}\angle P + \frac{1}{2}\angle S = 90°$

$\Rightarrow \quad \angle DPS + \angle DSP = 90°$

In ΔDPS,

$\angle SDP = 180° - (\angle DPS + \angle DSP) = 90°$

$\therefore \angle ADC = \angle SDP = 90°$

(vertically opposite $\angle S$)

Similarly,

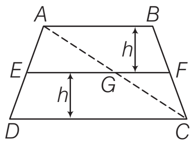
$\angle DAB = \angle ABC = \angle BCD = 90°$

Hence, $ABCD$ is a rectangle.

54. *(c)* Join AC. In ΔACD, $EG || DC$ and E and G are mid-points of AD and AC, respectively.

In ΔADC, $EG || DC$

$\therefore \quad EG = \frac{1}{2} DC = \frac{3}{2}$



Similarly, in ΔABC

$GF = \frac{1}{2} AB = 1$

$EF = EG + GF = 1 + \frac{3}{2} = \frac{5}{2}$

$\therefore$ Area of trapezium $= \frac{1}{2}$

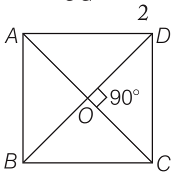
(sum of parallel sides × height)

Now, required ratio $= \dfrac{\text{Area of } ABFE}{\text{Area of } EFCD}$

$$= \frac{\frac{1}{2}\left(2 + \frac{5}{2}\right) \times h}{\frac{1}{2}\left(3 + \frac{5}{2}\right) \times h} = \frac{9}{11} \text{ or } 9 : 11$$

55. *(b)* Here, $\quad OD = \frac{BD}{2} = \frac{4.8}{2} = 2.4$

$\Rightarrow \quad OC = \frac{AC}{2} = \frac{1.4}{2} = 0.7$



Since, in rhombus diagonal bisect at 90°.

Then, in ΔODC, $OD^2 + OC^2 = CD^2$

$\Rightarrow \quad CD = \sqrt{OD^2 + OC^2}$

$= \sqrt{(2.4)^2 + (0.7)^2}$

$CD = \sqrt{6.25} \Rightarrow CD = 2.5 = \frac{5}{2}$

$\therefore$ Perimeter of rhombus

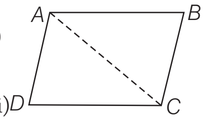
$= 4a = 4 \times \frac{5}{2} = 10$ cm

56. *(c)* Join AC.

Now, in ΔABC

$\because AB = BC$ (given)

$\therefore \angle BAC = \angle BCA$



...(i)

(angles opposite to equal side)

In ΔADC,

$\because \quad CD > AD$

$\therefore \quad \angle DAC > \angle DCA$ ...(ii)

(since in a triangle, angle opposite to greater side is bigger than the angle opposite to smaller side)

On adding Eqs. (i) and (ii), we get

$\angle BAD > \angle BCD$

57. *(b)* In ΔAPS and ΔDSR

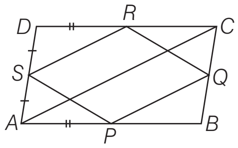
$AP = DR$ [$\because$ P and R are mid points]

$AS = DS$ [$\because$ S is the mid point]

and $SP = SR$ [$\because$ P or RS us a]

$\therefore \Delta APS \cong \Delta DRS$ [By SSS]

$\Rightarrow$ Area of ΔAPS = Area of ΔDSR



$\because \quad AS = SD$ and $AP = DR$

In ΔABC and ΔPBQ

$\angle ABC = \angle PBQ$ [common]

$\angle BAC = \angle BPQ$ [$\because PQ || AC$]

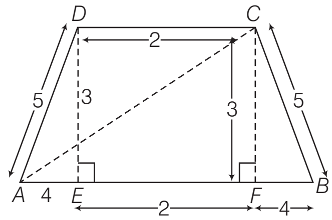
$\therefore \Delta ABC \sim \Delta PBQ$

$\dfrac{ar(\Delta ABC)}{ar(\Delta PBQ)} = \dfrac{AB^2}{PB^2} \Rightarrow \left[\dfrac{2PB}{PB}\right]^2$

$\therefore \quad ar(\Delta ABC) = 4\, ar(\Delta BPQ)$

Hence, only statement II is correct.

58. *(b)*



In ΔBCF,

By pythagoras theorem,

$(5)^2 = (3)^2 + (BF)^2 \Rightarrow BF = 4$ cm

$\therefore \quad AB = 2 + 4 + 4 = 10$ cm

Now in ΔACF, $AC^2 = CF^2 + FA^2$

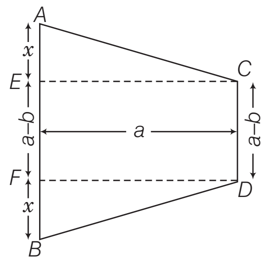
$\Rightarrow \quad AC^2 = 3^2 + 6^2$, $AC = \sqrt{45}$ cm

Similarly, $BD = \sqrt{45}$ cm

$\therefore$ Sum of diagonal $= 2 \times \sqrt{45}$

$= 2 \times 3\sqrt{5} = 6\sqrt{5}$ cm

59. *(d)* Since, they are symmetrically lying on horizontal plane.



$\therefore \quad AC = BD \therefore AE = BF = x$ (say)

Now, $\quad AB = (a - b) + 2x$

i.e. $\quad a + b = a - b + 2x$

$\Rightarrow \quad 2b = 2x$

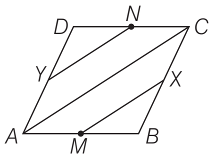
$\therefore \quad x = b$

Now, in ΔACE, $x^2 + a^2 = AC^2$

$\Rightarrow \quad AC^2 = b^2 + a^2$

$\therefore \quad AC = \sqrt{b^2 + a^2}$

60. *(b)* Given, $ABCD$ is a parallelogram. X and Y are mid-points of BC and AD, respectively. M and N are the mid-points of AB and CD, respectively.



Now, join AC.

In ΔABC, M and X are mid-points of AB and BC.

$\therefore \quad MX \parallel AC$ and $MX = \frac{1}{2} AC \quad$ ...(i)

In ΔADC, Y and N are mid-points of AD and CD.

$\therefore \quad YN \parallel AC$ and $YN = \frac{1}{2} AC \quad$ ...(ii)

From Eqs. (i) and (ii), we get

$MX \parallel YN$

So, statement I is not correct.

II Obviously, straight lines AC, BD, XY and MN meet at a point. So, statement II is correct.

61. *(c)* Given, each interior angle of a regular polygon = 135°

$\because$ Each exterior angle

= 180° − interior angle

= 180° − 135° = 45°

$\because$ Number of sides in the polygon (n)

$= \frac{360°}{\text{Each exterior angle}} = \frac{360°}{45°} = 8$

$\because$ Number of diagonals of a polygon

$= \frac{n(n-1)}{2} - n = \frac{8(8-1)}{2} - 8$

$= \frac{8 \times 7}{2} - 8$

$= 28 - 8 = 20$

Hence, the number of diagonals of a polygon is 20.

62. *(a)* $\because$ Each interior angle of a regular polygon of n side

$= \frac{(n-2) \times 180°}{n}$

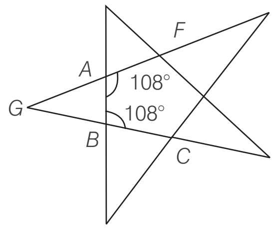
Here, $n = 5$

$\therefore$ Each interior angle of the pentagon

$= \frac{(5-2) \times 180°}{5} = \frac{3 \times 180°}{5}$

$= 3 \times 36° = 108°$

Since, GAF is a straight line.



$\therefore \quad \angle GAF = 180°$

$\Rightarrow \quad \angle GAB + \angle BAF = 180°$

$\Rightarrow \quad \angle GAB + 108° = 180°$

$\Rightarrow \quad \angle GAB = 180° - 108° = 72°$

Similarly,

$\angle GBA = 180° - 108° = 72°$

In ΔABG,

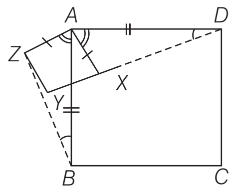
$\angle GAB + \angle ABG + \angle BGA = 180°$

$\Rightarrow \quad 72° + 72° + \angle BGA = 180°$

$\Rightarrow \quad \angle BGA = 180° - 144°, \quad \angle BGA = 36°$

63. *(c)* Here, $\angle DAB = \angle XAZ = 90°$

$\Rightarrow \angle DAB - \angle XAB = \angle XAZ - \angle XAB$



$\Rightarrow \quad \angle DAX = \angle BAZ$

Now, in ΔDAX and ΔBAZ,

$AD = AB$ (sides of squares)

$\Rightarrow \quad \angle DAX = \angle BAZ$ (proved earlier)

$\Rightarrow \quad AX = AZ$ (sides of square)

$\Rightarrow \quad \Delta DAX \cong \Delta BAZ$ (SAS criterial)

$\Rightarrow \quad BZ = DX$ (by CPCT)

and $\quad \angle ABZ = \angle ADX$ (by CPCT)

Hence, both statements are correct.

64. *(d)* Given as $\frac{PB}{PC} = \frac{1}{2} \quad$ ...(i)

$BQ \parallel DC \quad (\because AB \parallel DC)$

In ΔBPQ and ΔDPC

$\angle BPQ = \angle DPC$

($\because$ vertically opposite angles)

$\Rightarrow \quad \angle QBP = \angle PCD$

($\because$ alternate interior angle)

$\Rightarrow \Delta BPQ \sim \Delta DPC$ [by AA similarity]

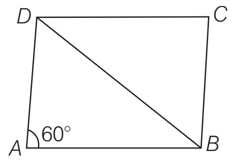
We know that area of two similar triangles are in the ratio of the square of the corresponding sides

$\frac{ar(\Delta BPQ)}{ar(\Delta DPC)} = \left(\frac{PB}{PC}\right)^2$

$\Rightarrow \frac{20}{ar(\Delta DPC)} = \left(\frac{1}{2}\right)^2 \quad$ [from Eq. (i)]

$\therefore \quad ar(\Delta DPC) = 80$ sq units

65. *(b)*



Let $AD = x$, then $AB = 2x$

using cosine rule in ΔABD, we have

$\cos A = \frac{AD^2 + AB^2 - BD^2}{2 \cdot AD \cdot AB}$

$\Rightarrow \cos 60° = \frac{x^2 + (2x)^2 - BD^2}{2.x.(2x)}$

$\Rightarrow \frac{1}{2} = \frac{5x^2 - BD^2}{4x^2}$

$\Rightarrow 2x^2 = 5x^2 - BD^2$

$\Rightarrow BD^2 = 3x^2$

$\Rightarrow BD = \sqrt{3}x = \sqrt{3}AD.$

66. *(b)* Given, each interior angle = 140°

Then, each exterior angle

$= (180° - 140°) = 40°$

$\therefore$ Number of sides

$= \frac{360°}{\text{Each exterior angle}} = \frac{360°}{40°} = 9$

Hence, the number of vertices of polygon is 9.

67. *(b)* Exterior angle of regular polygon

$= \frac{360°}{\text{Number of sides}}$

I. If exterior angle is 70°

Then, number of sides

$= \frac{360°}{70°} = \frac{36}{7}$ = A rational number

So, statement I is incorrect.

II. If number of sides ≥ 5, then

Exterior angle $= \frac{360°}{5} = 72°$

For number of sides ≥ 5, there is always an acute exterior angle.

So, statement II is correct.

68. *(d)* I. In regular polygon,

Sum of exterior angles = 360°

[which is always constant]

So, statements I is incorrect.

II. Sum of interior angles of m sides of regular polygon $= (m-2) \times 180°$

Sum of interior angles of m sides and n sides of regular polygon

$= (m-2) \times 180° + (n-2) \times 180°$

$= (m + n - 4) \times 180°$

So, statement II is incorrect.

Hence, neither I nor II is correct.

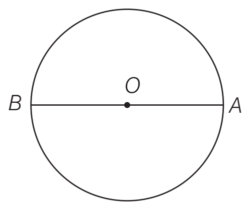
25

CIRCLE

Usually (3-6) questions have been asked from this chapter. Generally questions are asked from the topics related to theorem of circle, tangent to a circle and locus.

A circle is a curved figure consisting of all points in a plane that are at a fixed distance from a fixed point. The fixed point is called centre and the fixed distance is called radius of the circle.

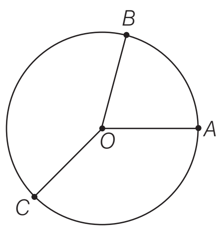
- Circles are simple closed curves which divide the plane into two regions : an interior and an exterior.
- In the given figure, O is the centre of circle, OA is the radius of the circle and AB is the diameter of the circle.



RADIUS

Radius is the fixed distance between the centre of the circle and the points lying on the circle.

Here, in the given figure OA, OB and OC all are having fixed distance, so they are known as radius of the circle. It is denoted by r.

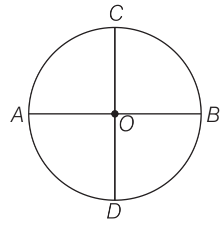


Diameter

Diameter is any line segment that passes through the centre of the circle and whose end points lie on the circle. The diameter of a circle is twice the radius. Here, AB and CD are the diameters of the circle.

$$AB = CD = 2(OA)$$
$$= 2(OB) = 2(OC) = 2(OD)$$

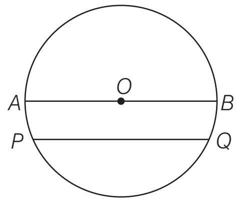
A circle can have an infinite number of diameters.



Chord

A chord is a line segment whose end points lie on the circle. A diameter is the longest chord in a circle.

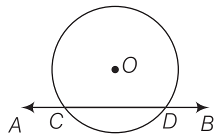
Here, in the given figure PQ is a chord and AB is the diameter of the circle which is the longest chord of that circle.



Secant

A line which intersect the circle at two points is called secant of the circle.

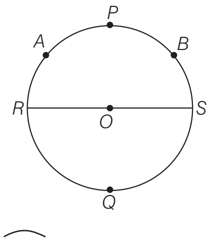
Here, in the figure AB is a secant cutting the circle at two distinct points as C and D.



Arc

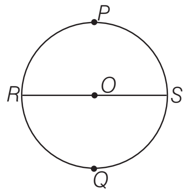
Any part of a circle between two points is called an arc of the circle. Any two points, say A and B of a circle divide it into two parts called arcs. The smaller arc APB is called minor arc and larger arc AQB is called major arc.

It is usually denoted by $\overset{\frown}{APB}$ and $\overset{\frown}{AQB}$.



Semi Circle

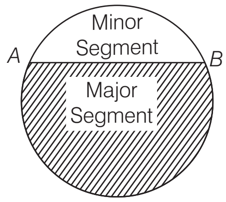
A diameter divides the circle into two equal parts. Each of the two is called a semi-circle.



In the above figure, $RPSR$ and $RQSR$ are semi-circles.

Segment

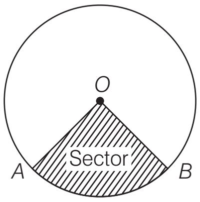
The area enclosed by an arc and its corresponding chord is called a segment of the circle.



- The segment containing the minor arc is called minor segment.
- The segment containing the major arc is called major segment.

Sector

The area enclosed by any two radii and the arc determined by the end points of the radii is called a sector of the circle.



Circumference

The length of the complete circle is called its circumference.

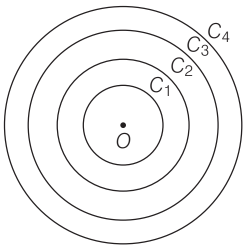
Circumference $= 2 \times \pi \times$ Radius of the circle

or

$= \pi \times$ Diameter of the circle

Concentric Circles

Two or more circles having the same centre are called concentric circles.

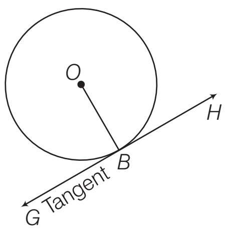


Here, in the given figure C_1, C_2, C_3 and C_4 are known as concentric circles as they have a common centre 'O'.

An infinite number of circles can be drawn with same centre.

Tangent

A tangent to a circle is a straight line that touches the circle at a single point.

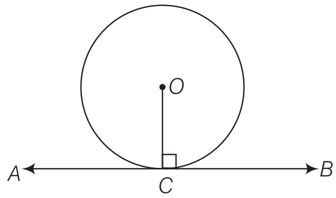


Here, in the given figure GBH is a tangent touching the circle at a single point B.

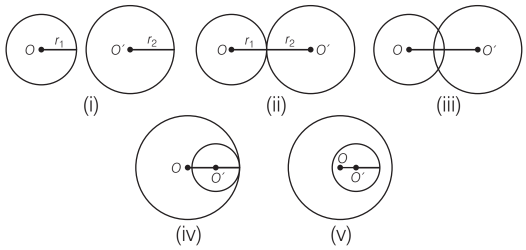
Length of tangent The distance between external point P from which tangent is drawn and the point of contact of the tangent is called length of tangent.

Properties of Tangent To a Circle

- Only two tangents can be drawn from a point outside the circle.
- Only one tangent can be drawn through a point lying on the circle.
- No tangent can be drawn from a point lying inside the circle.
- A tangent at any point of a circle is perpendicular to the radius through the point of contact.



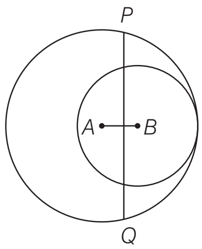
Relative Position of two Circles



Two circles will

(i) be disjoint, when $OO' > r_1 + r_2$.
(ii) be touching externally, when $OO' = r_1 + r_2$.
(iii) be intersecting when $OO' < r_1 + r_2$.
(iv) be touching internally, when $OO' = |r_2 - r_1|$
(v) one of the circle will lie inside the other, when $OO' < |r_2 - r_1|$.

EXAMPLE 1. Two circles with centres A and B touch each other internally, as shown in the figure given below. Their radii are 5 and 3 units, respectively. Perpendicular bisector of AB meets the bigger circle in P and Q. What is the length of PQ?



a. $2\sqrt{6}$ b. $\sqrt{34}$ c. $4\sqrt{6}$ d. $6\sqrt{2}$

Sol. c. Let PQ intersect AB at O. Join AP and AQ. Produce AB to intersect the circle at M.

Here, $AP = AM = 5$ units, $BM = 3$ units

$\therefore\ AB = AM - BM = 5 - 3 = 2$ units

PQ bisects AB perpendicularly.

$\therefore\ AO = OB = 1$ unit

Now, in right ΔAOP,

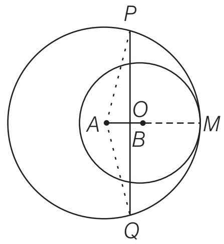
$$(PA)^2 = (OA)^2 + (OP)^2$$

$$\Rightarrow\ (5)^2 = (1)^2 + (OP)^2$$

$$\Rightarrow\ 24 = (OP)^2 \Rightarrow OP = 2\sqrt{6} \text{ units}$$

$$\therefore\ PQ = 2OP = 2 \times 2\sqrt{6} = 4\sqrt{6} \text{ units}$$

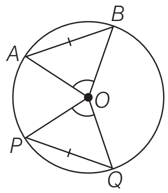
Hence, the length of PQ is $4\sqrt{6}$ units.



Important Theorems of Circles

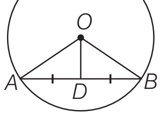
Theorem 1 If two arcs of a circle are congruent, then the corresponding chords are equal.

Theorem 2 Equal arcs (or chords) subtend equal angle at the centre. Here, if $AB = PQ$, then $\angle AOB = \angle POQ$.



Theorem 3 The perpendicular from the centre of a circle to a chord bisects the chord.

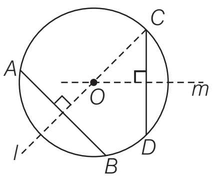
Here, if $OD \perp AB$, then $AD = DB$.



Theorem 4 The line joining the centre to the mid-point of a chord is perpendicular to the chord.

Here, if $AD = DB$, then $\angle ADO = \angle ODB = 90°$.

Theorem 5 The perpendicular bisectors of two chords of a circle intersect at its centre.

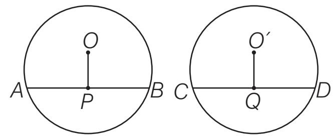


Here, AB, CD are the chords and l, m are perpendicular bisector of AB and CD. So, l and m meet at centre 'O'.

Theorem 6 There is one and only one circle passing through three non-collinear points.

- An infinite number of circles can be drawn to pass through a single point.
- An infinite number of circles can be drawn to pass through two given points.
- A unique circle can be drawn to pass through three given non-collinear points.

Theorem 7 Equal chords of congruent circles are equidistant from the corresponding centres.

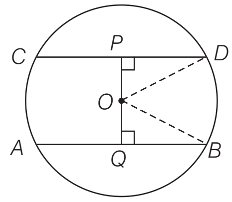


Here, if $AB = CD$, then $OP = O'Q$.

Conversely if $OP = O'Q$, then $AB = CD$.

Theorem 8 Equal chords of a circle are equidistant from the centre.

Here, if AB and CD are equal chords of circle, then $OP = OQ$.



Conversely, if $OP = OQ$, then also $AB = CD$

i.e. chords at equal distance from the centre are equal.

EXAMPLE 2. AB and CD are two parallel chords on the opposite sides of the centre of the circle. If $AB = 10$ cm, $CD = 24$ cm and the radius of the circle is 13 cm, then what is the distance between the chords?

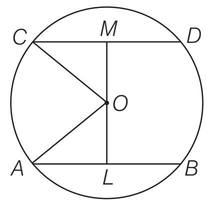
a. 10 cm b. 17 cm

c. 24 cm d. None of these

Sol. **b.** From O draw $OL \perp AB$ and $OM \perp CD$. Join OA and OC.

$$AL = \frac{1}{2} AB = \frac{10}{2} \text{cm} = 5 \text{ cm},$$

$$OA = 13 \text{ cm}$$



In right angled ΔOLA, by pythagoras theorem,

$$OL^2 = OA^2 - AL^2 = (13)^2 - (5)^2$$

$$= (169 - 25) = 144$$

$$\Rightarrow\ OL = \sqrt{144} = 12 \text{ cm}$$

Now, $CM = \frac{1}{2} \times CD = 12$ cm

and $OC = 13$ cm

In ΔOMC, [by pythagoras theorem]

$$OM^2 = OC^2 - CM^2$$

$$= (13)^2 - (12)^2$$

$$= (169 - 144) = 25$$

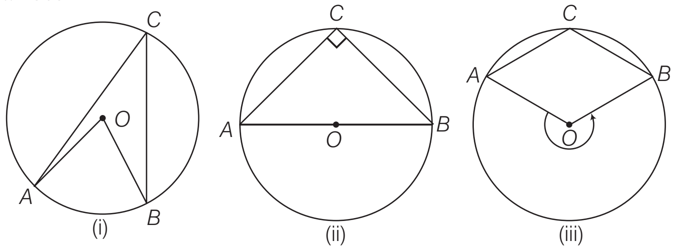
$$\Rightarrow\ OM = \sqrt{25} = 5 \text{ cm}$$

$$\therefore\ ML = OM + OL$$

$$= (5 + 12) \text{ cm}$$

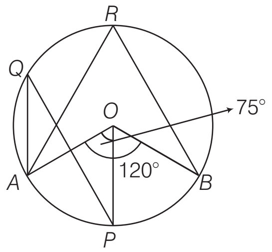
$$= 17 \text{ cm}$$

Theorem 9 The angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the circumference of circle. Here, three case arises.



In all the three cases $\angle AOB = 2\angle ACB$.

EXAMPLE 3 In the figure given below, if $\angle AOP = 75°$ and $\angle AOB = 120°$, then what is the value of $\angle AQP$?



a. 45° b. 37.5° c. 30° d. 22.5°

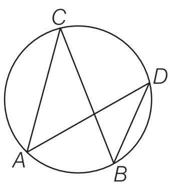
Sol. b. We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the circumference of a circle.

$$2\angle AQP = \angle AOP$$

$$\angle AQP = \frac{1}{2} \times \angle AOP = \frac{75°}{2} = 37.5°$$

Theorem 10 The angle in a semi-circle is a right angle.

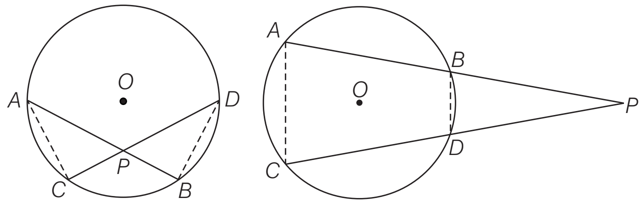
Theorem 11 Angles in the same segment of a circle are equal.



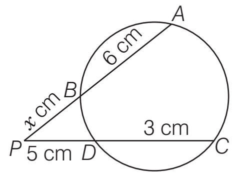
Here, $\angle ACB = \angle ADB$

Theorem 12 If two chords AB and CD of a circle intersect inside or outside the circle when produced at a point P

then, $$AP \times PB = DP \times PC$$



EXAMPLE 4. In the given figure, chords AB and CD of a circle intersect externally at P. If $AB = 6$ cm, $CD = 3$ cm and $PD = 5$ cm, then the measurement of PB is



a. 2.5 cm b. 4 cm c. 10 cm d. None of these

Sol. b. Let the measurement of PB be x.

$$PA \times PB = PC \times PD \Rightarrow (x + 6) \times x = 8 \times 5$$

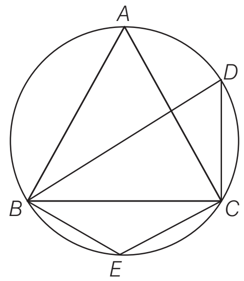
$$\Rightarrow x^2 + 6x - 40 = 0 \Rightarrow x^2 + 10x - 4x - 40 = 0$$

$$\Rightarrow x(x + 10) - 4(x + 10) = 0 \Rightarrow (x + 10)(x - 4) = 0$$

$$\Rightarrow x = 4, \text{ as } x \neq -10 \quad \text{[side cannot be negative]}$$

$$\therefore PB = 4 \text{ cm}$$

EXAMPLE 5. In the adjoining figure, ΔABC is an isosceles triangle with $AB = AC$ and $\angle ABC = 50°$. Then, the measure of $\angle BDC$ is



a. 80° b. 100° c. 40° d. 160°

Sol. a. Given, $AB = AC \Rightarrow \angle ACB = \angle ABC = 50°$

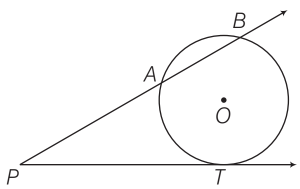
In ΔABC, $\angle BAC = 180° - (50° + 50°) = 80°$

[by angle sum property of a triangle]

$\therefore \quad \angle BDC = \angle BAC = 80°$

[angles lying in the same segment of the circle]

Theorem 13 Let PT be a line tangent to the circle at point T. From external point P, draw a secant which intersect the circle at two point A and B, then $PA \times PB = PT^2$



EXAMPLE 6. In a ΔABC, $AB = AC$. A circle through B touches AC at D and intersects AB at P. If D is the mid-point of AC, then which one of the following is correct?

a. $AB = 2AP$ b. $AB = 3AP$ c. $AB = 4AP$ d. $2AB = 5AP$

Sol. **c.** By using theorem,

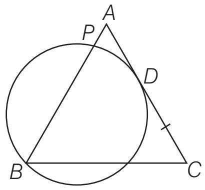
$$AB \times AP = AD^2$$

$$= \left(\frac{AC}{2}\right)^2 = \frac{1}{4}(AC)^2$$

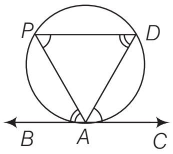
$$\Rightarrow \quad AB \times AP = \frac{1}{4}(AB)^2$$

$$[\because AC = AB \text{ given}]$$

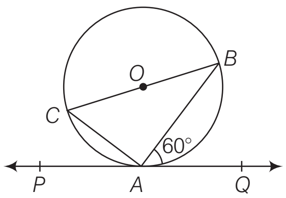
$$\Rightarrow \quad AB = 4AP$$



Theorem 14 Alternate Segment Theorem In the given figure B, if BAC is a tangent to a circle at point A and if AD is any chord, then $\angle DAC = \angle APD$ and $\angle PAB = \angle PDA$.



EXAMPLE 7. In the given figure, PAQ is the tangent. BC is the diameter of the circle. If $\angle BAQ = 60°$, then $\angle ABC$ is



a. 25° **b.** 30° **c.** 45° **d.** 50°

Sol. **b.** Since BC is the diameter of the circle.

$\therefore \quad \angle BAC = 90°$ $\quad [\because$ angle in a semi-circle is 90°]

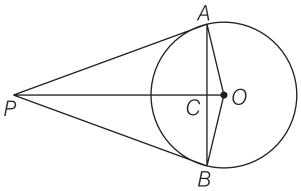
$\angle ACB = 60°$ $\quad [\because$ alternate angle theorem]

Now, $\quad \angle ACB + \angle BAC + \angle ABC = 180°$

$\Rightarrow \quad \angle ABC + 90° + 60° = 180°$

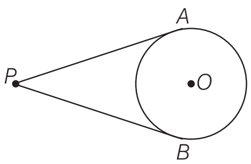
$\Rightarrow \quad \angle ABC = 180° - (90° + 60°) = 30°$

- **Theorem 15** If two tangents are drawn to a circle from an external point, then they subtend equal angles at the centre. These two tangents are equally inclined to the line segment joining the centre to that point. In the given figure, $\angle AOP = \angle BOP$ and $\angle APO = \angle BPO$

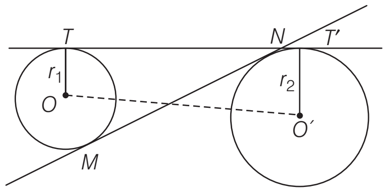


- **Theorem 16** Line joining centre and external point is perpendicular bisector of line joining point of contacts of tangents to the circle. In the above figure, $OP \perp AB$ and $AC = BC$
- **Theorem 17** The length of two tangents drawn from an external point to a circle are equal.

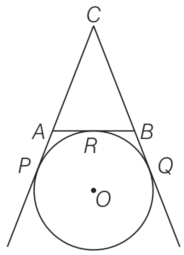
  Here, $\quad PA = PB$



- **Theorem 18** Tangents drawn at the end points of a diameter of a circle are always parallel.
- Length of the direct common tangent is $TT' = \sqrt{(OO')^2 - (r_1 - r_2)^2}$ and length of transverse common tangent is $MN = \sqrt{(OO')^2 - (r_1 + r_2)^2}$



EXAMPLE 8. In the given figure, CP and CQ are tangent to a circle with centre O. ARB is another tangent touching. The circle at R. If $CP = 11$ cm, $BC = 7$ cm length of BR is



a. 1 cm **b.** 2 cm
c. 4 cm **d.** 3 cm

Sol. **c.** Since, the length of two tangents drawn from external point are equal.

$\therefore \quad CQ = CP = 11$ cm and $BQ = BR$

Now, $\quad BQ = CQ - BC$

$= (11 - 7)$ cm $= 4$ cm

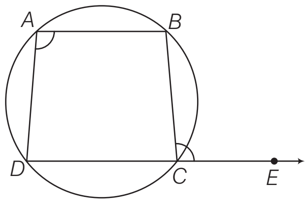
$\therefore \quad BR = BQ = 4$ cm

Cyclic Quadrilateral

A quadrilateral whose all vertices lie on a circle is called a cyclic quadrilateral.

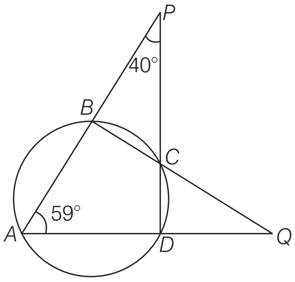
Properties of Cyclic Quadrilateral

- The sum of either pair of opposite angles of a cyclic quadrilateral is 180° i.e. the opposite angles of a cyclic quadrilateral are supplementary.
- If the sum of a pair of opposite angles of a quadrilateral is 180°, then quadrilateral is cyclic.
- The exterior angle, formed by producing a side of a cyclic quadrilateral is equal to the interior opposite angle.



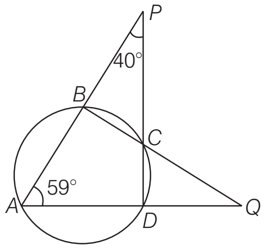
Here, $\quad \angle BAD = \angle BCE$

EXAMPLE 9. In the given figure, if $\angle PAQ = 59°$, $\angle APD = 40°$, then what is $\angle AQB$?



a. 19° b. 20° c. 22° d. 27°

Sol. c. Given, $\angle PAD = 59°$ and $\angle APD = 40°$



In ΔAPD, $\angle PAD + \angle APD + \angle ADP = 180°$

$\Rightarrow$ $59° + 40° + \angle ADP = 180° \Rightarrow \angle ADC = 81°$

Also, $\angle ABC + \angle ADC = 180°$

[property of cyclic quadrilateral]

$\Rightarrow$ $\angle ABC = 180° - 81° = 99°$

$\therefore$ $\angle CDQ = \angle ABC = 99°$ and $\angle QCD = \angle BAD = 59°$

In ΔCQD, $\angle CQD + \angle CDQ + \angle QCD = 180°$

$\Rightarrow$ $\angle CQD = 180° - 59° - 99° = 180° - 158° = 22°$

$\Rightarrow$ $\angle AQB = \angle CQD = 22°$

LOCUS

The path traced by a point moving under a given condition is called locus.

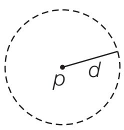
e.g. The locus of points (in a plane) equidistant from a fixed point (in a plane) is a circle with the fixed point as centre.

The plural of locus is loci.

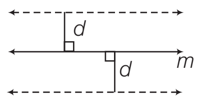
- The locus may be a line segment or lines, straight or curved.
- Every point which satisfies the given geometrical condition(s) lies on the locus and *vice-versa.*

Basic Locus Theorem

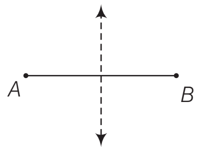
1. The locus of points at a fixed distance d from point p is a circle with the given point p as its centre and d as its radius.



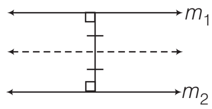
2. The locus of points at a fixed distance d from a line m is a pair of parallel lines at a distance of d from line m and located on either side of m.



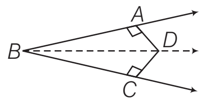
3. The locus of points equidistant from two points A and B is the perpendicular bisector of the line segment joining the two points.



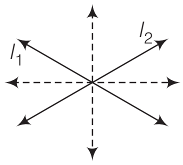
4. The locus of points equidistant from two parallel lines m_1 and m_2, is a line parallel to both m_1 and m_2 and midway between them.



5. The locus of points in the interior of an angle equidistant from the sides of an angle is the bisector of the angle.

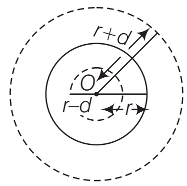


6. The locus of points equidistant from two intersecting lines l_1 and l_2 is a pair of bisectors that bisect the angles formed by l_1 and l_2.

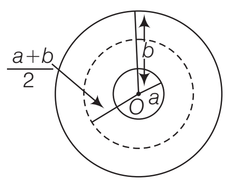


Locus in Some Standard Cases

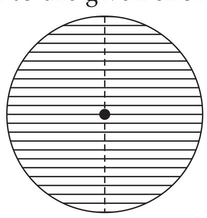
1. The loci of all points d units from a circle having a radius r units are two concentric circles with radii of $r - d$ and $r + d$ units respectively.



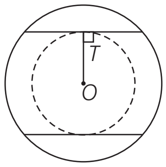
2. The locus of all points equidistant from two concentric circles is a third circle whose radius is the average of the radii of the given circles.



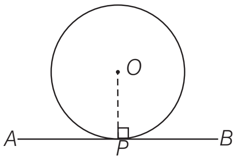
3. The locus of the mid-points of all parallel chords of a circle is the diameter of the circle, which is perpendicular to the given chords.



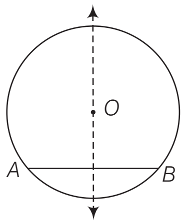
4. The locus of the mid-points of equal chords is a circle with the same centre as the given circle and radius equal to the distance from the centre of the given circle to the given chord.



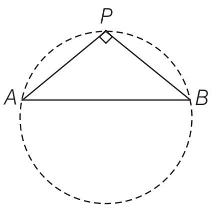
5. The locus of the centre of circle touching a given line at a given point is a line perpendicular to the given line through the given point.



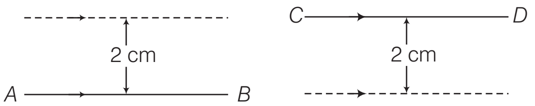
6. The locus of the centre of all circles passing through two given points is the perpendicular bisector of the line segment joining the two points.

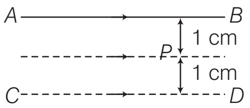


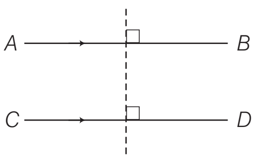
7. If A, B are fixed points, then the locus of a point P such that $\angle APB = 90°$ is the circle with AB as diameter.

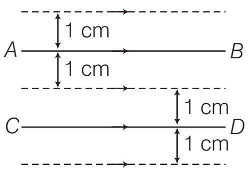


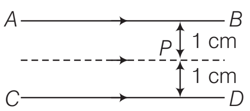
EXAMPLE 10. Given two parallel straight lines AB and CD which are 2 cm apart which of the following dotted lines may represent the locus of a point P such that it is equidistant from AB and CD ?

a. 

b. 

c. 

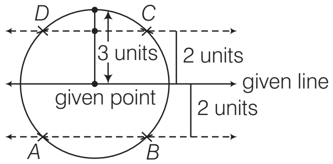
d. 

Sol. **b.** 

EXAMPLE 11. What is the number of points in a plane two units from a given line and three units from a given point of the line?

a. 1 **b.** 2 **c.** 3 **d.** 4

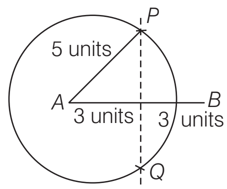
Sol. **d.** The answer will be 4 points. The dotted lines are first locus condition and the circle is the second locus condition. These two loci intersect at points marked with A, B, C and D.



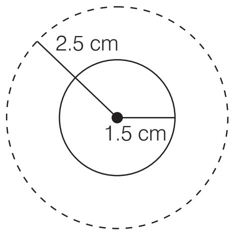
EXAMPLE 12. Two points A and B are 6 units apart. How many points are there that are equidistant from both A and B and also 5 units from A?

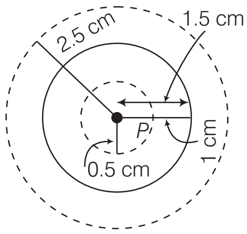
a. 0 **b.** 1 **c.** 2 **d.** 3

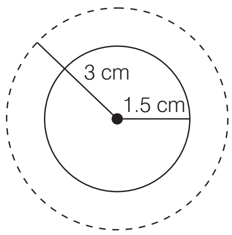
Sol. **c.** The answer will be 2 points. The dotted line is first locus condition and the circle is the second locus condition. These two loci intersect at two points marked with P and Q.

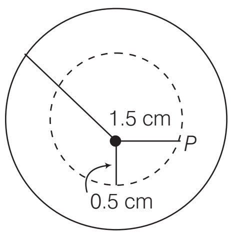


EXAMPLE 13. Given a circle with radius 1.5 cm. Which of the following dotted lines may represent the locus of a point P such that P is at a distance of 1 cm from the nearest point on the circle?

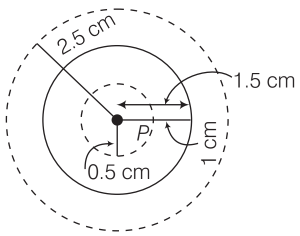
a. 

b. 

c. 

d. 

Sol. **b.**

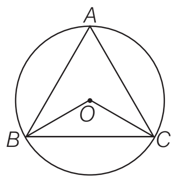


PRACTICE EXERCISE

1. In a circle with centre O and radius 5 cm, AB is a chord of length 8 cm. If $OM \perp AB$, then what is the length of OM?

(a) 4 cm (b) 5 cm
(c) 3 cm (d) None of these

2. An equilateral ΔABC is inscribed in a circle with centre O. Then, $\angle BOC$ is equal to



(a) 120° (b) 75° (c) 180° (d) 60°

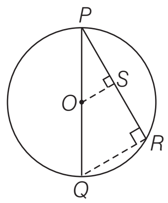
3. A square $ABCD$ is inscribed in a circle with centre O. Then, the angle subtended by each side of the square at the centre O is

(a) 120° (b) 180° (c) 45° (d) 90°

4. The points of concurrency of the perpendicular bisectors of the sides of a triangle is called its

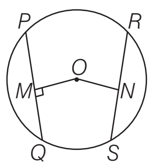
(a) incentre (b) circumcentre
(c) excentre (d) orthocentre

5. In the given figure, PQ is the diameter of a circle with centre at O. OS is perpendicular to PR. Then, OS is equal to



(a) $\frac{1}{4}QR$ (b) $\frac{1}{3}QR$ (c) $\frac{1}{2}QR$ (d) QR

6. In the given figure, OM and ON are the perpendiculars drawn on the chords PQ and RS. If $OM = ON = 6$ cm. Then,



(a) $PQ \geq RS$ (b) $PQ < RS$ (c) $PQ \leq RS$ (d) $PQ = RS$

7. The locus of the mid-points of all radii of a circle is a

(a) circle (b) parallelogram
(c) rhombus (d) square

8. The perpendicular bisectors of the sides of a triangle pass through the

(a) different point (b) more than 2 points
(c) same point (d) None of these

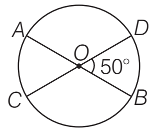
9. If a point P moves such that the sum of the squares of its distance from two fixed points A and B is a constant, then the locus of P is

(a) a circle
(b) a straight line
(c) an arbitrary curve
(d) the perpendicular bisector of AB

10. The locus of points equidistant from two fixed points is a straight line which

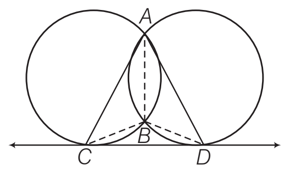
(a) is a right angles to the line joining the two fixed points
(b) bisects the line joining the two fixed points
(c) is the perpendicular bisector of the line joining the two fixed points
(d) None of the above

11. Diameter AB and CD of a circle intersect at O. If $m\angle BOD = 50°$, then $m\ \angle AOD$ is



(a) 50° (b) 180° (c) 130° (d) 310°

12. CD is a direct common tangent to two circles intersecting each other at A and B. Then, $\angle CAD + \angle CBD$ is equal to?



(a) 180° (b) 90° (c) 360° (d) 120°

13. In a circle of radius 17 cm, two parallel chords are drawn on opposite side of a diameter. The distance between the chords is 23 cm. If the length of one chord is 16 cm, then the length of the other chord is

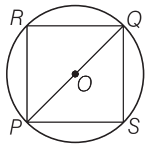
(a) 34 cm (b) 15 cm (c) 23 cm (d) 30 cm

14. If AB is a chord of a circle, P and Q are the two points on the circle different from A and B, then

(a) the angles subtended at P and Q by AB are always equal
(b) the sum of the angles subtended by AB at P and Q is always equal to two right angles
(c) the angles subtended by AB at P and Q are either equal or supplementary
(d) None of the above

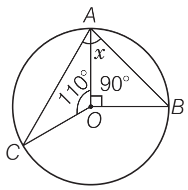
15. The locus of the centre of circles which pases through two given points is
(a) perpendicular to the line joining the given points at one of those points
(b) perpendicular bisector of the line joining the given points
(c) parallel to the line joining the given points
(d) None of the above

16. In the adjoining figure, POQ is the diameter of the circle, R and S are any two points on the circles. Then,



(a) $\angle PRQ > \angle PSQ$ (b) $\angle PRQ < \angle PSQ$
(c) $\angle PRQ = \angle PSQ$ (d) $\angle PRQ = \frac{1}{2}\angle PSQ$

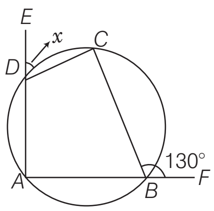
17. If O is the centre of the circle, then the value of x in the adjoining figure is



(a) 80° (b) 70° (c) 60° (d) 50°

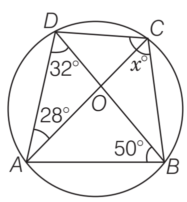
18. If S is a circle with centre C and P be a movable point outside S, then the locus of P such that the tangents from P to S are of constant length is
(a) the straight line CP
(b) the circle through P with centre at C
(c) a circle intersecting S
(d) a circle touching S

19. In the given figure A, B, C, D are the concyclic points. The value of x is



(a) 50° (b) 60° (c) 70° (d) 90°

20. If O is the centre of the circle, then x is

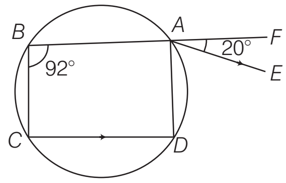


(a) 72° (b) 62° (c) 82° (d) 52°

21. Two circles touch each other internally. Their radii are 2 cm and 3 cm. The biggest chord of the outer circle which is outside the inner circle is of length
(a) $2\sqrt{2}$ cm (b) $3\sqrt{2}$ cm (c) $2\sqrt{3}$ cm (d) $4\sqrt{2}$ cm

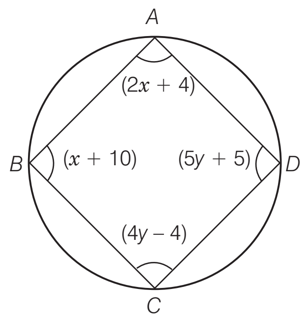
22. If two circles are such that the centre of one lies on the circumference of the other, then the ratio of the common chord of the two circles to the radius of any one of the circles is
(a) 2 : 1 (b) $\sqrt{3}$: 1 (c) $\sqrt{5}$: 1 (d) 4 : 1

23. In the given figure, $ABCD$ is a cyclic quadrilateral. AE is drawn parallel to CD and BA is produced. If $\angle ABC = 92°$ and $\angle FAE = 20°$, then $\angle BCD$ is equal to



(a) 88° (b) 98° (c) 108° (d) 72°

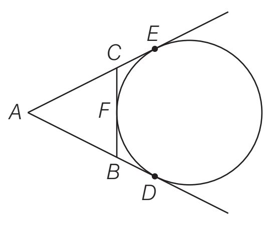
24. The values of $x + y$ in the figure is equal to



(a) 90° (b) 85° (c) 75° (d) 65°

25. Let A and B be two points. What is the locus of the point P such that $\angle APB = 90°$?
(a) The line AB itself
(b) The point P itself
(c) The circumference of the circle with AB as diameter
(d) The line perpendicular to AB and bisecting AB

26. In the adjoining figure AD, AE and BC are tangent to the circle at D, E, F respectively, then



(a) $AD = AB + BC + AC$ (b) $2AD = AB + BC + AC$
(c) $AD = \frac{1}{4}(AB + BC + AC)$ (d) $3AD = AB + BC + AC$

27. S_1 and S_2 are two circles on a plane with radii 4 cm, and 2 cm, respectively and the distance between their centres is 3 cm. Which one of the following statements is true?

(a) S_2 lies entirely within the circle S_1
(b) S_1 and S_2 touch each other internally
(c) S_1 and S_2 touch each other externally
(d) S_1 and S_2 intersect in two distinct points

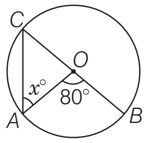
28. ACB is a tangent to a circle at C, CD and CE are chords such that $\angle ACE > \angle ACD$. If $\angle ACD = \angle BCE = 50°$, then

(a) $CD = CE$
(b) ED is not parallel to AB
(c) ED passes through the centre of the circle
(d) ΔCDE is a right angled triangle

29. A bicycle is running straight towards North. What is the locus of the centre of the front wheel of the bicycle whose diameter is d ?

(a) A line parallel to the path of the wheel of the bicycle at a height d cm
(b) A line parallel to the path of the wheel of the bicycle at a height $d/2$ cm
(c) A circle of radius $d/2$ cm
(d) A circle of radius d cm

30. If 'O' is the centre of circle, then x is equal to



(a) 80° (b) 60° (c) 40° (d) 20°

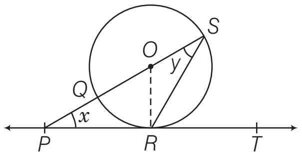
31. If two equal circles touch each other externally, the common tangent divides the line of centres in the ratio

(a) 1 : 1 (b) 2 : 1 (c) 1 : 2 (d) 3 : 2

32. With the vertices of a ΔABC as centre three circles are described, each touching the other two circle externally. If the sides of the triangle are 9 cm, 7 cm and 6 cm. Then, the radius of the circle are

(a) 4, 5, 2 (b) 4, 5, 6
(c) 3, 2, 3 (d) All equal to 3 cm

33. In the given figure PT touches the circle with centre O at R. Diameter SQ when produced meet PT at P. If $\angle SPR = x$ and $\angle QSR = y$, then $x + 2y$ is equal to

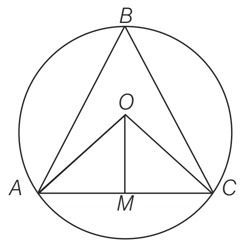


(a) 180° (b) 90°
(c) 135° (d) None of these

34. What is the locus of centres of circles which touch a given line at a given point?

(a) A line perpendicular to the given line, passing through the given point
(b) A line parallel to the given line
(c) A circle tangent to the given line at the given point
(d) A closed curve other than a circle

35. In the given figure, O is the centre of the circle, $OA = 3$ cm, $AC = 3$ cm and OM is perpendicular to AC. What is $\angle ABC$ equal to?



(a) 60° (b) 45°
(c) 30° (d) None of these

36. The distance between the centres of two circles having radii 4.5 cm and 3.5 cm, respectively is 10 cm. What is the length of the transverse common tangent of these circles?

(a) 8 cm (b) 7 cm
(c) 6 cm (d) None of these

37. ABC is an equilateral triangle inscribed in a circle with $AB = 5$ cm. Let the bisector of the angle, A meet BC in X and the circle in Y. What is the value of $AX \cdot AY$?

(a) 16 cm^2 (b) 20 cm^2 (c) 25 cm^2 (d) 30 cm^2

38. Two unequal circles are touching each other externally at P, APB and CPD are two secants cutting the circles at A, B, C and D. Which one of the following is correct?

(a) $ACBD$ is parallelogram (b) $ACBD$ is a trapezium
(c) $ACBD$ is a rhombus (d) None of these

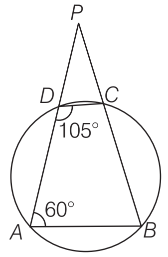
39. $ABCD$ is a quadrilateral, the sides of which touch a circle. Which one of the following is correct?

(a) $AB + AD = CB + CD$ (b) $AB : CD = AD : BC$
(c) $AB + CD = AD + BC$ (d) $AB : AD = CB : CD$

40. Let PAB be a secant to a circle intersecting at points A and B and PC is a tangent. Which one of the following is correct?

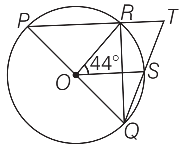
(a) The area of rectangle with PA, PB as sides is equal to the area of square with PC as sides
(b) The area of rectangle with PA, PC as sides is equal to the area of square with PB as sides
(c) The area of rectangular with PC, PB as sides is equal to the area of square with PA as side
(d) The perimeter of rectangle with PA, PB as sides is equal to the perimeter of square with PC as side

41. In the given figure, if $\angle BAD = 60°$, $\angle ADC = 105°$, then what is $\angle DPC$ equal to?



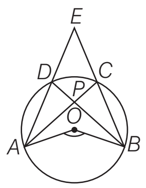
(a) 40° (b) 45° (c) 50° (d) 60°

42. In the given figure, PQ is a diameter of the circle whose centre is at O. If $\angle ROS = 44°$ and OR is a bisector of $\angle PRQ$, then what is the value of $\angle RTS$?



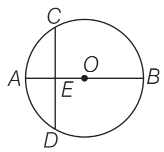
(a) 46° (b) 64°
(c) 69° (d) None of these

43. In the given figure, O is the centre of the circle. AC and BD intersect at P. If $\angle AOB = 100°$ and $\angle DAP = 30°$, what is $\angle APB$?



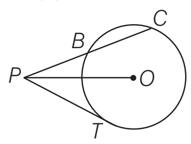
(a) 77° (b) 80° (c) 85° (d) 90°

44. In the given figure, AB is a diameter of a circle and CD is perpendicular to AB, if $AB = 10$ cm and $AE = 2$ cm, then what is the length of ED?



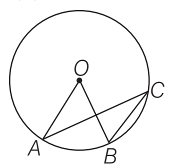
(a) 5 cm (b) 4 cm (c) $\sqrt{10}$ cm (d) $\sqrt{20}$ cm

45. In the given figure, PT is a tangent to a circle of radius 6 cm. If P is at a distance of 10 cm from the centre O and $PB = 5$ cm, then what is the length of the chord BC ?



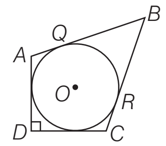
(a) 7.8 cm (b) 8 cm (c) 8.4 cm (d) 9 cm

46. In the given figure, $\angle AOB = 46°$, AC and OB intersect each other at right angles. What is the measure of $\angle OBC$? (O is the centre of the circle.)



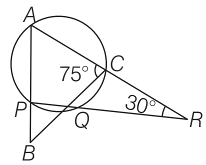
(a) 44° (b) 46° (c) 67° (d) 78.5°

47. In the given figure, a circle is inscribed in a quadrilateral $ABCD$. Given that, $BC = 38$ cm, $QB = 27$ cm, $DC = 25$ cm and AD is perpendicular to DC. What is the radius of the circle?



(a) 11 cm (b) 14 cm (c) 15 cm (d) 16 cm

48. In the figure given below, what is $\angle CBA$?



(a) 30° (b) 45° (c) 50° (d) 60°

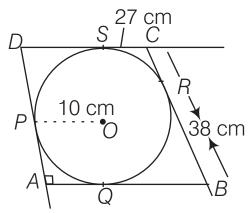
49. PQ is a common chord of two circles. APB is a secant line joining points A and B on the two circles. Two tangents AC and BC are drawn. If $\angle ACB = 45°$, then what is $\angle AQB$ equal to?

(a) 75° (b) 90° (c) 120° (d) 135°

50. $ABCD$ is concyclic quadrilateral. The tangents at A and C intersect each other at P. If $\angle ABC = 100°$, then what is $\angle APC$ equal to?

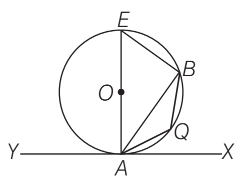
(a) 10° (b) 20° (c) 30° (d) 40°

51. In the given figure, quad. $ABCD$ is circumscribed, touching the circle at P, Q, R and S such that $\angle DAB = 90°$. If CS = 27 cm and CB = 38 cm and radius of the circle is 10 cm, then AB = ?



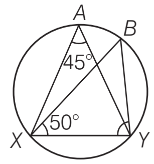
(a) 28 cm (b) 21 cm (c) 19 cm (d) 17 cm

52. In the figure given below, YAX is a tangent to the circle with centre O. If $\angle BAX = 70°$ and $\angle BAQ = 40°$, then what is $\angle ABQ$ equal to?



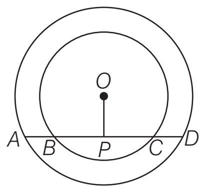
(a) 20° (b) 30°
(c) 35° (d) 40°

53. In the given figure, what is $\angle BYX$ equal to?



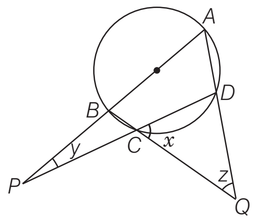
(a) 85° (b) 50° (c) 45° (d) 90°

54. In the given figure, AD is a straight line, OP perpendicular to AD and O is the centre of both circles. If $OA = 20$ cm, $OB = 15$ cm and $OP = 12$ cm, what is AB equal to?



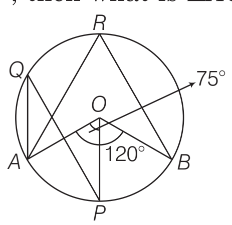
(a) 7 cm (b) 8 cm (c) 10 cm (d) 12 cm

55. In the given figure, if $\frac{x}{3} = \frac{y}{4} = \frac{z}{5}$, where $\angle DCQ = x$, $\angle BPC = y$ and $\angle DQC = z$, then what are the values of x, y and z, respectively ?



(a) 33°, 44° and 55° (b) 36°, 48° and 60°
(c) 39°, 52° and 65° (d) 42°, 56° and 70°

56. In the given figure, if $\angle AOP = 75°$ and $\angle AOB = 120°$, then what is $\angle AQP$?



(a) 45° (b) 37.5° (c) 30° (d) 22.5°

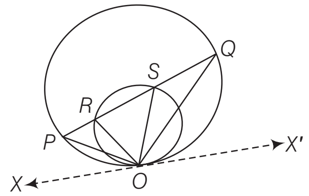
57. A circle of radius 5 cm has chord RS at a distance of 3 units from it. Chord PQ intersects with chord RS at T such that $TS = 1/3$ of RT. Find minimum value of PQ.

(a) $6\sqrt{3}$ (b) $4\sqrt{3}$ (c) $8\sqrt{3}$ (d) $2\sqrt{3}$

58. Two mutually perpendicular chords AB and CD intersect at P. $AP = 4$ cm, $PB = 6$ cm, $CP = 3$ cm. Find radius of the circle.

(a) $\frac{5\sqrt{5}}{2}$ cm (b) $\frac{3\sqrt{5}}{4}$ cm (c) $\frac{6\sqrt{5}}{2}$ cm (d) $\frac{2\sqrt{5}}{3}$ cm

59.



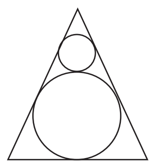
In the above figure which of the following holds good?

(a) $\angle SOQ = \angle ROP$ (b) $2\angle ROP = \angle SOR$
(c) $\angle POR = \angle ASO$ (d) $\angle QOX' = \angle SOR + \angle ROP$

60. Semi-circle C_1 is drawn with a line segment PQ as its diameter with centre at R. Semicircles C_2 and C_3 are drawn with PR and QR as diameters respectively, both C_2 and C_3 lying inside C_1. A full circle C_4 is drawn in such a way that it is tangent to all the three semicircles C_1, C_2 and C_3, C_4 lies inside C_1 and outside both C_2 and C_3. The radius of C_4 is

(a) $\frac{1}{3}PQ$ (b) $\frac{1}{6}PQ$ (c) $\frac{1}{\sqrt{2}}PQ$ (d) $\frac{1}{4}PQ$

61. Two circles are placed in an equilateral triangle as shown in the figure. What is the ratio of the area of the smaller circle to that of the equilateral triangle?



(a) $\pi : 36\sqrt{3}$ (b) $\pi : 18\sqrt{3}$ (c) $\pi : 27\sqrt{3}$ (d) $\pi : 42\sqrt{3}$

62. Consider the following statements :

I. The opposite angles of a cyclic quadrilateral are supplementary.

II. Angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Which one of the following is/are correct in respect of the above statements?

(a) Statement I $\Rightarrow$ Statement II
(b) Statement II $\Rightarrow$ Statement I
(c) Statement I $\Leftrightarrow$ Statement II
(d) Neither Statement I $\Rightarrow$ Statement II nor Statement II $\Rightarrow$ Statement I

63. Consider the following statements :

I. Let P be a point on a straight line L. Let Q, R and S be the points on the same plane containing the line L such that PQ, PR and PS are perpendicular to L. Then, there exists no triangle with vertices Q, R, S.

II. Let C be a circle passing through three distinct points D, E and F such that the tangent at D to the circle C is parallel to EF. Then, DEF is an isosceles triangle.

Which of the statement(s) given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

64. The incircle of a ΔABC touches the sides AB, BC and AC at the points P, Q, R, respectively, then which of the following statement is/are correct?

I. $AP + BQ + CR = PB + QC + RA$

II. $AP + BQ + CR = \frac{1}{2}$ (perimeter of ΔABC)

III. $AP + BQ + CR = 3(AB + BC + CA)$

Select the correct answer using the codes given below

(a) I, II and III are correct (b) Only I is correct
(c) II and III are correct (d) I and II are correct

65. Points X and Y are two different points on a circle. Point M is located so that line segment XM and line segment YM have equal length. Which of the following could be true?

I. M is the centre of the circle.

II. M is on arc XY III. M is outside of the circle.

(a) Only 1 (b) Only II (c) I and II (d) I, II and III

66. Which of the following statement(s) are correct?

I. A circle is the locus of points equidistant from a fixed point.

II. The locus of centre of circle rolling on the circumference of fixed circle is a circle concentric with fixed circle.

III. The locus of the points which are equidistant from three distinct points on a line, is a line parallel to the given line.

IV. If the bisectors of $\angle B$ and $\angle C$ of a quadrilateral $ABCD$ intersect in P, then P is equidistant from AB and CD.

Select the correct answer using the codes given below

(a) All are correct (b) Only IV
(c) Both I and II (d) I, II and IV

67. A square and a circle intersect at more than one point. Consider the following statements:

I If atleast two of the intersection points are on the vertices of the square then the square is cyclic.

II If there are exactly four intersection points then the square has less area than the circle.

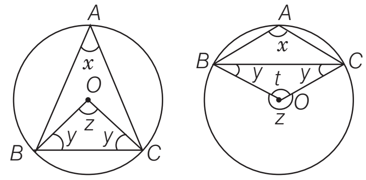
III If there are exactly four intersection point then the diameter of circle is equal to the diagonal of square.

IV If atleast two intersection points are on the vertices of square. Then it may be possible to have the sides of square equal to the diameter of circle.

Which of the statement (s) given above is/are correct ?

(a) Only I (b) Only IV
(c) Both I and II (d) II, III and IV

Directions (Q. Nos. 68-69) *BC is a chord with centre O. A is a point on an arc BC as shown in the figure below.*



68. What is the value of $\angle BAC + \angle OBC$ if A is the point on the major arc?

(a) 110° (b) 70° (c) 90° (d) 60°

69. What is the value of $\angle BAC - \angle OBC$ if A is the point on the minor arc?

(a) 90° (b) 180° (c) 130° (d) 50°

PREVIOUS YEARS' QUESTIONS

70. A circular ring with centre O is kept in the vertical position by two weightless thin strings TP and TQ attached to the ring at P and Q. The line OT meets the ring at E where as a tangential string at E meets TP and TQ at A and B, respectively. If the radius of the ring is 5 cm and OT = 13 cm, then what is the length of AB ? **2012 I**

(a) 10/3 cm (b) 20/3 cm (c) 10 cm (d) 40/3 cm

71. The locus of the mid-points of all equal chords in a circle is **2012 I**

(a) The circumference of the circle concentric with the given circle and having radius equal to the chords
(b) The circumference of the circle concentric with the given circle and having radius equal to the distance of the chords from the centre
(c) The circumference of the circle concentric with the given circle and having radius equal to half of the radius of the given circle
(d) The circumference of the circle concentric with the given circle and having radius equal to half of the distance of the chords from the centre

72. Consider the following statements

I. The locus of points which are equidistant from two parallel lines is a line parallel to both of them and drawn mid way between them.

II. The perpendicular distances of any point on this locus line from two original parallel lines are equal. Further, no point outside this locus line has this property.

Which of the statement(s) given above is/are correct? **2012 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

73. Consider a circle with centre at O and radius r. Points A and B lie on its circumference and a point M lies outside of it such that M, A and O lie on the same straight line. Then, the ratio of MA to MB is **2013 I**

(a) equal to 1 (b) equal to r
(c) greater than 1 (d) less than 1

74. Consider the following statements :

I. The perpendicular bisector of a chord of a circle does not pass through the centre of the circle.

II. The angle in a semi-circle is a right angle.

Which of the statement(s) given above is/are correct? **2013 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

75. The diameter of a circle with centre at C is 50 cm. CP is a radial segment of the circle. AB is a chord perpendicular to CP and passes through P. CP produced intersects the circle at D. If $DP = 18$cm, then what is the length of AB ? **2013 II**

(a) 24 cm (b) 32 cm
(c) 40 cm (d) 48 cm

76. Consider the following statements in respect of two chords XY and ZT of a circle intersecting at P.

I. $PX \cdot PY = PZ \cdot PT$

II. PXZ and PTY are similar triangles.

Which of the statement(s) given above is/are correct? **2013 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

77. In a ΔABC, $AB = BC = CA$. The ratio of the radius of the circumcircle to that of the incircle is **2014 I**

(a) 2 : 1 (b) 3 : 1
(c) 3 : 2 (d) None of these

78. AB and CD are two chords of a circle meeting externally at P. Then, which of the following is/are correct?

I. $PA \times PD = PC \times PB$

II. ΔPAC and ΔPDB are similar.

Select the correct answer using the codes given below **2014 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

79. For a triangle, the radius of the circumcircle is double the radius of the inscribed circle, then which one of the following is correct? **2014 II**

(a) The triangle is a right angled
(b) The triangle is an isosceles
(c) The triangle is an equilateral
(d) None of the above

80. If the chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Then, which one of the following is correct? **2014 II**

(a) $y(2z - y) = x^2$ (b) $y(2z - y) = 4x^2$
(c) $2y(2z - y) = x^2$ (d) $4y(2z - y) = x^2$

81. If the angle between the radii of a circle is 130°, then the angle between the tangents at the ends of the radii is **2015 I**

(a) 90° (b) 70° (c) 50° (d) 40°

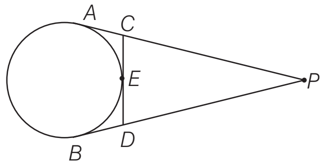
82. Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24 cm is tangent to the inner circle. The radius of the inner circle is **2015 I**

(a) 5 cm (b) 6 cm (c) 8 cm (d) 10 cm

83. AD is the diameter of a circle and AB is a chord. If $AD = 34$ cm, $AB = 30$ cm, then the distance of AB from the centre of the circle is **2015 I**

(a) 17 cm (b) 15 cm (c) 13 cm (d) 8 cm

84. From an external point P, tangents PA and PB are drawn to the circle as shown in the above figure. CD is the tangent to the circle at E. If $AP = 16$ cm, then the perimeter of the ΔPCD is equal to **2015 II**



(a) 24 cm (b) 28 cm (c) 30 cm (d) 32 cm

85. Chord CD intersects the diameter AB of a circle at right angle at a point P in the ratio 1:2. If diameter of circle is d, then CD is equal to **2015 II**

(a) $\frac{\sqrt{2}d}{3}$ (b) $\frac{2d}{3}$ (c) $\frac{2\sqrt{2}d}{3}$ (d) $\frac{2\sqrt{3}d}{3}$

86. The diameter of a wheel that makes 452 revolutions to move 2 km and 26 dm is equal to **2015 II**

(a) $1\frac{9}{22}$ m (b) $1\frac{13}{22}$ m (c) $2\frac{5}{11}$ m (d) $2\frac{7}{11}$ m

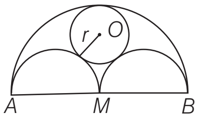
87. A boy is cycling such that the wheels of the cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, then the speed of the cycle is **2015 II**

(a) 15.5 km/h (b) 15.84 km/h (c) 16 km/h (d) 16.36 km/h

88. The two adjacent sides of a cyclic quadrilateral are 2 cm and 5 cm and the angle between them is 60°. If the third side is 3 cm, then the fourth side is of length **2015 II**

(a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

89. AB is a line segment of length $2a$, with M as mid-point. Semi-circles are drawn on one side with AM, MB and AB as diameter as shown in the above figure. A circle with centre O and radius r is drawn such that this circle touches all the three semi-circles. The value of r, is **2015 II**



(a) $\frac{2a}{3}$ (b) $\frac{a}{2}$ (c) $\frac{a}{3}$ (d) $\frac{a}{4}$

90. A tangent is drawn from an external point O to a circle of radius 3 units at P such that $OP = 4$ units. If C is the centre of the circle, then the sine of $\angle COP$ is **2016 I**

(a) 4/5 (b) 3/4 (c) 3/5 (d) 1/2

91. Let ABC be a triangle in which $AB = AC$. Let L be the locus of points X inside or on the triangle such that $BX = CX$.

Which of the following statements are correct?

I. L is a straight line passing through A and incentre of ΔABC is on L.

II. L is a straight line passing through A orthocentre of ΔABC is a point on L.

III. L is a straight line passing through A centroid of ΔABC is a point on L.

Select the correct answer using the code given below. **2016 I**

(a) I and II (b) II and III (c) I and III (d) I, II and III

92. In a circle of radius 2 units, a diameter AB intersects a chord of length 2 units perpendicularly at P. If $AP > BP$, then AP is equal to **2016 I**

(a) $(2 + \sqrt{5})$ units (b) $(2 + \sqrt{3})$ units
(c) $(2 + \sqrt{2})$ units (d) 3 units

93. Consider a circle with centre at C. Let OP, OQ denote respectively the tangents to the circle drawn from a point O outside the circle. Let R be a point on OP and S be a point on OQ such that $OR \times SQ = OS \times RP$. Which of the following statement(s) is/are correct?

I. If X is the circle with centre at O and radius OR and Y is the circle with centre at O and radius OS, then $X = Y$.

II. $\angle POC + \angle QCO = 90°$

Select the correct answer using the code given below. **2016 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

94. Consider a circle at O and radius 7 cm. Let QR be a chord of length 2 cm and let P be the mid-point of QR. Let CD be another chord of this circle passing through P such that $\angle CPQ$ is acute. If M is the mid-point of CD and $MP = \sqrt{24}$ *cm*, then which of the following statements are correct?

I. $\angle QPD = 135°$

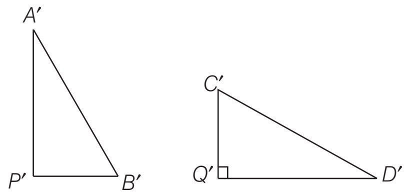
II. If $CP = m$ cm and $PD = n$ cm, then m and n are the roots of the quadratic equation $x^2 - 10x + 1 = 0$.

III. The ratio of the area of ΔOPR to the area of ΔOMP is $1 : 2\sqrt{2}$.

Select the correct answer using the code given below. **2016 I**

(a) I and II (b) II and III (c) I and III (d) I, II and III

95. Suppose chords AB and CD of a circle intersect at a point P inside the circle. Two right angled triangles $A'P'B'$ and $C'Q'D'$ are formed as shown in the figures below such that $A'P' = AP$, $B'P' = BP$, $C'Q' = CP$, $D'Q' = DP$ and $\angle A'P'B' = 90° = \angle C'Q'D'$.



Which of the following statements are not correct?

I. $A'P'B'$ and $C'Q'D'$ are similar triangles but need not be congruent.

II. $A'P'B'$ and $C'Q'D'$ are congruent triangles.

III. $A'P'B'$ and $C'Q'D'$ are triangles of same area.

IV. $A'P'B'$ and $C'Q'D'$ are triangles of same perimeter.

Select the correct answer using the code given below. **2016 I**

(a) II and III (b) I and III
(c) I, II and IV (d) I, II, III and IV

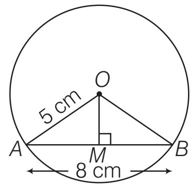
ANSWERS

1	c	2	a	3	d	4	b	5	c	6	d	7	a	8	c	9	a	10	c
11	c	12	a	13	d	14	c	15	b	16	c	17	a	18	b	19	a	20	c
21	d	22	b	23	c	24	d	25	c	26	b	27	d	28	a	29	b	30	c
31	a	32	a	33	b	34	a	35	c	36	c	37	c	38	b	39	c	40	a
41	b	42	d	43	b	44	b	45	a	46	c	47	b	48	d	49	d	50	b
51	b	52	b	53	a	54	a	55	b	56	b	57	b	58	a	59	a	60	b
61	c	62	d	63	c	64	d	65	d	66	d	67	d	68	c	69	a	70	b
71	b	72	c	73	d	74	b	75	d	76	c	77	a	78	b	79	c	80	d
81	c	82	a	83	d	84	d	85	c	86	b	87	b	88	a	89	c	90	c
91	d	92	b	93	c	94	a	95	c										

HINTS AND SOLUTIONS

1. *(c)* Given, $OA = 5$ cm, $AM = \frac{1}{2} \times AB = \frac{8}{2}$

$\Rightarrow \quad AM = 4$ cm



In right angled ΔOMA,

$OM^2 = OA^2 - AM^2$

[by pythagoras theorem]

$= 5^2 - 4^2 = 25 - 16 = 25 - 16 = 9$

$\Rightarrow \quad OM = 3$ cm

2. *(a)* $\because \angle BOC = 2\angle A = 2 \times 60°$

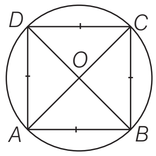
(Angle subtended by an arc at the center is double the angle subtended by it at any point on the circumference of circle)

$\Rightarrow \quad \angle BOC = 120°$

3. *(d)* A square has four equal side.

$\therefore$ Each side subtends the same angle at the centre O.

Let angle subtended be $x°$.



So, $\quad 4x° = 360°$

$x° = \frac{360°}{4} = 90°$

4. *(b)* Clearly, the point of concurrency of the perpendicular bisectors of the sides of a triangle is known as circumcentre.

5. *(c)* As, O is mid-point of diameter PQ and $\angle PRQ = 90°$ (angle in semi-circle)

So, $OS \parallel QR$ and $OS \perp PR$

$\therefore \quad \frac{PO}{PQ} = \frac{OS}{QR}$

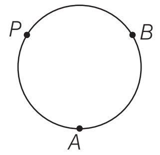
$\Rightarrow \quad \frac{OS}{QR} = \frac{1}{2} \Rightarrow OS = \frac{1}{2} QR$

6. *(d)* $PQ = RS$ as chords equidistant from the centre are equal.

7. *(a)* The locus of the mid-points of all radii of a circle is a circle.

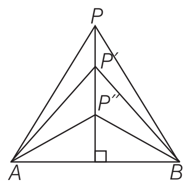
8. *(c)* The perpendicular bisectors of the sides of a triangle passes through the same point. The perpendicular bisectors are concurrent and point is called the circumcentre.

9. *(a)* Let P be the moving point.
Then, $PA^2 + PB^2 =$ constant



So, the locus of P is a circle.

10. *(c)* The locus of points equidistant from fixed points is a straight line which is the perpendicular bisector of the line joining the two fixed points.



Here, A, B are fixed point and P is locus.

11. *(c)* Given that, $\Rightarrow \angle BOD = 50°$

$\Rightarrow \quad \angle AOD = 180° - \angle BOD = 130°$

[linear pair]

12. *(a)* Here, $\angle CAB = \angle BCD$

[angles in alternate segments]

and $\quad \angle DAB = \angle CDB$

[angles in alternate segments]

$\Rightarrow \angle CAD = \angle CAB + \angle DAB$

$= \angle BCD + \angle CDB$

On adding $\angle CBD$ both sides, we get

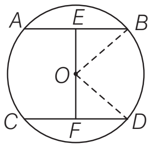
$\angle CAD + \angle CBD$

$= \angle BCD + \angle CDB + \angle CBD$

$= 180°$ [angles of a triangle]

13. *(d)* Here, $BE = \frac{1}{2} AB = \frac{16}{2}$ cm $= 8$ cm

$OB = OD = 17$ cm (radii)



In right angled ΔOEB,

$OE = \sqrt{OB^2 - BE^2}$

$= \sqrt{17^2 - 8^2} = \sqrt{289 - 64}$

$= \sqrt{225} = 15$ cm

$\therefore OF = EF - OE = (23 - 15) = 8$ cm

In right angled ΔOFD,

$FD = \sqrt{OD^2 - OF^2}$

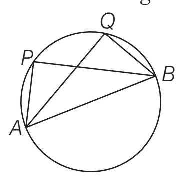
$= \sqrt{17^2 - 8^2} = \sqrt{289 - 64}$

$= \sqrt{225} = 15$ cm

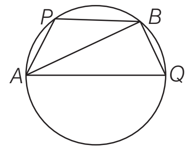
$\therefore \quad CD = 2FD = 30$ cm

14. *(c)* There are two possibilities

Case I When P and Q are on the same side of AB. In this case $\angle APB = \angle AQB$ (angle in the same segment).



Case II When P and Q are on the opposite of AB. In this case $PAQB$ is a cyclic quadrilateral.



So, $\angle APB + \angle AQB = 180°$

15. *(b)* Clearly, the locus of the centre of circles which passes through two given point is perpendicular bisector of the line joining the given points.

16. *(c)* As POQ is the diameter of the circle, therefore, $\widehat{PRQ}$ and $\widehat{PSQ}$ are two semicircles and $\angle PRQ = \angle PSQ$.

$\angle PRQ = \angle PSQ = 90°$

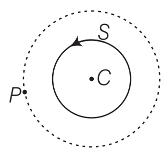
[angles in semicircle]

17. *(a)* $\angle COB = 360° - (110° + 90°) = 160°$

$\therefore \quad x = \angle CAB = \frac{1}{2} \angle COB$

$= \frac{1}{2} \times 160° = 80°$

18. *(b)* If S is a circle with centre C and P be a movable point outside S, then the locus of P such that the tangent from P to S are of constant length is the circle through P with centre at C.



19. *(a)* $\angle CBF = \angle CDA$

(angles in alternate segments)

$\Rightarrow \angle CDA = 130°$

$\angle CDA + x = 180°$ [linear pair]

$x = 180° - 130° = 50°$

Hence, the value of x is $50°$.

20. *(c)* $\angle ACB = \angle ADB = 32°$

[angle in same segment]

$\angle ACD = \angle ABD = 50°$

$\therefore \quad x = \angle BCD = \angle ACB + \angle ACD$

$= 32° + 50° = 82°$

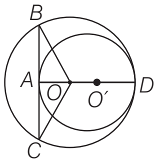
21. *(d)* Here, $OB = OD = 3$ cm and

$O'D = O'A = 2$ cm

$OO' = OD - O'D = 3 - 2 = 1$ cm

$\therefore\ OA = O'A - OO' = 2 - 1 = 1$ cm

In ΔOAB,



$AB = \sqrt{OB^2 - OA^2}$

[by pythagoras theorem]

$= \sqrt{3^2 - 1^2} = \sqrt{9-1}$

$= \sqrt{8}$cm

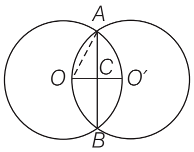
$\therefore$ Required length $= BC = 2AB$

$= 2\sqrt{8}$ cm $= 4\sqrt{2}$ cm

22. *(b)* Here, let O, O' be the centres of the circles.

As, the centre of each lies on the circumference of the other, therefore the two circles will have the same radius. Let the radius be r.

$\therefore\quad OC = O'C = \dfrac{r}{2}$



In right angled ΔOCA, by pythagoras theorem

$$\therefore\quad AC = \sqrt{OA^2 - OC^2} = \sqrt{r^2 - \frac{r^2}{4}} = \frac{\sqrt{3}}{2}r$$

$\Rightarrow\quad AB = 2AC = \sqrt{3}\,r$

$$\therefore\quad \frac{\text{Common chord}}{\text{Radius}} = \frac{AB}{OA} = \frac{\sqrt{3}r}{r}$$

Hence, ratio of common chord to the radius is $\sqrt{3} : 1$.

23. *(c)* $\because\ \angle B + \angle D = 180°$

(sum of two opposite angle in cyclic quadrilateral)

$\Rightarrow\quad \angle D = 180° - \angle B$

$= 180° - 92° = 88°$

$\Rightarrow\ \angle DAE = \angle D = 88°$ [alternate angle]

$\Rightarrow\ \angle FAD = 88° + 20° = 108°$

$\Rightarrow\quad \angle BCD = \angle FAD = 108°$

[angles in alternate segments]

$\Rightarrow\quad \angle BCD = 108°$

24. *(d)* As, $\angle B + \angle D = 180°$

and $\quad \angle A + \angle C = 180°$

[sum of two opposite angle in a cyclic quadrilateral]

$x + 10 + 5y + 5 = 180°$

$\Rightarrow\quad x + 5y = 165°\quad$ ...(i)

$\because\quad \angle A + \angle C = 180°$

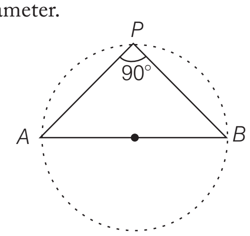
$\Rightarrow\quad 2x + 4 + 4y - 4 = 180°$

$\Rightarrow\quad 2x + 4y = 180°\quad$ ...(ii)

On solving, Eqs. (i) and (ii), we get x and y are 40° and 25°, respectively.

$\therefore\quad x + y = 40° + 25° = 65°$

25. *(c)* The locus of a point is the circumference of the circle with AB as diameter.



26. *(b)* As, the tangents drawn to a circle from a point outside it are equal. We have,

$AD = AE$, $BD = BF$ and $CE = CF$

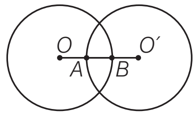
$\Rightarrow\ AD = AB + BD = AB + BF$

and $AD = AC + EC = AC + CF$

$\therefore\ 2AD = AB + AC + (BF + CF)$

$2AD = AB + AC + BC$

27. *(d)* $OB = 4$ cm



$O'A = 2$ cm, $OO' = 3$ cm

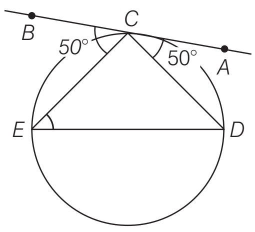
As, $\quad OO' \neq OB + O'A$

So, circle does not touch each other externally.

Also, $\quad OO' \neq OB - O'A$

So, circle does not touch internally, hence they intersect each other at two distinct points.

28. *(a)* Join ED, then



$\angle DEC = \angle ACD = 50°$

[angle in alternate segment]

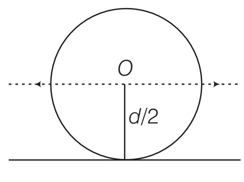
$\angle EDC = \angle BCE = 50°$

[angle in alternate segment]

$\therefore\ \angle DEC = \angle EDC$

So, $\quad CD = CE$

29. *(b)*



The locus of the centre of the front wheel of the cycle is a line parallel to the path of the wheel of the bicycle at a height $d/2$ cm.

30. *(c)* $\because\quad \angle OAC = \angle OCA$

[angle opposite to equal sides]

$\therefore\ 2\angle OAC = 80°$

[external angle of ΔOAC]

$$\angle OAC = \frac{80°}{2} = 40° \Rightarrow x = 40°$$

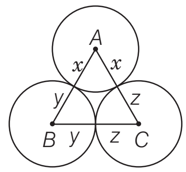
Hence, the value of x is 40°.

31. *(a)* Since, the direct common tangents of two circles divides the line joining their centres externally in the ratio of their radii. Here, both the circles being of equal radii. Hence, their ratio is 1 : 1.

32. *(a)* Let $AB = 9$ cm, $BC = 7$ cm,

$AC = 6$ cm

Let x, y and z be radii of circles with centre A, B, C.



$x + y = 9\quad$ ...(i)

$y + z = 7\quad$ ... (ii)

and $\quad z + x = 6\quad$ ...(iii)

On adding Eqs. (i), (ii) and (iii), we get

$2(x + y + z) = 22$

$\Rightarrow\quad (x + y + z) = 11\quad$ ...(iv)

On putting $x + y = 9$ in Eq. (iv), we get

$9 + z = 11 \Rightarrow z = 2$ cm

On putting value of z in Eq. (ii), we get

$y + 2 = 7 \quad\Rightarrow\quad y = 5$

Again, putting value of y in Eq. (i), we get

$x + 5 = 9 \quad\Rightarrow\quad x = 4$ cm

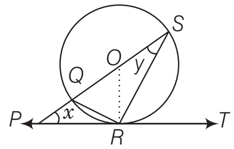
So, radii are 4 cm, 5 cm and 2 cm.

33. *(b)* $\angle QRS = 90°$ [angle in semi-circle]

and $\angle QRP = \angle QSR = y°$

[angle in alternate segments]

Also, $\quad \angle PRS = 90 + y°$



In ΔPRS,

$\angle SRP + \angle RPS + \angle PSR = 180°$

[$\because$ sum of all angles of triangle is 180°]

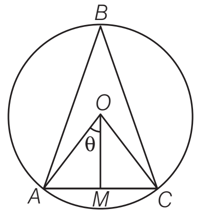
$\Rightarrow\quad (90 + y°) + x° + y° = 180°$

$\Rightarrow\quad x + 2y° = 90°$

34. *(a)* A line perpendicular to the given line, passing through the given point is the required locus.

35. *(c)* Given, $OA = 3$ cm and $AC = 3$ cm and $OA = OC = 3$ cm (radius of a circle)
In ΔAOC, all sides are equal.
$\therefore \Delta\, AOC$, is an equilateral triangle

$\Rightarrow \quad \angle AOC = 60°$



Hence, $\angle ABC = \frac{1}{2}\angle AOC$

[Angle subtended by an arc at the centre is double the angle subtended by it at any point on the circle]

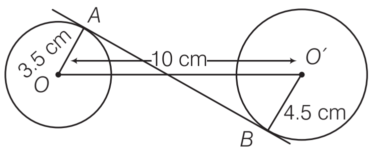
$= \frac{60°}{2} = 30°$

36. *(c)* From figure, length of the transverse common tangent of these circles

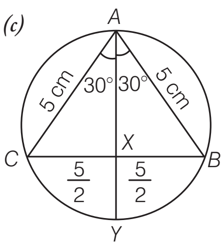
$= \sqrt{\text{(distance between the centres of circles)}^2 - \text{(sum of radius)}^2}$

$= \sqrt{10^2 - (4.5 + 3.5)^2} = \sqrt{10^2 - 8^2}$

$= \sqrt{(100 - 64)} = \sqrt{36} = 6$ cm



37. *(c)*



In equilateral triangle, angle bisector and median are same

$\therefore \quad CX = BX = \frac{BC}{2} = \frac{5}{2}$ cm

In ΔCXA, $\cos 30° = \frac{AX}{AC} \Rightarrow \frac{\sqrt{3}}{2} = \frac{AX}{5}$

$\Rightarrow \quad AX = \frac{\sqrt{3}}{2} \times 5 = \frac{5\sqrt{3}}{2}$ cm

AY and BC are the chord of circle.

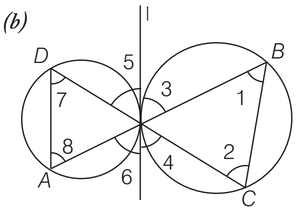
$\therefore \quad AX \cdot XY = BX \cdot XC$

$\Rightarrow \quad \frac{5\sqrt{3}}{2} \cdot XY = \frac{5}{2} \cdot \frac{5}{2} \Rightarrow XY = \frac{5}{2\sqrt{3}}$

$\therefore \quad AY \cdot AX = \left(\frac{5\sqrt{3}}{2} + \frac{5}{2\sqrt{3}}\right) \times \frac{5\sqrt{3}}{2}$

$= 25 \text{ cm}^2$

38. *(b)*



l is the common tangent of two circles

Then, $\angle 1 = \angle 4$ and $\angle 5 = \angle 8$
[angle in alternate segment]

But $\angle 4 = \angle 5$ [opposite angles]

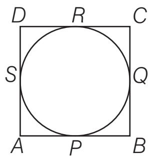
$\therefore \quad \angle 1 = \angle 8$

Similarly $\angle 2 = \angle 7$

But these are alternate angles

$\Rightarrow AD \parallel BC \Rightarrow ACBD$ is a trapezium.

39. *(c)* We know that, two tangents drawn from an external point to a circle are equal in length.



$\therefore \quad AP = AS \quad \ldots(\text{i})$

$BP = BQ \quad \ldots(\text{ii})$

$CR = CQ \quad \ldots(\text{iii})$

$DR = DS \quad \ldots(\text{iv})$

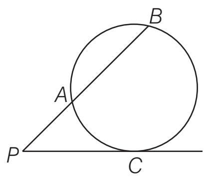
On adding Eqs. (i), (ii), (iii) and (iv), we get

$(AP + BP) + (CR + DR)$

$= (AS + DS) + (BQ + CQ)$

$\Rightarrow \quad AB + CD = AD + BC$

40. *(a)* If a secant to a circle intersect circle at points A and B and PC is a tangent to circle, then $PC^2 = PA \times PB$



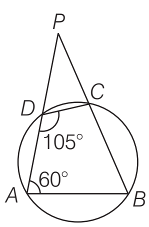
which is equivalent to saying that area of rectangle with PA and PB as sides is equal to the area of square with PC as sides.

41. *(b)* Given, $\angle BAD = 60°$ and $\angle ADC = 105°$

As $ABCD$ is a cyclic quadrilateral

$\therefore \angle DCP = \angle BAD = 60°$

and $\angle ADC + \angle CDP = 180°$ [linear line]



$\Rightarrow 105° + \angle CDP = 180°$

$\Rightarrow \quad \angle CDP = 75°$

In ΔCPD,

$\angle DCP + \angle CDP + \angle DPC = 180°$
[by angle sum property of a triangle]

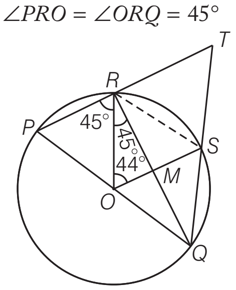
$\Rightarrow \quad 60° + 75° + \angle DPC = 180°$

$\Rightarrow \quad \angle DPC = 180° - 135° = 45°$

42. *(d)* $\angle PRQ = 90°$ as angle in a semi-circle is 90°.

Since, OR is a bisector of $\angle PRQ$.

$\therefore \quad \angle PRO = \angle ORQ = 45°$



Also, $\quad OP = OR \quad$ [radius]

$\therefore \quad \angle OPR = 45°$

in ΔORS, $\quad OR = OS$

$\Rightarrow \quad \angle ORS = \angle OSR$

in ΔORS, by angle sum property of a triangle,

$\angle ORS + \angle OSR + \angle ROS = 180°$

$2\angle ORS = 180° - 44°$

$2\angle ORS = 136°$

$\therefore \quad \angle ORS = \angle OSR = \frac{136°}{2} = 68°$

$\therefore \quad \angle MRS = 68° - 45° = 23°$

$\Rightarrow \quad \angle PRS = 90° + 23° = 113°$

By properties of cyclic quadrilateral,

$\angle PRS + \angle PQS = 180°$

$\Rightarrow \angle PQS = 180° - 113° = 67°$

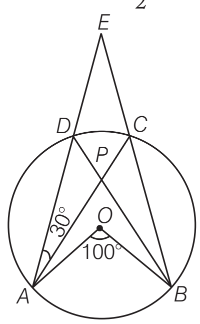
in ΔPTQ,

$\angle QPT + \angle PQT + \angle PTQ = 180°$
[by angle sum property of a triangle]

$\Rightarrow \angle PTQ = 180° - 45° - 67° = 68°$

$\therefore \quad \angle RTS = \angle PTQ = 68°$

43. *(b)* Since, $\angle ADB = \frac{1}{2}\angle AOB = 50°$



In ΔDPA, by angle sum property of a triangle

$\angle DAP + \angle ADP + \angle DPA = 180°$

$\Rightarrow \quad 30° + 50° + \angle DPA = 180°$

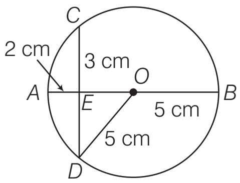
$\Rightarrow \quad \angle DPA = 100°$

Also, DPB is a straight line,

$\therefore \angle DPA + \angle APB = 180°$

$\Rightarrow \quad \angle APB = 180° - 100° = 80°$

44. *(b)* In ΔOED,



[by pythagoras theorem]

$$(OD)^2 = (DE)^2 + (EO)^2$$

$\Rightarrow \quad (5)^2 = (DE)^2 + (3)^2$

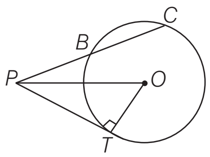
$\Rightarrow \quad (DE)^2 = 25 - 9 = 16$

$\Rightarrow \quad DE = 4$ cm

45. *(a)* Given, $PO = 10$ cm, $OT = 6$ cm and $PB = 5$ cm

In right angled ΔOTP,

$$(OP)^2 = (PT)^2 + (OT)^2$$



$\Rightarrow \quad (10)^2 = (PT)^2 + 6^2$

$\Rightarrow \quad PT^2 = 100 - 36 \Rightarrow PT^2 = 64$

$\Rightarrow \quad PT = \sqrt{64}$

$\therefore \quad PT = 8$ cm

By using theorem of circle,

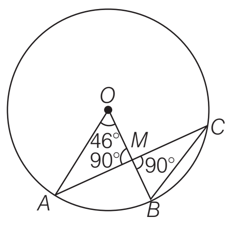
$(PT)^2 = PB \times PC$

$\Rightarrow \quad 8^2 = 5 \times (BC + PB)$

$\Rightarrow \quad 64 = 5(BC + 5) \Rightarrow 5BC = 39$

$\therefore \quad BC = 7.8$ cm

46. *(c)* Given, $\angle AOB = 46°$,



$\therefore \angle ACB = \frac{1}{2}\angle AOB = \frac{1}{2} \times 46° = 23°$

[angle subtended on the circumference is half of the angle subtend on centre.]

In ΔMCB,

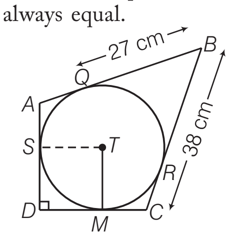
$\angle C + \angle B + \angle BMC = 180°$

[by angle sum property of a triangle]

$\Rightarrow \quad 23° + \angle B + 90° = 180°$

$\Rightarrow \quad \angle B = 67°$

47. *(b)* We know that, the tangents drawn to the circle from a point outside the circle are always equal.



$\therefore \quad BQ = BR = 27$ cm

$\Rightarrow \quad RC = BC - BR$

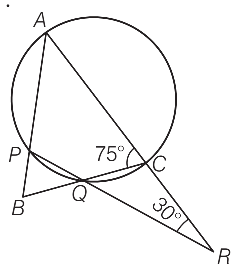
$= 38 - 27 = 11$ cm

$\therefore \quad RC = CM = 11$ cm

Now, $\quad DM = 25 - 11 = 14$ cm

$\therefore \quad ST = DM = 14$ cm

48. *(d)* We know that, the sum of opposite angles in a cyclic quadrilateral is always 180°.



$\angle ACQ + \angle APQ = 180°$

$\Rightarrow \quad 75° + \angle APQ = 180°$

$\Rightarrow \quad \angle APQ = 180° - 75° = 105°$

Since, $\quad \angle APQ + \angle BPQ = 180°$

[linear pair]

$\therefore \quad 105° + \angle BPQ = 180°$

$\Rightarrow \quad \angle BPQ = 180° - 105° = 75°$

Since, $\angle ACQ$ is an exterior angle of ΔRCQ.

$\therefore \quad \angle ACQ = \angle CRQ + \angle CQR$

$\Rightarrow \quad 75° = 30° + \angle CQR$

$\Rightarrow \quad \angle CQR = 45° = \angle PQB$

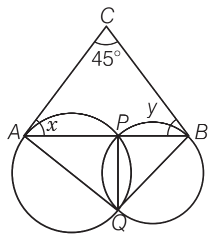
[vertically opposite angles]

In $\Delta BPQ, \angle B + \angle P + \angle Q = 180°$

[by angle sum property of a triangle]

$\Rightarrow \angle B + 75° + 45° = 180° \Rightarrow \angle B = 60°$

49. *(d)*



In ΔCAB, $45° + \angle x + \angle y = 180°$

[by angle sum property of triangle]

$\Rightarrow \quad \angle x + \angle y = 180° - 45° = 135° \quad \ldots$(i)

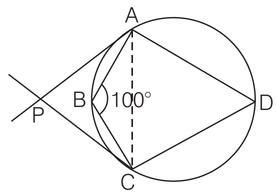
$\angle x = \angle AQP$ and $\angle y = \angle BQP$

[angles in alternate segment]

$\therefore \angle AQB = \angle AQP + \angle BQP$

$= \angle x + \angle y = 135°$

50. *(b)* We know that, the sum of opposite angles of a cyclic quadrilateral is always 180°.



$\therefore \quad \angle B + \angle D = 180°$

$\Rightarrow \quad 100 + \angle D = 180° \Rightarrow \angle D = 80°$

$\therefore \quad \angle ACP = \angle PAC = 80°$

[by theorem of alternate interior segment]

In ΔPAC,

$\angle P + \angle PAC + \angle PCA = 180°$

[by angle sum property of a triangle]

$\Rightarrow \quad \angle P + 80° + 80° = 180°$

$\Rightarrow \quad \angle P = 180° - 160° = 20°$

51. *(b)* Since, the length of tangents from an external point to a circle are equal

$\therefore \quad CR = CS = 27$ cm

$\therefore BR = (BC - CR) = 38 - 27 = 11$ cm

$\Rightarrow \quad BQ = BR = 11$ cm

join OQ, then $PAQO$ is a rectangle.

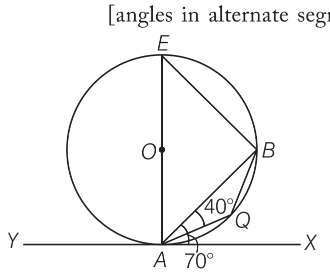
$\therefore \quad AQ = PQ = 10$ cm

So, $AB = (AQ + BQ) = (10 + 11) = 21$ cm

52. *(b)* Given, $\angle BAX = 70°$

and $\quad \therefore \angle BEA = 70°$

[angles in alternate segment]



Since, $AQBE$ is a cyclic quadrilateral.

$\therefore \quad \angle AEB + \angle AQB = 180°$

$\Rightarrow \quad \angle AQB = 180° - 70° = 110°$

In ΔABQ,

$\angle ABQ + \angle BAQ + \angle AQB = 180°$

$\Rightarrow \angle ABQ = 180° - (40° + 110°)$

$= 180° - 150° = 30°$

53. *(a)* We know that, the angles in the same segment of a circle are equal.

$\therefore\ \angle XBY = \angle XAY = 45°$

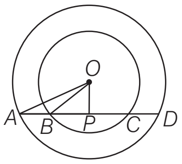
In ΔBXY, by angle sum property of a triangle

$\angle BXY + \angle XBY + \angle BYX = 180°$

$\Rightarrow \quad 50° + 45° + \angle BYX = 180°$ $\quad [\because \angle BXY = 50°]$

$\Rightarrow \quad \angle BYX = 180° - 95° = 85°$

54. *(a)* In right angled ΔOPB,



[by pythagoras theorem]

$OB^2 = OP^2 + BP^2$

$\Rightarrow \quad (15)^2 = (12)^2 + BP^2$

$\Rightarrow \quad 225 - 144 = BP^2$

$\Rightarrow \quad (BP)^2 = 81 \Rightarrow BP = 9$ cm

and in right angled ΔAOP, by pythagoras theoram,

$(OA)^2 = (OP)^2 + (AP)^2$

$\Rightarrow \quad (20)^2 = (12)^2 + (AP)^2$

$\Rightarrow \quad (AP)^2 = 400 - 144 = 256$

$\Rightarrow \quad AP = 16$ cm

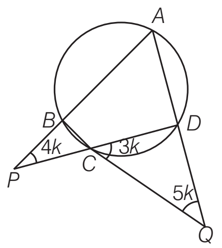
Hence, $AB = AP - BP = 16 - 9 = 7$ cm

55. *(b)* Given, $\dfrac{x}{3} = \dfrac{y}{4} = \dfrac{z}{5} = k$ [say]

$\therefore \quad x = 3k,\ y = 4k$ and $z = 5k$

Since, $\angle DCQ = \angle BCP = 3k$

[vertically opposite angle]



In ΔDCQ, by angle sum property of a triangle

$\angle CDQ = 180° - (3k + 5k) = 180° - 8k$

By properties of cyclic quadrilateral,

$\angle CDQ = \angle CBA = 180° - 8k$

$\Rightarrow \quad \angle PBC = 8k$

In ΔPBC, by angle sum property of a triangle

$\angle P + \angle B + \angle C = 180°$

$\therefore \quad 4k + 8k + 3k = 180°$

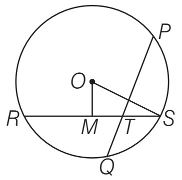
$\Rightarrow \quad k = \dfrac{180°}{15} \Rightarrow k = 12°$

$\therefore \quad x = 36°,\ y = 48°,\ z = 60°$

56. *(b)* We know that, angle subtended on the circumference is half of the angle subtended at the centre.

$\angle AQP = \dfrac{1}{2}\angle AOP = \dfrac{1}{2} 75° = 37.5°$

57. *(b)*



Given, $OM = 3$ cm, $OS = 5$ cm.

$MS = 4$ cm $= RM$

{Using pythagoras theorem}

$\Rightarrow \quad RS = 8$ cm

We know that, $TS = 1/3$ of RT

$\Rightarrow \quad TS = 1/4$ of RS, if $RS = 8$ cm

Then, $\quad TS = 2$ cm

Also, $\quad RT \times TS = PT \times TQ$

[When there are two intersecting chords, the product of the segments of one chord is equal to the product of the segments of other.]

$\Rightarrow \quad 6 \times 2 = PT \times TQ$

$\Rightarrow \quad PT \times TQ = 12$

Now, by $AM \geq GM$ inequality,

$(PT + TQ)/2 \geq \sqrt{(PT \times TQ)}$

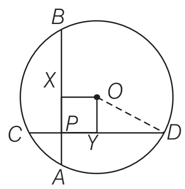
$(PT + TQ)/2 \geq \sqrt{12}$

$\Rightarrow \quad PT + TQ \geq 2\sqrt{12} \Rightarrow PQ \geq 2\sqrt{12}$

or $\quad PQ \geq 4\sqrt{3}$

$\therefore$ Minimum $PQ = 4\sqrt{3}$

58. *(a)* When 2 chords AB and CD intersect at P then



$AP \times PB = CP \times PD$

Hence,

$4 \times 6 = 3 \times PD$, Thus,

$PD = 8$ cm

Now $AB = AP + PB$

$= 10$ cm

and $CD = CP + PD$

Thus, $CD = 11$ cm

Consider the circle with center O.

Drop a perpendicular from O to chord AB and CD. This will bisect the chords at X and Y.

$\therefore \quad AX = \dfrac{AB}{2} = 5$ cm

and $\quad YD = \dfrac{CD}{2} = \dfrac{11}{2}$ cm

Now, $PX = AX - AP = (5 - 4) = 1$ cm

$OY = PX = 1$ cm

$[\because PXOY$ is a rectangle$]$

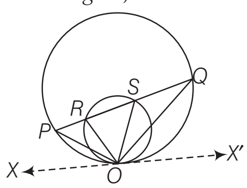
In right triangle OYD

$OD = \sqrt{OY^2 + YD^2}$

[by pythagoras theorem]

$= \sqrt{1^2 + \left(\dfrac{11}{2}\right)^2} = \dfrac{5\sqrt{5}}{2}$ cm

59. *(a)* In the figure,



Let $\angle POX = x°$, $\angle ROP = y°$

$\angle POX = \angle PQO = x$

and $\angle ROX = \angle RSO = x° + y°$

[angles in alternate segment]

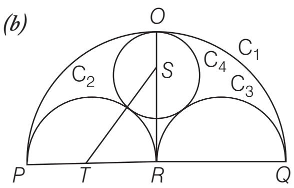
Now, in $\Delta SQO, \angle RSO$ is the exterior angle

$\therefore \quad \angle SQO + \angle SOQ = x° + y°$

$\Rightarrow \quad x + \angle SOQ = x° + y°$

$\Rightarrow \quad \angle SOQ = y = \angle ROP$

60. *(b)*



Assume that the radius of $C_4 = r$ and $PQ = k$

Now, $PR = k/2 = RQ = RO$

$\Rightarrow \quad RS = (k/2) - r, \quad RT = k/4$

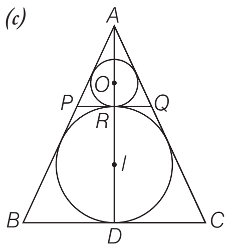
$ST = (k/4) + r$

Applying pythagoras theorem in triangle STR.

$\left(\dfrac{k}{4} + r\right)^2 = \left(\dfrac{k}{4}\right)^2 + \left(\dfrac{k}{2} - r\right)^2$

$\Rightarrow \quad r = k/6 = PQ/6$

61. *(c)*



In-radius of equilateral triangle of side

$a = \dfrac{a}{2\sqrt{3}}$,

Diameter of larger circle $= \dfrac{a}{\sqrt{3}}$

Let us say common tangent PQ touches the two circle at R, centre of smaller circle is I.

Now, PQ is parallel to BC. AR is perpendicular to PQ. Triangle PQR is also an equilateral triangle and AORID is a straight line.

$$AD = \frac{\sqrt{3}}{2}a,\ RD = \frac{a}{\sqrt{3}},\ AR = \frac{\sqrt{3}}{2}a - \frac{a}{\sqrt{3}}$$

$$AR = \frac{3a - 2a}{2\sqrt{3}} = \frac{a}{2\sqrt{3}},\ AR = \frac{1}{3}AD$$

Radius of smaller circle = 1/3 radius of larger circle

Radius of smaller circle

$$= \frac{1}{3}\cdot\frac{a}{2\sqrt{3}} = \frac{a}{6\sqrt{3}}$$

Area of smaller circle = πr^2

$$\pi\left(\frac{a}{6\sqrt{3}}\right)^2 = \frac{\pi a^2}{108}$$

$$\text{Area of triangle} = \frac{\sqrt{3}}{4}a^2$$

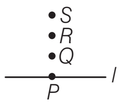
$$\text{Ratio} = \frac{\pi a^2}{108} : \frac{\sqrt{3}}{4}a^2,\quad \pi : 27\sqrt{3}$$

62. *(d)* I. It is true that the opposite angles of a cyclic quadrilateral are supplementary.

II. It is also true that the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.

Hence, both statements are individually true, but neither statements implies to each other.

63. *(c)* I. It is clear from the figure that points Q, S and R are in a straight line.

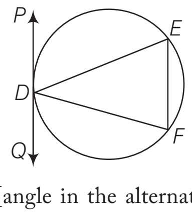


So, statement I is true.

II. Since, PQ is parallel to EF .

$\therefore \angle PDE = \angle DEF$ [alternate angle]

Also, $\angle PDE = \angle EFD$



[angle in the alternate segment of chord ED]

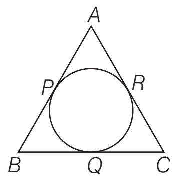
$\therefore \quad \angle DEF = \angle DFE$

$\Rightarrow \quad \Delta DEF$ is an isosceles triangle.

So, statement II is also true.

Hence, the statement I and II are correct.

64. *(d)* As, the tangents drawn from an external point to a circle are equal.



$\therefore \quad AP = AR \quad \ldots$(i)

$BQ = BP \quad \ldots$(ii)

and $CR = QC \quad \ldots$(iii)

On adding Eqs. (i), (ii) and (iii), we get

$$AP + BQ + CR = BP + QC + RA$$

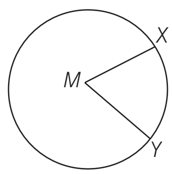
and perimeter of

$$\Delta ABC = AB + BC + CA$$
$$= (AP + PB) + (BQ + QC) + (CR + RA)$$
$$= (AP + BQ) + (BQ + CR) + (CR + AP)$$
$$= 2\,(AP + BQ + CR)$$

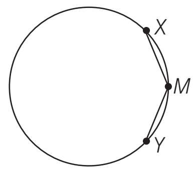
$\therefore \quad AP + BQ + CR = 1/2$ [perimeter of ΔABC]

Hence, the statement I and II are correct.

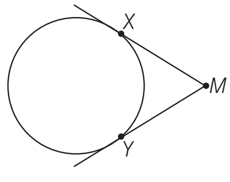
65. *(d)* I. Consider the possibility that M could be the centre of the circle, We know this could be true because M being the center point of the circle would make line XM and YM radii of the circle, which would mean that they are equal.



II. The statement presents us with the possiblity that point M lies somewhere on the arc of XY. Well, if point M rested exactly halfway between X and Y, then straight lines drawn from X to M and Y to M would certainly be equal.



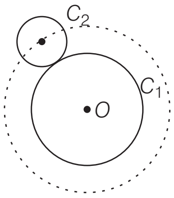
III. The statement presents us with the possibility that M lies somewhere outside of the circle. If two tangents are drawn from point M at X and Y they will be equal.



Hence, the statement I, II and III are correct.

66. *(d)* I. Circle is a locus of points which are equidistant from a fixed point.

II. Let C_1 be a fixed circle and C_2 be moving circle then locus of centre of C_2 is also a circle.



III. There does not exists any point which is equidistant from three distinct points on a line.

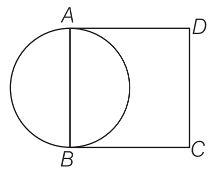
IV. If P lies on bisector of $\angle B$ then P is equidistant from AB and BC also, P lies on bisector of $\angle C$.

$\Rightarrow P$ is equidistant from BC and CD

$\Rightarrow P$ is equidistant from AB and CD.

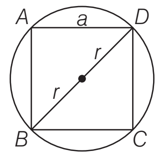
Hence, I, II, and IV are correct statement.

67. *(d)* I.



So, the square is not cyclic.

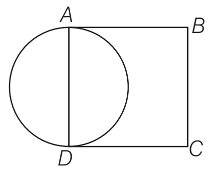
II.



Square has less area than circle.

III. From the above figure it is clear that the diameter of circle is equal to the diagonal of square. $\Rightarrow 2r = a\sqrt{2}$

IV.



So, it is possible that the sides of square is equal to diameter of circle.

Hence statement II, III and IV are correct.

68. *(c)* We observe that the arc BC makes $\angle BOC = z$ at the centre and $\angle BAC = x$ at a point on the circumference.

$\therefore \quad z = 2x$

In ΔOBC, we have

$\angle OBC + \angle OCB + \angle BOC = 180°$

$\Rightarrow \quad y + z + y = 180°$

$\Rightarrow \quad z + 2y = 180° \ \Rightarrow\ 2x + 2y = 180°$

$\Rightarrow \quad x + y = 90°$

$\Rightarrow \angle BAC + \angle OBC = 90°$

69. *(a)* In ΔOBC, we have

$\angle OBC + \angle OCB + \angle BOC = 180°$

$\Rightarrow \quad y + y + t = 180° \Rightarrow t = 180° - 2y$

Now, $z = 360° - t \Rightarrow z = 360° - (180° - 2y)$

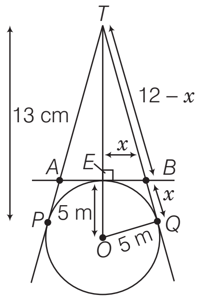
$\Rightarrow 2x = 180° + 2y \Rightarrow 2x - 2y = 180°$ $[\because z = 2x]$

$\Rightarrow \quad x - y = 90°$

$\Rightarrow \angle BAC - \angle OBC = 90°$

70. *(b)* In right angled ΔOTQ,

$$OT^2 = OQ^2 + TQ^2$$



$\Rightarrow (13)^2 = (5)^2 + (TQ)^2$

$\Rightarrow TQ^2 = 169 - 25 = 144 \Rightarrow TQ = 12\text{cm}$

In right angled ΔTEB,

$TB^2 = EB^2 + TE^2$

$\because (EB = BQ)$

$\Rightarrow (12 - x)^2 = BQ^2 + TE^2$

$\Rightarrow \quad 144 + x^2 - 24x = x^2 + (8)^2$

$\Rightarrow \quad 144 + x^2 - 24x = x^2 + 64$

$\Rightarrow \quad 24x = 80 \Rightarrow x = \dfrac{20}{6} = \dfrac{10}{3}$ cm

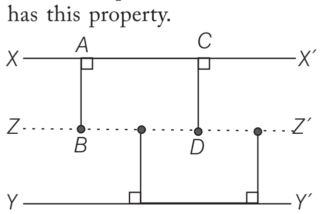
$\therefore \quad AB = 2EB = 2x = 2 \times \dfrac{10}{3} = \dfrac{20}{3}$ cm

Hence, the length of AB is $\dfrac{20}{3}$ cm.

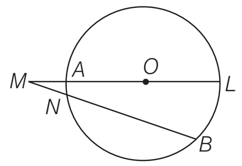
71. *(b)* The locus obtained is the circumference of the circle concentric with the given circle and having radius equal to the distance of the chords from the centre.

72. *(c)* Statements I and II are both correct, because the locus of points which are equidistant from two parallel lines is a line parallel to both of them and lie mid way between them.

Also, it is true that the perpendicular distances of any point on this locus line from two original parallel lines are equal. further, no point outside this locus line has this property.



73. *(d)* Since, secants LA and BN are intersecting at an exterior point M, then



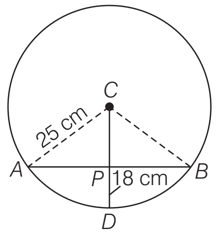
$LM \times AM = BM \times NM$

$\Rightarrow \quad \dfrac{MA}{MB} = \dfrac{MN}{LM} < 1$

74. *(b)* The perpendicular bisector of the chord of a circle always passes through the centre. So, statement I is wrong.

The angle in a semi-circle is a right angle. So, statement II is correct.

75. *(d)* Here, $CA = CD = CB = 25$ cm



and $DP = 18$ cm

In right angled ΔACP,

$CP = CD - PD = 25 - 18 = 7\text{cm}$

Now, $AC^2 = CP^2 + AP^2$ [by pythagoras theorem]

$\therefore$

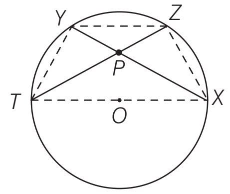
$AP = \sqrt{AC^2 - CP^2} = \sqrt{(25)^2 - (7)^2}$

$= \sqrt{625 - 49} = \sqrt{576} = 24$ cm

Similarly, $PB = 24$ cm

$\therefore \quad AB = AP + PB = 24 + 24 = 48$ cm

76. *(c)* When two chords of a circle are intersect internally, then they are divided in a proportion.



i.e. $PX \cdot PY = PZ \cdot PT$

In ΔPXZ and ΔPTY, $\angle ZPX = \angle YPT$ [vertically opposite angles]

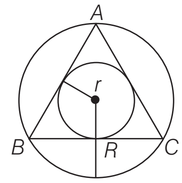
$\angle PZX = \angle PYT$ [angles in same segment]

$\angle PXZ = \angle PTY$ [angles in same segment]

$\therefore \quad \Delta PXZ \sim \Delta PTY$

Hence, the both statements are correct.

77. *(a)* In ΔABC, $AB = BC = AC$



Hence, ΔABC is equilateral triangle.

Let r be the radius of incircle and R be the radius of circumcircle.

Now, radius of incircle, $r = \dfrac{\text{side}}{2\sqrt{3}} = \dfrac{AB}{2\sqrt{3}}$

and radius of circumcircle,

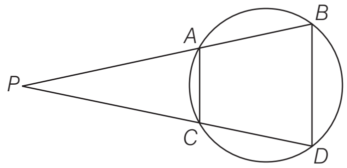
$$R = \frac{\text{side}}{\sqrt{3}} = \frac{AB}{\sqrt{3}}$$

So, the required ratio $= \dfrac{R}{r} = \dfrac{AB/\sqrt{3}}{AB/2\sqrt{3}}$

$= \dfrac{2}{1} = 2:1$

78. *(b)* I. AB and CD are chords when produced meet externally at P.

$\therefore AP \times BP = CP \times DP$



II. Since, $ABDC$ is a cyclic quadrilateral

$\angle PAC = \angle PDB$

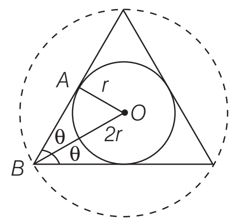
and $\angle PCA = \angle PBD$

Also, $\angle APC = \angle BPD$ [common]

$\therefore \Delta PAC \sim \Delta PDB$

So, statement II is true.

79. *(c)* Let $OA = r$ be the inradius of circle. Then, circumradius $= OB = 2r$



We know that, inradius is the perpendicular distance of centre O from side of triangle and circumradius OB bisect $\angle B$. Again, let $\angle OBA = \theta$

In right angled ΔOAB,

$\sin\theta = \dfrac{OA}{OB} = \dfrac{r}{2r}$

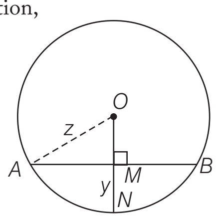
$\Rightarrow \sin\theta = \dfrac{1}{2} = \sin 30° \quad \left[\because \sin 30° = \dfrac{1}{2}\right]$

$\therefore \quad \theta = 30°$

Then, $\angle B = 2 \times \theta = 2 \times 30° = 60°$

Hence, given triangle is an equilateral triangle.

80. *(d)* Let O be the centre of circle and AB be the chord of an arc. According to the question,



Length of chord $AB = x$, Radius of circle $= OA = z$

and height of arc $MN = y$

Now, $AM = MB = \frac{AB}{2} = \frac{x}{2}$

[Perpendicular from the centre to a chord bisects the chord]

and $OM = ON - MN$

$= z - y$ [$\because ON = z$ (radius)]

In right angled ΔOMA,

$OA^2 = OM^2 + AM^2$ [by pythagoras theorem]

$\Rightarrow z^2 = (z - y)^2 + \left(\frac{x}{2}\right)^2$

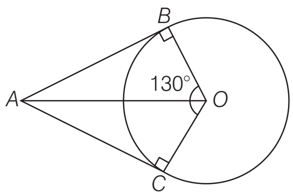
$\Rightarrow z^2 = z^2 + y^2 - 2yz + \frac{x^2}{4}$

[$\because (a - b)^2 = a^2 + b^2 - 2ab$]

$\Rightarrow 2yz - y^2 = \frac{x^2}{4} \Rightarrow 4(2yz - y^2) = x^2$

$\Rightarrow 4y(2z - y) = x^2$

81. *(c)* Given, AB and AC are the tangents of the circle.



and $OB = OC$ [radius of circle]

$\therefore \angle B = \angle C = 90°$

[radius is perpendicular to the tangent]

In quadrilateral $ABOC$,

$\angle ABO + \angle OCA + \angle BAC + \angle BOC = 360°$

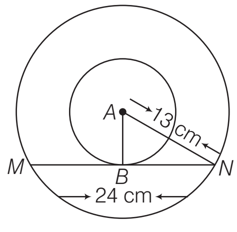
$\Rightarrow 90° + 90° + \angle BAC + 130° = 360°$

$\Rightarrow \angle BAC = 180° - 130° = 50°$.

Hence, the angle between the tangents at the ends of the radius is 50°.

82. *(a)* Given, $MN = 24$ cm and

$AN = \frac{26}{2}\text{cm} = 13\,\text{cm}$



Now, we draw a perpendicular bisector from A to MN, which meets MN at B.

Then, $MB = BN = \frac{MN}{2} = \frac{24}{2} = 12\,\text{cm}$

[$\because$ perpendicular from centre to the chord bisects the chord]

In right angled ΔABN,

$AN^2 = AB^2 + BN^2$

$\Rightarrow 13^2 = AB^2 + 12^2$

$\Rightarrow 169 = AB^2 + 144$

$\Rightarrow AB^2 = 169 - 144 = 25$

$\Rightarrow AB = 5\,\text{cm}$

$\therefore$ Radius of the inner circle is 5 cm.

83. *(d)* Given, $AD = 34\,\text{cm}$ and $AB = 30\,\text{cm}$

Let C be a point on AB such that $OC \perp AB$

$\Rightarrow OD = OB = 17$ cm

Also, $CB = \frac{AB}{2} = 15$ cm

[Perpendicular from centre to the chord bisects the chord]

$\Rightarrow OC = \sqrt{OB^2 - CB^2} = \sqrt{17^2 - 15^2}$

$= \sqrt{289 - 225} = \sqrt{64} = 8$ cm

84. *(d)* Since, $AP = BP; DE = BD$ and $AC = CE$

$\because$ Perimeter of ΔPCD

$= PC + PD + CD$

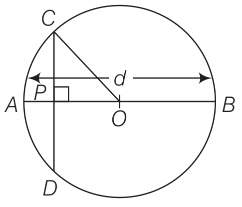
$= PC + PD + CE + DE$

$= (PC + CE) + (PD + DE)$

$= (PC + CA) + (PD + BD)$

$= PA + PB = 2PA = 2 \times 16 = 32$ cm

85. *(c)* Given, $AB = d$ is diameter of circle and CD intersect AB at P in the ratio 1 : 2.



$\therefore AP = \frac{d}{3}$ and $PB = \frac{2d}{3}$

$\Rightarrow OA = OC = OB = d/2$

$\left[\because \text{radius of circle} = \frac{1}{2}\text{ diameter}\right]$

In $\Delta OPC, OP = OA - AP = \frac{d}{2} - \frac{d}{3} = \frac{d}{6}$

Applying pythagoras theorem in ΔOPC,

$OC^2 = OP^2 + PC^2$

$\Rightarrow \left(\frac{d}{2}\right)^2 = \left(\frac{d}{6}\right)^2 + PC^2$

$\Rightarrow (PC)^2 = \frac{d^2}{4} - \frac{d^2}{36}$

$= \frac{9d^2 - d^2}{36} = \frac{8d^2}{36}$

$\Rightarrow PC = \frac{2\sqrt{2}\,d}{6} = \frac{\sqrt{2}d}{3}$

$\therefore CD = 2PC = \frac{2\sqrt{2}d}{3}$

86. *(b)* 1 km = 1000 m, 1 dm = 10 m

$\because$ 2 km and 26 dm

$= 2000 + 260 = 2260$ m

Let radius be r m.

Distance covered in revolution $= 2\pi r$

Distance covered in 452 revolution

$= 452 \times 2\pi r$

$\because 452 \times 2\pi r = 2260$

$\therefore r = \frac{2260 \times 7}{452 \times 2 \times 22} = \frac{35}{44}$

Diameter $= 2 \times \frac{35}{44} = \frac{35}{22} = 1\frac{13}{22}$ m

87. *(b)* Distance covered in 1 revolution

$= 2\pi\,(0.3)$ m $= 0.6\pi$ m

$\because$ Distance covered in 140 revolutions

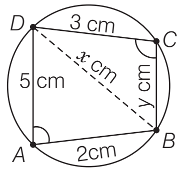
$= 140 \times 0.6\pi$ m

$\Rightarrow$ Speed the cycle

$= 140 \times 0.6 \times \frac{22}{7} \times 60\,\text{m/h}$

$= 15840$ m/h $= 15.84$ km/h

88. *(a)* Let $ABCD$ be cyclic quadrilateral.



in which $\angle A = 60°$,

$AB = 2$ cm, $AD = 5$ cm

Since, $\angle A + \angle C = 180°$

[property of cyclic quadrilateral]

$\Rightarrow 60° + \angle C = 180°$

$\Rightarrow \angle C = 180° - 60° = 120°$

In ΔABD, let BD be x cm.

$\cos 60° = \frac{2^2 + 5^2 - x^2}{2 \times 2 \times 5}$

[by cosine formula]

$\Rightarrow \frac{1}{2} = \frac{4 + 25 - x^2}{2 \times 2 \times 5} \Rightarrow 29 - x^2 = 10$

$\Rightarrow x^2 = 29 - 10 \Rightarrow x^2 = 19$

Again, in ΔBCD,

$\cos 120° = \frac{3^2 + y^2 - x^2}{2 \times 3 \times y}$

[by cosine formula]

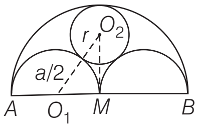
Let length of fourth side of cyclic quadrilateral be y cm.

$\Rightarrow -\frac{1}{2} = \frac{9 + y^2 - 19}{2 \times 3 \times y}$ [$\because x^2 = 19$]

$\Rightarrow -3y = 9 + y^2 - 19$

$\Rightarrow \quad y^2 + 3y - 10 = 0$
$\Rightarrow \quad y^2 + 5y - 2y - 10 = 0$
$\Rightarrow \quad y(y+5) - 2(y+5) = 0$
$\Rightarrow \quad (y+5)(y-2) = 0$
$\Rightarrow \quad y + 5 = 0$ or $y - 2 = 0$
$\therefore \quad y = 2$ cm

89. *(c)* Since, two circles touch each other externally if distance between their centres = Sum of their radii



$O_1O_2 = r + \frac{a}{2}$ and $O_2M = a - r$

$O_1M = \frac{a}{2}$

In right O_1MO_2

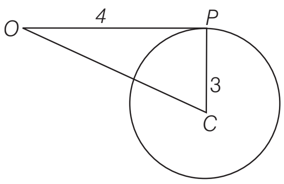
$(O_1O_2)^2 = (O_1M)^2 + (O_2M)^2$

$\Rightarrow \left(r + \frac{a}{2}\right)^2 = \frac{a^2}{4} + (a-r)^2$

$\Rightarrow \quad 3r = a$

$\therefore \quad r = \frac{a}{3}$

90. *(c)* Since, OP is tangent to the circle $OP \perp PC$.

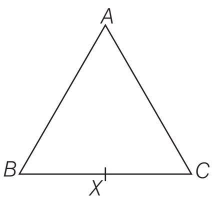


Here, ΔOPC is right angled triangle with $\angle P = 90°$

$OC = \sqrt{(3)^2 + (4)^2} = 5$ units

$\therefore \quad \sin(\angle COP) = \frac{3}{5}$.

91. *(d)* In ΔABC, $AB = AC$ i.e. ΔABC is isosceles triangle.



If L is the locus of points X inside or on the triangle, such that $BX = CX$. Then, L is a straight line passing through A.

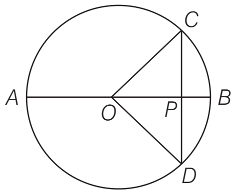
As, AX is bisecting the line BC, then it passes through the centroid.

As, $AB = AC$ in ΔABC, then AX is perpendicular to BC, hence it passes through the orthocentre.

In ΔABC, $AB = AC$ and $AX \perp BC$. So, AX is angle bisector of $\angle A$, hence it passes through the incentre. Hence, all statements are correct.

92. *(b)* Given, radius of a circle = 2 unit

$\therefore \quad CP = PD = \frac{CD}{2} = \frac{2}{2} = 1$ unit

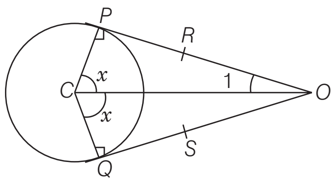


Now, $OP = OD^2 - PD^2$

$= \sqrt{4-1} = \sqrt{3}$ units

$\therefore \quad AP = AO + OP = (2 + \sqrt{3})$ units

93. *(c)* In the given figure, OP and OQ are tangents to the circle with centre C.



Then, $OP = OQ$ and

$\angle CPO = \angle CQO = 90°$

I. We have, $OR \times SQ = OS \times RP$

$\Rightarrow (OP - RP) \times SQ$

$= (OQ - SQ) \times PR$

$\Rightarrow (OP - RP) \times SQ$

$= (OP - SQ) \times PR$

$\Rightarrow \quad SQ = PR \Rightarrow OR = OS$

Hence, statement I is correct.

II. $\because \; \Delta OPC \cong \Delta OCQ$

In ΔOPC, $\quad 90° + x + \angle 1 = 180°$

$\Rightarrow \quad x + \angle 1 = 90°$

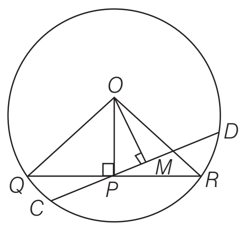
$\Rightarrow \angle POC + \angle QCO = x + \angle 1 = 90°$

Hence, statement II is correct.

94. *(a)* We have, $MP = \sqrt{24}$ cm,

$OQ = OR = 7$ cm

and $\quad QP = PR = \frac{QR}{2} = \frac{2}{2} = 1$



I. In ΔOPQ, $\quad OP = \sqrt{OQ^2 - QP^2}$

$= \sqrt{49 - 1} = \sqrt{48} = 4\sqrt{3}$ cm

In ΔOMP, $OP^2 = PM^2 + OM^2$

$\Rightarrow \quad OM = \sqrt{48 - 24} = \sqrt{24}$ cm

So, $\quad OM = PM = \sqrt{24}$ cm

$\therefore \quad \angle OPM = \angle POM = 45°$

Now, $\angle QPD = 90° + \angle OPM$

$= 90° + 45° = 135°$

Hence, statement I is correct.

II. Now, $\quad$ if $CP = m$ and $PD = n$, then $QP \times PR = CP \times PD$

$\Rightarrow \quad 1 \times 1 = m \times n \Rightarrow mn = 1$

Also, $CM = MD$

$\Rightarrow \quad CP + PM = PD - PM$

$\Rightarrow m + \sqrt{24} = n - \sqrt{24}$

$\Rightarrow \quad n - m = 2\sqrt{24}$

$\therefore \quad m + n = \sqrt{(m-n)^2 + 4mn}$

$= \sqrt{96 + 4} = 10$

So, statement II is also true

III. $\frac{\text{ar}(\Delta OPR)}{\text{ar}(\Delta OMP)} = \frac{\frac{1}{2} \times OP \times PR}{\frac{1}{2} \times OM \times PM}$

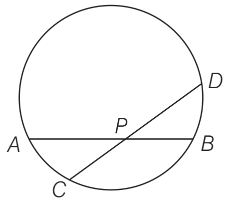
$= \frac{4\sqrt{3} \times 1}{\sqrt{24} \times \sqrt{24}} = \frac{1}{2\sqrt{3}}$

So, statement III is not true.

Hence, the statement I, II are correct.

95. *(c)* From the given circle,

$AP \times BP = CP \times PD \quad \ldots$(i)



Given that, $A'P' = AP$, $B'P' = BP$, $C'Q' = CP$ and $D'Q' = DP$

Now,

$\frac{\text{ar}(\Delta A'P'B')}{\text{ar}(\Delta C'Q'D')} = \frac{\frac{1}{2} \times A'P' \times P'B'}{\frac{1}{2} \times C'Q' \times Q'D'}$

$= \frac{AP \times BP}{CP \times DP} = \frac{AP \times BP}{AP \times BP} = 1$ [using (i)]

$\therefore A'P'B'$ and $C'Q'D'$ are triangles of same area.

So, statement III cannot be false.

Hence, only (c) is correct option.

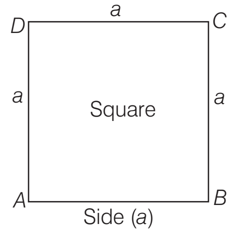
26

AREA AND PERIMETER OF PLANE FIGURES

Generally (7-9) questions have been asked from this chapter. Generally, questions are asked from the topics related to the triangles and circles.

Area of the plane figure is the amount of surface enclosed by its boundary. It is measured in square units. The perimeter of plane figure is total length of the sides enclosing the figure. Unit of Perimeter is same as the unit of sides of a given figure.

SQUARE



Let each side of a square be a unit.

(i) Perimeter of square = 4 (Side) = $4a$ units

(ii) Diagonal of square $= \sqrt{2} \times (Side) = a\sqrt{2}$ units

(iii) Area of square = Side × Side $= (a)^2$ sq units $= \frac{1}{2}$ (Diagonal)$^2 = \frac{1}{2} d^2$ sq units

EXAMPLE 1. The perimeter of a square is $2(2x + 4y)$. Then, the area is

a. $x^2 - 4xy + 4y^2$ b. $x^2 + 4y^2$ c. $x^2 - 4y^2$ d. $x^2 + 4xy + 4y^2$

Sol. d. Here, perimeter of a square $= 2(2x + 4y)$

$\because$ Side of a square $= \dfrac{\text{Perimeter of a square}}{4}$ $\therefore$ Side of a square $= \dfrac{2(2x + 4y)}{4} = x + 2y$

$\because$ Area of square = (Side)^2 $\therefore$ Area of square $= (x + 2y)^2 = x^2 + 4xy + 4y^2$

EXAMPLE 2. The side of a square exceeds the side of the another square by 4 cm and the sum of the areas of the two squares is 400cm^2. The dimensions of the squares are

a. 8 cm and 12 cm b. 6 cm and 10 cm c. 12 cm and 16 cm d. None of these

Sol. c. Let the side of a square be x cm. Then, side of another square $= (x + 4)$ cm.

Area of first square $= x^2$ cm^2 and area of second square $= (x + 4)^2$ cm^2

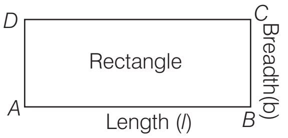
According to the question $\Rightarrow x^2 + (x + 4)^2 = 400 \Rightarrow x^2 + x^2 + 16 + 8x = 400 \Rightarrow 2x^2 + 8x - 384 = 0$

$$x^2 + 4x - 192 = 0 \Rightarrow x^2 + 16x - 12x - 192 = 0$$
$$\Rightarrow x(x+16) - 12(x+16) = 0 \Rightarrow (x-12)(x+16) = 0$$
$$\Rightarrow x - 12 = 0 \text{ or } x + 16 = 0$$
$$\therefore x = 12 \quad \text{[side cannot be negative]}$$

Hence, side of one square = 12 cm and side of another square = 12 + 4 = 16 cm

RECTANGLE

Let l and b be the length and breadth of a rectangle respectively, then

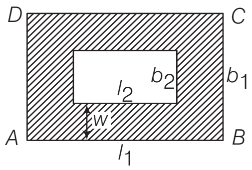


(i) Area of rectangle = Length × Breadth = $l \times b$

(ii) Perimeter of rectangle = 2 (Length + Breadth) $= 2(l + b)$ units

(iii) Diagonal of rectangle $= \sqrt{(\text{Length})^2 + (\text{Breadth})^2} = \sqrt{l^2 + b^2}$ units

(iv) Area of track (Shaded region) $= (l_1 b_1 - l_2 b_2)$ sq units



EXAMPLE 3. The area of a rectangular plot is 180 m^2. If its length is 18 m. Then, its perimeter is

a. 28 m b. 56 m c. 360 m d. None of these

Sol. b. Given, area of rectangular plot = 180 m^2

and length of the plot = 18 m

$$\therefore \text{Breadth} = \frac{\text{Area}}{\text{Length}} = \frac{180}{18} = 10 \text{ m}$$

Now, Perimeter of rectangular plot

= 2 (Length + Breadth) = 2 (18 + 10) = 56 m

Hence, the perimeter of rectangular plot is 56 m.

EXAMPLE 4. The area of the floor of a rectangular hall of length 40 m is 960 m^2. Carpets of size 6 m × 4 m are available. Then, how many carpets are required to cover the hall?

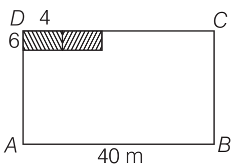
a. 20 b. 30 c. 40 d. 45

Sol. c. Given, area of the floor = 960 m^2

Area of one carpet = $6 \times 4 = 24$ m^2

Number of carpets required

$$= \frac{\text{Area of the floor}}{\text{Area of one carpet}} = \frac{960}{24} = 40$$



Hence, 40 carpets are required to cover the hall.

EXAMPLE 5. A lawn is in the shape of a rectangle of length 60 m and width 40 m. There is a footpath of uniform width 1m bordering the lawn. The area of the path is

a. 194 m^2 b. 196 m^2 c. 198 m^2 d. 200 m^2

Sol. b. Given, length of the outer rectangle = 60 m

and breadth of the outer rectangle = 40 m

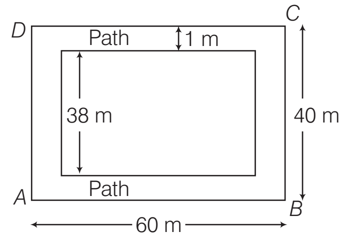
$\therefore$ Area of outer rectangle = $60 \times 40 = 2400$ m^2

$\because$ Width of path = 1m

$\therefore$ Length of the inner rectangle = 60 − (1 + 1) = 58 m

Breadth of the inner rectangle = 40 − (1 + 1) = 38 m

$\therefore$ Area of the inner rectangle = $58 \times 38 = 2204$ m^2



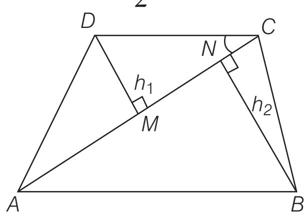
Area of the path = [Area of the outer rectangle] − [Area of the inner rectangle] = (2400 − 2204) = 196 m^2

Hence, the area of the path is 196 m^2.

QUADRILATERAL

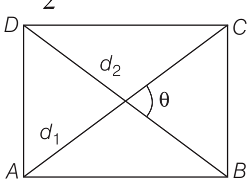
(i) Let $ABCD$ is a quadrilateral in which $DM = h_1$ and $BN = h_2$ are perpendiculars on diagonal AC from other two vertices B and D, then

$$\text{Area of the quadrilateral} = \frac{1}{2} \times \text{Diagonal} \times (h_1 + h_2)$$
$$= \frac{1}{2} \times AC \times (DM + BN) \text{ sq units}$$



(ii) Let d_1 and d_2 are two diagonals and θ is angle between them, then area of quadrilateral

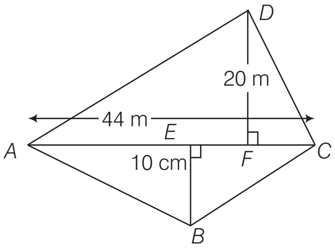
$$= \frac{1}{2} d_1 d_2 \sin\theta$$



EXAMPLE 6. In a quadrilateral *ABCD*, diagonal $AC = 44$ cm and the length of the perpendicular drawn from *B* and *D* to *AC* are 10 cm and 20 cm, respectively. The area of the quadrilateral is

a. 330 cm^2 b. 440 cm^2 c. 550 cm^2 d. 660 cm^2

Sol. d.



Area of quadrilateral

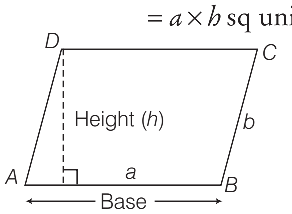
$$= \frac{1}{2} AC(h_1 + h_2) = \frac{1}{2}(44)(20+10) = \frac{1}{2} \times 44 \times 30 = 660 \text{ cm}^2$$

Hence, the area of the quadrilateral is 660 cm^2.

PARALLELOGRAM

Let adjacent sides of a parallelogram are *b* and *a* respectively and *h* is the corresponding altitude (height) of side *a*, then

(i) Area of the parallelogram = *(Base* × *Height)*
$= a \times h$ sq units



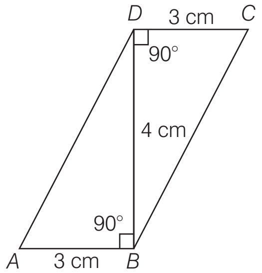
(ii) Perimeter of a parallelogram = 2 (Sum of adjacent side)
$= 2(a+b)$ units

Note Each diagonal of a parallelogram divides it into two triangles of equal area.

EXAMPLE 7. *ABCD* is a parallelogram as shown in figure, then its area is

a. 12 cm^2 b. 14 cm^2 c. 15 cm^2 d. 18 cm^2

Sol. a. Given, base = 3 cm and height = 4 cm



Area of parallelogram,

$ABCD$ = Base × height = $3 \times 4 = 12$ cm^2

EXAMPLE 8. In the parallelogram *ABCD*, $AB = 10$ cm. The altitude corresponding to sides *AB* and *AD* are respectively 7 cm and 8 cm. Then, *AD* is

a. 8.75 cm b. 8.95 cm c. 9 cm d. 9.25 cm

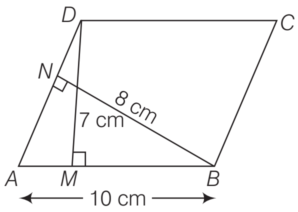
Sol. a. Area of parallelogram *ABCD*

= Base × Corresponding altitude

$\therefore$ Area of parallelogram

$ABCD = AB \times DM$

$= 10 \times 7 = 70$ cm^2 ...(i)



Also, area of parallelogram

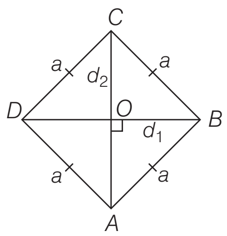
$ABCD = AD \times BN = 8AD$ [$\because BN = 8$ cm] ...(ii)

From Eqs. (i) and (ii), $8AD = 70$

$$\therefore \quad AD = \frac{70}{8} = 8.75 \text{ cm}$$

RHOMBUS

Let the length of each sides of a rhombus is *a* and length of both diagonals are d_1 and d_2, then



(i) Area of rhombus $= \frac{1}{2} \times d_1 \times d_2$ sq units

(ii) Side of rhombus $= \frac{1}{2}\sqrt{d_1^2 + d_2^2}$ units $\Rightarrow 4a^2 = d_1^2 + d_2^2$

(iii) Perimeter of rhombus = 4 × (Side) units

Note Diagonals of a rhombus bisect each other at right angles.

EXAMPLE 9. If the length of the diagonal of a rhombus is $(a+b)$ and its area is $\frac{a^2-b^2}{2}$ sq units, then the other diagonal is

a. $a+b$ b. $a-b$ c. $\frac{a-b}{2}$ d. $\frac{a+b}{2}$

Sol. b. Let second diagonal be *d* units.

Given, area of rhombus $= \frac{a^2-b^2}{2}$

$\because$ Area of rhombus $= \frac{1}{2} \times d_1 \times d_2$

$$\therefore \frac{1}{2} \times d_1 \times d_2 = \frac{a^2-b^2}{2} \Rightarrow \frac{1}{2}(a+b)\cdot d = \frac{a^2-b^2}{2}$$

[$\because d_1 = a+b$]

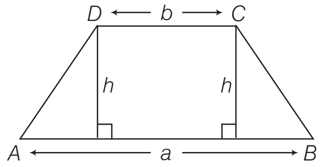
$$\Rightarrow (a+b)\,d = a^2 - b^2 \Rightarrow d = \frac{(a^2-b^2)}{a+b} = \frac{(a-b)(a+b)}{(a+b)}$$

$\therefore \quad d = (a-b)$ [$\because a^2 - b^2 = (a-b)(a+b)$]

TRAPEZIUM

Let the length of parallel sides of a trapezium are a and b and distance between them is h, then

Area of trapezium $= \frac{1}{2}$ (Sum of parallel sides) $\times$ (Distance between them)



$$= \frac{1}{2}(AB + CD) \times h \ = \frac{1}{2}(a+b) \times h \text{ sq units}$$

EXAMPLE 10. The difference between two parallel sides of a trapezium is 4 cm and the perpendicular distance between them is 19 cm. Find the lengths of the parallel sides, if the area of the trapezium is 475 cm^2.

a. 22 cm and 18 cm b. 25 cm and 21 cm
c. 29 cm and 25 cm d. 27 cm and 23 cm

Sol. d. Let the length of the parallel sides of the trapezium be a cm and b cm, respectively.

Then, according to the question, $(a-b) = 4$ ...(i)

$\therefore$ Area of trapezium $= \frac{1}{2} \times (a+b) \times h$

$$\frac{1}{2} \times (a+b) \times 19 = 475$$

$\Rightarrow$ $a + b = \frac{950}{19} = 50$

$\therefore$ $a + b = 50$ ...(ii)

On solving Eqs. (i) and (ii), we get $a = 27$ and $b = 23$

Thus, length of parallel sides are 27 cm and 23 cm.

SCALENE TRIANGLE

Let the sides of a triangle are a, b, c and h be the corresponding altitude to side a, then

(i) Perimeter of scalene triangle, $2s = (a+b+c)$ units

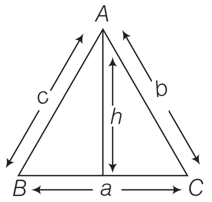
(ii) Semi-perimeter of scalene triangle, $s = \frac{a+b+c}{2}$ units

(iii) Area of triangle $= \sqrt{s(s-a)(s-b)(s-c)}$ sq units

where, $s = \frac{a+b+c}{2}$ [Heron's formula]

or area of triangle $= \frac{1}{2} \times a \times h$ sq units

(iv) The radius of incircle $= \frac{2(\text{Area})}{\text{Perimeter}}$



(v) The radius of circumcircle $= \frac{abc}{4(\text{Area})}$

EXAMPLE 11. The area of a triangle whose sides are 9 cm, 12 cm and 15 cm is

a. 45 cm^2 b. 54 cm^2 c. 56 cm^2 d. 64 cm^2

Sol. b. Here $a = 9$ cm , $b = 12$ cm and $c = 15$ cm

$$s = \frac{a+b+c}{2} = \frac{9+12+15}{2} = 18 \text{ cm}$$

Area of a triangle $= \sqrt{s(s-a)(s-b)(s-c)}$ [Heron's formula]

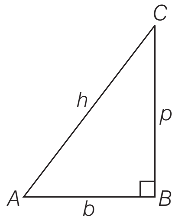
$$= \sqrt{18(18-9)(18-12)(18-15)}$$
$$= \sqrt{18 \times 9 \times 6 \times 3} = 54 \text{ cm}^2$$

Hence, the area of a triangle is 54 cm^2.

RIGHT ANGLED TRIANGLE

A figure bounded by three straight lines is called a triangle.

Let perpendicular, base and hypotenuse of a right angled triangle (ΔABC) be p, b and h, respectively, then



(i) Perimeter of right angled triangle $= AB + BC + CA = (b + p + h)$ units

(ii) Area of right angled triangle $= \frac{1}{2} \times \text{Base} \times \text{Altitude} = \frac{1}{2} \times b \times p$ sq units

EXAMPLE 12. The base of triangular field is three times its altitude. If the cost of cultivating the field at 50 per hectare be ₹ 675, then its base and height are

a. 900 m and 300 m b. 600 m and 300 m
c. 500 m and 200 m d. None of these

Sol. a. Area of the triangular field $= \frac{\text{Total cost}}{\text{Rate}}$

$$= \frac{675}{50} = 13.5 \text{ hec}$$
$$= (13.5 \times 10000) \text{ m}^2 = 135000 \text{ m}^2 \quad ...(i)$$

Let altitude of triangular field be x m
and base of field $= 3x$ m

Again, area of the field $= \frac{1}{2} \times$ Base $\times$ Altitude

$$= \frac{1}{2} \times 3x \times x = \frac{3x^2}{2} \quad \text{...(ii)}$$

From Eqs. (i) and (ii), we get

$$\frac{3x^2}{2} = 135000$$

$$\therefore \quad x^2 = \frac{135000 \times 2}{3} = 90000 \Rightarrow x = 300 \text{ m}$$

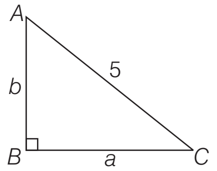
So, base $= 3x = 3 \times 300 = 900$ m
and altitude $= x = 300$ m

EXAMPLE 13. The perimeter of a right triangle is 12 cm. The hypotenuse is 5 cm. The other two sides and area of the triangle are

a. 3, 4 and 6 cm^2 b. 4, 3 and 12 cm^2
c. 6, 2 and 6 cm^2 d. None of these

Sol. a. Let other two sides of a right triangle be a cm and b cm such that $a > b$.

Given, perimeter of a right angled triangle = 12 cm



$$\therefore \quad a + b + 5 = 12 \Rightarrow a + b = 7 \text{ cm} \quad \text{...(i)}$$

Also, by pythagoras theorem,

$$a^2 + b^2 = 25$$

Also, $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$

$$(a-b)^2 = 2(a^2+b^2) - (a+b)^2$$

$$\Rightarrow \quad (a-b)^2 = 2(25) - (7)^2 = 50 - 49 = 1$$

$$\therefore \quad a - b = 1 \quad [\because a > b] \text{..(ii)}$$

On solving Eqs. (i) and (ii), we get

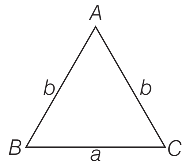
$$a = 4 \text{ cm and } b = 3 \text{ cm}$$

$\therefore$ Area of triangle $= \frac{1}{2} \times$ Base $\times$ Altitudes

$$\Rightarrow \quad \frac{1}{2} \times 3 \times 4 = 6 \text{ cm}^2$$

ISOSCELES TRIANGLE

Let sides of an isosceles triangle are a, b and b, then



(i) Perimeter of isosceles triangle

$$= a + b + b = a + 2b \text{ units}$$

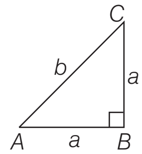
(ii) Area of isosceles triangle

$$= (s-b)(\sqrt{s(s-a)}) = \frac{1}{4} a\sqrt{4b^2 - a^2} \text{ sq units}$$

where, a = Base and b = Equal sides and
s = semi– perimeter

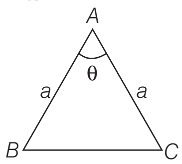
(iii) Area of a right isosceles triangle, in which equal sides from a right angle is given by

Area $= \frac{1}{2} a^2$ *sq units*



(iv) If θ is the angle between two equal sides of isosceles triangle, then

Area of triangle $= \frac{1}{2} a^2 \sin\theta$



EXAMPLE 14. The perimeter of an isosceles triangle is 32 cm while equal sides together measure 20 cm. Then, area of an isosceles triangle is

a. 48 cm^2 b. 84 cm^2 c. 44 cm^2 d. 41 cm^2

Sol. a. Let third side of an isosceles triangle be a cm.

Given, $b = 10$ cm

$\therefore$ Perimeter of isosceles triangle $= a + b + b = a + 2b$

$$\Rightarrow \quad 32 = a + 2 \times 10 = a + 20 \Rightarrow 32 - 20 = a$$

$$\therefore \quad a = 12 \text{ cm}$$

Now, area of an isosceles triangle

$$= \frac{1}{4} a\sqrt{4b^2 - a^2} = \frac{1}{4} \times 12\sqrt{4 \times 10^2 - 12^2}$$

$$= 3\sqrt{400 - 144} = 3 \times 16 = 48 \text{ cm}^2$$

EQUILATERAL TRIANGLE

Let a be the side of an equilateral triangle, then

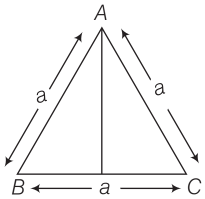
(i) Height (altitude) of an equilateral triangle $= \frac{\sqrt{3}}{2} a$

(ii) Area of an equilateral triangle $= \frac{\sqrt{3}}{4} a^2$

(iii) Perimeter of an equilateral triangle $= 3 \times$ Side $= 3a$

(iv) The radius of incircle $= \frac{a}{2\sqrt{3}}$

(v) The radius of circumcircle $= \dfrac{a}{\sqrt{3}}$



EXAMPLE 15. The perimeter of an equilateral triangle whose area is $4\sqrt{3}$ cm^2 is

a. 4 cm **b.** 3 cm **c.** 12 cm **d.** 7 cm

Sol. **c.** Area of an equilateral triangle $= \dfrac{\sqrt{3}}{4}(\text{Side})^2$

$\Rightarrow \dfrac{\sqrt{3}}{4}(\text{Side})^2 = 4\sqrt{3}, \ (\text{Side})^2 = 16$

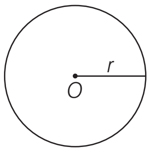
$\therefore$ Side = 4 cm

Hence, the perimeter of an equilateral triangle

$= 3 \times \text{Side} = 3 \times 4 = 12$ cm

CIRCLE

Let the radius of a circle be r, then



(i) Circumference of circle $= 2\pi r = \pi D$ units

[$\because D$ is diameter, $D = 2r$]

(ii) Area of circle $= \pi r^2$ sq units

(iii) Distance covered by a wheel in one revolution

= Circumference of the wheel

EXAMPLE 16. The circumference of a circle whose area is 24.64 m^2 is

a. 17.2 m **b.** 17.4 m **c.** 17.6 m **d.** 18.0 m

Sol. **c.** Let the radius of the circle be r m.

$\therefore \pi r^2 = 24.64$

$\Rightarrow \dfrac{22}{7} r^2 = 24.64 \Rightarrow r^2 = \dfrac{7 \times 24.64}{22}$

$\Rightarrow r = \sqrt{\dfrac{7 \times 24.64}{22}} = 2.8$

Thus, circumference $= 2\pi r = 2 \times \dfrac{22}{7} \times 2.8 = 17.6$ m

EXAMPLE 17. The diameter of the driving wheel of a car is 140 cm. Then, in order to keep the speed of 66 km/h, how many revolutions per minute must the wheel make?

a. 250 **b.** 275 **c.** 290 **d.** 295

Sol. **a.** Given, radius of the wheel = 70 cm = 0.7 m

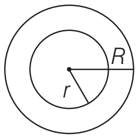
Circumference of the wheel $= 2\pi r = \left(2 \times \dfrac{22}{7} \times 0.7\right) = 4.4$ m

Distance to be covered in 1 min $= \dfrac{66 \times 1000}{60} = 1100$ m

$\therefore$ Number of revolutions per minute $= \dfrac{1100}{4.4} = 250$

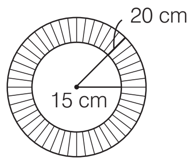
Circular Ring

If 'R' and 'r' be outer and inner radii of a ring, then the area of ring $= \pi(R^2 - r^2)$ sq units



EXAMPLE 18. The area of a ring whose outer and inner radii are respectively 20 cm and 15 cm is

a. 440 cm^2 **b.** 550 cm^2 **c.** 565 cm^2 **d.** 675 cm^2



Sol. **b.** Given, radius of outer circle $(R) = 20$ cm

Radius of inner circle $(r) = 15$ cm

$\therefore$ Area of the ring = [Area of outer circle − Area of inner circle]

$= (\pi R^2 - \pi r^2) = \pi(R^2 - r^2) = \dfrac{22}{7}(20^2 - 15^2)$

$= \dfrac{22}{7}(20 + 15)(20 - 15) = \dfrac{22}{7} \times 35 \times 5 = 550 \text{ cm}^2$

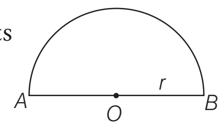
Semi-circle

A diameter divides a circle into two equal parts. Each of these two arcs is called semi-circle. If r is the radius of a circle, then

(i) Area of semi-circle $= \dfrac{1}{2}\pi r^2$ sq units

(ii) Perimeter of semi-circle

$= (\pi r + 2r) = (\pi r + D)$ units



EXAMPLE 19. If the perimeter of a semi-circular protractor is 36 cm, then its diameter is

a. 6 cm **b.** 7 cm **c.** 7.5 cm **d.** 14 cm

Sol. **d.** Let the radius of the protractor be r cm, then perimeter

$= (\pi r + 2r) = (\pi + 2)\, r = \dfrac{36}{7} r \Rightarrow 36 = \dfrac{36\, r}{7}$

$\therefore r = 7$ cm

Hence, diameter of the protractor $= 2r = 2 \times 7 = 14$ cm

Quadrant of a Circle

If r is the radius of a circle, then

(i) The perimeter of the quadrant

$$= \frac{1}{4}\text{(Circumference of a circle)} + 2r$$

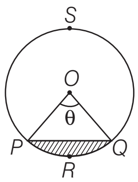
$$= \frac{1}{4}\times 2\pi r + 2r = \left(\frac{\pi}{2}+2\right) r \text{ units}$$

(ii) Area of the quadrant $= \frac{1}{4}\text{(Area of circle)} = \frac{\pi r^2}{4}$ sq units

If two diameters are perpendicular to each other, then they divides the circle into four quadrants.

Area of Sector

If θ be the angle at the centre of a circle of radius r, then



(i) Length of the arc $\overline{PQ} = \frac{2\pi r\,\theta}{360^\circ} = \frac{\pi r\,\theta}{180^\circ}$

(ii) Area of sector $OPRQO = \frac{\pi r^2\theta}{360^\circ}$

(iii) Area of segment $PRQP$

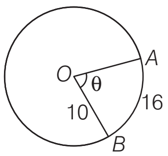
$$= \text{(Area of sector } OPRQO) - \text{Area of } \Delta OPQ$$

$$= \frac{\pi r^2\theta}{360^\circ} - \frac{1}{2} r^2 \sin\theta$$

(iv) Area of major segment $QPSQ$ = (Area of circle)
– (Area of minor segment $PRQP$)

EXAMPLE 20. The arc AB of the circle with centre at O and radius 10 cm has length 16 cm. What is the area of the sector bounded by the radii OA, OB and the arc AB?

a. 40π cm^2 b. 40 cm^2 c. 80 cm^2 d. 20π cm^2



Sol. c. Arc length $= 2\pi r\cdot\frac{\theta}{360^\circ}$

$$\Rightarrow \quad 16 = 2\pi r\cdot\frac{\theta}{360^\circ} \quad \Rightarrow \quad \frac{\theta}{360^\circ} = \frac{16}{2\pi r}$$

Now, area of sector $OABO = \pi r^2\cdot\frac{\theta}{360^\circ}$

$$= \pi r^2\cdot\frac{16}{2\pi r} = 8r = 8\times 10 = 80 \text{ cm}^2$$

REGULAR POLYGON

A polygon that has all sides equal and all interior angles equal are regular polygon.

Let, n = number of sides, a = length of side
r = radius of inscribed circle (or inradius)
R = radius of circumcircle

Area of polygon $= \frac{1}{4} n a^2 \cot\left(\frac{180^\circ}{n}\right)$

$$= n r^2 \tan\left(\frac{180^\circ}{n}\right) = \frac{1}{2} n R^2 \sin\left(\frac{360^\circ}{n}\right)$$

Perimeter of polygon $= n\times a = 2nr\sin\left(\frac{\pi}{n}\right)$

Inradius of polygon, $r = \frac{1}{2} a\cot\left(\frac{180^\circ}{n}\right)$

circumradius of polygon, $R = \frac{1}{2} a\,\text{cosec}\left(\frac{180^\circ}{n}\right)$

(i) Area of regular pentagon $= \frac{1}{4}\sqrt{5(5+2\sqrt{5})\,a^2}$ sq units

(ii) Area of regular hexagon $= \frac{3\sqrt{3}a^2}{2}$ sq units

(iii) Area of regular octagon $= 2(\sqrt{2}+1)\,a^2$ sq units

EXAMPLE 21. If the area of a regular hexagon is $96\sqrt{3}$ cm^2, then its perimeter is

a. 36 cm b. 48 cm c. 54 cm d. 64 cm

Sol. b. Given that, area of a regular hexagon $= 96\sqrt{3}$

$$\Rightarrow \quad \frac{3}{2}\sqrt{3}\,(\text{Side})^2 = 96\sqrt{3} \quad \Rightarrow \quad (\text{Side})^2 = 64 = (8)^2$$

$\therefore$ Side = 8 cm

Hence, perimeter of a regular hexagon

$$= 6\times(\text{Side}) = 6\times 8 = 48 \text{ cm}$$

Some Useful Results

- Angle inscribed by minute hand in 60 min = 360°
- Angle inscribed by hour hand in 12 h = 360°
- Angle inscribed by minute hand in 1 min = 6°
- If the length of a square/rectangle is increased decreased by x% and the breadth is increased decreased by y% , the net effect on the area is given by

$$\text{Net effect} = \left[\pm x \pm y + \frac{(\pm x)(\pm y)}{100}\right]\% .$$

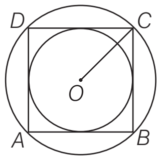
- If the length of a square/rectangle is increased/decreased by x% and the breadth is increased/decreased by y% the net effect on the area is given by
- $$\text{Net effect} = \left[\pm x \pm y + \frac{(\pm x)(\pm y)}{100}\right]\%.$$

Note (+) Increase
(–) Decrease

- If the side of a square/rectangle/triangle is doubled the area is increased by 300%, i.e. the area becomes four times of itself.
- If the radius of a circle is decreased by x%, then, area is decreased by $\left(-2x+\frac{x^2}{100}\right)$%. And, if radius is increased by x%, then area is increased by $\left(2x+\frac{x^2}{100}\right)$%.

- If a room of dimensions ($l \times b$) m is to be paved with square tiles, then
  (i) the side of the largest square tile = HCF of l and b
  (ii) the least number of tiles required
  $$= \frac{l \times b}{(HCF \text{ of } l \text{ and } b)^2}$$
- Area of a square inscribed in a circle of radius r is $2r^2$ and the side of a square inscribed in a circle of radius r is $\sqrt{2}r$.
- The area of the largest triangle inscribed in a semi-circle of radius r is r^2.

EXAMPLE 22. The ratio of the areas of the incircle and circumcircle of a square are



a. 1 : 1 b. 2 : 1 c. 1 : 2 d. 3 : 1

Sol. c. Let side $AB = BC = CD = AD = x$

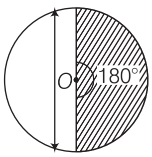
$\because$ Diagonal of square $= \sqrt{2}\,x$

$\therefore$ Radius of circumcircle $= \frac{\sqrt{2}x}{2} = \frac{x}{\sqrt{2}}$

$\therefore$ Radius of incircle $= \frac{x}{2}$

$\therefore$ Required ratio $= \left(\frac{\pi x^2}{4} : \frac{\pi x^2}{2}\right) = 2 : 4 = 1 : 2$

EXAMPLE 23. The minute hand of a clock is 14 cm long. The area of the face of the clock inscribed by the minute hand in 30 min is



a. 308 cm^2 b. 312 cm^2 c. 412 cm^2 d. 416 cm^2

Sol. a. Angle inscribed by the minute hand in 60 min = 360°

Angle inscribed by the minute hand in 30 min

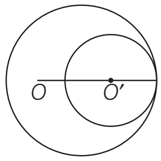
$$= \frac{360°}{60} \times 30 = 180°$$

So, $\theta = 180°$ and $r = 14$ cm.

Required area swept by minute hand in 30 min

$$= \frac{\theta}{360°} \times \pi r^2 = \frac{180°}{360°} \times \frac{22}{7} \times 14 \times 14$$
$$= \frac{1}{2} \times \frac{22}{7} \times 14 \times 14 = 308 \text{ cm}^2$$

EXAMPLE 24. Two circles touch internally. The sum of their areas is 116 π cm^2 and distance between their centres is 6 cm. Then, the radii of the circles are



a. 4 cm and 9 cm
b. 5 cm and 10 cm
c. 4 cm and 8 cm
d. 4 cm and 10 cm

Sol. d. Let the radius of outer circle be R and radius of inner circle = r cm

$\therefore$ Given, $\pi R^2 + \pi r^2 = 116\pi$, $R^2 + r^2 = 116$

If O and O' be the centre of these circles, then

$$OO' = (R - r)$$

Also, $(R - r) = 6$ [given]

So, $(R + r)^2 + (R - r)^2 = 2(R^2 + r^2)$

$\Rightarrow$ $(R + r)^2 = 2(116) - 36 = 196$

$\Rightarrow$ $R + r = \sqrt{196}$

So $R + r = 14$ ...(i)

$R - r = 6$ ...(ii)

On solving Eqs. (i) and (ii), we get

$r = 4$ cm and $R = 10$ cm

PRACTICE EXERCISE

1. The diagonal of a square field measures 50 m. The area of square field is
(a) 1250 m^2 (b) 1200 m^2 (c) 1205 m^2 (d) 1025 m^2

2. The length of a rectangle is 2 cm more than its breadth. The perimeter is 48 cm. The area of the rectangle (in cm^2) is
(a) 96 (b) 128 (c) 143 (d) 144

3. In a circle of radius 42 cm, an arc subtends an angle of 72° at the centre. The length of the arc is
(a) 52.8 cm (b) 53.8 cm (c) 72.8 cm (d) 79.8 cm

4. An isosceles right angle triangle has area $200\,cm^2$. The length of its hypotenuse is
(a) $15\sqrt{2}$ cm (b) $\frac{10}{\sqrt{2}}$ cm (c) $10\sqrt{2}$ cm (d) $20\sqrt{2}$ cm

5. With in a rectangular garden 10 m wide and 20 m long, we wish to pave a walk around the borders of uniform width so as to leave an area of $96\,m^2$ for flowers. The width of the walk is
(a) 1 m (b) 2 m (c) 2.5 m (d) 2.56 m

6. The least number of square slabs that can be fitted in a room 10.5 m long and 3 m wide, is
(a) 12 (b) 13 (c) 14 (d) 15

7. If the side of a square be increased by 50%, the percent increase in area is
(a) 50 (b) 100 (c) 125 (d) 150

8. The area of the largest circle that can be drawn inside a square of side 14 cm in length, is
(a) 84 cm^2 (b) 96 cm^2 (c) 104 cm^2 (d) 154 cm^2

9. If the radius of a circle is decreased by 50%, its area will decrease by
(a) 25% (b) 50% (c) 75% (d) 100%

10. The area of the circle whose circumference is equal to the perimeter of a square of side 11 cm is
(a) 154 cm^2 (b) 144 cm^2 (c) 134 cm^2 (d) 124 cm^2

11. A wire is in the form of a circle of radius 42 cm. It is bent into a square. The side of the square is
(a) 33 cm (b) 66 cm (c) 78 cm (d) 112 cm

12. A horse is tied to a pole with 28 m long string. The area which the horse can graze is equal to
(a) 246 m^2 (b) 2404 m^2 (c) 2464 m^2 (d) 2164 m^2

13. The area of ring is $418\,cm^2$. If the radius of the smaller circle is 6 cm. The radius of the bigger circle is
(a) 18 cm (b) 16 cm (c) 13 cm (d) 10 cm

14. The length of a rectangle is increased by 60%. By what per cent would the width have to be decreased to maintain the same area?
(a) $37\frac{1}{2}\%$ (b) 60% (c) 75% (d) 120%

15. The perimeter of a rectangular field is 240 m and the ratio between the length and breadth is 5 : 3. The area of the field is
(a) 33750 m^2 (b) 3375 m^2 (c) 3500 m^2 (d) 3950 m^2

16. The inner circumference of a circular park is 440 m. The track is 14 m wide. The diameter of the outer circle of the track is
(a) 168 m (b) 169 m (c) 144 m (d) 108 m

17. If the length and breadth of a rectangular plot are increased by 50% and 20% respectively, then the new area is how many times the original area?
(a) $\frac{5}{9}$ (b) 10 (c) $\frac{9}{5}$ (d) None of these

18. The area of an isosceles triangle, each of whose equal sides is 13 cm and whose base is 24 cm is
(a) 60 cm^2 (b) 55 cm^2 (c) 50 cm^2 (d) 40 cm^2

19. The difference between the sides at right angles in a right angled triangle is 14 cm. The area of the triangle is $120\,cm^2$. The perimeter of the triangle
(a) 68 cm (b) 64 cm (c) 60 cm (d) 58 cm

20. In a four sided field, the length of the longer diagonal is 128 m. The lengths of the perpendicular from the opposite vertices upon this diagonal are 22.7 m and 17.3 m. The area of the field is
(a) 2246 m^2 (b) 2460 m^2 (c) 2540 m^2 (d) 2560 m^2

21. The adjacent sides of a parallelogram are 36 cm and 27 cm in length. If the distance between the shorter sides is 12 cm. The distance between the longer sides is
(a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm

22. The area of the quadrilateral whose sides measures are 9 cm, 40 cm, 28 cm and 15 cm and in which the angle between the first two sides is a right angle is
(a) 206 cm^2 (b) 306 cm^2 (c) 356 cm^2 (d) 380 cm^2

23. A bicycle wheel makes 5000 revolutions in moving 11 km. The diameter of the wheel is
(a) 50 cm (b) 60 cm (c) 70 cm (d) 80 cm

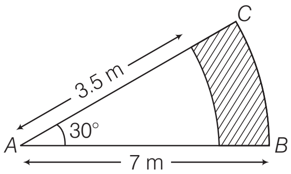
24. Two circles touch externally. The sum of their areas is $130\pi\, cm^2$ and the distance between their centres is 14 cm. The radii of the circles are

(a) 11 cm, 3 cm (b) 10 cm, 4 cm
(c) 9 cm, 5 cm (d) 8 cm, 6 cm

25. The minute hand of a clock is 12 cm long. The area of the face of the clock described by the minute hand in 35 min.

(a) 284 cm^2 (b) 294 cm^2 (c) 274 cm^2 (d) 264 cm^2

26. In the given figure, sectors of two concentric circles of radii 7 cm and 3.5 cm are shown. The area of the shaded region



(a) $\frac{77}{4}$ cm^2 (b) $\frac{77}{8}$ cm^2
(c) $\frac{77}{2}$ cm^2 (d) None of these

27. Of the two square fields, the area of one is 1 hectare, while the other one is broader by 2%. The difference in their areas is

(a) 104 m^2 (b) 200 m^2 (c) 204 m^2 (d) 404 m^2

28. If the diameter of the circle is increased by 100%, its area is increased by

(a) 100% (b) 200% (c) 300% (d) 400%

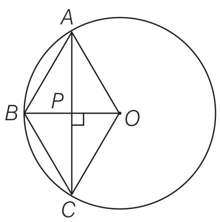
29. If the two parallel sides of a trapezium are 15 cm and 25 cm respectively and the distance between them is 7 cm, then the area of the trapezium is

(a) 105 cm^2 (b) 125 cm^2
(c) 140 cm^2 (d) None of these

30. The diagonals of a rhombus are 24 cm and 10 cm, respectively. The perimeter of the rhombus is

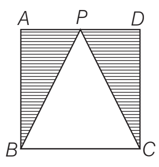
(a) 50 cm (b) 52 cm (c) 60 cm (d) 68 cm

31. In the given figure $OABC$ is a rhombus whose three vertices A, B, C lies on the circle of radius 10 cm. The area of rhombus is



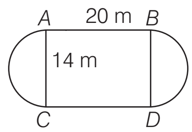
(a) $50\sqrt{3}$ cm^2 (b) $100\sqrt{3}$ cm^2 (c) $75\sqrt{3}$ cm^2 (d) $125\sqrt{3}$ cm^2

32. In the adjoining figure, $ABCD$ is a rectangle such that $AB = 2AD = a$. P is the mid-point of AD. The area of the shaded region is



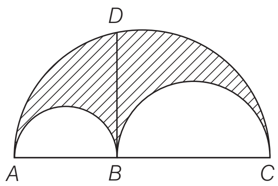
(a) $\frac{1}{3}a^2$ (b) $\frac{1}{2}a^2$ (c) $\frac{1}{4}a^2$ (d) a^2

33. A garden is in the form of a rectangle with semi-circular endson the either side as shown in the diagram below. The length and breadth of the rectangle are 20 m and 14 m, respectively. The cost of levelling the plot at ₹ 25



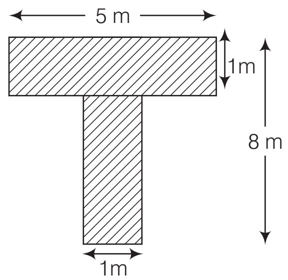
(a) ₹ 10850 (b) ₹ 5425 (c) ₹ 8510 (d) ₹ 4255

34. Consider the adjoining figure, let $AB = 4$ cm, $BC = 14$ cm, then the area (shaded) bounded by three semi-circles as shown in the adjoining figure in cm^2, is π times



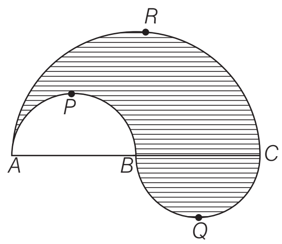
(a) 48 (b) 24 (c) 14 (d) 12

35. Area of shaded portion as shown in the figure is



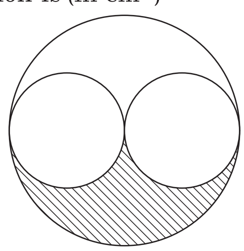
(a) 10 m^2 (b) 12 m^2 (c) 14 m^2 (d) 16 m^2

36. The boundary of the shaded region in the adjoining diagram consists of three semi-circular arc, the smaller two being equal. If the diameter of the large one is 10 cm, then the length of the boundary is



(a) 31 cm (b) 10π cm (c) 20π cm (d) 19π cm

37. In the adjoining figure, the larger circle with radius 4 cm is touched internally by two smaller circles which also touch each other externally at the centre O of the larger circle. The area of shaded region is (in cm^2)



(a) π (b) 2π (c) 3π (d) 4π

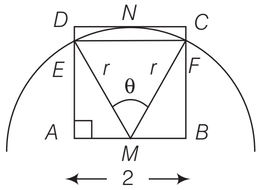
38. The lengths of two sides of a right angled triangle which contain the right angle are a and b, respectively. Three squares are drawn on the three sides of the triangle on the outer side. What is the total area of the triangle and the three squares?

(a) $2(a^2+b^2)+ab$ (b) $2(a^2+b^2)+2.5\,ab$
(c) $2(a^2+b^2)+0.5\,ab$ (d) $25\,(a^2+b^2)$

39. A grassy field has the shape of an equilateral triangle of side 6 m. A horse is tied to one of its vertices with a rope of length 4.2 m. The percentage of the total area of the field which is available for grazing is best approximated by

(a) 50% (b) 55% (c) 59% (d) 62%

40. From the mid-point of the side of a square of length 2 units, a circle of radius 2 units is circumscribed. The area of intersection of the square and the circle is



(a) $\frac{4}{3}\pi+3\sqrt{3}$ (b) $\pi+\sqrt{3}$ (c) $\frac{2}{3}\pi+\sqrt{3}$ (d) $\frac{2}{3}\pi+2\sqrt{3}$

41. In a circle with radius 2 cm, on its chord $2\sqrt{2}$ cm long taken as a diameter, another circle is constructed. The area of part of that circle, which is outside the greater circle is

(a) 1 cm^2 (b) 2 cm^2 (c) 6 cm^2 (d) 10 cm^2

42. If a rectangular park has length l, width w, area a and perimeter p, which equation describing this park must be true?

I. $wp-2w^2=2a$ II. $2a-2w^2-wp=0$
III. $p^2-8a=4\,(l^2+w^2)$

Select the correct answer using the codes given below

(a) I and II (b) II and III
(c) I and III (d) All of these

43. A regular polygon with number of sides 12 is inscribed in a circle of area 9π cm^2. If r is the radius of a circle inscribed in a polygon, then consider the following statements.

I. Radius of circumcircle is 3 cm.
II. Radius of incircle is 3cos 15°.
III. Ratio of area of circumcircle to area of polygon is $\pi:3$.

Which of the following statement(s) are correct?

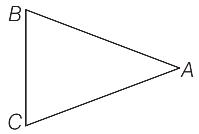
(a) I and II (b) II and III
(c) I and III (d) All of these

44. What is the area of rectangle R?

I. The length of rectangle R is twice the width.
II. The area of rectangle R is twice the perimeter.

(a) Statement I alone is sufficient but statement II alone is not sufficient.
(b) Statement II alone is sufficient but statement I alone is not sufficient.
(c) Both statements together are sufficient.
(d) Neither statement I nor II is sufficient.

Directions (Q. Nos. 45-46) *Answer the questions based on the following information.*
A cow is tethered at point A by a rope. Neither the rope nor the cow is allowed to enter the triangle ABC. $\angle BAC=30°$, $AB=AC=10$ m.



45. What is the area that can be grazed by the cow if the length of the rope is 8 m?

(a) $134\pi\frac{1}{3}$ m^2 (b) 121π m^2 (c) $132\,\pi$ m^2 (d) $\frac{176\pi}{3}$ m^2

46. What is the area that can be grazed by the cow if the length of the rope is 12 m?

(a) $133\pi\frac{1}{6}$ m^2 (b) 121π m^2 (c) $132\,\pi$ m^2 (d) $\frac{176\pi}{3}$ m^2

PREVIOUS YEARS' QUESTIONS

47. If the outer and inner diameters of a stone parapet around a well are 112 cm and 70 cm, respectively. Then, what is the area of the parapet? **2012 I**

(a) 264 cm^2 (b) 3003 cm^2
(c) 6006 cm^2 (d) 24024 cm^2

48. The area of a rectangle, whose one side is a is $2a^2$. What is the area of a square having one of the diagonals of the rectangle as side? **2012 I**

(a) $2a^2$ (b) $3a^2$ (c) $4a^2$ (d) $5a^2$

49. If the altitude of an equilateral triangle is $\sqrt{3}$ cm, then what is its perimeter? **2012 I**

(a) 3 cm (b) $3\sqrt{3}$ cm (c) 6 cm (d) $6\sqrt{3}$ cm

50. If the area of a rectangle whose length is 5 more than twice its width is 75 sq units. What is the perimeter of the rectangle? **2012 I**

(a) 40 units (b) 30 units
(c) 24 units (d) 20 units

51. A square, a circle and an equilateral triangle have same perimeter.
Consider the following statements

I. The area of square is greater than the area of the triangle.
II. The area of circle is less than the area of triangle.

Which of the statement(s) given above is/are correct? **2012 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

52. If the area of a circle is equal to the area of a square with side $2\sqrt{\pi}$ units, then what is the diameter of the circle? **2012 I**

(a) 1 unit (b) 2 units (c) 4 units (d) 8 units

53. In the ΔABC, the base BC is trisected at D and E. The line through D, parallel to AB, meets AC at F and the line through E parallel to AC meets AB at G. If EG and DF intersect at H, then what is the ratio of the sum of the area of parallelogram $AGHF$ and the area of the ΔDHE to the area of the ΔABC ? **2012 II**

(a) 1/2 (b) 1/3 (c) 1/4 (d) 1/6

54. If the area of a circle inscribed in an equilateral triangle is 154 cm^2, then what is the perimeter of the triangle? **2012 II**

(a) 21 cm (b) $42\sqrt{3}$ cm (c) $21\sqrt{3}$ cm (d) 42 cm

55. If the circumferences of two circles are in the ratio 2 : 3, then what is the ratio of their areas? **2012 II**

(a) 2 : 3 (b) 4 : 9 (c) 1 : 3 (d) 8 : 27

56. If the circumference of a circle is equal to the perimeter of square, then which one of the following is correct? **2012 II**

(a) Area of circle = Area of square
(b) Area of circle ≥ Area of square
(c) Area of circle > Area of square
(d) Area of circle < Area of square

57. The area enclosed between the circumferences of two concentric circles is 16π cm^2 and their radii are in the ratio 5 : 3. What is the area of the outer circle? **2012 II**

(a) 9π cm^2 (b) 16π cm^2 (c) 25π cm^2 (d) 36π cm^2

58. The perimeter of a rectangle is 82 m and its area is 400 m^2. What is the breadth of the rectangle? **2012 II**

(a) 18 m (b) 16 m (c) 14 m (d) 12 m

59. Consider the following statements

I. Area of a segment of a circle is less than area of its corresponding sector.
II. Distance travelled by a circular wheel of diameter $2d$ cm in one revolution is greater than $6d$ cm.

Which of the above statement(s) is/are correct? **2012 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

60. The minute hand of a watch is 2.5 cm long. The distance its extreme end transverses in 40 min is **2013 I**

(a) $10\pi/3$ cm (b) $3\pi/10$ cm
(c) 10/3 cm (d) 10 cm

61. The short and long hands of a clock are 4 cm and 6 cm long, respectively. Then, the ratio of distances travelled by tips of short hand in 2 days and long hand in 3 days is **2013 I**

(a) 4 : 9 (b) 2 : 9 (c) 2 : 3 (d) 1 : 27

62. The area of a square inscribed in a circle of radius 8 cm is **2013 I**

(a) 32 cm^2 (b) 64 cm^2 (c) 128 cm^2 (d) 256 cm^2

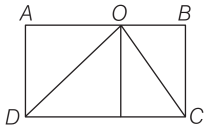
63. A circular water fountain 6.6 m in diameter is surrounded by a path of width 1.5 m. The area of this path (in m^2) is **2013 I**

(a) 13.62π (b) 13.15π
(c) 12.15π (d) None of these

64. If an isosceles right angled triangle has area 1sq unit, then what is its perimeter? **2013 I**

(a) 3 units (b) $2\sqrt{2}+1$ units
(c) $(\sqrt{2}+1)$ units (d) $2(\sqrt{2}+1)$ units

65. In the figure given below, the area of rectangle $ABCD$ is 100 cm^2, O is any point on AB and $CD = 20$ cm. Then, the area of ΔCOD is **2013 I**



(a) 40 cm^2 (b) 45 cm^2 (c) 50 cm^2 (d) 80 cm^2

66. One side of a parallelogram is 8.06 cm and its perpendicular distance from opposite side is 2.08 cm. What is the approximate area of the parallelogram? **2013 II**

(a) 12.56 cm^2 (b) 14.56 cm^2 (c) 16.76 cm^2 (d) 22.56 cm^2

67. The perimeter of a rectangle having area equal to 144 cm^2 and sides in the ratio 4 : 9 is **2013 II**

(a) 52 cm (b) 56 cm (c) 60 cm (d) 64 cm

68. What is the area between a square of side 10 cm and two inverted semi-circular, cross-section each of radius 5 cm inscribed in the square? **2013 II**

(a) 17.5 cm^2 (b) 18.5 cm^2
(c) 20.5 cm^2 (d) 21.5 cm^2

69. How many 200 mm lengths can be cut from 10 m of ribbon? **2013 II**

(a) 50 (b) 40 (c) 30 (d) 20

70. The area of an isosceles ΔABC with $AB = AC$ and altitude $AD = 3$ cm is 12 cm^2. What is its perimeter? **2013 II**

(a) 18 cm (b) 16 cm (c) 14 cm (d) 12 cm

71. If the area of an equilateral triangle is x and its perimeter is y, then which one of the following is correct? **2013 II**

(a) $y^4 = 432x^2$ (b) $y^4 = 216x^2$
(c) $y^2 = 432x^2$ (d) None of these

72. What is the area of a circle whose area is equal to that of a triangle with sides 7 cm, 24 cm and 25 cm? **2013 II**
(a) $80\,cm^2$ (b) $84\,cm^2$ (c) $88\,cm^2$ (d) $90\,cm^2$

73. If AB and CD are two diameters of a circle of radius r and they are mutually perpendicular, then what is the ratio of the area of the circle to the area of the ΔACD? **2014 I**
(a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{4}$ (d) 2π

74. A rectangle of maximum area is drawn inside a circle of diameter 5 cm. What is the maximum area of such a rectangle? **2014 I**
(a) $25\ cm^2$ (b) $12.5\ cm^2$
(c) $12\ cm^2$ (d) None of these

75. What is the area of the larger segment of a circle formed by a chord of length 5 cm subtending an angle of 90° at the centre? **2014 I**
(a) $\frac{25}{4}\left(\frac{\pi}{2}+1\right)cm^2$ (b) $\frac{25}{4}\left(\frac{\pi}{2}-1\right)cm^2$
(c) $\frac{25}{4}\left(\frac{3\pi}{2}+1\right)cm^2$ (d) None of these

76. Consider an equilateral triangle of side of unit length. A new equilateral triangle is formed by joining the mid-points of one, then a third equilateral triangle is formed by joining the mid-points of second. The process is continued. The perimeter of all triangles, thus formed is **2014 I**
(a) 2 units (b) 3 units (c) 6 units (d) Infinity

77. ABC is triangle right angled at A, $AB = 6$ cm and $AC = 8$ cm. Semi-circles drawn (outside the triangle) on AB, AC and BC as diameteres which enclose areas x, y and z square units, respectively. What is $x + y - z$ equal to? **2014 I**
(a) $48\ cm^2$ (b) $32\ cm^2$
(c) 0 (d) None of these

78. A square is inscribed in a circle of diameter $2a$ and another square is circumscribing circle. The difference between the areas of outer and inner squares is **2014 I**
(a) a^2 (b) $2a^2$ (c) $3a^2$ (d) $4a^2$

79. The area of a sector of a circle of radius 36 cm is $72\pi\ cm^2$ the length of the corresponding arc of the sector is **2014 I**
(a) π cm (b) 3π cm (c) 3π cm (d) 4π cm

80. What is the total area of three equilateral triangles inscribed in a semi-circle of radius 2 cm? **2014 I**
(a) $12\ cm^2$ (b) $\frac{3\sqrt{3}}{4}\ cm^2$ (c) $\frac{9\sqrt{3}}{4}\ cm^2$ (d) $3\sqrt{3}\ cm^2$

81. How many circular plates of diameter d be taken out of a square plate of side $2d$ with minimum loss of material? **2014 I**
(a) 8 (b) 6 (c) 4 (d) 2

82. The sides of a triangle are in the ratio $\frac{1}{2}:\frac{1}{3}:\frac{1}{4}$. If its perimeter is 52 cm, then what is the length of the smallest side? **2014 II**
(a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm

83. The sides of a triangle are 25 cm, 39 cm and 56 cm. The perpendicular from the opposite vertex on the side of 56 cm is **2014 II**
(a) 10 cm (b) 12 cm (c) 15 cm (d) 16 cm

84. The area of the largest triangle that can be inscribed in a semi-circle of radius r is **2015 I**
(a) r^2 (b) $2r^2$ (c) $3r^2$ (d) $4r^2$

85. The area of a rhombus with side 13 cm and one diagonal 10 cm will be **2015 I**
(a) $140\ cm^2$ (b) $130\ cm^2$ (c) $120\ cm^2$ (d) $110\ cm^2$

86. Four equal-sized maximum circular plates are cut off from a square paper sheet of area $784\ cm^2$. The circumference of each plate is **2015 I**
(a) 11 cm (b) 22 cm (c) 33 cm (d) 44 cm

87. From a circular piece of cardboard of radius 3 cm, two sectors of 40° each have been cut off. The area of the remaining portion is **2015 I**
(a) $11\ cm^2$ (b) $22\ cm^2$ (c) $33\ cm^2$ (d) $44\ cm^2$

88. Three equal circles each of diameter d are drawn on a plane in such a way that each circle touches the other two circles. A big circle is drawn in such a manner that it touches each of the small circles internally. The area of the big circle is **2015 I**
(a) πd^2 (b) $\pi d^2(2-\sqrt{3})^2$
(c) $\frac{\pi d^2(\sqrt{3}+1)^2}{2}$ (d) $\frac{\pi d^2(\sqrt{3}+2)^2}{12}$

89. The angles of a triangle are in the ratio 4 : 1 : 1. Then, the ratio of the largest side to the perimeter is **2015 I**
(a) $\frac{2}{3}$ (b) $\frac{1}{2+\sqrt{3}}$ (c) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (d) $\frac{2}{1+\sqrt{3}}$

90. Let a, b, c be the sides of a right triangle, where c is the hypotenuse. The radius of the circle which touches the sides of the triangle is **2015 I**
(a) $(a+b-c)/2$ (b) $(a+b+c)/2$
(c) $(a+2b+2c)/2$ (d) $(2a+2b-c)/2$

91. The ratio of the outer and inner perimeters of a circular path is 23 : 22. If the path is 5 m wide, the diameter of the inner circle is **2015 I**
(a) 55 m (b) 110 m (c) 220 m (d) 230 m

92. The sides of a triangular field are 41 m, 40 m and 9 m. The number of rose beds that can be prepared in the field if each rose bed, on an average, needs 900 cm^2 space, is **2015 I**

(a) 2000 (b) 1800 (c) 900 (d) 800

93. The diameter of a wheel that makes 452 revolutions to move 2 km and 26 dm is equal to **2015 II**

(a) $1\frac{9}{22}$m (b) $1\frac{13}{22}$m (c) $1\frac{5}{11}$m (d) $1\frac{7}{11}$m

94. From a rectangular sheet of sides 18 cm and 14 cm, a semi-circular portion with smaller side as diameter is taken out. Then, the area of the remaining sheet will be **2015 II**

(a) 98 cm^2 (b) 100 cm^2 (c) 108 cm^2 (d) 175 cm^2

95. A square and an equilateral triangle have equal perimeter. If the diagonal of the square is $12\sqrt{2}$cm, then the area of the triangle is **2015 II**

(a) $24\sqrt{2}$ cm^2 (b) $24\sqrt{3}$ cm^2 (c) $48\sqrt{3}$ cm^2 (d) $64\sqrt{3}$ cm^2

96. *ABCD* is a square. If the sides *AB* and *CD* are increased by 30%, sides *BC* and *AD* are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by **2015 II**

(a) 50% (b) 52% (c) 54% (d) 56%

97. The area of a trapezium is 336 cm^2. If its parallel sides are in the ratio 5 : 7 and the perpendicular distance between them is 14 cm, then the smaller of the parallel sides is **2015 II**

(a) 20 cm (b) 22 cm (c) 24 cm (d) 26 cm

98. The circumference of a circle is 100 cm. The side of the square inscribed in the circle is **2015 II**

(a) $50\sqrt{2}$ cm (b) $\frac{100}{\pi}$ cm (c) $\frac{50\sqrt{2}}{\pi}$ cm (d) $\frac{100\sqrt{2}}{\pi}$ cm

99. There are 437 fruit plants in an orchard planted in rows. The distance between any two adjacent rows is 2 m and the distance between any two adjacent plants is 2 m. Each row has the same number of plants. There is 1 m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of ₹ 100 per metre? **2015 II**

(a) ₹ 15600 (b) ₹ 16800 (c) ₹ 18200
(d) More information is required

100. A square is inscribed in a right triangle with legs x and y and has common right angle with the triangle. The perimeter of the square is given by **2015 II**

(a) $\frac{2xy}{x+y}$ (b) $\frac{4xy}{x+y}$ (c) $\frac{2xy}{\sqrt{x^2+y^2}}$ (d) $\frac{4xy}{\sqrt{x^2+y^2}}$

101. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p then the area of the polygon is **2015 II**

(a) $(p+n)r$ (b) $(2p-n)r$ (c) $\frac{pr}{2}$ (d) None of these

102. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times the area of the smaller ring, then n is equal to **2016 I**

(a) 4 (b) 6 (c) 8 (d) 10

103. The number of rounds that a wheel of diameter 7/11 m will make in traversing 4 km will be **2016 I**

(a) 500 (b) 1000 (c) 1700 (d) 2000

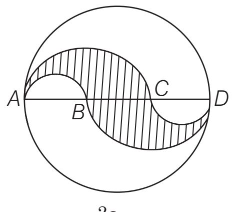
104. The base of an isosceles triangle is 300 units and each of its equal sides is 170 units. Then, the area of the triangle is **2016 I**

(a) 9600 sq units (b) 10000 sq units
(c) 12000 sq units (d) None of these

105. Four equal disc are placed such that each one touches two others. If the area of empty space enclosed by them is 150/847 cm^2, then the radius of each disc is equal to **2016 I**

(a) 7/6 cm (b) 5/6 cm (c) 1/2 cm (d) 5/11 cm

106. *AD* is the diameter of a circle with area 707 m^2 and $AB = BC = CD$ as shown in the figure above. All curves inside the circle are semicircles with their diameters on *AD*. What is the cost of levelling the shaded region at the rate of ₹ 63 per m^2? **2016 I**

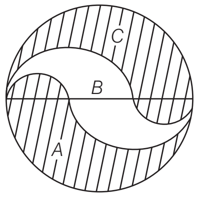


(a) ₹ 29700 (b) ₹ 22400 (c) ₹ 14847 (d) None of these

107. A rhombus is formed by joining mid-points of the sides of a rectangle in the suitable order. If the area of the rhombus is 2 sq units, then the area of the rectangle is **2016 I**

(a) $2\sqrt{2}$ sq units (b) 4 sq units
(c) $4\sqrt{2}$ sq units (d) 8 sq units

108. A circle of 3 m radius is divided into three areas by semi-circles of radii 1 m and 2 m as shown in the figure above. The ratio of the three areas *A*, *B* and *C* will be **2016 I**



(a) 2 : 3 : 2 (b) 1 :1 : 1 (c) 4 : 3 : 4 (d) 1 : 2 : 1

ANSWERS

1	*a*	2	*c*	3	*a*	4	*d*	5	*b*	6	*c*	7	*c*	8	*d*	9	*c*	10	*a*
11	*b*	12	*c*	13	*c*	14	*a*	15	*b*	16	*a*	17	*c*	18	*a*	19	*c*	20	*d*
21	*a*	22	*b*	23	*c*	24	*a*	25	*d*	26	*b*	27	*d*	28	*c*	29	*c*	30	*b*
31	*a*	32	*c*	33	*a*	34	*c*	35	*b*	36	*b*	37	*d*	38	*c*	39	*c*	40	*c*
41	*b*	42	*c*	43	*d*	44	*c*	45	*d*	46	*c*	47	*c*	48	*d*	49	*c*	50	*a*
51	*a*	52	*c*	53	*b*	54	*b*	55	*b*	56	*c*	57	*c*	58	*b*	59	*c*	60	*a*
61	*d*	62	*c*	63	*c*	64	*d*	65	*c*	66	*c*	67	*a*	68	*d*	69	*a*	70	*a*
71	*a*	72	*b*	73	*b*	74	*c*	75	*c*	76	*c*	77	*c*	78	*b*	79	*d*	80	*d*
81	*c*	82	*d*	83	*c*	84	*a*	85	*c*	86	*d*	87	*b*	88	*d*	89	*c*	90	*a*
91	*c*	92	*a*	93	*b*	94	*d*	95	*d*	96	*d*	97	*a*	98	*c*	99	*b*	100	*b*
101	*c*	102	*c*	103	*d*	104	*c*	105	*d*	106	*c*	107	*b*	108	*b*				

HINTS AND SOLUTIONS

1. *(a)* Area of square $= \frac{1}{2} \times (\text{Diagonal})^2$

$= \frac{1}{2} \times 50 \times 50$

$= 1250 \text{ m}^2$

2. *(c)* Let length of rectangle be x cm.

and breadth of rectangle $= (x - 2)$ cm

$\because$ Perimeter of rectangle $= 2(l + b)$

So, $2[x + (x - 2)] = 48$

$\Rightarrow \quad 4x - 4 = 48$

$\Rightarrow \quad x = \frac{52}{4} = 13$

$\therefore$ Length = 13 cm and breadth = 11 cm

Hence, area of rectangle

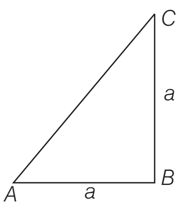
$= l \times b = 13 \times 11$

$= 143 \text{ cm}^2$

3. *(a)* Length of the arc $= \frac{2\pi r\theta}{360°}$

$= \frac{2 \times 22 \times 42 \times 72}{7 \times 360} = 52.8$ cm

4. *(d)* Area of an isosceles right triangle with side 'a' $= \frac{1}{2} a^2 = 200 \text{ cm}^2$



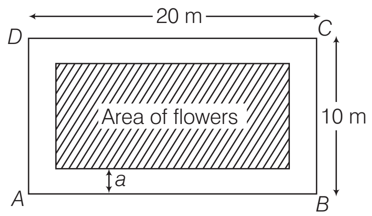
$\therefore$ Side $= a = 20$ cm

Hence, hypotenuse

$= \sqrt{a^2 + a^2}$

$= \sqrt{2}a = 20\sqrt{2}$ cm

5. *(b)* Let the width of the walk be 'a' m.



Given, length of outer rectangle = 20 m

and breadth of outer rectangle = 10 m

$\because$ Width of walk $= a$ m

$\therefore$ Length of inner rectangle

$= (20 - a - a) = (20 - 2a)$m

and breadth of inner rectangle

$= (10 - a - a) = (10 - 2a)$ m

Now, area of inner rectangle

$= (20 - 2a)(10 - 2a)$

$\therefore \quad (20 - 2a)(10 - 2a) = 96$ [given]

$\Rightarrow \quad 4a^2 - 60a + 104 = 0$

$\Rightarrow \quad a^2 - 15a + 26 = 0$

$\Rightarrow \quad (a - 13)(a - 2) = 0$

But $a \neq 13$, so $a = 2$ m

6. *(c)* Side of the greatest square tile

= GCM of the length and breadth of the room

= GCM of 10.5 and 3 is 1.5 m

Area of room $= 10.5 \times 3 \text{ m}^2$

$\therefore$ Number of tiles needed

$= \frac{l \times b}{(\text{H.C.F of } l \,\&\, b)^2}$

$= \frac{10.5 \times 3}{2.25} = 14$ tiles

7. *(c)* Let side of a square be a. Area of square $= a^2$

$\because$ The side of a square be increased by 50%, then new side $= a + \frac{a}{2} = \frac{3a}{2}$

New area $= \frac{9a^2}{4}$

Increase in area $= \frac{9a^2}{4} - a^2 = \frac{5a^2}{4}$

$\therefore$ Percent increase in area $= \frac{5a^2}{4a^2} \times 100$

$= 125\%$

Shortcut Method

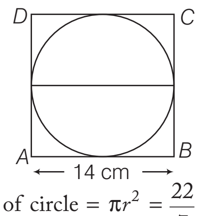
Side of square is increased by 50%.

$\therefore$ Net effect on area

$= \left[50 + 50 + \frac{50 \times 50}{100}\right]\% = 125\%$

8. *(d)* Given, diameter of circle = Side of square = 14 cm

$\therefore \quad r = 7$ cm



Area of circle $= \pi r^2 = \frac{22}{7} \times 7 \times 7$

$= 154 \text{ cm}^2$

9. *(c)* Let original radius of circle be r.

$\therefore$ Area of circle $= \pi r^2$

$\because$ Radius of a circle is decreased by 50%, then

New radius $= \frac{r}{2}$

New area of circle $= \dfrac{\pi r^2}{4}$

$\therefore$ Decrease in area

$$= \pi r^2 - \frac{\pi r^2}{4} = \frac{3\pi r^2}{4}$$

Decrease percent in area

$$= \left(\frac{3}{4}\pi r^2 \times \frac{1}{\pi r^2} \times 100\right)\% = 75\%$$

Short trick Area is decreased by

$$= \left(-2x + \frac{x^2}{100}\right) = \left(-2 \times 50 + \frac{(50)^2}{100}\right)$$

$$= \left(-100 + \frac{2500}{100}\right) = -75\%$$

Area is decreased by 75%.

10. *(a)* Perimeter of square

$$= 4 \times 11 = 44 \text{ cm}$$

$$2\pi r = 44 \text{ cm} \qquad \text{[by condition]}$$

$$\therefore \quad r = \frac{44}{2\pi} = 7 \text{ cm}$$

Thus, area of circle

$$= \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

11. *(b)* Circumference of circle

$$= 2\pi r = 2 \times \frac{22}{7} \times 42 = 264 \text{ cm}$$

$\therefore$ Length of wire = 264 cm

Wire is bent into a square.

$\therefore$ Perimeter of square = 264 cm

$\Rightarrow$ $4 \times$ Sides of square = 264

$\therefore$ Side of square $= \dfrac{264}{4} = 66$ cm

12. *(c)* Given, length of string = Radius of circle = 28 m

Area over which the horse can graze

$$= \pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464 \text{ m}^2$$

13. *(c)* $\because$ Area of ring $= \pi(R^2 - r^2)$

Here, $r = 6$ cm, $R = ?$

Area of ring $= 418 = \dfrac{22}{7}(R^2 - 6^2)$ [given]

$$\Rightarrow \quad R^2 - 36 = \frac{418 \times 7}{22}$$

$$\Rightarrow \quad R^2 = 133 + 36 = 169$$

$$\Rightarrow \quad R = \sqrt{169} = 13 \text{ cm}$$

14. *(a)* Let the length and breadth of rectangle be l and b, respectively.

$\because$ Area of rectangle $= l \times b$ sq units

$\therefore$ New length $= l + \dfrac{60}{100} \times l = \dfrac{8l}{5}$

New breadth $= a$

Then, $lb = \dfrac{8l}{5} \times a \quad \Rightarrow \quad a = \dfrac{5b}{8}$ [by condition]

Decrease percent

$$= \left[\left(b - \frac{5}{8}b\right) \times \frac{1}{b} \times 100\right]\%$$

$$= 37\frac{1}{2}\%$$

15. *(b)* Let the length and breadth of rectangular field be $5x$ and $3x$, respectively.

$l = 5x$ and $b = 3x$

Perimeter of rectangular field $= 2(l + b)$

$$2(5x + 3x) = 240$$

$$\Rightarrow \quad 8x = 120 \Rightarrow x = 15$$

$$\therefore \quad l = 5x = 5 \times 15 = 75 \text{ m}$$

$$\text{and} \quad b = 3x = 3 \times 15 = 45 \text{ m}$$

Now, area of rectangle $= l \times b$

$$= 75 \times 45 = 3375 \text{ m}^2$$

16. *(a)* Given, inner circumference

$$= 2\pi r = 440 \text{ m}$$

$$\Rightarrow \quad r = \frac{440}{2 \times 22} \times 7 = 70 \text{ m}$$

Width of track = 14 m

$\because$ Radius of outer circle = (70 + 14) m = 84 m

$\therefore$ Diameter of outer circle

$$= 2 \times 84 = 168 \text{ m}$$

17. *(c)* Let length and breadth of rectangular be x and y, respectively.

The original area $= xy$

New length $= x + x \times 50\%$

$$= x + \frac{x}{2} = \frac{3x}{2}$$

New breadth $= y + y \times 20\%$

$$= y + \frac{y}{5} = \frac{6y}{5}$$

New area $= \dfrac{3x}{2} \times \dfrac{6y}{5} = \dfrac{9}{5}xy$

$= \dfrac{9}{5}$ (original area)

Short trick Area is increased by

$$= 50 + 20 + \frac{50 \times 20}{100} = 80\%$$

New Area $= A + \dfrac{80}{100} \times A = \dfrac{9}{5}A$

18. *(a)* Let each equal side $= a = 13$ cm and base $b = 24$ cm

$\therefore$ Area of the isosceles triangle

$$= \frac{1}{4} b \cdot \sqrt{4a^2 - b^2}$$

$$= \left[\frac{1}{4} \times 24 \times \sqrt{4 \times 169 - 24 \times 24}\right]$$

$$= 60 \text{ cm}^2$$

19. *(c)* Let the sides containing right angle be x cm and $(x - 14)$ cm.

Now, area of right triangle

$$= \left[\frac{1}{2} x \times (x - 14)\right] \text{cm}^2$$

But area = 120 cm^2 [given]

$$\Rightarrow \quad \frac{1}{2}x(x - 14) = 120$$

$$\Rightarrow \quad x^2 - 14x - 240 = 0$$

$$\Rightarrow \quad (x - 24)(x + 10) = 0$$

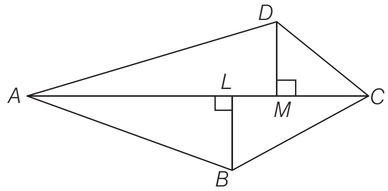
$$\Rightarrow \quad x \neq -10 \;\therefore\; x = 24$$

Other side = 24 − 14 = 10 cm

Hypotenuse $= \sqrt{24^2 + 10^2} = \sqrt{676}$ cm $= 26$ cm

$\therefore$ Perimeter = (24 + 10 + 26) = 60 cm

20. *(d)* Here, $AC = 128$ m, $BL = 22.7$ m, $DM = 17.3$ m



$\therefore$ Area of the field

$$= \frac{1}{2}[AC(BL + DM)]$$

$$= \frac{1}{2} \times 128\,(22.7 + 17.3)$$

$$= 64 \times 40 = 2560 \text{ m}^2$$

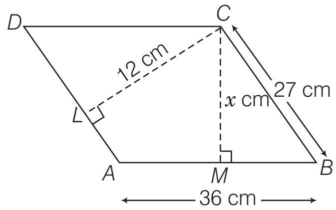
21. *(a)* Let distance between the longer sides be x cm.

Area of parallelogram

$$= AD \times LC = MC \times AB$$

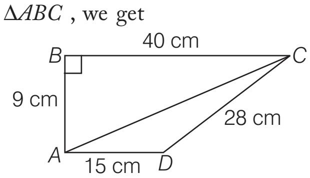
$$\therefore \quad 36 \times x = 27 \times 12$$

$$\Rightarrow \quad x = \frac{27 \times 12}{36} = 9$$



$\therefore$ Distance between the longer sides = 9 cm

22. *(b)* Applying pythagoras theorem in ΔABC, we get



$$9^2 + 40^2 = AC^2$$

$$\Rightarrow AC = \sqrt{1681} = 41 \text{ cm}$$

Area of $\Delta ABC = \dfrac{1}{2} \times 9 \times 40$ cm^2

$$= 180 \text{ cm}^2$$

In ΔACD,

$$s = \frac{41 + 28 + 15}{2} = \frac{84}{2} = 42 \text{ cm}$$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$
$= \sqrt{42(42-41)(42-28)(42-15)}$
$= \sqrt{42 \times 1 \times 14 \times 27}$
$= 14 \times 3 \times 3 = 126 \text{ cm}^2$

$\therefore$ Area of quadrilateral = Area of ΔABC + Area of ΔADC
$= 180 + 126 = 306 \text{ cm}^2$

23. *(c)* Distance covered in one revolution
$= \dfrac{11 \times 1000 \times 100}{5000} = 220$ cm

$\therefore$ The circumference of the wheel $= 220$ cm

Let the diameter be 'D'.

Then, $\pi D = 220 \Rightarrow \dfrac{22}{7} \times D = 220$

$\therefore \quad D = \dfrac{220 \times 7}{22} = 70$ cm

24. *(a)* Let radius of given circles be x cm and $(14 - x)$ cm.

$\therefore$ Sum of areas of circle
$= [\pi x^2 + \pi (14-x)^2]$
$130\pi = \pi x^2 + \pi(14-x)^2$ [by condition]
$\Rightarrow \quad 130 = 2x^2 - 28x + 196$
$\Rightarrow \quad x^2 - 14x + 33 = 0$
$\Rightarrow \quad (x-11)(x-3) = 0$
$\Rightarrow \quad x = 11$ or $x = 3$

So, the radii of circles are 11 cm and 3 cm.

25. *(d)* Angle inscribed by minute hand in 60 min = 360°.

Angle inscribed in 35 min $= \dfrac{360°}{60} \times 35$
$= 210°$

given, $r = 12$ cm

$\therefore$ Area swept by the minute-hand in 35 min
= Area of sector with $r = 12$ cm and $\theta = 210°$

$= \dfrac{22}{7} \times \left(12 \times 12 \times \dfrac{210}{360}\right) \text{cm}^2 = 264 \text{ cm}^2$

26. *(b)* Area of the shaded region
= (Area of sector with $r = 7$ cm, $\theta = 30°$) − (Area of sector with $r = 3.5$ cm, $\theta = 30°$)

$$= \left[\left(\frac{22}{7} \times 7 \times 7 \times \frac{30}{360}\right) - \left(\frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{30}{360}\right)\right] \text{cm}^2$$

$$= \left(\frac{77}{6} - \frac{77}{24}\right) \text{cm}^2 = \frac{77}{8} \text{ cm}^2$$

27. *(d)* Area of field = 1 hec = 10000 m^2

$\therefore$ side $= \sqrt{10000}$ m = 100 m

$\because$ Side of other field = 102% of 100 = 102

$\therefore$ Area of the field
$= 102 \times 102 = 10404 \text{ m}^2$

Thus, difference of area
$= (10404 - 10000) \text{ m}^2$
$= 404 \text{ m}^2$

28. *(c)* Let diameter of the circle be $2r$.

$\therefore$ Area of the circle $= \pi r^2$

Diameter is increased by 100%

New diameter $= 4r$

and new area $= 4\pi r^2$

$\because$ Increase in area $= 4\pi r^2 - \pi r^2 = 3\pi r^2$

$\therefore$ Increase percentage in area
$= \left(\dfrac{3\pi r^2}{\pi r^2} \times 100\right)\% = 300\%$

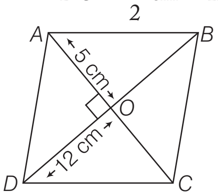
Short trick Area is increased by
$= \left(2x + \dfrac{x^2}{100}\right)\% = \left(2 \times 100 + \dfrac{100^2}{100}\right)\%$
$= (200 + 100)\% = 300\%$

29. *(c)* Area of trapezium = 1/2 (Sum of parallel sides) × Distance between them
$= 1/2(25 + 15) \times 7 = 140 \text{ cm}^2$

30. *(b)* Since, diagonal of rhombus bisect each other and perpendicular to each other.

$\therefore \quad AO = \dfrac{10}{2}$ cm = 5 cm,

$DO = \dfrac{24}{2}$ cm = 12 cm



$\therefore \quad AD = \sqrt{OA^2 + OD^2} = \sqrt{5^2 + 12^2}$
$= \sqrt{169} = 13$ cm

$\therefore$ Perimeter $= 4 \times 13 = 52$ cm

31. *(a)* Here, $OA = OB = OC = 10$ cm

Also, $\quad AC = 2CP$

and $\quad CP = \sqrt{OC^2 - OP^2} = \sqrt{10^2 - 5^2}$
$= \sqrt{75} = 5\sqrt{3}$ cm

As, $\quad AC = 2PC$

$\therefore \quad AC = 10\sqrt{3}$ cm

$\therefore$ Area of the rhombus $OABC$
$= \dfrac{1}{2} \times d_1 \times d_2 = \dfrac{1}{2} \times OB \times AC$
$= \dfrac{1}{2} \times 10 \times 10\sqrt{3} = 50\sqrt{3} \text{ cm}^2$

32. *(c)* Area of shaded region
= Area of ΔABP + Area of ΔPDC
$= \dfrac{1}{2} \times AB \times AP + \dfrac{1}{2} DC \times PD$
$= \dfrac{1}{2} \times AB \times AP + \dfrac{1}{2} AB \times PD$ $[\because AB = DC]$
$= \dfrac{1}{2} AB\,(AP + PD)$
$= \dfrac{1}{2} AB \times AD$ $[\because AD = (AP + PD)]$
$= \dfrac{1}{2} AB \times \dfrac{AB}{2} = \dfrac{1}{4} a^2$
$\left[\because AB = a \text{ and } AD = \dfrac{AB}{2}\right]$

33. *(a)* Area of figure = Area of rectangle + 2 (Area of semi-circular ends)
$= 20 \times 14 + 2 \times \dfrac{1}{2} \times \dfrac{22}{7} \times 7 \times 7$
$= 280 + 154 = 434 \text{ m}^2$

$\therefore$ Cost of levelling = ₹ (434×25)
= ₹ 10850

34. *(c)* Length of
$AC = AB + BC = 4 + 14 = 18$ cm

The area (shaded) bounded by three semi-circle = Area of semi-circle with AC as diameter − [(Area of semi-circle with diameter AB + Area of semi-circle with diameter BC)]

$= \dfrac{1}{2}\pi(9)^2 - \left[\dfrac{1}{2}\pi(7)^2 + \dfrac{1}{2}\pi(2)^2\right]$
$= \pi\dfrac{81}{2} - \dfrac{1}{2}[49\pi + 4\pi]$
$= \dfrac{\pi(81)}{2} - \dfrac{1}{2}[53\pi]$
$= \dfrac{\pi}{2}(81 - 53) = \dfrac{\pi(28)}{2} = 14\pi$

$\therefore$ Area of shaded region
= 14 times π

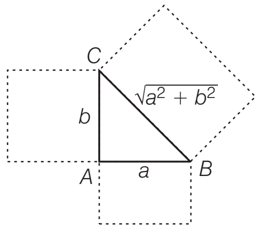
35. *(b)* Area of shaded region = Area of horizontal rectangle + Area of vertical rectangle
$= 5 \times 1 + (8-1) \times 1 = 5 + 7 = 12 \text{ m}^2$

36. *(b)* Length of boundary = Length of arc $(APB + BQC + ARC)$
$= \pi\left(\dfrac{AB}{2}\right) + \pi\left(\dfrac{BC}{2}\right) + \pi\left(\dfrac{AC}{2}\right)$
$= \pi\left(\dfrac{5}{2}\right) + \pi\left(\dfrac{5}{2}\right) + \pi(5)$
$= \dfrac{20\pi}{2} = 10\pi$ cm

37. *(d)* Radius of smaller circle = 2 cm

$\therefore$ Area of shaded region
$= \dfrac{1}{2}$[Area of larger circle − 2(Area of smaller circle)]
$= \dfrac{1}{2}[\pi 4^2 - 2(\pi) 2^2]$
$= \dfrac{1}{2}[16\pi - 8\pi]$
$= \dfrac{1}{2}(8\pi) = 4\pi \text{ cm}^2$

38. *(c)*

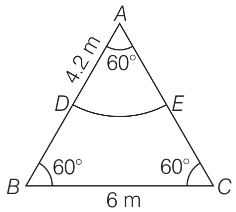


$\therefore$ Total area

$= a^2 + b^2 + (\sqrt{a^2+b^2})^2 + \frac{1}{2}ab$

$= 2(a^2+b^2) + 0.5ab$

39. *(c)* Suppose, a horse is tied at vertex A. Then, area available grazing field is ADE .



Now, area of sector $ADE = \frac{\pi r^2 \theta}{360°}$

$= \frac{22 \times (4.2)^2 \times 60°}{7 \times 360°} = 9.24 \text{ m}^2$

and area of equilateral

$\Delta ABC = \frac{\sqrt{3}}{4}(\text{Side})^2 = \frac{\sqrt{3}}{4} \times (6)^2$

$= 15.57$

$\therefore$ Required percentage $= \frac{9.24}{15.57} \times 100$

$= 59.34\% = 59\%$ (approx)

40. *(c)* Given, radius of circumcircle,

$r = 2$ units

side of square, $AB = 2$ units

Since, $EF = ME = MF = 2$ units

$\therefore MEF$ is an equilateral triangle

In ΔAEM, by pythagoras theorem

$AE^2 = EM^2 - AM^2 = (2)^2 - (1)^2$

$\therefore AE = \sqrt{3}$ units

Area of intersection region square and circle = Area of sector $MENF$ + Area of ΔAEM + Area of ΔFMB

$= \frac{\theta}{360°} \times \pi r^2 + \frac{1}{2} \times AE \times AM + \frac{1}{2} \times MB \times FB$

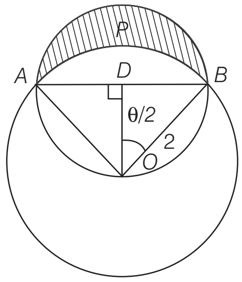
$= \frac{60°}{360°} \times \pi \times (2)^2 + \frac{1}{2} \times \sqrt{3} \times 1 + \frac{1}{2} \times \sqrt{3} \times 1$ $[\because FB = EA = \sqrt{3}]$

$= \frac{4\pi}{6} + \sqrt{3} = \frac{2}{3}\pi + \sqrt{3}$

41. *(b)* Given, radius, $r = 2$ cm

length of chord, $AB = 2\sqrt{2}$ cm

In ΔODB, $\sin\frac{\theta}{2} = \frac{BD}{OB}$



$\Rightarrow \sin\frac{\theta}{2} = \frac{\frac{2\sqrt{2}}{2}}{2} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\theta}{2} = 45°$

$\therefore \theta = 90°$

Area of segment $APBA$ = Area of sector − Area of ΔOAB

$= \frac{1}{4}.\pi r^2 - \frac{1}{2} \times OA \times OB$

$= \frac{1}{4} \times \pi \times (2)^2 - \frac{1}{2} \times 2 \times 2$

$= (\pi - 2) \text{ cm}^2$

Area of shaded region = Area of semicircle with AB as diameter − Area of segment $APBA$

$= \frac{1}{2} \times \pi \times (BD)^2 - (\pi - 2)$

$= \frac{1}{2} \times \pi \times (\sqrt{2})^2 - (\pi - 2) = 2 \text{ cm}^2$

42. *(c)* We have, $a = l \times w$ ...(i)

and $p = 2(l + w)$ ... (ii)

Putting the value of l from Eq. (i) in (ii), we get

$wp - 2w^2 = 2a$

Now, $p^2 - 8a = [2(l+w)]^2 - 8lw$

$= 4(l^2 + w^2 + 2lw) - 8lw$

$= 4l^2 + 4w^2 + 8lw - 8lw$

$= 4(l^2 + w^2)$

Hence, the statement I and III are correct.

43. *(d)* I. Let R be the radius of circumcircle

Area of circumcircle $= \pi R^2$

$\Rightarrow 9\pi = \pi R^2 \Rightarrow R = 3$ cm

II. $\frac{\text{Inradius of polygon}}{\text{circumradius of polygon}} = \frac{r}{R}$

$= \frac{\frac{1}{2}a \cot\left(\frac{180°}{n}\right)}{\frac{1}{2}a \operatorname{cosec}\left(\frac{180°}{n}\right)} = \cos\left(\frac{180°}{n}\right)$

$\Rightarrow r = R\cos\left(\frac{180}{12}\right)^\circ = 3\cos 15°$

III. Area of polygon $= \frac{1}{2}nR^2 \sin\left(\frac{360°}{n}\right)$

$= \frac{1}{2} \times 12 \times R^2 \sin\left(\frac{360°}{12}\right)$

$= 6R^2 \sin 30° = 3R^2$

$\therefore \frac{\text{Area of circumcircle}}{\text{Area of polygon}} = \frac{\pi R^2}{3R^2} = \frac{\pi}{3}$

or $\pi : 3$

Hence, all three statements are correct.

44. *(c)* $\because$ From statement I

length $= 2 \times$ width

$\therefore$ Area of rectangle $= 2 \times$ width $\times$ width

$= 2\,(\text{width})^2$

From statement II,

$\therefore$ Area of rectangle $= 2 \times$ Perimeter

$= 2 \times 2(\text{length} + \text{width})$

$= 4\,(\text{length} + \text{width})$

From I and II,

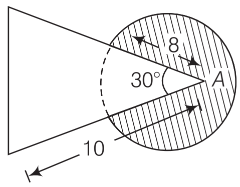
$2\,(\text{width})^2 = 4\,(2 \text{ width} + \text{width})$

$2\,(\text{width})^2 = 12$ width

$\Rightarrow$ width = 6 units.

So, both statements together are sufficient.

45. *(d)* The length of the rope is 8 m, then the cow will be able to graze an area equal to the area of the circle with radius = 8 m, subtracting from that the area of the sector of the same circle with angle 30°,



which is equal to $\pi(8)^2 - \frac{\theta}{360}\pi(8)^2$

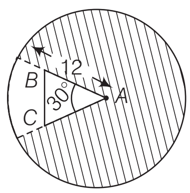
$= \pi \times 64 - \frac{30}{360}\pi \times 64 = \frac{176\pi}{3} \text{ m}^2$

46. *(c)* Since, the length of the rope is more than that of sides AB and AC,

$\therefore$ Required Area =

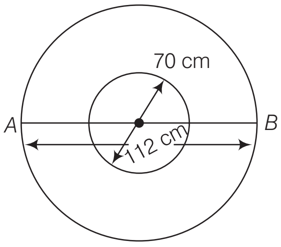
(area of the circle with radius 12) − (area of the sector of the same circle with angle 30°)

$= \pi(12)^2 - \frac{30}{360}\pi(12)^2 = 132\pi \text{ m}^2$



47. *(c)* ∵ Outer diameter = 112 cm
and inner diameter = 70 cm

∴ Required area $= \frac{1}{4}\pi(112^2 - 70^2)$



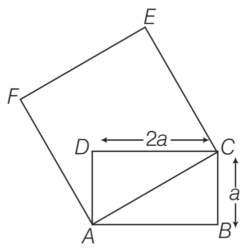
$= \frac{1}{4}(12544 - 4900)\pi$

$= \frac{1}{4} \times 7644 \times \frac{22}{7}$

$= \frac{1}{4} \times 24024 = 6006 \text{ cm}^2$

48. *(d)* Given that, Area of rectangle $= 2a^2 = l \times b$

$\Rightarrow l \times b = 2a^2 = l \times a \Rightarrow l = 2a$



Now, in ΔACD, by pythagoras theorem

$AC^2 = AD^2 + CD^2$

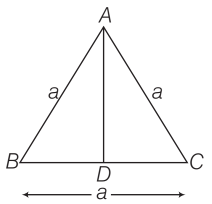
$= a^2 + 4a^2 = 5a^2$

∴ Side of square, $AC = a\sqrt{5}$ unit

Hence, area of the square $= (a\sqrt{5})^2$

$= 5a^2$ sq units

49. *(c)*



Height of equilateral triangle (AD)

$= \frac{\sqrt{3}}{2} \times \text{Side} \Rightarrow \sqrt{3} = \frac{\sqrt{3} \times \text{Side}}{2}$

$\Rightarrow 2\sqrt{3} = \sqrt{3} \times \text{Side}$

∴ Side $= \frac{2\sqrt{3}}{\sqrt{3}} = 2$ cm

∴ Perimeter of an equilateral triangle

$= 3a = 3 \times 2 = 6$ cm

50. *(a)* Let the width of the rectangle be x unit.

∴ Length $= (2x + 5)$ unit

According to the question,

Area $= x(2x + 5) \Rightarrow 75 = 2x^2 + 5x$

$\Rightarrow 2x^2 + 5x - 75 = 0$

$\Rightarrow 2x^2 + 15x - 10x - 75 = 0$

$\Rightarrow x(2x + 15) - 5(2x + 15) = 0$

$\Rightarrow (2x + 15)(x - 5) = 0$

∴ $x = 5$ and $\frac{-15}{2}$

Since, width cannot be negative.

∴ Width = 5 units

and length $= 2 \times 5 + 5 = 15$ units

∴ Perimeter of the rectangle

$= 2(15 + 5) = 40$ units

51. *(a)* Let the radius of circle is r and the side of a square is a, then by given condition,

$2\pi r = 4a \Rightarrow a = \frac{\pi r}{2}$

∴ Area of square $= \left(\frac{\pi r}{2}\right)^2 = \frac{\pi^2 r^2}{4}$

$= \frac{(3.14)^2 r^2}{4} = \frac{9.86 r^2}{4} = 2.46 r^2$

and area of circle $= \pi r^2 = 3.14 r^2$

Again, let the side of equilateral triangle be x.

Then, by given condition, $3x = 2\pi r$

$\Rightarrow x = \frac{2\pi r}{3}$

∴ Area of equilateral triangle $= \frac{\sqrt{3}}{4}x^2$

$= \frac{\sqrt{3}}{4} \times \frac{4\pi^2 r^2}{9} = \frac{\pi^2}{3\sqrt{3}} r^2 = 1.89 r^2$

Hence, Area of circle > Area of square > Area of equilateral triangle

Hence, only I statements is correct.

52. *(c)* Given that, Area of the circle = Area of the square = (Side)^2

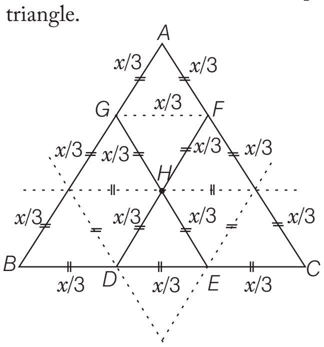
∴ $\pi r^2 = (2\sqrt{\pi})^2 \Rightarrow \pi r^2 = 4\pi$

$\Rightarrow r^2 = \frac{4\pi}{\pi} = 4$ ∴ $r = \sqrt{4} = 2$ units

∴ Diameter of circle $(d) = 2 \cdot r = 2 \cdot 2$

$= 4$ units

53. *(b)* Here, ΔABC forms an equilateral triangle.



where, $AGHF$ form a rhombus and ΔHDE is also an equilateral triangle.

∴ Area of rhombus = (Area of ΔAGF + Area of ΔGFH)

$= \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 + \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 = 2 \cdot \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2$

Now, area of $\Delta HDE = \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2$

and area of $\Delta ABC = \frac{\sqrt{3}}{4}x^2$

By given condition,

$$\frac{\text{Area of rhombus } AGHF + \text{Area of } \Delta HDE}{\text{Area of } \Delta ABC}$$

$$= \frac{2 \cdot \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 + \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2}{\frac{\sqrt{3}}{4}x^2}$$

$$= \frac{3 \cdot \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2}{\frac{\sqrt{3}}{4}x^2} = \frac{3}{9} = \frac{1}{3}$$

54. *(b)* We know that, the radius of a circle inscribed in a equilateral triangle $= a / 2\sqrt{3}$ where, a be the length of the side of an equilateral triangle.

Given that, area of a circle inscribed in an equilateral triangle = 154 cm^2

∴ $\pi\left(\frac{a}{2\sqrt{3}}\right)^2 = 154$

$\Rightarrow \left(\frac{a}{2\sqrt{3}}\right)^2 = \frac{154 \times 7}{22} = 7 \times 7 = (7)^2$

$\Rightarrow \frac{a}{2\sqrt{3}} = 7 \Rightarrow a = 14\sqrt{3}$ cm

∴ Perimeter of an equilateral triangle

$= 3a = 3(14\sqrt{3}) = 42\sqrt{3}$ cm

55. *(b)* Let the radii of two circles be r_1 and r_2, respectively.

Given, $\frac{\text{Circumference of Ist circle}}{\text{Circumference of IInd circle}} = \frac{2}{3}$

$\Rightarrow \frac{2\pi r_1}{2\pi r_2} = \frac{2}{3} \Rightarrow \frac{r_1}{r_2} = \frac{2}{3} \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$ ...(i)

∴ $\frac{\text{Area of Ist circle}}{\text{Area of IInd circle}} = \frac{\pi r_1^2}{\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$

or 4 : 9

56. *(c)* Let the radius of a circle be r and a be the length of the side of a square.

Given, circumference of a circle

= Perimeter of a square

$\Rightarrow 2\pi r = 4a \Rightarrow a = \frac{\pi}{2}r = 1.57r$

Now, area of the circle (A_c)

$= \pi r^2 = 3.14 r^2$

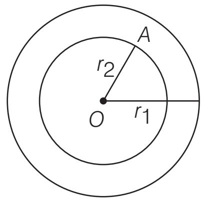
and area of the square (A_s)

$= a^2 = 2.4649\, r^2$

∴ Area of circle > Area of square

57. *(c)* Given that, ratio of their radii = 5 : 3

i.e. $r_1 : r_2 = 5:3$



$\Rightarrow \quad \dfrac{r_1}{r_2} = \dfrac{5}{3}$ ...(i)

Let, $r_1 = 5x$ and $r_2 = 3x$

Also, given that area enclosed between the circumferences of two concentric circles = $16\pi\,\text{cm}^2$

$\therefore \quad \pi(r_1^2 - r_2^2) = 16\pi$

$\Rightarrow (5x)^2 - (3x)^2 = 16$

$\Rightarrow \quad 25x^2 - 9x^2 = 16$

$\Rightarrow \quad 16x^2 = 16 \Rightarrow x^2 = 1$

$\therefore \quad x = 1$

$\therefore \quad r_1 = 5$ and $r_2 = 3$

$\therefore$ Area of the outer circle $= \pi r_1^2 = \pi(5)^2$

$= 25\,\pi\,\text{cm}^2$

58. *(b)* Given that, perimeter of a rectangle

= 82 m

$\therefore$ 2 (Length + Breadth) = 82 m

$\Rightarrow$ Length + Breadth = 41 m

$\Rightarrow \quad l + b = 41\,\text{m}$ ...(i)

Also, its area = 400 m^2

$\Rightarrow \quad l \cdot b = 400$ ...(ii)

Now, $(l-b)^2 = (l+b)^2 - 4lb$

$= (41)^2 - 4\,(400)$

$= 1681 - 1600 = 81$

$\therefore \quad l - b = 9$ ...(iii)

On solving Eqs. (i) and (iii), we get

$2l = 50 \Rightarrow l = 25$ m and $b = 16$ m

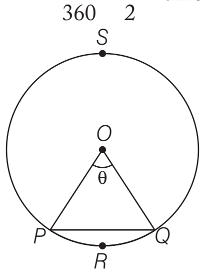
$\therefore$ Required breadth *(b)* = 16 m

59. *(c)* I. We know that, area of segment (*PRQP*)

= Area of sector (*OPRQO*)

– Area of $\Delta\,OPQ$

$= \dfrac{\pi r^2\theta}{360} - \dfrac{1}{2} r^2 \sin\theta$



So, the area of a segment of a circle is always less than area of its corresponding sector.

II. Distance travelled by a circular wheel of diameter $2d$ cm in one revolution

$= 2\pi \dfrac{(2d)}{2} = 2 \times 3.14 \times d = 6.28\,d$

which is greater than $6d$ cm.

Hence, both statements are correct.

60. *(a)* $\because$ Angle described in 60 min by minute hand of a clock = 360°

and angle described in 40 min by minute hand of a clock

$= \dfrac{360°}{60°} \times 40 = 240°$

$\therefore$ Required distance $= \dfrac{2\pi(2 \cdot 5) \times 240°}{360°}$

$= \dfrac{10\pi}{3}$ cm

61. *(d)* Given that, length of hour hand

= 4 cm

and length of minute hand = 6 cm

$\therefore$ Hour hand rotating in 1 day

$= 2 \times 360° = 720°$

$\therefore$ Hour hand rotating in 2 days

$= 2 \times 720° = 1440°$

Similarly,

Minute hand rotating in 1 day

$= 24 \times 360°$

$\therefore$ Minute hand rotating in 3 days

$= 72 \times 360°$

$\therefore$ Distance travelled by hour hand

$= \dfrac{2\pi(4) \times 1440°}{360°} = 32\pi$

and distance travelled by minute hand

$= \dfrac{2\pi(6) \times 72 \times 360°}{360}$

$= 6 \times 144\pi$ cm

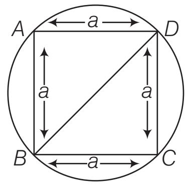
$\therefore$ Required ratio $= 32\pi : 6 \times 144\pi$

$= 1 : 27$

62. *(c)* Let side of a square be *a* cm.

Given that, radius of a circle = 8 cm

and diameter of a circle = 16 cm



$\because$ Length of a diagonal of a square

= Diameter of a circle

$\Rightarrow \quad a\sqrt{2} = 16$

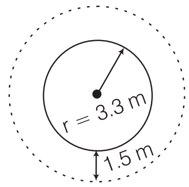
$\therefore \quad a = 8\sqrt{2}$ cm

$\therefore$ Area of square $ABCD = a^2 = (8\sqrt{2})^2$

$= 64 \times 2 = 128$ cm^2

63. *(c)* Area of the path = Area of circular water [fountain + path]

– Area of circular water fountain



$= \pi\,(3.3 + 1.5)^2 - \pi\,(3.3)^2$

$= [(4.8)^2 - (3.3)^2]\,\pi$

$= (23.04 - 10.89)\,\pi$

$= 12.15\,\pi$ m^2

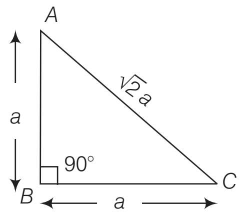
64. *(d)* Let the equal sides of isosceles right triangle be *a* cm.

Then $AB = BC = a$

In ΔABC, by pythagoras theorem

$AC^2 = AB^2 + BC^2 = 2AB^2$

$\Rightarrow \quad AC^2 = 2a^2 \quad \therefore \quad AC = \sqrt{2}a$



Area of $\Delta ABC = \dfrac{1}{2} \times a \times a$

$\Rightarrow \quad 1 = \dfrac{1}{2}a^2 \Rightarrow a = \sqrt{2}$

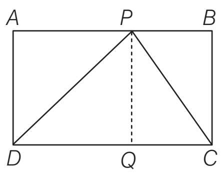
$\therefore$ Perimeter of $\Delta ABC = 2a + a\sqrt{2}$

$= 2\sqrt{2} + \sqrt{2} \cdot \sqrt{2}$

$= 2\,(\sqrt{2} + 1)$ units

65. *(c)* Given that, $CD = 20$ cm

and area of rectangle $ABCD = 100\,\text{cm}^2$



$\Rightarrow AD \times CD = 100$ cm^2

$\Rightarrow \quad AD \times 20 = 100$

$\therefore \quad AD = 5$ cm

$\therefore$ Area of $\Delta PDC = \dfrac{1}{2} \times PQ \times CD$

$= \dfrac{1}{2} \times 5 \times 20 = 5 \times 10 = 50$ cm^2

$[\because AD = PQ]$

66. *(c)* Area of parallelogram

= Base × Height

$= 8.06 \times 2.08 = 16.76$ cm^2

67. *(a)* Let $b = 4x$ and $l = 9x$

$\therefore$ Area of rectangle $= l \times b$

$144 = 4x \times 9x$

$\Rightarrow \quad x^2 = \frac{144}{36} \Rightarrow x^2 = 4 \therefore x = 2$

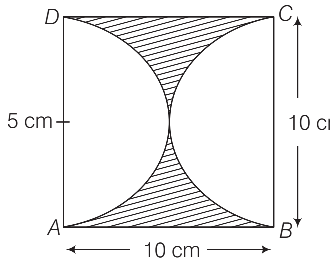
Now, $b = 4 \times 2 = 8$ cm and

$l = 9 \times 2 = 18$ cm

$\therefore$ Perimeter of rectangle $= 2(b + l)$

$= 2(8 + 18) = 2 \times 26 = 52$ cm

68. *(d)* Area between square and semi-circles = Area of square − 2 (Area of semi-circle)



$= (10)^2 - 2 \times \frac{1}{2} \times \frac{22}{7} \times 5)^2 = 100 - 78.5$

$= 21.5 \text{ cm}^2$

Hence, the area between the square and semi-circular cross-section is 21.5 cm^2.

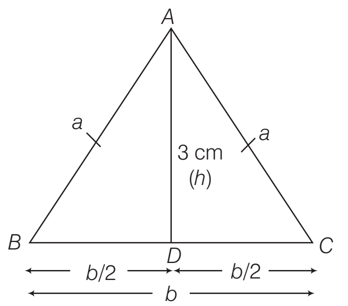
69. *(a)* 1 m = 1000 mm

$\therefore$ 10 m = 10000 mm

Number of 200 mm lengths that can be cut from 10 m of ribbon

$= \frac{10000}{200} = 50$

70. *(a)* Area of the $\Delta ABC = \frac{1}{2} \times b \times h$



$\Rightarrow \quad 12 = \frac{1}{2} \times b \times 3$

$\therefore \quad b = \frac{12 \times 2}{3} = 8$ cm

Here, $BD = CD = \frac{b}{2} = \frac{8}{2} = 4$ cm

In right angled ΔABD, by pythagoras theorem,

$AB = \sqrt{BD^2 + AD^2}$

$\Rightarrow \quad a = \sqrt{4^2 + 3^2} = \sqrt{16 + 9}$

$= \sqrt{25} = 5$ cm

Now, perimeter of an isosceles triangle

$= 2a + b = 2 \times 5 + 8$

$= 10 + 8 = 18$ cm

71. *(a)* Area of equilateral triangle

$= \frac{\sqrt{3}a^2}{4} = x \quad$ ...(i)

and perimeter $= 3a = y \Rightarrow a = \frac{y}{3}$ ...(ii)

Now, putting the value of a from Eq. (ii) in Eq. (i), we get

$\frac{\sqrt{3}\left(\frac{y}{3}\right)^2}{4} = x \quad \Rightarrow \quad x = \frac{\sqrt{3} \times y^2}{9 \times 4}$

$\Rightarrow \quad x = \frac{y^2}{3\sqrt{3} \times 4} \quad \Rightarrow \quad x = \frac{y^2}{12\sqrt{3}}$

$\Rightarrow \quad 12\sqrt{3}\, x = y^2$

On squaring both sides, we get

$y^4 = 432x^2$

72. *(b)* Here, $a = 7$ cm, $b = 24$ cm

and $c = 25$ cm

Semi-perimeter of triangle

$s = \frac{a + b + c}{2} = \frac{7 + 24 + 25}{2} = \frac{56}{2}$

$= 28$ cm

According to the question,

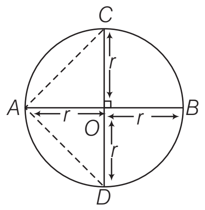
Area of circle = Area of triangle

$= \sqrt{s(s - a)(s - b)(s - c)}$

$= \sqrt{28(28 - 7)(28 - 24)(28 - 25)}$

$= \sqrt{28 \times 21 \times 4 \times 3} = \sqrt{7056} = 84 \text{ cm}^2$

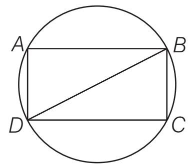
73. *(b)* Diameters of a circle intersect at the centre of circle.



Required ratio $= \frac{\text{Area of circle}}{\text{Area of } \Delta ACD}$

$= \frac{\pi r^2}{\frac{1}{2} \times 2r \times r} = \pi$

74. *(c)* Area of rectangle inscribed in a circle is maximum, when



Diameter of the circle = Diagonal of the rectangle

Now, let x and y be the length and breadth of the rectangle, respectively.

Now in ΔABD, by pythagoras theorem

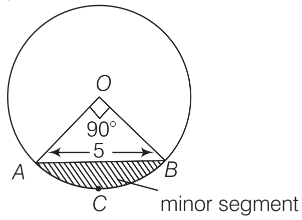
$AB^2 + AD^2 = (5)^2 \Rightarrow x^2 + y^2 = 25$

Since, they form pythagorian triplet,

$\therefore x = 4$ and $y = 3$

So, area of rectangle $= 3 \times 4 = 12 \text{ cm}^2$

75. *(c)* Let $AO = OB = r$ [radius of circle]



In ΔAOB, by pythagoras theorem,

$AB^2 = OA^2 + OB^2$

$\Rightarrow (5)^2 = r^2 + r^2 \quad [\because AB = 5 \text{ cm}]$

$\therefore \quad r^2 = \frac{25}{2}$ cm

Now, area of sector

$ACBOA = \frac{\theta}{360°} \times \pi r^2$

$= \frac{90°}{360°} \times \pi \times \frac{25}{2} = \frac{25\pi}{8} \text{ cm}^2$

Now, area of minor segment $ACBA$

= Area of sector $ACBOA$ − Area of triangle

$= \frac{25\pi}{8} - \frac{r^2}{2} = \frac{25\pi}{8} - \frac{25}{4}$

$= \left(\frac{25\pi - 50}{8}\right) \text{cm}^2$

Area of major segment = Area of circle − Area of minor segment

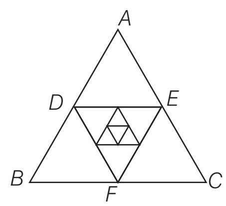
$= \pi r^2 - \left(\frac{25\pi - 50}{8}\right)$

$= \frac{25\pi}{2} - \frac{(25\pi - 50)}{8}$

$= \frac{100\pi - 25\pi + 50}{8} = \frac{75\pi + 50}{8}$

$= \frac{25}{8}(3\pi + 2) = \frac{25}{4}\left(\frac{3\pi}{2} + 1\right) \text{cm}^2$

76. *(c)*



Perimeters of triangles

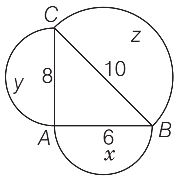
$= 3 + 3 \times \frac{1}{2} + 3 \times \frac{1}{4} + 3 \times \frac{1}{8} + \ldots$

$= 3\left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots\right] = 3 \times \frac{1}{1 - \frac{1}{2}}$

$= 6$ units

77. *(c)* In ΔABC, by pythagoras theorem,

$$BC^2 = AB^2 + AC^2$$



$$= 6^2 + 8^2 = 36 + 64 = 100 \text{ cm}$$

$\therefore \quad BC = 10$ cm

Radius of semi-circle $x = \dfrac{6}{2} = 3$

Now, area of semi-circle $= x = \dfrac{\pi(3)^2}{2}$

$$= \frac{9\pi}{2} \text{ cm}^2$$

Radius of semi-circle, $y = \dfrac{8}{2}$ cm

Area of semi-circle $= y = \dfrac{16\pi}{2}$ cm^2

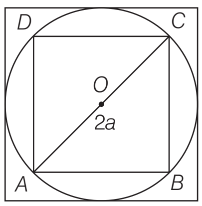
Radius of semi-circular,

$$= z = \frac{10}{2} = 5 \text{ cm}^2$$

Area of semi-circle $= z = \dfrac{25\pi}{2}$ cm^2

Now, $x + y - z = \dfrac{9\pi}{2} + \dfrac{16\pi}{2} - \dfrac{25\pi}{2} = 0$

78. *(b)*



For inscribed circle, Diameter of circle

$= $ Diagonal of square $= 2a$

Now, in ΔABC by pythagoras theorem,

$AB^2 + BC^2 = AC^2 \Rightarrow \quad 2AB^2 = 4a^2$

$AB^2 = 2a^2 \quad \Rightarrow \quad AB = \sqrt{2}a$

$\because \quad$ Area of inner square

$$= AB^2 = (\sqrt{2}a)^2 = 2a^2$$

For circumscribed square,

Diameter of a circle = Side of square

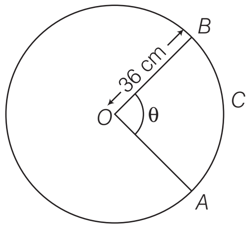
$\therefore$ Area of circumscribed square

$$= (2a)^2 = 4a^2$$

$\therefore$ Difference between areas of outer and inner squares

$$= 4a^2 - 2a^2 = 2a^2$$

79. *(d)* Given that, area of sector $= 72\pi$ cm^2



$$\Rightarrow \quad \frac{\pi r^2\theta}{360°} = 72\pi$$

$$\Rightarrow \quad \theta = \frac{72 \times 360}{36 \times 36} = 20°$$

Now, length of arc $ACB = \dfrac{\pi r\theta}{180°}$

$$= \frac{\pi \times 36 \times 20}{180} = 4\pi \text{ cm}$$

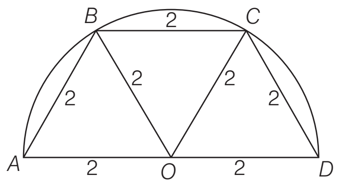
80. *(d)* Since, ΔAOB, ΔBOC and ΔCOD are equilateral.

$\therefore$ Side of an equilateral triangle

$= $ radius of semi-circle $= 2$ cm

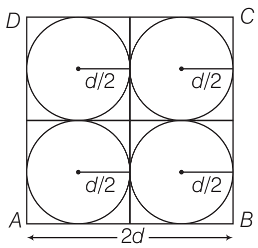
Now, total area of three equilateral triangles

$$= 3 \times \frac{\sqrt{3}}{4} (\text{Side})^2$$



$$= 3 \times \frac{\sqrt{3}}{4} \times 4 = 3\sqrt{3} \text{ cm}^2$$

81. *(c)*



From the figure it is clear that, 4 circular plates of diameter d can be made from.

square plate of side $2d$ with minimum loss of material.

82. *(d)* Given, sides of a triangle are in the ratio $= \dfrac{1}{2} : \dfrac{1}{3} : \dfrac{1}{4}$ i.e. $6 : 4 : 3$

Let the sides of a triangle be $6x$, $4x$ and $3x$, respectively.

We know that,

Perimeter of a triangle = Sum of all sides of a triangle

$\Rightarrow \quad 52 = 6x + 4x + 3x$

$\Rightarrow \quad 52 = 13x$

$\because \quad x = \dfrac{52}{13} = 4$

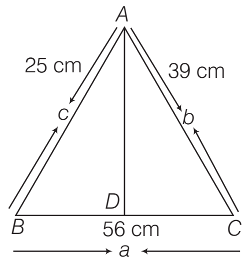
$\therefore$ Smallest side of a triangle

$$= 3x = 3 \times 4 = 12 \text{ cm}$$

83. *(c)* Let the three sides of a triangle be a, b and c, respectively.

$\therefore \quad a = 56$ cm, $b = 39$ cm

and $\quad c = 25$ cm



Perimeter of triangle, $2s = a + b + c$

$\Rightarrow \quad 2s = 25 + 39 + 56$

$\Rightarrow \quad s = 60$ cm

Now, area of triangle

$= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{60(60-25)(60-39)(60-56)}$

$= \sqrt{60 \times 35 \times 21 \times 4} = 420 \text{ cm}^2 \quad \ldots\text{(i)}$

Again area of triangle $= \dfrac{1}{2} \times$ Base $\times$ Altitude

$$= \frac{1}{2} \times BC \times AD \quad \ldots\text{(ii)}$$

From Eqs. (i) and (ii), we get

$$\frac{1}{2} \times BC \times AD = 420$$

$$\Rightarrow \quad \frac{1}{2} \times 56 \times AD = 420$$

$$\Rightarrow \quad AD = \frac{420}{28} = 15 \text{ cm}$$

Hence, the length of AD is 15 cm.

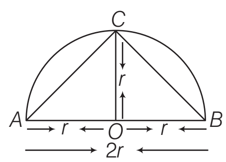
84. *(a)* Let the r be the radius of semi-circle.

In ΔABC,

$AO = OB = OC = r$ [radius of circle]

$\therefore$ Area of largest triangle

$$= \frac{1}{2} \times \text{Base} \times \text{Height}$$



$= \dfrac{1}{2} \times AB \times OC = \dfrac{1}{2} \times 2r \times r$

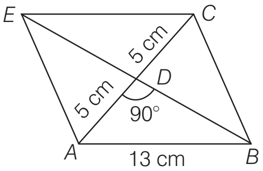
$[\because AB = OA + OB]$

$= r^2$

Hence, the area of the largest triangle is r^2.

85. *(c)* Given, a side of rhombus $(a) = 13$ cm and a diagonal of rhombus $(d_1) = 10$ cm

Now, in right angled ΔABD,



$$AB^2 = AD^2 + BD^2$$

$\Rightarrow \quad 13^2 = 5^2 + BD^2$

$\Rightarrow \quad BD^2 = 169 - 25 = 144$

$\because \quad BD = 12$ cm

$\therefore \quad EB = 2 \times BD = 2 \times 12 = 24$ cm

Length of second diagonal $(d_2) = 24$ cm

Now, area of rhombus $= \dfrac{1}{2} \times d_1 \times d_2$

$$= \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2$$

Shortcut Method

Here, $a = 13$ cm, $\quad d_1 = 10$ cm

We know that,

$$4a^2 = d_1^2 + d_2^2$$

$\Rightarrow \quad 4 \times 13^2 = 10^2 + d_2^2$

$\Rightarrow \quad 4 \times 169 = 100 + d_2^2$

$\Rightarrow \quad 676 - 100 = d_2^2$

$\Rightarrow d_2 = \sqrt{576} = 24$ cm

Now, area of rhombus $= \dfrac{1}{2} d_1 d_2$

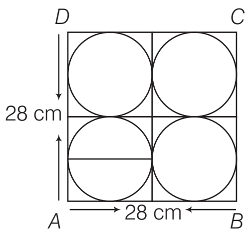
$$= \frac{1}{2} \times 10 \times 24 = 120 \text{ cm}^2$$

86. *(d)* Let the side of a square paper sheet be a cm.

Given, area of square paper sheet

$= 784$ cm^2

$\Rightarrow \quad a^2 = 784 \quad \Rightarrow \quad a = 28$ cm



$\because$ Diameter of one circle $= \dfrac{28}{2} = 14$ cm

and radius of one circle $= \dfrac{14}{2} = 7$ cm

$\therefore$ Circumference of each plate $= 2\pi r$

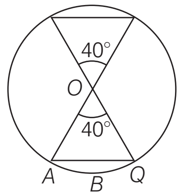
$$= 2\pi \times 7 = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

Hence, the circumference of each plate is 44 cm.

87. *(b)* Given, radius of circle, $r = 3$ cm

Area of remaining portion

= Area of circle − 2 (Area of sector $OABQO$).



$$= \pi r^2 - \frac{2\pi r^2 \theta}{360°}$$

$$= \pi r^2 \left[1 - \frac{2\theta}{360°}\right]$$

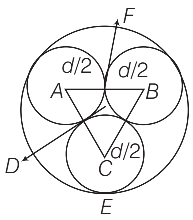
$$= \pi r^2 \left[1 - \frac{2 \times 40}{360°}\right]$$

$$= \frac{22}{7} \times 9 \left[1 - \frac{8}{36}\right]$$

$$= \frac{22}{7} \times 9 \left[1 - \frac{2}{9}\right] = \frac{22}{7} \times \frac{7}{9} \times 9 = 22 \text{ cm}^2$$

88. *(d)* Given, diameter of each circle $= d$

$\therefore$ Radius of each circle $= d/2$



$$CE = AF = BF = d/2$$

Here, ΔABC is an equilateral triangle.

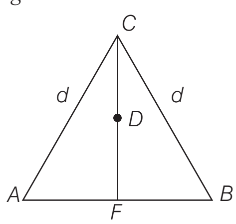
$\therefore$ Altitude of equilateral triangle

$$= \frac{\sqrt{3}}{2} \text{ (Side)} = \frac{\sqrt{3}}{2}(AB) = \frac{\sqrt{3}}{2} d$$

$[\because AB = AF + FB = d]$

Let D be the centroid of ΔABC

$\because$ Ratio of centroid of equilateral triangle = 2 : 1



$\therefore \quad$ Length of $DC = \dfrac{2}{3} \times \dfrac{\sqrt{3}}{2} d = \dfrac{\sqrt{3}}{3} d$

Now, radius of big circle,

$$R = DE = DC + CE = \frac{\sqrt{3}}{3} d + \frac{d}{2}$$

$$= d\left[\frac{\sqrt{3}}{3} + \frac{1}{2}\right] = \frac{d \times \sqrt{3}(2 + \sqrt{3})}{6}$$

$\therefore$ Area of big circle $= \pi R^2$

$$= \pi \left[d \times \frac{\sqrt{3}(2 + \sqrt{3})}{6}\right]^2$$

$$= \pi \times \left[\frac{d^2 \times 3(2 + \sqrt{3})^2}{36}\right]$$

$$= \frac{\pi d^2 (2 + \sqrt{3})^2}{12}$$

Hence, the area of big circle is

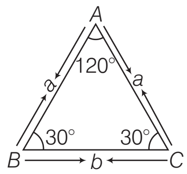
$$= \frac{\pi d^2 (2 + \sqrt{3})^2}{12}.$$

89. *(c)* Let the angles of a triangle be $4x$, x and x, respectively.

i.e. $\angle A = 4x$,

$\angle B = x \quad$ and

$\angle C = x$



$\because$ Sum of all angles of a triangle = 180°

$\therefore \angle A + \angle B + \angle C = 180°$

$\Rightarrow \quad 4x + x + x = 180° \Rightarrow 6x = 180°$

$\Rightarrow \quad x = 30°$

$\therefore \quad \angle A = 4x = 30 \times 4 = 120°$,

$\angle B = x = 30°$

and $\quad \angle C = x = 30°$

So, it is clear that, given triangle is isosceles triangle. Let the sides isosceles triangle be a, a and b, respectively.

$\therefore$ Perimeter of triangle

$$= a + a + b = 2a + b$$

Using sine rule, $\dfrac{b}{\sin 120°} = \dfrac{a}{\sin 30°}$

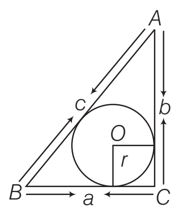
$\Rightarrow \dfrac{b}{\sqrt{3}/2} = \dfrac{a}{1/2} \Rightarrow \dfrac{2b}{\sqrt{3}} = 2a \Rightarrow a = \dfrac{b}{\sqrt{3}}$

Now, $\quad \dfrac{b}{2a + b} = \dfrac{b}{2 \times \dfrac{b}{\sqrt{3}} + b}$

$$= \frac{b}{\dfrac{2b + \sqrt{3}b}{\sqrt{3}}} = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

90. *(a)* Given, a, b and c be the sides of a right triangle, where c is the hypotenuse.

$\therefore \quad c^2 = a^2 + b^2 \quad$ ...(i)



and perimeter of triangle $2s = a + b + c$

$\therefore \quad s = \dfrac{a + b + c}{2}$

Area of triangle $= \dfrac{1}{2} \times BC \times AC$

$\Rightarrow \quad \Delta = \dfrac{1}{2} ab \quad$ ...(ii)

From Eq. (i), $c^2 = a^2 + b^2$

$\Rightarrow \quad c^2 = (a + b)^2 - 2ab$

$\Rightarrow \quad 2ab = (a+b)^2 - c^2$

$\Rightarrow \quad ab = \dfrac{(a+b)^2 - c^2}{2}$

Now, area of triangle $= \dfrac{ab}{2}$ [from Eq. (ii)]

$$\Delta = \frac{\dfrac{[(a+b)^2 - c^2]}{2}}{2}, \ \Delta = \frac{[(a+b)^2 - c^2]}{4}$$

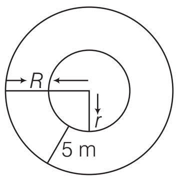
$\therefore$ Radius of circle, $r = \dfrac{\Delta}{s}$

$$= \frac{(a+b-c)(a+b+c)}{4 \times \left(\dfrac{a+b+c}{2}\right)} = \frac{(a+b-c)}{2}$$

Hence, the radius of circle is $\dfrac{(a+b-c)}{2}$

91. *(c)* Let the radius of outer circle and inner circle be R and r, respectively.

Perimeter of outer circle $= 2\pi R$ and perimeter of inner circle $= 2\pi r$



$\because \quad \dfrac{2\pi R}{2\pi r} = \dfrac{23}{22}$

$\Rightarrow \quad \dfrac{R}{r} = \dfrac{23}{22}$

From the figure, $R = 5 + r$

$\therefore \quad \dfrac{5+r}{r} = \dfrac{23}{22} \Rightarrow 110 + 22r = 23r$

$\Rightarrow \quad r = 110$ m

So, diameter of the inner circle is 220 m.

92. *(a)* Let the sides of a triangular field be a, b and c, respectively.

$\therefore \ a = 41$ m, $b = 40$ m and $c = 9$ m

Perimeter of triangular field $= a + b + c$

$\Rightarrow \quad 2s = a + b + c = (41 + 40 + 9)$

$\Rightarrow \quad 2s = 90 \Rightarrow s = 45$ m

$\because$ Area of triangular field

$= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{45(45-41)(45-40)(45-9)}$

$= \sqrt{45 \times 4 \times 5 \times 36}$

$= \sqrt{9 \times 5 \times 5 \times 4 \times 9 \times 4}$

$= 9 \times 5 \times 4 = 180 \text{ m}^2$

Given, area of each rose bed $= 900 \text{ cm}^2$

$= \dfrac{900}{100 \times 100} \text{ m}^2$

$\therefore$ Number of rose beds

$= \dfrac{\text{Area of triangular field}}{\text{Area of one rose bed}}$

$= \dfrac{180 \times 100 \times 100}{900} = 2000$

93. *(b)* 1 km = 1000 m 1 dm = 10 m

$\because$ 2 km and 26 dm

$= 2000 + 260 = 2260$ m

Let radius be r m.

Distance covered in revolution $= 2\pi r$

Distance covered in 452 revolution

$= 452 \times 2\pi r$

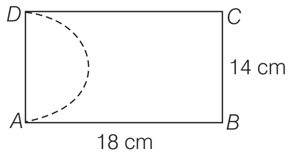
$\because \quad 452 \times 2\pi r = 2260$

$r = \dfrac{2260 \times 7}{452 \times 2 \times 22} = \dfrac{35}{44}$

Diameter $= 2 \times \dfrac{35}{44} = \dfrac{35}{22} = 1\dfrac{13}{22}$ m

94. *(d)* Area of remaining portion

= Area of rectangle – Area of semi-circle



$= 18 \times 14 - \dfrac{1}{2} \times \dfrac{22}{7} \times 7 \times 7$

$= 252 - 77 = 175 \text{ cm}^2$

95. *(d)* Let length of side of square and equilateral triangle be x and y, respectively.

Then, $\quad 4x = 3y \quad$ ...(i)

Given, diagonal of square $= 12\sqrt{2}$

$\Rightarrow \quad x\sqrt{2} = 12\sqrt{2}$

$\because \quad x = 12$ cm $\quad$ ...(ii)

$\therefore \quad y = \dfrac{4x}{3} = \dfrac{4}{3} \times 12$

$= 16$ cm

So, area of triangle $= \dfrac{\sqrt{3}}{4} \times 16 \times 16$

$= 64\sqrt{3} \text{ cm}^2$

96. *(d)* Let the side of square be x.

Area of square $= x^2$

Now, increased length

$= x\left(1 + \dfrac{30}{100}\right) = \dfrac{13x}{10}$

and increased breadth

$= x\left(1 + \dfrac{20}{100}\right) = \dfrac{12x}{10}$

Now, new area of rectangle

$= \dfrac{13x}{10} \times \dfrac{12x}{10} = \dfrac{156x^2}{100}$

Increased in area

$= \dfrac{156x^2}{100} - x^2 = \dfrac{56x^2}{100}$

Percentage increased in area

$= \dfrac{56x^2}{100x^2} \times 100$

$= 56\%$

Shortcut Method

Here, $x = 30\%$, $y = 20\%$

$\therefore$ Net effect $= \left[x + y + \dfrac{xy}{100}\right]\%$

$= \left[30 + 20 + \dfrac{30 \times 20}{100}\right] = 56\%$

Hence, area is increased by 56%.

97. *(a)* Area of trapezium

$= \dfrac{1}{2}$ (Sum of parallel side) $\times$ Distance between them

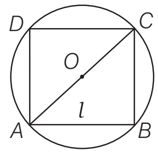
$\Rightarrow \quad \dfrac{1}{2}(5x + 7x) \times 14 = 336$

$\Rightarrow \quad 12x \times 7 = 336 \Rightarrow x = 4$

The length of smaller side

$= 4 \times 5 = 20$ cm

98. *(c)* We have, $2\pi r = 100$



$\therefore \quad 2r = \dfrac{100}{\pi}$ cm

Let the side of square be l cm.

$\therefore \quad l\sqrt{2} = \dfrac{100}{\pi} \Rightarrow l = \dfrac{50\sqrt{2}}{\pi}$ cm

99. *(b)* There are 437 fruits plants.

$\because \ 437 = 19 \times 23$

$\therefore$ There are 19 rows and 23 trees or there are 23 rows and 19 trees.

Given, the distance between any two adjacent plants is 2 m and the distance between any two adjacent rows is 2 m.

$\therefore$ Length of orchard

$= [1 + 22(2) + 1] = 46$ m

Breadth of orchard

$= [1 + 18(2) + 1] = 38$ m

$\therefore$ Perimeter of orchard

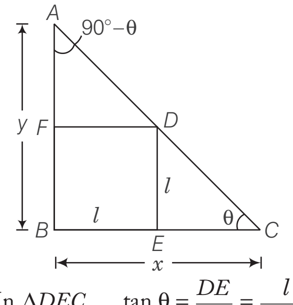
$= [(46 \times 2) + (38 \times 2)] = 168$ m

$\therefore$ Cost of fencing at the rate of

$=$ ₹ 100 per metre

$= 168 \times 100 =$ ₹ 16800

100. *(b)* Let length of the side of square be l.



In ΔDEC, $\quad \tan\theta = \dfrac{DE}{EC} = \dfrac{l}{x - l} \quad$ ...(i)

In ΔAFD, $\tan(90° - \theta) = \cot\theta = \dfrac{FD}{AF}$

$= \dfrac{l}{y - l} \quad$ ...(ii)

On multiplying Eqs. (i) and (ii), we get

$$1 = \frac{l^2}{(x-l)(y-l)}$$

$\Rightarrow \quad xy - xl - yl + l^2 = l^2$

$\because \quad l(x+y) = xy \quad \Rightarrow l = \dfrac{xy}{x+y}$

$\therefore$ Perimeter of square $= 4l = \dfrac{4xy}{x+y}$

101. *(c)* Let a be the side of regular polygon then, inradius, $r = \dfrac{a}{2\tan\left(\dfrac{\pi}{n}\right)}$

$\Rightarrow \quad a = 2r\tan\left(\dfrac{\pi}{n}\right)$

$\therefore$ Perimeter of polygon,

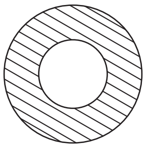
$$p = na = 2nr\tan\left(\frac{\pi}{n}\right)$$

$\Rightarrow \tan\left(\dfrac{\pi}{n}\right) = \dfrac{p}{2nr} \quad \ldots\text{(i)}$

Area of polygon $= nr^2\tan\left(\dfrac{\pi}{n}\right)$

$= nr^2 \times \dfrac{p}{2nr} = \dfrac{pr}{2} \quad$ [from Eq. (i)]

102. *(c)* Let the radius of smaller ring be r_1 and the radius of bigger ring be r.



Since, the smaller ring takes three full revolutions to roll over the circumference of bigger ring.

$\therefore \quad 2\pi r = 3(2\pi r_1) \Rightarrow r = 3r_1$

Area of pathway $= \pi r^2 - \pi r_1^2 = n(\pi r_1^2)$

$\Rightarrow \quad \pi r^2 = (n+1)\pi r_1^2$

$\Rightarrow \quad r^2 = (n+1)\dfrac{r^2}{9} \Rightarrow n = 9 - 1 = 8$

103. *(d)* Number of rounds that a wheel of diameter D will make in traversing a distance $= n(\pi D)$

Given that, distance = 4 km

$\Rightarrow \quad n(\pi D) = 4000\,\text{m}$

$\Rightarrow \quad n\left(\pi \times \dfrac{7}{11}\right) = 4000$

$\Rightarrow \quad n\left(\dfrac{22}{7} \times \dfrac{7}{11}\right) = 4000$

$\Rightarrow \quad n = 2000$

Hence, the number of rounds is 2000.

104. *(c)* Here, base = 300 units and side = 170 units

We know that,

Area of an isosceles triangle

$= \dfrac{\text{Base}}{4}\sqrt{4\,(\text{Side})^2 - (\text{Base})^2}$

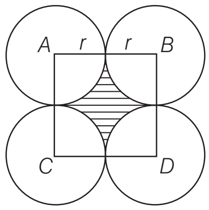
$= \dfrac{300}{4}\sqrt{4(170)^2 - (300)^2}$

$= 75\sqrt{4 \times 28900 - 90000}$

$= 75\sqrt{115600 - 90000} = 75\sqrt{25600}$

$= 75 \times 160 = 12000$ sq units

105. *(d)* Let the radius of the circle be r. Then, side of square $ABCD = 2r$



$\therefore$ Area of empty space = Area of square $- 4$ [Area of quadrant]

$\Rightarrow \quad \dfrac{150}{847} = (2r)^2 - 4\left[\dfrac{1}{4} \times \pi r^2\right]$

$\Rightarrow \quad \dfrac{150}{847} = \dfrac{6}{7}r^2$

$\Rightarrow \quad r^2 = \dfrac{25}{121} \qquad \left[\because \pi = \dfrac{22}{7}\right]$

$\Rightarrow \quad r = \dfrac{5}{11}$ cm

106. *(c)* Area of circle with diameter (AD)

$$= \frac{\pi(AD)^2}{4}$$

So, $\dfrac{\pi(AD)^2}{4} = 707$

$\Rightarrow \quad AD = \sqrt{\dfrac{707}{\pi} \times 4} = 2\sqrt{\dfrac{707}{\pi}}$ m

As, $AB = BC = CD = \dfrac{AD}{3} = \dfrac{2}{3}\sqrt{\dfrac{707}{\pi}}$ m

and $AC = 2AB = BD$

$= 2CD = \dfrac{4}{3}\sqrt{\dfrac{707}{\pi}}$ m

$\therefore$ Area of shaded portion = 2 [Area of semi-circle with AC as diameter − Area of semi-circle with AB as diameter]

$= \dfrac{2\pi(AC)^2}{8} - \dfrac{2\pi(AB)^2}{8}$

$= \dfrac{2\pi}{8}\left[\dfrac{16}{9} \cdot \dfrac{707}{\pi} - \dfrac{4}{9} \cdot \dfrac{707}{\pi}\right] = \dfrac{1}{3} \times 707\,\text{m}^2$

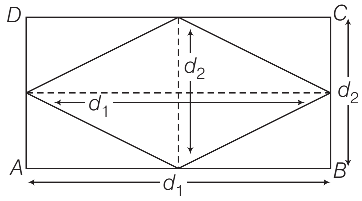
$\therefore$ Cost of levelling the shaded portion

$$= \frac{707}{3} \times 63 = ₹\,14847$$

107. *(b)* We know that, Area of rhombus

$= \dfrac{1}{2} \times$ product of diagonals

$= \dfrac{1}{2} \times d_1 \times d_2$



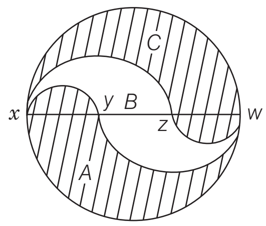
From the above figure,

Area of rectangle $= d_1 \times d_2$

$= 2$ (Area of rhombus)

$= 2 \times 2 = 4$ sq units

108. *(b)* In the given question, A radii of 3 m is divided in such a way that the radii of smaller semi-circle is 1 m and radii of bigger semi-circle is 2 m.



Area of shaded portion A

= Area of semi–circle of radii 3 m – Area of semi–circle of radii 2 m + Area of semi- circle of radii 1 m

$= \dfrac{1}{2}\pi(3)^2 - \dfrac{1}{2}\pi(2)^2 + \dfrac{1}{2}\pi(1)^2$

$= \dfrac{1}{2}\pi(9 - 4 + 1) = 3\pi$

Area of portion B

= 2 [Area of semi-circle of radii 2 m − Area of semi-circle of radii 1 m]

$= 2\left[\dfrac{1}{2}\pi(2)^2 - \dfrac{1}{2}\pi(1)^2\right] = 3\pi$

Similarly, area of shaded portion C

= Area of portion A

Hence, the ratio of areas A, B and C is 1 : 1 : 1.

27

SURFACE AREA AND VOLUME OF SOLIDS

Regularly (10-12) questions have been asked from this chapter. Questions from this chapter are direct formula based, so one must be through with all the formulae and their application. It is easy to score section in the exam.

In this chapter, we will study how to find the surface area and volume of solid figures, like parallelopiped, cube, cuboid, cylinder, cone, frustum of cone, sphere and hemisphere and the combination of solid figures.

SOLID FIGURES

The objects which occupy space (i.e. they have three dimensions) are called solid figures. The solid figures can be derived from the plane figures.

A solid figure has surface area and volume.

Surface Area of Solid Figure

Surface area of a solid body is the area of all of its surfaces together. It is always measured in square units. Surface area is also referred to as the total surface area. The area of four walls or the area excluding top and bottom is called curved surface area or lateral surface area.

Volume of Solid Figure

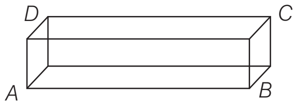
The measure of space occupied by a solid is called its volume. It is always measured in cubic units.

Cuboid

A figure which is surrounded by six rectangular surfaces, is called cuboid. It is also called rectangular parallelopiped.

- A cuboid has 8 corners, 12 edges, 6 faces and 4 diagonals.

- Volume of cuboid = Area of base × height = $l \times b \times h$
  where, l = length, b = breadth and h = height



- Total surface area of cuboid = $2(lb + bh + lh)$
- Diagonal of the cuboid = $\sqrt{l^2 + b^2 + h^2}$
- Lateral surface area or Area of four walls = $2(l + b)\,h$.

EXAMPLE 1. The volume of a cuboid is 880 cm^3, the area of its base is 88 cm^2. Then, its height is

a. 10 cm **b.** 12 cm **c.** 14 cm **d.** 16 cm

Sol. **a.** Here, volume of a cuboid $= 880\,cm^3$

Area of the base $= 88\,cm^2$

$$\text{Height} = \frac{\text{Volume of the cuboid}}{\text{Area of the base}} = \frac{880}{88} = 10\text{ cm}$$

EXAMPLE 2. The length of the longest rod that can be placed in a room 12 m long, 9 m broad and 8 m high is

a. 15 m **b.** 16 m
c. 17 m **d.** 17.5 m

Sol. **c.** Since, the length of the longest rod is the diagonal of the room

$$= \sqrt{l^2 + b^2 + h^2} = \sqrt{(12)^2 + (9)^2 + (8)^2}$$

$$= \sqrt{144 + 81 + 64} = \sqrt{289} = 17\text{ m}$$

EXAMPLE 3. The areas of three adjacent faces of a cuboid are x, y and z. If its volume is V, then which of the following is true?

a. $V = x^3 y^2 z^2$ **b.** $V^2 = xyz$
c. $V = \sqrt[3]{xyz}$ **d.** $V = \dfrac{x^2 y}{z}$

Sol. **b.** Let the dimensions of a cuboid be l, b and h respectively, then volume of cuboid, $V = lbh$

Also, $x = lb$, $y = bh$ and $z = hl$

$\therefore$ $V = lbh$

On squaring both sides, we get

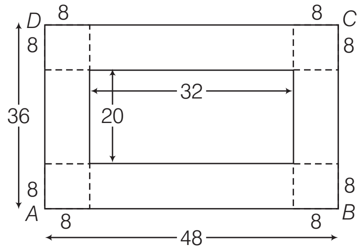
$$V^2 = l^2 b^2 h^2 = (lb)\cdot(bh)\cdot(hl) = xyz$$

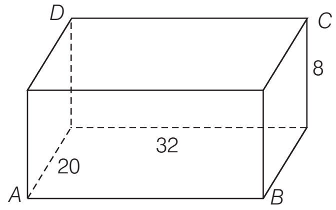
EXAMPLE 4. A metallic sheet is of rectangular shape with dimensions $48\,\text{cm} \times 36\,\text{cm}$ from each of its corners a square of 8 cm is cut-off. An open box is made of the remaining sheet, what is the volume of the box?

a. 13824 cm^3 **b.** 1728 cm^3
c. 5120 cm^3 **d.** None of these

Sol. **c.** Given length of metallic sheet = 48 cm

Breadth of metallic sheet = 36 cm





When square of side 8 cm is cut-off from each corner and the flaps turned up, we get an open box whose

Length $= 48 - (8 + 8) = 32$ cm

Breadth $= 36 - (8 + 8) = 20$ cm

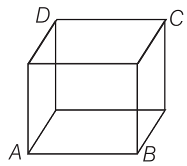
and height $= 8$ cm

$\therefore$ Volume of the box $= l \times b \times h = 32 \times 20 \times 8 = 5120\text{ cm}^3$

Cube

A cuboid whose length, breadth and height are same is called a cube.

- A cube has 6 surfaces, 12 edges, 8 corners and 4 diagonals.
- Cube is a special case of cuboid which has 6 square faces.
- Volume of a cube = $(side)^3$
- Total surface area of a cube $= 6 \times (side)^2$



- Diagonal of a cube $= \sqrt{3} \times$ edge
- Lateral surface area or area of four walls = $4 \times (\text{side})^2$

EXAMPLE 5. If the surface area of a cube is 96 cm^2, then, its volume is

a. 16 cm^3 **b.** 64 cm^3
c. 12 cm^3 **d.** 32 cm^3

Sol. **b.** Surface area of cube $= 6 \times a^2$, where a = Side of a cube

$$\Rightarrow \quad 6a^2 = 96 \;\Rightarrow\; a^2 = 16 \;\Rightarrow\; a = 4\text{ cm}$$

$\therefore$ Volume of the cube $= a^3 = 4^3 = 64\text{ cm}^3$

EXAMPLE 6. The surface area and the length of the diagonal of the cube, if the volume of a cube is 2197 cm^3, are

a. 1012 cm^2 and 21.516 cm **b.** 1024 cm^2 and 24.516 cm
c. 1014 cm^2 and 22.516 cm **d.** None of these

Sol. c. Volume of cube = $(\text{side})^3 = 2197 \text{ cm}^3$

$\Rightarrow$ Side of cube $= \sqrt[3]{\text{volume of a cube}} = 13 \text{ cm}$

$\Rightarrow$ Surface area of cube

$= 6\,(\text{side})^2 = 6\,(13)^2 = 6 \times 169 = 1014 \text{ cm}^2$

and length of the diagonal of the cube $= \sqrt{3} \times \text{side}$

$= \sqrt{3} \times 13 = 1.732 \times 13 = 22.516$ cm

EXAMPLE 7. How many 6 m cubes can be cut from a cuboid measuring $36 \text{ m} \times 15 \text{ m} \times 8 \text{ m}$?

a. 10 b. 15
c. 19 d. 20

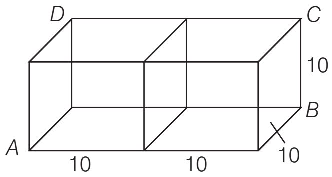
Sol. d. Volume of given cuboid $= (36 \times 15 \times 8) \text{ m}^3$

Volume of the cube to be cut $= (6 \times 6 \times 6) \text{ m}^3$

$\therefore$ Number of cubes that can be cut from the cuboid

$$= \frac{\text{Volume of the cuboid}}{\text{Volume of the cube}} = \frac{36 \times 15 \times 8}{6 \times 6 \times 6} = 20$$

EXAMPLE 8. Two cubes each of 10 cm edge are joined end-to-end. Then, the surface area of the resulting cuboid is



a. 100 cm^2 b. 1000 cm^2
c. 2000 cm^2 d. None of these

Sol. b. Length of the resulting cuboid = 10 + 10 = 20 cm

Breadth of resulting cuboid = 10 cm

Height of resulting cuboid = 10 cm

$\therefore$ Surface area of resulting cuboid $= 2\,(lb + bh + hl)$

$= 2[20 \times 10 + 10 \times 10 + 20 \times 10]$

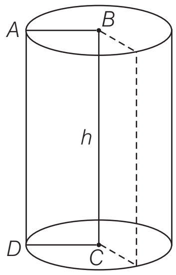
$= 2[200 + 100 + 200] = 2 \times 500 = 1000 \text{ cm}^2$

IMPORTANT FORMULAE

- Volume of prism = Area of base × height
- Lateral surface area of prism = Perimeter of base × height
- Volume of pyramid = 1/3 × Area of base × height
- Lateral surface area = 1/2 × Perimeter of base × slant height

Right Circular Cylinder

A right circular cylinder is a solid figure obtained by revolving the rectangle, say *ABCD* about its one side, say *BC*. Let base radius of right circular cylinder be r and its height be h. Then,



(i) Volume of the cylinder

= *(Area of base)* × *Height*

$= \pi r^2 h$ *cu units*

(ii) Curved surface area = Circumference of the base × Height

$= 2\pi rh$ *sq units*

(iii) Total surface area = Curved surface + Area of two ends

$= 2\pi rh + 2\pi r^2 = 2\pi r\,(h + r)$ sq units

EXAMPLE 9. The volume of a cylinder is $448\,\pi \text{ cm}^3$ and height 7 cm. Then, its lateral surface area and total surface area are

a. 349 cm^2 and 753.286 cm^2
b. 352 cm^2 and 754.286 cm^2
c. 353 cm^2 and 755.286 cm^2
d. None of the above

Sol. b. Given, volume of the cylinder $= 448\,\pi \text{ cm}^3$

and height of the cylinder = 6 cm

Let radius be r, then $\pi r^2 h = 448\,\pi$

$\therefore$ $r^2 = \dfrac{448\,\pi}{h\,\pi} = \dfrac{448}{7} = 64 \Rightarrow r = 8$ cm

$\therefore$ Lateral surface area of the cylinder $= 2\pi rh$

$= 2 \times \dfrac{22}{7} \times 8 \times 7 = 352 \text{ cm}^2$

Total surface area of the cylinder $= 2\pi\, r\,(h + r)$

$= 2 \times \dfrac{22}{7} \times 8\,(7 + 8)$

$= 2 \times \dfrac{22}{7} \times 8 \times 15 = \dfrac{5280}{7} \approx 754.286 \text{ cm}^2$

EXAMPLE 10. The radii of two cylinders are in the ratio 2 : 3 and their heights are in the ratio 5 : 3. Then, the ratio of their volumes is

a. 4 : 9 b. 16 : 25 c. 20 : 27 d. None of these

Sol. c. For first cylinder,

Let radius $= r_1$, height $= h_1$, volume $= V_1$

For second cylinder,

Let radius $= r_2$, height $= h_2$ and volume $= V_2$

Then, $\dfrac{r_1}{r_2} = \dfrac{2}{3}$ and $\dfrac{h_1}{h_2} = \dfrac{5}{3} \Rightarrow r_1 = \dfrac{2r_2}{3}$ and $h_1 = \dfrac{5h_2}{3}$

Required ratio of their volumes be $V_1 : V_2$.

$\Rightarrow$ $\pi r_1^2\, h_1 : \pi\, r_2^2\, h_2 \Rightarrow r_1^2 h_1 : r_2^2\, h_2$

$\Rightarrow$ $\dfrac{4}{9}\, r_2^2\, \dfrac{5h_2}{3} : r_2^2\, h_2 \Rightarrow \dfrac{20}{27} : 1 \Rightarrow 20 : 27$

EXAMPLE 11. A cylindrical bucket of diameter 28 cm and height 12 cm is full of water. The water is emptied into a rectangular tub of length 66 cm and breadth 28 cm. The height to which water rises in the tub is

a. 2 cm b. 4 cm c. 6 cm d. 8 cm

Sol. b. Volume of water in the bucket

$$= \pi r^2 h = \frac{22}{7} \times 14 \times 14 \times 12 = 7392 \text{ cm}^3$$

Let h be the height to which water rises in the tub.

$\therefore$ Volume of water in the tub $= 66 \times 28 \times h$ cm^3

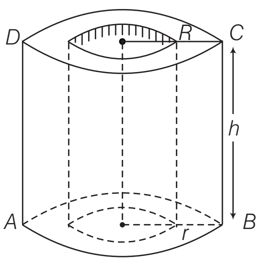
According to the question,

$$66 \times 28 \times h = 7392 \Rightarrow h = \frac{7392}{66 \times 28} = 4 \text{ cm}$$

Hence, the height to which water rises in the tub is 4 cm.

Hollow Cylinder

A hollow cylinder is one which is empty from inside and has some difference between the internal and external radius.



Let R and r be the external and internal radii of the hollow cylinder and h be its height. Then,

(i) Volume of hollow cylinder $= \pi (R^2 - r^2) h$ cu units

(ii) Curved surface area
$= 2\pi Rh + 2\pi rh = 2\pi (R + r) h$ sq units

(iii) Total surface area
$= 2\pi rh + 2\pi Rh + 2\pi (R^2 - r^2)$ sq units
$= 2\pi (R + r)(h + R - r)$ sq units

EXAMPLE 12. A hollow cylindrical tube, open at both ends is made of iron 1 cm thick. The volume of iron used in making the tube, if the external diameter is 12 cm and the length of tube is 70 cm, is

a. 2420 cm^3 b. 2520 cm^3
c. 2720 cm^3 d. 2900 cm^3

Sol. a. Here, external radius $(R_1) = 6$ cm

and internal radius $(R_2) = 6 - 1 = 5$ cm

Height of hollow cylinder $(h) = 70$ cm

$\therefore$ Volume of iron used in making tube

= External volume – Internal volume

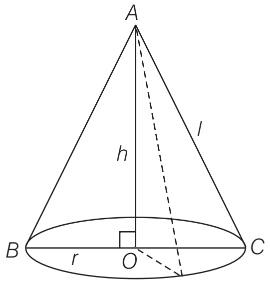
$= \pi R^2 h - \pi r^2 h = \pi h (R^2 - r^2)$

$= \frac{22}{7} \times 70 \times (36 - 25)$

$= 220 \times 11 = 2420$ cm^3

Right Circular Cone

A right circular cone is a solid, generated by the revolution of a right angled triangle about one of its sides containing the right angle. Let height of a right circular cone be h, slant height be l and its radius be r, Then,



(i) The slant height of the cone,
$$l = AC = \sqrt{r^2 + h^2} \text{ units}$$

(ii) Volume of cone $= \frac{1}{3} \pi r^2 h$ cu units

(iii) Curved surface area of cone $= \pi rl$ sq units

(iv) Total surface area of a cone = Curved surface area + Base area $= \pi r (l + r)$ sq units

EXAMPLE 13. How many metres of cloth 5 m wide will be required to make a conical tent, the radius of whose base is 7 m and whose height is 24 m?

a. 100 m b. 105 m c. 109 m d. 110 m

Sol. d. Given radius of base $(r) = 7$ m

and vertical height of tent $(h) = 24$ m

Slant height of the tent (l)

$$= \sqrt{r^2 + h^2} = \sqrt{24^2 + 7^2} = \sqrt{625} = 25 \text{ m}$$

Now, curved surface area $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550$ m^2

$\because$ Width of cloth = 5 m

$\therefore$ Length required to make conical tent

$$= \frac{550}{5} = 110 \text{ m}$$

EXAMPLE 14. The radius and the height of a right circular cone are in the ratio $5:12$. If its volume is 314 m^3, the slant height and the radius are

a. 12 m, 5 m b. 13 m, 4 m c. 1 m, 4 m d. 13 m, 5 m

Sol. d. Let radius of cone be $5x$ and the height of the cone be $12x$.

$\therefore$ Volume of cone $= 314 = \frac{1}{3} \pi r^2 h$

$\Rightarrow \frac{1}{3} \times 3.14 \times (5x)^2 \times (12x) = 314 \Rightarrow 314x^3 = 314$

$\Rightarrow x^3 = 1 \Rightarrow x = 1$

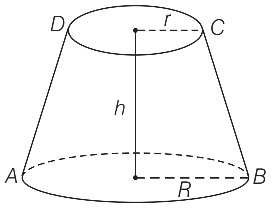
$\because$ Radius = 5 m and height = 12 m

$\therefore$ Slant height $= \sqrt{r^2 + h^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ m

Hence, the slant height and the radius are 5 m and 13 m.

Frustum of a Cone

If a right circular cone is cut-off by a plane parallel to the base of the cone, then the portion of the cone between the cutting plane and base of the cone is called the frustum of the cone. Let R, r be the radii of base and top $(R > r)$ of the frustum of a cone. Let height of frustum of a cone be h and slant height be l, then



(i) Slant height of frustum of right circular cone
$= BC = \sqrt{h^2 + (R-r)^2}$ units

(ii) Volume of frustum of right circular cone
$= \frac{\pi h}{3}[R^2 + r^2 + Rr]$ cu *units*

(iii) Curved (lateral) surface area of frustum of right circular cone $= \pi l (R + r)$ sq units

(iv) Total surface area of frustum of right circular cone
= Area of base + Area of top + Lateral surface area
$= \pi[R^2 + r^2 + l(R + r)]$ sq units

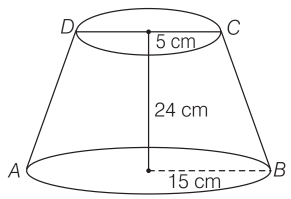
(v) Total surface area of bucket $= \pi[(R + r)l + r^2]$ sq unit
[$\because$ it is open at the bigger end.]

EXAMPLE 15. The radii of the ends of a bucket of height 24 cm are 15 cm and 5 cm. Then, its capacity is

a. 8000 cm^3 b. 8100 cm^3
c. 8171.43 cm^3 d. 8200.43 cm^3

Sol. c. Given, $h = 24$ cm, $R = 15$ cm and $r = 5$ cm

Capacity of bucket = Volume of frustum of a cone



$= \frac{\pi h}{3}[R^2 + r^2 + Rr]$

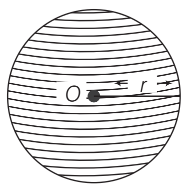
$= \frac{22}{7} \times \frac{24}{3}[(15)^2 + 5^2 + 15 \times 5]$

$= \frac{22}{7} \times 8[225 + 25 + 75]$

$= \frac{176}{7}(325) = 8171.43\ cm^3$

Sphere

The set of all points in space, which are equidistant from a fixed point, is called a sphere. Let radius of sphere be r, then



(i) Volume of sphere $= \frac{4}{3}\pi r^3$ cu units

(ii) Surface area of sphere $= 4\pi r^2$ sq units

(iii) Volume of a hollow sphere $= \frac{4}{3}\pi(R^3 - r^3)$ cu units
where, r = Inner radius and R = Outer radius

EXAMPLE 16. Given that the volume of a metal sphere is 38808 cm^3. Then, its radius and its surface area are

a. 7 cm and 616 cm^2 b. 21 cm and 5544 cm^2
c. 14 cm and 2464 cm^2 d. None of these

Sol. b. Given, volume of the metal sphere = 38808 cm^3

But volume of sphere $= \frac{4}{3}\pi r^3$

$\therefore \quad \frac{4}{3}\pi r^3 = 38808 \quad \Rightarrow \quad r^3 = 38808 \times \frac{3}{4} \times \frac{7}{22} = 9261$

$\Rightarrow \quad r = (9261)^{1/3} = 21$ cm

$\therefore$ Surface area of the sphere $= 4\pi r^2$

$= 4 \times \frac{22}{7} \times 21 \times 21$

$= 5544\ cm^2$

EXAMPLE 17. How many balls each of radius 2 cm can be made by melting a big ball whose radius is 8 cm.

a. 4 balls b. 16 balls c. 64 balls d. 128 balls

Sol. c. Given radius of big ball $(R) = 8$ cm

$\therefore$ Volume of big ball $= \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(8)^3$

Radius of small ball $(r) = 2$ cm

$\therefore$ Volume of small balls $= \frac{4}{3}\pi(2)^3$

$\therefore$ Required number of balls $= \frac{\text{Volume of big ball}}{\text{Volume of small ball}}$

$= \frac{\frac{4}{3}\pi(8)^3}{\frac{4}{3}\pi(2)^3} = \frac{8 \times 8 \times 8}{2 \times 2 \times 2} = 64$ balls

EXAMPLE 18. A copper sphere of diameter 18 cm is drawn into a wire of diameter 4 mm. Then, the length of the wire is

a. 243 m b. 343 m c. 443 m d. 972 m

Sol. a. Let the length of wire be h cm.

Volume of sphere = Volume of wire [by condition]

$\Rightarrow \quad \left(\frac{4}{3}\pi \times 9 \times 9 \times 9\right) = \left(\pi \times \frac{2}{10} \times \frac{2}{10} \times h\right)$

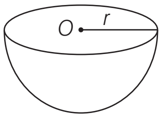
$\Rightarrow \quad \frac{h}{25} = 972 \quad \Rightarrow \quad h = (972 \times 25)$ cm

$= \frac{972 \times 25}{100} = 243$ m

Hence, the length of the wire is 243 m.

Hemisphere

A plane passing through the centre cuts the sphere in two equal parts, each part is called a hemisphere.



Let the radius of hemisphere be r, then

(i) Volume of hemisphere $= \frac{2}{3}\pi r^3$ cu units

(ii) Curved surface area of hemisphere $= 2\pi r^2$ sq units

(iii) Total surface area $= 2\pi r^2 + \pi r^2 = 3\pi r^2$ sq units

(iv) Volume of hemispherical shell $= \frac{2}{3}\pi(R^3 - r^3)$

where, R = outer radius, r = inner radius

EXAMPLE 19. The volumes of two hemispheres are in the ratio 8 : 27. What is the ratio of their radii?

a. 2 : 3 b. 3 : 2 c. 1 : 2 d. 2 : 1

Sol. a. Let volumes be V_1 and V_2.

$\therefore \quad V_1 : V_2 = 8 : 27 \Rightarrow \frac{2}{3}\pi r_1^3 : \frac{2}{3}\pi r_2^3 = 8 : 27$

$\Rightarrow \quad r_1^3 : r_2^3 = 8 : 27 \quad \Rightarrow \quad r_1 : r_2 = 2 : 3$

IMPORTANT FORMULAE

- If the side of a cube is increased by x%, then its volume is increased by $\left[\left(1 + \frac{x}{100}\right)^3 - 1\right] \times 100\%$.
- If the length, breadth and height of cuboid are made x, y and z times respectively, then its volume is increased by $(xyz - 1) \times 100\%$.
- If the length, breadth and height of a cuboid are increased by x%, y% and z% respectively, then its volume is increased by $\left[x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{(100)^2}\right]\%$.
- If the sides and diagonal of a cuboid are given, then the total surface area in terms of diagonal and sides is given by
  Total surface area = *(Sum of the sides)*^{2} − *(Diagonal)*^{2}.
- If the side of a cube is increased by x%, then the surface area is increased by $\left(2x + \frac{x^2}{100}\right)\%$.

PRACTICE EXERCISE

1. Three equal cubes are placed adjacently in a row. The ratio of total surface area of the new cuboid to that of the sum of the surface areas of the three cubes is

(a) 3 : 1 (b) 6 : 5 (c) 7 : 9 (d) 6 : 7

2. If side of a cube is increased by 12%, by how much percent does its volume increase?

(a) 35.2% (b) 40.5% (c) 45.0% (d) 42.4%

3. The curved surface area of a cylinder is 1320 cm^2 and its base has diameter 21 cm, then the height of the cylinder is

(a) 10 cm (b) 20 cm (c) 22 cm (d) 25 cm

4. A cylindrical vessel can hold 154 g of water. If the radius of its base is 3.5 cm and 1 cm^3 of water weights 1 g. The depth of the water is

(a) 2 cm (b) 3 cm
(c) 4 cm (d) 5 cm

5. The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3. The diameter of the pillar is

(a) 3 m (b) 6 m
(c) 7 m (d) 14 m

6. How many metres of cloth 50 m wide will be required to make a conical tent, the radius of whose base is 7 m and whose height is 24 m?

(a) 9 m (b) 11 m (c) 12 m (d) 13 m

7. The radius and height of a right circular cone are in the ratio of 5 : 12 and its volume is 2512 cm^3. The slant height of the cone is

(a) 24 cm (b) 25 cm (c) 26 cm (d) 27 cm

8. If the height of a cone is doubled, then its volume is increased by

(a) 100% (b) 200% (c) 300% (d) 400%

9. If the surface areas of two spheres are in the ratio of 4 : 25, then the ratio of their volumes is

(a) 2 : 25 (b) 4 : 75 (c) 8 : 125 (d) 16 : 125

10. A cone and a cylinder are of the same height. Their radii of the bases are in ratio of 2 : 1. The ratio of their volumes is

(a) 2 : 1 (b) 3 : 2 (c) 4 : 3 (d) 1 : 3

11. If the height and diameter of a right circular cylinder are 32 cm and 6 cm respectively, then the radius of the sphere whose volume is equal to the volume of the cylinder is

(a) 3 cm (b) 4 cm (c) 6 cm (d) 8 cm

12. Two solid spheres of gold having diameters 3 cm and 4 cm are molten and then cast into one big sphere of gold. If the radius of this sphere is x, then what is the value of x^3?
(a) 125 cm^3 (b) 15.625 cm^3
(c) 11.375 cm^3 (d) 9.875 cm^3

13. The total surface area of a cone, whose slant height is equal to the radius R of its base, is S. If A is the area of a circle of radius $2R$, then which one of the following is correct?
(a) $A = S$ (b) $A = 2S$ (c) $A = S/2$ (d) $A = 4S$

14. A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much distance will the cylinder roll in 5 s?
(a) 15 m (b) 1.5 m (c) 30 m (d) 3 m

15. 27 drops of water form a big drop of water. If the radius of each smaller drop is 0.2 cm, then what is the radius of the bigger drop?
(a) 0.4 cm (b) 0.6 cm (c) 0.8 cm (d) 1.0 cm

16. A circus tent is made of canvas and is in the form of a right circular cylinder and a right circular cone above it, the diameter and height of the cylindrical part of the tent are 126 m and 5 m, respectively. The total height of the tent is 21 m. Then, the cost of the canvas used for tent at the rate of ₹ 12 per m^2
(a) ₹ 14850 (b) ₹ 168200 (c) ₹ 178200 (d) ₹ 112000

17. A solid sphere of radius 6 cm is melted into a hollow cylinder of uniform thickness. If the external radius of the base of the cylinder is 5 cm and its height is 32 cm. The uniform thickness of the cylinder is
(a) 1.5 cm (b) 3 cm (c) 1.2 cm (d) 1 cm

18. Given a solid cylinder of radius 10 cm and length 1000 cm a cylindrical hole is made into it to obtain a cylindrical shell of uniform thickness and having volume equal to one-fourth of the original volume of the original cylinder. The thickness of the cylindrical shell is
(a) $5(\sqrt{5} - 2)$ cm (b) $7(2 - \sqrt{3})$ cm
(c) 10 cm (d) $5\sqrt{2}$ cm

19. A tent is of the shape of right circular cylinder upto a height of 3 m and then becomes a right circular cone with a maximum height of 13.5 m above the ground. The cost of painting the inner side of the tent at the rate of ₹ 2 per m^2, if the radius of the base is 14 m is
(a) ₹ 2048 (b) ₹ 2068 (c) ₹ 2008 (d) ₹ 2088

20. A container is in the form of a right circular cylinder surmounted by a hemisphere of the same radius 15 cm as the cylinder. If the volume of the container is $32400\,\pi\,cm^3$, then the height h of the container satisfies which one of the following ?
(a) 135 cm < h < 150 cm (b) 140 cm < h < 147 cm
(c) 145 cm < h < 148 cm (d) 139 cm < h < 145 cm

21. A conical flask of base radius r and height h is full of milk. The milk is now poured into a cylindrical flask of radius $2r$. What is the height to which the milk will rise in the flask?
(a) $\frac{h}{3}$ (b) $\frac{h}{6}$ (c) $\frac{h}{9}$ (d) $\frac{h}{12}$

22. The radius and height of a right circular cone are in the ratio 3 : 4 and its volume is $96\,\pi\,cm^3$. What is the lateral surface area?
(a) $24\,\pi\,cm^2$ (b) $36\,\pi\,cm^2$ (c) $48\,\pi\,cm^2$ (d) $60\,\pi\,cm^2$

23. What is the number of wax balls, each of radius 1 cm, that can be molded out of a sphere of radius 8 cm?
(a) 256 (b) 512 (c) 768 (d) 1024

24. 10 cylindrical pillars of a building have to be painted. The diameter of each pillar is 70 cm and the height is 4 m. What is the cost of painting at the rate of ₹ 5 per m^2 ?
(a) ₹ 400 (b) ₹ 440 (c) ₹ 480 (d) ₹ 500

25. The radii of the circular ends of a bucket of height 40 cm are of lengths 35 cm and 14 cm. What is the volume of the bucket?
(a) 60060 cm^3 (b) 70040 cm^3
(c) 80080 cm^3 (d) 80160 cm^3

26. If S is the total surface area of a cube and V is its volume, then which one of the following is correct?
(a) $V^3 = 216\,S^2$ (b) $S^3 = 216\,V^2$
(c) $S^3 = 6\,V^2$ (d) $S^2 = 36\,V^3$

27. A cylindrical tank 7 m in diameter, contains water to a depth of 4 m. What is the total area of wetted surface?
(a) 110.5 m^2 (b) 126.5 m^2
(c) 131.5 m^2 (d) 136.5 m^2

28. The radii of two cylinders are in the ratio 2 : 3 and their curved surface areas are in the ratio 5 : 3. What is the ratio of their volumes?
(a) 20 : 27 (b) 10 : 9 (c) 9 : 10 (d) 27 : 20

29. A cylindrical vessel of height 10 cm has base radius 60 cm. If d is the diameter of a spherical vessel of equal volume, then what is the value of d ?
(a) 30 cm (b) 60 cm (c) 90 cm (d) 120 cm

30. A hollow sphere of internal and external diameters 4 cm and 8 cm, respectively is melted into a cone of base diameter 8 cm. The height of the cone is
(a) 11 cm (b) 12 cm (c) 14 cm (d) 16 cm

31. If the diameter of a sphere is doubled, how does its surface area change?
(a) It increases two times (b) It increases three times
(c) It increases four times (d) It increases eight times

32. A cistern 6 m long and 4 m wide contains water to a depth of 1.25 m. What is the area of wetted surface?
(a) $40\ m^2$ (b) $45\ m^2$ (c) $49\ m^2$ (d) $73\ m^2$

33. The outer and inner diameters of a circular pipe are 6 cm and 4 cm, respectively. If its length is 10 cm, then what is the total surface area in cm^2?
(a) $35\ \pi$ (b) $110\ \pi$
(c) $150\ \pi$ (d) None of these

34. A toy is in the form of a cone mounted on a hemisphere such that the diameter of the base of the cone is equal to that of the hemisphere. If the diameter of the base of the cone is 6 cm and its height is 4 cm, what is the surface area in cm^2 of the toy ? (Take $\pi = 3.14$)
(a) 93.62 (b) 103.62 (c) 113.62 (d) 115.50

35. A solid cylinder of height 9 m has its curved surface area equal to one-third of the total surface area. What is the radius of the base?
(a) 9 m (b) 18 m (c) 27 m (d) 30 m

36. The volume of a sphere is 8 times that of another sphere. What is the ratio of their surface areas?
(a) 8 : 1 (b) 4 : 1 (c) 2 : 1 (d) 4 : 3

37. The dimensions of a field are $12\ m \times 10\ m$. A pit 5 m long, 4 m wide and 2 m deep is dug in one corner of the field and the Earth removed has been evenly spread over the remaining area of the field. The level of the field is raised by
(a) 30 cm (b) 35 cm (c) 38 cm (d) 40 cm

38. A cube of 9 cm edge is immersed completely in a rectangular vessel containing water. If the dimensions of base are 15 cm and 12 cm. Then, the rise in water level in the vessel is
(a) 4.05 cm (b) 4 cm (c) 3.5 cm (d) 3 cm

39. From a solid cube of edge 3 m, a solid of largest sphere is curved out. What is the volume of solid left?
(a) $(27 - 2.25\pi)\ m^3$ (b) $(27 - 4.5\pi)\ m^3$
(c) $2.25\pi\ m^3$ (d) $4.5\pi\ m^3$

40. A rectangular tank is $80 \times 40\ cm^3$. Water flows into it through a pipe of cross-section area $40\ cm^2$ at the speed of 10 km/h. The rise in the level of water in the tank in $\frac{1}{2}$ h is
(a) $\frac{3}{2}$ cm (b) $\frac{4}{3}$ cm (c) $\frac{5}{8}$ cm (d) 6 cm

41. A measuring jar of internal diameter 10 cm is partially filled with water. Four equal spherical balls of diameter 2 cm each are dropped in it and they sink down in the water completely. The change in the level of water in the jar is
(a) $\frac{16}{65}$ cm (b) $\frac{15}{16}$ cm (c) $\frac{16}{75}$ cm (d) None of these

42. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$ of the volume of the given cone, then the height above the base where the section is made, is
(a) 12 cm (b) 15 cm (c) 20 cm (d) 22 cm

43. From a wooden cylindrical block, whose diameter is equal to its height, a sphere of maximum possible volume is carved out. What is the ratio of the volume of the utilised wood to that of the wasted wood?
(a) 2 : 1 (b) 1 : 2 (c) 2 : 3 (d) 3 : 2

44. The base diameter of a right circular cylinder is 3 cm. There is a section making an angle of 30° with the cross section. What is its area?
(a) $\frac{9\pi}{4}\ cm^2$ (b) $\frac{3\sqrt{3}\pi}{2}\ cm^2$ (c) $\frac{9\pi}{8}\ cm^2$ (d) $\frac{9\sqrt{3}\pi}{8}\ cm^2$

45. A cone is inscribed in a hemisphere such that their bases are common. If C is the volume of the cone and H that of the hemisphere, then what is the value of $C : H$?
(a) 1 : 2 (b) 2 : 3 (c) 3 : 4 (d) 4 : 5

46. If the diameter of a wire is decreased by 10%, by how much per cent (approximately) will the length be increased to keep the volume constant?
(a) 5% (b) 17% (c) 20% (d) 23%

47. The diameter of a solid metallic right circular cylinder is equal to its height. After cutting out the largest possible solid sphere S from this cylinder, the remaining material is recast to form a solid sphere S_1. What is the ratio of the radius of sphere S to that of sphere S_1?
(a) $1 : 2^{\frac{1}{3}}$ (b) $2^{\frac{1}{3}} : 1$ (c) $2^{\frac{1}{3}} : 3^{\frac{1}{3}}$ (d) $3^{\frac{1}{2}} : 2^{\frac{1}{2}}$

48. A square has its side equal to the radius of a sphere. The square revolves round a side to generate a surface of total area S. If A is the surface area of the sphere, then which one of the following is correct?
(a) $A = 3S$ (b) $A = 2S$ (c) $A = S$ (d) $A < S$

49. A swimming pool is 24 m long and 15 m broad. When x number of men dive into the pool, the height of the water rises by 1 cm. If the average amount of water displaced by one man is $0.1\ m^3$, then what is the value of x?
(a) 36 (b) 72 (c) 108 (d) 360

50. Water is distributed to a town of 50000 inhabitants from a rectangular reservoir consisting of 3 equal compartments. Each compartment has length and breadth 200 m, 100 m, respectively and 12 m depth of water in the beginning. The allowance is 20 L per head per day. For how many days will the supply of water hold out?

(a) 240 days (b) 720 days (c) 800 days (d) 900 days

51. A right circular cylinder and a right circular cone have equal bases and equal volumes. But the lateral surface area of the right circular cone is 15/8 times the lateral surface area of the right circular cylinder. What is the ratio of radius to height of the cylinder?

(a) 3 : 4 (b) 9 : 4 (c) 15 : 8 (d) 8 : 15

52. The volume of a cuboid whose sides are in the ratio of 1 : 2 : 4 is same as that of a cube. What is the ratio of length of diagonal of cuboid to that of cube?

(a) $\sqrt{1.25}$ (b) $\sqrt{1.75}$ (c) $\sqrt{2}$ (d) $\sqrt{3.5}$

53. A field is 125 m long and 15 m wide. A tank $10\text{ m} \times 7.5\text{ m} \times 6\text{ m}$ was dug in it and the Earth, thus dug out was spread equally on the remaining field. The level of the field thus raised is equal to which one of the following?

(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm

54. If C_1 is a right circular cone with base radius r_1 cm and height h_1 cm and C_2 is a right circular cylinder with base radius r_2 cm and height h_2 cm and if $r_1 : r_2 = 1 : n$ (where n is a positive integer) and their volumes are equal, then which one of the following is correct?

(a) $h_1 = 3nh_2$ (b) $h_1 = 3n^2h_2$
(c) $h_1 = 3h_2$ (d) $h_1 = n^2h_2$

55. A right circular cone is cut by a plane parallel to its base in such a way that the slant heights of the original and the smaller cone thus obtained are in the ratio 2 : 1. If V_1 and V_2 are respectively the volumes of the original cone and of the new cone, then what is $V_1 : V_2$?

(a) 2 : 1 (b) 3 : 1 (c) 4 : 1 (d) 8 : 1

56. The surface area of a sphere is 616 cm^2. If its radius is changed so that the area gets reduced by 75%, then the radius becomes

(a) 1.6 cm (b) 2.3 cm (c) 2.5 cm (d) 3.5 cm

57. A sphere is inscribed in a cubical box such that the sphere is tangent to all six faces of the box. What is the ratio of the volume of the cubical box to the volume of sphere?

(a) 6π (b) 36π (c) $\frac{4\pi}{3}$ (d) $\frac{6}{\pi}$

58. From a solid cylinder of height 4 cm and radius 3 cm, a conical cavity of height 4 cm and of base radius 3 cm is hollowed out. What is the total surface area of the remaining solid?

(a) 15π cm^2 (b) 22π cm^2 (c) 33π cm^2 (d) 48π cm^2

59. The curved surface of a cylinder is 1000 cm^2. A wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used?

(a) 22 m (b) 20 m
(c) 18 m (d) None of these

60. In order to fix an electric pole along a roadside, a pit with dimensions 50 cm × 50 cm is dug with the help of a spade. The pit is prepared by removing Earth by 250 strokes of spade. If one stroke of spade removes 500 cm^3 of Earth, then what is the depth of the pit?

(a) 2 m (b) 1 m (c) 0.75 m (d) 0.5 m

61. A figure is formed by revolving a rectangular sheet of dimensions 7 cm × 4 cm about its length. What is the volume of the figure thus formed?

(a) 352 cm^3 (b) 296 cm^3
(c) 176 cm^3 (d) 616 cm^3

62. The diagonals of the three faces of a cuboid are x, y, z, respectively. What is the volume of the cuboid?

(a) $\frac{xyz}{2\sqrt{2}}$

(b) $\frac{\sqrt{(y^2+z^2)(z^2+x^2)(x^2+y^2)}}{2\sqrt{2}}$

(c) $\frac{\sqrt{(y^2+z^2-x^2)(z^2+x^2-y^2)(x^2+y^2-z^2)}}{2\sqrt{2}}$

(d) None of the above

63. The volume of a cone is equal to that of a sphere. If the diameter of base of cone is equal to the diameter of the sphere, what is the ratio of height of cone to the diameter of the sphere?

(a) 2 : 1 (b) 1 : 2 (c) 3 : 1 (d) 4 : 1

64. The length, breadth and height of a rectangular parallelopiped are in ratio 6 : 3 : 1. If the surface area of a cube is equal to the surface area of this parallelopiped, then what is the ratio of the volume of the cube to the volume of the parallelopiped?

(a) 1 : 1 (b) 5 : 4 (c) 7 : 5 (d) 3 : 2

65. A hollow hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 cm. What is the weight of the hemisphere (1 cm^3 of the metal weight 9 g)?

(a) 54 π g (b) 366 π g (c) 122 π g (d) 108 π g

66. Half of a large cylindrical tank open at the top is filled with water and identical heavy spherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius of a ball, what is the maximum number of balls that can be dropped?

(a) 12 (b) 24 (c) 36 (d) 48

67. A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the centre of one end, and the other end as its base. The volume of the cylinder, hemisphere and the cone are respectively in the ratio

(a) 2 : 3 : 2 (b) 3 : 2 : 1 (c) 3 : 1 : 2 (d) 1 : 2 : 3

68. A conical cavity is drilled in a circular cylinder of height 15 cm and base radius 8 cm. The height and the base radius of the cone are also same. Then the whole surface area of the remaining solid is

(a) 440π cm^2 (b) 240π cm^2 (c) 640π cm^2 (d) 960π cm^2

69. Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius 1 cm. Same possibilities are listed in the statements given below

I. The material is just sufficient to prepare 8 shots each of radius 0.5 cm.

II. A shot of radius 0.75 cm and a second shot of radius 0.8 cm can be prepared from the available material.

Which of the above statement is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

70. Consider the following statements:

I. The volume of a cuboid is the product of the lengths of its coterminous edges.

II. The surface area of a cuboid is twice the sum of the products of lengths of its coterminous edges taken two at a time.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

71. Consider the following statements :
The length of a side of a cube is 1 cm. Which of the following can be the distance between any two vertices?

I. 1 cm II. $\sqrt{2}$ cm III. $\sqrt{3}$ cm

Select the correct answer using the code given below.

(a) Only I (b) Only II (c) Only III (d) All of these

72. Consider the following statements.

I. When eight drops of water combine to form a single drop, the surface area of all the eight drops is greater than the surface area of big drop.

II. Square of volume of a spherical body is directly proportional to cube of its surface area.

III. A sphere occupies the biggest space but has the smallest surface area.

Which of the above statement(s) is/are correct?

(a) I and II (b) I and III (c) II and III (d) All of these

73. Consider the following statements:

I. The curved surface area of a right circular cone of base radius r and height h is given by $\pi r(\sqrt{h^2+r^2})$.

II. The right circular cone of base radius r and height h when cut opened along the slant height, forms a rectangle of length πr and breadth $\sqrt{h^2+r^2}$.

III. If a right circular cone is cut off by a plane parallel to the base of the cone, then its curved surface area is $\frac{\pi h}{3}[R^2+r^2+Rr]$.

IV. If a right triangle is being revolved about one of its edges. The surface generated by the hypotenuse is the lateral surface of the right circular cone.

Which of the above statement(s) are correct?

(a) I and II (b) I, II and III (c) I, III and IV (d) All of these

Directions (Q. Nos. 74-76) *Read the following information carefully and answer the given questions that follow.*

A cone of height 10 cm and radius 5 cm is divided into two parts by drawing a plane through the mid-point of its axis, parallel to its base.

74. What is the ratio of the volume of the original cone to the volume of the frustum left?

(a) $\frac{4}{3}$ (b) $\frac{7}{8}$ (c) $\frac{8}{7}$ (d) $\frac{9}{8}$

75. What is the area of the top circle of the frustum in square centimetre?

(a) 25π (b) $\frac{5\pi}{4}$ (c) $\frac{25\pi}{4}$ (d) None of these

76. What is the ratio of the curved surface area of the original cone and the curved surface area of the frustum?

(a) 3 : 1 (b) 3 : 2 (c) 4 : 1 (d) 4: 3

Directions (Q. Nos. 77-79) *Read the following information carefully and answer the given questions that follow.*

Let C be a right circular cone. It is given that the two ends of a frustum of C are of radii 3 cm and 6 cm and the height of the frustum is 9 cm.

77. What is the slant height of the given frustum?

(a) $3\sqrt{10}$ cm (b) $6\sqrt{10}$ cm (c) 12 cm (d) 15 cm

78. What is the height of the cone?

(a) 9 cm (b) 12 cm (c) 13.5 cm (d) 18 cm

79. What is the total surface area of the given frustum?
(a) $9\pi(2\sqrt{10}+5)\,cm^2$ (b) $9\pi(3\sqrt{10}+5)\,cm^2$
(c) $9\pi(3\sqrt{10}+4)\,cm^2$ (d) $27\pi(\sqrt{10}+1)\,cm^2$

PREVIOUS YEARS' QUESTIONS

80. What will be the cost to plaster the inner surface of a well 14 m deep and 4 m in diameter at the rate of ₹ 25 per m^2? 2012 I
(a) ₹ 4000 (b) ₹ 4200 (c) ₹ 4400 (d) ₹ 5400

81. What is the length of the uniform wire of diameter 0.4 cm that can be drawn from a solid sphere of radius 9 cm? 2012 I
(a) 243 m (b) 240 m (c) 60.75 m (d) 60 m

82. The total surface area of a cube is 150 cm^2. What is its volume? 2012 I
(a) 64 cm^3 (b) 81 cm^3 (c) 125 cm^3 (d) 160 cm^3

83. If the volume of a cube is 729 cm^3, then what is the length of its diagonal? 2012 I
(a) $9\sqrt{2}$ cm (b) $9\sqrt{3}$ cm (c) 18 cm (d) $18\sqrt{3}$ cm

84. The curved surface area of a right circular cone of radius 14 cm is 440 cm^2. What is the slant height of the cone? 2012 I
(a) 10 cm (b) 11cm (c) 12 cm (d) 13 cm

85. A large solid metallic cylinder whose radius and height are equal to each other is to be melted and 48 identical solid balls are to be recast from the liquid metal, so formed. What is the ratio of the radius of a ball to the radius of the cylinder? 2012 I
(a) 1 : 16 (b) 1 : 12
(c) 1 : 8 (d) 1 : 4

86. What are the dimensions (length, breadth and height, respectively) of a cuboid with volume 720 cm^3, surface area 484 cm^2 and the area of the base 72 cm^2? 2012 I
(a) 9, 8 and 10 cm (b) 12, 6 and 10 cm
(c) 18, 4 and 10 cm (d) 30, 2 and 12 cm

87. If the surface area of a sphere is 616 cm^2, then what is its volume? 2012 I
(a) 4312/3 cm^3 (b) 4102/3 cm^3
(c) 1257 cm^3 (d) 1023 cm^3

Directions (Q.Nos. 88-89) *Read the following information carefully and answer the given questions that follow.*
The areas of the ends of a frustum of a cone are P and Q, where $P < Q$ and H is its thickness.

88. What is the difference in radii of the ends of the frustum? 2012 II
(a) $\dfrac{\sqrt{Q}-\sqrt{P}}{\sqrt{\pi}}$ (b) $\dfrac{\sqrt{Q}-\sqrt{P}}{\pi}$
(c) $\sqrt{Q}-\sqrt{P}$ (d) None of these

89. What is the volume of the frustum? 2012 II
(a) $3H(P+Q+\sqrt{PQ})$ (b) $H(P+Q+\sqrt{PQ})$
(c) $H(P+Q+\sqrt{PQ})/3$ (d) $H(P+Q-\sqrt{PQ})/3$

90. Let the largest possible right circular cone and largest possible sphere be fitted into two cubes of same length. If C and S denote the volume of cone and volume of sphere, respectively. Then, which one of the following is correct? 2012 II
(a) $C=2S$ (b) $S=2C$ (c) $C=S$ (d) $C=3S$

91. 10 circular plates each of thickness 3 cm, are placed one above the other and a hemisphere of radius 6 cm is placed on the top just to cover the cylindrical solid. What is the volume of the solid so formed? 2012 II
(a) 264π cm^3 (b) 252π cm^3
(c) 236π cm^3 (d) None of these

92. A right circular metal cone (solid) is 8 cm high and the radius is 2 cm. It is melted and recast into a sphere. What is the radius of the sphere? 2012 II
(a) 2 cm (b) 3 cm (c) 4 cm (d) 5 cm

93. The volume of a cube is numerically equal to sum of its edges. What is the total surface area? 2012 II
(a) 12 sq units (b) 36 sq units
(c) 72 sq units (d) 144 sq units

94. The diameter of base of a right circular cone is 7 cm and slant height is 10 cm, then what is its lateral surface area? 2012 II
(a) 110 cm^2 (b) 100 cm^2 (c) 70 cm^2 (d) 49 cm^2

95. What is the height of a solid cylinder of radius 5 cm and total surface area is 660 cm^2? 2012 II
(a) 10 cm (b) 12 cm (c) 15 cm (d) 16 cm

96. If the ratio of the diameters of two spheres is $3:5$, then what is the ratio of their surface areas? 2012 II
(a) 9 : 25 (b) 9 : 10 (c) 3 : 5 (d) 27 : 125

97. What is the volume of the largest sphere that can be curved out of a cube of edge 3 cm? 2012 II
(a) 9π cm^3 (b) 6π cm^3 (c) 4.5π cm^3 (d) 3π cm^3

98. What is the quantity of cloth required to roll up to form a right circular tent whose base is of radius 12 m and height 5 m? 2013 I
(a) 40π m^2 (b) 60π m^2 (c) 78π m^2 (d) 156π m^2

99. The volume of the material of a hemispherical shell with outer and inner radii 9 cm and 7 cm, respectively is approximately 2013 I
(a) 808 cm^3 (b) 800 cm^3 (c) 816 cm^3 (d) 824 cm^3

100. The ratio of surface area to diameter of a sphere whose volume is 36π cm^3, is 2013 I
(a) 3π (b) 6π (c) 6 (d) None of these

101. A cylindrical tube open at both ends is made of metal. The internal diameter of the tube is 6 cm and length of the tube is 10 cm. If the thickness of the metal used is 1 cm, then the outer curved surface area of the tube is **2013 I**

(a) 140π cm^2 (b) 146.5π cm^2
(c) 70π cm^2 (d) None of these

102. From a solid wooden right circular cylinder, a right circular cone whose radius and height are same as the radius and height of the cylinder, respectively is carved out. What is the ratio of the volume of the utilised wood to that of the wasted wood? **2013 I**

(a) 1 : 2 (b) 2 : 1 (c) 2 : 3 (d) 1 : 3

103. If the heights and the areas of the base of a right circular cone and a pyramid with square base are the same, then they have **2013 I**

(a) same volume and same surface area
(b) same surface area but different volumes
(c) same volume but different surface areas
(d) different volumes and different surface areas

104. The volume of a right circular cone of height 3 cm and slant height 5 cm is **2013 I**

(a) 49.3 cm^3 (b) 50.3 cm^3
(c) 52 cm^3 (d) 53 cm^3

105. A bucket is of a height 25 cm. Its top and bottom radii are 20 cm and 10 cm, respectively. Its capacity (in L) is **2013 I**

(a) $17.5\pi/3$ (b) 17.5π (c) 20π (d) 25π

106. The height of a cylinder is 15 cm. The lateral surface area is 660 cm^2. Its volume is **2013 II**

(a) 1155 cm^3 (b) 1215 cm^3 (c) 1230 cm^3 (d) 2310 cm^3

107. The diameter of the Moon is approximately one-fourth of the diameter of the Earth. What is the ratio (approximate) of their volumes? **2013 II**

(a) 1 : 16 (b) 1 : 64 (c) 1 : 4 (d) 1 : 128

108. A conical cap has the base diameter 24 cm and height 16 cm. What is the cost of painting the surface of the cap at the rate of 70 paise per cm^2? **2013 II**

(a) ₹ 520 (b) ₹ 524 (c) ₹ 528 (d) ₹ 532

109. What is the whole surface area of a cone of base radius 7 cm and height 24 cm? **2013 II**

(a) 654 cm^2 (b) 704 cm^2 (c) 724 cm^2 (d) 964 cm^2

110. A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24 m. The height of the cylindrical portion is 11 m, while the vertex of the cone is 16 m above the ground. What is the area of the curved surface for conical portion? **2013 II**

(a) 3434/9 m^2 (b) 3431/8 m^2
(c) 3432/7 m^2 (d) 3234/7 m^2

111. If x is the curved surface area and y is the volume of a right circular cylinder, then which one of the following is correct? **2014 I**

(a) Only the ratio of the height to radius of the cylinder is independent of x
(b) Only the ratio of height to radius of the cylinder is independent of y
(c) Either (a) or (b)
(d) Neither (a) nor (b)

112. A cylinder is surmounted by a cone at one end, a hemisphere at the other end. The common radius is 3.5 cm, the height of the cylinder is 6.5 cm and the total height of the structure is 12.8 cm. The volume V of the structure lies between **2014 I**

(a) 370 cm^3 and 380 cm^3 (b) 380 cm^3 and 390 cm^3
(c) 390 cm^3 and 400 cm^3 (d) None of these

113. The diameter of the base of a cone is 6 cm and its altitude is 4 cm. What is the approximate curved surface area of the cone? **2014 I**

(a) 45 cm^2 (b) 47 cm^2 (c) 49 cm^2 (d) 51 cm^2

114. A drainage tile is a cylindrical shell 21 cm long. The inside and outside diameters are 4.5 cm and 5.1 cm, respectively. What is the volume of the clay required for the tile? **2014 I**

(a) 6.96π cm^3 (b) 6.76π cm^3
(c) 5.76π cm^3 (d) None of these

115. A cube has each edge 2 cm and a cuboid is 1 cm long, 2 cm wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54 cm^2. **2014 I**

Which one of the following is correct?

(a) Both cube and cuboid can be painted
(b) Only cube can be painted
(c) Only cuboid can be painted
(d) Neither cube nor cuboid can be painted

116. A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume of the original cone to the volume of the smaller cone? **2014 I**

(a) 4 : 1 (b) 8 : 1 (c) 2 : 1 (d) 6 : 1

117. If 64 identical small spheres are made out of big sphere of diameter 8 cm, then what is surface area of each small sphere? **2014 I**

(a) π cm^2 (b) 2π cm^2 (c) 4π cm^2 (d) 8π cm^2

118. The dimensions of a field are 15 m by 12 m. A pit 8 m long, 2.5 m wide and 2 m deep is dug in one corner of the field and the earth removed is evenly spread over the remaining area of the field. The level of the field is raised by **2014 I**

(a) 15 cm (b) 20 cm (c) 25 cm (d) $\frac{200}{9}$ cm

119. What is the diameter of the largest circle lying on the surface of a sphere of surface area 616 cm^2? 2014 I

(a) 14 cm (b) 10.5 cm (c) 7 cm (d) 3.5 cm

120. The volume of a hollow cube is $216x^3$. What is the surface area of the largest sphere which can be enclosed in it? 2014 I

(a) $18\pi x^2$ (b) $27\pi x^2$ (c) $36\pi x^2$ (d) $72\pi x^2$

Directions (Q. Nos. 121-122) *Read the following information carefully and answer the given questions that follow.*

A right angled triangle having hypotenuse 25 cm and legs in the ratio 3 : 4 is made to revolve about its hypotenuse. ($\pi = 3.14$)

121. What is the volume of the double cone so formed? 2014 II

(a) 3124 cm^3 (b) 3424 cm^3 (c) 3768 cm^3 (d) 3924 cm^3

122. What is the surface area of the double cone so formed? 2014 II

(a) 1101.2 cm^2 (b) 1111.4 cm^2 (c) 1310.4 cm^2 (d) 1318.8 cm^2

123. If the side of a cube is increased by 100%, then by what percentage is the surface area of the cube increased? 2014 II

(a) 150% (b) 200% (c) 300% (d) 400%

124. A cylinder circumscribes a sphere. What is the ratio of volume of the sphere to that of the cylinder? 2014 II

(a) 2 : 3 (b) 1 : 2 (c) 3 : 4 (d) 3 : 2

Directions (Q. Nos. 125-126) *Read the following information carefully and answer the given questions that follow.*

A toy is in the form of a cone mounted on the hemisphere with the same radius. The diameter of the base of the conical portion is 12 cm *and its height is* 8 cm.

125. What is the total surface area of the toy? 2014 II

(a) 132π cm^2 (b) 112π cm^2 (c) 96π cm^2 (d) 66π cm^2

126. What is the volume of the toy? 2014 II

(a) 180π cm^3 (b) 240π cm^3 (c) 300π cm^3 (d) 320π cm^3

127. The areas of the three adjacent faces of a cuboidal box are x, $4x$ and $9x$ sq unit. What is the volume of the box? 2014 II

(a) $6x^2$ cu units (b) $6x^{3/2}$ cu units
(c) $3x^{3/2}$ cu units (d) $2x^{3/2}$ cu units

128. What is the number of pairs of perpendicular planes in a cuboid? 2014 II

(a) 4 (b) 8
(c) 12 (d) None of these

129. Consider the following statements in respect of four spheres A, B, C and D having respective radii 6, 8, 10 and 12 cm.

I. The surface area of sphere C is equal to the sum of surface areas of spheres A and B.

II. The volume of sphere D is equal to the sum of volumes of spheres A, B and C. 2014 II

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

130. The diameter of a metallic sphere is 6 cm. The sphere is melted and drawn into a wire of uniform circular cross-section. If the length of the wire is 36 m, then what is its radius? 2014 II

(a) 0.1 cm (b) 0.01 cm (c) 0.001 cm (d) 1.0 cm

131. What is the maximum distance between two points of a cube of side 2 cm? 2014 II

(a) $\sqrt{3}$ cm (b) $2\sqrt{3}$ cm (c) $4\sqrt{3}$ cm (d) $2\sqrt{2}$ cm

132. The total outer surface area of a right circular cone of height 24 cm with a hemisphere of radius 7 cm upon its base is 2015 I

(a) 327π cm^2 (b) 307π cm^2
(c) 293π cm^2 (d) 273π cm^2

133. A sphere and a cube have same surface area. The ratio of square of their volumes is 2015 I

(a) $6 : \pi$ (b) $5 : \pi$ (c) 3 : 5 (d) 1 : 1

134. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume of the cone. The ratio of the height of the cone to the radius of its base is 2015 I

(a) 2 : 1 (b) 1 : 2 (c) 2 : 3 (d) 3 : 2

135. If the radius of a sphere is increased by 10%, then the volume will be increased by 2015 I

(a) 33.1% (b) 30% (c) 50% (d) 10%

136. If three metallic spheres of radii 6 cm, 8 cm and 10 cm are melted to form a single sphere, then the diameter of the new sphere will be 2015 I

(a) 12 cm (b) 24 cm (c) 30 cm (d) 36 cm

137. If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50%, then the volume of the cone 2015 I

(a) remains unaltered (b) decreases by 25%
(c) increases by 25% (d) increases by 50%

138. A rectangular paper of 44 cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper. The radius of the base of the cylinder so rolled is 2015 I

(a) 3.5 cm (b) 5 cm (c) 7 cm (d) 14 cm

139. The diagonals of three faces of a cuboid are 13, $\sqrt{281}$ and 20 linear units. Then, the total surface area of the cuboid is **2015 I**

(a) 650 sq units (b) 658 sq units
(c) 664 sq units (d) 672 sq units

140. A cylindrical vessel of radius 4 cm contains water. A solid sphere of radius 3 cm is lowered into the water until it is completely immersed. The water level in the vessel will rise by **2015 I**

(a) 1.5 cm (b) 2 cm (c) 2.25 cm (d) 4.5 cm

141. A rectangular block of wood having dimensions $3\text{m} \times 2\text{m} \times 1.75$ m has to be painted on all its faces. The layer of paint must be 0.1 mm thick. Paint comes in cubical boxes having their edges equal to 10 cm. The minimum number of boxes of paint to be purchased is **2015 I**

(a) 5 (b) 4 (c) 3 (d) 2

142. If the surface area of a cube is 13254 cm^2, then the length of its diagonal is **2015 II**

(a) $44\sqrt{2}$ cm (b) $44\sqrt{3}$ cm (c) $47\sqrt{2}$ cm (d) $47\sqrt{3}$ cm

143. A water tank, open at the top, is hemispherical at the bottom and cylindrical above it. The radius is 12 m and the capacity is $3312\,\pi\text{ m}^3$. The ratio of the surface areas of the spherical and cylindrical portions, is **2015 II**

(a) 3 : 5 (b) 4 : 5 (c) 1 : 1 (d) 6 : 5

144. A large water tank has the shape of a cube. If 128 m^3 of water is pumped out, the water level goes down by 2m. Then, the maximum capacity of the tank is **2015 II**

(a) 512 m^3 (b) 480 m^3 (c) 324 m^3 (d) 256 m^3

145. The areas of three mutually perpendicular faces of a cuboid are x, y, z. If V is the volume, then xyz is equal to **2015 II**

(a) V (b) V^2 (c) $2V$ (d) $2V^2$

146. How many right angled triangles can be formed by joining the vertices of a cuboid? **2015 II**

(a) 24 (b) 28 (c) 32 (d) None of these

147. Three rectangles R_1, R_2 and R_3 have the same area. Their lengths x_1, x_2 and x_3 respectively are such that $x_1 < x_2 < x_3$. If V_1, V_2 and V_3 are the volumes of the cylinders formed from the rectangles R_1, R_2 and R_3 respectively by joining the parallel sides along the breadth, then which one of the following is correct? **2015 II**

(a) $V_3 < V_2 < V_1$ (b) $V_1 < V_3 < V_2$
(c) $V_1 < V_2 < V_3$ (d) $V_3 < V_1 < V_2$

148. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive Y-axis. The cone is filled with water up to half of its height. The volume of water is **2015 II**

(a) $\frac{V}{8}$ (b) $\frac{V}{6}$ (c) $\frac{V}{3}$ (d) $\frac{V}{2}$

149. 30 metallic cylinders of same size are melted and cast in the form of cones having the same radius and height as those of the cylinders.

Consider the following statements

I. A maximum of 90 cones will be obtained.

II. The curved surface of the cylinder can be flattened in the shape of a rectangle but the curved surface of the cone when flattened has the shape of triangle.

Which one of the following statement(s) is/are correct in respect of the above? **2015 II**

(a) Both statement I and statement II are correct and statement II is the correct explanation of statement I
(b) Both statement I and statement II are correct and statement II is not the correct explanation of statement I
(c) statement I is correct but statement II is not correct
(d) statement I is not correct but statement II is correct

150. Water is filled in a container in such a manner that its volume doubles every 5 min. If it takes 30 min for the container to be full, in how much time will it be one-fourth full ? **2015 II**

(a) 7.5 min (b) 15 min (c) 20 min (d) 17.5 min

151. Consider the following statements:

I. If the height of a cylinder is doubled, the area of the curved surface is doubled.

II. If the radius of a hemispherical solid is doubled, its total surface area becomes fourfold.

Which of the statement(s) given above is/are correct? **2015 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

152. From the solid gold in the form of a cube of side length 1 cm, spherical solid balls each having the surface area $\pi^{1/3}$ cm^2 are to be made. Assuming that there is no loss of the material in process of making the balls, the maximum number of balls made will be **2015 II**

(a) 3 (b) 4 (c) 6 (d) 9

153. A hollow cylindrical drum has internal diameter of 30 cm and a height of 1 m. What is the maximum number of cylindrical boxes of diameter 10 cm and height 10 cm each that can be packed in the drum? **2015 II**

(a) 60 (b) 70 (c) 80 (d) 90

154. A pipe with square cross-section is supplying water to a cistern which was initially empty. The area of cross-section is 4 cm^2 and the nozzle velocity of water is 40m/s. The dimensions of the cistern $10\text{ m} \times 8\text{m} \times 6\text{m}$. Then, the cistern will be full in **2015 II**

(a) 9.5 h (b) 9 h
(c) 8 h and 20 min (d) 8 h

ANSWERS

1	*c*	2	*b*	3	*b*	4	*c*	5	*d*	6	*b*	7	*c*	8	*a*	9	*c*	10	*c*
11	*c*	12	*c*	13	*b*	14	*a*	15	*b*	16	*c*	17	*d*	18	*a*	19	*b*	20	*a*
21	*d*	22	*d*	23	*b*	24	*b*	25	*c*	26	*b*	27	*b*	28	*b*	29	*b*	30	*c*
31	*c*	32	*c*	33	*b*	34	*b*	35	*b*	36	*b*	37	*d*	38	*a*	39	*b*	40	*c*
41	*c*	42	*c*	43	*a*	44	*b*	45	*a*	46	*d*	47	*b*	48	*c*	49	*a*	50	*b*
51	*b*	52	*b*	53	*c*	54	*b*	55	*d*	56	*d*	57	*d*	58	*d*	59	*b*	60	*d*
61	*a*	62	*c*	63	*a*	64	*d*	65	*b*	66	*b*	67	*b*	68	*a*	69	*c*	70	*c*
71	*d*	72	*d*	73	*c*	74	*c*	75	*c*	76	*d*	77	*a*	78	*d*	79	*b*	80	*c*
81	*a*	82	*c*	83	*b*	84	*a*	85	*d*	86	*a*	87	*a*	88	*a*	89	*c*	90	*b*
91	*d*	92	*a*	93	*c*	94	*a*	95	*d*	96	*a*	97	*c*	98	*d*	99	*a*	100	*b*
101	*d*	102	*a*	103	*c*	104	*b*	105	*a*	106	*d*	107	*b*	108	*c*	109	*b*	110	*c*
111	*d*	112	*a*	113	*b*	114	*d*	115	*a*	116	*b*	117	*c*	118	*c*	119	*a*	120	*c*
121	*c*	122	*d*	123	*c*	124	*a*	125	*a*	126	*b*	127	*b*	128	*c*	129	*c*	130	*a*
131	*b*	132	*d*	133	*a*	134	*a*	135	*a*	136	*b*	137	*b*	138	*c*	139	*c*	140	*c*
141	*c*	142	*d*	143	*b*	144	*a*	145	*b*	146	*a*	147	*a*	148	*a*	149	*b*	150	*c*
151	*c*	152	*c*	153	*d*	154	*c*												

HINTS AND SOLUTIONS

1. *(c)* Let 'a' be the edge of three equal cubes. Surface area of a single cube

$$= 6\,(\text{Side})^2 = 6a^2$$

∴ Sum of the surface areas of the three cubes is $= 3\,(6a^2) = 18a^2$

Length of resulting cuboid $= 3a$

Breadth of resulting cuboid $= a$

Height of resulting cuboid $= a$

∴ Total surface area of cuboid

$$= 2\,(lb + bh + hl)$$
$$= 2\,(3a^2 + a^2 + 3a^2) = 14a^2$$

Required ratio $= \dfrac{14a^2}{18a^2} = \dfrac{7}{9}$, i.e. 7 : 9

2. *(b)* Here, $x = 12$,

According to the formula, percentage increase in volume

$$= \left[\left(1+\frac{x}{100}\right)^3 - 1\right]\times 100\%$$
$$= \left[\left(1+\frac{12}{100}\right)^3 - 1\right]\times 100\%$$
$$= \left[\left(1+\frac{3}{25}\right)^3 - 1\right]\times 100\%$$
$$= \left[\left(\frac{28}{25}\right)^3 - 1\right]\times 100\%$$
$$= \left[\frac{21952}{15625} - 1\right]\times 100\% = [1.405 - 1]\times 100\%$$
$$= 0.405 \times 100\% = 40.5\%$$

3. *(b)* Given, diameter of the base of the cylinder = 21 cm

∴ Radius $= \dfrac{21}{2}$ cm

As, curved surface area $= 2\pi rh = 1320$

$$\Rightarrow \quad 2\times\frac{22}{7}\times\frac{21}{2}\times h = 1320$$
$$\therefore \quad h = \frac{1320}{22\times 3} = 20\text{ cm}$$

4. *(c)* 1 cm^3 of water weighs 1 g.

∵ Volume of 154 g of water $= 154\text{ cm}^3$

∴ Volume of cylinder $= \pi r^2 h = 154\text{ cm}^3$

$$\Rightarrow \quad \frac{22}{7}(3.5)^2 \times h = 154$$
$$\Rightarrow \quad \frac{77h}{2} = 154$$
$$\Rightarrow \quad h = \frac{154\times 2}{77} = 4\text{ cm}$$

5. *(d)* Here, $2\pi rh = 264\text{ m}^2$

and $\quad \pi r^2 h = 924\text{ m}^3$ [given]

$$\therefore \quad \frac{\pi r^2 h}{2\pi rh} = \frac{924}{264}$$
$$\Rightarrow \quad \frac{r}{2} = \frac{7}{2} \Rightarrow r = 7\text{ m}$$

∴ Diameter of pillar

$$= r\times 2 = 7\times 2 = 14\text{ m}$$

6. *(b)* Given, $r = 7$ m, $h = 24$ m

Slant height of cone,

$$= \sqrt{r^2 + h^2} = \sqrt{24^2 + 7^2}$$
$$= \sqrt{576 + 49} = \sqrt{625} = 25\text{ m}$$

Curved surface area $= \pi rl$

$$= \frac{22}{7}\times 7\times 25 = 550\text{ m}^2$$

Width of cloth = 50 m

Length required $= \dfrac{550}{50} = 11$ m

7. *(c)* Let radius and height of a right circular cone be $5x$ and $12x$ respectively.

$$\therefore \quad r = 5x \text{ and } h = 12x$$

Volume of cone $= \dfrac{1}{3}\pi r^2 h = 2512$

$$\Rightarrow \quad \frac{1}{3}\times 3.14\times(5x)^2\times 12x = 2512$$
$$\Rightarrow \quad x^3 = \frac{2512}{314} = 8$$
$$\Rightarrow \quad x = 2$$

∴ Radius = 10 cm, height = 24 cm

∴ Slant height,

$$l = \sqrt{r^2 + h^2} = \sqrt{10^2 + 24^2}$$
$$= \sqrt{676} = 26\text{ cm}$$

8. *(a)* Let r and H be the radius and height of the cone respectively.

$\therefore$ Original volume $= \frac{1}{3}\pi r^2 h = V$

New radius $= r$ and new height $= 2h$

New volume $= \frac{1}{3}\pi r^2 \times 2h = 2V$

$\therefore$ Increase in volume $= 2V - V = V$

Increase percentage of volume

$$= \left(\frac{V}{V} \times 100\right)\% = 100\%$$

9. *(c)* Ratio of surface area

$$= \frac{4\pi r^2}{4\pi R^2} = \left(\frac{4}{25}\right) = \left(\frac{2}{5}\right)^2 \Rightarrow \frac{r}{R} = \frac{2}{5}$$

Now, ratio of volumes

$$= \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = \left(\frac{r}{R}\right)^3 = \left(\frac{2}{5}\right)^3 = \frac{8}{125}$$

$= 8 : 125$

10. *(c)* Let radius of cylinder $= x$ and radius of cone $= 2x$

Height of each $= h$

$\therefore$ Required ratio $= \dfrac{\text{Volume of cone}}{\text{Volume of cylinder}}$

$$= \frac{\frac{1}{3}\pi 4x^2 h}{\pi x^2 h} = \frac{4}{3} = 4:3$$

11. *(c)* Volume of sphere = Volume of cylinder

$\therefore \quad \frac{4}{3}\pi R^3 = \pi \times 3 \times 3 \times 32$

$\Rightarrow \quad R^3 = 3 \times 3 \times 3 \times 8 \Rightarrow R = 6$ cm

12. *(c)* Volume of first sphere $= \frac{4}{3}\pi\left(\frac{3}{2}\right)^3$

and volume of second sphere $= \frac{4}{3}\pi(2)^3$

$\therefore$ Volume of big sphere

$$= \frac{4}{3}\pi\left\{\left(\frac{3}{2}\right)^3 + (2)^3\right\} \text{ [by condition]}$$

$$\Rightarrow \frac{4}{3}\pi x^3 = \frac{4}{3}\pi\left\{\frac{27}{8} + 8\right\}$$

$$x^3 = 91/8 = 11.375 \text{ cm}^3$$

13. *(b)* Given, $l = R$

$\because$ Total surface area of cone,

$S = \pi R(R + l) = \pi R(R + R) = 2\pi R^2$

$\therefore$ Area of circle, $A = \pi(2R)^2 = 4\pi R^2$

$\Rightarrow \quad A = 2S$

14. *(a)* In one round, distance covered by cylinder = 60 cm

In one second, distance covered by cylinder $= 60 \times 5 = 300$ cm

In five seconds, distance covered by cylinder $= 300 \times 5 = 1500$ cm $= 15$ m

15. *(b)* Let the radius of big drop and small drop be R and r respectively.

By given condition,

$27 \times$ Volume of smaller drops = Volume of bigger drop

$\therefore \quad 27 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$

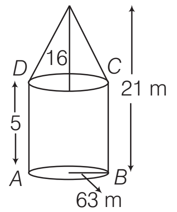
$\Rightarrow \quad 27 \times (0.2)^3 = R^3 \quad [\because r = 0.2 \text{ cm}]$

$\Rightarrow \quad (3 \times 0.2)^3 = R^3 \Rightarrow R = 0.6$ cm

16. *(c)* Common radius $= \frac{126}{2} = 63$ m

Curved surface area of cylindrical part

$= 2\pi rh$



$$= 2 \times \frac{22}{7} \times 63 \times 5 = 1980 \text{ m}^2$$

Height of cone $= 21 - 5 = 16$ m

Curved surface area of conical part $= \pi rl$

$$= \frac{22}{7} \times 63 \times \sqrt{(63)^2 + (16)^2}$$

$$= \frac{22}{7} \times 63 \times \sqrt{3969 + 256}$$

$$= \frac{22}{7} \times 63 \times 65 = 12870 \text{ m}^2$$

$\because$ Total surface area

$= 1980 + 12870 = 14850$ m^2

$\therefore$ Total cost of canvas used

$= 14850 \times 12 = ₹\ 178200$

17. *(d)* Given, radius of cylinder = 6 cm

Volume of sphere

$$= \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 6 \times 6 \times 6$$

$$= \frac{88 \times 2 \times 36}{7} \text{ cm}^3$$

Let r be its internal radius, R be its external radius, then material used to cast the cylinder

$$= \pi h(R^2 - r^2) = \frac{22}{7} \times 32(25 - r^2)$$

Hence, $\frac{22}{7} \times 32 \times (25 - r^2)$

$$= \frac{88 \times 2 \times 36}{7}$$

$$\Rightarrow (25 - r^2) = \frac{88 \times 2 \times 36 \times 7}{22 \times 32 \times 7} = 9$$

$\Rightarrow \quad r^2 = 25 - 9 = 16 \Rightarrow r = 4$ cm

Thickness of cylinder

$= R - r = 5 - 4 = 1$ cm

18. *(c)* Volume of original cylinder

$= [\pi \times (10)^2 \times 1000]$ cm^3

Volume of shell

$$= \frac{1}{4} \times \pi \times 100000 = 25000\pi \text{ cm}^3$$

Let r be the radius of shell, then

$\pi r^2 h = 25000\pi$

$\Rightarrow \quad r^2 = \frac{25000\pi}{\pi \times 1000} = 25$

$\Rightarrow \quad r = 5$ cm

Now, thickness $= 2r = 2 \times 5 = 10$ cm

$\Rightarrow \quad R^2 = 100 + 25 = 125$

$\Rightarrow \quad R = \sqrt{125}$

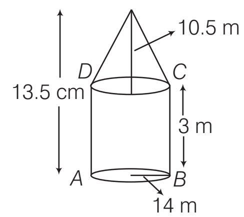
$\therefore$ Thickness of cylindrical shell

$= R - 10 = \sqrt{125} - 10$

$= 5\sqrt{5} - 5 \times 2 = 5(\sqrt{5} - 2)$ cm

19. *(b)* Curved surface area of cylinder

$$= 2 \times \frac{22}{7} \times 14 \times 3 = 264 \text{ m}^2$$



Height of cone $= 13.5 - 3 = 10.5$ m

Slant height of cone

$= \sqrt{(10.5)^2 + (14)^2} = \sqrt{110.25 + 196}$

$= \sqrt{306.65} = 17.5$ m

Curved surface area of cone $= \pi rl$

$$= \frac{22}{7} \times 14 \times 17.5 = 770 \text{ m}^2$$

Total area to be painted

$= 264 + 770 = 1034$ m^2

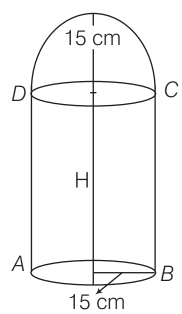
$\therefore$ Cost of painting $= 1034 \times 2 = ₹\ 2068$

20. *(a)* Let height of the cylinder be H.

By given condition,

Volume of hemisphere + Volume of cylinder = Volume of container

$\therefore \quad \frac{2}{3}\pi r^3 + \pi r^2 H = 32400\pi$



$\Rightarrow \frac{2}{3}\pi \times 3375 + \pi \times 225H = 32400\pi$

$\Rightarrow 2\pi \times 1125 + \pi \times 225H = 32400\pi$

On dividing both sides by 225 π, we get

$\Rightarrow$ $10 + H = 144 \Rightarrow H = 134$ cm

$\therefore$ Height of container

$= 15 + 134 = 149$ cm

21. *(d)* Since, milk is in a conical flask whose base radius and height are r and h, respectively.

$\therefore$ Volume of milk $= \frac{1}{3}\pi r^2 h$

Now, this milk is poured into a cylindrical flask whose base radius is $2r$. Let H be the height of cylindrical flask.

$\therefore$ Volume of milk

= Volume of cylindrical flask

$\Rightarrow \quad \frac{1}{3}\pi r^2 h = \pi(2r)^2 H$

$\Rightarrow \quad \frac{1}{3}\pi r^2 h = 4\pi r^2 H \Rightarrow H = \frac{h}{12}$

22. *(d)* Let radius and height of a cone be r and h.

$\because \quad \frac{r}{h} = \frac{3}{4} \Rightarrow h = \frac{4r}{3}$ ...(i)

Volume of cone $= \frac{1}{3}\pi r^2 h$

$\therefore \quad 96\pi = \frac{1}{3}\pi \times r^2 \times \frac{4r}{3}$ [from Eq. (i)]

$\Rightarrow \quad r^3 = \frac{96 \times 3 \times 3}{4} = 216$

$\Rightarrow \quad r = 6$ cm and $h = 8$ cm

$\therefore$ Lateral surface area $= \pi rl$ $[\because l^2 = r^2 + h^2]$

$= \pi r\sqrt{r^2 + h^2}$

$= \pi \times 6\sqrt{36 + 64} = 60\pi$ cm^2

23. *(b)* $\because$ Volume of wax balls

$= \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi$ cm^3

and volume of sphere $= \frac{4}{3}\pi(8)^3$ cm^3

$\therefore$ Required number of balls $= \dfrac{\frac{4}{3}\pi(8)^3}{\frac{4}{3}\pi}$

$= 512$

24. *(b)* Given, $r = \frac{70}{2}$ cm $= 0.35$ m

and $h = 4$ m

$\therefore$ Surface area of 10 cylinders

$= 10(2\pi rh)$

$= 10\left(2 \times \frac{22}{7} \times 0.35 \times 4\right) = 88$ m^2

$\therefore$ Total cost of 10 pillars painting at the rate of ₹ 5 per m^2

$= 88 \times 5 =$ ₹ 440

25. *(c)* Given,

$R = 35$ cm, $r = 14$ cm and $h = 40$ cm

$\therefore$ Volume of the bucket

$= \frac{\pi h}{3}(R^2 + r^2 + Rr)$

$= \frac{22 \times 40}{7 \times 3}(35^2 + 14^2 + 35 \times 14)$

$= \frac{880}{21}(1225 + 196 + 490)$

$= \frac{880}{21} \times 1911 = 80080$ cm^3

26. *(b)* Let side of a cube be a unit.

$\therefore$ Volume of cube, $V = a^3$

and total surface area of cube, $S = 6(a)^2$

$S^3 = 6^3(a^6) = 216V^2$

27. *(b)* Total area of wetted surface

= Curved surface area of cylinder + Area of base of cylinder

$= 2\pi rh + \pi r^2$

$= \pi[2 \times 3.5 \times 4 + (3.5)^2]$

$= \frac{22}{7}(28 + 12.25) = \frac{22}{7} \times 40.25$

$= 126.5$ m^2

28. *(b)* Let height h_1, radius r_1, area S_1 and volume V_1 of first cylinder.

Similarly, height h_2, radius r_2, area S_2 and volume V_2 of second cylinder.

By given condition, $\frac{r_1}{r_2} = \frac{2}{3}$ ...(i)

$\therefore \quad \frac{S_1}{S_2} = \frac{5}{3} \Rightarrow \frac{2\pi r_1 h_1}{2\pi r_2 h_2} = \frac{5}{3}$

$\Rightarrow \frac{h_1}{h_2} \times \frac{2}{3} = \frac{5}{3} \Rightarrow \frac{h_1}{h_2} = \frac{5}{2}$ ...(ii)

$\therefore \quad \frac{V_1}{V_2} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{r_1}{r_2}\right)^2\left(\frac{h_1}{h_2}\right)$

$= \left(\frac{2}{3}\right)^2\left(\frac{5}{2}\right) = \frac{10}{9}$ or, $10 : 9$

29. *(b)* By given condition,

Volume of cylindrical vessel = Volume of sphere

$\therefore \quad \pi r^2 h = \frac{4}{3}\pi R^3$

$\Rightarrow \quad (60)^2 \times 10 = \frac{4}{3}\left(\frac{d}{2}\right)^3$

$\Rightarrow \quad (60)^2 \times 10 = \frac{4}{3} \times \frac{d^3}{8}$

$\Rightarrow \quad d^3 = (60)^2 \times (60)$

$\therefore \quad d = 60$ cm

30. *(c)* Given, inner radius of hollow sphere

$(r) = \frac{4}{2} = 2$ cm

Outer radius of hollow sphere

$(R) = \frac{8}{2} = 4$ cm

Volume of metal of the sphere

$= \frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(4^3 - 2^3)$

$= \frac{4}{3}\pi \times 56$ cm^3

Radius of base of cone $(x) = \frac{8}{2} = 4$ cm

$\therefore \quad \frac{1}{3}\pi 4^2 h = \frac{4}{3}\pi \times 56$ [by condition]

$\Rightarrow \quad h = \dfrac{\frac{4}{3} \times 56 \times 3}{16} = 14$ cm

Hence, the height of cone is 14 cm.

31. *(c)* Surface area of sphere, $S_1 = 4\pi r^2$

If radius is $2r$, then surface area of sphere,

$S_2 = 4\pi(2r)^2 = 16\pi r^2$

$\therefore \quad S_2 = 4S_1$

Hence, it increases four times.

32. *(c)* Given, $l = 6$ m, $b = 4$ m

and $h = 1.25$ m

Area of wetted surface

$= 2(l \times h + b \times h) + l \times b$

$= 2(7.5 + 5) + 24$

$= 25 + 24 = 49$ m^2

33. *(b)* Given, $R = 3$ cm, $r = 2$ cm, $h = 10$ cm

Total surface area

$= 2\pi(R + r)(R + h - r)$

$= 2\pi(3 + 2)(3 + 10 - 2)$

$= 110\pi$ cm^2

34. *(b)* For conical part, $r = \frac{6}{2} = 3$ cm,

$h = 4$cm, $l = \sqrt{h^2 + r^2} = 5$ cm

Surface area of conical part $= \pi rl$

$= 3.14 \times 3 \times 5 = 47.1$ cm^2

For hemispherical part, $r = \frac{6}{2} = 3$ cm

Surface area of hemispherical part

$= 2\pi r^2 = 2 \times 3.14 \times 3 \times 3$

$= 56.52$ cm^2

$\therefore$ Surface area of toy $= 47.1 + 56.52$

$= 103.62$ cm^2

35. *(b)* Let r and h be the radius and height of the cylinder. Here, $h = 9$ m

By given condition,

$2\pi r \times h = \frac{2\pi r}{3}(h + r)$

$\Rightarrow \quad 9 = \frac{1}{3}(9 + r) \Rightarrow 27 = 9 + r$

$\therefore \quad r = 18$ m

36. *(b)* Let r_1 and r_2 be the radii of two spheres.

$$\because \quad \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{8}{1} \Rightarrow \frac{r_1}{r_2} = \frac{2}{1}$$

$\therefore$ Ratio of their surface areas $= \dfrac{4\pi r_1^2}{4\pi r_2^2}$

$$= \left(\frac{2}{1}\right)^2 = 4:1$$

37. *(d)* Area of the field = Length × Breadth

$= 12 \times 10 = 120 \text{ m}^2$

Area of the pit's surface $= 5 \times 4 = 20 \text{ m}^2$

Area on which the Earth is to be spread

$= 120 - 20 = 100 \text{ cm}^2$

Volume of Earth dug out

$= 5 \times 4 \times 2 = 40 \text{ cm}^3$

$\therefore$ Level of field raised $= \dfrac{40}{100} = \dfrac{2}{5}$ m

$= \dfrac{2}{5} \times 100 = 40$ cm

38. *(a)* Edge of cube = 9 cm

For rectangular vessel,

Length = 15 cm and breadth = 12 cm

Let the rise in the water level is x cm.

Volume of cube $= 9^3 = 9 \times 9 \times 9 \text{ cm}^3$

Volume of water risen in the vessel

$= 15 \times 12 \times x \text{ cm}^3$

$\Rightarrow \quad 15 \times 12 \times x = 9 \times 9 \times 9$

$\therefore \quad x = \dfrac{9 \times 9 \times 9}{15 \times 12} = \dfrac{81}{20} = 4.05$ cm

39. *(b)* The maximum diameter of a sphere in a cube is of 3 m.

$\therefore$ Volume of sphere, $V_1 = \dfrac{4}{3}\pi (1.5)^3$

$= 4.5\pi \text{ m}^3$

Volume of cube, $V_2 = (3)^3 = 27 \text{ m}^3$

and Volume of solid left $= V_2 - V_1$

$= (27 - 4.5\pi) \text{ m}^3$

40. *(c)* Length of water flowing in 1h = 10 km

$\therefore$ Length of the water flowing in $\dfrac{1}{2}$ h

$= 5 \text{ km} = 5000$ m

Area of pipe

$= 40 \text{ cm}^2 = \dfrac{40}{10000} \text{ m}^2 = \dfrac{1}{250} \text{ m}^2$

$\therefore$ Volume of water flowing in $\dfrac{1}{2}$ h

$= 5000 \times \dfrac{1}{250} = 20 \text{ m}^3$

Area of bottom of tank

$= 80 \times 40 = 3200 \text{ m}^2$

$\therefore$ Depth of water

$= \dfrac{\text{Volume}}{\text{Area}} = \dfrac{20}{3200} = \dfrac{1}{160}$ m

$= \dfrac{1}{160} \times 100 \text{ cm} = \dfrac{5}{8}$ cm

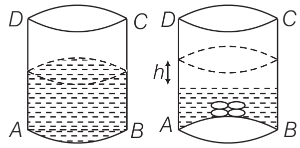
41. *(c)* Given radius of balls = 1 cm

Volume of each ball $= \dfrac{4}{3} \times \pi \times 1 \times 1 \times 1$

$= \dfrac{4}{3}\pi \text{ cm}^3$

$\therefore$ Volume of 4 balls

$= 4 \times \dfrac{4}{3}\pi = \dfrac{16}{3}\pi \text{ cm}^3$



Volume of water increased

= Volume of balls

$\Rightarrow$ Area of base × height $= \dfrac{16}{3}\pi$

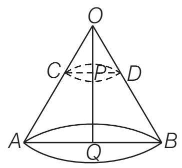
$\Rightarrow \quad \pi \times 5 \times 5 \times h = \dfrac{16}{3}\pi$

$\therefore \quad h = \dfrac{16}{3 \times 25} = \dfrac{16}{75}$ cm

42. *(c)* Given, height of the cone = 30 cm

$\therefore$ Volume of original cone (AOB)

$= \dfrac{1}{3}\pi R^2 H$



$= \dfrac{1}{3} \times \pi R^2 \times 30 = 10\pi R^2 \text{ cm}^3$

Volume of small cone OCD

$= \dfrac{1}{27}$ of volume cone AOB

$\Rightarrow \quad \dfrac{1}{3}\pi r^2 h = \dfrac{1}{27} \times 10\pi R^2$

$\Rightarrow h = \dfrac{10\pi R^2}{27} \times \dfrac{3}{\pi r^2} \Rightarrow h = \dfrac{10}{9}\left(\dfrac{R}{r}\right)^2$

$\because \quad \dfrac{R}{r} = \dfrac{30}{h} \qquad [\because \Delta OQB \sim \Delta OPD]$

$\therefore \quad h = \dfrac{10}{9}\left(\dfrac{30}{h}\right)^2 = \dfrac{10}{9} \times \dfrac{900}{h^2}$

$\Rightarrow h^3 = 10 \times 100$

$\Rightarrow \quad h = \sqrt[3]{1000} = 10$ cm

$\therefore$ Required height

$= H - h = 30 - 10 = 20$ cm

43. *(a)* Let r be the radius of cylindrical block, then height will be $2r$.

Volume of block $= \pi(r^2)(2r) = 2\pi r^3$

A sphere of maximum possible volume is curved out whose radius will be r.

$\because$ Volume of sphere $= \dfrac{4}{3}\pi r^3$

$\therefore$ Volume of utilised wood $= \dfrac{4}{3}\pi r^3$

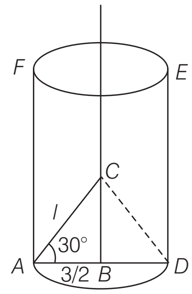
and volume of wasted wood

$= 2\pi r^3 - \dfrac{4}{3}\pi r^3 = \dfrac{6\pi r^3 - 4\pi r^3}{3} = \dfrac{2\pi r^3}{3}$

$\therefore$ Required ratio $= \dfrac{4}{3}\pi r^3 : \dfrac{2}{3}\pi r^3 = 2:1$

44. *(b)* Let the slant height of cone = l cm

In ΔABC,



$\cos 30° = \dfrac{AB}{AC} = \dfrac{3/2}{l} \qquad [\because AC = l \text{ cm}]$

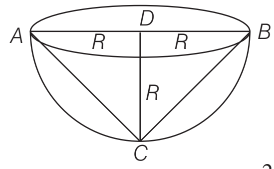
$\Rightarrow l = \dfrac{3/2}{\sqrt{3}/2} = \sqrt{3} \text{ cm} \left[\because \cos 30° = \dfrac{\sqrt{3}}{2}\right]$

$\therefore$ Area of cone $ACD = \pi rl$

$= \pi \times \dfrac{3}{2} \times \sqrt{3} = \dfrac{3\sqrt{3}\pi}{2} \text{ cm}^2$

45. *(a)* Volume of cone, $C = \dfrac{1}{3}\pi R^2 H$

$= \dfrac{1}{3}\pi R^3 \qquad [\because H = R]$



Volume of hemisphere, $H = \dfrac{2}{3}\pi R^3$

$\therefore C : H = \dfrac{1}{3}\pi R^3 : \dfrac{2}{3}\pi R^3 = 1:2$

46. *(d)* Volume of wire $= \pi r^2 h$

Now, new radius of the wire

$= \dfrac{r \times 90}{100} = \dfrac{9r}{10}$

Let new length of the wire be L.

$\therefore$ Volume of new wire $= \pi\left(\dfrac{9r}{10}\right)^2 \times L$

$= \dfrac{81}{100}\pi r^2 L$

By given condition, $\pi r^2 h = \dfrac{81}{100}\pi r^2 L$

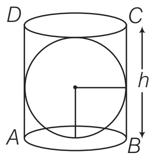
$\Rightarrow \quad L = \dfrac{100}{81} h$

Increases in length $= \frac{100}{81}h - h = \frac{19}{81}h$

Percentage increase $= \frac{19/81h}{h} \times 100\%$

$= 23.45\% = 23\%$ (approx)

47. *(b)* Let the height of cylinder be h.



Then, radius of cylinder $= \frac{h}{2}$

Also, radius of the sphere, $r = \frac{h}{2}$

$\because$ Volume of cylinder $= \pi\left(\frac{h}{2}\right)^2 h$

$= \frac{\pi h^3}{4}$

and volume of sphere.

$S = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi\left(\frac{h}{2}\right)^3 = \frac{\pi h^3}{6}$

$\because$ Volume of remaining material

$= \frac{\pi h^3}{4} - \frac{\pi h^3}{6} = \frac{\pi h^3}{12}$

$\therefore$ Volume of sphere in remaining material,

$S_1 = \frac{4}{3}\pi r_1^3$

By given condition, $\frac{4}{3}\pi r_1^3 = \frac{\pi h^3}{12}$

$\Rightarrow \quad r_1^3 = \frac{h^3}{16} \quad \Rightarrow \quad r_1 = \frac{h}{2 \cdot 2^{1/3}}$

Required ratio $= \frac{r}{r_1} = \frac{h/2}{h/2 \cdot 2^{1/3}} = 2^{1/3}$

$\Rightarrow \quad r : r_1 = 2^{1/3} : 1$

48. *(c)* Let side of square be x

$\therefore$ Radius of sphere $= x$

Surface area of sphere, $A = 4\pi x^2$

Since, square revolves round a side to generate a cylinder whose height and radius are x and x, respectively.

$\therefore \quad S = 2\pi x(x + x) = 4\pi x^2$

So, it is clear that, $A = S$.

49. *(a)* Given, length of pool = 24 m

and breadth of pool = 15 m

Rise in height of water

= 1 cm = 0.01 m

Volume of water displaced by x men

$= 24 \times 15 \times 0.01 = 3.6 \text{ m}^3$

But volume of water displaced by x men

$= 0.1x \text{ m}^3$

$\therefore 0.1x = 3.6 \Rightarrow x = \frac{3.6}{0.1} = 36$

50. *(b)* Given, length of one compartment

= 200 m

and breadth of one compartment

= 100 m

and height of water in one compartment

= 12 m

Volume of one compartment

$= 200 \times 100 \times 12$

$= 240000 \text{ m}^3$

$\therefore$ Volume of 3 compartments

$= 3 \times 240000 = 720000 \text{ m}^3$

$= 720000000$ L

$[\because 1 \text{ m}^3 = 1000 \text{ L}]$

Total requirement of water 50000 inhabitants in 1 day

$= 50000 \times 20 = 1000000$ L

$\therefore$ Required number of days

$= \frac{720000000}{1000000} = 720$ days

51. *(b)* Let r and h be the radius and height of the cone and r and H be the radius and height of the cylinder.

$\therefore$ Volume of cone $= \frac{1}{3}\pi r^2 h$

and volume of cylinder $= \pi r^2 H$

By given condition, $\frac{1}{3}\pi r^2 h = \pi r^2 H$

$\Rightarrow \quad h = 3H \quad \ldots$(i)

Lateral surface area of cone $= \pi rl$

and lateral surface area of cylinder

$= 2\pi rH$

By given condition, $\pi rl = \frac{15}{8} \times 2\pi rH$

$\Rightarrow \quad l = \frac{15}{4}H \quad \Rightarrow l^2 = \frac{225}{16}H^2$

$\Rightarrow r^2 + h^2 = \frac{225}{16}H^2 \quad [\because r^2 + h^2 = l^2]$

$\Rightarrow r^2 + 9H^2 = \frac{225}{16}H^2$ [from Eq. (i)]

$\Rightarrow \quad r^2 = \frac{81}{16}H^2 \quad \Rightarrow \frac{r^2}{H^2} = \frac{81}{16}$

$\Rightarrow \quad \frac{r}{H} = \frac{9}{4}$

$\therefore \quad r : H = 9 : 4$

52. *(b)* Let the sides of the cuboid be x, $2x$ and $4x$ and the side of the cube is y.

$\therefore$ Volume of cuboid $= x \times 2x \times 4x$

$= 8x^3$

and volume of cube $= y^3$

By given condition,

Volume of cuboid = Volume of cube

$\therefore 8x^3 = y^3 \Rightarrow y = 2x \quad \ldots$(i)

$\therefore$ Diagonal of cuboid

$= \sqrt{x^2 + 4x^2 + 16x^2} = \sqrt{21}x$

and diagonal of cube $= y\sqrt{3} = 2x\sqrt{3}$

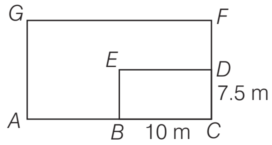
[from Eq. (i)]

Hence, the required ratio

$= \frac{\sqrt{21}x}{2x\sqrt{3}} = \sqrt{\frac{21}{4 \times 3}} = = \sqrt{1.75}$

53. *(c)* Area of tank,

$BCDE = 10 \times 7.5 = 75 \text{ m}^2$



Area of remaining field

$ABEDFGA = 125 \times 15 - 75$

$= 1800 \text{ m}^2$

Volume of Earth dug $= 10 \times 7.5 \times 6$

$= 450 \text{ m}^3$

By given condition, $1800 \times h = 450$

$\Rightarrow \quad h = \frac{1}{4} \text{ m} = \frac{1}{4} \times 100 \text{ cm} = 25 \text{ cm}$

54. *(b)* Let $r_1 = k$ and $r_2 = nk$

Since, $\quad V_1 = V_2$

$\therefore \quad \frac{1}{3}\pi r_1^2 h_1 = \pi r_2^2 h_2$

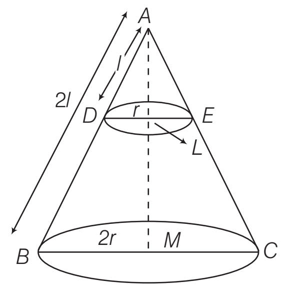
$\Rightarrow \quad \frac{1}{3}\pi k^2 \times h_1 = \pi n^2 k^2 h_2$

$\Rightarrow \quad h_1 = 3n^2 h_2$

55. *(d)* Let r be the radius, h the height and l the slant height of smaller cone

Given, $\frac{\text{slant height of original cone}}{\text{slant height of smaller cone}} = \frac{2}{1}$

Since, ΔABM and ΔADL are similar.



Therefore, radius and height of original cone will be $2r$ and $2h$ respectively.

$\therefore \quad \frac{V_1}{V_2} = \frac{\frac{1}{3}\pi \times (2r)^2 \times 2h}{\frac{1}{3}\pi r^2 h} = \frac{8r^2 h}{r^2 h} = 8$

$\therefore V_1 : V_2 = 8 : 1$

56. *(d)* By given condition,

Surface area of new sphere = 25% of 616

$4\pi r^2 = 154$

$\Rightarrow r^2 = \frac{154}{\frac{22}{7} \times 4} = \left(\frac{7}{2}\right)^2 \Rightarrow r = 3.5$ cm

57. *(d)* Let side of a cube be '*a*' unit.

Then, radius of sphere is $\frac{a}{2}$ unit.

$$\therefore \frac{\text{Volume of cube}}{\text{Volume of sphere}} = \frac{(a)^3}{\frac{4\pi}{3}\left(\frac{a}{2}\right)^3} = \frac{6}{\pi}$$

58. *(d)* Total surface area = Curved surface area of cylinder + Curved surface area of cone + Top surface area of cylinder

$= 2\pi rh + \pi rl + \pi r^2 = \pi[2rh + r^2 + rl]$

$= \pi[2 \times 3 \times 4 + 3^2 + 3\sqrt{3^2 + 4^2}]$

$= 48\pi \text{ cm}^2$

59. *(b)* Curved surface of a cylinder

$= 1000 \text{ cm}^2, \; 2\pi rh = 1000 \quad \ldots\text{(i)}$

Length of wire used in one round

= Perimeter of cylinder's base = $2\pi r$

$\therefore$ Number of rounds

$= \frac{\text{Height of cylinder}}{\text{Diameter of wire}} = \frac{h}{0.5}$

$\therefore$ Required length of wire

$= 2\pi r \times \frac{h}{0.5} = \frac{2\pi rh}{0.5}$

$= \frac{1000}{0.5} = 2000$ cm or 20 m

60. *(d)* Let h be the depth of the pit.

$\therefore$ Volume of Earth dug

$= 500 \times 250 \text{ cm}^3 = 125000 \text{ cm}^3$

But volume of pit $= 50 \times 50 \times h$

$\therefore \quad h = \frac{125000}{50 \times 50} = 50 \text{ cm} = 0.5 \text{ m}$

61. *(a)* Since, sheet is revolved about its length, threfore a cylinder is formed with $h = 7$ cm and $r = 4$ cm

$\therefore$ Volume of the figure, thus formed

$= \pi r^2 h = \frac{22}{7} \times 4 \times 4 \times 7 = 352 \text{ cm}^3$

62. *(c)* Let l, b and h be the sides of cuboid.

then, $\quad l^2 + b^2 = x^2 \quad \ldots\text{(i)}$

$b^2 + h^2 = y^2 \quad \ldots\text{(ii)}$

and $\quad h^2 + l^2 = z^2 \quad \ldots\text{(iii)}$

Adding Eq. (i), (ii) and (iii), we get

$2(l^2 + b^2 + h^2) = x^2 + y^2 + z^2$

$\Rightarrow l^2 + b^2 + h^2 = \frac{1}{2}(x^2 + y^2 + z^2) \quad \ldots\text{(iv)}$

From Eqs. (i), (ii), (iii) and (iv)

$h = \sqrt{\frac{y^2 + z^2 - x^2}{2}}, \; l = \sqrt{\frac{z^2 + x^2 - y^2}{2}}$

and $\quad b = \sqrt{\frac{x^2 + y^2 - z^2}{2}}$

Hence, volume of cuboid $= lbh$

$$= \sqrt{\frac{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}{2 \times 2 \times 2}}$$

$$= \frac{1}{2\sqrt{2}}\sqrt{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}$$

63. *(a)* Let the radius of cone and sphere be r and height of cone be h_1.

By given condition,

Volume of cone = Volume of sphere

$\therefore \; \frac{1}{3}\pi r^2 h_1 = \frac{4}{3}\pi(r)^3 \Rightarrow \frac{h_1}{2r} = \frac{2}{1}$

or, 2 : 1

64. *(d)* Let the length, breadth and height of a rectangular parallelopiped be $6x, 3x$ and x, respectively.

Also, let the side of a cube be a.

By given condition,

Surface area of a cube = Surface area of rectangular parallelopiped

$\Rightarrow 6(a)^2 = 2(6x \times 3x + 3x \times x + x \times 6x)$

$\Rightarrow \quad 6a^2 = 2(18x^2 + 3x^2 + 6x^2)$

$\Rightarrow \quad 6a^2 = 54x^2$

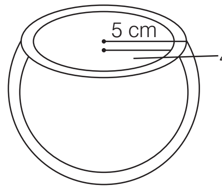
$\therefore \quad a = 3x$

Now,

$$\frac{\text{Volume of cube}}{\text{Volume of rectangular parallelopiped}} = \frac{a^3}{6x \times 3x \times x} = \frac{(3x)^3}{18x^3} = \frac{27x^3}{18x^3} = \frac{3}{2}$$

or, 3 : 2

65. *(b)* Given thickness of a metal = 1 cm



$\therefore$ Inner radius

= outer radius − thickness

$= 5 - 1 = 4$ cm

Volume of hemisphere $= \frac{2}{3}\pi(R^3 - r^3)$

$= \frac{2}{3}\pi(5^3 - 4^3) = \frac{2}{3}\pi(125 - 64)$

$= \frac{2}{3}\pi \times 61 \text{ cm}^3$

$\because$ Weight of 1 cm^3 of metal = 9 g

$\therefore$ Weight of all metal

$= 2/3\, \pi \times 61 \times 9 \text{ g} = 366\, \pi \text{ g}$

66. *(b)* Let the radius of ball be r.

$\because$ Radius of base of cylinder $= 4r$

and $\quad$ height of cylinder $= 4r$

$\therefore$ Volume of spherical ball $= \frac{4}{3}\pi r^3$

and volume of water

$= \frac{1}{2}\pi(4r)^2(4r) = 32\pi r^3$

Also, volume of remaining portion of cylinder $= 32\pi r^3$

Let number of spherical balls be n.

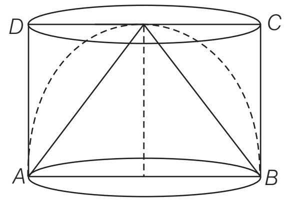
$\therefore \quad 32\pi r^3 = n \times \frac{4}{3}\pi r^3$

$\Rightarrow \quad n = 8 \times 3 = 24$

67. *(b)* From the information given in the question and the figure it is clear that

Radius of the hemisphere = radius of cone = height of cone = height of cylinder. Let it be r.

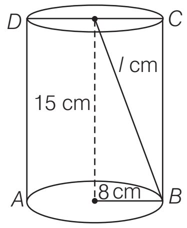
Then, ratio of volume of cylinder, hemisphere and cone.



$= \pi r^3 : \frac{2}{3}\pi r^3 : \frac{1}{3}\pi r^3$

$= 1 : \frac{2}{3} : \frac{1}{3} = 3 : 2 : 1$

68. *(a)* $l = \sqrt{15^2 + 8^2} = 17$ cm



Total surface area of the remaining solid

$= 2\pi rh + \pi r^2 + \pi rl$

$= 2\pi \times 8 \times 15 + \pi(8)^2 + \pi \times 8 \times 17$

$= 240\pi + 64\pi + 136\pi$

$= 440\pi \text{ cm}^2$

69. *(c)* Volume of spherical lead shot

$= \frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi \text{ cm}^3$

I. Volume of 8 shots

$= \frac{4}{3}\pi(0.5)^3 \times 8$

$= \frac{4}{3}\pi \text{ cm}^3$

II. Volume of both shots

$$= \frac{4}{3}\pi(0.75)^3 + \frac{4}{3}\pi(0.8)^3$$

$$= \frac{4}{3}\pi\left[\left(\frac{3}{4}\right)^3 + \left(\frac{4}{5}\right)^3\right]$$

$$= \frac{4}{3}\pi\left(\frac{27}{64} + \frac{64}{125}\right)$$

$$= \frac{4}{3}\pi\left(\frac{3375 + 4096}{8000}\right)$$

$$= \frac{4}{3}\pi\left(\frac{7471}{8000}\right) = \frac{4}{3}\pi(0.93)\text{ cm}^3$$

Hence, statement I, II are true.

70. *(c)* I. Coterminous edges are those who have same boundaries and for a cuboid, these may be considered as length, breadth and height.

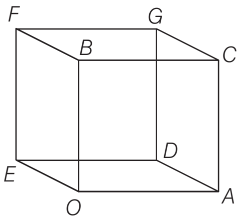
So, given statement is true

II. The surface area of a cuboid is twice the sum of the products of lengths of its coterminous edges taken two at a time.

Hence, both statements I and II are correct.

71. *(d)* I. The distance between vertices B and C is 1 cm.

II. The distance between A and B is $\sqrt{1^2 + 1^2} = \sqrt{2}$ cm



III. The distance between vertices B and D is $\sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ cm

Hence, the statements I and II, III are correct.

72. *(d)* I. It is true that the surface area of all the eight drops is greater than the surface area of big drop.

II. Let radius of spherical body be 'a'.

$\therefore$ Volume of sphere, $V = \frac{4}{3}\pi a^3$

$\therefore$ Surface area of sphere, $S = 4\pi a^2$

$$\Rightarrow \quad V^2 = \frac{16}{9}\pi^2 a^6 \propto S^3$$

III. It is true that sphere occupies biggest space but has smallest surface area. Hence, all statements are true.

73. *(c)* I. It is a correct statement.

II. The length may be πr but the breadth is always less than $\sqrt{h^2 + r^2}$ of formed rectangle,

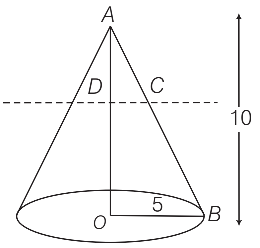
So, statement II is false.

Statement III and IV are correct.

Solution. (74-76)

As cone is divided into two parts by a plane through mid-point of its axis, therefore it will bisect the slant height.

$\Delta AOB \sim \Delta ADC$



$$\therefore \frac{DC}{OB} = \frac{AD}{AO} = \frac{1}{2} \Rightarrow DC = \frac{OB}{2}$$

$\therefore$ Radius of smaller cone will be half of original cone

74. *(c)* The volume of the original cone

$$= \frac{1}{3}\times\pi\times(5)^2\times 10 = \frac{250\pi}{3}$$

The volume of the frustum.

$$= \frac{250\pi}{3} - \frac{1}{3}\times\pi\left(\frac{5}{2}\right)^2\times 5$$

$$= \frac{250\pi}{3} - \frac{125\pi}{12} = \frac{875\pi}{12}$$

Thus, dividing the two volumes, we get $\frac{8}{7}$

75. *(c)* Area of the top circle of the frustum

$$= \pi\left(\frac{5}{2}\right)^2 = \frac{25\pi}{4}$$

76. *(d)* Let l be slant height of original cone.

Curved surface area of cone

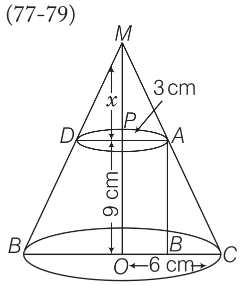
$= \pi\times 5\times l = 5\pi l$

Curved surface area of frustum

$$= 5\pi l - \pi\times\frac{5}{2}\times\frac{l}{2} = \frac{15}{4}\pi l$$

Thus, the ratio of the two will be $\frac{4}{3}$ or 4 : 3.

Solution. (77-79)



Here, $r_1 = 3$ cm, $r_2 = 6$ cm and $h = 9$ cm

77. *(a)* In ΔABC, $AC^2 = AB^2 + BC^2$

$AC^2 = 9^2 + (6-3)^2 = 81 + 9 = 90$

$\therefore AC = 3\sqrt{10}$ cm

78. *(d)* By using properties of similar triangle in ΔMPA and ΔMOC.

$$\frac{3}{6} = \frac{x}{x+9} \Rightarrow 3x + 27 = 6x$$

$\Rightarrow \quad 27 = 3x$

$\Rightarrow \quad x = 9$ cm

Hence, height of cone $= 9 + 9$

$= 18$ cm

79. *(b)* Total surface area of the frustum

$= \pi[R + r)l + r^2 + R^2]$

where, $l = \sqrt{h^2 + (R-r)^2}$

$= \pi[(6+3)\sqrt{81+9} + 9 + 36]$

$= \pi(9\sqrt{90} + 45)$

$= 9\pi(3\sqrt{10} + 5)\text{ cm}^2$

80. *(c)* Curved surface area of the well

$$= 2\pi rh = 2\times\frac{22}{7}\times 2\times 14 = 176\text{ m}^2$$

$\because$ Expense of getting per square metre plastered = ₹ 25

$\therefore$ Expense of 176 m^2 $= 176\times 25$

= ₹ 4400

81. *(a)* Given that,

Radius of sphere (r) = 9 cm = 0.09 m

and diameter of wire *(d)* = 0.4 cm

$\Rightarrow R = 0.2$ cm $= 0.002$ m

Given condition,

Volume of sphere = Volume of wire

$$\Rightarrow \frac{4}{3}\pi r^3 = \pi R^2 h \Rightarrow h = \frac{4}{3}\cdot\frac{r^3}{R^2}$$

$$= \frac{4}{3}\times\frac{0.09\times 0.09\times 0.09}{0.002\times 0.002} = 81\times = 243\text{ m}$$

82. *(c)* Total surface area of cube

$= 6\times(\text{Side})^2$

$\therefore \quad 150 = 6\times(\text{Side})^2$

$\Rightarrow \quad \text{Side}^2 = \frac{150}{6} = 25$

$\therefore \quad \text{Side} = \sqrt{25} = 5$ cm

$\therefore$ Volume of cube $= (\text{Side})^3$

$= 5\times 5\times 5 = 125\text{ cm}^3$

83. *(b)* Volume of cube $= (\text{Side})^3$

$\because \quad 729 = a^3 \Rightarrow a = 9$ cm

$\therefore$ Diagonal of cube $= \text{Side}\times\sqrt{3}$

$= 9\times\sqrt{3} = 9\sqrt{3}$ cm

84. *(a)* Curved surface area of right circular cone $= \pi r l$

$$\therefore 440 = \frac{22}{7}\times 14\times l \Rightarrow l = \frac{440\times 7}{22\times 14}$$

$= 10$ cm

85. *(d)* Volume of cylinder $= \pi r_1^2 h$

Volume of sphere $= \frac{4}{3}\pi r_2^3$

Number of spheres = 48

$$\therefore \quad \frac{\text{Volume of cylinder}}{\text{Volume of sphere}} = \frac{\pi r_1^2 h}{\frac{4}{3}\pi r_2^3}$$

$$\Rightarrow \quad \frac{\pi r_1^2 h}{\frac{4}{3}\pi r_2^3} = 48 \Rightarrow \frac{\pi r_1^3}{\frac{4}{3}\pi r_2^3} = 48$$

$$[\because r_1 = h]$$

$$\Rightarrow \frac{3}{4}\left(\frac{r_1}{r_2}\right)^3 = 48 \Rightarrow \left(\frac{r_1}{r_2}\right)^3 = \frac{48 \times 4}{3}$$

$$\Rightarrow \frac{r_1}{r_2} = (64)^{1/3} \Rightarrow \frac{r_1}{r_2} = \frac{4}{1} \Rightarrow \frac{r_2}{r_1} = \frac{1}{4}$$

Hence, the ratio of radius of ball to cylinder is 1 : 4.

86. *(a)* Volume of the cuboid = 720 cm^3

Area of base $= lb = 72$

Height of the cuboid

$$= \frac{\text{Volume of the cuboid}}{\text{Base area of the cuboid}}$$

$$= \frac{720}{72} = 10 \text{ cm}$$

Surface area of the cuboid

$$= 2(lb + bh + hl)$$

$$\Rightarrow \quad 484 = 2(72 + b \times 10 + l \times 10)$$

$$\Rightarrow 20(l + b) = 340 \Rightarrow l + b = 17$$

So, it is obvious that length, breadth and height of the cuboid is 9 cm, 8 cm and 10 cm.

87. *(a)* Surface area of the sphere

$$= 4\pi r^2$$

$$\Rightarrow \quad 616 = 4\pi r^2 \Rightarrow \pi r^2 = \frac{616}{4} = 154$$

$$\Rightarrow \quad r^2 = \frac{154 \times 7}{22} = 49$$

$$\because \quad r = \sqrt{49} = 7 \text{ cm}$$

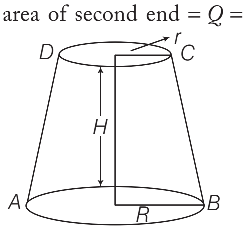
$$\therefore \quad \text{Volume of the sphere} = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times \frac{22}{7} \times 7 \times 7 \times 7 = \frac{4312}{3} \text{ cm}^3$$

88. *(a)* Given that,

Area of first end $= P = \pi r^2$

and area of second end $= Q = \pi R^2$



Given, $P < Q \Rightarrow r = \sqrt{\frac{P}{\pi}}$ and $R = \sqrt{\frac{Q}{\pi}}$

∴ Difference in radii of the ends of the frustum $= R - r$

$$= \sqrt{\frac{Q}{\pi}} - \sqrt{\frac{P}{\pi}} = \frac{\sqrt{Q} - \sqrt{P}}{\sqrt{\pi}}$$

89. *(c)* We know that,

Volume of frustum

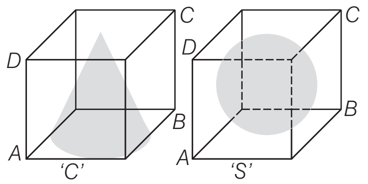
$$= \frac{\pi H}{3}(R^2 + r^2 + Rr)$$

$$= \frac{\pi}{3} H \left\{\left(\sqrt{\frac{Q}{\pi}}\right)^2 + \left(\sqrt{\frac{P}{\pi}}\right)^2 + \sqrt{\frac{Q}{\pi}} \cdot \sqrt{\frac{P}{\pi}}\right\}$$

$$= \frac{\pi H}{3}\left\{\frac{Q}{\pi} + \frac{P}{\pi} + \frac{\sqrt{PQ}}{\pi}\right\}$$

$$= \frac{H}{3}(P + Q + \sqrt{PQ})$$

90. *(b)* Let the side of both cubes be a, then the height and radius of a cone, $h = a$ and $r = \frac{a}{2}$



and radius of sphere $(R) = \frac{a}{2}$

$$\therefore \quad \text{Volume of cone } (C) = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi\left(\frac{a}{2}\right)^2 a \Rightarrow C = \frac{\pi a^3}{12} \quad \text{...(i)}$$

and volume of sphere $(S) = \frac{4}{3}\pi R^3$

$$= \frac{4}{3}\pi\left(\frac{a}{2}\right)^3 = \frac{\pi a^3}{6} \quad \text{... (ii)}$$

From Eqs. (i) and (ii), $S = 2C$

91. *(d)* If 10 circular plates each of thickness 3 cm, are placed one above the other, then it forms a cylinder with height (3× 10=30 cm) and a hemisphere of radius 6 cm is placed on the top just to cover the cylinder that means its radius is also 6 cm.

∴ Radius of hemi-sphere (R) = 6 cm

Radius of cylinder (r) = 6 cm

and height of cylinder (h) = 30 cm

∴ Volume of the solid = Volume of cylinder + Volume of hemisphere

$$= \pi r^2 h + \frac{2}{3}\pi R^3$$

$$= \pi(6)^2 \times 30 + \frac{2}{3}\pi(6)^3$$

$$= \pi \times 36 \times 30 + \frac{2}{3}\pi \times 216$$

$$= 1080\pi + 2\pi \times 72$$

$$= 1080\pi + 144\pi = 1224\pi \text{ cm}^3$$

92. *(a)* Given that, the height and radius of a right circular metal cone (solid) are 8 cm and 2 cm, respectively.

i.e. h = 8 cm and r = 2 cm

Let the radius of the sphere be R.

Then, by condition, $\frac{1}{3}\pi r^2 h = \frac{4}{3}\pi R^3$

$$\Rightarrow \quad 4 \times 8 = 4R^3 \Rightarrow R^3 = (2)^3$$

$$\therefore \quad R = 2$$

Hence, radius of the sphere = 2 cm

93. *(c)* Let the edge of a square be x. Then, its volume is x^3 and sum of its edges is $12x$.

Now, by condition, $x^3 = 12x$

$$\Rightarrow \quad x(x^2 - 12) = 0$$

$$\Rightarrow \quad x^2 = 12 \quad [\because x \neq 0]$$

$$\therefore \quad \text{Its total surface area} = 6x^2$$

$$= 6(12) = 72 \text{ sq units}$$

94. *(a)* Given that,

Diameter of a right circular cone = 7 cm

∴ Radius of a right circular cone $= \frac{7}{2}$ cm

and slant height of a right circular cone (l) = 10 cm

∴ Lateral surface area of a cone $= \pi r l$

$$= \frac{22}{7} \times \frac{7}{2} \times 10 = 11 \times 10 = 110 \text{ cm}^2$$

95. *(d)* Let the height and radius of solid cylinder be h and r cm, respectively.

Given that, r = 5 cm

and total surface area = 660 cm^2

$$\Rightarrow \quad 2\pi rh + 2\pi r^2 = 660$$

$$\Rightarrow \quad 2\pi r(h + r) = 660$$

$$\Rightarrow \quad (h + 5) = \frac{330}{5\pi} = \frac{330}{5} \times \frac{7}{22}$$

$$\Rightarrow \quad h = \frac{66 \times 7}{22} - 5 = 21 - 5 = 16$$

∴ Required height = 16 cm

96. *(a)* Let the diameters of two spheres be d_1 and d_2, respectively.

$$\because \quad d_1 : d_2 = 3:5$$

∴ Ratio of their surface areas

$$= \frac{4\pi r_1^2}{4\pi r_2^2} = \frac{(2r_1)^2}{(2r_2)^2} = \frac{d_1^2}{d_2^2}$$

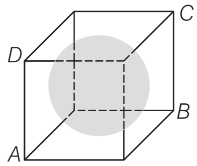
$$= \left(\frac{d_1}{d_2}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25} = 9 : 25$$

97. *(c)* From figure, it is clear that

Diameter of a sphere = Side of the cube = 3 cm

$$\because \quad \text{Radius} = \frac{3}{2} \text{ cm}$$

$\therefore$ Volume of the largest sphere



$= \dfrac{4}{3}\pi\,(\text{Radius})^3 = \dfrac{4}{3}\pi\left(\dfrac{3}{2}\right)^3$

$= \dfrac{4}{3}\pi \cdot \dfrac{27}{8} = \dfrac{9}{2}\pi = 4.5\pi \text{ cm}^3$

98. *(d)* Given that, radius of a right circular cone $(r) = 12$ m

Height of a right circular cone $(h) = 5$ m

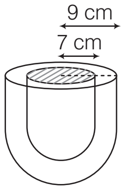
$\therefore$ Lateral height of cone $(l) = \sqrt{r^2 + h^2}$

$= \sqrt{144 + 25} = \sqrt{169} = 13$ m

$\therefore$ Required quantity of cloth to roll up to form a right circular tent $= \pi r l$

$= \pi \times (12) \times (13) = 156\ \pi\ \text{m}^2$

99. *(a)* Given that, Outer radius of hemispherical shell $(R) = 9$ cm



and inner radius of hemispherical shell $(r) = 7$ cm

$\therefore$ Volume of a hemispherical shell

$= \dfrac{2}{3}\pi\,(R^3 - r^3)$

$= \dfrac{2}{3} \times \dfrac{22}{7} \times (729 - 343)$

$= \dfrac{2}{3} \times \dfrac{22}{7} \times 386 = \dfrac{16984}{21} = 808.76$

$\approx 808 \text{ cm}^3$ (approx)

100. *(b)* Let r be the radius of the sphere.

Given that, volume of sphere $= 36\ \pi$

$\Rightarrow \quad \dfrac{4}{3}\pi r^3 = 36\ \pi \Rightarrow r^3 = 27 = (3)^3$

$\because \quad r = 3$ cm

$\therefore$ Diameter of sphere

$= 2r = 2\,(3) = 6$ cm

and surface area of sphere $= 4\pi r^2$

$= 4\ \pi\,(3)^2 = 36\ \pi \text{ cm}^2$

$\therefore$ Required ratio

$= \dfrac{\text{Surface area of sphere}}{\text{Diameter of sphere}} = \dfrac{36\ \pi}{6} = 6\pi$

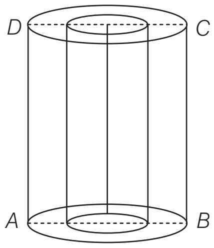
101. *(d)* Given that,

Internal diameter of the tube = 6 cm

$\therefore$ Internal radius $(r) = 3$ cm

Height of the tube $(h) = 10$ cm

Thickness of the metal = 1 cm



$\because$ Outer radius (R)

= Thickness of the metal + Internal radius

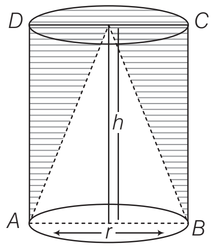
$= 1 + 3 = 4$ cm

$\therefore$ Outer curved surface area

$= 2\pi Rh = 2 \times \pi \times 4 \times 10 = 80\pi$

102. *(a)* Let the height and radius of right circular cylinder be h and r, respectively.

Then, volume of cylinder $= \pi r^2 h$



Volume of circular cone $= \dfrac{1}{3}\pi r^2 h$

$\therefore$ Required ratio

$= \dfrac{\text{Volume of utilised wood}}{\text{Volume of wasted wood}}$

$= \dfrac{\text{Volume of right circular cone}}{\text{Volume of right circular cylinder} - \text{Volume of right circular cone}}$

$= \dfrac{\frac{1}{3}\pi r^2 h}{\pi r^2 h - \frac{1}{3}\pi r^2 h} = \dfrac{\frac{1}{3}\pi r^2 h}{\frac{2}{3}\pi r^2 h} = \dfrac{1}{2} = 1:2$

103. *(c)* Since, volume of cone and pyramid $= \dfrac{1}{3} \times$ Base area $\times$ Height

Therefore, they have same volume but their surface areas are not same as nothing can be said about their slant heights.

104. *(b)* Given that, height of cone $(h) = 3$ cm

and slant height of cone $(l) = 5$ cm

Let r be the radius of cone.Then,

$l = \sqrt{r^2 + h^2} = 5 \Rightarrow r^2 + h^2 = 25$

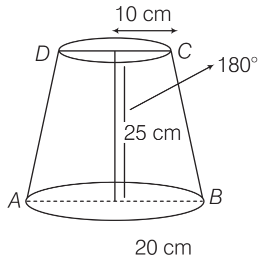
$\Rightarrow \quad r^2 = 25 - 9 = 16 \Rightarrow r = 4$ cm

$\therefore$ Volume of cone $= \dfrac{1}{3}\pi r^2 h$

$= \dfrac{1}{3} \times \dfrac{22}{7} \times 16 \times 3 = \dfrac{352}{7} = 50.3 \text{ cm}^3$

105. *(a)* Given that,

Height of bucket = 25 cm



Radius of top $(R) = 20$ cm

and radius of bottom $(r) = 10$ cm

$\therefore$ Capacity of bucket

$= \dfrac{\pi}{3} h\,(R^2 + r^2 + rh)$

$= \dfrac{\pi}{3} \times 25\,(400 + 100 + 200) \text{ cm}^3$

$= \dfrac{\pi}{3} \times 25 \times 700 \text{ cm}^3$

$= \dfrac{\pi}{3} \times \dfrac{175 \times 100}{1000}\ \text{L} = \dfrac{17.5\pi}{3}\ \text{L}$

106. *(d)* Given that, $h = 15$ cm

Let r be the radius of cylinder.

Lateral surface area of cylinder $= 2\pi rh$

$\Rightarrow \quad 2\pi rh = 660$ [given]

$\Rightarrow \quad \pi rh = 330$

$\Rightarrow \quad \dfrac{22}{7} \times r \times 15 = 330 \Rightarrow \dfrac{22}{7} \times r = 22$

$\because \quad r = 7$ cm

$\therefore$ Volume of cylinder $= \pi r^2 h$

$= \dfrac{22}{7} \times 49 \times 15$

$= 22 \times 7 \times 15 = 2310 \text{ cm}^3$

107. *(b)* Given that, the diameter of Moon is approximately one-fourth of the diameter of Earth.

If radius of Moon $= r$

Then, radius of Earth $= 4r$

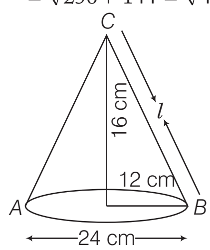
$\therefore \dfrac{\text{Volume of Moon}}{\text{Volume of Earth}} = \dfrac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi\,(4r)^3}$

$= \dfrac{r^3}{64\,r^3} = \dfrac{1}{64} = 1:64$

108. *(c)* Slant height, $l = \sqrt{h^2 + r^2}$

$= \sqrt{16^2 + 12^2}$

$= \sqrt{256 + 144} = \sqrt{400} = 20$ cm



Curved surface area $= \pi rl$

$$= \frac{22}{7} \times 12 \times 20 \text{ cm}^2$$

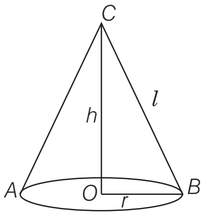
Cost of painting the surface of the cap

$$= \frac{22}{7} \times 12 \times 20 \times 0.70 = ₹\ 528$$

109. *(b)* Slant height, $l = \sqrt{h^2 + r^2}$

$$= \sqrt{(24)^2 + (7)^2} = \sqrt{576 + 49}$$

$$= \sqrt{625} = 25$$



Total surface area $= \pi r(l + r)$

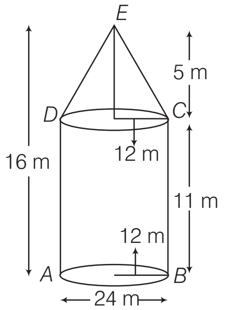
$$= \frac{22}{7} \times 7(25 + 7)$$

$$= \frac{22}{7} \times 7 \times 32 = 704 \text{ cm}^2$$

110. *(c)* Radius of cone, $r = \frac{24}{2} = 12$ cm

Slant height of the cone, $l = \sqrt{5^2 + 12^2}$

$$= \sqrt{25 + 144} = \sqrt{169} = 13 \text{ m}$$



Curved surface area for conical portion

$$= \pi rl = \frac{22}{7} \times 12 \times 13 = \frac{3432}{7} \text{ m}^2$$

111. *(d)* Curved surface area of cylinder

$= 2\pi rh = x$

Volume of cylinder $= \pi r^2 h = y$

$$\Rightarrow \quad \frac{2\pi rh}{\pi r^2 h} = \frac{x}{y} \Rightarrow r = \frac{2y}{x}$$

Also, $\quad h = \frac{x}{2\pi r}$

$$\therefore \quad \text{Required ratio} = \frac{h}{r} = \frac{\frac{x}{2\pi r}}{\frac{2y}{x}}$$

$$= \frac{x}{2\pi \cdot \frac{2y}{x}} \times \frac{x}{2y} = \frac{x^3}{8\pi y^2}$$

So, the ratio is not independent of x or y.

112. *(a)* Let common radius be r cm.

Then, height of cylinder $= h$

and height of cone $= h'$

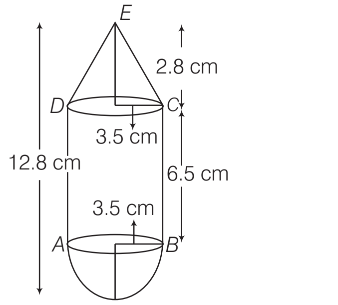
$\therefore$ Volume of the complete structure

$$= \frac{1}{3}\pi r^2 h' + \pi r^2 h + \frac{2}{3}\pi r^3$$

$$= \pi r^2 \left(\frac{h'}{3} + h + \frac{2}{3}r\right)$$

$$= \pi(3.5)^2 \left(\frac{2.8}{3} + 6.5 + \frac{2}{3} \times 3.5\right)$$

$$= \pi \times 3.5 \times 3.5 \times 9.76 = 375.86 \text{ cm}^3$$



Hence, volume (V) of the structure lies between 370 cm^3 and 380 cm^3.

113. *(b)* Given, radius of cone $= \frac{6}{2} = 3$ cm

and height of cone = 4 cm

Now, curved surface area $= \pi rl$

where, $l = \sqrt{r^2 + h^2} = \sqrt{3^2 + 4^2} = 5$ cm

$\therefore$ Curved surface area $= \pi \times 3 \times 5 = 15\pi$

$$= \frac{15 \times 22}{7} \approx 47 \text{ cm}^2$$

114. *(d)* Volume of clay required

$$= \pi\left[\left(\frac{5.1}{2}\right)^2 - \left(\frac{4.5}{2}\right)^2\right] \times 21$$

$$= \pi[(2.55)^2 - (2.25)^2] \times 21$$

$$= \pi(0.3 \times 4.8) \times 21 = 30.24\,\pi \text{ cm}^2$$

115. *(a)* Surface area of cube which can be painted $= 6\,(\text{Side})^2 = 6(2)^2 = 24 \text{ cm}^2$

Now, surface area of cuboid which can be painted

$$= 2(lb + bh + lh)$$

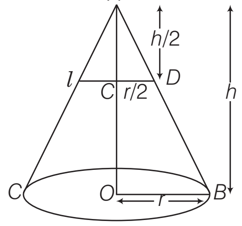
$$= 2(2 + 6 + 3) = 22 \text{ cm}^2$$

Total surface area of both cube and cuboid

$$= 22 + 24 = 46 \text{ cm}^2 < 54 \text{ cm}^2$$

Hence, both cube and cuboid can be painted.

116. *(b)* Let the cone is divided into two parts by a line l.



In ΔAOB and ΔACD, $\Delta AOB \sim \Delta ACD$

By basic proportionality theorem,

$$CD = \frac{r}{2}, \text{ since } AC = \frac{h}{2}$$

$$\text{Now, ratio} = \frac{\text{Volume of original cone}}{\text{Volume of smaller cone}}$$

$$= \frac{\frac{1}{3}\pi r^2 h}{\frac{1}{3}\pi\left(\frac{r}{2}\right)^2\left(\frac{h}{2}\right)} = \frac{8}{1}$$

$\therefore$ Required ratio = 8 : 1

117. *(c)* Volume of each small sphere

$$= \frac{\text{Volume of bigger sphere}}{\text{Number of small spheres}} = \frac{\frac{4}{3}\pi(4)^3}{64}$$

$$= \frac{4}{3} \times \frac{\pi \times 4 \times 4 \times 4}{64} = \frac{4}{3}\pi \text{ cm}^3$$

Let radius of small sphere be r'.

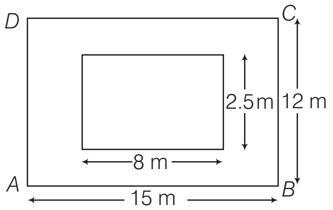
$$\therefore \quad \frac{4}{3}\pi r'^3 = \frac{4}{3}\pi \quad \Rightarrow r' = 1 \text{ cm}$$

Now, surface area of small sphere

$$= 4\pi r'^2 = 4\pi \text{ cm}^2$$

118. *(c)* Volume of Earth dug out

$$= 8 \times 2.5 \times 2 = 40 \text{ m}^3$$



Area where Earth is spread × Field level raised

= Volume of Earth dug out

$$\Rightarrow \quad [(12 \times 15) - (8 \times 2.5)] \times h = 40$$

$$\therefore \quad h = \frac{40}{180 - 20} = \frac{40}{160} = \frac{1}{4} \text{ m}$$

$$= \frac{100}{4} \text{ cm} = 25 \text{ cm}$$

119. *(a)* Given, surface area of sphere

$= 616 \text{ cm}^2$

$$\Rightarrow \quad 4\pi r^2 = 616 \Rightarrow r^2 = \frac{616 \times 7}{4 \times 22}$$

$$\Rightarrow \quad r^2 = 7 \times 7 \;\therefore\; r = 7 \text{ cm}$$

$\therefore$ Diameter of the largest circle lying on sphere

$= 2 \times r = 14$ cm

120. *(c)* Given, volume of cube $= 216x^3$

$(\text{Side})^3 = 216x^3 \Rightarrow \text{Side} = 6x$

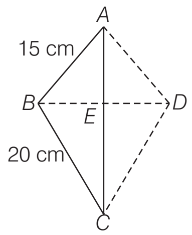
Since, sphere is enclosed in hollow cube.

$\therefore$ Diameter of sphere $= 6x$

Now, surface area of sphere $= 4\pi r^2$

$$= 4\pi\left(\frac{6x}{2}\right)^2 = 36\,\pi x^2$$

121. *(c)* Let ABC be a right angled triangle. Then, hypotenuse, $AC = 25$ cm



Let $\quad AB = 3x$ and $BC = 4x$

Thus, $AC^2 = AB^2 + BC^2$ [by pythagoras theorem]

$\Rightarrow \quad (25)^2 = (3x)^2 + (4x)^2$

$\Rightarrow \quad (25)^2 = 9x^2 + 16x^2$

$\Rightarrow \quad 25 = x^2 \Rightarrow x = 5$

$\therefore \quad AB = 15$ cm and $BC = 20$ cm

Now, ΔABC revolves about AC, so it forms two cones ABD and BCD.

Since, ΔAEB and ΔABC are similar.

$\therefore \quad \dfrac{BE}{BC} = \dfrac{AB}{AC} \Rightarrow \dfrac{BE}{20} = \dfrac{15}{25}$

$\Rightarrow \quad BE = \dfrac{15 \times 20}{25} = 12$ cm

So, radius of the base of cone

$= BE = 12$ cm

In right angled ΔAEB,

$AE = \sqrt{(AB)^2 - (BE)^2}$

$= \sqrt{(15)^2 - (12)^2}$

$= \sqrt{225 - 144} = \sqrt{81} = 9$ cm

So, height of cone $ABD = AE = 9$ cm

$\therefore \quad$ Height of cone $BCD = AC - AE$

$= 25 - 9 = 16$ cm

Now, volume of cone $ABD = \dfrac{1}{3}\pi r^2 h$

$= \dfrac{1}{3}\pi(12)^2 \times 9 = 432\pi \text{ cm}^3$

and volume of cone

$BCD = \dfrac{1}{3}\pi(12)^2 \times 16 = 768\pi \text{ cm}^3$

$\therefore$ Required volume of double cone

$= 432\pi + 768\pi = 1200\pi$

$= 1200 \times 3.14 \quad [\because \pi = 3.14]$

$= 3768 \text{ cm}^3$

122. *(d)* $\because$ Surface area of cone $ABD = \pi rl$

$= \pi \times 12 \times 15 = 180\pi \text{ cm}^2$

and surface area of cone

$BCD = \pi \times 12 \times 20 = 240\pi \text{ cm}^2$

$\therefore$ Required surface area of double cone

$= 180\pi + 240\pi = 420\pi$

$= 420 \times 3.14 = 1318.8 \text{ cm}^2$

123. *(c)* Let side of a cube be x.

Then, surface area of cube $= 6x^2$

If the side of cube is increased by 100%.

Then, new side of cube $= x + 100\%$ of x

$= x + x = 2x$

$\therefore \quad$ New surface area of cube $= 6(2x)^2$

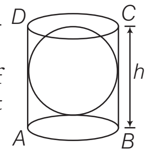
$= 6 \times 4x^2 = 24x^2$

Now, increase percentage in surface area

$= \dfrac{24x^2 - 6x^2}{6x^2} \times 100$

$= \dfrac{18}{6} \times 100 = 300\%$

124. *(a)* Let radius of the sphere be r.



Since, cylinder circumscribes a sphere.

$\therefore$ Radius of the base of cylinder $= r$ and height of cylinder $= 2r$

= Diameter of sphere

Now, volume of sphere $= \dfrac{4}{3}\pi r^3$

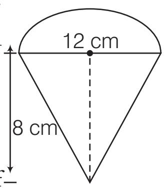
and volume of cylinder

$= \pi r^2 h = \pi r^2(2r) = 2\pi r^3$

$\therefore \quad$ Required ratio $= \dfrac{\frac{4}{3}\pi r^3}{2\pi r^3} = \dfrac{4}{3 \times 2}$

$= \dfrac{2}{3} = 2 : 3$

125. *(a)* Given, diameter of the base of the conical portion = 12 cm



$\therefore$ Radius of conical portion = 6 cm

Radius of hemisphere = 6 cm and height of conical portion = 8 cm

$\therefore \quad$ Slant height of conical portion

$= \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$

$[\because l = \sqrt{r^2 + h^2}]$

$= \sqrt{100} = 10$ cm

Now, total surface area of the toy

= Curved surface area of conical portion + Curved surface area of hemisphere

$= \pi rl + 2\pi r^2 = \pi(rl + 2r^2)$

$= \pi(6 \times 10 + 2 \times 6 \times 6)$

$= \pi(60 + 72) = 132\pi \text{ cm}^2$

126. *(b)* Volume of the toy

= Volume of conical portion + Volume of hemisphere

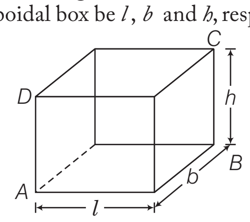
$= \dfrac{1}{3}\pi r^2 h + \dfrac{2}{3}\pi r^3$

$= \pi\left(\dfrac{1}{3}r^2 h + \dfrac{2}{3}r^3\right)$

$= \pi\left[\dfrac{1}{3} \times (6)^2 \times 8 + \dfrac{2}{3} \times (6)^3\right]$

$= \pi[96 + 144] = 240\pi \text{ cm}^3$

127. *(b)* Let length, breadth and height of a cuboidal box be l, b and h, respectively.



Given, areas of the three adjacent faces are x, $4x$ and $9x$ sq units.

Now, $lb = x$

[$\because$ area of rectangular face = length $\times$ breadth]

Similarly, $bh = 4x$ and $lh = 9x$

Now, $(lb) \cdot (bh) \cdot (lh) = (x) \cdot (4x) \cdot (9x)$

$\Rightarrow \quad (lbh)^2 = 36x^3 \Rightarrow lbh = \sqrt{36x^3}$

$\therefore \quad lbh = 6x^{3/2}$

Hence, volume of cuboidal box

$= lbh = 6x^{3/2}$ cu units

128. *(c)* In a cuboid, 4 perpendicular face pairs in bottom surface, 4 perpendicular face pairs in top surface and 4 perpendicular face pairs in vertical surface.

Hence, total perpendicular pairs are 12.

129. *(c)* I. Surface area of sphere A

$= 4\pi(6)^2 = 144\pi \text{ cm}^2$

Surface area of sphere $B = 4\pi(8)^2$

$= 256\pi \text{ cm}^2$

Surface area of sphere $C = 4\pi(10)^2$

$= 400\pi \text{ cm}^2$

Now, sum of surface area of spheres A and B

$= 144\pi + 256\pi = 400\pi \text{ cm}^2$

= Surface area of sphere C

Hence, statement I is correct.

II. $\because$ Volume of sphere D

$= \dfrac{4}{3}\pi(12)^3 = 2304\pi \text{ cm}^3$

Volume of sphere $A = \dfrac{4}{3}\pi(6)^3 \text{ cm}^3$

Volume of sphere $B = \dfrac{4}{3}\pi(8)^3 \text{ cm}^3$

and volume of sphere

$C = \dfrac{4}{3}\pi(10)^3 \text{ cm}^3$

Now, sum of volumes of spheres A, B and C

$= \dfrac{4}{3}\pi[6^3 + 8^3 + 10^3]$

$= \dfrac{4}{3}\pi[216 + 512 + 1000]$

$= 2304\pi \text{ cm}^3$ = Volume of sphere D

Hence, statement II is also correct.

130. *(a)* Given, diameter of a sphere, $d = 6$ cm

$\therefore$ Radius of a sphere, $r = \frac{d}{2} = \frac{6}{2} = 3$ cm

Let the radius of wire be R cm.

Also, given the length of wire, $H = 36$ m $= 3600$ cm

According to the question,

Volume of sphere = Volume of wire

$$\Rightarrow \frac{4}{3}\pi r^3 = \pi R^2 H$$

$$\Rightarrow \frac{4}{3} \times (3)^3 = R^2 \times 3600$$

$$\Rightarrow R^2 = \frac{4 \times 3^2}{3600} = \frac{(6)^2}{(60)^2}$$

$$\therefore R = \frac{6}{60} = \frac{1}{10} = 0.1 \text{ cm}$$

131. *(b)* Given, side of a cube = 2 cm

$\therefore$ Maximum distance between two points of a cube

= Length of diagonal

$= \sqrt{3} \times$ Side $= 2\sqrt{3}$ cm

132. *(d)* Let the slant height of cone be l cm.

Given, $h = 24$ cm and $r = 7$ cm

$$\therefore l^2 = h^2 + r^2$$

$$\Rightarrow l^2 = 24^2 + 7^2 = 576 + 49$$

$$\Rightarrow l^2 = 625 \Rightarrow l = 25 \text{ cm}$$

Total surface area = Curved surface area of cone + curved surface area of hemisphere

$= \pi rl + 2\pi r^2 = \pi r(l + 2r)$

$= \pi \times 7(25 + 2 \times 7) = 7\pi[25 + 14]$

$= 7\pi \times 39 = 273\pi$ cm^2

133. *(a)* Let the radius of the sphere and side of cube be r and a respectively.

$\therefore$ Total surface area of sphere

$(S_1) = 4\pi r^2$ sq units

and total surface area of cube

$(S_2) = 6a^2$ sq units

Now, according to the question,

$$S_1 = S_2 \Rightarrow 4\pi r^2 = 6a^2 \Rightarrow \frac{r^2}{a^2} = \frac{6}{4\pi} \quad \text{...(i)}$$

$\therefore$ Volume of sphere $(V_1) = \frac{4}{3}\pi r^3$ cu units

and volume of cube $(V_2) = a^3$ cu units

Now,

$$\left(\frac{V_1}{V_2}\right)^2 = \left(\frac{\frac{4}{3}\pi r^3}{a^3}\right)^2 = \frac{16}{9}\pi^2\left(\frac{r^2}{a^2}\right)^3$$

$$= \frac{16}{9}\pi^2\left(\frac{6}{4\pi}\right)^3 \quad \text{[from Eq. (i)]}$$

$$= \frac{16}{9}\pi^2 \times \frac{6 \times 6 \times 6}{4 \times 4 \times 4 \times \pi^3}$$

$$= \frac{6}{\pi} = 6 : \pi$$

Hence, the ratio of square of their volume is $6 : \pi$.

134. *(a)* Let radius of sphere and cone be r.

$\therefore$ Volume of sphere $V_1 = \frac{4}{3}\pi r^3$

and volume of cone $V_2 = \frac{1}{3}\pi r^2 h$

Now, according to the question,

$$V_1 = 2V_2 \Rightarrow \frac{4}{3}\pi r^3 = 2 \times \frac{1}{3}\pi r^2 h$$

$$\Rightarrow 2r^3 = r^2 h \Rightarrow 2r = h \Rightarrow \frac{h}{r} = \frac{2}{1}$$

$$\therefore h : r = 2 : 1$$

135. *(a)* Let the radius of sphere be r.

$\therefore$ Volume of sphere $= \frac{4}{3}\pi r^3$

According to the question,

If the radius of a sphere is increased by 10%.

Then, new radius $r' = r + r \times 10\%$

$$= r + \frac{r}{10} = \frac{11r}{10}$$

$\therefore$ New volume of sphere $= \frac{4}{3}\pi r'^3$

$$= \frac{4\pi}{3}\left(\frac{11r}{10}\right)^3 = \frac{4}{3}\pi \times \frac{1331r^3}{1000}$$

Increased volume

$$= \frac{4}{3}\pi \times \frac{1331}{1000}r^3 - \frac{4}{3}\pi r^3 = \frac{4}{3}\pi r^3 \times \frac{331}{1000}$$

Increased percentage

$$= \frac{\frac{4}{3}\pi r^3 \times \frac{331}{1000}}{\frac{4}{3}\pi r^3} \times 100\% = 33.1\%$$

136. *(b)* Since, three metallic spheres are melted to form a single sphere.

So, the sum of volume of three spherical solid sphere.

$$= \frac{4}{3}\pi(6)^3 + \frac{4}{3}\pi(8)^3 + \frac{4}{3}\pi(10)^3$$

$$= \frac{4}{3}\pi[6^3 + 8^3 + 10^3]$$

Let r be the radius of new sphere

$$\therefore \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(6^3 + 8^3 + 10^3)$$

$$\Rightarrow r^3 = 6^3 + 8^3 + 10^3$$

$$\Rightarrow r^3 = 216 + 512 + 1000$$

$$\Rightarrow r^3 = 1728$$

$$\Rightarrow r^3 = 12 \times 12 \times 12$$

$$\Rightarrow r = 12 \text{ cm}$$

$\therefore$ Diameter of the new sphere

$= 2r = 2 \times 12 = 24$ cm

137. *(b)* Let the radius and height of the cone be r and h, respectively.

$\therefore$ Initial volume of cone

$$(V) = \frac{1}{3}\pi r^2 h \quad \text{...(i)}$$

new height, $H = h + h \times 200\%$

$$= h + \frac{200h}{100} = 3h$$

and new radius, $R = r - r \times 50\%$

$$= r - \frac{50r}{100} = \frac{r}{2}$$

$\therefore$ New volume of cone $(V_2) = \frac{1}{3}\pi R^2 H$

$$= \frac{1}{3}\pi\left(\frac{r}{2}\right)^2 \times 3h = \frac{1}{3}\pi \times \frac{r^2}{4} \times 3h$$

$$= \frac{1}{3}\pi r^2 h \times \frac{3}{4} = \frac{3V}{4} \quad \text{[from Eq. (i)]}$$

Decrease in volume $= V - \frac{3V}{4} = \frac{V}{4}$

$\therefore$ Decrease percentage in volume

$$= \frac{\text{Decrease in Volume}}{\text{Initial Volume}} \times 100\%$$

$$= \frac{V/4}{V} \times 100\% = \frac{V}{4V} \times 100\% = 25\%$$

138. *(c)* Given a rectangular paper of 44 cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper.

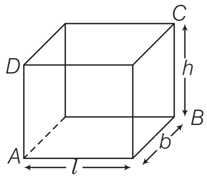
$\therefore$ Circumference of the base of cylinder $= 44$

i.e. $2\pi r = 44 \Rightarrow r = \frac{44}{2\pi} = \frac{44 \times 7}{2 \times 22}$

$\therefore r = 7$ cm

Hence, the radius of the base of the cylinder is 7 cm.

139. *(c)* Let the dimensions of cuboid be l, b and h respectively.



For first face

$$l^2 + b^2 = 13^2 \quad [\because l^2 + b^2 = d^2]$$

$$\Rightarrow l^2 + b^2 = 169 \quad \text{...(i)}$$

For second face

$$b^2 + h^2 = (\sqrt{281})^2$$

$$\Rightarrow b^2 + h^2 = 281. \quad \text{...(ii)}$$

For third face

$$h^2 + l^2 = 20^2 \Rightarrow h^2 + l^2 = 400 \quad \text{...(iii)}$$

On adding Eqs. (i), (ii) and (iii), we get

$$2(l^2 + b^2 + h^2) = 850$$

$$\Rightarrow l^2 + b^2 + h^2 = 425 \quad \text{...(iv)}$$

Subtracting Eq. (i) form Eq. (iv), we get

$h^2 = 256 \Rightarrow h = 16$ units

similarly, $b^2 = 25 \Rightarrow b = 5$ units

and $l^2 = 144 \Rightarrow l = 12$ units

$\therefore$ Total surface area of cuboid

$= 2(lb + bh + lh)$

$= 2(12 \times 5 + 5 \times 16 + 16 \times 12)$
$= 2(60 + 80 + 192)$
$= 2 \times 332 = 664$ sq units

Hence, the total surface area of cuboid is 664 sq units.

140. *(c)* Given, radius of vessel, $(r) = 4$ cm and radius of sphere $(R) = 3$ cm

Let height of vessel be h cm, then volume of vessel $= \pi r^2 h = \pi(4^2)h = 16\pi h$

and volume of sphere $= \frac{4}{3}\pi R^3 = 36\pi$

Let H be the rise in water level

$\therefore$ Volume of water displaced = volume of sphere

$\Rightarrow 16\pi H = 36\pi \Rightarrow H = \frac{36}{16} = 2.25$ cm

141. *(c)* Volume of paint required

$= 2 \times (3 \times 2 \times 0.0001) + 2 \times (2 \times 1.75 \times 0.0001) + 2 \times (3 \times 1.75 \times 0.0001)$

$= 0.0012 + 0.0007 + 0.00105 = 0.00295 \text{ m}^3$

Volume of cubical box $= 10^3$.

$= 1000 \text{ cm}^3 = 0.001 \text{ m}^3$

$\therefore$ Boxes required $= \frac{0.00295}{0.001} \approx 3.$

142. *(d)* Let the sides of a cube be l cm.

Then, $6l^2 = 13254$

$\Rightarrow \quad l^2 = \frac{13254}{6} = 2209 \Rightarrow l = 47$

$\therefore$ Length of diagonal $= l\sqrt{3} = 47\sqrt{3}$ cm

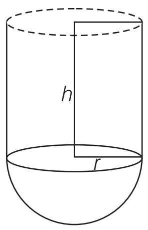
143. *(b)* Let the height of cylindrical portion be h.

Then, $\frac{2}{3}\pi r^3 + \pi r^2 h = 3312\pi$

$\Rightarrow \quad \pi r^2\left[\frac{2}{3}r + h\right] = 3312\pi$

$\Rightarrow 12 \times 12\left[\frac{2}{3} \times 12 + h\right] = 3312$

$\Rightarrow \quad 8 + h = 23 \Rightarrow h = 15$m



$\because \frac{\text{Surface area of hemisphere}}{\text{Surface area of cylinder}} = \frac{2\pi r^2}{2\pi rh}$

$= \frac{r}{h} = \frac{12}{15} = \frac{4}{5}$

$\therefore \quad r : h = 4 : 5$

144. *(a)* Let the length of water tank be lm.

volume of tank $= l \times l \times l = l^3$

volume of water tank after taking out water

$= l \times l \times (l - 2) = l^2(l - 2)$

$\therefore l^3 - 128 = l^2(l - 2)$

$\Rightarrow \quad 2l^2 = 128 \Rightarrow l^2 = 64 \Rightarrow l = 8$ m

So, volume of cubical tank

$= 8^3 = 512 \text{ m}^3.$

145. *(b)* Let the sides of the cuboid be l, b and h respectively.

Volume of cuboid,

$V = l \times b \times h \quad \ldots\text{(i)}$

Given, $x = lb$, $y = bh$ and $z = lh$

$\therefore \quad xyz = lb \times bh \times lh$
$= (lbh)^2 = V^2$ [using Eq. (i)]

146. *(a)* On each face, we can draw four right angled triangles by choosing one vertex to serve as the right angled and two adjacent vertices. Since there are six faces, you can form $6 \times 4 = 24$ triangles.

147. *(a)* Let the area of each rectangle be A.

Then, breadth of R_1, R_2 and R_3 are $\frac{A}{x_1}$, $\frac{A}{x_2}$ and $\frac{A}{x_3}$ respectively

Cylinder is formed by joining parallel side of breadth.

Here, length becomes height

and $2\pi r$ = breadth $= \frac{A}{x} \Rightarrow r = \frac{A}{2\pi x}$

$\therefore \quad V = \pi r^2 h = \pi \frac{A^2}{4\pi^2 x^2} \times x = \frac{A^2}{4\pi x}$

Now, V_1, V_2 and V_3 are $\frac{A^2}{4\pi x_1}, \frac{A^2}{4\pi x_2}$

and $\frac{A^2}{4\pi x_3}$ respectively

$\because \quad x_1 < x_2 < x_3 \Rightarrow \frac{1}{x_1} > \frac{1}{x_2} > \frac{1}{x_3}$

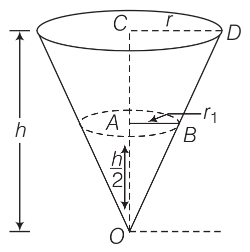
$\Rightarrow \quad \frac{A^2}{4\pi x_1} > \frac{A^2}{4\pi x_2} > \frac{A^2}{4\pi x_3}$

$\therefore \quad V_3 < V_2 < V_1$

148. *(a)* Since, ΔOAB and ΔOCD are similar

$\therefore \quad \frac{r}{r_1} = \frac{h}{\frac{h}{2}} = \frac{2}{1}$

We have, $V = \frac{1}{3}\pi r^2 h$



$\Rightarrow r^2 = \frac{3V}{\pi h} \Rightarrow r_1 = \frac{r}{2} \Rightarrow r_1^2 = \frac{r^2}{4}$

Hence, volume of water $= \frac{1}{3}\pi r_1^2 \frac{h}{2}$

$= \frac{1}{6}\pi h \frac{r^2}{4} = \frac{1}{24}\pi h \frac{3V}{\pi h} = \frac{3V}{24} = \frac{V}{8}$

149. *(b)* Let radius be r and height be h of the cylinder.

Then, $30 \cdot \pi r^2 h = \frac{1}{3}\pi r^2 h \times n$

$\Rightarrow \quad n = 30 \times 3 = 90$ cones

Statement I is correct. Statement II is also correct. But statement II is not correct explanation of statement of I.

150. *(c)* At 30 min, the container was full

$\therefore$ At 5 min before i.e. in 25 min, the container was half filled and at 20 min, container was one-fourth full.

151. *(c)* I. Area of curved surface $= 2\pi rh$

$\because \quad 2\pi r(2h) = 2 \cdot 2\pi rh$

$\therefore$ Curved surface area is also doubled.

It is correct.

II. Total surface area of hemisphere $= 3\pi r^2$

$\because \ 3\pi(2r)^2 = 3\pi 4r^2 = 4 \cdot 3 \cdot \pi r^2$
= 4 times the area

It is also correct.

152. *(c)* Let n spherical ball be made.

Then, $4\pi r^2 = \pi^{\frac{1}{3}}$

$\Rightarrow r^2 = \frac{\pi^{\frac{1}{3}}}{4\pi} = \frac{1}{4}\pi^{-\frac{2}{3}} \Rightarrow r^2 = \left(\frac{1}{2}\pi^{-\frac{1}{3}}\right)^2$

$\Rightarrow \quad r = \frac{1}{2}\pi^{-\frac{1}{3}}$

Given, $n \cdot \frac{4}{3}\pi\left(\frac{1}{2}\pi^{-\frac{1}{3}}\right)^3 = 1 \times 1 \times 1$

$\Rightarrow \quad n \cdot \frac{4}{3}\pi \times \frac{1}{8} \times \frac{1}{\pi} = 1 \Rightarrow \frac{n}{6} = 1$

$\therefore \quad n = 6$

153. *(d)* Let n cylindrical boxes be packed.

$\Rightarrow \quad n \times \pi \times 5 \times 5 \times 10$
$= \pi \times 15 \times 15 \times 100$

$\therefore \quad n = 90$ boxes

154. *(c)* We have, area of cross-section of pipe

$= 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$

and $4 \times 10^{-4} \times 40 \text{ m}^3$ water flows in 1 s.

Let cistern fill in t s.

$\Rightarrow \quad 4 \times 10^{-4} \times 40 \times t = 10 \times 8 \times 6$

$\Rightarrow \quad t = 30000 \text{ s} = \frac{30000}{60 \times 60}$ h

$= \frac{50}{6}$ h $= 8$ h and 20 min

28

STATISTICS

Usually (10-13) questions have been asked from this chapter. This chapter is very important from examination point of view and generally questions are asked from the graphical representation of data, measures of central tendency.

Statistics is the branch of Mathematics which deals with the collection, analysis and interpretation of numerical data. In this chapter, we shall study measures of central tendency i.e. mean, median and mode of ungrouped data and grouped data. Concept of cumulative frequency, the cumulative frequency distribution how to draw cumulative frequency curves (ogive) and graphical representation of data will also be discussed.

Collection of Data

Collection of data is the first step in statistics towards achieving the goal or conclusion.

On the basis of collection, data are of two types

1. **Primary data** The data collected actually in the process of investigation by the investigator is called primary data. It is original and first hand information.
2. **Secondary data** The data collected by someone and used by any other person known as secondary data.

Presentation of Data

Raw or Ungrouped data When the data presented is random and is not prepared according to some order, it is known as raw or ungrouped data. It does not give us a clear picture of the class.

Grouped data When the data is arranged in any manner like ascending or descending order etc., it is called grouped data. It can also be presented in the form of a table called frequency distribution table.

Class Intervals Class intervals are the groups in which all the observations are divided. Each class is bounded by two figures (numbers) which are called class limits. The figure on the left side of a class, is called its lower limit and that on the right side of a class, is called its upper limit.

Class mark It is the mid-point of the class interval.

i.e. Class mark $= \dfrac{\text{Lower class limit} + \text{Upper class limit}}{2}$

Range or a class size Difference between the upper limit and the lower limit of a class is called its class size.

Range = Upper limit – Lower limit

e.g. Range of the observations 4, 7, 8, 10, 12
$= 12 - 4 = 8$

Frequency of an observation The number of times an observation occurs is called its frequency.

Frequency distribution

The tabular arrangement of data, showing the frequency of each observation is called a frequency distribution. It is a method of presenting the data in a summarized form. Frequency distribution is also known as frequency table.

There are two types of frequency distribution which are as follows:

1. Discrete frequency distribution

A frequency distribution is called a discrete frequency distribution, if data are presented in a way such that exact measurements of the units are clearly shown.

Marks	Number of students (frequency)
40	7
60	3
80	3
100	2
Total	15

2. Continuous frequency distribution

Exclusive form A frequency distribution in which upper limit of each class is excluded and lower limit is included, is called an exclusive form.
e.g. In the class 0-10 of marks obtained by students, a student who has obtained 10 marks is not included in this class. It will be counted in the next class i.e 10-20.

Inclusive method In this method,the classes are so formed that the upper limit of a class is included in that class. The following example illustrates the method. In class 1000-1099, we include workers having wages between ₹ 1000 and ₹ 1099. If the income of a worker is exactly ₹ 1100, then it will be included in the next class 1100-1199.

Exclusive method		Inclusive method	
Wages (in ₹)	**Number of workers**	**Wages (in ₹)**	**Number of works**
1000-1100	125	1000-1099	125
1100-1200	150	1100-1199	150
1200-1300	200	1200-1299	200
1300-1400	250	1300-1399	250
1400-1500	175	1400-1499	175
1500-1600	100	1500-1599	100
Total	1000	**Total**	1000

It is clear from the above example that both the inclusive and exclusive methods give us the same class frequency although the class intervals are aparently different in the two cases.

In the above example on inclusive method, the difference between the lower limit of a class and upper limit of the preceding class is 1.

Therefore, we subtract $\frac{1}{2}$ from the lower limit and add $\frac{1}{2}$ to upper limit of each class to make it continuous.

The adjusted classes would be as follows:

Wages (in ₹)	Number of workers
999.5-1099.5	125
1099.5-1199.5	150
1199.5-1299.5	200
1299.5-1399.5	250
1399.5-1499.5	175
1499.5-1599.5	100

Cumulative frequency

If the frequency of first class interval is added to the frequency of second class and this sum is added to third class and so on, then frequencies so obtained are known as cumulative frequency.

EXAMPLE 1. Consider the table given below :

Marks	Number of students (frequency)	Cumulative frequency
0-10	13	13
10-20	7	20
20-30	5	25
30-40	4	29
40-50	1	30
50-60	7	37
60-70	3	40
70-80	4	44
80-90	5	49
90-100	1	50
Total	50	

Then, find the value of the following
(i) frequency of class 10-20. (ii) class size.
(iii) mid value of 60-70. (iv) total frequencies.

a. 10, 65, 50, 60
b. 7, 10, 65, 60
c. 50, 65, 10, 10
d. 7, 10, 65, 50

Sol. d. (i) Here, frequency of class 10-20 is 7.

(ii) Class size = Upper limit − Lower limit $= 30 - 20 = 10$

(iii) Mid value $= \frac{\text{Upper limit} + \text{Lower limit}}{2} = \frac{60 + 70}{2} = 65$

(iv) Total frequencies = 50

EXAMPLE 2. The class mark of the interval 12.5-17.5 is

a. 5
b. 12.5
c. 15
d. 17.5

Sol. c. Class mark $= \frac{\text{Lower limit} + \text{Upper limit}}{2}$

$$= \frac{12.5 + 17.5}{2} = \frac{30}{2} = 15$$

EXAMPLE 3. The class marks of a distribution are 54, 64, 74, 84, 94 and 104. Then, the class size is

a. 5 b. 10 c. 54 d. 104

Sol. b. Since, class size is the difference between the class marks of two adjacent classes.

$\therefore$ Class size $= 64 - 54 = 10$

MEASURES OF CENTRAL TENDENCY

An average or central value of a statistical series is the value of the variable which describes the charateristic of the entire distribution. The following are the five measure of central tendency

(i) Arithmetic mean or mean (ii) Geometric mean
(iii) Harmonic mean (iv) Median
(v) Mode

Out of these measures of central tendency. Arithmetic mean, median and mode are sometimes knows as measures of location.

Arithmetic Mean

The mean (or average) of number of observations is the sum of the values of all the observations divided by the total number of observations.

Mean = Sum of observations/Number of observation

Properties of Arithmetic Mean

1. If every observation is increased by a constant, then the mean of the observations, so obtained also increases by the same constant.
2. If every observation is decreased by a constant, then the mean of the observation, so obtained also decreases by the same constant.
3. If each observation is multiplied by a constant, then the mean of the resulting observations can be obtained by multiplying the mean by the same constant.
4. If each observation is divided by a constant, then the mean of the resulting observation can be obtained by dividing the mean by the same constant.
5. The average of natural numbers from 1 to n is $\frac{(n+1)}{2}$.
6. The average of odd numbers from 1 to n is $\left[\frac{\text{Last odd number}+1}{2}\right]$ and the average of even numbers from 1 to n is $\left[\frac{\text{Last even number}+2}{2}\right]$.

1. **Arithmetic mean of ungrouped or individual observations** If $x_1, x_2, x_3, \ldots, x_n$ are n observations, then

(i) Mean $(\bar{x}) = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i$

It is also called direct method.

(ii) Mean $(\bar{x}) = A + \frac{1}{n}\sum_{i=1}^{n} d_i$

where, A = Assumed mean and $d_i = x_i - A$.

It is also called shortcut method.

EXAMPLE 4. If the heights of 5 persons are 144 cm, 152 cm, 150 cm, 158 cm and 155 cm, respectively. Find the mean height.

a. 152.5 cm b. 150 cm c. 149.8 cm d. 151.8 cm

Sol. d. Mean height $= \frac{\text{Sum of the heights}}{\text{Number of persons}}$

$= \frac{144+152+150+158+155}{5} = \frac{759}{5} = 151.8$ cm

EXAMPLE 5. If the average of 6, 8, 5, 7, x and 4 is 7. Then, the value of x is

a. 10 b. 11 c. 12 d. 15

Sol. c. $\because$ Sum of observation $= 6+8+5+7+x+4 = 30+x$

and number of observation = 6

$\therefore$ Average $= \frac{30+x}{6} \Rightarrow 7 = \frac{30+x}{6}$

$\Rightarrow 42 = 30 + x \Rightarrow x = 42 - 30 = 12$

Hence, the value of x is 12.

EXAMPLE 6. The average of 27 observations is 35. If 5 is added to each observation, what will be the new mean?

a. 10 b. 20 c. 30 d. 40

Sol. d. Given, $\bar{x} = 35$ and $n = 27$

Sum of observation $= n\bar{x} = 27 \times 35 = 945$

and new total of observation $= 945 + 27 \times 5 = 1080$

$\therefore$ New mean $= \frac{1080}{27} = 40$

Shortcut Method

New mean = Previous mean + Number added to each term $= 35 + 5 = 40$

EXAMPLE 7. The mean of 53 observations is 18. If each observation is multiplied by 3. What will be the new mean?

a. 18 b. 36 c. 53 d. 54

Sol. d. Given, $\bar{x} = 18$ and $n = 53$

So, sum of observation $= n\bar{x} = 53 \times 18 = 954$

New total of observation $= 954 \times 3 = 2842$

$\therefore$ New mean $= \frac{2862}{53} = 54$

Shortcut Method

New mean = Previous mean × Constant multiplied to each term $= 18 \times 3 = 54$

2. **Mean of grouped or continuous observations** If $x_1, x_2, x_3, \ldots, x_n$ are n observations whose corresponding frequencies are $f_1, f_2, f_3, \ldots, f_n$, then

(i) **Direct Method**

(a) Mean, $\bar{x} = \dfrac{x_1 f_1 + x_2 f_2 + \ldots + x_n f_n}{f_1 + f_2 + \ldots + f_n}$

$$= \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = \frac{\Sigma f_i x_i}{\Sigma f_i}$$

(b) Mean, $\bar{x} = A + \dfrac{\sum_{i=1}^{n} f_i d_i}{\sum_{i=1}^{n} f_i} = A + \dfrac{\Sigma f_i d_i}{\Sigma f_i}$

where, A = Assumed mean and $d_i = x_i - A$.

(ii) **Step Deviation Method** $(\bar{x}) = A + \left(\dfrac{\Sigma f_i u_i}{\Sigma f_i}\right) \times h$

where, h = Difference between two consecutive class marks.

EXAMPLE 8. The arithmetic mean of the marks from the following table is

Marks	0-10	10-s20	20-30	30-40	40-50	50-60
Number of students	12	18	27	20	17	6

a. 20 b. 28 c. 2800 d. 100

Sol. b. We have,

Marks	Class mark, (x)	f	fx
0-10	5	12	60
10-20	15	18	270
20-30	25	27	675
30-40	35	20	700
40-50	45	17	765
50-60	55	6	330
Total		$\Sigma f = 100$	$\Sigma fx = 2800$

$$\therefore \quad \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{2800}{100} = 28$$

EXAMPLE 9. Find the mean wage from the data given below :

Wages (in ₹)	800	820	860	900	920	980	1000
Number of workers	7	14	19	25	20	10	5

a. 890 b. 890.5 c. 891.2 d. 100

Sol. c. Let the assumed mean be $A = 900$.
The given data can be written as under :

Wages (in ₹) (x_i)	Number workers (f_i)	$d = x_i - A$	$f_i d_i$
800	7	−100	−700
820	14	− 80	−1120
860	19	− 40	−760
900	25	0	0
920	20	20	400
980	10	80	800
1000	5	100	500
Total	$\Sigma f_i = 100$		$\Sigma f_i d_i = -880$

Here, $A = 900$

$$\therefore \quad \text{Mean} = A + \frac{\Sigma f_i d_i}{\Sigma f_i} = 900 + \frac{-880}{100} = 900 - 8.80 = 891.2$$

$\therefore$ Mean wage = ₹891.2

EXAMPLE 10. Find the mean of the following data.

x	15	25	35	45	55	65
f	4	28	15	20	17	16

a. 44.2 b. 42.4 c. 40.6 d. 41.6

Sol. d. Here, the values of class marks x_i are given, so let $A = 35$ and class width, h = difference between two consecutive class marks = 10, then

x_i	f_i	$u_i = \dfrac{x_i - a}{h}$	$f_i u_i$
15	4	$\dfrac{15-35}{10} = \dfrac{-20}{10} = -2$	$4 \times (-2) = -8$
25	28	$\dfrac{25-35}{10} = \dfrac{-10}{10} = -1$	$28 \times (-1) = -28$
35 = (A)	15	$\dfrac{35-35}{10} = \dfrac{0}{10} = -0$	$15 \times 0 = 0$
45	20	$\dfrac{45-35}{10} = \dfrac{10}{10} = 1$	$20 \times 1 = 20$
55	17	$\dfrac{55-35}{10} = \dfrac{20}{10} = 2$	$17 \times 2 = 34$
65	16	$\dfrac{65-35}{10} = \dfrac{30}{10} = 3$	$16 \times 3 = 48$
Total	$\Sigma f_i = 100$		$\Sigma f_i u_i = 66$

Thus, we have $\Sigma f_i u_i = 66$, $\Sigma f_i = 100$, $h = 10$

$$\therefore \quad \bar{x} = A + h\left(\frac{\Sigma f_i u_i}{\Sigma f_i}\right) = 35 + 10 \times \frac{66}{100}$$

$$= 35 + \frac{660}{100} = 35 + 6.6 \Rightarrow \bar{x} = 41.6$$

Weighted Arithmetic Mean

If corresponding weight of $x_1, x_2, \ldots, x_n$ are $w_1, w_2, \ldots, w_n$ respectively, then

Weighted arithmetic mean

$$= \frac{w_1 x_1 + w_2 x_2 + \ldots + w_n x_n}{w_1 + w_2 + \ldots + w_n} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

EXAMPLE 11. The weighted arithmetic mean of the first n natural numbers, the weights being the corresponding numbers, is

a. $\frac{n+1}{2}$ b. $\frac{n+2}{2}$ c. $\frac{2n+1}{3}$ d. None of these

Sol. c. First n natural numbers are $1, 2, 3, \dots, n$; whose corresponding weights are $1, 2, 3, \dots, n$, respectively.

$\therefore$ Weight arithmetic mean $= \frac{1\times1+2\times2+\dots+n\times n}{1+2+\dots+n}$

$$= \frac{1^2+2^2+\dots+n^2}{1+2+\dots+n} = \frac{n(n+1)(2n+1)}{\frac{6n(n+1)}{2}} = \frac{2n+1}{3}$$

$$\left[\begin{array}{l} \because \Sigma n^2 = \frac{n(n+1)(2n+1)}{6} \\ \Sigma n = \frac{n(n+1)}{2} \end{array}\right]$$

Combined Arithmetic Mean

If two sets of observations are given, then the combined mean of both the sets can be calculated by the following formula.

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1+n_2}$$

where, $\bar{x}$ = Mean of sets of observations
$\bar{x}_1$ = Mean of first set of observations
n_1 = Number of observations in first set
$\bar{x}_2$ = Mean of second set of observations
n_2 = Number of observations in second set

EXAMPLE 12. The average salary of male employees in a firm was ₹ 5200 and that of females was ₹ 4200. The mean salary of all the employees was ₹ 5000. The percentage of male and female employees are, respectively.

a. 80 and 20 b. 20 and 80 c. 60 and 40 d. 52 and 48

Sol. a. Let $x_1 = 5200$, $x_2 = 4200$ and $\bar{x} = 5000$

We know that, $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1+n_2}$

$\Rightarrow 5000(n_1+n_2) = 5200n_1 + 4200n_2 \Rightarrow \frac{n_1}{n_2} = \frac{4}{1}$

$\therefore$ The percentage of male employees in the firm

$$= \frac{4}{4+1}\times 100 = 80$$

and the percentage of female employees in the firm

$$= \frac{1}{4+1}\times 100 = 20$$

Geometric Mean

If a, G and b are in GP, then the geometric mean between a and b is, $G = \sqrt{ab}$.

EXAMPLE 13. What is the geometrical mean of the variate which takes values 210, 201, 202, 20, 12, 10, 2, 1 and 0?

a. 10 b. 9 c. 8 d. 0

Sol. d. The given variates are 210, 201, 202, 20, 12, 10, 2, 1 and 0.

$\therefore$ Geometric mean of given variates

$$= \sqrt[9]{210\times201\times202\times20\times10\times2\times1\times0} = \sqrt[9]{0} = 0$$

Harmonic Mean

The harmonic mean of n positive numbers $a_1, a_2, \dots, a_n$ is

$$\frac{1}{H} = \frac{1}{n}\left(\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+\dots+\frac{1}{a_n}\right)$$

where, H denotes the harmonic mean.

Median

After arranging the given data in ascending or descending order of magnitude, the value of the middle most observation is called the median of the data.

1. **Median of an individual series** If number of observations is n. Then, arrange the observations in ascending or descending order.

(i) If n is an odd number, then

$$\text{Median} = \text{Value of the} \left(\frac{n+1}{2}\right)\text{th observation}$$

(ii) If n is an even number, then

$$\text{Median} = \frac{\text{Value of the}\left(\frac{n}{2}\right)\text{th observation} + \text{Value of}\left(\frac{n}{2}+1\right)\text{th observation}}{2}$$

2. **Median of a discrete frequency series** First arrange the data in ascending or descending order and find cumulative frequency. Now, find $\frac{N}{2}$, where $N = \Sigma f_i$.

See the cumulative frequency just greater than $\frac{N}{2}$.

The corresponding value of x is median.

3. **Median of a continuous series** In this case, the class corresponding to the cumulative frequency just greater than $N/2$ is called the median class and the value of median is obtained by the following formula.

$$\text{Median} = l + \left(\frac{\frac{N}{2}-c}{f}\right)\times h$$

where, l = Lower limit of median class

f = Frequency of median class
h = Size of median class
c = Cumulative frequency of class before median class

EXAMPLE 14. From the data given, the median of the average deposit balance of saving for the branch during March 1982 is

Average deposit balance (in ₹)	Number of deposit
0 – 100	26
100-200	68
200-300	145
300-400	242
400-500	188
500-600	65
600-700	16

a. 356 b. 300 c. 56.2 d. 356.2

Sol. d.

Average deposit balance (in ₹)	f	Cumulative frequency (cf)
Less than 100	26	26
100-200	68	94
200-300	145	239
300-400	242	481
400-500	188	669
500-600	65	734
600-700	16	750

$$\frac{N}{2} = \frac{750}{2} = 375$$

The frequency just greater than 375 is 481.

$\therefore$ Median class is 300-400.

$$\text{Median} = l + \frac{\left(\frac{N}{2} - c\right)}{f} \times h = 300 + \frac{375 - 239}{242} \times 100$$

$$= 300 + 56.2 = 356.2$$

Mode

The data value which occurs maximum number of times in a data distribution is called mode.

1. **Mode of individual series** The value which is repeated maximum number of times is called mode of the series.

EXAMPLE 15. Find the mode for the following series 2.5, 2.3, 2.2, 2.2 2.4, 2.7, 2.7, 2.5, 2.3, 2.2, 2.6 and 2.2.

a. 2.2 b. 2.3
c. 2.7 d. 2.6

Sol. a. Arranging the data in the form of a frequency table, we have

Value	Frequency
2.2	4
2.3	2
2.4	1
2.5	2
2.6	1
2.7	2

We see that, the value 2.2 has the maximum frequency 4. So, the mode for the given series is 2.2.

2. **Mode of a discrete frequency series** In this case, mode is the value of the variate corresponding to the maximum frequency.

EXAMPLE 16. Compute the modal value for the following frequency distribution.

x	95	105	115	125	135	145	155	165	175
f	4	2	18	22	21	19	10	3	2

a. 115 b. 125 c. 22 d. 120

Sol. b. From the given table, it is clear that 125 has the highest frequency i.e. 22. Hence, modal value of the given frequency distribution is 125.

3. **Mode of a continuous series** The class which has maximum frequency is called modal class or group. The mode is given by the formula,

$$\text{Mode} = l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

where, l = Lower limit of modal class
h = Size of class interval
f_1 = Frequency of modal class
f_0 = Frequency of the class preceding the modal class
f_2 = Frequency of the class succeeding the modal class

EXAMPLE 17. The mode of the following distribution is

Class interval	Frequency
0-10	5
10-20	8
20-30	7
30-40	12
40-50	28
50-60	20
60-70	10
70-80	10

a. 46 b. 6.66
c. 46.67 d. None of these

Sol. **c.** Here, maximum frequency is 28. Thus, the class 40-50 is the modal class. Here, $f_1 = 28, f_0 = 12,\ f_2 = 20,\ l = 40$ and $h = 10$

$$\therefore \quad \text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 40 + \frac{10\,(28 - 12)}{(2 \times 28 - 12 - 20)}$$

$$= 40 + \frac{10 \times 16}{24} = 40 + 6.666 = 46.67 \text{ (approx)}$$

Relation between Mean, Median and Mode

The empirical relationship between the three measures of central tendency is given by

$$\boxed{\text{Mode} = 3\ (\text{Median}) - 2\ (\text{Mean})}$$

In case, mean = median = mode, then distribution is said to be symmetric distribution.

EXAMPLE 18. If in a frequency distribution, the mean and median are 20 and 21 respectively, then its mode is approximate by

a. 24 **b.** 23 **c.** 25 **d.** None of these

Sol. **b.** Here, mean = 20 and median = 21

Mode $= 3 \times \text{Median} - 2 \times \text{Mean} = 63 - 40 = 23$

Some Useful Formulae

If $x_1, x_2, \ldots, x_n$ are n observations with their mean M, then deviation d_i is given by $d_i = |x_i - M|$.

- Mean deviation for individual series is given by $\dfrac{\Sigma |d_i|}{n}$.
- Mean deviation for discrete series is given by $\dfrac{\Sigma f_i\, |d_i|}{\Sigma\, f_i}$.
- Standard deviation (σ) for ungrouped data is given by $\sigma = \sqrt{\dfrac{\Sigma\,|\,d_i\,|^2}{n}}$.
- Standard variation for grouped data is given by $\sigma = \sqrt{\dfrac{\Sigma f_i |d_i|^2}{n}}$
- Variance $= \sigma^2$
- Coefficient of variation $= \dfrac{\text{Standard deviation}}{\text{Mean}} \times 100$
- Coefficient of mean deviation $= \dfrac{\text{Mean deviation}}{\text{Mean}} \times 100$

EXAMPLE 19. The mean deviation of the data, 3, 5, 6, 7, 8, 10, 11, 14 is

a. 4 **b.** 3.25 **c.** 2.75 **d.** 2.4

Sol. **c.** Mean $= \dfrac{3+5+6+7+8+10+11+14}{8} = \dfrac{64}{8} = 8$

$$\therefore \quad \Sigma\,\delta = |\,3-8\,| + |\,5-8\,| + |\,6-8\,| + |\,7-8\,| + |\,8-8\,| + |\,10-8\,| + |\,11-8\,| + |\,14-8\,| = 22$$

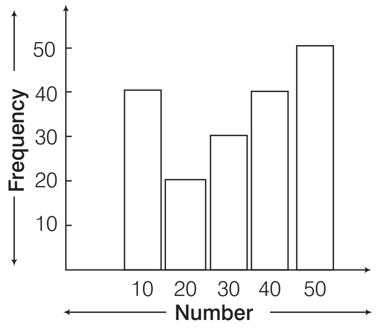
$$\therefore \quad \text{Mean deviation} = \frac{\Sigma\delta}{n} = \frac{22}{8} = 2.75$$

Graphical Representation of Data

There are many ways of picturing a frequency distribution of continuous type which are discussed below

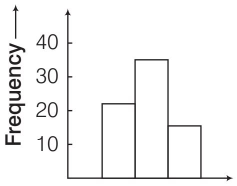
1. Bar Graph

In a bar graph, bars of uniform width are drawn with various heights. The heights of a bar represents the frequency of observation.



2. Histogram

A histogram is the graphical representation of a frequency distribution in the form of the rectangles with class intervals as bases, and the corresponding frequencies as height. There being no gap between any two consecutive rectangles.



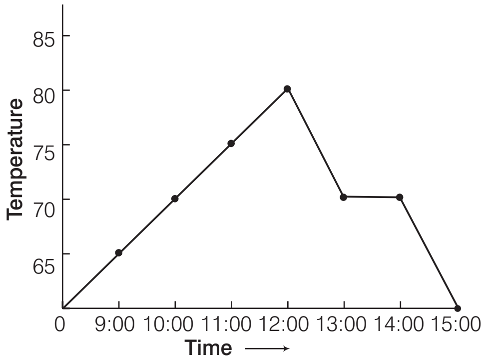
POINTS TO BE REMEMBERED

- A histogram consists of a set of adjacent rectangles, whose bases are equal to class sizes and height are equal to class frequencies.
- The total area of the histogram = Sum of areas of all rectangles. If the class intervals are of same size (width), then the area of histogram = NK, where K = Size of a class and N = Sum of frequences of all classes.
- If the frequency distribution is discontinuous (inclusive), change it to continuous (exclusive) and then construct a histogram.

3. Frequency polygon

A frequency polygon can be drawn by joining the mid-points of the respective tops of the rectangles of a histogram in case of equal class intervals.

A frequency polygon for a grouped data can also be drawn independently by plotting the mid-points of the all classes along X-axis and frequencies along Y-axis and joining the plotted points by straight line.



4. Ogive (Cumulative Frequency Curve)

When we plot the upper or lower class limits along X-axis and cumulative frequencies along Y-axis, And on joining them we get a curve called an ogive.

We have two types of ogive curves

(a) less than ogive (i.e. the rising curve)

It is the graph between upper limits and cumulative frequencies of a distribution.

(b) more than ogive (i.e. a falling curve)

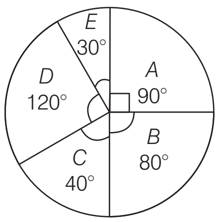
It is the graph between lower limits and cumulative frequencies of a distribution.

Note • If we have both ogive (i.e. less than type and more than type), then these two ogives intersect each other at a point.
• From this point, draw a perpendicular on X-axis, the point at which it cuts the X-axis gives the median.
i.e. the x-coordinate of intersection point gives the median.

Pie-Diagram (or Pie chart)

A type of graph in which a circle is divided into sectors such that each sector represents a proportion of the whole.

Thus, in a pie chart



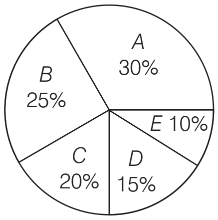
- Data is represented by sectors of a circle.
- Each part of data makes a certain central angle.
- Sum of all the angles of sector is 360°.
- Central angle $= \left(\dfrac{\text{Frequency} \times 360}{\text{Total frequency}}\right)^{\circ}$

EXAMPLE 20. In statistics, a suitable graph for representing the partitioning of total into subpart is

a. an ogive
b. a pictograph
c. a histogram
d. a pie chart

Sol. d. In statistics, a suitable graph for representing the partitioning of total into sub parts is a pie chart.

EXAMPLE 21. The following diagram show the expenditure of a family on various items A, B, C, D and E.



Study the diagram carefully and answer the following questions.

(i) The angle of pie diagram showing the expenditure incurred on item A is

a. 30° b. 35°
c. 108° d. None of these

(ii) Which two expenditures together will form an angle of 90° at the centre of pie diagram?

a. B and O b. C and A
c. D and E d. None of these

(iii) If the income of the family is ₹ 3000 per month, then expenditure of item C will be

a. ₹ 400 b. ₹ 500 c. ₹ 600 d. ₹ 800

Sol. (i) c. Angle for 100% = 360°

$\therefore$ Angle for expenditure of A = 30% of 360°

$$= \frac{30}{100} \times 360 = 108^{\circ}$$

(ii) c. $\because$ 100% = 360°

$\therefore$ An angle of 90° at the centre of pie diagram in percentage is $= \dfrac{90 \times 100}{360} = 25\%$

Expenditure of items D and E makes upto $(15 + 10) = 25\%$.

So, expenditure of D and E together will form an angle of 90° at the centre of pie diagram.

(iii) c. Expenditure on item C

$$= 20\% \text{ of ₹ } 3000 = \frac{20}{100} \times 3000 = ₹\ 600$$

PRACTICE EXERCISE

1. Frequency polygon can be drawn after drawing
(a) ogive (b) bar chart
(c) histogram (d) None of these

2. An ogive is used to determine
(a) mean (b) median (c) GM (d) HM

3. The mid-value of a class interval is 42. If the class size is 10, then the upper and lower limits of the class are
(a) 37.5 and 47.5 (b) 47 and 37
(c) 37 and 47 (d) 47.5 and 37.5

4. The actual lower class limits of the following classes 10-19, 20-29, 30-39 and 40-49 are
(a) 9.5, 19, 29 and 39.5 (b) 10, 20, 30 and 40
(c) 9.5, 19.5, 29.5 and 39.5 (d) 18.5, 28.5, 38.5 and 48.5

5. If the mean of five observations $x, x+2, x+4, x+6$ and $x+8$ is 11, then the mean of first three observations is
(a) 9 (b) 11
(c) 13 (d) None of these

6. The combined mean of three groups is 12 and the combined mean of first two groups is 3. If the first, second and third groups have 2, 3 and 5 items respectively, then mean of third group is
(a) 10 (b) 21
(c) 12 (d) 18

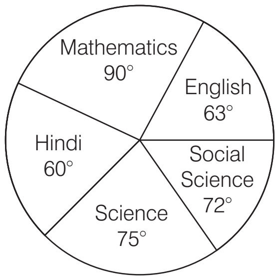
7. 10 is the mean of a set of 7 observations and 5 is the mean of a set of 3 observations. The mean of the combined set is given by
(a) 15 (b) 10 (c) 8.5 (d) 7.5

8. In a class of 50 students, 10 have failed and their average marks are 28. The total marks obtained by the entire class are 2800. The average marks of those who have passed are
(a) 43 (b) 53 (c) 63 (d) 70

9. Which of the following statements about the median is true?
(a) It is not affected by extreme values
(b) It can be found even, if some items are not known
(c) It is useful when the data cannot be measured quantitatively
(d) All of the above

10. The middle item of the series arranged in ascending or descending order is called
(a) mean (b) median
(c) mode (d) standard deviation

11. The following pie chart shows the marks obtained by a student in an examination who scored 540 marks in all. The subject in which the student scored 108 marks is



(a) Science (b) Hindi
(c) English (d) Social Science

12. A distribution consists of three components with frequencies 45, 40 and 15 having their means 2, 2.5 and 2 respectively. The mean of the combined distribution is
(a) 2.1 (b) 2.2 (c) 2.3 (d) 2.4

13. If the values $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{n}$ occur at frequencies 1, 2, 3, 4, 5, 6,..., n respectively, in a frequency distribution, then the mean is
(a) 1 (b) n (c) $\frac{1}{n}$ (d) $\frac{2}{n+1}$

14. If the geometric mean of three observations 40, 50 and x is 10, then the value of x is
(a) $\frac{1}{2}$ (b) 4 (c) 6 (d) 2

15. Suppose, X is some statistical variable with mean μ. Let $x_1, x_2, \ldots, x_n$ be its deviations from mean with the respective frequencies $f_1, f_2, \ldots, f_n$. What is the value of the sum $x_1 f_1 + x_2 f_2 + \ldots + x_n f_n$?
(a) 0 (b) 1 (c) μ (d) $\mu + 1$

16. Which one of the following statements is not correct with reference to a histogram?
(a) Frequency curve is obtained by joining the mid-points of the top of the adjacent rectangles with smooth curves
(b) Histogram is drawn for continuous data
(c) The height of the bar is proportional to the frequency of that class
(d) Mode of the distribution can be obtained from the histogram

17.

X	0	1	2	3	4
Frequency	4	*f*	9	*g*	4

The table above gives the frequency distribution of a discrete variable X with two missing frequencies. If the total frequency is 25 and the arithmetic mean of X is 2, then what is the value of the missing frequency f?

(a) 4 (b) 5 (c) 6 (d) 7

18. The yield of paddy per plot of one acre were obtained from a number of plots from two different districts in a state and are summarized in the following table.

Yield of paddy per plot in quintals	Number of plots in district A	Number of plots in district B
38.0-41.0	25	14
41.0-44.0	36	29
44.0-47.0	59	35
47.0-50.0	30	54
50.0-53.0	25	41

Which one of the following statements is correct?

(a) The mode for district A is higher than the mode for district B
(b) The mode for district B is higher than the mode for district A
(c) Both the distributions are symmetric
(d) Both the distributions have the same mean

19. Average score of 50 students in a class is 44. Later on it was found that the score 23 was incorrectly recorded as 73. The correct average score is

(a) 42 (b) 43 (c) 45 (d) 46

20. In the "less than" type of ogive the cumulative frequency is plotted against

(a) the lower limit of the concerned class interval
(b) the upper limit of the concerned class interval
(c) the mid value of the concerned class interval
(d) any value of the concerned class interval

21. A, B and C are three sets of values of x

A : 2, 3, 7, 1, 3, 2, 3
B : 7, 5, 9, 12, 5, 3, 8
C : 4, 4, 11, 7, 2, 3, 4

Select the correct statement from among the following

(a) Mean of A is equal to mode of C
(b) Mean of C is equal to median of B
(c) Median of B is equal to mode of A
(d) Median, mean and mode of A are same

22. A student obtains 75%, 80% and 85% marks in three subjects. If the marks of any other subject are added, then their average cannot be less than

(a) 60% (b) 65% (c) 70% (d) 80%

23. If the mean and median of a set of numbers are 8.9 and 9 respectively, then the mode will be

(a) 7.2 (b) 8.2 (c) 9.2 (d) 10.2

24. If every number of a finite set is increased by any number k, then the measure of central tendency should also increase by k. Which one of the following measures of central tendency does not have this property?

(a) Arithmetic mean
(b) Median
(c) Mid-range i.e. the arithmetic mean of the largest and the smallest numbers
(d) Geometric mean

25. If the median of the distribution (arranged in ascending order) 1, 3, 5, 7, 9, x, 15, 17, 19, 21 is 10, what is the value of x?

(a) 11 (b) 13 (c) $9 < x < 15$ (d) $9 \le x \le 15$

26. If the width of each of the ten classes in a frequency distribution is 2.5 and the lower class boundary of the lowest class is 5.1, then the upper class boundary of the highest class is

(a) 30.1 (b) 30 (c) 31.1 (d) 27.56

27. The average value of the median of 2, 8, 3, 7, 4, 6, 7 and the mode of 2, 9, 3, 4, 9, 6, 9 is

(a) 9 (b) 8 (c) 7.5 (d) 6

28. When 10 is subtracted from each of the given observations, the mean is reduced to 60%. If 5 is added to all the given observations, the mean will be

(a) 25 (b) 30 (c) 60 (d) 65

29. A data has highest value 120 and the lowest value 71. A frequency distribution in descending order with seven classes is to be constructed.

The limits of the second class interval shall be

(a) 77 and 78 (b) 78 and 85
(c) 85 and 113 (d) 113 and 120

30. If mean of y and $\frac{1}{y}$ is M, then what is the mean of y^3 and $\frac{1}{y^3}$?

(a) $\frac{M(M^2-3)}{3}$ (b) M^3
(c) $M^3 - 3$ (d) $M(4M^2 - 3)$

31. For the following frequency distribution

Class interval	0-5	5-10	10-15	15-20	20-25	25-30
Frequency	10	15	30	80	40	20

If m is the value of mode, then which one of the following is correct?

(a) $5 < m < 10$ (b) $10 < m < 15$
(c) $15 < m < 20$ (d) $20 < m < 25$

32. Square diagrams are drawn to represent the following data

Country	Pakistan	India	Myanmar	China
Labour Production (in ₹)	36	81	25	100

Using the scale 1 cm^2 = ₹ 25 what is the length of the representative square for India?

(a) 1.8 cm (b) 1.2 cm (c) 1 cm (d) 2 cm

33. The standard deviation of 7, 9, 11, 13, 15 is

(a) 2.4 (b) 2.5 (c) 2.7 (d) 2.82

34. The total number of cellphones sold for Motorola, Samsung and Sony was 45664. The number of cellphones sold for these companies were in the ratio 3:5:8 respectively. If these data were shown on a pie chart, calculate the angle represented by the number of cellphones sold by Motorola.

(a) 75° (b) 67.5° (c) 70° (d) 74.5°

35. If the population figures are given for each State of India, then the data can be classified as

(a) qualitative (b) quantitative
(c) chronological (d) geographical

36. Which one of the following represents statistical data?

(a) The names of all owners of shops located in a shopping complex
(b) A list giving the names of all States of India
(c) A list of all European countries and their respective capital cities
(d) The volume of a rainfall in certain geographical area, recorded every month for 24 consecutive months

37. Prime numbers are the numbers which comes in the table of 1 and itself only.

I. The mean of first seven prime numbers is greater than their median.
II. Mean is always greater than median.

Select the correct option from the options given below

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

38. Consider the following statements in respect of a histogram

I. The histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars.
II. The height of the rectangle is determined by the frequency of the class it represents.

Which of the statements given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

39. Consider the following statements in respect of histogram

I. Histogram is an equivalent graphical representation of the frequency distribution.
II. Histogram is suitable for continuous random variables, where the total frequency of an interval is evenly distributed over the interval.

Which of the statements given above is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

40. Examples of data are given below

I. Information on households collected by an investigator by door-to-door visits.
II. Data on the percentage of literates, sex-wise, for the different districts of a state collected from records of the census of India.
III. General information about families, collected by telephonic interviews.

Which one of the following in respect of the above is correct?

(a) I and II are primary data (b) I and III are primary data
(c) II and III are primary data (d) I, II and III are primary data

41. Consider the following types of data

I. Marks of students who appeared for a test of 100 marks.
II. Collar sizes of 200 shirts sold in a week.
III. Monthly incomes of 250 employees of a factory.

For which of the above data, mode is a suitable measure of central tendency?

(a) I and II (b) Only II (c) I and III (d) All of these

42. The cumulative frequency curve of a frequency distribution with 6 classes and total frequency 60 is a straight line. Consider the following statements

I. The first and the last classes have a frequency of 10 each.
II. Both the middle classes have a total frequency of 30.
III. The frequency distribution does not have a mode.

Which of the statements given above are correct?

(a) I and II (b) I and III (c) II and III (d) All of these

43. State which of the following variables are discrete?

I. Number of children in a family.
II. Wages of workers.
III. The ages of students.
IV. Weights of a set of a students.

Select the correct answer using the codes given below

(a) I and II (b) I, II and III
(c) All of these (d) None of these

Directions (Q. Nos. 44-45) *The number of goals scored by a soccer team in a season were as follows:*

Number of goals	Number of matches
0	3
1	6
2	2
3	x
4	5

44. If the mode is 1, the largest possible value of x is
(a) 2 (b) 5 (c) 3 (d) 6

45. If the median is 2, the smallest possible value of x is
(a) 2 (b) 5 (c) 3 (d) 6

Directions (Q. Nos. 46-47) *The itemwise expenditure of a non-government organisation for the year* 2008-2009 *is given below.*

Item	Expenditure (in ₹ lakh)
Salary of employees	6
Social welfare activities	7
Office contingency	3
Vehicle maint evance	4
Rent and hire charges	2.5
Miscellaneous expenses	1.5

The above data are represented by a pie diagram.

46. What is the sectorial angle of the largest sector?
(a) 120° (b) 105° (c) 90° (d) 85°

47. What is the difference in the sectorial angles of the largest and the smallest sectors?
(a) 90° (b) 85° (c) 82.5° (d) 77.5°

Directions (Q. Nos. 48-49) *The following table gives the frequency distribution of life length in hours of* 100 *electric bulbs having median life* 20 h.

Life of bulbs (in h)	Number of bulbs
8-13	7
13-18	x
18-23	40
23-28	y
28-33	10
33-38	2

48. What is the missing frequency of x?
(a) 31 (b) 27 (c) 24 (d) 14

49. What is the missing frequency of y?
(a) 27 (b) 24 (c) 14 (d) 11

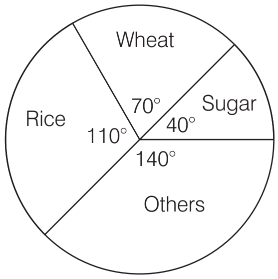
Directions (Q. Nos. 50-52) *The arithmetic mean, geometric mean and median of* 6 *positive numbers* a, a, b, b, c, c *where* $a < b < c$ *are* $\frac{7}{3}$, 2, 2 *respectively.*

50. What is the sum of the squares of all the six numbers?
(a) 40 (b) 42 (c) 45 (d) 48

51. What is the value of c ?
(a) 1 (b) 2 (c) 3 (d) 4

52. What is the mode?
(a) 1 (b) 2
(c) 1, 2 and 4 (d) None of these

Directions (Q. Nos. 53-55) *Study the following pie diagram and answer the questions.*



53. The percentage of wheat production is
(a) 35% (b) 70% (c) 19.44% (d) 20%

54. If the sugar production is 9000 kg, then the wheat production is
(a) 20000 kg (b) 24000 kg
(c) 15750 kg (d) None of these

55. If the total production is 180000 kg, the difference in sugar and wheat production is
(a) 10000 (b) 15000 (c) 20000 (d) None of these

Directions (Q. Nos. 56-58) *Consider the following frequency distribution*

Class interval	1-4	4-7	7-10	10-13	13-16	16-19
Frequency	6	30	40	16	4	4

56. What is the median of the data?
(a) 9.4 (b) 7.68 (c) 8.05 (d) 8.32

57. What is modal size of the given data?
(a) 8.14 (b) 7.88 (c) 7.62 (d) 8.48

58. The value of (Mean + Mode −2 Median) is equal to
(a) 0 (b) 0.2 (c) 0.4 (d) 0.1

PREVIOUS YEARS' QUESTIONS

59. Let the observations at hand be arranged in increasing order. Which one of the following measures will not be affected when the smallest and the largest observations are removed?
2012 I
(a) Mean (b) Median
(c) Mode (d) Standard deviation

60. Consider the following statements :

I. The data collected by the investigator to be used by himself are called primary data.

II. The data obtained from government agencies are called secondary data.

Which of the statement(s) given above is/are correct? **2012 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

61. Which one among the following statements is correct? **2012 I**

(a) Simple bar diagrams are those diagrams which show two characteristics of the data
(b) In pie diagrams, all the items are converted into angles
(c) A bar diagram is one in which data are shown in terms of bars
(d) Bar diagrams present data through length and breadth

62. Consider the following distribution :

Value of the variable	1	2	3	4	5
Frequency	3	f	6	5	3

For what value of f, is the arithmetic mean of the above distribution 3.1? **2012 I**

(a) 2 (b) 3 (c) 4 (d) 5

63. Which of the following pair(s) is/are correctly matched?

I. Weight of a person : Continuous variable

II. Educational qualification of the person : Attribute

(a) Only I (b) Only II **2012 II**
(c) Both I and II (d) Neither I nor II

64. The mean of 100 values is 45. If 15 is added to each of the first forty values and 5 is subtracted from each of the remaining sixty values, then the new mean becomes **2012 II**

(a) 45 (b) 48 (c) 51 (d) 55

65. Which one of the following relations for the numbers 10, 7, 8, 5, 6, 8, 5, 8 and 6 is correct?

(a) Mean = Median (b) Mean = Mode **2012 II**
(c) Mean > Median (d) Mean > Mode

66. In histogram the width of the bars is proportional to **2012 II**

(a) Frequency (b) Number of classes
(c) Class interval (d) None of these

67. Which one of the following statements is correct? **2013 I**

(a) A frequency polygon is obtained by connecting the corner points of the rectangles in a histogram
(b) A frequency polygon is obtained by connecting the mid-points of the tops of the rectangles in a histogram
(c) A frequency polygon is obtained by connecting the corner points of the class intervals in a histogram
(d) None of the above

68. Consider the following statements :

I. A frequency distribution condenses the data and reveals its important features.

II. A frequency distribution is an equivalent representation of original data.

Which of the statement(s) given above is/are correct? **2013 I**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Directions (Q. Nos. 69-71) *Read the following information carefully to answer the questions that follow.* **2013 I**

In a frequency distribution having class intervals 0-10, 10-20, 20-30 and 30-40 the respective frequencies are x, $x+8$, $x-2$ and $x-4$ and the arithmetic mean of the distribution is 17.8.

69. The value of x is

(a) 3 (b) 6 (c) 8 (d) 12

70. The median lies in which one of the following class intervals?

(a) 0-10 (b) 10-20 (c) 20-30 (d) 30-40

71. The modal class is

(a) 0-10 (b) 10-20 (c) 20-30 (d) 30-40

72. The mean of 7 observations is 7. If each observation is increased by 2, then the new mean is **2013 I**

(a) 12 (b) 10 (c) 9 (d) 8

73. There are 45 male and 15 female employees in an office. If the mean salary of the 60 employees is ₹ 4800 and the mean salary of the male employees is ₹ 5000, then the mean salary of the female employees is **2013 I**

(a) ₹ 4200 (b) ₹ 4500 (c) ₹ 5600 (d) ₹ 6000

Directions (Q. Nos. 74-75) *Read the following information carefully and answer the questions given below.* **2013 II**

The median of the following distribution is 14.4 and the total frequency is 20.

Class interval	0-6	6-12	12-18	18-24	24-30
Frequency	4	x	5	y	1

74. What is x equal to?

(a) 4 (b) 5 (c) 6 (d) 7

75. What is the relation between x and y?

(a) $2x = 3y$ (b) $3x = 2y$ (c) $x = y$ (d) $2x = y$

76. If m is the mean of p, q, r, s, t, u and v, then what is $(p-m)+(q-m)+(r-m)+(s-m)+(t-m)$ $+(u-m)+(v-m)$ equal to? **2013 II**

(a) 0 (b) s
(c) $\frac{(p+v)}{2}$ (d) None of these

77. The average of u, v, w, x, y and z is 10. What is the average of $u + 10, v + 20, w + 30, x + 40, y + 50$ and $z + 60$? **2013 II**

(a) 30 (b) 35 (c) 40 (d) 45

78. The mean of the following distribution is 18.

Class interval	Frequency
11-13	3
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

What is the value of f? **2014 II**

(a) 8 (b) 9 (c) 10 (d) 11

79. Consider the following statements pertaining to a frequency polygon of a frequency distribution of a continuous variable having seven class intervals of equal width.

I. The original frequency distribution can be reconstructed from the frequency polygon.

II. The frequency polygon touches the X-axis in its extreme right and extreme left.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

80. Consider the following in respect of variate which takes values 2, 2, 2, 2, 7, 7, 7 and 7.

I. The median is equal to mean.

II. The mode is both 2 and 7.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

81. Which of the following items of information is a good example of statistical data? **2014 II**

(a) A table of logarithms of numbers
(b) A list of names of 120 students of a class
(c) A list of annual incomes of the members of a club
(d) Holiday list of the offices of Government of India in the year 2013

82. The following table gives 'less than' type frequency distribution of income per day.

Income (in ₹) less than	Number of persons
1500	100
1250	80
1000	70
750	55
500	32
250	12

What is the modal class? **2014 II**

(a) 250-500 (b) 500-750
(c) 750-1000 (d) None of these

83. The geometric mean of $(x_1, x_2, x_3, \dots, x_n)$ is X and the geometric mean of $(y_1, y_2, y_3, \dots, y_n)$ is Y. Which of the following statement is/are correct?

I. The geometric mean of $(x_1y_1, x_2y_2, x_3y_3, \dots, x_ny_n)$ is XY.

II. The geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}\right)$ is $\frac{X}{Y}$.

Select the correct answer using the codes given below

(a) Only I (b) Only II **2014 II**
(c) Both I and II (d) Neither I nor II

84. Consider the following statements in respect of a discrete set of numbers.

I. The arithmetic mean uses all the data and is always uniquely defined.

II. The median uses only one or two numbers from the data and may not be unique.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

85. If x_i's are the mid-points of the class intervals of grouped data, f_i's are the corresponding frequencies and $\bar{x}$ is the mean, then what is $\Sigma f_i(x_i - \bar{x})$ equal to? **2014 II**

(a) 0 (b) – 1 (c) 1 (d) 2

86. When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the areas proportional to the frequencies of the respective class intervals, the graph so obtained is called **2014 II**

(a) bar diagram (b) frequency curve
(c) ogive (d) None of these

87. Consider the following data:

I. Number of complaints lodged due to road accidents in a state within a year for 5 consecutive years.

II. Budgetary allocation of the total available funds to the various items of expenditure.

Which of the above data is/are suitable for representation of a pie diagram? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

88. Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range

I. The cumulative frequency polygon gives an equivalent representation of frequency distribution table.

II. The cumulative frequency polygon is a closed polygon with one horizontal and one vertical side. The other sides have non-negative slope.

Which of the statement(s) given above is/are correct? **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

89. The class which has maximum frequency is known as **2014 II**

(a) median class (b) mean class
(c) modal class (d) None of these

90. Ten observations 6, 14, 15, 17, $x+1$, $2x-13$, 30, 32, 34 and 43 are written in ascending order. The median of the data is 24. What is the value of x? **2014 II**

(a) 15 (b) 18 (c) 20 (d) 24

Directions (Q. Nos. 91-94) *Consider the following frequency distribution.* **2015 I**

Class	Frequency
0-10	4
10-20	5
20-30	7
30-40	10
40-50	12
50-60	8
60-70	4

91. What is the mean of the distribution?

(a) 37.2 (b) 38.1 (c) 39.2 (d) 40.1

92. What is the median class?

(a) 20-30 (b) 30-40
(c) 40-50 (d) 50-60

93. What is the median of the distribution?

(a) 37 (b) 38 (c) 39 (d) 40

94. What is the mode of the distribution?

(a) 38.33 (b) 40.66
(c) 42.66 (d) 43.33

95. There are five parties A, B, C, D and E in an election. Out of total 100000 votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allocated for party E in the pie chart? **2015 I**

(a) 15° (b) 54° (c) 60° (d) 72°

96. If a variable takes discrete values $a+4$, $a-3.5$, $a-2.5$, $a-3$, $a-2$, $a+0.5$, $a+5$ and $a-0.5$, where $a>0$, then the median of the data set is **2015 II**

(a) $a-2.5$ (b) $a-1.25$ (c) $a-1.5$ (d) $a-0.75$

97. If each of n numbers $x_i = i$ $(i = 1, 2, 3, \dots, n)$ is replaced by $(i+1)x_i$, then the new mean is **2015 II**

(a) $\frac{n+3}{2}$ (b) $\frac{n(n+1)}{2}$
(c) $\frac{(n+1)(n+2)}{3n}$ (d) $\frac{(n+1)(n+2)}{3}$

98. The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to **2015 II**

(a) 7 (b) 14 (c) 35 (d) 38.5

99. The election result in which six parties contested was depicted by a pie chart. Party A had an angle 135° on this pie chart. If it secured 21960 votes, then how many valid votes in total were cast? **2016 I**

(a) 51240 (b) 58560 (c) 78320 (d) 87840

100. The mean and median of 5 observations are 9 and 8, respectively. If 1 is subtracted from each observation, then the new mean and the new median will respectively be **2016 I**

(a) 8 and 7 (b) 9 and 7
(c) 8 and 9
(d) Cannot be determined due to insufficient data

101. The age distribution of 40 children are as follows

Age (in years)	5-6	6-7	7-8	8-9	9-10	10-11
Number of children	4	7	9	12	6	2

Consider the following statements in respect of the above frequency distribution

I. The median of the age distribution is 7 yr.

II. 70% of the children are in the age group 6-9 yr.

III. The modal age of the children is 8 yr.

Which of the above statements are correct? **2016 I**

(a) I and II (b) II and III (c) I and III (d) I, II and III

102. Suppose $x_i = \lambda^i$ for $0 \le i \le 10$, where $\lambda > 1$. Which one of the following is correct? **2016 I**

(a) AM < Median (b) GM < Median
(c) GM = Median (d) AM = Median

103. Suppose $x_i = \frac{1}{i}$ for $i = 1, 2, 3, \dots, 11$. Which one of the following is not correct? **2016 I**

(a) AM > 1/6 (b) GM >1/6
(c) HM>1/6 (d) Median = HM

ANSWERS

1	*c*	2	*b*	3	*b*	4	*c*	5	*a*	6	*b*	7	*c*	8	*c*	9	*d*	10	*b*
11	*d*	12	*b*	13	*d*	14	*a*	15	*a*	16	*c*	17	*a*	18	*b*	19	*b*	20	*b*
21	*d*	22	*a*	23	*c*	24	*d*	25	*a*	26	*a*	27	*c*	28	*b*	29	*b*	30	*d*
31	*c*	32	*a*	33	*d*	34	*b*	35	*b*	36	*d*	37	*c*	38	*c*	39	*c*	40	*b*
41	*b*	42	*b*	43	*b*	44	*b*	45	*c*	46	*b*	47	*c*	48	*b*	49	*c*	50	*b*
51	*d*	52	*d*	53	*c*	54	*c*	55	*b*	56	*c*	57	*b*	58	*d*	59	*b*	60	*c*
61	*b*	62	*b*	63	*a*	64	*b*	65	*a*	66	*c*	67	*b*	68	*c*	69	*d*	70	*b*
71	*b*	72	*c*	73	*a*	74	*a*	75	*b*	76	*a*	77	*d*	78	*a*	79	*a*	80	*c*
81.	*c*	82	*b*	83	*c*	84	*c*	85	*a*	86	*d*	87	*c*	88	*a*	89	*c*	90	*c*
91	*a*	92	*b*	93	*c*	94	*d*	95	*b*	96	*b*	97	*d*	98	*d*	99	*b*	100	*a*
101	*b*	102	*c*	103	*d*														

HINTS AND SOLUTIONS

1. *(c)* Frequency polygon can be drawn by joining the mid-points of the respective tops in histogram.

2. *(b)* An ogive is used to determine the median.

3. *(b)* Let the lower limit be x. Then, the upper limit of class interval $= x + 10$

$\because$ mid-value = 42

$\therefore \quad \dfrac{x+(x+10)}{2} = 42 \Rightarrow 2x + 10 = 84$

$\Rightarrow \quad 2x = 74 \Rightarrow x = 37$

$\therefore$ Lower limit = 37

Upper limit = 37 + 10 = 47

4. *(c)* Lower class limits are obtained by subtracting 0.5 from the lower limit, so clearly 9.5, 19.5, 29.5 and 39.5 are the actual lower class limits.

5. *(a)* Here, sum of 5 observations

$$= x + (x+2) + (x+4) + (x+6) + (x+8)$$
$$= 5x + 20$$

and total number of observation = 5

$\therefore$ Mean of total observations

$$= \frac{\text{Sum of 5 observation}}{\text{Total number of observation}}$$
$$= \frac{5x+20}{5} = 11 \Rightarrow 5x + 20 = 55$$

$\Rightarrow 5x = 55 - 20 = 35 \Rightarrow x = 7$

$\therefore$ Mean of first three

$$= \frac{x+(x+2)+(x+4)}{3} = \frac{3x+6}{3}$$
$$= \frac{3\times 7+6}{3} = \frac{27}{3} = 9 \qquad (\because x = 7)$$

Hence, the mean of first three observation is 9.

6. *(b)* Total sum of items

$$= 2\times 12 + 3\times 12 + 5\times 12$$
$$= 24 + 36 + 60 = 120$$

Total sum of items of first two group

$$= (2+3)\times 3 = 15$$

Total sum of 5 items of group third

$$= 120 - 15 = 105$$

$\therefore$ Mean of third group $= \dfrac{105}{5} = 21$

Hence, the mean of third group is 21.

7. *(c)* Given, $n_1 = 7, \bar{x}_1 = 10$

and $n_2 = 3, \bar{x}_2 = 5$

$\because$ Combined mean $= \dfrac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

$$= \frac{7\times 10 + 3\times 5}{7+3} = \frac{85}{10} = 8.5$$

Hence, the mean of combined set is 8.5.

8. *(c)* Given, total number of students = 50

Total number of students passed

$$= 50 - 10 = 40$$

Total marks obtained by failed students

$$= 28 \times 10 = 280$$

$\therefore$ Total marks of 40 students

$$= 2800 - 280 = 2520$$

$\therefore$ Required average of passed students

$$= \frac{2520}{40} = 63$$

9. *(d)* Clearly, all the statements are true about median.

10. *(b)* Median is the middle item of the series arranged in ascending or descending order.

11. *(d)* Central angle for 540 marks = 360°.

$\therefore$ Central angle for 108 marks

$$= \frac{108}{540} \times 360° = 72°$$

Here, the student scored 108 marks in Social Science.

12. *(b)* Given, $n_1 = 45, \bar{x}_1 = 2, n_2 = 40$

$\bar{x}_2 = 2.5$ and $n_3 = 15, \bar{x}_3 = 2.$

$\therefore$ Required mean

$$= \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3}$$
$$= \frac{45\times 2 + 40\times 2.5 + 15\times 2}{45+40+15}$$
$$= \frac{90+100+30}{45+40+15} = \frac{220}{100} = 2.2$$

Hence, the mean of the combined distribution is 2.2.

13. *(d)* Here,

$$\Sigma f_i = (1+2+3+\ldots+n) = \frac{n(n+1)}{2}$$
$$\Sigma f_i \times x_i = \left(1\times 1 + 2\times\frac{1}{2} + 3\times\frac{1}{3} + \ldots + n\times\frac{1}{n}\right)$$
$$= (1+1+1+\ldots+n\text{ times}) = n$$
$$\therefore \text{Mean} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{n}{\frac{1}{2}n(n+1)} = \frac{2}{n+1}$$

14. *(a)* Geometric mean of 40, 50 and $x = (40 \times 50 \times x)^{1/3}$

$\Rightarrow \quad (40 \times 50 \times x)^{1/3} = 10$ (given)

$\Rightarrow \quad 40 \times 50 \times x = 10^3 \Rightarrow x = \dfrac{1000}{40 \times 50} = \dfrac{1}{2}$

Hence, the value of x is $\dfrac{1}{2}$.

15. *(a)* We know that the sum of deviation from their mean is zero.

$\therefore \; x_1 f_1 + x_2 f_2 + \cdots + x_n f_n = 0$

16. *(c)* The height of the bar is not proportional to the frequency of the class.

17. *(a)*

x	f	$f \times x$
0	4	0
1	f	f
2	9	18
3	g	$3g$
4	4	16
Total	$\Sigma f = 17 + f + g$	$\Sigma fx = 34 + f + 3g$

$\because \; \text{Mean} = \dfrac{\Sigma fx}{\Sigma f} = \dfrac{34 + f + 3g}{17 + f + g} \Rightarrow 2 = \dfrac{34 + f + 3g}{17 + f + g}$ $(\because \text{mean} = 2)$

$\Rightarrow \; 34 + 2f + 2g = 34 + f + 3g \Rightarrow f - g = 0$ …(i)

Given, $\quad \Sigma f = 25$

$\therefore \quad 17 + f + g = 25 \Rightarrow f + g = 25 - 17$

$\Rightarrow \quad f + g = 8$ …(ii)

On adding Eqs. (i) and (ii), we get $2f = 8 \Rightarrow f = 4$

On putting $f = 4$ Eq. (i), we get $4 - g = 0 \Rightarrow g = 4$

18. *(b)* For district A, as the class 44.0-47.0 has maximum frequency, so it is the modal class.

$\therefore \quad l = 44, f_1 = 59, f_0 = 36, f_2 = 30, \text{ and } \quad h = 3$

$\therefore \text{Mode} = l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 44 + \dfrac{59 - 36}{2 \times 59 - 36 - 30} \times 3$

$= 44 + \dfrac{23 \times 3}{52} = 44 + 1.33 = 45.33$

For district B, as the class 47.0-50.0 has the maximum frequency, so it is the modal class.

$\therefore l = 47, f_1 = 54, f_0 = 35, f_2 = 41$ and $h = 3$

$\therefore \text{Mode} = l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 47 + \dfrac{54 - 35}{2 \times 54 - 35 - 41} \times 3$

$= 47 + \dfrac{19 \times 3}{32} = 47 + 1.78 = 48.78$

Thus, the mode for district B is higher than the mode for district A.

19. *(b)* Given, average score of 50 students = 44

Total score = $44 \times 50 = 2200$

Correct score of 50 students = $(2200 - 73 + 23) = 2150$

$\therefore$ Correct average score $= \dfrac{2150}{50} = 43$

20. *(b)* In the less than type of ogive, the cumulative frequency is plotted against the upper limit of the concerned class interval.

21. *(d)* Mean of $A = \dfrac{2 + 3 + 7 + 1 + 3 + 2 + 3}{7} = \dfrac{21}{7} = 3$

Mode of $A = 3$, having maximum frequency.

Arrange data in ascending order, we have 1, 2, 2, 3, 3, 3, 7.

and median of $A = \left(\dfrac{7+1}{2}\right)$th value = 4th value = 3

Here, mean, median and mode are equal.

22. *(a)* Total marks in 3 subjects = $(75 + 80 + 85) = 240$ out of 300.

In any other subject the marks are atleast 0.

Average of 4 subjects = $\dfrac{240}{4} = 60\%$

$\therefore$ Average cannot be less than 60%.

23. *(c)* Given, mean = 8.9 and median = 9

$\because \quad \text{Mode} = (3 \times \text{Median}) - (2 \times \text{Mean})$

$= (3 \times 9 - 2 \times 8.9) = 27 - 17.8 = 9.2$

24. *(d)* If every number of a finite set is increased by any number k, then measure of central tendency should also increase by k.

But geometric mean does not have this property.

25. *(a)* Given distribution is $1, 3, 5, 7, 9, x, 15, 17, 19, 21$.

Number of terms = 10 (even)

$\therefore \text{Median} = \dfrac{\text{Value of } \dfrac{10}{2} \text{ th term} + \text{Value of} \left(\dfrac{10}{2} + 1\right) \text{th term}}{2}$

$10 = \dfrac{\text{Value of 5th term} + \text{Value of 6th term}}{2}$

$\Rightarrow \quad 10 = \dfrac{9 + x}{2} \Rightarrow 20 = 9 + x \Rightarrow x = 11$

26. *(a)* Required class boundary = Lower class boundary of lowest class + Width of class × Number of class

$= 5.1 + 2.5 \times 10 = 30.1$

27. *(c)* Arranging the observation in ascending order 2, 3, 4, 6, 7, 7, 8

$\therefore$ Number of items = 7

$\therefore \text{Median} = \left(\dfrac{n+1}{2}\right)$th value $= \left(\dfrac{7+1}{2}\right)$ = 4th value = 6

So, median = 6 and mode of 2, 9, 3, 4, 9, 6, 9 is 9.

$\therefore$ Average of median and mode $= \dfrac{1}{2}(6 + 9) = 7.5$

28. *(b)* Let the mean of observations $x_1, x_2, x_3, \ldots, x_n$ be $\bar{x}$.

On subtracting 10 from each, the mean would be $(\bar{x} - 10)$.

$(\bar{x} - 10) = 60\%$ of $\bar{x}$

$\Rightarrow \; (\bar{x} - 10) = \dfrac{60}{100}\bar{x} = \dfrac{3}{5}\bar{x} \Rightarrow \left(1 - \dfrac{3}{5}\right)\bar{x} = 10$

$\Rightarrow \quad \dfrac{2}{5}\bar{x} = 10 \Rightarrow \bar{x} = \dfrac{5 \times 10}{2} = 25$

Now, 5 is added to all the given observations, then new mean becomes $= \bar{x} + 5 = 25 + 5 = 30$

29. *(b)* Range of the data = 120 − 71 = 49

$\therefore$ Class size $= \dfrac{\text{Range}}{\text{Number of classes}} = \dfrac{49}{7} = 7$

The class are 71-78, 78-85, 85-92, 92-97 etc.

So, limits of second class interval are 78 and 85.

30. *(d)* Here, mean, $M = \dfrac{y + \dfrac{1}{y}}{2}$ ...(i)

New mean

$$= \frac{y^3 + \dfrac{1}{y^3}}{2} = \frac{\left(y + \dfrac{1}{y}\right)^3 - 3\left(y + \dfrac{1}{y}\right)}{2}$$

$$= \frac{\left(y + \dfrac{1}{y}\right)\left[\left(y + \dfrac{1}{y}\right)^2 - 3\right]}{2} \text{ [from Eq. (i)]}$$

$$= M[(2M)^2 - 3] = M(4M^2 - 3)$$

31. *(c)* Since, maximum frequency is 80, hence mode will be between 15 to 20.

32. *(a)* ₹ 25 = 1 cm^2 $\Rightarrow$ ₹ 1 = $\dfrac{1}{25}$ cm^2

$\therefore$ ₹ 81 = $\dfrac{81}{25}$ cm^2 = Area of square

Side of square = $\sqrt{\dfrac{81}{25}} = \dfrac{9}{5} = 1.8$ cm

33. *(d)* Given, $n = 5$

Mean $= \dfrac{7 + 9 + 11 + 13 + 15}{5} = \dfrac{55}{5} = 11$

$$\Sigma|d_i|^2 = |7 - 11|^2 + |9 - 11|^2 + |11 - 11|^2 + |13 - 11|^2 + |15 - 11|^2 = 40$$

$$\therefore \sigma = \sqrt{\frac{\Sigma d_i^2}{n}} = \sqrt{\frac{40}{5}} = 2\sqrt{2} = 2 \times 1.41$$

$= 2.82$

34. *(b)* Number of cellphones sold by Motorola $= \dfrac{3}{16} \times 45664$

Central angle of Motorola

$$= \left(\frac{\dfrac{3}{16} \times 45664}{45664} \times 360°\right) = \left(\frac{3}{16} \times 360°\right)$$

$= 67.5°$

35. *(b)* If the population figures are given for each State of India, then data can be classified as quantitative.

36. *(d)* The volume of rainfall in certain geographical area, recorded every month for 24 consecutive months.

37. *(c)* I. First seven prime numbers are 2, 3, 5, 7, 11, 13, 17.

Mean $= \dfrac{2 + 3 + 5 + 7 + 11 + 13 + 17}{7}$

$= \dfrac{58}{7} = 8.28$

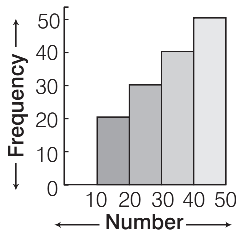
Median = Value of $\left(\dfrac{7+1}{2}\right)$th term

= Value of 4th term = 7

Since, prime numbers are always increasing, therefore here mean is always greater than median.

Hence, both statements are correct.

38. *(c)* Since, the histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars and also the height of the rectangle is determined by the frequency of the class it represents. Hence, both statements are correct in respect of a histogram.



39. *(c)* We know that, histogram is an equivalent graphical representation of the frequency distribution and is suitable for continuous random variables where the total frequency of an interval is evenly distributed over the interval.

Hence, both the given statements are correct.

40. *(b)* Primary data is original research that is obtained through first hand investigation. So, statements I and III are primary data.

41. *(b)* Collar sizes of 200 shirts sold in a week, here mode is a suitable measure of central tendency.

42. *(b)* Since, the frequency in a straight line, so we take all classes have equal frequency, i.e. 10.

I. It is true that first and last class have 10 frequency.

II. Both the middle classes have frequency, 10 + 10 = 20

III. Since, all have equal frequency, so we cannot determined the mode.

43. *(b)* A discrete frequency distribution is such a distribution in which data are presented in a way that exact measurements of the units are clearly shown. Clearly weights of a set of a students is continuous, while other three are discrete.

44. *(b)* Since, modal size of given data is 1 and its corresponding frequency is 6.

So, x cannot be equal to or more than 6.

Hence, the maximum value of x is 5.

45. *(c)*

Number of goals	Frequency	CF
0	3	3
1	6	9
2	2	11
3	x	$11 + x$
4	5	$16 + x$
Total	$N = 16 + x$	

Since, median is 2 and the corresponding CF is 11.

So, $\dfrac{N}{2}$ must be greater than 9.

$\Rightarrow \dfrac{16 + x}{2} > 9 \Rightarrow x > (18 - 16) \Rightarrow x > 2$

So, smallest possible value of $x = 3$.

46. *(b)* Here, total expenditure

= 6 + 7 + 3 + 4 + 2.5 + 1.5

= ₹ 24 lakh

Sectorial angle of largest sector

$= \dfrac{360°}{24} \times 7 = 105°$

47. *(c)* Difference in the sectorial angles of the largest and the smallest sectors

$= 105° - \dfrac{360°}{24} \times 1.5$

$= 105° - 22.5° = 82.5°$

48. *(b)* Here, $l = 18$, $\dfrac{N}{2} = 50$, $c = 7 + x$,

$f = 40$ and $h = 5$

$$\therefore \text{Median} = l + \frac{\left(\dfrac{N}{2} - c\right)}{f} \times h$$

$$= 18 + \frac{(50 - (7 + x))}{40} \times 5$$

$\therefore \quad 20 = 18 + \dfrac{5}{40}(43 - x)$ [$\because$ median = 20]

$\Rightarrow \quad 800 = 720 + 215 - 5x$

$\Rightarrow \quad 5x = 935 - 800$

$\Rightarrow 5x = 135 \Rightarrow x = \dfrac{135}{5} \Rightarrow x = 27$

Hence, the value of x is 27.

49. *(c)* We have,

$7 + x + 40 + y + 10 + 2 = 100$ ($\because x = 27$)

$\Rightarrow 86 + y = 100 \Rightarrow y = 100 - 86$

$\Rightarrow \quad y = 14$

Hence, the value of y is 14.

Solutions (Q. Nos. 50-52) Since, $a < b < c$

Therefore, series is in increasing order i.e. a, a, b, b, c, c

$\therefore$ Median

$$= \frac{\left(\dfrac{6}{2}\right)\text{th term} + \left(\dfrac{6}{2} + 1\right)\text{th term}}{2}$$

$$= \frac{\text{3rd term + 4th term}}{2}$$

$\Rightarrow 2 = \dfrac{b + b}{2} \Rightarrow b = 2$

Also, arithmetic mean

$$= \frac{a + a + b + b + c + c}{6}$$

$\Rightarrow \quad \dfrac{7}{3} = \dfrac{a + b + c}{3} \Rightarrow a + c + 2 = 7$

$\Rightarrow a + c = 7 - 2 = 5$ ...(i)

and geometric mean $= (a^2 \times b^2 \times c^2)^{1/6}$

$\Rightarrow 2 = (abc)^{1/3} \Rightarrow abc = 8 \Rightarrow ac = \frac{8}{2} = 4$ $[\because b = 2]$...(ii)

$\therefore$ From Eqs. (i) and (ii), we get, $a = 1$, $c = 4$ and $b = 2$

50. *(b)* Required sum $= 2(a)^2 + 2(b)^2 + 2(c)^2$
$= 2(1)^2 + 2(2)^2 + 2(4)^2 = 2 + 8 + 32 = 42$

51. *(d)* The value of c is 4.

52. *(d)* $\because$ Mode = 3 (median) − 2 (mean)
$= 3(2) - 2\left(\frac{7}{3}\right) = 6 - \frac{14}{3} = \frac{18-14}{3} = \frac{4}{3}$

53. *(c)* Percentage of wheat production $= \frac{70°}{360°} \times 100\% = 19.44\%$

Hence,the percentage of wheat production is 19.44%.

54. *(c)* If $40° = 9000$ kg $\Rightarrow 70° = \frac{9000 \times 70}{40} = 15750$ kg

Hence, the wheat production is 15750 kg.

55. *(b)* $360° = 180000$ kg

Difference in Sugar and wheat angle $= (70° - 40°) = 30°$

$\therefore$ Required production $= \frac{180000 \times 30}{360} = 15000$ kg

Hence, the difference in sugar and wheat production is 15000 kg.

Solutions (Q. Nos. 56-58)

Class interval	f_i	cf	x_i	$f_i x_i$
1-4	6	6	2.5	15.0
4-7	30	36	5.5	165.0
7-10	40	76	8.5	340.0
10-13	16	92	11.5	184.0
13-16	4	96	14.5	58.0
16-19	4	100	17.5	70.0
Total	$N = 100$			832.0

56. *(c)* We have, $\frac{N}{2} = \frac{100}{2} = 50$, so cf just greater than 50 is 76 which lies in class 7-10.

$\therefore$ Median class $= 7 - 10$, $l = 7$, $f = 40$, $c = 36$, $h = 3$

$$\text{Median} = l + \left(\frac{\frac{N}{2} - c}{f}\right) \times h = 7 + \left(\frac{50-36}{40}\right) \times 3 = 7 + \frac{42}{40} = 8.05$$

57. *(b)* The maximum frequency is 40, which lies in the class = 7 − 10

$\therefore$ Modal class $= 7 - 10 \Rightarrow l = 7$, $f_1 = 40$, $f_0 = 30$, $f_2 = 16$, $h = 3$

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$= 7 + \left(\frac{40-30}{2 \times 40 - 30 - 16}\right) \times 3 = 7 + \frac{30}{34} = 7 + 0.88 = 7.88$$

58. *(d)* Mean $= \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{832}{100} = 8.32$

$\therefore$ Mean + Mode − 2 Median $= 8.32 + 7.88 - 2 \times 8.05$
$= 16.2 - 16.1 = 0.1$

59. *(b)* In an increasing order arrangements of observations, the median will not be affected when the smallest and the largest observations are removed.

60. *(c)* Primary data are those data which is collected by investigator himself, while secondary data are those which is collected by other persons.

Hence, both statements are correct.

61. *(b)* In pie diagrams, all the items are converted into angles.

62. *(b)* Given that, the arithmetic mean of the above distribution =3.1

$$\Rightarrow \frac{\sum_{i=1}^{5} x_i f_i}{\sum_{i=1}^{5} f_i} = 3.1 \Rightarrow \frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + x_4 f_4 + x_5 f_5}{f_1 + f_2 + f_3 + f_4 + f_5} = 3.1$$

$$\Rightarrow \frac{(1)(3) + (2)(f) + (3)(6) + (4)(5) + (5)(3)}{3 + f + 6 + 5 + 3} = 3.1$$

$$\Rightarrow \frac{3 + 2f + 18 + 20 + 15}{17 + f} = 3.1 \Rightarrow 2f + 56 = 3.1f + 52.7$$

$$\Rightarrow 1.1f = 3.3 \Rightarrow f = 3$$

Hence, the value of f is 3.

63. *(a)* The weight of a person is continuous variable because the weight of the persons can be divide into different but certain range of intervals. But educational qualification of the person does not show the attributes. Hence, statement I is true and statement II is false.

64. *(b)* Let 100 value be $x_1, x_2, x_3, \ldots, x_{100}$

$\because$ Mean $= \frac{\text{Sum of values}}{\text{Total number of values}}$, $45 = \frac{x_1 + x_2 + x_3 + \cdots + x_{100}}{100}$

$\Rightarrow x_1 + x_2 + x_3 + \ldots + x_{100} = 4500$ ...(i)

Now, according to the question

$\therefore$ Sum of new value

$= (x_1 + 15 + x_2 + 15 + \ldots + x_{40} + 5) + x_{41} - 5 + x_{42} - 5 + \ldots + x_{100} - 5$

$\therefore$ New mean $= \frac{\text{Sum of new value}}{\text{Number of values}}$

$$= \frac{(x_1 + x_2 + x_3 + \ldots + x_{40}) + 40 \times 15 + (x_{41} + x_{42} + \ldots + x_{100}) - 5 \times 60}{100}$$

$$= \frac{(x_1 + x_2 + \ldots + x_{100}) + 600 - 300}{100} = \frac{4500 + 300}{100} \quad \text{[from Eq. (i)]}$$

$$= \frac{4800}{100} = 48$$

Hence, the new mean is 48.

65. *(a)* Given numbers are 10, 7, 8, 5, 6, 8, 5, 8 and 6.

First we arrange these numbers in ascending order.

5, 5, 6, 6, 7, 8, 8, 8, 10

Total term, $n = 9$ (odd)

Now,

(i) Mean $= \frac{5+5+6+6+7+8+8+8+10}{9} = \frac{63}{9} = 7$

(ii) Median $= \left(\frac{n+1}{2}\right)$th term $= \left(\frac{9+1}{2}\right)$th term = 5th term = 7

(iii) Mode = 8 (higher frequency term)

$\therefore$ Mean = Median

66. *(c)* In histogram the width of the bars is proportional to class interval.

67. *(b)* A frequency polygon can be drawn joining the mid-points of the respective tops of the rectangle of a histogram in the case of equal class intervals.

68. *(c)* A frequency distribution is a comprehensive way to classify raw data of a quantitative variable. It shows how different values of a variable are distributed in different classes along with their corresponding class frequencies. It is an equivalent representation of original data.

69. *(d)*

CI	x	f	f × x
0-10	5	x	$5x$
10-20	15	$x+8$	$15(x+8)$
20-30	25	$x-2$	$25(x-2)$
30-40	35	$x-4$	$35(x-4)$
Total		$\Sigma f = 4x+2$	$\Sigma fx = 80x-70$

$\because$ Mean $= \dfrac{\Sigma fx}{\Sigma f} \Rightarrow 17.8 = \dfrac{80x-70}{4x+2}$

$\Rightarrow 17.8(4x+2) = 80x - 70 \Rightarrow 71.2x + 35.6 = 80x - 70$

$\Rightarrow 8.8x = 105.6 \Rightarrow x = \dfrac{105.6}{8.8} = 12$

70. *(b)*

Class interval *(ci)*	0-10	10-20	20-30	30-40
Frequencies *(f)*	12	20	10	8
Cumulative frequency *(cf)*	12	32	42	50

Here, $N = 50 \Rightarrow \dfrac{N}{2} = 25$

The frequency just greater than 25 is 32.

So, median class is 10-20.

71. *(b)* Here, modal class is 10-20, because it has maximum frequency i.e. 20.

72. *(c)* Given that, mean of 7 observations = 7

$\Rightarrow$ New mean = Previous mean + Number added to each term $= 7 + 2 = 9$

73. *(a)* Given that, number of male employees $(M) = 45$

Number of female employees $(F) = 15$

Mean salary of male employees $(\bar{x}_M) =$ ₹ 5000

Total number of employees $= (M + F) = 45 + 15 = 60$

Mean salary of total employees $(\bar{x}_{MF}) =$ ₹ 4800

Let mean salary of female employees be $\bar{x}_F$.

$$\bar{x}_{MF} = \frac{M\bar{x}_M + F\bar{x}_F}{(M+F)} \Rightarrow 4800 = \frac{45 \times 5000 + 15 \times \bar{x}_F}{60}$$

$\Rightarrow 4800 \times 60 - 45 \times 5000 = 15 \times \bar{x}_F$

$\therefore \bar{x}_F = 4800 \times 4 - 3 \times 5000 = 300(16 \times 4 - 50)$

$= 300 \times 14 =$ ₹ 4200

Hence, the mean salary of female employees is ₹ 4200.

74. *(a)* Median $= l + \left(\dfrac{\dfrac{n}{2} - cf}{f}\right) \times h$

$$\Rightarrow 14.4 = 12 + \left[\frac{\dfrac{20}{2} - (4+x)}{5}\right] \times 6 = 12 + \frac{10 - 4 - x}{5} \times 6$$

$\Rightarrow 14.4 - 12 = \dfrac{6-x}{5} \times 6 \Rightarrow 2.4 = \dfrac{36-6x}{5}$

$\Rightarrow 12 = 36 - 6x \Rightarrow 6x = 24 \Rightarrow x = 4$

75. *(b)* We have, $4 + x + 5 + y + 1 = 20 \Rightarrow x + y = 10$

$\Rightarrow 4 + y = 10 \quad (\because x = 4) \Rightarrow y = 6$

Now, $\dfrac{x}{y} = \dfrac{4}{6} = \dfrac{2}{3} \Rightarrow 3x = 2y$

76. *(a)* $\because$ Mean $= \dfrac{\text{Sum of the observations}}{\text{Number of observations}}$

$\Rightarrow \dfrac{p+q+r+s+t+u+v}{7} = m$

$\Rightarrow p + q + r + s + t + u + v = 7m$ ...(i)

$\therefore (p-m) + (q-m) + (r-m) + (s-m) + (t-m) + (u-m) + (v-m)$

$= (p+q+r+s+t+u+v) - 7m$ [from Eq. (i)]

$= 7m - 7m = 0$

77. *(d)* $\because$ Average $= \dfrac{\text{Sum of the observations}}{\text{Number of observations}}$

$\Rightarrow \dfrac{u+v+w+x+y+z}{6} = 10 \quad (\because \text{average} = 10)$

$\Rightarrow u + v + w + x + y + z = 10 \times 6 = 60$ ...(i)

Now, sum of new observations

$u + 10 + v + 20 + w + 30 + x + 40 + y + 50 + z + 60$

$= (u + v + w + x + y + z) + 10 + 20 + 30 + 40 + 50 + 60$

$= 60 + 10 + 20 + 30 + 40 + 50 + 60 = 270$ [from Eq. (i)]

Now, average of $u + 10$, $v + 20$, $w + 30$, $x + 40$, $y + 50$ and $z + 60 = \dfrac{270}{6} = 45$

78. *(a)*

Class interval	x_i	f_i	$x_i f_i$
11-13	12	3	36
13-15	14	6	84
15-17	16	9	144
17-19	18	13	234
19-21	20	f	$20f$
21-23	22	5	110
23-25	24	4	96
Total		$\Sigma f_i = 40 + f$	$\Sigma x_i f_i = 704 + 20f$

Here, $\Sigma f_i = 40 + f$ and $\Sigma x f_i = 704 + 20f$ and mean, $\bar{x} = 18$

$\therefore$ Mean, $\bar{x} = \dfrac{\Sigma x_i f_i}{\Sigma f_i} \Rightarrow 18 = \dfrac{704 + 20f}{40 + f}$

$\Rightarrow 704 + 20f = 720 + 18f$

$\Rightarrow 20f - 18f = 720 - 704 \Rightarrow 2f = 16 \Rightarrow f = 8$

Hence, the value of f is 8.

79. *(a)* Frequency polygon is formed by joining the mid-points of histogram. The original frequency distribution can be reconstructed from frequency polygon. Frequency polygon does not touch the X-axis in its extreme right and extreme left.

Hence, only statement I is true.

80. *(c)* I. Mean of all observations $= \dfrac{2 \times 4 + 7 \times 4}{8} = \dfrac{8 + 28}{8} = 4.5$

For median, first we arrange the given observations in ascending order = 2, 2, 2, 2, 7, 7, 7, 7, $n = 8$ (even)

$$\therefore \quad \text{Median} = \frac{\frac{n}{2}\text{th term} + \left(\frac{n}{2}+1\right)\text{th term}}{2}$$

$$= \frac{\frac{8}{2}\text{th term} + \left(\frac{8}{2}+1\right)\text{th term}}{2} = \frac{4\text{th} + 5\text{th}}{2} = \frac{2+7}{2} = 4.5$$

II. Mode is both 2 and 7, since frequency of occurrence is same i.e. maximum frequency.

81. *(c)* A list of annual incomes of the members of a club is a good example of statistical data.

82. *(b)*

Income less than	Class interval	Number of persons	Frequency
1500	1250-1500	100	20
1250	1000-1250	80	10
1000	750-1000	70	15
750	500-750	55	23
500	250-500	32	20
250	0-250	12	12

Here, maximum frequency is 23.

So, the modal class is 500-750.

83. *(c)* Geometric mean of

$(x_1, x_2, x_3, \cdots, x_n) = (x_1 \cdot x_2 \cdots x_n)^{1/n} = X$

and geometric mean of $(y_1, y_2, y_3, \cdots, y_n)$

$$= (y_1 \cdot y_2 \cdots y_n)^{1/n} = Y$$

$\therefore$ Geometric mean of $(x_1 y_1, x_2 y_2, \ldots, x_n y_n)$

$$= (x_1 y_1 \cdot x_2 y_2 \cdots x_n y_n)^{1/n}$$

$$= (x_1 \cdot x_2 \cdots x_n)^{1/n} \times (y_1 \cdot y_2 \cdots y_n)^{1/n} = XY$$

Geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \ldots, \frac{x_n}{y_n}\right) = \left(\frac{x_1}{y_1} \cdot \frac{x_2}{y_2} \cdots \frac{x_n}{y_n}\right)^{1/n}$

$$= \frac{(x_1 \cdot x_2 \cdots x_n)^{1/n}}{(y_1 \cdot y_2 \cdots y_n)^{1/n}} = \frac{X}{Y}$$

84. *(c)* Arithmetic mean uses all the data and is always uniquely defined. Median uses only one or two numbers from the data and may not be uniques. $\left(\frac{n+1}{2}\right)$th term for odd n and half the sum of $\left(\frac{n}{2}+1\right)$th and $\frac{n}{2}$th terms for even n.

85. *(a)* If x_i's are the mid-points of the class intervals of grouped data, f_i's are the corresponding frequencies and $\bar{x}$ is the mean, then $\Sigma f_i (x_i - \bar{x}) = 0$

86. *(d)* When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the areas proportional to the frequencies of the respective class intervals, the graph so obtained is called histogram.

87. *(c)* Both statements I and II are suitable for representation of a pie diagram.

88. *(a)* Here, statement I is correct but statement II is not correct.

89. *(c)* The class which has maximum frequency is known as modal class.

90. *(c)* Given observations in ascending order are

6, 14, 15, 17, $x+1$, $2x-13$, 30, 32, 34 and 43.

Here, $n = 10$ (even)

$$\therefore \text{Median} = \frac{\text{Value of}\left(\frac{n}{2}\right)\text{th term} + \text{Value of}\left(\frac{n}{2}+1\right)\text{th term}}{2}$$

$$= \frac{\text{Value of}\left(\frac{10}{2}\right)\text{th term} + \text{Value of}\left(\frac{10}{2}+1\right)\text{th term}}{2}$$

$$= \frac{\text{Value of 5th term} + \text{Value of 6th term}}{2}$$

$$= \frac{x+1+2x-13}{2} = \frac{3x-12}{2}$$

But given, median = 24 $\Rightarrow \frac{3x-12}{2} = 24$

$\Rightarrow \quad 3x - 12 = 24 \times 2 = 48 \Rightarrow 3x = 48 + 12$

$\Rightarrow \quad 3x = 60 \Rightarrow x = 20$

Hence, the value of x is 20.

Solutions (Q. Nos. 91-94)

Class interval	Frequency (f)	Class mark (x)	f(x)	cf
0-10	4	5	20	4
10-20	5	15	75	9
20-30	7	25	175	16
30-40	10	35	350	26
40-50	12	45	540	38
50-60	8	55	440	46
60-70	4	65	260	50
Total	50		1860	

91. *(a)* Mean $= \frac{\Sigma fx}{\Sigma f} = \frac{1860}{50} = 37.2$

Hence, the value of mean is 37.2.

92. *(b)* Here, $N = 50$, Now, $\frac{N}{2} = \frac{50}{2} = 25$

which lies in the cumulative frequency corresponding class interval for cf 26 is 30-40.

93. *(c)* From the table, $l = 30, f = 10, C = 16$ and $h = 10$

$$\therefore \quad \text{Median} = l + \frac{\left(\frac{N}{2} - C\right)}{f} \times h$$

$$= 30 + \frac{(25-16)}{10} \times 10 = 30 + 9 = 39$$

94. *(d)* Modal class of the given data is 40-50, because it has largest frequency among the given classes of the data i.e. 12.

Here, $l = 40, f_1 = 12, f_0 = 10, f_2 = 8$ and $h = 10$

$$\therefore \text{Mode} = l + \left\{\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right\} \times h = 40 + \left\{\frac{12-10}{2 \times 12 - 10 - 8}\right\} \times 10$$

$$= 40 + \frac{2 \times 10}{24 - 18} = 40 + \frac{20}{6} = 40 + 3.33 = 43.33$$

Hence, the mode of given data is 43.33.

95. *(b)* Given, total number of votes = 100000

Party E get votes $= 100000 - (36000 + 24000 + 18000 + 7000)$

$= 100000 - 85000 = 15000$

$\therefore$ Angle allocated for party $E = \frac{15000}{100000} \times 360° = 54°$

96. *(b)* We have, discrete values $a+4$, $a-3.5$, $a-2.5$, $a-3$, $a-2$, $a+0.5$, $a+5$ and $a-0.5$

Now, ascending order is $a-3.5, a-3, a-2.5, a-2, a-0.5, a+0.5, a+4, a+5$.

$\therefore$ Median $= \dfrac{a-2+a-0.5}{2} = \dfrac{2a-2.5}{2} = a-1.25$

97. *(d)* We have, $x_i = i$

$$\text{Mean} = \frac{2x_1+3x_2+4x_3+\ldots+(n+1)x_n}{n}$$

$$= \frac{2\times1+3\times2+4\times3+\ldots+(n+1)\times n}{n}$$

$$= \frac{\Sigma n(n+1)}{n} = \frac{\Sigma n^2+\Sigma n}{n} = \frac{\dfrac{n(n+1)(2n+1)}{6}+\dfrac{n(n+1)}{2}}{n}$$

$$= \frac{n(n+1)}{2n}\left[\frac{2n+1}{3}+1\right] = \frac{(n+1)(n+2)}{3}$$

98. *(d)* Weighted AM $= \dfrac{1\times1+2\times2+3\times3+\ldots+10\times10}{10}$

$$= \frac{\dfrac{10\times11\times21}{6}}{10} = \frac{21\times11}{6} = \frac{77}{2} = 38.5$$

99. *(b)* Let the total votes be x.

Then, central angle of party $A = \dfrac{360°}{x}\times21960$

$\Rightarrow \quad 135° = \dfrac{360°}{x}\times21960 \Rightarrow x = \dfrac{360°\times21960}{135°} = 58560$

100. *(a)* $\because \dfrac{\text{Sum of 5 observations}}{\text{Number of observations}} = 9$

$\Rightarrow$ Sum of 5 observations $= 9\times5 = 45$

If 1 is subtracted from each observation, then

New mean of 5 observations $= \dfrac{\text{Sum of 5 observations} - 5}{5}$

$$= \frac{45-5}{5} = 8$$

Median of 5 observations $= \left(\dfrac{5+1}{2}\right)$th term $=$ 3rd term $= 8$

If 1 is subtracted from each observation, then

New median $= 8-1 = 7$

Hence, the new mean and median are 8 and 7, respectively.

101. *(b)*

Age (in years)	Number of children	Cumulative frequency
5-6	4	4
6-7	7	11
7-8	9	20
8-9	12	32
9-10	6	38
10-11	2	40
Total	$N = 40$	

I. Here, $N = 40 \Rightarrow \dfrac{N}{2} = 20$

Thus, 7-8 is the median class.

$\therefore \quad l = 7, f = 9, C = 11, \dfrac{N}{2} = 20$ and $h = 1$

$$\text{Median} = l + \frac{\dfrac{N}{2}-C}{f}\times h$$

$$= 7 + \frac{20-11}{9}\times1$$

$$= 7+1 = 8 \text{ yr}$$

Hence, statement I is incorrect.

II. Total number of children $(N) = 40$

Number of children in the age group 6-9 $= 7+9+12 = 28$

$\therefore$ Required percentage $= \dfrac{28}{40}\times100\% = 70\%$

Hence, statement II is correct.

III. $\because$ Modal group = 8-9,

$\therefore$ Modal age = 8 yr

Since, 9 is not included in this group.

Hence, statement III is correct.

102. *(c)* We have 11 observations as follow

$$1, \lambda, \lambda^2, \lambda^3, \lambda^4, \lambda^5, \lambda^6, \lambda^7, \lambda^8, \lambda^9, \lambda^{10}$$

Here, number of observations = 11 (odd)

$$\text{Median} = \left(\frac{11+1}{2}\right)\text{th term} = 6\text{th term} = \lambda^5 \quad \ldots(\text{i})$$

$$\therefore \quad \text{AM} = \frac{1+\lambda+\lambda^2+\ldots+\lambda^{10}}{11} = \left(\frac{\lambda^{11}-1}{11(\lambda-1)}\right)$$

and $\quad \text{GM} = (1\cdot\lambda\cdot\lambda^2\cdot\lambda^3\ldots\lambda^{10})^{\frac{1}{11}} = (\lambda^{1+2+3+\ldots+10})^{1/11}$

$$= (\lambda)^{55/11} = \lambda^5 \quad \ldots(\text{ii})$$

From Eqs. (i) and (ii), we get GM = Median

103. *(d)* We have, 11 observations as follows

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots, \frac{1}{11} \quad \text{(odd)}$$

$$\therefore \text{Median} = \left(\frac{11+1}{2}\right)\text{th term} = 6\text{th term} = \frac{1}{6} \quad \ldots(\text{i})$$

and $\quad \dfrac{1}{\text{HM}} = \dfrac{1}{11}\left(\dfrac{1}{1}+\dfrac{1}{1/2}+\dfrac{1}{1/3}+\dfrac{1}{1/4}+\ldots+\dfrac{1}{1/11}\right)$

$\Rightarrow \quad \dfrac{1}{\text{HM}} = \dfrac{1}{11}(1+2+3+\ldots+11)$

$\Rightarrow \quad \dfrac{1}{\text{HM}} = \dfrac{66}{11} \Rightarrow \text{HM} = \dfrac{1}{6}$

$\therefore \quad$ HM = Median [from Eq. (i)]

GENERAL ENGLISH

TREND ANALYSIS
(2016-2014)

S.No.	Chapter Name	2016 (I)	2015 (II)	2015 (I)	2014 (II)
1	Synonym	9	15	8	12
2	Antonym	-	10	7	8
3	One Word Substitution	-	-	5	-
4	Sentence Arrangement (S_1-S_6)	10	-	10	8
5	Sentence Arrangement (PQRS)	15	-	7	11
6	Spotting Error	15	15	20	25
7	Cloze Test	20	25	20	20
8	Suitable Word Selection	10	10	-	-
9	Question Based on Comprehension	21	20	23	16
10	Sentence Improvement	20	25	20	20
	Total	**120**	**120**	**120**	**120**

01

SPOTTING THE ERRORS

Generally, 15 to 20 questions based on spotting the errors are asked in CDS exam. In these type of questions, the given sentences are divided into three parts with each part marked as (a), (b) and (c). We have to choose that part as our answer which has error. If there is no error in any part, then we have to choose option (d) as the answer.

'Spotting the errors' or 'Syntax' is that part of English language, where the grammatical skills of the candidates are comprehensively tested. Therefore, it can be said to be the most important part of the language proficiency test.

It is an integrated grammar exercise, so it covers all the grammatical parts. *This includes*

- Parts of speech
- Number, gender, cases and degrees
- Confusing words
- Uses of tenses
- Non-finites
- Determiners and articles

All these areas are to be studied thoroughly to score high in the CDS examination.

FORMAT OF THE QUESTIONS

Each question consists of a complex sentence, which is divided into **three parts**. One of the parts may contain an **error**. The candidate has to spot the error and mark the incorrect option/part. In case, there is no error in the sentence, option (d), which stands for 'No error', needs to be selected.

An example of this type of question is given below.

Direction *Which part of the following sentence contains an error? In case there is no error, choose option (d).*

Question The train should arrive (a) at 7:30 in the morning (b) but it was almost an hour late (c) No error (d)

Ans. *(a)* In the given sentence, option (a) is given as 'The train should arrive' while the correct sentence is 'The train should have arrived'. Here 'should arrive' should be replaced by 'should have arrived'. So, **option (a)** has an error.

NOUN

A word that refers to a person, place, thing, event, substance or quality is called a Noun.

e.g. Ram, Delhi, Book, Lion, Street, etc.

Kinds of Noun

1. **Common Noun** It is a name given to every person or thing of the same class or kind.
   e.g. – Boy, girl, fan, chair, etc.
2. **Proper Noun** Details of a common noun are signified by proper noun.
   e.g. – Amit, Amita, Polar fan, Supreme chair, etc.
3. **Material Noun** It refers to a material or substance from which things are made such as silver, gold, iron, cotton, diamond and plastic.
4. **Collective Noun** Some nouns refer to groups of people (e.g. audience, team, committee, government etc). These are called as Collective Nouns.
5. **Abstract Noun** It is the name of a feeling, quality or a state.
   e.g. – *Feeling* Love, fear, hate, anger, respect, pleasure, etc.
   – *Quality* Strength, pitch, innocence, gluttony, judgement, obedience, beauty, etc.
   – *State* Weariness, mercy, birth, death, etc.

Number of Nouns

There are two kinds of nouns of number, which are as follow

Singular Noun It refers to one (single) person or thing.
e.g. – boy, girl, table, man, etc.

Plural Noun It refers to more than one person or things.
e.g. – boys, girls, tables, men, etc.

- Sometimes plurals are made by changing the inside vowel.

Singular	**Plural**
Man	Men
Woman	Women
Foot	Feet
Tooth	Teeth
Mouse	Mice

- Sometimes plurals are made by adding 's' to the principal word of a compound noun. e.g.

Singular	**Plural**
Brother-in-law	Brothers-in-law
Vice-Admiral	Vice-Admirals
Court Martial	Courts Martial
Commander-in-chief	Commanders-in-chief
Runner-up	Runners-up
Looker-on	Lookers-on

- *Some other examples are as follow*

Singular	**Plural**
Ox	Oxen
Datum	Data
Medium	Media
Stratum	Strata
Index	Indices
Formula	Formulae
Lacuna	Lacunae
Alumnus	Alumni
Corrigendum	Corrigenda
Erratum	Errata
Syllabus	Syllabi

- Nouns like sheep, deer, offspring, cod, fish, salmon, etc have the same form in both the numbers.

Rules of Noun

1. Articles are not used before material nouns.
   e.g. – *The* leather is used in making shoes. (✗)
   – Leather is used in making shoes. (✓)
2. Material nouns and abstract nouns are not used in plural.
   e.g. – *Cares* of the old is necessary. (✗)
   – *Care* of the old is necessary. (✓)
3. Proper nouns are sometimes used as common nouns.
   e.g. – Samudragupta is *the* Napoleon of India.
   – Kalidas is *the* Shakespeare of India.

   In these two sentences, the proper nouns Napolean and Shakespeare are used as common nouns.
4. Collective nouns usually take a singular verb and are substituted by a singular pronoun.
   e.g. – The jury *has* given *its* verdict.
   – The team *has* performed to *its* potential.
5. Collective nouns takes a plural verb and are substituted by a plural pronoun when the group members do not behave as a whole but take different directions.
   e.g. – The jury ~~is~~ / *are* divided in ~~its~~ / *their* decision.
   – The Ministry *are* much divided in *their* opinion regarding the foreign policy.
6. **Collective nouns**, even when they denote living beings, are considered to be of neuter gender.
   e.g. – India has a big army and *it* is divided into three major divisions.
7. Young children and animals are also referred to as neuter gender.
   e.g. – The baby started crying when *it* was hungry.
   – The lizard lost *its* tail when *it* was frightened.

8. When objects without life are personified, they are considered either masculine or feminine gender.

Masculine Gender is used for strength, violence, anger and vengeance. It is used with Sun, Death, Summer and Winter.

e.g. – Nature has taken *his* vengeance by claiming lives of 100 persons.
– The sun, with *his* brilliance, came out of the clouds.

These sentences emphasise the strong masculine traits of nature and the sun i.e. 'vengeance' and 'brilliance', so the masculine gender 'his' is used.

Feminine Gender is used for beauty, gentleness and gracefulness. It is used with Earth, Moon, Spring, Nature and even for Sun.

e.g. – Nature offers ~~*its*~~ / *her* lap to those who seek it.
– The sun, with *her* warmth, provided relief from the bitter cold.

In the above sentences, the tender feminine traits of nature and the sun are revealed, so the feminine gender (her) is used.

9. Units of counting, i.e., pair, dozen, score, hundred, thousand, etc, when used after numbers retain their singular form.

e.g. – Ten pair, Five dozen, Two score, Six hundred, Three thousand, One lakh, Ten crore, Eight million, Ten billion, etc.

e.g. – My friend bought two *dozen* / ~~*dozens*~~ eggs from the market.

Here, the quantity is mentioned before the dozen, so we have its singular form.

e.g. – My friend bought *dozens* of eggs from the market.

Here, the quantity is not mentioned in the above sentence, so we use the plural form 'dozens'. One more example is

e.g.
– Sunil Gavaskar scored *thousands* of runs in his cricketing career.

10. In a compound noun, a compound word is not used in plural if a noun does the work of an adjective.

e.g. – He is pursuing a two year/~~years~~ diploma course.
– He is a fifteen year/~~years~~ old boy.
– I have got a hundred rupee/~~rupees~~ note.
– He ran a five mile/~~miles~~ race.
– They went on a fifteen day/~~days~~ tour.

11. *The following nouns are used only in plural.*

- Names of instruments, which have two parts forming a kind of pair. Like scissors, spectacles, glasses, tongs, etc.

  e.g. – Scissors *are* / ~~*is*~~ used to cut.
  – My pair of spectacles *is* very expensive.

  In the above sentence, we use singular form 'is' as the word 'pair' is added before the word 'spectacles'.

- Certain articles of dress as trousers, shorts, jeans, etc.

  e.g. – Jeans *are* in vogue these days.

- Certain collective nouns (although they are singular in form). Police, cattle, gentry, peasantry, clergy, people, poultry, majority, artillery, infantry, etc.

  e.g. – Police *have* reached the crime scene.
  – Cattle *are* grazing in the field.

- Certain other nouns are thanks, assets, premises, alms, proceeds, contents, refreshments, orders, repairs, requirements, rations, statistics (*collection of data*), credentials, etc.

  e.g. – Court should make it mandatory for the ministers to declare their *assets*.
  – Alms *were* distributed among the beggars.

12. Some plural forms are commonly used in singular like Mathematics, Statistics, Physics, Economics (*All subjects*), Gymnastics, News, Innings, Series, Measles, Rabies, Mumps, Rickets, Summons, Names of Books, etc.

13. Uncountable nouns are used in the singular forms only. Indefinite article (a, an) is not used before them, nor are they (a, an) used with plural verbs. 'Much' is used in place of 'Many' for denoting plurality.

Some of the important nouns of this category are as follow

Advice, information, hair, luggage, business, work, word (promise), mischief, bread, scenery, abuse, vacation, evidence, employment, alphabet, poetry, furniture, baggage, trouble, fuel, wheat, rice, stationary, etc.

e.g. – (a) He gave me *an* information. (✗)
He gave me information. (✓)
– (b) You should be true to your *words*. (✗)
You should be true to your word. (✓)
– (c) He was punished for committing *many* mischiefs. (✗)
He was punished for committing much mischief. (✓)

Cases of Noun

There are three types of cases of noun, which are as follow

1. Nominative Case

A noun or pronoun is in the nominative case if it is used as the subject of a verb.

e.g. – *Sachin Tendulkar* scored a century.

Clue Put *Who* in active sentence and *What* in passive sentence before the verb to get nominative case.

2. Objective Case

A noun or pronoun is in the objective case if it is used as the object of a verb.

e.g. – Sachin Tendulkar scored *a century* (*Active*).
– A century was scored by *Sachin Tendulkar* (*Passive*).

Clue Put *What* in active sentence and *Whom* in the passive sentence after the subject and the verb to get the objective case.

3. Possessive Case

A noun is said to be in the possessive case if it denotes possession, authorship, origin, kind, etc.

e.g. – *Amit's* house is at the back-side.

How is Possessive Case Formed

i. By adding's to a singular noun.
e.g. – *Amit's, Donald's.*

ii. By adding's to plural nouns not ending in *s*.
e.g. – *Children's* school, *Men's* club.

iii. By adding only an apostrophe to a plural noun ending in *s*.
e.g. – *Boys', victims', pilgrims'*.

iv. By adding only an apostrophe to a singular noun when there are hissing sounds.
e.g. – *Jesus*' blessings, for *peace*' sake, for *conscience*' sake.

Rules of Possessive Case

i. In case of a compound noun, the possessive sign is attached only to the last word.
e.g. – My *brother-in-law's* marriage.
– The *Queen-of-England's* residence.

ii. When two or more nouns show joint possession, the apostrophe sign is put with the latter only.
e.g. – Dharmendra is *Sunny* and *Bobby's* father.

iii. The words his, hers, its, theirs, yours, ours are possessive and they are not written with the possessive sign.
e.g. – Neither did his efforts succeed nor ~~*your's*~~ / *yours*.

iv. The adverb 'else' combined with indefinite pronouns (somebody, anybody, etc) is expressed in possessive case as 'somebody else's' in place of 'somebody's else'.
e.g. – Is it your house? I thought it is ~~somebody's else~~ /somebody else's?

v. The words church, house, school, shop are often omitted after a possessive case.
e.g. – Yesterday I met my friend at *St John's.*

vi. The possessive case is chiefly used with the names of living things.

vii. The possessive case is also used with nouns denoting time, space or weight.
e.g. – *Stone's* throw away distance
– A *minute's* time
– A *day's* journey

Confusing Words

Word	*Meaning*	*Example*
Advice	Singular means an opinion or suggestion.	The teacher gave many *pieces of advice* to the students before the exam.
Advices	Plural (Advices) means information.	The invigilator gave *advices* to the students before the exam.
Cloth	Singular means unfinished product.	We give *cloth* to a tailor to get the *clothes* stitched.
Clothes	Plural means garments.	
Colour	Singular and plural means red, green, etc.	Yellow is my favourite *colour*.
Colours	(Plural only) implies 'true personality of someone'.	A realist sees true *colours* of people.
Force	Singular means strength.	Apply *force* to open the lid.
Forces	Plural means the military organisations for air, land and sea.	*Forces* landed at the disputed spot on time.
Content	Singular means satisfaction.	I am *content* with the *contents* of this book.
Contents	Plural means parts.	
Light	Singular means radiance.	There was a *light* on the ascetic's face.
Lights	Singular and plural means lamps.	Diwali is a festival of *lights*.
People	Singular is used when we are talking about masses.	*People* of India, *Peoples* of Europe.
Peoples	Plural means people belonging to different cultures and ethnicities.	
Practice	Singular means exercise of a profession.	It is compulsory for new doctors to *practice* in villages for 2 years in the beginning of their career.
Practices	(Both singular and plural) means habit.	
Custom	Singular means tradition.	One should respect one's *custom* and traditions.
Customs	Plural means a department.	The *customs* department seized illegal goods at the airport.

Words Denoting Groups

1. An *army* of soldiers.
2. An *alliance* of states, powers, etc.
3. An *assembly* of representatives.
4. An *attendance* of servants, persons.
5. An *audience* of listeners.
6. A *band* of musicians, followers.
7. A *batch* of pupils, candidates.
8. A *battery* of guns.
9. A *bench* of judges or magistrates.
10. A *block* of houses, buildings.
11. A *body* of men, soldiers, police, laws, etc.
12. A *brigade* of cavalry, infantry or artillery.
13. A *bundle* of hay.
14. A *bouquet* of flowers.
15. A *board* of trustees or directors.
16. A *caravan* of merchants.
17. A *code* of laws.
18. A *congress* of representatives.
19. A *century* of runs.
20. A *cloud* of locusts.
21. A *cluster* of islands.
22. A *constellation* of stars.
23. A *course* of lectures.
24. A *consignment* of goods.
25. A *catalogue* of books.
26. A *circle* of friends, acquaintances.
27. A *clan* of people.
28. A *clique* of persons (belonging to a body).
29. A *commonwealth* of bees.
30. A *concourse* of people.
31. A *confederacy* of persons, nations, states.
32. A *confederation* of persons, powers, states.
33. A *conference* of preachers, delegates.
34. A *congregation* of worshippers.
35. A *convoy* of ships.
36. A *corporation* of people.
37. A *corps* of soldiers, volunteers, police.
38. A *curriculum* of studies.
39. A *crew* of sailors.
40. A *herd* of cattle.
41. An *escort* of soldiers.
42. A *fraction* of people (engaged in politics).
43. A *family* of plants, languages.
44. A *federation* of states.

SPOTTING THE ERRORS **SET 1**

ERRORS OF NOUN

Directions (Q.Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. Order has been issued (a)/ for his transfer to another district (b)/ but he has not received them so far. (c)/ No error (d)

2. Although she has studied (a)/ English for almost a year (b)/ she is yet to learn the alphabets. (c)/ No error (d)

3. There are two scores of books (a)/ which are lying (b)/ unused in the library. (c)/ No error (d)

4. Children are prone (a)/ to making mischiefs (b)/ if they have nothing to do. (c)/ No error (d)

5. Sheeps are economically useful (a)/ and so they are reared (b)/ in the hills. (c)/ No error (d)

6. I have not gone through (a)/ the letter and so I am not aware (b)/ of its content. (c)/ No error (d)

7. I shall not attend the meeting (a)/ since I have many works to complete (b)/ within allotted time. (c)/ No error (d)

8. You should always be (a)/ true to your words (b)/ if you are to succeed in life. (c)/ No error (d)

9. It is a pity (a)/ that even five years old boys (b)/ are engaged in hazardous factories. (c)/ No error (d)

10. I gave him (a)/ two hundred rupees notes (b)/ for depositing. (c)/ No error (d)

11. It is not my business (a)/ to give an advice to those (b)/who are not sensible enough to deal with their own problems. (c)/ No error (d)

12. I don't think (a)/ it is your house (b)/ It is somebody's else. (c)/ No error (d)

13. She misplaced her spectacle (a)/ and is now feeling (b)/ great difficulty in studying. (c)/ No error (d)

14. Arabian Nights are (a)/ a collection of (b)/ very interesting episodes of adventure. (c)/ No error (d)

15. I hope to visit (a)/ my uncle only next year (b)/ during summer vacations. (c)/ No error (d)

16. Ration has run out (a)/ and the District Magistrate (b)/ has been informed. (c)/ No error (d)

17. The table's wood (a)/ is infested with mite (b)/ and I am likely to dispose it off. (c)/ No error (d)

18. I can't come to you now (a)/ because a lot of works (b)/ remains to be done. (c)/ No error (d)

19. A farmer was leading oxes (a)/ to his field for ploughing (b)/ early in the morning. (c)/ No error (d)

20. My sister-in-laws (a)/who live in Mumbai (b)/ have come to stay with us. (c)/ No error (d)

21. This article (a)/ is not available (b)/ in any of the shop in the market. (c)/No error (d)

22. The morale of the army (a)/ was high because the news (b)/ coming from the front are very encouraging. (c)/ No error (d)

23. A flock of lions (a)/ roamed about (b)/ fearlessly in the jungle. (c)/ No error (d)

24. I have done my best; (a)/ the whole thing is now (b)/ in the laps of God. (c)/ No error (d)

25. A trained gang of sailors (a)/ was employed (b)/ on the ship. (c)/ No error (d)

26. Interviews for (a)/ the posts of lecturers (b)/ will begin from Monday. (c)/ No error (d)

27. What (a)/ is the criteria (b)/ of selection in the examination? (c)/ No error (d)

28. Satyajit Ray, who conceived, co-authored (a)/ and directed a number of good films, was (b)/ one of India's most talented film-maker. (c)/ No error (d)

29. One of the most (a)/ widespread bad habit (b)/ is the use of tobacco. (c)/ No error (d)

30. The books (a)/ that you need (b)/ are kept on the table. (c)/ No error (d)

EXPLANATIONS

1. *(a)* When a person is assigned to a new station, he receives his 'orders'. This is a set of instructions, including where to be, when to be there etc. So, the correct sentence would be 'Orders have been issued ……… .'

2. *(c)* 'Alphabet' is the set of letters in a language. English alphabet has 26 letters from A-Z. In the given sentence, we will use 'alphabet' instead of 'alphabets'.

3. *(a)* When a quantity is given before 'score' like '2' or '5' etc, 'score' is not used in the plural form. Hence, we would use 'score' in place of 'scores' in the given sentence.

4. *(b)* The word 'mischief' does not have a plural form. Hence, 'mischief' would be used.

5. *(a)* The plural of 'sheep' is 'sheep'. In the given sentence, the word 'Sheeps' would be replaced by 'Sheep'.

6. *(c)* The word 'content' is used in its plural form 'contents' when it is used to mean 'something contained'. So, the given sentence will have 'contents' instead of 'content'.

7. *(b)* As 'work' is an uncountable noun we need to use 'much work' instead of 'many works'.

8. *(b)* 'True to your word' means 'promise'. In the given sentence, we would replace 'true to your words' by 'true to your word'.

9. *(b)* As we know that in a compound noun, a compound word is not used in plural if a noun does the work of an adjective.
In the given sentence, 'five years old' is an adjective of noun 'boys'. So, 'five year old' would be used instead of 'five years old'.

10. *(b)* In the given sentence, 'two hundred rupees' is used as an adjective of the noun 'notes'. So, it should be 'two hundred rupee'. (For explanation refer to Ans. 9).

11. *(b)* As 'advice' is an uncountable noun, the article 'an' would not be used before it.

12. *(c)* The possessive pronoun in the given sentence 'somebody's else' is incorrectly used. Its correct usage will be 'somebody else's'.

13. *(a)* The correct usage of 'spectacle' is 'spectacles'.

14. *(a)* The book 'Arabian Nights' is a singular noun. So, 'are' would be replaced by 'is'.

15. *(c)* The correct use of 'vacations' is 'vacation'.

16. *(a)* 'Rations have run out' should be used.

17. *(a)* 'The table's wood' does not seem appropriate. It should be replaced by 'The wood of the table' which makes sense.

18. *(b)* As 'work' is an uncountable noun, 'because a lot of work' would be used.

19. *(a)* The plural of 'ox' is 'oxen'. In the sentence, 'oxes' is used instead of 'oxen' which is incorrect.

20. *(a)* The plural of sister-in-law would be sisters-in-law. So, we would replace 'sister-in-laws' by 'sisters-in-law'.

21. *(c)* 'Any' refers to a number of things (here shops). So, we would replace 'shop' by 'shops'.

22. *(c)* 'News' is used as a singular noun. So, 'coming from the front is very encouraging' would be used.

23. *(a)* 'Pride' is the right word to refer to a group of lions. So, we would replace 'flock' by 'pride'.

24. *(c)* 'Laps' is incorrect. It should be changed to 'lap'.

25. *(a)* A group of sailors is called a crew. So, we would replace 'gang' by 'crew'.

26. *(b)* 'Posts' should be replaced by 'post' to make the sentence correct.

27. *(b)* 'Criteria' should be replaced by 'criterion' to make the sentence correct.

28. *(c)* 'Film-maker' should be changed to 'film-makers'.

29. *(b)* 'One of' takes a plural noun. So, we would change 'habit' to 'habits'.

30. *(d)* No error.

PRONOUN

Words used in place of nouns are called pronouns. They are used to avoid the repitition of nouns in a sentence. e.g. I saw a boy on the roof. *He* seemed to recognise me.

In this example, the pronoun 'he' is used instead of repeating the noun 'boy' which is underlined.

Kinds of Pronoun

Personal Pronouns

This pronoun refers to or is related to the words which are used in place of nouns referring to person.

Personal pronouns have the following characteristics

Ist Person : The one(s) speaking (I, me, my, mine, we, us, our, ours).

2nd Person : The one(s) spoken to (you, your, yours).

3rd Person : The one(s) spoken about (he, him, his, she, her, hers, it, its, they, their, theirs).

Persons	Nominative Case		Objective Case		Possessive Case	
	Singular	Plural	Singular	Plural	Singular	Plural
I Person	I	We	Me	Us	My/Mine	Our/Ours
II Person	You	You	You	You	Your/ Yours	Your/ Yours
III Person						
Male	He	They	Him	Them	His	Their/Theirs
Female	She		Her		Her/Hers	
Neuter	It		It		Its	

Rules of Personal Pronouns

1. A personal pronoun must be of the same number, gender and person as the noun for which it stands, i.e. antecedent should agree with the noun.
   e.g. – I am not one of those who believe everything ~~I~~ / *they see*.
   – Every man must love *his* / ~~*her*~~ / ~~*their*~~ country.
   – He is one of the best boys that *have* played here.
2. The component of the verb *to be (is, am, are, was, were, will, shall)*, when it is expressed by a pronoun, should be in the nominative case.
   To be means

Is	:	He, She, It
Am	:	I
Are	:	We, You, They
Was	:	I, He, She, It
Were	:	We, You, They
Will	:	You, He, She, It, They
Shall	:	I, We

   e.g. – It is ~~*me*~~ / *I*.
   – It will be *he* / ~~*him*~~ who is going to win.
   – Was it ~~*her*~~ / *she* who did it for you?
3. If a pronoun is used as the object of the verb or of a preposition, it should be in the objective case.
   e.g. – He was shouting at ~~*I*~~ / *me*.
   – He was teaching ~~*he*~~ / *him*.
   – The sweets are to be distributed among *you*, *him* and *me*.
4. Objective case is used after the following words *let, like, but,* etc.
   e.g. – It is no one else but ~~*he*~~ / *him* who has done the crime.
   – He likes *me*.
   – Let *him* come inside.
5. Words such as *as good as, as well as, as soon as, as beautiful as, as intelligent as, etc* are followed by nominative case.
   e.g. – When it comes to providing news, no one is as good as *he*.
   – In studies he is as good as *he*.
6. **2, 3, 1 *Rule*** The second person should come before the third and the third person before the first. This case applies to singular pronouns only.
   e.g. – *I, You and He* are good friends. (✗)
   – *You, He and I* are good friends. (✓)
7. **1, 2, 3 *Rule*** 2, 3, 1 becomes 1, 2, 3 when we are talking about plural pronouns.
   e.g. – *We, You and They* will go to the party.
   – *We, You and They* will take dinner at *our*/ *their* / *your* house.

   Exception When it comes to confessing something or committing a crime, 2, 3, 1 gets changed to 1, 2, 3 for all types of pronouns.
   e.g. – *I, you and he* will be punished for the crime.
8. When a pronoun stands for a collective noun, it must be in the neuter gender. But if the collective noun denotes separation or division, the pronoun used is plural.
   e.g. – The jury gave *its* decision unanimously.
   – The jury *were* divided in *thei*r opinion.
9. When two or more nouns are joined by 'and', the pronoun used would be plural.
   e.g. – Ram *and* Mohan went to their school.
   – Suresh *and* his family members have completed their work.

   Case I *Separate persons, the pronoun used for them must be plural.*

Case II *Same person, the pronoun used for them must be singular*.

[**Hint** If a single Article is used before the nouns, the verb and the pronoun are both singular, because the reference is to a singular person only].

e.g. – The Comptroller and Auditor General *has* / ~~*have*~~ submitted *his* / ~~*their*~~ report.

– The Chairman and the Managing Director ~~*has*~~ / *have* submitted ~~*his*~~ / *their* report.

10. When two or more singular nouns are joined by or, either-or or neither-nor, the pronoun and the verb should be singular.

e.g. – Either Raj or Amar *is* doing *his* duty.

11. Whenever one singular and one plural noun are used with either-or or neither-nor, the plural noun always comes second. In this case, the verb and the pronoun both become plural.

e.g. – Neither Amit nor his friends ~~*was*~~ / *were* present in *their* house.

12. While writing question tag, the subject and verb must be according to the main sentence.

e.g. – Our minister is intelligent, *isn't he*?

– The boys are not enjoying themselves, *are they*?

– They went to Delhi yesterday, *didn't they*?

Reflexive Pronouns

Reflexive pronouns are pronouns that refer back to the subject of the sentence. They end in-'self', as in singular form, or-'selves' in plural form.

e.g. – Myself, themselves, yourself, ourselves etc.

Rules of Reflexive Pronouns

1. The reflexive pronoun is used with the following words absent, avail, apply, enjoy, pride, resign, acquit, revenge, exert, adapt, adjust, etc.

e.g. – He absented *himself* from the class.

– He acquitted *himself* admirably in the meeting.

2. With the following words, reflexive pronouns are not used: bathe, break, feed, hide, turn, move, rest, qualify, stop, etc.

e.g. – We *bathed* in the river.

– We *fed* at the motel.

3. A reflexive pronoun cannot be used as a substitute for the subject.

e.g. – Amit and *I* / ~~*myself*~~ were present on the site.

– *I* / ~~*Myself*~~ will see to it that you do not get the job.

Demonstrative Pronouns

The pronouns that are used to point out the objects to which they refer are called demonstrative pronouns like these, that, those, such, it, this, etc.

e.g. – *That* is the book I was looking for.

Rules of Demonstrative Pronouns

1. Do not commit the error of omission by forgetting to use 'that' (for plural use 'those').

e.g. – The Mumbai Film Studio is *bigger than* Noida. (✗)

– The Mumbai Film Studio is bigger *than that* of Noida. (✓)

2. Pronoun 'it' comes before the phrase or clause to which it refers, whereas 'this' follows the phrase or clause it refers to.

e.g. – *This* is true that India has won the match. (✗)

– *It is* true that India has won the match. (✓)

Indefinite Pronouns

When a pronoun refers to a person or a thing in a general way, but not to any person or thing in particular, it is called indefinite pronoun like any, anyone, none, someone, everyone, everybody, one, etc.

Rules of Indefinite Pronouns

1. One, if used in a sentence, always repeats itself.

e.g. – *One* must respect ~~*his*~~ / *one's* country for ~~*his*~~ / *one's* sake.

– *One* must obey ~~*their*~~ / *one*'s elders.

2. When we are not talking specifically about females, only masculine gender is used.

e.g. – Everyone was getting ready for *his* / ~~*her*~~ *show.*

– Everyone of the Miss India contestants was getting ready for *her* show.

3. 'Either' is replaced by 'anyone' when we are talking about more than two persons or things. Same is the case with 'none' or 'neither'.

e.g. – I couldn't contact ~~*either*~~ / *anyone* of the three.

– *Anyon*e of the three can come in.

– None / ~~Neither~~ of his body parts is defective.

4. 'Each other' should be used in speaking of two persons or things, 'one another' in speaking of more than two persons or things.

e.g. – We should love *one another.*

Relative Pronouns

This pronoun refers or relates to some noun which comes after this pronoun. Besides, it acts as a conjunction also because it connects two sentences.

e.g. – Who, whom, which, where, etc.

– He is the boy *wh*o has topped the class.

Rules of Relative Pronouns

1. 'Who' is used for subject and 'Whom' for object.

e.g. – *Who* are you?

– *Who*m were you talking to?

– She is the girl ~~*who*~~ / *whom* I met in the train.

2. The relative pronoun *that* is preferred to 'who' or 'which' in the following cases

Case I After adjectives in the superlative degree.

e.g. – It is the best movie ~~which~~ / *that* I have ever seen.

– It is the best food ~~which~~ / *that* I have had for years.

Case II After the following words all, same, any, only, nothing, the only, etc.

e.g. – It is the same book ~~which~~ / *that* I saw in the market yesterday.

– All *that* glitters is not gold.

– It is not for nothing *that* he studied Psychology.

Case III After the interrogative pronouns, 'who' and 'what'.

e.g. – Who is the girl ~~which~~ / *that* comes in your dreams?

– What is it *that* you want?

Case IV After two antecedents, one denoting a person and the other denoting an animal or a thing.

e.g. – The man and his dog *that* had entered the school were turned out.

3. The relative pronoun should be placed as near as possible to its antecedent.

e.g. – The office was located in the heart of the town *which* had beautiful interior decoration. (✗)

– The office, *which* had beautiful interior decoration, was located in the center of the town. (✓)

4. The relative pronoun 'What' is used without any antecedent.

e.g. – I mean ~~that~~ / *what* I say.

– ~~That~~ / *What* cannot be cured must be endured.

5. 'Whose' is used to refer to persons only; 'of which' is used while referring to lifeless objects.

e.g. – I have a friend *whose* father is a doctor.

– I saw a watch, the dial *of which* was made of gold.

Interrogative Pronouns

The interrogative pronouns are used for making queries or asking questions. The pronoun 'who', 'what' and 'which' are used as interrogative pronouns.

Rules of Interrogative Pronouns

1. 'What' is used in broad sense, while 'Which' is used in the specific sense.

e.g. – *What* are you doing these days?

– *Which* institute have you joined for coaching?

2. 'Which' is used in place of 'who' and 'what' when we are referring to a choice between two or among more than two things or persons.

e.g. – Of the two brothers ~~who~~ / *which* is more intelligent?

– ~~Who~~ / *Which* is your mother in the crowd?

SPOTTING THE ERRORS **SET 2**

ERRORS OF PRONOUNS

Directions (Q. Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. The master did not know (a)/ who of the servants (b)/ broke the glass. (c)/ No error (d)

2. The ruling party stood (a)/ for implementation of the bill (b)/ and was ready to stake their political existence. (c)/ No error (d)

3. Wherever they go (a)/ the Indians easily adapt to (b)/ local circumstances. (c)/ No error (d)

4. It is not easy for anyone to command (a)/ respect from one's friends as well as critics (b)/ as Dr. Neil did for his integrity and honesty. (c)/ No error (d)

5. Mahatma Gandhi taught us (a)/ that one should respect (b)/ the religions of others as much as his own. (c)/ No error (d)

6. He, You and I (a)/ shall manage (b)/ this problem together. (c)/ No error (d)

7. Was it him (a)/ who got injured (b)/ in an accident this morning? (c)/ No error (d)

8. As soon as he (a)/ saw his mother (b)/ he ran to her and embraced. (c)/ No error (d)

9. Due to me being a newcomer (a)/ I was unable to get a house (b)/ suitable for my wife and me. (c)/ No error (d)

10. The audience (a)/ are requested (b)/ to be in its seats. (c)/ No error (d)

11. A scientist must follow (a)/ his hunches and his data (b)/ wherever it may lead. (c)/ No error (d)

12. The number of vehicles (a)/ plying on this road (a)/ is more than on the main road. (c)/ No error (d)

13. Being a destitute (a)/ I admitted him (b)/ to an old people's home. (c)/ No error (d)

14. One should make (a)/ his best efforts if one wishes to achieve (b)/ success in this organisation. (c)/ No error (d)

15. May I (a)/ know who you want (b)/ to see please? (c)/ No error (d)

16. Our is the only country (a)/ in the world that can boast of (b)/ unity in diversity. (c)/ No error (d)

17. Last summer he went (a)/ to his uncle's village (b)/ and enjoyed very much. (c)/ No error (d)

18. If I were him (a)/ I would have taught (b)/ those cheats a lesson. (c)/ No error (d)

19. Those sort of people (a)/ usually do not (b)/ earn fame in society. (c)/ No error (d)

20. Had I come (a)/ to know about his difficulties (b)/ I would have certainly helped. (c)/ No error (d)

21. One of them (a)/ forgot to take their bag (b)/ from the school. (c)/ No error (d)

22. Civil servants should (a)/ acquit efficiently (b)/ in the service of a common man. (c)/ No error (d)

23. Avail every chance that comes your way (a)/ lest you should (b)/ repent in the long run. (c)/ No error (d)

24. Let Raj and she (a)/ complete this job (b)/ as they like to do it. (c)/ No error (d)

25. It was with great difficulty (a)/ that each of the brothers (b)/ could get their share of property. (c)/ No error (d)

26. Do you know (a)/ whom all (b)/ are inside the room? (c)/ No error (d)

27. They that run (a)/ after fame and money (b)/ are likely to be disappointed. (c)/ No error (d)

28. One should (a)/ take care of one's belongings (b)/ himself. (c)/ No error (d)

29. God helps those (a)/ who helps (b)/ themselves. (c)/ No error (d)

30. He and her (a)/ went to the party (b)/ in the hotel. (c)/ No error (d)

EXPLANATIONS

1. *(b)* As the sentence refers to a choice among more than two persons (servants), 'which' will be used in place of 'who'.

2. *(c)* When a pronoun stands for a collective noun ('ruling party' in the sentence) it must be in the neuter gender. Hence, 'its' will be used in place of 'their'.

3. *(b)* As the sentence refers to a particular set of people (Indians), it will contain a reflexive pronoun 'themselves' after 'adapt'.

4. *(b)* In this sentence, the pronoun 'one's' is missing before the word 'critics'. Hence, we will add 'one's' before 'critics'.

5. *(c)* The given sentence has an indefinite pronoun 'one'. As per the rule, 'his' should be replaced with 'one's' to make the sentence grammatically correct.

6. *(a)* As per the rule, the second person should come before the third and the third person comes before the first. So, the correct order will be : 'You, he and I……'

7. *(a)* As per the rule, the component of the verb 'to be' (was) when expressed by a pronoun should be in the nominative case. Hence, 'him' in the sentence will be replaced by 'he'.

8. *(c)* The sentence is incomplete as it does not answer the question 'whom did he embrace?' So, we will add 'her' at the end of the sentence.

9. *(a)* The pronoun 'me' used in part (a) of the sentence is incorrect. It should be replaced by 'I'.

10. *(c)* As the sentence refers to living beings ('audience'), the pronoun 'its' will be replaced by 'their'.

11. *(c)* The pronoun 'it' is incorrectly used in the sentence. It should be replaced by 'they' as the sentence refers to 'his hunches' and 'his data'.

12. *(c)* The sentence does not have 'that' after 'than'. So, to make the sentence meaningful 'that' should be added after 'than'.

13. *(a)* The sentence does not convey the intended meaning as it has 'He' missing in the beginning. So, the correct sentence would start as 'He being a destitute ....'

14. *(b)* As the sentence begins with an Indefinite pronoun 'one', it should not have 'his' in the sentence. Hence, we replace 'his' by 'one's' to make the sentence correct.

15. *(b)* The pronoun 'who' would be replaced by 'whom' to make the sentence meaningful.

16. *(a)* The reflexive pronoun 'Ours' will be used in the sentence instead of 'Our' to make the sentence correct.

17. *(c)* The reflexive pronoun 'himself' would be used after the word 'enjoyed' to make the sentence meaningful.

18. *(a)* The verb 'were' when expressed by a pronoun, it should be in the nominative case. Hence, 'he' would be used instead of 'him'.

19. *(a)* 'Sort of' is incorrect. It should be replaced by 'sorts of' to make the sentence correct.

20. *(c)* The sentence is incomplete as it does not answer the question 'helped whom?'. So, we add 'him' at the end of the sentence.

21. *(b)* 'Their' should be replaced by 'his' to make the sentence correct.

22. *(b)* The pronoun 'themselves' should come after 'acquit'.

23. *(a)* 'Yourself of' needs to be added after 'Avail' to make the sentence correct.

24. *(a)* 'She' should be replaced by 'her' to make the sentence correct.

25. *(c)* 'Each of' takes a singular pronoun. So, 'their' should be replaced by 'his'.

26. *(b)* 'Whom' should be replaced by 'who'.

27. *(a)* 'They' needs to be replaced by 'Those'.

28. *(c)* 'Himself' should be replaced by 'oneself'.

29. *(b)* 'Helps' should be replaced by 'help' to make the sentence grammatically correct.

30. *(a)* 'Her' would be replaced by 'she' to make the sentence error free.

VERB

A verb is a word (such as jump, think, happen or exist) that is usually one of the main parts of a sentence and that expresses an action, an occurence, or a state of being.

Classification of Verbs

Verbs can be classified as following

1. Main Verbs

Main verbs have meanings related to actions, events and states. Most verbs in English are main verbs. e.g. go, show, exist, etc.

Main verbs can be divided into two categories; transitive and intransitive which are as follow

(i) Transitive Verbs

A transitive verb is a verb that can take a direct object.

e.g. She played the piano.

transitive verb   direct object

(ii) Intransitive Verbs

An Intransitive verb has two characteristics. *First*, it is an action verb expressing a double activity like arrive, go, lie, sneeze, sit, die, etc. *Second*, unlike a transitive verb, it will not have a direct object receiving the action.

Intransitive verbs have the pattern N + V (Noun + Verb). The clause is complete without anything else.

e.g. – John *smiled*.
– Nothing *happened*.

Here, 'smiled' and 'happened' are intransitive verbs.

Linking Verbs

Some main verbs are called Linking Verbs. These verbs are not followed by objects. Instead they are followed by phrases which give extra information about the subject. Linking verbs include appear, feel, look, seem sound, smell, taste, become, etc.

e.g. – A face *appeared* at the window.

In this sentence, 'appeared' is the linking verb and 'at the window' is the phrase.

2. Auxiliary Verbs

Auxiliary verbs can be divided into two categories; primary and modal auxiliary verbs, which are as follow

(i) Primary Auxiliary Verbs

Primary auxiliary verbs can further be divided as following

i. Verbs 'to be' : is, am, are, was, were, will be, shall be
ii. Verbs 'to have' : have, has, had
iii. Verbs 'to do' : do, does, did

(ii) Modal Auxiliary Verbs

Modal auxiliary verb denote the mood/mode of the subject. They are can, could, may, might, should, used to, need, dare, etc.

Tenses

There are three basic tenses : Present, Past and Future.

These can further be divided into the following segments

1. **Simple** The action is mentioned simply. Nothing is said about whether the action is complete or not.
2. **Continuous** The action is incomplete or is going on at the time of speaking.
3. **Perfect** The action is finished or complete with respect to a certain point of time.
4. **Perfect Continuous** The action is going on continuously over a long period of time and is yet to be finished.

Simple Present

(He, She, It, Single name — V_1 + s, es)

(*They, you, I, Plurals*—V_1)

- To express a habitual action.
  e.g. – He *goes* for a walk in the morning.
  – It *rains* in winter in Tamil Nadu.
  – He often *gets* late for dinner.
- To express universal truths.
  e.g. – The sun *rises* in the East.
  – Two and two *make* four.
- In exclamatory sentences beginning with 'here' and 'there', to express what is actually taking place in the present.
  e.g. – Here he *comes*!
  – There *goes* the train!
- To indicate a future event that is part of a plan or an arrangement.
  e.g. – PM *comes* to the town next month.
  – The Indian team *goes* to England this month.
- To introduce quotations.
  e.g. – Gita *says*, "Give your best and do not worry for the results".
  – Pope *says*, "A little knowledge is a dangerous thing".

Present Continuous

(*Is / Am / Are* + V_1 + *ing*)

- To express an action going on at the time of speaking.
  e.g. – I am *studying* in the class.

- To express a temporary action which may not be actually happening at the time of speaking.
  e.g. – I am *preparing* for the competition.
  – I am *working* on a project.
- It also represents future action or a definite arrangement in the near future.
  e.g. – I am *going* to Mumbai tomorrow.
  – He is *coming* tonight.
- When the reference is to a particular obstinate habit, the present continuous is used instead of present simple. An adverb, like always, constantly, etc, is also used.
  e.g. – It is no use scolding him, he ~~*always does*~~ / *is always doing* what is forbidden.
- *The following verbs are normally used in the present simple instead of present continuous*
  (a) **Verbs of Appearance** Look, appear, seem.
  (b) **Verbs of Emotion** Want, wish, desire, feel, like, love, hate, hope, prefer, etc.
  (c) **Verbs of Perception** See, hear, smell, taste, notice, recognise, etc.
  (d) **Verbs of Possession** Belong to, consist of, contain, own, etc.
  (e) **Verbs of Thinking** Agree, believe, consider, forget, imagine, know, mind, remember, etc.

Present Perfect *(Has / Have + V_3)*

- It is used to indicate completed activities in the immediate past. It is a mixture of present and past. It always implies a strong connection of past with the present.
  e.g. – He *has* just *gone* out.
- The most important point is that it is used with the past actions whose time is not given and not definite.
  e.g. – He *came* here. (✗)
  – He *has come* here. (✓)
  – He *came* here yesterday. (✓)
- It is used with the adverbs like ever, just, recently, already, yet, so far, of late, lately, by the time, for, etc.
  e.g. – He *has already* finished the work.
  – *Recently*, he *has* started working on a new project.
  – I *have just seen* that film.
  – It is the best book that I *have ever read*.
  – I *have known* him *for* twenty years.
  – He *has started* coming late *lately*.
  – He *hasn't paid* the bill *so far*.
- It can never be used with the words like last, ago, yesterday, before, back, formerly, fixed time, etc.
  e.g. – He *has come* here yesterday. (✗)
  – He *came* here yesterday. (✓)
  – India *has won* last year. (✗)
  – India *won* last year. (✓)

Present Perfect Continuous

(Have been/Has been + V_1 + ing)

- To express an action which began at sometime in the past and is still continuing.
  e.g. – He is *playing* since 8 o'clock. (✗)
  – He *has been playing* since 8 o'clock. (✓)
  – They *have been writing* since morning. (✓)

Simple Past (V_2)

- To indicate an action completed in the past at a definite time.
  e.g. – I *did* this. (✗)
  – I *have* done this. (✓)
  – I *did* this yesterday. (✓)
- Denoted by last, ago, yesterday, back, before, formerly, any fixed time, etc.
  e.g. – We *heard* a terrifying news *last* night.
  – They *celebrated* the occasion two days *ago*.
  – He *inherited* his father's business *after* his father's demise.
  – The train *didn't* arrive on time *yesterday*.
  – She *didn't* go there *in the morning*.
- To indicate past habits, indicated generally by the words like often, seldom, never, normally, generally, always, frequently, rarely, daily, used to, etc.
  e.g. – As a kid, I *often went* to school on foot.
  – My friend *frequently visited* his hometown in the past.
  – I *seldom wrote* a cheque even when there was balance in my account.
  – *Whenever* I *called* on him he *pretended* to be ill.
  – He *always carried* a stick when he *went* for a walk.
- After 'it is time'.
  e.g. – It is time Indian cricket team ~~*starts*~~ / *started* winning test matches.

> - The conjunction **since** denotes present time dating back to some event. It is therefore, followed by a Simple Past Tense and preceded by some form of Present Perfect Tense.
>   *e.g.* Many things have happened since I have left the school. (✗)
>   Many things have happened since I left the school. (✓)

Past Continuous (Was/Were + V_1 + *ing*)

- To denote an action going on at some time in the past.
  e.g. – When I went to his house, he *was playing*.
  – I *was studying* yesterday.
- For persistent habits in the past.
  e.g. – He *was* always *mooching* around.

Past Perfect (*Had* + V_3)

- To describe an action completed before a certain moment in the past.
  e.g. – I met him in New Delhi in 2000. I *had seen* him last five years before. (✓)
  – It *had rained* yesterday. (✗)
  – It *rained* yesterday. (✓)
  – My friend *had come* to visit me. (✗)
  – My friend *came* to visit me yesterday. (✓)
- Past perfect should be used only when we wish to say that one action got completed before the other started. It should never be used at all in any other sense.
  e.g. – I *went* to Mumbai. (✗)
  – I *had gone* to Mumbai (✗)
  – Ravi *had* walked two miles by lunch time. (✓)
  – I *had gone* to Mumbai when he *came* to meet me. (✓)

Past Perfect Continuous (*Had been* + V_1 + *ing*)

- To express an action that began before a certain point of time in the past and continued upto that time.
  e.g. – He *had been studying* for two hours when his girlfriend came.
  – Tendulkar *had been* playing for eleven years when his toe got injured.
- If there is Past tense in the Principal clause, it must be followed by a Past tense in the Dependent clause. In an Indirect narration, the Simple Past in the Dependent clause is changed to Past Perfect, if the Principal clause is in the Past tense.
  e.g. – He told me that he *intended* to start a business. (✗)
  – He told me that he *had intended* to start a business. (✓)
  – He hinted that he *tried* to save him. (✗)
  – He hinted that he *had tried* to save him. (✓)
- The exception to the above rule is if some universal, habitual or generally recognised fact is mentioned in the Dependent clause, the Present tense must be retained in all conditions.
  e.g. – He told me that the Earth *moves* round the Sun.
  – His illness convinced me that all men *are* mortal.

Simple Future (*Shall / Will* + V_1)

- To express an action that is still to take place.
  e.g. – I *shall go* for the preparation when I shall receive the call letter. (✗)
  – I *shall go* for the preparation when I *receive* the call letter. (✓)

Future Continuous (*Shall be / Will be* + V_1 + *ing*)

- To express an action as going on at some time in the future.
  e.g. – I *shall be earning* when I *shall* be 21. (✗)
  – I *shall be earning* when I am 21. (✓)

Future Perfect (*Shall have / Will have* + V_3)

- To indicate the completion of an action by a certain future time.
  e.g. – We *shall have completed* our syllabus by next month.
  – I *shall have done* this work by tomorrow.

Future Perfect Continuous

(Will have been / Shall have been + V_1 + *ing)*

- To indicate an action which is in progress over a period of time and will be in progress at a certain time in future.
  e.g. – Tendulkar *will have been playing* for India for 20 years when he completes the age of 35.
  – Time *will have been clocking* for ages in the coming moments.

Subject-Verb Agreement

1. If two subjects together express one idea, one being added to the other for the sake of emphasis or clarification, the verb is singular. No plurality is left to exist in such a case.
   e.g. – Slow and steady *wins* the race.
   – Bread and butter *is* essential for one's existence.
2. When the plural noun denotes some specific quantity, distance, time or amount considered as a whole, the verb *is* generally singular.
   e.g. – Six miles *is* not a long distance for me.
   – Ten lakh *is* equivalent to a million.
3. Two or more singular subjects connected by 'either-or', 'neither-nor', take a verb in singular (third person singular verb).
   e.g. – *Either* Vivek *or* Vimal *is* absent today.
   – He asked me if *either* of the applicants *was* suitable.
   – *Either* you *or* I *shall* / <u>*will*</u> go to the party.
4. When the subjects joined by 'or' or 'nor' are of different numbers, the verb must be plural and the plural subject must be placed next to the verb.
   e.g. – *Either* Amit *or* his parents *are* coming to the party.

5. Any noun qualified by 'each' or 'every' is followed by a singular verb. Even if two nouns so qualified are connected by 'and', the verb must still be singular.

e.g. – *Each* one of these boys *has* the potential to get selected.
– *Every* man and woman *was filled* with joy.
– *Every* day and each hour *teaches* us something.

6. Verb is according to the first subject when they are connected with 'and not', with 'as well as', in addition to, 'along with', 'besides', 'like', 'together', etc.

e.g. – Rahul *and not* his friend was absent.
– Amit, *like* his friends, is always late.
– He *as well as* you is a good boy.

7. When two nouns or pronouns are joined by 'not only....but also', the verb agrees with the second noun or pronoun.

e.g. – *Not only* the officer *but also* the soldiers *were* awarded.

8. If the subject is 'The number of', the singular verb is used and the noun is plural.

e.g. – The number of one-dayers played these days *has* / ~~*have*~~ led to the deterioration of the game.

9. A 'great many' is always followed by plural verb and a plural noun.

e.g. – A great many *students have* passed this year.
– A great many *fish are* there in the pond.

10. 'Many a' is always followed by a singular verb and a singular noun.

e.g. – Many a *soldier has* got medal this year.
– Many a *student has* passed this year with flying colours.

11. A singular or a plural verb is used with words as pains, a lot of, means, variety, plenty, rest, wages, according to the sense in which they are used.

e.g. – A large number of girls *were* absent on account of bad weather.
– The number of admissions *has* gradually fallen off.
– A variety of books on the subject *are* available.

12. A verb should agree with its subject and not with the complement. But in the case of sentence beginning with 'The', the verb is according to the predicate/complement.

e.g. – Our only guide *was* the stars.
– The stars *were* our only guide.

13. In a compound sentence, both auxiliary verbs and principal verbs should be mentioned separately if they differ in number, form or voice. In such cases, one verb cannot act for both the clauses.

e.g. – He has not and will not *marry* in near future. (✗)
– He has not *married* and will not *marry* in near future. (✓)
– She is *intelligent* but her sisters *dull*. (✗)
– She is *intelligent* but her sisters are *dull*. (✓)

14. Use of 'shall' and 'will'.

- To express simple future action 'shall' is used in the first person, and 'will' in the second and third person.

  e.g. – I *shall* come.
  – You *will* come.
  – He *will* come.

- Shall is used in the second and third persons to express Command, Promise, Threat, Determination.
- Will is used in the first person to express Willingness, Promise, Threat, Determination.

  e.g. – You *shall* not steal. (Command)
  – You *shall* have a holiday tomorrow. (Promise)
  – You *shall* be punished for this. (Threat)
  – You *shall* do it for your country. (Determination)
  – I *will* send you my book. (Willingness).
  – I *will* try to do better next time. (Promise).
  – I *will* punish you if you do that again. (Threat).
  – I *will* succeed or die in this attempt. (Determination).

- In asking questions 'shall' is used in the first person and 'will' in the third person. In the second person, 'shall' and 'will' are used according to the answer expected.

  e.g. – *Shall* we go?
  – We shall go.
  – *Will* he come tomorrow?
  – He will come tomorrow.
  – *Will* you do this for me?
  – I *will* do it for you.

UNREAL PAST/ SUBJUNCTIVE MOOD

A wish, desire, purpose, supposition contrary to fact or condition is expressed in subjunctive mood.

In subjunctive mood, 'were' and 'had' are used as the case may be. The sentence basically goes in the past tense.

e.g. – I wish I *had* a car.
– I wish I *had* not met him.

There are three types of conditional clauses. Each kind contains a different pair of tenses. Here are these

i. Present likely condition
e.g. – I *shall go* for the preparation when I *get* the call letter.
– You *will pass* if you *work hard*.

ii. Present unlikely condition
e.g. – If I *had* a house, I *would not have* rented yours.
– If I *were* there, I *would not have* let them go.

iii. Past condition
e.g. – If he *had* studied, he *would have* got the call letter last year itself.
– If she *had* brought money, she *could have* bought the jewellery.

• After, 'as if/as though'.
e.g. – He behaves *as if* he were the owner of this place.
– He came in looking *as though* he had seen a ghost.

THE INFINITIVE (To + V_1)

Rules of Infinitive

1. In negative sentences, 'dare not' and 'need not' are used without *to*.
e.g. – You dare not *to* leave India. (✗)
– You dare not leave India. (✓)
– How dare you fail in the exam? (✓)

2. Prepositions 'but' and 'except' take the infinitive without *to.*
e.g. – He did nothing *but* cry.
– There is no alternative *except* this offer.

3. Expressions like would rather, had rather, rather than, had better, as soon as, etc are followed by infinitive without to.
e.g. – I would rather *to* go for picnic. (✗)
– I would rather go for picnic. (✓)

4. The infinitive without *to* is used after Auxiliary verbs such as shall, will, can, may, did, should; but ought is an exception.
e.g. – I *should* go.
– I *ought to* go.

5. The *to* of one infinitive can be made to do duty for *to* of another infinitive in the sentence, provided that the verbs in the two infinitives are synonymous. If two separate ideas are better expressed by two infinitives, *to* of the latter infinitive should be omitted.
e.g. – He helped me *to* progress and prosper.
– It lies in my power *to* succeed or *to* fail.

6. Do not forget to use the preposition whenever the infinitive 'to' is made to qualify a noun.
e.g. – I have no pen to *write*. (✗)
– I have no pen to *write with*. (✓)

THE GERUND (V_1 + ing)

A verb which does the action of a noun.

e.g. – *Running* tap makes a lot of noise.
– *Smoking* isn't a good habit.

Rules of Gerund

1. *The following words are followed by gerund*
Avoid, dislike, enjoy, help (in the sense of avoid), mind, prevent, risk, stop, etc.
e.g. – I cannot *help* looking at you.
– I do not *mind* going there.

2. *The following phrases are followed by a gerund*
Accustomed to, fed-up with, habitual to, addicted to, is no good, is used to, looking forward to, tired of, is worth, with a view to, owing to, object to, given to, taken to, etc.
e.g. – I am *accustomed* to talking for hours.
– I am *fed up with* his useless accusing.

3. A gerund, and not an infinitive is used after such verbs and participles as are followed by their appropriate prepositions.

e.g. – desirous of	disqualify from
– refrain from	prevent from
– debar from	desist from
– restrain from	prohibit from
– dissuade from	abstain from
– intent on	bent on
– keen on	aim at
– confident of	insist on
– persist in	succeed in
– fond of	successful in
– justified in	hesitate in
– a hope of	fortunate in

4. The noun or pronoun governing a gerund must be in the possessive case.
e.g. – Please excuse *me* being late. (✗)
– Please excuse *my* being late. (✓)
– I remember *him* winning the race. (✗)
– I remember *his* winning the race. (✓)
– I like my *friend* coming on time. (✗)
– I like my *friend's* coming on time. (✓)

SPOTTING THE ERRORS **SET 3**

ERRORS OF VERB

Directions (Q.Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. Each of these players (a)/ have been warned (b)/ not to repeat the silly mistake. (c)/ No error (d)

2. Lime and soda (a)/ is (b)/ a digestive drink. (c)/ No error (d)

3. The mother as well as her children (a)/ were brought (b)/ to the police station for interrogation. (c)/ No error (d)

4. His benevolence and kindness (a)/ are (b)/ admired by his friends. (c)/ No error (d)

5. She never has and never will (a)/ allow her only son (b)/ to join politics. (c)/ No error (d)

6. Intelligence, as well as knowledge of the subject (a)/ are required to grasp (b)/ the meaning of the book. (c)/ No error (d)

7. Every word and every line (a)/ in the poems of Wordsworth (b)/ sings the blessings of nature. (c)/ No error (d)

8. So honestly he worked (a)/ that he was rewarded (b)/ by the chairman of the company. (c)/ No error (d)

9. Four miles (a)/ are not a long distance (b)/ for a young person like you. (c)/ No error (d)

10. No sooner he was brought (a)/ here than he began (b)/ to feel uneasy. (c)/ No error (d)

11. Never I have come across (a)/ a man (b)/ who is foolish to such an extent. (c)/ No error (d)

12. In old age none of the relatives (a)/ are prepared to come (b)/ to the help of the old and the sick. (c)/ No error (d)

13. It were the students (a)/ who wanted the teacher (b)/ to declare holiday. (c)/ No error (d)

14. So fast did he drive the motor car (a)/ that even the best driver (b)/ could not overtake him. (c)/ No error (d)

15. Everyone of the new nursing homes (a)/ coming up in the urban areas (b)/ need a lot of improvement. (c)/ No error (d)

16. He, like the other members (a)/ of his family, were left shelterless (b)/ as a result of flood in the town. (c)/ No error (d)

17. Two thirds of the majority (a)/ are needed to pass (b)/ the resolution for the impeachment of the President. (c)/ No error (d)

18. During freedom struggle (a)/ many a patriot (b)/ were filled with patriotism. (c)/ No error (d)

19. There are a dozen (a)/ of Geography books lying in the shelf of my personal library (b)/ and you can use them whenever you like. (c)/ No error (d)

20. The number of amendments of our Constitution (a)/ have been very large (b)/ during the last fifty years of independence. (c)/ No error (d)

21. When the dentist came in (a)/ my tooth was stopped aching (b)/ out of fear that I might loose my tooth. (c)/ No error (d)

22. One of his many (a)/ good traits that came to my mind (b)/ was his modesty. (c)/ No error (d)

23. I would like you (a)/ to complete this assignment (b)/ before you will leave for Mumbai. (c)/ No error (d)

24. The teacher was angry (a)/ when he found that (b)/ you are not there. (c)/ No error (d)

25. People have a right to criticise (a)/ but at the same time each of them (b)/ have to remember his duty also. (c)/ No error (d)

26. The period of twenty-five years (a)/ have passed (b)/ and still he is without a job. (c)/ No error (d)

27. No one in this world (a)/ can be able to work (b)/ continuously for 10 hours. (c)/ No error (d)

28. The chief idea of every traveller (a)/ is to see as many objects of interest (b)/ as he possibly could. (c)/ No error (d)

29. I could not (a)/ catch the train (b)/ as I was late. (c)/ No error (d)

30. A great many tourists (a)/ has visited India (b)/ last year. (c)/ No error (d)

EXPLANATIONS

1. *(b)* The phrase 'Each of' takes a singular verb. So, 'have' would be replaced by 'has'.
2. *(d)* There is no error in the sentence.
3. *(b)* The phrase 'as well as' takes the verb as per the noun before it. As 'mother' is singular, 'were' would be changed to 'was'.
4. *(b)* As per the rule, if two subjects together express one idea, one being added to the other for the sake of emphasis or classification, the verb is singular. Hence, 'are' in the sentence would be replaced by 'is'.
5. *(a)* The verb 'allowed' will be added after 'has' to make the sentence grammatically correct. This is because as per the rule that in a compound sentence both auxiliary verbs and principal verbs should be mentioned separately if they differ in number, form or voice.
6. *(b)* The phrase 'as well as' takes a singular verb. Hence, we would replace 'are' in the given sentence by 'is' to make it grammatically correct.
7. *(d)* The sentence is correct.
8. *(a)* The sentence should start with 'So honestly did he work ……' to make it grammatically correct.
9. *(b)* In the sentence 'four miles' is some specific distance considered as a whole. Hence, a singular verb 'is' would be used instead of 'are'.
10. *(a)* The correct usage of 'No sooner' in the sentence will be 'No sooner was he brought ……' .
11. *(a)* 'Never I have' in the given sentence should be replaced by 'Never have I' to make it grammatically correct.
12. *(b)* As per the rule, 'none of' takes a singular verb. Hence, we would replace 'are' by 'is' in the given sentence.
13. *(a)* As the students demanded the same thing i.e., declaring a holiday. So, the students would be taken as a singular entity. Therefore, 'were' in the sentence would be replaced by 'was'.
14. *(d)* The sentence is correct.
15. *(c)* 'Everyone of' uses a singular verb. Hence, the verb 'need' in the given sentence should be replaced by 'needs'.
16. *(b)* When using 'like', the verb in the sentence should agree with the subject of the sentence i.e. 'He'. Hence, 'were' in the given sentence would be replaced by 'was'.
17. *(b)* In the given 'majority' means a collection of people who have the same point of view (vote). Therefore, it will be considered as a single entity. Hence, It will take a singular verb. So, 'are' in the given sentence will be replaced by 'is'.
18. *(c)* 'Many a' takes a singular verb. So, 'were' would be replaced by 'was'.
19. *(a)* 'Is' would be used in place of 'are' as the sentence refers to one dozen ('a dozen').
20. *(b)* As the 'Constitution' is a single book, 'have' would be replaced by 'has'.
21. *(b)* 'Was' needs to be replaced by 'had' to convey the right meaning of the sentence.
22. *(c)* 'Was' should be replaced by 'is' as the sentence is in present tense.
23. *(c)* We need to remove 'will' from the sentence.
24. *(c)* As the sentence is in past tense, we need to replace 'are' by 'were'.
25. *(c)* 'Each of' uses a singular verb. So, we should replace 'have' by 'has' to make the sentence correct.
26. *(b)* '25 years' in the sentence is taken as a whole. So, we would use 'has' in place of 'have'.
27. *(b)* 'Can' needs to be replaced by 'would' to make the sentence meaningful.
28. *(c)* 'Could' needs to be replaced by 'can'.
29. *(d)* No error
30. *(b)* 'A great many' takes a plural verb. So, we would replace 'has' by 'have'.

ADJECTIVE

An adjective is a word used with a noun or a pronoun to add something to its meaning.

Kinds of Adjective

1. Proper Adjectives

They are derived from proper noun.

Proper Nouns	Proper Adjectives
India	Indian
China	Chinese
Turkey	Turkish
America	American
Shakespeare	Shakespearian

2. Possessive Adjectives

My, our, your, his, their, her, its are called possessive adjectives.

3. Adjectives of Quality

It shows the traits of a person or a thing.

e.g. – Flowers were plucked *fresh*. (✗)
– *Fresh* flowers were plucked. (✓)

Confusing Words

Word	*Meaning*	*Example*
Verbal	It pertains to words.	His *verbal* words spoken *orally* are more dangerous than his figures on paper.
Oral	It means mouth.	
Common	Shared by all concerned.	It is *common* to everyone that India and Pakistan do not share a very good *mutual* understanding.
Mutual	In relation to each other.	

4. Adjectives of Quantity

It shows how much of a thing is meant.

e.g. – Any, some, little, etc.

Confusing Words

Word	*Meaning*	*Example*
Little	'Little' has a negative meaning and it means hardly any.	He has *little* hope of recovery. (*i.e.*, he is not likely to recover).
A Little	'A little' has a positive meaning. It means some, though not much.	He has *a little* hope of recovery. (i.e. he may possibly recover).
The little	'The little' means not much but all there is.	*The little* money Bihar had has gone to Jharkhand.
Any	'Any' is used in negative or/and interrogative sentences.	I shall not buy *any* material from this shop.
Some	'Some' is used in affirmative sentences.	I shall buy *something* from this shop.

However, if the question is a request or a command, 'some' replaces *any*.

e.g. – Can I buy *something* from your shop?
– Can I have *some* money?
– Why don't you take *something*?

5. Adjectives of Number

It shows how many persons or things are meant or in what order a person or thing stands.

It is of three types, which are as follow

(a) Definite Numeral Adjective These adjectives denote exact number or order of persons/things.

e.g. – The first *three* benches of this class.

(b) Indefinite Numeral Adjective Few, many, less, more, some, any, etc.

- Use of *Less, Little* and *Fewer*

'Less' denotes quantity and 'fewer' denotes number.

e.g. – ~~Not less than~~ / *No fewer than* 10000 persons died in the Gujarat earthquake.

(c) Distributive Numeral Adjective Each and Every

- 'Each' is used in speaking of two or more things. The important point is that the things should be limited in number.
- 'Every' is used in speaking of more than two persons or things, where the things are not limited.

e.g. – ~~Each~~ / *Every* day is important for someone or the other.

Confusing Words

Word	*Meaning*	*Example*
Other	'Other' means second of the two.	Call the *other* boy who is with you.
Another	'Another' means additional one.	There is *another* boy also who wants to meet you.

6. Exclamatory Adjectives

It is used to express surprise.

e.g. – *What* an idea!
– *What* a piece of work!

7. Interrogative Adjectives

These adjectives are used to ask questions.

e.g. – *Which* picture do you like the most?

8. Demonstrative Adjectives

It points out which person or thing is meant.

e.g. – *This* boy is intelligent.

- The plural forms 'these' and 'those' are often wrongly used with the singular nouns 'kind of' and 'sort of'.

e.g. – *These* ~~sort~~ /*sorts* of questions are frequently asked in the exam.

Comparison of Adjectives

1. **Positive Degree** When only one case is there.
   e.g. – Meerut is a *big* city.
2. **Comparative Degree** When two cases are there.
   e.g. – Meerut is *bigger* than Ghaziabad.
   – She is *better* than anybody else in the school.
3. **Superlative Degree** When more than two cases are there.
   e.g. – Meerut is the *biggest* city of West Uttar Pradesh.
   – He is the *most corrupt* politician of all in the country.

Confusing Words

Word	*Meaning*	*Example*
Later/ Latter/ Latest/ Last	'Later' and 'latest' refer to time. 'Latter' and 'last' refer to position. Latter : Former (opposite) Last : First (opposite) Later : Earlier (opposite) Latest : Earliest (opposite)	Tempest was the *last*/~~*latest*~~ play of Shakespeare. What is the ~~*last*~~ /*latest* news? He came ~~*latter*~~ /*later* than me. The first half of the movie was very entertaining but the *latter* wasn't.
Elder/ Eldest	For blood relations only.	I am his *elder* brother.
Old/ Older/ Oldest	For both persons and things.	My friend is *older* than me.
Farther	Geographical distance (comparative degree). Its superlative is 'farthest'.	Mumbai is *farther* than Gwalior. (×) Mumbai is *farther* than Gwalior from Meerut. (✓)
Further	Besides (in addition to)	Kanyakumari is the *farthest* place in the South. (✓) I would like to add *further* meaning to the studies. (✓)
Nearest	It shows distance	The *nearest* post-office is *next* to the college.
Next	It shows position	

Rules of Degrees

1. Double comparative and double superlative are not used together.
   e.g. – She is the *most prettiest* girl. (✗)
   – She is the *prettiest* girl. (✓)
   – This boy is *more sweeter* than that boy. (✗)
   – This boy is *sweeter* than that boy. (✓)
2. The Adjective ending in –*er* (e.g. wiser) should be used as 'more wise' while comparing two qualities of the same person or thing.
   e.g. – He is ~~*wiser*~~ /*more wise* than strong.
   – He is ~~*more wise*~~ /*wiser* than his brother.
3. In comparative cases, 'other' is used with than.
   e.g. – He is *more* intelligent than his classmates. (✗)
   – He is *more* intelligent than his *other* classmates. (✓)
   – He is *stronger* than any person in the class. (✗)
   – He is *stronger* than any *other* person in the class. (✓)
   – This boy (*who has come from outside*) is *more* intelligent than any other boy in the class. (✓)
   – The Nile is *longer* than any river in the world. (✗)
   – The Nile is *longer* than any *other* river in the world. (✓)
4. In superlative cases, *other* is not used.
   e.g. – Samudragupta was the most powerful of all kings of his time.
5. Adjectives expressing qualities, that do not have different degrees, cannot be compared.
   e.g. – Perfect, complete, circular, finish, square, empty, impossible, enough, full, unique, wonderful, marvelous, excellent, ultimate.
   – This glass is *more* full than that glass. (✗)
   – I have had *more* than enough. (✗)
6. Similar things should be compared when we compare two things.
   e.g. – The climate of Dehradun is better *than* Meerut. (✗)
   – The climate of Dehradun is better *than* that of Meerut. (✓)
   – The pollution in Delhi is greater *than* any other city in India. (✗)
   – The pollution of Delhi is greater *than* that of any other city in India. (✓)
7. The comparative degree is generally followed by 'than', but the following comparative adjectives are followed by the preposition 'to'.
   e.g. – Superior, inferior, junior, senior, prefer, preferable, elder, younger, prior, etc.
   – He is senior *to* me.
   – I prefer tea *to* coffee.
   – The Jallianwala massacre happened prior *to* the Non-Cooperation Movement.
8. If there is a gradual increase, it is expressed with two comparatives and not with positives.
   e.g. – Indian fielding is getting *better and better* day-by-day.
   – He became *more and more* intelligent while studying.

9. When two adjectives qualify the same noun, both the adjectives should be represented in the same degree.

e.g. – Taj Mahal is the *most beautiful* and the most *sought-after* place in India.

– He is the *best* and *honest* minister of our time. (✗)

– He is the *best* and the *most honest* minister of our time. (✓)

10. The adjectives 'little' and 'few' are not made to qualify the nouns, 'quantity' and 'number'. Instead 'small' should be used to qualify these nouns.

e.g. – Would you please lend me ~~*a few*~~ /*small* number of books for a month?

– I asked him not to waste even ~~*a little*~~ /*small* quantity of food.

11. Do not say 'two first' for 'first two'.

e.g. – I saw only the *two first* episodes of the serial. (✗)

– I saw only the *first two* episodes of the serial. (✓)

12. When two adjectives in different degrees of comparison are used in the same sentence, each should be complete in itself.

e.g. – He is *as good* if not better than his brother. (✗)

– He is *as good as* if not better than his brother. (✓)

13. *Worth* + V_1 + *ing* is placed after the same noun it qualifies.

e.g. – Taj Mahal is a monument *worth visiting*.

– Computer is a commodity *worth buying.*

Confusing Words

Word	*Meaning*	*Example*
Hard	Difficult, tough	He studies *hard.* (i.e. works very hard in studies)
Hardly	Rarely, a little	He *hardly* studies. (i.e. he rarely studies)
Late	Delay	*Lately* he is coming *late* from the office.
Lately	Now-a-days	
Near	Close	Although he was *near* the truck, he *nearly* escaped.
Nearly	Almost	

SPOTTING THE ERRORS **SET 4**

ERRORS OF ADJECTIVE

Directions (Q. Nos. 1-20) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. There were (a)/ no less than fifty persons (b)/ present in the room. (c)/ No error (d)

2. Few remarks (a)/ that he made were (b)/ offensive to my friend. (c)/ No error (d)

3. It is a (a)/ worth watching documentary (b)/ and you must not miss it. (c)/ No error (d)

4. Of all the students (a)/ Rita was less worried (b)/ when the date for the annual examination was announced. (c)/ No error (d)

5. Even the most perfect person (a)/ in the world is said to have erred (b)/ when there was time to perform. (c)/ No error (d)

6. In the opinion of everyone (a)/ she is wiser (b)/ than beautiful. (c)/ No error (d)

7. The tiger is (a)/ as swift as (b)/ any animal. (c)/ No error (d)

8. He had to cut a sorry figure (a)/ when he realised that he had (b)/ no any money in his purse. (c)/ No error (d)

9. Of the three ministers (a)/ who do you think (b)/ is going to prove more successful? (c)/ No error (d)

10. She is the best (a)/ and beautiful girl (b)/ of our class. (c)/ No error (d)

11. I requested him (a)/ to lend me few books (b)/ that might help me in my studies. (c)/ No error (d)

12. He is the tallest (a)/ than anybody (b)/ in the school. (c)/ No error (d)

13. I was surprised (a)/ to see her speak (b)/ with somewhat anger. (c)/ No error (d)

14. My brother is elder (a)/ than me although (b)/ he looks younger. (c)/ No error (d)

15. Little care on your part (a)/ would have made you (b)/ more successful than your friend. (c)/ No error (d)

16. Privatisation offers the most ideal situation (a)/ for consumers because private sector (b)/ is very conscious of quality. (c)/ No error

17. She is better than (a)/ any girl that studies (b)/ in our institute. (c)/ No error (d)

18. The latest chapter of this novel (a)/ is the most comprehensive of all (b)/ the chapters in the book. (c)/ No error (d)

19. She was not punished (a)/ though she came (b)/ latter than I. (c)/ No error (d)

20. Neither she is intelligent (a)/ nor hard working (b)/ and still she expects to secure first class. (c)/ No error (d)

21. It is all the more better (a)/ if you work (b)/ in my company. (c)/ No error (d)

22. It was bitter cold (a)/ so we preferred (b)/ not to go out that morning. (c)/ No error (d)

23. No animal (a)/ is as sacred to the Hindus (b)/ as the cow is. (c)/ No error (d)

24. There is no name (a)/ more glorious than Sardar Patel (b)/ in the history of India. (c)/ No error (d)

25. It very often happens (a)/ that a man who talks most (b)/ does less. (c)/ No error (d)

26. From all accounts I learn that (a)/ he is the best and honest member (b)/ of the new cabinet. (c)/ No error (d)

27. The flood situation (a)/ this year is worst than (b)/ that prevailed in the last year. (c)/ No error (d)

28. Geometry and Drawing (a)/ are more easier than (b)/ Geography and Social Studies. (c)/No error (d)

29. Of all the friends (a)/ I have had (b)/ he is the most helpful and less arrogant. (c)/ No error (d)

30. My uncle forbade me (a)/ not to go through (b)/ the contents of his letter. (c)/ No error (d)

EXPLANATIONS

1. *(b)* 'No fewer than' should be replaced by 'no less than' to make the sentence meaningful.

2. *(a)* 'A few' would be used in the sentence instead of 'Few'.

3. *(b)* The usage of 'worth watching documentary' is incorrect. The correct usage will be 'documentary worth watching.'

4. *(b)* 'Least worried' should be replaced by 'less worried' to make the sentence meaningful.

5. *(a)* 'Most perfect' should be replaced by 'perfect' to make the sentence meaningful.

6. *(b)* 'Wiser' in the given sentence should be replaced by 'more wise'.

7. *(c)* To make the sentence grammatically correct, we need to add 'other' after the word 'any'.

8. *(c)* The word 'any' in the given sentence is not needed. So, we will delete 'any' from the sentence.

9. *(c)* 'More successful' is going to be replaced by 'most successful' to make the sentence meaningful.

10. *(b)* 'Beautiful' should be changed to 'most beautiful' to make the sentence grammatically correct.

11. *(b)* 'Few' means 'nothing'. So, we will use 'a few' in the given sentence to make it meaningful.

12. *(b)* 'Of all' is used instead of 'than anybody' to make the sentence meaningful.

13. *(c)* 'Somewhat' does not make sense with 'anger'. So, we would use 'some' instead of 'somewhat'.

14. *(b)* 'Elder to me' would be the correct usage.

15. *(a)* 'Little' should be replaced by 'A little'.

16. *(a)* Adjectives like 'ideal' do not have degrees. So, we would remove 'most' from the sentence.

17. *(b)* 'Other' should be added after 'any' to make the sentence grammatically correct.

18. *(a)* 'Latest' should be replaced by 'last' to make the sentence meaningful.

19. *(c)* 'Latter' should be changed to 'later' to make the sentence correct.

20. *(a)* 'Neither is she intelligent' should be replaced by 'Neither she is intelligent'.

21. *(a)* 'More' does not go with 'better'. So, we remove 'all the more' from the sentence.

22. *(a)* We need to change 'bitter' to 'bitterly' to make the sentence meaningful.

23. *(a)* We should add 'other' after 'No' to make the sentence grammatically correct.

24. *(a)* We should add 'other' after 'no' to make the sentence grammatically correct.

25. *(c)* 'Less' should be replaced by 'least' to make the sentence correct.

26. *(b)* 'Honest' needs to be changed to 'most honest' because as per the rule all the adjectives in a sentences should be of the same degree.

27. *(b)* When two things are compared the comparative degree is used. So, we would change 'worst' to 'worse'.

28. *(b)* 'More' does not go with 'easier'. So, we would remove it.

29. *(c)* 'Less' should be replaced by 'least' to make the sentence correct.

30. *(b)* As two negative words cannot go together. So, we would remove 'not'. ('Forbade' is also a negative word)

ARTICLES

Articles are the members of the determiners family. 'A, An and The' are articles. Articles are used before nouns.

e.g. – My brother is *a* businessman.
– An apple *a* day keeps the doctor away.

Here, 'a' is an article which is used before the noun (businessman) and 'an' is an article which is used before the noun 'apple'.

Indefinite Articles

- Indefinite article 'A' is used before the words starting with the consonants.
- Indefinite article 'An' is used with the words starting with the vowel sound. (a, e, i, o, u)

e.g. – *An* umbrella is kept there.
– *A* European lives in our colony.
– Meerut has *a* university.
– *An* honest and hardworking person always succeeds in life.
– *An* hour is left.
– *An* MA, *An* SP, *An* LLB

Use of Indefinite Articles

(a) Before a common noun in singular to suggest the sense of one.
e.g. – Twelve inches make *a* foot.
– There is *a* fan in the room.
– *A* man is standing on the road.

(b) Before a common noun in singular to suggest the vague sense of a certain or any.
e.g. – *A* book is kept there.

(c) To make a common noun of a proper noun.
In this case, both definite and/or indefinite article can be used according to the case.
e.g. – Ajay is *a* Sherlok Holmes of our class.
– Samudragupta was *the* Napoleon of India.

(d) In the sense of each, every or per.
e.g. – The doctor advised the patient to take the medicine twice *a* day.
– He studies for two hours *a* day.

(e) In exclamations before singular countable nouns.
e.g. – What *a* shot!
– What *a* beautiful girl!

Definite Article

Use of Definite Article 'The'

(a) When we speak of a particular person or thing or one already referred to.
e.g. – *The* book is kept there.
– *The* CD of that movie is available in *the* market.
– *The* book you mentioned is available in *the* store.
– *The* pleasant weather is inviting us outside.

(b) Before the names of physical features in Geography.
e.g. – *The* Himalayas, *The* Gangas, *The* Indian Ocean, etc.

- However, if the reference is to a Single Island or a Single Mountain, 'the' is not used.

e.g. – *The* Kanchenjunga is the highest peak of the Himalayas in India. (✗)
– Kanchenjunga is the highest peak of the Himalayas in India. (✓)
– *The* North Andaman is a part of the Andaman group of islands. (✗)
– North Andaman is a part of the Andaman group of islands. (✓)

(c) Before common nouns which are names of things unique or of their kind.
e.g. – *The* Sun, *The* Moon, *The* Earth, *The* Sky.

(d) Before the names of countries which show federation.
e.g. – *The* United States of America, *The* United Kingdom, *The* Republic of Korea, *The* Republic of South Africa, etc.

Exception The Netherlands, The Hague

(e) Before the names of religious and mythological books, newspapers, magazines, journals, etc.
e.g. – *The* Bible, *The* India Today, *The* Times of India, etc

(f) Before Superlatives.
e.g. – I am *the* best.
– He is *the* cleverest.

(g) As an adverb with a comparative.
e.g. – *The* more, *the* merrier.
– *The* sooner, *the* better.
– *The* higher you go, *the* colder it is.

(h) Before terms denoting Nationality, Community and Castes.
e.g. – *The* Australians, *The* Indians, *The* Hindus, etc

(i) Before historic events.
e.g. – *The* Ist Battle of Panipat, *The* Revolt of 1857, *The* Quit India Movement, *The* Gulf War, etc.

(j) Before musical instruments.
e.g. – I can play *the* flute very well.
– Ustad Amjad Ali Khan plays *the* sarod.

(k) Before an adjective when it represents a class of a people.
e.g. – *The* rich should help *the* poor.
– *The* old should be respected by *the* young.

(l) Before a unit of measurement.
e.g. – Cloth is sold by *the* metre.
– Bananas are sold by *the* dozen.

(m) Before a common noun to give the force of superlative.
e.g. – He is *the* man.
– This is *the* thing to do.

(n) Before the adjectives 'same' and 'whole' and after the adjectives 'all' and 'both'.
e.g. – He is *the same* person that I saw yesterday.
– *The whole* class performed splendidly.
– *All the* boys passed with flying colours.
– *Both the* friends were present there.

(o) Before a common noun to give it the meaning of an abstract noun.
e.g. – *The* moralist in Gandhi revolted against injustice.
– *The* judge in him prevailed upon the father and he sentenced his son to death.

(p) Before various Cups and Trophies.
e.g. – Sri Lanka has won *the* Asia Cup.
– Almost all countries participate in *the* Olympics.

(q) Before comparative degree in case of a choice.
e.g. – He is *the* stronger of the two friends.

Omission of Articles

(a) Before proper nouns.
e.g. – *The* New Delhi is the capital of India. (✗)

(b) Before material noun and abstract noun used in general sense.
e.g. – The building is made up of *the* brick and *the* stone. (✗)
– The building is made up of brick and stone. (✓)
– I have *the* love for you. (✗)
– I have love for you. (✓)
– *The* care of the old is necessary. (✗)
– Care of the old is necessary. (✓)

- However, if the material noun and abstract noun show some specific cases, article should be used.

e.g. – *The* stone of Kota is of good quality.
– *The* love I have for you is unconditional and infinite.

(c) Before a noun used in its widest sense.
e.g. – *The* science is a subject based on facts. (✗)
– Science is a subject based on facts. (✓)
– *The* man is immortal. (✗)
– Man is immortal. (✓)
– *The* tiger is the national animal of India. (✗)
– Tiger is the national animal of India. (✓)

(d) Before the noun following 'kind of'.
e.g. – What kind of *a* person you are? (✗)
– What kind of person you are? (✓)
– What kind of *a* book is it? (✗)
– What kind of book is it? (✓)

(e) Before uncountable nouns.
e.g. – He gave me *advice*.
– He passed on *information* to me.

(f) Before school, college, home, church, temple, sea, work, bed, table, hospital, market, prison, court.

However, when their purpose is thought of rather than the actual building, 'the' is used.
e.g. – I go to *church* every Sunday.
– I go to *bed* early these days.

(g) Before names of diseases.
e.g. – *AIDS* can be prevented using safety measures.
– *Cancer* can be treated if detected early.

(h) Before regular meals except when preceded by an Adjective.
e.g. – I usually take *breakfast* at 9 o'clock.
– We should take *dinner* atleast two hours before the sleeping time to avoid indigestion.
– They gave us a sumptuous *lunch*.

(i) Before modes of travel.
e.g. – He will go by *air*.
– Journey by *road* takes time.

(j) Before names of relations like uncle, mother, father, etc.
e.g. – *Uncle* will come tomorrow.
– *Father* is working on a project.

(k) Before adjectives used as nouns signifying a language or colours.
e.g. – He doesn't know Hindi, but he knows *English.*
– I like *red and blue*.

(l) When two or more descriptive adjectives qualifying the same noun are connected by 'and', the Article is used before the first adjective only.
e.g. – It is *a* Hindi and English Dictionary. (Here, the dictionary is one).

(m) When two or more adjectives qualify different nouns, expressed or understood, the Article is used before each adjective.
e.g. – *The* inner and *the* outer wall were both strongly defended.
– He possesses *a* black dog and *a* white bitch.

SPOTTING THE ERRORS **SET 5**

ERRORS OF ARTICLES

Directions (Q. Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. The road (a)/ to famous monument (b)/ passes through a forest. (c)/ No error (d)

2. Our Housing Society comprises of (a)/ six blocks and (b)/ thirty three flats in an area of about thousand sq. metres. (c)/ No error (d)

3. Now that she is living in her own flat, (a)/ she cleans the windows (b)/ twice a week in the summer and once a week in the winter. (c)/ No error (d)

4. With little imagination and enterprise, (a)/ the tournament could have been transformed (b)/ into a major attraction. (c)/ No error (d)

5. These display (a)/ the (b)/ remarkable variety. (c)/ No error (d)

6. If you have faith in Almighty (a)/ everything will turn out (b)/ to be all right. (c)/ No error (d)

7. According to the Bible (a)/ it is meek and humble (b)/ who shall inherit the Earth. (c)/ No error (d)

8. On my request (a)/ Lalit introduced me to his friend (b)/ who is singer and a scientist. (c)/ No error (d)

9. This town isn't very well known (a)/ and there isn't much to see (b)/ so a few tourists come here. (c)/ No error (d)

10. He took to (a)/ reading Times (b)/ for better knowledge of the facts. (c)/ No error (d)

11. The accelerating pace of life in our metropolitan city (a)/ has had the tremendous effect (b)/ on the culture and life-style of the people. (c)/ No error (d)

12. Both the civilians (a)/ and armymen (b)/ joined the First World War. (c)/ No error (d)

13. The school is (a)/ within hundred yards (b)/ from my house. (c)/ No error (d)

14. The majority of the computer professionals recommends (a)/ that effective measures (b)/ should be taken against software piracy. (c)/ No error (d)

15. The famous Dr. Chandra (a)/ is only dentist (b)/ in our village. (c)/ No error (d)

16. This candidate lacks (a)/ an experience (b)/ otherwise he is well qualified. (c)/ No error (d)

17. The person I met (a)/ in the theatre (b)/ was the playwright himself. (c)/ No error (d)

18. The interviewer asked me (a)/ if I knew that Kalidas was the greater (b)/ than any other poet. (c)/ No error (d)

19. The reason we have not been able to pay income tax (a)/ is due to fact (b)/ that we did not receive pay on time. (c)/ No error (d)

20. Even now when I see the spot (a)/ I am reminded of an unique incident (b)/ that took place several years ago. (c)/ No error (d)

21. My father (a)/was (b)/ an honest and God fearing man. (c)/ No error (d)

22. Gangotri has special significance (a)/ for the devout as it is considered as (b)/ the abode of King Bhagirath who brought the Ganga down to Earth. (c)/ No error (d)

23. The recent study has indicated that there (a)/ is perceptible change in (b)/ the attitudes of the people. (c)/ No error (d)

24. He was fascinated by insects (a)/ and the more he studied their habits (b)/ greater was his fascination. (c)/ No error (d)

25. With little patience (a)/ you will be able to (b)/ cross this hurdle. (c)/ No error (d)

26. My friends insisted (a)/ that I should see the movie (b)/ from beginning to the end. (c)/ No error (d)

27. Both optimists and pessimists contribute to society; (a)/ the optimist invents the aeroplane (b)/ the pessimist parachute. (c)/ No error (d)

28. A friend (a)/ in need is (b)/ the friend indeed. (c)/ No error (d)

29. Amitabh had (a)/ the cameo (b)/ in the movie. (c)/ No error (d)

30. The former cricketer (a)/ donated a huge amount (b)/ in charity. (c)/ No error (d)

EXPLANATIONS

1. *(b)* The Article 'the' should be used before 'famous' as the sentence refers to a particular thing i.e. 'a famous monument'.
2. *(c)* Article 'a' is used before a common noun to suggest the sense of 'one'. Therefore, 'a' would be used before 'thousand sq. metres' to suggest that it refers to one thousand square metres.
3. *(c)* The season e.g., summer, winter, etc are abstract nouns. So, articles are not used with them. So, we would remove 'the' before 'summer' and 'winter' and change them to 'summers' and 'winters'.
4. *(a)* 'Little' means 'none or negligible'. Thus, we should use article 'a' before 'little' as 'a little' means 'some'.
5. *(b)* In the given sentence, 'the' used with 'remarkable' is incorrect. Instead of 'the' we should use the article 'a'.
6. *(a)* In the given sentence, the article 'the' would come before 'almighty'. This is done because 'almighty' in the given sentence refers to 'God'.
7. *(b)* The given sentence implies to represent a class of people i.e., 'the meek' and 'the humble'. Therefore, 'the' should be used before the adjectives 'meek' and 'humble'.
8. *(c)* The article 'a' should precede the word 'singer'.
9. *(c)* 'A few' means some and 'few' means 'hardly any'. As per the sentence, 'few' should be used instead of 'a few'.
10. *(b)* In the given sentence, 'Times' is the name of a newspaper. Therefore, it should be written as 'the Times' .
11. *(b)* The sentence is incorrect as 'the' is used before the adjective 'tremendous'. So, to correct the sentence we would use 'a' instead of 'the' before 'tremendous'.
12. *(b)* As per the rule, 'the' should come before an adjective which represents a class of people. Therefore, we should add 'the' before 'armymen'.
13. *(b)* In the given sentence, the article 'a' should come before 'hundred yards' to convey the meaning that the school is within one hundred yards from the house.
14. *(a)* The verb 'recommends' is used incorrectly. It should be replaced by 'recommend' to make the sentence grammatically correct.
15. *(b)* The sentence wants to emphasise the fact that Dr. Chandra is the only dentist in the village. So, we would use 'the' before the word 'only' to convey the desired meaning.
16. *(b)* As per the rule, articles are omitted before abstract nouns used in general sense. In the given sentence, 'experience' is an abstract noun. So, it should not be preceded by 'an'.
17. *(d)* The sentence is correct.
18. *(b)* 'The' should be removed before 'greater' to correct the sentence.
19. *(b)* 'The' should be added before the word 'fact'.
20. *(b)* 'An' should be replaced by 'a' to make the sentence grammatically correct.
21. *(d)* No error
22. *(a)* The article 'a' would come before 'special'.
23. *(b)* The article 'a' would come before 'perceptible'.
24. *(c)* The article 'the' would come before the word 'greater'.
25. *(a)* As 'little' means negligible. So, we would add 'a' before 'little'.
26. *(c)* Article 'the' would come before the word 'beginning'.
27. *(c)* 'The' should be added before 'parachute'.
28. *(c)* 'A' should come in place of 'the' to make the sentence correct.
29. *(b)* 'The' should be replaced by 'a' to convey the right meaning of the sentence.
30. *(d)* No error.

ADVERB

An adverb tells more about a verb, an adjective or another adverb.

e.g. – He is running *fast*.

Here 'fast' is the adverb which tells how he is running.

Kinds of Adverb

1. **Adverb of Time** Answers the question when, for how long or how often a certain thing happens.
   e.g. – I shall meet you *tomorrow.*
   – We were late by *two hours.*
   – The newspaper arrives *daily*.
2. **Adverb of Place** Answers the question where.
   e.g. – I shall meet you *in the market.*
   – I shall meet you *there.*
   – We were studying *in the institute.*
3. **Adverb of Manner** Answers the question how.
   e.g. – He is sitting *quietly.*
   – He is dancing *like Hritik Roshan.*
4. **Adverb of Frequency** It tells us how often an action takes place.
   e.g. – Delhi Police is with you, *always.*
5. **Adverb of Degree or Quantity** These tell us how much or in what degree or to what extent.
   e.g. – We have studied *enough.*
   – There is *something* fishy out here.
   – There is *little* doubt of his success.

Rules of Adverbs

1. **So and Too** They should not be used without their co-relatives 'that' and 'to'.
   e.g. – He is *so* rich (✗)
   – He is *so* rich that he can buy anything. (✓)
   – He is *too* intelligent (✗)
   – He is *too* intelligent to pass any exam. (✓)
2. **Difference between very and much**
   (a) 'Very' is used in positive degree and 'much' is used in comparative degree.
   e.g. – He is *much* intelligent. (✗)
   – He is *very* intelligent. (✓)
   – He is *much* intelligent than his brother. (✓)
   (b) 'Very' is used with V_1 + ing.
   'Much' is used with V_3.
   e.g. – The match became ~~much~~ / *very* interesting.
   – The crowd became *much* / ~~very~~ interested in the match.
3. Some words retain their form when they become adverbs : Fast, first, next, back, ill, better, best, etc.
4. Adverbs ending in -ly form the comparative by adding 'more' and superlative by adding 'most'.
   e.g. – This work is *more beautifully* done than that work.
   – The scenery of Kashmir is *most lovely* of all.
5. Adverbs of manner, place and time are generally placed after the verb or after the object of the verb.
   e.g. – We visited Kashmir *last year*.
   – He is talking *on and on*.
   – He is sitting *quietly*.
6. **MPT RULE** If adverbs of manner, place and time are used in a single sentence, then the sequence followed is that of *MPT.*
   e.g. – I read the book *yesterday* meticulously at home. (✗)
   – I read the book meticulously at home *yesterday*. (✓)
7. When an adverb modifies an adjective or another adverb, the adverb comes before it.
   e.g. – Her dress was ready *nearly.* (✗)
   – Her dress was *nearly ready*. (✓)
8. Adverbs of frequency such as always, ever, never, seldom, frequently, etc, are always placed before the verb they modify.
   e.g. – He *sometimes* comes late.
   – She *always* looks beautiful.
9. 'Enough' is both an Adjective and an Adverb. As an adverb, it is always placed after the adjective it modifies. As an adjective it is placed before the noun.
   e.g. – He is *enough* intelligent. (✗)
   – He is intelligent *enough* to win the competition. (✓)
   – He has *enough* money to spend.

   In the last sentence, 'enough' is used as an adjective.
10. 'Only' and 'even' should be placed immediately before the word intended to modify.
    e.g. – I worked *only* two sums. (✗)
    – I *only* worked two sums. (✓)
11. An adverb can be placed at the beginning of a sentence, when it is intended to qualify, not any word in particular, but sentence as a whole.
    e.g. – *Fortunately*, he was not present at that time.
    – *Interestingly*, the PM went in the metro.
12. 'Ever' is sometimes misused for 'never'. 'Seldom or never' and 'seldom if ever' are both correct, but 'seldom or ever' is incorrect. In the same manner, 'little if anything' is correct, but 'little or anything' is wrong.
    e.g. – He *seldom* or *never* passes in the examination with fair means.
    – He *seldom if ever* parties.

13. 'Else' should be followed by 'but', and not by 'than'.
e.g. – It is nothing *else* ~~*than*~~ / *but* his carelessness which has led to his failure.
– Aishwarya Rai is nothing *else but* a statue of beauty.

14. 'Rather' and 'Fairly' can mean moderately. But 'fairly' is used with favourable adjectives and adverbs while 'rather' is used with unfavourable adjectives and adverbs.
e.g. – She is *fairly* intelligent but my sister is *rather* stupid.
– He did *fairly* well in her exams but his sister did *rather* badly.
– He grew up in *rather* unusual circumstances.

15. 'Rather' can also be used when we are correcting something that we have just said.
e.g. – The process is not a circle but *rather* a spiral.

16. 'Rather' is also used in case of preference–would rather, had rather, rather than are used to express preference.
e.g. – I *would rather* study than sleep.
– I prefer getting up early *rather than* sleeping late.

17. We use 'rather' before verbs that introduce our thoughts and feelings, in order to express our opinion politely, especially when a different opinion has been expressed.
e.g. – I *rather* think that he was telling the truth.
– I *rather* like the decorative effect.

18. Adverb 'as' should be used to introduce predicative of the verbs such as regard, describe, define, treat, view, know.
e.g. – I regard him *as* my elder brother.
– The newspapers described the situation *as* horrible.
– Biology is defined *as* the study of nature.

19. Adverb 'as' should not be used to introduce predicative of the verbs such as name, elect, think, consider, call, appoint, make, choose.
e.g. – He was *considered* the best dancer of his time.
– He was *appointed* Governor by the President.
– Saurav Ganguly was *chosen* the captain.
– He was called *dynamic* by his mates.

20. Negative Adverbs should not be used with the words that are already negative in sense. So two negatives in the same sentence should be avoided. Seldom, nowhere, never, nothing, hardly, scarcely, neither, barely, rarely are some of the Adverbs. The verbs in such category are 'deny', 'forbid', while the conjunctions are 'unless', 'until', 'lest' and 'both'.
e.g. – *No one scarcely* practises all the exercises. (✗)
– *Scarcely anyone* practises all the exercises. (✓)
– I *rarely* went to meet ~~*nobody*~~ /*anybody* in my childhood.
– She *hardly* knows *anything*/ ~~*nothing*~~ about me.
– He does *nothing* without *ever*/ ~~*never*~~ consulting me.
– He has denied that he was ~~*not*~~ going there.
– Walk steadily, lest you should ~~*not*~~ fall.
– Both of them are not coming. (✗)
– Neither of them is coming. (✓)

21. Consider the following cases

(a) 'Coward', 'miser', 'niggard', 'rogue' are Nouns. 'Cowardly', 'miserly', 'niggardly', 'roguish' are Adjectives.
e.g. – An officer is trained never to fight ~~*cowardly*~~ /*in a cowardly manner.*
– Although he seems brave, he is actually a *coward*.
– It was a sheer *cowardly* act of violence.

(b) 'Fast' retains its form in both Adjective and Adverb.
e.g. – He is *fast* (Adjective).
– He is running *fast* (Adverb).

(c) 'Direct' and 'Directly' are adverbs. Direct means *straight* and Directly means *at once.*
e.g. – Don't stop anywhere, return home *direct*.
– Don't stop now, return home *directly*.

(d) 'Manly', 'masterly', 'slovenly', 'monthly', 'weekly', 'sickly', 'friendly' are Adjectives and should not be confused with Adverbs.
e.g. – He is earning fifty thousand rupees ~~*monthly*~~ /*a month.*
– He is a *friendly* old man.

22. The use of 'never' for 'not' is incorrect, because 'never' means *not ever*.
e.g. – I *never* remember having met him. (✗)
– I do *not* remember ever having met him. (✓)
– We met the other day, but he *never* referred to the matter. (✗)
– We met the other day, but he did *not* refer to the matter. (✓)

23. No sentence should begin with 'Due to'. It must be used after some form of the verb 'to be'.
e.g. – *Due to* bad weather, the match was abandoned. (✗)
– It was *due to* bad weather, the match was abandoned. (✓)

24. **Cent-per cent** It should be hundred per cent.

25. **Do the needful** It should be do what is necessary.

SPOTTING THE ERRORS **SET 6**

ERRORS OF ADVERB

Directions (Q. Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. Firstly, you should think (a)/ over the meaning of the words (b)/ and then use them. (c)/ No error (d)

2. The driver tried his best (a)/ to avert the accident by bringing the car (b)/ to a suddenly stop. (c)/No error (d)

3. The Sunshine hotel was fully equipped (a)/ to offer leisure stay (b)/ to its clients. (c)/ No error (d)

4. The technician reminded them (a)/ to have a thoroughly cleaning (b)/ of the machine after each use. (c)/ No error (d)

5. I am (a)/ much glad (b)/ that you have won the trophy. (c)/ No error (d)

6. He is too coward (a)/ to make it (b)/ happen. (c)/ No error (d)

7. People invent new machines (a)/ when they think (b)/ different. (c)/ No error (d)

8. A man entered the tavern (a)/ and asked for some bread and cheese (b)/ with a decided foreign accent. (c)/ No error (d)

9. Watch how careful (a)/ the sparrow knits the straws (b)/ into one another to form a nest. (c)/ No error (d)

10. On hearing the news (a)/ he went directly (b)/ to the Manager's room. (c)/ No error (d)

11. They reached home (a)/ safely (b)/ although they started late. (c)/ No error (d)

12. It is the duty of every citizen (a)/ to do his utmost to defend (b)/ the hardly won freedom of the country. (c)/ No error (d)

13. The principal was (a)/ enough kind to (b)/ grant me scholarship. (c)/ No error (d)

14. It is nothing else (a)/ than foolishness (b)/ that led to his downfall. (c)/ No error (d)

15. The tried travellers were bundled off (a)/ to the nearby cop house till anyone (b)/ could come and vouch for their credentials. (c)/ No error (d)

16. I advised my brother to engage two coolies instead of one (a)/ because the luggage was too much heavy (b)/ for a single coolie to handle. (c)/ No error (d)

17. There is no one else (a)/ whom I esteem (b)/ than your father. (c)/ No error (d)

18. Although I was (a)/ in Delhi last month (b)/ I never met him. (c)/ No error (d)

19. When I got (a)/ home I was (b)/ too exhausted. (c)/ No error (d)

20. I did not know hardly (a)/ anyone in the city (b)/ and so I felt lonely. (c)/ No error (d)

21. My mother works (a)/ very quicker than (b)/ I at embroidery. (c)/ No error (d)

22. She is sure a great singer (a)/ and no other singer (b)/ is a match for her. (c)/ No error (d)

23. It is better to be frugal (a)/ but don't be miser (b)/ in giving alms. (c)/ No error (d)

24. I refused to (a)/ accompany him because (b)/ I was so tired. (c)/ No error (d)

25. The student came to the classroom (a)/ lately and was punished (b)/ by the teacher. (c)/ No error (d)

26. He looks full of energy (a)/ today because he (b)/ soundly slept last night. (c)/ No error (d)

27. She had barely nothing (a)/ to eat when she came (b)/ to me last month. (c)/ No error (d)

28. It had been too cold (a)/ the whole month and we preferred (b)/ to stay in the plains. (c)/ No error (d)

29. The quickly brown fox (a)/ jumped over (b)/ the lazy dog. (c)/ No error (d)

30. He slowly (a)/ went to open (b)/ the lock. (c)/ No error (d)

EXPLANATIONS

1. *(a)* 'Firstly' is used to introduce a first point or reason. It does not make sense in the given sentence. So, we would use 'First' instead of 'Firstly'.
2. *(c)* 'Suddenly stop' in the sentence does not make sense. Hence, we would change it to 'sudden stop' to make the sentence meaningful.
3. *(b)* 'Leisure' as an adjective and does not make sense in the sentence. It should be changed to its adverb form i.e. 'leisurely'.
4. *(b)* 'Thoroughly cleaning' is incorrect. It should be 'thorough cleaning'.
5. *(b)* 'Much' is used in comparative degree and 'very' is used in positive degree. So, we should use 'very' in place of 'much' in the given sentence.
6. *(d)* The sentence is correct.
7. *(c)* 'Different' in the sentence does not convey the right meaning of the sentence. It must be changed to 'differently'.
8. *(c)* 'Decided' is incorrect and does not make sense with the sentence. It should be changed to its adverb form 'decidedly'.
9. *(a)* 'Careful' in the given sentence should be changed to 'carefully' to convey the right meaning of the sentence.
10. *(b)* 'Direct' means straight and 'directly' means 'at once'. Hence, we would use 'direct' in the sentence instead of 'directly'.
11. *(d)* The sentence is correct.
12. *(c)* 'Hardly won freedom' means 'negligibly won freedom' and does not make sense in the sentence. It should be changed to 'hard won freedom'.
13. *(b)* 'Enough kind' does not make sense. It should be changed to 'kind enough'.
14. *(b)* 'Than' used in the sentence is incorrect. It should be changed to 'but'.
15. *(b)* 'Anyone' should be replaced by 'someone' to make the sentence meaningful.
16. *(b)* 'Too' and 'much' are not used together. It should be 'too heavy'.
17. *(b)* 'More' should be used after 'esteem' to make the sentence meaningful.
18. *(c)* 'Never' in the given sentence should be replaced by 'did not' to make the sentence meaningful.
19. *(c)* 'Too' should be replaced by 'very' to make the sentence grammatically correct.
20. *(a)* 'Did not know' should be removed and 'knew' should be added after 'hardly' to correct the sentence.
21. *(b)* 'Very quicker' should be replaced by 'quicker' to make the sentence correct.
22. *(a)* 'Sure' needs to be changed to its adverb form 'surely' to make the sentence correct.
23. *(b)* 'Miser' needs to be changed to 'miserly' to make the sentence grammatically correct.
24. *(c)* 'So' needs to be replaced by 'very' to correct the sentence.
25. *(b)* 'Lately' does not convey the right meaning. It should be changed to 'late'.
26. *(c)* 'Soundly slept' needs to be changed to 'slept soundly' to make the sentence meaningful.
27. *(a)* As per the rule, two negative words cannot be used together in a sentence. In the given sentence, 'barely' and 'nothing' are negative words. So, we would replace 'nothing' by 'anything'.
28. *(a)* 'Too' needs to be replaced by 'very' to make the sentence grammatically correct.
29. *(a)* 'Quickly ' does not make sense. So, we replace it by its adjective form i.e. 'quick'.
30. *(d)* No error.

CONJUNCTIONS

A conjunction is a part of speech that connects words, phrases, clauses or sentences.

e.g. – and, but, while, though, although, lest etc.

Rules of Conjunctions

1. 'Scarcely' and 'Hardly' should be followed by when and not by than.
   e.g. – *Scarcely* had I started for the institute, *when/* ~~*than*~~ the rain started.
   – *Hardly* had he arrived, *when* he had to leave again.
2. 'No sooner' is followed by than and not by when.
   e.g. – *No sooner* had I left, *than* / ~~*when*~~ the rain started.
   – *No sooner* did he arrive, ~~*then*~~ / *than* he had to leave.
3. 'Seldom or never' and 'seldom if ever' are both correct, but 'seldom or ever' is incorrect.
   e.g. – He *seldom or never* goes to see movies in a theatre.
4. 'Either-or', 'Neither-nor', 'not only-but also', 'both-and', 'whether-or' etc., should be followed by the same parts of speech or of the same function.
   e.g. – He *neither* agreed to my proposal *nor* to his. (✗)
   – He agreed *neither* to my proposal *nor* to his. (✓)
   – *Neither* he helps his mother by money *nor* by other means. (✗)
   – He helps his mother *neither* by money *nor* by other means. (✓)
   – He helps his mother *not only* by money *but also* by other means. (✓)
5. The conjunctions 'though' (or although) and 'but' do the work of setting one statement against another by way of oppositions or contrast and therefore, the correlative of 'though' is 'yet' or a comma (,).
   e.g. – *Though* he worked hard *yet* he could not top the class.
6. After the adjective 'other' (which is regarded as a kind of comparative), the only word that can be correctly used for contrasting one thing with another is 'than'. The prepositions *from, but, except* in such a connection are wrong.
   e.g. – He had no *other* option ~~*but*~~ / *than* to fight.
   – He had *another* reason ~~*from*~~ / *than* what he professed.
   – She had no other claim to the post ~~*except*~~ / *than* her good looks.
7. Conjunction 'that' is not used in the following cases
   (a) Direct narration.
   e.g. – He said, "I am smart".
   (b) Indirect speech, if the sentence is interrogative.
   e.g. – He asked who he was.
8. In a 'not only......but also' sentence, the verb should agree with the noun or pronoun mentioned second, because this is the part being emphasised.
   e.g. – *Not only* the teacher *but also* the students are enjoying themselves.
9. 'Such as' is used to denote a category, whereas 'such that' emphasises the degree of something by mentioning its consequences.
   e.g. – Yuvraj played *such* an innings *as* played by the best batsman.
   – Yuvraj played *such* an innings *that* it took the match away from the opposition.
10. 'Both' is followed by *and* not by *as well as*. Besides, both has positive sense and cannot be used in negative sentences.
    e.g. – *Both* Amit ~~*as well as*~~ / *and* his friends are coming.
    – *Both* Amit *and* his friends are *not* coming. (✗)
    – Neither Amit nor his friends are coming. (✓)
11. 'Unless' means if not and therefore, it should not be used in a sentence or clause which is already negative.
    e.g. – *Unless* you *do not* work hard, you will fail. (✗)
    – *Unless* you work hard, you will fail. (✓)
12. 'Lest' expresses a negative sense and therefore cannot be used with not. 'Should' is always used with 'lest'.
    e.g. – Walk steadily, *lest* you should *not* fall. (✗)
    – Walk steadily, *lest* you should fall. (✓)
13. When 'suppose' is used in the beginning of a sentence to denote a command or a request, it is not followed by *if*.
    e.g. – *Suppose if* you are caught, what will happen to your parents? (✗)
    – *Suppose you* are caught, what will happen to your parents? (✓)
14. Avoid the error of using 'than' for 'from' after the adjective *different.*
    e.g. – He took a different role ~~*than*~~ / *from* the ones he has been doing for long.
15. 'Nothing else' should be followed by *but,* not by *than.*
    e.g. – It is nothing else *but*/~~*than*~~ your carelessness, that you have failed in the exam.
16. The use of Present tense after *as if* and *as though* should be avoided. [Subjunctive Mood].
    e.g. – He looks as if he *suspects* something. (✗)
    – He looks as if he *suspected* something. (✓)

SPOTTING THE ERRORS **SET 7**

ERRORS OF CONJUNCTION

Directions (Q. Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. Although they listen to me (a)/ but their actions (b)/ prove otherwise. (c)/ No error (d)
2. He treats (a)/ us as (b)/ slaves. (c)/ No error (d)
3. Most of the girls are doing (a)/ their post graduation because (b)/ they may get good husbands. (c)/ No error (d)
4. Such was his pronunciation (a)/ as (b)/ I could not understand him. (c)/ No error (d)
5. He asked (a)/ that who (b)/ I was. (c)/ No error (d)
6. I am interested (a)/ in such books (b)/ that are interesting. (c)/ No error (d)
7. Each member of the alliance (a)/ agrees to take such action (b)/ that it deems necessary. (c)/ No error (d)
8. She looked at him (a)/ in such distress (b)/ as he had to look away. (c)/ No error (d)
9. This film is interesting (a)/ and the previous one (b)/ was boring. (c)/ No error (d)
10. It is difficult to know (a)/ whether (b)/ you are selected or not. (c)/ No error (d)
11. He has no chance (a)/ than to start (b)/ his own business. (c)/ No error. (d)
12. They had hardly finished (a)/ their meals that at once (b)/ they resumed their duty. (c)/ No error (d)
13. I don't know whether (a)/ Raj is equally (b)/ good as Vimal. (c)/ No error (d)
14. He (a)/ will return (b)/ on either Monday or Tuesday. (c)/ No error (d)
15. He is (a)/ not honest and not (b)/ truthful. (c)/ No error (d)
16. He has no other business (a)/ but to play (b)/ with computers. (c)/ No error (d)
17. Be smart (a)/ not only in dress (b)/ and also in action. (c)/ No error (d)
18. Hardly had I reached the airport (a)/ where I learned about (b)/ the powerful bomb explosion. (c)/ No error (d)
19. My book has been missing (a)/ from my room (b)/ till yesterday. (c)/ No error (d)
20. The manager of the bank was busy; (a)/ so he asked them to come and see him (b)/ between two to three in the afternoon. (c)/ No error (d)
21. No sooner did the Sun set (a)/ than we had ours dinner (b)/ and went to sleep. (c)/ No error (d)
22. Gunjan asked me (a)/ what was (b)/ the time. (c)/ No error (d)
23. Rohit took a different route (a)/ than the one (b)/ he usually takes for going to office. (c)/ No error (d)
24. Komal looks as if (a)/ she wants (b)/ to go home. (c)/ No error (d)
25. What would you do (a)/ if you were (b)/ marooned on an island? (c)/ No error (d)
26. Both Nilakshi (a)/ as well as Shivani (b)/ are going to school. (c)/ No error (d)
27. Koel is not only (a)/ lazy but also (b)/ insane. (c)/ No error (d)
28. Rakshit seldom (a)/ or ever (b)/ goes to play outside. (c)/ No error (d)
29. Priya is fair (a)/ but her child (b)/ are dark coloured. (c)/ No error (d)
30. Simran had (a)/ another reason to declare her assets (b)/ from what she said. (c)/ No error (d)

EXPLANATIONS

1. *(b)* The correlative of 'Although' is 'yet' and not 'but'. Hence, we should remove 'but' and use 'yet' in place of it.
2. *(d)* The sentence is correct.
3. *(b)* As the sentence starts with 'Most of', 'because' will not be used. We would replace it by 'so that'.
4. *(b)* As per the rule, 'such…that' is used for mentioning consequences. Hence, we would use 'that' in place of 'as'.
5. *(b)* As per the rule, 'that' is not used in Indirect speech if the sentence is interrogative. Hence, we would remove 'that' from the sentence.
6. *(b)* The conjunction 'such' is not needed in the sentence. We need to delete it from the sentence.
7. *(d)* The sentence is correct.
8. *(c)* In the given sentence, 'that' would be used in place of 'as'. (For explanation refer to Ans. 4)
9. *(b)* As a comparison is made in the given sentence, we would use 'but' in place of 'and'.
10. *(d)* The sentence is correct.
11. *(b)* We would use 'but' in place of 'than' as 'but' is used for the work of setting one statement ('He has no chance') against another ('to start his own business').
12. *(b)* 'That at once' needs to be deleted from the sentence and should be replaced by 'when'.

13. *(b)* In the given sentence, 'as' should be used in place of 'equally' as 'equally' is an adverb.

14. *(c)* 'on either' is not the correct usage. As per the rule 'Either …or' should be followed by same parts of speech. Hence, the sentence should be : 'He will return either on Monday or on Tuesday.'

15. *(b)* Instead of using 'not' two times in the given sentence, we should use 'neither…nor'. So, part (b) would become 'neither honest nor'.

16. *(b)* As per the rule, after 'other' the only words that can be correctly used for contrasting one thing with another is 'than'. Hence, we should use 'than' in place of 'but'.

17. *(c)* The correct usage is 'not only …but also'. Hence, we would use 'but' in place of 'and'.

18. *(b)* As per the rule, 'hardly' should be followed by 'when'. Hence, we would use 'when' in place of 'where'.

19. *(c)* 'Till' does not make sense in the sentence. It should be changed to 'since'.

20. *(c)* 'Between' takes 'and' and not 'to'. So, we would replace 'to' by 'and' in the sentence.

21. *(b)* 'Ours' should be replaced by 'our' to make the sentence correct.

22. *(d)* No error

23. *(b)* 'Than' should be replaced by 'from' to make the sentence correct.

24. *(b)* As per the rule, the use of present tense should be avoided after 'as if'. Hence, we should change 'wants' to 'wanted' to make the sentence correct.

25. *(d)* No error

26. *(b)* As per the rule, 'both' is followed by 'and' and not by 'as well as'. So, we would replace 'as well as' by 'and' to make the sentence correct.

27. *(d)* No error

28. *(b)* Usage of 'seldom or ever' is incorrect.

29. *(c)* As 'child' is singular, we would replace 'are' by 'is' to make the sentence correct.

30. *(c)* As per the rule, 'from' should not be used with 'another'. So, we would replace it by 'than' to correct the sentence.

PREPOSITION

Preposition is a word that connects a noun, pronoun to another word, especially to a verb, another noun or an adjective.

Some Important Prepositions

1. 'In' is used for bigger places (towns, cities, countries) while 'at' is used for smaller places.
   e.g. – I live *at* Shastri Nagar in Meerut.
2. **In / Into** 'In' is used in speaking of things at rest. 'Into' is used in speaking of things in motion.
   e.g. – He is shopping *in* the market.
   – He jumped *into* the well.
   – He is falling *in* love.
   – He is *in* the office.
   – The snake crawled *into* the hole.
   – The cup broke-off *into* a hundred pieces.
3. 'On' denotes position, 'upon' denotes movement.
   e.g. – The cat is *on* the table.
   – The cat pounced *upon* the mouse.
4. 'With' denotes the 'instrument' and 'by' denotes the 'agent'.
   e.g. – The letter was written *by* him *with* his pen.
   – The music was generated ~~*by*~~ /*with* a guitar.
   – The murder was committed *by* him *with* a pistol.
   – The ball was hit *by* the batsman *with* his bat.
5. 'Ago' refers to past time while 'before' denotes precedence between two events.
   e.g. – Long *ago,* there was a king named Rama.
   – Ram existed *before* Mahabharata was fought.
   – He came *before* me.
   – The train arrived *before* the scheduled time.
   – India achieved independence 69 years *ago.*
6. 'Above' and 'below' merely denote position while 'over' and 'under' also carry a sense of covering or movement.
   e.g. – We live *below* the roof.
   – Sky is *above* us.
   – Train is running *under* the bridge.
   – The train is standing *below* the bridge.
   – The bird is flying *over* the pond.
   – A wire is passing *above* the building.
   – I was wearing two sweaters *under* the jacket.
   - 'Under' is used before a noun to indicate that a person or thing is being affected by something or is going through a particular process.
     e.g.
     – I'm rarely *under* pressure and my co-workers are always nice to me.
   - 'Under' can mean junior in ranks.
     e.g. – He is *under* me.
   - If something happens 'under' a particular person or government, it happens when that person or government is in power.
     e.g. – There will be no new taxes *under* his leadership.

- If someone does something 'under' a particular name, he uses that name instead of his real name.
  e.g. – The patient was registered *under* a false name.
- 'Beneath' has the same meaning as 'under', but it is better to use it for abstract meanings.
  e.g. – *Beneath* the festive mood, there is an underlying apprehension.
  – Everybody thought that she was marrying *beneath* her.
  – Many find themselves having to take jobs far *beneath* them.

7. Difference between 'on time', 'in time' and 'in good time'.
   - 'On time' signifies absolutely right time, neither before nor after.
     e.g. – The flight is *on time*.
   - 'In time' means you are not late for the event.
     e.g. – I arrived just *in time* for my flight.
   - 'In good time' means with comfortable margin.
     e.g. – I arrived at the airport *in good time*.

8. Difference between at the beginning/at the end and in the beginning/in the end
   - 'At the beginning' means literally at the beginning.
     e.g. – India scored fast *at the beginning* of the match.
   - 'At the end' means literally at the end.
     e.g. – *At the end* of the book, you'll find the bibliography.
   - 'In the beginning' (or at first) means in the early stage. It implies that later there was a change.
     e.g. – Sachin was nervous *in the beginning,* later he settled down.
   - 'In the end' (or at last) means eventually/after sometime.
     e.g. – At first he was scared, but *in the end* he started enjoying.

9. No preposition is placed after the following verbs when they are used in active voice. Order, request, reach, attack, resemble, emphasise, accompany, discuss, investigate, comprise, enter (come into), flee (a place), join, affect, board, etc.
   e.g. – Our forces attacked ~~on~~ the enemy fort.
   – We reached ~~at~~ the station on time.
   – We ordered ~~for~~ a cup of tea.
   – He resembles ~~to~~ his father.
   – He accompanies ~~with~~ her wherever she goes.
   – The police are investigating ~~into~~ the case.
   – The teacher emphasised ~~on~~ morality.
   – I don't want to discuss ~~about~~ the problem with you.

10. Omit 'to' after verb of communication such as advise, tell, ask, beg, command, encourage, request, inform, order.
    e.g. – I advised ~~to~~ him to study hard.
    – I commanded ~~to~~ him to leave.
    – I ordered ~~to~~ him to bring me something to eat.

11. 'Till' is used in particular time while 'until' is used for indefinite time.
    Only 'until' is used at the starting of a sentence.
    e.g. – We shall work *until* we fell down.
    – We shall work *till* sunset.
    – ~~*Till*~~ / *Until* 30, he was a bachelor.

12. 'Till' is used for time while 'upto' for place.
    e.g. – We shall work *till* 5 pm.
    – We walked ~~*till*~~/*upto* the station.

13. The same preposition should not be used with two words unless it is appropriate to each of them.
    e.g. – It is different and inferior *to* the other. (✗)
    – It is different *from* and inferior *to* the other. (✓)
    – Her dress does not add but detract *from* her appearance. (✗)
    – Her dress does not add *to* but detracts *from* her appearance. (✓)

14. 'Since' and 'from' are used before a noun or phrase denoting some point of time but whereas 'since' is preceded by a verb in some perfect tense, 'from' is used with other tenses except the perfect tense. *For* refers to a period of time, not to a point of time, and should not be replaced by *since* or *from*.
    e.g. – I haven't done anything *since* yesterday.
    – He has been here *since* nine o'clock.
    – I started my shop *from* 1st January.
    – I shall start work *from* July.
    – He will join the office *from* tomorrow.
    – I have been practising *for* ten days.

15. Regarding the phrases of time, *morning, afternoon* and *evening* are preceded by the preposition 'in' whereas *dawn, daybreak, noon, midday* and *midnight* are preceded by the preposition 'at'. Besides, when these time phrases are qualified by 'last' or 'next', they are not preceded by any preposition.
    e.g. – I like to roam around *in* the evening.
    – I'll see you *at* night.
    – The sun is hottest *at* midday.
    – I met him *last* evening.

16. **Across / Through**
    e.g. – Walk *across* a road and pass *through* a tunnel.

17. **Between/Among** 'Between' is used while referring to two persons/things whereas 'Among' is used for more than two.
e.g. – *Between* the two of you, who is stronger?
– The sweets are to be distributed *among* ten friends.

18. **Beside/Besides** 'Beside' means by the side of, whereas 'Besides' means in addition to.
e.g. – *Besides* eating, he is also watching T.V.
– You were sitting *beside* him.

19. **From/Between** 'From' is normally used with to/till, whereas 'Between' is used with and.
e.g. – He works *from* nine *to* six (or nine till six).
– The meeting was scheduled to be held *between* 2 pm *and* 3 pm.

20. **Within/In** 'Within' means before the end of time, whereas 'In' means at the end of time.
e.g. – He will return *in* five minutes.
– He will return *within* five minutes.

SPOTTING THE ERRORS **SET 8**

ERRORS OF PREPOSITION

Directions (Q. Nos. 1-30) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. The widely publicised manifesto (a)/ of the new party is not (b)/ much different than ours. (c)/ No error (d)

2. I was taken with surprise (a)/ when I saw (b)/ the glamorous Appu Ghar. (c)/ No error (d)

3. Man needs security (a)/ and leisure (b)/ of free thinking. (c)/ No error (d)

4. This watch is (a)/ superior and more expensive (b)/ than that. (c)/ No error (d)

5. It was apparent for everyone (a)/ present that if the patient did not receive (b)/ medical attention fast he would die. (c)/ No error (d)

6. He knows very well (a)/ what is expected from him (b)/ but he is not able to fulfil all the expectations. (c)/ No error (d)

7. My brother has (a)/ ordered for (b)/ a new book. (c)/ No error (d)

8. That Brutus, who was his trusted friend (a)/ had attacked on him (b)/ caused heart break to Julius Caesar. (c)/ No error (d)

9. Bhuvan was (a)/ blind with (b)/ one eye. (c)/ No error (d)

10. The doctor attended (a)/ to (b)/ the patient very quietly. (c)/ No error (d)

11. I was shocked to hear (a)/ that his father (b)/ died of an accident. (c)/ No error (d)

12. I must start at dawn (a)/ to reach the station (b)/ in time. (c)/ No error (d)

13. None could dare (a)/ to encroach (b)/ on his rights. (c)/ No error (d)

14. The father brought the sweets (a)/ and distributed them (b)/ between his five children. (c)/ No error (d)

15. Raman developed the habit (a)/ for sleeping late (b)/ when he was staying in the hostel. (c)/ No error (d)

16. It is the duty of every right thinking citizen (a)/ to try to make (b)/ the whole world a happier place to live. (c)/ No error (d)

17. The top-ranking candidates (a)/ will be appointed in senior jobs (b)/ in good companies. (c)/ No error (d)

18. My niece has been married (a)/ with (b)/ the richest man of the town. (c)/ No error (d)

19. The venue of examination (a)/ is one mile (b)/ further up the hill. (c)/ No error (d)

20. The doctor referred the patient (a)/ for the OPD (b)/ without examining him. (c)/ No error (d)

21. On a holiday Madhu prefers (a)/ reading than going (b)/ out visiting friends. (c)/ No error (d)

22. People who are (a)/ averse with (b)/ hard work generally do not succeed in life. (c)/ No error (d)

23. Vishal is one year junior (a)/ than (b)/ Madan in our office. (c)/ No error (d)

24. They walked (a)/ besides (b)/ each other in silence. (c)/ No error (d)

25. Our Mathematics teacher often emphasises (a)/ on the need (b)/ for a lot of practice. (c)/ No error (d)

26. Please put away (a)/ the candle before (b)/ you leave. (c)/ No error (d)

27. All the doctors were puzzled (a)/ on the strange symptoms (b)/ reported by the patient. (c)/ No error (d)

28. The detective says that (a)/ there is no chance for finding (b)/ the person who wrote these letters. (c)/ No error (d)

29. In urban society the social circle is limited (a)/ with the family but in the village (b)/ it encompasses the entire village. (c)/ No error (d)

30. Being most loquacious among her brothers and sisters (a)/ she related a good many tales (b)/ in each breath. (c)/ No error (d)

EXPLANATIONS

1. *(c)* In formal writing, 'different from' is generally preferred, to 'different than'. Hence, we would replace 'than' by 'from'.

2. *(a)* The preposition 'with' should be replaced by 'by' to make the sentence correct.

3. *(c)* 'Security' and 'leisure' are needed 'for' free thinking. Hence, we would replace 'of' by 'for'.

4. *(b)* The preposition 'to' must follow the word 'superior' in the given sentence to make it grammatically correct.

5. *(a)* The preposition 'for' in part (a) of the sentence does not make sense. It should be replaced by 'to' to make the sentence correct.

6. *(b)* The preposition 'of' should be used instead of 'from' in the given sentence.

7. *(b)* 'For' in part (b) of the sentence is not needed. It should be deleted.

8. *(b)* The preposition 'on' in the sentence is not needed. We need to delete it to convey the correct meaning of the sentence.

9. *(b)* 'Blind in one eye' is the correct usage. Hence, we should replace 'with' by 'in'.

10. *(d)* No error

11. *(c)* 'Of' is used when the cause is a disease. In the given sentence, we should use 'in' in place of 'of'.

12. *(d)* The sentence is correct.

13. *(c)* 'Upon' is the right conjunction that should be used with 'rights'. Hence, we replace 'on' by 'upon'.

14. *(c)* As per the rule, 'among' should be used when more than two persons are involved. Hence, we replace 'between' in the given sentence by 'among'.

15. *(b)* 'For sleeping late' does not make sense in the given sentence. It should be replaced with 'of sleeping late'.

16. *(c)* The sentence is not complete unless we add 'in' after 'live'.

17. *(b)* The correct usage is 'appointed to'. Hence, we would replace 'in' by 'to'.

18. *(b)* As per the right usage, 'you are married to someone' and not 'with someone'. Hence, we replace the preposition 'with' by 'to'.

19. *(a)* The preposition 'of' needs to be replaced by 'for' to make the sentence grammatically correct.

20. *(a)* 'For' is incorrect. It should be replaced by 'to' to make the sentence gramatically correct.

21. *(b)* 'Than' should be replaced by 'to' to correct the sentence.

22. *(b)* 'With' is not used with 'averse'. So, we replace it by 'to'.

23. *(b)* 'Than' is not the right preposition that should be used with 'junior'. It should be replaced by 'to'.

24. *(b)* 'Besides' should be replaced by 'beside' to correct the sentence.

25. *(b)* We need to remove 'on' to rectify the sentence.

26. *(a)* 'Put away' does not convey the right meaning. It should be changed to 'put off'.

27. *(b)* 'On' should be replaced by 'at' in order to make the sentence correct.

28. *(b)* 'For' should be replaced by 'of' to make the sentence correct.

29. *(b)* 'With' should be replaced by 'to' to rectify the sentence.

30. *(a)* 'Among' should be replaced by 'of' to rectify the sentence.

SPOTTING THE ERRORS

COMPLETE EXERCISE

Directions (Q. Nos 1-111) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

1. One of the most (a)/ widespread bad habit (b)/ is the use of tobacco. (c)/ No error (d)

2. Recently I visited Kashmir (a)/ and found the sceneries (b)/ to be marvellous. (c)/ No error (d)

3. All the furnitures have been (a)/ sent to the new house (b)/ located in a village. (c)/ No error (d)

4. The crowd of angry students (a)/ ordered the (b)/ closing of shops. (c)/ No error (d)

5. They left (a)/ their luggages (b)/ at the railway station. (c)/ No error (d)

6. The bus could not (a)/ ascend the steep hill (b)/ because it was in the wrong gears. (c)/ No error (d)

7. The Indian force (a)/ drove away (b)/ the Chinese. (c)/ No error (d)

8. His mouth watered (a)/ when he saw (b)/ a bouquet of grapes. (c)/ No error (d)

9. My brother-in-laws (a)/ who live in Mumbai have come (b)/ to stay with us. (c)/ No error (d)

10. These kind of shirts (a)/ are rather expensive (b)/ for him to buy. (c)/ No error (d)

11. Those sort of people (a)/ usually do not (b)/ earn fame in society. (c)/ No error (d)

12. Being a very (a)/ hot day I (b)/ remained indoors. (c)/ No error (d)

13. Had I come (a)/ to know about his difficulties (b)/ I would have certainly helped. (c)/ No error (d)

14. One of them (a)/ forgot to take their bag (b)/ from the school. (c)/ No error (d)

15. Mr. Sharma, our representative,(a) / he will attend the meeting (b)/ on our behalf. (c)/ No error (d)

16. If the teacher is good, (a)/ the students will respond (b)/ positively to them. (c)/ No error (d)

17. It is not difficult to believe that a man (a)/ who has lived in this city for a long time (b)/ he will never feel at home anywhere else in the world. (c)/ No error (d)

18. Each girl was (a)/ given a bunch of flowers (b)/ which pleased her very much. (c)/ No error (d)

19. As it was Rajan's (a)/ first interview, he dressed him (b)/ in his most formal suit. (c)/ No error (d)

20. Gopal and myself (a)/ will take care of (b)/ the function on Sunday. (c)/ No error (d)

21. During freedom struggle (a)/ many a patriot (b)/ were filled with patriotism. (c)/ No error (d)

22. There are a dozen (a)/ of Geography books lying in the shelf of my personal library (b)/ and you can use them whenever you like. (c)/ No error (d)

23. Ramesh has agreed (a)/ to marry with the girl (b)/ of his parents's choice. (c)/ No error (d)

24. Just to the North of India (a)/ is the Himalayas (b)/ that were once impregnable. (c)/ No error (d)

25. She disappeared (a)/ and found dead (b)/ near a well outside the village. (c)/ No error (d)

26. I had been (a)/ to Delhi last week (b)/ to visit my friend. (c)/ No error (d)

27. Either she or you (a)/ is to blame (b)/ for the mismanagement of the domestic affairs. (c)/ No error (d)

28. It were the children (a)/ that caused a lot of problem to their parents (b)/ during the long bus journey. (c)/ No error (d)

29. He will be likely (a)/ to leave for the United States (b)/ last year to visit his brother. (c)/ No error (d)

30. The soldiers along with the commander (a)/ was court-martialled (b)/ for defying the orders. (c)/ No error (d)

31. The pity is that (a)/ no sooner he had left the place (b)/ than the fire broke out. (c)/ No error (d)

32. When he was arriving (a)/ the party was (b)/ in full swing. (c)/ No error (d)

33. She was not punished (a)/ though she came (b)/ latter than I. (c)/ No error (d)

34. She is the best (a)/ and beautiful girl (b)/ of our class. (c)/ No error (d)

35. My notes are superior (a)/ than yours although I have prepared (b)/ them in a hurry. (c)/ No error (d)

36. Of all other my neighbours (a)/ he is the kindest (b)/ and the most considerate. (c)/ No error (d)

37. The works of Shakespeare (a)/ are more famous (b)/ than any other English dramatist. (c)/ No error (d)

38. It is all the more better (a)/ if you work (b)/ in my company. (c)/ No error (d)

39. A little quantity of sugar (a)/ is required to meet (b)/ the present demands. (c)/ No error (d)

40. The Dean wrote that he constituted a committee of experts (a)/ comprising five members (b)/ before the next meeting took place. (c)/ No error (d)

41. He doesn't need (a)/ your help because (b)/ he is too intelligent. (c)/ No error (d)

42. I can't help to sneeze as (a)/ I got drenched yesterday (b)/ and have a bad cold. (c)/ No error (d)

43. I have lived (a)/ from the hand to the mouth (b)/ for all these fifty years though nobody knows it. (c)/ No error (d)

44. As soon as the teacher entered (a)/ everyone fell (b)/ in a silence. (c)/ No error (d)

45. As he had taken only a few sips (a)/ there was still little water (b)/ left in the glass. (c)/ No error (d)

46. To perform this experiment (a)/ drop little sugar (b)/ into a glass of water. (c)/ No error (d)

47. It is written in Gita (a)/ that God incarnates himself (b)/ in times of trouble. (c)/ No error (d)

48. Troy was taken by Greeks; (a)/ this formed the basis of a story (b)/ which has become famous. (c)/ No error (d)

49. These facts make it very clear (a)/ that he had hand in the murder (b)/ though he still pleads innocence. (c)/ No error (d)

50. A nationwide survey has brought up an (a)/ interesting finding (b)/ regarding infant mortality rate in India. (c)/ No error (d)

51. When I got (a)/ home I was (b)/ too exhausted. (c)/ No error (d)

52. Both he as well as his friend (a)/ worked in close harmony (b)/ on the same project. (c)/ No error (d)

53. I rarely found something (a)/ in the movie (b)/ that is worth remembering. (c)/ No error (d)

54. You have (a)/ acted nobler (b)/ than all of us. (c)/ No error (d)

55. Don't stop (a)/ anywhere. Go home (b)/ directly. (c)/ No error (d)

56. He has no time (a)/ to read magazines (b)/ and no desire neither. (c)/ No error (d)

57. He has not seldom (a)/ visited his parents (b)/ since he left this place. (c)/ No error (d)

58. It was much hot (a)/ yesterday and we (b)/ didn't go out. (c)/ No error (d)

59. I meet him often (a)/ near(b)/ the Town Hall. (c)/ No error (d)

60. I told her (a)/ as blunt as I could (b)/ but she was not convinced. (c)/ No error (d)

61. No sooner did the sun rise (a)/ when we took a hasty breakfast (b)/ and resumed the journey. (c)/ No error (d)

62. Because he is physically strong (a)/ therefore he was selected (b)/ for the school boxing team. (c)/ No error (d)

63. The reason for his failure (a)/ is because (b)/ he did not work hard. (c)/ No error (d)

64. Arjun asked him (a)/ that which the way was (b)/ to the post office. (c)/ No error (d)

65. Unless you do not listen to his advice (a)/ I am not going (b)/ to help you. (c)/ No error (d)

66. How do you say (a)/ that neither he or Raj has qualified (b)/ in the examination ? (c)/ No error (d)

67. We are not sure (a)/ if he is coming (b)/ to the party. (c)/ No error (d)

68. Sooner than he had arrived (a)/ his friends arranged a reception in his honour (b)/ in the best hotel in the town. (c)/ No error (d)

69. Mrs Dhaka went to Delhi (a)/ because she might (b)/ see Mrs Rai. (c)/ No error (d)

70. Neither he gave him no money (a)/ nor he helped him (b)/ in any way. (c)/ No error (d)

71. The Monk loved riding and hunting (a)/ and refused to conform by rules and regulations (b)/ of the ancient monastic order. (c)/ No error (d)

72. A man who always connives (a)/ on the faults of his children (b)/ is their worst enemy. (c)/ No error (d)

73. I do not understand (a)/ why (b)/ he is so angry at me. (c)/ No error (d)

74. I am hearing a lot (a)/ about the problem (b)/ of AIDS these days. (c)/ No error (d)

75. Because of his innocence (a)/ he can not distinguish (b)/ a cheat for an honest person. (c)/ No error (d)

76. After opening the door we entered (a)/ into the room (b)/ next to the kitchen. (c)/ No error (d)

77. As the meeting was (a)/ about to end (b)/ he insisted to ask several questions. (c)/ No error (d)

78. The watchman was kind enough (a)/ to inform us about the conspiracy (b)/ but declined to name the person behind it. (c)/ No error (d)

79. The captain and his wife (a)/ were invited for the cultural function (b)/ at my home. (c)/ No error (d)

80. The engineer came out to a novel solution (a)/ which may even reduce (b)/ daily energy consumption. (c)/ No error (d)

81. We are meeting today afternoon (a)/ to discuss the matter (b)/ and reach a compromise. (c)/ No error (d)

82. Either Ram or (a)/ you is responsible (b)/ for this action. (c)/ No error (d)

83. The student flatly denied (a)/ that he had copied (b)/ in the examination hall. (c)/ No error (d)

84. By the time you arrive tomorrow (a)/ I have finished (b)/ my work. (c)/ No error (d)

85. The speaker stressed repeatedly on (a)/ the importance of improving (b)/ the condition of the slums. (c)/ No error (d)

86. The captain with the members of his team (a)/ are returning (b)/ after a fortnight. (c) /No error (d)

87. After returning from (a)/ an all-India tour (b)/ I had to describe about it. (c)/ No error (d)

88. The teacher asked his students (a)/ if they had gone through (b)/ either of three chapters included in the prescribed text. (c)/ No error (d)

89. Although they are living in the country (a)/ since they were married (b)/ they are moving to the town. (c)/ No error (d)

90. Do you know (a)/ how old were you (b)/ when you came here? (c)/ No error (d)

91. Whenever a person losses anything (a)/ the poor folks around him (b)/ are suspected. (c)/ No error (d)

92. Still impressive is that (a)/ we achieve this selective attention (b)/ through our latent ability to lip read. (c)/ No error (d)

93. As I entered the famous gallery (a)/ my attention was at once drawn to the large sculpture in the corner. (c)/ No error (d)

94. Everyday before (a)/ I start work for my livelihood (b)/ I do my prayer. (c)/ No error (d)

95. Pooja went to her friend's house at the appointed hour ; but (a)/ she was told (b)/ that her friend left half an hour earlier. (c) No error (d)

96. Rekha is (a)/ enough old (b)/ to get married. (c)/ No error (d)

97. As far as I am concerned, (a)/ I shall do everything (b)/ possible to help you. (c)/ No error (d)

98. The person in the seat of justice (a)/ should be absolutely partial (b)/ and not treat his nearest and dearest with favour. (c)/ No error (d)

99. Let us congratulate him (a)/ for his success (b)/ in the examination. (c)/ No error (d)

100. Many people prefer to travel (a)/ by the road (b)/ because it is less expensive. (c)/ No error (d)

101. She was beside herself in joy (a)/ when she came to know (b)/ that she had been selected for the job. (c)/ No error (d)

102. Mother tongue is as natural (a)/ for the development of a man's mind (b)/ as mother's milk is for the development of the infant's body. (c)/ No error (d)

103. The Prime Minister as well as his secretary were expected to (a)/ arrive in Chennai (b)/ on Saturday morning. (c)/ No error (d)

104. The speaker was (a)/ not only slow (b)/ but also inaudible as well. (c)/ No error (d)

105. The crowd surged forward (a)/ to have a glimpse (b)/ of their favourite leader. (c)/ No error (d)

106. There is a distinctive possibility (a)/ that he will leave the job (b)/ once the investigation is over. (c)/ No error (d)

107. Many a star (a)/ are (b)/ twinkling in the sky. (c)/ No error (d)

108. We discussed the problem (a)/ so thoroughly that (b)/ I found it easy to work it out. (c)/ No error (d)

109. He hesitated to accept the post (a)/ as he did not think (b)/ that the salary would not be enough for a man with a family of three. (c)/ No error (d)

110. Have you gone through (a)/either of these three chapters (b)/ that have been included in this volume? (c)/ No error (d)

111. I am learning English (a)/ for ten years (b)/ without much effect. (c)/ No error (d)

QUESTIONS FROM CDS EXAM (2012-2016)

Directions (Q. Nos 1-167) *Which part of the given sentences has an error? In case, there is no error, choose option (d).*

2012 (I)

1. These are the ideas and ideals (a)/ which have shaped (b)/ our economic thought in the past. (c)/ No error (d)

2. India's problems are not similar with (a)/ those of other countries (b)/ in several ways. (c)/ No error (d)

3. He had lost a ring in the sand and (a)/ I helped him search for it, (b)/ but it was like a look for a needle in a haystack. (c)/ No error (d)

4. The Ganges and (a)/ it's tributaries constitute (b)/ one of the largest river-systems in the world. (c)/ No error (d)

5. The sudden change (a)/ of place (b)/ effected her health. (c)/ No error (d)

6. There are a number of people (a)/ of every class and nationality (b)/ who doubts the truth of his statement. (c)/ No error (d)

7. I like this book because the writer has explained (a)/ the reasons (b)/ of his failure truly. (c)/ No error (d)

8. She is very weak in the subject (a)/ and does not understand things (b)/ though the teacher explains her repeatedly. (c)/ No error (d)

9. The speaker from the Fifth Avenue, (a)/ who was a rich banker's wife, (b)/ was simple and compassionate. (c)/ No error (d)

10. There was no any piece of paper (a)/ in my pocket (b)/ as I had expected. (c)/ No error (d)

11. Neither the teacher (a)/ or the student (b)/ is keen on joining the dance. (c) No error (d)

12. My neighbour Deepak (a)/ is a person (b)/ that will help anyone. (c)/ No error (d)

13. I'll ask that man (a)/ which of the roads (b)/ are the one we want. (c)/ No error (d)

14. Now we have banks (a)/ and people deposit there money there, (b)/ and draw it out by cheques. (c)/ No error (d)

15. Apart government agencies, (a)/ a number of private organisations too (b)/ have been making use of satellites. (c)/ No error (d)

16. What sort of a drug is this (a)/ that no one seems to be able to predict its long-term effects (b)/ with any certainty? (c)/ No error (d)

17. You will lose (a)/ your dog (b)/ if you did not tie it up. (c)/ No error (d)

18. In view of the fact that almost all varieties of rural games and sports (a)/ are fast gaining national importance it is desired (b)/ that the rules of such games are strictly adhered to. (c)/ No error (d)

19. More than one (a)/ workman (b)/ was killed. (c)/ No error (d)

20. The parties disagreed (a)/ on the two first clauses (b)/ in the agreement. (c)/ No error (d)

2012 (II)

21. The scientist was seemed (a)/ to be excited (b)/ over the result of his experiment. (c)/ No error (d)

22. The student could not answer the teacher (a)/ when he was asked to explain (b)/ why he was so late that day. (c)/ No error (d)

23. John could not come (a)/ to school (b)/ as he was ill from cold. (c)/ No error (d)

24. Though she has aptitude in Mathematics (a)/ I won't allow her to take it up as a subject of study for the Master's Degree (b)/ because I know the labour involved will tell upon her health. (c)/ No error (d)

25. I am not familiar with (a)/ all the important places in this town, (b)/ although I have been living here since two years. (c)/ No error (d)

26. If I would be a millionaire, (a)/ I would not be wasting my time (b)/ waiting for a bus. (c)/ No error (d)

27. Until you begin to make a better use of your time, (a)/ I shall not stop (b)/ finding fault in you. (c)/ No error (d)

28. Neither of the two boys (a)/ is sensible (b)/ enough to do this job. (c)/ No error (d)

29. They left (a)/ their luggages (b)/ at the railway station.(c)/ No error (d)

30. You will get (a)/all the informations (b)/if you read this booklet carefully. (c)/No error (d)

31. She sang (a)/very well, (b)/ isn't it ? (c)/No error (d)

32. He is working (a)/ in a bank in New Delhi (b)/ for the past several months. (c)/ No error (d)

33. There is no question (a)/ of my failing (b)/ in the examination. (c)/ No error (d)

34. He is going everyday (a)/ for a morning walk (b)/ with his friends and neighbours. (c)/ No error (d)

35. Her relatives could not explain to us (a)/ why did not she come for the wedding (b)/ as she was expected. (c)/ No error (d)

36. He was prevented to accept the assignment (a)/ because he was a government employee (b)/ and as such barred from accepting such assignments. (c)/ No error (d)

37. If you repeat this mistake, (a)/ I will inform to your father (b)/ and do not blame me then. (c)/ No error (d)

38. Lieutenant Anand was short and muscular (a)/ with shoulders that bulged impressively (b)/ against his smart uniform. (c)/ No error (d)

2013 (I)

39. I should do (a)/ the same (b)/ If I were in your place. (c)/ No error (d)

40. He has been suffering (a)/ with fever (b)/ for the last six weeks. (c)/ No error (d)

41. The examination begins (a)/ from Monday (b)/ next week. (c)/ No error (d)

42. My father says (a)/ that one should always be sincere (b)/ to his duties. (c)/ No error (d)

43. There has been (a)/ a number of railway accidents (b)/ during the last month. (c)/ No error (d)

44. Inspite of all efforts to eradicate malaria (a)/ it still prevalent (b)/ in many parts of India. (c)/ No error (d)

45. It is only three days ago (a)/ that (b)/ he has arrived. (c)/ No error (d)

46. He has lost (a)/ all what (b)/ I gave him. (c)/ No error (d)

47. I have (a)/ no news from him (b)/ for a long time. (c)/ No error (d)

48. Mahatma Gandhi's entire life (a)/ was one unrelenting experiment (b)/ with truth. (c)/ No error (d)

49. As the thieves ran out of the bank (a)/ they got into the gateway car (b)/ which was waiting with its engine running. (c)/ No error (d)

50. He denied that he had not stolen my purse, (a)/ though I was quite sure (b)/ that he had. (c)/ No error (d)

51. The media of films has been accepted by all (a)/ as the most powerful force (b)/ that influences the younger generation. (c)/ No error (d)

52. The French Embassy employs him (a)/ regularly (b)/ as he knows to speak French. (c)/ No error (d)

53. How is it that neither your friend Mahesh (a)/ nor his brother Ramesh (b)/ have protested against this injustice? (c)/ No error (d)

2013 (II)

54. Lack of winter rains (a)/ have delayed the sowing of (b)/ wheat crop in this area.(c)/ No error (d)

55. The teacher let the boy off (a)/ with a warning though he (b)/ was convinced with his guilt. (c)/ No error (d)

56. Our first trip was the most interesting one, (a)/ but our second one, (b)/ was even more interesting. (c)/ No error (d)

57. He has been going to the office (a)/ for a year now, (b)/ and he even can't understand its working. (c)/ No error (d)

58. He boasts of having visited Europe many times (a)/ but he can speak neither English (b)/ nor he can speak French. (c)/ No error (d)

59. Whenever possible, one should avail the opportunity (a)/ that come one's way (b)/ if one wants to achieve success in life. (c)/ No error (d)/

60. When my friends came to visit us (a)/ at the railway station (b)/ they left some of their luggages. (c)/ No error (d)

61. As an officer (a)/ he not only was competent (b)/ but also honest. (c)/ No error (d)

62. If you will come tomorrow (a)/ we can go to the market (b)/ and do our own shopping together. (c)/ No error (d)

63. If we exercise regularly (a)/ we will be (b)/ more healthier. (c)/ No error (d)

64. News travel (a)/ very fast today (b)/ due to advancement in technology. (c)/ No error (d)

65. The Chairman made it clear at the meeting (a)/ that he will not step down (b)/ from his position as chairman. (c)/ No error (d)

66. We had (a)/ lot of difficulty (b)/ in finding the way here. (c)/ No error (d)

67. Just as he was driving along the road, (a)/ a bus pulled up and the driver asked him (b)/ if he has seen a briefcase on the road. (c)/ No error (d)

68. Experience has taught me (a)/ not to ignore any man, high or low, (b)/ not to ignore anything, great or small. (c)/ No error (d)

69. I have spent (a)/ most of my money, (b)/ so I can travel only by bus. (c)/ No error (d)

70. When he asked me as to why (a)/ I had not finished my work in time, (b)/ I felt confused. (c)/ No error (d)

71. The Foreign Minister said (a)/ there was no use to criticise the policy of non-alignment (b)/ which had stood the test of time. (c)/ No error (d)

72. The train should arrive at (a)/ 7:30 in the morning (b)/ but it was almost an hour late. (c)/ No error (d)

2014 (I)

73. He asked her that (a)/ whether she knew (b)/ what had happened last week when she was on leave. (c)/ No error (d)

74. Until you do not go to the station (a)/ to receive him (b)/ I can hardly feel at ease. (c)/ No error (d)

75. I did not know where they were going (a)/ nor could I understand (b)/ why had they left so soon.(c)/ No error (d)

76. The distinguished visitor said that he had great pleasure to be with us for some time (a)/ and that the pleasure was all the greater (b)/ because his visit afforded him an opportunity to study the working of an institution of such eminence as ours. (c)/ No error (d)

77. Please convey (a)/ my best wishes (b)/ back to your parents. (c)/ No error (d)

78. The call of the seas (a)/ have always (b)/ found an echo in me. (c)/ No error (d)

79. Hardly I had left home for Mumbai (a)/ when my son who is settled in Kolkata arrived (b)/ without any prior information. (c)/ No error (d)

80. Now, it can easily be said (a)/ that the population of this city is greater (b)/ than any other city in India. (c)/ No error (d)

81. It is difficult to explain (a)/ why did Rajagopalachari resigned (b)/ from the Congress in 1940. (c)/ No error (d)

82. The boss reminded them of the old saying (a)/ that honesty was the best policy (b)/ and told them that they had better be honest in their work. (c)/ No error (d)

83. 'Gulliver's Travels' are (a)/ the most fascinating adventure story (b)/ that I have ever read. (c)/ No error (d)

84. The teenager reassured his father at the station (a)/ "Don't worry, dad' (b)/ I will pull on very nicely at the hostel." (c)/ No error (d)

85. By the way he's behaving (a)/ he'll soon spill the beans (b)/ I'm afraid. (c)/ No error (d)

86. Most of the developing countries find it (a)/ difficult to cope up with the problems (b)/ created by the sudden impact of technological progress. (c)/ No error (d)

87. People blamed him (a)/ for being (b)/ a coward person. (c)/ No error (d)

88. We swam up to the drowning man, caught hold of his clothes (a)/ before he could go down again (b)/ and pulled him out, safe to the shore. (c)/ No error (d)

89. Meena was so tired (a)/ that she could not hardly (b)/ talk to the guests for a few minutes. (c)/No error (d)

90. If I was knowing (a)/ why he was absent, (b)/ I would have informed you. (c)/ No error (d)

91. He goes (a)/ to office (b)/ by foot. (c)/ No error (d)

92. The hundred-rupees notes (a)/ that he gave them for the goods bought from them looked genuine (b)/ but later they reliably learnt that the notes were all counterfeit. (c)/ No error (d)

2014 (II)

93. He went to England to work as a doctor (a)/ but returned back (b)/ as he could not endure the weather there. (c)/ No error (d)

94. She inquired whether (a)/ anyone (b)/ has seen (c)/ her baby. No error (d)

95. When I went (a)/ outdoor (b)/ I found frost everywhere. (c)/ No error (d)

96. These are (a)/ his (b)/ conclusion remarks. (c)/ No error (d)

97. The shopkeeper offered either to exchange (a)/ the goods (b)/ or refund the money. (c)/No error (d)

98. Churchill was (a)/ one of the greatest (b)/ war leaders. (c)/ No error (d)

99. We should keep (a)/ such people (b)/ at an arm's length. (c)/ No error (d)

100. He did not know (a)/ as much as (b)/ he claimed he knew. (c)/ No error (d)

101. That was very dangerous (a)/: you might (b)/ have been killed. (c)/ No error (d)

102. My friend (a)/ is going (b)/ to a movie (c)/ every week. No error (d)

103. They sit (a)/ at the window (b)/ and watch the traffic. (c)/ No error (d)

104. I started early (a)/ for the station lest I (b)/ should not miss the train. (c)/ No error (d)

105. I wanted to see (a)/ that whether they (b)/ had actually read the notes. (c)/ No error (d)

106. They made him treasurer (a)/ because they considered (b)/ him as honest and efficient. (c)/ No error (d)

107. Having finished the paper early (a)/ he came out of the hall (b)/ almost an hour before the bell rang. (c)/ No error (d)

108. The (a)/ young man (b)/ had no manner. (c)/ No error (d)

109. No news (a)/ are (b)/good news. (c)/ No error (d)

110. The work involved (a)/ is almost impossible (b)/ to cope with. (c)/ No error (d)

111. There is (a)/ no place (b)/ in this compartment. (c)/ No error (d)

112. Shakespeare (a)/ is greater than (b)/ any poet. (c)/ No error (d)

113. I should (a)/ have preferred (b)/ to go by myself. (c)/ No error (d)

114. The minister announced (a)/ compensation for (b)/ the victims from the accident. (c)/No error (d)

115. The Australian team (a)/ losed the match (b)/ yesterday. (c)/ No error (d)

116. He told us (a)/ that (b)/ he has not read the book. (c)/ No error (d)

117. The composition contained (a)/ even not less (b)/ than twenty mistakes. (c) /No error (d)

2015 (I)

118. The reason for (a)/ his failure is because (b)/ he did not work hard. (c)/No error (d)

119. Food as well as water (a)/ is necessary (b)/ for life. (c)/ No error (d)

120. India is larger than (a)/ any democracies (b)/ in the world. (c)/No error (d)

121. The Judge heard the arguments (a)/ of the lawyers and found (b)/ that the boy was innocent. (c)/No error (d)

122. I have been living (a)/in Delhi (b)/ from 1965. (c)/No error (d)

123. All scientists agree (a)/ that there should be (b)/ a total ban on nuclear explosions. (c)/No error (d)

124. Such books (a)/ which you read (b)/ are not worth reading. (c)/No error (d)

125. Tagore was (a)/ one of the greatest poet (b)/ that ever lived. (c)/No error (d)

126. You may please (a)/ apply for an advance of salary (b)/ to cover costs of transport. (c)/No error (d)

127. The taxi that will take the family to Haridwar (a)/ had to be ready (b)/ at six the next morning. (c)/ No error (d)

128. Employees are expected to (a)/ adhere the rules (b)/ laid down by the management. (c)/ No error (d)

129. The owner of the horse (a)/ greedily ask (b)/ too high a price. (c)/ No error (d)

130. I convinced (a)/ him to (b)/ see the play. (c)/ No error (d)

131. Some man (a)/ are born (b)/ great. (c)/ No error (d)

132. We must sympathise (a)/ for others (b)/ in their troubles. (c)/ No error (d)

133. My detailed statement (a)/ is respectively (b)/ submitted. (c)/ No error (d)

134. I am waiting (a)/ for my friend (b)/ since this morning. (c)/ No error (d)

135. He is representing (a)/ my constituency (b)/ for the last five years. (c)/ No error (d)

136. If he hears (a)/ of your conduct (b)/ he is to be unhappy. (c)/ No error (d)

137. No sooner he appeared (a)/ on the stage than the people (b)/ began to cheer loudly. (c)/ No error (d)

2015 (II)

138. Of all those involved (a)/ with the accident (b)/ none was seriously injured. (c)/ No error (d)

139. Radar equipments (a)/ that is to be used (b)/ for ships must be installed carefully. (c)/ No error (d)

140. New types of electrical circuits (a)/ has been developed (b)/ by our engineers. (c)/ No error (d)

141. Recently I visited Kashmir (a)/ and found the sceneries (b)/ to be marvellous. (c)/ No error (d)

142. It is of primary importance (a)/ in swimming to learn (b)/ to breathe properly. (c)/ No error (d)

143. When the party was over, (a)/ he looked around for the girl (b)/ who had come with him. (c)/ No error (d)

144. After we were driving for miles (a)/ on the winding road (b)/ I was suddenly sick. (c)/ No error (d)

145. The forecast was for fair (a)/ and warm weather (b)/ and the day dawned dark and chill. (c)/ No error (d)

146. To write, to speak or to act (a)/ seem (b)/ very easy. (c)/ No error (d)

147. I have not had tea (a)/ since (b)/ two days. (c)/ No error (d)

148. Beside (a)/ his mother, he has two aunts (b)/ who stay with him. (c)/ No error (d)

149. This photograph (a)/ appears to be (b)/ the best of the two. (c)/ No error (d)

150. Either the operator (a)/or the foreman are (b)/ to be blamed for the accident. (c)/ No error (d)

151. The article offers (a)/ good advice to (b)/ whomever must accept it. (c)/ No error (d)

152. She dislikes (a)/ you being most organised (b)/ than she is. (c)/ No error (d)

2016 (I)

153. This hardly won freedom (a)/ should not be lost (b)/ so soon.(c)/ No error (d)

154. I tried to meet the person (a)/ whom you said (b)/ was looking for me. (c)/ No error (d)

155. We looked after the thief (a)/ but he was nowhere (b)/ to be found. (c)/ No error (d)

156. I hoped that the train (a)/ will arrive on time, (b)/ but it did not. (c)/ No error (d)

157. Their all belongings (a)/ were lost (b)/ in the fire. (c)/ No error (d)

158. He was in the temper (a)/ and refused (b)/ to discuss the matter again. (c)/ No error (d)

159. The decorations in your house (a)/ are similar (b)/ to his house. (c)/ No error (d)

160. Despite of the increase in air fares, (a)/ most people still prefer (b)/ to travel by plane. (c)/ No error (d)

161. He told the boys that (a)/ if they worked hard, (b)/ they will surely pass. (c)/ No error (d)

162. I shall write (a)/ to you (b)/ when I shall reach Chennai. (c)/ No error (d)

163. Neither of these two documents (a)/ support your claim (b)/ on the property. (c)/ No error (d)

164. He is school teacher, (a)/ but all his sons (b)/ are doctors. (c)/ No error (d)

165. His grandfather (a)/ had told him that to smoke (b)/ was a bad habit. (c)/ No error (d)

166. My book, which (a)/ I gave it to you yesterday, (b)/ is very interesting. (c)/ No error (d)

167. I am entirely agreeing with you (a)/ but I regret (b)/ I can't help you. (c)/ No error (d)

ANSWERS

Complete Exercise

1	*b*	2	*b*	3	*a*	4	*a*	5	*b*	6	*c*	7	*a*	8	*c*	9	*a*	10	*a*
11	*a*	12	*a*	13	*d*	14	*b*	15	*b*	16	*c*	17	*c*	18	*c*	19	*b*	20	*a*
21	*c*	22	*a*	23	*b*	24	*b*	25	*b*	26	*d*	27	*b*	28	*a*	29	*a*	30	*b*
31	*b*	32	*a*	33	*c*	34	*b*	35	*b*	36	*a*	37	*c*	38	*a*	39	*a*	40	*a*
41	*c*	42	*a*	43	*b*	44	*c*	45	*b*	46	*b*	47	*a*	48	*a*	49	*b*	50	*a*
51	*c*	52	*a*	53	*a*	54	*b*	55	*c*	56	*c*	57	*a*	58	*a*	59	*a*	60	*b*
61	*b*	62	*a*	63	*b*	64	*b*	65	*a*	66	*b*	67	*d*	68	*a*	69	*b*	70	*a*
71	*b*	72	*b*	73	*c*	74	*a*	75	*c*	76	*b*	77	*c*	78	*b*	79	*b*	80	*a*
81	*a*	82	*b*	83	*d*	84	*b*	85	*d*	86	*b*	87	*c*	88	*c*	89	*a*	90	*b*
91	*a*	92	*c*	93	*d*	94	*c*	95	*c*	96	*b*	97	*d*	98	*b*	99	*b*	100	*b*
101	*a*	102	*c*	103	*a*	104	*c*	105	*b*	106	*a*	107	*b*	108	*c*	109	*b*	110	*b*
111	*a*																		

Questions from CDS Exam (2012-16)

1	*b*	2	*a*	3	*b*	4	*b*	5	*b*	6	*c*	7	*d*	8	*d*	9	*d*	10	*a*
11	*b*	12	*c*	13	*c*	14	*b*	15	*a*	16	*d*	17	*c*	18	*a*	19	*c*	20	*b*
21	*a*	22	*d*	23	*c*	24	*a*	25	*c*	26	*a*	27	*c*	28	*d*	29	*b*	30	*b*
31	*c*	32	*a*	33	*b*	34	*a*	35	*b*	36	*a*	37	*b*	38	*d*	39	*a*	40	*b*
41	*b*	42	*c*	43	*a*	44	*b*	45	*c*	46	*b*	47	*b*	48	*b*	49	*c*	50	*a*
51	*a*	52	*c*	53	*c*	54	*b*	55	*c*	56	*a*	57	*c*	58	*c*	59	*b*	60	*c*
61	*b*	62	*a*	63	*c*	64	*a*	65	*b*	66	*b*	67	*c*	68	*c*	69	*b*	70	*a*
71	*b*	72	*a*	73	*a*	74	*a*	75	*c*	76	*c*	77	*c*	78	*b*	79	*a*	80	*c*
81	*b*	82	*b*	83	*a*	84	*d*	85	*a*	86	*d*	87	*c*	88	*a*	89	*b*	90	*a*
91	*c*	92	*a*	93	*b*	94	*c*	95	*b*	96	*c*	97	*c*	98	*d*	99	*c*	100	*c*
101	*b*	102	*b*	103	*d*	104	*c*	105	*b*	106	*c*	107	*b*	108	*c*	109	*b*	110	*c*
111	*b*	112	*c*	113	*c*	114	*c*	115	*b*	116	*c*	117	*b*	118	*b*	119	*d*	120	*b*
121	*d*	122	*c*	123	*c*	124	*a*	125	*b*	126	*a*	127	*b*	128	*b*	129	*b*	130	*c*
131	*a*	132	*b*	133	*b*	134	*a*	135	*a*	136	*c*	137	*a*	138	*b*	139	*a*	140	*b*
141	*b*	142	*a*	143	*d*	144	*a*	145	*c*	146	*a*	147	*b*	148	*a*	149	*c*	150	*b*
151	*c*	152	*b*	153	*a*	154	*b*	155	*a*	156	*b*	157	*a*	158	*a*	159	*c*	160	*a*
161	*c*	162	*c*	163	*b*	164	*a*	165	*b*	166	*b*	167	*a*								

EXPLANATIONS

Complete Exercise

1. *(b)* The phrase 'one of the' takes a plural noun. So, 'bad habit' should be replaced by 'bad habits'.

2. *(b)* The word 'scenery' is always used in a singular form. Hence, in the given sentence, 'scenery' should be used in place of 'sceneries'.

3. *(a)* The word 'furniture' is always used in a singular form. So, part (a) would become 'All the furniture has been.'

4. *(a)* 'Mob' is a 'group of people with one common thought of criminal consequence'. Hence, in the given sentence, 'mob' should be used in place of 'crowd'.

5. *(b)* The word 'luggage' is always used in a singular form. The correct sentence would be 'They left their luggage……'

6. *(c)* In this sentence, 'gears' is used incorrectly. It should be changed to 'gear'.

7. *(a)* The sentence means to convey the message that the Indian Military drove away the Chinese. To convey this meaning, 'force' needs to be changed to 'forces'.

8. *(c)* 'Bouquet' is used with flowers. For grapes, we use 'bunch'.

9. *(a)* The plural of 'brother-in-law' is 'brothers-in-law'.

10. *(a)* The sentence refers to a number of shirts. So, the word 'kind' must be replaced by 'kinds'.

11. *(a)* As the sentence refers to a number of persons i.e. 'people', 'sorts' will be used instead of 'sort'.

12. *(a)* The pronoun 'It' comes before the phrase or clause to which it refers. So, 'It' will come at the starting of the sentence.

13. *(d)* No error.

14. *(b)* As the sentence refers to one person i.e. 'One of them' so it would have a singular pronoun. Hence, we would replace 'their' by 'his' to make the sentence correct.

15. *(b)* The pronoun 'he' in the sentence is not needed. So, we would remove 'he'.

16. *(c)* The sentence refers to one teacher i.e. singular form. Hence, it should accompany a singular pronoun. Therefore, we will replace 'them' by 'him'.

17. *(c)* The pronoun 'he' in the sentence is not needed. So, we would remove 'it'.

18. *(c)* 'Each girl' means a number of girls. Hence, the pronoun used for it will be plural. So, 'them' would replace 'her' in the sentence.

19. *(b)* This sentence would contain a reflexive pronoun. Hence, 'him' in the sentence would be replaced by 'himself'.

20. *(a)* The pronoun 'myself' used in the sentence is erroneous. Instead of 'myself', 'I' would be used.

21. *(c)* As per the rule, 'many a' takes a singular verb. So, in the given sentence we would replace 'were' by 'was' to make it grammatically correct.

22. *(a)* In the given sentence, 'dozen of geography books' is considered as a single entity. Hence, the verb 'are' in the sentence would be replaced by 'is'.

23. *(b)* 'With' is not needed as 'to marry the girl' makes sense. Hence, we would remove 'with'.

24. *(b)* 'The Himalayas' are a mountain range and hence, should take a plural verb. Therefore, 'is' in the given sentence would be replaced by 'are'.

25. *(b)* In the given sentence, 'was' would be used after 'and'.

26. *(d)* No error.

27. *(b)* As per the rule, the verb should agree with the pronoun 'you'. So, 'is' would be changed to 'are'.

28. *(a)* 'The children' in the given sentence did a common thing (action) of troubling their parents. So, 'the children' would be taken as a single entity (singular) and hence would take a singular verb i.e. 'was'. So, 'was' would replace 'were'.

29. *(a)* As the sentence states an event in the past tense (last year), 'will be' should be replaced by 'was'.

30. *(b)* In the sentences with 'along with', the verb should agree with the first subject. So, 'was' would be chsnged to 'were'.

31. *(b)* 'He had left' needs to be changed to 'had he left' to make the sentence grammatically correct.

32. *(a)* 'Was arriving' is not correct as per the sentence structure and tense. It should be change to past tense i.e., 'When he arrived'.

33. *(c)* 'Latter' is not the correct word to be used as per the sentence. Hence, we would use 'later' instead of it.

34. *(b)* As per the rule, when two adjectives qualify the same noun, both the adjectives should be represented in the same degree. So, we would change 'beautiful' in the given sentence to 'the most beautiful'.

35. *(b)* As per the rule 'superior' is followed by preposition 'to'. So, we would replace 'than' by 'to'.

36. *(a)* The word 'other' in the given sentence is not needed. So, we will remove it.

37. *(c)* 'Those of' would be added before 'any other' dramatist.

38. *(a)* To make the sentence meaningful, we would remove 'all the more' from the sentence.

39. *(a)* As per the rule, the adjectives 'little' and 'few' are not made to qualify the nouns 'quantity' and 'number'. Hence, we would use 'A small' instead of 'A little'.

40. *(a)* The sentence refers to a past event. Hence, the verb 'constituted' should be changed to 'had constituted'.

41. *(c)* 'Too' is not the correct word to be used with 'intelligent' in the given sentence. Hence, we would use 'very' instead of 'too'.

42. *(a)* We should remove 'to' and change 'sneeze' to 'sneezing' to make the sentence correct.

43. *(b)* 'The hand to the mouth' is incorrect. The correct phrase/idiom is 'hand to mouth' which means 'to have just enough money to live on and nothing extra'.

44. *(c)* As per the rule, articles are not used before abstract nouns. 'silence' in the given sentence is an abstract noun. Hence, we would remove 'a' before 'silence'.

45. *(b)* 'Little' should be replaced by 'some'.

46. *(b)* 'Little sugar' means 'no sugar'. Hence, to make sense, we would add 'a' before 'little'.

47. *(a)* 'Gita' is a religious book of the Hindus. So, as per the rule, it should be written as 'the Gita'.

48. *(a)* 'Greeks' in the sentence does not make sense. It should be written as 'the Greeks' as it intends to mention 'people of Greece' in the given sentence.

49. *(b)* To make the sentence correct, the article 'a' should be added before 'hand'.

50. *(a)* 'Brought up' means 'to rear'. Here, it should be changed to 'brought out' which means 'to make (it) public'.

51. *(c)* 'Very' should be used instead of 'too' in the given sentence.

52. *(a)* As per the rule 'as well as' is not used in place of 'and'. Hence, we replace 'as well as' by 'and' to make the sentence correct.

53. *(a)* 'Something' is used in a positive sense. It should be replaced by 'anything' in the given sentence to convey the right meaning.

54. *(b)* 'Nobler' does not make sense. It should be replaced by 'more nobly' in the given sentence.

55. *(c)* 'Direct' means 'straight' and 'directly' means 'at once'. Therefore, we must use 'direct' instead of 'directly' in the given sentence.

56. *(c)* 'Neither' should be replaced by 'either' to make the sentence meaningful.

57. *(a)* 'Not' in the given sentence is not needed. As per the rule two negatives 'not' and 'seldom' are not used in a single sentence. Hence, we would remove 'not'.

58. *(a)* 'Much' should be replaced by 'very' in the sentence to make it meaningful.

59. *(a)* As per the rule it should be 'often meet him' instead of 'meet him often'.

60. *(b)* 'Blunt' needs to be replaced by the adverb 'bluntly' to make the sentence meaningful.

61. *(b)* 'No sooner' is followed by 'than' and not by 'when'. Hence, we would replace 'when' by 'than'.

62. *(a)* The word 'because' is not needed in the sentence. Hence, we would delete that.

63. *(b)* 'Because' is not needed in the sentence as the sentence starts with stating the reason for failure. We should replace 'because' by 'that' to make the sentence correct.

64. *(b)* 'That' is not used in the indirect speech when the sentence is interrogative. Hence, we would remove 'that'.

65. *(a)* As per the rule 'unless' is not used with negatives like 'not'. Hence, we would remove 'do not' from the sentence.

66. *(b)* 'Neither' is used with 'nor'. Hence, we would replace 'or' by 'nor' in the sentence.

67. *(d)* No error.

68. *(a)* Part (a) is incorrect. It should be rephrased as 'No sooner had he arrived than ....'

69. *(b)* 'Because' is the incorrect conjunction used in the sentence. It should be replaced by 'so that'.

70. *(a)* 'No' in the sentence should be removed as it is not needed in the sentence.

71. *(b)* The preposition 'by' is incorrect. It should be replaced by 'to' to make the sentence meaningful.

72. *(b)* The word 'connive' is used with 'at'. Hence, we would replace 'on' by 'at' to make the sentence grammatically correct.

73. *(c)* The right usage is 'angry with' and not 'angry at'. Hence, we replace 'at' by 'with' in the sentence.

74. *(a)* 'Am hearing' should be changed to 'hear'.

75. *(c)* 'Distinguish' uses the preposition 'from'. Hence, we would replace 'for' by 'from'.

76. *(b)* As per the rule, no preposition is used after 'enter', so we would remove 'into' from part (b).

77. *(c)* 'To ask' should be replaced by 'on asking' to make the sentence grammatically correct.

78. *(b)* 'Inform of' means the information about something and 'inform about' means the information with the details. Hence, we would use 'of' instead of 'about'.

79. *(b)* 'Invited to' is the correct usage. Hence, we would not use 'for'.

80. *(a)* 'Came out with' is the correct usage. Hence, we would replace 'to' by 'with' in the given sentence.

81. *(a)* 'Today' should not be used. It should be replaced by 'this'.

82. *(b)* The verb in the sentence must agree with 'you'. Hence, we would change 'is' to 'are'.

83. *(d)* The sentence is correct.

84. *(b)* 'I have finished' is incorrect as per the sentence structure. It should be changed to 'I will have finished.'

85. *(d)* The sentence is correct.

86. *(b)* The verb in the sentence should agree with the noun 'captain'. Hence, we should change 'are' to 'is'.

87. *(c)* The word 'about' is not needed in the sentence. Hence, we would remove 'about'.

88. *(c)* 'Either of' is used for two things. Hence, we would use 'any of' in the given sentence.

89. *(a)* 'Had been' should be used in place of 'are' to convey the correct meaning of the sentence.

90. *(b)* 'How old were you' should be changed into 'how old you were' to make the sentence correct.

91. *(a)* 'Losses' is the incorrect word. It should be changed to 'loses'.

92. *(c)* 'Through' is not the right word. It should be replaced by 'with'.

93. *(d)* The sentence is correct.

94. *(c)* 'Say' should be used in place of 'do' as 'do' is not used with 'prayer'.

95. *(c)* We need to add 'had' before 'left' to make the sentence grammatically correct.

96. *(b)* 'Enough old' should be changed to 'old enough'.

97. *(d)* The sentence is correct.

98. *(b)* We should use 'impartial' in place of 'partial'.

99. *(b)* 'For' should be replaced by 'on' to convey the correct meaning of the sentence.

100. *(b)* 'By the road' is incorrect. We should use 'by road'.

101. *(a)* 'In' in the given sentence needs to be replaced by 'with'.

102. *(c)* 'The' in the part (c) of the sentence should be replaced by 'an' as the next word 'infant's' starts with a vowel.

103. *(a)* As per the given sentence, the verb 'were' should be replaced by 'was' to make the sentence grammatically correct.

104. *(c)* 'As well' is not needed in the sentence. Hence, we delete it.

105. *(b)* The right phrase to use is 'catch a glimpse'. Hence, we replace 'have' by 'catch' in the given sentence.

106. *(a)* We need to use 'distinct' in place of 'distinctive' to make the sentence meaningful.

107. *(b)* 'Many a' uses a singular verb. Hence, we replace 'are' in the given sentence by 'is'.

108. *(c)* 'Work it out' does not convey the right meaning it should be changed to 'work out'.

109. *(b)* 'Did not think' needs to be replaced by 'thought' to make the sentence correct.

110. *(b)* 'Either' is used two things. For more than two, we use 'any'. Hence, we would replace 'either' by 'any' to make the sentence correct.

111. *(a)* 'Am' needs to be replaced by 'have been' as the task in the sentence ('learning English') is continuous.

Questions from CDS Exam (2012-2016)

1. *(b)* Second part of the sentence is in the past tense and talks about what happened in the past so in part (b) 'had' should replace 'have'.
2. *(a)* 'With' should be replaced by 'to'.
3. *(b)* There is no need to use 'for' with search. 'Search' itself implies to search for lost ring in the sentence.
4. *(b)* 'It's' should be replaced by 'its' to make the sentence meaningful.
5. *(b)* For changing the place 'change in place' is better usage than 'change of place'. As later one shows the shift or movement.
6. *(c)* 'A number of people' is plural and verb-subject agreement should be there in part (c) of the sentence so 'doubt' will replace 'doubts'.
7. *(d)* No error
8. *(d)* No error
9. *(d)* No error
10. *(a)* In this part 'any' is a redundant word and makes the sentence grammatically wrong. So, we remove 'any'.
11. *(b)* Neither –nor combination is a standard usage. So, we would replace 'or' by 'nor'.
12. *(c)* 'That' should be replaced by 'who'.
13. *(c)* Use of one with 'are' is not acceptable in this part. 'Is' should replace by 'are'.
14. *(b)* 'There' after the word 'deposit' should be changed to 'their' to convey the correct meaning.
15. *(a)* 'Apart from' is to be used . Use of apart does not make any sense here.
16. *(d)* No error
17. *(c)* 'Do not' should be used in place of 'did not' to make the sentence correct.
18. *(a)* 'The' should be added before varieties to make the sentence correct.
19. *(c)* 'Was' should be changed to 'were' as more than one workman is involved.
20. *(b)* 'Two first' is inappropriate. A proper sequence of 'first two' should be used to make the sentence meaningful.
21. *(a)* Remove 'was'. Its use is superfluous.
22. *(d)* No error
23. *(c)* It should be 'suffering from cold' or 'ill with cold'.
24. *(a)* Use 'for' in place of 'in'.
25. *(c)* Use 'for' in place of 'since'.
26. *(a)* Use 'were' in place of 'would be'.
27. *(c)* Use 'with' in place of 'in' as 'find fault with somebody' is used.
28. *(d)* No error
29. *(b)* Use 'luggage' in place of 'luggages'. Luggage is an uncountable noun and does not have a plural form.
30. *(b)* Use 'information' in place of 'informations'. Information is an uncountable noun and does not have a plural form.
31. *(c)* Part (c) should be changed to 'didn't she?'
32. *(a)* Use 'has been' in place of 'is' as the sentence mentions a continuous activity (working in a bank).
33. *(b)* 'Of my failing' should be replaced by 'of me failing'.
34. *(a)* Use 'goes' in place of 'is going'.
35. *(b)* Use 'she did not' in place of 'did not she' as the sentence is not an interrogative sentence.
36. *(a)* Use 'from accepting' in place of 'to accept'.
37. *(b)* Remove 'to'. Its use is superfluous.
38. *(d)* No error
39. *(a)* Use 'would' in place of 'should'.
40. *(b)* Use 'from' in place of 'with' as suffer from something is used.
41. *(b)* Use 'on' in place of 'from' as 'on' is used to show a day or date.
42. *(c)* Use 'one's in place of 'his'.
43. *(a)* Use 'have' in place of 'has'. 'A number of' always takes a plural noun and a plural verb.
44. *(b)* It would be 'it is still prevalent'.
45. *(c)* Remove 'has'.
46. *(b)* Use 'that' in place of 'what'.
47. *(b)* Use 'of' in place of 'from'.
48. *(b)* Use 'an' in place of 'one'.
49. *(c)* Part (c) needs to be changed to 'which was left with its engine on'.
50. *(a)* Remove 'not' to make the sentence meaningful.
51. *(a)* 'Media' should be changed to 'medium'.
52. *(c)* We would replace 'knows to' by 'can'.
53. *(c)* Use 'has' in place of 'have' as neither takes a singular verb.
54. *(b)* Use 'has' instead of 'have'.
55. *(c)* Use 'of' instead of 'with' as the verb 'convinced' always takes the preposition 'of' after it.
56. *(a)* We should use 'much' in place of 'the most'.
57. *(c)* 'Even' should be replaced by 'yet'.
58. *(c)* Remove 'he can speak'. It is superfluous.
59. *(b)* Replace 'come' by 'comes' because the word 'opportunity' is singular.
60. *(c)* Replace 'luggages' by 'items of luggage'.
61. *(b)* Use 'was' after 'he'. Part (b) would become 'he was not only competent'.
62. *(a)* Remove 'will'. In the conditional sentences, where both the acts are to take place in future the clause with 'if' is in present indefinite tense.
63. *(c)* Remove 'more'. 'More' is not used with an adjective in its comparative degree.
64. *(a)* Use 'travels' as the noun, 'News' is singular.
65. *(b)* Replace 'will' by 'would' as the first clause is in the past tense.
66. *(b)* Use 'a' before 'lot of'.
67. *(c)* Replace 'has' by 'had' because the first clause is in the past indefinite tense.

68. *(c)* Remove the words 'not to ignore' and use 'nor'.

69. *(b)* Use 'the' before 'most'. When the preposition 'of' is used after a superlative degree adjective like most, best, greatest, the article 'the' is used before it.

70. *(a)* Remove 'as to'.

71. *(b)* Replace 'there was no use' by 'it was of no use' or 'it was useless'.

72. *(a)* Replace 'should' by 'was to'.

73. *(a)* Delete 'that'. 'that' is not used after asked.

74. *(a)* Delete 'do not'. Do not is not used with until as both are negatives.

75. *(c)* Replace 'why had they' with 'why they had'.

76. *(c)* Add 'that of' before 'ours'.

77. *(c)* Remove 'back'.

78. *(b)* Use 'has' in place of 'have'.

79. *(a)* Replace 'Hardly I had' with 'Hardly had I'.

80. *(c)* Add 'that of' after than.

81. *(b)* Delete 'did'.

82. *(b)* Write 'is' in place of 'was' as it is a proverb.

83. *(a)* Use 'is' in place of 'are' as singular verb is used after the name of books.

84. *(d)* No error

85. *(a)* Replace 'behaving' with 'behaving in'.

86. *(d)* No error

87. *(c)* Remove 'person' after 'coward'.

88. *(a)* Use 'got hold' in place of 'caught hold'.

89. *(b)* Remove 'not' before 'hardly'.

90. *(a)* Use 'I knew' in place of 'I was knowing'.

91. *(c)* Use 'on foot' in place of 'by foot'.

92. *(a)* It should be 'hundred rupee notes' in place of 'hundred rupee notes'.

93. *(b)* Use of 'back' is superfluous here.

94. *(c)* Use 'had seen' in place of 'has seen' as the sentence requires past perfect.

95. *(b)* 'Outdoor' should be replaced by 'outdoors'.

96. *(c)* Replace 'conclusion' with 'concluding'.

97. *(c)* Use 'to' before 'refund' as infinitive is required here.

98. *(d)* No error

99. *(c)* Remove 'an' as it is not required here.

100. *(c)* We would use 'he had claimed' in place of 'he claimed' as here two past events are given, so prior should be in past perfect.

101. *(b)* Use, 'could' in place of 'might' as here is speculation about something that didn't happen in the sentence.

102. *(b)* Use 'goes' in place of 'is going' as habitual verbs are not used in progressive forms.

103. *(d)* No error.

104. *(c)* Remove 'not' as 'lest' contains a negative meaning.

105. *(b)* 'Whether' or 'that' both are connectives and we can't use them simultaneously. So, we would remove 'that'.

106. *(c)* Remove 'as' because 'consider' takes no preposition.

107. *(b)* Use 'had come' in place of 'came' as here former event should be in past perfect tense.

108. *(c)* Use 'manners' in place of 'manner' as it is always used in plural.

109. *(b)* Use 'is' in place of 'are' as 'news' is always treated as singular.

110. *(c)* Remove 'with' as 'cope' takes a preposition only when an object comes after it.

111. *(b)* Use 'room' in place of 'place'.

112. *(c)* Use 'other' before 'poet' because here comparative degree is used.

113. *(c)* Use 'going' in place of 'to go' as 'prefer' always agree with gerund.

114. *(c)* Use 'of' in place of 'from' as 'victim' takes the preposition 'of'.

115. *(b)* Use 'lost' in place of 'losed'.

116. *(c)* Use 'had' in place of 'has' as sentence is in Indirect narration.

117. *(b)* Use of 'even' is not required. Remove 'even'.

118. *(b)* 'Because' is superfluous here. Starting of the sentence is with 'reason' so using 'because' is improper here.

119. *(d)* No error

120. *(b)* Use 'any other democracy'.

121. *(d)* No error

122. *(c)* Use 'since' in place of 'from'.

123. *(c)* Replace 'total' by 'complete'. 'Total' is used to show the sum of individual but 'complete' means 'thorough'.

124. *(a)* Replace 'Such' by 'The'. 'Such' is used with the noun that has already been stated earlier for a particular trait.

125. *(b)* Replace 'poet' by 'poets'. Phrase 'one of the' agrees with plural noun and singular verb after it.

126. *(a)* Use of 'please' is inappropriate here. There is no mention of a request as such in the sentence.

127. *(b)* Use 'has to be ready' in place of 'had to be ready'

128. *(b)* Use preposition 'to' after adhere.

129. *(b)* 'Ask' should be changed to 'asked'.

130. *(c)* 'Watch' should be used in place of 'see'.

131. *(a)* 'Some' always takes a plural number noun after it. Use 'men' in place of 'man'.

132. *(b)* 'Sympathise' is followed by 'with' and not by 'for'.

133. *(b)* 'Respectively' is inappropriate usage. Use 'respectfully' in place of it.

134. *(a)* 'Have been waiting' should be used in place of 'am waiting'.

135. *(a)* For a continued action of past to present, we use present perfect continuous tense. So, use 'has been' in place of 'is'.

136. *(c)* Use 'will' in place of 'is to' because the sentence reflects a futuristic approach.

137. *(a)* Part (a) has error. 'No sooner had he appeared' should be used.

138. *(b)* 'With' should be replaced by 'in'.

139. *(a)* 'Equipment' is not used in plural form.

140. *(b)* 'Has' should be replaced by 'have' as the subject of the verb is plural.

141. *(b)* 'Sceneries' is wrong usage as 'scenery' is always singularly used.

142. *(a)* 'Utmost' should be used in place of 'primary' as the importance of breathing is always most important rather than primary or secondary.

143. *(d)* No error

144. *(a)* Use 'we had driven for miles' in place of 'we were driving for miles'

145. *(c)* 'And' should be replaced by 'but' as the sentence has two contradictory statements.

146. *(a)* 'Or' should be replaced by 'and' because 'to write, to speak and to act' is the combination of actions that happens together.

147. *(b)* 'Since' should be replaced by 'for' because 'two days' defines the time period.

148. *(a)* 'Beside' should be replaced by 'besides' because beside means 'next to' and 'besides' means 'except or in addition to'.

149. *(c)* 'The best' should be replaced by 'the better' because there is a comparison between two objects (photographs).

150. *(b)* 'Are' should be replaced by 'is' because conjunction 'either or' connects two subjects, but the verb depends on the nearer subject. Here, 'foreman' is nearer subject which is singular.

151. *(c)* 'Whomever must accept it' should be replaced by 'those who accept it' as 'must' shows the compulsion which is wrong to use in this context.

152. *(b)* Use 'you for being more organised' Use of than indicates that comparative degree is required here.

153. *(a)* 'Hardly' should be replaced by 'hard' as hardly does not convey the right meaning.

154. *(b)* 'Who' should replace 'whom' in the sentence as 'who' represents nominative case and 'whom' is used for objective case.

155. *(a)* 'Looked after' is not to be used for 'thief ', here 'looked' should be followed by 'for'.

156. *(b)* 'Will' should be replaced by 'would' as the sentence is in past tense.

157. *(a)* 'All' will come before 'their'.

158. *(a)* 'The' should be removed before 'temper' as 'temper' is abstract noun and abstract noun is not used with articles.

159. *(c)* 'To that of his house' (Case of Attribute) should be used in the sentence.

160. *(a)* 'Of' should be removed as 'despite' is not followed by a preposition.

161. *(c)* 'Will' should be replaced with 'would' as the reporting verb is in past tense.

162. *(c)* 'Shall' should be removed from this part of sentence because when the condition is in present, then result should be in future.

163. *(b)* 'Supports' should be used in place of 'support' as the subjective case (Neither) is singular.

164. *(a)* Article 'a' should be used before 'school teacher' as a countable noun is always preceded by an article.

165. *(b)* 'To smoke' should be replaced by 'smoking' because here, use of gerund is more appropriate than infinitive.

166. *(b)* 'It to' should be removed from the sentence.

167. *(a)* 'Agreeing' is wrong use of 'agree', so it should be replaced with 'agree'. Also, 'am' is inappropriate. (I entirely agree with you, but I regret I can't help you.)

02

VOCABULARY

Vocabulary comprises one of the most scoring bunch of questions in CDS exam. A sound vocabulary is always anticipated to crack the questions based on synonyms and antonyms. The following chapter gives you an edge in mastering the words that have importance in the examination.

Vocabulary is a broad concept in itself. One can enhance one's language skills by acquiring a good hold over **vocabulary.** To score high in the CDS exam (English Paper), a candidate should improve his/her vocabulary skills. It helps one in solving questions of Comprehension, Cloze Test, Antonyms and Synonyms etc.

Here, we illustrate some steps to enrich vocabulary.

Step I **Identify the Word** Whenever we come across a new word in a sentence while reading a textbook, newspaper or a magazine, we should look up its meaning. This is the best way to enhance one's vocabulary. Suppose you come across a word 'Antique' and you don't know its meaning, you consult a dictionary and find its meaning which is 'old and often valuable'.

Step II **Identify the Antonyms of that Word** Along with the meaning of a word, its antonyms (words with opposite meaning) should also be taken into consideration. A good knowledge of words and their antonyms is very beneficial from the examination point of view. e.g. Opposite of Antique is modern.

Step III **Identify the Synonyms of that Word** The knowledge of words that are similar or closer in meaning to a word is very useful. It makes a student efficient enough to have a strong sense of the language. Example: 'Antique' can be replaced by Traditional or Ancient.

Step IV **Form a Proper Sentence** This is one of the most important parts in vocabulary building. It serves to stimulate memory by recalling the words as and when needed, apart from making the proper sense and the use of words clear. e.g. To be more familiar with a word, we should use it in sentence form. For 'Antique' a proper sentence is 'People love to purchase antique items.'

A

Aback Taken by surprise
Synonyms Surprised, thrown off guard
Antonyms Relax, Contended
- *Everyone was taken aback by Sachin's decision to quit.*

Abandon To leave something and never return to it
Synonyms Desert, Leave
Antonyms Continue, Carry on
- *Railways has abandoned their outer signal.*

Abase To humiliate
Synonyms Degrade, Disregard, Dishonour
Antonyms Regard, Honour, Respect
- *Abasing someone is immoral.*

Abashed Make someone feel embarrassed or ashamed
Synonyms Embarrass, Humiliate
Antonyms Unabashed, Undaunted
- *He is abashed of his own mistakes.*

Abate To make or become less strong
Synonyms Weaken, Lessen
Antonyms Strengthen, Intensify
- *We waited for the wind to abate.*

Abbreviate To shorten
Synonyms To abridge, To curtail
Antonyms Lengthen, Enhance
- *The voluminous book was abbreviated for the convenience of the students.*

Abdicate To give up power
Synonyms Relinquish, Renounce
Antonyms Accept
- *The old king abdicated the throne.*

Aberrant straying from the right or normal way
Synonyms Deviant, atypical
Antonyms Normal, Usual
- *Aberrant behaviour can be a sign of rabies in an animal.*

Abet To encourage someone to do wrong
Synonyms Assist, Incite, Encourage
Antonyms Demotivate, Prevent
- *She abetted the thief in the robbery.*

Abeyance A state of not happening or being used at present
Synonyms Abandon, Suspension, Discontinuation
Antonyms Continuation, Resumption
- *The old generator has been in abeyance for six months.*

Abhor To feel hatred or dislike
Synonyms Detest, Loathe
Antonyms Like, Admire
- *The world would be like heaven if all the people abhor none.*

Abide To accept something in accordance with
Synonyms Obey, follow
Antonyms Flout, Reject
- *Citizens have to abide by the rules.*

Abnegate To give-up; renunciation
Synonyms Discard, Reject
Antonym Accept
- *Abnegating superstitions is advantageous.*

Abound To exist in large numbers or amounts
Synonyms Plenty, Suffice
Antonyms Scarce, Scanty
- *Kiwis abound in New Zealand.*

Abrasive Showing little concern for feeling of others
Synonyms Rude, Annoying, Unfriendly
Antonyms Pleasant, Friendly
- *Abrasive behaviour of employees may prove harmful to a firm.*

Abrogate To end a law, agreement or custom formally
Synonyms Abandon, Abort
Antonyms Institute, Introduce
- *Our country should abrogate outdated laws.*

Abstain Withhold or Refrain
Synonyms Avoid, Cease
Antonyms Do, Continue
- *It is worth while to abstain from intoxicants.*

Abstruse Difficult to understand, obscure.
Synonyms Esoteric, Perplexing
Antonyms Clear, Obvious
- *You are not the only one who finds Enstein's theory abstruse.*

Absurd Ridiculous, Unreasonable
Synonyms Foolish, Ridiculous
Antonyms Reasonable, Genuine
- *Political parties indulge in absurd arguments before the election.*

Abut To border upon
Synonyms Adjoin, Lie next to, Adjacent
Antonyms Far, Opposite
- *Our land abuts a nature preserve.*

Abysmal extremely poor or bad
Synonyms Awful, Terrible
Antonyms Good, Pleasant
- *The quality of her work is abysmal.*

Accede To agree
Synonyms Consent, Acceptance
Antonyms Disagree, Refusal, Denial
- *The business contract between the two parties was acceded successfully.*

Accentuate To emphasise or to make noticeable
Synonyms Highlight, Hype
Antonyms Shadowed, Downtrodden
- *People often shout to accentuate their opinion.*

Accessible Easy to obtain, approachable
Synonyms Achievable, Acquiresome
Antonyms Remote, Distant
- *Everything is accessible with the Internet.*

Accessory A thing which can be added to something else in order to make it more useful, versatile or attractive
Synonyms Adornment, Retrofit
Antonym Subsidiary
- *Cellular phones are incomplete without the accessories.*

Acclaim Public approval and praise
Synonyms Praise, Applaud, Cheer
Antonyms Criticise
- *Sardar Patel was an acclaimed leader.*

Accolade An award or an expression of praise.
Synonyms Appreciation, Honour, Award
Antonym Criticism
- *Getting success is a great accolade.*

Accord Be harmonious or consistent
Synonyms Concord, Agreement
Antonyms Disagree, Contrast
- *The board of directors could not reach an accord in the annual meeting.*

Accost Approach and address angrily or aggressively
Synonyms Annoy, Confront
Antonyms Aid, Help
- *On the mistake of the son, father accosted him.*

Accrue To increase in number or amount
Synonyms To collect, To accumulate
Antonyms Disperse, Dwindle
- *Crossing for a single run accrued the score of the team.*

Adept Skilful
Synonyms Expert, Efficient
Antonyms Unskilled, Inapt
- *It seems that he is adept in computers.*

Adjourn Temporary breaking-off
Synonyms Suspend, Interrupt
Antonyms Carry out, Advance
- *He has adjourned his journey.*
- *The court is adjourned for the day.*

Adjunct Something joined or added to another thing but is not an essential part of it
Synonyms Supplement, Addition
Antonyms Subtraction, Lessening
- *The witness of the case has adjuncted a new twist in it.*

Adjure To urge solemnly
Synonym Request
Antonym Answer
- *On the continuous adjuring of students, a picnic was arranged.*

Admonish To warn
Synonyms Scold, Reprove
Antonyms Allow, Compliment
- *The teacher admonished the student for his insolent behaviour.*

Adorn Make more beautiful or attractive
Synonyms To embellish, To decorate
Antonyms Malign, Deface
- *The temple is adorned with flowers.*

Adroit Very skilful
Synonyms Expert, Proficient
Antonyms Unskilled, Incompetent
- *The showroom needs an adroit mechanic.*

Afflict Affect adversely
Synonyms Suffer, Bother
Antonyms Comfort, Aid
- *The flood has greatly afflicted the crops in this village.*

Affluence Having a lot of money
Synonyms Wealth, Prosperity
Antonyms Scarcity, Poverty
- *Generally, affluent fathers have spoilt kids.*

Affront An action or remark that causes outrage or offence
Synonyms Insult, Offence
Antonyms Honour, Compliment
- *Poor dressing sense often causes affront.*

Aggrandize Increase power, status or wealth of
Synonyms Exalt, Boost
Antonyms Abase, Degrade
- *Its a movie that aggrandizes the bad guys.*

Aggravate To make a problem worse
Synonyms Worsen, Compound
Antonyms Soothe, Calm
- *The symptoms were aggravated by drinking alcohol.*

Agog very eager or curious to hear or see something
Synonyms Eager, Impatient
Antonyms Reluctant, Uninterested
- *He was all agog on hearing the news of his promotion.*

Altercation A noisy argument or disagreement, especially in public
Synonyms Quarrel, Bickering
Antonyms Agreement, Harmony
- *A general political talk should not lead to an altercation.*

Altruism Disinterested and selfless concern for the well-being of others.
Synonyms Benevolence, Humanitarianism
Antonyms Greediness, Meanness
- *Mother Teresa is known for her altruism.*

Amalgamate To combine to form a larger group
Synonyms To merge, Combine
Antonyms Separate, Disjoin
- *Hutchison and Essar group amalgamated to form Hutchison-Essar.*

Ambiguous Open to more than one interpretation, not having one obvious meaning
Synonyms Unclear, Confusing
Antonyms Clear, Obvious
- *Ambiguous answers must be removed.*

Ameliorate Making a situation better, less painful
Synonyms Mitigate, Improve
Antonyms Worsen, Aggravate
- *Government grant is ameliorating the situation in the territory.*

Amenable Open and responsive to suggestions
Synonyms Compliant, Manageable, Persuadable
Antonyms Stubborn, Rigid, Non-compliant
- *A better way to resolve the problems is being amenable.*

Amicable Friendly behaviour of a person
Synonyms Friendly, Good-natured
Antonyms Unfriendly, Hostile
- *Noble people are always amicable.*

Annul To make something legally void
Synonyms Cancel, Abolish, Invalidate
Antonyms Accept, Validate
- *The contract was annulled by the second party.*

Anomaly Deviation from the standard
Synonyms Oddity, Peculiarity
Antonyms Conformity, Normality
- *We do not publish cheap quality books as it is an anomaly to our policy.*

Antagonism A strong feeling of dislike or hatred
Synonyms Hate, Prejudice
Antonyms Love, Affection
- *Man cannot be an antagonist as he has to live in the society.*

Antipathy A strong feeling of dislike
Synonyms Aversion, Dislike
Antonyms Affinity, Cordiality
- *Pakistan's antipathy is open to India.*

Antithesis The direct or exact opposite
Synonyms Counterpart, Converse
Antonym Same
- *It seems that he has decided to be in antithesis of my opinion.*

Aphorism A short, wise and true statement
Synonyms Adage, Maxim
Antonyms Nonsense, Absurdity
- *"Honesty is the best policy" is a very practical aphorism.*

Aplomb Confidence and style
Synonyms Assurance, Poise
Antonyms Gaucheness, Discomposure
- *Continuous failure had a great effect on his aplomb.*

Apocryphal Well-known but probably not true
Synonyms Fictitious, Made-up
Antonyms Authentic, Real
- *Existence of God is an apocryphal fact to the people world wide.*

Apogee Most successful part of something
Synonyms The top, Apex
Antonyms Bottom, Base
- *Romance is the apogee of a relationship.*

Appease to make someone pleased or less angry by giving or saying something they desire
Synonyms To pacify, Placate
Antonyms Annoy, Irritate
- *His appeasing behaviour is always appreciated.*

Append To add something to the end of a writing
Synonyms Add, Attach
Antonyms Disjoin, Detach
- *It is always advisible to append the hints to a mathematical problem.*

Apportion To divide something among people
Synonyms Distribute, Allocate
Antonyms Keep, Withhold
- *The property of the deceased man was apportioned between his two sons.*

B

Babble To talk or say something in a quick, confused, excited or silly way
Synonyms Chatter, Bumble
Antonyms Quiet, Sense
- *He seems to be babbling.*

Badger To try to make someone do something by asking them many times
Synonyms Pester, Bother, Torment
Antonyms Aid, Delight
- *The peon had to be badgered to get the form signed by the principal.*

Baleful Full of evil intentions, menacing
Synonyms Destructive, Malignant
Antonyms Good, Helping, Promising
- *His baleful behaviour was strange.*

Banal Trite; something boring ; ordinary and not original
Synonyms Common place, Trite, Boring, Dull
Antonyms Entertaining, Original
- *I hate the places that seem banal.*

Bane A cause of great distress or annoyance.
Synonyms Ruin, Destruction
Antonyms Blessing, Boon, Advantage
- *Traffic congestion and air pollution are major banes for Delhi and Delhiites.*

Bashful Tending to feel uncomfortable with other people and be embarrassed easily; shy
Synonyms Diffident, Modest, Meek, Coy, Nervous
Antonyms Open, Confident
- *She feels bashful in my company.*

Berate To criticize or scold severely
Synonyms Lash out, Tear into, Abuse
Antonyms Praise, Compliment
- *Father berated his son for his mistakes.*

Bereavement The situation you are in when a close friend or a family member has just died
Synonyms Death, Loss
Antonym Happiness
- *Bereavement caused by my father's death is immeasurable.*

Bestow To give or confer honour to someone
Synonyms Award, Give, Grant, Present
Antonyms Deprive, Refuse, Take
- *A lot of awards are bestowed upon him.*

Bigotry Intolerance towards those who hold different opinion from oneself
Synonyms Fanaticism, Prejudice
Antonyms Tolerance, Impartiality
- *A deeply ingrained bigotry prevented her from even considering the arguments.*

Bizarre Strange and difficult to explain
Synonyms Strange, Weird
Antonyms Explainable, Normal
- *The bizarre events taking place in the deserted house led the people to assume that it was haunted.*

Blabber Talk foolishly
Synonyms Chatter, Babble
Antonym Sense
- *Blabbering is what one can expect from fools.*

Bohemian A socially unconventional person, especially an artist or a writer
Synonym Non-conformist
Antonym Conformist
- *The Bohemian attitude is considered rebellious.*

Bolster To support or strengthen
Synonyms Strengthen, Reinforce
Antonyms Discourage, Undermine
- *More money is needed to bolster the industry.*

Boor A person who is rude and does not consider other people's feelings
Synonyms Lout, Rogue
Antonyms Civilised, Decent, Modest
- *Terrorists are nothing but boors.*

Bovine Relating to or affecting cattle, looking or acting like a cow
Synonyms Cow-like, Cattle-like
- *She stared at us with a stupid bovine expression.*

Brag To speak proudly of what you have done or what you own.
Synonyms Swagger, Boast
Antonym Be modest
- *He was bragging about his success.*

Brash Showing too much confidence and too little respect
Synonyms Arrogant, Brazen
Antonyms Diffident, Meek
- *A brash man has less friends.*

Bumble To speak or move in a confused way
Synonyms Lurch, Stumble
Antonyms Efficient, Expert
- *Bumbling persons create a doubt to security.*

Bungle To do something badly or unsuccessfully
Synonyms Mishandle, Mismanage
Antonym Succeed
- *He has bungled the whole work.*

Buoyant Happy and confident
Synonyms Happy, Joyous
Antonyms Unhappy, Sad
- *He was very buoyant about the visit to Agra.*

Burgeon To grow or develop quickly
Synonyms Expand, Swell
Antonyms Shrink, Contract
- *Terrorism is burgeoning across the border.*

Burly A large and strong person
Synonyms Tawny, Gigantic
Antonyms Lean, Thin
- *There must be a burly man for the role of a demon.*

Bustling If a place is bustling, it is full of busy activity
Synonyms Dashing, Scurrying
Antonyms Quiet, Inactive, Dormant
- *The house, usually bustling with activity, was strangely silent.*

C

Cabal A group of people who secretly work together
Synonyms Clique, Faction
Antonym Individual
- *He was assassinated by a cabal of aides.*

Cache A hidden store of provision, weapons, treasure; to hide weapons or other things
Synonyms Hoard, Store
Antonyms Discard, Remove
- *People must create a cache of medicines for emergency.*

Cajole To persuade someone to do something by coaxing or flattery
Synonyms Persuade, Coax
Antonyms Dissuade, Discourage
- *He knows how to cajole people into doing what he wants.*

Calamity An event that brings terrible loss, lasting distress or severe affliction
Synonyms Tragedy, Catastrophe
Antonyms Blessing, Godsend
- *The great calamity was brought to rest.*

Callous Showing or having an insensitive and cruel disregard for others.
Synonyms Insensitive; Unsympathetic
Antonyms Sensitive, Sympathetic
- *People often think that doctors are callous.*

Calumny A false accusation
Synonyms Defamation, Slander
Antonyms Eulogy, Praise
- *He is a victim of calumny.*

Camouflage To disguise
Synonyms Hide, Conceal
Antonyms Reveal, Show
- *The camouflaged players arrived at the airport.*

Canard A false, report or story
Synonyms Tale, Story
Antonym Truth
- *The newspaper was sued for publishing a canard about a celebrity.*

Canny Very clever and able to make intelligent decisions.
Synonyms Clever, Shrewd
Antonyms Uncanny, Daft
- *Rohan is a canny card player.*

Cantankerous Bad tempered, argumentative and uncooperative
Synonyms Bad tempered, Uncooperative
Antonyms Affable, Good-natured
- *Children are by nature cantankerous.*

Capacious Having a lot of space
Synonyms Spacious, Open
Antonyms Small, Cramped
- *A bunglow is a capacious house to live-in.*

Cardinal Of the greatest importance, fundamental.
Synonyms Significant, Fundamental
Antonym Unimportant
- *Cardinal facts of the case are hidden.*

Careen To go forward quickly while moving from side to side
Synonyms Lurch, Rock
Antonyms Crawl, Creep
- *I saw the duck careening through the ponds.*

Castigate To criticise severely
Synonyms Rebuke, Chide, Scold
Antonyms Admire, Praise
- *Odd behaviour of a person is a matter of castigation.*

Catalyst Stimulus; a person who causes change by his presence
Synonyms Impetus, Incentive
Antonyms Inhibitor, Preventer
- *Birbal was a catalyst in the court of Akbar.*

Catapult To suddenly put someone into an important position; to propel
Synonyms Excel, Marshal
Antonyms To recede, Decline
- *Someone cannot be catapulted to be the chief straight way.*

Charismatic Possessing spiritual grace; inspiring
Synonyms Charming, Fascinating
Antonyms Offensive, Frightening
- *Modi is a charismatic leader.*

Chaste Morally pure or decent
Synonyms Decent, Pure
Antonyms Impure, Indecent
- *The chaste conduct of the austere person commands respect.*

Cherubic Good natured
Synonyms Innocent, Angelic
Antonyms Demonic, Devilish
- *His behaviour reveals his approach to be cherubic.*

Chide To express mild disapproval of someone, to scold someone gently
Synonyms Rebuke, Scold
Antonyms Admire, Praise
- *Mother chided the son for his rude behaviour with the guests.*

Chronic Happening or existing frequently or most of the time
Synonyms Persistent, Long standing
Antonyms Temporary, Mild
- *She suffers from chronic pain in her knees.*

Churlish Rude, unfriendly and unpleasant
Synonyms Arrogant, ill-mannered
Antonyms Gracious, Polite
- *Churlish behaviour is his trademark.*

Clout Power and influence
Synonyms Sway, Power
Antonym Powerless
- *Hitler was a man of great clout.*

Coalesce To grow together or unite into one; to fuse
Synonyms Fuse, Join
Antonyms Split, Breakup
- *The ice-masses coalesced into a glacier over time.*

Cogent Very clear and easy for the mind to accept and believe
Synonyms Convincing, Compelling
Antonyms Vague, Unconvincing
- *His ideas were cogent and sound.*

Conciliate To end a disagreement or someone's anger by acting in a friendly way or to slightly change your opinion
Synonyms Appease, Placate
Antonym Provoke
- *The nagging child was conciliated by the toys.*

Concomitant Naturally accompanying or associated with something
Synonyms Linked, Associated
Antonyms Disassociated, Unlinked
- *Loss of memory is a concomitant of old age.*

Concussion Temporary unconsciousness or confusion caused by a blow on the head
Synonym Violent shaking
Antonym Consciousness
- *The accident caused the man a severe concussion.*

Condone To accept behaviour that is morally wrong
Synonyms Accept, Allow
Antonyms Condemn, Punish
- *Parents always condone the fault of children.*

Contentious Causing or likely to cause disagreement
Synonyms Controversial, Argumentative
Antonym Agreeable
- *The contentious issue may obstruct the development.*

Contort To twist or bend out of normal shape
Synonyms Distort, Twist
Antonyms Straighten, Smooth
- *He contorted the instrument by rough handling.*

Contrive To invent or make something in a clever or unusual way
Synonyms Create, Manufacture
Antonyms Destroy, Ruin
- *The Defence Ministry contrived a plan to tackle cross border terrorism.*

Conundrum A confusing and difficult question or problem
Synonyms Dilemma, Quandary
Antonym Easy-way
- *Competitive exams make you face conundrums.*

Convene To call together for a meeting or activity
Synonyms Summon, Call
Antonyms Disperse, Leave
- *We convened at the hotel for a seminar.*

Convivial (Of an event or atmosphere) Friendly, lively and enjoyable
Synonyms Jovial, Pleasant
Antonyms Sad, Unhappy
- *The President of the club arranged a convivial cocktail party.*

Copious Ample, producing much
Synonyms Plentiful, Abudant
Antonyms Scarce, Meager
- *The storm produced a copious amount of rain.*

D

Dabble Take part in an activity in a casual way.
Synonyms Tinker, Dally, Trifle
Antonym Take seriously
- *Rita dabbled in many things before she got married.*

Dainty Small and graceful
Synonyms Elegant, Petite
Antonyms Crude, Ugly
- *The house looks dainty and beautiful.*

Dank Unpleasantly moist and cold
Synonyms Wet, Damp
Antonyms Arid, Dry
- *He shivered as he entered the dank room.*

Deadlock A situation involving opposing parties, in which no progress can be made; stalemate
Synonyms Gridlock, Dilemma
Antonyms Solution, Agreement
- *The mediator will help the opposing parties end the deadlock so the contract can be signed.*

Debacle A complete failure; a crushing defeat
Synonyms Fiasco, Failure
Antonyms Success, Accomplishment
- *He faced a debacle in yesterday's game.*

Debase To adulterate, to make poor in quality or of less value
Synonyms Degrade, Devalue
Antonyms Upgrade, Enhance
- *Debased commodities are sold in the market.*

Debauch To destroy or damage something so that it is no longer considered good or moral.
Synonyms Abase, Corrupt
Antonyms Ennoble, Uplift
- *Western culture has debauched the moral fabric of our society.*

Debilitate To make someone very weak and infirm
Synonyms Cripple, Disable
Antonyms Strengthen, Enable
- *The virus debilitates the body's immune system.*

Decadence Having low moral standards and behaviour
Synonyms Corruption, Debauchery
Antonyms Ascent, Decency
- *Decadent people are not valued.*

Decimate To kill a large number of (something) or to reduce very heavily
Synonyms Annihilate, Exterminate
Antonyms Bear, Build, Create
- *Population of endangered animals have been decimated.*

Decrepit Worn out or ruined because of age or neglect
Synonyms Dilapidated, Battered
Antonyms Firm, Healthy
- *The building was a decrepit sample of bricks now.*

Defalcate To steal or misuse funds entrusted to one
Synonyms Embezzle, Loot, Filch
Antonyms Appropriate, Receive
- *Public funds are defalcated by leaders.*

Demure (of a woman or her behaviour) Modest; shy; reserved
Synonyms Meek, Bashful
Antonyms Brazen, Shameless
- *Her demure nature gets everybody's attention.*

Deplore To feel or express strong condemnation of something
Synonyms Abhor, Denounce
Antonyms Praise, Commend
- *Public deplored the casual steps taken by the administration for the safety of women.*

Depravity The state of being morally bad, or an action that is morally bad.
Synonyms Perversion, Criminality
Antonyms Goodness, Uprightness
- *People were shocked by the depravity of her actions.*

Deprecate To criticise or express disapproval of something
Synonyms Belittle, Detract
Antonyms Approve, Commend
- *Everybody deprecated the death of the charitable man.*

Deride To laugh at someone or something in a way that shows you think they are stupid or are of no value.
Synonyms Ridicule, Mock, Disdain
Antonyms Admire, Approve
- *He was derided at for his strange ways.*

Desperado A desperate or reckless person, especially a criminal
Synonyms Bandit, Villain
Antonym Civilised
- *Law must punish the desperado.*

Despot A ruler who has total power and often uses it in cruel and unfair ways.
Synonyms Tyrant, Oppressor
Antonym Democrat
- *Hitler was one of the biggest despots.*

Desultory Lacking a plan, purpose or enthusiasm
Synonyms Aimless, Chaotic
Antonyms Organised, Systematic
- *He wandered around, cleaning up in a desultory way.*

Detract Dimish the worth or value of something
Synonyms Belittle, Decrease
Antonyms Commend, Compliment
- *Numerous errors in the book detracted the reader's attention.*

Devious Showing a skilful use of underhand tactics to achieve goals
Synonyms Unfair, Fraudulent
Antonyms Honest, Fair
- *The minister was a devious politician.*

Devolve 1. Transfer of power to a lower level
2. Pass into a worse state; degenerate
Synonyms Delegate, Pass on
Antonyms Centralize, Improve
- *In a democratic system, power is devolved to the local level.*

Diabolic Extremely evil
Synonyms Cruel, Atrocious
Antonyms Kind, Moral
- *The police quickly mobilised to track down the diabolical serial killer.*

Diffident Modest or shy because of a lack of self-confidence
Synonyms Bashful, Meek
Antonyms Bold, Confident
- *He is too diffident to work in a company.*

Disapprobation Strong disapproval, typically on moral grounds
Synonyms Deprecation, Disapproval
Antonyms Approval, Approbation
- *Sherry was used to constant disapprobation of critics.*

Disconcert to make someone upset; unsettle
Synonyms Perplex, Baffle, Bewilder
Antonyms Assist, Calm
- *The whole experience had disconcerted him.*

E

Earmark Designate funds or resources for a particular purpose, procure
Synonyms Attribute, Designate
Antonym Disallocate
- *The government earmarked a huge package for agriculture*

Ebullient Very happy and enthusiastic; exuberant
Synonyms Cheerful, Exuberant
Antonyms Unhappy, Depressed
- *The man seems to be ebullient on his success.*

Edifice A large impressive building; a system that has been established for a long time
Synonyms Monument, Building
- *The glass edifice is an architectural wonder.*

Effeminate Womanish; Feminine, Unmanly
Synonyms Effete, Unmanly
Antonyms Masculine, Manly
- *He had a high and somewhat effeminate voice.*

Effete Weak and Powerless
Synonyms Unmanly, Effeminate
Antonyms Manly, Powerful
- *His effete body looks pale.*

Effusive Expressing gratitude, approval or pleasure in a way that shows very strong feeling
Synonyms Gushing, Unrestrained
Antonyms Restrained, Reserved
- *The coach was effusive in praising Tendulkar.*

Emaciated Very thin and weak, usually because of illness or extreme hunger
Synonyms Thin, Skeletal
Antonyms Chubby, Fat
- *Chronic diseases make one emaciated.*

Embodiment someone or something that represents a quality or an idea exactly
Synonyms Personification, Incarnation
Antonym Exclusion
- *Mother Teresa was often regarded as the embodiment of selfless devotion to others.*

Enervate to make someone feel weak and without energy
Synonyms Debilitate, Devitalise
Antonyms Strengthen, Energise
- *We were enervated by the lengthy discussion.*

Enjoin To instruct or urge someone to do something or behave in a particular way
Synonyms Urge, Command, Insist
Antonyms Obey, follow
- *The boss enjoined the workers to do the best.*

Ensconce To make yourself very comfortable or safe in a place or position
Synonyms Settle, Install
Antonyms Unsettle, Exhibit
- *He ensconced in his new abode there.*

Entree Admittance; the right to enter something
Synonyms Entry, Ingress
Antonyms Exit, Refusal
- *Entree to the country club is through sponsorship.*

Entropy Lack of order or predictability, gradual decline into disorder
Synonyms Break up, Collapse
Antonyms Improvement, Order
- *The mishandling of the situation led to entropy.*

Ephemeral Existing only for a short time
Synonyms Fleeting, Transient
Antonyms Enduring, Lasting
- *Fame in the show business is ephemeral.*

Epoch A long period of time, especially one in which there are new advances and great changes
Synonyms Era, Span, Age
- *The development of the steam engine marked an important epoch in the history of industry.*

Equitable Treating everyone fairly and in the same way.
Synonyms Unbiased, Reasonable
Antonyms Partial, Biased
- *He is fighting for a more equitable distribution of funds.*

Excruciating Intensely painful
Synonyms Acute, Agonising
Antonyms Painless, Calm, Easy
- *The Uttrakhand disaster was an excruciating experience for the victims.*

Expatiate Speak or write in detail
Synonyms Expound, Lecture
Antonyms Compress, Abridge
- *It is useless to expatiate upon the beauty of nature to one who is blind.*

F

Facade The principal front of a building that faces on to a street or open space
Synonyms Front, Elevation, Frontage
Antonyms Rear, Back
- *The gallery's 18th century facade attracted the visitors.*

Facetious Treating serious issues deliberately with inappropriate humour
Synonyms Flippant, Frivolous
Antonyms Serious, Formal
- *The facetious boy was chided.*

Facile 1. Ignoring the true complexities of an issue; superficial 2. A success which is easily achieved
Synonyms Superficial, Hasty
Antonyms Thorough, Profound
- *This problem needs more than just a facile solution.*

Factitious Artificial, not natural
Synonyms Fake, Bogus
Antonyms Natural, Real
- *His explanations were all factitious.*

Fallacious Based on a mistaken belief; misleading
Synonyms Erroneous, False
Antonyms Genuine, True
- *Someone is spreading fallacious information.*

Fastidious Very attentive to and concerned about accuracy and details; very concerned about matters of cleanliness
Synonyms Painstaking, Meticulous
Antonyms Careless, Sloppy
- *His culinary skills are fastidious.*

Fatuous Stupid, not correct or not carefully thought about
Synonyms Silly, Foolish
Antonyms Sensible, Intelligent
- *All his reasons appeared fatuous to me.*

Feckless Lacking initiative or strength of character; irresponsible
Synonyms Useless, Worthless
Antonyms Responsible, Competent
- *Country does not need feckless people.*

Fecund Very fertile
Synonyms Fertile, Fruitful
Antonyms Unproductive, Infertile
- *The soil of the plains is fecund.*

Feisty Having or showing exuberance, strong determination and lack of fear
Synonyms Courageous, Gutsy
Antonyms Cowardly, Dull
- *Mountaineering needs feisty people.*

Felicity 1. Intense happiness
2. The ability to find appropriate expressions for one's thoughts
Synonyms Bliss, Delight, Eloquence
Antonyms Sorrow, Unhappiness
- *Felicity is a bliss to be enjoyed.*

Fester (of a problem or negative feeling) become worse or more intense, especially through long term neglect or indifference
Synonyms Smoulder, Aggravate
Antonyms Flourish, Grow
- *It is better to express your anger than let it fester inside you.*

Fetid Smelling extremely bad, Foul
Synonyms Smelly, Putrid
Antonyms Aromatic, Perfumed
- *The carcass of the dog has made the surroundings fetid.*

Fidelity Honest or lasting support, Loyalty, Faithfulness
Synonyms Loyalty, Constancy
Antonyms Dishonesty, Infidelity
- *Friendship survives on fidelity.*

Fiend 1. Cruel, inhuman
2. An enthusiastic or devotee of a particular thing
Synonyms Barbarian, Ogre
Antonyms Angel, God
- *His hands were trembling as if he was some sort of fiend.*

Flack Strong criticism or opposition
Synonyms Criticism, Censure
Antonyms Appraisal, Praise
- *Dowry should be a matter of flack.*

Flagrant Too bad to be ignored
Synonyms Heinous, Shameless
Antonyms Magnificent, Wonderful
- *The killing of innocent villagers is an example of flagrant ways of Maoists.*

Foment To cause trouble to develop
Synonyms Incite, Instigate
Antonyms Deter, Discourage
- *He was accused of fomenting violence.*

Fortitude Courage in pain or adversity
Synonyms Courage, Bravery
Antonyms Cowardice, Fear
- *Fortitude makes you a winner.*

Frenzied Madly excited or uncontrolled
Synonyms Wild, Frantic
Antonyms Controlled, Calm
- *The office was a scene of frenzied activity this morning.*

G

Gainsay To deny or contradict a fact or statement
Synonyms Dispute, Oppose
Antonyms Accept, Confirm
- *There is no gainsaying the fact that they have built a great building.*

Gall Bold and impudent behaviour
Synonyms Impudence, Insolence
Antonyms Cordiality, Politeness
- *After borrowing my car, he had the gall to complain about its seats.*

Gallant 1. Brave; heroic
2. (Of a man) polite and kind towards woman
Synonyms Valiant, Unafraid
Antonyms Timid, Rude
- *Although she lost, she made a gallant effort.*

Garble To make words or messages unclear and difficult to understand
Synonyms Muddle, Jumble
Antonyms Obvious, Clear
- *The terrified child gave a garbled account of the incident to the police.*

Garish Too bright or colourful
Synonyms Gaudy; Bright
Antonyms Dull, Unflashy
- *The decoration looked garish.*

Gauche Awkward and uncomfortable with other people, specially because young and lacking in experience
Synonyms Awkward, Gawky
Antonyms Elegant, Sophisticated
- *She had grown from a gauche teenager to a self assured young woman.*

Genial Friendly and cheerful
Synonyms Affable, Cordial
Antonyms Hostile, Unfriendly
- *His genial outpour surprised me.*

Ghastly Unpleasant and shocking
Synonyms Terrible, Horrible
Antonyms Pleasant, Charming
- *The scene of the crime looks ghastly.*

Giddy Having a sensation of whirling and a tendency to fall or stagger.
Synonyms Dizzy, Light-headed
Antonyms Steady, Sensible
- *He was walking giddily.*

Gingerly In a careful or cautious manner
Synonyms Warily, Cautiously
Antonyms Carelessly, Rashly
- *He did the work gingerly.*

Gratify To please someone or to satisfy a wish or need
Synonyms Please , Gladden
Antonyms Dissatisfy, Displease
- *I am gratified by his words.*

Grimace To make an expression of pain, strong dislike, etc in which the face twists in an ugly way
Synonyms Scowl, Frown
Antonyms Smile
- *After falling down, he started to stand up grimacing with pain.*

Grisly Causing horror or disgust
Synonyms Gruesome, Ghastly
Antonyms Pleasant, Attractive
- *The grisly figure made the child cry.*

Grubby Covered with dirt
Synonyms Filthy, Mucky, Grimy
Antonyms Clean, Tidy
- *One must clean off grubby hands before eating.*

Gruff (person's voice) Low and unfriendly
Synonyms Rough, Hoarse
Antonyms Soft, Mellow
- *He speaks in a gruff way, but is really kind.*